

The Zcash Arboretum

Foundations, deployed protocol, and frontier designs

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Introduction

The Zcash Arboretum is a non-normative guide to the mathematics, cryptography, and engineering of the Zcash protocol. It covers the foundations of Halo 2 and Orchard, deployed consensus and wallet protocols, and designs being explored beyond them. The protocol specification, applicable ZIPs, and consensus rules remain authoritative.

The order is layered. The *Math*, *Crypto*, and *Halo 2* Guides construct the foundations. The *Consensus*, *Ironwood*, and *Wallet* Guides explain the deployed system and its boundaries. The remaining parts examine the unbuilt FlyClient bridge, shielded assets, and threshold authorization.

Three reading paths cover most uses. For prerequisites, begin with the first three parts; readers new to proof systems may start with the worked example that closes the Halo 2 Guide. For the life of a shielded payment, read *Ironwood*, then *Wallet* and *Consensus*. For proposed changes, read the relevant frontier part only after its lower-layer dependencies. Each part restarts its own section numbering so that citations agree with the separately published volume.

Part I

Math Guide: Mathematical foundations for Halo 2 and Zcash Orchard

Abstract

This volume of *The Zcash Arboretum* develops the mathematics underlying the cryptography of Zcash from first principles, assuming only elementary algebra and calculus. It builds a single tower: the notation of sets, functions, and relations in which every later claim is written and spoken; the integers and their remainders modulo a prime, where an extended Euclidean algorithm turns division into multiplication by an inverse; the abstract structures—groups, rings, and fields—that name the laws this arithmetic obeys; polynomials over a field, whose scarcity of roots lets two enormous polynomials be compared at a single random point; the finite fields, their possible sizes, and their cyclic multiplicative structure; the number theory of squares, square roots, and the discrete logarithm; linear algebra and the inner products behind short algebraic receipts for long lists of secrets; the roots of unity that furnish evaluation domains and the fast Fourier transform; elliptic curves, whose point groups have no known subexponential discrete-log attack when chosen appropriately; and finally the probability, asymptotics, and model of computation required to state—and mean—a claim of 128-bit security.

The series consumes three artefacts of this tower: the Pasta curve cycle, constructed in §10; the polynomial machinery on which Halo 2 rests, developed in §5 and §9; and the security-statement language, fixed in §11. A first reading takes the sections in order; a reader after one artefact may enter at its section and follow its backward references down the tower.

1 Notation: sets, functions, and relations

Open the specification of a modern cryptographic protocol — a signature scheme, say, or a scheme for proving a statement without revealing why it is true — and try to read a single line of it aloud. Before any mathematics has happened, the page presents three alphabets at once: double-struck capitals such as $\mathbb{Z}$ and $\mathbb{F}_q$ naming entire systems of numbers, Greek letters such as λ and ρ naming parameters and challenges, and a thicket of operator symbols — $\oplus$, $\parallel$, $\xleftarrow{\$}$, $[k]P$ — each with a fixed meaning and a spoken name. A reader who cannot pronounce a line cannot discuss it with a colleague, and a reader who mistakes $\subseteq$ for $\in$, or swaps the order of two quantifiers, has misread the claim itself.

This section fixes the small stock of set-theoretic and logical language in which the whole series is written, and it closes by paying the opening debt in full: three reference tables listing every recurring symbol with its spoken form (§1.5). A reader comfortable with elementary mathematics may skim the prose and keep the tables to hand.

1.1 Sets

A *set* is a collection of distinct objects, its *elements*; one writes $x \in A$ for membership and $x \notin A$ for its negation. A set may be given by listing, $\{a, b, c\}$, or by *set-builder* notation $\{x \in A : P(x)\}$ (“the x in A such that $P(x)$ ”). One writes $A \subseteq B$ when every element of A lies in B , and $A = B$ exactly when $A \subseteq B$ and $B \subseteq A$ — so proving two sets equal means proving two inclusions. The usual operations are union $A \cup B$, intersection $A \cap B$, difference $A \setminus B$, and the empty set $\emptyset$. The *Cartesian product* is $A \times B = \{(a, b) : a \in A, b \in B\}$, and A^n is the set of n -tuples over A . For a finite set A , $|A|$ denotes its number of elements. The standard number systems are the naturals $\mathbb{N}$, the integers $\mathbb{Z}$, the rationals $\mathbb{Q}$, the reals $\mathbb{R}$, and the complex numbers $\mathbb{C}$, with $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$; we construct the underlying arithmetic of the prime field $\mathbb{F}_p$ in §2, certify it as a field—and first call it $\mathbb{F}_p$ —in §4, and treat general finite fields $\mathbb{F}_q$ in §6.

1.2 Functions

A function $f: A \rightarrow B$ assigns to each $a \in A$ a unique value $f(a) \in B$; A is the *domain* and B the *codomain*. It is *injective* (one-to-one) if $f(a) = f(a')$ implies $a = a'$; *surjective* (onto) if every $b \in B$ equals $f(a)$ for some $a \in A$; and *bijective* if both. The *composition* of $f: A \rightarrow B$ and $g: B \rightarrow C$ is $(g \circ f)(a) = g(f(a))$; composition is associative, $h \circ (g \circ f) = (h \circ g) \circ f$, both sides sending a to $h(g(f(a)))$. The *identity* on A is $\text{id}_A: A \rightarrow A$, $\text{id}_A(a) = a$.

Theorem 1.1 (Inverse functions). *A function $f: A \rightarrow B$ has a two-sided inverse $f^{-1}: B \rightarrow A$, meaning $f^{-1} \circ f = \text{id}_A$ and $f \circ f^{-1} = \text{id}_B$, if and only if f is bijective. When it exists, the inverse is unique.*

Proof. Suppose a two-sided inverse g exists. If $f(a) = f(a')$, applying g gives $a = g(f(a)) = g(f(a')) = a'$, so f is injective; and every $b \in B$ satisfies $b = f(g(b))$, so f is surjective. Conversely, suppose f is bijective. Each $b \in B$ has at least one preimage by surjectivity and at most one by injectivity, so we may define $g(b)$ to be the unique $a \in A$ with $f(a) = b$; then $g(f(a)) = a$ and $f(g(b)) = b$ hold by construction. Finally, if g and g' are both two-sided inverses, associativity of composition gives $g = g \circ \text{id}_B = g \circ (f \circ g') = (g \circ f) \circ g' = \text{id}_A \circ g' = g'$; the inverse is unique, and we write it f^{-1} . $\square$

For $S \subseteq A$ the *image* is $f(S) = \{f(a) : a \in S\}$, and for $T \subseteq B$ the *preimage* is $f^{-1}(T) = \{a \in A : f(a) \in T\}$, defined whether or not f is invertible. The overloading of f^{-1} is harmless: when the inverse function exists, the preimage of T under f is exactly the image of T under f^{-1} .

1.3 Relations and equivalences

A *binary relation* on A is a subset $R \subseteq A \times A$; one writes $a \sim b$ for $(a, b) \in R$. It is an *equivalence relation* if it is *reflexive* ($a \sim a$), *symmetric* ($a \sim b \Rightarrow b \sim a$), and *transitive* ($a \sim b$ and $b \sim c$ imply $a \sim c$). The *equivalence class* of a is $[a] = \{x \in A : x \sim a\}$, and the set of distinct classes is the *quotient* $A/\sim$.

Proposition 1.2 (Classes partition the set). *Let $\sim$ be an equivalence relation on a set A . The distinct equivalence classes partition A : they are nonempty, pairwise disjoint, and their union is A .*

Proof. Reflexivity gives $a \in [a]$ for every $a \in A$, so each class is nonempty and every element lies in some class; the union of the classes is therefore A . For disjointness, suppose the classes $[a]$ and $[b]$ share an element x , and let $y \in [a]$ be arbitrary. From $y \sim a$ and $a \sim x$ (the latter by symmetry from $x \sim a$), transitivity gives $y \sim x$; combining with $x \sim b$ gives $y \sim b$, so $y \in [b]$. Thus $[a] \subseteq [b]$, and exchanging the roles of a and b gives the reverse inclusion; two classes that meet coincide. $\square$

We use this construction twice in the present volume: to form the integers modulo n (§2) and the cosets of a subgroup (§3).

A second kind of relation orders rather than identifies. A *partial order* on A is a relation $\leq$ that is reflexive, transitive, and *antisymmetric*: $a \leq b$ and $b \leq a$ imply $a = b$. Four partial orders recur in the series: the usual $\leq$ on $\mathbb{Z}$; inclusion $\subseteq$ on the subsets of a fixed set, whose antisymmetry is the two-inclusions rule of §1.1; divisibility $|$ on $\mathbb{N}$ (§2.2); and the *prefix* relation on finite sequences, under which a sequence precedes every sequence that extends it. Two elements are *comparable* if $a \leq b$ or $b \leq a$, and a partial order under which every two elements are comparable is a *total* (or *linear*) order. The order $\leq$ on $\mathbb{Z}$ is total; the other three are not: $\{1\}$ and $\{2\}$ are incomparable under inclusion, as are 2 and 3 under divisibility and the one-term sequences (1) and (2) under prefix.

Every partial order has a *strict companion*, $a < b$ meaning $a \leq b$ and $a \neq b$; the two determine each other.

Proposition 1.3 (Strict and non-strict orders). *If $\leq$ is a partial order on A , its strict companion $<$ is irreflexive ($a < a$ holds for no a) and transitive. Conversely, if $<$ is irreflexive and transitive, then “ $a < b$ or $a = b$ ” defines a partial order on A whose strict companion is $<$.*

Proof. Irreflexivity of the companion is the clause $a \neq a$. For transitivity let $a < b < c$; then $a \leq c$ because $\leq$ is transitive, and $a = c$ would give $a \leq b \leq a$, hence $a = b$ by antisymmetry, contradicting $a < b$. Conversely, the relation “ $a < b$ or $a = b$ ” is reflexive by construction and transitive by cases: if either link is an equality the other link carries the conclusion, and otherwise transitivity of $<$ does. It is antisymmetric, since $a \neq b$ with both $a \leq b$ and $b \leq a$ would mean $a < b$ and $b < a$, whence $a < a$, against irreflexivity. Its strict companion is $<$ again: “($a < b$ or $a = b$) and $a \neq b$ ” reduces to $a < b$, which already excludes $a = b$ by irreflexivity. $\square$

A strict order $<$ is a *strict total order* when its non-strict companion is total, equivalently when for all a, b exactly one of $a < b$, $a = b$, $b < a$ holds (*trichotomy*): at least one holds by totality, and any two together would give $a < a$, by substitution or by transitivity. The usual $<$ on $\mathbb{Z}$ is the model. A function $f: A \rightarrow B$ between ordered sets is *monotone* (order-preserving) if $a \leq a'$ implies $f(a) \leq f(a')$, and *strictly increasing* if $a < a'$ implies $f(a) < f(a')$. The one further property of an order that the volume assumes, the well-ordering of $\leq$ on $\mathbb{N}$, opens §2.1.

1.4 Logic and proof conventions

We employ the symbols $\forall$ (“for all”), $\exists$ (“there exists”), $\exists!$ (“there exists a unique”), and “:” or “s.t.” for “such that”. Negation swaps the quantifiers: $\neg(\forall x P(x))$ is $\exists x \neg P(x)$, and $\neg(\exists x P(x))$ is $\forall x \neg P(x)$. The order of unlike quantifiers matters: $\forall x \exists y P(x, y)$ (the y may depend on x) is weaker than $\exists y \forall x P(x, y)$ (one y works for all x).

The following conventions are fixed for the entire series. We prove an implication $P \Rightarrow Q$ directly (assume P , derive Q), by contraposition (assume $\neg Q$, derive $\neg P$), or by contradiction; we prove a statement $\forall x P(x)$ by taking an arbitrary x , and $\exists x P(x)$ by exhibiting a witness. We prove statements about all natural numbers by induction.

1.5 Symbols and how to say them

The debt incurred at the opening of this section falls due here. The series draws on three alphabets: blackboard-bold letters for the standard number systems and structures, Greek letters, and a handful of relational and operator symbols. Tables 1–3 collect these symbols and the complete Greek alphabet, with common English pronunciations, spelled approximately for reading mathematics aloud. These vary between speakers and regions; the alternatives shown are not exhaustive. A capitalised syllable carries the stress, and a slash separates alternatives. These are English letter names, not a guide to pronouncing Greek words. Each symbol is defined where it is first used; the notes here are reminders, not definitions.

2 The integers and modular arithmetic

How does one divide in a finite world? The proof systems documented in this series do their arithmetic not with the unbounded integers of school algebra but with the finitely many remainders left upon division by a fixed, enormous prime. Every step a prover or verifier takes is an addition, a multiplication, or—the delicate case—a division of such remainders. Adding and multiplying remainders is easy to imagine: compute in $\mathbb{Z}$, then discard multiples of the modulus. But what

Symbol	Name	Said	Typical role in the series
$\mathbb{N}$	blackboard N	“natural numbers”	the naturals
$\mathbb{Z}$	blackboard Z	“the integers”	the integers (German <i>Zahlen</i>)
$\mathbb{Q}$	blackboard Q	“the rationals”	the rationals (<i>quotients</i>)
$\mathbb{R}$	blackboard R	“the reals”	the real numbers
$\mathbb{C}$	blackboard C	“the complexes”	the complex numbers
$\mathbb{F}_q$	blackboard F	“eff-cue”	the finite field of q elements
$\mathbb{G}$	blackboard G	“group G”	an abstract group
$\mathbb{E}$	blackboard E	“expectation”	expectation of a random variable (§11)
$\mathbb{P}$	blackboard P	“the Pallas curve”	the Pallas group (§10; protocol-spec notation)

Table 1: Blackboard-bold letters. “Blackboard bold” names the double-struck style itself; in speech one usually says only the letter or the system it denotes. Elliptic curves are written italic E or calligraphic $\mathcal{E}$, never blackboard bold. The squared $\mathbb{P}^2$ of §10 is unrelated: it denotes the projective plane.

can it mean to divide, when “one half” names no remainder? The answer developed in this section is that division is multiplication by a *modular inverse*; that a remainder possesses an inverse exactly when a certain greatest common divisor equals one; and that an algorithm already in Euclid’s *Elements*, suitably extended, computes the inverse efficiently.

The integers $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$ are the arithmetical bedrock on which every algebraic and cryptographic structure in this monograph ultimately rests. The finite number systems underlying elliptic curves, the scalar arithmetic of Halo 2, and the polynomial commitments of Orchard (short values that bind a prover to entire polynomials, which a verifier then tests by evaluation) all reduce, in the end, to computations with congruence classes of integers. This section develops the theory of the integers from a small set of assumptions and arrives at the integers modulo n —the set $\mathbb{Z}/n\mathbb{Z}$ of congruence classes—together with its units and the algorithms that make arithmetic with them effective. We keep the treatment constructive throughout: every existence statement comes, wherever possible, with an algorithm that produces the object in question, since cryptography requires not merely that inverses and representatives exist but that one can compute them efficiently.

2.1 Well-ordering and the integers

We take the basic algebraic properties of $\mathbb{Z}$ for granted: addition and multiplication are associative and commutative, and multiplication distributes over addition; the integers 0 and 1 are identities for addition and multiplication respectively; every integer a has a negative $-a$; there are no zero divisors (if $ab = 0$ then $a = 0$ or $b = 0$); and $\mathbb{Z}$ carries a total order $\leq$ compatible with the arithmetic. The single deep property we assume axiomatically is the following.

Theorem 2.1 (Well-ordering principle). *Every nonempty subset $S \subseteq \mathbb{N} = \{0, 1, 2, \dots\}$ of the natural numbers has a least element: there exists $m \in S$ such that $m \leq s$ for all $s \in S$.*

The well-ordering principle is logically equivalent to the principle of mathematical induction; we use both freely. It is also the engine of this section: nearly every existence result below is, at bottom, an application of well-ordering, in which one exhibits a nonempty set of natural numbers and extracts its minimum. Four instances map the road ahead. The remainder in the division algorithm is the least element of the set $\{a - bk : k \in \mathbb{Z}, a - bk \geq 0\}$; the Euclidean algorithm terminates because a strictly decreasing sequence of nonnegative integers cannot be infinite; the Bézout identity extracts the least positive element of a set of integer combinations; and existence in the fundamental theorem of arithmetic proceeds by minimal counterexample.

Lower	Capital	Name	Said	Lower	Capital	Name	Said
α	A	alpha	AL-fa	ν	N	nu	nyoo / noo
β	B	beta	BEE-ta / BAY-ta	ξ	Ξ	xi	ksigh / zigh
γ	Γ	gamma	GAM-a	$\omicron$	O	omicron	OM-i-kron / oh-MY-kron
δ	Δ	delta	DEL-ta	π	Π	pi	pie
ϵ, ε	E	epsilon	EP-si-lon / ep-SIGH-lun	ρ	P	rho	roe
ζ	Z	zeta	ZEE-ta / ZAY-ta	σ	Σ	sigma	SIG-ma
η	H	eta	EE-ta / AY-ta	τ	T	tau	tow / taw
θ	Θ	theta	THEE-ta / THAY-ta	υ	Y	upsilon	UP-si-lon / yoop-SIGH-lun
ι	I	iota	eye-OH-ta	ϕ, φ	Φ	phi	fie / fee
κ	K	kappa	KAP-a	χ	X	chi	kigh
λ	Λ	lambda	LAM-da	ψ	Ψ	psi	sigh / psigh / see / psee
μ	M	mu	mew / moo	ω	Ω	omega	OH-mig-a / oh-MAY-ga

Table 2: The 24 letters of the Greek alphabet in lower and upper case, with common English readings. For β , ζ , η , and θ , the EE forms are usual in British English; the AY forms are common in American English. The spellings “ksigh”, “zigh”, “kigh”, and “psigh” rhyme with “eye”; the k in “kigh” is hard, and the th in theta is as in “thin”. For τ , “tow” rhymes with “cow” and “taw” with “paw”. Greek-language readings differ: the Greek name for ζ , for example, is approximately “ksee”. Capitals reuse the same names: Δ (“delta”), Σ (“sigma”; the summation sign, and the name of the Σ -protocol family of later volumes), Π (“pi”; the product sign, and a conventional name in those volumes for a protocol or problem), Φ , Ψ , Ω , Θ , Λ .

2.2 Divisibility

Definition 2.2. Let $a, b \in \mathbb{Z}$. One says that a divides b , written $a \mid b$, if there exists an integer $c \in \mathbb{Z}$ with $b = ac$. In this case a is called a *divisor* or *factor* of b , and b is a *multiple* of a . If no such c exists one writes $a \nmid b$.

Several immediate consequences of the definition deserve to be recorded; we shall use them constantly and usually without explicit citation.

Proposition 2.3. Let $a, b, c, d \in \mathbb{Z}$. Then:

1. $a \mid a$ (reflexivity), $1 \mid a$, and $a \mid 0$.
2. If $a \mid b$ and $b \mid c$ then $a \mid c$ (transitivity).
3. If $a \mid b$ and $a \mid c$ then $a \mid (bx + cy)$ for all $x, y \in \mathbb{Z}$; in particular $a \mid (b \pm c)$.
4. If $a \mid b$ and $b \neq 0$ then $|a| \leq |b|$.
5. If $a \mid b$ and $b \mid a$ then $a = \pm b$.
6. $0 \mid b$ if and only if $b = 0$.

Proof. (1) is immediate: $a = a \cdot 1$, $a = 1 \cdot a$, $0 = a \cdot 0$. For (2), write $b = ae$ and $c = bf$; then $c = a(ef)$, so $a \mid c$. For (3), write $b = ae$ and $c = af$; then $bx + cy = a(ex + fy)$. For (4), write $b = ac$ with $b \neq 0$, so $c \neq 0$ and hence $|c| \geq 1$; then $|b| = |a||c| \geq |a|$. For (5), if either is 0 then by (6) so is the other and the claim holds; otherwise (4) gives $|a| \leq |b|$ and $|b| \leq |a|$, so $|a| = |b|$ and $a = \pm b$. For (6), $0 \mid b$ means $b = 0 \cdot c = 0$; conversely $0 \mid 0$. $\square$

The relation $\mid$ is thus reflexive and transitive; it is a partial order in the sense of §1.3 when restricted to the natural numbers (where (5) becomes $a = b$), but on all of $\mathbb{Z}$ it fails to be antisym-

Symbol	Said	Meaning
$\in, \notin$	"in", "not in"	set membership
$\subseteq, \subset$	"subset of"	(proper) inclusion
$\cup, \cap$	"union", "intersect"	set union and intersection
$A \setminus B$	"A minus B"	set difference
$\times$	"times" / "cross"	Cartesian product
$, \nmid$	"divides", "does not divide"	divisibility
$\leq, <$	"precedes"	partial order, strict companion
$\equiv$	"congruent to"	congruence, qualified (mod n)
$a \bmod n$	"a mod n"	remainder of division by n
$:=, =:$	"is defined as"	definitional equality
$\mapsto$	"maps to"	the action of a function
$g \circ f$	"g after f"	function composition
$\lfloor x \rfloor, \lceil x \rceil$	"floor", "ceiling"	round down, round up
$\log x, \log_2 x, \log_g h$	"log", "log base two", "log base g"	natural, base-2, and discrete (base g) logarithm
$\langle a, b \rangle$	"inner product"	inner product; $\langle g \rangle$ the subgroup generated by g
$\oplus$	"x-or"	bitwise exclusive or
$x \parallel y$	"x concat y"	concatenation of bit or byte strings
$x \leftarrow D, x \xleftarrow{\$} S$	"drawn from"	sampling ($\$$ marks uniform); also assignment
$[k]P$	"k times P"	scalar multiplication of a point (§10)
P^*	"P star"	canonical bit-encoding of a point or element
$\cdot$	"dot"	product; argument placeholder, as in $F(\cdot)$
$ A $	"size of A" / "order of A"	cardinality

Table 3: Relational and operator symbols. The symbol $\oplus$ is read "x-or"; every use in the series is the bitwise exclusive or. The encoding P^* is protocol-spec notation, defined in later volumes.

metric precisely because of signs. The *units* of $\mathbb{Z}$ —the integers dividing 1—are exactly ± 1 , and two integers that divide each other are called *associates*; in $\mathbb{Z}$ the associates of a are $\pm a$.

2.3 The division algorithm

Divisibility is the special case of exact division. The fundamental fact that makes the arithmetic of $\mathbb{Z}$ tractable is that one can always divide *with remainder*, and that the quotient and remainder are uniquely determined once a convention for the remainder's range is fixed.

Theorem 2.4 (Division algorithm). *Let $a \in \mathbb{Z}$ and $b \in \mathbb{Z}$ with $b > 0$. Then there exist unique integers q (the quotient) and r (the remainder) such that*

$$a = bq + r, \quad 0 \leq r < b.$$

Proof. Existence. Consider the set

$$S = \{a - bk : k \in \mathbb{Z}, a - bk \geq 0\} \subseteq \mathbb{N}.$$

First, $S \neq \emptyset$. If $a \geq 0$ then $a = a - b \cdot 0 \in S$. If $a < 0$ then, since $b \geq 1$, taking $k = a$ gives $a - ba = a(1 - b) \geq 0$ (a product of a nonpositive and a nonpositive factor), so $a - ba \in S$. Thus S is a nonempty subset of $\mathbb{N}$, and by the well-ordering principle (Theorem 2.1) it has a least element $r = a - bq$ for some $q \in \mathbb{Z}$. By construction $r \geq 0$. If $r \geq b$, then $r - b = a - b(q + 1) \geq 0$ would lie in S and be strictly smaller than r , contradicting minimality. Hence $0 \leq r < b$.

Uniqueness. Suppose $a = bq + r = bq' + r'$ with $0 \leq r, r' < b$. Then $b(q - q') = r' - r$, so $b \mid (r' - r)$. But $-b < r' - r < b$, and the only multiple of b strictly between $-b$ and b is 0. Hence $r' = r$, and then $b(q - q') = 0$ with $b \neq 0$ forces $q = q'$. $\square$

Remark 2.5. The hypothesis $b > 0$ is only for definiteness. For general $b \neq 0$ one obtains unique q, r with $a = bq + r$ and $0 \leq r < |b|$, by applying the theorem to $|b|$ and adjusting the sign of q . Note that the remainder convention $0 \leq r < |b|$ differs from the truncation behaviour of many programming languages, where the remainder of a negative dividend may be negative; for the number-theoretic results below the convention above is the appropriate one.

We denote the remainder r produced by the division algorithm by $a \bmod b$, and the quotient q by $\lfloor a/b \rfloor$ when $b > 0$ (the *floor* of the rational number a/b). With this notation, $b \mid a$ holds precisely when $a \bmod b = 0$.

2.4 The greatest common divisor and the Euclidean algorithm

Definition 2.6. A *common divisor* of integers a and b is an integer d with $d \mid a$ and $d \mid b$. The *greatest common divisor* of a and b , not both zero, is the largest such d ; it is denoted $\gcd(a, b)$. By convention $\gcd(0, 0) = 0$. Integers a and b are *coprime* (or *relatively prime*) if $\gcd(a, b) = 1$.

The greatest common divisor is well defined when $(a, b) \neq (0, 0)$: the set of common divisors is nonempty (it contains 1) and bounded above (every common divisor of, say, the nonzero a has absolute value at most $|a|$ by Proposition 2.3(4)), so it has a largest element, which is positive since 1 is a common divisor. We establish a much stronger characterisation shortly: the gcd is not merely the numerically largest common divisor but is divisible by every common divisor. The key computational engine is the following observation.

Lemma 2.7. Let $a, b \in \mathbb{Z}$. For any $q \in \mathbb{Z}$,

$$\gcd(a, b) = \gcd(b, a - qb).$$

In particular, if $a = bq + r$ then $\gcd(a, b) = \gcd(b, r)$, and $\gcd(a, 0) = |a|$.

Proof. It suffices to show the two pairs (a, b) and $(b, a - qb)$ have exactly the same common divisors; equal sets of common divisors have equal greatest elements (and the same coprimality status). If $d \mid a$ and $d \mid b$, then $d \mid (a - qb)$ by Proposition 2.3(3), so d is a common divisor of b and $a - qb$. Conversely, if $d \mid b$ and $d \mid (a - qb)$, then $d \mid ((a - qb) + qb) = a$, so d is a common divisor of a and b . The two sets of common divisors coincide. Finally $\gcd(a, 0) = |a|$ because the common divisors of a and 0 are exactly the divisors of a , the largest of which in absolute value is $|a|$ (and $|a| > 0$ when $a \neq 0$; the case $a = 0$ is the convention $\gcd(0, 0) = 0$). $\square$

Iterating the lemma with the division algorithm yields the oldest nontrivial algorithm in mathematics.

Theorem 2.8 (Euclidean algorithm). Let a, b be integers with $b \neq 0$. Define a sequence of remainders by $r_0 = |a|$, $r_1 = |b|$, and, as long as $r_i \neq 0$, let $r_{i+1} = r_{i-1} \bmod r_i$, so that

$$r_{i-1} = r_i q_i + r_{i+1}, \quad 0 \leq r_{i+1} < r_i.$$

The sequence $r_1 > r_2 > \dots$ of positive remainders is strictly decreasing, so some $r_{n+1} = 0$; the last nonzero remainder satisfies $r_n = \gcd(a, b)$.

Proof. The remainders $r_1, r_2, \dots$ are nonnegative integers and, while nonzero, strictly decreasing because $r_{i+1} = r_{i-1} \bmod r_i < r_i$. A strictly decreasing sequence of nonnegative integers cannot be infinite (again by well-ordering: otherwise the set of values would have no least element), so

the algorithm terminates with some $r_{n+1} = 0$ and $r_n > 0$. By Lemma 2.7, each step preserves the gcd:

$$\gcd(a, b) = \gcd(r_0, r_1) = \gcd(r_1, r_2) = \cdots = \gcd(r_n, r_{n+1}) = \gcd(r_n, 0) = r_n,$$

using $\gcd(|a|, |b|) = \gcd(a, b)$ (signs do not affect common divisors). $\square$

Example 2.9. Compute $\gcd(1071, 462)$:

$$1071 = 2 \cdot 462 + 147,$$

$$462 = 3 \cdot 147 + 21,$$

$$147 = 7 \cdot 21 + 0.$$

The last nonzero remainder is 21, so $\gcd(1071, 462) = 21$. Indeed $1071 = 21 \cdot 51$ and $462 = 21 \cdot 22$, with $\gcd(51, 22) = 1$.

Remark 2.10. The Euclidean algorithm is efficient. The number of division steps for inputs $a > b > 0$ is $O(\log b)$; the worst case occurs at consecutive Fibonacci numbers (Lamé’s theorem), where every quotient q_i equals 1 except the final one (which equals 2) and the remainders shrink as slowly as possible. With schoolbook arithmetic on k -bit inputs the whole computation costs $O(k^2)$ bit operations. This efficiency is what renders gcd-based cryptographic operations—above all modular inversion, below—practical at the sizes used in Halo 2 and Orchard.

2.5 The Bézout identity and the extended Euclidean algorithm

The Euclidean algorithm computes the gcd, but the back-substitution of its equations yields something more powerful: the gcd is an *integer linear combination* of the inputs. This is among the most useful facts in elementary number theory.

Theorem 2.11 (Bézout’s identity). *Let $a, b \in \mathbb{Z}$, not both zero, and let $d = \gcd(a, b)$. Then there exist integers $u, v \in \mathbb{Z}$ with*

$$au + bv = d.$$

Moreover d is the smallest positive integer expressible in the form $ax + by$ with $x, y \in \mathbb{Z}$, and the set of all such combinations is exactly the set of multiples of d :

$$\{ax + by : x, y \in \mathbb{Z}\} = d\mathbb{Z} = \{kd : k \in \mathbb{Z}\}.$$

Proof. Let $I = \{ax + by : x, y \in \mathbb{Z}\}$. This set contains a and b and is closed under addition and under multiplication by any integer; concretely, if $s = ax_1 + by_1$ and $t = ax_2 + by_2$ lie in I , then for any $m, n \in \mathbb{Z}$ one has

$$ms + nt = a(mx_1 + nx_2) + b(my_1 + ny_2) \in I.$$

Since a, b are not both zero, I contains a positive element (one of $\pm a, \pm b$), so the set $I \cap \mathbb{N}_{>0}$ of positive elements of I is nonempty. By well-ordering it has a least element $g = au + bv > 0$.

We claim that g divides every element of I . Indeed, let $c \in I$ and write $c = gq + r$ with $0 \leq r < g$ by the division algorithm. Then $r = c - gq \in I$ because I is closed under the operations just described. By minimality of g among positive elements of I , and $0 \leq r < g$, necessarily $r = 0$. Hence $g \mid c$ for every $c \in I$; in particular $g \mid a$ and $g \mid b$, so g is a common divisor of a and b . Conversely, any common divisor e of a and b divides $au + bv = g$ by Proposition 2.3(3). Therefore g is a common divisor that is divisible by every common divisor; being positive, it is in particular the *greatest* common divisor, so $g = d$. The displayed equation $au + bv = d$ follows, d is the least positive element of I , and $I = g\mathbb{Z} = d\mathbb{Z}$ because every element of I is a multiple of g , as shown, and conversely every multiple $kg = a(ku) + b(kv)$ lies in I . $\square$

Remark 2.12. The proof yields a characterisation of the gcd that is independent of the order $\leq$: $d = \gcd(a, b)$ is, up to sign, the unique common divisor of a and b that is divisible by every common divisor. This is the formulation that survives in settings where “largest” may make no sense but “divisible by every common divisor” still does; later sections adopt it when they leave the integers. We call a nonempty subset $I \subseteq \mathbb{Z}$ closed under addition and under multiplication by arbitrary integers an *ideal*; the content of Bézout’s identity is that every ideal of $\mathbb{Z}$ has the form $d\mathbb{Z}$ for a single generator d . Ideals recur in section 4, where they drive a general construction whose prototype is the passage from integers to congruence classes carried out later in this section.

The Bézout coefficients u, v are not unique. If $au + bv = d$, then for every $k \in \mathbb{Z}$,

$$a\left(u + k\frac{b}{d}\right) + b\left(v - k\frac{a}{d}\right) = d,$$

since the added terms $k\frac{ab}{d}$ and $-k\frac{ab}{d}$ cancel; and these are all the solutions. To see this, note first that a/d and b/d are coprime: a common divisor e of both makes de a common divisor of a and b , so $de \mid au + bv = d$ by Proposition 2.3(3), forcing $e = \pm 1$. Now suppose $au' + bv' = d$. Subtracting $au + bv = d$ and dividing by $d > 0$ gives $\frac{a}{d}(u' - u) = -\frac{b}{d}(v' - v)$, so b/d divides $\frac{a}{d}(u' - u)$; by Proposition 2.15(1) below (whose proof rests only on Bézout’s identity, already established), b/d divides $u' - u$, say $u' = u + k\frac{b}{d}$ with $k \in \mathbb{Z}$. Substituting back gives $b(v' - v + k\frac{a}{d}) = 0$, so $v' = v - k\frac{a}{d}$ when $b \neq 0$; and when $b = 0$ we have $u' = u$, $a/d = \pm 1$, and $k = (v - v')\frac{a}{d}$ places (u', v') in the family. To compute a particular pair (u, v) efficiently, we augment the Euclidean algorithm so that it carries each remainder as an explicit combination of a and b .

Theorem 2.13 (Extended Euclidean algorithm). *There is an algorithm that, on input $a, b \in \mathbb{Z}$ not both zero, outputs $d = \gcd(a, b)$ together with integers u, v satisfying $au + bv = d$, using the same number of division steps as the Euclidean algorithm.*

Proof. Alongside the remainder sequence of Theorem 2.8, maintain two auxiliary sequences (s_i) and (t_i) with the invariant

$$r_i = as_i + bt_i \quad \text{for all } i. \quad (*)$$

Initialise (taking $a, b \geq 0$ for clarity; we restore signs at the end)

$$r_0 = a, s_0 = 1, t_0 = 0; \quad r_1 = b, s_1 = 0, t_1 = 1,$$

so $(*)$ holds for $i = 0, 1$. At step $i \geq 1$, having $r_{i-1} = r_i q_i + r_{i+1}$ from the division algorithm, set

$$s_{i+1} = s_{i-1} - q_i s_i, \quad t_{i+1} = t_{i-1} - q_i t_i.$$

Then

$$as_{i+1} + bt_{i+1} = (as_{i-1} + bt_{i-1}) - q_i (as_i + bt_i) = r_{i-1} - q_i r_i = r_{i+1},$$

so induction preserves $(*)$. When the algorithm terminates with $r_{n+1} = 0$ and $r_n = d = \gcd(a, b)$, the invariant for $i = n$ reads $d = as_n + bt_n$; output $u = s_n, v = t_n$. If $b = 0$, then $a \neq 0$ and termination is immediate with $n = 0$: no division step runs, the last nonzero remainder is $r_0 = a = \gcd(a, 0)$ (Lemma 2.7), and the invariant at $i = 0$ already gives the output $(d, u, v) = (a, 1, 0)$. If $b \neq 0$, termination and the value of d are exactly as in Theorem 2.8. In either case each step does only a constant amount of extra work. For general inputs, run the algorithm on $(|a|, |b|)$ to obtain $|a|u' + |b|v' = d$ (recall $\gcd(|a|, |b|) = \gcd(a, b)$) and output $u = \operatorname{sgn}(a)u', v = \operatorname{sgn}(b)v'$, so that $au + bv = d$. $\square$

In practice it is convenient to lay the computation out as a table with columns (r_i, q_i, s_i, t_i) , computing each new row from the previous two.

Example 2.14. Run the extended algorithm on $a = 1071$, $b = 462$ (continuing Example 2.9). The quotients are $q_1 = 2, q_2 = 3, q_3 = 7$.

i	r_i	s_i	t_i
0	1071	1	0
1	462	0	1
2	147	1	-2
3	21	-3	7
4	0	22	-51

For instance, row 2 is row 0 minus $q_1 = 2$ times row 1: $s_2 = 1 - 2 \cdot 0 = 1$, $t_2 = 0 - 2 \cdot 1 = -2$, and indeed $1071 \cdot 1 + 462 \cdot (-2) = 1071 - 924 = 147$. The last nonzero remainder is $r_3 = 21 = \gcd(1071, 462)$, with

$$1071 \cdot (-3) + 462 \cdot 7 = -3213 + 3234 = 21.$$

Thus $(u, v) = (-3, 7)$. The final row $(s_4, t_4) = (22, -51) = (b/d, -a/d)$ up to sign records the homogeneous solution: $1071 \cdot 22 + 462 \cdot (-51) = 0$.

A first, decisive application of Bézout's identity is Euclid's lemma, the property of primes that ultimately forces unique factorisation.

Proposition 2.15 (Euclid's lemma). *Let $a, b, c \in \mathbb{Z}$.*

1. *If $c \mid ab$ and $\gcd(c, a) = 1$, then $c \mid b$.*
2. *If $\gcd(a, b) = 1$, $a \mid n$, and $b \mid n$, then $ab \mid n$.*
3. *If $\gcd(a, b) = 1$ and $\gcd(a, c) = 1$, then $\gcd(a, bc) = 1$.*

Proof. (1) By Bézout there are u, v with $cu + av = 1$. Multiply by b : $b = cub + avb = c(ub) + (ab)v$. Since $c \mid ab$, c divides both terms on the right, hence $c \mid b$.

(2) Write $n = ak$. Since $b \mid n = ak$ and $\gcd(b, a) = 1$, part (1) gives $b \mid k$, say $k = bm$. Then $n = a(bm) = (ab)m$, so $ab \mid n$.

(3) By Bézout, $ax_1 + by_1 = 1$ and $ax_2 + cy_2 = 1$ for suitable integers. Multiplying,

$$1 = (ax_1 + by_1)(ax_2 + cy_2) = a \underbrace{(ax_1x_2 + x_1cy_2 + by_1x_2)}_{\in \mathbb{Z}} + bc(y_1y_2),$$

which exhibits 1 as an integer combination of a and bc . Hence $\gcd(a, bc)$ divides 1, so it equals 1. $\square$

2.6 Primes and the fundamental theorem of arithmetic

Definition 2.16. An integer $p > 1$ is *prime* if its only positive divisors are 1 and p . An integer $n > 1$ that is not prime is *composite*; thus n is composite iff $n = ab$ for some integers $1 < a, b < n$. The integer 1 is a *unit* and is by convention neither prime nor composite.

The exclusion of 1 from the primes is not arbitrary: it is exactly what makes factorisation into primes unique, since otherwise one could insert arbitrarily many factors of 1. The decisive property of primes is the prime case of Euclid's lemma.

Lemma 2.17 (Euclid's lemma for primes). *Let p be prime and $a, b \in \mathbb{Z}$. If $p \mid ab$ then $p \mid a$ or $p \mid b$. More generally, if $p \mid a_1a_2 \cdots a_k$ then $p \mid a_i$ for some i .*

Proof. The only positive divisors of p are 1 and p , so $\gcd(p, a) \in \{1, p\}$. If $\gcd(p, a) = p$ then $p \mid a$ and the claim holds; otherwise $\gcd(p, a) = 1$, and Proposition 2.15(1) gives $p \mid b$. The general statement follows by induction on k : from $p \mid a_1(a_2 \cdots a_k)$ it follows that $p \mid a_1$ or $p \mid a_2 \cdots a_k$, and the inductive hypothesis handles the latter. $\square$

Theorem 2.18 (Fundamental theorem of arithmetic). *Every integer $n > 1$ can be written as a product of primes,*

$$n = p_1 p_2 \cdots p_k,$$

and this factorisation is unique up to the order of the factors. Equivalently, there is a unique expression

$$n = p_1^{e_1} p_2^{e_2} \cdots p_r^{e_r}, \quad p_1 < p_2 < \cdots < p_r \text{ prime, } e_i \geq 1.$$

Proof. Existence. Suppose, for contradiction, that some integer > 1 is not a product of primes, and let n be the least such (well-ordering). Then n is not prime (a prime is trivially a product of one prime), so $n = ab$ with $1 < a, b < n$. By minimality, each of a, b is a product of primes; concatenating these factorisations expresses n as a product of primes, a contradiction. Hence every $n > 1$ factors into primes.

Uniqueness. Suppose

$$p_1 p_2 \cdots p_k = q_1 q_2 \cdots q_\ell$$

are two factorisations of the same integer into primes. The argument proceeds by induction on k . If $k = 0$ the product is the empty product 1, forcing $\ell = 0$ as well (a nonempty product of primes is > 1). For the inductive step, p_1 divides the right-hand product, so by Lemma 2.17 $p_1 \mid q_j$ for some j . Since q_j is prime, its only divisor > 1 is itself, so $p_1 = q_j$. Cancelling this common factor (legitimate as $\mathbb{Z}$ has no zero divisors) yields

$$p_2 \cdots p_k = q_1 \cdots \widehat{q_j} \cdots q_\ell,$$

where the hat denotes omission. By the inductive hypothesis these two factorisations agree up to order, with $k - 1 = \ell - 1$; restoring $p_1 = q_j$ shows the original factorisations agree up to order. Collecting equal primes into prime powers gives the canonical form, whose uniqueness is immediate from the uniqueness of the multiset of primes—the factors counted with multiplicity, without regard to order. $\square$

Remark 2.19. Both halves of the theorem are needed and have different characters. Existence is a statement about the order structure (well-ordering) and would hold in any reasonable factorisation; uniqueness rests squarely on Euclid’s lemma, hence on Bézout, hence on the division algorithm. There are number systems with a division-like structure failing Euclid’s lemma in which factorisation into irreducibles is *not* unique; the integers avoid this pathology precisely because they admit division with remainder, from which the whole chain above flows.

2.7 Congruences and residue classes

The development now passes from divisibility to the arithmetic of remainders. Fix an integer $n \geq 1$, the *modulus*.

Definition 2.20. Let $n \geq 1$ and $a, b \in \mathbb{Z}$. The integer a is *congruent to b modulo n* , written

$$a \equiv b \pmod{n},$$

if $n \mid (a - b)$, equivalently if a and b leave the same remainder upon division by n .

The two formulations agree: if $a = nq_1 + r_1$ and $b = nq_2 + r_2$ with $0 \leq r_1, r_2 < n$, then $a - b = n(q_1 - q_2) + (r_1 - r_2)$, and $n \mid (a - b)$ iff $n \mid (r_1 - r_2)$ iff $r_1 = r_2$ (since $|r_1 - r_2| < n$).

Proposition 2.21. Fix $n \geq 1$. Congruence modulo n is an equivalence relation on $\mathbb{Z}$ that is moreover compatible with addition and multiplication: if $a \equiv a' \pmod{n}$ and $b \equiv b' \pmod{n}$, then

$$a + b \equiv a' + b' \pmod{n}, \quad ab \equiv a'b' \pmod{n}.$$

Consequently $a^k \equiv (a')^k \pmod{n}$ for all $k \geq 0$, and one may substitute congruent values freely in any polynomial expression with integer coefficients.

Proof. Reflexivity ($n \mid 0$), symmetry ($n \mid (a - b) \Rightarrow n \mid (b - a)$), and transitivity ($n \mid (a - b)$ and $n \mid (b - c)$ give $n \mid (a - c)$ by Proposition 2.3(3)) are immediate. For compatibility, write $a - a' = ns$ and $b - b' = nt$. Then

$$(a + b) - (a' + b') = n(s + t),$$

and

$$ab - a'b' = ab - a'b + a'b - a'b' = (a - a')b + a'(b - b') = n(sb + a't),$$

so both $n \mid ((a + b) - (a' + b'))$ and $n \mid (ab - a'b')$. The statement about powers and polynomial expressions follows by induction from the multiplicative case. $\square$

Because congruence is an equivalence relation, it partitions $\mathbb{Z}$ into disjoint *congruence classes* (also called *residue classes*).

Definition 2.22. The *congruence class* (or *residue class*) of a modulo n is

$$[a]_n = \{b \in \mathbb{Z} : b \equiv a \pmod{n}\} = \{a + kn : k \in \mathbb{Z}\} = a + n\mathbb{Z}.$$

We write the set of all congruence classes as

$$\mathbb{Z}/n\mathbb{Z} = \{[a]_n : a \in \mathbb{Z}\}.$$

By the division algorithm every integer is congruent modulo n to exactly one of $0, 1, \dots, n - 1$, namely its remainder; hence the n classes $[0]_n, [1]_n, \dots, [n - 1]_n$ are distinct and exhaust $\mathbb{Z}/n\mathbb{Z}$. The set $\{0, 1, \dots, n - 1\}$ is the standard *system of representatives*, called the *least nonnegative residues*.

2.8 Units modulo n and modular inverses

Division is the inverse of multiplication, so the opening problem of the section comes down to a multiplicative question about congruences: for which a does the congruence $ax \equiv 1 \pmod{n}$ admit a solution? The Euclidean machinery characterises the solvable cases completely and computes the solutions.

Definition 2.23. Let $n \geq 1$. A class $[a]_n \in \mathbb{Z}/n\mathbb{Z}$ is a *unit* (or is *invertible*) if there exists an integer x with $ax \equiv 1 \pmod{n}$. Any class $[x]_n$ with this property is the (*modular*) *inverse* of $[a]_n$, written $[a]_n^{-1}$; one also calls x a modular inverse of a modulo n . We write the set of units as

$$(\mathbb{Z}/n\mathbb{Z})^\times = \{[a]_n : [a]_n \text{ is a unit}\}.$$

The definition is well posed on classes: by Proposition 2.21, if $a' \equiv a$ and $x' \equiv x \pmod{n}$ then $a'x' \equiv ax \pmod{n}$, so whether $ax \equiv 1 \pmod{n}$ holds depends only on the classes $[a]_n$ and $[x]_n$, never on the representatives chosen. The definite article in “the inverse” is justified by the uniqueness part of the next theorem—the payoff of the section.

Theorem 2.24. *Let $n \geq 1$ and $a \in \mathbb{Z}$. The class $[a]_n$ is a unit in $\mathbb{Z}/n\mathbb{Z}$ if and only if $\gcd(a, n) = 1$. When $\gcd(a, n) = 1$, the inverse is unique, and the extended Euclidean algorithm computes a representative: if $au + nv = 1$, then $[a]_n^{-1} = [u]_n$.*

Proof. Suppose $\gcd(a, n) = 1$. By Bézout (Theorem 2.11) there are u, v with $au + nv = 1$. Reducing modulo n , $au \equiv 1 \pmod{n}$, so $[u]_n$ is an inverse of $[a]_n$. The extended Euclidean algorithm (Theorem 2.13) produces such a u explicitly.

Conversely, if $[a]_n$ is a unit, say $ax \equiv 1 \pmod{n}$, then $ax - 1 = nk$ for some k , i.e. $ax - nk = 1$. Any common divisor of a and n divides the left side, hence divides 1; thus $\gcd(a, n) = 1$.

Uniqueness: if $ax \equiv 1$ and $ax' \equiv 1 \pmod{n}$, then $x \equiv x(ax') \equiv (xa)x' \equiv x' \pmod{n}$ —a sandwich using only the associativity and commutativity of integer multiplication together with Proposition 2.21—so $[x]_n = [x']_n$. $\square$

Example 2.25. Consider the computation of $[17]_{43}^{-1}$. The extended Euclidean algorithm on $(43, 17)$:

$$\begin{aligned} 43 &= 2 \cdot 17 + 9, \\ 17 &= 1 \cdot 9 + 8, \\ 9 &= 1 \cdot 8 + 1, \\ 8 &= 8 \cdot 1 + 0, \end{aligned}$$

so $\gcd(43, 17) = 1$. Back-substituting:

$$1 = 9 - 8 = 9 - (17 - 9) = 2 \cdot 9 - 17 = 2(43 - 2 \cdot 17) - 17 = 2 \cdot 43 - 5 \cdot 17.$$

Hence $-5 \cdot 17 \equiv 1 \pmod{43}$, and $[17]_{43}^{-1} = [-5]_{43} = [38]_{43}$. Check: $17 \cdot 38 = 646 = 15 \cdot 43 + 1$, so $17 \cdot 38 \equiv 1 \pmod{43}$.

Counting the units will matter as much as finding them: the count defined next carries structural weight in later sections, and we fix it here so that those sections can cite a definition rather than re-derive one.

Definition 2.26 (Euler totient). *Euler’s totient function $\varphi(n)$ is the number of integers a with $1 \leq a \leq n$ and $\gcd(a, n) = 1$. Equivalently, $\varphi(n) = |(\mathbb{Z}/n\mathbb{Z})^\times|$.*

The two counts agree: the classes of $1, 2, \dots, n$ run exactly once through all of $\mathbb{Z}/n\mathbb{Z}$ (the class of n being $[0]_n$), and by Theorem 2.24 the class $[a]_n$ is a unit precisely when $\gcd(a, n) = 1$.

Example 2.27. We have $\varphi(1) = 1$, and $\varphi(p) = p - 1$ for prime p : for $1 \leq a < p$ the positive divisor $\gcd(a, p)$ of p is 1 or p , and $p \nmid a$ rules out p , so every one of the $p - 1$ nonzero classes is a unit. By direct enumeration, $\varphi(12) = |\{1, 5, 7, 11\}| = 4$.

The cheque presented at the head of the section is now cashed. To divide by a in the finite world of remainders modulo n is to multiply by the inverse class $[a]_n^{-1}$; Theorem 2.24 says exactly when that inverse exists—precisely when $\gcd(a, n) = 1$, hence for every nonzero class when the modulus is prime—and the extended Euclidean algorithm computes it in $O(\log n)$ division steps (Remark 2.10). The laws verified along the way—associativity, commutativity, distributivity, identities, negatives, and now inverses—have been stated concretely, for the integers and their congruence classes alone. They are instances of patterns that pervade mathematics; Sections 3 and 4 isolate those patterns, give them their standard names, and re-read $\mathbb{Z}/n\mathbb{Z}$ in their light. Nothing proved there revises anything proved here.

3 Groups

Two parties who have never met wish to agree on a secret value while every message they exchange is read by an eavesdropper. One solution begins with a public set carrying a single public operation and a public starting element g . Each party privately chooses a count of repetitions—say a and b —combines g with itself that many times, and publishes the result. Each then takes the *other's* published element and applies their own secret count of repetitions to it. If repetition is well behaved, both arrive at the element produced by ab repetitions of g , and so share a secret, while the eavesdropper has seen only g and the two published values. Three demands on the operation are implicit. It must *compose*: “ a repetitions, then b more” must be unambiguous however the intermediate steps are bracketed. It must *invert*: repetition counts should obey an arithmetic, with an identity and the possibility of undoing a step, not a mere accumulation. And it must *hide* repetition: recovering the secret count from the published value must be hard. The third demand is a computational matter taken up in later volumes; the first two are algebra, and this section isolates exactly the structure that supplies them.

That structure is the *group*, the foundational abstraction of modern algebra. A group axiomatises the bare minimum needed to discuss symmetry and invertible composition: one associative operation with an identity and inverses. Almost every object met later in this series is a group—the integers modulo a prime under addition, the nonzero residues under multiplication, the points of an elliptic curve, the symmetries of a finite field. This section develops group theory from first principles and proves its central structural results in full: Lagrange’s theorem, the classification of cyclic groups, the quotient of an abelian group by a subgroup, and the basic theory of homomorphisms and isomorphisms. The payoff for the opening problem is the cyclic-group theory together with Lagrange’s theorem: in any finite group, the arithmetic of repetition counts is precisely the modular arithmetic of section 2, the modulus being the order of the repeated element—a divisor of the group’s order.

3.1 The group axioms

The first task is to make “combining two elements” precise.

Definition 3.1 (Binary operation). Let S be a set. A *binary operation* on S is a function

$$*: S \times S \rightarrow S, \quad (a, b) \mapsto a * b.$$

Closure—that $a * b$ again lies in S —is built into the very statement that $*$ maps *into* S . We name it explicitly nonetheless, because for a subset $H \subseteq S$ the question of whether $*$ restricts to a binary operation on H is exactly the question of whether H is *closed* under $*$; this question drives the theory of subgroups below (§3.4).

Definition 3.2 (Group). A *group* is a pair $(G, *)$ consisting of a set G and a binary operation $*$ on G satisfying three axioms.

(G1) *Associativity*. For all $a, b, c \in G$, $(a * b) * c = a * (b * c)$.

(G2) *Identity*. There exists $e \in G$ such that $e * a = a * e = a$ for all $a \in G$.

(G3) *Inverses*. For each $a \in G$ there exists $b \in G$ such that $a * b = b * a = e$, where e is an identity as in (G2).

A set with an operation satisfying only (G1) is a *semigroup*; if it also satisfies (G2) it is a *monoid*. A group is thus a monoid in which every element is invertible.

The number of elements of G , written $|G|$, is the *order* of the group; if $|G|$ is finite, G is a *finite group*, and otherwise an *infinite group*. We frequently say “the group G ”, leaving the operation understood.

The axioms assert the *existence* of an identity and of inverses, but not, on their face, uniqueness. Uniqueness is a theorem—and a useful one, since it is what licenses the notations e and a^{-1} .

Proposition 3.3. *Let $(G, *)$ be a group.*

1. *The identity element is unique.*
2. *Each $a \in G$ has a unique inverse, denoted a^{-1} .*
3. *(Cancellation.) For all $a, x, y \in G$: if $a * x = a * y$ then $x = y$, and if $x * a = y * a$ then $x = y$.*
4. *For all $a, b \in G$, $(a^{-1})^{-1} = a$ and $(a * b)^{-1} = b^{-1} * a^{-1}$.*

Proof. (1) Suppose e and e' both satisfy (G2). Using that e' is an identity, $e = e * e'$; using that e is an identity, $e * e' = e'$. Hence $e = e'$.

(2) Suppose b and b' are both inverses of a . Then

$$b = b * e = b * (a * b') = (b * a) * b' = e * b' = b',$$

using (G2), the inverse property of b' , associativity, and the inverse property of b .

(3) Suppose $a * x = a * y$. Multiplying on the left by a^{-1} and reassociating,

$$x = (a^{-1} * a) * x = a^{-1} * (a * x) = a^{-1} * (a * y) = (a^{-1} * a) * y = y.$$

The right-cancellation law is proved symmetrically, multiplying on the right by a^{-1} .

(4) Since $a * a^{-1} = a^{-1} * a = e$, the element a is an inverse of a^{-1} ; by (2) it is *the* inverse, so $(a^{-1})^{-1} = a$. For the product formula, compute

$$(a * b) * (b^{-1} * a^{-1}) = a * (b * b^{-1}) * a^{-1} = a * a^{-1} = e,$$

and symmetrically $(b^{-1} * a^{-1}) * (a * b) = e$; by (2), $(a * b)^{-1} = b^{-1} * a^{-1}$. The order reversal is the “socks–shoes” rule: the inverse of putting on socks, then shoes, is removing shoes, then socks. $\square$

Two notational conventions dominate. *Multiplicative* notation writes the operation as juxtaposition ab (or $a \cdot b$), the identity as 1, the inverse as a^{-1} , and powers as $a^n = \underbrace{a \cdots a}_{n \text{ factors}}$ for $n \geq 1$, with $a^0 = 1$ and $a^{-n} = (a^{-1})^n$. *Additive* notation, reserved almost exclusively for commutative groups, writes $a + b$, the identity as 0, the inverse as $-a$, and na for the n -fold sum. The exponent laws

$$a^m a^n = a^{m+n}, \quad (a^m)^n = a^{mn}$$

hold for all integers m, n , proved by induction together with Proposition 3.3. Henceforth ab denotes $a * b$, and we drop the symbol $*$ unless clarity demands it.

The power notation presumes something the axioms do not literally grant: that an n -fold product needs no brackets at all. Axiom (G1) speaks only of three factors. The gap is closed once and for all by the next proposition, and thereafter products are written without brackets and powers without comment.

Proposition 3.4 (Generalised associativity). *Let $a_1, \dots, a_n$, $n \geq 1$, be elements of a group—or of any set with an associative operation. Every way of bracketing the product $a_1 a_2 \cdots a_n$ yields the same*

element, namely the left-normed product $L_n = (\cdots ((a_1 a_2) a_3) \cdots) a_n$. In particular a^n is a single well-defined element, however the n factors are combined.

Proof. By strong induction on n . For $n \leq 2$ there is only one bracketing, and for $n = 3$ the two bracketings agree by (G1). Let $n \geq 3$ and assume the claim for all shorter products. Any bracketed product P of $a_1, \dots, a_n$ has an outermost multiplication, splitting it as $P = QR$ with Q a bracketed product of $a_1, \dots, a_k$ and R one of $a_{k+1}, \dots, a_n$, for some $1 \leq k < n$. The induction hypothesis evaluates both halves: $Q = L_k$, and R is the left-normed product of $a_{k+1}, \dots, a_n$. If $k = n - 1$, then $R = a_n$ and $P = L_{n-1} a_n = L_n$ directly. If $k < n - 1$, write $R = R' a_n$ with R' the left-normed product of $a_{k+1}, \dots, a_{n-1}$; then (G1) and the induction hypothesis, applied to the bracketed product $L_k R'$ of the $n - 1$ elements $a_1, \dots, a_{n-1}$, give

$$P = L_k(R' a_n) = (L_k R') a_n = L_{n-1} a_n = L_n.$$

Every bracketing therefore equals L_n . □

Definition 3.5 (Abelian group). A group $(G, *)$ is *abelian* (or *commutative*) if $a * b = b * a$ for all $a, b \in G$; otherwise it is *non-abelian*.

The term honours Niels Henrik Abel. In an abelian group products may be freely reordered, and the socks–shoes formula $(ab)^{-1} = b^{-1}a^{-1} = a^{-1}b^{-1}$ loses its order-sensitivity. Convention reserves additive notation for abelian groups precisely because $+$ connotes commutativity.

3.2 First examples

Example 3.6. The pair $(\mathbb{Z}, +)$ is an infinite abelian group: addition is associative and commutative, the identity is 0, and the inverse of n is $-n$. Likewise $(\mathbb{Q}, +)$, $(\mathbb{R}, +)$, and $(\mathbb{C}, +)$ are abelian groups. By contrast $(\mathbb{Z}, \cdot)$ is *not* a group: the integer 2 has no multiplicative inverse in $\mathbb{Z}$.

Example 3.7. The nonzero rationals $\mathbb{Q}^\times = \mathbb{Q} \setminus \{0\}$ form an abelian group under multiplication, with identity 1 and inverse $1/a$; likewise $\mathbb{R}^\times$ and $\mathbb{C}^\times$. Zero must be excluded because it has no multiplicative inverse. The pattern is general: for any field F —the reader has met $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$; the abstract notion is defined in section 4—the nonzero elements $F^\times = F \setminus \{0\}$ form an abelian group under multiplication. Indeed, this is part of the very definition of a field.

Congruence classes furnish the central finite examples. Recall from section 2 the set $\mathbb{Z}/n\mathbb{Z}$ of residue classes $[a]_n$ and define operations on classes through representatives,

$$[a]_n + [b]_n = [a + b]_n, \quad [a]_n [b]_n = [ab]_n.$$

By Proposition 2.21 the right-hand sides do not depend on the representatives chosen, so both operations are well defined, and each law of integer arithmetic—associativity, commutativity, the identities $[0]_n$ and $[1]_n$, negatives—passes to classes by computing on representatives. (The full structural statement, that these operations make $\mathbb{Z}/n\mathbb{Z}$ a commutative ring, is Theorem 4.6 in section 4; the present section uses only the fragments just listed.) Addition makes all of $\mathbb{Z}/n\mathbb{Z}$ a group, as Example 3.9 below records. For $n > 1$, multiplication does not— $[0]_n$ has no inverse—but section 2 identified exactly which classes are invertible: the *units*, the classes $[a]_n$ with $\gcd(a, n) = 1$ (Theorem 2.24). The units form a group.

Proposition 3.8. Under multiplication of classes, the set $(\mathbb{Z}/n\mathbb{Z})^\times$ of units of $\mathbb{Z}/n\mathbb{Z}$ is an abelian group. Its order is $|(\mathbb{Z}/n\mathbb{Z})^\times| = \varphi(n)$, Euler’s totient of Definition 2.26: the number of integers a with $1 \leq a \leq n$ and $\gcd(a, n) = 1$.

Proof. Multiplication of classes is associative and commutative with identity $[1]_n$, as derived above from Proposition 2.21 by computing on representatives, and $[1]_n$ is a unit (it is its own inverse). The operation is closed on units: if $[a]_n$ and $[b]_n$ are units with inverses $[a]_n^{-1}$ and $[b]_n^{-1}$, then $[ab]_n$ has inverse $[b]_n^{-1}[a]_n^{-1}$, since $(ab)(b^{-1}a^{-1}) \equiv 1 \pmod{n}$ for any representatives of the inverse classes. Every unit has an inverse by definition, and that inverse is itself a unit (it is invertible, with inverse the original class). Hence $(\mathbb{Z}/n\mathbb{Z})^\times$ is an abelian group. By Theorem 2.24 its elements are exactly the classes $[a]_n$ with $\gcd(a, n) = 1$ and $1 \leq a \leq n$, of which there are $\varphi(n)$ by Definition 2.26. $\square$

Example 3.9. The pair $(\mathbb{Z}/n\mathbb{Z}, +)$ is an abelian group of order n : the discussion above derived associativity, commutativity, the identity $[0]_n$, and negatives from Proposition 2.21, and the classes number exactly n (the least nonnegative residues, Definition 2.22). The invertible classes form the abelian group $(\mathbb{Z}/n\mathbb{Z})^\times$ of order $\varphi(n)$ under multiplication (Theorem 2.24 and Proposition 3.8). When $n = p$ is prime, every nonzero class is a unit (Example 2.27), so $(\mathbb{Z}/p\mathbb{Z})^\times$ has order $p - 1$ —this is Example 3.7 specialised to $\mathbb{Z}/p\mathbb{Z}$, which section 4 shows to be a field and denotes $\mathbb{F}_p$.

Example 3.10. The one-element set $\{e\}$, with its only possible operation $e * e = e$, is the *trivial group*. The group $\mathbb{Z}/2\mathbb{Z} = \{0, 1\}$ is the smallest nontrivial group. A subtler small example is the *Klein four-group* $V = \{e, a, b, c\}$, in which every element composed with itself gives e and the product of any two distinct nonidentity elements is the third; the verification that this table satisfies the axioms is routine. The group V is abelian of order 4.

Although V and $\mathbb{Z}/4\mathbb{Z}$ both have order 4, they are structurally different: in V every element combined with itself returns the identity, while in $\mathbb{Z}/4\mathbb{Z}$ the class 1 must be added to itself four times to reach 0. The language that makes “structurally different” precise is that of *isomorphism* (§3.8); in its terms, V is not isomorphic to $\mathbb{Z}/4\mathbb{Z}$, because V has no element of order 4 in the sense defined next.

3.3 The order of an element

Definition 3.11 (Order of an element). Let G be a group and $a \in G$. The *order* of a , written $\text{ord}(a)$, is the least positive integer n with $a^n = e$, if such an n exists; a then has *finite order*. If no positive power of a equals e , then a has *infinite order* and one writes $\text{ord}(a) = \infty$.

The word “order” now has two meanings—the order $|G|$ of a group and the order $\text{ord}(a)$ of an element. The clash is deliberate and is reconciled by the cyclic-group theory below: $\text{ord}(a)$ equals the order of the smallest subgroup of G containing a (Proposition 3.21).

Proposition 3.12. Let a have finite order $d = \text{ord}(a)$, and let $m, k \in \mathbb{Z}$.

1. $a^m = e$ if and only if $d \mid m$.
2. $a^m = a^k$ if and only if $m \equiv k \pmod{d}$.
3. The distinct powers of a are exactly $e = a^0, a^1, \dots, a^{d-1}$.

Proof. (1) If $d \mid m$, write $m = dq$; then $a^m = (a^d)^q = e^q = e$. Conversely, suppose $a^m = e$ and divide with remainder (Theorem 2.4): $m = dq + r$ with $0 \leq r < d$. Then

$$e = a^m = (a^d)^q a^r = a^r,$$

so r is a nonnegative integer smaller than d with $a^r = e$. Minimality of d among *positive* exponents forces $r = 0$, whence $d \mid m$.

(2) We have $a^m = a^k$ if and only if $a^{m-k} = e$ (multiply by a^{-k} and use the exponent laws), which by (1) holds if and only if $d \mid (m - k)$, i.e. $m \equiv k \pmod{d}$ (Definition 2.20).

(3) By (2) the powers $a^0, a^1, \dots, a^{d-1}$ are pairwise distinct, their exponents being pairwise incongruent modulo d ; and every power a^m equals $a^{m \bmod d}$, again by (2). $\square$

The final step of part (1) deserves to be isolated, because it is the recurring device of this section: to show that every witness of some property is a multiple of the *least positive* witness m , take an arbitrary witness t , divide $t = mq + r$ with $0 \leq r < m$, show that the remainder r is itself a witness, and conclude $r = 0$ by minimality. It is the same shape as the proof of Bézout's identity (Theorem 2.11), transplanted from the integers into a group, and it reappears twice below: for the least positive power of g lying in a subgroup (Theorem 3.24) and for the least positive element of a subgroup of $\mathbb{Z}$ (Proposition 3.26).

Example 3.13. In $(\mathbb{Z}/n\mathbb{Z}, +)$, read the powers of Definition 3.11 additively: the k -fold sum of the class 1 is the class k , which is 0 exactly when $n \mid k$, so $\text{ord}(1) = n$. More generally $\text{ord}(a) = n / \gcd(a, n)$ for any class a , by Corollary 3.23 below read additively. In $\mathbb{C}^\times$ the element i has order 4: its successive powers are

$$i^1 = i, \quad i^2 = -1, \quad i^3 = -i, \quad i^4 = 1.$$

In $(\mathbb{Z}, +)$ every nonzero element has infinite order: no nonzero multiple of a nonzero integer is 0.

3.4 Subgroups

Definition 3.14 (Subgroup). A subset $H \subseteq G$ of a group $(G, *)$ is a *subgroup*, written $H \leq G$, if H is itself a group under the restriction of $*$ to $H \times H$: explicitly, H is nonempty, closed under $*$, contains the identity e of G , and contains a^{-1} whenever it contains a . Every group has the *trivial* subgroup $\{e\}$ and G itself; a subgroup other than G is *proper*.

Two small points keep this definition honest. First, the identity of a subgroup necessarily coincides with the identity of G : if $e_H \in H$ satisfies $e_H * e_H = e_H$, then cancelling e_H in G against $e_H * e = e_H$ (Proposition 3.3(3)) gives $e_H = e$. Second, inverses computed in H agree with those computed in G , by uniqueness of inverses in G . Nothing about H , in other words, depends on whether we regard it as a group in its own right or as a subset of G .

Checking all the group axioms for a subset would be wasteful, since associativity is inherited for free. A single condition suffices.

Proposition 3.15 (One-step subgroup test). *Let G be a group and $H \subseteq G$. Then $H \leq G$ if and only if*

1. H is nonempty, and
2. for all $a, b \in H$, the element ab^{-1} lies in H .

Proof. If $H \leq G$, then H is nonempty and, for $a, b \in H$, it contains b^{-1} and hence ab^{-1} ; both conditions hold.

Conversely, suppose (1) and (2) hold. Pick $a \in H$ by (1); taking $b = a$ in (2) gives $e = aa^{-1} \in H$. For any $b \in H$, taking the pair (e, b) in (2) gives $b^{-1} = eb^{-1} \in H$, so H is closed under inverses. Finally, for $a, b \in H$ we now know $b^{-1} \in H$, and applying (2) to the pair (a, b^{-1}) gives $a(b^{-1})^{-1} = ab \in H$, so H is closed under the operation. Associativity is inherited from G , so H is a group. $\square$

Remark 3.16. For a *finite* nonempty subset H , closure under the operation alone forces $H \leq G$. Indeed, given $a \in H$, closure puts all the powers $a, a^2, a^3, \dots$ in H ; since H is finite they cannot all be distinct, so $a^i = a^j$ for some $i < j$, giving $a^{j-i} = e \in H$. Then $a^{-1} = a^{j-i-1} \in H$: this is a positive power of a when $j - i \geq 2$, and when $j - i = 1$ we have $a = e$, whose inverse e already lies in H . Thus for finite subsets, closure under multiplication is the only condition to verify.

Example 3.17. The even integers $2\mathbb{Z} = \{2k : k \in \mathbb{Z}\}$ form a subgroup of $(\mathbb{Z}, +)$: the set is nonempty, and $2k - 2l = 2(k - l)$ is again even (the subgroup test, written additively). More generally $n\mathbb{Z} = \{nk : k \in \mathbb{Z}\} \leq \mathbb{Z}$ for every $n \geq 0$, by the same computation—and these are *all* the subgroups of $\mathbb{Z}$, as Proposition 3.26 will show.

Proposition 3.18. *The intersection $\bigcap_{i \in I} H_i$ of any family of subgroups $H_i \leq G$ is a subgroup of G .*

Proof. The identity e lies in every H_i , so the intersection is nonempty. If a and b lie in the intersection, then for each i both lie in H_i , so $ab^{-1} \in H_i$ by Proposition 3.15; hence ab^{-1} lies in the intersection, and the one-step test applies. $\square$

Proposition 3.18 is what makes “the smallest subgroup containing a given set” well defined: the intersection of all subgroups containing the set is itself a subgroup containing the set, and it is contained in every other one.

Definition 3.19 (Generated subgroup). For a subset $S \subseteq G$, the *subgroup generated by S* , written $\langle S \rangle$, is the intersection of all subgroups of G containing S . By Proposition 3.18 it is a subgroup, and by construction it is the smallest subgroup of G containing S .

Concretely, $\langle S \rangle$ consists of all finite products $s_1^{\varepsilon_1} s_2^{\varepsilon_2} \cdots s_k^{\varepsilon_k}$ with $s_i \in S$ and $\varepsilon_i \in \{+1, -1\}$, the empty product being e . Indeed, the set of such products contains S , contains e , and is closed under multiplication (concatenate) and under inversion (reverse the factors and negate the exponents, by the socks–shoes rule), so it is a subgroup containing S ; conversely, every subgroup containing S is closed under these operations and so contains every such product. Hence the two descriptions coincide.

3.5 Cyclic groups

Definition 3.20 (Cyclic group). A group G is *cyclic* if there exists $g \in G$ with

$$G = \langle g \rangle = \{g^k : k \in \mathbb{Z}\};$$

such an element g is a *generator* of G .

The displayed description matches Definition 3.19: the set $\{g^k : k \in \mathbb{Z}\}$ contains g , and it is closed under products and inverses by the exponent laws, so it is a subgroup containing g ; conversely any subgroup containing g must contain all its powers. Hence $\langle g \rangle = \{g^k : k \in \mathbb{Z}\}$: for a single element g , the general generated-subgroup construction and the naive one agree. Every cyclic group is abelian, since $g^m g^n = g^{m+n} = g^n g^m$.

Proposition 3.21. *Let $G = \langle g \rangle$ be cyclic.*

1. *If g has infinite order, the powers g^k ($k \in \mathbb{Z}$) are pairwise distinct, so G is infinite.*
2. *If g has finite order d , then $G = \{e, g, g^2, \dots, g^{d-1}\}$ has exactly d elements.*

In both cases $|\langle g \rangle| = \text{ord}(g)$.

Proof. (1) Suppose $g^i = g^j$ with $i \neq j$; without loss of generality $i > j$. Then $g^{i-j} = e$ with $i - j > 0$, contradicting infinite order. So the powers are pairwise distinct and G is infinite.

(2) This is Proposition 3.12(3): the distinct powers of g are exactly $g^0, g^1, \dots, g^{d-1}$, and these exhaust $G = \{g^k : k \in \mathbb{Z}\}$. $\square$

Example 3.22. The group $(\mathbb{Z}, +)$ is infinite cyclic, generated by 1 and also by -1 . The group $(\mathbb{Z}/n\mathbb{Z}, +)$ is cyclic of order n , generated by 1 (Example 3.13). The unit group $(\mathbb{Z}/8\mathbb{Z})^\times = \{1, 3, 5, 7\}$ is *not* cyclic: each of its four elements squares to 1 ($3^2 = 9 \equiv 1$, $5^2 = 25 \equiv 1$, $7^2 = 49 \equiv 1 \pmod{8}$), so no element has order 4. By contrast $(\mathbb{Z}/5\mathbb{Z})^\times = \{1, 2, 3, 4\}$ is cyclic, generated by 2: its powers run

$$2^1 \equiv 2, \quad 2^2 \equiv 4, \quad 2^3 \equiv 3, \quad 2^4 \equiv 1 \pmod{5},$$

exhausting the group.

The order of a power of g is determined by a gcd, and the formula tells us exactly which powers are again generators.

Corollary 3.23. *Let g have finite order n and let $k \in \mathbb{Z}$. Then*

$$\text{ord}(g^k) = \frac{n}{\gcd(n, k)}.$$

In particular g^k generates $\langle g \rangle$ if and only if $\gcd(n, k) = 1$, so a cyclic group of order n has exactly $\varphi(n)$ generators.

Proof. Set $d = \gcd(n, k)$ and write $k = dk'$, $n = dn'$ with $\gcd(n', k') = 1$. For $m \geq 1$,

$$(g^k)^m = g^{km} = e \iff n \mid km \iff dn' \mid dk'm \iff n' \mid k'm \iff n' \mid m,$$

using Proposition 3.12(1) for the first equivalence and Euclid's lemma (Proposition 2.15(1), with $\gcd(n', k') = 1$) for the last. The least positive such m is $n' = n/d$, which is therefore $\text{ord}(g^k)$.

For the second claim, $\langle g^k \rangle \subseteq \langle g \rangle$ always, and by Proposition 3.21 the subgroup $\langle g^k \rangle$ has $\text{ord}(g^k) = n/\gcd(n, k)$ elements; it equals the n -element set $\langle g \rangle$ exactly when $\gcd(n, k) = 1$. Since the powers $g^1, g^2, \dots, g^n$ run through each element of $\langle g \rangle$ exactly once (Proposition 3.12(3)), the generators correspond to the $k \in \{1, \dots, n\}$ with $\gcd(n, k) = 1$, of which there are $\varphi(n)$ by Definition 2.26. $\square$

The subgroups of a cyclic group can be classified completely. The theorem below is used constantly in the sequel; its part (3) is the origin of the “one subgroup per divisor” picture of $\mathbb{Z}/n\mathbb{Z}$.

Theorem 3.24 (Subgroups of cyclic groups). *Let $G = \langle g \rangle$ be a cyclic group.*

1. *Every subgroup of G is cyclic.*
2. *If G is infinite, its subgroups are exactly $\langle g^m \rangle$ for $m \geq 0$; distinct $m \geq 0$ give distinct subgroups, and for $m \geq 1$ the subgroup $\langle g^m \rangle$ is isomorphic to $(\mathbb{Z}, +)$.*
3. *If $|G| = n$, then for each positive divisor $d \mid n$ there is exactly one subgroup of G of order d , namely $\langle g^{n/d} \rangle$, and these are all the subgroups of G . Subgroups of G thus correspond bijectively to positive divisors of n .*

Proof. (1) Let $H \leq G$. If $H = \{e\}$, then $H = \langle e \rangle$ is cyclic. Otherwise H contains some $g^t \neq e$, so $t \neq 0$; since H also contains $(g^t)^{-1} = g^{-t}$, it contains a power of g with positive exponent. Let m be the least positive integer with $g^m \in H$ (well-ordering, Theorem 2.1). We claim $H = \langle g^m \rangle$. The inclusion $\supseteq$ holds by closure. For $\subseteq$, take any element of H ; it is some g^t , since $H \subseteq G = \langle g \rangle$. Divide $t = mq + r$ with $0 \leq r < m$; then

$$g^r = g^{t-mq} = g^t (g^m)^{-q} \in H,$$

and minimality of m forces $r = 0$ —the division-minimality device again. Hence $g^t = (g^m)^q \in \langle g^m \rangle$.

(2) Let G be infinite. Then g has infinite order, for otherwise G would be finite by Proposition 3.21(2). By part (1) every subgroup is $\{e\} = \langle g^0 \rangle$ or $\langle g^m \rangle$ with $m \geq 1$ least positive, and conversely each $\langle g^m \rangle$ is a subgroup. For distinctness, note that for $m \geq 1$ the positive exponents t with $g^t \in \langle g^m \rangle$ are exactly the positive multiples of m : the elements of $\langle g^m \rangle$ are the powers g^{mk} , and $g^t = g^{mk}$ forces $t = mk$ because distinct exponents give distinct powers (Proposition 3.21(1)). The least such exponent recovers m , so distinct $m \geq 1$ give distinct subgroups, and each differs from $\{e\}$ because $g^m \neq e$. Finally, for $m \geq 1$ the element g^m has infinite order (if $(g^m)^t = g^{mt} = e$ then $mt = 0$, so $t = 0$), so $\langle g^m \rangle$ is an infinite cyclic group and is therefore isomorphic to $(\mathbb{Z}, +)$ by Theorem 3.43(1) below. No circularity arises: the proof of that theorem uses only Propositions 3.12 and 3.21.

(3) Let $|G| = n$, so $\text{ord}(g) = n$ by Proposition 3.21.

Existence. Let $d \mid n$ be a positive divisor. Since n/d divides n , we have $\gcd(n, n/d) = n/d$, so Corollary 3.23 gives

$$\text{ord}(g^{n/d}) = \frac{n}{\gcd(n, n/d)} = \frac{n}{n/d} = d,$$

and $\langle g^{n/d} \rangle$ is a subgroup of order d by Proposition 3.21.

Uniqueness. Let $H \leq G$ with $|H| = d$. If $d = 1$ then $H = \{e\} = \langle g^n \rangle = \langle g^{n/1} \rangle$, since $g^n = e$. Otherwise, as in part (1), $H = \langle g^m \rangle$ for the least positive m with $g^m \in H$, and Propositions 3.12 and 3.21 together with Corollary 3.23 give

$$d = |H| = \text{ord}(g^m) = \frac{n}{\gcd(n, m)}, \quad \text{so} \quad \gcd(n, m) = \frac{n}{d}.$$

In particular n/d divides m , say $m = (n/d)m'$; then $g^m = (g^{n/d})^{m'}$, so $H = \langle g^m \rangle \subseteq \langle g^{n/d} \rangle$. Both sides have exactly d elements, so $H = \langle g^{n/d} \rangle$.

All subgroups arise. Any $H \leq G$ equals some $\langle g^m \rangle$ (or $\{e\}$), whose order $n/\gcd(n, m)$ is a positive divisor d of n ; by uniqueness $H = \langle g^{n/d} \rangle$. The assignment $d \mapsto \langle g^{n/d} \rangle$ is therefore surjective onto the set of subgroups, and it is injective because subgroups of different orders are distinct. This is the asserted bijection. $\square$

Remark 3.25. Specialised to $G = \mathbb{Z}/n\mathbb{Z}$ with generator 1 (written additively), Theorem 3.24(3) reads: for each divisor $d \mid n$ there is exactly one subgroup of order d , namely

$$\langle n/d \rangle = \{0, n/d, 2n/d, \dots, (d-1)n/d\},$$

itself cyclic of order d , hence isomorphic to $\mathbb{Z}/d\mathbb{Z}$ by the classification below. For $n = 12$ the six subgroups correspond to the divisors 1, 2, 3, 4, 6, 12, and inclusion between subgroups mirrors divisibility between the corresponding divisors: the subgroup lattice of $\mathbb{Z}/12\mathbb{Z}$ is the divisor lattice of 12, with chains such as $1 \mid 2 \mid 4$, $1 \mid 2 \mid 6$, and $3 \mid 6$.

The classification promised in Example 3.17 now follows; it is Theorem 3.24(2) read additively for $G = (\mathbb{Z}, +) = \langle 1 \rangle$, but the direct argument is short enough to give in full.

Proposition 3.26. *Every subgroup of $(\mathbb{Z}, +)$ has the form $n\mathbb{Z}$ for a unique integer $n \geq 0$.*

Proof. Let $H \leq \mathbb{Z}$. If $H = \{0\}$, then $H = 0\mathbb{Z}$. Otherwise H contains a nonzero integer and, being closed under negation, contains a positive one; let n be its least positive element (Theorem 2.1). Closure under addition and negation gives $n\mathbb{Z} \subseteq H$. Conversely, for any $h \in H$, divide $h = nq + r$ with $0 \leq r < n$; then $r = h - nq \in H$, and minimality of n forces $r = 0$, so $h \in n\mathbb{Z}$. Hence $H = n\mathbb{Z}$. For uniqueness, note n is recovered from H as its least positive element (and $n = 0$ exactly for $H = \{0\}$), so distinct $n \geq 0$ give distinct subgroups. $\square$

3.6 Cosets and Lagrange's theorem

A subgroup $H \leq G$ slices G into translated copies of itself, and counting the slices is the mechanism underlying Lagrange's theorem: partition the group into blocks, exhibit a bijection between each block and H , and multiply the block count by the block size. The blocks are the cosets.

Definition 3.27 (Cosets and index). Let $H \leq G$ and $a \in G$. The *left coset* of H by a is

$$aH = \{ah : h \in H\},$$

and the *right coset* is $Ha = \{ha : h \in H\}$. In additive notation one writes $a + H = \{a + h : h \in H\}$. The set of left cosets is denoted G/H , and the *index* of H in G is $[G : H] = |G/H|$, the number of left cosets.

Lemma 3.28 (Cosets partition the group). *Let $H \leq G$.*

1. *Each $a \in G$ lies in aH ; the left cosets cover G .*
2. *Two left cosets are either equal or disjoint: $aH = bH$ if and only if $a^{-1}b \in H$, and otherwise $aH \cap bH = \emptyset$.*
3. *Every left coset has the same cardinality as H : the map $h \mapsto ah$ is a bijection from H onto aH .*

The same statements hold for right cosets, with $ba^{-1} \in H$ as the criterion in (2).

Proof. (1) Since $e \in H$, we have $a = ae \in aH$.

(2) First suppose the cosets meet: $ah_1 = bh_2$ for some $h_1, h_2 \in H$. Then $a^{-1}b = h_1h_2^{-1} \in H$. Conversely, suppose $a^{-1}b = h \in H$, so $b = ah$. Then every element $bh' \in bH$ equals $a(hh') \in aH$, giving $bH \subseteq aH$; and since $b^{-1}a = h^{-1} \in H$, the same argument with the roles of a and b exchanged gives $aH \subseteq bH$. Hence $aH = bH$. Combining the two directions: if two cosets share even one element they coincide, so distinct cosets are disjoint.

(3) The map $\lambda_a : H \rightarrow aH, h \mapsto ah$, is surjective by the definition of aH and injective by left cancellation (Proposition 3.3(3)): $ah = ah'$ implies $h = h'$. Hence $|aH| = |H|$.

For right cosets, the same three arguments apply with sides exchanged; in (2) the criterion becomes $ba^{-1} \in H$. $\square$

Theorem 3.29 (Lagrange's theorem). *Let G be a finite group and $H \leq G$. Then $|H|$ divides $|G|$, and*

$$|G| = [G : H] |H|.$$

Proof. By Lemma 3.28, the distinct left cosets of H partition G : they cover G by part (1) and are pairwise disjoint by part (2). Since G is finite there are finitely many of them, say $[G : H] = k$,

and each has exactly $|H|$ elements by part (3). Summing the sizes of the blocks of the partition,

$$|G| = \sum_{\text{distinct cosets } aH} |aH| = k |H| = [G : H] |H|. \quad \square$$

Three corollaries follow in quick succession; the third is the theorem this section owes to the rest of the volume.

Corollary 3.30. *In a finite group G , the order of every element divides $|G|$.*

Proof. For $a \in G$, the cyclic subgroup $\langle a \rangle$ has order $\text{ord}(a)$ by Proposition 3.21 (the order is finite, since $\langle a \rangle \subseteq G$ is finite), and Theorem 3.29 makes $|\langle a \rangle|$ a divisor of $|G|$. $\square$

Corollary 3.31. *If G is a finite group of order N , then $a^N = e$ for every $a \in G$.*

Proof. Let $d = \text{ord}(a)$. By Corollary 3.30, $d \mid N$, say $N = dm$; then $a^N = (a^d)^m = e^m = e$. $\square$

Corollary 3.32 (Euler's theorem; Fermat's little theorem). *Let $n \geq 1$ and $a \in \mathbb{Z}$ with $\gcd(a, n) = 1$. Then*

$$a^{\varphi(n)} \equiv 1 \pmod{n}.$$

In particular, for a prime p and a not divisible by p ,

$$a^{p-1} \equiv 1 \pmod{p},$$

and $a^p \equiv a \pmod{p}$ holds for all integers a .

Proof. Apply Corollary 3.31 to the group $(\mathbb{Z}/n\mathbb{Z})^\times$, which has order $\varphi(n)$ (Proposition 3.8). For $\gcd(a, n) = 1$ the class $[a]_n$ is a unit (Theorem 2.24), so $[a]_n^{\varphi(n)} = [1]_n$, which is the first congruence. For $n = p$ prime, $\varphi(p) = p - 1$ (Example 2.27), and $p \nmid a$ gives $\gcd(a, p) = 1$, yielding the second congruence. Multiplying it by a gives $a^p \equiv a \pmod{p}$; and when $p \mid a$ both sides are congruent to 0, so the last congruence holds for all a . $\square$

Remark 3.33. The congruences of Corollary 3.32 are proved here, once; section 7 restates them under the heading *Fermat's little theorem and Euler's theorem* and develops their cryptographic role there, citing this proof rather than repeating it.

Corollary 3.34. *Every group G of prime order p is cyclic, hence isomorphic to $\mathbb{Z}/p\mathbb{Z}$, and has no proper nontrivial subgroups.*

Proof. Pick $a \in G$ with $a \neq e$ (possible since $p \geq 2$). The subgroup $\langle a \rangle$ has at least two elements, and by Theorem 3.29 its order divides p ; a divisor of p exceeding 1 equals p , so $\langle a \rangle = G$ and G is cyclic. The isomorphism with $\mathbb{Z}/p\mathbb{Z}$ is Theorem 3.43 below. Finally, any subgroup has order 1 or p by Lagrange, i.e. is $\{e\}$ or G . $\square$

3.7 Quotients of abelian groups

Cosets do more than count: for an abelian group they themselves form a group. This single construction underlies the quotient rings of section 4 and, through them, $\mathbb{Z}/n\mathbb{Z}$ and the polynomial quotients used throughout the series; it is all of quotient theory the volume requires.

Proposition 3.35 (Quotient of an abelian group). *Let A be an abelian group, written additively, and let $H \leq A$. The operation*

$$(a + H) + (b + H) = (a + b) + H$$

on the coset set A/H is well defined and makes A/H an abelian group—the quotient group of A by H —with identity $H = 0 + H$ and inverse $-(a + H) = (-a) + H$. If A is finite, then

$$|A/H| = [A : H] = \frac{|A|}{|H|}.$$

Proof. The only point of substance is that the operation is well defined: its output is stated in terms of chosen representatives a and b , and we must check that replacing them by other representatives of the same cosets leaves the result unchanged. Suppose then that $a + H = a' + H$ and $b + H = b' + H$. By Lemma 3.28(2), written additively, this means $a' - a \in H$ and $b' - b \in H$. Then

$$(a' + b') - (a + b) = (a' - a) + (b' - b) \in H,$$

since H is closed under addition; applying Lemma 3.28(2) once more, $(a' + b') + H = (a + b) + H$. The operation is well defined. This representative-independence check is the template for every coset-wise construction in the volume; it recurs, in the same words, for quotient-ring multiplication in section 4.

The axioms now pass coset-wise from A : for associativity,

$$((a + H) + (b + H)) + (c + H) = ((a + b) + c) + H = (a + (b + c)) + H = (a + H) + ((b + H) + (c + H));$$

commutativity follows the same way from commutativity in A ; the coset $0 + H = H$ is an identity; and $(-a) + H$ is an inverse of $a + H$. Hence A/H is an abelian group. When A is finite, the number of cosets is $[A : H] = |A|/|H|$ by Theorem 3.29—the same partition-into-equal-blocks count as before. $\square$

The construction is not new to this volume in disguise: for $A = (\mathbb{Z}, +)$ and $H = n\mathbb{Z}$, Definition 2.22 exhibits each residue class as $[a]_n = a + n\mathbb{Z}$, a coset of $n\mathbb{Z} \leq \mathbb{Z}$, and the coset addition above is exactly addition of classes. The notation $\mathbb{Z}/n\mathbb{Z}$, introduced in section 2 as a bare name for the set of classes, is thus an instance of the general quotient notation A/H , and $(\mathbb{Z}/n\mathbb{Z}, +)$ is the quotient group $\mathbb{Z}/n\mathbb{Z}$ in the sense just constructed.

Remark 3.36. For a non-abelian group G the same construction succeeds exactly for those subgroups $N \leq G$ whose left and right cosets coincide, $gN = Ng$ for all $g \in G$; these are the *normal* subgroups. Commutativity makes every subgroup of an abelian group normal, which is why no extra hypothesis was needed in Proposition 3.35. The present volume deliberately needs only the abelian case, and we do not develop the general theory.

3.8 Homomorphisms and isomorphisms

Groups are compared through the maps that respect their operations.

Definition 3.37 (Group homomorphism). Let $(G, *)$ and $(G', \circ)$ be groups. A function $\phi: G \rightarrow G'$ is a *group homomorphism* if

$$\phi(a * b) = \phi(a) \circ \phi(b) \quad \text{for all } a, b \in G.$$

The set of all homomorphisms from G to G' is denoted $\text{Hom}(G, G')$.

Proposition 3.38. Let $\phi: G \rightarrow G'$ be a homomorphism, and write e, e' for the identities of G and G' .

1. $\phi(e) = e'$.
2. $\phi(a^{-1}) = \phi(a)^{-1}$ for all $a \in G$.
3. $\phi(a^n) = \phi(a)^n$ for all $a \in G$ and $n \in \mathbb{Z}$.
4. The image $\phi(G)$ is a subgroup of G' .

Proof. (1) From $e = e * e$ we get $\phi(e) = \phi(e) \circ \phi(e)$; cancelling $\phi(e)$ in G' (Proposition 3.3(3)) gives $e' = \phi(e)$.

(2) Applying ϕ to $a * a^{-1} = e = a^{-1} * a$ and using (1),

$$\phi(a) \circ \phi(a^{-1}) = e' = \phi(a^{-1}) \circ \phi(a),$$

so $\phi(a^{-1})$ is an inverse of $\phi(a)$; by uniqueness of inverses, $\phi(a^{-1}) = \phi(a)^{-1}$.

(3) For $n \geq 0$, induct: the case $n = 0$ is (1), and $\phi(a^{n+1}) = \phi(a^n * a) = \phi(a)^n \circ \phi(a) = \phi(a)^{n+1}$. For $n < 0$, write $n = -m$ with $m > 0$ and combine with (2): $\phi(a^{-m}) = \phi((a^{-1})^m) = \phi(a^{-1})^m = (\phi(a)^{-1})^m = \phi(a)^{-m}$.

(4) The image is nonempty, containing $e' = \phi(e)$. For $x = \phi(a)$ and $y = \phi(b)$ in $\phi(G)$,

$$x \circ y^{-1} = \phi(a) \circ \phi(b)^{-1} = \phi(a * b^{-1}) \in \phi(G),$$

using (2); the one-step subgroup test (Proposition 3.15) concludes. $\square$

Definition 3.39 (Kernel and image). The *kernel* and *image* of a homomorphism $\phi: G \rightarrow G'$ are

$$\ker(\phi) = \{a \in G : \phi(a) = e'\} = \phi^{-1}(\{e'\}), \quad \text{im}(\phi) = \phi(G) = \{\phi(a) : a \in G\}.$$

Proposition 3.40. For any homomorphism $\phi: G \rightarrow G'$, the kernel is a subgroup, $\ker(\phi) \leq G$, and ϕ is injective if and only if $\ker(\phi) = \{e\}$.

Proof. The kernel contains e by Proposition 3.38(1), so it is nonempty; and for $a, b \in \ker(\phi)$,

$$\phi(ab^{-1}) = \phi(a) \circ \phi(b)^{-1} = e' \circ (e')^{-1} = e',$$

so $ab^{-1} \in \ker(\phi)$; the one-step test gives $\ker(\phi) \leq G$.

If ϕ is injective and $\phi(a) = e' = \phi(e)$, then $a = e$; the kernel is trivial. Conversely, suppose $\ker(\phi) = \{e\}$ and $\phi(a) = \phi(b)$. Then $\phi(ab^{-1}) = \phi(a) \circ \phi(b)^{-1} = e'$, so $ab^{-1} \in \ker(\phi) = \{e\}$, forcing $ab^{-1} = e$, i.e. $a = b$. $\square$

Kernels are exactly the subgroups by which one may quotient: for abelian G , Proposition 3.35 forms $G/\ker(\phi)$ directly, and in general the kernel satisfies the normality condition of Remark 3.36—for $n \in \ker(\phi)$ and $g \in G$, one computes $\phi(gng^{-1}) = \phi(g) \circ e' \circ \phi(g)^{-1} = e'$, so conjugating the kernel by g lands back in the kernel, whence $g\ker(\phi) = \ker(\phi)g$.

Definition 3.41 (Isomorphism). An *isomorphism* is a bijective homomorphism $\phi: G \rightarrow G'$; when one exists, the groups are *isomorphic*, written $G \cong G'$. An isomorphism from G to itself is an *automorphism*. An injective homomorphism is a *monomorphism*; a surjective one is an *epimorphism*.

Proposition 3.42. *The set-theoretic inverse of an isomorphism is an isomorphism, and $\cong$ is an equivalence relation on any collection of groups.*

Proof. Let $\phi: G \rightarrow G'$ be an isomorphism, with inverse function ϕ^{-1} (which exists since ϕ is a bijection, Theorem 1.1). For $x, y \in G'$, write $x = \phi(a)$ and $y = \phi(b)$; then

$$\phi^{-1}(x \circ y) = \phi^{-1}(\phi(a * b)) = a * b = \phi^{-1}(x) * \phi^{-1}(y),$$

so ϕ^{-1} is a homomorphism, and it is a bijection; hence an isomorphism. For the equivalence relation: reflexivity is witnessed by the identity automorphism id_G ; symmetry by the inverse just constructed; and transitivity because a composite of homomorphisms is a homomorphism (apply the defining property twice) and a composite of bijections is a bijection. $\square$

Isomorphic groups are “the same group” for every group-theoretic purpose: any property expressible in terms of the operation alone—orders of elements, the lattice of subgroups, commutativity—transfers across an isomorphism. One therefore classifies groups *up to isomorphism*, and the classification of cyclic groups is the model result of this kind: up to isomorphism there is exactly one cyclic group of each order.

Theorem 3.43 (Classification of cyclic groups). *Every cyclic group is isomorphic to exactly one of the following:*

1. *the infinite cyclic group $(\mathbb{Z}, +)$, if it is infinite;*
2. *the finite cyclic group $(\mathbb{Z}/n\mathbb{Z}, +)$, if it has order n .*

In particular, two cyclic groups are isomorphic if and only if they have the same order.

Proof. Let $G = \langle g \rangle$ and define

$$\phi: \mathbb{Z} \rightarrow G, \quad \phi(k) = g^k.$$

The exponent law $g^{k+l} = g^k g^l$ says exactly that ϕ is a homomorphism from $(\mathbb{Z}, +)$ to G , and ϕ is surjective because every element of G is a power of g .

Case 1: g has infinite order. Distinct exponents give distinct powers (Proposition 3.21(1)), so ϕ is injective, hence a bijective homomorphism, and $\mathbb{Z} \cong G$.

Case 2: $\text{ord}(g) = n$. By Proposition 3.12(1),

$$\ker(\phi) = \{k \in \mathbb{Z} : g^k = e\} = n\mathbb{Z}.$$

Define the induced map

$$\bar{\phi}: \mathbb{Z}/n\mathbb{Z} \rightarrow G, \quad \bar{\phi}([k]_n) = g^k.$$

It is well defined: if $[k]_n = [l]_n$ then $n \mid (k - l)$, so $g^k = g^l$ by Proposition 3.12(2). It is a homomorphism: $\bar{\phi}([k]_n + [l]_n) = \bar{\phi}([k+l]_n) = g^{k+l} = \bar{\phi}([k]_n) \bar{\phi}([l]_n)$. It is surjective because ϕ is. And it is injective by Proposition 3.40: if $\bar{\phi}([k]_n) = g^k = e$, then $n \mid k$ (Proposition 3.12(1)), so $[k]_n = [0]_n$; the kernel of $\bar{\phi}$ is trivial. Hence $\mathbb{Z}/n\mathbb{Z} \cong G$.

Distinctness. Isomorphic groups have equal order, an isomorphism being in particular a bijection. The group $\mathbb{Z}$ is infinite and the groups $\mathbb{Z}/n\mathbb{Z}$ for $n = 1, 2, 3, \dots$ have the pairwise distinct finite orders n , so no two of the listed groups are isomorphic, and a cyclic group is isomorphic to exactly one of them—the one matching its order. The final claim follows: cyclic groups of equal order are isomorphic to the same listed group, hence to each other (Proposition 3.42), and isomorphic groups have equal order. $\square$

The cheque presented at the head of the section can now be cashed. Fix any finite group G and a public element $g \in G$, and let $n = \text{ord}(g)$. Associativity makes “ a repetitions of g ” unambiguous—the element g^a , however bracketed, by generalised associativity (Proposition 3.4)—and the exponent laws give

$$(g^a)^b = g^{ab} = (g^b)^a,$$

so the two parties of the opening protocol, each applying a secret repetition count to the other’s published value, really do arrive at a common secret element. Repetition counts obey an arithmetic, not a mere accumulation: by the classification theorem the subgroup $\langle g \rangle$ is a copy of $\mathbb{Z}/n\mathbb{Z}$, so exponents add, negate, and matter only modulo n (Proposition 3.12)—the modular arithmetic of section 2, transplanted into the exponent. Lagrange’s theorem locates the modulus: n divides $|G|$, and $a^{|G|} = e$ for every $a \in G$ (Corollary 3.31); in the unit groups $(\mathbb{Z}/n\mathbb{Z})^\times$ this specialises to Euler’s congruence $a^{\varphi(n)} \equiv 1 \pmod{n}$ (Corollary 3.32). Composition and inversion, the first two demands of the opening problem, are thereby supplied in full generality. The third demand—that publishing g^a should *hide* a —is not a theorem of algebra at all: it is a statement about computation, and it fails in some groups (in $(\mathbb{Z}/n\mathbb{Z}, +)$, recovering a from ag is a single modular division) while appearing to hold in others. Making the demand precise, and building the groups for which it is credible, occupies the sections that follow and the volumes above them.

4 Rings and fields

An arithmetic circuit—the object a proof system ultimately asks a verifier to check—is a network of gates, each of which either *adds* or *multiplies* two previously computed values. A verifier must therefore calculate in a number system where both operations make sense at once and interact through the familiar laws of school algebra; and at certain moments—inverting a random challenge, cancelling a common factor from both sides of an identity—the verifier must *divide* as well. In what kind of structure can one do both, and when can one also divide? The structures of genuine cryptographic interest all support two interacting operations obeying the arithmetic laws of $\mathbb{Z}$: the integers themselves, the rationals, the residue classes modulo n , polynomials, matrices. The abstract capture of this pattern is the *ring*; the key special case in which division by every nonzero element is possible is the *field*. Fields are the natural arena for linear algebra and for the theory of polynomials, and the finite fields $\mathbb{F}_p$ and $\mathbb{F}_q$ are the computational substrate of essentially all the elliptic-curve and proof-system machinery later in the monograph.

The section also discharges a debt. Section 2 verified, concretely and one law at a time, that congruence classes admit an arithmetic, and promised that the verified laws would receive their structural names in due course. They receive them here: $\mathbb{Z}/n\mathbb{Z}$ is re-read as a ring, indeed as the quotient of the ring $\mathbb{Z}$ by the ideal $n\mathbb{Z}$, and the theorem that $\mathbb{Z}/p\mathbb{Z}$ is a field is stated in the language that the rest of the series uses. As a standing convention we assume from §3 only the theory of *abelian* groups; additive structure is written additively with identity 0, multiplicative structure multiplicatively with identity 1.

4.1 Rings

Definition 4.1 (Ring). A *ring* is a set R equipped with two binary operations $+$ and $\cdot$ such that:

1. $(R, +)$ is an abelian group: addition is associative and commutative, there is an element 0 with $a + 0 = a$ for all a , and every a has an *additive inverse* $-a$ with $a + (-a) = 0$;
2. multiplication is associative: $a \cdot (b \cdot c) = (a \cdot b) \cdot c$;

3. there is a *multiplicative identity* 1 with $1 \cdot a = a \cdot 1 = a$ for all a ;
4. multiplication distributes over addition on both sides: $a \cdot (b + c) = a \cdot b + a \cdot c$ and $(a + b) \cdot c = a \cdot c + b \cdot c$.

The ring is *commutative* if moreover $a \cdot b = b \cdot a$ for all $a, b \in R$. We write ab for $a \cdot b$, let multiplication bind tighter than addition, and call 0 the *zero* and 1 the *unity* of the ring.

Remark 4.2. Conventions in the literature vary. Some authors omit requirement (3), saying “ring with unity” for the notion above and “rng” for the weaker one. In this monograph *every ring has a 1*, and the structure-preserving maps between rings—the ring homomorphisms of theorem 4.21 below—are required to preserve it. Moreover “ring” frequently means “commutative ring” in the examples that matter to us; commutativity is always stated explicitly when a result depends on it.

The distributive law is the only axiom coupling the two operations, and a surprising amount of everyday algebra follows from it alone. The following identities are used constantly and, after this proposition, without comment.

Proposition 4.3 (Elementary ring arithmetic). *Let R be a ring and $a, b \in R$. Then:*

1. $0 \cdot a = a \cdot 0 = 0$;
2. $(-a)b = a(-b) = -(ab)$;
3. $(-a)(-b) = ab$;
4. for all integers $m, n \in \mathbb{Z}$, $(ma)(nb) = (mn)(ab)$, where na denotes the n -fold additive multiple ($0a = 0$, $(n+1)a = na + a$, and $(-n)a = -(na)$, as for any abelian group, §3).

Proof. (1) From $0 = 0 + 0$ and distributivity, $0 \cdot a = (0 + 0) \cdot a = 0 \cdot a + 0 \cdot a$; adding $-(0 \cdot a)$ to both sides leaves $0 = 0 \cdot a$. The computation for $a \cdot 0$ is symmetric.

(2) By distributivity and (1), $ab + (-a)b = (a + (-a))b = 0 \cdot b = 0$, so $(-a)b$ is the additive inverse of ab , that is, $(-a)b = -(ab)$; likewise $ab + a(-b) = a(b + (-b)) = 0$ gives $a(-b) = -(ab)$.

(3) Applying (2) twice, $(-a)(-b) = -(a(-b)) = -(-(ab)) = ab$, the last step because negation is an involution in the abelian group $(R, +)$.

(4) First fix a, b and show $a(nb) = n(ab)$ for $n \geq 0$ by induction: the case $n = 0$ is (1), and $a((n+1)b) = a(nb + b) = a(nb) + ab = n(ab) + ab = (n+1)(ab)$ by distributivity and the inductive hypothesis. For $n < 0$ write $n = -m$ with $m > 0$; then $a(nb) = a(-(mb)) = -(a(mb)) = -(m(ab)) = n(ab)$ by (2). The same argument in the first factor gives $(ma)c = m(ac)$ for all c , and combining,

$$(ma)(nb) = m(a(nb)) = m(n(ab)) = (mn)(ab),$$

the final equality being the usual bookkeeping for iterated multiples in an abelian group, which we omit. $\square$

Remark 4.4 (The zero ring). Nothing in theorem 4.1 forces $1 \neq 0$. If $1 = 0$ in a ring R , then every $a \in R$ satisfies $a = 1 \cdot a = 0 \cdot a = 0$ by theorem 4.3(1), so $R = \{0\}$: the *zero ring* (or *trivial ring*), a legitimate ring in which both operations are the only possible ones. Whenever the zero ring must be excluded we impose the assumption $1 \neq 0$ explicitly; the definition of a field below builds $1 \neq 0$ in as a standing assumption.

Example 4.5 (Basic rings). 1. The integers $\mathbb{Z}$ form the prototype commutative ring; much of ring theory consists of isolating which features of $\mathbb{Z}$ survive in generality.

2. The rationals $\mathbb{Q}$, the reals $\mathbb{R}$, and the complexes $\mathbb{C}$ are commutative rings—indeed fields, in the sense of §4.4.

3. The residue classes $\mathbb{Z}/n\mathbb{Z} = \{\bar{0}, \bar{1}, \dots, \overline{n-1}\}$ for $n \geq 1$ —in this section’s examples we abbreviate the class $[a]_n$ of Definition 2.22 to $\bar{a}$ when the modulus is clear—under the induced operations form a commutative ring whose additive group is the cyclic group of order n (§3); this is theorem 4.6 below. We write $\mathbb{Z}/n\mathbb{Z}$, or $\mathbb{Z}_n$ when no confusion threatens. The case $n = 1$ gives the zero ring of theorem 4.4.
4. The *polynomial ring* $R[x]$ over a commutative ring R consists of the polynomials $a_0 + a_1x + \dots + a_dx^d$ with coefficients $a_i \in R$, added coefficient-wise and multiplied by expanding and collecting powers of x ; it is a commutative ring with the same 0 and 1 as R (the constant polynomials). The theory of polynomials over a field is developed in its own section later in the volume.
5. The *matrix ring* $M_n(R)$ of $n \times n$ matrices over a commutative ring R , $n \geq 1$, is a ring under matrix addition and multiplication. It is *not* commutative for $n \geq 2$ and $R \neq \{0\}$, even when R is commutative: in $M_2(\mathbb{Z})$,

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \text{but} \quad \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

This is the first important noncommutative example.

6. The *product ring* $R \times S$ of two rings carries the componentwise operations, with zero $(0, 0)$ and identity $(1, 1)$; likewise any finite product $R_1 \times \dots \times R_k$.
7. The zero ring $\{0\}$.

The third example is the one this monograph computes in, and it is the one whose ring structure was promised rather than named in §2. We now discharge that promise.

Theorem 4.6. *Let $n \geq 1$. The operations*

$$[a]_n + [b]_n := [a + b]_n, \quad [a]_n \cdot [b]_n := [ab]_n$$

are well defined and make $\mathbb{Z}/n\mathbb{Z}$ a commutative ring with additive identity $[0]_n$ and multiplicative identity $[1]_n$. It has exactly n elements.

Proof. The right-hand sides are defined in terms of chosen representatives a, b , so we must check that different choices give the same class. This is exactly the compatibility half of Proposition 2.21: if $a' \equiv a$ and $b' \equiv b \pmod{n}$, then $a' + b' \equiv a + b$ and $a'b' \equiv ab \pmod{n}$, so $[a' + b']_n = [a + b]_n$ and $[a'b']_n = [ab]_n$. The operations are therefore well defined on classes.

Every ring axiom is now inherited from the corresponding law of $\mathbb{Z}$ by passing to classes. For instance, distributivity:

$$\begin{aligned} [a]_n([b]_n + [c]_n) &= [a]_n[b + c]_n = [a(b + c)]_n = [ab + ac]_n \\ &= [ab]_n + [ac]_n = [a]_n[b]_n + [a]_n[c]_n, \end{aligned}$$

using only the definitions and the distributive law of the integers. Associativity and commutativity of both operations follow by the identical pattern. The class $[0]_n$ is an additive identity and $[1]_n$ a multiplicative one, since $[a + 0]_n = [a]_n$ and $[1 \cdot a]_n = [a]_n$; and $-[a]_n = [-a]_n$ because $[a]_n + [-a]_n = [a + (-a)]_n = [0]_n$.

Finally, by the division algorithm every integer is congruent modulo n to exactly one of $0, 1, \dots, n-1$ (the least nonnegative residues, Definition 2.22 and the discussion following it), so the classes $[0]_n, \dots, [n-1]_n$ are distinct and exhaust $\mathbb{Z}/n\mathbb{Z}$: the ring has exactly n elements. $\square$

4.2 Units, zero divisors, and integral domains

Two features of the integers now deserve names of their own. First, $\mathbb{Z}$ enjoys *cancellation*: a product of nonzero integers is nonzero, so a nonzero common factor may be struck from both sides of an equation. Second, division *within* $\mathbb{Z}$ is nonetheless rare: only ± 1 divide 1. The next two definitions abstract these phenomena—units capture the elements one can divide by, zero divisors the obstruction to cancelling.

Definition 4.7 (Unit). An element u of a ring R is a *unit* (or is *invertible*) if there exists $v \in R$ with $uv = vu = 1$. Such a v is unique: if $uv = vu = 1$ and $uv' = v'u = 1$, then $v = v \cdot 1 = v(uv') = (vu)v' = 1 \cdot v' = v'$. It is called the *inverse* of u and written u^{-1} . The set of units of R is denoted $R^\times$.

Proposition 4.8. For any ring R , the set $R^\times$ of units is a group under the multiplication of R , called the group of units (or multiplicative group) of R .

Proof. The identity 1 is a unit (it is its own inverse), so $1 \in R^\times$. For closure, let $u, w \in R^\times$; then

$$(uw)(w^{-1}u^{-1}) = u(ww^{-1})u^{-1} = u \cdot 1 \cdot u^{-1} = 1,$$

and symmetrically $(w^{-1}u^{-1})(uw) = 1$, so uw is a unit with $(uw)^{-1} = w^{-1}u^{-1}$. Multiplication in $R^\times$ is associative because it is associative in R . Finally each u^{-1} is itself a unit, its inverse being u ; so every element of $R^\times$ has an inverse in $R^\times$, and the group axioms are verified. $\square$

- Example 4.9** (Units). 1. The units of the integers are $\mathbb{Z}^\times = \{1, -1\}$: if $uv = 1$ in $\mathbb{Z}$ then $|u||v| = 1$ forces $|u| = 1$.
2. In $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$ every nonzero element is a unit: $\mathbb{Q}^\times = \mathbb{Q} \setminus \{0\}$, $\mathbb{R}^\times = \mathbb{R} \setminus \{0\}$, $\mathbb{C}^\times = \mathbb{C} \setminus \{0\}$. Commutative rings with $1 \neq 0$ in which *every* nonzero element is a unit are precisely the fields of §4.4; this is their defining feature. Both qualifiers are needed: the zero ring of theorem 4.4 satisfies the unit condition vacuously, and noncommutative rings satisfying it—the *division rings*—exist but are not fields.
3. In $\mathbb{Z}/n\mathbb{Z}$, the class $\bar{a}$ is a unit if and only if $\gcd(a, n) = 1$, by Theorem 2.24; thus $(\mathbb{Z}/n\mathbb{Z})^\times$ consists of the residue classes coprime to the modulus, and its order is Euler’s totient $\varphi(n)$ (Definition 2.26).
4. In the matrix ring, $M_n(R)^\times$ is the group of invertible $n \times n$ matrices. Over a field it is the *general linear group* GL_n , the matrices of nonzero determinant; the determinant belongs to linear algebra, treated later in the volume.

Definition 4.10 (Zero divisor). A nonzero element a of a ring R is a *left zero divisor* if $ab = 0$ for some nonzero $b \in R$, and a *right zero divisor* if $ba = 0$ for some nonzero b . In a commutative ring the two notions coincide and we say simply *zero divisor*. A nonzero element that is neither is called *regular* (or *cancellable*).

The terminology “cancellable” is justified by the following equivalence: zero divisors are exactly the obstruction to dividing out a common factor.

Proposition 4.11. *A nonzero element a of a ring R is not a left zero divisor if and only if it is left-cancellable: $ab = ac$ implies $b = c$ for all $b, c \in R$.*

Proof. Suppose a is not a left zero divisor and $ab = ac$. Then $a(b - c) = ab - ac = 0$, using theorem 4.3(2) to expand $a(b + (-c))$; since a is nonzero and not a left zero divisor, the factor $b - c$ must be 0, so $b = c$. Conversely, suppose a is a left zero divisor, say $ab = 0$ with $b \neq 0$. Then $ab = 0 = a \cdot 0$ while $b \neq 0$: cancellation fails. $\square$

Remark 4.12. A unit is never a zero divisor: if $u \in R^\times$ and $ub = 0$, then $b = 1 \cdot b = (u^{-1}u)b = u^{-1}(ub) = u^{-1} \cdot 0 = 0$. The converse fails in general—in $\mathbb{Z}$ every nonzero element is a non-zero-divisor, yet only ± 1 are units—but every nonzero non-zero-divisor is a unit in a finite commutative ring, by the injectivity-implies-surjectivity argument that proves theorem 4.29 below.

Example 4.13 (Zero divisors). 1. In $\mathbb{Z}/6\mathbb{Z}$, $\bar{2} \cdot \bar{3} = \bar{6} = \bar{0}$, so $\bar{2}$ and $\bar{3}$ are zero divisors. In general a nonzero class $\bar{a} \in \mathbb{Z}/n\mathbb{Z}$ is a zero divisor precisely when $d := \gcd(a, n) > 1$: in that case $a \cdot (n/d) = (a/d) \cdot n \equiv 0 \pmod{n}$ while $n/d \neq \bar{0}$ (as $0 < n/d < n$); and when $d = 1$ the class is a unit (Example 4.9(3)), hence no zero divisor by theorem 4.12. Thus in $\mathbb{Z}/n\mathbb{Z}$ every nonzero element is either a unit or a zero divisor—a dichotomy revisited from a structural angle by theorem 4.29.

2. In a product ring $R \times S$ with both factors nonzero, $(1, 0)(0, 1) = (0, 0)$: product rings always have zero divisors.
3. In $M_2(\mathbb{R})$, the diagonal matrices $\text{diag}(1, 0)$ and $\text{diag}(0, 1)$ multiply to the zero matrix; more generally every nonzero *singular* matrix—the standard name for a square matrix that is not invertible—is a zero divisor there.

Definition 4.14 (Integral domain). An *integral domain* (or simply *domain*) is a commutative ring R with $1 \neq 0$ and no zero divisors: $ab = 0$ implies $a = 0$ or $b = 0$. Equivalently, by theorem 4.11, a domain is a nonzero commutative ring in which every nonzero element is cancellable.

The residue rings sort themselves neatly against this definition, and the sorting is governed by primality.

Proposition 4.15. *Let $n \geq 2$. The ring $\mathbb{Z}/n\mathbb{Z}$ has no zero divisors if and only if n is prime; that is, $\mathbb{Z}/n\mathbb{Z}$ is an integral domain if and only if n is prime.*

Proof. If $n = ab$ is composite with $1 < a, b < n$, then $[a]_n[b]_n = [n]_n = [0]_n$ with both factors nonzero (their representatives lie strictly between 0 and n), so $\mathbb{Z}/n\mathbb{Z}$ has zero divisors. Conversely let $n = p$ be prime and suppose $[a]_p[b]_p = [0]_p$, that is, $p \mid ab$. Euclid’s lemma for primes (Lemma 2.17) gives $p \mid a$ or $p \mid b$, that is, $[a]_p = [0]_p$ or $[b]_p = [0]_p$. Since $p \geq 2$ we also have $[1]_p \neq [0]_p$, so $\mathbb{Z}/p\mathbb{Z}$ is an integral domain. $\square$

For prime p the ring $\mathbb{Z}/p\mathbb{Z}$ is in fact far better than a domain: every nonzero class is invertible, so it is a *field*—a term defined in §4.4, where the statement is proved as theorem 4.26.

Example 4.16 (Integral domains). The integers $\mathbb{Z}$ form an integral domain—the example the name commemorates. Every field is a domain (theorem 4.28 below). The ring $\mathbb{Z}/n\mathbb{Z}$ is a domain exactly when n is prime (theorem 4.15; the composite case is witnessed concretely by Example 4.13(1)). If R is a domain, so is the polynomial ring $R[x]$: let f and g be nonzero of degrees m and n —the largest indices carrying nonzero coefficients f_m and g_n , the *leading coefficients*. In fg

the coefficient of x^{m+n} is $\sum_{i+j=m+n} f_i g_j$; every term with $i > m$ or $j > n$ has a vanishing factor, and $i \leq m, j \leq n, i + j = m + n$ force $i = m, j = n$, so the coefficient equals $f_m g_n$, nonzero because R is a domain. Hence $fg \neq 0$. Non-domains include $\mathbb{Z}/6\mathbb{Z}$, product rings with two nonzero factors, and the matrix rings $M_n(R)$ for $n \geq 2$ (Example 4.13).

4.3 Ideals and quotient rings

We next carry quotient formation from groups to rings. For an abelian group and any subgroup, the cosets themselves form a group (Proposition 3.35); the ring-theoretic analogue asks for the subsets of a ring by which one can quotient while keeping *both* operations, and the answer is the notion of an *ideal*. The treatment here is deliberately brief: we need ideals chiefly to construct $\mathbb{Z}/n\mathbb{Z}$ conceptually and to indicate where the finite fields $\mathbb{F}_q$ come from; a fuller development belongs to commutative algebra.

Definition 4.17 (Ideal). Let R be a commutative ring. An *ideal* of R is a subset $I \subseteq R$ such that

1. I is a subgroup of the additive group $(R, +)$: $0 \in I$, and $a - b \in I$ whenever $a, b \in I$; and
2. I *absorbs* multiplication by arbitrary ring elements: $ra \in I$ whenever $r \in R$ and $a \in I$.

In noncommutative rings one distinguishes left, right, and two-sided ideals according to the side on which absorption holds; commutative rings suffice for this volume, and there the three notions coincide.

Despite containing 0 and being closed under the ring's operations in the sense above, an ideal is almost never a subring: if an ideal I contains 1—or, by the same absorption argument, any unit u , since then $1 = u^{-1}u \in I$ —then $r = r \cdot 1 \in I$ for every $r \in R$, so $I = R$. The only ideal containing a unit is the whole ring. This little observation will carry real weight: it is the reason a field has no ideals other than the two trivial ones, and hence no interesting quotients.

Example 4.18 (Ideals). 1. In any commutative ring R , the subsets $\{0\}$ and R are ideals—the *trivial* and *improper* ideals respectively.

2. For $a \in R$, the *principal ideal generated by a* is

$$(a) := aR = \{ar : r \in R\},$$

the set of multiples of a ; absorption holds because $r'(ar) = a(r'r)$. In $\mathbb{Z}$, the principal ideal $(n) = n\mathbb{Z}$ is the set of multiples of n . In fact *every* ideal of $\mathbb{Z}$ is principal: an ideal $I \neq \{0\}$ contains a nonzero element and hence (closing under negation) a positive one, so it contains a least positive element n by well-ordering; for any $a \in I$ the division algorithm gives $a = qn + r$ with $0 \leq r < n$, whence $r = a - qn \in I$ by absorption and subgroup closure, forcing $r = 0$ by minimality; so $I = n\mathbb{Z}$. This sharpens the observation of Remark 2.12, where ideals of $\mathbb{Z}$ first appeared as the sets of Bézout combinations.

3. For $a_1, \dots, a_k \in R$, the *ideal generated by $a_1, \dots, a_k$* is $(a_1, \dots, a_k) = \{r_1 a_1 + \dots + r_k a_k : r_i \in R\}$, the smallest ideal containing them all.

An integral domain in which every ideal is principal, as in (2), is called a *principal ideal domain* (PID).

Definition 4.19 (Quotient ring operations). Let I be an ideal of a commutative ring R . Since I is a subgroup of the abelian group $(R, +)$, the additive cosets $a + I$ form the quotient group R/I under coset addition $(a + I) + (b + I) = (a + b) + I$ (Proposition 3.35). Define a multiplication

of cosets by

$$(a + I)(b + I) := ab + I.$$

Proposition 4.20. *With the operations of theorem 4.19, the set R/I is a commutative ring—the quotient ring of R by I —with zero $0 + I = I$ and identity $1 + I$. In particular the coset multiplication is well defined.*

Proof. Well-definedness is the point at which absorption is needed. Let $a' = a + s$ and $b' = b + t$ be other representatives of the same cosets, with $s, t \in I$. Then

$$a'b' = (a + s)(b + t) = ab + at + sb + st,$$

and each of at, sb, st lies in I by the absorption property; hence $a'b' - ab \in I$ and $a'b' + I = ab + I$. The product coset is therefore independent of the representatives chosen.

The ring axioms now pass to cosets exactly as in the proof of theorem 4.6. For instance, distributivity reads

$$\begin{aligned} (a + I)((b + I) + (c + I)) &= (a + I)((b + c) + I) = a(b + c) + I \\ &= (ab + ac) + I = (a + I)(b + I) + (a + I)(c + I), \end{aligned}$$

and associativity and commutativity of both operations follow by the same pattern from the corresponding laws in R . The coset $I = 0 + I$ is the additive identity, $1 + I$ the multiplicative identity, and $-(a + I) = (-a) + I$. $\square$

One more definition names the maps under which ring structure is preserved; we need it to say precisely in what sense the quotient construction generalises the passage from $\mathbb{Z}$ to $\mathbb{Z}/n\mathbb{Z}$.

Definition 4.21 (Ring homomorphism). A *ring homomorphism* between rings R and S is a map $\phi: R \rightarrow S$ with

$$\phi(a + b) = \phi(a) + \phi(b), \quad \phi(ab) = \phi(a)\phi(b), \quad \phi(1) = 1$$

for all $a, b \in R$. Its *kernel* is $\ker \phi = \{a \in R : \phi(a) = 0\}$.

The kernel of a ring homomorphism out of a commutative ring is always an ideal: it is a subgroup of $(R, +)$ because ϕ is in particular a homomorphism of additive groups, and it absorbs because $a \in \ker \phi$ gives $\phi(ra) = \phi(r)\phi(a) = \phi(r) \cdot 0 = 0$.

Example 4.22. Take $R = \mathbb{Z}$ and $I = n\mathbb{Z}$. The cosets $a + n\mathbb{Z}$ are literally the residue classes $[a]_n$ of Definition 2.22, and the coset operations of theorem 4.19 are exactly residue arithmetic: the quotient construction recovers the ring $\mathbb{Z}/n\mathbb{Z}$ of Example 4.5(3), and is the conceptual origin of modular arithmetic. The same construction with a different ring produces the general finite fields: quotients $\mathbb{F}_p[x]/(f)$ of a polynomial ring by the principal ideal of an *irreducible* polynomial f of degree k —irreducibility, the polynomial analogue of primality, is defined in §5—furnish the finite fields $\mathbb{F}_q$ with $q = p^k$ elements. We record the recipe here and take it up in §6; the curves of Halo 2 and Orchard work over *prime* fields, so the prime case $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$ carries the main line of development.

Remark 4.23. Structurally, then, $\mathbb{Z}/n\mathbb{Z}$ is the quotient of the ring $\mathbb{Z}$ by the ideal $n\mathbb{Z} = \{kn : k \in \mathbb{Z}\}$, and the map

$$\pi: \mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}, \quad \pi(a) = [a]_n,$$

is a surjective ring homomorphism with kernel $n\mathbb{Z}$. This is the prototype of every quotient construction used in the volume's algebra: one starts from a ring, singles out an ideal of elements

to be “declared zero”, and computes with cosets. On notation: we abbreviate $[a]_n$ to $\bar{a}$ (as in the examples above) or drop the decoration altogether, writing plain a , whenever the modulus is clear from context; $\mathbb{Z}/n\mathbb{Z}$ and $\mathbb{Z}_n$ are used interchangeably; and $\mathbb{Z}_n$ always means this ring, never the n -adic integers (which do not appear in this series).

4.4 Fields

Definition 4.24 (Field). A *field* is a commutative ring F with $1 \neq 0$ in which every nonzero element is a unit. Equivalently, a field is a set F with two operations $+$ and $\cdot$ such that $(F, +)$ is an abelian group with identity 0, the nonzero elements $(F \setminus \{0\}, \cdot)$ form an abelian group with identity 1, and multiplication distributes over addition. Unwinding the definition: in a field one may add, subtract, multiply, and divide by any nonzero element, with $a/b := ab^{-1}$. The group $F^\times = (F \setminus \{0\}, \cdot)$ is the *multiplicative group* of the field. (Two points of the equivalence are not direct transcriptions. Passing from the first formulation to the second, one must show $F \setminus \{0\}$ is closed under multiplication; that is theorem 4.28 below. Passing back, the ring axioms must be checked for products involving 0: distributivity forces $0 \cdot a = a \cdot 0 = 0$ by the computation of theorem 4.3(1), whereupon every product with a zero factor vanishes and the associativity, commutativity, and identity laws extend from $F \setminus \{0\}$ to all of F ; and $1 \neq 0$ holds because the group $F \setminus \{0\}$ contains its identity 1.)

Definition 4.25 (Subfield and extension field). A subset $F \subseteq K$ of a field K is a *subfield* when it contains 0 and 1 and is itself a field under the operations of K restricted to F . When F is a subfield of K , the larger field K is called an *extension field* (or simply an *extension*) of F ; the two phrases name one relation viewed from its two ends. More generally, an injective ring homomorphism $F \hookrightarrow K$ between fields exhibits K as an extension of the copy of F inside it, and we permit ourselves the usual abuse of identifying F with that copy.

The finite world of §2 already contains the series’ most important fields, and the unit criterion proved there identifies them at once.

Corollary 4.26. *If p is prime, then every nonzero class in $\mathbb{Z}/p\mathbb{Z}$ is a unit, so $\mathbb{Z}/p\mathbb{Z}$ is a field; it is denoted $\mathbb{F}_p$. More generally, $\mathbb{Z}/n\mathbb{Z}$ is a field if and only if n is prime.*

Proof. Let p be prime and $[a]_p \neq [0]_p$, so that $p \nmid a$. The only positive divisors of p are 1 and p , so $\gcd(a, p) = 1$, and Theorem 2.24 makes $[a]_p$ a unit. Since $[1]_p \neq [0]_p$ for $p \geq 2$, the ring $\mathbb{Z}/p\mathbb{Z}$ is a field. Conversely, if $n = ab$ is composite with $1 < a, b < n$, then $[a]_n \neq [0]_n$ but $\gcd(a, n) = a > 1$, so $[a]_n$ is not a unit (Theorem 2.24 again) and the ring is not a field; and $n = 1$ gives the zero ring, which is not a field because a field has $1 \neq 0$ by definition. $\square$

Example 4.27 (Fields). 1. The rationals $\mathbb{Q}$, the reals $\mathbb{R}$, and the complexes $\mathbb{C}$ are fields. The integers are not: 2 has no inverse in $\mathbb{Z}$.

2. For p prime, $\mathbb{F}_p := \mathbb{Z}/p\mathbb{Z}$ is a field (theorem 4.26; theorem 4.29 below recovers the same fact abstractly). It is called the *prime field* of characteristic p —theorem 4.30 names the invariant—and is fundamental to every later cryptographic construction in the series. Its finite extensions $\mathbb{F}_q$ with $q = p^k$ elements exist for every prime power, arising from the quotient recipe of Example 4.22, and are developed in §6.

3. The subset $\mathbb{Q}(i) = \{a + bi : a, b \in \mathbb{Q}\} \subseteq \mathbb{C}$ is a field: the inverse of a nonzero $a + bi$ is $(a - bi)/(a^2 + b^2)$, which again has rational coordinates. More generally the *number fields*,

obtained by adjoining to $\mathbb{Q}$ roots of polynomial equations with rational coefficients, are fields.

4. For any field F , the *rational functions* $F(x) = \{f/g : f, g \in F[x], g \neq 0\}$ form a field under the usual arithmetic of fractions—the field of fractions of the polynomial ring $F[x]$.

Proposition 4.28. *Every field is an integral domain.*

Proof. Let F be a field. Commutativity and $1 \neq 0$ hold by definition. Suppose $ab = 0$ with $a \neq 0$. Then a is a unit, and multiplying by its inverse,

$$b = 1 \cdot b = (a^{-1}a)b = a^{-1}(ab) = a^{-1} \cdot 0 = 0.$$

Hence F has no zero divisors. □

The converse fails: $\mathbb{Z}$ is a domain but not a field (Example 4.16). Finiteness, however, closes the gap entirely, and does so by a counting device worth naming, for it recurs. Call it the *injectivity-implies-surjectivity argument*: to invert an element a of a finite structure, show that multiplication by a is injective, conclude that it is surjective because an injective map from a finite set to itself must be onto, and read off a preimage of 1. The same device proves the claim of theorem 4.12 that in a finite commutative ring every nonzero non-zero-divisor is a unit.

Theorem 4.29. *Every finite integral domain is a field.*

Proof. Let R be a finite integral domain and $a \in R$ nonzero. Consider the multiplication map

$$\mu_a: R \rightarrow R, \quad \mu_a(x) = ax.$$

It is injective: if $ax = ay$, then $a(x - y) = 0$, and since R is a domain and $a \neq 0$ this forces $x - y = 0$, that is, $x = y$ (equivalently, a is cancellable by theorem 4.11). An injective map from a finite set to itself is surjective: the image $\mu_a(R)$ has $|R|$ distinct elements and sits inside R , so it is all of R . In particular 1 lies in the image, so $ab = 1$ for some $b \in R$; commutativity gives $ba = 1$ as well, so $b = a^{-1}$ and a is a unit. Since R is a domain, $1 \neq 0$, and every nonzero element has just been shown invertible: R is a field. □

Applied to $\mathbb{Z}/p\mathbb{Z}$ —a finite domain by theorem 4.15—the theorem recovers theorem 4.26. The two proofs differ instructively. The pigeonhole argument merely asserts that each inverse *exists*; the extended Euclidean algorithm of §2 *computes* it, in $O(\log p)$ division steps. Cryptography needs both: the abstract statement to reason with, and the algorithm to run.

4.5 The characteristic

Every ring receives a canonical map from the integers, and the behaviour of that map is a fundamental invariant: it measures how much of $\mathbb{Z}$ survives inside the ring.

Definition 4.30 (Characteristic). Let R be a ring, and for $n \in \mathbb{Z}$ let $n \cdot 1$ denote the n -fold additive multiple of 1 (so $n \cdot 1 = 1 + \dots + 1$ with n summands for $n > 0$, $0 \cdot 1 = 0$, and $(-n) \cdot 1 = -(n \cdot 1)$). The *characteristic* $\text{char}(R)$ is the least positive integer n with $n \cdot 1 = 0$, if such an n exists, and 0 otherwise.

An equivalent formulation is often more useful. The map

$$\iota: \mathbb{Z} \rightarrow R, \quad \iota(n) = n \cdot 1,$$

is a ring homomorphism: additivity is the bookkeeping of iterated multiples in the abelian group $(R, +)$, multiplicativity is theorem 4.3(4) with $a = b = 1$ —namely $\iota(m)\iota(n) = (m \cdot 1)(n \cdot 1) = (mn)(1 \cdot 1) = \iota(mn)$ —and $\iota(1) = 1$. Its kernel is an ideal of $\mathbb{Z}$, hence of the form $n\mathbb{Z}$ for a unique $n \geq 0$ by Example 4.18(2), and $\text{char}(R)$ is exactly this nonnegative generator: $\text{char}(R) = 0$ precisely when ι is injective, so that a faithful copy of $\mathbb{Z}$ sits inside R , and $\text{char}(R) = n > 0$ when the kernel is $n\mathbb{Z}$.

Example 4.31 (Characteristics). The rings $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$ all have characteristic 0: no positive multiple of 1 vanishes. For the residue rings, $\text{char}(\mathbb{Z}/n\mathbb{Z}) = n$: the multiple $n \cdot \bar{1} = \bar{n} = \bar{0}$ vanishes, and no smaller positive multiple does, since $k \cdot \bar{1} = \bar{k} \neq \bar{0}$ for $0 < k < n$. In particular $\text{char}(\mathbb{F}_p) = p$.

Theorem 4.32. *The characteristic of an integral domain—in particular, of a field—is either 0 or a prime number.*

Proof. Let R be an integral domain with $\text{char}(R) = n > 0$; we show n is prime. First $n \neq 1$: otherwise $1 \cdot 1 = 1 = 0$, making R the zero ring (theorem 4.4) and contradicting $1 \neq 0$ in a domain. Suppose $n = ab$ with $1 < a, b < n$. By theorem 4.3(4),

$$0 = n \cdot 1 = (ab)(1 \cdot 1) = (a \cdot 1)(b \cdot 1),$$

so $a \cdot 1 = 0$ or $b \cdot 1 = 0$ since R is a domain. Either case exhibits a positive integer smaller than n whose multiple of 1 vanishes, contradicting the minimality of $n = \text{char}(R)$. Hence n admits no such factorisation, and being greater than 1, it is prime. $\square$

Remark 4.33 (The prime subfield). Let F be a field. If $\text{char}(F) = 0$, the homomorphism ι embeds a copy of $\mathbb{Z}$ in F , and dividing—possible in a field—a copy of $\mathbb{Q}$ as well. If $\text{char}(F) = p$, the image of ι is a copy of $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$. The smallest subfield so obtained is called the *prime subfield* of F ; every field is thus an extension of $\mathbb{Q}$ or of some $\mathbb{F}_p$, and the dichotomy between characteristic zero and characteristic p pervades the subject. The fields $\mathbb{F}_q$ underlying the cryptography treated in this series all have prime characteristic p . There the “freshman’s dream” identity $(a + b)^p = a^p + b^p$ actually holds, and gives rise to the Frobenius endomorphism; both are developed in the finite-fields section (theorem 6.13).

The question that opened the section is now answered in full. A verifier that must add and multiply computes in a ring; the laws verified one at a time for $\mathbb{Z}/n\mathbb{Z}$ in §2 are precisely the ring axioms, and the construction that produced them is the quotient of $\mathbb{Z}$ by the ideal $n\mathbb{Z}$. A verifier that must also divide computes in a field, and theorem 4.26 says exactly which moduli provide one: the primes. The payoff theorem, theorem 4.29, closes the circle with a purely structural guarantee: in a finite world there is no daylight between the absence of zero divisors and the presence of division—any nonzero finite commutative ring in which nonzero elements never multiply to zero is automatically a field. The prime fields $\mathbb{F}_p$ certified by these results are the ground on which the volume now builds: polynomials, linear algebra, roots of unity, and elliptic curves are all developed over them in the sections that follow.

5 Polynomials over a field

Throughout this section F denotes a field. Consider a problem that sounds impossible. Two parties each hold a polynomial of enormous degree—millions of coefficients, say, the encoding

of an entire computation—and wish to convince themselves that the two polynomials are equal. Comparing coefficient lists costs as much as reading the computation itself. Could a *single* spot check suffice: pick one value α at random, evaluate both polynomials there, and declare them equal if the two field elements agree? The astonishing answer is yes—with an error probability so small that no adversary can exploit it—and this section develops exactly the algebra that makes the answer honest.

Polynomials are the engine behind finite fields, error-correcting codes, and—most importantly here—the polynomial commitment schemes and *arithmetisation* (the encoding of a computation as identities between polynomials) underlying Halo 2 and Zcash Orchard. The single most exploited fact in that edifice is elementary: a nonzero polynomial of degree d over a field has at most d roots. From this one fact the section extracts Lagrange interpolation and the Schwartz–Zippel principle, the probabilistic heart of every interactive proof we eventually construct. The only prerequisite is the field theory of the preceding section.

5.1 The polynomial ring

One point of principle governs everything that follows, so we state it before the first definition: *a polynomial is not a function*. It is a formal expression—at bottom, a finite list of coefficients—which one may *evaluate* to obtain a function, but which in general carries more information than that function. Over the finite fields of cryptographic interest the distinction is not pedantic but essential, and the closing subsections of this section (§5.9) return to it at length.

Definition 5.1 (Polynomial). A *polynomial* over F in the *indeterminate* X is a formal expression

$$f = a_0 + a_1X + a_2X^2 + \cdots + a_nX^n = \sum_{i=0}^n a_iX^i,$$

where $n \in \mathbb{N} = \{0, 1, 2, \dots\}$ and the *coefficients* $a_0, \dots, a_n$ lie in F . Formally, f is identified with its coefficient sequence $(a_0, a_1, a_2, \dots)$, in which only finitely many entries are nonzero. Two polynomials are equal precisely when their coefficient sequences agree term by term:

$$\sum_i a_iX^i = \sum_i b_iX^i \iff a_i = b_i \text{ for every } i \in \mathbb{N}.$$

The symbol X is an *indeterminate* (a formal variable): it is not an element of F and stands for nothing in particular; it is merely a placeholder organising the coefficients. The set of all polynomials over F in X is denoted $F[X]$.

Remark 5.2 (The sequence model). The fully rigorous rendering of theorem 5.1 takes $F[X]$ to be the set of all functions $a: \mathbb{N} \rightarrow F$ that are *eventually zero*: there is some N with $a(i) = 0$ for all $i > N$. The expression $\sum_i a_iX^i$ is then merely suggestive notation for the sequence $(a_0, a_1, \dots)$. We use the two viewpoints interchangeably; the notation is justified because the arithmetic of these sequences, defined next, is exactly the arithmetic the notation suggests.

Definition 5.3 (Ring operations on $F[X]$). Let $f = \sum_i a_iX^i$ and $g = \sum_i b_iX^i$ lie in $F[X]$, with the shorter coefficient sequence zero-padded so that both run over the same index set. Their *sum* and *product* are

$$f + g = \sum_{i \geq 0} (a_i + b_i) X^i, \quad fg = \sum_{k \geq 0} c_k X^k \quad \text{where} \quad c_k = \sum_{i+j=k} a_i b_j = \sum_{i=0}^k a_i b_{k-i}.$$

The coefficient c_k is the *convolution* of the two coefficient sequences. Both defining sums are finite—only finitely many a_i and b_j are nonzero, hence only finitely many c_k are nonzero—so the sum and the product are again polynomials.

The convolution formula is exactly what multiplying out $(\sum_i a_i X^i)(\sum_j b_j X^j)$ and collecting like powers of X gives, using $X^i \cdot X^j = X^{i+j}$. Stated as a convolution, however, it makes no appeal to substituting a value for X : it is a purely formal manipulation of coefficient sequences. The formal/functional distinction announced above is thereby kept intact at the level of the operations themselves.

The verification that these operations obey the ring axioms (theorem 4.1) is a single device applied over and over: to prove an identity between polynomials, compute the coefficient of X^m on each side as a finite sum over index decompositions, then reindex or regroup that sum. No evaluation is ever involved. The associativity step below is the archetype.

Theorem 5.4 ($F[X]$ is a commutative ring). *With the operations of theorem 5.3, the set $F[X]$ is a commutative ring whose additive identity is the zero polynomial $0 = (0, 0, 0, \dots)$ and whose multiplicative identity is the constant polynomial $1 = (1, 0, 0, \dots)$. Moreover the map $c \mapsto (c, 0, 0, \dots)$ embeds F into $F[X]$ as a subring, the constant polynomials; in this sense $F \subseteq F[X]$.*

Proof. Additive structure. Addition is coordinatewise, so its commutativity and associativity are inherited directly from the corresponding laws of the additive group of F , the sequence $(0, 0, \dots)$ is an additive identity, and $(-a_0, -a_1, \dots)$ is an additive inverse of $(a_0, a_1, \dots)$. Hence $(F[X], +)$ is an abelian group.

Commutativity of multiplication. The coefficient of X^k in fg is $\sum_{i+j=k} a_i b_j$; reindexing the sum by $(i, j) \mapsto (j, i)$ and using commutativity of multiplication in F ,

$$\sum_{i+j=k} a_i b_j = \sum_{i+j=k} b_i a_j,$$

which is the coefficient of X^k in gf .

Associativity of multiplication. Write $h = \sum_l c_l X^l$. The coefficient of X^m in $(fg)h$ is

$$\sum_{k+l=m} \left(\sum_{i+j=k} a_i b_j \right) c_l = \sum_{i+j+l=m} a_i b_j c_l,$$

and expanding $f(gh)$ instead produces $\sum_{i+j+l=m} a_i (b_j c_l)$, the same symmetric triple sum by associativity in F . Hence $(fg)h = f(gh)$ coefficient by coefficient.

Distributivity. The coefficient of X^k in $f(g + h)$ is $\sum_{i+j=k} a_i (b_j + c_j)$, which distributivity in F splits as $\sum_{i+j=k} a_i b_j + \sum_{i+j=k} a_i c_j$: the coefficient of X^k in $fg + fh$. The other distributive law follows from this one and commutativity.

Identity. With $1 = (1, 0, 0, \dots)$, the k -th coefficient of $1 \cdot f$ is $\sum_{i+j=k} 1[i = 0] a_j = a_k$, where $1[i = 0]$ denotes the coefficient sequence of 1 (equal to 1 at index 0 and 0 elsewhere); so $1 \cdot f = f$.

Embedding. Let $\iota(c) = (c, 0, 0, \dots)$. The map ι preserves sums coordinatewise, and $\iota(c)\iota(d)$ has k -th coefficient cd for $k = 0$ and 0 otherwise, so $\iota(c)\iota(d) = \iota(cd)$; also $\iota(1) = 1$. Distinct constants give distinct sequences, so ι is an injective ring homomorphism (theorem 4.21) and its image is a subring of $F[X]$ isomorphic to F . $\square$

Remark 5.5 (Scalar multiplication). Multiplying $f = \sum_i a_i X^i$ by a constant $c \in F$, viewed as the constant polynomial $\iota(c)$, gives the *scalar multiple* $cf = \sum_i (ca_i) X^i$. Together with addition, this makes $F[X]$ a *vector space* over F —a structure studied systematically in the linear-algebra section later in the volume; informally, a system closed under addition and under scaling by field elements, in which the monomials $1, X, X^2, X^3, \dots$ play the role of an infinite coordinate system (a *basis*). This linear structure is relied on constantly, for instance whenever we count polynomials of bounded degree.

5.2 Degree

Definition 5.6 (Degree, leading coefficient, monic). Let $f = \sum_i a_i X^i \in F[X]$ be nonzero. The *degree* $\deg f$ is the largest index n with $a_n \neq 0$; the coefficient a_n is the *leading coefficient* $\text{lc}(f)$, the term $a_n X^n$ is the *leading term*, and a_0 is the *constant term*. A nonzero polynomial is *monic* if its leading coefficient is 1. The constant polynomials are the elements of F embedded by theorem 5.4; the nonzero ones have degree 0. The zero polynomial is assigned $\deg 0 = -\infty$, with the conventions $-\infty < n$, $-\infty + n = -\infty$, and $-\infty + (-\infty) = -\infty$ for every $n \in \mathbb{Z}$. Polynomials of degree 1, 2, and 3 are called *linear*, *quadratic*, and *cubic* respectively.

The convention $\deg 0 = -\infty$ is not pedantry: it makes the degree formulas below hold without exceptions, which streamlines every later induction on degree. Induction on degree is thereby set up as a reusable device for the rest of the volume.

Proposition 5.7 (Degree of sums and products). For all $f, g \in F[X]$,

$$\deg(f + g) \leq \max\{\deg f, \deg g\}, \quad (1)$$

with equality whenever $\deg f \neq \deg g$, and

$$\deg(fg) = \deg f + \deg g. \quad (2)$$

Proof. If either polynomial is zero, both statements reduce to the $-\infty$ conventions of theorem 5.6. So let f and g be nonzero, of degrees m and n with leading coefficients a_m and b_n .

Sum. The coefficient of X^k in $f + g$ is $a_k + b_k$, which vanishes for $k > \max\{m, n\}$; this is (1). If $m > n$, the coefficient of X^m is $a_m + 0 = a_m \neq 0$, giving equality (symmetrically for $n > m$). When $m = n$, cancellation $a_m + b_m = 0$ can lower the degree, which is why the sum bound is only an inequality.

Product. The coefficient $c_{m+n} = \sum_{i+j=m+n} a_i b_j$ has only the surviving term $i = m, j = n$: terms with $i > m$ have $a_i = 0$, and terms with $i < m$ force $j > n$, so $b_j = 0$. Hence $c_{m+n} = a_m b_n$, and the same argument shows $c_k = 0$ for $k > m+n$. Since F is a field it has no zero divisors (theorem 4.28), so $a_m b_n \neq 0$; therefore $\deg(fg) = m + n$ and $\text{lc}(fg) = \text{lc}(f) \text{lc}(g)$. $\square$

The product formula (2) is the first place the field hypothesis does real work: it required $a_m b_n \neq 0$, that is, the absence of zero divisors in F . This single fact has sweeping consequences for everything that follows, beginning with the next three corollaries.

Corollary 5.8 ($F[X]$ is an integral domain). The ring $F[X]$ has no zero divisors: if $fg = 0$, then $f = 0$ or $g = 0$.

Proof. If f and g are both nonzero, then $\deg(fg) = \deg f + \deg g \geq 0$ by (2), so $fg \neq 0$. Contrapositively, $fg = 0$ forces $f = 0$ or $g = 0$. $\square$

Corollary 5.9 (Cancellation). Let $f, g, h \in F[X]$ with $h \neq 0$. If $fh = gh$, then $f = g$.

Proof. From $fh = gh$ we get $(f - g)h = 0$; since $h \neq 0$, theorem 5.8 forces $f - g = 0$. $\square$

Proposition 5.10 (Units of $F[X]$). *A polynomial $f \in F[X]$ is a unit—has a multiplicative inverse in $F[X]$ —if and only if it is a nonzero constant. Thus $F[X]^\times = F^\times$.*

Proof. A nonzero constant c has the inverse $c^{-1} \in F \subseteq F[X]$. Conversely, $fg = 1$ gives $\deg f + \deg g = \deg 1 = 0$ by (2); since degrees of nonzero polynomials are nonnegative, $\deg f = \deg g = 0$, and f is a nonzero constant. $\square$

Definition 5.11 (Associates). Two polynomials $f, g \in F[X]$ are *associates* if $f = ug$ for some unit $u \in F^\times$, that is, if they differ only by a nonzero scalar factor.

Every nonzero polynomial f has exactly one monic associate, obtained by dividing through by its leading coefficient: $\text{lc}(f)^{-1}f$. Choosing the monic representative is the standard way to pin down *the* greatest common divisor or *the* irreducible factor uniquely, and we adopt this normalisation below.

5.3 Division with remainder

The arithmetic of $F[X]$ parallels that of the integers $\mathbb{Z}$ with remarkable fidelity: greatest common divisors, a Euclidean algorithm, a Bézout identity, primes (here called irreducibles), and unique factorisation all reappear. The source of the parallel is a single statement, division with remainder, in which *degree* plays the role that absolute value plays in $\mathbb{Z}$.

Theorem 5.12 (Division algorithm). *Let $f, g \in F[X]$ with $g \neq 0$. There exist unique polynomials $q, r \in F[X]$ such that*

$$f = qg + r \quad \text{and} \quad \deg r < \deg g.$$

We call q the quotient and r the remainder, and write $q = f \text{ div } g$ and $r = f \text{ mod } g$.

Proof. Existence. Fix g , of degree n with leading coefficient b_n , and induct on $\deg f$. If $\deg f < n$ —including $f = 0$ with $\deg f = -\infty$ —take $q = 0$ and $r = f$. Otherwise let $m = \deg f \geq n$, with leading coefficient a_m , and form

$$f_1 = f - \frac{a_m}{b_n} X^{m-n} g.$$

The subtracted term has degree m and leading coefficient $(a_m/b_n)b_n = a_m$, so it cancels the leading term of f : either $f_1 = 0$ or $\deg f_1 < m$. By the induction hypothesis $f_1 = q_1 g + r$ with $\deg r < n$, and then $f = (q_1 + (a_m/b_n)X^{m-n})g + r$. The field hypothesis enters visibly here: division by the leading coefficient b_n requires b_n^{-1} to exist.

Uniqueness. Suppose $f = qg + r = q'g + r'$ with both remainders of degree $< n$. Then $(q - q')g = r' - r$. If $q \neq q'$, then $\deg((q - q')g) = \deg(q - q') + n \geq n$ by (2), whereas $\deg(r' - r) \leq \max\{\deg r', \deg r\} < n$ by (1)—a contradiction. Hence $q = q'$, and then $r = f - qg = r'$. $\square$

Remark 5.13 (Polynomial long division). The existence proof is constructive, and it is precisely *polynomial long division*: repeatedly eliminate the current leading term by subtracting an appropriate monomial multiple of g . Each step strictly lowers the degree of the running remainder, so the process terminates after at most $\deg f - \deg g + 1$ steps when $\deg f \geq \deg g$, and takes no steps when $f = 0$ or $\deg f < \deg g$. The device is reusable in both of its guises: as an algorithm, and as the leading-term-elimination pattern for inductions on degree.

For a concrete instance, divide $f = X^3 + 2X + 1$ by $g = X^2 + 1$ over $\mathbb{Q}$. Subtracting $X \cdot g = X^3 + X$ leaves $X + 1$, whose degree 1 is already below $\deg g = 2$; hence $q = X$ and $r = X + 1$, and indeed

$$X^3 + 2X + 1 = X(X^2 + 1) + (X + 1).$$

One elimination step suffices here.

Definition 5.14 (Divisibility). For $f, g \in F[X]$, we say g *divides* f , written $g \mid f$, if $f = qg$ for some $q \in F[X]$; equivalently, for $g \neq 0$, if the remainder $f \bmod g$ is 0. In that case g is a *divisor* (or *factor*) of f .

5.4 Roots and the factor theorem

The destination of this subsection is the central counting bound of the whole section, stated at once so that the reader knows what the machinery is for: *a nonzero polynomial of degree d over a field has at most d distinct roots* (theorem 5.18 below). Everything the volume later builds on polynomials—the cyclicity of the multiplicative group of a finite field, Lagrange interpolation, the Schwartz–Zippel principle—is load-bearing on this one fact. To reach it we must first say what a root is, and that requires making substitution precise: this is the first place where a polynomial gives rise to a function.

Definition 5.15 (Evaluation and roots). Let $f = \sum_{i=0}^n a_i X^i \in F[X]$ and $\alpha \in F$. The *evaluation* of f at α is the field element

$$f(\alpha) = \sum_{i=0}^n a_i \alpha^i \in F,$$

obtained by substituting α for the indeterminate X . An element $\alpha \in F$ is a *root* (or *zero*) of f if $f(\alpha) = 0$.

Proposition 5.16 (Evaluation is a ring homomorphism). Fix $\alpha \in F$. The evaluation map $\text{ev}_\alpha: F[X] \rightarrow F, f \mapsto f(\alpha)$, satisfies

$$(f + g)(\alpha) = f(\alpha) + g(\alpha), \quad (fg)(\alpha) = f(\alpha)g(\alpha), \quad 1(\alpha) = 1$$

for all $f, g \in F[X]$; that is, ev_α is a ring homomorphism.

Proof. Additivity is immediate from the coordinatewise definition of addition. For multiplicativity, with $c_k = \sum_{i+j=k} a_i b_j$ as in theorem 5.3,

$$(fg)(\alpha) = \sum_k c_k \alpha^k = \sum_k \sum_{i+j=k} a_i b_j \alpha^{i+j} = \left(\sum_i a_i \alpha^i \right) \left(\sum_j b_j \alpha^j \right) = f(\alpha)g(\alpha),$$

the middle equality because expanding the product of the two finite sums and grouping terms by $k = i + j$ produces exactly the double sum. The formal reason evaluation respects the algebra is that the convolution defining polynomial multiplication collapses to ordinary multiplication in F via $\alpha^i \alpha^j = \alpha^{i+j}$. Finally $1(\alpha) = 1$ by direct substitution. $\square$

Theorem 5.17 (Remainder theorem; factor theorem). Let $f \in F[X]$ and $\alpha \in F$.

1. (Remainder theorem.) The remainder of f upon division by the linear polynomial $X - \alpha$ is the constant $f(\alpha)$: there is $q \in F[X]$ with $f = q(X - \alpha) + f(\alpha)$.
2. (Factor theorem.) The element α is a root of f if and only if $(X - \alpha) \mid f$.

Proof. (1) Divide f by $X - \alpha$ (theorem 5.12): since $\deg(X - \alpha) = 1$, the remainder has degree < 1 , hence is a constant $r \in F$. Evaluating $f = q(X - \alpha) + r$ at α via theorem 5.16 gives $f(\alpha) = q(\alpha)(\alpha - \alpha) + r = r$.

(2) By (1) and the uniqueness of the remainder, $(X - \alpha) \mid f$ holds exactly when the remainder $f(\alpha)$ is 0, that is, when α is a root of f . $\square$

The factor theorem is an immediate but powerful consequence of the division algorithm: it converts a statement about *values* (a root) into a statement about *divisibility* (a linear factor). Iterating that conversion proves the promised bound.

Theorem 5.18 (At most d roots). *A nonzero polynomial $f \in F[X]$ of degree d has at most d distinct roots in F .*

Proof. We induct on $d = \deg f$. If $d = 0$, then f is a nonzero constant, which evaluates to itself everywhere and so has $0 \leq d$ roots. Let $d \geq 1$. If f has no root in F the claim holds; otherwise let α be a root. By the factor theorem, $f = (X - \alpha)q$, and $\deg q = d - 1$ by (2). Now let $\beta \neq \alpha$ be any other root of f . Evaluating with theorem 5.16,

$$0 = f(\beta) = (\beta - \alpha)q(\beta);$$

since $\beta - \alpha \neq 0$ and F has no zero divisors (theorem 4.28), we get $q(\beta) = 0$. Every root of f other than α is therefore a root of q , and by the induction hypothesis q has at most $d - 1$ distinct roots. Hence f has at most $(d - 1) + 1 = d$. $\square$

Remark 5.19 (Why a field is needed). Both ingredients of the proof—the absence of zero divisors and the degree bookkeeping—are essential. Over a ring with zero divisors the bound fails outright: in $\mathbb{Z}/8\mathbb{Z}$ the quadratic $X^2 - 1$ has the four roots $\bar{1}, \bar{3}, \bar{5}, \bar{7}$, since $1^2 = 1$, $3^2 = 9 \equiv 1$, $5^2 = 25 \equiv 1$, and $7^2 = 49 \equiv 1 \pmod{8}$ —a degree-2 polynomial with four roots. This is why the polynomial methods underlying Halo 2 require a field, and indeed a finite field $\mathbb{F}_p$ or $\mathbb{F}_q$, beneath them.

Corollary 5.20 (Few points determine a low-degree polynomial). *Let $f, g \in F[X]$ both have degree at most d . If $f(\alpha) = g(\alpha)$ for $d + 1$ distinct values $\alpha \in F$, then $f = g$ as polynomials.*

Proof. The difference $h = f - g$ has $\deg h \leq \max\{\deg f, \deg g\} \leq d$ by (1) and vanishes at $d + 1$ distinct points. Were h nonzero, it would be a polynomial of degree at most d with $d + 1$ distinct roots, contradicting theorem 5.18. Hence $h = 0$, that is, $f = g$. $\square$

The corollary is the deterministic skeleton of two constructions developed later in this section: Lagrange interpolation (§5.7), its constructive converse, and the Schwartz–Zippel principle (§5.10), its probabilistic refinement. The root bound is thus positioned as the load-bearing fact behind the polynomial identity testing and interpolation machinery used by the proving system.

5.5 Greatest common divisors

Because $F[X]$ supports division with remainder, the entire theory of greatest common divisors from elementary number theory (§2) transfers verbatim from $\mathbb{Z}$ to $F[X]$, with degree replacing absolute value as the measure that decreases. We run through the transfer explicitly, both because the statements are used constantly and because the polynomial Euclidean algorithm is itself a tool of the trade.

Definition 5.21 (Greatest common divisor). Let $f, g \in F[X]$, not both zero. A *greatest common divisor* of f and g is a polynomial d such that

1. $d \mid f$ and $d \mid g$; and

2. every common divisor c of f and g satisfies $c \mid d$.

A greatest common divisor is determined only up to multiplication by a unit; the unique *monic* one (theorem 5.23) is singled out and denoted $\gcd(f, g)$. When $\gcd(f, g) = 1$, the polynomials f and g are called *coprime* (or *relatively prime*). Since every polynomial divides 0, we have $\gcd(f, 0) = \text{lc}(f)^{-1}f$ for $f \neq 0$.

Lemma 5.22 (Euclidean step). *Let $f, g \in F[X]$ with $g \neq 0$, and let $f = qg + r$ be the division of theorem 5.12. Then the pairs (f, g) and (g, r) have exactly the same common divisors; in particular, their greatest common divisors coincide.*

Proof. If $c \mid f$ and $c \mid g$, then $c \mid (f - qg) = r$; so c is a common divisor of (g, r) . Conversely, if $c \mid g$ and $c \mid r$, then $c \mid (qg + r) = f$. The two sets of common divisors are therefore equal, hence so are their maximal elements under divisibility. $\square$

Theorem 5.23 (Euclidean algorithm; Bézout's identity). *Let $f, g \in F[X]$, not both zero. Then $\gcd(f, g)$ exists and is unique as a monic polynomial, and there exist $s, t \in F[X]$ with*

$$\gcd(f, g) = sf + tg. \quad (3)$$

Concretely: set $r_0 = f$ and $r_1 = g$, and while $r_i \neq 0$ divide

$$r_{i-1} = q_i r_i + r_{i+1}, \quad \deg r_{i+1} < \deg r_i.$$

The degrees strictly decrease, so some $r_{N+1} = 0$; the last nonzero remainder r_N , made monic, is $\gcd(f, g)$.

Proof. Termination. If $r_1 = 0$, the algorithm stops at once with $N = 0$ and $r_N = f \neq 0$. Otherwise the degrees $\deg r_1 > \deg r_2 > \dots$ form a strictly decreasing sequence of nonnegative integers, which cannot continue forever; so some $r_{N+1} = 0$ with $r_N \neq 0$.

The gcd property. Theorem 5.22, applied at each division, preserves the set of common divisors:

$$\begin{aligned} \{\text{common divisors of } (f, g)\} &= \{\text{common divisors of } (r_1, r_2)\} \\ &= \dots = \{\text{common divisors of } (r_N, 0)\}, \end{aligned}$$

and the last set is exactly the set of divisors of r_N , since every polynomial divides 0. Thus r_N divides both f and g , and every common divisor of f and g divides r_N : the monic associate $\text{lc}(r_N)^{-1}r_N$ is a monic greatest common divisor.

Uniqueness. Two monic greatest common divisors divide each other, so they differ by a unit $u \in F^\times$ (theorem 5.10); comparing leading coefficients of monic polynomials forces $u = 1$.

Bézout. We show by induction on i that each remainder r_i is an $F[X]$ -linear combination of f and g . The bases $r_0 = 1 \cdot f + 0 \cdot g$ and $r_1 = 0 \cdot f + 1 \cdot g$ are trivial, and the recurrence $r_{i+1} = r_{i-1} - q_i r_i$ expresses r_{i+1} as a combination once r_{i-1} and r_i are. Hence $r_N = s_0 f + t_0 g$ for some $s_0, t_0 \in F[X]$; dividing through by $\text{lc}(r_N)$ yields (3). $\square$

Remark 5.24 (Extended Euclidean algorithm). Carrying the linear-combination coefficients along with the remainders, exactly as in the integer case (Theorem 2.13), produces s and t explicitly alongside the gcd. The standard application is inversion in a quotient $F[X]/(m)$ for irreducible m : if $\gcd(f, m) = 1$, then $sf + tm = 1$, so s is the inverse of f modulo m . This is how one divides in the extension fields $\mathbb{F}_q$ constructed in section 6. The polynomial version matters both because it parallels the integer one exactly and because it computes gcds of the polynomials a proof system

manipulates. A caveat on deployed practice: the curves of Halo 2 and Orchard are defined over prime fields $\mathbb{F}_p$, not extension fields, and field inversion there uses the integer extended Euclidean algorithm (Theorem 2.13) or Fermat's little theorem—not the polynomial algorithm in $F[X]/(m)$.

Corollary 5.25 (Euclid's lemma in $F[X]$). *Let $p, f, g \in F[X]$.*

1. *If $\gcd(p, f) = 1$ and $p \mid fg$, then $p \mid g$.*
2. *If p is irreducible and $p \mid fg$, then $p \mid f$ or $p \mid g$.*

Here irreducible is the polynomial analogue of prime, made precise in theorem 5.26 below.

Proof. (1) Multiplying the Bézout identity $sp + tf = 1$ by g gives $spg + tfg = g$. The polynomial p divides both terms on the left—the second because $p \mid fg$ —hence divides g .

(2) Suppose $p \nmid f$. The gcd $\gcd(p, f)$ is a monic divisor of p that is not the monic associate of p (otherwise $p \mid f$). By irreducibility the only monic divisors of p are 1 and its monic associate, so $\gcd(p, f) = 1$ and part (1) applies. $\square$

5.6 Unique factorisation

Definition 5.26 (Irreducible polynomial). A polynomial $p \in F[X]$ is *irreducible over F* if $\deg p \geq 1$ and p admits no factorisation $p = gh$ with both $\deg g \geq 1$ and $\deg h \geq 1$; equivalently, p is nonconstant and its only factorisations are the trivial ones, (unit) $\times$ (associate of p). A nonconstant polynomial that is not irreducible is *reducible*.

Remark 5.27 (Irreducibility is relative to the base field). The polynomial $X^2 + 1$ is irreducible over $\mathbb{R}$ (it has no real root, and a real quadratic factors precisely when it has a root, by the test below), yet splits as $(X - i)(X + i)$ over $\mathbb{C}$, and factors as $(X + 1)^2$ over $\mathbb{F}_2$, by direct expansion in characteristic 2. Degree-1 polynomials are irreducible over every field. A useful *rootlessness test*: a polynomial of degree 2 or 3 is irreducible over F if and only if it has no root in F , because any nontrivial factorisation of a quadratic or cubic must include a linear factor, which by the factor theorem (theorem 5.17) supplies a root. The test fails from degree 4 onward: a product of two irreducible quadratics has no roots yet is reducible.

Theorem 5.28 ($F[X]$ is a unique factorisation domain). *Every nonconstant $f \in F[X]$ factors as*

$$f = c p_1 p_2 \cdots p_k$$

with $c = \text{lc}(f) \in F^\times$ a constant and each p_i monic irreducible. The factorisation is unique up to order: if also $f = c' q_1 \cdots q_l$ with $c' \in F^\times$ and each q_j monic irreducible, then $c = c'$, $k = l$, and the q_j are a reordering of the p_i .

Proof. The proof is templated on the fundamental theorem of arithmetic (Theorem 2.18), with degree in place of absolute value.

Existence, by induction on $\deg f \geq 1$. If f is irreducible, then $f = \text{lc}(f) \cdot p_1$ with p_1 its monic associate. If $f = gh$ is reducible with $1 \leq \deg g, \deg h < \deg f$, the induction hypothesis factors g and h , and multiplying the two factorisations—collecting the two constants into one—factors f .

Uniqueness. Comparing leading coefficients in the two factorisations gives $c = c' = \text{lc}(f)$, since monic factors contribute leading coefficient 1 by (2). Cancelling the constants (theorem 5.9) leaves $p_1 \cdots p_k = q_1 \cdots q_l$; we induct on k . For $k = 1$ the left side is irreducible, so $l = 1$ and $p_1 = q_1$. For $k \geq 2$: p_1 divides $q_1 \cdots q_l$, so iterating Euclid's lemma (theorem 5.25(2)) gives $p_1 \mid q_j$ for

some j . A monic irreducible dividing a monic irreducible equals it: $q_j = p_1 h$ forces $\deg h = 0$ by irreducibility of q_j , and comparing leading coefficients gives $h = 1$. Cancel $p_1 = q_j$ from both sides (theorem 5.9). If $l = 1$, the cancellation leaves $p_2 \cdots p_k = 1$, which is impossible: the left side is a product of $k - 1 \geq 1$ nonconstant polynomials and so has degree at least 1 by (2), while $\deg 1 = 0$. Hence $l \geq 2$, and the induction hypothesis, applied to $p_2 \cdots p_k = \prod_{j' \neq j} q_{j'}$, concludes $k = l$ and matches the remaining factors. $\square$

Remark 5.29 (The structural reason: $F[X]$ is a PID). The deeper explanation of unique factorisation is structural. The degree function and theorem 5.12 make $F[X]$ a *Euclidean domain*—an integral domain with division with remainder relative to a size measure—and every Euclidean domain is a *principal ideal domain* (PID): each ideal (theorem 4.17) of $F[X]$ is $(g) = \{qg : q \in F[X]\}$ for a single generator g , namely any nonzero element of least degree in a nonzero ideal (the zero ideal is generated by 0), by the same division-and-minimality argument that proved every ideal of $\mathbb{Z}$ principal (Example 4.18(2)). The generator of the ideal $\{sf + tg : s, t \in F[X]\}$ is a gcd of f and g , and Bézout’s identity (3) is the concrete shadow of this principality. The standard chain

$$\text{Euclidean} \implies \text{PID} \implies \text{UFD}$$

has thus been walked explicitly for $F[X]$, ending at theorem 5.28.

5.7 Lagrange interpolation

The corollary “few points determine a low-degree polynomial” (theorem 5.20) says that a polynomial of degree at most d is pinned down by its values at $d + 1$ points. The present subsection proves the constructive converse: *any* $d + 1$ prescribed values at distinct points are attained by exactly one polynomial of degree at most d , and there is an explicit formula for it. This is the algebraic backbone of secret sharing, of Reed–Solomon error-correcting codes, and of the polynomial encodings in Halo 2.

Theorem 5.30 (Lagrange interpolation). *Let $x_0, \dots, x_d \in F$ be distinct (the nodes) and let $y_0, \dots, y_d \in F$ be arbitrary targets. There exists a unique $f \in F[X]$ with $\deg f \leq d$ and $f(x_i) = y_i$ for all i , given explicitly by*

$$f(X) = \sum_{i=0}^d y_i \ell_i(X), \quad \text{where} \quad \ell_i(X) = \prod_{j \neq i} \frac{X - x_j}{x_i - x_j}$$

are the Lagrange basis polynomials for the nodes.

Proof. The ℓ_i are genuine polynomials. Each denominator $\prod_{j \neq i} (x_i - x_j)$ is a product of nonzero field elements, the nodes being distinct, hence is an invertible constant; so ℓ_i is a polynomial, of degree exactly d (a product of d linear factors divided by a nonzero constant).

The delta property. Writing δ_{im} for the Kronecker symbol (equal to 1 if $i = m$ and 0 otherwise),

$$\ell_i(x_m) = \delta_{im} \quad (0 \leq i, m \leq d) : \tag{4}$$

at $m = i$ every factor $(x_i - x_j)/(x_i - x_j)$ is 1, while at $m \neq i$ the factor with $j = m$ has numerator $x_m - x_m = 0$. Consequently $f(x_m) = \sum_i y_i \ell_i(x_m) = y_m$ for each m , and $\deg f \leq d$ by (1), each summand having degree at most d .

Uniqueness. Two interpolants of degree at most d agree at the $d + 1$ distinct nodes, hence are equal by theorem 5.20. $\square$

Example 5.31 (A worked interpolation over $\mathbb{Q}$). Find the polynomial of degree at most 2 with $f(0) = 1, f(1) = 3$, and $f(2) = 7$. The basis polynomials for the nodes 0, 1, 2 are

$$\begin{aligned}\ell_0 &= \frac{(X-1)(X-2)}{(0-1)(0-2)} = \frac{(X-1)(X-2)}{2}, \\ \ell_1 &= \frac{(X-0)(X-2)}{(1-0)(1-2)} = -X(X-2), \\ \ell_2 &= \frac{(X-0)(X-1)}{(2-0)(2-1)} = \frac{X(X-1)}{2}.\end{aligned}$$

Hence

$$f = 1 \cdot \ell_0 + 3 \cdot \ell_1 + 7 \cdot \ell_2 = \frac{1}{2}(X^2 - 3X + 2) + 3(2X - X^2) + \frac{7}{2}(X^2 - X) = X^2 + X + 1,$$

and indeed $f(0) = 1, f(1) = 3, f(2) = 7$.

Remark 5.32 (The evaluation isomorphism). Fix $d + 1$ distinct nodes and let $V = \{f \in F[X] : \deg f \leq d\}$. In the language of the linear-algebra section later in the volume, V is a $(d + 1)$ -dimensional vector space over F with basis $1, X, \dots, X^d$, and the evaluation map

$$\text{ev}: V \rightarrow F^{d+1}, \quad f \mapsto (f(x_0), \dots, f(x_d)),$$

is F -linear. Its surjectivity is exactly the existence half of theorem 5.30, and its injectivity exactly the uniqueness half; so ev is a linear isomorphism $V \cong F^{d+1}$. The Lagrange basis $\{\ell_i\}$ is precisely the basis of V matched to the standard coordinates of F^{d+1} , by the delta property (4): the polynomial with prescribed values $(y_0, \dots, y_d)$ has coordinates $(y_0, \dots, y_d)$ in the basis $\{\ell_i\}$. This switching between a polynomial's *coefficient representation* and its *evaluation representation* is the conceptual core of fast polynomial arithmetic via the fast Fourier transform, treated with the roots of unity later in the volume, and of the evaluation-domain encodings used throughout Halo 2.

Remark 5.33 (The vanishing polynomial). The companion object to the Lagrange basis is the *vanishing polynomial* of the node set $\{x_0, \dots, x_d\}$,

$$Z(X) = \prod_{i=0}^d (X - x_i),$$

the unique monic polynomial of degree $d + 1$ vanishing at every node. A polynomial g vanishes on the whole node set if and only if $Z \mid g$. One direction is immediate from theorem 5.16. For the other, apply the factor theorem repeatedly: $g(x_0) = 0$ gives $g = (X - x_0)g_1$; evaluating at $x_1 \neq x_0$ gives $0 = (x_1 - x_0)g_1(x_1)$, so $g_1(x_1) = 0$ since F has no zero divisors, and $g_1 = (X - x_1)g_2$; continuing through the remaining nodes strips off one linear factor per node. The repetition is legitimate precisely because the nodes are distinct—the linear factors $X - x_i$ are pairwise coprime. The pair (Lagrange basis, vanishing polynomial) is exactly the data a proof system manipulates when arguing that a committed polynomial agrees with prescribed values on an evaluation domain.

5.8 Multiplicity and the formal derivative

A root can be a root “several times over”: the quadratic $X^2 - 2X + 1 = (X - 1)^2$ vanishes at 1 doubly. Counting roots correctly requires making that idea precise, and detecting it requires a derivative—which we define with no limits at all.

Definition 5.34 (Multiplicity of a root). Let $f \in F[X]$ be nonzero and let $\alpha \in F$ be a root of f . The *multiplicity* $\text{mult}_\alpha(f)$ is the largest integer $m \geq 1$ with $(X - \alpha)^m \mid f$. A root of multiplicity

1 is *simple*; a root of multiplicity at least 2 is *repeated* (or *multiple*).

The largest such m exists: the powers $(X - \alpha)^m$ have strictly increasing degrees, so only finitely many can divide the fixed f , and at least one does by the factor theorem. An equivalent characterisation is often handier: $m = \text{mult}_\alpha(f)$ exactly when $f = (X - \alpha)^m g$ with $g(\alpha) \neq 0$. Indeed, writing $f = (X - \alpha)^m g$ with m maximal forces $g(\alpha) \neq 0$, else the factor theorem would split a further factor $X - \alpha$ off g ; conversely, if $f = (X - \alpha)^m g$ with $g(\alpha) \neq 0$, then $(X - \alpha)^{m+1} \nmid f$ would give $(X - \alpha)^m g = (X - \alpha)^{m+1} h$, and cancelling (theorem 5.9) $g = (X - \alpha) h$, making α a root of g .

The count of roots *with* multiplicity still respects the degree. The proof rests on a small device worth isolating, since it recurs.

Lemma 5.35 (Coprime factors multiply). *Let $f_1, \dots, f_r \in F[X]$ be pairwise coprime ($\gcd(f_i, f_j) = 1$ for $i \neq j$) and suppose each f_i divides f . Then $f_1 f_2 \cdots f_r \mid f$.*

Proof. First let $r = 2$, and write $f = f_1 u = f_2 v$. Bézout (3) gives $s f_1 + t f_2 = 1$, so

$$f = s f_1 f + t f_2 f = s f_1 (f_2 v) + t f_2 (f_1 u) = f_1 f_2 (s v + t u),$$

and $f_1 f_2 \mid f$. For $r > 2$, induct: by hypothesis $f_1 \cdots f_{r-1} \mid f$, and the product $f_1 \cdots f_{r-1}$ is coprime to f_r —a nonconstant common divisor would have a monic irreducible factor (theorem 5.28), which by Euclid’s lemma (theorem 5.25(2)) would divide some f_i with $i < r$ as well as f_r , contradicting $\gcd(f_i, f_r) = 1$. Now apply the case $r = 2$. $\square$

Proposition 5.36 (Counting roots with multiplicity). *Let $f \in F[X]$ be nonzero of degree d , with all its distinct roots $\alpha_1, \dots, \alpha_r \in F$ of multiplicities $m_1, \dots, m_r$. Then*

$$f = (X - \alpha_1)^{m_1} \cdots (X - \alpha_r)^{m_r} g$$

for some $g \in F[X]$ with no roots in F , and $\sum_i m_i \leq d$. The number of roots counted with multiplicity is thus at most d , with equality if and only if f splits into linear factors over F .

Proof. We first check that the powers $(X - \alpha_i)^{m_i}$ are pairwise coprime. Each $X - \alpha_i$ is monic irreducible (degree 1), and by unique factorisation the monic divisors of $(X - \alpha_i)^{m_i}$ are exactly the powers $(X - \alpha_i)^e$ with $0 \leq e \leq m_i$. For $i \neq j$ the only monic polynomial that is simultaneously a power of $X - \alpha_i$ and of $X - \alpha_j$ is 1, since $X - \alpha_i = X - \alpha_j$ would force $\alpha_i = \alpha_j$; so $\gcd((X - \alpha_i)^{m_i}, (X - \alpha_j)^{m_j}) = 1$.

Each $(X - \alpha_i)^{m_i}$ divides f by the definition of multiplicity, so theorem 5.35 gives $f = (X - \alpha_1)^{m_1} \cdots (X - \alpha_r)^{m_r} g$. Maximality of each m_i gives $g(\alpha_i) \neq 0$: a further factor $X - \alpha_i$ inside g would raise the multiplicity. And g has no other roots either: a root β of g is a root of f , hence one of the α_i . Taking degrees with (2),

$$m_1 + \cdots + m_r + \deg g = d,$$

so $\sum_i m_i \leq d$, with equality precisely when $\deg g = 0$, that is, when f is a constant times a product of linear factors. $\square$

Definition 5.37 (Formal derivative). The *formal derivative* is the map $D: F[X] \rightarrow F[X]$ defined on $f = \sum_{i=0}^n a_i X^i$ by

$$f' = Df = \sum_{i=1}^n i a_i X^{i-1} = a_1 + 2a_2 X + \cdots + n a_n X^{n-1},$$

where the coefficient $i a_i$ means a_i added to itself i times in F —that is, a_i multiplied by the image of the integer i under the canonical map $\mathbb{Z} \rightarrow F$ (theorem 4.30). No limits are involved: the definition is purely formal and algebraic, valid over any field, and it is introduced for one purpose—to detect repeated roots without any appeal to analysis.

Proposition 5.38 (Rules of formal differentiation). *For all $f, g \in F[X]$, $c \in F$, $\alpha \in F$, and $m \geq 1$:*

1. (Linearity.) $(f + g)' = f' + g'$ and $(cf)' = cf'$.
2. (Product rule.) $(fg)' = f'g + fg'$.
3. (Power rule.) $((X - \alpha)^m)' = m(X - \alpha)^{m-1}$.

Proof. (1) is immediate from the coordinatewise, visibly linear definition of D .

(2) Both sides are *bilinear* in the pair (f, g) : by part (1), each is linear in f with g fixed and linear in g with f fixed. It therefore suffices to verify the identity on monomials $f = X^a$, $g = X^b$, since every polynomial is a linear combination of monomials and the general case follows by expanding $f = \sum_a u_a X^a$, $g = \sum_b v_b X^b$ and applying the monomial case termwise. On monomials,

$$(X^a X^b)' = (X^{a+b})' = (a + b) X^{a+b-1} = a X^{a-1} X^b + X^a b X^{b-1} = (X^a)' X^b + X^a (X^b)',$$

with the convention that a term carrying the integer coefficient 0 (the case $a = 0$ or $b = 0$) is the zero polynomial.

(3) Induct on m . The base is $(X - \alpha)' = 1$. For the step, the product rule gives

$$((X - \alpha)^m)' = ((X - \alpha)(X - \alpha)^{m-1})' = (X - \alpha)^{m-1} + (X - \alpha)(m-1)(X - \alpha)^{m-2} = m(X - \alpha)^{m-1}. \quad \square$$

Part (3) of the next theorem must exhibit a repeated root inside an extension field that is not given in advance; the following lemma manufactures the required extension. It is the quotient recipe of Example 4.22 carried out for polynomial rings, and the proof is the argument that made $\mathbb{Z}/p\mathbb{Z}$ a field (theorem 4.26), transposed from $\mathbb{Z}$ to $F[X]$.

Lemma 5.39 (Adjoining a root). *Let $h \in F[X]$ be irreducible. Then the quotient ring $K = F[X]/(h)$ is a field; the map $a \mapsto a + (h)$ embeds F into K as a subfield, so that K is an extension of F ; and the class $\alpha = X + (h)$ is a root of h in K .*

Proof. The principal ideal $(h) = hF[X]$ is an ideal of the commutative ring $F[X]$ (Example 4.18(2)), so K is a commutative ring under the coset operations (theorem 4.20). Its identity is not zero: $1 + (h) = 0 + (h)$ would say $h \mid 1$, impossible since a factorisation $1 = hk$ has degree $\deg h + \deg k = 0$ by (2), while $\deg h \geq 1$.

Every nonzero class is a unit. Let $f + (h) \neq 0 + (h)$, i.e. $h \nmid f$. The gcd $\gcd(h, f)$ is then a monic divisor of the irreducible h other than its monic associate, hence equals 1, exactly as in the proof of theorem 5.25(2); Bézout's identity (3) supplies $s, t \in F[X]$ with $sf + th = 1$, and reducing modulo (h) gives $(s + (h))(f + (h)) = 1 + (h)$. So K is a field.

The map $\iota(a) = a + (h)$ on constants preserves sums, products, and 1 directly from the coset operations of theorem 4.19, and it is injective: $\iota(a) = 0$ with $a \neq 0$ would make h divide a nonzero constant, again impossible by degrees. We identify F with its image, making K an extension field of F .

Finally, write $h = \sum_i c_i X^i$ and compute in K :

$$h(\alpha) = \sum_i \iota(c_i) (X + (h))^i = \left(\sum_i c_i X^i \right) + (h) = h + (h) = 0 + (h),$$

so α is a root of h in K . □

Theorem 5.40 (Derivative test for repeated roots; separability). *Let $f \in F[X]$ be nonzero and $\alpha \in F$.*

1. *The element α is a repeated root of f —a root of multiplicity at least 2—if and only if $f(\alpha) = 0$ and $f'(\alpha) = 0$.*
2. *If f has a repeated root in F , then $\gcd(f, f') \neq 1$.*
3. *Call f separable if it has no repeated root in any field containing F . Then f is separable if and only if $\gcd(f, f') = 1$ in $F[X]$.*

Proof. (1) Let α be a root of multiplicity $m \geq 1$ and write $f = (X - \alpha)^m g$ with $g(\alpha) \neq 0$. The product and power rules give

$$f' = m(X - \alpha)^{m-1}g + (X - \alpha)^m g' = (X - \alpha)^{m-1} [mg + (X - \alpha)g'].$$

If $m = 1$, the first factor is 1 and evaluating at α gives $f'(\alpha) = 1 \cdot g(\alpha) = g(\alpha) \neq 0$: a simple root of f is not a root of f' . If $m \geq 2$, then $(X - \alpha)^{m-1}$ divides f' and vanishes at α , so $f'(\alpha) = 0$. Together with the factor theorem for $f(\alpha) = 0$, this proves the equivalence.

(2) A repeated root $\alpha \in F$ is a common root of f and f' by (1), so the factor theorem puts the common factor $X - \alpha$ into both, and $(X - \alpha) \mid \gcd(f, f')$.

(3) The key observation is that *gcds are stable under field extension*: for an extension field K of F (theorem 4.25), the Euclidean algorithm of theorem 5.23 run on $f, f' \in F[X]$ uses only field operations on their coefficients, which lie in F ; run inside $K[X]$ on the same inputs it produces the identical quotients and remainders. Hence the gcd computed in $K[X]$ equals the gcd computed in $F[X]$.

Suppose $\gcd(f, f') = 1$ in $F[X]$, and let $K \supseteq F$ be any extension. By stability, $\gcd(f, f') = 1$ in $K[X]$ as well, so by (2) applied over K , the polynomial f has no repeated root in K : f is separable.

Conversely, suppose $d = \gcd(f, f')$ is nonconstant; we exhibit a repeated root in some extension. Let h be a monic irreducible factor of d (theorem 5.28), and let $K = F[X]/(h)$ be the extension field supplied by theorem 5.39, in which h has a root α . The polynomial h divides d , which divides both f and f' ; evaluating the factorisations at α (theorem 5.16) gives $f(\alpha) = f'(\alpha) = 0$. By (1), applied over K , the element α is a repeated root of f in K : f is not separable. $\square$

The stability observation in the proof deserves emphasis as a technique: it licenses testing separability—the absence of repeated roots in *every* extension of the base field at once—by a gcd computation carried out entirely in the base field.

Remark 5.41 (The pitfall of positive characteristic). The integer coefficient i in $f' = \sum_i i a_i X^{i-1}$ is interpreted in F , and may vanish there. Over a field of characteristic p —where $p = 0$ in F , p prime, the setting of $\mathbb{F}_p$ and $\mathbb{F}_q$ (theorem 4.30)—the derivative of X^p is $p X^{p-1} = 0$. Thus $f = X^p - 1$ over $\mathbb{F}_p$ has $f' = 0$ and $\gcd(f, f') = \gcd(f, 0) = f \neq 1$; and indeed

$$X^p - 1 = (X - 1)^p \quad \text{in } \mathbb{F}_p[X],$$

a factorisation the finite-fields section establishes through the freshman's-dream identity $(a + b)^p = a^p + b^p$ of characteristic p (theorem 6.13, applied to $(X + (-1))^p$ together with $(-1)^p = -1$ in every $\mathbb{F}_p$): a single root 1 of multiplicity p . This characteristic- p behaviour is a structural feature to respect when working over the finite fields of Orchard, not a pathology to avoid.

5.9 Polynomials versus polynomial functions

The opening of this section insisted that a polynomial is not a function. The debt of that insistence now falls due: we make the relationship precise and show exactly when it is safe—and when it is not—to conflate the two.

Definition 5.42 (Polynomial function). Every $f \in F[X]$ induces a *polynomial function*

$$\tilde{f}: F \rightarrow F, \quad \alpha \mapsto f(\alpha).$$

The assignment $f \mapsto \tilde{f}$ is a ring homomorphism from $F[X]$ to the ring F^F of *all* functions $F \rightarrow F$ under pointwise operations ($(u + v)(\alpha) = u(\alpha) + v(\alpha)$ and $(uv)(\alpha) = u(\alpha)v(\alpha)$), by theorem 5.16 applied at every point.

Proposition 5.43 (When formal equals functional). Let $f, g \in F[X]$.

1. If F is infinite, then $\tilde{f} = \tilde{g}$ as functions implies $f = g$ as polynomials. Equivalently, $f \mapsto \tilde{f}$ is injective, and polynomials over an infinite field may be identified with polynomial functions.
2. If F is finite, the map is not injective. Over $\mathbb{F}_p$, the nonzero polynomial $X^p - X$ induces the zero function, since Fermat's little theorem (Corollary 3.32) gives $\alpha^p = \alpha$ for every $\alpha \in \mathbb{F}_p$.

Proof. (1) Let $h = f - g$, so that $\tilde{h}$ is the zero function: h vanishes at every point of the infinite set F . Were h nonzero of degree d , it would have at most d roots (theorem 5.18), yet it has infinitely many—a contradiction. Hence $h = 0$.

(2) Let F be finite, with $q = |F|$ elements, and consider the *vanishing-polynomial witness*

$$w(X) = \prod_{\alpha \in F} (X - \alpha),$$

a monic polynomial of degree q —in particular nonzero as a formal object—which vanishes at every point of F by construction. Thus $\tilde{w} = \tilde{0}$ while $w \neq 0$: the map $f \mapsto \tilde{f}$ identifies distinct polynomials. Over $F = \mathbb{F}_p$ the witness specialises to $X^p - X$: by Fermat's little theorem every element of $\mathbb{F}_p$ is a root of the monic degree- p polynomial $X^p - X$, so theorem 5.36 factors it as $\prod_{\alpha \in \mathbb{F}_p} (X - \alpha)$ —the p distinct linear factors exhaust the degree—and $X^p - X$ induces the zero function despite having degree $p \geq 1$. $\square$

Remark 5.44 (Why the formal viewpoint is the correct one). Part (2) of theorem 5.43 is precisely why polynomials are defined as coefficient sequences and not as functions. Cryptography works over finite fields, where the functional view loses information: the polynomials X and X^p induce the same function on $\mathbb{F}_p$, yet have degrees 1 and p . And the *degree* of a polynomial—a property of the formal object, invisible to the induced function over a finite field—is what the security of polynomial commitment schemes rests on. A prover commits to a low-degree *formal* polynomial; the verifier tests it by *evaluation*; and the gap between agreement-as-function at the tested points and identity to the claimed formal polynomial is exactly what the Schwartz–Zippel lemma, next, quantifies.

5.10 The Schwartz–Zippel lemma

Only one preliminary is needed to state the payoff theorem. To draw α *uniformly at random* from a finite nonempty set S means to select each element with equal likelihood $1/|S|$; the *probability* of an event is then the fraction of elements of S producing it, so that $\Pr[f(\alpha) = 0] = |\{\alpha \in S : f(\alpha) = 0\}|/|S|$. The systematic language of probability is developed later in the volume; this counting case is all the theorem needs.

Theorem 5.45 (Univariate Schwartz–Zippel). Let $f \in F[X]$ be nonzero of degree at most d , let

$S \subseteq F$ be finite and nonempty, and let α be drawn uniformly at random from S . Then

$$\Pr[f(\alpha) = 0] \leq \frac{d}{|S|}. \quad (5)$$

More generally, if $f, g \in F[X]$ are distinct, each of degree at most d , then $\Pr[f(\alpha) = g(\alpha)] \leq d/|S|$.

Proof. By theorem 5.18, the nonzero f has at most d roots in F , hence at most d in the subset S . Under the uniform draw,

$$\Pr[f(\alpha) = 0] = \frac{|\{\alpha \in S : f(\alpha) = 0\}|}{|S|} \leq \frac{d}{|S|}.$$

For the general form, apply this to $h = f - g$, which is nonzero and has degree at most d by (1), noting that $f(\alpha) = g(\alpha)$ exactly when $h(\alpha) = 0$. $\square$

Remark 5.46 (The lemma stated plainly, and used). Inequality (5) formalises the slogan: *two low-degree polynomials that agree at many points must be equal*. The deterministic version is theorem 5.20—agreement at $d + 1$ points forces equality—and Schwartz–Zippel is its probabilistic refinement: a *single* random evaluation point detects unequal polynomials of degree at most d , except with probability at most $d/|S|$. With $|S|$ exponentially large—taking $S = \mathbb{F}_q$ with $q \approx 2^{254}$, the 255-bit fields underlying Orchard—and d polynomially bounded, the failure probability is astronomically small; the term *negligible* receives its precise asymptotic meaning later in the volume. This is the foundational soundness argument for polynomial identity testing, and hence for the polynomial commitment schemes and arithmetisation checks of Halo 2: to verify a purported polynomial identity, evaluate both sides at a random challenge and compare.

Remark 5.47 (Multivariate generalisation). The full Schwartz–Zippel lemma extends to polynomials in several indeterminates $f \in F[X_1, \dots, X_n]$: a nonzero polynomial of total degree d —the largest sum of exponents $i_1 + \dots + i_n$ over the monomials with nonzero coefficient—vanishes at a uniformly random point of S^n with probability at most $d/|S|$. The univariate case proved here is both the base case of the multivariate inductive proof and the version most directly used: the later volumes invoke only the univariate lemma directly. Where an identity in several random challenges arises—for instance a permutation argument drawing a pair of challenges (β, γ) —the multivariate bound applies and follows from the univariate lemma applied once per variable.

The cheque drawn in the section’s opening can now be cashed in full. Two parties hold polynomials f and g of degree at most d over a 255-bit field $\mathbb{F}_q$ —even with d in the millions, they agree on a uniformly random challenge $\alpha \in \mathbb{F}_q$ with probability at most $d/q \approx d \cdot 2^{-254}$ unless they are literally the same formal polynomial. One evaluation apiece, one comparison of field elements, and the enormous coefficient lists never change hands: that is the trade the proving system makes on every identity it checks, and the entire cost of the bargain is the elementary fact with which the section began—a nonzero polynomial of degree d over a field has at most d roots.

6 Finite fields

A processor stores a number in a word of fixed width. A 64-bit word can hold exactly 2^{64} distinct values, and the hardware’s arithmetic wraps around on overflow: what the machine natively computes is addition and multiplication modulo 2^{64} . For a verifier that must add, multiply, *and divide*, this built-in arithmetic falls at the last hurdle: in $\mathbb{Z}/2^{64}\mathbb{Z}$ no even class has an inverse, since a class is a unit only when its representative is coprime to the modulus (Theorem 2.24). The question this section answers is the natural one to ask next. Among all finite arithmetics—finite sets closed under an addition and a multiplication obeying the ring laws—which ones allow division by *every* nonzero element? In the language of §4: what are the finite fields, what sizes do they come in, and what structure do they carry?

Part of the answer is already in hand. The residue ring $\mathbb{Z}/n\mathbb{Z}$ is a field exactly when n is prime (theorem 4.26), giving one finite field $\mathbb{F}_p$ for every prime p ; and the quotient recipe of Example 4.22 promised further examples of prime-power size p^k . This section completes the picture with two theorems. The first, theorem 6.9, is a prohibition: *no other sizes occur*—every finite field has exactly p^k elements for some prime p , so there is no field with 6, 10, or 12 elements. This constrains cardinalities, not binary representation widths: fields of size 2^k exist for every positive k . The second, theorem 6.17, is the section’s payoff and one of the most consequential facts in the volume: the multiplicative group $\mathbb{F}_q^\times$ is *cyclic*—the powers of a single well-chosen element sweep out every nonzero value. Cyclicity is what makes discrete logarithms meaningful and what guarantees the roots of unity on which the polynomial machinery of later sections runs.

The route is as follows. We first make the prime field $\mathbb{F}_p$ computational, with two inversion algorithms and the square-and-multiply technique that powers one of them. We then determine the possible sizes of a finite field through the characteristic and the prime subfield, meet the Frobenius map $x \mapsto x^p$ —the symmetry peculiar to characteristic p —and finally prove cyclicity by a counting argument of Gauss.

6.1 The prime field and two inversion routes

The field $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$ certified by theorem 4.26 is the workhorse of the entire series, and the first order of business is to compute in it. Addition, subtraction, and multiplication are residue arithmetic as in §2; the operation that needs an algorithm is inversion. As §4 stressed, an existence proof and an algorithm are different assets, and cryptography needs both. For inversion in $\mathbb{F}_p$ there are two standard routes, with different characters.

Construction 6.1 (Inversion by the extended Euclidean algorithm). Given a with $0 < a < p$, run the extended Euclidean algorithm (Theorem 2.13) on the pair (a, p) . Since p is prime and $p \nmid a$, the gcd is 1, and the algorithm outputs integers u, v with $au + pv = 1$. Reducing modulo p gives $au \equiv 1 \pmod{p}$, that is,

$$a^{-1} = [u]_p \quad \text{in } \mathbb{F}_p.$$

The cost is $O(\log p)$ division steps (Theorem 2.13). This is the standard general-purpose method; a worked instance is Example 2.25, which computed $[17]_{43}^{-1} = [38]_{43}$ by exactly this route.

The second route rests on Fermat’s little theorem and on a technique for fast exponentiation that is used throughout cryptography, so we record the technique first, in the generality it deserves.

Construction 6.2 (Square-and-multiply). Let a be an element of any structure with an associative multiplication, and let $e \geq 1$ have binary expansion $e = \sum_i e_i 2^i$ with digits $e_i \in \{0, 1\}$. Squaring repeatedly produces the values

$$a^{2^0} = a, \quad a^{2^{i+1}} = (a^{2^i})^2,$$

and multiplying together the factors indexed by the nonzero digits gives

$$a^e = \prod_{i: e_i=1} a^{2^i}.$$

The computation uses $\lfloor \log_2 e \rfloor$ squarings and at most $\lfloor \log_2 e \rfloor$ further multiplications— $O(\log e)$ multiplications in all, against the $e - 1$ multiplications of the naive product $a \cdot a \cdots a$. For the

255-bit exponents of later sections the difference is that between a few hundred multiplications and a count that no conceivable computer could finish.

Remark 6.3. Written additively, the same algorithm computes the multiple $e \cdot P$ of a group element P by repeated doubling and selective adding. Under the name *double-and-add* it is the algorithm for scalar multiplication on elliptic curves (theorem 10.20), where the group operation is written $+$; the two names denote one algorithm in two notations.

Construction 6.4 (Inversion by Fermat's little theorem). Let $a \in \mathbb{F}_p$ be nonzero. Fermat's little theorem (Corollary 3.32) gives $a^{p-1} = 1$, hence $a \cdot a^{p-2} = 1$, and so

$$a^{-1} = a^{p-2},$$

computed by square-and-multiply (theorem 6.2) with exponent $p - 2$, at a cost of $O(\log p)$ modular multiplications when $p > 2$. For $p = 2$, the only nonzero element is 1, whose inverse is 1; the exponent is zero and no multiplication is needed.

Remark 6.5 (Constant-time inversion). The two routes are asymptotically comparable, but they differ in a respect invisible to asymptotics and central to practice. In theorem 6.4 the sequence of operations performed—which steps square, which multiply—depends only on the binary digits of the *public* exponent $p - 2$, never on the secret input a ; the method is *constant-time-friendly*. The extended Euclidean route, by contrast, performs a sequence of divisions whose lengths and quotients depend on a itself, so its running time and memory-access pattern can leak information about a to an observer timing the computation. Side-channel-resistant implementations therefore often prefer the Fermat route despite its slightly larger constant factor.

6.2 Characteristic, prime subfield, and prime-power order

We turn to the sizes question: which cardinalities can a finite field have? The tool is the characteristic of section 4.5. Recall the canonical homomorphism

$$\iota: \mathbb{Z} \rightarrow F, \quad \iota(n) = n \cdot 1_F,$$

whose kernel is an ideal $n\mathbb{Z}$ of $\mathbb{Z}$; the characteristic $\text{char}(F)$ is its nonnegative generator (Definition 4.30).

Theorem 6.6. *Every finite field F has prime characteristic: $\text{char}(F) = p$ for some prime p . In particular $\text{char}(\mathbb{F}_p) = p$.*

Proof. The map ι cannot be injective, since its domain $\mathbb{Z}$ is infinite and its codomain F is finite; two distinct integers $m < n$ must satisfy $\iota(m) = \iota(n)$, and then $n - m$ is a nonzero element of $\ker \iota$. Hence $\ker \iota = n\mathbb{Z}$ with $n > 0$, that is, $\text{char}(F) > 0$; and the characteristic of a field, being nonzero, is prime by Theorem 4.32. For $F = \mathbb{F}_p$ itself, Example 4.31 computed $\text{char}(\mathbb{Z}/p\mathbb{Z}) = p$ directly. $\square$

The characteristic locates a canonical copy of $\mathbb{F}_p$ inside every field of characteristic p . Remark 4.33 described this copy informally; we now make it precise, beginning with the notion of a subfield.

Definition 6.7 (Subfield; prime subfield). A *subfield* of a field F is a subset $K \subseteq F$ containing 0 and 1 and closed under addition, negation, multiplication, and inversion of nonzero elements; with the restricted operations, K is itself a field. The *prime subfield* of F is the intersection of

all subfields of F —equivalently, the smallest subfield of F , and equivalently again the subfield generated by 1_F .

Two words on the equivalences. The intersection of any family of subfields is again a subfield, since each defining closure property survives intersection; and the intersection of *all* subfields is contained in every subfield, hence is the smallest one. That this smallest subfield is exactly what 1_F generates—the closure of $\{1_F\}$ under the field operations—follows from the classification below, whose proof exhibits the prime subfield inside the closure of $\{1_F\}$.

Proposition 6.8 (Classification of prime subfields). *Let F be a field. If $\text{char}(F) = p > 0$, the prime subfield of F is isomorphic to $\mathbb{F}_p$. If $\text{char}(F) = 0$, it is isomorphic to $\mathbb{Q}$.*

Proof. Suppose first that $\text{char}(F) = p > 0$, so that $\ker \iota = p\mathbb{Z}$. Define

$$\bar{\iota}: \mathbb{F}_p = \mathbb{Z}/p\mathbb{Z} \rightarrow F, \quad \bar{\iota}([a]_p) = a \cdot 1_F.$$

The map is well defined and injective at a stroke: $a \cdot 1_F = b \cdot 1_F$ holds if and only if $(a - b) \cdot 1_F = 0$, i.e. $a - b \in \ker \iota = p\mathbb{Z}$, i.e. $[a]_p = [b]_p$. It is a ring homomorphism because ι is one (section 4.5) and $\bar{\iota}$ merely re-reads ι on residue classes. Its image $K = \{a \cdot 1_F : a \in \mathbb{Z}\}$ is therefore a subring of F isomorphic to the field $\mathbb{F}_p$, and K is in fact a subfield: for $[a]_p \neq [0]_p$ the computation $\bar{\iota}([a]_p) \bar{\iota}([a]_p^{-1}) = \bar{\iota}([1]_p) = 1_F$ exhibits the inverse of each nonzero element of K inside K . Finally, K lies in every subfield: a subfield contains 1_F , hence all its additive multiples $a \cdot 1_F$ by closure under addition and negation. So K is contained in the intersection of all subfields and is itself a subfield, forcing K to be that intersection: the prime subfield is $K \cong \mathbb{F}_p$.

Now suppose $\text{char}(F) = 0$, so that ι is injective and $b \cdot 1_F \neq 0$ for every nonzero integer b . Define

$$j: \mathbb{Q} \rightarrow F, \quad j\left(\frac{a}{b}\right) = (a \cdot 1_F)(b \cdot 1_F)^{-1}.$$

The map is well defined: if $a/b = c/d$ with $b, d \neq 0$, then $ad = bc$ in $\mathbb{Z}$, so $(a \cdot 1_F)(d \cdot 1_F) = (b \cdot 1_F)(c \cdot 1_F)$ by applying ι , and multiplying both sides by $(b \cdot 1_F)^{-1}(d \cdot 1_F)^{-1}$ gives $(a \cdot 1_F)(b \cdot 1_F)^{-1} = (c \cdot 1_F)(d \cdot 1_F)^{-1}$. That j preserves sums, products, and 1 is the usual arithmetic of fractions, transported by the homomorphism ι and commutativity. Injectivity is immediate: $j(a/b) = 0$ forces $a \cdot 1_F = 0$, hence $a = 0$. The image of j is thus a subfield of F isomorphic to $\mathbb{Q}$; and it lies in every subfield K , since K contains 1_F , hence every $a \cdot 1_F$, hence—being closed under inversion and multiplication—every $(a \cdot 1_F)(b \cdot 1_F)^{-1}$. As before the image must equal the intersection of all subfields, and the prime subfield is a copy of $\mathbb{Q}$. $\square$

The size constraint now falls out of a counting device worth naming, for it recurs whenever a field sits over a subfield: *vector-space counting over the prime subfield*. Every field F of characteristic p is a vector space over its prime subfield $\mathbb{F}_p$: vectors are the elements of F , vector addition is the field addition, and scalar multiplication $c \cdot x$ for $c \in \mathbb{F}_p \subseteq F$ is the field multiplication, so the vector-space axioms are instances of the ring axioms of F . The proof below uses exactly two facts about vector spaces: a space with a finite spanning set has a basis, and coordinates with respect to a basis are unique. Both are stated and proved in the linear-algebra section (section 8), whose development is independent of the present theorem, so no circularity arises.

Theorem 6.9 (Finite fields have prime-power order). *Every finite field F has cardinality $|F| = p^k$ for some prime p and integer $k \geq 1$, where $p = \text{char}(F)$.*

Proof. By theorem 6.6 the characteristic of F is a prime p , and by theorem 6.8 the prime subfield of F is a copy of $\mathbb{F}_p$, with which we identify it. As explained above, F is then a vector space over

$\mathbb{F}_p$. It is spanned by the finite set F itself, so it has a finite basis $e_1, \dots, e_k$; and $k \geq 1$ because $1 \neq 0$ makes F nonzero. Every element of F is thus

$$c_1 e_1 + c_2 e_2 + \dots + c_k e_k$$

for a *unique* tuple $(c_1, \dots, c_k)$ with each $c_i \in \mathbb{F}_p$: existence because the e_i span, uniqueness because coordinates with respect to a basis are unique (section 8). The assignment $(c_1, \dots, c_k) \mapsto \sum_i c_i e_i$ is therefore a bijection from $\mathbb{F}_p^k$ to F , and counting tuples gives $|F| = p \cdot p \cdots p = p^k$. $\square$

Fields of every prime-power size do exist, and are unique; we state the fact in full, but prove only the part the series stands on.

Theorem 6.10 (Existence and uniqueness of $\mathbb{F}_q$). *For every prime power $q = p^k$ there exists a field with exactly q elements, and any two finite fields of the same cardinality are isomorphic. One constructs such a field as the quotient $\mathbb{F}_p[x]/(f)$ of the polynomial ring by the ideal generated by an irreducible polynomial f of degree k —the recipe recorded in Example 4.22.*

Remark 6.11. The theorem is deliberately left unproved, and the omission costs the series nothing. The fields on the critical path of Halo 2 and Orchard are *prime* fields, the case $q = p$ with $k = 1$, and there the theorem is already proved: existence is theorem 4.26, and uniqueness is immediate from the classification, since a field with p elements has order $p^k = p$ with $k = 1$ by theorem 6.9, so it equals its own prime subfield, a copy of $\mathbb{F}_p$ (theorem 6.8). The general case $k \geq 2$ would require a development of polynomial factorisation over field extensions on which nothing later in the series relies, so we state it for orientation only. *Convention, fixed henceforth:* the letter $q = p^k$ denotes a prime power, and $\mathbb{F}_q$ a field with q elements, of characteristic p ; results are stated for general $\mathbb{F}_q$ whenever the proof is identical to the prime case, and the reader focused on Halo 2 may read $q = p$ throughout.

6.3 The Frobenius endomorphism

Characteristic p is not merely a bookkeeping invariant; it changes what algebra is true. The most famous instance is the schoolroom error $(a + b)^p = a^p + b^p$, false over $\mathbb{Q}$ and $\mathbb{R}$ for every $p \geq 2$, which in characteristic p becomes a theorem.

The proof expands $(a + b)^p$ by the binomial theorem, so we first fix its coefficients and their two readings.

Lemma 6.12 (Binomial theorem; the coefficients count subsets). *Let R be a commutative ring. For all $a, b \in R$ and every integer $n \geq 0$,*

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k},$$

where the integer coefficient $\binom{n}{k}$ acts as the additive multiple $\binom{n}{k} \cdot x = x + \dots + x$, equals the number of k -element subsets of an n -element set, and satisfies

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

Proof. Expand the product $(a + b)^n = (a + b) \cdots (a + b)$ by distributivity without collecting terms: the result is a sum of 2^n products, one for each way of choosing, from every factor, either its a or its b . By commutativity such a product equals $a^k b^{n-k}$ exactly when a was chosen from k of the n factors, and it is determined by the set of those k factors; so the coefficient of $a^k b^{n-k}$ is the number of k -element subsets of the n factors, and collecting the terms gives the displayed sum.

For the formula, count the sequences of k distinct elements of an n -element set in two ways. Choosing the entries in turn, there are n choices for the first, $n - 1$ for the second, and $n - k + 1$ for the k -th: $n(n - 1) \cdots (n - k + 1) = n!/(n - k)!$ sequences. Alternatively, each such sequence is a k -element subset together with an ordering of it, and a k -element set has $k!$ orderings — the same count with k in place of n — so there are $k!$ sequences per subset, and the number of subsets is $n!/(k!(n - k)!)$. $\square$

Lemma 6.13 (Freshman’s dream). *Let R be a commutative ring of prime characteristic p . Then for all $a, b \in R$,*

$$(a + b)^p = a^p + b^p,$$

and more generally $(a + b)^{p^n} = a^{p^n} + b^{p^n}$ for every $n \geq 1$.

Proof. By the binomial theorem (theorem 6.12),

$$(a + b)^p = \sum_{i=0}^p \binom{p}{i} a^i b^{p-i},$$

where the integer coefficient $\binom{p}{i}$ acts as the additive multiple $\binom{p}{i} \cdot x = x + \cdots + x$. We claim that $p \mid \binom{p}{i}$ for $0 < i < p$. From the factorial identity of the same lemma

$$\binom{p}{i} i! (p - i)! = p!$$

the prime p divides the right-hand side. It does not divide $i! (p - i)!$: that product is a product of integers all smaller than p , and if p divided the product it would divide one of the factors by Euclid’s lemma for primes (Lemma 2.17), which is impossible. Applying Euclid’s lemma to the left-hand side instead, p must divide $\binom{p}{i}$, say $\binom{p}{i} = pm$.

Now let $0 < i < p$ and consider the corresponding term. Writing $\iota(n) = n \cdot 1_R$ for the canonical homomorphism of section 4.5, the coefficient acts as multiplication by $\iota(\binom{p}{i}) = \iota(pm) = \iota(p) \iota(m) = 0 \cdot \iota(m) = 0$, since $\text{char}(R) = p$ means exactly $\iota(p) = p \cdot 1_R = 0$. Every middle term of the binomial expansion therefore vanishes, and the terms $i = p$ and $i = 0$ leave $(a + b)^p = a^p + b^p$.

The general case follows by induction on n , the base case $n = 1$ being the identity just proved. For $n \geq 2$, apply the base case to the elements $a^{p^{n-1}}$ and $b^{p^{n-1}}$:

$$(a + b)^{p^n} = ((a + b)^{p^{n-1}})^p = (a^{p^{n-1}} + b^{p^{n-1}})^p = a^{p^n} + b^{p^n},$$

using the inductive hypothesis in the middle step. $\square$

The identity says that in characteristic p , raising to the p -th power respects addition—and it obviously respects multiplication. A map respecting both operations is a homomorphism, and a homomorphism from a structure to itself is called an *endomorphism*; this one has a name of its own.

Definition 6.14 (Frobenius endomorphism). Let F be a field of characteristic $p > 0$. The *Frobenius map* of F is

$$\text{Frob}: F \rightarrow F, \quad \text{Frob}(x) = x^p.$$

Proposition 6.15. *Let F be a field of characteristic p . The Frobenius map is a ring homomorphism $F \rightarrow F$ fixing the prime subfield pointwise. If F is finite, Frob is an automorphism of F —a bijective*

ring homomorphism of F onto itself.

Proof. Multiplicativity is commutativity and associativity alone: $(xy)^p = x^p y^p$. Additivity is the freshman's dream (Lemma 6.13), and $1^p = 1$; so Frob is a ring homomorphism.

Injectivity holds for a reason worth isolating, since it recurs: *every ring homomorphism between fields is injective*. Indeed, if $\phi: F \rightarrow F'$ had $\phi(x) = 0$ for some $x \neq 0$, then

$$1 = \phi(1) = \phi(x^{-1}x) = \phi(x^{-1})\phi(x) = \phi(x^{-1}) \cdot 0 = 0$$

in F' , contradicting $1 \neq 0$ in a field. In particular Frob is injective.

The prime subfield is fixed pointwise. By theorem 6.8 its elements are the values $\bar{\iota}([a]_p) = a \cdot 1_F$, and since $\bar{\iota}$ is a ring homomorphism,

$$\text{Frob}(\bar{\iota}([a]_p)) = \bar{\iota}([a]_p)^p = \bar{\iota}([a]_p^p) = \bar{\iota}([a^p]_p) = \bar{\iota}([a]_p),$$

the last step because $a^p \equiv a \pmod{p}$ for all integers a (Corollary 3.32).

Finally, let F be finite. An injective map from a finite set to itself is surjective—the injectivity-implies-surjectivity device of section 4.4—so Frob is a bijective ring homomorphism of F onto itself, an automorphism. $\square$

The iterates $\text{Frob}^n(x) = x^{p^n}$ are again homomorphisms, as the general case of the freshman's dream states directly. On the prime field $\mathbb{F}_p$ itself the Frobenius map is the identity; on larger fields of characteristic p it is a genuine, structure-preserving symmetry, and theorem 6.24 below records where it re-enters the cryptographic story.

6.4 Cyclicity of the multiplicative group

The multiplicative group $\mathbb{F}_q^\times$ of a finite field is a finite abelian group of order $q - 1$: every one of the $q - 1$ nonzero elements is invertible, precisely because $\mathbb{F}_q$ is a field. Finite abelian groups can in general be quite far from cyclic—the unit group $(\mathbb{Z}/8\mathbb{Z})^\times$ of Example 3.22 has order 4 with no element of order 4—so it is a genuinely strong fact that $\mathbb{F}_q^\times$ is always cyclic. The proof is a counting argument of Gauss, and its arithmetic engine is an identity about the totient function φ . Everything we use about φ is its *counting definition* (Definition 2.26): $\varphi(n)$ is the number of integers a with $1 \leq a \leq n$ and $\gcd(a, n) = 1$. No further totient theory is needed or assumed.

Lemma 6.16 (Gauss's divisor-sum identity). *For every integer $n \geq 1$,*

$$\sum_{d|n} \varphi(d) = n,$$

the sum running over the positive divisors d of n .

Proof. Partition the set $\{1, 2, \dots, n\}$ according to the value of $\gcd(a, n)$. For each a the gcd is a positive divisor d of n , so the classes

$$C_d = \{a : 1 \leq a \leq n, \gcd(a, n) = d\}, \quad d | n,$$

are disjoint and cover $\{1, \dots, n\}$; hence $n = \sum_{d|n} |C_d|$.

We claim $|C_d| = \varphi(n/d)$. A member of C_d is divisible by d , so it has the form $a = db$ with $1 \leq b \leq n/d$; we show that, for such a ,

$$\gcd(db, n) = d \iff \gcd(b, n/d) = 1.$$

For the forward direction, suppose $e = \gcd(b, n/d) > 1$; then de divides both $db = a$ and $d \cdot (n/d) = n$, so de is a common divisor exceeding d and $\gcd(a, n) \geq de > d$. For the converse, suppose

$\gcd(b, n/d) = 1$. The integer d is a common divisor of a and n , so d divides $g = \gcd(a, n)$ by the strongest-common-divisor property (Remark 2.12); write $g = dc$ with $c \geq 1$. From $dc \mid db$ we get $c \mid b$, and from $dc \mid n = d(n/d)$ we get $c \mid n/d$ (cancel d in each divisibility); so c divides $\gcd(b, n/d) = 1$, forcing $c = 1$ and $g = d$.

The members of C_d are therefore exactly the integers db with $1 \leq b \leq n/d$ and $\gcd(b, n/d) = 1$, and by the counting definition of the totient (Definition 2.26) there are precisely $\varphi(n/d)$ of them. Summing,

$$n = \sum_{d \mid n} \varphi\left(\frac{n}{d}\right) = \sum_{d \mid n} \varphi(d),$$

the second equality because $d \mapsto n/d$ is a bijection of the set of positive divisors of n onto itself. $\square$

The theorem now follows from a beautiful squeeze. We count the elements of each possible order twice over—once inside the group, once through the totient—and Gauss's identity leaves the two counts no room to differ.

Theorem 6.17 (Finite subgroups of $F^\times$ are cyclic). *Let F be any field and let $G \leq F^\times$ be a finite subgroup of the multiplicative group, of order N . Then G is cyclic. In particular, for a finite field $\mathbb{F}_q$ the whole multiplicative group $\mathbb{F}_q^\times$ is cyclic of order $q - 1$.*

Proof. For each divisor $d \mid N$, let

$$\psi(d) = |\{a \in G : \text{ord}(a) = d\}|$$

count the elements of G of order exactly d . Every element of the finite group G has finite order dividing $N = |G|$ (Corollary 3.30), so the sets counted by the $\psi(d)$ partition G and

$$\sum_{d \mid N} \psi(d) = N. \quad (6)$$

The key claim is that *for each $d \mid N$, either $\psi(d) = 0$ or $\psi(d) = \varphi(d)$* . Suppose $\psi(d) > 0$ and choose $a \in G$ of order d . The powers $a^0, a^1, \dots, a^{d-1}$ are d distinct elements of G (Proposition 3.12(3)), and each satisfies

$$(a^j)^d = (a^d)^j = 1^j = 1,$$

so each is a root in F of the polynomial $X^d - 1 \in F[X]$. That polynomial has degree d , so it has *at most* d roots in F (Theorem 5.18); the d distinct powers of a therefore account for *all* of its roots. Now take any $b \in G$ of order d . Then $b^d = 1$, so b is a root of $X^d - 1$ and hence $b = a^j$ for some $0 \leq j < d$. By Corollary 3.23,

$$\text{ord}(a^j) = \frac{d}{\gcd(d, j)},$$

which equals d exactly when $\gcd(j, d) = 1$. The elements of order d in G are thus precisely the powers a^j with $0 \leq j < d$ and $\gcd(j, d) = 1$, and by the counting definition of the totient there are $\varphi(d)$ of them: $\psi(d) = \varphi(d)$. This proves the claim, and with it the inequality $\psi(d) \leq \varphi(d)$ for every $d \mid N$.

Summing the inequality over the divisors of N and comparing (6) with Gauss's identity (Lemma 6.16),

$$N = \sum_{d \mid N} \psi(d) \leq \sum_{d \mid N} \varphi(d) = N.$$

The two ends are equal, so the inequality is an equality term by term: $\psi(d) = \varphi(d)$ for *every* divisor $d \mid N$. In particular

$$\psi(N) = \varphi(N) \geq 1,$$

since $a = 1$ always witnesses $\gcd(1, N) = 1$ in the totient count. Hence G contains an element g of order N ; its powers form a subgroup of G with N distinct elements (Proposition 3.12(3)), which must be all of G . So $G = \langle g \rangle$ is cyclic.

For the final statement, let $F = \mathbb{F}_q$ be finite. The multiplicative group $\mathbb{F}_q^\times$ has $q - 1$ elements—every nonzero element of a field is a unit—and is a finite subgroup of itself, so it is cyclic of order $q - 1$. $\square$

The equality $\psi(d) = \varphi(d)$ established along the way is worth extracting: it was proved for subgroups of a field's multiplicative group, but it holds in any cyclic group, by the same gcd computation and with no field in sight.

Corollary 6.18 (Element-order census in cyclic groups). *A cyclic group G of order N contains, for each divisor $d \mid N$, exactly $\varphi(d)$ elements of order d , and no elements of any other order. In particular G has exactly $\varphi(N)$ generators.*

Proof. Write $G = \langle g \rangle$ with $\text{ord}(g) = N$; the elements of G are the powers g^j , $0 \leq j < N$, each appearing once (Proposition 3.12(3)). By Corollary 3.23, $\text{ord}(g^j) = N / \gcd(N, j)$, which is always a divisor of N ; and $\text{ord}(g^j) = d$ if and only if $\gcd(j, N) = N/d$. By the gcd computation in the proof of Lemma 6.16 (applied with $n = N$ and divisor N/d), the exponents $j \in \{1, \dots, N\}$ with $\gcd(j, N) = N/d$ are exactly $j = (N/d)t$ with $1 \leq t \leq d$ and $\gcd(t, d) = 1$, of which there are $\varphi(d)$ by Definition 2.26. The generators are the elements of order N , numbering $\varphi(N)$. $\square$

Definition 6.19 (Primitive root). A generator of the cyclic group $\mathbb{F}_q^\times$ is called a *primitive root* (or *primitive element*) of $\mathbb{F}_q$. Equivalently, an element $g \in \mathbb{F}_q^\times$ is primitive if and only if $\text{ord}(g) = q - 1$.

Existence is theorem 6.17, and the census refines it: $\mathbb{F}_q$ has exactly $\varphi(q - 1)$ primitive roots (theorem 6.18), always at least one.

Remark 6.20. The name has a second reading, developed later in the volume: a primitive root of $\mathbb{F}_q$ is precisely a primitive $(q - 1)$ -th root of unity in the sense of theorem 9.3, where the identification is restated once roots of unity have been defined in general. The two vocabularies—group-theoretic “generator of $\mathbb{F}_q^\times$ ” and polynomial-flavoured “primitive $(q - 1)$ -th root of unity”—name the same elements.

Example 6.21 (Primitive roots of $\mathbb{F}_7$). The group $\mathbb{F}_7^\times$ has order 6. Testing $g = 3$, the successive powers modulo 7 are

$$3^1 = 3, \quad 3^2 = 2, \quad 3^3 = 6, \quad 3^4 = 4, \quad 3^5 = 5, \quad 3^6 = 1,$$

which run through all of $\{1, \dots, 6\}$: the element 3 is a primitive root of $\mathbb{F}_7$. By contrast 2 is not: its powers are 2, 4, 1, so $\text{ord}(2) = 3 < 6$. The full roster of primitive roots consists of the powers 3^j with $\gcd(j, 6) = 1$ (Corollary 3.23), namely $j = 1$ and $j = 5$: the primitive roots of $\mathbb{F}_7$ are $3^1 = 3$ and $3^5 = 5$, and there are $\varphi(6) = 2$ of them, as the census predicts.

Cyclicity settles, in one stroke, how many solutions the equation $x^n = 1$ has in a finite field—the question on which the evaluation domains of later sections turn.

Corollary 6.22 (Roots of unity in $\mathbb{F}_q$). *Let $\mathbb{F}_q$ be a finite field and $n \geq 1$. The solutions of $x^n = 1$ in $\mathbb{F}_q^\times$ number exactly $\gcd(n, q - 1)$, and they form the cyclic subgroup of $\mathbb{F}_q^\times$ of that order. In particular, when $n \mid q - 1$ the equation has exactly n solutions.*

Proof. Fix a primitive root g (theorem 6.17) and write each element of $\mathbb{F}_q^\times$ uniquely as $x = g^j$ with $0 \leq j \leq q-2$ (Proposition 3.12(3)). Then $x^n = g^{jn}$, and $g^{jn} = 1$ holds if and only if $(q-1) \mid jn$ (Proposition 3.12(1)). Set $d = \gcd(n, q-1)$ and write $q-1 = dm$, $n = dn'$ with $\gcd(m, n') = 1$. Then

$$(q-1) \mid jn \iff dm \mid jdn' \iff m \mid jn' \iff m \mid j,$$

the last step by Euclid's lemma (Proposition 2.15(1), with $\gcd(m, n') = 1$). The qualifying exponents in $\{0, \dots, q-2\}$ are therefore the multiples of $m = (q-1)/d$, namely $j = tm$ for $t = 0, 1, \dots, d-1$: exactly d of them. The corresponding solutions $x = (g^m)^t$ are exactly the powers of g^m , which by Corollary 3.23 has order $(q-1)/\gcd(q-1, m) = (q-1)/m = d$; so the solution set is the cyclic subgroup $\langle g^m \rangle$ of order $d = \gcd(n, q-1)$. When $n \mid q-1$ we have $d = n$, giving exactly n solutions. $\square$

Remark 6.23 (Two-adicity and proof-system fields). The corollary is precisely what renders a curve such as Pallas or Vesta suitable for Halo 2: one chooses the field size q so that $q-1$ is divisible by a large power of 2, and the corollary then *guarantees* a multiplicative subgroup of 2-power order—the evaluation domain on which the number-theoretic transform, and hence efficient polynomial arithmetic, operates. The construction of these domains and the transform itself are the business of section 9; the curves are introduced in section 10.

Remark 6.24 (Cyclicity and Frobenius in cryptography). The section's two structural discoveries each carry cryptographic weight. First, the cyclicity of $\mathbb{F}_q^\times$ (theorem 6.17) underlies the discrete-logarithm problem—writing every nonzero element as g^j invites the question of recovering j , and the presumed hardness of doing so is a foundational assumption of later volumes—and, through theorem 6.22, guarantees the existence of the roots of unity used in polynomial commitment schemes. Second, the Frobenius endomorphism (theorem 6.14) is not merely abstract: on an elliptic curve over $\mathbb{F}_p$ it induces an endomorphism of the curve whose characteristic polynomial $T^2 - tT + p$ encodes the number of curve points as $N = p + 1 - t$, where t is the Hasse trace—tying the size of the cryptographic group directly to the field-theoretic Frobenius. Both connections are taken up in section 10.

The question that opened the section is now answered in full. A finite arithmetic in which every nonzero element inverts exists only at prime-power sizes—theorem 6.9 forbids every other cardinality—and at each admissible size $q = p^k$ there is exactly one field, up to isomorphism (theorem 6.10). The machine's own wrap-around arithmetic is instructive here: 2^{64} is a prime-power size, so a field with 2^{64} elements exists, but it is *not* the ring $\mathbb{Z}/2^{64}\mathbb{Z}$ of word arithmetic—whose even classes have no inverses—but the polynomial-quotient construction of theorem 6.10, with a different multiplication altogether. The proof systems of this series instead take the prime route: a prime p of roughly 255 bits, the field $\mathbb{F}_p$, inversion by theorem 6.1 or theorem 6.4. The payoff theorem then delivers the structure that everything later rides on: the multiplicative group of the chosen field is a single cycle, generated by one element, with exactly $\varphi(d)$ elements of each order $d \mid q-1$ and a guaranteed subgroup of every size dividing $q-1$. Polynomials over these fields, the subject that the volume develops next into evaluation domains, fast transforms, and commitment schemes, will draw on that guarantee constantly.

7 Number theory for cryptography

An implementation that transmits a point of an elliptic curve—a pair (x, y) of field elements satisfying an equation $y^2 = x^3 + ax + b$, as constructed in section 10—need not send both coordinates. For a given x the equation determines y^2 exactly, hence determines y up to sign, so wire formats send x together with a single bit selecting one of the two candidates, and the transmission halves in size. The receiver is then left with two number-theoretic tasks: decide whether $x^3 + ax + b$ is a square modulo p at all—if not, no point has this x -coordinate—and, if it is, *compute* a square root

of it. Both tasks must run in the time of a few modular exponentiations, on numbers hundreds of bits long. The same constructions make a complementary demand in the opposite direction: they publish powers g^x of a known group element while the exponent x is a secret key, so exponentiation must be easy to perform and yet infeasible to *invert*. This section assembles the classical number theory behind both demands: the totient and the Chinese Remainder Theorem, Fermat's little theorem and Euler's theorem in their role as workhorses of modular exponentiation, multiplicative orders and primitive roots, quadratic residues with the Euler criterion, square-root extraction modulo a prime, and finally the discrete logarithm problem together with the reason *prime* group order is so important for cryptographic hardness.

Throughout, n and m denote positive integers and p denotes a prime. The section builds directly on the arithmetic of section 2, the group theory of section 3, and the cyclicity theorem of section 6; several statements proved there reappear here, restated in the arithmetic language in which cryptography uses them, with proofs reduced to citations.

7.1 Euler's totient function and the Chinese Remainder Theorem

Recall from section 2 that a class $[a]_n \in \mathbb{Z}/n\mathbb{Z}$ is a unit if and only if $\gcd(a, n) = 1$ (Theorem 2.24), that the units form a finite abelian group $(\mathbb{Z}/n\mathbb{Z})^\times$ (Proposition 3.8), and that Euler's totient $\varphi(n)$ counts them: $\varphi(n) = |(\mathbb{Z}/n\mathbb{Z})^\times|$ (Definition 2.26). In particular $\varphi(p) = p - 1$ for a prime p (Example 2.27), so $\mathbb{F}_p^\times$ has order $p - 1$. These counts govern everything that follows: the totient is the exponent in Euler's theorem, the order of the group whose cyclic structure the next subsections exploit, and—through the orders of elliptic-curve groups—a quantity that the parameter choices of Halo 2 and Orchard are built around. What section 2 did not provide is a way to *compute* $\varphi(n)$ beyond direct enumeration. Two facts close the gap: the totient's value on prime powers, and its behaviour on coprime factors.

Proposition 7.1 (Totient of a prime power). *For a prime p and an integer $k \geq 1$,*

$$\varphi(p^k) = p^k - p^{k-1} = p^k \left(1 - \frac{1}{p}\right).$$

Proof. Among $1, \dots, p^k$, we claim the integers *not* coprime to p^k are exactly the multiples of p . If $p \mid a$ then $p \mid \gcd(a, p^k)$, so the gcd exceeds 1. Conversely, if $g = \gcd(a, p^k) > 1$, then g has a prime factor ℓ ; from $\ell \mid g \mid p^k$ and Euclid's lemma for primes (Lemma 2.17) we get $\ell = p$, and $\ell \mid g \mid a$ gives $p \mid a$. The multiples of p in the range are $p, 2p, \dots, p^{k-1} \cdot p$, of which there are p^{k-1} ; subtracting them from the p^k integers in the range leaves $\varphi(p^k) = p^k - p^{k-1}$. $\square$

The behaviour on coprime factors is a consequence of a structural theorem that will be used again, in a quite different role, at the end of the section: a residue modulo a product of two coprime moduli carries exactly the same information as a *pair* of residues, one modulo each factor. The natural home for the statement is the language of rings (section 4): recall the product ring $R \times S$ with componentwise operations (Example 4.5), and call a bijective ring homomorphism (Definition 4.21) a *ring isomorphism*, mirroring Definition 3.41 for groups.

Theorem 7.2 (Chinese Remainder Theorem). *If $\gcd(m, n) = 1$, then the map*

$$\mathbb{Z}/mn\mathbb{Z} \longrightarrow \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}, \quad [a]_{mn} \longmapsto ([a]_m, [a]_n)$$

is a ring isomorphism. It restricts to a group isomorphism

$$(\mathbb{Z}/mn\mathbb{Z})^\times \cong (\mathbb{Z}/m\mathbb{Z})^\times \times (\mathbb{Z}/n\mathbb{Z})^\times.$$

Proof. The map is well defined: if $[a]_{mn} = [a']_{mn}$ then $mn \mid a - a'$, hence $m \mid a - a'$ and $n \mid a - a'$, so the two components agree. It is a ring homomorphism because each component is: reduction modulo m (and modulo n) sends sums to sums, products to products, and 1 to 1 (Theorem 4.6), and the product ring's operations are componentwise.

For injectivity it suffices, since the map is in particular a homomorphism of additive groups, to check that the kernel is trivial (Proposition 3.40). Suppose $[a]_m = [0]_m$ and $[a]_n = [0]_n$, that is, $m \mid a$ and $n \mid a$. Since $\gcd(m, n) = 1$, Euclid's lemma (Proposition 2.15(2)) gives $mn \mid a$, i.e. $[a]_{mn} = [0]_{mn}$.

Both sides have exactly mn elements (Theorem 4.6 for each factor), and an injective map between finite sets of equal size is bijective: the mn distinct inputs have mn distinct images, which must exhaust the codomain. Hence the map is a ring isomorphism.

For the multiplicative statement, a ring isomorphism ϕ matches units with units: if $uv = 1$ then $\phi(u)\phi(v) = \phi(1) = 1$, and the same applies to ϕ^{-1} . The units of the product ring are exactly the pairs of units, $(u, v)^{-1} = (u^{-1}, v^{-1})$ existing precisely when both components are invertible. Restricting ϕ to unit groups therefore gives a bijection $(\mathbb{Z}/mn\mathbb{Z})^\times \rightarrow (\mathbb{Z}/m\mathbb{Z})^\times \times (\mathbb{Z}/n\mathbb{Z})^\times$ that respects multiplication: a group isomorphism. $\square$

Corollary 7.3 (Multiplicativity and the product formula). *If $\gcd(m, n) = 1$ then $\phi(mn) = \phi(m)\phi(n)$. Hence if $n = \prod_i p_i^{k_i}$ is the prime factorisation of n , then*

$$\phi(n) = \prod_i (p_i^{k_i} - p_i^{k_i-1}) = n \prod_{p \mid n} \left(1 - \frac{1}{p}\right),$$

the last product running over the distinct primes dividing n .

Proof. The unit-group isomorphism of Theorem 7.2 is in particular a bijection, and the cardinality of a product of two sets is the product of the cardinalities, so $\phi(mn) = \phi(m)\phi(n)$.

For the product formula, the prime powers $p_i^{k_i}$ are pairwise coprime: a common divisor exceeding 1 of two of them would have a prime factor dividing both, which by Euclid's lemma for primes (Lemma 2.17) would equal both p_i and p_j . Moreover each $p_i^{k_i}$ is coprime to the product of the others, by repeated application of Proposition 2.15(3). Induction on the number of factors therefore extends multiplicativity to $\phi(n) = \prod_i \phi(p_i^{k_i})$, and Proposition 7.1 evaluates each factor:

$$\phi(n) = \prod_i (p_i^{k_i} - p_i^{k_i-1}) = \prod_i p_i^{k_i} \left(1 - \frac{1}{p_i}\right) = n \prod_{p \mid n} \left(1 - \frac{1}{p}\right). \quad \square$$

As a sanity check, the product formula gives $\phi(12) = 12(1 - \frac{1}{2})(1 - \frac{1}{3}) = 4$, agreeing with the direct enumeration $\{1, 5, 7, 11\}$ of Example 2.27. Together with Gauss's divisor-sum identity $\sum_{d \mid n} \phi(d) = n$, proved as Lemma 6.16, this completes the totient's basic theory.

7.2 Fermat's little theorem and Euler's theorem

The congruences of this subsection were proved once, as Corollary 3.32, where they fell out of Lagrange's theorem in three lines. We restate them here in the arithmetic form in which cryptography uses them; the proofs are citations, not repetitions.

Theorem 7.4 (Euler's theorem). *If $\gcd(a, n) = 1$, then*

$$a^{\phi(n)} \equiv 1 \pmod{n}.$$

Proof. The hypothesis says that $[a]_n$ is a unit (Theorem 2.24), i.e. an element of the finite group $(\mathbb{Z}/n\mathbb{Z})^\times$, whose order is $\varphi(n)$ (Definition 2.26). By Corollary 3.31—every element g of a finite group of order N satisfies $g^N = e$, a consequence of Lagrange’s theorem—we get $[a]_n^{\varphi(n)} = [1]_n$, which is the stated congruence. This is Corollary 3.32 verbatim. $\square$

Corollary 7.5 (Fermat’s little theorem). *Let p be prime. If $p \nmid a$, then $a^{p-1} \equiv 1 \pmod{p}$. Moreover $a^p \equiv a \pmod{p}$ for every integer a .*

Proof. For $p \nmid a$ we have $\gcd(a, p) = 1$, so Theorem 7.4 with $n = p$ and $\varphi(p) = p - 1$ (Example 2.27) gives the first congruence; multiplying it by a gives $a^p \equiv a \pmod{p}$. When $p \mid a$ both a^p and a are congruent to 0, so the second congruence holds for all a . $\square$

Remark 7.6 (Cryptographic use of Euler and Fermat). Euler’s theorem underlies RSA: if $ed \equiv 1 \pmod{\varphi(n)}$, say $ed = 1 + k\varphi(n)$, then

$$(a^e)^d = a^{1+k\varphi(n)} = a \cdot (a^{\varphi(n)})^k \equiv a \pmod{n}$$

whenever $\gcd(a, n) = 1$: raising to the e -th power is undone by raising to the d -th, which is what makes one exponent a public encryption key and the other its private inverse. The elliptic-curve cryptography of Orchard instead exploits Fermat’s little theorem in the field $\mathbb{F}_p$: every nonzero x satisfies $x^{p-1} = 1$, so inversion is computable as $x^{-1} = x^{p-2}$ (theorem 6.4), and Fermat is the engine of the Euler criterion for square roots developed below.

7.3 Multiplicative order and primitive roots

Definition 7.7 (Multiplicative order). Let $\gcd(a, n) = 1$. The *multiplicative order* of a modulo n , written $\text{ord}_n(a)$, is the least positive integer k with $a^k \equiv 1 \pmod{n}$. Such a k exists because $a^{\varphi(n)} \equiv 1$ by Theorem 7.4.

The quantity $\text{ord}_n(a)$ is exactly the order of $[a]_n$ as an element of the group $(\mathbb{Z}/n\mathbb{Z})^\times$, so the order theory of section 3 applies verbatim: $a^m \equiv 1 \pmod{n}$ if and only if $\text{ord}_n(a) \mid m$ (Proposition 3.12(1))—in particular $\text{ord}_n(a) \mid \varphi(n)$, by Euler’s theorem—and

$$\text{ord}_n(a^j) = \frac{\text{ord}_n(a)}{\gcd(j, \text{ord}_n(a))}$$

(Corollary 3.23).

Definition 7.8 (Primitive root). A *primitive root* modulo n is an element g with $\text{ord}_n(g) = \varphi(n)$; equivalently, g generates $(\mathbb{Z}/n\mathbb{Z})^\times$ as a cyclic group, so that $\{g^0, g^1, \dots, g^{\varphi(n)-1}\}$ lists all the units.

Primitive roots need not exist for every modulus. The unit group $(\mathbb{Z}/8\mathbb{Z})^\times = \{1, 3, 5, 7\}$ has order $\varphi(8) = 4$, yet every one of its elements squares to 1 (Example 3.22): the element orders are 1 for the identity and 2 for each of 3, 5, 7, so no element has order 4 and no primitive root modulo 8 exists. Indeed the unit group is a copy of the Klein four-group of Example 3.10: every element is its own inverse, and the product of two distinct non-identity elements a and b is neither e (which would force $b = a^{-1} = a$) nor a nor b (which would force $b = e$ or $a = e$), so it is the third. The decisive case for cryptography is $n = p$ prime, where the multiplicative group is always cyclic.

Corollary 7.9 ($\mathbb{F}_p^\times$ is cyclic). *For every prime p , the group $\mathbb{F}_p^\times = (\mathbb{Z}/p\mathbb{Z})^\times$ is cyclic of order $p - 1$. In particular primitive roots modulo p exist, and there are exactly $\varphi(p - 1)$ of them; Example 6.21 exhibits the two primitive roots 3 and 5 of $\mathbb{F}_7$.*

Proof. The group $\mathbb{F}_p^\times$ is a finite subgroup, of order $p - 1$, of the multiplicative group of the field $\mathbb{F}_p$, hence cyclic by Theorem 6.17. A generator g has order $p - 1$, and the powers g^k that again generate are those with $\gcd(k, p - 1) = 1$ (Corollary 3.23), numbering $\varphi(p - 1)$. $\square$

Remark 7.10 (Cyclicity and prime-order groups). Cyclicity is why the prime-order groups of elliptic-curve cryptography behave exactly like $\mathbb{Z}/q\mathbb{Z}$ under addition once a generator is fixed: a cyclic group of order q is isomorphic to $(\mathbb{Z}/q\mathbb{Z}, +)$ via $g^k \leftrightarrow k$ (Theorem 3.43). For completeness we record, without proof and without needing it later, the full classification: primitive roots modulo n exist exactly when $n \in \{1, 2, 4, p^k, 2p^k\}$ for an odd prime p .

7.4 Quadratic residues and the Euler criterion

We turn to squares modulo a prime. They control whether a curve point with a given x -coordinate exists—the first of the two tasks posed at the head of the section—and they are the objects from which square roots in $\mathbb{F}_p$ will be extracted.

Definition 7.11 (Quadratic residue). Let p be an odd prime and a an integer with $p \nmid a$. We say a is a *quadratic residue* modulo p (a QR) if the congruence $x^2 \equiv a \pmod{p}$ has a solution, and a *quadratic non-residue* (a QNR) otherwise.

Proposition 7.12 (Exactly half are residues). *For an odd prime p , exactly $\frac{p-1}{2}$ of the nonzero residues modulo p are quadratic residues and $\frac{p-1}{2}$ are non-residues. The squaring map $x \mapsto x^2$ is exactly two-to-one on $\mathbb{F}_p^\times$.*

Proof. Consider $\sigma: \mathbb{F}_p^\times \rightarrow \mathbb{F}_p^\times$, $\sigma(x) = x^2$; its image is exactly the set of quadratic residues. For $x, y \in \mathbb{F}_p^\times$,

$$\sigma(x) = \sigma(y) \iff x^2 - y^2 = 0 \iff (x - y)(x + y) = 0 \iff y = \pm x,$$

the middle equivalence because a field has no zero divisors (Proposition 4.28). Since p is odd, $x \neq -x$ for $x \neq 0$ (otherwise $2x = 0$ with 2 and x nonzero), so every fibre of σ over its image has exactly two elements. Counting, $|\text{im } \sigma| = \frac{p-1}{2}$, and the complement—the non-residues—has the same size. $\square$

Theorem 7.13 (Euler's criterion). *Let p be an odd prime and $p \nmid a$. Then*

$$a^{\frac{p-1}{2}} \equiv \begin{cases} 1 \pmod{p} & \text{if } a \text{ is a quadratic residue,} \\ -1 \pmod{p} & \text{if } a \text{ is a quadratic non-residue.} \end{cases}$$

Proof. First, the value is always ± 1 . Fermat's little theorem (Corollary 7.5) gives $(a^{(p-1)/2})^2 = a^{p-1} \equiv 1$, and the only square roots of 1 modulo a prime are ± 1 : from $x^2 \equiv 1$ we get $p \mid (x - 1)(x + 1)$, so $p \mid x - 1$ or $p \mid x + 1$ by Euclid's lemma for primes (Lemma 2.17), i.e. $x \equiv \pm 1$. Hence $a^{(p-1)/2} \equiv \pm 1 \pmod{p}$.

If a is a residue, say $a \equiv x^2$ with $p \nmid x$, then $a^{(p-1)/2} \equiv x^{p-1} \equiv 1$, again by Fermat. Thus every one of the $\frac{p-1}{2}$ quadratic residues (Proposition 7.12) is a root of the polynomial $t^{(p-1)/2} - 1$ over $\mathbb{F}_p$. That polynomial has degree $\frac{p-1}{2}$, so it has *at most* $\frac{p-1}{2}$ roots (Theorem 5.18); the residues therefore account for all of them. A non-residue a is consequently not a root, yet still satisfies $a^{(p-1)/2} \equiv \pm 1$; the remaining possibility is $a^{(p-1)/2} \equiv -1$. $\square$

The standard notation for quadratic-residue status is the Legendre symbol.

Definition 7.14 (Legendre symbol). For an odd prime p and an integer a , the *Legendre symbol* is

$$\left(\frac{a}{p}\right) = \begin{cases} 0 & \text{if } p \mid a, \\ 1 & \text{if } a \text{ is a quadratic residue modulo } p, \\ -1 & \text{if } a \text{ is a quadratic non-residue modulo } p. \end{cases}$$

Euler's criterion (Theorem 7.13) then reads, uniformly,

$$\left(\frac{a}{p}\right) \equiv a^{\frac{p-1}{2}} \pmod{p},$$

so the symbol is computable by a single modular exponentiation—an $O(\log p)$ affair by square-and-multiply (theorem 6.2). The criterion also shows that the symbol is *completely multiplicative* in a :

$$\left(\frac{ab}{p}\right) = \left(\frac{a}{p}\right) \left(\frac{b}{p}\right), \quad \text{since} \quad (ab)^{\frac{p-1}{2}} = a^{\frac{p-1}{2}} b^{\frac{p-1}{2}}.$$

Both observations carry weight in the worked example that follows, which verifies a constant deployed in Orchard.

Example 7.15 (13 is a non-square in both Pasta fields). The Pasta hash-to-curve construction used by Orchard—a procedure, taken up in later volumes, that maps arbitrary byte strings to curve points and requires a fixed non-square constant in the base field—fixes $Z = -13$ as a verifiably non-square constant in both base fields. Let p and q be the Pasta primes of section 10; both satisfy $p \equiv q \equiv 1 \pmod{4}$. We verify the non-squareness here, working modulo p ; the computation modulo q is identical.

First, -1 is a square modulo any prime $\equiv 1 \pmod{4}$: the exponent $\frac{p-1}{2}$ is even, so Euler's criterion gives

$$\left(\frac{-1}{p}\right) \equiv (-1)^{\frac{p-1}{2}} = 1.$$

By complete multiplicativity,

$$\left(\frac{-13}{p}\right) = \left(\frac{-1}{p}\right) \left(\frac{13}{p}\right) = \left(\frac{13}{p}\right),$$

so -13 is a non-square precisely when 13 is. Whether 13 is a square is settled by one Euler-criterion exponentiation in each field, a computation any computer-algebra system reproduces in milliseconds:

$$13^{\frac{p-1}{2}} \equiv p-1 \equiv -1 \pmod{p}, \quad 13^{\frac{q-1}{2}} \equiv q-1 \equiv -1 \pmod{q},$$

whence $\left(\frac{13}{p}\right) = \left(\frac{13}{q}\right) = -1$. Thus 13 , and with it $Z = -13$, is a quadratic non-residue modulo both Pasta primes, exactly as the hash-to-curve design requires.

Example 7.16 (5 is a non-square in the Pallas base field). The same one-line test settles a second constant on which later volumes lean, this time in the curve equation itself. Pallas is the curve $y^2 = x^3 + 5$ over $\mathbb{F}_p$ (section 10), and the arithmetic circuits of Halo 2 encode its point at infinity as the coordinate pair $(0, 0)$ (theorem 10.13). The encoding is unambiguous only if no genuine point has x -coordinate 0, i.e. only if $y^2 = 5$ has no solution in $\mathbb{F}_p$: the question is whether 5 is a square in the *base* field $\mathbb{F}_p$ of Pallas, not in its scalar field $\mathbb{F}_q$. Euler’s criterion decides it in one exponentiation:

$$5^{\frac{p-1}{2}} \equiv p-1 \equiv -1 \pmod{p},$$

so $\left(\frac{5}{p}\right) = -1$. The constant 5 is a quadratic non-residue modulo the Pallas base-field prime, and no point of Pallas has $x = 0$. (The same computation modulo q gives $5^{(q-1)/2} \equiv -1 \pmod{q}$, so Vesta has no such point either.) That no point has $y = 0$ is a different fact—5 is not a cube in $\mathbb{F}_p$ —which theorem 10.26 reads off from the oddness of the group order; the two together are the hypotheses of theorem 10.14.

7.5 Square roots modulo a prime

Deciding that a is a residue is one matter; *producing* an x with $x^2 \equiv a$ is another, and the point decompression of the section’s opening needs the production, not just the decision: the receiver must reconstruct the coordinate y from y^2 . How hard this is depends on $p \bmod 4$.

Proposition 7.17 (The $p \equiv 3 \pmod{4}$ shortcut). *Let $p \equiv 3 \pmod{4}$ be prime and let a be a quadratic residue modulo p with $p \nmid a$. Then*

$$x \equiv a^{\frac{p+1}{4}} \pmod{p}$$

is a square root of a , and the two square roots of a are $\pm x$.

Proof. Since $p \equiv 3 \pmod{4}$, the quantity $\frac{p+1}{4}$ is a positive integer. Compute

$$x^2 = a^{\frac{p+1}{2}} = a^{\frac{p-1}{2}} \cdot a \equiv \left(\frac{a}{p}\right) \cdot a = a \pmod{p},$$

using Euler’s criterion (Theorem 7.13) and the hypothesis that a is a residue, so $\left(\frac{a}{p}\right) = 1$. The roots come in the pair $\pm x$ by the two-to-one property of squaring (Proposition 7.12). $\square$

Remark 7.18 (Cost and the Pasta primes). The single-exponentiation shortcut requires $p \equiv 3 \pmod{4}$, which is one reason cryptographers often favour such primes. For $p \equiv 5 \pmod{8}$ a closed-form variant (Atkin’s formula) still exists. In general the cost of square-root extraction grows with the 2-adic valuation s of $p-1$: writing $p-1 = 2^s d$ with d odd, the Tonelli–Shanks algorithm below performs up to s corrective iterations. The Pasta primes have $s = 32$, a value chosen deliberately for FFT-friendliness (theorem 9.10), so their implementations use Tonelli–Shanks.

When $p \equiv 1 \pmod{4}$ one writes $p-1 = 2^s d$ with d odd and $s \geq 2$. The algorithm operates inside the subgroup of $\mathbb{F}_p^\times$ of order 2^s —the unique such subgroup, since $\mathbb{F}_p^\times$ is cyclic (Corollary 7.9 and Theorem 3.24(3))—where the obstruction to the naive exponentiation trick lives, and clears that obstruction one power of two at a time.

Theorem 7.19 (Tonelli–Shanks). *Let p be an odd prime, $p-1 = 2^s d$ with d odd, and let a be a*

quadratic residue with $p \nmid a$. Given any quadratic non-residue z modulo p , one can compute a square root of a modulo p using $O(s^2 + \log p)$ modular multiplications.

Proof. Write $H \leq \mathbb{F}_p^\times$ for the unique cyclic subgroup of order 2^s . Set $c = z^d$. We claim c generates H . By Euler's criterion (Theorem 7.13) $z^{(p-1)/2} \equiv -1$, so

$$c^{2^{s-1}} = z^{2^{s-1}d} = z^{\frac{p-1}{2}} \equiv -1 \neq 1, \quad c^{2^s} = z^{p-1} \equiv 1$$

by Fermat (Corollary 7.5); hence $\text{ord}_p(c)$ divides 2^s but not 2^{s-1} , forcing $\text{ord}_p(c) = 2^s$ and $\langle c \rangle = H$.

Initialise

$$x = a^{\frac{d+1}{2}}, \quad t = a^d, \quad m = s$$

(the exponent $\frac{d+1}{2}$ is an integer because d is odd). We maintain three invariants:

1. $x^2 = at$;
2. the current c has order exactly 2^m ;
3. the order of t divides 2^{m-1} .

Initially (1) holds since $x^2 = a^{d+1} = a \cdot a^d$; (2) was just proved; and (3) holds because $t^{2^{s-1}} = a^{(p-1)/2} \equiv 1$ by Euler's criterion (Theorem 7.13), a being a residue.

Now iterate. If $t = 1$, then $x^2 = a$ by (1) and we are done. Otherwise let $2^i = \text{ord}_p(t)$, found as the least $i \geq 1$ with $t^{2^i} = 1$ by repeatedly squaring t ; invariant (3) gives $1 \leq i \leq m-1$. Set

$$b = c^{2^{m-i-1}},$$

and update

$$x \leftarrow xb, \quad t \leftarrow tb^2, \quad c \leftarrow b^2, \quad m \leftarrow i.$$

The invariants survive. For (1), $(xb)^2 = x^2b^2 = atb^2$, which is a times the new t . For (2), the old c has order 2^m , so $\text{ord}_p(b) = 2^m / \gcd(2^m, 2^{m-i-1}) = 2^{i+1}$ (Corollary 3.23), and the new $c = b^2$ has order 2^i —exactly 2 to the new m . For (3), both the old t and b^2 have order exactly 2^i , so $t^{2^{i-1}}$ and $(b^2)^{2^{i-1}}$ each have order 2; the only element of order 2 in $\mathbb{F}_p^\times$ is -1 , because the square roots of 1 are ± 1 (as in the proof of Theorem 7.13); hence

$$(tb^2)^{2^{i-1}} = t^{2^{i-1}}(b^2)^{2^{i-1}} = (-1)(-1) = 1,$$

so the new t has order dividing 2^{i-1} , which is 2^{m-1} for the new m .

Each iteration strictly decreases m , from s through a descending chain of nonnegative integers, so after at most s iterations the order of t reaches $2^0 = 1$, i.e. $t = 1$, and x is the desired root. For the cost: locating i and computing b each take at most $m \leq s$ squarings, so the loop costs $O(s)$ multiplications per iteration and $O(s^2)$ in total; the three initial exponentiations, with exponents smaller than p , cost $O(\log p)$ multiplications each by square-and-multiply (theorem 6.2). $\square$

The one ingredient the theorem presupposes is a quadratic non-residue z , and here—for the only time in this section—randomness enters. A uniformly random element of $\mathbb{F}_p^\times$ is a non-residue with probability exactly $\frac{1}{2}$, since precisely half the elements qualify (Proposition 7.12), and each candidate is checked by one Euler-criterion exponentiation (Definition 7.14), i.e. $O(\log p)$ modular multiplications. A handful of random trials therefore finds z almost immediately, and implementations simply hard-code one non-residue per field, verified once—as the constant $Z = -13$ of Example 7.15 is. No deterministic polynomial-time method for finding a non-residue is invoked anywhere in this volume.

Remark 7.20 (Which root, and detecting failure). In every case the square roots of a residue come in a pair $\pm x$ (Proposition 7.12), so a *canonical* root must be pinned down by a convention—for instance “choose the root whose least nonnegative representative is even”, as elliptic-curve point encodings do; the compressed point’s extra bit selects between the two. If a is *not* a residue, both algorithms above detect the failure, in different ways. The shortcut of Proposition 7.17 still outputs a candidate $a^{(p+1)/4}$, and that candidate fails the final check $x^2 \equiv a$ —one squaring suffices to detect failure. The variable-time Tonelli–Shanks algorithm just given cannot complete its update on a non-residue: Euler’s criterion gives $a^{(p-1)/2} \equiv -1$, so $t = a^d$ satisfies $t^{2^{s-1}} \equiv -1$ and has order exactly 2^s , and already the first search for the least $i \geq 1$ with $t^{2^i} = 1$ returns $i = m$, which invariant (3) of the proof of Theorem 7.19 forbids and for which the update exponent 2^{m-i-1} is undefined. This variant can report “not a residue” the moment $i = m$ occurs; equivalently it may verify $a^{(p-1)/2} \equiv 1$ up front, at the same $O(\log p)$ cost as the initial exponentiations.

7.6 The discrete logarithm problem

The second demand of the section’s opening—exponentiation easy forward, infeasible backward—now takes centre stage. We phrase it for a generic cyclic group, written multiplicatively, because in Orchard the group in question is the group of points of an elliptic curve, written additively, of large prime order; every statement below transfers by renaming the operation.

Definition 7.21 (Discrete logarithm problem). Let $G = \langle g \rangle$ be a finite cyclic group of order N with generator g . Given $h \in G$, the *discrete logarithm* of h to base g , written $\log_g h$, is the unique residue $x \in \mathbb{Z}/N\mathbb{Z}$ with $g^x = h$. The *discrete logarithm problem* (DLP) is to compute x from (G, g, h) .

Existence and uniqueness of $\log_g h$ are the content of the classification of cyclic groups: the map $x \mapsto g^x$ is a group isomorphism $(\mathbb{Z}/N\mathbb{Z}, +) \xrightarrow{\sim} G$ (Theorem 3.43, whose map $\bar{\phi}$ this is), so it has a well-defined inverse, and $\log_g$ is that inverse. The two directions could not differ more in cost. Computing g^x from x takes $O(\log N)$ group operations by square-and-multiply (theorem 6.2); computing x from g^x is the DLP, for which no efficient algorithm is known in well-chosen groups. This asymmetry—a bijection cheap in one direction and expensive in the other—is the one-wayness on which elliptic-curve cryptography rests.

Remark 7.22 (Generic complexity and group choice). Two *generic* algorithms solve the DLP in any group of order N , using nothing but the group operation. *Baby-step giant-step* is deterministic: set $m = \lceil \sqrt{N} \rceil$ and write the unknown as $x = im + j$ with $0 \leq i, j < m$; tabulate the “baby steps” g^j for all j , then walk the “giant steps” $h(g^{-m})^i$ for $i = 0, 1, 2, \dots$ until one equals a tabulated g^j , at which point $hg^{-im} = g^j$ gives $x = im + j$. Both the table and the walk are $O(\sqrt{N})$, so the cost is $O(\sqrt{N})$ time and space. *Pollard’s rho* (not developed here) is a randomised alternative with the same $O(\sqrt{N})$ expected running time—on a heuristic analysis, made explicit in section 10—but only $O(1)$ space, and is what attackers would run in practice. In the other direction, Shoup’s theorem shows that $\Omega(\sqrt{N})$ group operations are *necessary* for any generic algorithm when N is prime, so for generic attacks the square-root cost is exact. Reaching a λ -bit security level—no attack cheaper than 2^λ operations—therefore requires $N \approx 2^{2\lambda}$. This is why the Pallas and Vesta scalar fields have prime order roughly 255 bits long, targeting roughly 128-bit security (section 10). The qualifier *generic* matters: in $(\mathbb{Z}/p\mathbb{Z})^\times$ sub-exponential index-calculus attacks exist, which exploit the ring structure of the integers behind the group, and this is why elliptic-curve groups—where no such attack is known—permit much smaller parameters than multiplicative groups of comparable security.

The next theorem makes the phrase “well-chosen” precise: if the group order N is composite with small prime factors, the DLP falls apart into small pieces, and the Chinese Remainder Theorem—proved in this section for a purely arithmetic purpose—reassembles the pieces into the secret.

Theorem 7.23 (Pohlig–Hellman reduction). *Let $G = \langle g \rangle$ be cyclic of order $N = \prod_{i=1}^r q_i^{e_i}$, the q_i distinct primes. The discrete logarithm in G is computable by solving discrete logarithms in subgroups of prime order q_i — e_i of them for each i —and combining the answers with the Chinese Remainder Theorem, at a total cost of*

$$O\left(\sum_{i=1}^r e_i(\log N + \sqrt{q_i})\right)$$

group operations. When N has a large prime factor $q_{\max}$ the $\sqrt{q_{\max}}$ term dominates; when N is smooth—a product of primes bounded polynomially in $\log N$ —every term is polynomial in $\log N$ and the DLP is easy. Corollary 7.24 draws the design consequence.

Proof. Let $h = g^x$ with x the unknown. The reduction has two stages.

Stage (a): splitting across coprime factors. For each i set $N_i = N/q_i^{e_i}$ and project:

$$g_i = g^{N_i}, \quad h_i = h^{N_i}.$$

By Corollary 3.23, $\text{ord}(g_i) = N / \gcd(N, N_i) = N/N_i = q_i^{e_i}$, so g_i generates the unique subgroup of G of order $q_i^{e_i}$ (Theorem 3.24(3)). Moreover $h_i = g^{xN_i} = g_i^x$, and since g_i has order $q_i^{e_i}$, the exponent matters only modulo $q_i^{e_i}$ (Proposition 3.12(2)): writing $x_i = x \bmod q_i^{e_i}$,

$$\log_{g_i} h_i = x_i.$$

Solving the r smaller DLPs yields $x \bmod q_i^{e_i}$ for every i ; since the prime powers $q_i^{e_i}$ are pairwise coprime, the Chinese Remainder Theorem (Theorem 7.2, iterated across the factors as in the proof of Corollary 7.3) determines x modulo N uniquely. Each projection costs $O(\log N)$ operations by square-and-multiply.

Stage (b): lifting a prime power digit by digit. Fix one prime power q^e (dropping the subscript i), with $\gamma = g_i$ of order q^e and $\eta = h_i = \gamma^{x'}$, $x' = x_i$. Write x' in base q :

$$x' \equiv x_0 + x_1q + \cdots + x_{e-1}q^{e-1} \pmod{q^e}, \quad 0 \leq x_j < q.$$

The element $\delta = \gamma^{q^{e-1}}$ has order q (Corollary 3.23 again). Raise η to the power q^{e-1} : every digit beyond the first is multiplied by a multiple of q^e in the exponent and vanishes, leaving

$$\eta^{q^{e-1}} = \gamma^{x'q^{e-1}} = \delta^{x_0},$$

a single DLP in the group $\langle \delta \rangle$ of prime order q , which baby-step giant-step (Remark 7.22) solves in $O(\sqrt{q})$ operations. With $x_0, \dots, x_{j-1}$ in hand, strip them off and repeat: the element $\eta_j = \eta \gamma^{-(x_0 + x_1q + \cdots + x_{j-1}q^{j-1})}$ equals γ raised to $x_jq^j + (\text{multiples of } q^{j+1})$, so

$$\eta_j^{q^{e-1-j}} = \delta^{x_j},$$

and digit x_j is again one order- q DLP. Each of the e digits costs one such DLP plus $O(\log N)$ operations for the exponentiations.

Summing over the digits of all the prime-power factors gives $\sum_i e_i$ order- q_i DLPs at $O(\sqrt{q_i})$ apiece plus $O(\log N)$ bookkeeping for each, which is the stated bound. $\square$

Corollary 7.24 (Why prime order matters). *If N has a small prime factor q , an attacker can recover the discrete logarithm modulo q in $O(\log N + \sqrt{q})$ group operations— $O(\log N)$ to project into the order- q subgroup and $O(\sqrt{q})$ for the small DLP there—leaking part of the secret x . Security therefore requires N to have a large prime factor; the safest and simplest choice is to make N itself a large prime, so that G has no proper nontrivial subgroups and the Pohlig–Hellman reduction offers no purchase at all. This is precisely why the elliptic-curve groups used in Halo 2 and Orchard have large prime order.*

Proof. For the first claim, run stage (a) of Theorem 7.23 for the single factor q : the projections $g^{N/q}, h^{N/q}$ cost $O(\log N)$ operations, and the resulting DLP in a group of order q costs $O(\sqrt{q})$ by baby-step giant-step—yielding $x \bmod q$. For the second, if N is prime then G has no subgroups besides $\{e\}$ and G (Corollary 3.34), so the reduction’s only output is the original problem, and the generic $\Omega(\sqrt{N})$ floor of Remark 7.22 stands at full height. $\square$

Remark 7.25 (Cofactors and small-subgroup attacks). In practice an elliptic curve over $\mathbb{F}_p$ has group order $h \cdot q$, where q is a large prime and h is a small *cofactor*, and cryptographic operations stay confined to the subgroup of prime order q . If a protocol neglects to check that incoming points lie in that subgroup (or fails to clear the cofactor), an adversary can supply a point of low order and run Pohlig–Hellman in the small factor to extract information about the secret modulo h —a real-world *small-subgroup attack*. The Pasta curves of Halo 2 have cofactor $h = 1$: their group orders are themselves prime (the primes recorded in section 10), which eliminates this class of attacks at the level of the group itself rather than by protocol-level checks.

The two demands with which the section opened are now both met. A receiver handed a compressed point (x, bit) decides in one modular exponentiation whether $x^3 + ax + b$ is a square—Euler’s criterion—and, when it is, extracts a root by Proposition 7.17 or by Tonelli–Shanks (Theorem 7.19), the transmitted bit selecting between the pair $\pm y$ (Remark 7.20); for the Pasta fields, with their deliberately large 2-adic valuation $s = 32$, Tonelli–Shanks is the deployed route. On the security side, exponentiation runs in $O(\log N)$ operations while its inverse, the discrete logarithm, costs any generic attacker $\Omega(\sqrt{N})$ —a floor that composite group orders would undermine through Pohlig–Hellman, and that the prime, roughly 255-bit orders of the Pasta groups preserve in full, with no small subgroup left to attack. The structure of $\mathbb{Z}/q\mathbb{Z}$ for a large prime q thus supplies both the functionality—a field of scalars acting on the group—and the security of the constructions built on it. The curves themselves, and the primes p and q that this section has treated as black boxes, are the business of section 10.

8 Linear algebra over a field

Begin with the engineering problem that shapes the whole section. A prover holds a list of n secret numbers—the intermediate values of some large computation, the *wire values* of an *arithmetic circuit* in the vocabulary of later volumes, with n in the millions, say—and must hand a verifier a *receipt* for the list: a single short object that pins the list down, to be opened or argued about later. Two demands make the problem interesting. The receipt must be *one* element of some fixed algebraic structure, however large n grows; and receipts must *add*—whoever holds the receipts for two lists, together with two scaling factors, must be able to compute the receipt for the scaled sum of the lists without ever seeing the lists themselves. The construction that meets both demands is disarmingly brief: fix n public elements $G_1, \dots, G_n$ of a suitable group, and issue for the list $(a_1, \dots, a_n)$ the single group element $[a_1]G_1 + \dots + [a_n]G_n$. Whether this can possibly work—what “receipts add” should mean exactly, how one group element can absorb n secrets, and what price in ambiguity the compression exacts—are questions of *linear algebra*, and answering them is the business of this section.

Linear algebra is the study of *vector spaces*—collections of objects that can be added together and scaled by elements of a field—together with the structure-preserving maps between them. It

is the computational backbone of the cryptography this monograph serves: commitments to vectors, the linear constraints that encode arithmetic circuits, the polynomial-evaluation arguments at the heart of the inner-product argument, and the Pedersen-style multiscalar multiplications of Halo 2 and Orchard are all, at bottom, statements about linear maps and (bi)linear forms over a finite field. The section develops the theory from the ground up, assuming only the notion of a field (section 4), and then specialises to the two settings that matter here: the coordinate space F^n , in practice taken over a finite field $\mathbb{F}_q$, and the “mixed” pairing between vectors of scalars and vectors of group elements.

The section also settles two debts contracted below it. Remark 5.5 promised that the linear structure of $F[X]$ —closure under addition and scaling, with the monomials as an infinite coordinate system—would be studied systematically in due course; the study happens here. And the proof of Theorem 6.9 borrowed, against a promissory note, exactly two facts about vector spaces: that a space with a finite spanning set has a basis, and that coordinates with respect to a basis are unique. The first is proved in Theorem 8.17; the second follows from linear independence, as recorded in Definition 8.14. These arguments use nothing from the theory of finite fields, so the note is honoured without circularity. Throughout, F denotes an arbitrary field with identities 0 and 1; the field axioms of Definition 4.24 are used freely and without citation.

8.1 Vector spaces and subspaces

Definition 8.1 (Vector space). Let F be a field. A *vector space over F* (an *F -vector space*) is a set V equipped with two operations, *vector addition* $+: V \times V \rightarrow V$ and *scalar multiplication* $\cdot: F \times V \rightarrow V$, such that for all $u, v, w \in V$ and all $a, b \in F$:

1. $(V, +)$ is an abelian group: addition is associative and commutative, there is a *zero vector* $0 \in V$ with $v + 0 = v$, and every v has an *additive inverse* $-v$ with $v + (-v) = 0$;
2. scalar multiplication is compatible with the multiplication of F : $a \cdot (b \cdot v) = (ab) \cdot v$;
3. the identity of F acts as the identity: $1 \cdot v = v$;
4. scalar multiplication distributes over vector addition: $a \cdot (u + v) = a \cdot u + a \cdot v$;
5. scalar multiplication distributes over scalar addition: $(a + b) \cdot v = a \cdot v + b \cdot v$.

The elements of V are called *vectors*, the elements of F *scalars*. We write av for $a \cdot v$, and we allow the scalar $0 \in F$ and the vector $0 \in V$ to share a symbol whenever context distinguishes them.

A few consequences of the axioms recur so often that we record them once and, thereafter, use them without comment.

Proposition 8.2. Let V be an F -vector space, $v \in V$, and $a \in F$. Then:

1. $0 \cdot v = 0$;
2. $a \cdot 0 = 0$;
3. $(-1) \cdot v = -v$;
4. if $av = 0$, then $a = 0$ or $v = 0$.

Proof. (1) By distributivity over scalar addition, $0 \cdot v = (0 + 0) \cdot v = 0 \cdot v + 0 \cdot v$; adding $-(0 \cdot v)$ to both sides leaves $0 = 0 \cdot v$.

(2) Likewise $a \cdot 0 = a \cdot (0 + 0) = a \cdot 0 + a \cdot 0$, and cancelling in the abelian group $(V, +)$ gives $a \cdot 0 = 0$.

(3) Using the identity axiom, distributivity, and (1),

$$v + (-1) \cdot v = 1 \cdot v + (-1) \cdot v = (1 + (-1)) \cdot v = 0 \cdot v = 0,$$

so $(-1) \cdot v$ is the additive inverse of v ; by uniqueness of inverses in a group, $(-1) \cdot v = -v$.

(4) Suppose $av = 0$ and $a \neq 0$. Since F is a field, a^{-1} exists, and by the identity and compatibility axioms

$$v = 1 \cdot v = (a^{-1}a) \cdot v = a^{-1} \cdot (av) = a^{-1} \cdot 0 = 0,$$

the last step by (2). □

Example 8.3 (The coordinate space F^n). The single most important example is the *coordinate space*

$$F^n = \{ (x_1, \dots, x_n) : x_i \in F \},$$

the set of n -tuples of scalars, under componentwise operations:

$$(x_1, \dots, x_n) + (y_1, \dots, y_n) = (x_1 + y_1, \dots, x_n + y_n), \quad a \cdot (x_1, \dots, x_n) = (ax_1, \dots, ax_n).$$

The zero vector is $(0, \dots, 0)$, and every axiom of theorem 8.1 follows from the corresponding field axiom applied in each coordinate. The degenerate case $n = 0$ is the *trivial* (or *zero*) vector space $\{0\}$, consisting of the empty tuple alone. As a standing convention we write vectors of F^n in boldface and think of them as columns, $\mathbf{x} = (x_1, \dots, x_n)^T$.

Example 8.4 (Further examples). 1. *Polynomials*. The polynomials $F[X]$ form an F -vector space under addition and the scalar multiplication of Remark 5.5. So does the subset

$$F[X]_{<n} = \{f \in F[X] : f = 0 \text{ or } \deg f < n\}$$

of polynomials of degree less than n : sums and scalar multiples of such polynomials again have degree less than n (or are zero), by Proposition 5.7, so the operations restrict and the axioms are inherited. This space is exactly the setting in which polynomial commitment schemes operate.

2. *Functions*. For any set S , the set F^S of all functions $S \rightarrow F$ is an F -vector space under pointwise operations: $(f + g)(s) = f(s) + g(s)$ and $(af)(s) = af(s)$. Example 8.3 is the special case $S = \{1, \dots, n\}$, a tuple being a function of its index.
3. *Matrices*. The set $F^{m \times n}$ of $m \times n$ arrays of scalars is an F -vector space under entrywise addition and scaling—the case of (2) in which S is the grid of index pairs (i, j) .
4. *Field extensions*. If F is a subfield of a larger field K (§4), then K is an F -vector space: vector addition is the addition of K , and scalar multiplication is the multiplication of K restricted to pairs in $F \times K$, so the axioms are instances of the field axioms of K . The dimension of this vector space—in the sense of theorem 8.17 below—is called the *degree* of the extension, written $[K : F]$. For instance $\mathbb{C}$ is an $\mathbb{R}$ -vector space of dimension 2, with basis $1, i$, so $[\mathbb{C} : \mathbb{R}] = 2$; and the finite field $\mathbb{F}_q$ with $q = p^k$ is a k -dimensional $\mathbb{F}_p$ -vector space, $[\mathbb{F}_q : \mathbb{F}_p] = k$: this is precisely the structure on which the proof of Theorem 6.9 rested.

A vector space often contains smaller vector spaces sitting inside it, sharing its operations; the solution sets of homogeneous linear equations, the spans of the next subsection, and the kernels and images of linear maps are all of this kind.

Definition 8.5 (Subspace). A subset $W \subseteq V$ of an F -vector space V is a (linear) *subspace* if W is itself a vector space under the operations inherited from V ; equivalently—as the next proposition makes precise—if and only if W is nonempty and closed under the vector-space operations.

Proposition 8.6 (Subspace criterion). A subset $W \subseteq V$ is a subspace if and only if

1. $0 \in W$;
2. W is closed under addition: $u, v \in W$ implies $u + v \in W$; and
3. W is closed under scalar multiplication: $a \in F$ and $v \in W$ imply $av \in W$.

Equivalently, W is a subspace if and only if $W \neq \emptyset$ and $au + bv \in W$ for all $a, b \in F$ and $u, v \in W$.

Proof. If W is a subspace, then (1)–(3) hold: closure (2) and (3) is what it means for the inherited operations to be operations *on* W , and the zero vector of W is the zero vector of V , since a zero vector 0_W of W satisfies $0_W + 0_W = 0_W$, and cancelling in V gives $0_W = 0$.

Conversely, suppose (1)–(3) hold. Closure makes the two operations well defined on W . The additive inverse of any $v \in W$ is $(-1)v \in W$, by (3) and Proposition 8.2(3). Every remaining axiom of theorem 8.1 is a universally quantified identity—associativity, commutativity, compatibility, the two distributive laws, the action of 1—and each holds for all elements of V , in particular for all elements of W . Hence W is a vector space.

For the two-scalar form: if W is a subspace, then $au + bv \in W$ by (2) and (3) combined, and $W \ni 0$ is nonempty. Conversely, choosing $(a, b) = (1, 1)$ recovers (2), choosing $(a, b) = (a, 0)$ recovers (3) (using Proposition 8.2(1) to identify $0v$), and $0 = 0v \in W$ for any v in the nonempty W , which is (1). $\square$

Example 8.7. In F^3 the set $W = \{(x_1, x_2, x_3) : x_1 + x_2 + x_3 = 0\}$ is a subspace: it contains 0, and if the coordinates of x and y each sum to 0, then those of $ax + by$ sum to $a \cdot 0 + b \cdot 0 = 0$. By contrast $\{(x_1, x_2, x_3) : x_1 + x_2 + x_3 = 1\}$ is *not* a subspace: it does not contain 0. The same computation shows that the solution set of any single homogeneous linear equation $c_1x_1 + \cdots + c_nx_n = 0$ (the c_i fixed scalars) is a subspace of F^n , and the solution set of a finite *system* of such equations, being an intersection of subspaces, is a subspace by the next proposition. These solution sets return shortly in structural guise, as the kernels of linear maps (theorem 8.18).

Proposition 8.8 (Intersections and sums). Let V be an F -vector space.

1. The intersection $\bigcap_{i \in I} W_i$ of an arbitrary family of subspaces of V is a subspace.
2. The sum $W_1 + \cdots + W_k = \{w_1 + \cdots + w_k : w_j \in W_j\}$ of finitely many subspaces is a subspace, and it is the smallest subspace of V containing every W_j .

The union of two subspaces is in general not a subspace.

Proof. (1) Each W_i contains 0, so the intersection does. If u, v lie in every W_i and $a, b \in F$, then $au + bv \in W_i$ for every i by Proposition 8.6, hence $au + bv$ lies in the intersection, which is therefore a subspace by the same criterion.

(2) The sum contains $0 = 0 + \cdots + 0$, and it contains each W_j (take every other summand 0). For closure, take two elements $\sum_j w_j$ and $\sum_j w'_j$ with $w_j, w'_j \in W_j$, and scalars a, b ; regrouping,

$$a \sum_j w_j + b \sum_j w'_j = \sum_j (aw_j + bw'_j),$$

and $aw_j + bw'_j \in W_j$ since W_j is a subspace, so the combination lies in the sum. Finally, any subspace U containing every W_j contains each summand w_j of any element of the sum, hence contains the element itself by closure under addition; so the sum is contained in U , and is the smallest such subspace.

For the union, take $V = F^2$ (any field has $|F| > 1$, since $1 \neq 0$) and the two axes $\{(x, 0) : x \in F\}$ and $\{(0, y) : y \in F\}$ —subspaces by the criterion. Their union contains $(1, 0)$ and $(0, 1)$ but not the sum $(1, 1) = (1, 0) + (0, 1)$, since $(1, 1)$ lies on neither axis; closure under addition fails. $\square$

8.2 Linear combinations, span, and linear independence

The subspaces met so far were carved out by equations. The complementary way to produce a subspace is to build one up from chosen vectors, taking everything their repeated addition and scaling can reach.

Definition 8.9 (Linear combination and span). Let $S \subseteq V$. A *linear combination* of vectors in S is any finite sum

$$a_1v_1 + a_2v_2 + \cdots + a_kv_k, \quad a_i \in F, v_i \in S.$$

The *span* of S , written $\text{span}(S)$ or $\langle S \rangle$, is the set of all linear combinations of finitely many vectors of S ; by convention $\text{span}(\emptyset) = \{0\}$. The set S *spans* (or *generates*) V if $\text{span}(S) = V$, and V is *finitely generated* (synonymously, *finite-dimensional*) if some finite set spans it.

Proposition 8.10. For any $S \subseteq V$, the span $\text{span}(S)$ is the smallest subspace of V containing S : it is a subspace, it contains S , and it is contained in every subspace of V that contains S .

Proof. The zero vector is the empty linear combination (or, for nonempty S , the combination $0v$), and a sum of two linear combinations of vectors in S , or a scalar multiple of one, is again such a linear combination—concatenate the lists and rescale the coefficients. By Proposition 8.6, $\text{span}(S)$ is a subspace, and it contains each $v \in S$ as the combination $1 \cdot v$. If W is any subspace containing S , then closure under addition and scaling forces every linear combination of elements of S into W , by induction on the number of summands; hence $\text{span}(S) \subseteq W$. $\square$

Definition 8.11 (Linear independence). A finite list of vectors $v_1, \dots, v_k \in V$ is *linearly independent* if the only way to express the zero vector as a linear combination of the list is with all coefficients zero:

$$a_1v_1 + \cdots + a_kv_k = 0 \implies a_1 = \cdots = a_k = 0.$$

Otherwise the list is *linearly dependent*, and a relation $\sum_i a_iv_i = 0$ with some $a_i \neq 0$ is called a *dependence relation*. An arbitrary—possibly infinite—set $S \subseteq V$ is linearly independent if every finite list of distinct vectors from S is linearly independent.

Remark 8.12. Linear independence captures *non-redundancy*: no vector of an independent list contributes anything the others already provide. In the smallest cases the definition unwinds to familiar statements. A single vector v is independent if and only if $v \neq 0$ (Proposition 8.2(4)). Two vectors are dependent if and only if one is a scalar multiple of the other. In general, a list is dependent exactly when some vector in it lies in the span of the others—the content of the next lemma.

Lemma 8.13 (Dependence and redundancy). *A list $v_1, \dots, v_k$ with $k \geq 1$ is linearly dependent if and only if some v_j is a linear combination of the others. Moreover, if $v_1, \dots, v_k$ is independent but $v_1, \dots, v_k, w$ is dependent, then $w \in \text{span}(v_1, \dots, v_k)$, and the coefficients expressing w are unique.*

Proof. Suppose first that $v_j = \sum_{i \neq j} a_i v_i$. Then

$$\sum_{i \neq j} a_i v_i + (-1) v_j = 0$$

is a dependence relation, its coefficient -1 on v_j being nonzero (as $1 \neq 0$ in a field). Conversely, given a dependence relation $\sum_i a_i v_i = 0$ with $a_j \neq 0$, the field supplies a_j^{-1} , and solving for v_j gives $v_j = -a_j^{-1} \sum_{i \neq j} a_i v_i$, a linear combination of the others.

For the second statement, take a dependence relation $\sum_i a_i v_i + bw = 0$ for the extended list. The coefficient b cannot be 0: otherwise the relation would be a nontrivial relation among $v_1, \dots, v_k$ alone, contradicting their independence. Hence $w = -b^{-1} \sum_i a_i v_i \in \text{span}(v_1, \dots, v_k)$. For uniqueness, suppose $w = \sum_i c_i v_i = \sum_i c'_i v_i$; subtracting, $\sum_i (c_i - c'_i) v_i = 0$, and independence forces $c_i = c'_i$ for every i . $\square$

8.3 Bases, dimension, and linear maps

Spanning says a list reaches everything; independence says it carries no dead weight. A list with both properties is a coordinate system.

Definition 8.14 (Basis). *A basis of V is a linearly independent set B that spans V . Every $v \in V$ is then a *unique finite* linear combination of elements of B : existence because B spans, uniqueness because two expressions subtract to a relation among finitely many distinct elements of the independent set (as in Lemma 8.13). For a finite basis $B = (b_1, \dots, b_n)$, write*

$$v = c_1 b_1 + \dots + c_n b_n, \quad [v]_B = (c_1, \dots, c_n),$$

and call the scalars c_i the *coordinates* of v in B .

Example 8.15 (The two bases in constant use). The *standard basis* of F^n is $e_1, \dots, e_n$, where e_i has a 1 in position i and 0 elsewhere. Both properties reduce to reading off coordinates: the combination $\sum_i c_i e_i$ is the tuple $(c_1, \dots, c_n)$, so every vector is such a combination (spanning) and only the zero combination gives 0 (independence). The *monomial basis* of $F[X]_{<n}$ is $1, X, \dots, X^{n-1}$: a polynomial of degree less than n is, by construction, a linear combination of these monomials, and independence is the fact that polynomials are equal exactly when their coefficient sequences agree (Definition 5.1).

That all bases of a given space have the same length is not obvious; it rests on the following exchange principle, which lets an independent list consume a spanning list one slot at a time.

Lemma 8.16 (Steinitz exchange). *If $v_1, \dots, v_m$ are linearly independent in V and $w_1, \dots, w_n$ span V , then $m \leq n$.*

Proof. We prove by induction on $k = 0, 1, \dots, m$ that, after a suitable renumbering of the w_j , the hybrid list

$$v_1, \dots, v_k, w_{k+1}, \dots, w_n$$

spans V . The case $k = 0$ is the hypothesis on the w_j .

For the induction step, suppose the hybrid list at stage $k < m$ spans V . In particular it expresses v_{k+1} :

$$v_{k+1} = \sum_{i \leq k} a_i v_i + \sum_{j > k} c_j w_j.$$

Some coefficient c_j with $j > k$ must be nonzero: were they all zero, v_{k+1} would be a linear combination of $v_1, \dots, v_k$, and Lemma 8.13 would make the list $v_1, \dots, v_{k+1}$ dependent, contradicting the independence of the v_i . (This step silently requires a slot $j > k$ to exist, i.e. $k < n$; if instead $k = n$, the hybrid list is $v_1, \dots, v_n$ and the displayed equation already yields the same contradiction.) Renumber so that $c_{k+1} \neq 0$, and solve for the displaced vector:

$$w_{k+1} = c_{k+1}^{-1} \left(v_{k+1} - \sum_{i \leq k} a_i v_i - \sum_{j > k+1} c_j w_j \right).$$

Every vector of the stage- k list therefore lies in the span of the stage- $(k+1)$ list $v_1, \dots, v_{k+1}, w_{k+2}, \dots, w_n$ —the v_i and the remaining w_j trivially, w_{k+1} by the display—so by Proposition 8.10 the stage- $(k+1)$ list spans V , completing the induction.

Each step trades one w for one v , so the process can run to $k = m$ only if there are at least m slots to trade: $m \leq n$. $\square$

Theorem 8.17 (Dimension). *Let V be a finitely generated F -vector space. Then V has a basis, and any two bases of V have the same number of elements. This common number is the dimension $\dim_F V$ (written $\dim V$ when the field is clear). In particular $\dim_F F^n = n$ and $\dim_F F[X]_{<n} = n$.*

Proof. Existence. Let $w_1, \dots, w_n$ be a finite spanning list. While the list is dependent, Lemma 8.13 exhibits some w_j in the span of the others; discarding it leaves the span unchanged, since any combination using w_j can be rewritten without it (substitute and regroup, Proposition 8.10). Each discard shortens the list, so after at most n discards the process halts at an independent spanning list—a basis. (If V is the zero space, the process discards everything and halts at the empty list, a basis of $\{0\}$ with $\text{span}(\emptyset) = \{0\}$; the dimension is 0.)

Invariance. Every basis is finite: if a basis contained more than the n elements of a finite spanning list, any $n+1$ of its elements would contradict Lemma 8.16. Let $b_1, \dots, b_m$ and $b'_1, \dots, b'_n$ be bases. Applying Lemma 8.16 with the first list as the independent one and the second as the spanning one gives $m \leq n$; exchanging the roles gives $n \leq m$. Hence $m = n$.

The two named spaces have the explicit bases of Example 8.15, each of length n . $\square$

With coordinates in hand, the maps that respect the linear structure come next; they are the section's real subject, for the Pedersen-style commitments developed in the series are such maps.

Definition 8.18 (Linear map, kernel, image). A function $T: V \rightarrow W$ between F -vector spaces is *linear* (an *F -linear map*) if

$$T(au + bv) = aT(u) + bT(v) \quad \text{for all } a, b \in F \text{ and } u, v \in V.$$

Its *kernel* and *image* are

$$\ker T = \{v \in V : T(v) = 0\} \subseteq V, \quad \text{im } T = \{T(v) : v \in V\} \subseteq W.$$

A bijective linear map is an *isomorphism*.

Three basic facts follow at once from the subspace criterion and linearity, and we verify them here once. First, taking $a = b = 0$ gives $T(0) = 0$, so $0 \in \ker T$ and $0 \in \text{im } T$. Second, both kernel and image are subspaces: if $u, v \in \ker T$ then $T(au + bv) = aT(u) + bT(v) = 0$, and if

$T(u), T(v) \in \text{im } T$ then $aT(u) + bT(v) = T(au + bv) \in \text{im } T$; Proposition 8.6 applies to each. Third, T is injective if and only if $\ker T = \{0\}$: if $T(u) = T(v)$, then $T(u - v) = 0$ by linearity, so a trivial kernel forces $u = v$; conversely an injective T sends only 0 to $T(0) = 0$. We also note that for fixed V and W the set $\text{Hom}(V, W)$ of all linear maps $V \rightarrow W$ is itself an F -vector space under the pointwise operations of Example 8.4(2)—a sum or scalar multiple of linear maps is linear, so $\text{Hom}(V, W)$ is a subspace of the space of all functions $V \rightarrow W$.

Theorem 8.19 (Rank–nullity). *Let $T: V \rightarrow W$ be a linear map with V finite-dimensional. Then*

$$\dim V = \dim \ker T + \dim \text{im } T.$$

Proof. The kernel is a subspace of the finite-dimensional V , and is itself finite-dimensional: an independent list in $\ker T$ has length at most $\dim V$ by Lemma 8.16 (tested against a basis of V , which spans), so a longest independent list $u_1, \dots, u_k$ in $\ker T$ exists, and it spans $\ker T$ —any $u \in \ker T$ outside its span would extend it to a longer independent list by Lemma 8.13. Thus $u_1, \dots, u_k$ is a basis of $\ker T$.

Extend it to a basis of V : while the current independent list fails to span V , adjoin any vector outside its span—independence is preserved by Lemma 8.13—and the process terminates, again because independent lists cannot exceed $\dim V$ in length (Lemma 8.16 against any basis of V). Write the resulting basis as $u_1, \dots, u_k, v_1, \dots, v_r$, so that $\dim V = k + r$.

We claim $T(v_1), \dots, T(v_r)$ is a basis of $\text{im } T$; the theorem follows, since then $\dim \text{im } T = r$ and $\dim V = k + r = \dim \ker T + \dim \text{im } T$.

Spanning. Any element of $\text{im } T$ is $T(v)$ for some $v = \sum_i a_i u_i + \sum_j c_j v_j$, and linearity with $T(u_i) = 0$ gives $T(v) = \sum_j c_j T(v_j)$.

Independence. Suppose $\sum_j c_j T(v_j) = 0$. By linearity $T(\sum_j c_j v_j) = 0$, so $\sum_j c_j v_j \in \ker T$, and expressing this kernel element in the kernel basis gives $\sum_j c_j v_j = \sum_i a_i u_i$ for some scalars a_i . Rearranged, $\sum_i a_i u_i - \sum_j c_j v_j = 0$ is a linear relation among the combined basis of V , whose independence forces every coefficient—in particular every c_j —to vanish. $\square$

Rank–nullity is the dimension-counting engine of the section. It shows at once that a linear map from a higher-dimensional space into a lower-dimensional one must have a nontrivial kernel—the image cannot have dimension exceeding the target’s, so the kernel absorbs the difference—and distinct inputs must therefore collide. That forced collision is exactly what makes vector commitments compressing and only *computationally* binding (Remark 8.29).

8.4 Bilinear forms and the inner product on F^n

The maps considered so far are linear in a single argument. Commitment and proof systems are pervaded by expressions linear in *each* of two arguments—above all the inner product between two vectors. Over $\mathbb{R}$ such products come wrapped in geometry: lengths, angles, positivity. Over the finite fields of cryptographic interest there is no ordering, hence no “positive” and no length; what survives, and what the applications actually use, is the bare algebra. The appropriate level of generality is therefore the *bilinear form*.

Definition 8.20 (Bilinear form). *A bilinear form on an F -vector space V is a function $\beta: V \times V \rightarrow F$ that is linear in each argument separately:*

$$\beta(au + a'u', v) = a\beta(u, v) + a'\beta(u', v), \quad \beta(u, bv + b'v') = b\beta(u, v) + b'\beta(u, v'),$$

for all $u, u', v, v' \in V$ and $a, a', b, b' \in F$. The form is *symmetric* if $\beta(u, v) = \beta(v, u)$ for all u, v , and *nondegenerate* if $\beta(u, v) = 0$ for all v implies $u = 0$.

Relative to a fixed basis $b_1, \dots, b_n$ of V , a bilinear form is captured by finitely many scalars: expanding both arguments in the basis and applying bilinearity termwise, $\beta(u, v) = \sum_{i,j} \beta(b_i, b_j) [u]_i [v]_j$, so β is determined by—and represented by—the unique $n \times n$ array M with entries $M_{ij} = \beta(b_i, b_j)$, via $\beta(u, v) = [u]_B^T M [v]_B$ in matrix shorthand. We record the representation for orientation but make no further use of it; only the special case with M the identity array matters below.

Definition 8.21 (Standard inner product on F^n). The *standard inner product* (or *dot product*) on F^n is

$$\langle a, b \rangle = \sum_{i=1}^n a_i b_i = a^T b, \quad a = (a_1, \dots, a_n), \quad b = (b_1, \dots, b_n).$$

Bilinearity holds because each term $a_i b_i$ is linear in each factor, and symmetry because multiplication in F commutes; the representing array in the standard basis is the identity ($\langle e_i, e_j \rangle$ is 1 if $i = j$ and 0 otherwise). The form is nondegenerate: if $\langle a, b \rangle = 0$ for every b , then in particular $a_i = \langle a, e_i \rangle = 0$ for each i , so $a = 0$.

Remark 8.22 (No positivity over finite fields). Over $\mathbb{R}$ the standard inner product is *positive definite*: $\langle a, a \rangle = \sum_i a_i^2 \geq 0$, with equality only at 0. Positivity is what yields the Euclidean norm $\|a\| = \sqrt{\langle a, a \rangle}$ and the Cauchy–Schwarz inequality $|\langle a, b \rangle| \leq \|a\| \|b\|$ —the geometric superstructure of real inner products. Over a finite field $\mathbb{F}_q$ there is no ordering compatible with the arithmetic, so “ ≥ 0 ” is meaningless, and the superstructure collapses: there exist nonzero *isotropic* vectors $a \neq 0$ with $\langle a, a \rangle = 0$. Over $\mathbb{F}_5$, for instance, the vector $a = (1, 2)$ has

$$\langle a, a \rangle = 1^2 + 2^2 = 1 + 4 = 5 = 0$$

in $\mathbb{F}_5$. What survives the collapse are the algebraically robust properties: bilinearity, symmetry, and nondegeneracy. These are all that the arguments in proof systems rely on, which is why the loss of positivity costs the applications nothing.

Nondegeneracy is not a consolation prize; it powers a dictionary between vectors and the scalar-valued linear maps on them.

Proposition 8.23 (Functionals via the inner product). For each $a \in F^n$, the map $b \mapsto \langle a, b \rangle$ is a linear map $F^n \rightarrow F$ (a linear functional). The assignment

$$F^n \longrightarrow \text{Hom}(F^n, F), \quad a \longmapsto \langle a, \cdot \rangle,$$

is an isomorphism of F -vector spaces onto the dual space $\text{Hom}(F^n, F)$ of all linear functionals on F^n . In particular both spaces have dimension n .

Proof. Each $\langle a, \cdot \rangle$ is linear in b by bilinearity, and the assignment itself is linear in a for the same reason, mapping into the vector space $\text{Hom}(F^n, F)$ of theorem 8.18. It is injective: if $\langle a, \cdot \rangle$ is the zero functional, then $\langle a, b \rangle = 0$ for every b , and nondegeneracy (theorem 8.21) gives $a = 0$, so the kernel is trivial. It is surjective: given a linear functional φ , set $a_i = \varphi(e_i)$ and $a = (a_1, \dots, a_n)$; then for every $b = \sum_i b_i e_i$, linearity of φ gives

$$\varphi(b) = \sum_i b_i \varphi(e_i) = \sum_i a_i b_i = \langle a, b \rangle,$$

so $\varphi = \langle a, \cdot \rangle$. A bijective linear map is an isomorphism, and $\dim \text{Hom}(F^n, F) = \dim F^n = n$ follows since an isomorphism carries a basis to a basis (its inverse is linear, and both directions preserve spanning and independence). $\square$

Remark 8.24 (The dictionary in use). Proposition 8.23 says that “a linear functional on F^n ” and “a vector in F^n ” are interchangeable: every linear way of producing one scalar from a vector is an inner product against a fixed vector, and conversely. The instance to keep in mind is polynomial evaluation. Fix a point $z \in F$; the map sending a polynomial to its value at z is linear in the coefficients, so it must be an inner product against some fixed vector—and indeed, for $p \in F[X]_{<n}$ with coefficient vector $c = (c_0, c_1, \dots, c_{n-1})$,

$$p(z) = c_0 + c_1 z + \dots + c_{n-1} z^{n-1} = \langle c, (1, z, z^2, \dots, z^{n-1}) \rangle.$$

Evaluating a *committed* polynomial at a challenge point is therefore an inner-product claim about the committed coefficient vector; this is precisely the shape of statement an inner-product argument proves (theorem 8.32).

8.5 The mixed inner product with group elements

The commitments used in Halo 2 and Orchard pair a vector of *scalars* with a vector of *group elements*. We isolate the algebra here, working over an abstract group; the group of actual cryptographic interest—the points of an elliptic curve—is constructed in the elliptic-curve section later in the volume, and only its abelian-group structure is needed for everything proved below, so the results transfer verbatim.

Construction 8.25 (Scalar multiplication by $\mathbb{F}_r$ on a prime-order group). Let $(\mathbb{G}, +)$ be a finite abelian group of prime order r , written additively with identity O . For an integer $a \in \mathbb{Z}$ and $G \in \mathbb{G}$, write $[a]G$ for the a -fold multiple of G —the additive reading of the powers of §3: $[0]G = O$, $[a]G = G + \dots + G$ with a summands for $a > 0$, and $[-a]G = -([a]G)$. The brackets keep the scalar visually separate from the group element. The multiple $[a]G$ depends only on the residue of a modulo r : the order of G divides $|\mathbb{G}| = r$ (Corollary 3.30), and $[a]G = [b]G$ whenever $a \equiv b \pmod{\text{ord}(G)}$ (Proposition 3.12(2), read additively), so congruence modulo r suffices. The operation therefore *descends* to scalars in the field $\mathbb{F}_r = \mathbb{Z}/r\mathbb{Z}$ (theorem 4.26): for $a \in \mathbb{F}_r$ and $G \in \mathbb{G}$, define $[a]G$ using any integer representative of a . It is this descended operation—group elements scaled by elements of a *field*—that the rest of the section uses.

Proposition 8.26 ($\mathbb{G}$ as an $\mathbb{F}_r$ -vector space). *Let $\mathbb{G}$ be an abelian group of prime order r , with the scalar multiplication of theorem 8.25. Then $\mathbb{G}$ is an $\mathbb{F}_r$ -vector space of dimension 1; any element $G \neq O$ is a basis, and every element of $\mathbb{G}$ is $[a]G$ for a unique $a \in \mathbb{F}_r$.*

Proof. The vector-space axioms are the standard laws of multiples. For integers a, b , the identities $[a + b]G = [a]G + [b]G$ and $[a]([b]G) = [ab]G$ are the exponent laws of §3 in additive notation, $[1]G = G$ is the definition, and $[a](G + H) = [a]G + [a]H$ holds in any *abelian* group: for $a > 0$ it is the rearrangement of a summands $G + H$ into a summands G followed by a summands H , legitimate by commutativity and associativity (formally, induction on a), and the cases $a \leq 0$ follow by negating. Each law respects reduction modulo r by theorem 8.25, so the four scalar axioms of theorem 8.1 hold with scalars in $\mathbb{F}_r$, and $(\mathbb{G}, +)$ is an abelian group by hypothesis: $\mathbb{G}$ is an $\mathbb{F}_r$ -vector space.

Now let $G \neq O$. By Lagrange’s theorem (Theorem 3.29, via Corollary 3.30) the order of G divides the prime r , and it exceeds 1 since G is not the identity; hence $\text{ord}(G) = r$. The r multiples $[0]G, [1]G, \dots, [r-1]G$ are then pairwise distinct (Proposition 3.12(2), additively: $[a]G = [b]G$ forces $a \equiv b \pmod{r}$), and $\mathbb{G}$ has exactly r elements, so the multiples exhaust $\mathbb{G}$. Thus $\{G\}$ spans; and it is independent, since $[a]G = O$ forces $a = 0$ in $\mathbb{F}_r$, again because $\text{ord}(G) = r$. A one-element basis gives $\dim_{\mathbb{F}_r} \mathbb{G} = 1$, and the uniqueness of a in $[a]G$ is the uniqueness of coordinates in a basis (theorem 8.14). $\square$

Definition 8.27 (Mixed inner product, multiscalar multiplication). Let $G = (G_1, \dots, G_n) \in \mathbb{G}^n$ be a fixed tuple of group elements and $a = (a_1, \dots, a_n) \in \mathbb{F}_r^n$ a vector of scalars. Their *mixed inner product*—in computational contexts a *multiscalar multiplication* (MSM), in cryptographic ones a *Pedersen-style inner product*—is the group element

$$\langle a, G \rangle = \sum_{i=1}^n [a_i] G_i \in \mathbb{G}.$$

We reuse the angle-bracket notation of theorem 8.21 deliberately; the type of the second argument—scalar vector or vector of group elements—determines which product is meant.

Proposition 8.28 (Bilinearity of the mixed product). Fix $G \in \mathbb{G}^n$. The map $a \mapsto \langle a, G \rangle$ is $\mathbb{F}_r$ -linear from $\mathbb{F}_r^n$ to $\mathbb{G}$:

$$\langle ca + c'a', G \rangle = [c]\langle a, G \rangle + [c']\langle a', G \rangle \quad \text{for all } c, c' \in \mathbb{F}_r \text{ and } a, a' \in \mathbb{F}_r^n.$$

Symmetrically, for fixed $a \in \mathbb{F}_r^n$ the map $G \mapsto \langle a, G \rangle$ is a group homomorphism $\mathbb{G}^n \rightarrow \mathbb{G}$, and is $\mathbb{F}_r$ -linear when $\mathbb{G}^n$ is viewed as an $\mathbb{F}_r$ -vector space (componentwise, as in Example 8.3 with $\mathbb{G}$ in place of F). Thus $\langle \cdot, \cdot \rangle: \mathbb{F}_r^n \times \mathbb{G}^n \rightarrow \mathbb{G}$ is bilinear over $\mathbb{F}_r$.

Proof. Both statements are the vector-space laws of $\mathbb{G}$ (Proposition 8.26) applied componentwise. For the first,

$$\langle ca + c'a', G \rangle = \sum_i [ca_i + c'a'_i] G_i = \sum_i ([c][a_i]G_i + [c'] [a'_i]G_i) = [c] \sum_i [a_i]G_i + [c'] \sum_i [a'_i]G_i,$$

using $[a + b]G = [a]G + [b]G$ and $[ab]G = [a]([b]G)$ in each component and then regrouping the sum in the abelian group $\mathbb{G}$. For the second, linearity in G follows by the symmetric computation on the other slot of the defining sum, using $[a](G + H) = [a]G + [a]H$ componentwise. $\square$

Remark 8.29 (Why the mixed product is the right abstraction). Read $\langle a, G \rangle$ as a *commitment* to the scalar vector a under the public “basis” G —the receipt of the section’s opening problem. Proposition 8.28 is then exactly the *homomorphic* property that the problem demanded: the commitment to $ca + c'a'$ is the $[c]$ -, $[c']$ -scaled combination of the individual commitments, computable by anyone holding those commitments and the scalars c, c' , without knowledge of a or a' . Provided the G_i are chosen so that no party can exhibit a nontrivial relation $\sum_i [a_i]G_i = O$ —a computational hypothesis, the *discrete-logarithm hardness assumption*, made precise in later volumes of this series—the commitment is *binding*: the map $a \mapsto \langle a, G \rangle$ is computationally injective, in the sense that no feasible computation produces two vectors with the same image. Genuinely injective it cannot be once $n \geq 2$: the map is linear from the n -dimensional $\mathbb{F}_r^n$ into the one-dimensional $\mathbb{G}$ (Proposition 8.26), so by rank-nullity (Theorem 8.19) its kernel has dimension at least $n - 1$. Collisions exist in abundance; binding is the claim that they cannot be *found*, a computational rather than information-theoretic statement. The classical Pedersen commitment to a single value a with randomness ρ is the special case $n = 2$, with $H \neq O$,

$$[a]G + [\rho]H = \langle (a, \rho), (G, H) \rangle,$$

which is moreover *perfectly hiding*: for fixed a , as ρ ranges uniformly over $\mathbb{F}_r$, the term $[\rho]H$ ranges uniformly over $\mathbb{G}$ (it is a bijection of $\mathbb{G}$, by unique representation in the basis $\{H\}$), so the commitment’s distribution is uniform and carries no information about a .

8.6 How this machinery appears in commitment and proof systems

We close by drawing the threads together. Each point below is developed rigorously in later volumes of this series; the remarks here are descriptive, but the linear-algebraic content behind them is already proved in full.

Remark 8.30 (Witnesses, constraints, and assignments). An arithmetic circuit over $\mathbb{F}_q$ becomes, after *arithmetisation*—the encoding of a computation as a system of polynomial constraints on the wire values—a collection of equations that the vector $z \in \mathbb{F}_q^N$ of all wire values must satisfy. The *linear* part of such a system—the wiring (or “copy”) constraints of the PLONK family, which assert that certain wires carry equal values—consists of homogeneous linear equations in the entries of z , organised in practice as arrays of coefficients (one row per equation) acting on the assignment vector. Satisfying the linear constraints confines z to a subspace of $\mathbb{F}_q^N$ (Example 8.7), or, when public inputs fix some entries, to a translate $z_0 + W$ of one—an *affine* subspace. The multiplicative constraints are handled separately, by the polynomial machinery of the surrounding sections. Rank-nullity (Theorem 8.19) quantifies the degrees of freedom that remain after the linear constraints are imposed: the dimension of the witness space.

Remark 8.31 (Commitments as linear maps). A Pedersen vector commitment $a \mapsto \langle a, G \rangle$ is a linear map $\mathbb{F}_r^n \rightarrow \mathbb{G}$ (Proposition 8.28), and its linearity is the source of every algebraic manipulation a verifier performs: taking known linear combinations of commitments, folding two commitment keys G, G' into $G + [x]G'$ under a challenge x , and checking claimed openings by re-evaluating the same linear map. A polynomial commitment is the identical map applied to a coefficient vector, and an *evaluation* of the committed polynomial at a point z is the scalar inner product $\langle c, (1, z, \dots, z^{n-1}) \rangle$ (Remark 8.24).

Remark 8.32 (The inner-product relation). The core statement an inner-product argument proves has the shape: “I know vectors $a, b \in \mathbb{F}_r^n$ such that

$$P = \langle a, G \rangle + \langle b, H \rangle + [\langle a, b \rangle]U$$

for public $G, H \in \mathbb{G}^n$ and $U \in \mathbb{G}$.” Three of this section’s constructions meet in the one equation: two mixed inner products $\langle a, G \rangle$ and $\langle b, H \rangle$, committing to the witness vectors; one scalar inner product $\langle a, b \rangle$, the value being argued; and the bilinearity of all three, which is what permits prover and verifier to *fold* a length- n instance into a length- $n/2$ one and recurse. The argument’s soundness ultimately reduces to the nondegeneracy of these forms together with the hardness of finding kernel elements of the commitment map—exactly the linear-algebraic and computational facts assembled above.

The cheque drawn at the head of the section is now cashed. The receipt for a list $(a_1, \dots, a_n)$ of secrets is the mixed inner product $\langle a, G \rangle = \sum_i [a_i]G_i$: one group element, whatever the size of n . “Receipts add” means precisely that the receipt map is linear, and that is the payoff theorem, Proposition 8.28. One element can absorb n secrets only because the map compresses an n -dimensional space into a one-dimensional one, and rank-nullity prices the compression exactly: a kernel of dimension at least $n - 1$, an unavoidable reservoir of collisions that Remark 8.29 tames computationally rather than absolutely.

In summary: vector spaces, bases and dimension, linear maps with their kernels and images governed by rank-nullity, bilinear forms, and the two flavours of inner product—scalar with scalar, scalar with group—constitute the linear-algebraic vocabulary for the cryptography ahead. Pedersen-style commitments are linear maps, and their polynomial evaluation openings assert inner products; nondegeneracy and dimension counting supply the relevant algebra. Other commitment constructions, such as hash-based commitments, need not be linear. The next section applies the machinery to the polynomial spaces $F[X]_{<n}$ over structured evaluation domains, and the section on elliptic curves constructs the group $\mathbb{G}$ that the mixed inner product has so far taken as abstract.

9 Roots of unity and evaluation domains

Run a computation for n steps and record, at each step, the value it holds in each of its registers. The record is a table: one column per register, one row per step, every entry an element of a finite field. A proof system of the kind underlying Halo 2 and Zcash Orchard convinces a verifier that such a table obeys its rules—that every row follows from the one before it—without the verifier replaying the computation. Its basic move is to treat each column as the value table of a *polynomial*: a function given by its values at n fixed points is, by Lagrange interpolation (theorem 5.30), the same data as a polynomial of degree less than n , and statements about the table become statements about polynomials—checkable all at once, at a single random point, by the Schwartz–Zippel principle (theorem 5.45). For this plan to be workable two demands must be met. The n evaluation points must be *structured*, so that “every row” conditions, “next row” shifts, and the bookkeeping polynomials attached to the point set stay cheap; and converting between a polynomial’s values and its coefficients—and multiplying two polynomials—must cost far less than the n^2 field operations of the schoolbook methods, or provers could never handle tables of practical size. Both demands are met by one choice of point set: the powers of a *root of unity*. Nearly every operation such a proof system performs—committing to a polynomial, checking that two polynomials agree, enforcing that a constraint vanishes everywhere it should, multiplying or interpolating efficiently—is ultimately a computation indexed by the finite multiplicative subgroup those powers form. This section develops the theory of these subgroups and delivers, in the fast Fourier transform, the $O(n \log n)$ arithmetic promised above.

Throughout, $\mathbb{F}$ denotes a field. When $\mathbb{F}$ is to be finite, the notation $\mathbb{F}_q$ denotes the field with q elements (so q is a prime power, theorem 6.9); when we intend a prime field specifically we write $\mathbb{F}_p$. The multiplicative group of nonzero elements is $\mathbb{F}^\times = \mathbb{F} \setminus \{0\}$ (Example 3.7). We reserve the letter n for the order of the domain we shall construct; throughout, $n \geq 1$.

The route is as follows. We first define roots of unity and the primitive ones among them, and settle exactly when a finite field contains them: the condition is $n \mid q - 1$, a direct consequence of the cyclicity of $\mathbb{F}_q^\times$ proved in §6. We then fix the *evaluation domain* H , the subgroup those roots form, and work out its toolkit: the vanishing polynomial $X^n - 1$ and its cosets, the Lagrange basis in closed form, and the evaluation/interpolation isomorphism in matrix form. The isomorphism, read as a matrix, is the number-theoretic transform; the convolution theorem turns polynomial multiplication into entrywise multiplication of value vectors; and the fast Fourier transform computes the transform in $O(n \log n)$ operations. A closing synthesis assembles the pieces into the reason these domains form the substrate of polynomial proof systems.

9.1 Roots of unity

The starting point is the equation $X^n = 1$. Its solutions in a field are the roots of unity of order dividing n .

Definition 9.1 (Root of unity). Let $\mathbb{F}$ be a field and $n \in \mathbb{N}_{>0}$. An element $\omega \in \mathbb{F}$ is an n -th root of unity if $\omega^n = 1$. We write

$$\mu_n(\mathbb{F}) = \{ \omega \in \mathbb{F} : \omega^n = 1 \}$$

for the set of all n -th roots of unity in $\mathbb{F}$.

Note that $0 \notin \mu_n(\mathbb{F})$, since $0^n = 0 \neq 1$; every n -th root of unity is therefore a unit, i.e. lies in $\mathbb{F}^\times$. The set $\mu_n(\mathbb{F})$ is not merely a set; it is a group.

Proposition 9.2. *The set $\mu_n(\mathbb{F})$ is a finite subgroup of $\mathbb{F}^\times$ of order at most n .*

Proof. The identity 1 satisfies $1^n = 1$, so $1 \in \mu_n(\mathbb{F})$. If $\omega, \zeta \in \mu_n(\mathbb{F})$, then $(\omega\zeta)^n = \omega^n\zeta^n = 1 \cdot 1 = 1$ because $\mathbb{F}^\times$ is abelian, so $\omega\zeta \in \mu_n(\mathbb{F})$; and $(\omega^{-1})^n = (\omega^n)^{-1} = 1^{-1} = 1$, so $\omega^{-1} \in \mu_n(\mathbb{F})$. Hence $\mu_n(\mathbb{F})$ is a subgroup of $\mathbb{F}^\times$ (theorem 3.14). For the order bound, every element of $\mu_n(\mathbb{F})$ is a root of the polynomial $X^n - 1 \in \mathbb{F}[X]$, which is nonzero of degree n and so has at most n roots in $\mathbb{F}$ (theorem 5.18). Thus $|\mu_n(\mathbb{F})| \leq n$. $\square$

The bound $|\mu_n(\mathbb{F})| \leq n$ need not be attained. Whether it is attained is precisely the question of the existence of a *primitive* n -th root of unity.

Definition 9.3 (Primitive root of unity). An element $\omega \in \mathbb{F}$ is a *primitive n -th root of unity* if $\omega^n = 1$ and $\omega^k \neq 1$ for every integer k with $1 \leq k < n$. Equivalently, ω is a primitive n -th root of unity if its multiplicative order $\text{ord}(\omega)$ (theorem 3.11) equals n .

Here $\text{ord}(\omega)$ is the least positive integer d with $\omega^d = 1$; it exists for any $\omega \in \mu_n(\mathbb{F})$ because n itself is such an exponent and the positive integers are well-ordered. The two formulations in theorem 9.3 agree: $\omega^k \neq 1$ for $1 \leq k < n$ together with $\omega^n = 1$ states exactly that the least positive exponent killing ω is n .

Remark 9.4 (Primitive roots revisited). The identification promised in theorem 6.20 can now be stated. A primitive root of a finite field $\mathbb{F}_q$ —a generator of the cyclic group $\mathbb{F}_q^\times$, in the sense of theorem 6.19—is an element of multiplicative order $q - 1$, which by theorem 9.3 is precisely a primitive $(q - 1)$ -th root of unity. The group-theoretic vocabulary “generator of $\mathbb{F}_q^\times$ ” and the polynomial-flavoured vocabulary “primitive $(q - 1)$ -th root of unity” name the same elements.

Lemma 9.5 (Order divides exponent). Let $\omega \in \mathbb{F}^\times$ have finite order $d = \text{ord}(\omega)$. Then for any integer m we have $\omega^m = 1$ if and only if $d \mid m$. In particular, ω is an n -th root of unity if and only if $d \mid n$, and ω is a primitive n -th root of unity if and only if $d = n$.

Proof. The first claim is theorem 3.12(1), read in the abelian group $\mathbb{F}^\times$: if $d \mid m$, write $m = dk$, so $\omega^m = (\omega^d)^k = 1$; conversely, division with remainder $m = qd + r$, $0 \leq r < d$, gives $1 = \omega^m = \omega^r$, and minimality of d forces $r = 0$. The remaining clauses instantiate it: $\omega^n = 1 \iff d \mid n$ is the case $m = n$, and primitivity is the case $d = n$ by theorem 9.3. $\square$

The lemma’s constant companion is the order-of-a-power formula in a cyclic group, $\text{ord}(g^k) = \text{ord}(g) / \gcd(k, \text{ord}(g))$, proved as Corollary 3.23; we use the pair repeatedly below as the standard device for computing orders.

Proposition 9.6. If $\omega \in \mathbb{F}$ is a primitive n -th root of unity, then

$$\mu_n(\mathbb{F}) = \langle \omega \rangle = \{1, \omega, \omega^2, \dots, \omega^{n-1}\}$$

is a cyclic group of order exactly n , and these n powers are pairwise distinct.

Proof. The powers $1, \omega, \dots, \omega^{n-1}$ are pairwise distinct: if $\omega^i = \omega^j$ with $0 \leq i \leq j \leq n - 1$, then $\omega^{j-i} = 1$ with $0 \leq j - i < n$, so primitivity (theorem 9.5, with $d = n$) forces $j - i = 0$. Thus $\langle \omega \rangle$ has exactly n elements, all of which lie in $\mu_n(\mathbb{F})$ since $(\omega^k)^n = (\omega^n)^k = 1$. But $|\mu_n(\mathbb{F})| \leq n$ by theorem 9.2, so the inclusion $\langle \omega \rangle \subseteq \mu_n(\mathbb{F})$ is an equality of n -element sets. $\square$

Over the complex numbers a primitive n -th root of unity always exists, namely $\omega = e^{2\pi i/n}$: this is the classical picture of n equally spaced points on the unit circle, with the primitive roots sitting at the angles $2\pi k/n$ for $\gcd(k, n) = 1$. Over a finite field existence is more delicate, and it is exactly the finite case that matters for cryptography.

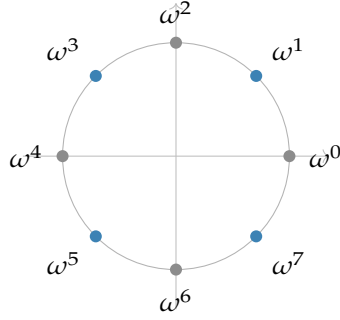


Figure 1: The eight 8-th roots of unity in $\mathbb{C}$, equally spaced on the unit circle at angles $2\pi k/8$, with $\omega = e^{2\pi i/8}$. The four primitive ones ($k = 1, 3, 5, 7$, exactly the k coprime to 8; marked in colour) each generate the whole cycle; the grey points $\omega^0 = 1$, $\omega^2 = i$, $\omega^4 = -1$, $\omega^6 = -i$ have orders 1, 4, 2, 4 and generate proper subgroups. Over a finite field the same group structure survives with the geometry stripped away—the picture to keep is the cycle, not the circle.

9.2 Existence over finite fields: the condition $n \mid q - 1$

The decisive structural fact about finite fields is that their multiplicative group $\mathbb{F}_q^\times$ is cyclic of order $q - 1$ (theorem 6.17); the existence theorem for roots of unity rests on it.

Theorem 9.7 (Existence and count of primitive n -th roots of unity). *Let $\mathbb{F}_q$ be a finite field and $n \in \mathbb{N}_{>0}$ with $\gcd(n, q) = 1$ (automatic when $n < p$, where $p = \text{char } \mathbb{F}_q$; equivalently, in characteristic p , exactly the condition $p \nmid n$). Then:*

1. $\mathbb{F}_q$ contains a primitive n -th root of unity if and only if $n \mid q - 1$.
2. When $n \mid q - 1$, the group $\mu_n(\mathbb{F}_q)$ is cyclic of order exactly n , and the number of primitive n -th roots of unity in $\mathbb{F}_q$ is $\varphi(n)$, where φ is Euler's totient function (theorem 2.26).

Proof. Let g be a generator of $\mathbb{F}_q^\times$, which is cyclic of order $q - 1$ by theorem 6.17.

($\Rightarrow$) Suppose ω is a primitive n -th root of unity in $\mathbb{F}_q$. Then $\omega \in \mathbb{F}_q^\times$ has order n by theorem 9.3, and by Lagrange's theorem the order of an element divides the group order $q - 1$ (Corollary 3.30). Hence $n \mid q - 1$.

($\Leftarrow$) Suppose $n \mid q - 1$, say $q - 1 = nm$. Set $\omega = g^m$. Then $\omega^n = g^{mn} = g^{q-1} = 1$, so $\omega \in \mu_n(\mathbb{F}_q)$. Its order is

$$\text{ord}(\omega) = \text{ord}(g^m) = \frac{\text{ord}(g)}{\gcd(m, \text{ord}(g))} = \frac{q-1}{\gcd(m, q-1)} = \frac{q-1}{m} = n,$$

using the order-of-a-power formula (Corollary 3.23) together with $m \mid q-1$ (so $\gcd(m, q-1) = m$). Thus ω is a primitive n -th root of unity, and by theorem 9.6, $\mu_n(\mathbb{F}_q) = \langle \omega \rangle$ is cyclic of order exactly n .

Count. In the cyclic group $\mu_n(\mathbb{F}_q) = \langle \omega \rangle$ of order n , the element ω^k has order $n / \gcd(k, n)$, again by Corollary 3.23. This equals n precisely when $\gcd(k, n) = 1$, and the number of $k \in \{0, 1, \dots, n-1\}$ with $\gcd(k, n) = 1$ is by definition $\varphi(n)$ (theorem 2.26). Hence there are exactly $\varphi(n)$ primitive n -th roots of unity.

(A side note on the coprimality hypothesis. The condition $\gcd(n, q) = 1$ is what guarantees that $X^n - 1$ is separable— $\gcd(X^n - 1, nX^{n-1}) = 1$ when $n \neq 0$ in $\mathbb{F}_q$ —hence has n distinct roots in a splitting field over $\mathbb{F}_q$ (theorem 5.40); that all n of them lie in $\mathbb{F}_q$ itself is exactly the condition $n \mid q - 1$. If $p = \text{char } \mathbb{F}_q$ divided n , then $X^n - 1$ would have repeated roots and $\mu_n(\mathbb{F}_q)$ would be strictly smaller. But $n \mid q - 1$ already forces $\gcd(n, q) = 1$, since $\gcd(q - 1, q) = 1$, so the hypothesis is in fact subsumed.) $\square$

Remark 9.8. The proof shows that for a finite field the clean statement is simply: $\mathbb{F}_q$ has a primitive n -th root of unity $\iff n \mid q - 1$, with no separate coprimality side-condition, because $n \mid q - 1$ automatically makes n coprime to q . We stated the condition $\gcd(n, q) = 1$ explicitly only to flag the role of the characteristic.

Example 9.9. Take $\mathbb{F}_q = \mathbb{F}_p$ with $p = 7$. Here $q - 1 = 6$, and 3 is a generator of $\mathbb{F}_7^\times$: its successive powers are 3, 2, 6, 4, 5, 1, all six nonzero residues (Example 6.21). For $n = 3 \mid 6$, set $m = 6/3 = 2$ and $\omega = 3^2 = 2$. Indeed $2^1 = 2, 2^2 = 4, 2^3 = 8 \equiv 1$, so $\omega = 2$ has order 3 and $\mu_3(\mathbb{F}_7) = \{1, 2, 4\}$. There are $\varphi(3) = 2$ primitive cube roots of unity, namely 2 and 4. By contrast $4 \nmid 6$, so $\mathbb{F}_7$ has no primitive 4-th root of unity: $X^4 - 1 = (X^2 - 1)(X^2 + 1)$, and $X^2 + 1$ is irreducible over $\mathbb{F}_7$ because it has no root there, -1 not being a square modulo 7.

Remark 9.10 (Two-adicity and FFT-friendly primes). For the fast algorithms below it is desirable that n be a power of two, or at least *smooth*—a product of small primes, ideally with many factors of two. Theorem 9.7 shows this is possible exactly when $2^k \mid q - 1$. The largest k with $2^k \mid q - 1$ is called the *two-adicity* of the field. Designers deliberately choose primes p with high two-adicity—so that $p - 1 = 2^k \cdot c$ for large k —as scalar fields in proof systems precisely so that large smooth evaluation domains exist. Both Pasta primes—those underlying the Pallas and Vesta curves of section 10—satisfy $2^{32} \mid p - 1$, with two-adicity exactly 32 (the primes are specified in §5.4.9.6, “Pallas and Vesta”, of the Zcash protocol specification), so each Pasta field contains primitive 2^k -th roots of unity for every $k \leq 32$, furnishing power-of-two evaluation domains of every size up to 2^{32} ; this is what makes the entire machinery of this section available to Halo 2 and Orchard.

9.3 The evaluation domain H and its vanishing polynomial

We now fix the central object of this section.

Definition 9.11 (Evaluation domain). Let $\mathbb{F}$ be a field containing a primitive n -th root of unity ω . The *evaluation domain* of order n is the multiplicative subgroup

$$H = \langle \omega \rangle = \{1, \omega, \omega^2, \dots, \omega^{n-1}\} = \mu_n(\mathbb{F}) \subseteq \mathbb{F}^\times,$$

a cyclic group of order n by theorem 9.6. We index its elements as $\omega^0, \omega^1, \dots, \omega^{n-1}$, and abbreviate ω^i by h_i when convenient. Both $\mathbb{F}$ and ω remain fixed for the rest of the section.

Because H is a multiplicative *group*, it is closed under products and inverses, and $1 \in H$. These elementary facts are exactly what make H substantially more useful than an arbitrary set of n interpolation points: cosets, products, and shifts of H interact algebraically. The single most important polynomial attached to H is the one that vanishes precisely on it.

Definition 9.12 (Vanishing polynomial). The *vanishing polynomial* (or *zerofier*) of H is

$$Z_H(X) = \prod_{h \in H} (X - h) \in \mathbb{F}[X].$$

On a generic n -point set the vanishing polynomial (theorem 5.33) is an unstructured product of n linear factors. On the group H it collapses.

Theorem 9.13 (Closed form and factorisation of the vanishing polynomial). With $H = \mu_n(\mathbb{F})$

of order n generated by a primitive n -th root of unity ω ,

$$Z_H(X) = X^n - 1 = \prod_{i=0}^{n-1} (X - \omega^i).$$

Moreover Z_H is separable: it has n distinct roots and $\gcd(Z_H, Z'_H) = 1$.

Proof. Every $h \in H$ satisfies $h^n = 1$, hence is a root of $X^n - 1$. The n elements of H are distinct (theorem 9.6), so $X^n - 1$ has n distinct roots, namely all of H , and factors as $X^n - 1 = c \prod_{h \in H} (X - h)$ for some constant $c \in \mathbb{F}^\times$ by iterating the factor theorem (theorem 5.36, with the degree leaving no room for further factors). Comparing leading coefficients—both $X^n - 1$ and the product are monic—gives $c = 1$, whence $X^n - 1 = \prod_{h \in H} (X - h) = Z_H(X)$.

For separability, the formal derivative is $Z'_H(X) = nX^{n-1}$ (theorem 5.37). The existence of the n distinct roots already forces $\text{char } \mathbb{F} \nmid n$: if $\text{char } \mathbb{F} = \ell$ divided n , say $n = \ell m$, the freshman's dream (theorem 6.13) would give $X^n - 1 = (X^m - 1)^\ell$, a polynomial with at most $m < n$ distinct roots—a contradiction. Hence $n \neq 0$ in $\mathbb{F}$, so nX^{n-1} is nonzero at every element of H , since each is nonzero. Thus it has no root in common with $X^n - 1$, so $\gcd(Z_H, Z'_H) = 1$ and Z_H is separable (theorem 5.40). $\square$

The contraposition through the freshman's dream is worth isolating as a device: it converts a *root count* (n distinct n -th roots of unity exist) into a *characteristic condition* ($\text{char } \mathbb{F} \nmid n$), and thereby licenses dividing by n in $\mathbb{F}$ —which the closed forms below do constantly.

The closed form $Z_H(X) = X^n - 1$ is the workhorse of polynomial proof systems: testing whether a polynomial f vanishes on all of H reduces to checking divisibility by $X^n - 1$, which one can verify by exhibiting a quotient $t(X)$ with $f(X) = t(X)(X^n - 1)$. Moreover, evaluating Z_H at any point costs a single exponentiation X^n , not n multiplications. We record two structural consequences.

Proposition 9.14 (Cosets and shifts). *Let $\gamma \in \mathbb{F}^\times$ and let $\gamma H = \{\gamma h : h \in H\}$ be the corresponding coset (theorem 3.27). Then the polynomial vanishing on γH is*

$$Z_{\gamma H}(X) = \prod_{h \in H} (X - \gamma h) = X^n - \gamma^n.$$

In particular, distinct cosets of H in $\mathbb{F}^\times$ have vanishing polynomials $X^n - c$ for distinct constants $c = \gamma^n$, and the cosets of H partition $\mathbb{F}^\times$ (theorem 3.28) into translates of H , on each of which X^n is constant.

Proof. Substitute $Y = X/\gamma$:

$$\prod_{h \in H} (X - \gamma h) = \gamma^n \prod_{h \in H} (Y - h) = \gamma^n (Y^n - 1) = \gamma^n ((X/\gamma)^n - 1) = X^n - \gamma^n,$$

using theorem 9.13 for the middle equality. The constant is $c = \gamma^n$, and

$$\gamma_1 H = \gamma_2 H \iff \gamma_1/\gamma_2 \in H \iff (\gamma_1/\gamma_2)^n = 1 \iff \gamma_1^n = \gamma_2^n,$$

so distinct cosets give distinct constants c . $\square$

Lemma 9.15 (Finite geometric series). *Let $\mathbb{F}$ be a field, $\zeta \in \mathbb{F}$ with $\zeta \neq 1$, and $n \geq 1$. Then*

$$\sum_{i=0}^{n-1} \zeta^i = \frac{\zeta^n - 1}{\zeta - 1}.$$

Proof. Write $S = 1 + \zeta + \zeta^2 + \dots + \zeta^{n-1}$. Multiplying by ζ advances every term one slot:

$$\zeta S = \zeta + \zeta^2 + \dots + \zeta^{n-1} + \zeta^n.$$

In the difference $\zeta S - S$, the terms $\zeta, \zeta^2, \dots, \zeta^{n-1}$ appear in both lines and cancel; only the two extremes survive:

$$(\zeta - 1)S = \zeta S - S = \zeta^n - 1.$$

Since $\zeta \neq 1$, the factor $\zeta - 1$ is invertible, and dividing by it yields the claim. $\square$

Proposition 9.16 (Sum of roots of unity). *If $n > 1$ then $\sum_{i=0}^{n-1} \omega^i = 0$, and more generally for any integer k ,*

$$\sum_{i=0}^{n-1} \omega^{ki} = \begin{cases} n & \text{if } n \mid k, \\ 0 & \text{if } n \nmid k. \end{cases}$$

Proof. If $n \mid k$, then $\omega^{ki} = (\omega^n)^{ki/n} = 1$ for each i , so the sum is n . If $n \nmid k$, set $\zeta = \omega^k \neq 1$ (since ω has order n , we have $\omega^k = 1 \iff n \mid k$ by theorem 9.5), and theorem 9.15 gives

$$\sum_{i=0}^{n-1} \zeta^i = \frac{\zeta^n - 1}{\zeta - 1} = \frac{(\omega^n)^k - 1}{\zeta - 1} = \frac{1 - 1}{\zeta - 1} = 0,$$

the denominator being nonzero because $\zeta \neq 1$. The first claim is the case $k = 1$, valid for $n > 1$ since then $n \nmid 1$. $\square$

Remark 9.17 (Rotation makes the vanishing inevitable). For $n \nmid k$ there is also a proof with no formula at all. The n terms of $\sum_i \omega^{ki}$ run through the subgroup $\langle \zeta \rangle$ generated by $\zeta = \omega^k$, each element appearing equally often; multiplying the sum by ζ merely rotates that cycle one notch—each term advances to the next, the last wrapping around to the first—so the sum satisfies $\zeta S = S$. A quantity invariant under scaling by something other than 1 can only be zero: $(\zeta - 1)S = 0$ with $\zeta \neq 1$ forces $S = 0$. The balanced cycle of fig. 1 is this argument made visible.

Remark 9.18 (Why “orthogonality”). Call two vectors in $\mathbb{F}^n$ *orthogonal* when their standard inner product (theorem 8.21) vanishes. For each frequency k let $v_k = (1, \omega^k, \omega^{2k}, \dots, \omega^{(n-1)k})$ be its vector of powers, and let $\bar{v}_k = (1, \omega^{-k}, \omega^{-2k}, \dots, \omega^{-(n-1)k})$ be the same construction run with ω^{-1} in place of ω : the *mirror* of v_k , walking the cycle in the opposite direction, in the role the complex conjugate plays classically (on the unit circle $\bar{z} = z^{-1}$). Then $\langle \bar{v}_k, v_{k'} \rangle = \sum_i \omega^{i(k'-k)}$, which theorem 9.16 evaluates to 0 for $k \neq k'$ and to n for $k = k'$: distinct frequency vectors are orthogonal, and each pairs with its own mirror to n . (Which factor carries the mirror is a convention: with the roles exchanged the exponent is $k-k'$, and the criterion $n \mid k'-k$ is symmetric, so the verdict is the same; the order here matches the product $\bar{V}V$ computed in theorem 9.29.) This is the finite-field counterpart of the orthogonality of the exponentials $\theta \mapsto e^{2\pi i k \theta}$ in classical Fourier analysis, the balanced trip around the cycle replacing the integral over the circle. It is the engine of the inverse-transform proof below (theorem 9.29): there the v_k appear as the columns of the transform matrix, so in the inverse sum every wrong frequency’s contribution rides on one of these vanishing pairings and cancels, while the right one survives n -fold and the $1/n$ rescales it.

9.4 The Lagrange basis on H

Attention now passes from the group H to the polynomials we study on it. Since H has exactly n points, values on H pin down polynomials of degree less than n (theorem 5.20); the basis adapted to this situation is the Lagrange basis of §5.7, which on the group H acquires a closed form it has on no generic point set.

Definition 9.19. Let $\mathbb{F}[X]_{<n}$ denote the $\mathbb{F}$ -vector space of polynomials of degree strictly less than n , together with the zero polynomial. It has the monomial basis $\{1, X, \dots, X^{n-1}\}$ (theorem 8.14), so $\dim_{\mathbb{F}} \mathbb{F}[X]_{<n} = n$ (theorem 8.17).

Definition 9.20. For each $i \in \{0, 1, \dots, n-1\}$, the i -th *Lagrange basis polynomial* for $H = \{\omega^0, \dots, \omega^{n-1}\}$ is

$$L_i(X) = \prod_{\substack{j=0 \\ j \neq i}}^{n-1} \frac{X - \omega^j}{\omega^i - \omega^j} \in \mathbb{F}[X]_{<n},$$

the basis polynomial ℓ_i of theorem 5.30 for the nodes $\omega^0, \dots, \omega^{n-1}$.

Proposition 9.21 (Interpolation property). *The polynomials $L_0, \dots, L_{n-1}$ satisfy the Kronecker-delta conditions*

$$L_i(\omega^j) = \delta_{ij} = \begin{cases} 1 & i = j, \\ 0 & i \neq j, \end{cases}$$

they form a basis of $\mathbb{F}[X]_{<n}$, and they give the unique interpolant: for any data $(v_0, \dots, v_{n-1}) \in \mathbb{F}^n$, the polynomial

$$f(X) = \sum_{i=0}^{n-1} v_i L_i(X)$$

is the unique element of $\mathbb{F}[X]_{<n}$ with $f(\omega^i) = v_i$ for all i .

Proof. Each L_i is well defined and of degree $n-1$ because the nodes ω^j are distinct (theorem 9.6), so the denominators $\omega^i - \omega^j$ with $j \neq i$ are nonzero. The delta conditions, the interpolation $f(\omega^j) = v_j$, and the uniqueness are theorem 5.30 (with (4)) applied to these nodes with $d = n-1$; uniqueness rests, as there, on the root count: two interpolants in $\mathbb{F}[X]_{<n}$ differ by a polynomial of degree $< n$ with n distinct roots, which is the zero polynomial (theorem 5.18). It remains to see that the L_i form a basis of $\mathbb{F}[X]_{<n}$. They are linearly independent: if $\sum_i c_i L_i = 0$, evaluation at ω^j yields $c_j = 0$ for every j (theorem 8.11). Being n independent vectors in the n -dimensional space $\mathbb{F}[X]_{<n}$ (theorem 9.19), they form a basis (theorem 8.17). $\square$

The root-count step deserves a name, for it recurs: to prove two polynomials of degree $< n$ equal, show they agree on all n points of H . We use this *root-count device* for the uniqueness above, for theorem 9.24, and for Example 9.25 below.

On a generic point set the Lagrange polynomials admit no closed form simpler than the defining product. On the group H they collapse to a compact expression involving the vanishing polynomial.

Theorem 9.22 (Closed form of the Lagrange basis on H). *For $H = \mu_n(\mathbb{F})$ with primitive root ω ,*

and for each i ,

$$L_i(X) = \frac{\omega^i}{n} \cdot \frac{X^n - 1}{X - \omega^i} = \frac{\omega^i (X^n - 1)}{n (X - \omega^i)}.$$

In particular, dividing out one linear factor and scaling yields all n Lagrange polynomials from the single polynomial $Z_H(X) = X^n - 1$.

Proof. Fix i . From theorem 9.13, $Z_H(X) = \prod_{j=0}^{n-1} (X - \omega^j)$, so

$$\frac{Z_H(X)}{X - \omega^i} = \prod_{\substack{j=0 \\ j \neq i}}^{n-1} (X - \omega^j),$$

which is exactly the numerator of L_i in theorem 9.20. It remains to evaluate the denominator $\prod_{j \neq i} (\omega^i - \omega^j)$. Writing $Z_H = (X - \omega^i)g$ with $g = \prod_{j \neq i} (X - \omega^j)$, the product rule (theorem 5.38) gives $Z'_H = g + (X - \omega^i)g'$, so

$$Z'_H(\omega^i) = g(\omega^i) = \prod_{\substack{j=0 \\ j \neq i}}^{n-1} (\omega^i - \omega^j)$$

—the derivative at a simple root equals the product of the differences to the other roots, the computation already made for simple roots in the proof of theorem 5.40. Concretely $Z'_H(X) = nX^{n-1}$, so

$$Z'_H(\omega^i) = n (\omega^i)^{n-1} = n \omega^{in} \omega^{-i} = n \cdot 1 \cdot \omega^{-i} = \frac{n}{\omega^i},$$

using $\omega^{in} = (\omega^n)^i = 1$. Therefore

$$L_i(X) = \frac{Z_H(X)/(X - \omega^i)}{Z'_H(\omega^i)} = \frac{(X^n - 1)/(X - \omega^i)}{n/\omega^i} = \frac{\omega^i (X^n - 1)}{n (X - \omega^i)},$$

as claimed. $\square$

Remark 9.23 (Barycentric form). The closed form renders the *barycentric* interpolation formula on H especially clean. Writing $f(X) = \sum_i v_i L_i(X)$ and substituting theorem 9.22,

$$f(X) = \frac{X^n - 1}{n} \sum_{i=0}^{n-1} \frac{\omega^i v_i}{X - \omega^i}.$$

Evaluating f at a point $z \notin H$ thus costs one evaluation of $z^n - 1$ and n terms of the sum, with no per-point polynomial reconstruction. A verifier can use this formula when the values v_i are known, as for public-input polynomials; a commitment to hidden values does not by itself permit this evaluation.

Proposition 9.24 (Lagrange polynomials sum to one). *The Lagrange basis polynomials for H satisfy $\sum_{i=0}^{n-1} L_i(X) = 1$ identically.*

Proof. The polynomial $g(X) = \sum_i L_i(X) - 1$ lies in $\mathbb{F}[X]_{<n}$ and vanishes at every ω^j , because $\sum_i L_i(\omega^j) = \sum_i \delta_{ij} = 1$ (theorem 9.21). By the root-count device—a polynomial of degree $< n$ with n distinct roots is zero (theorem 5.18)— $g \equiv 0$. $\square$

Example 9.25 (The all-ones value vector). The constant polynomial 1 interpolates the data $v_i = 1$ for all i ; theorem 9.24 is the statement $\sum_i 1 \cdot L_i = 1$. Likewise the values of X on H are $v_i = \omega^i$, and, for $n \geq 2$, theorem 9.21 gives $\sum_i \omega^i L_i(X) = X$: both sides have degree $< n$ and agree on all of H , so the root-count device applies. We invoke these identities constantly when rewriting monomials in the Lagrange basis.

9.5 Evaluation and interpolation as mutually inverse linear maps

An element of $\mathbb{F}[X]_{<n}$ admits two natural descriptions: by its n coefficients, or by its n values on H . Both are coordinate systems on the same n -dimensional vector space, and passage between them is a linear isomorphism—the specialisation to H of the evaluation isomorphism met for general nodes in theorem 5.32.

Definition 9.26 (Evaluation map). The *evaluation map* on H is the $\mathbb{F}$ -linear map (theorem 8.18)

$$\text{ev}_H: \mathbb{F}[X]_{<n} \longrightarrow \mathbb{F}^n, \quad \text{ev}_H(f) = (f(\omega^0), f(\omega^1), \dots, f(\omega^{n-1})).$$

Linearity is immediate from $(af + bg)(\omega^i) = af(\omega^i) + bg(\omega^i)$, an instance of theorem 5.16.

Theorem 9.27 (Evaluation and interpolation are mutually inverse). The *evaluation map* ev_H is a vector-space isomorphism. Its inverse is the interpolation map

$$\text{int}_H: \mathbb{F}^n \longrightarrow \mathbb{F}[X]_{<n}, \quad \text{int}_H(v_0, \dots, v_{n-1}) = \sum_{i=0}^{n-1} v_i L_i(X).$$

That is, $\text{int}_H \circ \text{ev}_H = \text{id}_{\mathbb{F}[X]_{<n}}$ and $\text{ev}_H \circ \text{int}_H = \text{id}_{\mathbb{F}^n}$.

Proof. Both maps are linear and act between spaces of the same finite dimension n , so verifying one composition would suffice (a one-sided inverse makes ev_H injective, hence surjective by rank-nullity, theorem 8.19); we check both for clarity. For $v \in \mathbb{F}^n$, the polynomial $\text{int}_H(v) = \sum_i v_i L_i$ has value $\sum_i v_i L_i(\omega^j) = v_j$ at ω^j by theorem 9.21, so $\text{ev}_H(\text{int}_H(v)) = v$. For $f \in \mathbb{F}[X]_{<n}$, the polynomial $\text{int}_H(\text{ev}_H(f)) = \sum_i f(\omega^i) L_i$ is, by the uniqueness clause of theorem 9.21, the unique degree- $< n$ polynomial taking the values $f(\omega^i)$ on H —which is f itself. Hence the two maps are inverse; in particular ev_H is bijective and linear, i.e. an isomorphism (theorem 8.18). $\square$

In coordinates, ev_H is given by a matrix, which we name here because it is the bridge to the Fourier transform. The notation is worth fixing once, since §8 did not need it: an $n \times n$ matrix over $\mathbb{F}$ is a doubly indexed family $A = (A_{ik})_{0 \leq i, k \leq n-1}$ of elements of $\mathbb{F}$, acting on a vector $c \in \mathbb{F}^n$ by $(Ac)_i = \sum_k A_{ik} c_k$ —the general form, written out through the standard basis (Example 8.15), of a linear map $\mathbb{F}^n \rightarrow \mathbb{F}^n$. The product of two matrices is defined so that $(AB)c = A(Bc)$, which forces $(AB)_{ij} = \sum_k A_{ik} B_{kj}$; the *identity matrix* I , with entries δ_{ik} , acts as the identity map; and A is *invertible* if some matrix A^{-1} satisfies $A^{-1}A = AA^{-1} = I$.

Definition 9.28 (Vandermonde / DFT matrix). Let $f(X) = \sum_{k=0}^{n-1} c_k X^k$ have coefficient vector $c = (c_0, \dots, c_{n-1})$. Then $\text{ev}_H(f)$ has i -th entry $f(\omega^i) = \sum_k c_k \omega^{ik}$. Thus, relative to the monomial basis on $\mathbb{F}[X]_{<n}$ and the standard basis on $\mathbb{F}^n$, the *Vandermonde matrix*

$$V = (\omega^{ik})_{0 \leq i, k \leq n-1} = \begin{pmatrix} 1 & 1 & 1 & \dots & 1 \\ 1 & \omega & \omega^2 & \dots & \omega^{n-1} \\ 1 & \omega^2 & \omega^4 & \dots & \omega^{2(n-1)} \\ \vdots & & & & \vdots \\ 1 & \omega^{n-1} & \omega^{2(n-1)} & \dots & \omega^{(n-1)^2} \end{pmatrix}$$

—rows indexed by the evaluation point ω^i , columns by the power k —represents ev_H , so that $\text{ev}_H(f) = Vc$. We call V the (discrete Fourier) transform matrix on H .

Theorem 9.29 (Invertibility and the inverse transform). *The matrix V of theorem 9.28 is invertible, with*

$$(V^{-1})_{k,i} = \frac{1}{n} \omega^{-ik}.$$

Equivalently, $V^{-1} = \frac{1}{n} \bar{V}$, where $\bar{V} = (\omega^{-ik})_{i,k}$ is the transform matrix built from ω^{-1} (itself a primitive n -th root of unity). Consequently the coefficient and value representations are related by the mutually inverse formulas

$$c_k = \frac{1}{n} \sum_{i=0}^{n-1} f(\omega^i) \omega^{-ik}, \quad f(\omega^i) = \sum_{k=0}^{n-1} c_k \omega^{ik}.$$

Proof. First note that $\omega^{-1} = \omega^{n-1}$ has order n by Corollary 3.23 (with $\gcd(n-1, n) = 1$), so $\bar{V}$ is again a matrix of the shape in theorem 9.28. We show $\bar{V}V = nI$. The (k, k') entry of $\bar{V}V$ is

$$\sum_{i=0}^{n-1} \omega^{-ik} \omega^{ik'} = \sum_{i=0}^{n-1} \omega^{i(k'-k)}.$$

By theorem 9.16 this sum is n if $n \mid (k' - k)$ and 0 otherwise; for $k, k' \in \{0, \dots, n-1\}$ we have $n \mid (k' - k) \iff k = k'$. Hence $\bar{V}V = nI$, and the same computation with the roles of the two factors exchanged gives $V\bar{V} = nI$. Since $n \neq 0$ in $\mathbb{F}$ (theorem 9.13), we may divide by n : $V^{-1} = \frac{1}{n} \bar{V}$. The component formulas write out the matrix identities $\text{ev}_H(f) = Vc$ and $c = V^{-1} \text{ev}_H(f)$. $\square$

Remark 9.30. Theorem 9.27 and theorem 9.29 are two faces of one statement. The abstract face: evaluation on H is an isomorphism whose inverse is Lagrange interpolation. The concrete face: that isomorphism is the matrix V , and its inverse is, up to the scalar $1/n$, the same matrix with ω replaced by ω^{-1} . The near-symmetry between a transform and its inverse—differing only by the substitution $\omega \mapsto \omega^{-1}$ and a scale $1/n$ —is the algebraic reason the fast inverse transform of theorem 9.37 costs exactly as much as the forward one.

9.6 The number-theoretic transform and convolution

The map ev_H , read as the matrix V , is the discrete Fourier transform, but over a finite field rather than over $\mathbb{C}$. Because it involves no analytic structure—no complex exponentials, no convergence—it is called the *number-theoretic transform*.

Definition 9.31 (Number-theoretic transform). Let ω be a primitive n -th root of unity in $\mathbb{F}$. The *number-theoretic transform* (NTT) sends the coefficient vector of a polynomial to its value vector on H : it is the evaluation map ev_H written in coordinates. Explicitly, the NTT of $a = (a_0, \dots, a_{n-1}) \in \mathbb{F}^n$ is $\hat{a} = (\hat{a}_0, \dots, \hat{a}_{n-1})$ with

$$\hat{a}_j = \sum_{k=0}^{n-1} a_k \omega^{jk}, \quad j = 0, \dots, n-1,$$

each entry being the value at ω^j of the polynomial with coefficients a . The *inverse number-theoretic transform* is interpolation—the recovery of the coefficients from the values—and by theorem 9.29 it too is a single explicit sum:

$$a_k = \frac{1}{n} \sum_{j=0}^{n-1} \hat{a}_j \omega^{-jk}.$$

The NTT enjoys the same convolution property as the classical DFT, and this is the mechanism underlying fast polynomial multiplication. Recall that the coefficients of a product are a *convolution*: with $f = \sum_i a_i X^i$ and $g = \sum_j b_j X^j$, the coefficient of X^k in fg is $\sum_{i+j=k} a_i b_j$ (theorem 5.3). The *cyclic* variant folds indices modulo n : for $a, b \in \mathbb{F}^n$, the cyclic convolution $a * b \in \mathbb{F}^n$ has entries

$$(a * b)_k = \sum_{\substack{0 \leq i, j \leq n-1 \\ i+j \equiv k \pmod{n}}} a_i b_j, \quad k = 0, \dots, n-1.$$

Theorem 9.32 (Convolution theorem). *For $f, g \in \mathbb{F}[X]_{<n}$ with $\deg f + \deg g < n$, let $c = fg$ be their product (of degree $< n$). Then for every $h \in H$,*

$$c(h) = f(h) g(h),$$

i.e. pointwise multiplication of value vectors corresponds to polynomial multiplication. Equivalently, writing $a, b \in \mathbb{F}^n$ for the coefficient vectors of f, g (padded with zeros up to length n),

$$\widehat{a * b}_j = \hat{a}_j \hat{b}_j \quad (j = 0, \dots, n-1),$$

so the NTT turns convolution into entrywise multiplication.

Proof. The first identity is the statement that evaluation at h is a ring homomorphism $\mathbb{F}[X] \rightarrow \mathbb{F}$ (theorem 5.16): $(fg)(h) = f(h) g(h)$. Since $\deg(fg) < n$, interpolation recovers the product $c = fg$ from its values on the n -point set H (theorem 9.21), so the entrywise product of the value vectors uniquely determines c .

For the convolution statement the plan is: (i) multiplication in $\mathbb{F}[X]/(X^n - 1)$ is cyclic convolution of coefficient vectors; (ii) evaluation on H is a ring isomorphism from that quotient onto $\mathbb{F}^n$ with entrywise operations; (iii) the degree bound rules out wrap-around, so the cyclic product is the genuine one. For (i), work in the quotient ring $\mathbb{F}[X]/(X^n - 1)$ (theorem 4.19 and theorem 4.20, applied to the ideal of multiples of $X^n - 1$, theorem 4.17). Every class has a unique representative of degree $< n$, by division with remainder (theorem 5.12); and multiplying two such representatives and reducing replaces each X^k with $k \geq n$ by X^{k-n} , since $X^n \equiv 1$. The coefficient of X^k in the reduced product of the representatives with coefficient vectors a and b is therefore $\sum_{i+j \equiv k \pmod{n}} a_i b_j = (a * b)_k$: multiplication in $\mathbb{F}[X]/(X^n - 1)$ is cyclic convolution of coefficient vectors.

Now consider the map $[p] \mapsto \text{ev}_H(p)$ from $\mathbb{F}[X]/(X^n - 1)$ to $\mathbb{F}^n$, where $\mathbb{F}^n$ is a commutative ring under entrywise addition and multiplication (the ring laws hold in each coordinate because they hold in $\mathbb{F}$), with multiplicative identity $(1, \dots, 1)$. The map is well defined: two representatives of a class differ by a multiple of $X^n - 1$, which vanishes on H (theorem 9.13). It is a ring homomorphism (theorem 4.21), because evaluation at each point of H is one (theorem 5.16) and the operations on $\mathbb{F}^n$ are entrywise. And it is bijective: classes correspond to their degree- $< n$ representatives, on which ev_H is a bijection by theorem 9.27. Hence it is a ring isomorphism $\mathbb{F}[X]/(X^n - 1) \xrightarrow{\sim} \mathbb{F}^n$, and it sends the product of the classes of f and g —coefficient vector $a * b$ —to the entrywise product of $\hat{a}$ and $\hat{b}$. This is the displayed identity. Finally, when $\deg f + \deg g < n$, the genuine product fg already has degree $< n$, so no folding occurs, the cyclic and ordinary products coincide, and the statement computes fg itself. $\square$

Remark 9.33 (Computational significance). Theorem 9.32 reduces multiplying two degree- $< n/2$ polynomials—a quadratic-cost operation by the schoolbook method—to: transform both (NTT), multiply the value vectors entrywise (n multiplications), and transform back (inverse NTT). If the NTT runs in $O(n \log n)$ operations, the whole multiplication costs $O(n \log n)$. The fast Fourier transform supplies that conditional.

Example 9.34 (The convolution theorem in $\mathbb{F}_{17}$). In $\mathbb{F}_{17}$, the element $\omega = 4$ has $\omega^2 = 16 = -1$, so it is a primitive 4-th root of unity with domain $H = \{1, 4, 16, 13\}$. To multiply $f = 1 + 2X$ and $g = 3 + X$, pad both coefficient vectors to length 4 and transform: $\hat{a} = (3, 9, 16, 10)$ and $\hat{b} = (4, 7, 2, 16)$, the values of f and of g on H . Multiplying entrywise gives $(12, 12, 15, 7)$, the values of fg on H , and the inverse transform returns $(3, 7, 2, 0)$: the coefficients of $fg = 3 + 7X + 2X^2$, agreeing with the schoolbook product. At $n = 4$ nothing is saved; the point is the shape of the pipeline, which becomes decisive once the transforms themselves cost $O(n \log n)$ (§9.7).

9.7 The fast Fourier transform

Computing $\hat{a} = Va$ directly from theorem 9.28 costs $\Theta(n^2)$ field multiplications: each of the n outputs is a sum of n products. The fast Fourier transform (FFT)—here a number-theoretic FFT—computes the same vector in $O(n \log n)$ operations by a divide-and-conquer recursion, provided n is a power of two (or, more generally, smooth). This is where the requirement that H have smooth order, theorem 9.10, becomes indispensable.

Theorem 9.35 (Cooley–Tukey radix-2 step). *Suppose $n = 2m$ and ω is a primitive n -th root of unity in $\mathbb{F}$. Given $f(X) = \sum_{k=0}^{n-1} a_k X^k$, split its coefficients by parity into*

$$f_{\text{even}}(Y) = \sum_{k=0}^{m-1} a_{2k} Y^k, \quad f_{\text{odd}}(Y) = \sum_{k=0}^{m-1} a_{2k+1} Y^k,$$

so that $f(X) = f_{\text{even}}(X^2) + X f_{\text{odd}}(X^2)$. Then $\eta := \omega^2$ is a primitive m -th root of unity, and for $j = 0, \dots, m-1$,

$$f(\omega^j) = f_{\text{even}}(\eta^j) + \omega^j f_{\text{odd}}(\eta^j), \quad (7)$$

$$f(\omega^{j+m}) = f_{\text{even}}(\eta^j) - \omega^j f_{\text{odd}}(\eta^j). \quad (8)$$

Thus the length- n NTT of a reduces to two length- m NTTs (of the even and odd coefficient subsequences, over the domain $\langle \eta \rangle$) plus $O(n)$ extra operations.

Proof. The identity $f(X) = f_{\text{even}}(X^2) + X f_{\text{odd}}(X^2)$ regroups the sum $\sum_k a_k X^k$ into even-index and odd-index terms; in the odd terms one factor of X is pulled out, leaving powers X^{2k} . That $\eta = \omega^2$ is a primitive m -th root of unity follows from theorem 9.5 and the order-of-a-power formula (Corollary 3.23): $\text{ord}(\omega^2) = n / \gcd(2, n) = n/2 = m$, using $2 \mid n$.

It remains to evaluate. For (7), put $X = \omega^j$; then $X^2 = \omega^{2j} = \eta^j$, so $f(\omega^j) = f_{\text{even}}(\eta^j) + \omega^j f_{\text{odd}}(\eta^j)$. For (8), put $X = \omega^{j+m}$; then $X^2 = \omega^{2j+2m} = \omega^{2j} \omega^n = \eta^j \cdot 1 = \eta^j$ (since $2m = n$ and $\omega^n = 1$), while the linear factor is $\omega^{j+m} = \omega^j \omega^m = -\omega^j$. Here $\omega^m = -1$, because ω^m has order $n / \gcd(m, n) = n/m = 2$ (Corollary 3.23), and -1 is the unique element of order 2 in a field: $x^2 = 1$ with $x \neq 1$ forces $(x-1)(x+1) = 0$ and so $x = -1$; and $-1 \neq 1$, since $\text{char } \mathbb{F} \nmid n$ (theorem 9.13) with n even forces $\text{char } \mathbb{F} \neq 2$. Substituting, $f(\omega^{j+m}) = f_{\text{even}}(\eta^j) - \omega^j f_{\text{odd}}(\eta^j)$. $\square$

The half-domain trick just used— $\omega^m = -1$ whenever $n = 2m$, so the second half of the outputs differs from the first only by a sign on the odd part—recurs in every radix-2 argument over an evaluation domain; it is worth remembering alongside the theorem.

Remark 9.36 (The butterfly). Equations (7)–(8) constitute the *butterfly*: from the two subtransform outputs $E_j = f_{\text{even}}(\eta^j)$ and $O_j = f_{\text{odd}}(\eta^j)$ one forms the *twiddle* $t_j = \omega^j O_j$ and reads off the two outputs $E_j + t_j$ and $E_j - t_j$. Pictorially, E_j and O_j sit as two nodes on the left; each is carried to both outputs on the right, the two E_j -edges with weight 1 and the two O_j -edges with weights $+\omega^j$ and $-\omega^j$; the pair of crossing edges forms the $\times$ that names the operation. Each butterfly costs one multiplication (the twiddle) and two additions, and there are $n/2$ of them per level.

Theorem 9.37 ($O(n \log n)$ cost of the FFT). *Let $n = 2^t$ and let $T(n)$ be the number of field additions and multiplications the radix-2 recursion of theorem 9.35 uses to compute a length- n NTT. Then*

$$T(n) = 2T(n/2) + cn \quad (n > 1), \quad T(1) = O(1),$$

for a constant c , and therefore $T(n) = O(n \log n)$. The inverse NTT has the same cost, since it is the forward NTT with ω replaced by ω^{-1} , followed by scaling each entry by $1/n$ (theorem 9.29).

Proof. The recursion is exactly theorem 9.35: a length- n transform calls two length- $n/2$ transforms (on the even and odd coefficient subsequences) and combines them with $n/2$ butterflies of $O(1)$ operations each, for the cn overhead. Unrolling, with $n = 2^t$,

$$T(2^t) = 2T(2^{t-1}) + c2^t.$$

Divide by 2^t and set $S(t) = T(2^t)/2^t$: then $S(t) = S(t-1) + c$, so $S(t) = S(0) + ct = O(1) + ct$, and $T(2^t) = 2^t S(t) = O(2^t) + ct2^t = O(t2^t)$. With $t = \log_2 n$ this is $T(n) = O(n \log n)$. (Equivalently: the recursion tree has $\log_2 n$ levels, each performing $\Theta(n)$ total work across all subproblems at that level.) The inverse-transform claim is theorem 9.29: the inverse is the same Cooley–Tukey recursion with ω^{-1} in place of ω (also a primitive n -th root of unity), plus a final pass of n divisions by n , all within $O(n \log n)$. $\square$

Remark 9.38 (Mixed radix and smoothness). The hypothesis that enables fast transforms is that n be *smooth*—a product of small primes—and, crucially, that the field actually *contain* a primitive n -th root of unity, i.e. $n \mid q-1$ (theorem 9.7). A power-of-two n dividing $q-1$ is the optimal case: pure radix-2 with the cleanest butterfly.

Example 9.39 (A length-4 transform). Take $n = 4$ with primitive 4-th root ω (so $\omega^2 = -1$ and $\omega^4 = 1$, and $\eta = \omega^2 = -1$ is the primitive 2nd root). For $f = a_0 + a_1X + a_2X^2 + a_3X^3$, the two length-2 subtransforms over the domain $\{1, -1\}$ —each itself a single butterfly, with twiddle $\eta^0 = 1$ —are

$$E = (a_0 + a_2, a_0 - a_2), \quad O = (a_1 + a_3, a_1 - a_3),$$

and the butterflies give

$$\hat{a}_0 = E_0 + \omega^0 O_0, \quad \hat{a}_2 = E_0 - \omega^0 O_0, \quad \hat{a}_1 = E_1 + \omega^1 O_1, \quad \hat{a}_3 = E_1 - \omega^1 O_1.$$

(Note the output indexing: outputs j and $j+m$ pair up, per (7)–(8).) This uses 2 subtransforms of 2 additions each plus 2 butterflies of 2 additions each ($n/2 = 2$ per level, theorem 9.36)—together 2 twiddle multiplications and 8 additions—versus 16 coefficient-products and 12 additions for the direct 4×4 matrix multiply; and the gap widens as n grows.

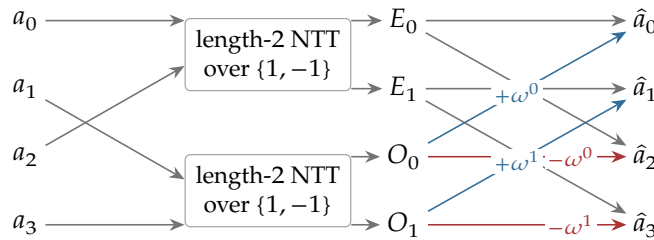


Figure 2: The length-4 transform of theorem 9.39 as a circuit. The inputs split by parity into two length-2 subtransforms, whose outputs E_0, E_1 and O_0, O_1 feed the two butterflies of theorem 9.36: each E_j is carried to outputs $\hat{a}_j$ and $\hat{a}_{j+2}$ with weight 1, each O_j with weights $+\omega^j$ and $-\omega^j$ (eq. (7)–eq. (8)); the crossing pair of weighted edges is the $\times$ that names the butterfly.

9.8 The role of smooth multiplicative domains in polynomial proof systems

This section closes by assembling the pieces into the reason these domains form the substrate of Halo 2 and Orchard. A proof system of the polynomial type encodes a computation as a system of polynomial identities and convinces a verifier that those identities hold. The evaluation domain $H = \mu_n$ is where the encoding lives—the n -step table of the section’s opening is laid out on the n points of H —and every property proved above is load-bearing.

Remark 9.40 (The roles played by H , Z_H , the Lagrange basis, and the FFT).

- **Witnesses as evaluations on H .** A prover lays out the trace of a computation as the values of one or more polynomials on the n points of H . By theorem 9.27 this is the same data as a polynomial of degree $< n$; the prover may move freely between the value (Lagrange) representation, natural for stating constraints row by row, and the coefficient representation, natural for committing and for low-degree testing. The change of representation is the (inverse) NTT.
- **Constraints as divisibility by Z_H .** “Constraint polynomial C holds on every row” means C vanishes on all of H , which by theorem 9.13 means exactly that $Z_H = X^n - 1$ divides C . The prover demonstrates this by exhibiting a quotient t with $C(X) = t(X)(X^n - 1)$. The inexpensive closed form $Z_H(X) = X^n - 1$ —one exponentiation to evaluate—is what makes this check cheap at a random point. Theorem 9.14 extends the same technique to cosets γH , used to separate quotient and remainder domains.
- **Lagrange selectors.** The Lagrange polynomials L_i of theorem 9.20 are *selectors*: L_i is 1 on row i and 0 elsewhere (theorem 9.21), so boundary and instance constraints (e.g. “the first row equals the public input”) become $L_0 \cdot (\dots)$. The closed form and the barycentric formula (theorem 9.22, theorem 9.23) let the verifier evaluate these at a random challenge in $O(n)$ or even $O(\log n)$ field operations rather than reconstructing polynomials.
- **Group structure for permutations and shifts.** Because H is a cyclic group (theorem 9.11), multiplication by ω is the cyclic “next-row” shift $\omega^i \mapsto \omega^{i+1}$. Constraints relating a row to its neighbour become $f(\omega X)$ versus $f(X)$; permutation arguments—the heart of Halo 2’s copy constraints—rest on the action of the group on itself and on its cosets. An arbitrary set of n points offers none of this.
- **Smoothness for speed.** Provers manipulate polynomials of size n (often after blowing the domain up by a small constant factor for low-degree testing). Every transform between representations, every multiplication, every coset evaluation is an NTT. By theorem 9.37 these cost $O(n \log n)$ instead of $O(n^2)$ —the difference between feasible and infeasible at the sizes used in practice—*provided* n is smooth and $\mathbb{F}_q$ contains a primitive n -th root of unity. By theorem 9.7 that requires $n \mid q - 1$; by theorem 9.10 this is why high-two-adicity primes are chosen for the scalar fields of the Pallas and Vesta curves (section 10).

In summary: the existence condition $n \mid q - 1$ (theorem 9.7) supplies the group H ; the group supplies the clean vanishing polynomial (theorem 9.13), the closed-form Lagrange basis (theorem 9.22), and the evaluation/interpolation isomorphism (theorem 9.27); and smoothness of n supplies the $O(n \log n)$ FFT (theorem 9.37). Together these discharge both demands of the section’s opening—a structured point set on which an execution trace can live, and $O(n \log n)$ multiplication and interpolation on it—and they are exactly the ingredients a polynomial proof system requires, which is why such systems rest on smooth multiplicative evaluation domains and on the fields that contain them.

10 Elliptic curves

The multiplicative groups of the preceding sections harbour a structural weakness. As theorem 7.22 recorded, the discrete logarithm in $\mathbb{F}_p^\times$ does not resist attack at the generic square-root cost: the *index-calculus* family of algorithms exploits the fact that residues come from ordinary integers, and integers factor into primes, to solve the problem in subexponential time. The practical consequence is severe. To push the cheapest known attack on $\mathbb{F}_p^\times$ beyond 2^{128} operations, the prime p must run to *thousands* of bits, and every exponentiation drags that bulk along. What cryptography wants instead is a group of roughly 2^{255} elements for which no attack faster than the square-root bound is known—a group without known exploitable arithmetic behind its elements, where the generic $O(\sqrt{N})$ attacks of theorem 7.22 remain the best known. This section constructs the standard supplier of such groups.

An *elliptic curve* is, concretely, the solution set of a certain cubic equation in two variables, together with one extra “point at infinity”. Its notable feature is that this solution set carries the structure of an abelian group, defined by a strikingly geometric rule; and that inverting scalar multiplication in this group is hard—the discrete logarithm problem in a well-chosen elliptic-curve group appears to require exponential time. The combination of clean algebraic structure with a hard computational problem is exactly what cryptography requires.

Throughout the section we work over a field K whose characteristic (theorem 4.30) is neither 2 nor 3. In the applications K is a finite prime field $\mathbb{F}_p$ with p a large prime, but the basic theory is cleaner over a general field, and we specialise when needed. The reader should keep in mind the two *Pasta curves*

$$E_p : y^2 = x^3 + 5 \quad \text{over } \mathbb{F}_p, \quad E_q : y^2 = x^3 + 5 \quad \text{over } \mathbb{F}_q,$$

where p and q are the two 255-bit primes

$$\begin{aligned} p &= 2^{254} + 0x224698fc094cf91b992d30ed00000001, \\ q &= 2^{254} + 0x224698fc0994a8dd8c46eb2100000001. \end{aligned}$$

These are the primes that theorem 6.23 and theorem 7.18 handled as black boxes; here they finally acquire their curves. The constants are the deployed ones: the primes and the curve equations $y^2 = x^3 + 5$ are specified in §5.4.9.6 (“Pallas and Vesta”) of the Zcash protocol specification. The curve E_p , defined over $\mathbb{F}_p$, is called *Pallas*; the curve E_q , defined over $\mathbb{F}_q$, is called *Vesta*. Together they form a *cycle*: the number of $\mathbb{F}_p$ -points of Pallas equals q , and the number of $\mathbb{F}_q$ -points of Vesta equals p —a coincidence of design, not chance, whose meaning unfolds across the section. The equation $y^2 = x^3 + 5$ serves as the running example throughout, with a toy curve over $\mathbb{F}_{11}$ developed in parallel for hand computation.

10.1 Weierstrass equations

Definition 10.1 (Short Weierstrass equation). Let K be a field of characteristic $\neq 2, 3$, and let $a, b \in K$. The *short Weierstrass equation* with coefficients a, b is the equation

$$y^2 = x^3 + ax + b. \tag{9}$$

The elements a and b are the *Weierstrass coefficients*.

Over a field of characteristic $\neq 2, 3$, an invertible affine change of variables always reduces the most general cubic of this shape, $y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6$, to the form (9): completing the square in y (possible since $2 \neq 0$) removes the xy and y terms but changes the quadratic coefficient to $a_2 + a_1^2/4$; translating with the substitution $x = X - (a_2 + a_1^2/4)/3$ (possible since $3 \neq 0$) removes the X^2 term. This is precisely why characteristics 2 and 3 are excluded: those

cases require the longer Weierstrass form, and they are not relevant to the Pasta curves, which live over large prime fields.

Not every choice of a, b gives a usable curve. We must exclude the cubics whose graph has a self-intersection or a cusp, because the group law defined below breaks down at such a singular point.

Definition 10.2 (Discriminant and nonsingularity). The *discriminant* of the short Weierstrass equation (9) is

$$\Delta = -16(4a^3 + 27b^2) \in K. \quad (10)$$

The equation, and the cubic $f(x) = x^3 + ax + b$, is *nonsingular* (or *smooth*) if $\Delta \neq 0$, equivalently $4a^3 + 27b^2 \neq 0$. (The factor -16 is a convention inherited from the general Weierstrass form; over our fields it is nonzero and affects nothing.)

The quantity $4a^3 + 27b^2$ is exactly the obstruction to f having distinct roots. The following lemma makes this explicit by a computation reusable for any depressed cubic—a cubic with no square term—and is the engine of the smoothness criterion.

Lemma 10.3 (Discriminant of a depressed cubic). Let $f(x) = x^3 + ax + b = (x - r_1)(x - r_2)(x - r_3)$ with r_1, r_2, r_3 in some field containing K . Then

$$\prod_{i < j} (r_i - r_j)^2 = -(4a^3 + 27b^2).$$

Proof. Expanding $(x - r_1)(x - r_2)(x - r_3)$ and comparing coefficients with $x^3 + ax + b$ gives Vieta's relations for the depressed cubic:

$$r_1 + r_2 + r_3 = 0, \quad r_1r_2 + r_1r_3 + r_2r_3 = a, \quad r_1r_2r_3 = -b.$$

Differentiating $f(x) = \prod_i (x - r_i)$ by the product rule and evaluating at a root kills every term but one: $f'(r_i) = \prod_{j \neq i} (r_i - r_j)$. Multiplying the three evaluations pairs each factor $(r_i - r_j)$ with $(r_j - r_i)$:

$$\prod_i f'(r_i) = \prod_{i < j} (r_i - r_j)(r_j - r_i) = (-1)^3 \prod_{i < j} (r_i - r_j)^2.$$

It remains to compute $\prod_i f'(r_i) = \prod_i (3r_i^2 + a)$. Expanding the product and collecting the elementary symmetric functions of the squares r_i^2 ,

$$\prod_i (3r_i^2 + a) = 27 \prod_i r_i^2 + 9a \sum_{i < j} r_i^2 r_j^2 + 3a^2 \sum_i r_i^2 + a^3.$$

Vieta's relations evaluate each symmetric function: with $e_1 = 0, e_2 = a, e_3 = -b$,

$$\sum_i r_i^2 = e_1^2 - 2e_2 = -2a, \quad \sum_{i < j} r_i^2 r_j^2 = e_2^2 - 2e_1 e_3 = a^2, \quad \prod_i r_i^2 = e_3^2 = b^2.$$

Substituting, $\prod_i f'(r_i) = 27b^2 + 9a^3 - 6a^3 + a^3 = 4a^3 + 27b^2$, and the two displays together give $\prod_{i < j} (r_i - r_j)^2 = -(4a^3 + 27b^2)$. $\square$

Proposition 10.4 (Smoothness, four ways). Let $f(x) = x^3 + ax + b \in K[x]$. The following are equivalent.

1. The quantity $4a^3 + 27b^2$ of theorem 10.2 is nonzero.

2. The cubic f has no repeated root in the algebraic closure $\bar{K}$ —the smallest extension field of K in which every nonconstant polynomial over K factors into linear factors.
3. The polynomials f and f' are coprime in $\bar{K}[x]$.
4. There is no point $(x_0, y_0) \in \bar{K}^2$ satisfying the curve equation $y^2 = f(x)$ together with the two partial-derivative equations $2y = 0$ and $f'(x) = 0$.

Proof. Equivalence (1) $\Leftrightarrow$ (2): over $\bar{K}$ the monic cubic factors as $f = (x - r_1)(x - r_2)(x - r_3)$, and theorem 10.3 gives $\prod_{i < j} (r_i - r_j)^2 = -(4a^3 + 27b^2)$. The left side vanishes exactly when two roots coincide, so f has a repeated root if and only if $4a^3 + 27b^2 = 0$.

Equivalence (2) $\Leftrightarrow$ (3): this is the derivative test of theorem 5.40, applied over $\bar{K}$ where f splits: a repeated root of f is a common root of f and f' , hence a common factor $(x - \alpha)$, and conversely a nontrivial $\gcd(f, f')$ has a root in $\bar{K}$ which is then a repeated root of f .

Equivalence (4) $\Leftrightarrow$ (2): a singular point of the affine curve $y^2 = f(x)$ is a point where both partial derivatives of $g(x, y) = y^2 - f(x)$ vanish, namely $\partial g / \partial y = 2y = 0$ and $\partial g / \partial x = -f'(x) = 0$. Since $\text{char}(K) \neq 2$, the first equation forces $y_0 = 0$; the curve equation then gives $f(x_0) = 0$, which combined with $f'(x_0) = 0$ says x_0 is a repeated root of f (theorem 5.40(1)). Conversely a repeated root x_0 of f yields the singular point $(x_0, 0)$. $\square$

Definition 10.5 (Elliptic curve and rational points). An *elliptic curve* over K is a short Weierstrass equation (9) with nonzero discriminant, $4a^3 + 27b^2 \neq 0$. For any field extension $L \supseteq K$, the set of L -rational affine points is

$$E^{\text{aff}}(L) = \{ (x, y) \in L \times L : y^2 = x^3 + ax + b \},$$

and the set of L -rational points is

$$E(L) = E^{\text{aff}}(L) \cup \{O\},$$

where O is a single formal symbol called the *point at infinity*. When $L = K$ one writes simply E for the curve and $E(K)$ for its rational points.

Remark 10.6 (The Pasta curves are smooth). For the Pasta curves $a = 0$ and $b = 5$, so $4a^3 + 27b^2 = 27 \cdot 25 = 675 = 3^3 \cdot 5^2$. The curve is therefore nonsingular whenever the characteristic does not divide 675, i.e. whenever the prime is neither 3 nor 5; the Pasta primes are enormous, so this is automatic. More generally, any curve $y^2 = x^3 + b$ with $b \neq 0$ is nonsingular away from characteristics 2 and 3.

10.2 The point at infinity, geometrically

The extra point O is not decoration; it is the identity element of the group to come, and it has an honest geometric home. To see it, view the affine plane as sitting inside the *projective plane*. A point of the projective plane $\mathbb{P}^2(K)$ is an equivalence class $[X : Y : Z]$ of nonzero triples $(X, Y, Z) \in K^3 \setminus \{0\}$, where $(X, Y, Z) \sim (\lambda X, \lambda Y, \lambda Z)$ for every $\lambda \in K^\times$. The affine points (x, y) embed as $[x : y : 1]$; the points with $Z = 0$ form the *line at infinity*, the extra “directions”. Homogenising the Weierstrass equation (9) by setting $x = X/Z$, $y = Y/Z$ and clearing denominators gives the *projective Weierstrass equation*

$$Y^2Z = X^3 + aXZ^2 + bZ^3. \quad (11)$$

Setting $Z = 0$ in (11) forces $X^3 = 0$, hence $X = 0$, leaving the *single* projective point $[0 : 1 : 0]$. This is the point at infinity O . It lies infinitely far up in the vertical direction, and every vertical

line $x = \text{const}$ passes through it. The reader may safely picture O as one point pinned at the top (and bottom) of the y -axis, where all vertical lines meet; the projective equation above is the licence for that picture.

10.3 The chord-and-tangent group law: geometry

We now describe the procedure for “adding” two points of $E(K)$. The construction is purely geometric, and it rests on a single fact: a line meets a cubic curve in exactly three points, counted with multiplicity. The point at infinity is engineered precisely so that the counting is always exact.

Definition 10.7 (The group law, geometric form). Let $P, Q \in E(K)$. Define $P + Q$ as follows.

- If $P = O$, set $P + Q = Q$; if $Q = O$, set $P + Q = P$.
- Otherwise $P = (x_1, y_1)$ and $Q = (x_2, y_2)$ are affine. Draw the line ℓ through P and Q ; if $P = Q$, let ℓ be the tangent line to the curve at P . The line ℓ meets the curve in a third point R' , counted with multiplicity, where a vertical ℓ has third point O . Define $P + Q$ to be the *reflection of R' across the x -axis*: if $R' = (x_3, -y_3)$ then $P + Q = (x_3, y_3)$, and if $R' = O$ then $P + Q = O$.

The reflection step deserves a comment, because it is what converts “the three intersection points of a line sum to a fixed thing” into a genuine group law with O as identity. The underlying rule is this: *the three intersection points of any line with the curve add up to O* . (Stated here as a guiding picture; every use of it below is discharged by an explicit computation with the formulas of §10.4.) Reflection across the x -axis sends an affine point (x, y) to $(x, -y)$, which—as the next subsection confirms—is its additive inverse; and it sends O to O . To convert the geometry into formulas one must (i) find the third intersection point of a line with the cubic and (ii) handle the degenerate cases. The key algebraic device for (i) is Vieta’s relation between the roots of a cubic and its coefficients, already deployed in theorem 10.3.

10.4 Explicit affine formulas

Fix an elliptic curve $E : y^2 = x^3 + ax + b$ over K , and let $P = (x_1, y_1)$ and $Q = (x_2, y_2)$ denote affine points.

Definition 10.8 (Negation). Define

$$-O = O, \quad -(x_1, y_1) = (x_1, -y_1).$$

The reflected point $(x_1, -y_1)$ is again on the curve, since the curve equation depends on y only through y^2 . Geometrically, the line through P and $-P$ is the vertical line $x = x_1$, which meets the curve at P , $-P$, and O .

Construction 10.9 (Chord addition). Suppose $P, Q \neq O$ and $x_1 \neq x_2$. The line through P and Q is $y = \lambda x + \mu$ with

$$\lambda = \frac{y_2 - y_1}{x_2 - x_1}, \quad \mu = y_1 - \lambda x_1.$$

Substituting the line into the curve equation, $(\lambda x + \mu)^2 = x^3 + ax + b$, and collecting terms gives

$$g(x) := x^3 - \lambda^2 x^2 + (a - 2\lambda\mu)x + (b - \mu^2) = 0.$$

Both x_1 and x_2 are roots of the monic cubic g : each point lies on the line and on the curve, so $g(x_i) = f(x_i) - (\lambda x_i + \mu)^2 = f(x_i) - y_i^2 = 0$. Since $x_1 \neq x_2$, factoring out $(x - x_1)(x - x_2)$ (theorem 5.17, applied twice) leaves a monic linear factor $x - x_3$ with $x_3 \in K$: the x -coordinate of the third intersection point R' . By Vieta, the sum of the roots equals the negative of the x^2 -coefficient—so $x_1 + x_2 + x_3 = \lambda^2$, and the unknown root is read off with no factoring at all. Hence

$$x_3 = \lambda^2 - x_1 - x_2, \quad y_3 = \lambda(x_1 - x_3) - y_1, \quad (12)$$

and $P + Q = (x_3, y_3)$. The y -formula already incorporates the reflection across the x -axis: the third intersection point is $(x_3, \lambda x_3 + \mu)$, and reflecting gives $y_3 = -(\lambda x_3 + \mu) = -(\lambda x_3 + y_1 - \lambda x_1) = \lambda(x_1 - x_3) - y_1$.

Construction 10.10 (Tangent doubling). Suppose $P = Q = (x_1, y_1)$ with $y_1 \neq 0$. Implicit differentiation of $y^2 = x^3 + ax + b$ yields the tangent slope: from $2y dy = (3x^2 + a) dx$,

$$\lambda = \frac{3x_1^2 + a}{2y_1}.$$

With $\mu = y_1 - \lambda x_1$, substituting $y = \lambda x + \mu$ into the curve equation produces the same monic cubic $g(x) = x^3 + ax + b - (\lambda x + \mu)^2$ as in theorem 10.9, and here x_1 is a *double* root. The slogan “tangency counts the intersection twice” is a one-line check: $g(x_1) = f(x_1) - y_1^2 = 0$, and

$$g'(x_1) = 3x_1^2 + a - 2\lambda(\lambda x_1 + \mu) = 3x_1^2 + a - 2\lambda y_1 = 0$$

by the choice of λ , so x_1 is a repeated root of g (theorem 5.40(1)). Factoring $(x - x_1)^2$ out of g leaves a monic linear factor $x - x_3$ with $x_3 \in K$, and Vieta now reads $x_1 + x_1 + x_3 = \lambda^2$, hence

$$x_3 = \lambda^2 - 2x_1, \quad y_3 = \lambda(x_1 - x_3) - y_1, \quad [2]P = (x_3, y_3). \quad (13)$$

The remaining cases are exactly those in which the line is vertical.

- If $P = O$, then $P + Q = Q$; symmetrically if $Q = O$.
- If $x_1 = x_2$ but $y_1 = -y_2$ (so $Q = -P$, including the subcase $y_1 = y_2 = 0$), the line is vertical, the third point is O , and $P + Q = O$.
- If $P = Q$ and $y_1 = 0$, the tangent is vertical (the slope formula in theorem 10.10 has zero denominator), so $[2]P = O$. These are exactly the points of order two, studied in section 10.9.

Every case is covered: either an input is O ; or both inputs are affine with $x_1 \neq x_2$ (use (12)); or both are affine with $x_1 = x_2$, in which case either $y_1 = -y_2$ (result O) or $y_1 = y_2 \neq 0$ (use (13)).

Proposition 10.11 (Closure). *For all $P, Q \in E(K)$, the point $P + Q$ computed by the case analysis above lies in $E(K)$. In particular $E(K)$ is closed under $+$.*

Proof. If either input is O , the result is the other input, which lies in $E(K)$ by hypothesis; if the result is O , it lies in $E(K)$ by definition. Otherwise (x_3, y_3) is produced by (12) or (13), and in both constructions x_3 was exhibited as a root of the cubic $g(x) = x^3 + ax + b - (\lambda x + \mu)^2$ —as the third root beside x_1, x_2 in theorem 10.9, and beside the double root x_1 in theorem 10.10. The equation $g(x_3) = 0$ states precisely that the third intersection point $(x_3, \lambda x_3 + \mu) = (x_3, -y_3)$ satisfies the curve equation; and since the curve equation is invariant under $y \mapsto -y$, the reflected point (x_3, y_3) is on the curve too, so $P + Q \in E^{\text{aff}}(K) \subseteq E(K)$. For rationality, observe that all coordinates are rational expressions in x_1, x_2, y_1, y_2, a whose denominators are $x_2 - x_1$ or $2y_1$, nonzero in the cases where each formula is invoked; so $x_3, y_3 \in K$. $\square$

Remark 10.12 (Complete versus incomplete addition). The chord formula (12) is *incomplete*: it fails whenever $x_1 = x_2$ —the doubling and inverse cases—and an implementation must branch into the case analysis above. *Complete* addition formulas, valid for *all* input pairs including $P = Q$ and $P = -Q$, exist for the curves used here, at the cost of more field operations per addition; theorem 10.13 exhibits the deployed set and theorem 10.14 proves it complete. The distinction is load-bearing in later volumes. Halo 2’s elliptic-curve gadgets provide complete addition and use incomplete gates only where their callers justify the exclusions; the Sinsemilla hash of Orchard deliberately uses the *incomplete* formulas nondeterministically—an exceptional case does not violate the constraints but leaves the affected output unconstrained (the prover may choose the intermediate slope freely), which the specification models by letting the hash return $\perp$ and weakening the proved statement accordingly; its Merkle path validity check is even permitted to pass on the resulting sentinel values. Safety rests not on the proof system but on Theorem 5.4.4 of the Zcash protocol specification: any input reaching an exceptional case of the incomplete addition immediately yields a nontrivial discrete-logarithm relation among the Sinsemilla generators, so under discrete-log hardness on Pallas (theorem 10.31) such inputs are infeasible to find, for honest and adversarial provers alike.

An implementation branches; a circuit cannot. In a proof system the group law is checked by a fixed list of polynomial identities in the input, output, and auxiliary cells of a row, evaluated on every row alike, so the case analysis of this subsection has to be folded into algebra: each branch must appear as a factor that vanishes exactly when the branch is not taken. The deployed gate achieves this with five auxiliary cells, four of them “inverse-or-zero” hints that let a polynomial test a field element for zero.

Construction 10.13 (Complete addition as deployed). Let $E : y^2 = x^3 + b$ over $\mathbb{F}_p$ with $p > 3$, and encode the point at infinity as the pair $(0, 0)$, so that every element of $E(\mathbb{F}_p)$ is a pair of field elements. For $z \in \mathbb{F}_p$ let $\text{inv}_0(z)$ be z^{-1} if $z \neq 0$ and 0 if $z = 0$. For encoded inputs $P = (x_p, y_p)$ and $Q = (x_q, y_q)$ the prover fills five auxiliary cells,

$$\alpha = \text{inv}_0(x_q - x_p), \quad \beta = \text{inv}_0(x_p), \quad \gamma = \text{inv}_0(x_q), \quad \delta = \begin{cases} \text{inv}_0(y_q + y_p) & \text{if } x_q = x_p, \\ 0 & \text{otherwise,} \end{cases}$$

$$\lambda = \begin{cases} (y_q - y_p) \alpha & \text{if } x_q \neq x_p, \\ 3x_p^2 (2y_p)^{-1} & \text{if } x_q = x_p \text{ and } y_p \neq 0, \\ 0 & \text{otherwise,} \end{cases}$$

and an output pair (x_r, y_r) : it is Q if $x_p = 0$; else P if $x_q = 0$; else $(0, 0)$ if $x_q = x_p$ and $y_q = -y_p$; else $x_r = \lambda^2 - x_p - x_q$ and $y_r = \lambda(x_p - x_r) - y_p$. The verifier accepts the row iff the twelve

identities

$$(x_q - x_p)((x_q - x_p)\lambda - (y_q - y_p)) = 0, \quad (C1)$$

$$(1 - (x_q - x_p)\alpha)(2y_p\lambda - 3x_p^2) = 0, \quad (C2)$$

$$x_px_q(x_q - x_p)(\lambda^2 - x_p - x_q - x_r) = 0, \quad (C3a)$$

$$x_px_q(x_q - x_p)(\lambda(x_p - x_r) - y_p - y_r) = 0, \quad (C3b)$$

$$x_px_q(y_q + y_p)(\lambda^2 - x_p - x_q - x_r) = 0, \quad (C3c)$$

$$x_px_q(y_q + y_p)(\lambda(x_p - x_r) - y_p - y_r) = 0, \quad (C3d)$$

$$(1 - x_p\beta)(x_r - x_q) = 0, \quad (C4a)$$

$$(1 - x_p\beta)(y_r - y_q) = 0, \quad (C4b)$$

$$(1 - x_q\gamma)(x_r - x_p) = 0, \quad (C5a)$$

$$(1 - x_q\gamma)(y_r - y_p) = 0, \quad (C5b)$$

$$(1 - (x_q - x_p)\alpha - (y_q + y_p)\delta)x_r = 0, \quad (C6a)$$

$$(1 - (x_q - x_p)\alpha - (y_q + y_p)\delta)y_r = 0 \quad (C6b)$$

hold in $\mathbb{F}_p$. These are, in this order, the twelve constraints of the gate `complete` addition in the `halo2_gadgets` crate (`src/ecc/chip/add.rs`, `create_gate`), each carrying there one further factor, a per-row cell equal to 1 on the rows where the gate applies and 0 elsewhere, which switches the list on or off; the cell assignments are those of `assign_region` in the same file, and the circuit field is the Pallas base field $\mathbb{F}_p$.

Proposition 10.14 (The gate is complete and sound). *Suppose no affine point of $E(\mathbb{F}_p)$ has $x = 0$ and none has $y = 0$ —for $y^2 = x^3 + b$, that b is a non-square and $-b$ a non-cube in $\mathbb{F}_p$. Then for all $P, Q \in E(\mathbb{F}_p)$, encoded as in theorem 10.13:*

1. (completeness) *the prover's cells satisfy (C1)–(C6b), and the output pair encodes $P + Q$;*
2. (soundness) *every $(\lambda, \alpha, \beta, \gamma, \delta, x_r, y_r) \in \mathbb{F}_p^7$ satisfying (C1)–(C6b) has (x_r, y_r) equal to the encoding of $P + Q$.*

This is what complete means for a gate: the twelve identities are satisfiable for every input pair—doublings, inverse pairs, and the identity included—and the output they admit is unique, so they are the graph of the total function $+: E(\mathbb{F}_p) \times E(\mathbb{F}_p) \rightarrow E(\mathbb{F}_p)$, with no case left to a branch outside the constraint system.

Proof. Under the hypothesis an encoded pair has $x = 0$ iff it is O , and two affine points with the same x -coordinate have $y_q = \pm y_p$ with $y_p \neq 0$, so they are equal or inverse and not both. One observation drives every case: for $z \in \mathbb{F}_p$ and any cell value h , the factor $1 - zh$ equals 1 when $z = 0$, whatever h is, while for $z \neq 0$ the honest choice $h = z^{-1}$ makes it 0. A constraint of the shape $(1 - zh)t = 0$ therefore forces $t = 0$ exactly when $z = 0$ and is discharged by the honest hint otherwise. Five cases exhaust the input pairs.

Case $P = O$. Then $x_p = 0$, so (C4a)–(C4b) force $(x_r, y_r) = (x_q, y_q)$, the encoding of $O + Q = Q$; that is soundness. The prover outputs Q by the first clause, and its cells satisfy the rest: (C1) reads $x_q(x_q\lambda - y_q) = 0$ and holds for $\lambda = y_q \text{ inv}_0(x_q)$; (C2) and (C3a)–(C3d) carry a factor x_p or $2y_p\lambda - 3x_p^2$, both zero; (C5a)–(C5b) are discharged by $\gamma = x_q^{-1}$ if $Q \neq O$, and read $(x_r, y_r) = (0, 0) = (x_p, y_p)$ if $Q = O$; and (C6a)–(C6b) have first factor $1 - x_q\alpha - 0$, which is 0 for $\alpha = x_q^{-1}$ when $Q \neq O$, while for $Q = O$ the other factor $x_r = y_r = 0$ vanishes.

Case $Q = O, P \neq O$. Now $x_q = 0 \neq x_p$, so (C5a)–(C5b) force $(x_r, y_r) = (x_p, y_p)$, the encoding of P : soundness. The prover outputs P by the second clause, with $\alpha = (-x_p)^{-1}$ and $\lambda = (-y_p)\alpha = y_p/x_p$; then (C1) reads $(-x_p)(-x_p\lambda + y_p) = 0$ and holds; (C2) and (C6a)–(C6b) carry the factor $1 - (x_q - x_p)\alpha = 0$, the latter with $\delta = 0$; (C3a)–(C3d) carry $x_q = 0$; and (C4a)–(C4b) are discharged by $\beta = x_p^{-1}$.

Case P, Q affine with $x_q \neq x_p$. Constraint (C1) forces λ to be the chord slope $(y_q - y_p)/(x_q - x_p)$, and since $x_p x_q (x_q - x_p) \neq 0$, (C3a)–(C3b) force (x_r, y_r) to be the chord formula (12)—the encoding of $P + Q$, which is affine because $x_p \neq x_q$. That is soundness. The prover's λ is that slope and its output that formula, by the fourth clause; (C3c)–(C3d) contain the same two vanishing factors; (C2), (C4a)–(C4b), (C5a)–(C5b) are discharged by the honest α, β, γ ; and (C6a)–(C6b) have first factor $1 - 1 - 0 = 0$, since $\delta = 0$.

Case $Q = -P, P$ affine. Here $x_q = x_p$ and $y_q + y_p = 0$, so both hint-bearing products in (C6a)–(C6b) are 0 whatever α and δ are, and the constraints force $x_r = y_r = 0$: the encoding of $P + (-P) = O$, which is soundness. The prover outputs $(0, 0)$ by the third clause; (C1) and (C3a)–(C3b) carry $x_q - x_p = 0$, and (C3c)–(C3d) carry $y_q + y_p = 0$; (C2), whose first factor is now 1, needs $2y_p\lambda = 3x_p^2$, which the honest $\lambda = 3x_p^2(2y_p)^{-1}$ supplies ($y_p \neq 0$); and (C4a)–(C5b) are discharged by $\beta = \gamma = x_p^{-1}$.

Case $Q = P$ affine. Again $x_q = x_p$ fixes $(x_q - x_p)\alpha = 0$, so (C2) forces λ to be the tangent slope $3x_p^2/(2y_p)$, and since $x_p x_q (y_q + y_p) = 2x_p^2 y_p \neq 0$, (C3c)–(C3d) force (x_r, y_r) to be the doubling formula (13)—the encoding of $[2]P$, affine because $y_p \neq 0$. That is soundness. The prover's λ is that slope and its output that formula, by the fourth clause; (C1) and (C3a)–(C3b) carry $x_q - x_p = 0$; (C4a)–(C5b) are discharged by $\beta = \gamma = x_p^{-1}$; and (C6a)–(C6b) have first factor $1 - 0 - 2y_p\delta = 0$ for the honest $\delta = (2y_p)^{-1}$. $\square$

Example 10.15 (The gate on the toy curve). Pallas satisfies both hypotheses of theorem 10.14: 5 is a non-square in $\mathbb{F}_p$ (theorem 7.16) and -5 is not a cube (theorem 10.26). The toy curve $y^2 = x^3 + 5$ over $\mathbb{F}_{11}$ of theorem 10.38 satisfies neither— $5 = 4^2$ gives the points $(0, \pm 4)$, and $(8, 0)$ has order 2—which makes it the right place to watch the gate work and to see the hypotheses earn their keep. For each input pair considered below, a script determines every output pair (x_r, y_r) that *some* assignment of the five auxiliary cells in $\mathbb{F}_{11}$ makes the twelve identities accept—the search over $\alpha, \beta, \gamma, \delta$ factored by which identities mention each: β only (C4), γ only (C5), α and δ only (C2) and (C6)—so soundness is checked against all provers, not only the honest one. On the nine points $O, (4, \pm 5), (5, \pm 3), (6, \pm 1), (10, \pm 2)$ with $x, y \neq 0$, all 81 ordered pairs—eight doublings, eight inverse pairs, seventeen involving O —admit exactly one output, the encoding of the true sum. Three of them, worked: for $[2](5, 3)$ the honest cells are $\lambda = 3 \cdot 5^2 \cdot 6^{-1} = 9 \cdot 2 = 7$ and $\delta = 6^{-1} = 2$, giving $(x_r, y_r) = (49 - 10, 7(5 - 6) - 3) = (6, 1)$, the value of (13); for the inverse pair $(5, 3) + (5, 8)$ the prover supplies the same $\lambda = 7$ but $\delta = \text{inv}_0(11) = 0$, and (C6a)–(C6b) force $(0, 0)$; for $O + (5, 3)$ the cells are $\alpha = 5^{-1} = 9, \lambda = 3 \cdot 9 = 5, \beta = 0$, and (C4a)–(C4b) force $(5, 3)$. Off the admissible set the gate fails as the proof predicts. With $P = (0, 4)$ and $Q = (5, 3)$ the identities admit only the output $(5, 3)$ —the coordinate $x_p = 0$ is read as O —whereas $P + Q = (10, 9)$ (theorem 10.38); the pair $(0, 4) + (0, 7)$ admits no output at all, (C4b) demanding $y_r = 7$ and (C5b) $y_r = 4$; and $(8, 0) + (8, 0)$ admits none either, since (C2) reduces to $3 \cdot 8^2 = 0$, false in $\mathbb{F}_{11}$. On Pallas none of the three inputs exists, and the gate is complete.

10.5 Identity, inverses, commutativity, and associativity

The destination of the whole construction is the following theorem.

Theorem 10.16 (Group law). *Let E be an elliptic curve over a field K . The set $E(K)$ with the chord-and-tangent operation $+$ of theorem 10.7 is an abelian group with identity O .*

Closure is theorem 10.11. The next proposition supplies every remaining axiom except one.

Proposition 10.17 (Identity, inverses, commutativity). *The operation $+$ on $E(K)$ has the following properties.*

1. *The point O is a two-sided identity: $P + O = O + P = P$ for all P .*
2. *Every P has the two-sided inverse $-P$ of theorem 10.8: $P + (-P) = O$.*
3. *The operation is commutative: $P + Q = Q + P$.*

Proof. (1) is the first clause of theorem 10.7. (2): for $P = (x, y)$ with $y \neq 0$, the line through P and $-P = (x, -y)$ is vertical, so its third point is O and $P + (-P) = O$ by the vertical-line case; for $y = 0$ one has $P = -P$ and $[2]P = O$ directly by the vertical-tangent case; and $O + (-O) = O + O = O$. (3): the chord through P and Q does not depend on the order in which the two points are named, and neither does the tangent in the doubling case; algebraically, the formulas (12) are symmetric under the swap $(x_1, y_1) \leftrightarrow (x_2, y_2)$ —the slope λ is unchanged, and $x_3 = \lambda^2 - x_1 - x_2$ is symmetric in x_1, x_2 . $\square$

The one axiom not yet established is *associativity*, $(P + Q) + R = P + (Q + R)$, and it is the deep and surprising part of the theory: the definition gives no reason why reflecting third intersection points should be an associative operation, and a brute-force verification through the explicit formulas splinters into a large number of coordinate cases, each demanding lengthy rational-function identities. We sketch the conceptual proof, which exhibits the reason associativity holds.

Proof of associativity (sketch). The key tool is the classical *Cayley–Bacharach theorem* for plane cubics: if two cubics with no common component meet in nine points counted with intersection multiplicity, then any third cubic passing through eight of them with the required multiplicities also passes through the ninth. To prove $(P + Q) + R = P + (Q + R)$, one writes both sides via the chord-and-tangent construction and arranges the relevant intersections among the curve E and two degenerate cubics, each a product of three suitably chosen lines. Cayley–Bacharach forces the two candidate sums to coincide; its intersection-multiplicity formulation includes tangencies and coincident points. Translating “reflect across the x -axis” into “the three points of any line sum to O ” renders the bookkeeping uniform.

A complete, case-free treatment identifies $E(K)$ with the degree-zero part $\text{Pic}^0(E)$ of the divisor class group of the curve via the map $P \mapsto [P] - [O]$, a bijection that intertwines the two operations; the group law on $\text{Pic}^0(E)$ is manifestly associative, because it is inherited from addition of divisors modulo principal divisors. Either route yields a genuine proof; both lie beyond the elementary scope of this volume, so we take associativity as established. $\square$

Remark 10.18 (The notation $[n]P$). For $n \in \mathbb{Z}$ and $P \in E(K)$, we write $[n]P$ for the n -fold sum $P + \cdots + P$, with $[0]P = O$ and $[-n]P = -([n]P)$. This makes the abelian group $E(K)$ a $\mathbb{Z}$ -module: integers act on points. The map $n \mapsto [n]P$ is the *scalar multiplication* of theorem 10.20—not any coordinate-wise product.

10.6 Projective and Jacobian coordinates

The affine formulas (12) and (13) each require a field *inversion*, by $x_2 - x_1$ or $2y_1$. In a large prime field an inversion costs roughly as much as dozens of multiplications, whether computed by the extended Euclidean algorithm (theorem 6.1) or as z^{p-2} by Fermat (theorem 6.4). When a long chain of group operations arises, as in the scalar multiplication of the next subsection, it pays to defer all inversions to the very end by carrying points in a redundant coordinate system whose group law uses only additions, subtractions, and multiplications.

Remark 10.19 (Inversion-free coordinate systems). Two redundant systems serve this purpose. In *(homogeneous) projective coordinates*, a triple $[X : Y : Z]$ with $Z \neq 0$ represents the affine point $(X/Z, Y/Z)$ on the projective curve (11), with $O = [0 : 1 : 0]$. In *Jacobian coordinates*, a triple $(X : Y : Z)$ with $Z \neq 0$ represents $(X/Z^2, Y/Z^3)$, under the equivalence $(X, Y, Z) \sim (\lambda^2 X, \lambda^3 Y, \lambda Z)$; the curve equation becomes $Y^2 = X^3 + aXZ^4 + bZ^6$. In either system the addition and doubling formulas can be rewritten with no divisions at all, at the cost of more multiplications per operation; a long chain of group operations then defers every inversion to a single final recovery of affine coordinates—compute Z^{-1} once, then the needed powers of it by multiplication. Jacobian coordinates usually yield the faster doubling and are the standard choice in cryptographic libraries, including the implementation Halo 2 uses: the `pasta_curves` crate carries its curve points in Jacobian coordinates. For curves $y^2 = x^3 + b$ with $a = 0$, the term aZ^4 in the Jacobian doubling formulas vanishes, making doubling especially cheap—one reason the Pasta curves take $a = 0$.

10.7 Scalar multiplication and double-and-add

The fundamental operation of elliptic-curve cryptography is *scalar multiplication*: given a point P and an integer $n \geq 0$, compute

$$[n]P = \underbrace{P + P + \cdots + P}_{n \text{ times}}.$$

Computing this by $n - 1$ successive additions is infeasible when n has hundreds of bits. Instead one uses the binary expansion of n , exactly as in the square-and-multiply technique of theorem 6.2; as theorem 6.3 promised, the additive rendition of that algorithm is the following.

Definition 10.20 (Double-and-add). If $n = 0$, return O . Otherwise write $n = \sum_{i=0}^{k-1} n_i 2^i$ with digits $n_i \in \{0, 1\}$ and top digit $n_{k-1} = 1$. The *left-to-right double-and-add* algorithm computes $[n]P$ as follows: set $R \leftarrow P$; for $i = k - 2$ down to 0 , set $R \leftarrow [2]R$, and if $n_i = 1$ set $R \leftarrow R + P$; return R .

Proposition 10.21. For $n \geq 1$, *double-and-add* returns $[n]P$ using exactly $k - 1$ doublings and at most $k - 1$ additions, where $k = \lfloor \log_2 n \rfloor + 1$ is the bit length of n . For $n = 0$, it returns O without a group operation.

Proof. The case $n = 0$ is immediate. Let $n \geq 1$, and let R_j denote the value of R after bits $n_{k-1}, \dots, n_j$ have been incorporated. We claim $R_j = \lfloor n/2^j \rfloor P$, by downward induction on j . At initialization, $R_{k-1} = P = \lfloor n/2^{k-1} \rfloor P$ because the top bit is 1. Assuming $R_{j+1} = \lfloor n/2^{j+1} \rfloor P$, the next iteration computes

$$[2]R_{j+1} + [n_j]P = [2\lfloor n/2^{j+1} \rfloor + n_j]P = \lfloor n/2^j \rfloor P,$$

where the last step is the digit identity $2\lfloor n/2^{j+1} \rfloor + n_j = \lfloor n/2^j \rfloor$: stripping the last bit of $\lfloor n/2^j \rfloor$ and re-appending it recovers the number. At $j = 0$ the invariant reads $R_0 = [n]P$, which is what the algorithm returns. For the counts, the loop has exactly $k - 1$ iterations, each performing one doubling and at most one addition. $\square$

Scalar multiplication by an n -bit scalar therefore costs $O(n)$ group operations— $O(n)$ field operations up to constant factors, or $O(n \log^2 p)$ bit operations with schoolbook field arithmetic. This efficiency of the *forward* map, contrasted with the apparent hardness of the inverse problem (§10.11), is the foundation of the cryptography.

10.8 The group structure of $E(\mathbb{F}_p)$

Now specialise $K = \mathbb{F}_p$, the prime field with p elements (p prime, $p > 3$). Since $\mathbb{F}_p$ is finite, so is $E(\mathbb{F}_p)$: each of the p values of x admits at most two values of y (theorem 5.18, applied to $y^2 - f(x)$ as a polynomial in y), so with O there are at most $2p + 1$ points. By theorem 10.16, the set $E(\mathbb{F}_p)$ is thus a *finite abelian group*. One of the central theorems of the subject governs its order.

Theorem 10.22 (Hasse). *Let E be an elliptic curve over $\mathbb{F}_p$. Write*

$$\#E(\mathbb{F}_p) = p + 1 - t,$$

which defines the integer t , called the trace of Frobenius. Then

$$|t| \leq 2\sqrt{p};$$

equivalently, $\#E(\mathbb{F}_p)$ lies in the Hasse interval $[p + 1 - 2\sqrt{p}, p + 1 + 2\sqrt{p}]$.

Proof (sketch). The count has an exact character-sum expression. For fixed x , the number of $y \in \mathbb{F}_p$ with $y^2 = f(x)$ is $1 + \left(\frac{f(x)}{p}\right)$ when $f(x) \neq 0$ —two solutions for a nonzero square, none for a non-square (theorem 7.12)—and exactly $1 = 1 + \left(\frac{0}{p}\right)$ when $f(x) = 0$, using the Legendre symbol of theorem 7.14 with the convention $\left(\frac{0}{p}\right) = 0$. Summing over x ,

$$\#E^{\text{aff}}(\mathbb{F}_p) = \sum_{x \in \mathbb{F}_p} \left(1 + \left(\frac{x^3 + ax + b}{p}\right)\right) = p + \sum_{x \in \mathbb{F}_p} \left(\frac{x^3 + ax + b}{p}\right),$$

so $t = -\sum_x \left(\frac{x^3 + ax + b}{p}\right)$. Heuristically, as x ranges over $\mathbb{F}_p$ the value $f(x)$ is a nonzero square about half the time (two points), a non-square about half the time (none), and zero occasionally (one point); the p summands ± 1 behave like a random walk, and Hasse's bound $|t| \leq 2\sqrt{p}$ is exactly square-root cancellation in this character sum.

The conceptual proof realises t as the trace of the p -power *Frobenius endomorphism* $\pi: (x, y) \mapsto (x^p, y^p)$ of the curve—the curve-level shadow of the field automorphism of theorem 6.14. One shows that π satisfies the characteristic equation $\pi^2 - [t]\pi + [p] = 0$ in the endomorphism ring of E , and that the associated quadratic form (the degree) is positive definite; the discriminant condition $t^2 - 4p \leq 0$ for the complex roots of the characteristic polynomial then yields $|t| \leq 2\sqrt{p}$. The full argument requires the theory of isogenies and the Weil pairing, beyond the present scope. $\square$

Hasse's theorem says the order of $E(\mathbb{F}_p)$ is *very close* to p : the deviation from $p + 1$ is at most $2\sqrt{p}$, a relative error of order $p^{-1/2}$. For cryptography this is crucial twice over. It guarantees with no brute-force count that the group is large—about p elements, so about 255 bits for the Pasta curves—and it anchors the point-counting algorithms (Schoof–Elkies–Atkin) that compute the *exact* order in polynomial time, which is how curve designers certify their group orders.

For the cryptographic application one wants $E(\mathbb{F}_p)$ to contain a large subgroup of *prime* order. The general structure theory of $E(\mathbb{F}_p)$ as an abelian group is not needed in this series, because the Pasta curves are designed so that $\#E(\mathbb{F}_p)$ is itself prime: a group of prime order is cyclic with no proper nontrivial subgroups (theorem 3.34), so $E(\mathbb{F}_p)$ is cyclic of prime order—the optimal case.

10.9 Torsion, the cofactor, and prime-order subgroups

Definition 10.23 (Order of a point and torsion). The *order* of a point $P \in E(\mathbb{F}_p)$ is its order as a group element (theorem 3.11): the least integer $m \geq 1$ with $[m]P = O$. It exists and divides $\#E(\mathbb{F}_p)$, by Lagrange’s theorem (theorem 3.29 via theorem 3.30). For an integer $m \geq 1$, the *m-torsion subgroup* is

$$E[m] = \{P \in E(\overline{\mathbb{F}_p}) : [m]P = O\},$$

the points over the algebraic closure killed by m ; its rational part is $E(\mathbb{F}_p)[m] = E[m] \cap E(\mathbb{F}_p)$.

Definition 10.24 (Cofactor). Suppose $\#E(\mathbb{F}_p) = h \cdot r$, where r is the (large) prime to be used and h is the remaining factor. The integer h is the *cofactor*, and r is the order of the *prime-order subgroup*: a point P of order r generates a cyclic subgroup $\langle P \rangle \cong \mathbb{Z}/r\mathbb{Z}$, and this subgroup is where the cryptography lives.

A nontrivial cofactor $h > 1$ is undesirable: it admits *small-subgroup* points, of order dividing h , which can leak information or break protocol assumptions—the small-subgroup attacks of theorem 7.25. Implementations therefore either *clear the cofactor*, multiplying incoming points by h , or use prime-order curves with $h = 1$. The Pasta curves have cofactor 1: the orders $\#E_p(\mathbb{F}_p) = q$ and $\#E_q(\mathbb{F}_q) = p$ are themselves prime, so $E_p(\mathbb{F}_p) \cong \mathbb{Z}/q\mathbb{Z}$ and $E_q(\mathbb{F}_q) \cong \mathbb{Z}/p\mathbb{Z}$ are cyclic of prime order and no cofactor clearing is ever needed. This is one of the principal design advantages of the Pasta cycle for proof systems.

Proposition 10.25 (Two-torsion). *The affine points of order 2 in $E(\mathbb{F}_p)$ are exactly the points $(x_0, 0)$ with $f(x_0) = x_0^3 + ax_0 + b = 0$. Hence $E(\mathbb{F}_p)$ has 0, 1, or 3 points of order 2 according as f has 0, 1, or 3 roots in $\mathbb{F}_p$; and $\#E(\mathbb{F}_p)$ is even if and only if f has a root in $\mathbb{F}_p$.*

Proof. A point $P = (x, y)$ has order 2 iff $P = -P$ and $P \neq O$, i.e. $y = -y$, i.e. $y = 0$ (since $\text{char}(\mathbb{F}_p) \neq 2$); the curve equation then forces $f(x) = 0$. Conversely each root x_0 of f in $\mathbb{F}_p$ gives the order-2 point $(x_0, 0)$.

Next, the number N of roots of f in $\mathbb{F}_p$ can only be 0, 1, or 3: two roots force a third. Suppose f has distinct roots $r_1 \neq r_2$ in $\mathbb{F}_p$. The factor theorem (theorem 5.17) gives $f = (x - r_1)h$; evaluating at r_2 yields $0 = (r_2 - r_1)h(r_2)$, so $h(r_2) = 0$, and a second application factors the monic cubic completely, $f = (x - r_1)(x - r_2)(x - r_3)$ with $r_3 \in \mathbb{F}_p$ —explicitly $r_3 = -r_1 - r_2$ by Vieta, the depressed cubic having no x^2 term. Nonsingularity says f has no repeated root even over $\overline{\mathbb{F}_p}$ (theorem 10.4(2)), so $r_3 \notin \{r_1, r_2\}$, and f has exactly three roots (theorem 5.18 caps the count). Hence $N \in \{0, 1, 3\}$, matching the trichotomy of the statement. Together with O the order-2 points form the rational 2-torsion subgroup $E(\mathbb{F}_p)[2]$, which is accordingly trivial, $\mathbb{Z}/2\mathbb{Z}$, or $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.

For the parity claim, consider negation $P \mapsto -P$, an *involution*—a self-inverse map—of the finite set $E(\mathbb{F}_p)$. Its fixed points are exactly O and the points $(x_0, 0)$; every other point pairs off with its distinct negative into a 2-element orbit. Counting the set by orbits,

$$\#E(\mathbb{F}_p) \equiv 1 + N \pmod{2},$$

where $N \in \{0, 1, 3\}$ is the number of roots of f in $\mathbb{F}_p$. Hence $\#E(\mathbb{F}_p)$ is even iff N is odd, i.e. iff $N \geq 1$, i.e. iff f has a root in $\mathbb{F}_p$. (The device is worth remembering: an involution reads off a group’s cardinality modulo 2 from its fixed points alone.) $\square$

Remark 10.26 (No two-torsion on the Pasta curves). For $y^2 = x^3 + 5$ over $\mathbb{F}_p$, the curve has a point of order 2 iff $x^3 + 5 = 0$ has a solution, i.e. iff -5 is a cube in $\mathbb{F}_p$. For the Pasta primes the order $\#E_p(\mathbb{F}_p) = q$ is odd (it is prime), so there are *no* rational 2-torsion points; consistently, -5 is not a cube in $\mathbb{F}_p$.

10.10 Base fields, scalar fields, and the Pasta cycle

Step back from the geometry and ask what a machine actually does when it works on $E(\mathbb{F}_p)$. Every operation of the group law—the slope $\lambda = (y_2 - y_1)(x_2 - x_1)^{-1}$, the squarings, and the inversions needed to add and to double—is arithmetic in $\mathbb{F}_p$, performed with exactly the prime-field algorithms of section 6.1 (extended-Euclid or Fermat inversion). The field $\mathbb{F}_p$ is called the *base field* (or coordinate field) of the curve: it is where the coordinates live.

Scalars live elsewhere. By theorem 10.22 the group $E(\mathbb{F}_p)$ is a finite abelian group of order $N = \#E(\mathbb{F}_p)$ close to p ; for cryptography one selects a curve whose order has a large prime factor r , and keys and signatures reside in the subgroup of order r (theorem 10.24). A scalar multiplication $P \mapsto [s]P$ on a point of order r depends only on $s \bmod r$, since $[r]P = O$ (theorem 3.12); scalars therefore live modulo r —and since r is prime, they form a *field* (theorem 4.26).

Definition 10.27 (Base field and scalar field). For an elliptic curve used in cryptography, with a prime-order subgroup of order r :

- the *base field* $\mathbb{F}_p$ is the field over which the curve is defined and in which point coordinates lie;
- the *scalar field* $\mathbb{F}_r = \mathbb{Z}/r\mathbb{Z}$ is the field of scalars by which one multiplies points, i.e. the integers modulo the subgroup order r .

Both are prime fields, but in general $p \neq r$.

Remark 10.28 (Two fields at once, and the Pasta cycle). A typical elliptic-curve operation manipulates two distinct prime fields simultaneously: the exponent arithmetic runs in $\mathbb{F}_r$ while the coordinate arithmetic runs in $\mathbb{F}_p$. Halo 2’s Pallas and Vesta curves (the “Pasta” pair) exploit this deliberately. They form a *cycle of curves*: the base field of one is the scalar field of the other and vice versa,

$$\#E_p(\mathbb{F}_p) = q, \quad \#E_q(\mathbb{F}_q) = p,$$

so the scalar field of Pallas is $\mathbb{F}_q$, the base field of Vesta, and symmetrically (§5.4.9.6 of the Zcash protocol specification). This lets a proof system express the verifier of one curve’s arithmetic *natively* in the field of the other, enabling the recursive proof composition at the heart of Halo 2. Moreover both primes are chosen $\equiv 1 \pmod{2^{32}}$, so by theorem 6.22 each field contains a multiplicative subgroup of order 2^{32} , furnishing the high-order roots of unity that the number-theoretic transform of section 9 needs for fast polynomial multiplication.

10.11 The elliptic-curve discrete logarithm problem

The section’s opening problem now returns in its proper habitat. The discrete logarithm problem of theorem 7.21 was stated for an abstract cyclic group; the elliptic-curve instance reads as follows, in additive notation.

Definition 10.29 (ECDLP). Let $P \in E(\mathbb{F}_p)$ be a point of large prime order r , generating $G = \langle P \rangle$. The *elliptic-curve discrete logarithm problem* (ECDLP) is: given P and a point $Q \in G$, find the unique integer $n \in \{0, 1, \dots, r-1\}$ with $Q = [n]P$. One writes $n = \log_P Q$.

The map $n \mapsto [n]P$ is a group isomorphism $\mathbb{Z}/r\mathbb{Z} \rightarrow G$ (theorem 3.43), so a discrete logarithm always exists and is unique; the entire content of the problem is *computational*. The forward map is fast by double-and-add (theorem 10.21); no efficient algorithm is known to invert it on a well-chosen curve. What is known is the generic pair of attacks, which transfer verbatim from theorem 7.22 to the additive setting.

Proposition 10.30 (Square-root attacks). *Baby-step giant-step solves the ECDLP in a group of prime order r in $O(\sqrt{r})$ group operations and $O(\sqrt{r})$ space. Pollard’s rho method matches the $O(\sqrt{r})$ bound in expectation with only $O(1)$ space, under a modelling assumption on its pseudo-random walk that is stated explicitly in the proof and is unproved for every concrete walk.*

Proof. For baby-step giant-step, set $m = \lceil \sqrt{r} \rceil$ and write the unknown logarithm as $n = im + j$ with $0 \leq i, j < m$; since $m^2 \geq r$, every $n \in \{0, \dots, r-1\}$ has such a decomposition. Tabulate the *baby steps* $[j]P$ for $j = 0, \dots, m-1$ (a table of m points, built with $m-1$ additions), compute $S = [m]P$ once, and walk the *giant steps* $Q - [i]S$ for $i = 0, 1, 2, \dots$, testing each against the table. At $i = \lfloor n/m \rfloor$ the walk hits $Q - [im]P = [n - im]P = [j]P$, a table entry; the collision reveals i and j , hence $n = im + j$. Both the table and the walk cost $O(m) = O(\sqrt{r})$ group operations, and the table holds $O(\sqrt{r})$ points.

Pollard’s rho eliminates the table, at the price of an explicitly heuristic analysis. The algorithm takes a pseudo-random walk on G whose every position is maintained in the form $[a]P + [b]Q$ with known coefficients a, b , started from such a combination with random coefficients; the walk’s next step depends only on the current point, so once the walk revisits a point it cycles forever. A repeat is therefore detected with $O(1)$ memory: run a second copy of the walk at double speed, and the two copies meet at a common point inside the cycle within a constant factor of the first repeat time. *Modelling assumption:* the successive positions of the walk behave like independent uniform samples from G . No proof of this is known for any concrete walk—the assumption is what “well-designed” means, and experiment supports it—and the rest of the argument is conditional on it. Under the assumption, the first k positions are pairwise distinct with probability at most $e^{-k(k-1)/(2r)}$ by the birthday bound—the forward reference theorem 11.18 is quantitative machinery from the closing section—so the number T of steps to the first repeat satisfies $\Pr[T > k] \leq e^{-k(k-1)/(2r)}$. The tail-sum formula (theorem 11.33) converts the tail bound into an expectation bound: with $m = \lceil \sqrt{r} \rceil$ and $r \geq 4$,

$$\mathbb{E}[T] = \sum_{k \geq 1} \Pr[T \geq k] \leq m \sum_{j \geq 0} \Pr[T > jm] \leq m \left(1 + \sum_{j \geq 1} e^{-j^2/4}\right) = O(\sqrt{r}),$$

where the middle step groups the indices k into blocks of length m , bounding each block by its first term, and the last uses $jm(jm-1) \geq j^2 m(m-1) \geq j^2(r-\sqrt{r}) \geq j^2 r/2$ for $j \geq 1$, so that $\Pr[T > jm] \leq e^{-j^2/4} \leq e^{-j/4}$, a convergent geometric tail. A repeat presents one group element with two representations, $[a]P + [b]Q = [a']P + [b']Q$. If $b' \not\equiv b \pmod{r}$, this rearranges to $[a-a']P = [b'-b]Q = [(b'-b)n]P$, whence

$$n \equiv (a-a')(b'-b)^{-1} \pmod{r},$$

the inverse existing because r is prime. If instead $b' \equiv b$, then $[a]P = [a']P$ forces $a \equiv a'$: the two representations coincide, the collision reveals nothing, and the walk is restarted with fresh random starting coefficients; that such degenerate collisions are rare is a further part of the heuristic, borne out in practice. Both algorithms run in time exponential in the bit length $\log_2 r$. $\square$

Remark 10.31 (Best known attacks on well-chosen curves). For well-chosen curves, no unrestricted classical attack faster than $O(\sqrt{r})$ is known; this is not an unconditional lower bound for all classical algorithms. In the classical generic-group model, however, Shoup’s matching lower bound is proved (theorem 7.22). For $r \approx 2^{254}$ —the 255-bit Pasta orders—this is about 2^{127} operations, infeasible. Curves with special structure can be weaker, and the known weak classes are avoided by design. *Anomalous curves*, those with $\#E(\mathbb{F}_p) = p$, admit a polynomial-time attack through the formal group. The MOV/Frey–Rück attack transfers the ECDLP into a finite-field discrete logarithm when the *embedding degree*—the least $k \geq 1$ with $r \mid p^k - 1$, i.e. the degree (theorem 8.4; the dimension of the extension over its base) of the smallest extension of $\mathbb{F}_p$ whose multiplicative group contains a subgroup of order r —is small, where index calculus

applies. Cryptographically secure curves, the Pasta curves included, are chosen with a large prime subgroup order, a large embedding degree, and $\#E(\mathbb{F}_p) \neq p$, defeating every known subexponential attack.

Remark 10.32 (The index-calculus contrast). The contrast with the multiplicative groups $\mathbb{F}_p^\times$ is the economic heart of the subject. In $\mathbb{F}_p^\times$ the index-calculus method solves the discrete logarithm in subexponential time $L_p[1/3]$ —shorthand for $2^{\Theta(\lambda^{1/3}(\log \lambda)^{2/3})}$ operations in the bit length $\lambda = \log_2 p$, a notation defined precisely in theorem 11.60 of the closing section—exploiting the integer arithmetic behind the residues (theorem 7.22); those groups must therefore be made very large. On a well-chosen elliptic curve no index calculus is known—the points offer no analogue of “factoring into small primes”—so the generic $O(\sqrt{r})$ bound stands, and much smaller groups suffice: about 256 bits for 128-bit security, versus thousands of bits for $\mathbb{F}_p^\times$. Smaller groups mean faster arithmetic and shorter keys, which is why elliptic curves dominate modern protocols, including Halo 2.

10.12 The j -invariant and the GLV endomorphism

Two final pieces of structure round out the theory: a single quantity that classifies curves up to isomorphism, and an “extra symmetry” available exactly when that quantity vanishes—which the Pasta curves arrange on purpose.

Definition 10.33 (j -invariant). The j -invariant of the curve $y^2 = x^3 + ax + b$ (with $4a^3 + 27b^2 \neq 0$) is

$$j = 1728 \frac{4a^3}{4a^3 + 27b^2} \in K,$$

where $1728 = 12^3$ is a normalising constant of historical origin.

Proposition 10.34. Two elliptic curves over $\bar{K}$ are isomorphic over $\bar{K}$ if and only if they have the same j -invariant. The admissible coordinate changes preserving short Weierstrass form are $(x, y) \mapsto (u^2x, u^3y)$ for $u \in \bar{K}^\times$, under which $(a, b) \mapsto (u^4a, u^6b)$; the combination j is invariant.

Proof (sketch). A direct computation shows that the only isomorphisms between short Weierstrass curves fixing O are the scalings $(x, y) \mapsto (u^2x, u^3y)$. Substituting $x = u^{-2}x'$, $y = u^{-3}y'$ into $y^2 = x^3 + ax + b$ and clearing u^6 gives $y'^2 = x'^3 + u^4ax' + u^6b$, which is the stated action on coefficients. Under it, numerator and denominator of $4a^3/(4a^3 + 27b^2)$ both scale by u^{12} , so j is unchanged. Conversely, given two curves with equal j -invariants one solves for a suitable u transforming one coefficient pair into the other, with separate easy cases $j = 0$ (both a vanish) and $j = 1728$ (both b vanish). $\square$

Remark 10.35 (The Pasta curves have $j = 0$). For $y^2 = x^3 + b$ one has $a = 0$, so $j = 0$; both Pasta curves have $j = 0$. Curves with $j = 0$ are special: they admit an extra automorphism of order 3, which underlies the efficient endomorphism described next. (At the other extreme, $b = 0$ gives $j = 1728$ and an order-4 automorphism.) The vanishing $j = 0$ is deliberate in the Pasta design: it is what makes the Gallant–Lambert–Vanstone (GLV) speedup available, roughly halving the doubling count in scalar multiplication.

Proposition 10.36 (The GLV endomorphism for $y^2 = x^3 + b$). Suppose $\mathbb{F}_p$ contains a primitive cube root of unity ω , i.e. $p \equiv 1 \pmod{3}$ (theorem 6.22), so $\omega \in \mathbb{F}_p$ satisfies $\omega^2 + \omega + 1 = 0$. On

$E : y^2 = x^3 + b$, the map

$$\phi(x, y) = (\omega x, y), \quad \phi(O) = O,$$

is a group endomorphism of $E(\mathbb{F}_p)$. Let $G = \langle P \rangle$ be a subgroup of prime order r with $r \equiv 1 \pmod{3}$ and $\phi(G) \subseteq G$. Then ϕ acts on G as multiplication by a scalar: there is $\lambda \in \mathbb{Z}/r\mathbb{Z}$ with $\lambda^2 + \lambda + 1 \equiv 0 \pmod{r}$ and

$$\phi(Q) = [\lambda]Q \quad \text{for all } Q \in G.$$

For the Pasta curves the hypothesis $\phi(G) \subseteq G$ is automatic, since $E(\mathbb{F}_p)$ itself is the prime-order group G ; and $r \equiv 1 \pmod{3}$ holds for both orders.

Proof. The map lands on the curve. If $y^2 = x^3 + b$ then $y^2 = (\omega x)^3 + b$, because $\omega^3 = 1$; so $(\omega x, y) \in E(\mathbb{F}_p)$.

The map is a homomorphism. We check $\phi(P_1 + P_2) = \phi(P_1) + \phi(P_2)$ against the case analysis of §10.4. The O cases are trivial, and $\phi(-P) = -\phi(P)$ since negation touches only y ; in particular the inverse case $\phi(P) + \phi(-P) = O = \phi(O)$ is respected, and ϕ preserves the case split, since it preserves equality of x -coordinates. For a chord $(x_1 \neq x_2)$, the slope transforms as

$$\lambda' = \frac{y_2 - y_1}{\omega x_2 - \omega x_1} = \omega^{-1} \lambda = \omega^2 \lambda,$$

using $\omega^{-1} = \omega^2$. Then, by (12) and $\omega^4 = \omega$,

$$x'_3 = \lambda'^2 - \omega x_1 - \omega x_2 = \omega^4 \lambda^2 - \omega x_1 - \omega x_2 = \omega (\lambda^2 - x_1 - x_2) = \omega x_3,$$

$$y'_3 = \lambda'(\omega x_1 - x'_3) - y_1 = \omega^2 \lambda \cdot \omega(x_1 - x_3) - y_1 = \lambda(x_1 - x_3) - y_1 = y_3,$$

so $\phi(P_1) + \phi(P_2) = (\omega x_3, y_3) = \phi(P_1 + P_2)$. The tangent case runs identically with $\lambda' = 3(\omega x_1)^2 / (2y_1) = \omega^2 \lambda$ in (13).

The relation $\phi^2 + \phi + \text{id} = 0$. Let $P = (x, y)$ be affine; we show $P + \phi(P) + \phi^2(P) = O$ directly from the formulas of §10.4.

First let $x \neq 0$. The points $P = (x, y)$, $\phi(P) = (\omega x, y)$, $\phi^2(P) = (\omega^2 x, y)$ share their y -coordinate and have pairwise distinct x -coordinates, since $1, \omega, \omega^2$ are distinct and $x \neq 0$. The chord formula (12) applies to $P + \phi(P)$, with slope $\lambda = (y - y) / (\omega x - x) = 0$:

$$x_3 = 0 - x - \omega x = -(1 + \omega)x = \omega^2 x, \quad y_3 = 0 \cdot (x - x_3) - y = -y,$$

using $1 + \omega + \omega^2 = 0$. Hence $P + \phi(P) = (\omega^2 x, -y) = -\phi^2(P)$ (theorem 10.8), and $P + \phi(P) + \phi^2(P) = O$.

If instead $x = 0$, the three points coincide: $\phi(P) = \phi^2(P) = P$. Here $y \neq 0$, since $x = y = 0$ on the curve would force $b = 0$, which nonsingularity $27b^2 \neq 0$ forbids; so the doubling formula (13) applies, with $a = 0$:

$$\lambda = \frac{3 \cdot 0^2 + 0}{2y} = 0, \quad x_3 = 0 - 2 \cdot 0 = 0, \quad y_3 = 0 \cdot (0 - 0) - y = -y,$$

so $[2]P = (0, -y) = -P$, hence $P + \phi(P) + \phi^2(P) = [3]P = O$ in this case too. (The three points $P, \phi(P), \phi^2(P)$ are exactly the intersections of the horizontal line through P with the curve, the roots of $X^3 = y^2 - b$ being $x, \omega x, \omega^2 x$; the computation just performed is the “three points of a line sum to O ” picture made rigorous, including the triple intersection at $x = 0$.)

With $\phi(O) = O$ the identity holds on all of $E(\mathbb{F}_p)$; as an identity of endomorphisms, $\phi^2 + \phi + \text{id} = 0$. Incidentally $\phi^3 = \text{id}$, since $\omega^3 = 1$; and whenever $\#E(\mathbb{F}_p) > 3$ —as always at cryptographic sizes, the Pasta curves included— ϕ has order exactly 3, because at most two affine points, $(0, \pm y_0)$ with $y_0^2 = b$, share the coordinate $x = 0$, so some point of $E(\mathbb{F}_p)$ has $x \neq 0$ and ϕ moves it. The proviso is not vacuous: $y^2 = x^3 + 4$ over $\mathbb{F}_7$ has only the three points $O, (0, 2), (0, 5)$, and there $\phi = \text{id}$.

Action on G as a scalar. The subgroup $G \cong \mathbb{Z}/r\mathbb{Z}$ is cyclic with generator P , and $\phi(G) \subseteq G$ by hypothesis, so $\phi(P) = [\lambda]P$ for some $\lambda \in \mathbb{Z}/r\mathbb{Z}$; then for any $Q = [m]P \in G$, the homomorphism property gives $\phi(Q) = [m]\phi(P) = [m\lambda]P = [\lambda]Q$. (This is the statement $\text{Hom}(\mathbb{Z}/r\mathbb{Z}, \mathbb{Z}/r\mathbb{Z}) \cong \mathbb{Z}/r\mathbb{Z}$: an endomorphism of a cyclic group is determined by, and equal to multiplication by, the image of a generator.) Applying the endomorphism relation to P ,

$$[\lambda^2 + \lambda + 1]P = O \implies \lambda^2 + \lambda + 1 \equiv 0 \pmod{r},$$

since P has order r . Consistently, this congruence is solvable exactly when $\mathbb{F}_r$ contains a primitive cube root of unity, i.e. when $r \equiv 1 \pmod{3}$ (theorem 6.22 again)—the hypothesis of the statement. $\square$

Remark 10.37 (How GLV halves the doubling count). Computing ϕ costs a single field multiplication $x \mapsto \omega x$, negligible against a group operation. One therefore decomposes a scalar $n \in \mathbb{Z}/r\mathbb{Z}$ as

$$[n]P = [n_1]P + [n_2]\phi(P), \quad n \equiv n_1 + n_2\lambda \pmod{r},$$

where n_1, n_2 can each be taken only about $\frac{1}{2} \log_2 r$ bits long—such a pair is found by lattice reduction in the lattice of pairs (c_1, c_2) with $c_1 + c_2\lambda \equiv 0 \pmod{r}$. The two half-length scalar multiplications are then run *simultaneously*, sharing their doublings (Straus–Shamir’s trick): one double-and-add loop of half the length, adding $P, \phi(P)$, or both according to the bits of n_1 and n_2 . The result costs about half the doublings of a single full-length double-and-add—a substantial saving, and its availability is one reason the Pasta curves take the form $y^2 = x^3 + b$ over primes $p \equiv 1 \pmod{3}$.

10.13 A worked toy example

The entire machinery now runs on a curve small enough for hand computation.

Example 10.38 (The curve $y^2 = x^3 + 5$ over $\mathbb{F}_{11}$). Take $p = 11$: then $p > 3$, and $4 \cdot 0 + 27 \cdot 5^2 = 675 \equiv 4 \not\equiv 0 \pmod{11}$, so the curve $E : y^2 = x^3 + 5$ is nonsingular over $\mathbb{F}_{11}$ (theorem 10.5). Squaring the residues $0, 1, \dots, 10$ yields the squares $\{0, 1, 3, 4, 5, 9\}$, so the nonzero quadratic residues are $\{1, 3, 4, 5, 9\}$. Tabulating $f(x) = x^3 + 5 \pmod{11}$ and reading off the y -values:

x	$x^3 \pmod{11}$	$f(x) = x^3 + 5$	y with $y^2 = f(x)$
0	0	5	± 4 ($4^2 = 16 \equiv 5$)
1	1	6	none ($6 \notin \text{QR}$)
2	8	2	none
3	5	10	none
4	9	3	± 5 ($5^2 = 25 \equiv 3$)
5	4	9	± 3
6	7	1	± 1
7	2	7	none
8	6	0	0
9	3	8	none
10	10	4	± 2

Counting: the five values $x \in \{0, 4, 5, 6, 10\}$ contribute two points each (10 points), $x = 8$ contributes the single point $(8, 0)$, and O completes the census:

$$\#E(\mathbb{F}_{11}) = 10 + 1 + 1 = 12.$$

Hasse’s bound checks out: $p + 1 = 12$, so the trace is $t = 0$, and indeed $|t| = 0 \leq 2\sqrt{11} \approx 6.63$. The point $(8, 0)$ has order 2, consistent with theorem 10.25 and the even order 12; here $-5 \equiv 6$ is a cube, namely $8^3 = 512 \equiv 6$.

An addition. Let $P = (0, 4)$ and $Q = (5, 3)$. Since $x_1 = 0 \neq 5 = x_2$, the chord formula (12) applies:

$$\lambda = \frac{3 - 4}{5 - 0} = (-1) \cdot 5^{-1} = (-1) \cdot 9 = -9 \equiv 2 \pmod{11},$$

using $5^{-1} = 9$ (check: $5 \cdot 9 = 45 \equiv 1$). Then

$$x_3 = \lambda^2 - x_1 - x_2 = 4 - 0 - 5 = -1 \equiv 10, \quad y_3 = \lambda(x_1 - x_3) - y_1 = 2(0 - 10) - 4 = -24 \equiv 9,$$

so $P + Q = (10, 9)$. The result is on the curve: $9^2 = 81 \equiv 4$ and $10^3 + 5 = 1005 \equiv 4 \pmod{11}$, as required.

A doubling. Take $P = (0, 4)$. With $a = 0$, formula (13) gives

$$\lambda = \frac{3x_1^2 + a}{2y_1} = \frac{0}{8} = 0, \quad x_3 = \lambda^2 - 2x_1 = 0, \quad y_3 = \lambda(x_1 - x_3) - y_1 = -4 \equiv 7,$$

so $[2](0, 4) = (0, 7) = -(0, 4)$. Hence $[3](0, 4) = O$: the point $(0, 4)$ has order 3.

The GLV endomorphism. Since $11 \equiv 2 \pmod{3}$, the field $\mathbb{F}_{11}$ contains *no* primitive cube root of unity (theorem 6.22 requires $3 \mid p - 1$), so the GLV map is not available over $\mathbb{F}_{11}$. The Pasta primes do satisfy $p \equiv q \equiv 1 \pmod{3}$, so GLV applies there. For a small GLV-friendly illustration take instead $p = 13 \equiv 1 \pmod{3}$, where $\omega = 3$ satisfies $3^2 + 3 + 1 = 13 \equiv 0 \pmod{13}$; then $\phi(x, y) = (3x, y)$ is a nontrivial endomorphism of $y^2 = x^3 + b$ over $\mathbb{F}_{13}$.

10.14 Pallas and Vesta assembled

Every property developed in this section was chosen with the closing example in mind. Here the pieces assemble.

Example 10.39 (The Pasta curves). For the running example $y^2 = x^3 + 5$, with p and q the 255-bit primes of the section opening:

- *Pallas*, over $\mathbb{F}_p$: the order is $\#E_p(\mathbb{F}_p) = q$, a prime, so the group is cyclic, $E_p(\mathbb{F}_p) \cong \mathbb{Z}/q\mathbb{Z}$ (theorem 3.34), with cofactor $h = 1$ (theorem 10.24). The trace is $t = p + 1 - q$, an 87-bit (negative) integer, far inside the Hasse bound $2\sqrt{p} \approx 2^{128}$ of theorem 10.22.
- *Vesta*, over $\mathbb{F}_q$: the order is $\#E_q(\mathbb{F}_q) = p$, a prime, so $E_q(\mathbb{F}_q) \cong \mathbb{Z}/p\mathbb{Z}$, again with cofactor 1.
- Both curves have $j = 0$ (theorem 10.35), and both primes satisfy $p \equiv q \equiv 1 \pmod{3}$, so each curve admits the order-3 endomorphism $\phi(x, y) = (\omega x, y)$ of theorem 10.36 and supports the GLV speedup of theorem 10.37.
- Both fields have two-adicity 32 in the sense of theorem 9.10: 2^{32} divides $p - 1$ and $q - 1$, so each field carries the multiplicative subgroups of order 2^{32} promised by theorem 10.28—the evaluation domains of the fast polynomial arithmetic of section 9.
- The cycle property $\#E_p(\mathbb{F}_p) = q$ and $\#E_q(\mathbb{F}_q) = p$ means the scalar field of each curve is the base field of the other (theorem 10.27): the base field of Pallas is $\mathbb{F}_p$ and its scalar field is $\mathbb{F}_q$, matching the base field of Vesta, and symmetrically. This is exactly what lets recursive proof systems chain proofs across the two curves without expensive non-native field arithmetic (theorem 10.28). The cycle is a deployed fact: the `pasta_curves` crate wires the scalar field of Pallas to be $\mathbb{F}_q$ and that of Vesta to be $\mathbb{F}_p$, as specified in §5.4.9.6 of the Zcash protocol specification.

The discrete logarithm problem in $E_p(\mathbb{F}_p)$ and $E_q(\mathbb{F}_q)$ is believed to require about 2^{127} operations— $\sqrt{r}$ for the 255-bit orders $r \approx 2^{254}$ (theorem 10.30)—with no faster attack known (theorem 10.31), just below the 128-bit security level targeted by Halo 2 and Zcash Orchard.

The cheque from the section’s opening is hereby cashed. The demand was a roughly 255-bit group for which no attack faster than the square-root bound is known, and each Pasta curve supplies one: a cyclic group of prime 255-bit order (theorem 10.22 and theorem 3.34), carrying a fast forward map (theorem 10.21, accelerated by theorem 10.37) whose inversion—the ECDLP—resists everything known except the generic $O(\sqrt{r})$ attacks of theorem 10.30. Where $\mathbb{F}_p^\times$ needs thousands of bits to outrun index calculus, the curve group needs 255; and the two groups even interlock as a cycle, each one’s scalars living in the other’s coordinates. Later volumes build on precisely this pair.

11 Probability, asymptotics, and computation

A deployed signature scheme is typically advertised as offering “128-bit security”. The question this section exists to answer is: what, precisely, does that phrase assert, and about whom? It cannot assert that forgery is impossible. A forger may simply guess the secret key: keys are finite strings, so the guess succeeds with some positive probability, and a forger with astronomical luck wins outright. Any honest guarantee is therefore a statement of *probability* — a bound on the chance that the forger wins. Nor can it speak of one forger only: a guarantee against yesterday’s attack says nothing about tomorrow’s, so the bound must quantify over *every* attacker whose resources fall within stated limits. Making the sentence “every efficient forger succeeds with at most a stated probability” precise takes three intertwined languages: the language of *probability*, which permits reasoning about random keys, random challenges, and the probability that an attack succeeds; the language of *asymptotics*, which describes how resources and success probabilities scale; and the language of *computation*, which fixes what an efficient algorithm is and what it means for a problem to be hard. This section develops all three from elementary foundations and closes by cashing the opening cheque: the final subsection states exactly what “128-bit security” asserts, and about whom. The only prerequisites are familiarity with finite sets, sums, and elementary calculus — limits, the exponential function, and, for the closing subsection’s density-defined laws, the improper Riemann integral; everything else is constructed here.

Until the closing subsection, all probability spaces are finite. This is no real restriction for the cryptographic material: keys, messages, group elements, bounded-length messages, and the random coins of an algorithm with a fixed running-time bound live in finite sets, and finite spaces need no measure theory. Unbounded retry loops need the extension below even when they terminate with probability one. The closing subsection (§11.7) then extends the theory by exactly the rungs later volumes need — countable additivity, continuity along monotone events, a waiting-time toolkit, and laws defined by densities — declaring its one measure-theoretic debt in place (Remark 11.31).

11.1 Finite probability spaces and events

The basic object is a finite set of outcomes together with an assignment of weights summing to one.

Definition 11.1 (Finite probability space). A *finite probability space* is a pair (Ω, Pr) where Ω is a nonempty finite set, called the *sample space*, and $\text{Pr}: \Omega \rightarrow [0, 1]$ is a function, called the *probability mass function*, satisfying the normalisation condition

$$\sum_{\omega \in \Omega} \text{Pr}[\omega] = 1.$$

An element $\omega \in \Omega$ is an *outcome* (or *elementary event*).

Definition 11.2 (Event). An *event* is a subset $A \subseteq \Omega$. Its *probability* is

$$\Pr[A] = \sum_{\omega \in A} \Pr[\omega],$$

with the convention $\Pr[\emptyset] = 0$ (an empty sum). The event A *occurs* on outcome ω if $\omega \in A$.

Because Ω is finite, every subset is an event. This convenience is worth flagging: on an uncountable sample space the general theory specifies a family of admissible events together with its probability assignment, rather than automatically admitting all subsets; finiteness spares us that machinery, and Remark 11.31 marks the one place this series touches it. The following elementary facts hold, and we use them freely.

Proposition 11.3 (Basic axioms). For any finite probability space $(\Omega, \Pr)$ and events $A, B \subseteq \Omega$:

1. $0 \leq \Pr[A] \leq 1$, $\Pr[\emptyset] = 0$, and $\Pr[\Omega] = 1$.
2. (Complement.) $\Pr[\Omega \setminus A] = 1 - \Pr[A]$.
3. (Monotonicity.) If $A \subseteq B$ then $\Pr[A] \leq \Pr[B]$.
4. (Inclusion–exclusion for two events.) $\Pr[A \cup B] = \Pr[A] + \Pr[B] - \Pr[A \cap B]$.
5. (Finite additivity.) If $A_1, \dots, A_n$ are pairwise disjoint then $\Pr[\bigcup_{i=1}^n A_i] = \sum_{i=1}^n \Pr[A_i]$.

Proof. Every claim follows by manipulating the defining sums. For (1), each summand $\Pr[\omega]$ lies in $[0, 1]$ and the total is 1 by normalisation; an event’s probability is a partial sum of nonnegative terms, hence lies in $[0, 1]$. For (2), the sets A and $\Omega \setminus A$ partition Ω , so $\Pr[A] + \Pr[\Omega \setminus A] = \sum_{\omega \in \Omega} \Pr[\omega] = 1$. For (3), splitting the sum over B gives $\Pr[B] = \sum_{\omega \in A} \Pr[\omega] + \sum_{\omega \in B \setminus A} \Pr[\omega] \geq \Pr[A]$, the second term being a sum of nonnegative numbers. For (5), disjointness means each ω in the union lies in exactly one A_i , so the double sum $\sum_i \sum_{\omega \in A_i} \Pr[\omega]$ counts each $\omega \in \bigcup_i A_i$ exactly once. For (4), write $A \cup B = A \sqcup (B \setminus A)$ (disjoint), so by (5) $\Pr[A \cup B] = \Pr[A] + \Pr[B \setminus A]$; also $B = (A \cap B) \sqcup (B \setminus A)$, giving $\Pr[B \setminus A] = \Pr[B] - \Pr[A \cap B]$. Substituting yields the claim. $\square$

The single most important example, ubiquitous in cryptography, is the uniform distribution.

Definition 11.4 (Uniform distribution). Let S be a nonempty finite set. The *uniform distribution* on S is the probability space $(S, \Pr)$ with $\Pr[\omega] = 1/|S|$ for every $\omega \in S$. The notation $x \leftarrow S$ (or $x \overset{\$}{\leftarrow} S$) means that x is sampled according to the uniform distribution on S . Under the uniform distribution, for any event $A \subseteq S$,

$$\Pr[A] = \frac{|A|}{|S|}.$$

Thus, for uniform sampling, probability is exactly the ratio of favourable outcomes to total outcomes, recovering the classical counting notion of probability.

11.2 Conditional probability and independence

Frequently we learn that one event has occurred and must update the probability of another accordingly. Conditioning captures this.

Definition 11.5 (Conditional probability). Let A, B be events with $\Pr[B] > 0$. The *conditional probability of A given B* is

$$\Pr[A \mid B] = \frac{\Pr[A \cap B]}{\Pr[B]}.$$

When $\Pr[B] = 0$ the quantity is left undefined.

Conditioning on B restricts the sample space to B and renormalises so that B has probability 1: the map $\omega \mapsto \Pr[\{\omega\} \mid B]$ is itself a probability mass function on B , and $\Pr[A \mid B]$ is the probability it assigns to $A \cap B$. Two immediate and indispensable consequences follow.

Proposition 11.6 (Chain rule and total probability). Let $A_1, \dots, A_n$ be events with $\Pr[A_1 \cap \dots \cap A_{n-1}] > 0$. Then

$$\Pr[A_1 \cap \dots \cap A_n] = \Pr[A_1] \Pr[A_2 \mid A_1] \Pr[A_3 \mid A_1 \cap A_2] \dots \Pr[A_n \mid A_1 \cap \dots \cap A_{n-1}].$$

Moreover, if $B_1, \dots, B_k$ partition Ω (pairwise disjoint with union Ω) and each $\Pr[B_i] > 0$, then for any event A ,

$$\Pr[A] = \sum_{i=1}^k \Pr[A \mid B_i] \Pr[B_i].$$

Proof. The chain rule follows by telescoping: each factor $\Pr[A_j \mid A_1 \cap \dots \cap A_{j-1}]$ equals $\Pr[A_1 \cap \dots \cap A_j] / \Pr[A_1 \cap \dots \cap A_{j-1}]$ by theorem 11.5, and the product collapses to $\Pr[A_1 \cap \dots \cap A_n] / 1$. (Monotonicity ensures that all the intermediate intersections have positive probability, so every factor is defined.) For total probability, the events $A \cap B_i$ are pairwise disjoint with union A (since the B_i partition Ω), so finite additivity and $\Pr[A \cap B_i] = \Pr[A \mid B_i] \Pr[B_i]$ give $\Pr[A] = \sum_i \Pr[A \cap B_i] = \sum_i \Pr[A \mid B_i] \Pr[B_i]$. $\square$

Independence is the formal statement that one event carries no information about another.

Definition 11.7 (Independence of events). Two events A, B are *independent* if

$$\Pr[A \cap B] = \Pr[A] \Pr[B].$$

A family $\{A_i\}_{i \in I}$ of events is *mutually independent* if for every finite subset $J \subseteq I$,

$$\Pr\left[\bigcap_{i \in J} A_i\right] = \prod_{i \in J} \Pr[A_i].$$

Remark 11.8. When $\Pr[B] > 0$, independence of A and B is equivalent to $\Pr[A \mid B] = \Pr[A]$: conditioning on B does not change the probability of A . Mutual independence is strictly stronger than pairwise independence. The standard counterexample: toss two fair coins, and let A be “the first is heads”, B “the second is heads”, and C “the two coins differ”. Each pair is independent, yet $\Pr[A \cap B \cap C] = 0 \neq \frac{1}{8} = \Pr[A] \Pr[B] \Pr[C]$, so the three are not mutually independent.

11.3 Random variables and expectation

A random variable assigns a value to each outcome; it summarises a random experiment by a number (or a vector, a string, a group element).

Definition 11.9 (Random variable). Let $(\Omega, \Pr)$ be a finite probability space and T any set. A (T -valued) *random variable* is a function $X: \Omega \rightarrow T$. When $T \subseteq \mathbb{R}$, we call X *real-valued*. For $t \in T$,

$$\Pr[X = t] = \Pr[\{\omega \in \Omega : X(\omega) = t\}],$$

and more generally $\Pr[X \in B]$ denotes the probability that X takes a value in $B \subseteq T$. The function $t \mapsto \Pr[X = t]$ on the finite image of X is the *distribution* (or *law*) of X .

Definition 11.10 (Independent random variables). Random variables $X_1, \dots, X_n$ (with values in sets $T_1, \dots, T_n$) are *mutually independent* if for all $t_1 \in T_1, \dots, t_n \in T_n$,

$$\Pr[X_1 = t_1, \dots, X_n = t_n] = \prod_{i=1}^n \Pr[X_i = t_i].$$

Definition 11.11 (Expectation). Let $X: \Omega \rightarrow \mathbb{R}$ be a real-valued random variable on a finite space. Its *expectation* (or *mean*) is

$$\mathbb{E}[X] = \sum_{\omega \in \Omega} X(\omega) \Pr[\omega] = \sum_{t \in \text{im}(X)} t \cdot \Pr[X = t],$$

where $\text{im}(X)$ denotes the (finite) image of X . The two expressions agree by grouping outcomes according to their X -value.

A frequently used bridge between the two notions is the following: the expectation of the indicator of an event is exactly its probability.

Definition 11.12 (Indicator). For an event A , its *indicator random variable* $1_A: \Omega \rightarrow \{0, 1\}$ takes the value $1_A(\omega) = 1$ if $\omega \in A$ and 0 otherwise.

Proposition 11.13 (Indicator expectation). For any event A , $\mathbb{E}[1_A] = \Pr[A]$.

Proof. By theorem 11.11, $\mathbb{E}[1_A] = \sum_{\omega} 1_A(\omega) \Pr[\omega] = \sum_{\omega \in A} \Pr[\omega] = \Pr[A]$, since the indicator vanishes off A . $\square$

The principal property of expectation is its linearity, which holds with no independence assumption whatsoever; this absence of any independence hypothesis is what makes expectation so much more tractable than probability.

Theorem 11.14 (Linearity of expectation). Let $X, Y: \Omega \rightarrow \mathbb{R}$ be random variables on a finite space and $a, b \in \mathbb{R}$. Then

$$\mathbb{E}[aX + bY] = a \mathbb{E}[X] + b \mathbb{E}[Y].$$

More generally, for random variables $X_1, \dots, X_n$ and scalars $a_1, \dots, a_n$,

$$\mathbb{E}\left[\sum_{i=1}^n a_i X_i\right] = \sum_{i=1}^n a_i \mathbb{E}[X_i].$$

This holds whether or not the X_i are independent.

Proof. Directly from the definition,

$$\begin{aligned}\mathbb{E}[aX + bY] &= \sum_{\omega \in \Omega} (aX(\omega) + bY(\omega)) \Pr[\omega] \\ &= a \sum_{\omega} X(\omega) \Pr[\omega] + b \sum_{\omega} Y(\omega) \Pr[\omega] \\ &= a \mathbb{E}[X] + b \mathbb{E}[Y],\end{aligned}$$

splitting the finite sum and extracting the constants. The n -term statement follows by induction on n . $\square$

A one-line consequence converts an expectation bound into a probability bound; it underlies every averaging (“heavy-row”) argument in the series.

Proposition 11.15 (Markov’s inequality). *Let $X \geq 0$ be a real-valued random variable and $a > 0$. Then $\Pr[X \geq a] \leq \mathbb{E}[X]/a$.*

Proof. Every outcome contributes nonnegatively to $\mathbb{E}[X]$, and each outcome with $X(\omega) \geq a$ contributes at least $a \Pr[\omega]$; hence $\mathbb{E}[X] \geq a \Pr[X \geq a]$. $\square$

11.4 The union bound and a birthday calculation

The payoff of this subsection, stated up front, is the square-root law that sets the output lengths of cryptographic hash functions: among roughly $\sqrt{N}$ uniform samples from a set of size N , two are likely to coincide, so an n -bit hash resists collision-finding only up to about $2^{n/2}$ work, and output lengths are chosen twice the desired security level. The engine of the calculation is the most frequently used inequality in cryptographic proofs: the union bound. It is crude, it requires no independence, and it almost always suffices.

Theorem 11.16 (Union bound / Boole’s inequality). *For any events $A_1, \dots, A_n$ in a finite probability space,*

$$\Pr\left[\bigcup_{i=1}^n A_i\right] \leq \sum_{i=1}^n \Pr[A_i].$$

Proof. Disjointify the union into “first occurrence” events: $B_1 = A_1$ and $B_i = A_i \setminus (A_1 \cup \dots \cup A_{i-1})$ for $i \geq 2$. The B_i are pairwise disjoint, $B_i \subseteq A_i$, and $\bigcup_i B_i = \bigcup_i A_i$. By finite additivity and monotonicity (theorem 11.3),

$$\Pr\left[\bigcup_i A_i\right] = \Pr\left[\bigcup_i B_i\right] = \sum_i \Pr[B_i] \leq \sum_i \Pr[A_i]. \quad \square$$

The disjointification device — replacing events by pairwise disjoint pieces with the same union — recurs throughout the section: the “first occurrence” form above proves the union bound, and a telescoping form for increasing chains proves the continuity theorem of §11.7. In each case it turns a subadditivity or limit claim into an exact additivity computation. The complementary inequality for conjunctions is equally useful and follows by taking complements.

Corollary 11.17 (Union bound for the good event). *If each event A_i holds except with probability at most ε_i (that is, $\Pr[\Omega \setminus A_i] \leq \varepsilon_i$), then all of them hold simultaneously except with probability at*

most $\sum_i \varepsilon_i$:

$$\Pr\left[\bigcap_{i=1}^n A_i\right] \geq 1 - \sum_{i=1}^n \varepsilon_i.$$

Proof. By De Morgan, $\Omega \setminus \bigcap_i A_i = \bigcup_i (\Omega \setminus A_i)$. Apply the union bound to the complements and take the complement of both sides. $\square$

We now derive the *birthday bound*, which governs how soon collisions appear when sampling uniformly.

Proposition 11.18 (Birthday bound). *Sample $x_1, \dots, x_k$ independently and uniformly from a set of size N , and let C be the event that some two of them collide, i.e. $x_i = x_j$ for some $i < j$. Then*

$$\Pr[C] \leq \binom{k}{2} \frac{1}{N} = \frac{k(k-1)}{2N},$$

and, on the other side,

$$\Pr[C] = 1 - \prod_{i=0}^{k-1} \left(1 - \frac{i}{N}\right) \geq 1 - e^{-k(k-1)/(2N)}.$$

In particular a collision becomes likely once $k = \Theta(\sqrt{N})$.

Proof. Upper bound. For $i < j$ let A_{ij} be the event $x_i = x_j$. Since x_i and x_j are independent and uniform, $\Pr[A_{ij}] = \sum_v \Pr[x_i = v] \Pr[x_j = v] = N \cdot (1/N)^2 = 1/N$. Now $C = \bigcup_{i < j} A_{ij}$, a union of $\binom{k}{2}$ events, one per two-element subset of the indices (theorem 6.12), so the union bound (theorem 11.16) gives $\Pr[C] \leq \binom{k}{2}/N$.

Lower bound. The complement $\bar{C}$ is the event that all k samples are distinct. If $k > N$, the pigeonhole principle forces a collision, so $\Pr[\bar{C}] = 0$; the product also vanishes, through its factor $1 - N/N$, so the stated equality holds and the exponential bound is trivial. Let then $k \leq N$, and for $0 \leq i \leq k$ let D_i be the event that the first i samples are pairwise distinct ($D_0 = D_1 = \Omega$, and $D_k = \bar{C}$). We show $\Pr[D_i] = \prod_{j=0}^{i-1} (1 - j/N)$ by induction on $i \leq k$; every factor is positive, since $j \leq k-1 \leq N-1$, so in particular each D_i has positive probability and conditioning on it is legitimate (theorem 11.5). For the step, the events pinning the first i samples to specific distinct values partition D_i , each with probability $N^{-i} > 0$ by independence and uniformity; given any one of them, the next sample avoids the i values taken with probability $1 - i/N$, again by independence and uniformity. Averaging over the cells (total probability, theorem 11.6) gives $\Pr[D_{i+1} | D_i] = 1 - i/N$, and $\Pr[D_{i+1}] = \Pr[D_{i+1} | D_i] \Pr[D_i]$ completes the induction. Taking $i = k$ yields the stated equality, in which every factor is nonnegative, so the inequality $1 - x \leq e^{-x}$ applies factor by factor. (For $0 \leq x < 1$ this inequality follows from Bernoulli's inequality, $(1 - x/n)^n \geq 1 - x$ for $n > x$, by letting $n \rightarrow \infty$ in the limit definition of e^{-x} ; for $x \geq 1$ the left side is nonpositive.) Thus

$$\Pr[\bar{C}] = \prod_{i=0}^{k-1} \left(1 - \frac{i}{N}\right) \leq \prod_{i=0}^{k-1} e^{-i/N} = \exp\left(-\frac{1}{N} \sum_{i=0}^{k-1} i\right) = e^{-k(k-1)/(2N)},$$

summing the exponents via $\sum_{i=0}^{k-1} i = k(k-1)/2$. Taking complements gives $\Pr[C] = 1 - \Pr[\bar{C}] \geq 1 - e^{-k(k-1)/(2N)}$. $\square$

Example 11.19. With $N = 365$ and $k = 23$ — the classical birthday party — the exact product gives $\Pr[C] = 0.5073$, already a majority chance, and the proposition brackets it as $0.5000 \leq$

$\Pr[C] \leq 0.6932$. The threshold is visible at small scale too: with $N = 16$, the fifth sample (just past $\sqrt{16} = 4$) tips the odds, $\Pr[C] = 0.5001$ at $k = 5$.

Remark 11.20. The threshold $k \approx \sqrt{N}$ is the reason an n -bit hash function (so $N = 2^n$) offers only about $2^{n/2}$ collision resistance: an adversary expects a collision after roughly $2^{n/2}$ random evaluations — for $N = 2^{256}$, that is $\sqrt{2^{256}} = 2^{128}$. For this reason one chooses output lengths twice the desired security level. The same square-root phenomenon recurs in random-walk discrete-log attacks (Pollard’s rho, theorem 10.30), so designers size the curves used in Halo 2 and Orchard so that $\sqrt{N}$ is itself astronomically large.

11.5 Statistical distance

Comparing distributions — for example a real key-generation procedure against an idealised uniform one — requires a notion of how far apart two distributions are. The appropriate notion for “no statistical test can distinguish them well” is statistical distance.

Definition 11.21 (Statistical distance). Let P and Q be probability distributions on the same finite set S (i.e. probability mass functions $P, Q: S \rightarrow [0, 1]$). Their *statistical distance* (or *total variation distance*) is

$$\Delta(P, Q) = \frac{1}{2} \sum_{s \in S} |P(s) - Q(s)|.$$

For random variables X, Y with values in S , $\Delta(X, Y)$ denotes the statistical distance of their distributions.

The factor $\frac{1}{2}$ makes Δ range over $[0, 1]$ and gives it a clean interpretation as the maximum advantage of a distinguisher, which part (3) of the next theorem establishes.

Theorem 11.22 (Properties of statistical distance). Let P, Q, R be distributions on a finite set S . Then:

1. $0 \leq \Delta(P, Q) \leq 1$; and $\Delta(P, Q) = 0$ if and only if $P = Q$.
2. (Metric.) Δ is symmetric and satisfies the triangle inequality $\Delta(P, R) \leq \Delta(P, Q) + \Delta(Q, R)$.
3. (Variational characterisation.) $\Delta(P, Q) = \max_{A \subseteq S} (P(A) - Q(A)) = \max_{A \subseteq S} |P(A) - Q(A)|$, where $P(A) = \sum_{s \in A} P(s)$.
4. (Data processing / post-processing.) For any (possibly randomised) function f applied to a sample, $\Delta(f(P), f(Q)) \leq \Delta(P, Q)$. In particular no algorithm can increase statistical distance.

Proof. (1) Nonnegativity is clear. For the upper bound, split S into $S^+ = \{s : P(s) \geq Q(s)\}$ and $S^- = S \setminus S^+$. Since $\sum_s P(s) = \sum_s Q(s) = 1$, we have $\sum_s (P(s) - Q(s)) = 0$, so $\sum_{s \in S^+} (P(s) - Q(s)) = \sum_{s \in S^-} (Q(s) - P(s)) =: D \geq 0$. Hence $\Delta(P, Q) = \frac{1}{2} (\sum_{S^+} (P - Q) + \sum_{S^-} (Q - P)) = D$. As $D = \sum_{S^+} (P(s) - Q(s)) \leq \sum_{S^+} P(s) \leq 1$, we get $\Delta \leq 1$. If $\Delta = 0$ then every $|P(s) - Q(s)| = 0$, so $P = Q$; the converse is immediate.

(2) Symmetry is evident. For the triangle inequality, $|P(s) - R(s)| \leq |P(s) - Q(s)| + |Q(s) - R(s)|$ termwise by the triangle inequality on $\mathbb{R}$; summing over s and halving gives the claim.

(3) With D as above, take $A = S^+$: then $P(A) - Q(A) = \sum_{S^+} (P - Q) = D = \Delta(P, Q)$, so the maximum is at least Δ . Conversely, for any $A \subseteq S$, $P(A) - Q(A) = \sum_{s \in A} (P(s) - Q(s)) \leq \sum_{s \in A \cap S^+} (P(s) - Q(s)) \leq D = \Delta$, since dropping the negative terms only increases the sum and extending from $A \cap S^+$ to all of S^+ increases it further. Thus the maximum over A of $P(A) - Q(A)$ is

exactly Δ ; replacing A by its complement swaps the sign, giving the same value for $\max_A |P(A) - Q(A)|$.

(4) First take f deterministic, $f: S \rightarrow U$. For $u \in U$, $f(P)(u) = P(f^{-1}(u))$. Then

$$\Delta(f(P), f(Q)) = \frac{1}{2} \sum_u |P(f^{-1}(u)) - Q(f^{-1}(u))| \leq \frac{1}{2} \sum_u \sum_{s \in f^{-1}(u)} |P(s) - Q(s)| = \Delta(P, Q),$$

using $|\sum_s a_s| \leq \sum_s |a_s|$ on each fibre, and that the fibres $f^{-1}(u)$ partition S . A randomised f is a deterministic function of the sample s together with independent fresh coins r ; apply the deterministic case to the joint variable (s, r) , noting that the two joint distributions $P \times U_r$ and $Q \times U_r$ have the same statistical distance $\Delta(P, Q)$, the shared coin distribution contributing nothing. $\square$

Remark 11.23 (Distinguishing interpretation). Part (3) states that $\Delta(P, Q)$ is exactly the best advantage of any test — even a computationally unbounded one — that, given a single sample, must guess whether it came from P or Q : the optimal test outputs “ P ” on the set $A = S^+$, and its advantage $\Pr_P[A] - \Pr_Q[A]$ equals $\Delta(P, Q)$. Part (4) is what makes statistical distance compose in proofs: applying the same procedure to two close inputs keeps the outputs close. If $\Delta(P, Q) \leq \varepsilon$, then P and Q are interchangeable in any analysis at the cost of an additive ε in every probability. When ε is negligible (theorem 11.61 below), we call P and Q *statistically indistinguishable*.

11.6 Uniform sampling and the bias of modular reduction

Cryptography requires uniform elements of finite sets throughout: a uniform scalar in $\mathbb{Z}/q\mathbb{Z}$ (section 2), a uniform field element of $\mathbb{F}_p$ (section 6), a uniform point in a group. The most tractable source of randomness, however, is a stream of independent uniform bits. We must therefore convert uniform bitstrings into uniform elements of a target set S , and we must understand the bias the conversion introduces.

If $|S| = 2^\ell$ is a power of two, the conversion is exact: read ℓ uniform bits as a number in $\{0, \dots, 2^\ell - 1\}$ and index into S . The nontrivial case arises when $|S|$ is not a power of two — most importantly when $S = \mathbb{Z}/q\mathbb{Z}$ for a prime q , the scalar field of an elliptic curve (section 10). A common and elementary method is *reduce-modulo*: sample a long uniform integer and reduce it mod q . The result is not perfectly uniform — some residues receive one more preimage than others — but the bias is negligible provided the integer is long enough. The next proposition makes this precise.

Proposition 11.24 (Bias of modular reduction). *Let $q \geq 2$ be an integer and let $K = 2^L$ for a nonnegative integer L . Sample U uniformly from $\{0, 1, \dots, K - 1\}$ (equivalently, read L uniform bits as an integer), and set $R = U \bmod q \in \{0, 1, \dots, q - 1\}$. Let $\mathcal{U}_q$ denote the uniform distribution on $\{0, \dots, q - 1\}$. Then the statistical distance of R from uniform satisfies*

$$\Delta(R, \mathcal{U}_q) \leq \frac{q}{K} = \frac{q}{2^L}.$$

Consequently, if $q < 2^n$ and one takes $L = n + s$ bits, then $\Delta(R, \mathcal{U}_q) < 2^{-s}$.

Proof. Write $K = mq + t$ with $m = \lfloor K/q \rfloor$ and remainder $t = K \bmod q$, so $0 \leq t < q$. For a residue $r \in \{0, \dots, q - 1\}$, the number of integers in $\{0, \dots, K - 1\}$ congruent to r modulo q is $m + 1$ if $r < t$ and m if $r \geq t$. Hence

$$\Pr[R = r] = \begin{cases} (m + 1)/K, & r < t, \\ m/K, & r \geq t. \end{cases}$$

Comparing with the target probability $1/q$ and using $m/K \leq 1/q \leq (m+1)/K$ (since $mq \leq K \leq (m+1)q$),

$$|\Pr[R = r] - \frac{1}{q}| \leq \frac{m+1}{K} - \frac{m}{K} = \frac{1}{K} \quad \text{for every } r.$$

Therefore

$$\Delta(R, \mathcal{U}_q) = \frac{1}{2} \sum_{r=0}^{q-1} |\Pr[R = r] - \frac{1}{q}| \leq \frac{1}{2} \cdot q \cdot \frac{1}{K} = \frac{q}{2K} \leq \frac{q}{K}.$$

(The computation actually yields the sharper bound $q/(2K)$; q/K is the usual stated form.) For the final claim, $q < 2^n$ and $K = 2^{n+s}$ give $q/K < 2^n/2^{n+s} = 2^{-s}$. $\square$

Example 11.25. Reduce $L = 6$ uniform bits modulo $q = 5$. Here $K = 64 = 12 \cdot 5 + 4$, so the four residues $r < 4$ receive 13 preimages each and the residue $r = 4$ receives 12; the statistical distance from uniform works out to exactly $1/80$, within the guarantee 2^{-3} of the proposition (as $q < 2^3$ and $s = 6 - 3 = 3$).

Remark 11.26 (Sampling scalars in practice). The consequence to remember: to sample a scalar mod a ~ 256 -bit prime q with statistical distance below 2^{-128} from uniform, draw $256 + 128 = 384$ uniform bits and reduce. The bias 2^{-s} , with the slack s set to the security parameter, is negligible (theorem 11.61), so the reduce-modulo sample is statistically indistinguishable from a perfectly uniform scalar, and by the data-processing inequality (theorem 11.22(4)) substituting one for the other anywhere costs at most 2^{-128} per use. An alternative, *rejection sampling* — draw $\lceil \log_2 q \rceil$ bits, accept if the result is $< q$, else retry — gives *exactly* uniform output at the cost of a variable but (with overwhelming probability) small number of retries; for this series' purposes the negligible bias of reduce-modulo is acceptable and simpler to analyse. The Orchard specification employs exactly this reduce-modulo sampling when deriving scalars from hash outputs: its ToScalar reduces the full 512-bit PRFexpand output modulo the 255-bit group order — a slack of 257 bits, comfortably beyond the 128 of the worked example (Zcash protocol specification §4.2.3 and §5.4.2).

11.7 Beyond finite spaces: countable additivity, limits, and densities

Everything so far lives on a finite sample space, and for the cryptographic material of the upper volumes that is enough: keys, challenges, and transcripts are finite strings. Two uses need more. A process watched for as long as it takes — a random walk pursued until it first hits a barrier — has no finite horizon, and a waiting time measured in continuous units has no finite sample space at all. This subsection extends the finite theory by exactly the rungs those uses require: countable additivity, continuity along monotone events, a toolkit for waiting times under stopping rules, laws defined by densities, and a toolkit of clocks and tail bounds built on them.

Definition 11.27 (Countable probability space). A *countable probability space* is a pair $(\Omega, \Pr)$ in which Ω is a countable set and $\Pr: \Omega \rightarrow [0, 1]$ satisfies $\sum_{\omega \in \Omega} \Pr[\omega] = 1$; the sum is independent of the enumeration order because its terms are nonnegative. Events are arbitrary subsets $A \subseteq \Omega$, with $\Pr[A] = \sum_{\omega \in A} \Pr[\omega]$, and Definitions 11.2–11.11 carry over with “finite” read as “countable” (a random variable’s image may now be countably infinite), with one exception: theorem 11.4 does not — equal weights cannot sum to 1 on a countably infinite set, so the uniform distribution remains a strictly finite notion. Expectation additionally requires its defining sum to converge absolutely.

Proposition 11.28 (Countable additivity). *Let $(\Omega, \Pr)$ be a countable probability space and*

$A_1, A_2, \dots$ pairwise disjoint events. Then

$$\Pr\left[\bigcup_{n \geq 1} A_n\right] = \sum_{n \geq 1} \Pr[A_n],$$

and in particular the basic axioms of theorem 11.3 hold unchanged.

Proof. Every outcome of the union lies in exactly one A_n , so both sides sum the same nonnegative terms $\Pr[\omega]$, grouped differently; a series of nonnegative terms may be summed in any order, or in groups, without changing its value. $\square$

The free regrouping of nonnegative series invoked here — reorder, regroup, exchange double sums, recount by multiplicity — is the workhorse of every infinite-space argument in this subsection: it proves countable additivity above, the tail-sum formula below (where each term is counted by its multiplicity), and the exchange step in Wald’s identity.

Theorem 11.29 (Continuity along monotone events). *Let $\Pr$ be a probability assignment, on any sample space, defined for a family of events closed under complement and countable union, and countably additive on it. If $A_1 \subseteq A_2 \subseteq \dots$ is an increasing chain of events, then*

$$\Pr\left[\bigcup_{n \geq 1} A_n\right] = \lim_{n \rightarrow \infty} \Pr[A_n],$$

and dually $\Pr[\bigcap_n B_n] = \lim_n \Pr[B_n]$ for a decreasing chain $B_1 \supseteq B_2 \supseteq \dots$. On a countable space the hypothesis holds automatically, by theorem 11.28.

Proof. Disjointify by telescoping: set $D_1 = A_1$ and $D_n = A_n \setminus A_{n-1}$ for $n \geq 2$. The D_n are pairwise disjoint, $\bigcup_{k \leq n} D_k = A_n$, and $\bigcup_n D_n = \bigcup_n A_n$. Countable additivity gives

$$\Pr\left[\bigcup_n A_n\right] = \sum_{n \geq 1} \Pr[D_n] = \lim_{n \rightarrow \infty} \sum_{k=1}^n \Pr[D_k] = \lim_{n \rightarrow \infty} \Pr[A_n],$$

the final step by finite additivity. The decreasing case follows by complementation. $\square$

Definition 11.30 (Infinite sequence of independent trials). Fix a finite outcome set T and a distribution p on T . An *infinite sequence of independent trials* with law p is a sequence of random variables $X_1, X_2, \dots$ such that every prefix $(X_1, \dots, X_n)$ consists of n mutually independent p -distributed trials in the sense of theorem 11.10 — equivalently, the prefix’s law is that of the finite probability space T^n carrying the product weights $\Pr[(t_1, \dots, t_n)] = p(t_1) \cdots p(t_n)$. An event is *finitely determined* if membership in it is decided by some fixed finite prefix; its probability is the finite-space value. An event of the form “some prefix satisfies P ” is the union of an increasing chain of finitely determined events, and theorem 11.29 assigns it the limit of the prefix probabilities.

Remark 11.31 (What is taken as given). Theorem 11.30 presumes that a single, countably additive probability assignment on infinite sequences exists that restricts correctly to every finite prefix at once. That existence statement is the Kolmogorov extension theorem, and its proof belongs to measure theory, which lies off this series’ critical path: the space of infinite sequences over at least two possible outcomes is uncountable, so it escapes theorem 11.27, and the extension specifies the admissible events. We therefore take it — with its countable additivity, hence theorem 11.29 — as a foundation stone rather than a theorem. On infinite-sequence spaces, only finitely determined events and their monotone limits appear in this series, and for those the values are forced by the

finite theory together with continuity; the density-defined laws of theorem 11.32 carry the analogous debt, their interval and joint-event probabilities being taken to cohere as the stated integrals. The independent families of theorem 11.40 extend that debt in two ways: joint events of several density-defined quantities are taken to be iterated integrals, in whichever order of integration is convenient — the exchange of order for nonnegative integrands is the Fubini–Tonelli theorem, likewise measure-theoretic — and the gap sequences of theorem 11.44 are infinite families, of which only finite prefixes enter any event this series uses, except through countable unions over the values of an arrival count — $N(t)$ of theorem 11.44, or the count X of theorem 11.51, each event “count = x ” being a joint event of finitely many gaps. For any set B of values, the event “count $\in B$ ” — among them the event that the count is finite, and the tail events of theorem 11.49 — is taken to have probability $\sum_{x \in B} \Pr[\text{count} = x]$, by the countable additivity assumed above.

Definition 11.32 (Law defined by a density). A real-valued random quantity X has *density* $f: \mathbb{R} \rightarrow \mathbb{R}_{\geq 0}$ if $\int_{-\infty}^{\infty} f(t) dt = 1$ and

$$\Pr[a \leq X \leq b] = \int_a^b f(t) dt \quad \text{for all } a \leq b;$$

improper Riemann integrals suffice throughout this series. The *distribution function* is $F(t) = \Pr[X \leq t] = \int_{-\infty}^t f$, so $\Pr[X > t] = 1 - F(t)$, and single points carry probability zero. The expectation is $\mathbb{E}[X] = \int_{-\infty}^{\infty} t f(t) dt$ when this integral converges absolutely. Two density-defined quantities X, Y are *independent* if $\Pr[X \in I, Y \in J] = \Pr[X \in I] \Pr[Y \in J]$ for all intervals I, J ; probabilities of joint events are then taken to factor through iterated integrals of the product $f_X(s)f_Y(t)$.

11.7.1 The waiting-time toolkit: tail sums, geometric trials, and Wald’s identity

Waiting times are the extension’s chief dividend. The three short results of this toolkit — used by later volumes’ extractor and race analyses — follow from countable expectation and finitely determined events alone. One convention first: for a variable with values in $\mathbb{N} \cup \{\infty\}$, the defining sum of the expectation has nonnegative terms, and we permit the value $+\infty$ when it diverges.

Lemma 11.33 (Tail-sum formula). *Let T be a random variable with values in $\mathbb{N} \cup \{\infty\}$ and $\Pr[T = \infty] = 0$. Then $\mathbb{E}[T] = \sum_{k \geq 1} \Pr[T \geq k]$, either side being infinite exactly when the other is.*

Proof. Expanding $\Pr[T \geq k] = \sum_{m \geq k} \Pr[T = m]$ and regrouping the doubly indexed nonnegative series, each term $\Pr[T = m]$ is counted once for every $k \leq m$, that is, m times; nonnegative series may be regrouped freely. $\square$

Proposition 11.34 (Geometric waiting time). *In an infinite sequence of independent trials, each succeeding with probability $\varepsilon > 0$, let T be the index of the first success. Then $\Pr[T \geq k] = (1 - \varepsilon)^{k-1}$ and $\mathbb{E}[T] = 1/\varepsilon$.*

Proof. The event $T \geq k$ says the first $k-1$ trials all fail — a finitely determined event of probability $(1 - \varepsilon)^{k-1}$; in particular $\Pr[T = \infty] = \lim_k (1 - \varepsilon)^{k-1} = 0$ by continuity (theorem 11.29). The tail-sum formula then gives $\mathbb{E}[T] = \sum_{k \geq 1} (1 - \varepsilon)^{k-1} = 1/\varepsilon$, a geometric series. $\square$

Proposition 11.35 (Wald’s identity). *Let $X_1, X_2, \dots$ be an infinite sequence of independent trials with common law, each $X_i \geq 0$ with finite mean, and let T be a stopping rule: a random index such that, for every k , whether $T = k$ is decided by the first k trials alone. If $\mathbb{E}[T] < \infty$, then*

$$\mathbb{E}\left[\sum_{i=1}^T X_i\right] = \mathbb{E}[T] \cdot \mathbb{E}[X_1].$$

Proof. Write $S = \sum_{i=1}^T X_i$ and $S_k = \sum_{i=1}^k X_i$, so that $S = S_k$ on the event $T = k$. The hypothesis $\mathbb{E}[T] < \infty$ forces $\Pr[T = \infty] = 0$, by the convention preceding theorem 11.33. The variable S takes countably many values — on $\{T = k\}$ it agrees with S_k , whose values run over a finite set, the trials having finitely many outcomes — and each event $\{S = s\}$ is the disjoint countable union $\bigcup_k \{T = k, S_k = s\}$ of finitely determined events, so its probability is assigned by countable additivity (theorem 11.31). By $\mathbb{E}[S]$ we mean $\sum_s s \Pr[S = s]$, the expectation of this countably supported distribution, a sum of nonnegative terms.

Expanding each $\Pr[S = s]$ and regrouping the doubly indexed nonnegative series,

$$\mathbb{E}[S] = \sum_s s \sum_{k \geq 1} \Pr[T = k, S_k = s] = \sum_{k \geq 1} \mathbb{E}[S_k 1_{T=k}],$$

each inner expectation living on the finite prefix space of the first k trials (theorem 11.30), where linearity (theorem 11.14) expands it as $\sum_{i=1}^k \mathbb{E}[X_i 1_{T=k}]$. Regrouping the nonnegative double series once more,

$$\mathbb{E}[S] = \sum_{i \geq 1} \sum_{k \geq i} \mathbb{E}[X_i 1_{T=k}].$$

Fix i and let $n \geq i$. The events $T = k$ (for $k \leq n$), $T > n$, and $T \geq i$ are all decided by the first n trials, and on that prefix space the pointwise identity $1_{T \geq i} = \sum_{k=i}^n 1_{T=k} + 1_{T > n}$ holds, so linearity gives

$$\sum_{k=i}^n \mathbb{E}[X_i 1_{T=k}] = \mathbb{E}[X_i 1_{T \geq i}] - \mathbb{E}[X_i 1_{T > n}].$$

The trials take finitely many values, so $X_i \leq M$ for a constant M , and $0 \leq \mathbb{E}[X_i 1_{T > n}] \leq M \Pr[T > n] \rightarrow 0$ as $n \rightarrow \infty$: the terms $\Pr[T \geq k]$ of the convergent series $\mathbb{E}[T]$ tend to zero. Hence $\sum_{k \geq i} \mathbb{E}[X_i 1_{T=k}] = \mathbb{E}[X_i 1_{T \geq i}]$.

Now the crux factorisation. The event $T \geq i$ — the negation of $T \leq i - 1$ — is decided by the first $i - 1$ trials, so its indicator is a function $g(t_1, \dots, t_{i-1})$ of the first $i - 1$ trial values, and on the prefix space of the first i trials the expectation splits along the product weights of theorem 11.30: with p the common law,

$$\mathbb{E}[X_i 1_{T \geq i}] = \sum_{t_1, \dots, t_i} g(t_1, \dots, t_{i-1}) t_i p(t_1) \cdots p(t_i) = \Pr[T \geq i] \cdot \mathbb{E}[X_1],$$

the double sum factoring into the sum over $(t_1, \dots, t_{i-1})$, which totals $\Pr[T \geq i]$, times $\sum_{t_i} t_i p(t_i) = \mathbb{E}[X_1]$. Assembling the pieces and applying the tail-sum formula (theorem 11.33),

$$\mathbb{E}[S] = \sum_{i \geq 1} \mathbb{E}[X_1] \Pr[T \geq i] = \mathbb{E}[X_1] \cdot \mathbb{E}[T]. \quad \square$$

11.7.2 The clock toolkit: exponential clocks, Poisson counts, and the Chernoff method

The second dividend is continuous time. A block-discovery process is modelled by *clocks*: each party’s next block arrives after a random delay, a delay does not remember how long it has already run, and two parties race. A light client that samples a chain needs a tail bound — how likely is

an adversary to have found cL blocks while the honest chain found L ? — and a wallet that spaces its network queries needs a delay law with a prescribed mean and no memory. This unit builds the clocks, counts their arrivals, and proves the tail bounds those uses invoke, in the forms they invoke them. Everything reduces to density-defined laws (theorem 11.32), elementary calculus, and Markov's inequality with a free parameter.

Definition 11.36 (Exponential law). A random quantity X is *exponential with rate $\lambda > 0$* , written $X \sim \text{Exp}(\lambda)$, if it has the density $f(t) = \lambda e^{-\lambda t}$ for $t \geq 0$ and $f(t) = 0$ for $t < 0$.

Lemma 11.37 (Tail, mean, and memorylessness). Let $X \sim \text{Exp}(\lambda)$. Then $\Pr[X > t] = e^{-\lambda t}$ for all $t \geq 0$, $\mathbb{E}[X] = 1/\lambda$, and

$$\Pr[X > s + t \mid X > s] = \Pr[X > t] \quad \text{for all } s, t \geq 0,$$

conditional probability being the ratio of theorem 11.5: having waited without an arrival teaches nothing about the remaining wait. Conversely, a nonnegative quantity with $\Pr[X > t] = e^{-\lambda t}$ for all $t \geq 0$ is exponential with rate λ .

Proof. The density integrates to 1, and $\Pr[X > t] = 1 - \int_0^t \lambda e^{-\lambda s} ds = e^{-\lambda t}$. Integrating by parts,

$$\mathbb{E}[X] = \int_0^\infty t \lambda e^{-\lambda t} dt = [-te^{-\lambda t}]_0^\infty + \int_0^\infty e^{-\lambda t} dt = \frac{1}{\lambda}.$$

Since $X > s + t$ implies $X > s$, the conditional probability is $e^{-\lambda(s+t)}/e^{-\lambda s} = e^{-\lambda t}$. For the converse, the events $X > a$ and $X > b$ ($a \leq b$) differ by $a < X \leq b$, so $\Pr[a < X \leq b] = e^{-\lambda a} - e^{-\lambda b} = \int_a^b \lambda e^{-\lambda t} dt$; and the point a carries probability at most $e^{-\lambda(a-\delta)} - e^{-\lambda a}$ for every $\delta > 0$, hence zero, so $\Pr[a \leq X \leq b]$ is the same integral. $\square$

Remark 11.38 (Where exponential clocks come from). The exponential law is the continuous limit of the geometric waiting time (theorem 11.34). If trials succeed with a small probability ε and each occupies time ε/λ , the first success arrives at time $\varepsilon T/\lambda$, and for integer-valued T

$$\Pr[\varepsilon T/\lambda > t] = \Pr[T \geq \lfloor \lambda t/\varepsilon \rfloor + 1] = (1 - \varepsilon)^{\lfloor \lambda t/\varepsilon \rfloor} \rightarrow e^{-\lambda t} \quad (\varepsilon \rightarrow 0),$$

because $(1 - \varepsilon)^{1/\varepsilon} \rightarrow e^{-1}$ and the floor alters the exponent by less than one. Whether a proof of work fits this picture — each candidate independent of the last — is a question about the trials, which the volumes on consensus take up.

Example 11.39 (A jittered schedule). A wallet that must poll each of n addresses about once a day waits $\Delta \sim \text{Exp}(1/T)$ between polls, with $T = 86400/n$ seconds: $\mathbb{E}[\Delta] = T$ by theorem 11.37, one expected poll per address per day. For $n = 3$, $T = 28800$ seconds, eight hours; the gap exceeds a full day with probability $e^{-3} = 0.0498$, and exceeds twice its mean with probability $e^{-2} = 0.1353$ whatever T is. By memorylessness, an observer who has seen no poll for a while has learnt nothing about when the next one comes.

Two clocks race, and clocks are restarted; both need joint events of several density-defined quantities, which theorem 11.32 provides only for pairs.

Definition 11.40 (Independent family of density-defined quantities). Density-defined quantities $X_1, \dots, X_n$ with densities $f_1, \dots, f_n$ form an *independent family* if $\Pr[X_1 \in I_1, \dots, X_n \in I_n] = \prod_i \Pr[X_i \in I_i]$ for all intervals $I_1, \dots, I_n$. As in theorem 11.32, the probability of a joint event $(X_1, \dots, X_n) \in R$, for a region $R \subseteq \mathbb{R}^n$ cut out by finitely many inequalities, is then taken to be

the iterated integral of $f_1(s_1) \cdots f_n(s_n)$ over R , in whichever order of integration is convenient (theorem 11.31). In particular a quantity that is a function of one block of the family is independent of a quantity that is a function of a disjoint block: integrate the two blocks separately. A sequence $X_1, X_2, \dots$ is an independent family if every finite prefix is one, in the manner of theorem 11.30.

Lemma 11.41 (Integrating out one quantity). *Let U be a density-defined quantity, with density f_U , that is a function of the first k members of an independent family $X_1, \dots, X_n$, and write $W = (X_{k+1}, \dots, X_n)$. Let R be a region in the (U, W) -coordinates whose section $\{s : (s, w) \in R\}$ is an interval for every fixed value w of W . Then*

$$\Pr[(U, W) \in R] = \int_{-\infty}^{\infty} f_U(s) \Pr[W \in R_s] ds, \quad R_s = \{w : (s, w) \in R\},$$

the inner probability being a joint event of $X_{k+1}, \dots, X_n$ alone.

Proof. Integrate the product density over the first k coordinates first, holding w fixed: the result is the probability that U lies in the interval $\{s : (s, w) \in R\}$, which by theorem 11.32 equals $\int f_U(s) 1[(s, w) \in R] ds$. What remains is the integral of $f_{k+1} \cdots f_n$ against this function of w ; exchanging the order and integrating over w first leaves, for each s , the inner probability $\Pr[W \in R_s]$. $\square$

Lemma 11.42 (Density of a sum). *Let U, V be independent density-defined quantities, nonnegative (their densities vanish on the negative axis). Then $U + V$ is density-defined, with density the convolution $(f_U * f_V)(x) = \int_0^x f_U(u) f_V(x - u) du$ for $x \geq 0$.*

Proof. For $0 \leq a \leq b$, the event $a \leq U + V \leq b$ is the region $\{(u, v) : u, v \geq 0, a \leq u + v \leq b\}$, so

$$\Pr[a \leq U + V \leq b] = \int_0^{\infty} f_U(u) \int_{\max(a-u, 0)}^{\max(b-u, 0)} f_V(v) dv du = \int_0^{\infty} f_U(u) \int_{\max(a, u)}^{\max(b, u)} f_V(x - u) dx du,$$

substituting $x = u + v$ in the inner integral. Exchanging the order, the region becomes $a \leq x \leq b$, $0 \leq u \leq x$, and the right-hand side is $\int_a^b (f_U * f_V)(x) dx$. With $a = 0$ and $b = \infty$ the left-hand side is 1, so $f_U * f_V$ is a density. $\square$

Lemma 11.43 (Two-clock race). *Let $X \sim \text{Exp}(\lambda)$ and $Y \sim \text{Exp}(\mu)$ be independent. Then, for every $u \geq 0$,*

$$\Pr[X < Y, X \leq u] = \frac{\lambda}{\lambda + \mu} (1 - e^{-(\lambda + \mu)u}),$$

symmetrically with the roles of (X, λ) and (Y, μ) exchanged, and $\Pr[X = Y] = 0$. Consequently $\Pr[X < Y] = \lambda / (\lambda + \mu)$; the winning time $\min(X, Y)$ is exponential with rate $\lambda + \mu$; and the winner is independent of the winning time,

$$\Pr[X < Y, \min(X, Y) \leq u] = \Pr[X < Y] \cdot \Pr[\min(X, Y) \leq u] \quad \text{for all } u \geq 0.$$

Proof. The event is the region $0 \leq s \leq u, t > s$ in the (X, Y) -plane. Integrating over t first (theorem 11.40),

$$\Pr[X < Y, X \leq u] = \int_0^u \lambda e^{-\lambda s} \Pr[Y > s] ds = \int_0^u \lambda e^{-(\lambda + \mu)s} ds = \frac{\lambda}{\lambda + \mu} (1 - e^{-(\lambda + \mu)u}).$$

The diagonal $s = t$ has inner integral zero, so $\Pr[X = Y] = 0$, and $u = \infty$ gives $\Pr[X < Y]$. On the event $X < Y$ the winning time is X , so the displayed formula is $\Pr[X < Y, \min(X, Y) \leq u]$; adding its mirror for $Y < X$ gives $\Pr[\min(X, Y) \leq u] = 1 - e^{-(\lambda+\mu)u}$, whence $\min(X, Y)$ is exponential with rate $\lambda + \mu$ by the characterisation in theorem 11.37, and the formula is exactly the product $\Pr[X < Y] \cdot \Pr[\min(X, Y) \leq u]$. Differencing at two values of u extends the factorisation from half-lines to intervals. $\square$

Definition 11.44 (Exponential clock process). An exponential clock of rate λ — a Poisson process, once theorem 11.47 is in hand — is an independent family $G_1, G_2, \dots$ of $\text{Exp}(\lambda)$ quantities, the gaps. Its arrival times are the partial sums $S_n = G_1 + \dots + G_n$, with $S_0 = 0$, and its arrival count up to time $t \geq 0$ is $N(t) = \sup\{n : S_n \leq t\}$, the number of arrivals in $[0, t]$, a value in $\mathbb{N} \cup \{\infty\}$ that theorem 11.47 shows finite with probability one. The event $N(t) = n$ is $S_n \leq t < S_{n+1}$, a joint event of the first $n + 1$ gaps.

Lemma 11.45 (Arrival-time density). The n -th arrival time S_n of an exponential clock of rate λ has density $f_n(s) = \lambda^n s^{n-1} e^{-\lambda s} / (n-1)!$ for $s \geq 0$.

Proof. Induction on n : f_1 is the exponential density, and $S_{n+1} = S_n + G_{n+1}$ with S_n a function of the first n gaps, hence independent of G_{n+1} (theorem 11.40), so theorem 11.42 gives

$$f_{n+1}(s) = \int_0^s \frac{\lambda^n u^{n-1} e^{-\lambda u}}{(n-1)!} \lambda e^{-\lambda(s-u)} du = \frac{\lambda^{n+1} e^{-\lambda s}}{(n-1)!} \int_0^s u^{n-1} du = \frac{\lambda^{n+1} s^n e^{-\lambda s}}{n!}. \quad \square$$

Definition 11.46 (Poisson law). A random variable K with values in $\mathbb{N}$ is Poisson with mean $\nu > 0$ if $\Pr[K = k] = e^{-\nu} \nu^k / k!$ for every $k \geq 0$. The weights sum to 1 by the exponential series $e^\nu = \sum_{k \geq 0} \nu^k / k!$, and $\mathbb{E}[K] = \sum_{k \geq 1} k e^{-\nu} \nu^k / k! = \nu e^{-\nu} \sum_{k \geq 1} \nu^{k-1} / (k-1)! = \nu$, so the parameter is indeed the mean.

An arrival count is a function of the gaps of a clock family, so it lives on no finite or countable space, and the expectation of theorem 11.11 does not literally apply to it. One convention serves every such count in this unit — $N(t)$ above and the count X of theorem 11.51 below. Let X be a variable with values in $\mathbb{N} \cup \{\infty\}$ that is a function of the gaps, each event $X = x$ with x finite being a joint event of finitely many gaps, with the probability of theorem 11.40. Events $X \in B$, for sets B of finite values, have the probabilities $\sum_{x \in B} \Pr[X = x]$ (theorem 11.31), and when X is finite with probability 1 its mean is defined by the law,

$$\mathbb{E}[X] = \sum_{x \geq 0} x \Pr[X = x],$$

a sum of nonnegative terms, permitted to be $+\infty$ as in the convention preceding theorem 11.33. On a finite or countable space this sum is the second expression of theorem 11.11, the expectation regrouped by value (theorem 11.27), so the two definitions agree whenever both apply.

Theorem 11.47 (Poisson arrival counts). For an exponential clock of rate λ and every $t \geq 0$, the arrival count $N(t)$ is Poisson with mean λt :

$$\Pr[N(t) = n] = e^{-\lambda t} \frac{(\lambda t)^n}{n!}, \quad n = 0, 1, 2, \dots$$

In particular $N(t)$ is finite with probability 1 and $\mathbb{E}[N(t)] = \lambda t$: a rate- λ clock delivers λ arrivals per

unit time on average.

Proof. For $n = 0$ the event is $G_1 > t$, of probability $e^{-\lambda t}$. For $n \geq 1$ the event $S_n \leq t < S_n + G_{n+1}$ is, in the (S_n, G_{n+1}) -coordinates, the region $s \leq t, g > t - s$, whose section at fixed g is the interval $(t - g, t]$; the quantity S_n is a function of the first n gaps with the density of theorem 11.45, so theorem 11.41 gives

$$\Pr[N(t) = n] = \int_0^t \frac{\lambda^n s^{n-1} e^{-\lambda s}}{(n-1)!} \Pr[G_{n+1} > t - s] ds = \frac{\lambda^n e^{-\lambda t}}{(n-1)!} \int_0^t s^{n-1} ds = e^{-\lambda t} \frac{(\lambda t)^n}{n!}.$$

The event $N(t) < \infty$ is the disjoint union of the events $N(t) = n$, so its probability is the sum of the weights (theorem 11.31), which is 1 (theorem 11.46); and the mean — the sum over the law, by the convention above — is the one computed in theorem 11.46. $\square$

The tail bounds now. Markov's inequality applied to X itself decays only like $1/a$; applied to e^{tX} it decays exponentially, and the free parameter t is then tuned to the threshold. That is the whole of the Chernoff method.

Definition 11.48 (Moment generating function). For a real-valued random variable X with finite or countable image — one on a finite or countable space, or an arrival count of a clock family in the sense of the convention preceding theorem 11.47, restricted to its finite values (the value ∞ has probability zero there), whose law is countably additive over its values: $\Pr[X \in B] = \sum_{x \in B} \Pr[X = x]$ for every set B of values (theorem 11.31) — and a real t , the *moment generating function* is

$$M_X(t) = \sum_x e^{tx} \Pr[X = x] \in (0, \infty],$$

the sum running over the finite values of X ; its terms are positive, so it either converges or is $+\infty$. On a countable space it is $\mathbb{E}[e^{tX}]$, the expectation regrouped by value, and $M_X(0) = \sum_x \Pr[X = x] = 1$.

Theorem 11.49 (The Chernoff method). *Let X be as above, a real, and $t > 0$ with $M_X(t) < \infty$. Then*

$$\Pr[X \geq a] \leq e^{-ta} M_X(t),$$

and hence $\Pr[X \geq a] \leq \inf_t e^{-ta} M_X(t)$, the infimum over the $t > 0$ at which M_X is finite.

Proof. The events $X = x$ are disjoint, so $\Pr[X \geq a] = \sum_{x \geq a} \Pr[X = x]$ by countable additivity (theorem 11.28, or the clause of theorem 11.48 for a clock family). Since $t > 0$, each term with $x \geq a$ satisfies $\Pr[X = x] \leq e^{t(x-a)} \Pr[X = x]$, and summing gives $\Pr[X \geq a] \leq e^{-ta} \sum_x e^{tx} \Pr[X = x]$, the terms with $x < a$ being nonnegative. This is Markov's inequality (theorem 11.15) applied to e^{tX} , written on the law of X . $\square$

Corollary 11.50 (Chernoff bound for a Poisson variable). *Let K be Poisson with mean ν and let $a > \nu$. Then $M_K(t) = \exp(\nu(e^t - 1))$ for every real t , and*

$$\Pr[K \geq a] \leq e^{a-\nu} \left(\frac{\nu}{a}\right)^a.$$

In particular, for $\nu = \mu L$ and $a = cL$ with $c > \mu$, the bound reads $e^{L(c-\mu)} (c/\mu)^{-cL}$.

Proof. Expanding, $M_K(t) = \sum_k e^{tk} e^{-\nu} \nu^k / k! = e^{-\nu} \sum_k (\nu e^t)^k / k! = e^{-\nu} e^{\nu e^t}$. The exponent $\phi(t) = -ta + \nu(e^t - 1)$ of the Chernoff bound has $\phi'(t) = -a + \nu e^t$ and $\phi''(t) = \nu e^t > 0$, so it is minimised at $e^t = a/\nu$, a positive t exactly when $a > \nu$; there $\phi = -a \log(a/\nu) + a - \nu$, with $\log$ the natural logarithm — base-free log is natural throughout, base two being written $\log_2$ — and exponentiating gives the bound. Substituting $\nu = \mu L$ and $a = cL$ gives $e^{cL - \mu L} (\mu/c)^{cL}$. $\square$

Theorem 11.51 (Negative-binomial arrival count). *Let two exponential clocks be independent — their gaps together form one independent family — with rates λ , the honest clock, and μ , the adversary clock, and let X be the number of adversary arrivals strictly before the L -th honest arrival, for a fixed $L \geq 1$. Then X is finite with probability 1, with the negative binomial law*

$$\Pr[X = m] = \binom{m+L-1}{m} p^L q^m, \quad p = \frac{\lambda}{\lambda + \mu}, \quad q = \frac{\mu}{\lambda + \mu}, \quad m = 0, 1, 2, \dots,$$

of mean $Lq/p = L\mu/\lambda$, and, for every real t with $\mu e^t < \lambda + \mu$,

$$M_X(t) = \left(\frac{\lambda}{\lambda + \mu - \mu e^t} \right)^L;$$

with the honest rate normalised to 1 this is $M_X(t) = (1/(1 + \mu - \mu e^t))^L$. Only the ratio μ/λ enters.

Proof. Write S_L for the L -th honest arrival time, a function of the honest gaps with the density f_L of theorem 11.45, and N_A for the adversary's arrival count. The event $X = m$ differs from $N_A(S_L) = m$ only on the events $S_L = A_1 + \dots + A_m$ and $S_L = A_1 + \dots + A_{m+1}$, with A_i the adversary gaps; each of these, in the coordinates $(S_L, A_1, \dots, A_{m+1})$, has a single point as its section at fixed adversary gaps, hence probability zero (theorem 11.32), so $\Pr[X = m] = \Pr[N_A(S_L) = m]$. The latter event is the region $A_1 + \dots + A_m \leq S_L < A_1 + \dots + A_{m+1}$, whose section at fixed adversary gaps is an interval. By theorem 11.41 and theorem 11.47,

$$\begin{aligned} \Pr[X = m] &= \int_0^\infty f_L(s) \Pr[N_A(s) = m] ds = \frac{\lambda^L \mu^m}{(L-1)! m!} \int_0^\infty s^{m+L-1} e^{-(\lambda+\mu)s} ds \\ &= \frac{\lambda^L \mu^m}{(L-1)! m!} \cdot \frac{(m+L-1)!}{(\lambda + \mu)^{m+L}}, \end{aligned}$$

using $\int_0^\infty s^k e^{-\alpha s} ds = k!/\alpha^{k+1}$ for $\alpha > 0$, which k integrations by parts establish. Rearranging the factorials gives the binomial coefficient and the powers of p and q .

For the generating function, set $x = qe^t$, so that $0 \leq x < 1$ exactly when $\mu e^t < \lambda + \mu$. Then $M_X(t) = p^L \sum_{m \geq 0} \binom{m+L-1}{m} x^m$, and we claim the series is $(1-x)^{-L}$, by induction on L . For $L = 1$ it is the geometric series. For the step, write $T_L(n) = \sum_{m=0}^n \binom{m+L-1}{m} x^m$ for the partial sums. Pascal's rule $\binom{m+L}{m} = \binom{m+L-1}{m-1} + \binom{m+L-1}{m}$ — sort the m -subsets of $m+L$ items, which the coefficients count (theorem 6.12), by whether they contain a distinguished item — gives, on subtracting $x T_{L+1}(n)$ from $T_{L+1}(n)$ term by term,

$$(1-x) T_{L+1}(n) = T_L(n) - \binom{n+L}{n} x^{n+1}.$$

Hence $T_{L+1}(n) \leq T_L(n)/(1-x) \leq (1-x)^{-L-1}$ by the induction hypothesis: the partial sums $T_{L+1}(n)$ increase and are bounded, so the series converges, and its terms $\binom{n+L}{n} x^n$ tend to zero. Letting $n \rightarrow \infty$ in the display gives $(1-x) \sum_{m \geq 0} \binom{m+L}{m} x^m = (1-x)^{-L}$, the claim for $L+1$. Hence $M_X(t) = (p/(1-qe^t))^L$, which is the displayed form after multiplying numerator and denominator by $\lambda + \mu$. At $t = 0$ this gives $\sum_m \Pr[X = m] = 1$; the event $X < \infty$ is the disjoint

union of the events $X = m$, so its probability is this sum (theorem 11.31), and X is finite with probability 1. For the mean — the sum over the law, by the convention preceding theorem 11.47 — $m \binom{m+L-1}{m} = L \binom{m+L-1}{m-1}$, so $\mathbb{E}[X] = Lq p^L \sum_{k \geq 0} \binom{k+L}{k} q^k = Lq p^L (1-q)^{-L-1} = Lq/p$ by the same series with $L+1$ in place of L . $\square$

Remark 11.52 (The coin-sequence form). The same law arises without clocks. In an infinite sequence of independent trials (theorem 11.30) succeeding with probability p , the number of failures strictly before the L -th success equals m exactly when the $(m+L)$ -th trial succeeds and, among the first $m+L-1$, precisely m fail: $\binom{m+L-1}{m}$ orders, one per choice of the m failing positions (theorem 6.12), each of probability $p^{L-1} q^m \cdot p$. The two-clock race and memorylessness (theorem 11.43 and theorem 11.37) are what identify the clock model's attribution sequence — which clock fires next — with such a coin sequence of success probability $\lambda/(\lambda+\mu)$; later volumes carry out that identification.

Theorem 11.53 (Chernoff bound for the arrival count). *In the setting of theorem 11.51 with honest rate 1 and adversary rate μ , for every $c > \mu$,*

$$\Pr[X \geq cL] \leq \left[\left(\frac{1+c}{1+\mu} \right)^{1+c} \left(\frac{\mu}{c} \right)^c \right]^L,$$

the Chernoff bound optimised at $e^t = c(1+\mu)/(\mu(1+c))$. The base in brackets is strictly less than 1, so the bound decays exponentially in L .

Proof. The count X is finite with probability one with values in $\mathbb{N}$ (theorem 11.51) and has a law countably additive over its values (theorem 11.31), so theorem 11.49 applies: with theorem 11.51, $\Pr[X \geq cL] \leq e^{-tcL} M_X(t) = e^{L\phi(t)}$ with $\phi(t) = -tc - \log(1+\mu - \mu e^t)$, for $0 < t < \log((1+\mu)/\mu)$. On that interval

$$\phi'(t) = -c + \frac{\mu e^t}{1+\mu - \mu e^t}, \quad \phi''(t) = \frac{\mu e^t (1+\mu)}{(1+\mu - \mu e^t)^2} > 0,$$

so ϕ' is strictly increasing (positive derivative) and vanishes at most once, namely where $\mu e^t (1+\mu) = c(1+\mu)$, that is, at $e^{t^*} = c(1+\mu)/(\mu(1+c))$; this lies in the interval because $c > \mu$ gives $e^{t^*} > 1$ and $c/(1+c) < 1$ gives $e^{t^*} < (1+\mu)/\mu$. Thus ϕ decreases before t^* and increases after it, and $\phi(t^*)$ is its minimum on the interval. At t^* , $1+\mu - \mu e^{t^*} = (1+\mu)/(1+c)$, so $e^{\phi(t^*)} = (e^{t^*})^{-c} (1+c)/(1+\mu)$, which is the bracketed base. Finally $\phi(0) = 0$ and ϕ decreases on $[0, t^*]$, since $\phi'(0) = \mu - c < 0$, so $\phi(t^*) < 0$: the base is below 1. $\square$

Example 11.54 (The Poisson expression is not a bound for the arrival count). With the honest rate 1, the arrival count X has mean μL , the same as the Poisson variable of theorem 11.50 with $\nu = \mu L$, which counts adversary arrivals in a window of fixed length L ; the two laws differ because the window here ends at the random time S_L . Substituting one for the other is tempting and wrong. At $\mu = 1/2$ and $c = 3/4$, the exact tail $\Pr[X \geq cL]$ (summing theorem 11.51), the bound of theorem 11.53, and the Poisson expression $e^{L(c-\mu)} (c/\mu)^{-cL}$ are

L	exact tail	clock bound	Poisson expression
40	$5.05 \cdot 10^{-2}$	$2.53 \cdot 10^{-1}$	$1.15 \cdot 10^{-1}$
200	$1.09 \cdot 10^{-4}$	$1.04 \cdot 10^{-3}$	$2.00 \cdot 10^{-5}$
400	$8.23 \cdot 10^{-8}$	$1.08 \cdot 10^{-6}$	$4.00 \cdot 10^{-10}$

At $L = 200$ and $L = 400$ the Poisson expression lies *below* the exact tail, so it is not an upper bound for the arrival count; its base per unit of L is 0.9473, against 0.9662 for the valid bound. A bound stated for the wrong law is not a bound.

Later volumes build block-discovery and extraction analyses directly on this base: block clocks are the exponential clocks above, their races and arrival counts the lemmas of this unit, block-attribution sequences are infinite sequences of independent trials, extractor running times are the waiting times of the previous unit (the geometric waiting time and Wald's identity), and the limit arguments instantiate theorem 11.29.

11.8 Asymptotic notation

The discussion now turns from probability to growth rates. To address efficiency and hardness, we must compare how functions of an integer parameter grow as that parameter tends to infinity. The Bachmann–Landau notation makes such comparisons precise. Throughout, functions map $\mathbb{N}$ (or a cofinite subset of it) to the nonnegative reals $\mathbb{R}_{\geq 0}$, and “for sufficiently large n ” means “for all n beyond some threshold n_0 ”.

Definition 11.55 (Big-O, Omega, Theta, little-o, little-omega). Let $f, g: \mathbb{N} \rightarrow \mathbb{R}_{\geq 0}$ with g eventually positive.

- $f = O(g)$ (read “ f is big-O of g ”, an asymptotic upper bound) if there exist constants $c > 0$ and n_0 with $f(n) \leq c g(n)$ for all $n \geq n_0$.
- $f = \Omega(g)$ (asymptotic lower bound) if there exist $c > 0$ and n_0 with $f(n) \geq c g(n)$ for all $n \geq n_0$; equivalently, $f = \Omega(g)$ if and only if $g = O(f)$.
- $f = \Theta(g)$ (tight bound) if $f = O(g)$ and $f = \Omega(g)$; i.e. there are $c_1, c_2 > 0$ and n_0 with $c_1 g(n) \leq f(n) \leq c_2 g(n)$ for $n \geq n_0$.
- $f = o(g)$ (read “little-o”, strictly smaller order) if for every $c > 0$ there is n_0 with $f(n) \leq c g(n)$ for all $n \geq n_0$; equivalently $\lim_{n \rightarrow \infty} f(n)/g(n) = 0$.
- $f = \omega(g)$ (strictly larger order) if $g = o(f)$; equivalently $\lim_{n \rightarrow \infty} f(n)/g(n) = \infty$.

Remark 11.56. The “=” in “ $f = O(g)$ ” is traditional but denotes set membership: $O(g)$ is the class of all functions bounded above by a constant multiple of g , and $f = O(g)$ means $f \in O(g)$. One never reads such equations right to left. We follow the customary abuse of notation.

Example 11.57. The polynomial $3n^2 + 10n + 7$ is $\Theta(n^2)$: for $n \geq 1$ one has $3n^2 \leq 3n^2 + 10n + 7 \leq 20n^2$, with equality on the right at $n = 1$, so the constant 20 is exact there. Also $n^2 = o(n^3)$, since $n^2/n^3 = 1/n \rightarrow 0$; $\log_2 n = o(n)$; $n^{100} = o(2^n)$ (any polynomial is little-o of any exponential with base > 1); and $n! = \omega(2^n)$. A cautionary pair: $2^n = o(2^{n+1})$ is *false*, but $2^n = \Theta(2^{n+1})$ is *true*, since $2^{n+1} = 2 \cdot 2^n$ is within a constant factor.

Proposition 11.58 (Useful closure facts). Let $f_1 = O(g_1)$ and $f_2 = O(g_2)$. Then $f_1 + f_2 = O(\max(g_1, g_2)) = O(g_1 + g_2)$ and $f_1 f_2 = O(g_1 g_2)$. The same closure statements hold with O replaced uniformly by Ω or Θ . Moreover $O(\cdot)$ is transitive: $f = O(g)$ and $g = O(h)$ imply $f = O(h)$.

Proof. Choose witnesses (c_1, n_1) and (c_2, n_2) for the two hypotheses and let $n_0 = \max(n_1, n_2)$. For $n \geq n_0$, $f_1(n) + f_2(n) \leq c_1 g_1(n) + c_2 g_2(n) \leq (c_1 + c_2) \max(g_1(n), g_2(n))$, and $\max(g_1, g_2) \leq g_1 + g_2 \leq 2 \max(g_1, g_2)$, so the two right-hand forms agree up to a constant; this proves the sum rule. For products, $f_1(n) f_2(n) \leq c_1 c_2 g_1(n) g_2(n)$. Transitivity: $f \leq c g$ and $g \leq c' h$ eventually give $f \leq cc' h$ eventually. The Ω statements follow by the equivalence $f = \Omega(g) \Leftrightarrow g = O(f)$, and Θ by combining the two. $\square$

11.9 Polynomial, exponential, and negligible functions

The payoff of this subsection is a closure theorem (theorem 11.63): the class of *negligible* functions — those shrinking faster than the reciprocal of every polynomial — is stable under exactly the operations that security proofs perform, addition and multiplication by polynomials, which is why “negligible” is the right formalisation of “ignorable”. We first classify the growth rates relevant to cryptography. Fix a distinguished variable $\lambda \in \mathbb{N}$, the *security parameter* (discussed in detail in §11.11); informally, larger λ means more security and larger keys. Resources (running time, key length) and success probabilities are functions of λ .

Definition 11.59 (Polynomial and super-polynomial). A function $f: \mathbb{N} \rightarrow \mathbb{R}_{\geq 0}$ is *polynomially bounded*, written $f = \text{poly}(\lambda)$, if there is a constant c (and a threshold) with $f(\lambda) \leq \lambda^c + c$ for all large λ ; equivalently, $f(\lambda) = O(\lambda^c)$ for some constant c . It is *super-polynomial* if it grows faster than every polynomial, i.e. $f(\lambda) = \omega(\lambda^c)$ for every constant c (equivalently, $\lambda^c = o(f(\lambda))$ for all c).

Definition 11.60 (Exponential and sub-exponential). A function f is *exponential* if $f(\lambda) = 2^{\Omega(\lambda)}$; the prototypical case is $f(\lambda) = 2^{c\lambda}$ for a constant $c > 0$. A super-polynomial function with $f(\lambda) = 2^{o(\lambda)}$ is *sub-exponential*; examples are $\lambda^{\log_2 \lambda} = 2^{(\log_2 \lambda)^2}$ and the index-calculus running time $L_p[1/3] = 2^{\Theta(\lambda^{1/3}(\log \lambda)^{2/3})}$ in the bit length $\lambda = \log_2 p$ (theorem 7.22). Every exponential function is super-polynomial; the converse fails by these examples.

The central cryptographic notion is that of a negligible function. A negligible probability may be ignored because no efficient (polynomially many) repetition of an experiment can amplify it to a noticeable chance — the closure theorem below makes this precise.

Definition 11.61 (Negligible function). A function $\nu: \mathbb{N} \rightarrow \mathbb{R}_{\geq 0}$ is *negligible*, written $\nu = \text{negl}(\lambda)$, if for every positive constant c there exists λ_0 such that

$$\nu(\lambda) < \frac{1}{\lambda^c} \quad \text{for all } \lambda \geq \lambda_0.$$

Equivalently, $\nu(\lambda) = o(\lambda^{-c})$ for every constant $c > 0$; equivalently, $\lambda^c \nu(\lambda) \rightarrow 0$ as $\lambda \rightarrow \infty$ for every constant c . A function is *non-negligible* if it is not negligible, i.e. $f(\lambda) \geq \lambda^{-c}$ for infinitely many λ , for some constant c ; it is *noticeable* if there exist c and λ_0 with $f(\lambda) \geq \lambda^{-c}$ for all $\lambda \geq \lambda_0$ (noticeable implies non-negligible, but not conversely); it is *overwhelming* if $1 - f$ is negligible.

Example 11.62. The functions $2^{-\lambda}$, $2^{-\sqrt{\lambda}}$, and $\lambda^{-\log \lambda}$ are negligible. For $2^{-\lambda}$: given any c , the product $2^{-\lambda} \lambda^c = \lambda^c / 2^\lambda$ tends to 0 (a polynomial over an exponential), meeting the definition. By contrast, $1/\lambda^{100}$ is *not* negligible — taking $c = 101$ shows $1/\lambda^{100} > 1/\lambda^{101}$ for all $\lambda \geq 2$ — it is noticeable, even though it tends to 0. This is the key subtlety: tending to zero does not suffice; a negligible function must beat every inverse polynomial.

Proposition 11.63 (Closure properties of negligible functions). *Let ν, μ be negligible and let p be a polynomially bounded function. Then:*

1. $\nu + \mu$ is negligible (the class is closed under addition, hence under any fixed finite sum).
2. $p \cdot \nu$ is negligible (a polynomial times a negligible function is negligible).

3. If $f(\lambda) \leq \nu(\lambda)$ for all large λ , then f is negligible (domination by a negligible function).
4. Summing $p(\lambda)$ functions, each bounded by a single negligible function ν , yields a negligible function; in particular a union bound over polynomially many negligible events is negligible.

Proof. The mechanism throughout is an exponent shift: to beat λ^{-c} , invoke negligibility at a strictly larger exponent, so that the surplus absorbs the combining factor. The same shift is the standard mechanism behind hybrid and union-bound steps in security proofs.

(1) Fix c . Applying the definition at the exponent $c + 1$, there is a threshold beyond which $\nu(\lambda), \mu(\lambda) < \lambda^{-(c+1)} \leq \frac{1}{2}\lambda^{-c}$ (the latter once $\lambda \geq 2$); adding the two bounds gives $\nu(\lambda) + \mu(\lambda) < \lambda^{-c}$ for all large λ .

(2) Suppose $p(\lambda) \leq \lambda^d$ for large λ (and some constant d). Fix c . By negligibility applied at the exponent $c + d$, $\nu(\lambda) < \lambda^{-(c+d)}$ for large λ , whence $p(\lambda)\nu(\lambda) < \lambda^d\lambda^{-(c+d)} = \lambda^{-c}$ for large λ . Hence $p\nu$ is negligible.

(3) Immediate: if $f \leq \nu$ eventually and $\nu < \lambda^{-c}$ eventually, then $f < \lambda^{-c}$ eventually, for every c .

(4) Write the sum $\Sigma(\lambda) = \sum_{i=1}^{p(\lambda)} \nu_i(\lambda)$, where each $\nu_i \leq \nu$ by hypothesis (in security proofs all the ν_i arise from one bound). Then $\Sigma(\lambda) \leq p(\lambda)\nu(\lambda)$, which is negligible by (2); conclude by (3). $\square$

Remark 11.64 (Why negligible is the threshold of security). Suppose an adversary wins a game with negligible probability $\nu(\lambda)$. If it repeats the attack $p(\lambda)$ times (polynomially many, the most an efficient adversary can afford), the probability that some repetition succeeds is, by the union bound, at most $p(\lambda)\nu(\lambda)$ — still negligible by theorem 11.63(2). Thus negligible success probability remains negligible under any efficient amplification: it is genuinely ignorable. Conversely, a noticeable success probability $\geq \lambda^{-c}$ can be boosted to a constant by λ^c repetitions. The negligible/non-negligible split is the formal backbone of the phrase “the scheme is secure except with negligible probability”: security asserts a negligible advantage, and an attack means a non-negligible one. (A merely non-negligible — not noticeable — success probability is boosted to a constant only on the infinitely many λ where the inverse-polynomial bound holds, which already suffices to contradict a security definition.)

11.10 Algorithms, running time, and PPT

It remains to make “efficient algorithm” and “infeasible” precise. We adopt the standard model: algorithms are Turing machines (equivalently, up to polynomial factors, programs in any reasonable model of computation), inputs and outputs are finite binary strings, and the measured resource is the number of elementary steps as a function of input length. Write $\{0, 1\}^*$ for the set of all finite binary strings and $|x|$ for the length of $x \in \{0, 1\}^*$.

Definition 11.65 (Deterministic polynomial-time algorithm). An algorithm M runs in (*deterministic*) *polynomial time* if there is a polynomial p such that, on every input $x \in \{0, 1\}^*$, M halts within at most $p(|x|)$ steps and produces an output. The class of decision problems solvable by such machines is P .

Determinism is insufficient for cryptography: key generation and sampling require randomness. We model randomness by granting the machine an auxiliary tape of independent uniform random bits (the “coins”).

Definition 11.66 (Probabilistic algorithm and PPT). A *probabilistic* (or *randomised*) algorithm M is a Turing machine with an additional read-only *random tape* initialised with independent

uniformly random bits $r \in \{0,1\}^{\mathbb{N}}$. For a fixed input x , the output $M(x)$ is a random variable over the choice of r ; write $M(x;r)$ to display the coins explicitly, and then $M(x;\cdot)$ is a deterministic function. The algorithm runs in *probabilistic polynomial time* (it is *PPT*) if there is a polynomial p such that, for every input x and every choice of coins r , M halts within $p(|x|)$ steps. (Bounding the time for every r , not merely in expectation, is the standard “strict” notion and ensures M ever reads only $p(|x|)$ coins.)

Remark 11.67 (Non-uniformity and circuits). There are two standard ways to formalise an efficient adversary. The *uniform* model is a single PPT machine that works for all input lengths, as above. The *non-uniform* model allows a separate algorithm (equivalently, a Boolean circuit) of size $\text{poly}(\lambda)$ for each λ , which may be hard-wired with an arbitrary polynomial-length “advice” string depending on λ . Non-uniform adversaries are at least as powerful as uniform ones and are often more convenient in reduction proofs, because the advice can encode a “best” attack. The asymptotic theory of this section is identical for both models; security definitions in the Halo 2/Orchard setting typically state security against non-uniform PPT adversaries, the more conservative choice.

With efficiency pinned down, we can state the strong hardness notion used in cryptography. The guiding principle is the Cobham–Edmonds thesis: “efficiently computable” means “computable in polynomial time”. Feasibility and the infeasibility notion below are useful opposite regimes, but they are not logical negations: intermediate success probabilities can fall into neither class.

Definition 11.68 (Computational infeasibility). A computational task parameterised by the security parameter λ is (*computationally*) *feasible* if there is a PPT algorithm solving it with overwhelming success probability. It is *infeasible* if for every PPT algorithm A the probability that A solves it is negligible in λ :

$$\Pr[A \text{ solves the task on parameter } \lambda] = \text{negl}(\lambda),$$

where the probability is over the task’s randomness and A ’s coins. Equivalently: no efficient algorithm succeeds with non-negligible probability.

Remark 11.69 (Reading a hardness assumption). Cryptographic assumptions are infeasibility statements. For instance, the discrete-logarithm assumption in a family of groups $\{\mathbb{G}_\lambda\}$ of order $q(\lambda)$ asserts: for every PPT A ,

$$\Pr[A(g, g^x) = x] = \text{negl}(\lambda),$$

the probability taken over a uniformly chosen generator g of $\mathbb{G}_\lambda$, a uniform $x \leftarrow \mathbb{Z}/q\mathbb{Z}$, and A ’s coins. The entire edifice of Halo 2 and Orchard rests on such asymptotic infeasibility assumptions; choosing a parameter size does not prove them. Separately, concrete analysis asks how much work the best known attacks need at one fixed size. Generic discrete-log attacks take $\Theta(\sqrt{q})$ group operations by the birthday/rho phenomenon of theorem 11.20; absent a faster structure-specific attack, an order near $2^{2\lambda}$ therefore targets about λ bits of generic-group security. The Pasta orders are just above 2^{254} , giving roughly 2^{127} generic work. The asymptotic language permits the terms “negligible” and “polynomial”; concrete estimates remain conditional on the attacks presently known.

11.11 The security parameter

The section closes by making explicit the parameter that threads through every preceding definition, and by paying its opening debt.

Definition 11.70 (Security parameter). The *security parameter* is a positive integer $\lambda \in \mathbb{N}$ supplied to all algorithms of a cryptographic scheme, conventionally in unary as the string 1^λ (so that “polynomial in the input length” coincides with “polynomial in λ ”). It governs simultaneously:

- the *sizes* of objects — key lengths and encodings, each $\text{poly}(\lambda)$; in a generic discrete-log instantiation one commonly chooses a group order near $2^{2\lambda}$;
- the *running time* of honest algorithms, which must be $\text{poly}(\lambda)$; and
- the *security level*: every PPT adversary’s advantage must be $\text{negl}(\lambda)$.

Remark 11.71 (Why unary, and what λ buys). Passing 1^λ (a string of λ ones, of length λ) rather than the integer λ in binary (length $\log_2 \lambda$) is a deliberate convention: it forces “polynomial time in the input length” to mean polynomial in λ itself, so an honest key generator may take time $\text{poly}(\lambda)$ to produce λ -bit keys. In a scheme satisfying the security definition, increasing λ raises honest cost only polynomially while every PPT adversary’s success remains negligible; in common concrete families, the best known attack cost grows exponentially. Asymptotic security is the statement that this gap holds for all sufficiently large λ ; the practitioner then fixes a single value (commonly $\lambda = 128$, targeting 2^{128} -operation attacks) and checks the concrete estimates against that target — the just-over- 2^{254} Pallas/Vesta group orders give roughly 2^{127} generic group operations (section 10; Zcash protocol specification §5.4.9.6), hash outputs of ≥ 256 bits per the birthday bound of theorem 11.18, and sampling slack of at least 128 extra bits per theorem 11.24 (the deployed ToScalar of theorem 11.26 reduces 512 bits, a slack of 257). Every “negligible” in the remainder of the series is implicitly a function of this λ .

The opening cheque can now be cashed, with its two layers kept separate. Asymptotic infeasibility (theorem 11.68) quantifies over every probabilistic polynomial-time forger — uniform or hard-wired with advice (theorem 11.67) — and requires its success to be negligible in λ , a bound that no polynomial number of repetitions amplifies to a noticeable chance (theorem 11.64). A concrete “128-bit” estimate instead says that the best known applicable attack at a fixed parameter size costs on the order of 2^{128} operations; it is an attack estimate, not a theorem about every possible algorithm. Pasta’s just-over- 2^{254} group orders give roughly 2^{127} generic group operations and are conventionally placed near that target. The forger of the opening paragraph may still guess a key and win with astronomical luck; the formal promise is negligible success against every PPT forger, conditional on the stated cryptographic assumptions.

Part II

Crypto Guide: Cryptographic primitives from scratch

Abstract

This volume of *The Zcash Arboretum* assembles the cryptographic toolbox: the primitives from which a shielded protocol is built, each defined, constructed, and proved against a named enemy. It assumes the *Math Guide*, whose groups, finite fields, elliptic curves, probability, and model of computation are used throughout without re-derivation.

Every guarantee here is a claim about an adversary, and every security definition is a game: a challenger, an efficient adversary, and an advantage that must be negligible. A proof is a reduction that turns any winning adversary into a solver for a long-attacked computational problem, with the reduction's loss accounted for down to a concrete bit-security budget. Under that discipline the volume denies, in turn: the cryptanalyst who would recover the exponent behind a public group element, and with it every key; the forger by coincidence, who would collide two documents into one digest or replay a digest outside its role; the peeker and the equivocator on either side of a sealed envelope, who would read a commitment early or open it to a different value; the oracle distinguisher, who would tell a short secret key from blind chance; the owner of the channel, who would read, splice, or counterfeit what passes over it; the woman in the middle, who would terminate both ends of a “secure” conversation at her own desk; the forger of signatures, who would exhibit an authorisation the signer never made; the cheating prover and the curious verifier, who would certify a falsehood or squeeze a transcript for the witness it hides; and the teller of lies about sets, who would prove membership in a collection that never recorded it. What survives their attentions—hash functions, commitments, pseudorandom functions, authenticated encryption, key agreement, signatures, zero-knowledge arguments, and Merkle trees—is the toolbox the deployed protocol spends.

1 The provable-security model

Every claim this volume makes ends, once unwound, in a sentence about an enemy. No forger produces a valid signature on a message the signer never signed; no observer tells an encryption of m_0 from an encryption of m_1 ; no prover opens one commitment to two different values. Before any such sentence can be a theorem, the enemy must be a mathematical object: something with a precisely delimited repertoire, about which a universally quantified statement can be proved. The preceding volume supplied the algebraic raw material—groups in which discrete logarithms appear hard, finite fields, elliptic curves—but an object is not yet a guarantee. This section builds the adversary, and around it the framework in which “the scheme is secure” becomes a claim with mathematical content.

The central methodological idea is the following. One almost never knows how to prove unconditionally that a cryptographic scheme is secure; such a proof would typically imply $P \neq NP$ —that problems whose solutions can be checked in polynomial time are not all solvable in polynomial time—and a great deal more. Cryptography therefore adopts a *relativised* standard of proof. One isolates a small number of computational problems—factoring, discrete logarithm, decisional Diffie–Hellman, and the like—that have resisted decades of attack, and *reduces* the security of the schemes to the conjectured hardness of those problems. A proof of security is then a proof of an implication: *if* the underlying problem is hard, *then* the scheme is secure. Equivalently, in its contrapositive and more operationally useful form: *any* adversary that breaks the scheme can be mechanically transformed into an algorithm that solves the underlying hard problem. Since no such algorithm is believed to exist, one concludes that no such adversary exists.

To render any of this rigorous, we must pin down four things: what counts as a “scheme”, parametrised by a notion of input size; what counts as an “adversary”; what it means for an

adversary to “break” a scheme, measured by a quantity called its *advantage*; and what it means for one problem to reduce to another. We develop these in turn, followed by the two idealisations under which proofs proceed (the standard model and the random oracle model), and finally the translation between the clean asymptotic language and the concrete “ n -bit security” numbers that practitioners actually quote.

1.1 Adversaries and the security parameter

The vocabulary of efficiency and smallness is the *Math Guide*’s, fixed in its closing section; we recall it here in the form this volume uses and extend it with the refinements that are specifically cryptographic.

The *security parameter* $\lambda \in \mathbb{N}$ is a positive integer that controls the sizes of all objects in a cryptographic scheme—key lengths, group orders, output lengths—and is conventionally supplied to every algorithm in unary, written 1^λ , the string of λ ones, so that an algorithm running in time polynomial in the length of its input automatically receives time polynomial in λ (*Math Guide*, §“The security parameter”). Every scheme in this monograph is accordingly not a single object but an infinite *family* of objects indexed by λ .

Two questions immediately arise: why a family, and why unary.

The reason for a *family* is that security is inherently asymptotic. There is no such thing as an absolutely unbreakable scheme with fixed finite key length: an adversary with enough time can always enumerate a finite key space. What we can hope for is that as λ grows, the resources required to break the scheme grow far faster than the resources required to use it. The honest parties’ costs (key generation, encryption, signing) should grow *slowly* in λ —polynomially—while the best attack’s cost should grow *quickly*—super-polynomially, ideally exponentially. The security parameter is the dial that separates these two growth rates, and security is a statement about the limit $\lambda \rightarrow \infty$.

The reason for *unary* encoding, 1^λ rather than the $\lceil \log_2 \lambda \rceil$ -bit binary representation of λ , is a bookkeeping convenience that aligns two notions of “polynomial time”. Complexity theory measures running time as a function of input *length*. If λ came in binary it would have length $\Theta(\log \lambda)$, and an algorithm “polynomial in its input length” would get only $\text{poly}(\log \lambda)$ steps—not even enough to write down a λ -bit key. Padding the parameter to length λ ensures that “polynomial in the input length” coincides with the cryptographically meaningful “polynomial in λ ”. This is purely a normalisation; nothing of substance depends on it.

Remark 1.1. Because everything is indexed by λ , every quantity we discuss is really a *function* of λ : the adversary’s success probability, its running time, the key length. A single number—“the advantage is at most 2^{-128} ”—either fixes a concrete λ or describes the function’s behaviour for the λ of interest. Keeping the dependence on λ explicit, at least mentally, prevents nearly every confusion in this subject.

Having fixed how scheme sizes scale, we fix what an attacker may do. An algorithm A is *probabilistic polynomial-time* (PPT) if it is a probabilistic Turing machine—one equipped with a read-only tape of independent uniformly random bits, its *random tape*—and there is a polynomial p such that on every input x , and for every setting of the random tape, A halts within $p(|x|)$ steps: the strict, worst-case notion, exactly as fixed in the *Math Guide* (§“Algorithms, running time, and PPT”). Write $y \leftarrow A(x)$ for the random variable obtained by running A on x with freshly sampled randomness, and $y := A(x; r)$ for the deterministic output when the random tape is fixed to the string r . The honest algorithms and the adversary alike are modelled as PPT machines.

Two of the modelling choices folded into that sentence warrant comment.

Why polynomial time? Polynomial time is the standard mathematical abstraction of “feasible computation”. It is robust—closed under composition, and invariant across all reasonable deterministic computational models (the strong Church–Turing thesis)—which makes it a clean class to quantify over. Crucially, we require the *honest* parties to run in polynomial time too; a

scheme is useful only if it can be used efficiently. The security claim is then an asymmetry between a polynomial-time honest world and a quantified statement that *no* polynomial-time attack succeeds.

Why probabilistic? Randomness is not a luxury but a necessity in cryptography. Deterministic encryption leaks whether two ciphertexts encrypt the same message; key generation must sample secrets unpredictably. We therefore grant the adversary randomness too, both for fairness and because a deterministic adversary is the special case where the random tape is ignored.

The adversary class itself admits a strengthening that the *Math Guide* mentions (§“Algorithms, running time, and PPT”) and this volume adopts as standard.

Definition 1.2 (Non-uniform PPT). A *non-uniform* PPT adversary is a family $\{A_\lambda\}_{\lambda \in \mathbb{N}}$ of probabilistic circuits (or, equivalently, Turing machines each supplied with an *advice string* z_λ of length $\text{poly}(\lambda)$) such that a single polynomial in λ bounds the size of A_λ . The advice may depend arbitrarily on λ but not on the particular inputs drawn at run time.

Remark 1.3. The distinction between uniform adversaries (a single machine that reads 1^λ and works for all λ) and non-uniform ones (a separate circuit per λ , or a uniform machine with per- λ advice) is a genuine one. Non-uniformity grants the adversary free precomputed advice tailored to each parameter size—for example, a short hint about the group of the day. Security against non-uniform adversaries is the stronger and now-standard requirement, partly because non-uniform models compose more cleanly under reduction (one may “hard-wire” a fixed good choice of randomness or auxiliary input). All definitions below can be read in either flavour; we state hardness assumptions against non-uniform adversaries when it matters and otherwise leave the choice implicit.

Remark 1.4 (Expected versus strict polynomial time, and quantum adversaries). Two refinements recur. First, some definitions allow *expected* polynomial-time adversaries, bounding the average rather than worst-case running time; this matters for certain extraction arguments in zero-knowledge, and we flag it where it arises. Second, for post-quantum security one replaces PPT by *quantum* polynomial-time (QPT)—polynomial-size circuits of quantum gates followed by a measurement, a model this volume does not formalise; the consequences for the assumptions this volume relies on are taken up under “The quantum caveat: Shor’s algorithm” in the next section. The structural definitions of this section (advantage, indistinguishability, reduction) are agnostic to which class of adversaries one quantifies over; only the underlying hardness assumptions change.

1.2 Distribution ensembles and indistinguishability

The most basic thing an adversary can attempt is to *tell two situations apart*—a real protocol transcript from a simulated one, an encryption of m_0 from an encryption of m_1 , a pseudorandom string from a truly random one. The most basic security goal is that it cannot. To formalise “look the same” we first formalise the two situations as *ensembles* of probability distributions, then define three grades of sameness.

Definition 1.5 (Probability ensemble). A *probability ensemble* is a sequence $X = \{X_\lambda\}_{\lambda \in \mathbb{N}}$ of probability distributions (equivalently, random variables), one for each value of the security parameter, where X_λ is supported on binary strings whose length is bounded by a polynomial in λ . There is often an additional index—an input a —written $\{X_\lambda(a)\}$; for clarity we suppress it unless needed.

The natural metric on a single pair of distributions is the statistical distance of the *Math Guide* (§“Statistical distance”), which we recall in the slightly wider generality this volume needs: supports there are finite, while ensembles as just defined live on countable sets of strings.

Definition 1.6 (Statistical distance). Let X, Y be discrete random variables over a common finite or countable set S . Their *statistical distance* (or total variation distance) is

$$\Delta(X, Y) := \frac{1}{2} \sum_{s \in S} |\Pr[X = s] - \Pr[Y = s]|.$$

Proposition 1.7 (Variational characterisation). For X, Y over S ,

$$\Delta(X, Y) = \max_{T \subseteq S} (\Pr[X \in T] - \Pr[Y \in T]) = \max_D (\Pr[D(X) = 1] - \Pr[D(Y) = 1]),$$

where the last maximum ranges over all (even computationally unbounded, deterministic) decision functions $D : S \rightarrow \{0, 1\}$. In particular, no procedure whatsoever—efficient or not—can distinguish X from Y with advantage exceeding $\Delta(X, Y)$.

Proof. The set form is the variational characterisation in the statistical-distance theorem of the *Math Guide* (§“Statistical distance”), whose argument is unchanged over a countable S . Only the reformulation over decision functions is new: a deterministic D is the indicator of the set $T = D^{-1}(1)$, so the maximum over D equals the maximum over T ; a randomised D is a convex combination of deterministic ones and cannot do better. $\square$

Proposition 1.8 (Properties of statistical distance). The statistical distance is a metric on probability distributions: $0 \leq \Delta(X, Y) \leq 1$, it is symmetric, $\Delta(X, Y) = 0$ iff X and Y are identically distributed, and it satisfies the triangle inequality $\Delta(X, Z) \leq \Delta(X, Y) + \Delta(Y, Z)$. Moreover it never increases under (possibly randomised) post-processing: for any function—indeed any randomised algorithm— f ,

$$\Delta(f(X), f(Y)) \leq \Delta(X, Y).$$

Proof. These are the metric and data-processing parts of the statistical-distance theorem of the *Math Guide* (§“Statistical distance”), again with the argument unchanged over a countable S . $\square$

The weakest of the three grades below needs the asymptotic vocabulary of the *Math Guide* (§“Polynomial, exponential, and negligible functions”), which we recall in one breath. A function $f : \mathbb{N} \rightarrow \mathbb{R}_{\geq 0}$ is *negligible*, written $f = \text{negl}(\lambda)$, if for every $c \in \mathbb{N}$ there exists λ_0 such that $f(\lambda) < \lambda^{-c}$ for all $\lambda \geq \lambda_0$ —it decays faster than the reciprocal of every polynomial. A function is *non-negligible* if it is not negligible; it is *noticeable* if there exists c with $f(\lambda) \geq \lambda^{-c}$ for all sufficiently large λ (noticeable functions are non-negligible, but not conversely); it is *overwhelming* if $1 - f$ is negligible. The class of negligible functions is closed under addition and under multiplication by any polynomial—the closure property, proved there, on which the hybrid argument of the next subsection depends.

Definition 1.9 (Perfect, statistical, and computational indistinguishability). Let $X = \{X_\lambda\}$ and $Y = \{Y_\lambda\}$ be probability ensembles.

- The ensembles X and Y are *perfectly indistinguishable*, written $X \equiv Y$, if X_λ and Y_λ are *identically distributed* for every λ ; equivalently $\Delta(X_\lambda, Y_\lambda) = 0$ for all λ .
- The ensembles X and Y are *statistically indistinguishable*, written $X \approx_s Y$, if their statistical distance is negligible:

$$\Delta(X_\lambda, Y_\lambda) = \text{negl}(\lambda).$$

- The ensembles X and Y are *computationally indistinguishable*, written $X \approx_c Y$, if for every PPT algorithm D (the *distinguisher*), the function

$$\text{Adv}_{X,Y}^{\text{dist}}(D, \lambda) := |\Pr[D(1^\lambda, X_\lambda) = 1] - \Pr[D(1^\lambda, Y_\lambda) = 1]|$$

is negligible in λ . The probabilities are over the draw from the ensemble and over D 's internal randomness.

These three notions sit in a strict hierarchy. The implications are immediate from the definitions and the variational characterisation; standard examples witness the strictness. The second example needs two primitives, which we define here in the form the volume uses throughout; write U_n for the uniform distribution on $\{0,1\}^n$. A *one-way function* is a polynomial-time computable $h: \{0,1\}^* \rightarrow \{0,1\}^*$ that no PPT algorithm inverts on a random input: for every PPT A ,

$$\Pr[h(A(1^\lambda, h(x))) = h(x)] = \text{negl}(\lambda) \quad (x \leftarrow U_\lambda).$$

A *pseudorandom generator* (PRG) is a polynomial-time computable $G: \{0,1\}^\lambda \rightarrow \{0,1\}^{2\lambda}$ whose output on a uniform seed is computationally indistinguishable from uniform, $\{G(U_\lambda)\} \approx_c \{U_{2\lambda}\}$; the doubling stretch is a normalisation. Whether one-way functions exist is open, which is why the second strictness below is stated conditionally.

Proposition 1.10 (Hierarchy). *For all ensembles X, Y :*

$$X \equiv Y \implies X \approx_s Y \implies X \approx_c Y.$$

The first implication is strict: there exist ensembles that are statistically but not perfectly indistinguishable. The second is strict provided one-way functions exist: under that assumption there exist ensembles that are computationally but not statistically indistinguishable.

Proof. If $X \equiv Y$ then $\Delta(X_\lambda, Y_\lambda) = 0$, which is negligible, so $X \approx_s Y$. If $X \approx_s Y$, then for any—in particular any PPT—distinguisher D , Proposition 1.7 gives $\text{Adv}_{X,Y}^{\text{dist}}(D, \lambda) \leq \Delta(X_\lambda, Y_\lambda) = \text{negl}(\lambda)$, so $X \approx_c Y$.

For strictness of the first implication, let X_λ be uniform on $\{0,1\}^\lambda$ and let Y_λ be uniform on $\{0,1\}^\lambda \setminus \{0^\lambda\}$. They are not identically distributed (the point 0^λ has probabilities $2^{-\lambda}$ and 0), so not perfectly indistinguishable, yet $\Delta(X_\lambda, Y_\lambda) = 2^{-\lambda} = \text{negl}(\lambda)$, so they are statistically indistinguishable.

For strictness of the second, assume one-way functions exist; then a PRG exists—the Håstad–Impagliazzo–Levin–Luby theorem, together with the routine lemma that extends any stretch to the doubling one, both used without proof—and furnishes an ensemble: let G be a PRG, let $X_\lambda := G(U_\lambda)$, and let $Y_\lambda := U_{2\lambda}$. By definition of a PRG, $X \approx_c Y$. But X_λ is supported on at most 2^λ strings out of $2^{2\lambda}$, so a set T equal to the image of G has $\Pr[X_\lambda \in T] = 1$ while $\Pr[Y_\lambda \in T] \leq 2^\lambda / 2^{2\lambda} = 2^{-\lambda}$; hence $\Delta(X_\lambda, Y_\lambda) \geq 1 - 2^{-\lambda}$, which is not negligible. Thus $X \approx_c Y$ but $X \not\approx_s Y$. $\square$

Remark 1.11 (The price and the prize of computational indistinguishability). The strictness example is instructive. Statistically the PRG output and a random string are almost as far apart as two distributions can be—distance nearly 1—because the PRG image is a vanishing fraction of the codomain. An unbounded distinguisher simply checks membership in the image and wins. The entire content of computational indistinguishability is that *finding* that image membership test is infeasible. This is the bargain at the heart of cryptography: we trade the impossible (information-theoretic security with short keys) for the merely conjectured-hard (computational security), and in exchange we obtain the ability to encrypt long messages under short keys, build

public-key primitives, and so on. Statistical and perfect security, when attainable, are unconditional and need no hardness assumption; computational security buys vastly more functionality at the cost of resting on an assumption.

1.3 The hybrid argument

Indistinguishability would be of little use if every application had to be argued from scratch. Two lemmas make it composable: efficient post-processing cannot create a distinguishing edge, and a polynomially long chain of pairwise-indistinguishable steps cannot accumulate one. The first is a reduction in miniature—the template for every reduction in this volume; the second, the hybrid argument, is the workhorse of the entire proof literature.

Lemma 1.12 (Closure under efficient post-processing). *If $X \approx_c Y$ and M is any PPT algorithm, then the ensembles $\{M(1^\lambda, X_\lambda)\}$ and $\{M(1^\lambda, Y_\lambda)\}$ satisfy $\{M(X_\lambda)\} \approx_c \{M(Y_\lambda)\}$.*

Proof. Suppose some PPT distinguisher D separates $\{M(X_\lambda)\}$ from $\{M(Y_\lambda)\}$ with advantage $\delta(\lambda)$. Define $D' := D \circ M$, i.e. $D'(1^\lambda, z) := D(1^\lambda, M(1^\lambda, z))$. Since M and D are both PPT, so is D' . Then

$$\text{Adv}_{X,Y}^{\text{dist}}(D', \lambda) = |\Pr[D(M(X_\lambda)) = 1] - \Pr[D(M(Y_\lambda)) = 1]| = \delta(\lambda).$$

By $X \approx_c Y$ the left side is negligible, so δ is negligible. $\square$

Many proofs must compare two ensembles that are far apart but connected by a chain of pairwise-close intermediate ensembles. The hybrid argument turns such a chain into a single conclusion. It is essentially the triangle inequality, but with a refinement—the uniformity hypothesis—that explains why the number of hybrids must be polynomial and that careless statements omit.

Lemma 1.13 (Hybrid lemma). *Let $t = t(\lambda)$ be polynomially bounded, and let $H_\lambda^0, H_\lambda^1, \dots, H_\lambda^t$ be a sequence of distributions (the hybrids). Suppose that for every PPT distinguisher D the neighbouring advantages*

$$\alpha_i(D, \lambda) := |\Pr[D(H_\lambda^i) = 1] - \Pr[D(H_\lambda^{i+1}) = 1]|$$

are uniformly negligible: a single negligible function v (depending on D) satisfies $\alpha_i(D, \lambda) \leq v(\lambda)$ for all $i \in \{0, \dots, t(\lambda) - 1\}$ simultaneously. Equivalently, $\max_{0 \leq i < t(\lambda)} \alpha_i(D, \lambda) = \text{negl}(\lambda)$, i.e. $\alpha_{i(\lambda)}(D, \lambda)$ is negligible along every index function $i(\lambda)$ —arbitrary, not merely efficiently computable, since the maximising index need not be efficient. Then $\{H_\lambda^0\}$ and $\{H_\lambda^t\}$ are computationally indistinguishable, i.e. for every PPT D , $|\Pr[D(H_\lambda^0) = 1] - \Pr[D(H_\lambda^t) = 1]| = \text{negl}(\lambda)$.

Proof. Fix a PPT distinguisher D and let v be the uniform bound the hypothesis supplies for D . By the triangle inequality applied to the telescoping sum,

$$|\Pr[D(H_\lambda^0) = 1] - \Pr[D(H_\lambda^t) = 1]| = \left| \sum_{i=0}^{t-1} (\Pr[D(H_\lambda^i) = 1] - \Pr[D(H_\lambda^{i+1}) = 1]) \right| \leq \sum_{i=0}^{t-1} \alpha_i(D, \lambda),$$

and the hypothesis bounds the sum by $t(\lambda) \cdot v(\lambda)$, which is negligible since t is a polynomial (closure of the negligible class under polynomial multiplication). Hence $\{H^0\} \approx_c \{H^t\}$. $\square$

Uniformity in i is essential, and no argument derives it from the weaker hypothesis that each α_i is negligible for every *fixed* i : there the threshold beyond which the i -th term is small can depend on i , and the dependence cannot be removed. A countermodel makes this concrete: take $t(\lambda) = \lambda + 1$ and let H_λ^i be the point mass at 0 if $i \leq \lambda$ and the point mass at 1 otherwise. For

every fixed i the advantage $\alpha_i(D, \lambda)$ is nonzero only at $\lambda = i$, hence negligible; yet H_λ^0 and $H_\lambda^{t(\lambda)}$ are the point masses at 0 and 1, perfectly distinguishable.

Applications usually obtain the conclusion by a stronger, structural route. Suppose the chain arises from a *single* underlying assumption $X \approx_c Y$: neighbouring hybrids H_λ^i and H_λ^{i+1} differ only in that one designated component is drawn from X_λ in the former and from Y_λ in the latter, and each hybrid is efficiently samplable given $(1^\lambda, i)$ and a sample of that component. Then the *random-index* distinguisher D' for the pair (X, Y) , on challenge z , samples i uniformly from $\{0, \dots, t-1\}$, builds the hybrid with z in the designated position, and runs D ; writing $p_i := \Pr[D(H_\lambda^i) = 1]$, its distinguishing advantage equals $|\frac{1}{t} \sum_{i=0}^{t-1} (p_i - p_{i+1})| = |p_0 - p_t|/t$. Since D' is a single PPT machine, the assumption makes its advantage negligible, and $|p_0 - p_t| \leq t(\lambda) \cdot \text{Adv}(D') = \text{negl}(\lambda)$. Note that this argument bounds D 's total advantage directly, bypassing the lemma's uniform per-step hypothesis rather than discharging it; the lemma itself covers the general case in which no such common structure is available.

Remark 1.14. The hybrid argument is precisely where the “polynomially many” constraint earns its keep. A negligible per-step advantage multiplied by a polynomial number of steps remains negligible; a negligible per-step advantage multiplied by, say, 2^λ steps need not. This is why constructions that would require exponentially many hybrids do not yield security proofs, and why the closure of negligibility under polynomial factors is not a technicality but the load-bearing wall of the whole edifice.

1.4 Security as a game

We can now state what it means to “break” a scheme. The modern idiom expresses every security definition as a *game* (or experiment) played between a *challenger* and an adversary $\mathcal{A}$. The challenger, following a fixed protocol, sets up the scheme, answers whatever queries the adversary may pose, and at the end declares the adversary a winner or loser by outputting a single bit. Security says that no PPT adversary wins by more than it could by trivial guessing.

Definition 1.15 (Security game and advantage). A *security game* (or *experiment*) G is a PPT interactive algorithm, the challenger, that takes 1^λ , interacts with an adversary $\mathcal{A}$, and finally outputs a bit $G^\mathcal{A}(\lambda) \in \{0, 1\}$, where 1 denotes “adversary wins”. The *advantage* of $\mathcal{A}$ is a measure of how much better than trivial its winning probability is; its precise form depends on the type of game:

- *Distinguishing (indistinguishability) games*, where the challenger secretly chooses a uniform bit $b \in \{0, 1\}$ and $\mathcal{A}$ outputs a guess b' ; the adversary wins iff $b' = b$. Here

$$\text{Adv}_G(\mathcal{A}, \lambda) := |\Pr[G^\mathcal{A}(\lambda) = 1] - \frac{1}{2}| = \frac{1}{2}|\Pr[b' = 1 \mid b = 1] - \Pr[b' = 1 \mid b = 0]|.$$

Trivial guessing achieves winning probability $1/2$, hence advantage 0.

- *Search (unforgeability/inversion) games*, where the adversary must produce a specific kind of object—a forgery, a preimage, a discrete logarithm—and wins iff it succeeds. Here the baseline winning probability is (essentially) 0, and one simply sets

$$\text{Adv}_G(\mathcal{A}, \lambda) := \Pr[G^\mathcal{A}(\lambda) = 1].$$

Definition 1.16 (Security relative to a game). A scheme is *secure* with respect to the game G if for every PPT adversary $\mathcal{A}$,

$$\text{Adv}_G(\mathcal{A}, \lambda) = \text{negl}(\lambda).$$

Two remarks on the shape of this definition are in order. First, the quantifier order is essential: *for every* PPT adversary, the advantage is negligible. For each fixed adversary there may be a different negligible bound; the claim is universal over the entire class. Second, the use of negl on the right is what makes the definition asymptotic and clean: it is robust to the polynomial slop that reductions introduce, and it captures “the adversary does no better than the trivial strategy, up to an amount that vanishes faster than any inverse polynomial”.

Example 1.17 (Pseudorandomness, as a game). To anchor the abstraction with machinery already on the table, consider the distinguishing game for a pseudorandom generator $G : \{0, 1\}^\lambda \rightarrow \{0, 1\}^{2\lambda}$ —the primitive that witnessed strictness in Proposition 1.10:

1. The challenger samples $b \leftarrow \{0, 1\}$ uniformly. If $b = 0$ it samples a seed $s \leftarrow \{0, 1\}^\lambda$ and sends $z := G(s)$; if $b = 1$ it sends a uniform $z \leftarrow \{0, 1\}^{2\lambda}$.
2. The adversary A outputs a guess b' ; the game outputs 1 iff $b' = b$.

The generator is secure iff $\text{Adv}(A, \lambda) = |\Pr[b' = b] - \frac{1}{2}|$ is negligible for every PPT A . Here the game packaging and the ensemble packaging coincide: the two situations are the fixed ensembles $\{G(U_\lambda)\}$ and $\{U_{2\lambda}\}$, and the displayed identity in Definition 1.15 shows the game advantage of A is exactly half its distinguishing advantage in the sense of Definition 1.9—so the game is secure iff the ensembles are computationally indistinguishable. The game form earns its keep beyond this case: when the adversary chooses part of the challenge or queries oracles adaptively, the interaction no longer collapses to a comparison of two fixed ensembles, and the game is the more general packaging—the shape the security definitions of the later sections take.

Remark 1.18 (Game-hopping). Once we phrase security as a game, proofs become *sequences of games*. One starts from the real security game and rewrites it through a series of games $G_0, G_1, \dots, G_k$, each differing from the last by a small, justifiable change, until reaching a final game in which the adversary provably has zero (or trivially negligible) advantage. Each hop is justified by exactly one of three moves: the two games are identical unless some negligible-probability “bad” event occurs, so their winning probabilities differ by at most $\Pr[\text{bad}]$ (on the complement of that event the two runs coincide); or distinguishing the two games would break a hardness assumption (a reduction, below); or the change is purely syntactic. Bounding the total advantage is then a telescoping sum across hops—a structured cousin of the hybrid argument (Lemma 1.13). This discipline keeps otherwise sprawling proofs honest and auditable.

1.5 Reductions

The engine of provable security is the reduction. A *reduction* is an efficient transformation that converts a successful adversary against a scheme into a successful algorithm against a problem believed to be hard. Its existence proves the implication “ Π is hard $\Rightarrow$ X is secure” by establishing the contrapositive “ X is broken $\Rightarrow$ Π is solved”.

Definition 1.19 (Computational problem and hardness). A *computational problem* Π consists of a PPT instance generator $\text{Inst}(1^\lambda)$ producing a problem instance together with any side information, and a relation deciding when a candidate output *solves* the instance. The *success probability* of an algorithm $\mathcal{B}$ is

$$\text{Succ}_\Pi(\mathcal{B}, \lambda) := \Pr[\mathcal{B}(1^\lambda, \text{Inst}(1^\lambda)) \text{ solves the instance}].$$

The problem Π is *hard* if $\text{Succ}_\Pi(\mathcal{B}, \lambda) = \text{negl}(\lambda)$ for every PPT $\mathcal{B}$ (for a search problem), or if every PPT $\mathcal{B}$ decides the instance with at most negligible advantage, $\text{Adv}(\mathcal{B}, \lambda) = \text{negl}(\lambda)$ in the sense of Definition 1.15 (for a decision problem).

Definition 1.20 (Black-box reduction). Let X be a scheme with security game G_X and Π a computational problem. A (black-box) reduction from breaking X to solving Π is a PPT oracle algorithm $\mathcal{R}$ with the following per-adversary guarantee: for every adversary A , writing $\varepsilon_X(\lambda) := \text{Adv}_{G_X}(A, \lambda)$ for its advantage, the algorithm $\mathcal{R}^A$ —which runs A as a subroutine, simulating A 's game internally—solves Π with success probability

$$\text{Succ}_{\Pi}(\mathcal{R}^A, \lambda) \geq f(\varepsilon_X(\lambda), \lambda),$$

where f is a function fixed by the reduction (not by A) that maps every non-negligible ε_X to a non-negligible bound. The running time of $\mathcal{R}$ is measured with each call to A counted as one step, so “PPT” bounds $\mathcal{R}$'s own work, and hence its number of calls to A , by a polynomial in λ ; the running time of $\mathcal{R}^A$ is then polynomial whenever A 's is, however many times $\mathcal{R}$ runs A . How far f degrades ε_X is the reduction's loss, discussed below. The reduction is *black-box* because $\mathcal{R}$ uses A only through its input/output interface, making no use of A 's code.

The structure of a reduction is always the same and is worth stating as a recipe. To prove “if Π is hard then X is secure”:

1. Assume toward the contrapositive a PPT adversary A that breaks X with non-negligible advantage ε_X .
2. Construct a PPT algorithm $\mathcal{R}$ that, given a random instance of Π , *embeds* that instance into a simulated security game for A . The simulation must be faithful: the view $\mathcal{R}$ presents to A must follow (perfectly, statistically, or computationally) the distribution of the real game, so that A 's advantage carries over.
3. Run A . From whatever A does to win its game, $\mathcal{R}$ *extracts* a solution to the Π -instance.
4. Conclude that $\mathcal{R}$ solves Π with non-negligible probability, contradicting the hardness of Π . Therefore no such A exists, and X is secure.

A complete, genuine reduction illustrates the point.

Theorem 1.21 (Hard-core security of the Diffie–Hellman-style key, schematically). Let $\Pi = \text{CDH}$ be the computational Diffie–Hellman problem in a cyclic group $\mathbb{G} = \langle g \rangle$ of prime order q , the setting of the Math Guide (§“The discrete logarithm problem”): given (g, g^a, g^b) for uniform $a, b \in \mathbb{Z}/q\mathbb{Z}$, compute g^{ab} . Consider the “key-recovery” game G_X for a key-exchange scheme in which the adversary, given the public transcript (g^a, g^b) , wins iff it outputs the shared key g^{ab} . If CDH is hard, then no PPT adversary wins G_X with noticeable probability.

Proof. Suppose A is a PPT adversary that, on input a transcript (g^a, g^b) , outputs g^{ab} with probability $\varepsilon_X(\lambda) = \text{Adv}_{G_X}(A, \lambda)$. Construct $\mathcal{R}$ solving CDH. On input a CDH instance $(g, A, B) = (g, g^a, g^b)$:

1. The reduction $\mathcal{R}$ forms the simulated transcript (A, B) and feeds it to A as the public transcript of G_X .
2. Because (a, b) are uniform in the CDH instance *and* in the real game, the view of A follows *identically* the distribution of its view in G_X (the simulation is perfect; no loss in faithfulness).
3. When A outputs a candidate key K , $\mathcal{R}$ outputs K as its CDH answer.

Since the simulated and real views are identically distributed, $\mathcal{A}$ outputs $K = g^{ab}$ with exactly probability $\varepsilon_X(\lambda)$; whenever it does, $\mathcal{R}$ returns the correct CDH solution. Hence

$$\text{Succ}_{\text{CDH}}(\mathcal{R}, \lambda) = \varepsilon_X(\lambda).$$

The running time of $\mathcal{R}$ is that of $\mathcal{A}$ plus a constant amount of group bookkeeping, hence polynomial. If ε_X were noticeable, $\mathcal{R}$ would solve CDH with noticeable probability, contradicting its hardness. Therefore $\varepsilon_X(\lambda)$ is not noticeable, as claimed. In fact, since $\mathcal{R}$ is PPT and CDH is hard (Definition 1.19, search branch), $\varepsilon_X(\lambda) = \text{Succ}_{\text{CDH}}(\mathcal{R}, \lambda) = \text{negl}(\lambda)$. $\square$

This example is deliberately *tight*: the reduction loses nothing, calls the adversary once, and converts advantage ε_X into success probability exactly ε_X . Most reductions are not so fortunate, which leads to the quantitative anatomy of a reduction.

Definition 1.22 (Reduction loss and tightness). Consider a reduction $\mathcal{R}$ that transforms an adversary running in time t with advantage ε into a Π -solver running in time $t' = t + t_{\mathcal{R}}$ with success probability ε' . The *security loss* is the factor

$$L(\lambda) := \frac{\varepsilon/t}{\varepsilon'/t'} = \frac{\varepsilon}{\varepsilon'} \cdot \frac{t'}{t},$$

the ratio of the solver's work-per-success to the adversary's work-per-success. A reduction is *tight* if $L(\lambda) = O(1)$ (a small constant, ideally 1), and *loose* if $L(\lambda)$ grows with λ (e.g. $L = q_H$, the number of random-oracle queries (Definition 1.26 below), or $L = q_s$, the number of signature queries, or $L = \lambda$). Two common sources of loss are: an *advantage gap* $\varepsilon' \ll \varepsilon$ (the solver succeeds far less often than the adversary, e.g. because the reduction must *guess* where to embed its instance among q queries, succeeding only with probability $1/q$); and a *time blow-up* $t' \gg t$ (the reduction runs the adversary many times, as in rewinding/forking arguments).

Remark 1.23 (Why loss matters concretely). Asymptotically, loss is invisible: a polynomial loss factor turns a negligible advantage into a negligible advantage and a hard problem into security, without qualification. Concretely, however, loss carries a cost. Suppose a scheme reduces to a problem believed to offer 128 bits of security, but the reduction loses a factor $L = 2^{40}$ (not unusual: $q_H \approx 2^{60}$ random-oracle queries times a forking-lemma square-root loss can be much worse). Then the *proof* guarantees only $128 - 40 = 88$ bits of security for the scheme. To restore a 128-bit guarantee one must enlarge the group so that the *underlying* problem offers 168 bits—larger keys, slower operations. A tight reduction lets the scheme inherit the full strength of the assumption at the smallest parameters; a loose one imposes a permanent cost on every user. This is why tightness is a first-class design goal in modern cryptography, not a theoretical nicety.

Remark 1.24 (Rewinding and the forking lemma, in brief). A recurrent source of large loss is *rewinding*: running the adversary, recording its behaviour, then restarting it from a saved state with fresh randomness on some branch to obtain a second, correlated transcript. From two transcripts that share a prefix but diverge afterward one can often extract a secret (e.g. two valid signatures on the same commitment yield the signing key, or two accepting proof transcripts yield a witness). The *forking lemma*, stated as the general forking lemma under “Security in the random oracle model and the forking lemma”, quantifies the success probability of obtaining such a pair: if the adversary succeeds with probability ε after at most q random-oracle queries whose answers are drawn from a set of size h , the forked run succeeds with probability at least $\varepsilon(\varepsilon/q - 1/h)$, i.e. on the order of ε^2/q . The square in ε is a quadratic security loss: to keep ε^2/q noticeable one needs ε noticeably large, roughly halving the bit-security extracted. Such arguments—central to the analyses of Schnorr-type and Fiat–Shamir signatures that this volume builds toward—are the archetype of loose but still valid reductions.

1.6 The standard model and the random oracle model

Reductions reduce security to assumptions. *Which* assumptions one is willing to make defines the *model* in which a proof resides. Two models predominate.

Definition 1.25 (Standard model). A proof resides in the *standard model* if it assumes only the computational hardness of explicitly stated problems (factoring, discrete logarithm, the decisional Diffie–Hellman problem of the next section, and so on) and grants the adversary no idealised access to any component of the scheme. Every primitive, including every hash function, is a concrete algorithm whose code the adversary may inspect and exploit.

Definition 1.26 (Random oracle model). A proof resides in the *random oracle model* (ROM) if it idealises one or more hash functions as a *random oracle*: a publicly accessible function O that, on each distinct input, returns an *independent, uniformly random* output of the appropriate length, consistently answering repeated queries with the same value. The oracle is available to the honest parties, to the adversary, and—crucially—to the reduction. No party holds the “code” of O ; it can only be queried.

The random oracle model is an idealisation: real hash functions are fixed, public algorithms, not truly random functions. The motivation for adopting it is that the reduction gains two capabilities that the standard model withholds and that render many otherwise-intractable proofs tractable.

Remark 1.27 (The powers of the random oracle: observability and programmability). In the ROM the reduction *simulates* the oracle for the adversary, and this yields two additional capabilities. *Observability*: the reduction sees every query the adversary makes to O . Since the adversary, to use a hash value meaningfully, must query it, the reduction learns the adversary’s intermediate computations—for instance the preimage the adversary is about to hash. *Programmability*: the reduction may choose each oracle answer on the fly, subject only to the answers appearing uniform and consistent. It can therefore *plant* its hard-problem instance inside an oracle response—e.g. set $O(x) := B$ where B is part of a CDH challenge—so that the adversary’s success necessarily reveals the solution. Neither capability exists for a concrete standard-model hash function, which fixes its outputs in advance and hides its internal queries.

Example 1.28 (The benefit of the ROM: full-domain hash, sketched). In the “full domain hash” signature, the signer signs a message m as $\sigma = H(m)^d \bmod N$, where (N, d) is an RSA private key—a modulus $N = pq$ with secret prime factors and a secret exponent d paired with a public e so that $x^{ed} \equiv x \pmod{N}$ for x coprime to N , by Euler’s theorem (*Math Guide*, §“Fermat’s little theorem and Euler’s theorem”); thus $x \mapsto x^e$ is a permutation of the unit group $(\mathbb{Z}/N\mathbb{Z})^\times$ that only the holder of d can invert—and H is a hash function mapping messages into $(\mathbb{Z}/N\mathbb{Z})^\times$. (Restricting to units costs nothing: a nonzero non-unit of $\mathbb{Z}/N\mathbb{Z}$ shares a factor with N , so stumbling on one factors N . Hash functions appear here informally, as this sketch needs them; their formal syntax and security games are the business of the next-but-one section.) To prove unforgeability one reduces to RSA inversion: given a challenge y (find y^d), the reduction programs the oracle so that for a random subset of messages $H(m) := y \cdot r_m^e$ (embedding the challenge) and for the rest $H(m) := r_m^e$ (where it knows the “signature” r_m); with each r_m drawn uniformly from $(\mathbb{Z}/N\mathbb{Z})^\times$, the programmed values are distributed exactly as honest oracle answers on H ’s range. When the forger outputs a forgery on a message where the reduction embedded the challenge, the reduction extracts y^d . This proof has *no known standard-model analogue* for the plain scheme: it relies essentially on programming H . The cost is a security loss (the reduction must guess which message the forger attacks, incurring a factor q_H unless cleverly optimised)—connecting back to Definition 1.22.

Remark 1.29 (The status of the ROM: heuristic, not theorem). A proof in the ROM is a proof about an *idealised* world: deploying the scheme instantiates $\mathcal{O}$ with a concrete hash function, and the proof does *not* formally transfer—a ROM proof rules out all attacks that treat the hash as a black box, while offering no guarantee against attacks on the specific structure of the chosen hash. The case for trusting it runs in three steps: prove the scheme secure against a perfect random oracle; instantiate $\mathcal{O}$ with a hash believed to have no exploitable structure; and observe that any remaining attack must exploit structure the random oracle lacks, so would be a direct break of the presumed structure-free hash. The gap is nonetheless a theorem: Canetti, Goldreich, and Halevi exhibit contrived schemes provably secure in the ROM yet insecure under *every* efficiently computable instantiation, so no concrete hash soundly instantiates the oracle for all schemes. No natural deployed scheme is known to suffer this fate, and the working stance of this volume is the practical one: a ROM proof is strong positive evidence, not a guarantee, and a standard-model proof is preferred when one is available at comparable cost. One caveat looks forward. A quantum adversary (Remark 1.4) queries the oracle *in superposition*, a single query addressing a weighted combination of all inputs at once; in the resulting quantum random oracle model (QROM) the two powers of Remark 1.27 become subtler and classical ROM theorems do not automatically carry over, so post-quantum claims for arguments made non-interactive by hashing the transcript (the Fiat–Shamir transform, developed under “The Fiat–Shamir transform: from interactive to non-interactive”) must be made in that model.

Remark 1.30 (Relatives: CRS, generic-group, and algebraic-group models). Between and beyond these poles lie further idealisations relevant to the proof systems this volume targets. The *common reference string* (CRS) model posits a trusted public string sampled from a prescribed distribution, used by many succinct proof systems. The *generic group model* (GGM) idealises the group dual to the manner in which the ROM idealises a hash: the adversary may only perform group operations through an oracle on opaque “handles”, never exploiting the bit representation of group elements; it is the natural setting for lower bounds on discrete-log-type problems. The *algebraic group model* (AGM) is an intermediate, milder idealisation requiring the adversary to “explain” every group element it outputs as a known combination of its inputs. These models trade realism for provability along the same axis as the ROM, and the constructions ahead invoke them by name where they require them.

1.7 Concrete security and the bit-security budget

The asymptotic language—“negligible”, “polynomial”, “for all sufficiently large λ ”—is mathematically clean but operationally silent. It establishes that a scheme is eventually secure; it does not establish whether *this* deployment with *these* parameters resists an adversary with *that* budget. Concrete (or exact) security supplies the missing numbers.

Definition 1.31 (Concrete security: (t, ϵ) -security). Fix the security parameter to a concrete value (so all sizes are fixed). A scheme is (t, ϵ) -secure for a game G if every adversary running in time at most t has advantage $\text{Adv}_G(A) \leq \epsilon$. More refined statements bound the advantage as an explicit function of the adversary’s resources, e.g. time t , number of queries q , and memory s : $\text{Adv}_G(A) \leq f(t, q, s)$.

Definition 1.32 (n -bit security). A scheme (or problem) is commonly said to offer n bits of security when the cheapest known attack achieving constant success probability (say at least $\frac{1}{2}$) costs on the order of 2^n elementary operations. A proof may justify the stronger, explicit claim that every adversary running in time t has advantage at most about $t/2^n$, but that linear advantage bound is not implied merely by the absence of a faster known attack. Thus “128-bit

security” is a concrete estimate, relative to a specified attack model and state of knowledge, that breaking the scheme costs roughly 2^{128} work.

The relationship to the asymptotic picture is the following: one chooses the security parameter λ *precisely so that* the resulting concrete scheme delivers the desired number of bits of security. In the cleanest case the two coincide, $n = \lambda$, but in general one must account for the actual cost of the best attack on the underlying problem.

Example 1.33 (Why $\lambda \neq n$ for discrete logarithm). For a generic-group or hash-based primitive, doubling the output length doubles the bit security: a hash with $2n$ -bit output offers n -bit collision resistance—by the birthday bound (*Math Guide*, §“The union bound and a birthday calculation”)—and $2n$ -bit preimage resistance, so the parameter and the security level track simply. For discrete-logarithm-based primitives the accounting differs and instructs. In a group of prime order $q \approx 2^\ell$, generic attacks (baby-step/giant-step, Pollard’s rho) solve discrete logarithm in time $\approx \sqrt{q} = 2^{\ell/2}$ (*Math Guide*, §“The discrete logarithm problem”). Hence a prime-order elliptic-curve group whose order is 256 bits offers only about 128 bits of security—one halves the bit length of the group order to obtain the generic security level. For discrete logarithm in a finite-field multiplicative group, sub-exponential index-calculus attacks additionally force the ambient field modulus into the thousands of bits for 128-bit security. This is precisely why elliptic-curve groups, for which no comparable sub-exponential classical attack is known, are preferred: they deliver n -bit security at $\ell \approx 2n$ rather than thousands of bits. The single number n conceals a problem-specific translation from all parameter sizes to the security level. The working groups of this volume are priced by exactly this arithmetic: the Pasta curves assembled in the *Math Guide* (§“Pallas and Vesta assembled”) have prime order close to 2^{254} , so the square-root rule prices generic discrete logarithm there at about 2^{127} operations; the next section audits them criterion by criterion under “Instantiation on elliptic curves; the Pasta curves”.

Remark 1.34 (Reading a bit-security budget end to end). Combining the preceding components yields the practitioner’s parameter-selection recipe. Decide the target security level n (commonly 128). Identify the underlying hard problem and the cost of its best attack as a function of the parameter (e.g. $2^{\ell/2}$ for elliptic-curve discrete log), and set the parameter so the problem itself offers n bits (here $\ell = 2n = 256$). Then subtract the reduction’s security loss (Definition 1.22): if the proof loses a factor $L = 2^k$ (that is, k bits), the problem must offer $n + k$ bits for the *scheme* to offer n , so one enlarges the parameter accordingly—or, with a tight reduction, one need not. Finally, sanity-check against the ROM caveat (Remark 1.29): a ROM proof bounds only black-box attacks on the hash, so the concrete hash must additionally exhibit no exploitable structure. The clean asymptotic theorem “the scheme is secure if the problem is hard” thus unfolds, through the quantitative machinery of this section, into a concrete choice of bytes.

Remark 1.35 (A note on what security does and does not promise). A caution that frames everything above is in order. A proof of security is a proof *relative to a model*: a definition of the goal (the game), a class of adversaries (PPT, possibly non-uniform or quantum), and a set of assumptions (the hard problems, and any idealisation such as the ROM). It guarantees nothing outside that model. Side-channel leakage (timing, power), faulty randomness, implementation bugs, broken assumptions, and goals the game did not capture all lie beyond its reach. The value of the framework is not that it renders schemes unconditionally safe—nothing can—but that it renders the *conditions* of safety explicit, falsifiable, and modular, so that progress on assumptions and attacks translates directly into recalibrated, auditable guarantees. Everything this monograph builds rests on this discipline.

2 Hardness assumptions

Every secret in this volume is an exponent. A spending key, the trapdoor of a commitment, the nonce of a signature—each lives in $\mathbb{Z}/q\mathbb{Z}$ and faces the world only through a group element g^x .

The adversary of this section is the patient cryptanalyst who collects those elements from the public record and asks for nothing more than the exponent behind one of them: an adversary who computes discrete logarithms recovers signing keys from verification keys, opens and re-opens commitments at will, and strips the blinding from every rerandomised key the later sections construct. What she can *forge* is everything; what she must be *denied* is a single inversion. This section assembles the family of assumptions that deny it: the discrete logarithm problem itself, the ladder of Diffie–Hellman variants above it, the generic-attack bounds that calibrate how hard the problem can possibly be, the composite-order failure mode that squanders that hardness, the elliptic curves on which the remainder of the volume instantiates the family, and the quantum caveat under which the whole edifice stands. Throughout, the lodestar is a single number: the security parameter λ , read as the bit-length of security demanded. The recurring conclusion is that to obtain λ bits of security against the best generic attacks the group must have *prime* order q with $\log_2 q \geq 2\lambda$. Notation is fixed once: $G = \langle g \rangle$ denotes a cyclic group of prime order q , written multiplicatively until §2.6 switches to the additive notation of elliptic curves.

2.1 The discrete logarithm problem

The stage is a finite cyclic group. Recall from the *Math Guide* (§“The discrete logarithm problem”) that a group G is *cyclic* if $G = \langle g \rangle = \{g^k : k \in \mathbb{Z}\}$ for some *generator* g ; written additively, as we shall write elliptic curves, this reads $G = \{kg : k \in \mathbb{Z}\}$. If $|G| = q$ then $\mathbb{Z}/q\mathbb{Z} \cong G$ via $k \mapsto g^k$, an isomorphism trivial to evaluate forwards—repeated squaring costs $O(\log k)$ group operations—but conjecturally infeasible to invert. That asymmetry *is* the discrete logarithm problem.

Definition 2.1 (Discrete logarithm). Let $G = \langle g \rangle$ be a cyclic group of order q , written multiplicatively. For $h \in G$, the *discrete logarithm* of h to the base g , denoted $\log_g h$, is the unique residue $x \in \mathbb{Z}/q\mathbb{Z}$ with $g^x = h$. Existence follows from $G = \langle g \rangle$ and $\text{ord}(g) = q$; uniqueness because $g^x = g^{x'}$ forces $g^{x-x'} = 1$, whence $q \mid x - x'$.

Definition 2.2 (Discrete logarithm problem, DLP). Fix a family $\{G_\lambda\}$ of cyclic groups, where $G_\lambda = \langle g_\lambda \rangle$ has *prime* order q_λ with $\log_2 q_\lambda = \Theta(\lambda)$, together with a sampler that on input 1^λ outputs a description of $(G_\lambda, g_\lambda, q_\lambda)$. The game $\text{DLOG}^A(\lambda)$ runs as follows.

1. The challenger runs the sampler, draws $x \xleftarrow{\$} \mathbb{Z}/q_\lambda\mathbb{Z}$ uniformly, and sends $(G_\lambda, g_\lambda, q_\lambda, h = g_\lambda^x)$ to the adversary A .
2. The adversary outputs $x' \in \mathbb{Z}/q_\lambda\mathbb{Z}$.
3. The game outputs 1 iff $g_\lambda^{x'} = h$.

The *DLP assumption* for the family states that for every PPT A ,

$$\Pr[\text{DLOG}^A(\lambda) = 1] \in \text{negl}(\lambda).$$

The order is restricted to be prime for reasons that §2.5 makes emphatic: in prime order every non-identity element is a generator, there are no proper nontrivial subgroups through which information can leak, and the problem cannot be split across factors of the order. For now we record the convenience that prime order affords immediately.

Proposition 2.3 (Random self-reducibility of DLP). *In a cyclic group $G = \langle g \rangle$ of prime order q , the*

discrete logarithm problem is random self-reducible: an algorithm solving a fraction ε of instances, over uniformly random h , converts into one solving any fixed instance with probability ε per trial. Consequently worst-case and average-case hardness coincide.

Proof. Let A succeed on a uniform target with probability ε . Given a fixed challenge $h = g^x$, draw $r \xleftarrow{\$} \mathbb{Z}/q\mathbb{Z}$ and form $h' = h \cdot g^r = g^{x+r}$. Addition modulo q is a bijection and r is uniform, so h' is uniform in G and independent of h ; hence $A(h')$ returns $x + r \bmod q$ with probability ε , and subtracting r recovers x . Each candidate is checked by testing $g^x \stackrel{?}{=} h$, so failures are recognised. Fresh randomness r per trial makes trials independent: t of them succeed with probability $1 - (1 - \varepsilon)^t$, overwhelming once $t = \omega(\varepsilon^{-1} \log \lambda)$. $\square$

Random self-reducibility is the formal licence to speak of “the” hardness of DLP in a group: an average-case solver also solves every fixed instance after rerandomisation. This does not literally make every instance difficult—the logarithm of the identity, for one, is immediate.

2.2 Diffie–Hellman: computational and decisional

The discrete logarithm problem asks to invert exponentiation. Many protocols—the Diffie–Hellman key exchange and ElGamal encryption are the historical motivations—need only a weaker-looking guarantee about the *product* of two exponents. Suppose Alice publishes g^a and Bob publishes g^b ; their shared secret is g^{ab} , which each computes from the other’s public value and their own exponent. The eavesdropper sees g, g^a, g^b and wants g^{ab} .

Definition 2.4 (Computational Diffie–Hellman, CDH). With notation as in Definition 2.2, the game $\text{CDH}^\Lambda(\lambda)$ runs as follows: the challenger draws $a, b \xleftarrow{\$} \mathbb{Z}/q\mathbb{Z}$ independently and sends (g, g^a, g^b) ; the adversary outputs $z \in G$; the game outputs 1 iff $z = g^{ab}$. The *CDH assumption* states that for every PPT A ,

$$\Pr_{a,b}[\text{CDH}^\Lambda(\lambda) = 1] \in \text{negl}(\lambda).$$

The computational guarantee delivers the shared key but not its secrecy: an adversary may be unable to *produce* g^{ab} yet perfectly able to *distinguish* it from a random group element, and that alone already breaks, say, the indistinguishability of an ElGamal ciphertext. The decisional variant excludes it.

Definition 2.5 (Decisional Diffie–Hellman, DDH). The *decisional Diffie–Hellman problem* is to distinguish two distributions over G^3 ,

$$\mathcal{D}_{\text{dh}} = (g^a, g^b, g^{ab}), \quad \mathcal{D}_{\text{rand}} = (g^a, g^b, g^c),$$

with a, b, c independent and uniform in $\mathbb{Z}/q\mathbb{Z}$; this is the distinguishing game of Definition 1.15, the challenger handing over a sample of one or the other according to its hidden bit. The *DDH assumption* states that every PPT distinguisher D has negligible advantage

$$\text{Adv}_{\text{ddh}}(D) = \left| \Pr_{a,b}[D(g^a, g^b, g^{ab}) = 1] - \Pr_{a,b,c}[D(g^a, g^b, g^c) = 1] \right| \in \text{negl}(\lambda).$$

A triple (g^a, g^b, g^c) with $c \equiv ab \pmod{q}$ is a *Diffie–Hellman triple*; the assumption says no efficient distinguisher recognises such triples.

Remark 2.6 (DDH admits a random self-reduction). Rerandomisation works for DDH as it did for DLP, though the correct map takes some care. Given a target triple $(u, v, w) = (g^a, g^b, g^c)$, choose $\alpha, \gamma \xleftarrow{\$} (\mathbb{Z}/q\mathbb{Z})^\times$ and $\beta, \delta \xleftarrow{\$} \mathbb{Z}/q\mathbb{Z}$, and form

$$(u', v', w') = (u^\alpha g^\beta, v^\gamma g^\delta, w^{\alpha\gamma} u^{\alpha\delta} v^{\beta\gamma} g^{\beta\delta}).$$

Writing $A = \alpha a + \beta$, $B = \gamma b + \delta$, and $E = c - ab$, the new exponents are A , B , and $AB + \alpha\gamma E$, with A and B uniform and independent. A DH triple ($E = 0$) therefore becomes a fresh uniform DH triple; a non-DH triple becomes uniform conditional on being non-DH, because $\alpha\gamma E$ is uniform in $(\mathbb{Z}/q\mathbb{Z})^\times$. The latter distribution lies at statistical distance (Definition 1.6) $1/q$ from $\mathcal{D}_{\text{rand}}$, whose third component completes a DH triple by accident with probability $1/q$. A noticeable distinguishing advantage on random instances thus transfers to every fixed triple, up to this negligible term.

2.3 The assumption hierarchy

The three problems form a chain. Write $A \leq B$ for “ A reduces to B ”: an oracle solving B yields an efficient solver for A ; equivalently, B is at least as hard as A , so that if B is easy then A is easy; equivalently again, the assumption “ A is hard” is the *stronger* commitment, since it implies “ B is hard”. The reductions of this subsection give

$$\text{DDH} \leq \text{CDH} \leq \text{DLP},$$

ordering the problems by increasing hardness and the assumptions by decreasing strength: assuming DDH hard is the strongest commitment, assuming DLP hard the weakest.

Theorem 2.7 (DLP is at least as hard as CDH). *If DLP is solvable in PPT, then CDH is solvable in PPT. Equivalently, the CDH assumption implies the DLP assumption.*

Proof. Suppose A solves DLP. Given a CDH instance (g, g^a, g^b) , run A on (g, g^a) to recover a , then compute $(g^b)^a = g^{ab}$ by fast exponentiation. The cost is one DLP call plus $O(\log q)$ group operations, and only one of the two exponents ever needed inverting. $\square$

Theorem 2.8 (CDH is at least as hard as DDH). *If CDH is solvable in PPT with non-negligible success probability, then DDH is solvable in PPT with non-negligible advantage; if the success probability is noticeable, rerandomisation drives the advantage overwhelmingly close to 1. Equivalently, the DDH assumption implies the CDH assumption.*

Proof. Suppose A solves CDH with probability ε . Build a distinguisher D : on input (g^a, g^b, z) , call $A(g, g^a, g^b)$ to obtain a candidate w , and output 1 iff $w = z$. On a Diffie–Hellman triple, $z = g^{ab}$, so D outputs 1 whenever A succeeds—probability ε . On a random triple, $z = g^c$ with c uniform and independent of a, b , while w depends only on (g^a, g^b) ; the event $w = z$ therefore has probability exactly $1/q$. Hence

$$\text{Adv}_{\text{ddh}}(D) \geq \varepsilon - \frac{1}{q},$$

non-negligible whenever ε is, since $1/q$ is negligible.

For the amplified form let ε be noticeable. The distinguisher D' rerandomises its input t times independently by the map of Remark 2.6, runs D with fresh coins on each copy, and outputs 1 iff *some* copy answered 1. A DH input becomes t independent uniform DH triples, each accepted with probability ε , so D' accepts with probability at least $1 - (1 - \varepsilon)^t$. A sample of $\mathcal{D}_{\text{rand}}$ fails to be a DH triple except with probability $1/q$, and then each copy’s third component is uniform over

the $q - 1$ non-completing values given the first two, so each copy is accepted with probability at most $1/(q - 1)$ and D' accepts with probability at most $1/q + t/(q - 1)$. With $t = \omega(\varepsilon^{-1} \log \lambda)$, polynomial because ε is noticeable, the advantage of D' is $1 - \text{negl}(\lambda)$. $\square$

Example 2.9 (DDH can be easy while CDH and DLP are hard). Some elliptic-curve groups carry an efficiently computable, non-degenerate *bilinear pairing* $e: G \times G \rightarrow G_T$ with $e(g^a, g^b) = e(g, g)^{ab}$. On such a group DDH is *easy*: to test whether (g^a, g^b, z) is a DH triple, check

$$e(g^a, g^b) \stackrel{?}{=} e(g, z),$$

which holds iff $\log_g z \equiv ab \pmod{q}$. Yet CDH—and a fortiori DLP—may remain hard, because the pairing reveals g^{ab} only inside the *target* group G_T , never as an element of G . Such groups, called *gap groups*, separate DDH from CDH and found pairing-based cryptography. The moral is that DDH is a genuinely stronger assumption: choosing a “DDH group” means choosing a group that rules out pairings of this convenient kind on G itself.

Remark 2.10 (The converses are open). The reverse reductions are not known in general. Whether CDH implies DLP—whether the ability to compute g^{ab} can be leveraged to extract a —is the long-standing Diffie–Hellman versus discrete log question; partial results of den Boer and Maurer establish equivalence when a suitable auxiliary group of *smooth* order—an order that is a product of small primes—is available, but no unconditional equivalence is known. In the other direction DDH may be strictly easier than CDH, as Example 2.9 shows. Practice therefore assumes exactly the strength a protocol requires: signatures and key exchange often need only CDH, while ElGamal-style encryption needs DDH for its ciphertexts to hide the plaintext.

2.4 Generic algorithms and the square-root barrier

How hard can the discrete logarithm problem possibly be? In a group with no exploitable structure beyond the group law, the best known algorithms are *generic*: they manipulate group elements only through multiplication, inversion, and equality testing, never inspecting the bit-representation of an element. Shoup’s idealisation makes the class precise: a random injective *labelling* $\sigma: \mathbb{Z}/q\mathbb{Z} \rightarrow S$ encodes the element g^k as the opaque string $\sigma(k)$, and the algorithm may only ask an oracle for $\sigma(i+j)$ given $\sigma(i)$ and $\sigma(j)$, and compare labels for equality. Two generic algorithms achieve $O(\sqrt{q})$ group operations; a matching lower bound then shows that, generically, nothing does better.

Baby-step giant-step

Proposition 2.11 (Baby-step giant-step, BSGS). *In a cyclic group of order q , any discrete logarithm can be computed in $O(\sqrt{q})$ group operations and $O(\sqrt{q})$ stored group elements, deterministically and unconditionally.*

Proof. Let $m = \lceil \sqrt{q} \rceil$. Since $m^2 \geq q$, every $x \in \{0, \dots, q-1\}$ decomposes as

$$x = i \cdot m + j, \quad 0 \leq i < m, \quad 0 \leq j < m,$$

and the target relation $g^x = h$ becomes $g^j = h \cdot (g^{-m})^i$. *Baby steps*: compute and store the table $\{(j, g^j) : 0 \leq j < m\}$, indexed by group element in a hash map— $m - 1$ multiplications. *Giant steps*: set $u = g^{-m}$ (one inversion plus one exponentiation, $O(\log q)$ operations) and for $i = 0, 1, 2, \dots$ compute $h \cdot u^i$ incrementally, looking each value up in the table. The decomposition guarantees some pair (i, j) matches, yielding $x = im + j$ after at most m giant steps. Total: $O(\sqrt{q})$ operations and $O(\sqrt{q})$ memory. $\square$

Remark 2.12 (Memory is the bottleneck). The $O(\sqrt{q})$ memory of BSGS, not its time, is what bites in practice: at the 128-bit security level no attacker can store $\sqrt{q} \approx 2^{128}$ table entries. Pollard’s rho method matches the time bound with negligible memory, which is why rho, rather than BSGS, is the realistic generic attack.

Pollard’s rho

Proposition 2.13 (Pollard’s rho, sketch). *In a cyclic group of prime order q there is a randomised algorithm computing discrete logarithms in expected $O(\sqrt{q})$ group operations and $O(1)$ storage.*

Proof. Sketch. Walk pseudo-randomly on G , keeping with each visited element its known representation $g^\alpha h^\beta$. Partition G into three roughly equal sets S_1, S_2, S_3 by an easily computed function of the element’s label, and from $P = g^\alpha h^\beta$ step to

$$f(P) = \begin{cases} h \cdot P = g^\alpha h^{\beta+1}, & P \in S_1, \\ P^2 = g^{2\alpha} h^{2\beta}, & P \in S_2, \\ g \cdot P = g^{\alpha+1} h^\beta, & P \in S_3, \end{cases}$$

each step updating $(\alpha, \beta) \in (\mathbb{Z}/q\mathbb{Z})^2$ explicitly at $O(1)$ cost. The sequence eventually cycles—a tail leading into a loop, the shape of the letter ρ . Modelling f as a random function on q elements, the birthday bound (*Math Guide*, §“The union bound and a birthday calculation”) produces a collision $P_i = P_j, i \neq j$, within $O(\sqrt{q})$ steps in expectation. A collision gives $g^{\alpha_i} h^{\beta_i} = g^{\alpha_j} h^{\beta_j}$, hence $g^{\alpha_i - \alpha_j} = h^{\beta_j - \beta_i}$; writing $h = g^x$,

$$\alpha_i - \alpha_j \equiv x(\beta_j - \beta_i) \pmod{q},$$

and if $\beta_j \not\equiv \beta_i$ the difference is invertible modulo the prime q , so $x \equiv (\alpha_i - \alpha_j)(\beta_j - \beta_i)^{-1} \pmod{q}$. Floyd’s tortoise-and-hare cycle detection, or Brent’s improvement, finds the collision with $O(1)$ memory, never storing the trajectory. The exceptional case $\beta_j \equiv \beta_i$ occurs with probability $O(1/q)$; re-randomising the start point handles it. $\square$

Remark 2.14 (Cost, not time, and constant factors). Pollard rho parallelises almost perfectly via the van Oorschot–Wiener method of *distinguished points*: each of P processors walks independently and reports to a shared table only the points whose label satisfies a cheap predicate, say a run of leading zero bits; two walks that ever meet coincide from then on and are detected at the next distinguished point, so the processors achieve $\sqrt{q}/P$ wall-clock time with modest communication. The security-relevant measure is therefore the *cost* of the attack—the total operation count $\sqrt{q}$, read as the attacker’s budget—not the time on any single machine. A further constant-factor saving applies in groups admitting the $\pm$ automorphism $P \mapsto -P$, as elliptic curves do: identifying P with $-P$ halves the effective search space and lowers the expected count from $\sqrt{\pi q/2}$ to $\sqrt{\pi q/4}$, a factor $\sqrt{2}$ that leaves the $\Theta(\sqrt{q})$ scaling untouched. The constant is the birthday integral: a random walk on N points is still collision-free after k steps with probability $\prod_{i < k} (1 - i/N) \approx e^{-k^2/2N}$ (the product in the birthday calculation cited above), and summing over k gives an expected first-collision index of $\int_0^\infty e^{-k^2/2N} dk = \sqrt{\pi N/2}$.

The generic lower bound

The two attacks are not merely the best known; up to constants they are the best *possible* for any generic algorithm. This is Shoup’s theorem, the formal statement of the square-root barrier.

Theorem 2.15 (Shoup’s generic lower bound). *Let q be prime. Any generic algorithm that makes at most n queries to the group oracle outputs the discrete logarithm of a uniformly random challenge with probability at most*

$$\frac{\binom{n+2}{2} + 1}{q} = O\left(\frac{n^2}{q}\right).$$

In particular, constant success probability requires $n = \Omega(\sqrt{q})$ queries.

Proof. Sketch—the linear-polynomial technique. In the idealisation, the challenge logarithm is an indeterminate X drawn uniformly from $\mathbb{Z}/q\mathbb{Z}$. By induction on the queries, every element the algorithm forms is σ of a *known linear polynomial* $a_i + b_i X \pmod{q}$ whose coefficients the algorithm chose. Treat X formally: as long as the polynomials produced are pairwise distinct as functions of X , the labels seen are consistent with X taking any value, so the algorithm has learnt nothing and can only guess, succeeding with probability $1/q$. Information leaks only through an accidental collision: two distinct polynomials agreeing at the secret x , that is, $(a_i - a_j) + (b_i - b_j)x \equiv 0 \pmod{q}$. For a fixed pair this nonzero linear equation has at most one root modulo the prime q —the degree-one case of the Schwartz–Zippel lemma (*Math Guide*, §“The Schwartz–Zippel lemma”)—so it holds with probability at most $1/q$ over uniform x . With n queries plus the two inputs g and h there are at most $\binom{n+2}{2}$ pairs, and the union bound (*Math Guide*, §“The union bound and a birthday calculation”) caps the collision probability at $\binom{n+2}{2}/q$. Adding the $1/q$ guess yields the bound; solving $n^2/q = \Omega(1)$ gives $n = \Omega(\sqrt{q})$. $\square$

Corollary 2.16 (Why $\log_2 q \geq 2\lambda$). *To force every generic attacker to spend at least 2^λ group operations— λ bits of security against generic attacks—the prime group order must satisfy*

$$\sqrt{q} \gtrsim 2^\lambda \iff \log_2 q \geq 2\lambda.$$

Proof. Propositions 2.11 and 2.13 show that $\Theta(\sqrt{q})$ operations suffice for a generic attacker, and Theorem 2.15 shows no generic attacker does asymptotically better. Equating the attacker’s cost $\sqrt{q}$ with the target work factor 2^λ gives $q \approx 2^{2\lambda}$; demanding at least this much is $\log_2 q \geq 2\lambda$. $\square$

Example 2.17 (The 128-bit target). For $\lambda = 128$ the corollary demands a prime group order with $\log_2 q \geq 256$. This explains the standardised 256-bit elliptic curves: NIST P-256 (secp256r1) has a 256-bit field and a 256-bit group order, so $\sqrt{q} \approx 2^{128.0}$. The Pasta curves of §2.6, with 255-bit orders just above 2^{254} and about 126 bits of security, sit just below this class. Contrast the multiplicative group $\mathbb{F}_p^\times$: there the sub-exponential index calculus of Remark 2.18 forces $\log_2 p$ into the thousands of bits for the same λ . Elliptic curves earn their keep precisely because no sub-exponential, generic-beating algorithm is known for them.

Remark 2.18 (Why the barrier is special to elliptic curves). The square-root barrier governs *generic* groups only; a concrete representation can leak structure that beats it. The multiplicative group $\mathbb{F}_p^\times$ is the cautionary case: *index calculus* exploits the factorisation of integers into small primes—smooth numbers—to solve DLP in sub-exponential time

$$\exp(O((\log p)^{1/3}(\log \log p)^{2/3})),$$

far below $\sqrt{p}$. No analogue is known for the points of a well-chosen elliptic curve over $\mathbb{F}_p$: there is no useful notion of a “small” point over which to factor. This non-generic gap is why elliptic-curve groups can live at 256 bits where $\mathbb{F}_p^\times$ must reach about 3072 bits for the same security. It also means the rule $\log_2 q \geq 2\lambda$ protects against generic attacks *only*; a curve must additionally be screened against the special-purpose attacks of Definition 2.24.

2.5 Pohlig–Hellman and the necessity of large prime order

The square-root barrier priced the attack at $\sqrt{q}$ for *prime* q . If the group order factors, the discrete logarithm splits into independent pieces the size of the factors, and the difficulty collapses to that of the *largest prime* dividing the order. This is the Pohlig–Hellman reduction—the reason Definition 2.2 insisted on prime order.

Theorem 2.19 (Pohlig–Hellman). *Let $G = \langle g \rangle$ have order $N = \prod_{i=1}^r p_i^{e_i}$ with the p_i distinct primes. Then a discrete logarithm in G can be computed in*

$$O\left(\sum_{i=1}^r e_i (\log N + \sqrt{p_i})\right)$$

group operations. The term $\sqrt{p_{\max}}$ dominates, where $p_{\max}$ is the largest prime dividing N .

Proof. We compute $x = \log_g h$ modulo each prime power $p_i^{e_i}$ and recombine by the Chinese Remainder Theorem, the moduli being pairwise coprime with product N .

Reduction to prime-power order. Fix $p = p_i$, $e = e_i$, and put $g_i = g^{N/p^e}$, $h_i = h^{N/p^e}$. Then g_i has order exactly p^e , and $\log_{g_i} h_i \equiv x \pmod{p^e}$: raising to N/p^e kills the part of x coprime to p 's power and leaves $x \bmod p^e$. It therefore suffices to solve DLP in $\langle g_i \rangle$ of order p^e .

Reduction to prime order, digit by digit. Write the unknown in base p :

$$x \equiv x_0 + x_1 p + \cdots + x_{e-1} p^{e-1} \pmod{p^e}, \quad 0 \leq x_k < p.$$

Let $\gamma = g_i^{p^{e-1}}$, an element of order exactly p . Raising h_i to the power p^{e-1} gives

$$h_i^{p^{e-1}} = g_i^{x p^{e-1}} = \gamma^{x_0},$$

because every higher digit contributes a multiple of p^e to the exponent and annihilates γ . Thus x_0 is a single discrete logarithm in a group of prime order p , solvable in $O(\sqrt{p})$ operations by BSGS or rho (Propositions 2.11 and 2.13). Having x_0 , set $h_i^{(1)} = h_i \cdot g_i^{-x_0}$; its logarithm is divisible by p , and the same step at exponent p^{e-2} yields x_1 . Iterating recovers all e digits with e prime-order DLPs of cost $O(\sqrt{p})$ each, plus $O(e \log N)$ operations of exponentiation. Summing over i and applying the CRT gives the stated bound. $\square$

Example 2.20 (A composite-order trap). Suppose, contrary to good practice, a group has order $N = 2^{256} - 1$. The order factors completely into

$$\begin{aligned} 2^{256} - 1 = & 3 \cdot 5 \cdot 17 \cdot 257 \cdot 641 \cdot 65537 \cdot 274177 \cdot 6700417 \cdot 67280421310721 \\ & \cdot 59649589127497217 \cdot 5704689200685129054721, \end{aligned}$$

a product of small primes—it is *smooth*. Despite N having 256 bits, the largest prime factor is only about $2^{72.3}$, far below 2^{128} , so Pohlig–Hellman with rho in the largest subgroup solves DLP in roughly $\sqrt{p_{\max}} \approx 2^{36.1}$ operations—about 2^{36} , nowhere near the apparent 128-bit level. A 256-bit order is necessary but *not* sufficient: it must be (almost) prime.

Corollary 2.21 (Design rule for the group order). *For DLP-based security at level λ , the group order must be divisible by a prime q with $\log_2 q \geq 2\lambda$, and the cryptographic group must be, or be restricted to, the subgroup of order q . When the ambient group has order $N = h \cdot q$ with small cofactor h and large prime q , one works in the order- q subgroup; clearing the cofactor—multiplying incoming ele-*

ments by h , or checking subgroup membership—prevents small-subgroup attacks that would otherwise leak $x \bmod h$ via Pohlig–Hellman on the order- h part.

Remark 2.22 (Prime order, revisited). Here is the deeper justification for “prime order” in Definition 2.2. Prime order q makes Pohlig–Hellman vacuous (the only factor is q itself), makes every non-identity element a generator (no element lands in a weak proper subgroup), and removes cofactor bookkeeping entirely. The Pasta curves below are engineered to have prime order: each group is its own prime-order subgroup, with cofactor 1.

2.6 Instantiation on elliptic curves; the Pasta curves

We now fix the concrete groups the remainder of the volume uses. Recall from the *Math Guide* (§“Elliptic curves”, with the group law in §“The group structure of $E(\mathbb{F}_p)$ ”) that an elliptic curve over $\mathbb{F}_p$, for $p > 3$ prime, in short Weierstrass form is

$$E: y^2 = x^3 + ax + b, \quad a, b \in \mathbb{F}_p, \quad 4a^3 + 27b^2 \neq 0,$$

and that its set of $\mathbb{F}_p$ -points,

$$E(\mathbb{F}_p) = \{(x, y) \in \mathbb{F}_p^2 : y^2 = x^3 + ax + b\} \cup \{O\},$$

equipped with chord-and-tangent addition and the point at infinity O as identity, forms a finite abelian group. The discrete logarithm problem in this group—given P and $Q = [x]P$, find x —is the *elliptic-curve discrete logarithm problem* (ECDLP), stated in its own habitat in the *Math Guide* (§“The elliptic-curve discrete logarithm problem”); CDH and DDH transcribe verbatim with multiplicative notation replaced by the additive scalar multiplication $x \mapsto [x]P$, giving ECCDH and ECDDH. The group size is pinned down by the Hasse bound, likewise recalled.

Theorem 2.23 (Hasse bound, recalled). For E over $\mathbb{F}_p$ (*Math Guide*, §“The group structure of $E(\mathbb{F}_p)$ ”),

$$|\#E(\mathbb{F}_p) - (p + 1)| \leq 2\sqrt{p}, \quad \text{i.e.} \quad \#E(\mathbb{F}_p) = p + 1 - t, \quad |t| \leq 2\sqrt{p},$$

where t is the trace of Frobenius. The group order is thus $p + 1$ up to a deviation of order $\sqrt{p}$.

The parameter recipe for λ -bit security follows: choose p of about 2λ bits, so that $\#E(\mathbb{F}_p) \approx p \approx 2^{2\lambda}$, and insist by Corollary 2.21 that $\#E(\mathbb{F}_p)$ be prime—cofactor 1. One then assumes ECDLP, ECCDH, and ECDDH at the level Corollary 2.16 predicts, *provided* the curve avoids the known structural attacks that beat the generic bound. The screening criteria are as follows.

Definition 2.24 (Secure-curve criteria). A curve $E/\mathbb{F}_p$ of prime order $q = \#E(\mathbb{F}_p)$ is admissible for λ -bit DLP security if:

1. (*Size / generic.*) The order satisfies $\log_2 q \geq 2\lambda$, so Pollard rho costs at least 2^λ operations (Corollary 2.16).
2. (*Prime order.*) The order q is prime, so Pohlig–Hellman is vacuous and the cofactor is 1 (Corollary 2.21).
3. (*Anti-MOV / large embedding degree.*) The multiplicative order k of p modulo q —the *embedding degree*—is large. The MOV/Frey–Rück attack uses a pairing to embed $E(\mathbb{F}_p)$ into $\mathbb{F}_{p^k}^\times$ and runs the sub-exponential index calculus of Remark 2.18 there; the threat is real only for small k , as on *supersingular* curves—over a prime field $p > 3$, those with

trace $t = 0$, so that $\#E(\mathbb{F}_p) = p + 1$ divides $p^2 - 1$ and $k \leq 2$. A curve with $t \neq 0$ is *ordinary*.

4. (*Anti-Smart / not anomalous*.) The order avoids $q = p$: on an *anomalous* curve, $\#E(\mathbb{F}_p) = p$ with trace $t = 1$, the Smart–Sato–Araki–Semaev attack solves ECDLP in polynomial time.
5. (*Twist security*.) The *quadratic twist* E' —the curve $dy^2 = x^3 + ax + b$ for a fixed nonsquare $d \in \mathbb{F}_p$; every $x \in \mathbb{F}_p$ is the x -coordinate of a point of E or of E' —should also have large prime order, so that computation cannot be pushed onto a weak twist by a fault or an invalid-point injection.

Definition 2.25 (The Pasta curves). The *Pasta* curves are a pair, *Pallas* and *Vesta*, in short Weierstrass form

$$y^2 = x^3 + 5$$

(so $a = 0$, $b = 5$, a j -invariant-0 shape), defined over two 255-bit primes p and q arranged in a *cycle*:

$$\#Pallas = q \text{ (Pallas defined over } \mathbb{F}_p), \quad \#Vesta = p \text{ (Vesta defined over } \mathbb{F}_q).$$

The order of each curve equals the field of definition of the other. Both primes are congruent to 1 mod 2^{32} , giving their multiplicative groups large 2-adic valuation for fast FFTs, and each curve has cofactor 1: the whole group is of prime order. The primes themselves are assembled digit by digit in the *Math Guide* (§“Pallas and Vesta assembled”), and the field/scalar bookkeeping of the cycle in §“Base fields, scalar fields, and the Pasta cycle” there.

Remark 2.26 (Why a cycle, and what it buys). The defining property— $\#Pallas = q$, the base field of Vesta, and $\#Vesta = p$, the base field of Pallas—makes (Pallas, Vesta) an *amicable*, or cyclic, pair: one curve’s scalar field is the other’s coordinate field. A proof system can therefore use Pallas-arithmetic to verify statements about Vesta-arithmetic and vice versa without non-native field emulation; this is the mechanism behind recursive proof composition in Halo 2, whose recursion alternates between the two curves. The cycle is an *efficiency* property, not a security one: it does not enter the hardness assumptions, and each curve is screened independently against Definition 2.24.

Proposition 2.27 (The Pasta curves against the criteria). *The curves Pallas and Vesta satisfy criteria (1)–(4) of Definition 2.24 at $\lambda \approx 126$. Criterion (5) fails for both curves—neither twist has prime order, and the Pallas twist falls below the SafeCurves 2^{100} twist-attack cost threshold—but the failure is rendered moot by mandatory point validation.*

Proof. Sketch, criterion by criterion.

(1) Each order is a 255-bit prime just above 2^{254} , so $\log_2 q \approx 254.0$ and the bare generic cost is $\sqrt{q} \approx 2^{127.0}$. Plain Pollard rho expects about $\sqrt{\pi q/2} \approx 2^{127.3}$ operations; the $\pm$ automorphism speedup of Remark 2.14 lowers this to $\sqrt{\pi q/4} \approx 2^{126.8}$; and the order-6 automorphism group of these j -invariant-0 curves—the factor $\sqrt{m}$ for an order- m automorphism group of Remark 2.28, here an extra $\sqrt{3}$ —lowers it to $\sqrt{\pi q/12} \approx 2^{126.0}$. Hence “about 126 bits” of security, the same range as the 128-bit curves of Example 2.17.

(2) Both orders are prime by construction: cofactor 1, Pohlig–Hellman vacuous.

(3) Both curves are ordinary: the trace $t = p + 1 - q$ is odd, so $t \neq 0$. The embedding degree k of each is enormous. For Pallas, $q - 1 = 2^{32} \cdot 3^2 \cdot 1709 \cdot 24859 \cdot C$ with C a 194-bit composite,

and $p^{(q-1)/32} \not\equiv 1$ and $p^{(q-1)/\ell} \not\equiv 1 \pmod{q}$ for $\ell \in \{3, 1709, 24859\}$, so the order k of p modulo q is a multiple of $2^{28} \cdot 3^2 \cdot 1709 \cdot 24859 > 2^{56}$; for Vesta, $p - 1 = 2^{32} \cdot 3 \cdot 463 \cdot C'$, and $q^{(p-1)/4} \not\equiv 1$, $q^{(p-1)/\ell} \not\equiv 1 \pmod{p}$ for $\ell \in \{3, 463\}$ make k a multiple of $2^{31} \cdot 3 \cdot 463 > 2^{41}$. Either extension field is astronomically beyond any feasible index-calculus target; MOV/Frey–Rück does not apply.

(4) Neither curve is anomalous: $\#Pallas = q \neq p$ and $\#Vesta = p \neq q$, so the trace $t = p + 1 - q \neq 1$ and the Smart attack does not apply.

(5) Twist security is the criterion the pair does not meet. The Pasta search deliberately ignored it—the curves were generated with the search flag `--ignoretwist` of the `amicable.sage` search recorded in the README of the `zcash/pasta` repository—and neither twist has prime order. Each $x \in \mathbb{F}_p$ supplies two affine points in total, on E or on E' (one on each when $x^3 + b = 0$), so $\#E + \#E' = 2(p + 1)$ and $\#E' = 2(p + 1) - \#E = p + 1 + t$; hence $\#Pallas' = 2(p + 1) - q$ and $\#Vesta' = 2(q + 1) - p$, which factor as

$$\begin{aligned}\#Pallas' &= 3^2 \cdot 7 \cdot 8191 \cdot 85021 \cdot 11540602760389 \cdot P_{176}, \\ \#Vesta' &= 3^3 \cdot 7 \cdot 13 \cdot 2851 \cdot 30097 \cdot P_{217},\end{aligned}$$

with P_{176} a 176-bit prime and P_{217} a 217-bit prime. The best attack on the Pallas twist therefore costs about $\sqrt{P_{176}} \approx 2^{87.6}$, below the SafeCurves 2^{100} twist-security threshold; the Vesta twist clears the threshold at about $2^{108.2}$. The failure is moot in our setting: twist attacks require an implementation that accepts an attacker-supplied x -coordinate without validation (an x -only ladder) or that skips its on-curve checks, and the deployed arithmetic does neither. The `pasta_curves` crate carries points in full coordinates and has no x -only code path; its deserialisation (`from_bytes`) reconstructs y by taking the square root of $x^3 + 5$, failing exactly when x lies on the twist, where $x^3 + 5$ is a non-square, and even `from_bytes_unchecked` delegates to the checked path because a compressed encoding cannot skip the curve check, while points built from raw coordinates are gated on `is_on_curve`. This is the behaviour protocol specification § 5.4.9.6 mandates: its decoding map returns $\perp$ (failure) when $x^3 + b$ has no square root, and an implementation must not assume that the root exists or that the encoding lies on the curve. Orchard deserialises its points through this routine, and cofactor 1 leaves no small subgroup on the curve itself. Invalid-curve and twist-point injection are therefore excluded by *validation*, not by twist structure.

The standard generic analysis governs, and the security level is about 126 bits. $\square$

Remark 2.28 (j -invariant 0 and endomorphisms). The shape $y^2 = x^3 + b$ gives j -invariant 0, and with it an extra automorphism of order 3 (*Math Guide*, §“The j -invariant and the GLV endomorphism”): for ω a primitive cube root of unity in $\mathbb{F}_p$, the cheap endomorphism $\phi(x, y) = (\omega x, y)$ acts on the group as multiplication by a fixed scalar ζ with $\zeta^2 + \zeta + 1 \equiv 0 \pmod{q}$ (the λ of the *Math Guide*’s proposition), enabling the GLV speedup for honest scalar multiplication. The same endomorphism hands the attacker a slightly larger automorphism group, and with it the constant-factor Pollard-rho speedup used in Proposition 2.27—a factor $\sqrt{m}$ for an order- m automorphism group—but that is a small constant and does not change the $\Theta(\sqrt{q})$ scaling. Beyond that constant, the extra automorphism is not known to weaken ECDLP; the danger lies only in small embedding degree and in anomalousness, both excluded above.

2.7 The quantum caveat: Shor’s algorithm

Every assumption above holds the adversary to classical PPT computation. Against a large fault-tolerant quantum computer the picture changes qualitatively, and honesty requires stating exactly what breaks and what survives.

Theorem 2.29 (Shor, informal; stated without proof). *There is a quantum algorithm that, given*

a cyclic group $G = \langle g \rangle$ of order q with efficient group operations and a target $h = g^x$, outputs x in time polynomial in $\log q$, succeeding with overwhelming probability. Consequently DLP—and with it CDH and DDH—falls in quantum polynomial time in any group, elliptic-curve groups included.

We do not prove the theorem; the shape of the algorithm is nevertheless worth recording. Discrete logarithm is an instance of the *hidden subgroup problem* over $(\mathbb{Z}/q\mathbb{Z})^2$. The function $f(\alpha, \beta) = g^\alpha h^{-\beta} = g^{\alpha-x\beta}$ is constant exactly on the cosets of

$$H = \{(\alpha, \beta) : \alpha \equiv x\beta \pmod{q}\} = \langle (x, 1) \rangle \subseteq (\mathbb{Z}/q\mathbb{Z})^2,$$

a line of slope x . The algorithm prepares a uniform superposition over $(\mathbb{Z}/q\mathbb{Z})^2$, evaluates f into a second register, and applies the quantum Fourier transform—the reversible change of basis into the characters of $(\mathbb{Z}/q\mathbb{Z})^2$, the homomorphisms $\chi_{u,v}(\alpha, \beta) = e^{2\pi i(u\alpha+v\beta)/q}$ into the unit circle; measurement then samples characters trivial on H , i.e. pairs (u, v) with $ux + v \equiv 0 \pmod{q}$, and a handful of samples reveal the slope x by classical linear algebra modulo q . The whole procedure uses $O(\text{poly}(\log q))$ qubits and gates, and it touches the group only through the black-box map $(\alpha, \beta) \mapsto g^\alpha h^{-\beta}$, so it is agnostic to representation: elliptic-curve DLP falls exactly as $\mathbb{F}_p^\times$ DLP does, and integer factoring falls to the order-finding variant.

Remark 2.30 (No contradiction with the square-root barrier). Shoup’s lower bound (Theorem 2.15) assumes the generic model for *classical* algorithms making sequential, individual oracle queries. Shor’s algorithm is non-generic in exactly the relevant sense: it queries the group function in superposition and exploits the periodicity the Fourier transform exposes, abilities the classical generic model does not grant. The two results coexist: the barrier bounds classical generic attackers; Shor is a quantum, non-generic attacker.

Remark 2.31 (What Shor breaks and what survives). Shor’s algorithm breaks the entire discrete-log family and integer factoring in polynomial time: RSA, finite-field Diffie–Hellman, and elliptic-curve cryptography are all quantum-broken. What it does *not* break:

- *Symmetric primitives and hash functions.* Grover’s algorithm searches a space of size 2^n in about $2^{n/2}$ quantum queries—a quadratic, not exponential, speedup—so a cipher with a 2λ -bit key, or an $n = 2\lambda$ -bit hash against preimages, retains λ bits of *quantum* security. Generic quantum collision finding instead costs $\Theta(2^{n/3})$ queries, so in the query metric an output of about 3λ bits is needed for λ -bit quantum collision security—though the algorithm needs comparably much quantum memory, and whether it beats classical parallel collision search in realistic cost models is debated. Doubling therefore suffices for key search and preimages, but for collisions only under that contested cost reading.
- *Post-quantum assumptions.* Hardness assumptions based on neither discrete logarithms nor factoring are not known to fall to Shor and underlie post-quantum cryptography.

The discrete-log assumptions of this volume are therefore adopted under the standing premise that no cryptographically relevant quantum computer exists. Within that premise, and against all classical attacks, the Pasta curves deliver the approximately 126-bit security of Proposition 2.27—the ground on which the Halo 2 constructions to come are built.

2.8 Standing assumptions

Remark 2.32 (Standing assumptions for the sequel). We collect the premises the rest of the volume relies on. For each Pasta curve $C \in \{\text{Pallas}, \text{Vesta}\}$ with prime order q_C and generator g_C :

1. ECDLP is hard in $\langle g_C \rangle$: no classical PPT adversary finds x from $[x]g_C$ with non-negligible probability—concretely, every known attack costs at least 2^{126} operations (Proposition 2.27).

2. ECCDH and, where needed, ECDDH hold in $\langle g_C \rangle$ (Definitions 2.4 and 2.5, transcribed additively).
3. The adversary is classical and probabilistic polynomial-time (Remark 2.31).

By the hierarchy of §2.3, assuming ECDDH is the strongest commitment and implies the rest; each later construction invokes, by name, the weakest assumption it requires.

3 Hash functions and the random oracle model

The adversary of this section forges by *coincidence*. She prepares two contracts—one benign, one ruinous—whose digests agree, has the benign one signed, and produces the ruinous one in court; or she takes a digest minted for one purpose, say a key-derivation output, and replays it in another role, say a commitment, exploiting the fact that both are interchangeable bit strings; or she simply walks a pseudo-random path through the digest space until, square-root-of-the-space steps later, two inputs collide. Denying her these coincidences is the business of the hash function: a function compressing arbitrary data into a short fixed-length tag. It underlies digital signatures, commitment schemes, message authentication, key derivation, proof-of-work, Merkle trees, and the Fiat–Shamir transform behind Halo 2’s non-interactive arguments—so many purposes for one primitive because an *ideal* hash acts like a deterministic source of fresh randomness keyed by its input. This section builds the primitive from the ground up: its syntax; the three classical security notions and the generic birthday attack that bounds them; the random oracle model; domain separation; hashing into a finite field and onto an elliptic curve; the arithmetisation-friendly hash Poseidon that Orchard evaluates inside its circuit; and finally a register of the transparent layer’s SHA-256-derived hashes.

3.1 Syntax

Notation for strings is fixed once. The set of all finite binary strings is $\{0, 1\}^*$; the strings of length exactly n form $\{0, 1\}^n$; the length of a string x is $|x|$; and concatenation is written $x \parallel y$.

Definition 3.1 (Hash function). A *hash function* with output length $n \in \mathbb{N}$ is a function

$$H: \{0, 1\}^* \rightarrow \{0, 1\}^n.$$

Elements of the codomain are called *digests*, *hashes*, or *tags*. The function is *compressing* on a domain $D \subseteq \{0, 1\}^*$ if D contains strings longer than n ; when the domain is all of $\{0, 1\}^*$, the function is overwhelmingly many-to-one.

A single fixed function such as SHA-256 is what practice deploys, but a foundational wrinkle appears the moment one tries to say what its security *means*. For any fixed compressing H there *exist* inputs $x \neq x'$ with $H(x) = H(x')$ —the pigeonhole principle guarantees it—and therefore a tiny program that simply outputs one such hardcoded pair also exists. The asymptotic statement “no efficient adversary finds a collision” is consequently *false* for every fixed compressing function: the hardcoding program is efficient and always succeeds. (Writing that program down is another matter—nobody knows a collision for SHA-256—but existence suffices to falsify the universally quantified claim.) The standard remedy indexes the function by a key.

Definition 3.2 (Keyed hash family). A *keyed hash family* (or *hash function ensemble*) is a function

$$H: \mathcal{K} \times \{0, 1\}^* \rightarrow \{0, 1\}^n,$$

where $\mathcal{K}$ is a finite key space. A key $s \in \mathcal{K}$ is sampled *publicly* at random, fixed, and we write $H_s(\cdot) := H(s, \cdot)$ for the family member in force.

The key is *not* secret. Its role is to index which member of the family the adversary must attack: a hardcoded collision for one member is useless against a freshly sampled other, so “no efficient adversary finds a collision for a random member” is a meaningful asymptotic statement.

Remark 3.3 (Convention: keyed definitions, unkeyed deployments). The convention of this volume follows the standard resolution of the mismatch between theory and practice. Security notions are stated for the keyed family, where the definitions are clean; every concrete construction we deploy is unkeyed; and the key is mentally identified with “the choice of standard”—the sampling of s happened, once and in public, when the function was designed. No formal theorem bridges the two readings; the gap is the same one the random oracle model of §3.4 inhabits, and we flag it rather than hide it.

Definition 3.4 (Extendable-output function). An *extendable-output function* (XOF) is a primitive producing a digest of caller-chosen length: syntactically a function

$$H: \{0, 1\}^* \times \mathbb{N} \rightarrow \{0, 1\}^*, \quad |H(x, \ell)| = \ell,$$

whose outputs are *prefix-consistent*: for $\ell \leq \ell'$, the string $H(x, \ell)$ is a prefix of $H(x, \ell')$, so a longer request extends a shorter one as a stream.

The canonical XOFs are SHAKE128 and SHAKE256 from the SHA-3 family. An XOF is the natural tool wherever a *controlled number* of uniform bits is needed rather than a fixed digest—above all when hashing into a finite field or onto a curve, where the output must be reduced modulo a prime; §3.6 returns to this.

3.2 Security notions: preimage, second-preimage, and collision resistance

Each of the three classical security notions is a game between a challenger and a PPT adversary $\mathcal{A}$, in the template of Definition 1.15; the keyed family H is secure in a given sense if every PPT adversary wins the corresponding game with probability negligible in the security parameter λ . All sizes are functions of λ : the output length $n = n(\lambda)$, the key length, and a polynomially bounded challenge input length $m = m(\lambda)$ from which the games sample their random challenge $x \xleftarrow{\$} \{0, 1\}^m$. Recall that a function $\varepsilon(\lambda)$ is negligible, $\varepsilon \in \text{negl}(\lambda)$, if it decays faster than the reciprocal of every polynomial (*Math Guide*, §“Polynomial, exponential, and negligible functions”; recalled in §1.2). The three notions form a strict hierarchy, and no single definition of a “good” hash exists; which notion an application needs is a per-application question that Remark 3.10 takes up.

Definition 3.5 (Preimage resistance). The game $\text{PRE}_H^{\mathcal{A}}(\lambda)$ runs as follows.

1. The challenger samples $s \xleftarrow{\$} \mathcal{K}$ and $x \xleftarrow{\$} \{0, 1\}^m$, computes $y := H_s(x)$, and sends (s, y) to the adversary.
2. The adversary outputs $x' \in \{0, 1\}^*$.
3. The game outputs 1 iff $H_s(x') = y$.

The family H is *preimage resistant* (or *one-way*) if for every PPT A ,

$$\text{Adv}_H^{\text{pre}}(A, \lambda) := \Pr[\text{PRE}_H^A(\lambda) = 1] \in \text{negl}(\lambda).$$

Note the winning condition: the adversary, given the key and the digest of a random message, must output *any* string hashing to y . Recovering the original x is not required; any preimage wins.

Definition 3.6 (Second-preimage resistance). The game $\text{SEC}_H^A(\lambda)$ runs as follows.

1. The challenger samples $s \xleftarrow{\$} \mathcal{K}$ and $x \xleftarrow{\$} \{0, 1\}^m$, and sends (s, x) to the adversary.
2. The adversary outputs $x' \in \{0, 1\}^*$.
3. The game outputs 1 iff $x' \neq x$ and $H_s(x') = H_s(x)$.

The family H is *second-preimage resistant* if for every PPT A ,

$$\text{Adv}_H^{\text{sec}}(A, \lambda) := \Pr[\text{SEC}_H^A(\lambda) = 1] \in \text{negl}(\lambda).$$

Here the challenger *gives* the adversary a specific random message; the adversary must produce a different message colliding with it.

Definition 3.7 (Collision resistance). The game $\text{COL}_H^A(\lambda)$ runs as follows.

1. The challenger samples $s \xleftarrow{\$} \mathcal{K}$ and sends s to the adversary.
2. The adversary outputs a pair (x, x') .
3. The game outputs 1 iff $x \neq x'$ and $H_s(x) = H_s(x')$.

The family H is *collision resistant* if for every PPT A ,

$$\text{Adv}_H^{\text{col}}(A, \lambda) := \Pr[\text{COL}_H^A(\lambda) = 1] \in \text{negl}(\lambda).$$

The three games differ in *who chooses what*. In the preimage game the challenger chooses the message and reveals only its digest; in the second-preimage game the challenger chooses the message and reveals it; in the collision game the adversary chooses *both* messages and need only make them collide. More adversarial choice means an easier game and hence a stronger security demand on the function; collision resistance, granting the most choice, is the strongest of the three.

Proposition 3.8 (Hierarchy of notions). *Let H be a keyed hash family that is compressing, with challenge input length $m > n$. Then:*

1. *collision resistance implies second-preimage resistance, with no loss; and*
2. *second-preimage resistance implies preimage resistance up to an additive loss of 2^{n-m} , which is negligible—so the implication holds outright—whenever H is sufficiently compressing, $m - n = \omega(\log \lambda)$.*

Neither implication reverses in general.

Proof. (1) From an adversary A against second-preimage resistance, construct $\mathcal{B}$ against collision resistance: on input the key s , the reduction $\mathcal{B}$ samples $x \xleftarrow{\$} \{0, 1\}^m$ itself, runs $x' \leftarrow A(s, x)$ —

the simulated view is exactly that of SEC, since $\mathcal{B}$ samples x from the same distribution as the challenger—and outputs the pair (x, x') . Whenever A succeeds, $x' \neq x$ and $H_s(x') = H_s(x)$, which is precisely a collision. Hence $\text{Adv}_H^{\text{col}}(\mathcal{B}, \lambda) \geq \text{Adv}_H^{\text{sec}}(A, \lambda)$: the reduction is tight, and if the left side is negligible so is the right.

(2) Toward the contrapositive, let A be an inverter; we build a second-preimage finder $\mathcal{B}$. On input (s, x) , the reduction computes $y := H_s(x)$, runs $x' \leftarrow A(s, y)$ —again a perfect simulation of PRE, since x is uniform and y is its digest—and outputs x' . Whenever A succeeds, $H_s(x') = y = H_s(x)$; what remains is to argue $x' \neq x$ with good probability. Fix A 's coins, so that its output on each digest is a single string $x' = A(s, y)$. An input x can satisfy $x = A(s, H_s(x))$ only if x is the output assigned to its own digest, so each of the 2^n digests contributes at most one such input, and at most 2^n inputs in all are “fixed points” of the composed map. Over uniform $x \xleftarrow{\$} \{0, 1\}^m$, therefore, $\Pr[x' = x] \leq 2^n / 2^m = 2^{n-m}$; averaging over A 's coins preserves the bound. Hence

$$\text{Adv}_H^{\text{sec}}(\mathcal{B}, \lambda) \geq \text{Adv}_H^{\text{pre}}(A, \lambda) - 2^{n-m}.$$

The loss is *additive*, not multiplicative: nothing bounds $\Pr[x' \neq x]$ conditioned on success, because A 's successes might concentrate on small fibres where returning x itself is likely. Sufficient compression, $m - n = \omega(\log \lambda)$, makes 2^{n-m} negligible and the implication unconditional.

Non-reversal is witnessed by Example 3.9 below. $\square$

Example 3.9 (Preimage-resistant but not collision-resistant). Take any one-way family H and define $H'_s(x) := H_s(\bar{x})$, where $\bar{x}$ deletes the last bit of x . The family H' inherits one-wayness: a preimage challenge (s, y) for H is distributed exactly as one for H' at one bit greater challenge length, and an H' -inverter's output x'' satisfies $H_s(\overline{x''}) = y$, so deleting its last bit hands the reduction a preimage under H . Yet the pair $x \parallel 0, x \parallel 1$ is an immediate collision for every x , found without any computation. Hence preimage resistance does not imply collision resistance. The other separations are constructed similarly; the implications of Proposition 3.8 are genuinely one-directional.

Remark 3.10 (Which application needs which notion). Password storage—keeping a digest of each password so that a stolen table does not reveal them—is the natural home of *preimage* resistance: the thief holds digests, and any preimage lets her log in. (The notion alone is not the whole story there: passwords are guessable, so deployments hash each together with a unique random string, its *salt*, through a deliberately slow function, against offline guessing.) A signature scheme that signs $H(m)$, where an adversary may later try to substitute a forged document for the honestly chosen m , needs *second-preimage* resistance: the message is fixed and honest, and the attacker must collide with it. A scheme in which the *signer herself* may cheat—preparing two contracts with the same digest, signing the benign one, presenting the malicious one—needs full *collision* resistance, because there the adversary controls both messages. Commitment schemes, Merkle trees, and the Fiat–Shamir transform all fall into this last category, which is why collision resistance is the headline property of a cryptographic hash.

3.3 The birthday bound on generic collision finding

How hard can collision finding be? The generic attack—evaluate the hash on many inputs and wait for a coincidence—succeeds far sooner than intuition suggests, and its cost is governed by the birthday calculation of the *Math Guide*.

Lemma 3.11 (Birthday bound for hashing). *Let $H: \{0, 1\}^* \rightarrow \{0, 1\}^n$ and $N = 2^n$. Evaluate H on q distinct inputs, in the idealised model where the q digests behave as independent uniform samples*

from $\{0, 1\}^n$, and let C be the event that two digests coincide. Then

$$\Pr[C] = 1 - \prod_{i=0}^{q-1} \left(1 - \frac{i}{N}\right),$$

with the two-sided estimate

$$1 - e^{-q(q-1)/(2N)} \leq \Pr[C] \leq \frac{q(q-1)}{2N} \leq \frac{q^2}{2N},$$

and the lower bound gives $\Pr[C] \geq 1/2$ once $q \geq 1.18\sqrt{N}$. A collision therefore appears with constant probability after about $\sqrt{N} = 2^{n/2}$ evaluations.

Proof. The exact product and both estimates are the birthday-bound proposition of the *Math Guide* (§“The union bound and a birthday calculation”), instantiated with set size $N = 2^n$ and sample count $k = q$; we do not re-derive them. Only the half-probability threshold is new. Setting the exponent of the lower bound equal to $-\ln 2$ gives $q(q-1) = 2N \ln 2$, i.e.

$$q \approx \sqrt{2 \ln 2} \sqrt{N} \approx 1.18\sqrt{N},$$

the constant being $\sqrt{2 \ln 2} = 1.1774 \dots$; for $q \geq 1.18\sqrt{N}$ the exponential falls below $1/2$ and $\Pr[C] \geq 1/2$. $\square$

Remark 3.12 (Reading the birthday bound). Three consequences deserve emphasis.

The square-root law. An n -bit digest yields only $n/2$ bits of collision security in the sense of Definition 1.32. For the conventional 128-bit collision security one must take $n = 256$ —which is why SHA-256 (and not “SHA-128”, which does not exist) is the workhorse digest size, and why the transcript hashes of Fiat–Shamir-style protocols must carry at least 256-bit output.

The attack is practical. The naive attack stores all $q \approx 2^{n/2}$ digests and sorts— $2^{n/2}$ memory, the same obstruction that made baby-step giant-step unrealistic in Remark 2.12. But collision finding, like the discrete-logarithm walk of Proposition 2.13, admits cycle finding: iterate $x \mapsto H(x)$ and detect the cycle with Floyd’s or Brent’s algorithm, in the manner of Pollard’s rho, or run the van Oorschot–Wiener parallel collision search across many machines. These achieve the same $2^{n/2}$ time in essentially constant memory, so the birthday attack is a genuine budget item, not a thought experiment.

Preimages cost the square. The best generic attack against preimage and second-preimage resistance is brute force: about 2^n evaluations to hit a prescribed digest. Those notions therefore enjoy full n -bit generic security—twice the collision level—because the attacker cannot exploit birthday coincidences among self-chosen pairs; the collision game is the only one that pays the square-root discount.

3.4 The random oracle model

The model section introduced the random oracle model as one of the two idealisations under which proofs proceed (Definition 1.26), stated its two powers informally (Remark 1.27), and fixed its status as a heuristic with known limits (Remark 1.29). With hash syntax now in hand we can define the idealised object precisely and *prove* the powers.

Definition 3.13 (Random oracle). A *random oracle* with output length n is a function $O: \{0, 1\}^* \rightarrow \{0, 1\}^n$ drawn uniformly at random from the set of all such functions. Equivalently—and this is the form every proof uses—the oracle is realised *lazily*: a table, initially empty, is maintained; on query x , if x is in the table the stored value is returned, and

otherwise a fresh $O(x) \xleftarrow{\$} \{0,1\}^n$ is sampled, recorded, and returned.

Three features characterise the object: outputs on distinct inputs are independent and uniform; outputs on repeated inputs are consistent; and access is *only* by querying—the oracle has no description shorter than its (exponentially long) truth table. A scheme lives in the random oracle model (ROM) when every party, honest or adversarial, interacts with a single shared O in place of a concrete hash, and security is proved relative to the random choice of O ; on deployment a concrete hash—SHA-256, say, or BLAKE2b—*instantiates* the oracle.

Proposition 3.14 (The two powers of the ROM). *In a ROM security proof, the reduction, which simulates O for the adversary, gains two capabilities that no concrete hash affords.*

1. *Observability (extractability). The reduction sees every input the adversary queries. Since querying is the only way to learn anything about $O(x)$, if the adversary’s output depends on $O(x)$ then the reduction holds x .*
2. *Programmability. On a not-yet-queried input x , the reduction may choose the response $O(x)$ to be any value of its liking, provided the value is distributed uniformly; the simulation remains perfect, and the reduction can thereby plant a challenge—a value it must invert, say—inside an oracle answer.*

Proof. Both capabilities are immediate from the lazy realisation of Definition 3.13, because in the simulation the reduction *is* the table-keeper. Every query arrives at the reduction before an answer exists, which is observability. And on a fresh input the honest table-keeper would sample the answer uniformly and independently of the adversary’s view so far; a reduction that instead inserts any value *with that same distribution* produces a view identical to the honest one, while the table keeps repeated queries consistent. That is programmability. $\square$

Remark 3.15 (Why Fiat–Shamir needs the ROM). The paradigmatic consumer of both powers at once is the Fiat–Shamir transform—the compiler that renders Halo 2’s interactive protocol non-interactive, developed in this volume under “From identification to signature via the Fiat–Shamir transform” and in full generality under “The Fiat–Shamir transform: from interactive to non-interactive”. The transform replaces the verifier’s random challenge e by $e = O(\text{transcript so far})$. Soundness of the resulting non-interactive argument is proved by *rewinding*: the reduction observes (capability 1) the transcript the prover commits to, then *reprograms* (capability 2) the oracle to return a fresh independent challenge on that same transcript, extracting two accepting transcripts that share a prefix and, from them, a witness—the forking technique previewed in Remark 1.24. Neither step is available against a concrete hash, whose outputs the reduction can neither observe selectively nor rewrite. This is the proof that lives *only* in the ROM, and the reason the model is indispensable to the cryptography this monograph targets.

3.5 Domain separation and personalisation

One physical hash serves many purposes in a deployed protocol: it derives keys, commits to notes, tags spent notes, seeds Fiat–Shamir challenges. Digests, however, are interchangeable bit strings, and an output produced for one purpose can be *replayed* as another unless the purposes are cryptographically insulated. Domain separation is the discipline that insulates them.

Definition 3.16 (Domain separation). Let H be a hash modelled as a random oracle, and let $\{\tau_i\}_{i \in I}$ be a set of distinct, *prefix-free* domain tags (also called labels or personalisation strings), one per intended use: no τ_i is a prefix of another—as when all tags have equal length, or when

each is length-prefixed. The *use- i hash* is

$$H^{(i)}(x) := H(\tau_i \| x).$$

Proposition 3.17 (Domain separation yields independent oracles). *If H is a random oracle and the tags $\{\tau_i\}$ are prefix-free, then the family $\{H^{(i)}\}_{i \in I}$ is distributed exactly as a collection of independent random oracles. Consequently a digest obtained from $H^{(i)}$ is—except with the negligible probability of a generic collision—useless against any use $j \neq i$.*

Proof. Prefix-freeness makes the domains $\tau_i \| \{0, 1\}^*$ pairwise disjoint: a string with prefix τ_i cannot also have prefix τ_j unless one tag is a prefix of the other. Inputs queried through different $H^{(i)}$ are therefore never equal. A random oracle assigns independent uniform values to distinct inputs, so restricting H to the disjoint slices yields, per slice, a uniformly random function, independent across slices—precisely a collection of independent random oracles. The replay conclusion follows: the value $H^{(i)}(x)$ reveals nothing about any $H^{(j)}(\cdot)$ with $j \neq i$. $\square$

Three idioms implement the discipline in deployed practice; Zcash uses all three.

1. *Context tagging.* Bind an ASCII string naming the use into every call—prepended, as in Definition 3.16, or in any framing that keeps the tag unambiguously delimited from the data. The Orchard tag inventory is of this kind: the strings `z.cash:Orchard-cv`, `z.cash:Orchard-NoteCommit`, and `z.cash:Orchard-CommitIvk` separate the value commitment, note commitment, and key commitment uses of one group-hash (protocol specification §§ 5.4.8.3–5.4.8.4), and `z.cash:Orchard-gd` separates diversified key generation (protocol specification § 5.4.1.6). The deployed framing mirrors the definition’s rather than copying it: the curve hash of §3.7 *appends* its domain tag after the message, closed by a byte carrying the tag’s length (protocol specification § 5.4.9.8); the length byte makes the tag self-delimiting, so distinct tags still induce disjoint effective inputs—all the proof of Proposition 3.17 uses. The tags reach that hash by two routes, and the commitment tags by both: `-cv` and `-gd` serve directly as its domain tag, while each commitment tag, suffixed `-M`, is hashed as the *message* under the fixed domain tag `z.cash:SinsemillaQ` to derive one of the commitment’s two base points and, suffixed `-r`, serves directly as the domain tag deriving the other (protocol specification §§ 5.4.1.9 and 5.4.8.4).
2. *Native personalisation.* BLAKE2b’s parameter block carries a dedicated 16-byte personalisation field that is mixed into the initialisation vector, so distinct personalisations separate the primitive’s effective inputs before a single message byte is processed—the field for which Zcash favours BLAKE2 for its oracles. With BLAKE2b modelled as a random oracle of the pair (personalisation, message), the fixed-length field is a prefix-free tag and Proposition 3.17 makes the resulting domains independent; a concrete BLAKE2b instantiation does not make that independence a fact “by construction”. The deployed personalisations include `Zcash_ExpandSeed` for key expansion (protocol specification § 5.4.2), `Zcash_OrchardKDF` for the note-encryption key-derivation function (KDF, the object of §6.6; protocol specification § 5.4.5.6), and `Zcash_Orchardock` for the outgoing cipher key (protocol specification § 5.4.2)—all exactly 16 bytes.
3. *Suffix and framing bits in the SHA-3 family.* FIPS 202 reserves domain-separation suffix bits appended to the message before padding: 01 for SHA-3 hashing and 1111 for the SHAKE XOFs. SHA3-256 and SHAKE256 share the same underlying permutation, yet the framed inputs differ, so their effective inputs are disjoint—though their finite digests can still coincide by chance.

The unifying requirement behind all three idioms is that distinct uses induce *disjoint effective inputs* to the underlying primitive. An ad hoc tag that happens to be a prefix of another silently merges two domains and reopens the replay attack—which is why Definition 3.16 demands prefix-freeness rather than mere distinctness. The Merkle trees of the final section apply the same discipline *within* a single structure, tagging tree layers to foreclose a forgery that confuses leaves with internal nodes; see “Layer and domain separation” there.

3.6 Hashing to a field element

The primitives ahead—commitments over the Pasta curves, Fiat–Shamir challenges, key derivations—repeatedly need to hash a message *to a field element*: a near-uniform element of $\mathbb{F}_p$ rather than a bit string. The obstacle is arithmetic: a hash emits uniform *bits*, and reducing n uniform bits modulo a prime p is not uniform on $\mathbb{F}_p$, because 2^n is not a multiple of p —some residues receive one more preimage than others. The remedy is to hash *wide* and reduce.

Construction 3.18 (Expand-then-reduce). Let p be a prime of $L = \lceil \log_2 p \rceil$ bits, and fix a XOF, or a domain-separated hash, `expand`, producing arbitrary-length near-uniform output. To map a message m to $\mathbb{F}_p$, with a domain tag τ and a security slack of k bits (commonly $k = 128$):

1. compute $b \leftarrow \text{expand}(\tau \parallel m, L + k)$, a string of $L + k$ near-uniform bits;
2. interpret b as an integer $B \in \{0, \dots, 2^{L+k} - 1\}$ and output $B \bmod p \in \mathbb{F}_p$.

Proposition 3.19 (Near-uniformity of expand-then-reduce). *If `expand` is a random oracle, the output of Construction 3.18 lies within statistical distance (Definition 1.6) at most*

$$\frac{p}{2^{L+k+1}} \leq 2^{-k}$$

of the uniform distribution on $\mathbb{F}_p$. With $k = 128$, no algorithm whatsoever—efficient or not—distinguishes the output from a uniform field element with advantage better than 2^{-128} .

Proof. The integer B is uniform on $\{0, \dots, K - 1\}$ with $K = 2^{L+k}$, and the bias of reducing a uniform B modulo p is exactly the situation of the modular-reduction proposition of the *Math Guide* (§“Uniform sampling and the bias of modular reduction”): each residue’s probability differs from $1/p$ by at most $1/K$, because the preimage counts $\lfloor K/p \rfloor$ and $\lceil K/p \rceil$ differ by one, and summing over the p residues gives statistical distance at most $p/(2K) = p/2^{L+k+1}$ —the sharper form proved there. Since $p < 2^L$, this is less than $2^{-(k+1)} \leq 2^{-k}$. The final claim is the variational characterisation of statistical distance (Proposition 1.7): no decision procedure exceeds the distance. $\square$

The slack is not optional. With $k = 0$ —a single L -bit block reduced modulo p —the bias can be macroscopic: for p just above a power of two, $K = 2^L$ is barely twice p , the residues beyond the wrap-around point receive only one preimage while the rest receive two, and each deprived residue carries as little as half its uniform mass. The k extra bits are exactly what drives this imbalance to negligibility.

Deployed practice packages the construction. The IETF hash-to-curve specification (RFC 9380) defines `hash_to_field( $m, count$ )`, producing `count` field elements by calling an expansion routine—`expand_message_xmd` over a fixed-output hash, or `expand_message_xof` over a XOF—with a *mandatory* domain-separation tag, and reducing each chunk of

$$L_{\text{RFC}} = \lceil (\lceil \log_2 p \rceil + k) / 8 \rceil \text{ bytes}$$

modulo p , where k is the suite’s target security level in bits. The RFC’s own suites for 255-bit fields—the curve25519 family—set $k = 128$, hence 48-byte chunks. The Zcash suite for Pallas and Vesta is more generous: it takes $k = 256$, reducing each element from a full 64-byte BLAKE2b-512 digest, so the slack of Proposition 3.19 is realised with room to spare (the `hash_to_field` routine expands with BLAKE2b in the XMD pattern under the suite tag `{tag}-{curve}_XMD:BLAKE2b_SSWU_R0_` and reduces two 64-byte chunks each via `from_uniform_bytes`; protocol specification § 5.4.9.8). This is the concrete recipe for mapping identities to field elements, and for producing the field elements that the next subsection feeds to a curve. Halo 2’s deployed Fiat–Shamir transcript uses the same wide-output-then-reduce principle: each challenge is the 64-byte digest of a BLAKE2b state personalised with `Hal02-Transcript`, reduced by `from_uniform_bytes`; the order in which that state absorbs messages and squeezes challenges is the *Halo 2 Guide*’s, §“The complete Halo 2 protocol”.

3.7 Hashing to a curve point

The volume ahead also needs to hash *onto an elliptic curve*: given a message, produce a point of $E(\mathbb{F}_p)$ that behaves like a uniform group element—the raw material for commitment generators whose discrete logarithms nobody knows. Two requirements pull against each other. The output should be near-uniform in the relevant subgroup, so that the map can stand in for a random oracle into the group in security arguments; and the map must run in *constant time*, because its input is often secret-adjacent.

The naive approach fails the second requirement instructively. Hash the message to a candidate $x \in \mathbb{F}_p$ (Construction 3.18); test whether $g(x) = x^3 + ax + b$ is a quadratic residue, so that a y -coordinate exists; if not, increment x and retry. This *try-and-increment* loop is correct and roughly uniform, but the number of trials depends on the message: about half of all x fail the residue test, since the squares are half of $\mathbb{F}_p^\times$ (*Math Guide*, §“Quadratic residues and the Euler criterion”, whose Legendre symbol $\left(\frac{g(x)}{p}\right)$ is exactly the test), so the running time leaks information about m —an avenue for side-channel attacks. Constant-time hashing to a curve requires a map that succeeds on the *first try*, for every input.

Construction 3.20 (Simplified Shallue–van de Woestijne–Ulas map). Let $E: y^2 = g(x) = x^3 + ax + b$ over $\mathbb{F}_p$ with $ab \neq 0$, and choose, as RFC 9380 requires, a constant $Z \in \mathbb{F}_p$ such that Z is a nonsquare, $Z \neq -1$, the polynomial $g(X) - Z$ is irreducible over $\mathbb{F}_p$, and $g(b/(Za))$ is a square. (The RFC selects Z mechanically, as the candidate of smallest absolute value meeting these conditions—a nothing-up-my-sleeve constant with no hidden design freedom.) The map $\text{SWU}: \mathbb{F}_p \rightarrow E(\mathbb{F}_p)$ sends u to a point as follows. Compute the two candidate x -coordinates

$$x_1 = \frac{-b}{a} \left(1 + \frac{1}{Z^2 u^4 + Z u^2}\right), \quad x_2 = Z u^2 x_1,$$

with the fallback $x_1 = b/(Za)$ when the denominator $Z^2 u^4 + Z u^2$ vanishes (as at $u = 0$). Set $x = x_1$ if $g(x_1)$ is a square and $x = x_2$ otherwise; let y be a square root of $g(x)$ (*Math Guide*, §“Square roots modulo a prime”), its sign fixed by a convention—the deployed map matches the parity of y to that of u —and output (x, y) .

The construction contains no trial loop, hence runs in constant time; what needs proof is that one of the two candidates always works.

Proposition 3.21 (Correctness of the branch selection). *Under the parameter conditions of Construction 3.20, the map is well-defined for every $u \in \mathbb{F}_p$: the selected value $g(x)$ is a square, so $\text{SWU}(u)$*

is a genuine point of $E(\mathbb{F}_p)$.

Proof sketch. Suppose first that $D = Z^2u^4 + Zu^2 \neq 0$. The key algebraic identity is

$$g(x_2) = Z^3u^6 g(x_1),$$

a polynomial identity in u : substitute $x_2 = Zu^2x_1$ into g and simplify using the formula for x_1 (we take the computation as given). If $g(x_1) = 0$, the first candidate already works, with $y = 0$. Otherwise apply multiplicativity of the Legendre symbol (*Math Guide*, §“Quadratic residues and the Euler criterion”):

$$\left(\frac{g(x_2)}{p}\right) = \left(\frac{Z^3u^6}{p}\right) \left(\frac{g(x_1)}{p}\right) = \left(\frac{Z}{p}\right) \left(\frac{(Zu^3)^2}{p}\right) \left(\frac{g(x_1)}{p}\right) = \left(\frac{Z}{p}\right) \left(\frac{g(x_1)}{p}\right),$$

since $Z^2u^6 = (Zu^3)^2$ is a nonzero square, contributing $+1$ (note that $D \neq 0$ forces $u \neq 0$). Because Z is a nonsquare, $\left(\frac{Z}{p}\right) = -1$, so the two symbols are opposite and the branch selection picks the square. If $D = 0$, the prescribed fallback has $g(x_1) = g(b/(Za))$, a square by the last condition on Z . Thus every input produces a valid affine point. $\square$

A single SWU evaluation is not the end of the pipeline: the map is not surjective onto $E(\mathbb{F}_p)$, and its image is not uniformly covered—it reaches only part of the curve, with a lumpy distribution. The standardised pipeline repairs both defects.

Construction 3.22 (The hash-to-curve pipeline). The RFC 9380 map `hash_to_curve`: $\{0, 1\}^* \rightarrow E(\mathbb{F}_p)$ composes four steps:

1. hash to the field: `hash_to_field`($m, 2$) produces two independent near-uniform field elements u_0, u_1 (§3.6);
2. map each through SWU: $Q_0 = \text{SWU}(u_0)$, $Q_1 = \text{SWU}(u_1)$;
3. sum on the curve: $R = Q_0 + Q_1$. Summing two independent SWU images provably repairs the lumpiness: the sum lies within negligible statistical distance of uniform on $E(\mathbb{F}_p)$, the theorem on which the pipeline’s random-oracle claim rests;
4. clear the cofactor: multiply R by $h = \#E(\mathbb{F}_p)/r$ to land in the prime-order subgroup of order r where the cryptography occurs.

The result is a constant-time hash from messages to group elements which, under the hash model and suite conditions of RFC 9380, can instantiate a random oracle into the group. For the Pasta curves the last step is vacuous—their cofactor is 1 (Definition 2.25).

Remark 3.23 (The isogeny detour forced by j -invariant 0). For the Pasta curves one step needs a detour. The SWU formula for x_1 divides by a , and its numerator carries the factor $-b$: with $a = 0$ the formula is undefined, while $b = 0$ makes x_1 identically zero with $g(x_1) = 0$, collapsing the map to the constant point $(0, 0)$. The construction therefore requires $ab \neq 0$ —and Pallas and Vesta have $a = 0$ (the j -invariant-0 shape $y^2 = x^3 + 5$ of Definition 2.25; curves with $b = 0$, $j = 1728$, are excluded symmetrically). The obstruction is intrinsic, not notational: the j -invariant is an isomorphism invariant, and a Weierstrass change of variables rescales (a, b) to (u^4a, u^6b) (*Math Guide*, §“The j -invariant and the GLV endomorphism”), so *no* model of these curves attains $ab \neq 0$. The standardised remedy evaluates simplified SWU on an *isogenous* auxiliary curve $E': y^2 = x^3 + a'x + b'$ with $a'b' \neq 0$, then transports the result through an *isogeny* $E' \rightarrow E$: a nonconstant map between curves given by polynomial coordinate formulas and respecting the group law. The deployed implementation applies a degree-3 isogeny, and—because an isogeny is a group homomorphism—sums $Q_0 + Q_1$ on E' first and transports the single sum (protocol specification § 5.4.9.8).

The culminating application ties together collision resistance, the random-oracle idealisation, and hashing to a curve.

Construction 3.24 (Nothing-up-my-sleeve generators). To generate public generators $G_1, \dots, G_k$ of a vector commitment (the Pedersen vector commitment of Construction 4.22 in the next section) so that no one knows any discrete-logarithm relation among them, derive each as

$$G_i = \text{hash_to_curve}(\tau \parallel i)$$

from a fixed public domain tag τ —often a human-readable protocol name—and the index i . Because the seed is a transparent, auditable string with no hidden parameters, no party could have engineered a backdoor relation in advance. This is exactly how Halo 2 produces the generators of its Pedersen vector and inner-product commitments (the latter Construction 4.38): every generator G_i is $\text{hash_to_curve}(\text{"Halo2-Parameters"})$ applied to a five-byte message—a zero byte, then the four-byte little-endian encoding of i —the leading zero keeping the G_i disjoint from the two auxiliary generators the same tag derives from the one-byte messages 1 and 2, and Orchard’s value commitment derives its two bases from the tag `z.cash:Orchard-cv` with the one-byte messages `"v"` and `"r"`.

Remark 3.25 (Why NUMS generators are binding-safe). Knowledge of a discrete-logarithm relation $\sum_i a_i G_i = 0$ with not all a_i zero would let a committer open one commitment two ways, destroying the binding property that the next section defines and relies on. The NUMS derivation forecloses this: the generators are outputs of a function modelled as a random oracle into the group, so they are distributed as independent uniform group elements, and finding a discrete-logarithm relation among independent uniform elements *reduces* to computing a discrete logarithm on E , hard under the standing assumptions of Remark 2.32; the reduction, which embeds its challenge in every generator, is Theorem 4.23 of the next section. Hence no efficient party, the scheme designer included, can know such a relation.

3.8 Arithmetisation-friendly hashing: Poseidon

The hashes of this section so far are *bit-oriented*: SHA-2 and BLAKE2 shuffle and XOR 32- and 64-bit words. Inside an arithmetic circuit over a large prime field $\mathbb{F}_p$ —the setting of proof systems such as Halo 2, whose cost model, developed in the *Halo 2 Guide*, §“From computation to a grid of constraints”, prices each field multiplication as a constraint—such a hash is prohibitively expensive: every Boolean operation must be re-expressed through field arithmetic and range checks, and the tens of thousands of resulting constraints dominate the circuit. Arithmetisation-friendly hashes invert the design: the round function is a handful of *low-degree field operations*, native to the circuit, incurring few constraints. Poseidon (Grassi, Khovratovich, Rechberger, Roy, and Schofnegger, 2019) is the instance Orchard employs, and it arrives via a mode of operation worth defining in general.

Construction 3.26 (The sponge construction). Fix a permutation $f: \mathbb{F}_p^t \rightarrow \mathbb{F}_p^t$ on a state of t field elements. Split the state into a *rate* of r elements—the only part input is added into and output is read from—and a *capacity* of $c = t - r$ elements, never accessed directly: the security reserve. To hash, *absorb* the message into the rate, r elements at a time, applying f between blocks; then *squeeze* output from the rate, applying f between reads. Collision and preimage resistance hold up to approximately $\sqrt{p^c}$ work, as for a random sponge: the birthday bound of Lemma 3.11 applied to the p^c capacity states the adversary cannot see.

Construction 3.27 (The Poseidon permutation). The permutation f is a sequence of rounds, each comprising three steps:

1. *add round constants*: add a fixed vector in $\mathbb{F}_p^t$ to the state;
2. *S-box*: apply $x \mapsto x^\alpha$ to state elements, where $\alpha \geq 3$ is the smallest integer with $\gcd(\alpha, p-1) = 1$, so that the power map is a bijection on $\mathbb{F}_p$; over the Pasta fields $\alpha = 5$, because $3 \mid p-1$ there rules out $\alpha = 3$;
3. *mix*: multiply the state by a fixed maximum-distance-separable (MDS) matrix $M \in \mathbb{F}_p^{t \times t}$ —one whose every square submatrix is invertible, giving the fastest possible diffusion of any element’s change across the whole state.

The cost-saving principle (the HADES strategy) distinguishes two round types: a few *full* rounds at the start and end apply the S-box to all t elements, while in the many *partial* rounds between them the S-box touches a single element—the MDS mix still diffusing it across the state. Round counts are chosen to stand against the known algebraic (Gröbner-basis) attacks.

Remark 3.28 (Circuit cost). Inside a circuit, the only non-linear operation in all of Poseidon is $x \mapsto x^5$, verified by a couple of multiplication gates per S-box; the round-constant additions and the MDS multiplication are linear and almost free. A 2-to-1 Poseidon hash costs a few hundred constraints, against tens of thousands for a bit-oriented hash—the difference between a practical circuit and an impractical one.

Example 3.29 (The Orchard Poseidon instance). Orchard fixes $t = 3$ (rate 2, capacity 1), $\alpha = 5$, and 8 full plus 56 partial rounds, over the Pallas base field, at the 128-bit security level—the parametrisation named P128Pow5T3 in the deployed source (protocol specification § 5.4.1.10). It is used in *constant-input-length* mode as a two-input hash:

$$\text{PoseidonHash}(x, y) = f([x, y, 2^{65}])_0,$$

the first component of the state after a single permutation of the input padded by the fixed capacity value $2^{65} = 2 \cdot 2^{64}$, which encodes the input length 2 into the capacity so that no message padding is required (the `ConstantLength` domain sets the initial capacity element to $L \cdot 2^{64}$). Keyed by its first argument, this hash serves as the pseudorandom function

$$F_{\text{nk}}(\rho) = \text{PoseidonHash}(\text{nk}, \rho)$$

—the PRF vocabulary is owned by “Pseudorandom functions and permutations” later in this volume—deployed as $\text{PRF}^{\text{nfOrchard}}$ (protocol specification § 5.4.2), from which Orchard constructs the nullifier—a construction developed in the *Ironwood Guide*, § “The nullifier, and the loop back to minting”; this volume supplies only the PRF. One contrast deserves note: unlike the Pedersen and Sinsemilla hashes constructed in the next section, whose collision resistance reduces to a clean assumption (discrete log), Poseidon’s security rests on cryptanalytic confidence in its round function—the same epistemic footing as SHA-256, not the same as a reduction.

3.9 The transparent layer’s hashes

The Orchard shielded protocol hashes with BLAKE2b, Sinsemilla, and Poseidon. The transparent layer, inherited from Bitcoin, hashes with SHA-256 and two functions derived from it, none needing machinery beyond §3.2 and the birthday bound of §3.3; this register fixes their names, domains, and deployed uses for the volumes above.

- *RIPEMD-160*. A 160-bit iterated (Merkle–Damgård, the construction of §5.3) hash, $\{0, 1\}^* \rightarrow \{0, 1\}^{160}$, older than Bitcoin and inherited with its address format, whose collision

resistance sits at the 80-bit birthday bound of Lemma 3.11. Zcash uses it only inside hash160 below and the script opcodes `OP_RIPEMD160` and `OP_HASH160`.

- *The address hash* `hash160` $:=$ `RIPEMD160` $\circ$ `SHA256`. A pay-to-public-key-hash (P2PKH) address carries the 20-byte hash160 of the 33-byte compressed encoding of an ECDSA public key (the scheme is registered in §8.10), and a pay-to-script-hash (P2SH) address the hash160 of a *redeem script*, the spending condition the spender later reveals and satisfies (protocol specification § 5.6.1.1). The P2PKH output script—the condition a spend of the output must satisfy—is `OP_DUP OP_HASH160 <h> OP_EQUALVERIFY OP_CHECKSIG`, which recomputes the hash from the public key the spender reveals; wallets compute it through the `zcash_transparent` crate, and a partially created transaction (PCZT), the multi-party construction format of the *Wallet Guide*, §“Partially created transactions (PCZT)”, carries on each transparent input preimage maps keyed by RIPEMD-160, SHA-256, hash160, and hash256 digests.
- *Double SHA-256* (*SHA-256d*, or `hash256`). The composition $x \mapsto \text{SHA256}(\text{SHA256}(x))$ (protocol specification § 5.4.1.1), whose outer application also closes the length-extension property of plain SHA-256 (§5.3). It hashes block headers; its leading four bytes are the checksum that Base58Check, the string encoding of transparent addresses, appends to the raw address bytes before converting to base 58 (protocol specification § 5.6.1.1) and the payload checksum in every peer-to-peer message header; and it is the legacy transaction identifier: the txid of a version-4-or-earlier transaction is SHA-256d of its serialisation (protocol specification § 7.1.1), whereas a version-5 txid is the BLAKE2b digest tree of ZIP 244.

The toolbox now holds its first primitive in full: a hash with three calibrated security notions, an idealisation that licenses proofs, a discipline that insulates its many uses, and maps carrying its output into fields, onto curves, and inside circuits. The next section spends this capital immediately: commitments—the sealed envelopes of the protocol—are built from exactly the hashes, generators, and hardness assumptions now on the table.

4 Commitment schemes

Two adversaries stalk this section, one on each side of a sealed envelope. Before the envelope is opened, the adversary is the *peeker*: the receiver, who holds the sealed commitment and tries to distinguish which of two candidate messages it contains—to read the bid before the auction closes, the vote before the tally. After the envelope has been delivered, the adversary is the *equivocator*: the sender, who tries to open the same commitment to a message other than the one sealed inside—to change the bid once the rival’s is public. A commitment scheme is the cryptographic sealed, tamper-evident envelope that defeats both: once delivered, the sender can no longer alter the message (*binding*), yet the recipient cannot read it until it is unsealed (*hiding*).

The primitive is the natural next step after hashing: the two constructions at the heart of this section, Pedersen and Sinsemilla, need nothing beyond the discrete-logarithm hardness of §2 and the nothing-up-my-sleeve generators that §3 showed how to derive, and nearly everything built later consumes them. Classical applications set the scene: coin-flipping over a telephone (each party commits to a bit before either reveals), zero-knowledge proofs (the prover commits to its secrets and proves statements about the sealed values), and verifiable secret sharing. Centrally for this volume, the *polynomial* commitments of §4.9 are the cryptographic engine of modern succinct argument systems, including the one deployed in Orchard: the value commitments checked by binding signatures, the note commitments accumulated in Merkle trees, and the witness polynomials of the Halo 2 proof system are all commitments in the precise sense defined here.

The plan of the section follows the two adversaries. We first fix the syntax of a commitment scheme and the three models of who may be trusted to set it up. We then define hiding and binding as games in the tradition of §1.4, each in three quantitative flavours—perfect, statistical,

computational—and prove that the two perfections are jointly unattainable: every scheme must choose which adversary to defeat unconditionally. The constructions that follow are ordered by increasing algebraic richness: hash-based commitments (opaque but transparent and plausibly post-quantum), the Pedersen commitment and its vector variant (perfectly hiding, computationally binding under discrete log, and—decisively—linear), Sinsemilla (a commitment whose internal operations are curve operations, built to be cheap inside a proof), then polynomial commitment schemes, contrasting the pairing-based KZG family with the inner-product family that Zcash deploys, and the bridge from commitments to arguments. A closing subsection returns to the classical application named above—secret sharing—and builds its verifiable form from coefficient commitments, for the threshold signatures of the *FROST Guide*.

4.1 The sealed envelope

The envelope metaphor repays being taken slowly, because each of its physical properties has an exact formal counterpart. The sender writes a message on a card, seals it in the envelope, and hands the envelope over. From that moment the receiver *possesses* the commitment: whatever the message was, it is now fixed—the sender cannot swap the card, because the envelope is in the receiver’s hands and any tampering would show. Yet the receiver *learns* nothing: the envelope is opaque. Later, at a time of the protocol’s choosing, the sender opens the envelope—in the formalism, transmits the message together with the randomness used to seal it—and the receiver checks that the revealed message is consistent with the envelope received earlier.

The two guarantees are directed at different parties and different times. Hiding protects the *sender* against a curious *receiver* during the window between commit and reveal; binding protects the *receiver* against a dishonest *sender* at reveal time. A scheme that is hiding but not binding is an envelope with a false bottom: private, but worthless as evidence. A scheme that is binding but not hiding is a glass envelope: tamper-evident, but no secret survives it. The games of §4.3 formalise exactly these two failure modes, and the theorem of §4.4 shows that against computationally unbounded adversaries one of the two protections must give way.

4.2 Syntax

Definition 4.1 (Commitment scheme). A *commitment scheme* Π for a message space $\mathcal{M}$ and randomness space $\mathcal{R}$ is a triple of algorithms (Setup, Commit, Open):

- The algorithm Setup(1^λ) is PPT (§1.1) and outputs *public parameters* pp , which implicitly determine the message space $\mathcal{M}_{pp}$ and randomness space $\mathcal{R}_{pp}$ and are an implicit input to the remaining algorithms.
- The algorithm Commit($pp, m; r$), for $m \in \mathcal{M}_{pp}$ and $r \in \mathcal{R}_{pp}$, is deterministic given r and outputs a *commitment* c . We write Commit(pp, m), without an explicit r , for the probabilistic algorithm that samples $r \xleftarrow{\$} \mathcal{R}_{pp}$ uniformly and returns r alongside c .
- The algorithm Open(pp, c, m, r) is deterministic and outputs a bit: 1 (accept) or 0 (reject).

Correctness requires that honestly produced commitments open: for all pp in the support of Setup(1^λ), all $m \in \mathcal{M}_{pp}$, and all $r \in \mathcal{R}_{pp}$,

$$\text{Open}(pp, \text{Commit}(pp, m; r), m, r) = 1.$$

The pair (m, r) is called the *opening* (or *decommitment*) of c . Every scheme in this volume is *canonically opening*: Open(pp, c, m, r) merely recomputes Commit($pp, m; r$) and checks equality with c . This is the adopted convention unless stated otherwise. For a canonically opening

scheme correctness holds automatically, and a valid opening of c is precisely a pair (m, r) with $\text{Commit}(pp, m; r) = c$ —a reformulation the proofs below use constantly.

Remark 4.2 (The two phases, and a third concern). A commitment scheme is used in two temporally separated phases. In the *commit phase* the sender transmits c and nothing else; in the *reveal phase* the sender transmits the opening (m, r) and the receiver runs *Open*. The two security properties attach to the two phases: hiding constrains what the receiver can learn *after the commit phase*, while binding constrains what the sender can do *in the reveal phase*. A third concern precedes both: the *setup phase* that produces pp . Who runs *Setup*, and whether that party can be trusted, proves decisive for some of the constructions ahead (Remark 4.3).

Remark 4.3 (Three setup models). Commitment schemes divide by how much trust their parameters demand.

1. *Common reference string (CRS)*. A trusted party runs *Setup* and is assumed to *erase* the randomness behind it. The parameters have internal structure, and knowledge of the discarded randomness—the *trapdoor*—would break the scheme. Of the schemes treated in this section, only KZG (§4.9) genuinely resides here.
2. *Uniform random string (URS), or transparent setup*. The parameters consist only of public coins: uniformly random group elements with no known relations among them. There is no trapdoor and no trusted party—anyone can re-derive the parameters from a public seed. This is the regime of the Pedersen and inner-product commitments below. A Pedersen generator derived by hashing to the curve (Construction 3.24) is transparent even though the textbook presentation phrases *Setup* as sampling and discarding a secret.
3. *Plain*. No parameters at all, or only a standardised hash function—the home of the hash-based commitment below.

Transparent setup is highly desirable: a flawed or subverted trusted setup can silently destroy binding, as the KZG trapdoor demonstrates concretely in §4.9.

4.3 Hiding and binding

Both properties are games in the sense of Definition 1.15. The hiding game arms the peeker: she chooses the two candidate messages herself, receives a commitment to one of them, and must say which.

Definition 4.4 (Hiding game and hiding). Let Π be a commitment scheme and $\mathcal{A} = (\mathcal{A}_1, \mathcal{A}_2)$ an adversary. For $b \in \{0, 1\}$, the game $\text{Hide}_{\Pi}^{\mathcal{A}}(\lambda, b)$ runs as follows.

1. The challenger runs $pp \leftarrow \text{Setup}(1^\lambda)$.
2. The adversary $\mathcal{A}_1(pp)$ outputs two messages $m_0, m_1 \in \mathcal{M}_{pp}$ and state st .
3. The challenger samples $r \xleftarrow{\$} \mathcal{R}_{pp}$ and computes $c = \text{Commit}(pp, m_b; r)$.
4. The adversary $\mathcal{A}_2(st, c)$ outputs a bit b' , which is the output of the experiment.

The *hiding advantage* of $\mathcal{A}$ is

$$\text{Adv}_{\Pi}^{\text{hide}}(\mathcal{A}, \lambda) := \left| \Pr[\text{Hide}_{\Pi}^{\mathcal{A}}(\lambda, 0) = 1] - \Pr[\text{Hide}_{\Pi}^{\mathcal{A}}(\lambda, 1) = 1] \right|.$$

The scheme is *computationally hiding* if the advantage is negligible (§1.2) for every PPT A ; *statistically hiding* if it is negligible even for computationally unbounded A ; and *perfectly hiding* if it is exactly 0 for every adversary and every λ .

Proposition 4.5 (Distributional characterisation of hiding). *A commitment scheme Π is perfectly hiding if and only if for every pp in the support of Setup and every pair $m_0, m_1 \in M_{pp}$, the random variables*

$$\text{Commit}(pp, m_0; r) \quad \text{and} \quad \text{Commit}(pp, m_1; r), \quad r \xleftarrow{\$} \mathcal{R}_{pp},$$

are identically distributed. It is statistically hiding if the statistical distance (Definition 1.6) between them is negligible. (Only the perfect clause is an equivalence: parameters of negligible Setup probability could exhibit large distance without granting any adversary noticeable advantage.)

Proof. Fix pp and m_0, m_1 , and write D_m for the distribution of $\text{Commit}(pp, m; r)$ over uniform r . By the variational characterisation of statistical distance (Proposition 1.7), $\Delta(D_{m_0}, D_{m_1})$ equals exactly the maximum advantage, over all decision procedures—efficient or not—of telling D_{m_0} from D_{m_1} . An adversary who outputs the pair (m_0, m_1) on parameters pp and applies the optimal test therefore achieves hiding advantage $\Delta(D_{m_0}, D_{m_1})$ whenever Setup produces pp ; conversely no adversary can exceed the expectation of this quantity over pp . Advantage exactly 0 for all adversaries is thus equivalent to $\Delta = 0$ —identical distributions—for every attainable pp and every pair, and negligible statistical distance yields negligible advantage for unbounded adversaries. This coincidence of the game formulation with the distance formulation is precisely why statistical hiding was *defined* through the unbounded-adversary game. $\square$

The binding game arms the equivocator. He is not handed an honest commitment; he manufactures his own, together with two openings.

Definition 4.6 (Binding game and binding). For a commitment scheme Π and adversary A , the game $\text{Bind}_{\Pi}^A(\lambda)$ runs as follows.

1. The challenger runs $pp \leftarrow \text{Setup}(1^\lambda)$.
2. The adversary $A(pp)$ outputs a commitment c and two openings (m_0, r_0) and (m_1, r_1) .
3. The game outputs 1 iff $m_0 \neq m_1$ and $\text{Open}(pp, c, m_0, r_0) = \text{Open}(pp, c, m_1, r_1) = 1$.

The *binding advantage* of A is

$$\text{Adv}_{\Pi}^{\text{bind}}(A, \lambda) := \Pr[\text{Bind}_{\Pi}^A(\lambda) = 1],$$

a search game in the sense of Definition 1.15. The scheme is *computationally*, *statistically*, or *perfectly binding* according as the advantage is negligible for every PPT adversary, negligible for every unbounded adversary, or exactly 0 for every adversary and every λ .

For a canonically opening scheme the binding game has a useful collision reformulation: for fixed pp , two openings collide precisely when $\text{Commit}(pp, m_0; r_0) = \text{Commit}(pp, m_1; r_1)$ with $m_0 \neq m_1$. Perfect binding then says that distinct messages never share a commitment value at all—the message m is a deterministic function of c . Perfect binding is thus equivalent to the map $c \mapsto m$ being well defined on all attainable commitments, a formulation the counting argument of Remark 4.9 exploits.

Remark 4.7 (The asymmetry of the quantifiers). The two games quantify over different objects. Hiding quantifies over what can be *extracted* from a single honestly produced commitment: the

challenger seals the envelope, and the adversary only looks. Binding quantifies over what can be *manufactured*: the adversary builds an ambiguous commitment of its own choosing, under no obligation to follow Commit honestly. A scheme can be strong in one respect and weak in the other, and the two properties actively pull in opposite directions—the more commitment values a message can take (good for hiding), the more opportunities for two messages to share one (bad for binding). The next subsection makes this tension a theorem.

4.4 Incompatibility of perfect hiding and perfect binding

Theorem 4.8 (No scheme is perfectly hiding and perfectly binding). *Let Π be a canonically opening commitment scheme with $|\mathcal{M}_{pp}| \geq 2$. If Π is perfectly hiding, then it is not perfectly binding: indeed, a computationally unbounded adversary wins the binding game with probability 1. Hence no nontrivial commitment scheme is simultaneously perfectly hiding and perfectly binding.*

Proof. Fix any pp in the support of $\text{Setup}(1^\lambda)$ and any two distinct messages $m_0, m_1 \in \mathcal{M}_{pp}$. For a message m , let

$$C_m := \{ \text{Commit}(pp, m; r) : r \in \mathcal{R}_{pp} \}$$

be the set of attainable commitments to m , and let D_m be the distribution of $\text{Commit}(pp, m; r)$ for uniform r . Perfect hiding makes D_{m_0} and D_{m_1} identical (Proposition 4.5); identical distributions have identical supports, so $C_{m_0} = C_{m_1}$. Pick any c in this common support. By definition of the supports there exist $r_0, r_1 \in \mathcal{R}_{pp}$ with

$$\text{Commit}(pp, m_0; r_0) = c = \text{Commit}(pp, m_1; r_1),$$

and then $(c, (m_0, r_0), (m_1, r_1))$ wins the binding game: both openings are valid by canonical opening, and $m_0 \neq m_1$. The double opening *exists* for every attainable pp ; an unbounded adversary *finds* it by enumerating $\mathcal{R}_{pp}$, so it wins Bind with probability 1, giving $\text{Adv}_{\Pi}^{\text{bind}} = 1 \neq 0$. $\square$

Remark 4.9 (The impossibility as a counting fact). Information-theoretically the theorem is a statement about how many bits the commitment carries. Perfect binding forces the map $c \mapsto m$ to be well defined, so c determines m completely: the commitment carries at least $\log_2 |\mathcal{M}|$ bits of information about the message. Perfect hiding forces the distribution of c to be independent of m : the commitment carries *zero* bits. The two demands are compatible only when $\log_2 |\mathcal{M}| = 0$, that is, $|\mathcal{M}| = 1$ —when there is nothing to commit to. Every real scheme therefore chooses which property to make perfect (or statistical) and settles for the other being computational. The Pedersen commitment of §4.6 is perfectly hiding and computationally binding; the hash-based commitment of §4.5, read in the random oracle model, is computationally binding (from collision resistance) and computationally hiding.

Remark 4.10 (Two regimes, two failure modes). The choice is not merely aesthetic; it decides *how* the scheme fails if its assumption falls. If binding is only computational, an adversary with enough power—one able to compute discrete logarithms, say—could *retroactively change* a committed value, and the soundness of any protocol built on top degrades to a computational assumption. If hiding is only computational, such an adversary could *retroactively learn* a committed value: privacy degrades the same way, and everlasting secrecy is lost the day the assumption breaks, even for commitments published long before. Which failure is worse is a judgement about the application: whether integrity or long-term confidentiality is the greater concern. A ledger that publishes its commitments forever has reason to care about the second failure mode; a protocol whose soundness guards live funds has reason to care about the first.

4.5 Hash-based commitments

The simplest construction seals the envelope with a hash. Recall from §3 the two properties it needs: a hash function $H: \{0,1\}^* \rightarrow \{0,1\}^n$ is *collision resistant* (Definition 3.7) if no PPT adversary finds $x \neq x'$ with $H(x) = H(x')$ except with negligible probability, and *preimage resistant* (one-way) if, given $H(x)$ for random x of appropriate length, no PPT adversary finds any x' with $H(x') = H(x)$ except with negligible probability.

Construction 4.11 (Hash commitment). Let $H: \{0,1\}^* \rightarrow \{0,1\}^n$ be a hash function. The *hash commitment* has message space $\mathcal{M} = \{0,1\}^*$ and randomness space $\mathcal{R} = \{0,1\}^\ell$ for a randomness length $\ell = \ell(\lambda)$ (for example $\ell = 2n$). Let $\text{enc}(m, r)$ be an injective, self-delimiting encoding of the pair—for example a canonical length encoding of m , followed by m , followed by the fixed-length string r . Define

$$\text{Commit}(m; r) := H(\text{enc}(m, r)),$$

opened canonically. There are no nontrivial public parameters beyond the choice of H —the scheme lives in the plain model of Remark 4.3.

Proposition 4.12 (Binding of the hash commitment). *If H is collision resistant, the hash commitment is computationally binding.*

Proof. Suppose $\mathcal{A}$ wins the binding game: it outputs c with valid openings (m_0, r_0) , (m_1, r_1) and $m_0 \neq m_1$, so $H(\text{enc}(m_0, r_0)) = c = H(\text{enc}(m_1, r_1))$. Injectivity of enc makes the two hash inputs distinct strings—this is exactly what the self-delimiting encoding buys; without it, a boundary shift could present distinct pairs as the same string. The adversary has therefore produced an H -collision, and the reduction that runs $\mathcal{A}$ and outputs the pair $(\text{enc}(m_0, r_0), \text{enc}(m_1, r_1))$ wins the collision game with exactly the binding advantage of $\mathcal{A}$, in essentially the same time. $\square$

Proposition 4.13 (Hiding of the hash commitment, in the ROM). *Model H as a random oracle (Definition 3.13). For messages of length $\text{poly}(\lambda)$ and randomness length $\ell = \omega(\log \lambda)$, the hash commitment is computationally hiding: an adversary making at most q oracle queries has hiding advantage at most $q \cdot 2^{-\ell}$.*

Proof. The challenge commitment is $c = H(\text{enc}(m_b, r))$ for uniform $r \in \{0,1\}^\ell$. Because the oracle's outputs are independent and uniform, c is a fresh uniform string—carrying no information about b —unless the adversary queries the oracle at the exact point $\text{enc}(m_b, r)$. Before any such query the randomness r is information-theoretically hidden: the adversary's view is independent of it. Each of the q queries therefore hits that point with probability at most $2^{-\ell}$, and the union bound (*Math Guide*, §“The union bound and a birthday calculation”) caps the probability of any hit by $q \cdot 2^{-\ell}$. Conditioned on no hit, the adversary's view is identically distributed in the two worlds $b = 0$ and $b = 1$, so the advantage is at most $q \cdot 2^{-\ell}$, negligible for $\ell = \omega(\log \lambda)$ and polynomial q . $\square$

Remark 4.14 (Why the randomness is essential). Dropping r is fatal. The deterministic commitment $c = H(m)$ fails hiding for every low-entropy message space: to distinguish a commitment to m_0 from one to m_1 , the peeker simply hashes her two candidates and compares. Nothing about collision resistance or one-wayness prevents this—the attack never inverts anything. The salt r contributes ℓ uniformly random bits, so the hash input is unpredictable even when the message

is one of two known values. The same phenomenon renders deterministic encryption insecure for low-entropy plaintexts, and the same cure—randomisation—applies.

Remark 4.15 (Strength profile). The hash commitment has two virtues. It is *transparent*: no trusted setup, indeed no parameters, and no algebraic structure for a subverted setup to poison. And it is *plausibly post-quantum*: generic quantum collision finding takes $\Theta(2^{n/3})$ queries against an n -bit hash (Remark 2.31), so increasing the output length compensates for the loss. Its weakness is the flip side of its opacity: the commitment is an unstructured bit string. One cannot add two hash commitments and obtain a commitment to the sum of the messages; no algebraic relation on commitments reflects any relation on the committed values. The Pedersen construction repairs exactly this defect, at the price of resting on discrete log.

4.6 The Pedersen commitment

We change algebraic scenery. Let $\mathbb{G}$ be a cyclic group of prime order q , written *additively*: for a point $P \in \mathbb{G}$ and scalar $a \in \mathbb{F}_q$, the scalar multiple is $[a]P$, with $[0]P = O$ the identity, and $a \mapsto [a]P$ is a group homomorphism $\mathbb{F}_q \rightarrow \mathbb{G}$. Concretely, $\mathbb{G}$ is the group of $\mathbb{F}_p$ -points of a prime-order elliptic curve (*Math Guide*, §“The group structure of $E(\mathbb{F}_p)$ ”), and in this volume the Pallas curve of §2.6; but the only structural facts the constructions use are that $\mathbb{G}$ is isomorphic to $\mathbb{F}_q$ as a group and that discrete logarithms in $\mathbb{G}$ are hard. Formally, a *group generator* $\mathcal{G}$ on input 1^λ outputs $(\mathbb{G}, q, G)$ with q a $\Theta(\lambda)$ -bit prime and G a generator, and the DLog assumption for $\mathcal{G}$ states that no PPT adversary, given $(\mathbb{G}, q, G, [a]G)$ for uniform $a \xleftarrow{\$} \mathbb{F}_q$, outputs a except with negligible probability—Definition 2.2, transcribed additively as in §2.6.

Construction 4.16 (Pedersen commitment). The algorithm $\text{Setup}(1^\lambda)$ runs $(\mathbb{G}, q, G) \leftarrow \mathcal{G}(1^\lambda)$, samples $s \xleftarrow{\$} \mathbb{F}_q^\times$, sets $H := [s]G$, outputs $pp = (\mathbb{G}, q, G, H)$, and *discards* s . The message and randomness spaces are $\mathcal{M} = \mathcal{R} = \mathbb{F}_q$, and

$$\text{Commit}(pp, m; r) := [m]G + [r]H,$$

opened canonically.

The parameters are two generators G and H whose relative discrete logarithm $s = \log_G H$ is unknown to *all* parties, the committer included. Everything hinges on this one unknown scalar: knowledge of s breaks binding (Remark 4.19), while perfect hiding, as the proof below shows, does not even require ignorance of s . In deployment the sampled-and-discarded s of the textbook presentation is replaced by deriving H through hashing to the curve (Construction 3.24): nobody ever *holds* s , and the setup is transparent (Remark 4.3).

Theorem 4.17 (Perfect hiding). *The Pedersen commitment is perfectly hiding.*

Proof. Fix any attainable $pp = (\mathbb{G}, q, G, H)$; since $s \in \mathbb{F}_q^\times$, the point H is not O . Because $\mathbb{G}$ has prime order, the nonzero H generates $\mathbb{G}$, so $r \mapsto [r]H$ is a bijection $\mathbb{F}_q \rightarrow \mathbb{G}$. For uniform r , therefore, $[r]H$ is uniform on $\mathbb{G}$ —and so is its translate $[m]G + [r]H$, for *every* message m : adding the fixed element $[m]G$ permutes the group and leaves the uniform distribution invariant. The commitment distribution is uniform on $\mathbb{G}$ regardless of m , so for any m_0, m_1 the two distributions coincide exactly, and every adversary’s advantage is 0 by Proposition 4.5. $\square$

Theorem 4.18 (Computational binding under DLog). *If the DLog assumption holds for $\mathcal{G}$, the Pedersen commitment is computationally binding. Concretely, from any adversary A that wins the*

binding game with probability ε one constructs $\mathcal{B}$ computing discrete logarithms with probability $(1 - 1/q) \varepsilon \geq \varepsilon - 1/q$, in essentially the same running time.

Proof. First the algebra. A double opening gives $[m_0]G + [r_0]H = [m_1]G + [r_1]H$ with $m_0 \neq m_1$, so

$$[m_0 - m_1]G = [r_1 - r_0]H.$$

The case $r_1 = r_0$ is impossible: it would force $[m_0 - m_1]G = O$, and since G has order q this means $m_0 \equiv m_1 \pmod{q}$, contradicting $m_0 \neq m_1$ in $\mathbb{F}_q$. With $r_1 - r_0$ invertible modulo the prime q , the relation solves for the hidden discrete logarithm:

$$s = \log_G H = (m_0 - m_1)(r_1 - r_0)^{-1} \bmod q.$$

A double opening is the discrete log of H .

Now the reduction. On a DLog challenge $(\mathbb{G}, q, G, Q)$, the algorithm $\mathcal{B}$ sets $H := Q$, hands $pp = (\mathbb{G}, q, G, Q)$ to $\mathcal{A}$, and, when $\mathcal{A}$ returns a double opening, outputs $(m_0 - m_1)(r_1 - r_0)^{-1} \bmod q = \log_G Q$. One distribution gap needs accounting: in the real scheme $H = [s]G$ with $s \xleftarrow{\$} \mathbb{F}_q^\times$, so H is uniform on $\mathbb{G} \setminus \{O\}$, whereas the challenge $Q = [a]G$ has a uniform on all of $\mathbb{F}_q$, so Q is uniform on all of $\mathbb{G}$. The two distributions differ only on the event $Q = O$, of probability $1/q$ —their statistical distance is exactly $1/q$ —and on that event $\mathcal{A}$ cannot win anyway (with $H = O$ the relation above forces $m_0 = m_1$). Conditioned on $Q \neq O$, the simulation is perfect, so

$$\text{Adv}^{\text{dlog}}(\mathcal{B}) = \left(1 - \frac{1}{q}\right) \text{Adv}_{\Pi}^{\text{bind}}(\mathcal{A}) \geq \varepsilon - \frac{1}{q},$$

and $\mathcal{B}$'s overhead over $\mathcal{A}$ is a constant number of field operations. $\square$

The same embed-and-extract pattern, scaled up to many generators, proves the vector generalisation (Theorem 4.23).

Remark 4.19 (Equivocation with the trapdoor). Any party who knows $s = \log_G H$ can open a Pedersen commitment to *any* message. Given $c = [m]G + [r]H$ and a target m' , set

$$r' := r + (m - m')s^{-1} \bmod q;$$

then $[m']G + [r']H = [m']G + [r]H + [(m - m')]G = c$. This trapdoor property is not a defect of the construction—it is precisely why discarding s in Setup is mandatory, and it is a feature the theory exploits deliberately: zero-knowledge simulators use exactly this equivocation to open commitments consistently with a transcript produced out of order. The same phenomenon priced the setup models of Remark 4.3: a commitment scheme with a trapdoor is only as binding as the claim that nobody holds it.

Theorem 4.20 (Additive homomorphism). *For all $m_0, m_1, r_0, r_1 \in \mathbb{F}_q$,*

$$\text{Commit}(m_0; r_0) + \text{Commit}(m_1; r_1) = \text{Commit}(m_0 + m_1; r_0 + r_1),$$

and more generally, for all scalars $\alpha, \beta \in \mathbb{F}_q$,

$$[\alpha] \text{Commit}(m_0; r_0) + [\beta] \text{Commit}(m_1; r_1) = \text{Commit}(\alpha m_0 + \beta m_1; \alpha r_0 + \beta r_1).$$

Proof. Expand and regroup, using commutativity of $+$ in $\mathbb{G}$, the identity $[a]P + [b]P = [a + b]P$, and the linearity $[\alpha]([m]G + [r]H) = [\alpha m]G + [\alpha r]H$ of scalar multiplication:

$$\begin{aligned} [\alpha]([m_0]G + [r_0]H) + [\beta]([m_1]G + [r_1]H) &= [\alpha m_0]G + [\beta m_1]G + [\alpha r_0]H + [\beta r_1]H \\ &= [\alpha m_0 + \beta m_1]G + [\alpha r_0 + \beta r_1]H. \end{aligned} \quad \square$$

Remark 4.21 (Value balancing: the homomorphism deployed). The homomorphism is what makes Pedersen commitments the right tool for *hidden arithmetic*, and Zcash’s value balancing is the canonical deployment. Each transaction value v_i is committed as $C_i = [v_i]G + [r_i]H$; a transaction balances iff the sum of its input values equals the sum of its output values. By Theorem 4.20, the verifier can check this *without learning any v_i* : the value components cancel in $\sum C_{\text{in}} - \sum C_{\text{out}}$ precisely when the value sums agree *modulo the group order q* , in which case the difference equals $[r]H$ for the net randomness r —a commitment to 0. Cancellation alone certifies balance only modulo q —value sums differing by a multiple of q cancel too—and the deployed protocol closes the gap with range checks carried inside the proofs: the values of the notes (the shielded value records) spent and created by each Action (Orchard’s combined spend-and-output unit) are proved to lie in $\{0, \dots, 2^{64} - 1\}$ (the Action statement, protocol specification § 4.18.4; enforced by the note-commitment gadget’s ValueCanonicity gate), the committed net value is the difference of two such values, and the number of summands is bounded, so the sums cannot wrap (§ 4.14). The prover does not reveal r ; it proves *knowledge* of r by using $[r]H$ as the verification key of a signature over the transaction—the *binding signature*, treated with the completed balance argument in § 8.9. In deployed Orchard the instantiation is $\text{ValueCommit}(v; \text{rcv}) = [v]V + [\text{rcv}]R$ with hashed-to-curve generators V, R (protocol specification § 5.4.8.3); each Action publishes a commitment to its *net* value change, and the verifier folds the published commitments with a commitment to the transparent value balance to obtain the binding verification key. The homomorphism turns an arithmetic relation on hidden values into a group equation on public commitments—a template the rest of the volume reuses repeatedly.

4.7 Pedersen vector commitments

A single group element can seal far more than a single scalar.

Construction 4.22 (Pedersen vector commitment). On input 1^λ and a length $n = \text{poly}(\lambda)$, Setup outputs $(\mathbb{G}, q)$ together with generators $G_1, \dots, G_n, H \xleftarrow{\$} \mathbb{G} \setminus \{O\}$, sampled so that no nontrivial discrete-log relation among them is known to any party—in practice each generator is derived from a public seed by hashing into the curve, rejecting the identity if it occurs, a nothing-up-my-sleeve procedure (Construction 3.24) that makes the setup transparent. The message space is $\mathbb{F}_q^n$, the randomness space $\mathbb{F}_q$, and for $\mathbf{m} = (m_1, \dots, m_n)$,

$$\text{Commit}(pp, \mathbf{m}; r) := \sum_{i=1}^n [m_i]G_i + [r]H = \langle \mathbf{m}, \mathbf{G} \rangle + [r]H,$$

where $\langle \mathbf{m}, \mathbf{G} \rangle := \sum_i [m_i]G_i$ is the multiscalar (mixed) inner product of the message vector with the generator vector $\mathbf{G} = (G_1, \dots, G_n)$. Opening is canonical.

Theorem 4.23 (Properties of the vector commitment). *The Pedersen vector commitment is perfectly hiding, and computationally binding under the DLog assumption in $\mathbb{G}$.*

Proof. Hiding. Exactly as in Theorem 4.17: for uniform r the blinding term $[r]H$ is uniform on $\mathbb{G}$, so the commitment is uniform on $\mathbb{G}$ regardless of the message vector.

Binding. A double opening with $\mathbf{m} \neq \mathbf{m}'$ gives $\langle \mathbf{m}, \mathbf{G} \rangle + [r]H = \langle \mathbf{m}', \mathbf{G} \rangle + [r']H$, that is, a *nontrivial discrete-log relation*

$$\sum_{i=1}^n [m_i - m'_i] G_i + [r - r'] H = O$$

with not all coefficients zero. The reduction $\mathcal{B}$ must extract a single logarithm from a relation among $n+1$ generators, and it does so by embedding its challenge in *every* generator. On challenge $Q = [s]G$, it samples $u_i, w_i \xleftarrow{\$} \mathbb{F}_q$ for each i and $u_H, w_H \xleftarrow{\$} \mathbb{F}_q$, sets

$$G_i := [u_i]G + [w_i]Q, \quad H := [u_H]G + [w_H]Q,$$

and resamples any pair whose generator lands on O . Each result is uniform on $\mathbb{G} \setminus \{O\}$ —the $[u_i]G$ term alone makes the pre-rejection point uniform on $\mathbb{G}$ —so the simulated parameters follow the real distribution. Crucially, the parameters reveal nothing about the w -coordinates: for every candidate value of w_i there is exactly one u_i consistent with the observed G_i , so conditioned on the adversary’s entire view the w_i and w_H remain independent and uniform. Write the recovered relation as $\sum_i [a_i]G_i + [a_H]H = O$ with coefficient vector $(a_1, \dots, a_n, a_H) \neq 0$. Substituting the embeddings and collecting the G - and Q -components gives

$$c_0 + c_1 s \equiv 0 \pmod{q}, \quad c_0 = \sum_i a_i u_i + a_H u_H, \quad c_1 = \sum_i a_i w_i + a_H w_H.$$

Since the coefficient vector is nonzero and the w -coordinates are uniform and independent of it, c_1 is uniform on $\mathbb{F}_q$, hence nonzero with probability $1 - 1/q$; in that case $s = -c_0 c_1^{-1} \pmod{q}$, and $\mathcal{B}$ outputs the challenge logarithm. Altogether $\text{Adv}^{\text{dlog}}(\mathcal{B}) \geq (1 - 1/q) \text{Adv}^{\text{bind}}(\mathcal{A})$. $\square$

Remark 4.24 (Compression and the cardinality trade-off). The vector commitment is *compressing*: n field elements (plus the randomness) map to a single group element, exponentially shorter than the message for large n . Compression is incompatible with statistical binding for a counting reason: there are q^n message vectors and only q commitment values, so colliding openings exist in overwhelming abundance. This is exactly why binding is only computational—an unbounded adversary finds a colliding opening by solving the underlying discrete-log relation—and it is the trade-off of Theorem 4.8 made visible at the level of cardinalities: a compressing commitment *cannot* determine its message, so it might as well hide it perfectly.

Remark 4.25 (Linearity survives). The vector commitment remains additively homomorphic in both message and randomness, componentwise: $\text{Commit}(m; r) + \text{Commit}(m'; r') = \text{Commit}(m + m'; r + r')$, by the same regrouping as Theorem 4.20. This linearity, applied to a *committed* vector against *public* coefficient vectors, is the foundation of the inner-product arguments of §4.9, which prove statements about $\langle m, b \rangle$ for public b while revealing only a single committed group element.

4.8 Sinsemilla: an algebraic hash-based commitment

Between the opaque hash commitment of §4.5 and the fully linear Pedersen commitment sits an intermediate species: *algebraic hash functions*, collision-resistant hashes whose internal operations are elliptic-curve group operations. The point of the species is proof cost. A zero-knowledge circuit that must verify a hash evaluation pays for every internal operation; when those operations are curve additions in the very field the proof system works over, the verification is cheap, while collision resistance still reduces to the discrete logarithm. Sinsemilla, deployed in Zcash Orchard, is the archetype: a Pedersen-style hash over a sequence of message chunks. We build up to it through its ancestor.

Construction 4.26 (Pedersen hash). Split the message M into k chunks $m_1, \dots, m_k$, each interpreted as a scalar in a bounded range, and fix k independent generators $P_1, \dots, P_k \in \mathbb{G}$ with

unknown mutual discrete logarithms. Define

$$\text{PedersenHash}(M) := \sum_{j=1}^k [m_j]P_j \in \mathbb{G}.$$

This is the unblinded Pedersen vector commitment map used as a hash. The output is a group element; composing with a fixed encoding of group elements as bit strings yields an ordinary hash.

Proposition 4.27 (Collision resistance of the Pedersen hash). *Suppose no efficient algorithm finds a nontrivial relation $\sum_j [a_j]P_j = O$ with not all $a_j = 0$ and each $|a_j|$ within the allowed chunk range—hardness that follows from DLog when the P_j are independent uniform generators. Then PedersenHash is collision resistant.*

Proof. A collision $M \neq M'$, with chunk decompositions (m_j) and (m'_j) , gives $\sum_j [m_j - m'_j]P_j = O$ with some coefficient $m_j - m'_j \neq 0$ —a nontrivial relation with coefficients in the allowed range. The reduction from relation-finding to DLog is the every-generator embedding of Theorem 4.23: set each $P_j = [u_j]G + [w_j]Q$ for the challenge Q , and read the unknown logarithm off the recovered relation. $\square$

The Pedersen hash needs one independent generator per chunk, which is costly for long inputs: the generator table grows linearly with the maximum message length. Sinsemilla reorganises the computation into an iterated two-operand form that reuses a small fixed generator table—shared by all chunk positions—while preserving the discrete-log collision reduction.

Construction 4.28 (Sinsemilla hash). Split the message M into pieces $m_1, \dots, m_k$, each of fixed width $w \leq 32$ bits, and identify a piece $v \in \{0,1\}^w$ with its integer value, bits read little-endian (the specification’s LEBS2IP). The public parameters are a base point $Q \in \mathbb{G}$ and a table $S: \{0,1\}^w \rightarrow \mathbb{G}$ assigning to each possible piece value its own generator, both obtained by hashing to the curve. Write $\text{GroupHash}(D, M)$ for the hash-to-curve pipeline of Construction 3.22 applied to the byte string M under the domain-separation tag built from the byte string D —for Pallas the RFC 9380 suite tag $D \parallel \text{"-pallas_XMD:BLAKE2b_SSWU_R0_"} of §3.6 (protocol specification § 5.4.9.8)—and write $\text{LE}_{32}(j)$ for the four-byte little-endian encoding of an integer $0 \leq j < 2^{32}$. The table is$

$$S(v) := \text{GroupHash}(\text{"z.cash:SinsemillaS"}, \text{LE}_{32}(v)), \quad v \in \{0,1\}^w,$$

with 2^w entries shared by every instance, and the base point is personalised (§3.5) to the enclosing use—a byte string D naming it, such as a note-commitment or Merkle-hash domain—by taking that string as the message:

$$Q := \text{GroupHash}(\text{"z.cash:SinsemillaQ"}, D).$$

Both derivations are nothing-up-my-sleeve (Construction 3.24), so no party knows any discrete-log relation among $\{S(v)\}_v \cup \{Q\}$ (Remark 3.25). The deployed instantiation takes $w = 10$ (the specification’s k) over Pallas: the sinsemilla crate fixes the two personalisation strings and derives Q in `HashDomain::new`; the table ships precomputed as `SINSEMILLA_S`, and the crate’s test `sinsemilla_s` re-derives every entry from `u32::to_le_bytes` of the index—exactly LE_{32} ; the circuit loads that table as its lookup; protocol specification § 5.4.1.9.

The hash iterates over the pieces with a running accumulator $\text{Acc} \in \mathbb{G}$:

$$\text{Acc}_0 := Q, \quad \text{Acc}_j := \text{Acc}_{j-1} + (\text{Acc}_{j-1} + S(m_j)) = [2]\text{Acc}_{j-1} + S(m_j),$$

and outputs the point $\text{SinsemillaHashToPoint}(M) := \text{Acc}_k$. The step costs two *incomplete* curve additions—first $\text{Acc}_{j-1} + S(m_j)$, then the result plus Acc_{j-1} ; no doubling is ever performed, and the addition formula is undefined when its operands are equal or are mutual negatives (*Math Guide*, §“Elliptic curves”)—and this pair of additions is the operation the circuit verifies in very few constraints. Unrolling the recurrence gives the closed form

$$\text{SinsemillaHashToPoint}(M) = \sum_{j=1}^k [2^{k-j}]S(m_j) + [2^k]Q :$$

the integer coefficients are powers of two determined by position alone, and the message enters only through *which* generator $S(m_j)$ each piece selects.

Proposition 4.29 (Collision resistance of Sinsemilla). *Model the table entries $S(v)$ and the base point Q as independent uniform generators (the nothing-up-my-sleeve hash-to-curve derivation covers Q as well as the table). Then any collision in $\text{SinsemillaHashToPoint}$, on inputs for which its incomplete additions are defined, yields a nontrivial discrete-log relation among $\{S(v)\}_v \cup \{Q\}$, and any input on which an incomplete addition is undefined yields one directly; either way a discrete logarithm follows via the every-generator embedding of Theorem 4.23 extended to Q . The point hash is thus collision resistant under the DLog assumption in $\mathbb{G}$, the exceptional cases of incomplete addition covered by the same reduction rather than excluded.*

Proof sketch. Let $M \neq M'$ collide. Suppose first that they have the same piece count k . Subtracting the unrolled forms cancels the $[2^k]Q$ terms and leaves

$$\sum_{j=1}^k [2^{k-j}](S(m_j) - S(m'_j)) = O.$$

Regroup over the distinct table generators: for each value $v \in \{0, 1\}^w$, the net coefficient on $S(v)$ is

$$c_v = \sum_{j: m_j=v} 2^{k-j} - \sum_{j: m'_j=v} 2^{k-j},$$

and the relation reads $\sum_v [c_v] S(v) = O$. Each c_v is a difference of two subset-sums of the distinct powers $2^{k-1}, \dots, 2^0$, hence an integer with $|c_v| < 2^k$. For *admissible* message lengths—those with 2^k below the group order q , a finite design-time check—such a coefficient vanishes modulo q only if it vanishes as an integer; and $c_v = 0$ forces, by uniqueness of binary expansions, the two subset-sums to be equal as sets of powers, that is, the positions carrying value v in M and in M' to coincide. Coincidence for every v would force $M = M'$; since $M \neq M'$, some $c_v \neq 0$ and the relation is nontrivial.

If instead the piece counts differ, say $k > k'$, the Q -constants do not cancel: the relation acquires the coefficient $2^k - 2^{k'}$ on Q , nonzero and below the group order for admissible lengths, so the relation is again nontrivial. In either case, embedding a DLog challenge in every table generator *and* in Q recovers the challenge logarithm exactly as in Proposition 4.27.

Finally, the exceptional cases. An incomplete addition is undefined only when its operands are equal or are mutual negatives, so an input reaching one presents a coincidence $[\alpha]\text{Acc}_{j-1} + S(m_j) = O$ with $\alpha \in \{-1, 1, 2\}$. Substituting the unrolled form of Acc_{j-1} makes this an explicit relation among Q and the table generators, with coefficient $\alpha 2^{j-1} \neq 0$ on Q and all magnitudes

at most 2^k —below the group order for admissible lengths. An input that merely triggers an exceptional case therefore hands over a nontrivial relation itself, and the same embedding extracts the challenge logarithm from it; the protocol specification proves exactly this for the deployed instantiation (§ 5.4.1.9). $\square$

The deployed hash differs from the point hash in two ways. First, the deployed `SinsemillaHash` applies the Pallas coordinate extractor to `SinsemillaHashToPoint( $M$ )`, returning its x -coordinate rather than the point itself. Collision resistance survives the extraction: equal extracted x -coordinates mean the two points are equal or are negatives of one another, and either signed equality yields the same kind of nontrivial generator relation as in the proof above. Second, the final piece is zero-padded to the piece width—the `Pad` iterator of the `sinsemilla` crate pads the message bits to a multiple of the piece width before chunking—so two bit strings of different lengths within the same final piece present identical piece sequences, and the deployed hash claims collision resistance only *between inputs of a fixed length for a given personalisation*. This is exactly the property the protocol specification requires and no more (protocol specification § 5.4.1.9: collision resistance is demanded “between inputs of fixed length, for a given personalization input”); each personalisation domain hashes inputs of a single length, as the Merkle-tree hash `MerkleCRH` of §10 illustrates. The variable-length argument in the proof above applies to the piece-aligned point presentation, where the piece sequence is the message.

The step from hash to commitment is the Pedersen blinding term, verbatim.

Construction 4.30 (`SinsemillaCommit`). Fix one further independent generator H' , hashed to the curve with its own domain string, so that no discrete-log relation between H' and $\{S(v)\}_v \cup \{Q\}$ is known. Define, for $r \xleftarrow{\$} \mathbb{F}_q$,

$$\text{SinsemillaCommit}(m; r) := \text{SinsemillaHashToPoint}(m) + [r]H'.$$

Hiding is inherited from Pedersen verbatim: by the argument of Theorem 4.17, $[r]H'$ is uniform on $\mathbb{G}$ for uniform r , so the commitment distribution is independent of m , and the scheme is perfectly hiding. Binding rests on discrete log: two valid openings to distinct messages yield either a `SinsemillaHashToPoint` collision—excluded by Proposition 4.29—or a nontrivial discrete-log relation between H' and the hash generators.

Orchard’s note commitment `NoteCommit` instantiates exactly this map: the orchard crate derives it through the `sinsemilla` crate’s `CommitDomain::commit` (protocol specification § 5.4.8.4). Its key commitment `CommitLvk` composes the same map with x -coordinate extraction—the specification’s `SinsemillaShortCommit`—returning a base-field element rather than a point.

Remark 4.31 (Why `Sinsemilla` exists: cost inside a proof). `Sinsemilla`’s advantage is not speed outside a circuit—ordinary hashes are faster there—but verification cost *inside* one. In a SNARK (§9) the prover re-expresses every computational step as polynomial constraints; a bit-oriented hash such as SHA-256, all Boolean choppings and modular additions foreign to the proof field, costs tens of thousands of constraints per evaluation, whereas `Sinsemilla`’s per-piece cost is a handful of curve operations chosen to lie in the very field the proof system already works over. The result is a collision-resistant hash and commitment—used for note commitments and Merkle trees in Orchard—simultaneously cheap to prove and grounded in the same discrete-log hardness as the surrounding protocol. The presentation here is the conceptual core; the *Ironwood Guide* takes up the concrete Orchard instantiation—its per-use personalisation strings and the in-circuit handling of incomplete addition.

4.9 Polynomial commitment schemes

The final refinement commits not to a value or a vector but to a *function*—a polynomial—and supports proving statements about its values. This is the primitive on which the succinct argument systems of §9 stand.

Definition 4.32 (Polynomial commitment scheme). Fix a field $\mathbb{F}$ and a degree bound d . A *polynomial commitment scheme* (PCS) is a tuple $(\text{Setup}, \text{Commit}, \text{Open}, \text{Eval}, \text{Verify})$:

- The algorithm $\text{Setup}(1^\lambda, d)$ outputs public parameters pp supporting degree at most d ;
- the algorithm $\text{Commit}(pp, f; r)$ outputs a commitment C to a polynomial $f \in \mathbb{F}[X]_{\leq d}$, optionally with hiding randomness r ;
- the algorithm $\text{Open}(pp, C, f, r)$ outputs a bit, verifying that C commits to the *whole* polynomial f ;
- the algorithm $\text{Eval}(pp, f, r, z)$, for an evaluation point $z \in \mathbb{F}$, outputs a claimed value $y = f(z)$ together with an *evaluation proof* π ;
- the algorithm $\text{Verify}(pp, C, z, y, \pi)$ outputs a bit, checking the evaluation proof against the commitment.

Beyond the hiding and binding of the underlying commitment (Definitions 4.4 and 4.6), a PCS must satisfy *evaluation binding*: no PPT prover can produce a commitment C , a point z , and two accepting proofs for distinct claimed values $y \neq y'$ at z , except with negligible probability. Stronger schemes additionally satisfy *extractability* (proof of knowledge): from any successful prover one can extract an actual polynomial f of degree at most d consistent with all the evaluations it gets accepted.

Remark 4.33 (Rigidity, and why evaluation binding is separate). Why should one short commitment support probing at arbitrary points? Because polynomials are *rigid*: a nonzero polynomial of degree at most d has at most d roots (*Math Guide*, §“The Schwartz–Zippel lemma”), so two distinct polynomials of degree at most d agree on at most d points. Answering a different value at even a single point is the behaviour of a prover holding a *different polynomial*—there is no wiggle room between functions. But a warning is in order: an accepting evaluation proof is *not* an opening of C , so evaluation binding does *not* follow from the binding of Commit . It is a separate cryptographic requirement, and each construction below proves it from its own assumption—a strong Diffie–Hellman assumption for KZG, discrete log via a folding extractor for the inner-product scheme. The central design problem of a PCS is making the evaluation proof π short and fast to verify; committing to the coefficient list with any old commitment is easy, and useless.

KZG: evaluation in the exponent

The pairing-based scheme of Kate, Zaverucha, and Goldberg achieves constant-size proofs by hiding a secret evaluation point inside the parameters.

Construction 4.34 (KZG commitment). Let $\mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T$ be groups of prime order $|\mathbb{F}|$ equipped with a nondegenerate bilinear pairing $e: \mathbb{G}_1 \times \mathbb{G}_2 \rightarrow \mathbb{G}_T$ (Example 2.9), with generators $G \in \mathbb{G}_1$ and $\hat{G} \in \mathbb{G}_2$. The algorithm Setup samples a secret $\tau \xleftarrow{\$} \mathbb{F}$, publishes the *structured reference string*

$$pp = ([\tau^0]G, [\tau^1]G, \dots, [\tau^d]G, \hat{G}, [\tau]\hat{G}),$$

and discards τ . The commitment to $f = \sum_{i=0}^d a_i X^i$ is evaluation in the exponent:

$$\text{Commit}(pp, f) := \sum_{i=0}^d [a_i]([\tau^i]G) = [f(\tau)]G,$$

computable from pp by anyone, without knowledge of τ .

Construction 4.35 (KZG evaluation proof). To prove $f(z) = y$, the prover uses the factor theorem: z is a root of $f(X) - y$, so the quotient

$$q(X) := \frac{f(X) - y}{X - z} \in \mathbb{F}[X]_{\leq d-1}$$

is a genuine polynomial *precisely because* $f(z) - y = 0$. The proof is the commitment to the quotient, $\pi := [q(\tau)]G$, computed from the reference string. The verifier checks the single pairing equation

$$e(C - [y]G, \hat{G}) \stackrel{?}{=} e(\pi, [\tau]\hat{G} - [z]\hat{G}),$$

which, by bilinearity, is exactly the in-the-exponent form of the polynomial identity $f(\tau) - y = q(\tau)(\tau - z)$.

Proposition 4.36 (KZG evaluation binding, sketched). *Evaluation binding of KZG rests on the d -strong Diffie–Hellman assumption in the bilinear group: given the powers $[\tau^i]G$, it is infeasible to produce $[(\tau - z)^{-1}]G$ for any z of one’s choosing.*

Proof idea. Two accepting proofs π, π' for distinct values $y \neq y'$ at the same point z satisfy the two pairing equations; subtracting them (in the exponent) gives $\pi - \pi' = [(y' - y)(\tau - z)^{-1}]G$, so the forger’s output yields

$$[(\tau - z)^{-1}]G = [(y' - y)^{-1}](\pi - \pi'),$$

exactly the element the assumption declares infeasible. $\square$

Remark 4.37 (The KZG trade). What KZG buys: commitments and evaluation proofs of *constant* size (one group element each) and *constant* verification time (two pairings), all independent of the degree d . What it costs is twofold. First, the setup is trusted in the strongest sense: the secret τ is a global trapdoor. Anyone who learns τ can forge accepting evaluation proofs for *false* statements—for any commitment C , any point z , and any desired value y , the element $\pi := [(\tau - z)^{-1}](C - [y]G)$ passes the pairing check, no knowledge of any polynomial required. This is the promised demonstration that a subverted setup silently destroys binding (Remark 4.3); evaluation binding survives only if τ is provably destroyed, in practice by a multi-party *powers-of-tau* ceremony whose trapdoor stays secret as long as at least one participant is honest. Second, the construction requires pairing-friendly curves, a more delicate and less efficient setting than the ordinary curves of §2.6. Binding is computational either way, resting on the bilinear assumption—consistent with Theorem 4.8.

The inner-product alternative

The transparent alternative commits to the coefficient vector with the Pedersen vector commitment and replaces the pairing check by a recursive argument.

Construction 4.38 (IPA-style polynomial commitment). Let $G = (G_0, \dots, G_d)$ and H be transparent generators with unknown mutual discrete logs, derived from a public seed (Construction 4.22). Commit to $f = \sum_{i=0}^d a_i X^i$ by the Pedersen vector commitment of its coefficient vector $\mathbf{a} = (a_0, \dots, a_d)$:

$$\text{Commit}(pp, f; r) := \sum_{i=0}^d [a_i]G_i + [r]H = \langle \mathbf{a}, G \rangle + [r]H.$$

No pairings, no trusted setup.

Evaluation is an inner product: writing $\mathbf{z} = (1, z, z^2, \dots, z^d)$ for the public vector of powers of the query point,

$$f(\mathbf{z}) = \sum_{i=0}^d a_i z^i = \langle \mathbf{a}, \mathbf{z} \rangle,$$

so proving an evaluation reduces to proving an inner-product relation about a committed vector against a public vector. The *inner-product argument* (IPA) proves it recursively: each round folds the two halves of the coefficient vector—and of the generator vector—into vectors of half the dimension by taking a random linear combination dictated by a verifier challenge, sending two group elements per round (the cross-term commitments L_j and R_j) to account for the mixed terms; after $\log_2(d+1)$ rounds a single scalar remains, and the verifier checks one final equation tying it to the folded commitment and the folded generators.

Proposition 4.39 (Properties of the IPA-based PCS). *The inner-product polynomial commitment has: transparent setup (public-coin parameters, no trapdoor); evaluation proofs of $O(\log d)$ group elements; and $O(d)$ -group-operation verification, batchable across proofs. Its binding and evaluation binding reduce to the discrete logarithm assumption in $\mathbb{G}$.*

Proof idea. Binding of the commitment itself is Theorem 4.23. For evaluation binding and extractability, one shows the folding protocol is *tree special-sound*: from a suitable tree of accepting transcripts, branching over the verifier’s challenge at each round, an extractor works back up the recursion and computes a coefficient vector $\mathbf{a}$ consistent with the commitment and the claimed inner product—a notion this sketch uses only informally and §9 defines precisely. Two distinct extracted openings of the same commitment would yield a nontrivial discrete-log relation among G, H , contradicting DLog by the every-generator embedding. The $O(\log d)$ proof size is the halving recursion; the $O(d)$ verification is the final multiscalar multiplication over the unfolded generators. $\square$

Remark 4.40 (KZG versus IPA, and the Halo lineage). The two families mark out the design space. KZG gives constant proof size and constant verification, at the price of a structured trusted setup and pairing-friendly curves; the IPA gives transparent setup over a plain curve and rests on nothing beyond discrete log, at the price of logarithmic proofs and linear-time verification. The Halo 2 proof system deployed in Zcash Orchard builds on the inner-product family precisely to avoid trusted setup, and recovers practical verification cost through *batching*: the deployed verifier folds the final multiscalar multiplications of many proofs into one shared multiexponentiation—each proof’s deferred check is scaled by a fresh random factor and accumulated, and a single combined check is evaluated once. Batching amortises the per-proof group operations across a batch, though per-proof verifier work remains linear in d . The Halo lineage’s accumulation technique extends the same deferral across *recursively composed* proofs, one proof attesting to another’s deferred check—the capability the Pasta cycle of Remark 2.26 was designed to serve—but deployed Zcash does not use recursion: each Orchard proof is verified (in batch) directly.

Remark 4.41 (Hiding variants). As stated, the KZG commitment is binding but not hiding: it is a deterministic function of f , so committing twice to the same polynomial is visible, and a low-entropy polynomial can be found by search—the failure of Remark 4.14 in new clothing. That is acceptable when the polynomial encodes only public data. Whenever it encodes secrets—as the witness polynomials of a zero-knowledge proof do—randomness is required: KZG is augmented with a blinding summand $[r][\gamma]G$ using an extra setup element $[\gamma]G$, while the IPA commitment of Construction 4.38 already carries its $[r]H$ and is perfectly hiding by Theorem 4.23. The resulting regime is the perfect-hiding/computational-binding profile of Pedersen (Theorems 4.17 and 4.18), lifted from field elements to polynomials.

From commitments to arguments

The polynomial commitment is where the section’s main line hands over to the proof systems of §9, and the hand-over can be stated in one paragraph. A polynomial commitment scheme turns an information-theoretic proof object—a polynomial identity holding over a large field—into a succinct cryptographic argument. The prover commits to its witness polynomials; the verifier challenges at random points; evaluation proofs then convince the verifier that the committed polynomials satisfy the required identities, without the polynomials ever being revealed. Soundness is supplied by Schwartz–Zippel (*Math Guide*, §“The Schwartz–Zippel lemma”): a nonzero polynomial of low degree is almost surely nonzero at a random point, so an identity that holds at the challenge almost surely holds identically. Binding and succinctness are supplied by the commitment: the prover is locked to its polynomials *before* seeing the challenge, and the verifier handles only short commitments and proofs in place of the polynomials themselves. How this compilation is carried out in full—and what “argument of knowledge” precisely means—is the business of §9, whose SNARK compilation subsection develops the evaluation-argument role sketched here.

4.10 Secret sharing

The section opened by naming verifiable secret sharing among the classical applications of commitments; it closes by constructing it, because the threshold signatures of the *FROST Guide* stand on it and nothing below that volume otherwise provides it. The setting is a prime field $\mathbb{F}_q$ —in the sequel the scalar field of the group $\mathbb{G}$ of §4.6—and the tool is Lagrange interpolation (*Math Guide*, §“Lagrange interpolation”): a polynomial of degree at most $t - 1$ over $\mathbb{F}_q$ is determined by its values at any t distinct points, and any t prescribed values at distinct points are attained by exactly one such polynomial.

Construction 4.42 (Shamir t -of- n secret sharing). Fix a threshold t and a party count n with $1 \leq t \leq n < q$. To share a secret $s \in \mathbb{F}_q$, the dealer samples coefficients $a_1, \dots, a_{t-1} \xleftarrow{\$} \mathbb{F}_q$ independently and uniformly and forms

$$f(X) := s + a_1X + a_2X^2 + \dots + a_{t-1}X^{t-1} \in \mathbb{F}_q[X], \quad f(0) = s,$$

a polynomial of degree at most $t - 1$ with constant term the secret. Party $i \in \{1, \dots, n\}$ receives the *share* $s_i := f(i)$, sent privately; the identifiers $1, \dots, n$ are distinct nonzero elements of $\mathbb{F}_q$ because $n < q$. *Reconstruction* from the shares of any set I of t parties is interpolation at 0:

$$s = f(0) = \sum_{i \in I} \lambda_{I,i} s_i, \quad \lambda_{I,i} := \prod_{j \in I, j \neq i} \frac{j}{j - i},$$

where $\lambda_{I,i} = \ell_i(0)$ is the Lagrange basis polynomial for the nodes I evaluated at 0, computable from the identifiers alone.

Correctness of reconstruction is the interpolation theorem: the t points $(i, s_i)_{i \in I}$ determine a unique polynomial of degree at most $t - 1$, which is f , and the Lagrange formula $f(X) = \sum_{i \in I} s_i \ell_i(X)$ evaluated at $X = 0$ gives the displayed sum. The coalition may be chosen after the shares are dealt, since the weights depend only on I .

Theorem 4.43 (Perfect secrecy below the threshold). *Let I be any set of at most $t - 1$ identifiers. For every secret $s \in \mathbb{F}_q$, the shares $(s_i)_{i \in I}$ of Construction 4.42 are jointly uniform on $\mathbb{F}_q^{|I|}$. In particular their distribution does not depend on s : fewer than t shares are independent of the secret, and an adversary of unbounded power holding them has advantage exactly 0 at telling any two candidate secrets apart.*

Proof. It suffices to treat $|I| = t - 1$, since fewer shares are a projection of these. Fix s and I , and consider the map

$$\Sigma_s: \mathbb{F}_q^{t-1} \rightarrow \mathbb{F}_q^{t-1}, \quad (a_1, \dots, a_{t-1}) \mapsto (f(i))_{i \in I},$$

from the dealer's coefficient vector to the coalition's shares. It is injective: two coefficient vectors with the same shares give two polynomials of degree at most $t - 1$ that agree at the t distinct points 0 (where both equal s) and I , hence are equal by uniqueness of interpolation, so the coefficient vectors coincide. It is surjective: for any target $(y_i)_{i \in I}$, Lagrange interpolation through the t points $(0, s)$ and $(i, y_i)_{i \in I}$ produces a polynomial of degree at most $t - 1$ with constant term s , whose remaining coefficients are a preimage. So Σ_s is a bijection—for every s —and it carries the uniform distribution on coefficient vectors to the uniform distribution on $\mathbb{F}_q^{t-1}$. The share vector is therefore uniform whatever the secret, and for any $s_0 \neq s_1$ the two share distributions coincide exactly; by the variational characterisation of statistical distance (Proposition 1.7) no procedure, efficient or not, distinguishes them. $\square$

The theorem is the perfect-hiding half of a commitment (Proposition 4.5 in different clothing): the coalition's view is a uniformly random point of $\mathbb{F}_q^{t-1}$, consistent with every candidate secret through exactly one polynomial each. What Shamir sharing lacks is the binding half. Nothing stops a dishonest dealer from handing out values that lie on no common polynomial of degree at most $t - 1$, so that different coalitions of t parties reconstruct different “secrets”—and a party receiving a share cannot tell. Verifiable secret sharing repairs this by having the dealer commit to the polynomial.

Construction 4.44 (Feldman verifiable secret sharing). Let $(\mathbb{G}, q, G)$ be as in §4.6, so that $\mathbb{F}_q$ is the scalar field of $\mathbb{G}$. The dealer shares s as in Construction 4.42, with $f(X) = \sum_{k=0}^{t-1} a_k X^k$ and $a_0 = s$, and additionally *broadcasts* the coefficient commitments

$$\Phi_k := [a_k]G \in \mathbb{G}, \quad k = 0, \dots, t-1,$$

the coefficients in the exponent. Party i , on receiving its private share s_i , accepts it iff

$$[s_i]G \stackrel{?}{=} \sum_{k=0}^{t-1} [i^k] \Phi_k.$$

An honest share passes: since $a \mapsto [a]G$ is a homomorphism $\mathbb{F}_q \rightarrow \mathbb{G}$, $\sum_k [i^k] \Phi_k = \sum_k [a_k i^k]G = [f(i)]G$.

Proposition 4.45 (What the Feldman check certifies). *Because G generates the prime-order group $\mathbb{G}$, the map $a \mapsto [a]G$ is a bijection $\mathbb{F}_q \rightarrow \mathbb{G}$, so the broadcast vector $(\Phi_0, \dots, \Phi_{t-1})$ —however the dealer chose it—determines a unique polynomial $f^*(X) := \sum_k a_k^* X^k$ with $a_k^* = \log_G \Phi_k$, of degree at most $t - 1$. Party i 's check passes iff $s_i = f^*(i)$. Consequently every coalition of t parties holding accepted shares reconstructs the same polynomial f^* and the same secret $s^* = f^*(0) = \log_G \Phi_0$, and a party handed an inconsistent share detects it alone and with certainty. The check certifies nothing about the value of s^* , which the dealer chooses freely.*

Proof. Both claims are injectivity of $a \mapsto [a]G$. The right-hand side of the check equals $[f^*(i)]G$ by the homomorphism, so the equation $[s_i]G = [f^*(i)]G$ holds iff $s_i = f^*(i)$. Accepted shares are thus evaluations of the one polynomial f^* , and t of them interpolate to it by uniqueness. $\square$

In the language of this section, the vector $(\Phi_k)_k$ is a *perfectly binding* commitment to the polynomial f —the coefficient list committed entry by entry under the bijection $a \mapsto [a]G$ —and the share check is an evaluation check against it, verifiable by anyone because it is linear in the coefficients. Perfect binding costs hiding, as Theorem 4.8 demands.

Remark 4.46 (The leak, and where it is harmless). Feldman sharing publishes $\Phi_0 = [s]G$. Secrecy below the threshold is no longer perfect: from the broadcast alone an unbounded adversary computes $s = \log_G \Phi_0$. For an efficient coalition of $t - 1$ parties the leak is exactly $[s]G$ and nothing more: its entire view—shares and broadcast—is computable from $[s]G$ and $t - 1$ uniformly random field elements, because the shares are uniform by Theorem 4.43 and the Φ_k are then determined by interpolating in the exponent through $(0, [s]G)$ and the points $(i, [s_i]G)$. Recovering s from that view is the discrete-logarithm problem (Definition 2.2). The leak is acceptable exactly when s is itself a discrete-logarithm secret key whose public key $[s]G$ is published anyway; then Φ_0 is the public key and nothing new escapes. When the secret must stay perfectly hidden, Pedersen verifiable secret sharing replaces each Φ_k by the Pedersen commitment $[a_k]G + [b_k]H$ of Construction 4.16, restoring perfect hiding at the price of computational binding—off this volume's path and not constructed here. Threshold signing is the consumer: in the *FROST Guide* each signer holds a Shamir share of a signing key, the coefficient commitments broadcast during key generation are Feldman's, and the Lagrange weights $\lambda_{i,i}$ recombine per-signer contributions into a single signature.

5 Pseudorandom functions, block ciphers, and format-preserving encryption

The adversary of this section sits before a black box. She may submit any input she likes and read off the answer; she may let each query depend on everything the box has said so far; and when she has asked all she cares to ask, she must announce a single bit: does a short secret key drive the box, or blind chance? If no efficient strategy tells the difference, then λ bits of key have bought a perfect counterfeit of an object too large to exist in the world—a uniformly random function—and everything downstream that needs random-looking values on demand can draw them from that key. If she *can* tell, forgery follows at once. Key a Merkle–Damgård hash by naive concatenation and she extends an authentication tag she was shown onto a message nobody authenticated, without ever learning the key. Derive the diversifiers behind Orchard's shielded addresses by any map she can predict or invert, and she links every address of one wallet to its siblings. One adversary—the *oracle distinguisher*—thus carries the whole section, and every definition in it is the same distinguishing game at a different strength: pseudorandom functions against random functions, pseudorandom permutations against random permutations, strong pseudorandom permutations against a two-sided oracle. The switching lemma prices the substitution of one ideal object for the other; AES supplies the deployed permutation; keyed

BLAKE2 supplies the PRFs this section inventories; and format-preserving encryption closes the section by manufacturing, from a PRF that is not invertible, the keyed bijection on an 11-byte format that Orchard’s address derivation demands.

Notation is that of the provable-security model. Strings of length ℓ form $\{0,1\}^\ell$, all finite strings form $\{0,1\}^*$, and $|x|$ is the length of x ; for a finite set S , the notation $x \xleftarrow{\$} S$ draws x uniformly from S , independently of everything else. Adversaries are PPT machines in the security parameter λ (§1.1), a negligible function is $\text{negl}(\lambda)$ (recalled in §1.2), and every security notion below is a distinguishing game in the sense of Definition 1.15, scored by the difference

$$\text{Adv}(A) = |\Pr[A \text{ outputs } 1 \text{ in } \text{Exp}_1] - \Pr[A \text{ outputs } 1 \text{ in } \text{Exp}_0]|$$

between the two experiments the game distinguishes. One device remains to be formalised: the black box itself.

Definition 5.1 (Oracle access). For a function f , the notation A^f denotes an adversary granted *oracle access* to f : a black box to which A may submit inputs x of its choice, receiving $f(x)$ in a single computation step, and from which it learns nothing about f beyond the answers to the queries it makes. The adversary’s running time always bounds the number of its oracle queries, which is hence polynomial in λ .

5.1 Pseudorandom functions and permutations

Pseudorandomness is measured against an ideal object, which must come first: the “truly random function” has to be made precise before anything can be said to imitate it.

Definition 5.2 (Random function). Let $\text{Func}(D, R)$ denote the set of all functions from a finite set D to a finite set R . A *random function* from D to R is an element $F \xleftarrow{\$} \text{Func}(D, R)$, chosen uniformly at random. Equivalently, the values $\{F(x)\}_{x \in D}$ are independent and uniformly distributed over R .

There are $|R|^{|D|}$ functions in $\text{Func}(D, R)$, an astronomically large set for the domains of interest: for $D = R = \{0,1\}^{128}$ the count is $2^{128 \cdot 2^{128}}$. No such function can be stored or transmitted—its truth table is the only description it has, and the table does not fit in the universe. The essential property of a pseudorandom function is that it replaces this object by a short key while preserving its observable behaviour. The clean mental model of the ideal object is *lazy sampling*, exactly as for the random oracle of Definition 3.13: the oracle keeps a table, initially empty; on a fresh query x it samples $F(x) \xleftarrow{\$} R$, records the pair, and returns it; on a repeated query it returns the recorded value. This produces exactly the distribution of Definition 5.2 while only ever materialising the points actually queried.

Definition 5.3 (Keyed function). A *keyed function* is a deterministic, efficiently computable map

$$F: \{0,1\}^\lambda \times \{0,1\}^{\ell(\lambda)} \rightarrow \{0,1\}^{m(\lambda)}, \quad (k, x) \mapsto F_k(x),$$

where the input length ℓ and output length m are polynomially bounded functions of the key length λ . Fixing the key k yields the function $F_k: \{0,1\}^{\ell(\lambda)} \rightarrow \{0,1\}^{m(\lambda)}$.

Definition 5.4 (Pseudorandom function). Let F be a keyed function. The distinguishing game $\text{PRF}_F^A(\lambda)$ runs as follows.

1. The challenger samples a bit $b \xleftarrow{\$} \{0, 1\}$. If $b = 1$ it samples $k \xleftarrow{\$} \{0, 1\}^\lambda$ and sets $O := F_k$; if $b = 0$ it sets $O := R$ for $R \xleftarrow{\$} \text{Func}(\ell(\lambda), m(\lambda))$, realised by lazy sampling, where $\text{Func}(\ell, m) := \text{Func}(\{0, 1\}^\ell, \{0, 1\}^m)$.
2. The adversary $A(1^\lambda)$, receiving the security parameter in unary to convey the parameter size, queries O adaptively.
3. The adversary outputs a bit b' ; the game outputs 1 iff $b' = b$.

The family F is a *pseudorandom function* (PRF) if for every PPT adversary A ,

$$\text{Adv}_F^{\text{prf}}(A, \lambda) := \left| \Pr_{k \xleftarrow{\$} \{0, 1\}^\lambda} [A^{F_k}(1^\lambda) = 1] - \Pr_{R \xleftarrow{\$} \text{Func}(\ell(\lambda), m(\lambda))} [A^R(1^\lambda) = 1] \right| \in \text{negl}(\lambda).$$

The two probabilities are the adversary's output distributions in the worlds $b = 1$ and $b = 0$, so this difference equals twice the bit-guessing advantage $|\Pr[\text{PRF}_F^A(\lambda) = 1] - \frac{1}{2}|$ of Definition 1.15; the factor of two is immaterial to negligibility, and the difference form is the one every bound below is stated in.

Stated informally: an adversary handed a black box that is either F_k for a secret random key or a freshly sampled truly random function, and allowed to query it adaptively, cannot tell which it holds. Two features of the definition carry the weight. First, the algorithm defining F is public; only the sampled key is hidden. The adversary obtains oracle access to the resulting keyed instance F_k through Definition 5.1—never the key, only its input–output behaviour. Second, the comparison object is the gigantic, unstorable random function; the PRF compresses it into λ bits while remaining indistinguishable to every efficient observer.

That last qualifier is not removable: pseudorandomness is an inherently computational notion. Information theory cannot afford the trade. As k ranges over its 2^λ values, the keyed function F_k takes at most 2^λ distinct values in $\text{Func}(D, R)$ —a vanishing fraction of all $|R|^{|D|}$ functions—so the two distributions of Definition 5.4 are statistically almost as far apart as distributions can be, and an *unbounded* adversary could in principle distinguish them. Security holds only against computationally bounded distinguishers, in precisely the manner of the PRG separation in the proof of Proposition 1.10.

Remark 5.5 (Necessity of adaptivity). The adversary chooses its queries; it need not commit to them in advance, and later queries may depend on earlier answers. This *adaptive* access is the strongest reasonable model and the one downstream constructions rely on. A weaker non-adaptive definition, in which the adversary fixes all queries before seeing any answer, is genuinely weaker—families exist that pass every non-adaptive test yet fall to an adaptive distinguisher—and is insufficient for many applications.

A block cipher is not merely a keyed function but a keyed *permutation*: each key must yield an invertible map, or decryption would be impossible. The relevant ideal object is therefore a random permutation.

Definition 5.6 (Keyed permutation). A keyed function $F: \{0, 1\}^\lambda \times \{0, 1\}^\ell \rightarrow \{0, 1\}^\ell$ with equal input and output length is a *keyed permutation* if for every key k the map F_k is a bijection of $\{0, 1\}^\ell$, and both F_k and F_k^{-1} are efficiently computable given k . The set of all permutations of $\{0, 1\}^\ell$ is $\text{Perm}(\ell)$, of cardinality $(2^\ell)!$.

Definition 5.7 (Pseudorandom permutation). Let F be a keyed permutation. The game $\text{PRP}_F^\lambda(\lambda)$ is the game of Definition 5.4 with the ideal world replaced: for $b = 0$ the challenger sets $O := P$ for $P \xleftarrow{\$} \text{Perm}(\ell)$, a uniformly random permutation. The family F is a *pseudorandom permutation* (PRP) if for every PPT adversary A ,

$$\text{Adv}_F^{\text{PRP}}(A, \lambda) := \left| \Pr_k[A^{F_k}(1^\lambda) = 1] - \Pr_{P \xleftarrow{\$} \text{Perm}(\ell)}[A^P(1^\lambda) = 1] \right| \in \text{negl}(\lambda).$$

Definition 5.8 (Strong pseudorandom permutation). A keyed permutation F is a *strong pseudorandom permutation* (SPRP) if no PPT adversary distinguishes even when granted oracle access to *both* the permutation and its inverse: for every PPT A ,

$$\text{Adv}_F^{\text{SPRP}}(A, \lambda) := \left| \Pr_k[A^{F_k, F_k^{-1}}(1^\lambda) = 1] - \Pr_{P \xleftarrow{\$} \text{Perm}(\ell)}[A^{P, P^{-1}}(1^\lambda) = 1] \right| \in \text{negl}(\lambda).$$

The distinction between a PRP and an SPRP is exactly the availability of a decryption oracle. A block cipher deployed where the attacker can obtain decryptions of chosen ciphertexts must be an SPRP; this is the natural security target for a block cipher, and the one the next subsection adopts.

From the outside, the two ideal objects are nearly the same thing. Query a length-preserving oracle on q distinct inputs. If the oracle is a random function $R \in \text{Func}(\ell, \ell)$, the answers are q independent uniform strings, and output collisions occur. If the oracle is a random permutation $P \in \text{Perm}(\ell)$, the answers are q *distinct* uniform strings, and no collision can occur. This is the *only* statistical difference visible to any observer, and it is the entire content of the switching lemma.

Lemma 5.9 (PRP/PRF switching lemma). *Let $\ell \geq 1$ and let A be any adversary—even a computationally unbounded one—making at most q queries to its oracle. Then*

$$\left| \Pr_{P \xleftarrow{\$} \text{Perm}(\ell)}[A^P = 1] - \Pr_{R \xleftarrow{\$} \text{Func}(\ell, \ell)}[A^R = 1] \right| \leq \frac{q(q-1)}{2^{\ell+1}}.$$

We use the lemma as a statement only. Its proof is a standard game-hopping argument (Remark 1.18) bounding the probability of the single distinguishing event just identified—an output collision among the q answers—and the bound is the birthday-type count $\binom{q}{2}/2^\ell$ familiar from Lemma 3.11, whose underlying calculation is the *Math Guide*’s (§“The union bound and a birthday calculation”).

The consequence is a licence: a secure PRP on a large domain is also a secure PRF whenever the number of queries stays well below the birthday bound $2^{\ell/2}$, because the adversary’s total advantage against the function world exceeds its advantage against the permutation world by at most $q(q-1)/2^{\ell+1}$. For AES, with $\ell = 128$, the switching term $q^2/2^{129}$ is negligible for any realistic query budget. This is the licence by which the FF1 construction of §5.4 treats AES—a permutation—as the PRF its round function requires.

5.2 Block ciphers and AES

Definition 5.10 (Block cipher). A *block cipher* with block length b and key length κ is a keyed permutation

$$E: \{0, 1\}^\kappa \times \{0, 1\}^b \rightarrow \{0, 1\}^b,$$

together with its inverse $D = E^{-1}$ satisfying $D_k(E_k(x)) = x$ for all k, x . Here E_k is *encryption* and D_k *decryption* under key k . The security goal is that E be a strong pseudorandom permutation (Definition 5.8).

A block cipher is a concrete, fast, fixed-parameter instantiation of the SPRP abstraction, and the fixing of parameters changes the language of its security claims. The asymptotic PRP definition speaks of a family indexed by λ ; a real block cipher fixes b and κ once and for all— $b = 128$ and $\kappa \in \{128, 192, 256\}$ for AES—and offers *concrete* security in the resource-explicit style of Definition 1.31: an explicit function of the adversary’s resources bounds the advantage of the best known attacks.

Definition 5.11 (Concrete (t, q, ε) -security). A block cipher E is a (t, q, ε) -secure PRP if every adversary running in time at most t and making at most q oracle queries has $\text{Adv}_E^{\text{PRP}}(A) \leq \varepsilon$; analogously for SPRP and PRF security.

Construction 5.12 (AES). The Rijndael cipher of Daemen and Rijmen, standardised as AES in FIPS 197, has block length $b = 128$ bits and key lengths $\kappa \in \{128, 192, 256\}$, with $N_r \in \{10, 12, 14\}$ rounds respectively. Structurally it is a *substitution-permutation network* (SPN): the 128-bit state, viewed as a 4×4 array of bytes—elements of $\mathbb{F}_{2^8}$ —is first whitened by an initial round-key XOR, then passes through N_r rounds built from four layers:

1. **SubBytes**, a nonlinear byte substitution—the *S-box*—built on field inversion $a \mapsto a^{-1}$ in $\mathbb{F}_{2^8}$ (composed with a fixed affine map), chosen so that every input difference maps to any given output difference with only small probability, and every linear approximation of it holds with probability close to one half;
2. **ShiftRows**, a transposition of bytes within the array;
3. **MixColumns**, an invertible column-wise linear mix; and
4. **AddRoundKey**, the XOR of a 128-bit round key that a key schedule expands from k .

The first $N_r - 1$ rounds compose all four layers in this order; the final round omits MixColumns. The two permutation layers together guarantee that after two rounds every output byte depends on every input byte. Each layer—the initial whitening included—is individually invertible, so E_k is a permutation as Definition 5.10 requires, and decryption applies the inverses in reverse order.

Remark 5.13 (Confusion, diffusion, and the status of AES security). The SPN design is the modern embodiment of Shannon’s two principles. The nonlinear S-box supplies *confusion*: making the relationship between key and ciphertext as complex as possible. The ShiftRows and MixColumns layers supply *diffusion*: spreading the influence of each input bit over the whole output. Iterating the two for ten or more rounds is what gives AES its empirical resistance to all known attacks. No proof that AES is a PRP exists; like every practical block cipher, its security is a *conjecture*, corroborated by decades of failed cryptanalysis but resting on no reduction to a cleaner problem. Every theorem downstream therefore takes “ E is a PRP/SPRP” as an assumption and reasons from it—the same epistemic footing on which Remark 1.29 placed the concrete hash functions.

5.3 PRFs from hash functions: length extension and keyed BLAKE2

Block ciphers are one source of PRFs; cryptographic hash functions are another, often more convenient one, being unkeyed and ubiquitous. Here the oracle distinguisher does not merely

distinguish—she forges. The naive keying of a hash H by prepending the key,

$$F_k(m) = H(k \parallel m),$$

is unsafe for the iterated Merkle–Damgård hashes (MD5, SHA-1, SHA-256)—which pad the message into blocks, fold the blocks one at a time into a fixed-width chaining state through a compression function, and emit the final state—because of the *length-extension attack*: from $H(k \parallel m)$ and $|m|$ alone, the adversary computes $H(k \parallel m \parallel \text{pad} \parallel m')$ for a suffix m' of her choosing, *without knowing* k . The reason is structural: a Merkle–Damgård digest is exactly the internal chaining state after absorbing $k \parallel m$, so the published output is a resumable snapshot of the iteration, and the attacker simply continues absorbing blocks from where the honest computation stopped. Used as a message authentication code (MAC)—a keyed function whose output tag certifies a message, defined under “Message authentication codes” in the next section—the naive construction thus surrenders a valid tag on $m \parallel \text{pad} \parallel m'$, a message the key holder never authenticated.

The classical repair is HMAC, the nested double call

$$\text{HMAC}_k(m) = H((k \oplus \text{opad}) \parallel H((k \oplus \text{ipad}) \parallel m))$$

for two fixed pad constants: the inner hash absorbs the message under one key-derived value, and the outer hash re-keys the fixed-length result, so the emitted digest is no longer a resumable chaining state. Bellare’s analysis shows that HMAC is a PRF whenever the underlying compression function is, up to a birthday term $q^2/2^\ell$ in the chaining-state length ℓ ; as a PRF it is in particular a secure message authentication code, and it is the building block of the standardised key-derivation function HKDF, noted in passing under “Key-derivation functions” in the next section. This summary is all we need, because the shielded protocol uses HMAC nowhere: its hash of choice admits direct keying.

Modern hash functions of the sponge family (Construction 3.26) and of the HAIFA family—the Merkle–Damgård variant whose compression function also takes the count of bits hashed so far, the lineage of BLAKE2—do not suffer length extension and so admit a far simpler keying. The BLAKE2 design is built to be keyed *natively*: the construction loads the key length into its parameter block and prepends the key, padded to a full block, as the first message block, and its compression function mixes in a block counter and finalisation flags. (Its successor BLAKE3 is natively keyed too, by a different mechanism: the key words replace the initial state, and a dedicated flag marks the keyed mode.) One simply writes

$$F_k(m) = \text{BLAKE2}(m; \text{key} = k)$$

and obtains a PRF directly—no nested double call. No length-extension attack remains to repair: the finalisation flag, XORed into the state for the last block, means the emitted digest is *not* the raw chaining value and cannot be used to resume the computation.

Remark 5.14 (Zcash’s use of keyed and personalised BLAKE2). The Zcash protocol uses BLAKE2b and BLAKE2s pervasively, exploiting two parameters of the BLAKE2 design: the *key*, for PRF and MAC use as above, and the *personalisation* string, a domain-separation tag baked into the initial state—16 bytes for BLAKE2b, 8 bytes for BLAKE2s. Personalisation makes the same input hashed for two purposes—nullifier derivation versus note-commitment derivation, say—yield independent-looking outputs, so each use site is effectively an independent PRF: the practical realisation of the independent-oracle-per-context idiom that Proposition 3.17 proved for tagged random oracles, obtained here before a single message byte is processed and cheaper and cleaner than HMAC, because BLAKE2 provides keying and domain separation as first-class features. The deployed inventory spans both widths. On the BLAKE2b side, the key-expansion PRF $\text{PRF}^{\text{expand}}$ is BLAKE2b-512 personalised with `Zcash_ExpandSeed` (protocol specification §“Pseudo Random Functions”), evaluated on the spending key to derive Orchard’s key tree; the note-encryption KDF is personalised with `Zcash_OrchardKDF` (§“Orchard Key Derivation”)

and the outgoing cipher key with `Zcash_Orchardock` (§“Pseudo Random Functions”)—all 16-byte tags. On the BLAKE2s side, the nullifier PRF and fixed-base generators of Sapling—the shielded protocol preceding Orchard—use 8-byte personalisations such as `Zcash_nf` and `Zcash_PH` (§“Pseudo Random Functions” and §“Pedersen Hash Function”).

5.4 Format-preserving encryption and FF1

All constructions so far operate on bit strings of a length the cipher’s block size dictates. The deployed need that closes this section does not. Orchard derives the *diversifiers* behind its shielded addresses—one 11-byte value per address, many addresses per spending key—by encrypting a sequential address index with a keyed bijection on the set of 11-byte strings, so that the published diversifiers of one wallet look like unrelated random values rather than the consecutive integers they conceal. An adversary who could invert the map, or merely distinguish it from a random permutation of the format, would link a wallet’s addresses to one another. What is needed is precisely a block cipher whose “block” is an exotic finite set; the classical commercial need—encrypting a 16-digit card number into a 16-digit card number so that legacy databases, field-length constraints, and checksum formats continue to accept it—is the same problem. This is the problem *format-preserving encryption* (FPE) solves.

Definition 5.15 (Format-preserving encryption). Let $\mathcal{X}$ be a finite set, the *format*—for example $\mathcal{X} = \{0, 1, \dots, 9\}^{16}$, the 16-digit strings. A *format-preserving encryption scheme* on $\mathcal{X}$ is a keyed permutation $\mathcal{E}: \{0, 1\}^k \times \mathcal{X} \rightarrow \mathcal{X}$ such that for every key k the map $\mathcal{E}_k: \mathcal{X} \rightarrow \mathcal{X}$ is a bijection with efficiently computable inverse $\mathcal{D}_k$. Security requires $\mathcal{E}$ to be a (strong) pseudorandom permutation on $\mathcal{X}$ —Definition 5.7 with $\text{Perm}(\mathcal{X})$ in place of $\text{Perm}(\{0, 1\}^\ell)$: no efficient adversary distinguishes $\mathcal{E}_k$ from a uniformly random permutation of $\mathcal{X}$, even with access to the inverse.

The defining feature is that the domain $\mathcal{X}$ is *arbitrary*—in particular $|\mathcal{X}|$ need not be a power of two, so a plain block cipher does not apply. What is required is a keyed pseudorandom *bijection* on a set whose size is, say, 10^{16} . Determinism is forced, and it separates FPE from the randomised encryption modes of the next section: the format $\mathcal{X}$ has no room to store a per-message random value, so FPE is the deterministic, format-preserving analogue of a block cipher on the alphabet of $\mathcal{X}$. It necessarily leaks equality of plaintexts, as any deterministic cipher must; this is the unavoidable cost of preserving format with no expansion, and it is acceptable for the tokenisation use cases FPE targets—and immaterial for diversifier derivation, where each index is encrypted once.

Formats reduce to integers. Any format of the shape “strings of length ν over an alphabet of radix ρ ” is in bijection with $\{0, 1, \dots, \rho^\nu - 1\}$ by reading the string as a base- ρ numeral, so it suffices to build a keyed pseudorandom permutation of $\mathbb{Z}/M\mathbb{Z}$ for an arbitrary modulus $M = \rho^\nu$. The FF1 construction below splits $M = a \cdot b$ into two factors of nearly equal size and runs a *Feistel network* whose two halves live in $\mathbb{Z}/a\mathbb{Z}$ and $\mathbb{Z}/b\mathbb{Z}$. The Feistel network is the structural device that turns a round function—not necessarily invertible—into a permutation; we present the classical bitwise case first, then the modular generalisation FF1 uses.

Definition 5.16 (Balanced Feistel network). Let $F_1, \dots, F_r: \{0, 1\}^w \rightarrow \{0, 1\}^w$ be arbitrary *round functions*. The r -round Feistel network $\Psi[F_1, \dots, F_r]$ is the map on $\{0, 1\}^{2w}$ defined as follows: write the input as (L_0, R_0) with $L_0, R_0 \in \{0, 1\}^w$, iterate

$$L_i = R_{i-1}, \quad R_i = L_{i-1} \oplus F_i(R_{i-1}), \quad i = 1, \dots, r,$$

and output (L_r, R_r) .

Proposition 5.17 (Feistel is a bijection regardless of the round functions). *For any round functions $F_1, \dots, F_r$ —injective or not—the map $\Psi[F_1, \dots, F_r]$ is a bijection of $\{0, 1\}^{2w}$, inverted by running the rounds backwards:*

$$R_{i-1} = L_i, \quad L_{i-1} = R_i \oplus F_i(L_i), \quad i = r, \dots, 1.$$

Proof. It suffices to show each round is invertible, the composition of bijections being a bijection. The i -th round sends $(L_{i-1}, R_{i-1}) \mapsto (R_{i-1}, L_{i-1} \oplus F_i(R_{i-1}))$. Given the output (L_i, R_i) , one recovers $R_{i-1} = L_i$ directly, and then $L_{i-1} = R_i \oplus F_i(R_{i-1}) = R_i \oplus F_i(L_i)$, using that XOR is its own inverse. The recovery uses only *forward* evaluations of F_i —at no point is an inverse of the round function required, which is the crucial point permitting a permutation to be built out of a one-way-ish PRF. The displayed backward recursion is exactly this inversion applied round by round. $\square$

This proposition is the structural heart of format-preserving encryption: it manufactures a *keyed bijection* from a keyed function that is not itself a bijection. In FF1 the round functions derive from a PRF—AES iterated in cipher-block-chaining fashion, as the construction below specifies—so the network is invertible without ever inverting the PRF: AES is only ever evaluated in the forward direction, in encryption and decryption alike.

Bijectivity is free; pseudorandomness is not. The Luby–Rackoff theorem is the classical statement that enough Feistel rounds over a PRF yield a pseudorandom permutation.

Theorem 5.18 (Luby–Rackoff). *Let the round functions $F_1, \dots, F_r$ be independent PRFs on $\{0, 1\}^w$. Against an adversary making q queries, the Feistel network Ψ is:*

1. *with $r = 3$ rounds, a pseudorandom permutation (forward queries only); and*
2. *with $r = 4$ rounds, a strong pseudorandom permutation (forward and inverse queries);*

in each case with distinguishing advantage $O(q^2/2^w)$ above the PRF advantage of the round functions.

We use the theorem as a statement, without proof, but the shape of the argument is worth recording, because it explains both the birthday term and the round counts. One first replaces each PRF round function by a truly random function, at the cost of r PRF advantage terms. For three rounds one then shows that, unless two of the at most q queries collide in an intermediate Feistel value—a birthday event of probability $O(q^2/2^w)$ —the middle round’s random function is evaluated at distinct inputs, so it produces independent uniform values that perfectly mask the relationship between input and output halves, and the network’s answers are distributed exactly as a random *function*’s—*independent and uniform on $\{0, 1\}^{2w}$* . A final application of the switching lemma (Lemma 5.9), at a further cost of $q(q-1)/2^{2w+1}$, carries the comparison from the random function to the random permutation; the term is absorbed by the same $O(q^2/2^w)$ bound. The fourth round adds the symmetry needed to resist inverse queries as well, upgrading PRP to SPRP. The full argument is a transcript analysis bounding the probability of any internal collision; the leftover term is the stated birthday bound.

Remark 5.19 (Why FF1 uses ten rounds rather than four). Theorem 5.18 gives security only up to $q \approx 2^{w/2}$ queries, and in the FPE setting w is tiny—one half of a 16-digit number—so the birthday bound $q^2/2^w$ is weak. The standardised FF1 therefore prescribes *ten* rounds rather than the minimal four, as a security margin against the more powerful attacks known for small-domain Feistel networks, which can sometimes beat the generic birthday bound. The classification of this design claim is worth making explicit: the round count itself is *specified*—NIST SP 800-38G fixes ten rounds—while a security proof matching the margin is an *open problem*; the analysis that exists (Luby–Rackoff and its small-domain refinements) does not certify the full strength the margin

is hoped to buy. What the round count does *not* affect is invertibility, which Proposition 5.17 guarantees for any round count and any round function.

Construction 5.20 (The FF1 mode of NIST SP 800-38G). The FF1 mode instantiates an *unbalanced, modular, tweaked* Feistel network of ten rounds to build a PRP on $\mathbb{Z}/M\mathbb{Z}$ with $M = \rho^v$, the radix- ρ numeral reading of a length- v string. It supports a *tweak* T : a non-secret string that selects, in effect, one permutation from a family indexed by T , so that the same key encrypts the same plaintext differently under different tweaks—useful for binding a token to its context. Its operation:

1. **Split.** Set $u = \lfloor v/2 \rfloor$ and $v = v - u$. Split the plaintext numeral string X into a left part A of u symbols and a right part B of v symbols, with integer values in $\{0, \dots, \rho^u - 1\}$ and $\{0, \dots, \rho^v - 1\}$ respectively. The two halves live in the moduli $a = \rho^u$ and $b = \rho^v$; the network is unbalanced when v is odd.
2. **Round function from a PRF.** For each round $i = 0, 1, \dots, 9$, assemble a byte string Q encoding the tweak T , the round index i , and the current right half B ; prepend a fixed parameter block P encoding the radix, the lengths, and the tweak length; and apply $\text{PRF}(P \parallel Q)$, where PRF is AES in CBC-MAC mode: set $Y_0 = 0^{128}$, iterate $Y_j = \text{AES}_k(Y_{j-1} \oplus X_j)$ over the 16-byte blocks X_j of $P \parallel Q$, and output the final block Y_m (SP 800-38G, Algorithm 6). The iteration chains AES through an evolving state exactly as the Merkle–Damgård iteration of §5.3 chains a compression function; emitting the final state is safe *here* because the PRF’s inputs form a *prefix-free* set: the leading block P encodes the lengths of everything that follows, so two distinct inputs either differ already in their first block or agree on all lengths, and in neither case is one a proper prefix of the other. Prefix-free inputs are precisely the setting in which plain CBC-MAC is proved to be a PRF (a result of Petrank and Rackoff), and no length-extension query can arise. Further AES calls then expand the result to a long enough digest, which is read as an integer y .
3. **Modular Feistel step.** Update the halves by the modular analogue of Definition 5.16:

$$C = (A + y) \bmod m, \quad A \leftarrow B, \quad B \leftarrow C,$$

where the modulus m alternates between $a = \rho^u$ and $b = \rho^v$ with the parity of the round, so that the sum lands in the correct half’s domain; modular addition replaces XOR.

4. **Output.** After ten rounds, concatenate the final A and B and render the result back as a length- v radix- ρ string.

Decryption runs the rounds in reverse, replacing modular addition by modular subtraction, $A \leftarrow (C - y) \bmod m$, exactly mirroring the backward recursion of Proposition 5.17. One quantitative detail of step 2 deserves note: the standard sizes the expanded digest at least four bytes beyond what the larger half requires, so the integer y ranges over at least 2^{32} multiples of the modulus, and the bias of the reduction $y \bmod m$ away from uniform is bounded by the modular-reduction bias of the *Math Guide* (§“Uniform sampling and the bias of modular reduction”) at below 2^{-33} per round.

Proposition 5.21 (FF1 is a keyed bijection on $\mathcal{X}$). *For every key k and tweak T , the FF1 map $\mathcal{E}_{k,T}: \mathcal{X} \rightarrow \mathcal{X}$ is a bijection of the format set $\mathcal{X}$, and $\mathcal{D}_{k,T}$ is its two-sided inverse.*

Proof. Each round is the map $(A, B) \mapsto (B, (A + y) \bmod m)$, where $y = y(i, T, B)$ depends only on the round index, the tweak, and the right half B —*not* on A . Fix B , i , and T ; then y is a constant

and $A \mapsto (A + y) \bmod m$ is a translation of $\mathbb{Z}/m\mathbb{Z}$, hence a bijection, inverted by subtracting the same constant. The full round is therefore invertible: from the output (B, C) recover B directly, recompute $y = y(i, T, B)$ by a *forward* PRF evaluation, and set $A = (C - y) \bmod m$. Each even-indexed round is a bijection of $\mathbb{Z}/a\mathbb{Z} \times \mathbb{Z}/b\mathbb{Z}$ onto $\mathbb{Z}/b\mathbb{Z} \times \mathbb{Z}/a\mathbb{Z}$ and each odd-indexed round a bijection back, the two products coinciding when ν is even; the composition of the ten rounds, an even number, is a bijection of $\mathbb{Z}/a\mathbb{Z} \times \mathbb{Z}/b\mathbb{Z}$. The numeral map $(A, B) \mapsto A \cdot b + B$ identifies $\mathbb{Z}/a\mathbb{Z} \times \mathbb{Z}/b\mathbb{Z}$ with $\mathbb{Z}/M\mathbb{Z}$ as a bijection of *sets only*, not of groups: the factors satisfy $\gcd(a, b) = \rho^{\min(u, v)} > 1$, so the Chinese remainder theorem does not apply, and only bijectivity is used. Transporting through this map and the numeral bijection $\mathcal{X} \cong \mathbb{Z}/M\mathbb{Z}$, the composition $\mathcal{E}_{k, T}$ is a bijection of $\mathcal{X}$. Running the rounds backward with subtraction inverts it on both sides, so $\mathcal{D}_{k, T} = \mathcal{E}_{k, T}^{-1}$. $\square$

Remark 5.22 (Bijectivity from algebra, security from the PRF). The construction separates its two obligations cleanly, and the separation is why Feistel is the standard route to FPE. Invertibility comes for free from the structure: Propositions 5.17 and 5.21 hold for *any* round function whatsoever, including a broken or constant one. The PRF—AES—is responsible not for invertibility but for *pseudorandomness*: making $\mathcal{E}_{k, T}$ look like a uniformly random permutation of $\mathcal{X}$ in the sense of Definition 5.15, per the Luby–Rackoff *heuristic*—Theorem 5.18 adapted, beyond what it literally proves, to the modular, multi-round, tweaked setting.

Remark 5.23 (The domain-size caveat, and the single Orchard use). Two facts complete the picture, one cautionary and one deployed.

The domain-size caveat is real. The published SP 800-38G requires $\rho^\nu \geq 100$ and recommends $\rho^\nu \geq 10^6$ —its draft Revision 1 would make 10^6 mandatory—because for very small domains an attacker can recover or distinguish the keyed permutation by exhaustively cataloguing input–output pairs, and known message-recovery attacks on small-domain FPE (Bellare–Hoang–Tessaro and successors) erode the security margin. For the moderately large domains FPE is meant for, and with its ten rounds, FF1 is the NIST-standardised, AES-backed realisation of a keyed pseudorandom bijection on an arbitrary finite format.

The single use of FF1 in Orchard is the diversifier derivation with which this subsection opened: FF1-AES256 maps an address index to the 11-byte diversifier embedded in a shielded address, so that one spending key yields many mutually unlinkable addresses. The deployed instance (protocol specification §“Pseudo Random Permutations”) fixes radix $\rho = 2$, length $\nu = 88$, and the empty tweak, and reads the 11-byte index as a little-endian bit string, so the domain is the 2^{88} strings of 88 bits—comfortably beyond the caveat above. The address built on the diversifier, and the case for a keyed permutation rather than a hash in this role, are the business of the *Ironwood Guide* (§“Diversified addresses”).

The toolbox has gained its keyed compartment: one distinguishing game at three strengths, a standard permutation conjectured to win it, a natively keyed hash that turns domain-separated digests into independent PRFs, and a Feistel construction that bends a PRF into a bijection on any finite format. The next section spends these tools on the channel itself: symmetric encryption, authentication, and the authenticated-encryption composition that carries every shielded note, each built by running a PRF in a mode that converts its indistinguishability into secrecy or integrity.

6 Symmetric encryption, AEAD, and key derivation

The adversary of this section owns the channel. Every ciphertext passes through her hands, and she attacks on two fronts at once. As an eavesdropper she tries to *distinguish*: to tell an encryption of one note plaintext from an encryption of another, or to recognise that two ciphertexts hide the same message. As a forger she tries to *modify*: a stream cipher’s ciphertext is XOR-malleable, so flipping a ciphertext bit flips the corresponding plaintext bit without any key material at all; she

can also shift bytes between a packet’s public header and its encrypted body, replay a ciphertext under a different context, or manufacture a wholly new ciphertext that the receiver accepts. She is patient enough to exploit a single reused pad—two ciphertexts under one pad hand her the XOR of the plaintexts—and mathematically literate enough to exploit a key that is merely high-entropy rather than uniform. This section arms the honest parties against all of it, one guarantee at a time: confidentiality from the one-time pad through the nonce-based stream cipher ChaCha20 and the chosen-plaintext game; integrity from message authentication codes through universal hashing and Carter–Wegman; their composition as authenticated encryption with associated data, culminating in the ChaCha20-Poly1305 scheme that Zcash’s note-encryption layer deploys; and finally key derivation, the manufacture of the uniform key that every theorem in between takes as its hypothesis.

Throughout, an *adversary* is a probabilistic algorithm A ; the notation $y \leftarrow A(x)$ denotes running A on input x and assigning its output to y , and $x \xleftarrow{\$} S$ denotes sampling x uniformly from the finite set S . Every adversary in the computational definitions is probabilistic polynomial time (PPT) in the security parameter λ , and $\text{negl}(\lambda)$ denotes a negligible function (*Math Guide*, §“Polynomial, exponential, and negligible functions”; recalled in §1.2).

The one-time pad and EAV security. The information-theoretic baseline is the *one-time pad*: to encrypt $m \in \{0,1\}^L$ under a uniformly random key $k \in \{0,1\}^L$ used exactly once, output $c = m \oplus k$. Since $m \oplus k$ is uniform whatever m is, the ciphertext distribution is independent of the message, and even an unbounded eavesdropper learns nothing—Shannon’s *perfect secrecy*. The price is severe on both counts the name advertises: the key must be as long as the message, and it must never be reused, for two ciphertexts under one pad reveal $c \oplus c' = m \oplus m'$ —the *two-time-pad* failure, which the adversary of the opening paragraph collects for free. The computational relaxation is *EAV security* (security against an eavesdropper): an adversary who chooses two equal-length messages m_0, m_1 and sees a single ciphertext $\text{Enc}_k(m_b)$ for a hidden uniform bit b guesses b with advantage at most $\text{negl}(\lambda)$ —the one-ciphertext, no-oracle restriction of the chosen-plaintext game formalised in §6.2. Replacing the truly random pad by the output of a pseudorandom generator stretched from a short seed—the *pseudo-one-time pad*—achieves EAV security with a fixed-length key, at the cost of resting on a pseudorandomness assumption.

6.1 Stream ciphers and ChaCha20

A *stream cipher* is the practical instantiation of the pseudo-one-time pad: a keyed, seekable generator producing a *keystream* that XORs into the plaintext. To encrypt many messages under one key without violating the one-time-pad no-reuse rule, modern stream ciphers take an additional public input called a *nonce* (number used once) and behave like a pseudorandom function of the nonce, in the sense of Section 5.

Definition 6.1 (Nonce-based stream cipher). A *nonce-based stream cipher* is a deterministic function

$$\text{KS} : \mathcal{K} \times \mathcal{N} \times \mathbb{N} \rightarrow \{0,1\}^*$$

producing, for key k , nonce ν , and length L , a keystream $\text{KS}(k, \nu, L) \in \{0,1\}^L$. Encryption of $m \in \{0,1\}^L$ is $\text{Enc}_k(\nu, m) = m \oplus \text{KS}(k, \nu, |m|)$ with ciphertext (ν, c) ; decryption recomputes the keystream and XORs it off. The security goal is that for distinct nonces the keystreams are jointly indistinguishable from independent uniform strings, provided no nonce repeats under a fixed key.

The stream cipher Zcash deploys—inside the AEAD of §6.5, in the note-encryption layer—is ChaCha20, designed by Bernstein and standardised in RFC 8439. Its core is a reversible mixing of four 32-bit words built only from modular addition, XOR, and rotation (an “ARX” design),

which avoids data-dependent table lookups and hence the timing side channels that table-based ciphers must engineer around.

Construction 6.2 (ChaCha20). The ChaCha20 state is a 4×4 matrix of 32-bit words, viewed as a vector $(x_0, \dots, x_{15}) \in (\mathbb{Z}/2^{32}\mathbb{Z})^{16}$; the symbol $+$ denotes addition modulo 2^{32} , $\oplus$ XOR, and $x \lll r$ left rotation by r bits. The *quarter-round* $QR(a, b, c, d)$ updates four words by

$$\begin{aligned} a &+= b; & d &\oplus= a; & d &\lll= 16; \\ c &+= d; & b &\oplus= c; & b &\lll= 12; \\ a &+= b; & d &\oplus= a; & d &\lll= 8; \\ c &+= d; & b &\oplus= c; & b &\lll= 7. \end{aligned}$$

The state is initialised from the 128-bit constant “expand 32-byte k” (words x_0 – x_3), a 256-bit key (x_4 – x_{11}), a 32-bit block counter (x_{12}), and a 96-bit nonce (x_{13} – x_{15}). One *double round* applies QR to the four columns and then to the four diagonals; the 20 rounds of ChaCha20 are ten double rounds. Writing X for the initial state and $R(X)$ for the state after the rounds, the 64-byte keystream block is the *feed-forward* sum $R(X) + X$ (word-wise, modulo 2^{32}), serialised in little-endian order.

Remark 6.3 (Feed-forward and counter mode). The rounds R form a bijection—each quarter-round is invertible, being a composition of invertible word operations—so without the final addition the block function would be an invertible permutation, and an adversary holding a keystream block could run the rounds backwards. The feed-forward sum $R(X) + X$ destroys invertibility; it is the standard Davies–Meyer-style move for turning a permutation into a one-way mixing function. Incrementing the block counter x_{12} produces successive 64-byte keystream blocks, so $KS(k, \nu, L)$ is the concatenation of the blocks for counters $0, 1, 2, \dots$; this *counter mode* is what makes the cipher seekable and parallelisable. The security status is a conjecture, in the same heuristic position as every concrete cipher of Section 5: ChaCha20 is *conjectured* to be a secure PRF of the pair (nonce, counter), and no attack better than brute force over the 256-bit key is known. Reusing a (k, ν) pair reproduces the same keystream and re-creates the two-time-pad failure; nonce uniqueness is a hard requirement, not a hygiene recommendation.

6.2 Chosen-plaintext security

EAV security models an adversary who sees a single ciphertext. The adversary of this section is stronger: she can influence which messages get encrypted—by triggering payments, injecting plaintext into a session, or simply guessing—and she observes many ciphertexts. The corresponding notion is security against *chosen-plaintext attack* (CPA), modelled by granting her oracle access to the encryption algorithm, in the game template of Definition 1.15.

Definition 6.4 (CPA security). For a symmetric encryption scheme Π with key generator Gen , adversary $\mathcal{A}$, and bit $b \in \{0, 1\}$, the game $\text{CPA}_{\Pi}^{\mathcal{A}}(\lambda, b)$ runs as follows.

1. The challenger samples $k \leftarrow \text{Gen}(1^\lambda)$ and gives $\mathcal{A}$ oracle access to $\text{Enc}_k(\cdot)$.
2. At some point $\mathcal{A}$ submits a challenge pair (m_0, m_1) of equal-length messages and receives $c^* \leftarrow \text{Enc}_k(m_b)$.
3. The adversary may continue querying the oracle and finally outputs a bit b' , which is the output of the game.

The scheme Π is CPA-secure if for every PPT $\mathcal{A}$,

$$\text{Adv}_{\Pi}^{\text{cpa}}(\mathcal{A}, \lambda) := |\Pr[\text{CPA}_{\Pi}^{\mathcal{A}}(\lambda, 0) = 1] - \Pr[\text{CPA}_{\Pi}^{\mathcal{A}}(\lambda, 1) = 1]| \in \text{negl}(\lambda).$$

The first consequence of giving the adversary an encryption oracle is a structural one: encryption cannot be a function of the message alone.

Proposition 6.5 (Determinism precludes CPA security). *No scheme whose Enc is deterministic and stateless is CPA-secure.*

Proof. The adversary queries the oracle on two distinct messages $m_0 \neq m_1$, recording $c_0 = \text{Enc}_k(m_0)$ and $c_1 = \text{Enc}_k(m_1)$. Decryption correctness forces $c_0 \neq c_1$: a single ciphertext cannot decrypt to both messages. She submits the challenge (m_0, m_1) , receives $c^* = \text{Enc}_k(m_b)$, and outputs 0 if $c^* = c_0$ and 1 otherwise. Since Enc is deterministic, $c^* = c_b$, and since $c_0 \neq c_1$ the comparison identifies b : the guess is always correct and the advantage is 1. $\square$

Consequently a CPA-secure scheme must be randomised or nonce-based. A nonce-based stream cipher used with a fresh nonce per encryption—random, or drawn from a counter—attains CPA security under the PRF conjecture for its keystream generator, because each encryption is then effectively a fresh one-time pad. We record the notion here and defer the routine PRF-based proof to the AEAD analysis (Theorem 6.20), where confidentiality and integrity are treated together.

Remark 6.6 (Semantic security). Goldwasser and Micali’s original notion, *semantic security*, demands that whatever an efficient adversary can compute about the plaintext from the ciphertext, she could have computed without the ciphertext, from the message length alone. Semantic security is provably equivalent to the indistinguishability notions above; we use indistinguishability throughout because it is far easier to manipulate in reductions. The content is the same: the ciphertext is computationally useless for learning anything about the message beyond its length.

6.3 Message authentication codes

CPA security protects confidentiality and says nothing about integrity, and for the XOR-based stream ciphers above the gap is not hypothetical: flipping a bit of the ciphertext flips the corresponding plaintext bit, so the channel-owning adversary can convert a payment of one amount into a payment of another without ever touching the key. Detecting tampering requires a second primitive, keyed like the cipher but aimed at the forger rather than the eavesdropper: the *message authentication code*. Here we build this symmetric, shared-key form of authentication, the form that the AEAD needs.

Definition 6.7 (Message authentication code). A *message authentication code* (MAC) is a pair $(\text{Mac}, \text{Vrfy})$ with key space $\mathcal{K}$: algorithm $\text{Mac}_k(m)$ outputs a tag t , and $\text{Vrfy}_k(m, t) \in \{0, 1\}$ accepts (1) or rejects (0). Correctness requires $\text{Vrfy}_k(m, \text{Mac}_k(m)) = 1$ for all k, m . When Mac is deterministic, canonical verification recomputes the tag and checks equality.

Definition 6.8 (Existential and strong unforgeability). For a MAC Π and adversary $\mathcal{A}$, the game $\text{Forge}_{\Pi}^{\mathcal{A}}(\lambda)$ runs as follows.

1. The challenger samples $k \xleftarrow{\$} \mathcal{K}$ and gives $\mathcal{A}$ oracle access to $\text{Mac}_k(\cdot)$; let Q be the set of messages queried.

2. The adversary outputs a pair (m^*, t^*) ; the game outputs 1 iff $\text{Vrfy}_k(m^*, t^*) = 1$ and $m^* \notin Q$.

The MAC is *existentially unforgeable under chosen-message attack* (EUF-CMA) if for every PPT A ,

$$\text{Adv}_{\Pi}^{\text{forge}}(A, \lambda) := \Pr[\text{Forge}_{\Pi}^A(\lambda) = 1] \in \text{negl}(\lambda).$$

A *strongly unforgeable* (sUF-CMA) MAC additionally forbids producing a new valid tag on an already-queried message: the adversary wins on (m^*, t^*) as long as the oracle never returned that exact *pair*. We write $\text{Adv}_{\Pi}^{\text{s-forge}}(A, \lambda)$ for the advantage in this variant.

The MAC Zcash uses, Poly1305, is not a pseudorandom-function MAC but a *one-time, universal-hash* MAC: it provides information-theoretic security per nonce and is extremely fast, at the cost of a key that authenticates only a single message. The enabling abstraction is a hash family whose *differences* are unpredictable.

Definition 6.9 (ε -almost- Δ -universal hash family). Let $(G, +)$ be a finite abelian group. A family of functions $\{h_r : \mathcal{X} \rightarrow G\}_{r \in \mathcal{R}}$ is ε -almost- Δ -universal (ε -A Δ U) if for all distinct $x, x' \in \mathcal{X}$ and every $\delta \in G$,

$$\Pr_{r \xleftarrow{\$} \mathcal{R}} [h_r(x) - h_r(x') = \delta] \leq \varepsilon.$$

Taking $\delta = 0$ recovers ordinary ε -universality—low collision probability—while the Δ condition controls *additive differences*, which is what makes the one-time MAC below secure. With $G = \{0, 1\}^n$ under XOR this is the classical almost-XOR-universality; Poly1305 takes $G = \mathbb{Z}/2^{128}\mathbb{Z}$ under addition modulo 2^{128} .

The Carter–Wegman paradigm turns any A Δ U family into a one-time MAC by masking the hash value with a fresh uniform pad.

Theorem 6.10 (Carter–Wegman one-time MAC). Let $\{h_r\}$ be ε -A Δ U into $(G, +)$. Define a MAC with key (r, s) , where $r \xleftarrow{\$} \mathcal{R}$ and the pad $s \xleftarrow{\$} G$, by $\text{Mac}_{r,s}(m) = h_r(m) + s$. If (r, s) authenticates at most one message, then any adversary—even a computationally unbounded one—who sees one message–tag pair (m, t) forges a valid pair (m^*, t^*) with $m^* \neq m$ with probability at most ε .

Proof. The adversary observes $t = h_r(m) + s$ and must output (m^*, t^*) with $m^* \neq m$ and $t^* = h_r(m^*) + s$. Subtracting the two tag equations eliminates the pad: the forgery succeeds iff

$$h_r(m^*) - h_r(m) = t^* - t.$$

Condition on the adversary’s view. The pad s is uniform and independent of r , so given the single observed tag t , every value of $h_r(m)$ is equally consistent—each candidate value determines a unique s —and the observation therefore leaks nothing about r : the posterior on r remains uniform on $\mathcal{R}$. The adversary’s difference $\delta := t^* - t$ and distinct pair (m, m^*) are thus fixed independently of r , and the A Δ U property gives $\Pr[h_r(m^*) - h_r(m) = \delta] \leq \varepsilon$. $\square$

The MAC Poly1305 instantiates the paradigm with a polynomial-evaluation hash over the prime field $\mathbb{F}_p$ for $p = 2^{130} - 5$ (prime-field arithmetic as in the *Math Guide*, §“The prime field and two inversion routes”).

Construction 6.11 (Poly1305). Fix the prime $p = 2^{130} - 5$ and work in $\mathbb{F}_p$. The key is a pair (r, s) : r is a 128-bit value subjected to a fixed bit-clamping (the top four bits of bytes 3, 7, 11, 15 and

the bottom two bits of bytes 4, 8, 12 are forced to zero), and s is a 128-bit pad. To authenticate a message, split it into ℓ blocks $c_1, \dots, c_\ell$ of at most 16 bytes; encode each block as an integer in $[0, 2^{128})$ and add 2^{8b} , where b is the block's byte length—this sets a 1 bit just above the block's bytes, making every coefficient nonzero and the block encoding injective in both content and byte length. Interpreting r and each c_i as elements of $\mathbb{F}_p$, the hash is the polynomial evaluation

$$h_r(m) = \sum_{i=1}^{\ell} c_i r^{\ell-i+1} \in \mathbb{F}_p,$$

computed by Horner's rule as $(\dots((c_1 r + c_2) r + c_3) r \dots) r$. The tag is

$$\text{Mac}_{r,s}(m) = ((h_r(m) \bmod p) + s) \bmod 2^{128},$$

the field hash reduced and added to the pad modulo 2^{128} .

Proposition 6.12 (Poly1305 is almost- Δ -universal). *Restricted to messages of at most L bytes—hence at most $\ell_{\max} = \lceil L/16 \rceil$ blocks—the reduced Poly1305 hash family $\{m \mapsto h_r(m) \bmod p\}$, read into $\mathbb{Z}/2^{128}\mathbb{Z}$, is ε -almost- Δ -universal with respect to addition modulo 2^{128} , with*

$$\varepsilon \leq \frac{8 \ell_{\max}}{2^{106}}.$$

The loss over the ideal $\ell_{\max} \cdot p^{-1}$ accounts for the clamping of r —which restricts r to a subset of size 2^{106} —and for the final reduction modulo 2^{128} .

Proof. Sketch. Fix two distinct messages m, m' with block encodings (c_i) and (c'_i) , padded to a common length $\ell \leq \ell_{\max}$ by leading zero coefficients without changing their values. The block encoding guarantees that $m \neq m'$ yields distinct coefficient vectors: genuine coefficients are nonzero, so differing block counts set a nonzero coefficient against a zero padding one, while equal block counts leave two blocks differing in content or in final-block byte length, hence in encoding. The difference of hashes is

$$h_r(m) - h_r(m') = \sum_{i=1}^{\ell} (c_i - c'_i) r^{\ell-i+1} =: P(r) \in \mathbb{F}_p,$$

a nonzero polynomial in r of degree at most ℓ . Now count the ways a target difference $\delta \in \mathbb{Z}/2^{128}\mathbb{Z}$ can be hit. The reduced hash values lie in $[0, p)$, so their integer difference lies in the interval $(-p, p)$, which has length $2p < 8 \cdot 2^{128}$; each residue class modulo 2^{128} therefore contains at most $\lceil 2p/2^{128} \rceil = 8$ candidate integer differences u . For each such u , the equation $P(r) = u$ in $\mathbb{F}_p$ has at most ℓ roots—a polynomial of degree $\leq \ell$ over a field has at most ℓ roots (*Math Guide*, §“Roots and the factor theorem”)—so at most 8ℓ values of r produce the difference δ . The clamped r is uniform over a set of 2^{106} values (clamping zeroes 22 of the 128 bits), so it lands on one of those roots with probability at most $8\ell/2^{106} \leq 8\ell_{\max}/2^{106}$. $\square$

The bound is comfortably small in practice: for messages of 16 384 bytes ($\ell_{\max} = 1024$ blocks) it gives $\varepsilon \leq 8 \cdot 1024/2^{106} = 2^{-93}$, against the ideal $\ell_{\max}/p \approx 2^{-120}$ —the clamping and reduction losses cost 27 bits of the margin and leave it enormous. Combining Proposition 6.12 with the Carter–Wegman Theorem 6.10: Poly1305 with a fresh, uniformly random pad s per message—uniformity, not mere unpredictability, is what the theorem's pad-elimination step uses—is a one-time MAC with forgery probability at most $8\ell_{\max}/2^{106}$ per verification attempt. In the AEAD below, ChaCha20 itself generates the one-time key (r, s) pseudorandomly from the nonce—uniform once the cipher is idealised as a random function—so the per-nonce one-time guarantee is exactly what is needed.

6.4 Authenticated encryption with associated data

We now combine the two guarantees. The composite goal is *authenticated encryption*: the adversary should be unable to learn anything about plaintexts (CPA security) *and* unable to produce any new ciphertext that decrypts to anything other than the distinguished rejection symbol $\perp$, a value distinct from every message (ciphertext integrity). Real protocols additionally require ciphertexts to bind to public context—packet headers, version bytes, an ephemeral public key—that travels in the clear but must be authenticated; this is the *associated data*.

Definition 6.13 (AEAD scheme). An *authenticated encryption with associated data* (AEAD) scheme is a pair (Enc, Dec) with key space $\mathcal{K}$, nonce space $\mathcal{N}$, and associated-data space $\mathcal{D}$: algorithm $\text{Enc}_k(\nu, a, m)$ returns a ciphertext c , and $\text{Dec}_k(\nu, a, c)$ returns a message or $\perp$. Correctness: $\text{Dec}_k(\nu, a, \text{Enc}_k(\nu, a, m)) = m$ for all k, ν, a, m . The scheme authenticates but does not encrypt the associated data a .

Definition 6.14 (Ciphertext integrity, INT-CTXT). For an AEAD scheme Π and adversary $\mathcal{A}$, the game $\text{Ctxt}_{\Pi}^{\mathcal{A}}(\lambda)$ runs as follows.

1. The challenger samples $k \xleftarrow{\$} \mathcal{K}$ and gives $\mathcal{A}$ two oracles: an encryption oracle $\text{Enc}_k(\cdot, \cdot, \cdot)$, recording the set Q of returned triples (ν, a, c) , and a verification oracle that on (ν, a, c) returns 1 if $\text{Dec}_k(\nu, a, c) \neq \perp$ and 0 otherwise. The adversary must never repeat a nonce to the encryption oracle (the *nonce-respecting* model); verification queries may reuse nonces freely.
2. The game outputs 1 iff some verification query $(\nu^*, a^*, c^*) \notin Q$ is answered 1.

The scheme Π has *ciphertext integrity* if for every PPT $\mathcal{A}$, $\Pr[\text{Ctxt}_{\Pi}^{\mathcal{A}}(\lambda) = 1] \in \text{negl}(\lambda)$.

Definition 6.15 (Authenticated encryption). An AEAD scheme is *secure* (is an AEAD) if it is both CPA-secure and has ciphertext integrity (Definition 6.14), where the CPA game of Definition 6.4 is run with $k \xleftarrow{\$} \mathcal{K}$ and extended so that the adversary supplies the nonce and associated data of every oracle query and of the challenge, subject to the *nonce-respecting* restriction—a restriction on the adversary, not a property of the scheme—that no nonce is used twice across oracle queries and challenge. The restriction is necessary: an AEAD may encrypt deterministically given (ν, a, m) —ChaCha20-Poly1305 does—and then querying the oracle at (ν, a, m_0) and submitting the challenge (ν, a, m_0, m_1) would identify the bit by comparing ciphertexts.

The standard route from a CPA-secure cipher and a strongly unforgeable MAC to an AEAD is *encrypt-then-MAC*: encrypt the message, compute a MAC over the ciphertext together with the associated data and nonce, and append the tag; decryption verifies the tag *before* decrypting, and any tag failure returns $\perp$. The contrasting orders—MAC-then-encrypt and encrypt-and-MAC—do *not* generically achieve authenticated encryption; encrypt-then-MAC does.

Theorem 6.16 (Encrypt-then-MAC yields authenticated encryption). Let $\Pi_E = (\text{Enc}, \text{Dec})$ be a CPA-secure encryption scheme and $\Pi_M = (\text{Mac}, \text{Vrfy})$ a strongly unforgeable (sUF-CMA) MAC,

with independent keys k_E, k_M . Define the composite scheme Π' by

$$\text{Enc}'_{k_E, k_M}(\nu, a, m) = (c, t), \quad c \leftarrow \text{Enc}_{k_E}(\nu, m), \quad t \leftarrow \text{Mac}_{k_M}(\nu \| a \| c),$$

with Dec' returning $\perp$ unless $\text{Vrfy}_{k_M}(\nu \| a \| c, t) = 1$, in which case it returns $\text{Dec}_{k_E}(\nu, c)$. Then Π' is a secure AEAD: it is CPA-secure and has ciphertext integrity.

Proof. Ciphertext integrity. Suppose a PPT A making at most q_v verification queries wins the game: some fresh query $(\nu^*, a^*, (c^*, t^*)) \notin Q$ decrypts successfully, i.e. $\text{Vrfy}_{k_M}(\nu^* \| a^* \| c^*, t^*) = 1$. Each encryption-oracle call on (ν, a, m) returned (c, t) and corresponds to one MAC query on the string $w = \nu \| a \| c$ returning tag t . Because the framing $\nu \| a \| c$ parses uniquely (the component lengths are fixed or prepended), the freshness of (ν^*, a^*, c^*) means the MAC oracle never produced the string-tag pair (w^*, t^*) with $w^* := \nu^* \| a^* \| c^*$: either w^* was never queried, or it was queried and returned a different tag. Either way (w^*, t^*) is a fresh valid pair—a *strong* forgery against Π_M . The reduction $\mathcal{B}$ runs A , answering encryption queries with its own MAC oracle and a self-chosen k_E . Verification queries $\mathcal{B}$ cannot answer, lacking k_M , so it guesses: it samples a uniform index $i \leq q_v$ in advance, answers each fresh verification query before the i -th with 0 (queries on triples in Q are answered 1, correctly), halts at the i -th fresh query, and outputs that query's (w^*, t^*) as its forgery. On the event that A wins and i hits her *first* successful fresh query—the guess is right with probability $1/q_v$ —every earlier 0 answer was correct, the simulation is perfect, and $\mathcal{B}$ wins, so

$$\Pr[\text{Ctxt}_{\Pi'}^A(\lambda) = 1] \leq q_v \cdot \text{Adv}_{\Pi_M}^{\text{s-forge}}(\mathcal{B}, \lambda) \in \text{negl}(\lambda),$$

the factor q_v being polynomial.

CPA security. The tag is computed from c (together with k_M, ν, a , and the MAC's own coins) and so carries no information about the challenge bit beyond what c already carries. Formally, the reduction $\mathcal{B}'$ samples k_M itself, forwards A 's oracle and challenge queries to its own Π_E oracle to obtain the c -components, and appends tags it computes with k_M . This simulates Π' perfectly, so $\text{Adv}_{\Pi'}^{\text{cpa}}(A, \lambda) = \text{Adv}_{\Pi_E}^{\text{cpa}}(\mathcal{B}', \lambda) \in \text{negl}(\lambda)$. $\square$

Remark 6.17 (Why the ordering matters). Encrypt-then-MAC lets the receiver reject forged ciphertexts *without decrypting*, and this is what blocks chosen-ciphertext and padding-oracle attacks: a rejected ciphertext never reaches the decryption logic, so the decryption logic cannot leak. By contrast, MAC-then-encrypt verifies only after decrypting, exposing the decryption routine to attacker-chosen ciphertexts—the historical source of a long line of padding-oracle breaks. Ciphertext integrity is also exactly the property that upgrades CPA security to chosen-ciphertext (CCA) security, the CPA game of Definition 6.4 with a decryption oracle added that refuses only the challenge ciphertext: an authenticated-encryption scheme is automatically CCA-secure, because that oracle is useless to the adversary—every ciphertext she did not legitimately obtain decrypts to $\perp$ except with negligible probability.

6.5 The ChaCha20-Poly1305 AEAD

We now assemble the concrete scheme of RFC 8439, the AEAD by which Zcash encrypts every shielded note ciphertext and outgoing ciphertext (protocol specification, §“Symmetric Encryption”). It is an encrypt-then-MAC composition of the ChaCha20 stream cipher with the Poly1305 one-time MAC, with one economical twist: the cipher derives *both* the keystream and the one-time MAC key from the single (k, ν) input.

Construction 6.18 (ChaCha20-Poly1305). Fix a 256-bit key k and a 96-bit nonce ν . Encryption of a message m with associated data a proceeds as follows.

1. *One-time MAC key.* Run ChaCha20 with key k , nonce ν , and block counter 0; take the first 32 bytes of the keystream block as the Poly1305 one-time key (r, s) —clamp the first 16 bytes to form r ; the next 16 are s . Discard the rest of this block.
2. *Encrypt.* Run ChaCha20 with key k and nonce ν starting at block counter 1, producing the keystream KS_1 (the keystream of Remark 6.3 taken from counter 1 onward), and set $c = m \oplus \text{KS}_1(k, \nu, |m|)$.
3. *Authenticate.* Form the Poly1305 input by concatenating: a padded with zeros to a 16-byte boundary, c padded with zeros to a 16-byte boundary, the 8-byte little-endian length of a , and the 8-byte little-endian length of c . Compute the tag $t = \text{Poly1305}_{r,s}(\text{that input})$.

The ciphertext is (c, t) . Decryption recomputes (r, s) and the tag from (k, ν, a, c) , compares it to t in *constant time*, returns $\perp$ on mismatch, and otherwise outputs $m = c \oplus \text{KS}_1(k, \nu, |c|)$.

Remark 6.19 (Length encoding prevents splicing). The trailing 8-byte lengths of a and c are essential. Without them, the zero-padding that aligns a and c to 16-byte boundaries would let an adversary shift bytes between the associated data and the ciphertext, or pad-extend one of them, while leaving the Poly1305 input string unchanged—breaking the binding between a and c even though every authenticated byte is intact. Appending the explicit lengths makes the MAC input an injective encoding of the pair (a, c) , the same injectivity that the per-block length encoding supplies within one message in the $\Delta\Delta\text{U}$ analysis (Proposition 6.12).

Theorem 6.20 (Security of ChaCha20-Poly1305). *Model ChaCha20 as a pseudorandom function $F_k(\nu, j)$ from (nonce, block counter) pairs to 64-byte blocks. In the nonce-respecting model, ChaCha20-Poly1305 is a secure AEAD. Concretely, consider a PPT adversary making q encryption queries and q_v verification queries, each of whose Poly1305 inputs—padded associated data, padded ciphertext, and the length block—comprises at most $\ell_{\max}$ sixteen-byte blocks, $\ell_{\max} = \lceil |a|/16 \rceil + \lceil |c|/16 \rceil + 1$. Her CPA advantage is at most twice the best PRF-distinguishing advantage against F at comparable cost, and she forges—wins the INT-CTXT game—with probability at most*

$$\text{Adv}_F^{\text{prf}}(\mathcal{B}, \lambda) + \frac{8q_v \ell_{\max}}{2^{106}},$$

where $\mathcal{B}$ is a PRF distinguisher of comparable cost.

Proof. Sketch. Replace F_k by a truly random function Φ from (nonce, counter) pairs to 64-byte blocks. Any noticeable change in the adversary’s success probability yields a PRF distinguisher of comparable cost—it simulates the whole game using its function oracle—so each replacement costs one $\text{Adv}_F^{\text{prf}}$ term; the CPA game is played twice (once per challenge bit), whence the factor two there. We analyse the idealised scheme.

Confidentiality. Since the adversary is nonce-respecting, every encryption query uses a fresh ν , and the keystream blocks $\Phi(\nu, 1), \Phi(\nu, 2), \dots$ are uniform and independent of everything else in her view; the ciphertext $c = m \oplus \text{KS}$ is a genuine one-time pad and reveals nothing about m beyond its length. The tag is a deterministic function of c, a, ν and the (independent) key (r, s) , and adds no information. The idealised CPA advantage is therefore zero.

Integrity. For each nonce ν , the block $\Phi(\nu, 0)[0:32]$ is uniform and independent across distinct nonces; clamping its first half samples r from Poly1305’s prescribed restricted set, while the second half s stays uniform, so the resulting one-time keys are independent across nonces. Because counter 0 is reserved for this key while encryption consumes counters ≥ 1 , each one-time key is also independent of the keystream that produced its ciphertext. Per nonce, the setting is exactly the Carter–Wegman one-time MAC of Theorem 6.10 instantiated with the ε - $\Delta\Delta\text{U}$ Poly1305 hash of Proposition 6.12, $\varepsilon \leq 8\ell_{\max}/2^{106}$. Consider any single fresh verification query (ν^*, a^*, c^*, t^*) . If

ν^* was never used for encryption, the pad s for ν^* is uniform and entirely unknown, so the tag is correct with probability 2^{-128} . If ν^* equals the nonce of some encryption query—nonce reuse is allowed in *verification* queries, only encryption is nonce-respecting—then the adversary has seen exactly one tag under that (r, s) , and freshness of (a^*, c^*) makes this a forgery on a new message under a one-time key, succeeding with probability at most ε by Theorem 6.10. The union bound over the q_v verification queries (*Math Guide*, §“The union bound and a birthday calculation”) gives forgery probability at most $8q_v\ell_{\max}/2^{106}$ in the idealised scheme; adding back the PRF-switching cost yields the stated bound. $\square$

Remark 6.21 (Single key, two roles). The scheme reuses one 256-bit key for both encryption and authentication, seemingly violating the independent-keys hypothesis of Theorem 6.16. Reserving block counter 0 for the Poly1305 key and counters ≥ 1 for the keystream is exactly what restores independence: once the cipher is idealised as a truly random function—the switch the proof of Theorem 6.20 pays for with a PRF-distinguishing term—outputs at distinct counters are independent, so (r, s) is independent of the keystream and the two-key analysis applies. For the PRF itself the independence is computational, not literal. This is a recurring design pattern—domain separation by a counter or label, the same discipline as §3.5, lets one master key safely play several roles.

Remark 6.22 (Wrong-key rejection and key commitment). The upper volumes consume this AEAD in a *trial-decryption* pattern: a receiver detects the notes addressed to it by attempting decryption of every ciphertext under each of its keys, relying on decryption under a wrong key to return $\perp$ (protocol specification, §“Symmetric Encryption”). For *honestly produced* ciphertexts the scheme delivers this. Fix a ciphertext $(c, t) = \text{Enc}_k(\nu, a, m)$ and a key k' independent of k , and idealise the cipher under k and under k' as independent random functions, as in the proof of Theorem 6.20. The one-time key (r', s') that decryption under k' derives from block $(\nu, 0)$ is then uniform and independent of (c, t) : the pad s' alone makes the recomputed tag match t with probability exactly 2^{-128} , and even conditioned on one exposed tag under (r', s') , Theorem 6.10 bounds acceptance by the AΔU slack $8\ell_{\max}/2^{106}$ of Proposition 6.12, with $\ell_{\max}$ as in Theorem 6.20. Decryption under an independent key therefore rejects except with probability at most $2^{-128} + 8\ell_{\max}/2^{106}$ per attempt, plus two PRF-distinguishing terms for the two idealisations.

The guarantee stops at honest senders, and the boundary deserves care. CPA security and INT-CTXT are *single-key* notions: nothing in Definition 6.15 constrains a ciphertext crafted by a *malicious sender* who knows several keys. The scheme is in fact not *key-committing*—the separate notion demanding that no ciphertext verify under two distinct keys—and a sender who chooses keys k_1, k_2 herself can craft a single (c, t) that decrypts validly, to different plaintexts, under both: with (r_i, s_i) the one-time key each k_i derives, the hash h_{r_i} of Construction 6.11 is linear in the blocks of c , so fixing t , leaving two blocks of c free, and solving over $\mathbb{F}_p$ the two linear equations that pin the hash under r_i to $t - s_i$ (retrying until both solutions encode valid blocks) makes both keys accept (c, t) —the multi-key collision of Len, Grubbs, and Ristenpart’s partitioning-oracle attack. The upper volumes’ trial-decryption arguments therefore never rest on the tag alone against adversarial senders: their note-plaintext consistency checks—recomputing the note commitment from the decrypted plaintext and comparing it against the commitment the transaction carries—are what close this gap.

6.6 Key-derivation functions

Theorem 6.20 assumes a uniformly random 256-bit key. In practice keys may come from sources that are *high-entropy but not uniform*: a Diffie–Hellman shared secret is a structured group encoding, not a uniform bit string, and a hardware noise source may spread its unpredictability unevenly across its bits. (A passphrase is a different, usually *low-entropy* source, and needs a deliberately costly password KDF rather than the machinery of this subsection.) The adversary is owed nothing here—if the honest parties feed a non-uniform key into a scheme proved secure

for uniform keys, the proof simply does not apply, and structure in the key is structure she may exploit. A *key-derivation function* (KDF) converts an appropriate source secret into one or more pseudorandom keys. The right entropy measure is worst-case, not average-case.

Definition 6.23 (Min-entropy). The *min-entropy* of a random variable X is

$$H_\infty(X) = -\log_2 \max_x \Pr[X = x];$$

equivalently, the best chance of guessing X in one try is $2^{-H_\infty(X)}$. Given side information Z , the *conditional min-entropy* is

$$\tilde{H}_\infty(X | Z) = -\log_2 \mathbb{E}_{z \leftarrow Z} \max_x \Pr[X = x | Z = z],$$

so that $2^{-\tilde{H}_\infty(X|Z)}$ is the best chance of guessing X from Z . Recall the *statistical distance* $\Delta(X, Y) = \frac{1}{2} \sum_u |\Pr[X = u] - \Pr[Y = u]|$ of Definition 1.6 (*Math Guide*, §“Statistical distance”); we say X is ϵ -close to uniform on a set S if $\Delta(X, U_S) \leq \epsilon$.

Definition 6.24 (Key-derivation function and its security). A *key-derivation function* is a function

$$\text{KDF} : \mathcal{X} \times \mathcal{I} \times \{0, 1\}^* \times \mathbb{N} \rightarrow \{0, 1\}^*$$

taking a source secret (a sample of a random variable Σ over $\mathcal{X}$), a non-secret *salt* $\text{slt} \in \mathcal{I}$, an *info* or context string ctx , and a desired output length L . The security goal must specify the source model. For an arbitrary source of conditional min-entropy at least γ given the adversary’s side information, an extractor-style guarantee uses a salt sampled uniformly and independently of the source and requires the pair $(\text{slt}, \text{KDF}(\Sigma, \text{slt}, \text{ctx}, L))$ to be computationally indistinguishable from (slt, U_L) , even given ctx and that side information. No fixed deterministic unsalted function can provide this guarantee for *every* high-min-entropy source; an unsalted KDF therefore rests on a narrower premise, such as input keying material that is already computationally pseudorandom, or that is hard to query in a random-oracle model.

The notion a KDF targets is that of a seeded randomness extractor whose output need only look uniform.

Definition 6.25 (Strong computational extractor). A keyed function $\text{Ext} : \mathcal{I} \times \mathcal{X} \rightarrow \{0, 1\}^n$ is a *strong* (γ, ϵ) -computational extractor if for every source Σ over $\mathcal{X}$ of conditional min-entropy at least γ given auxiliary information Z , and a uniformly random salt $\text{slt} \xleftarrow{\$} \mathcal{I}$, the pair $(\text{slt}, \text{Ext}_{\text{slt}}(\Sigma))$ is computationally indistinguishable from (slt, U_n) , even given Z , with distinguishing advantage at most ϵ . “Strong” means the salt is revealed to the distinguisher.

Information theory can realise the notion unconditionally; we state the classical result *without proof*, since the deployed mechanism below rests on random-oracle modelling instead.

Theorem 6.26 (Leftover hash lemma; stated without proof). Let $\{g_r : \mathcal{X} \rightarrow \{0, 1\}^n\}$ be a 2^{-n} -universal hash family with a uniform public seed R . For any X with $H_\infty(X) \geq \gamma$,

$$\Delta((R, g_R(X)), (R, U_n)) \leq \frac{1}{2} \sqrt{2^{n-\gamma}}.$$

Thus investing $\gamma \geq n + 2 \log_2(1/\epsilon)$ bits of min-entropy buys n output bits that are ϵ -close to uniform—

| a strong (γ, ϵ) -extractor in the statistical, not merely computational, sense.

HKDF, in passing. The standardised general-purpose KDF is Krawczyk’s HKDF (RFC 5869), built from HMAC (§5.3) in an *extract-then-expand* architecture: a salted HMAC call first distils suitable non-uniform input keying material into a short pseudorandom key (the extractor stage, justified along leftover-hash lines), and a counter-chained sequence of HMAC calls then expands that key, under a context string, up to RFC 5869’s limit of 255 hash-output-length blocks (the expander stage, justified by HMAC’s PRF security). The resulting KDF is a standard choice across many protocol source models and output lengths, though its precise guarantee still depends on assumptions about the source and about HMAC. Zcash does not use it—indeed the shielded protocol uses HMAC nowhere: neither the orchard nor the `zcash_note_encryption` crate carries an HMAC dependency (HMAC enters the wallet stack through the transparent BIP-32 derivation and BIP-39 mnemonic layers and, transitively, through the optional Tor client, never through a shielded crate)—because its requirements are narrower and a single hash call meets them in the random-oracle analysis.

The Zcash KDF. Orchard derives the note-encryption key by one personalised BLAKE2b call:

$$K_{\text{enc}} = \text{KDF}(\text{sharedSecret}, \text{epk}) = \text{BLAKE2b-256}(\text{"Zcash_OrchardKDF"}; \text{sharedSecret} \parallel \text{epk}),$$

where `sharedSecret` is the Diffie–Hellman shared point of the note-encryption key agreement, `epk` is the sender’s ephemeral public key, and the quoted string occupies BLAKE2b’s 16-byte personalisation field (the function `kdf_orchard` hashes the serialised shared point followed by the ephemeral-key bytes with a 32-byte output; protocol specification, §“Orchard Key Derivation”). The analysis models this personalised hash as a random oracle: by the domain-separation principle of §3.5 (Proposition 3.17, via the native-personalisation idiom catalogued there), the personalisation makes it an oracle independent of every other hash use in the protocol, and the derived key is uniform in the adversary’s view *unless* she queries that oracle at the hidden `sharedSecret`—a query whose argument she must first compute from the two public keys, an instance of the computational Diffie–Hellman problem of §2.2. Folding `epk` into the input binds the derived key to this particular ephemeral, the role the context string plays in HKDF. No salt is used because the claim is this narrower Diffie–Hellman-plus-random-oracle one—the unsalted premise anticipated in Definition 6.24—not extraction from every high-min-entropy source.

Remark 6.27 (The role of the KDF in the larger protocol). In the Zcash note-encryption layer, a Diffie–Hellman exchange on the Pallas curve produces a shared group element, and the single BLAKE2b call above condenses that element into the symmetric key fed to ChaCha20-Poly1305. The two halves of this section meet exactly here: key derivation manufactures the uniform key that the AEAD security theorem (Theorem 6.20) takes as its hypothesis, and the chain from a non-uniform shared secret to an authenticated ciphertext is complete. Section 7 assembles that chain end to end as hybrid public-key encryption.

7 Key agreement

Two parties who have never met share no secret, and the adversary of this section intends to keep it that way. She owns the channel between them. As an eavesdropper she records the two group elements they exchange and tries to *distinguish* the key they derive from a random string—a key she can distinguish is a key whose every later use leaks. As an active attacker she does better than listen: she intercepts each party’s contribution, substitutes her own, and runs one honest-looking session with each victim, so that both “secure channels” terminate at her desk and she reads and rewrites every message in transit—forging, as we shall prove, without violating any hardness assumption at all. And on a public ledger her seat is permanent: every note ciphertext

is published for ever, and she may scan them all at leisure, holding each against every address she can enumerate. This section builds the machinery that denies her all three postures: the Diffie–Hellman protocol atop the exponentiation asymmetry of Section 2; the static/ephemeral taxonomy that turns an interactive exchange into a one-shot flow; the key-agreement scheme abstraction that the deployed code programs against; passive security, which we prove equal to the DDH advantage *exactly*; the man-in-the-middle attack and the three external mechanisms of authentication; the elliptic-curve instantiation with its validation duties; the hybrid composition with a key-derivation function and an AEAD; key privacy, which keeps a ciphertext silent about its recipient; and the deployed endpoint, shielded-note delivery as hybrid public-key encryption carried in band on the ledger.

7.1 The Diffie–Hellman protocol

The raw material is the asymmetry established in Section 2. In a cyclic group $G = \langle g \rangle$ of order q , the map

$$\exp_g : \mathbb{Z}/q\mathbb{Z} \rightarrow G, \quad x \mapsto g^x,$$

is a group isomorphism—it carries addition of exponents to multiplication of elements—that is cheap to evaluate, at $O(\log q)$ group operations by repeated squaring, but believed infeasible to invert in the groups this volume uses: inverting it is precisely the discrete logarithm problem (Definition 2.2). Everything in this section depends only on the group structure and the hardness of certain problems in it, never on the presentation of the group; the reference instances are a large prime-order subgroup of $\mathbb{F}_p^\times$ and the elliptic-curve point groups of §7.6. Diffie and Hellman’s observation is that the asymmetry alone already lets two strangers manufacture a shared secret in public.

Construction 7.1 (Diffie–Hellman key exchange). Fix public parameters (G, g, q) with $G = \langle g \rangle$ of order q , known to all parties (and to the adversary).

1. Alice samples $a \xleftarrow{\$} \mathbb{Z}/q\mathbb{Z}$ uniformly, computes $A = g^a$, and sends A to Bob.
2. Bob samples $b \xleftarrow{\$} \mathbb{Z}/q\mathbb{Z}$ uniformly, computes $B = g^b$, and sends B to Alice.
3. Alice computes $k_A = B^a$; Bob computes $k_B = A^b$.

The value $k_A = k_B$ is the *shared secret*; the transmitted elements A and B are the *public shares* (public keys), and the exponents a and b are the *private keys*, never sent.

Proposition 7.2 (Correctness of Diffie–Hellman). *In Construction 7.1 both parties compute the same group element, $k_A = k_B = g^{ab}$.*

Proof. Unwinding the definitions,

$$k_A = B^a = (g^b)^a = g^{ba} = g^{ab} = (g^a)^b = A^b = k_B,$$

where $ba = ab$ because integer multiplication is commutative, and exponents may be read modulo q because g has order q . $\square$

The shared secret g^{ab} is thus a well-defined function of the transcript (A, B) : it can be computed by anyone who knows *one* of the two private keys, and—conjecturally—by no one who knows only the public shares. Making that conjecture precise, and seeing exactly how much it buys, is the business of §7.4. First, a structural precaution.

Remark 7.3 (Small subgroups and the prime-order rule). If G has small subgroups, an adversary can exploit them: by substituting for a public share an element of small order she confines the victim’s computed secret to a small subgroup and learns the victim’s exponent modulo a small factor of q —the Pohlig–Hellman leakage of Corollary 2.21 in protocol form. The remedy is the standing rule of Section 2: prefer G of *prime* order q . Then the only subgroups are $\{1\}$ and G itself, every non-identity element is a generator, and no nontrivial confinement is possible. When the natural ambient group does not have prime order— $\mathbb{F}_p^\times$ has composite order $p - 1$ —one works inside a prime-order subgroup and has each recipient *validate* incoming elements as members: check $B^q = 1$ and $B \neq 1$ before exponentiating. The elliptic-curve analogue of this validation appears in Remark 7.19.

7.2 Static and ephemeral keys

Nothing in Construction 7.1 says how long a key pair lives, and the protocol changes character with the answer.

Definition 7.4 (Static and ephemeral keys). A Diffie–Hellman key pair (x, g^x) is *static* (long-term) if it is generated once and reused across many sessions, typically published and bound to an identity; it is *ephemeral* if it is generated freshly for a single session and discarded immediately afterwards.

Three protocol shapes result, and all three matter to what follows.

- *Ephemeral–ephemeral* (DHE). Both parties use fresh keys. Once a and b are erased, no one—the parties included—can recover g^{ab} : this is *forward secrecy*, the guarantee that a later compromise of all stored long-term state cannot decrypt recorded past sessions, because the only secrets that ever existed were the deleted ephemerals.
- *Static–ephemeral*. The recipient Bob holds a static pair $(b, B = g^b)$, with B known in advance; Alice contributes a fresh ephemeral (a, A) . The shared secret g^{ab} then requires *no interaction from Bob at agreement time*: Alice computes $g^{ab} = B^a$ from the published B , attaches A , and sends everything in a single flow; Bob recovers $g^{ab} = A^b$ whenever he next comes online. This is exactly the shape of hybrid encryption (§7.7) and of shielded-note delivery (§7.9): sender online once, recipient possibly offline, no round trip.
- *Static–static*. Both keys are long-term, so g^{ab} is a fixed function of the two identities—the same in every session. There is no forward secrecy, and the never-varying secret must be combined with fresh per-session randomness (a nonce) before it may key any symmetric scheme; using it directly would reuse one key across all time.

The static–ephemeral case is the conceptual bridge from key exchange, an interactive protocol, to public-key encryption, a one-shot algorithm. A static public key $B = g^b$ functions exactly as an encryption public key: anyone can derive a secret shared with its owner by sending a single ephemeral share. The rest of the section develops that bridge until, by §7.9, it carries deployed traffic.

7.3 The key-agreement scheme abstraction

Deployed code does not manipulate exponents; it programs against an interface. We therefore package the non-interactive (static–ephemeral) pattern as a scheme with named algorithms, so that later constructions—and the deployed note-encryption machinery—can invoke key agreement without caring which group sits underneath.

Definition 7.5 (Key-agreement scheme). A (non-interactive) *key-agreement scheme* over a public parameter set pp consists of a private-key space $\mathcal{S}$, a public-key space $\mathcal{P}$, a shared-secret space $\mathcal{K}$, and three algorithms:

- $\text{KeyGen}(\text{pp}) \rightarrow s$, randomised, outputting a private key $s \in \mathcal{S}$;
- $\text{DerivePublic}(\text{pp}, s) \rightarrow p$, deterministic, mapping a private key to its public key $p \in \mathcal{P}$;
- $\text{Agree}(\text{pp}, s, p') \rightarrow k$, deterministic, mapping one’s own private key and a counterparty’s public key to a shared secret $k \in \mathcal{K}$.

Correctness demands that for all pp and all s, t in the support of $\text{KeyGen}(\text{pp})$,

$$\text{Agree}(\text{pp}, s, \text{DerivePublic}(\text{pp}, t)) = \text{Agree}(\text{pp}, t, \text{DerivePublic}(\text{pp}, s)). \quad (1)$$

Construction 7.6 (Diffie–Hellman as a key-agreement scheme). Fix $\text{pp} = (G, g, q)$ with $G = \langle g \rangle$ of prime order q . The scheme $\text{DH}_{G,g}$ takes $\mathcal{S} = \mathbb{Z}/q\mathbb{Z}$ (or $(\mathbb{Z}/q\mathbb{Z}) \setminus \{0\}$, to avoid the degenerate case whose public key is the identity), $\mathcal{P} = \mathcal{K} = G$, and

$$\text{KeyGen}: a \xleftarrow{\$} \mathcal{S}, \quad \text{DerivePublic}(\text{pp}, a) = g^a, \quad \text{Agree}(\text{pp}, a, B) = B^a.$$

Correctness is exactly Proposition 7.2: $\text{Agree}(\text{pp}, s, g^t) = (g^t)^s = g^{ts} = g^{st} = (g^s)^t = \text{Agree}(\text{pp}, t, g^s)$.

Remark 7.7 (The deployed interface generalises the base point). The Zcash note-encryption machinery programs against precisely this abstraction, with one generalisation: the deployed DerivePublic takes the base point as an *explicit argument* rather than fixing a protocol-wide generator, so that Sapling and Orchard derive keys over per-address *diversified bases* (protocol specification §“Key Agreement”; the `zcash_note_encryption` crate’s `Domain` trait’s `ka_derive_public`, `ka_agree_enc`, and `ka_agree_dec` operate on abstract associated types, and Orchard’s implementation of `derive_public` in the `orchard` crate receives the base g_d as a parameter). Those bases are put to work in §7.9, and the address machinery that produces them is detailed in the *Ironwood Guide* (§“Diversified addresses”). What matters here is that the interface still invokes an abstract key agreement and never inspects the underlying group—which is why the same construction has transported across three groups, from Sprout’s Curve25519 to Sapling’s Jubjub to Orchard’s Pallas.

Remark 7.8 (The abstraction boundary, and what falls outside it). Any concrete scheme satisfying equation (1) can be swapped in without changing the surrounding logic: a prime-field group, an elliptic curve, or a post-quantum non-interactive key agreement such as the CSIDH group action. A *key-encapsulation mechanism* (KEM), by contrast, cannot satisfy the correctness equation at all: encapsulation requires the recipient’s public key before it can produce its ciphertext, so no counterparty-independent DerivePublic exists. The static–ephemeral one-flow *pattern* does generalise to any KEM—the encapsulation ciphertext plays the role of the ephemeral public key—but at the cost of changing the interface the surrounding logic programs against.

7.4 Passive security and the Diffie–Hellman assumptions

The weakest adversary is the eavesdropper: she reads the channel but does not alter it, and her entire view of an honest session is the transcript (g^a, g^b) . Both formalisations of her task were laid down in §2.2. The *computational* problem is to produce the shared secret: writing $\text{Adv}_{\text{cdh}}(\mathcal{A})$

for the success probability of A in the CDH game of Definition 2.4,

$$\text{Adv}_{\text{cdh}}(A) = \Pr_{a,b}[A(g^a, g^b) = g^{ab}].$$

The *decisional* problem is to distinguish (g^a, g^b, g^{ab}) from (g^a, g^b, g^c) with advantage $\text{Adv}_{\text{ddh}}(A)$ (Definition 2.5). The hierarchy $\text{DDH} \leq \text{CDH} \leq \text{DLP}$ was established in §2.3 (Theorems 2.7 and 2.8), together with the pairing-based separation of DDH from CDH (Example 2.9). The distinction is not pedantry. Hardness of CDH asserts only that the *whole* secret cannot be computed; keying a symmetric cipher requires that the secret be indistinguishable from a uniform group element, so that no individual bit or predicate of it leaks. The following classical break shows the gap is real in the most familiar group.

Remark 7.9 (A concrete DDH break: the Legendre symbol). Take $G = \mathbb{F}_p^\times$, the *full* multiplicative group, with generator g . The Legendre symbol $\left(\frac{h}{p}\right)$ is efficiently computable by Euler’s criterion $h^{(p-1)/2} \bmod p$ and is multiplicative (*Math Guide*, §“Quadratic residues and the Euler criterion”). A generator of $\mathbb{F}_p^\times$ is a nonsquare, so $\left(\frac{g^a}{p}\right) = (-1)^a$: the symbol of a public share reveals the *parity* of its exponent. The parities of a and b determine the parity of ab , hence $\left(\frac{g^{ab}}{p}\right) = (-1)^{ab}$ is determined by the two observed symbols. The distinguisher computes the predicted symbol of the shared secret from (g^a, g^b) , compares it with the symbol of the third component, and outputs “Diffie–Hellman” on a match. On a genuine triple the match is certain; on a random triple the third component’s symbol is an independent uniform sign, matching with probability $1/2$. The distinguishing advantage is the constant $1/2$ —even though CDH in $\mathbb{F}_p^\times$ is believed hard. The CDH/DDH gap therefore needs no pairings; the oldest group in the subject exhibits it. The cure is the prime-order rule of Remark 7.3: in a subgroup of odd prime order q , every element h is a square ($h = (h^{(q+1)/2})^2$), the symbol is identically $+1$, and the leak vanishes. This is one of the two reasons—the other being Pohlig–Hellman—that DDH is only ever assumed in prime-order groups.

With the assumption in place, passive security is not merely implied by DDH; it *is* DDH, advantage for advantage. We state the game in the volume’s tradition and prove the identification.

Definition 7.10 (Passive security of a key-agreement scheme). For a key-agreement scheme Σ over pp and an adversary A , the game $\text{KA-IND}_\Sigma^A(\lambda)$ runs as follows.

1. The challenger samples $s, t \leftarrow \text{KeyGen}(\text{pp})$ independently and computes the transcript $(\text{DerivePublic}(\text{pp}, s), \text{DerivePublic}(\text{pp}, t))$.
2. It sets $k_0 = \text{Agree}(\text{pp}, s, \text{DerivePublic}(\text{pp}, t))$, samples $k_1 \xleftarrow{\$} \mathcal{K}$ uniformly, draws a hidden bit $\beta \xleftarrow{\$} \{0, 1\}$, and sends the transcript together with k_β .
3. The adversary outputs a bit β' .

The advantage is $\text{Adv}_\Sigma^{\text{ka-ind}}(A) = |\Pr[\beta' = 1 \mid \beta = 0] - \Pr[\beta' = 1 \mid \beta = 1]|$; the scheme is *passively secure* if every PPT adversary’s advantage is negligible in λ .

Theorem 7.11 (Passive security of Diffie–Hellman). *For the scheme $\text{DH}_{G,g}$ of Construction 7.6 with KeyGen uniform on $\mathbb{Z}/q\mathbb{Z}$, every adversary A satisfies*

$$\text{Adv}_{\text{DH}_{G,g}}^{\text{ka-ind}}(A) = \text{Adv}_{\text{ddh}}(A),$$

identically and with running time preserved exactly. Under the DDH assumption the shared secret of an honest ephemeral–ephemeral session is therefore pseudorandom given the transcript: a passive adversary learns nothing about it that she could not have guessed about a random group element.

Proof. The two games are the same game. In KA-IND for $\text{DH}_{G,g}$, the transcript is (g^a, g^b) with a, b uniform and independent; the real key is $k_0 = g^{ab}$; and the uniform alternative $k_1 \xleftarrow{\$} G$ is exactly g^c for uniform independent c , because $\exp_g$ is a bijection. The challenge (g^a, g^b, k_β) is thus verbatim a sample of $\mathcal{D}_{\text{dh}}$ or $\mathcal{D}_{\text{rand}}$ according to β —the DDH distinguishing game of Definition 2.5. The identity map converts each adversary into the other, so the advantages coincide and no running time is added. $\square$

Remark 7.12 (Why pseudorandomness, not mere unpredictability). Every use of a Diffie–Hellman secret in this volume feeds it into a key-derivation function to produce a symmetric key. What should the group deliver to that function? Hardness of CDH guarantees the secret is hard to produce *in full*, but not that it lacks efficiently predictable structure: the Legendre leak of Remark 7.9 is exactly such structure. Hardness of DDH guarantees the secret is as good as a random group element (Theorem 7.11)—no predicate of it is efficiently computable from the transcript. In practice one often assumes only CDH and models the hash-based KDF as a random oracle: the derived key is then uniform unless the adversary queries the oracle at the hidden shared secret, a route made precise in Theorem 7.24. Either way the design goal is the same: a derived symmetric key indistinguishable from uniform, because that is the hypothesis every theorem of Section 6 consumes.

7.5 Active attacks and the role of authentication

Passive security is the most the bare protocol can offer, and the gap between it and what a channel needs is best exhibited as a construction. The adversary now controls the wire.

Definition 7.13 (Man-in-the-middle attack on unauthenticated Diffie–Hellman). An active adversary $\mathcal{M}$ controlling the channel between Alice and Bob runs two independent Diffie–Hellman sessions, one with each victim. She samples $m \xleftarrow{\$} \mathbb{Z}/q\mathbb{Z}$ and computes g^m . When Alice sends $A = g^a$ intended for Bob, $\mathcal{M}$ intercepts it and forwards g^m to Bob in its place; when Bob replies $B = g^b$, she intercepts it and forwards g^m to Alice. Alice computes $k_A = (g^m)^a = g^{am}$, which $\mathcal{M}$ also computes as A^m ; symmetrically Bob computes $k_B = (g^m)^b = g^{bm}$, which $\mathcal{M}$ computes as B^m .

Proposition 7.14 (The attack defeats unauthenticated Diffie–Hellman). *After the attack of Definition 7.13, Alice and Bob each hold a secret they believe is shared with the other but is in fact shared with $\mathcal{M}$; and $\mathcal{M}$, knowing both g^{am} and g^{bm} , can decrypt, read, modify, and re-encrypt every subsequent message in each direction, transparently to both parties. No assumption on G —not even ideal hardness of the discrete logarithm problem—prevents this.*

Proof. Correctness of Diffie–Hellman (Proposition 7.2), applied once per session, gives agreement between Alice and $\mathcal{M}$ on g^{am} and between Bob and $\mathcal{M}$ on g^{bm} . The adversary knows her own exponent m and received both A and B , so she computes both secrets directly—no problem instance is ever inverted, which is why no hardness assumption is relevant. Neither victim verifies the *origin* of the group element they received; each transcript is a syntactically valid run of Construction 7.1, so nothing in either party’s view distinguishes the attacked execution from an honest one. With k_A and k_B in hand, $\mathcal{M}$ decrypts traffic from Alice under k_A , re-encrypts it under k_B for Bob, and symmetrically in reverse, editing at will in between. $\square$

The lesson deserves stating structurally: *passive security concerns the secrecy of the shared secret; active security additionally requires the authenticity of the public shares*. Confidentiality of a channel is worthless if the channel is established with the wrong party. The bare protocol (Construction 7.1) provides no authentication—nothing in g^a or g^b testifies to who sent it—so the remedy must bind each public share to its sender’s identity, letting the recipient reject a substituted share. Every such binding is *external* to the key exchange itself.

Remark 7.15 (Three mechanisms supply authentication). All deployed remedies fall into three families, each external to the bare exchange.

1. *Signed Diffie–Hellman*. Each party signs its ephemeral share with a long-term signing key whose verification key is certified—by a certificate authority (a third party whose signature on the key the parties already trust) or by trust-on-first-use (accepting the first key seen and rejecting later changes). The adversary of Proposition 7.14 cannot forge Alice’s signature on g^m , so her substitution is rejected. Digital signatures are the subject of Section 8; this composition is the structure of authenticated TLS.
2. *Static keys that are themselves authenticated*. If Bob’s static public key $B = g^b$ reaches Alice authentically—a trusted directory, a pinned key—then a static-ephemeral exchange toward B already authenticates Bob to Alice: only the holder of b can complete it. Note the direction: this authenticates the *recipient*, not the sender. It is the model relevant to public-key encryption, and to shielded-note delivery, where the recipient’s address *is* an authenticated public key (§7.9).
3. *Out-of-band verification*. The parties compare a short fingerprint of the exchanged public keys over a channel that is authentic even if low-bandwidth—digits read aloud on a call, a QR code scanned in person.

In every case the key-agreement security goal remains the passive guarantee of Theorem 7.11; authentication is a separable layer composed around the exchange, not a property of the group operation.

7.6 Elliptic-curve Diffie–Hellman

The development so far is deliberately group-agnostic; we now instantiate it on the groups the deployed protocol actually uses. Recall from the *Math Guide* (§“Elliptic curves”) that an elliptic curve E over a field F of characteristic neither 2 nor 3 is given in short Weierstrass form $y^2 = x^3 + ax + b$ with nonzero discriminant, and that its F -rational points $E(F)$ form a finite abelian group under chord-and-tangent addition with the point at infinity O as identity (*Math Guide*, §“The group structure of $E(\mathbb{F}_p)$ ”). The group is written additively, so exponentiation $g \mapsto g^x$ becomes *scalar multiplication* $P \mapsto [x]P$, computable in $O(\log x)$ point additions by double-and-add. Elliptic-curve Diffie–Hellman is the transliteration.

Construction 7.16 (ECDH). Fix an elliptic curve E/F , a base point $P \in E(F)$ of large prime order q generating $G = \langle P \rangle$, and cofactor $h = |E(F)|/q$. In the abstraction of Definition 7.5:

$$\text{KeyGen}: a \xleftarrow{\$} \mathbb{Z}/q\mathbb{Z}, \quad \text{DerivePublic}(a) = [a]P, \quad \text{Agree}(a, Q) = [a]Q.$$

The shared secret of parties with private keys a and b is $[a]([b]P) = [ab]P = [b]([a]P)$. The associated hard problems are the discrete-logarithm, CDH, and DDH definitions of Section 2 (Definitions 2.2, 2.4, and 2.5) read with $[x]P$ in place of g^x —the problems ECDLP (*Math Guide*, §“The elliptic-curve discrete logarithm problem”), ECCDH, and ECDDH of §2.6.

Proposition 7.17 (Correctness of ECDH). *The scheme of Construction 7.16 satisfies the key-agreement correctness condition (1).*

Proof. Scalar multiplication is cyclic-group exponentiation in additive dress: the group is abelian, and $\mathbb{Z}$ acts on $\langle P \rangle$ through $\mathbb{Z}/q\mathbb{Z}$ because $[q]P = O$, so $[x]([y]P) = [xy]P$ for all integers x, y . Hence $\text{Agree}(a, [b]P) = [ab]P = [ba]P = \text{Agree}(b, [a]P)$, which is condition (1). $\square$

Remark 7.18 (Why elliptic curves). Well-chosen elliptic-curve groups admit only the generic $O(\sqrt{q})$ attacks of §2.4, whereas index calculus solves the discrete logarithm in $\mathbb{F}_p^\times$ in subexponential time (Remark 2.18); a 256-bit curve therefore delivers the security level that a multi-thousand-bit prime field requires (Example 2.17). That efficiency, together with the availability of curves engineered for cheap in-circuit arithmetic, is why the Pallas curve of §2.6 underlies Orchard.

Remark 7.19 (Subgroup and curve validation). A recipient computing $[a]Q$ on an attacker-supplied point Q must guard against two pitfalls, the elliptic-curve forms of the validation duty in Remark 7.3.

1. *Invalid-curve attack.* The received coordinates may satisfy not E 's equation but that of a different, weaker curve which shares the same addition formulas (the chord and tangent slopes of the *Math Guide*, §“Explicit affine formulas”, involve a but never b); scalar multiplication then runs happily in the wrong group, whose order may be smooth. The implementation must check the curve equation on every incoming point.
2. *Small-subgroup attack.* When the cofactor $h > 1$, an adversary submits a point Q of small order dividing h , confining $[a]Q$ to a tiny subgroup and learning a modulo a small factor. The defences are *cofactor clearing*—multiply incoming points by h —or choosing a curve with $h = 1$.

Prime-order curves such as Pallas sidestep the cofactor issue entirely: with $h = 1$ there is no small subgroup to hit, and only the on-curve check remains. The deployed arithmetic enforces exactly that check at deserialisation, as documented in Proposition 2.27, whose compressed encoding cannot even represent a point off the curve.

7.7 Hybrid public-key encryption

Three obstacles separate the Diffie–Hellman shared group element from a usable encryption scheme. First, the shared secret is a *structured group element*—a curve point with coordinates satisfying an equation—not a uniform string of key bits. Second, the exchange by itself transports no message: it manufactures a secret, and stops. Third, we want the static–ephemeral non-interactive flow of §7.2, so that the recipient need not be online. All three are resolved by one composition: key agreement for the secret, a key-derivation function for uniformity, an authenticated cipher for the message—*hybrid public-key encryption* (HPKE). The two symmetric ingredients were built in Section 6; we recall them in the specialised shape the composition needs.

Definition 7.20 (KDF, recalled and specialised). A *key-derivation function* here is a deterministic function

$$\text{KDF}: \mathcal{K} \times \{0, 1\}^* \rightarrow \{0, 1\}^\ell,$$

mapping input keying material $k \in \mathcal{K}$ together with a context string to an ℓ -bit key. This specialises Definition 6.24 (§6.6) by omitting the salt and fixing the output length ℓ . For Theorem 7.24 we model the function as a random oracle (§3.4): distinct inputs then have independent uniform outputs, and security follows provided the adversary never queries the oracle at the hidden Diffie–Hellman secret and context. This is a source-specific random-oracle

guarantee, not a generic claim that a deterministic function extracts uniform bits from every high-min-entropy source.

Definition 7.21 (AEAD, recalled). An *authenticated encryption scheme with associated data* with key space $\{0,1\}^\ell$ —matching the KDF output—is a pair (Enc, Dec) with $\text{Enc}_K(N, ad, m) = c$ and $\text{Dec}_K(N, ad, c) \in \{m\} \cup \{\perp\}$, where N is a nonce, ad is associated data authenticated but not encrypted, and $\perp$ denotes rejection (Definition 6.13). Correctness: $\text{Dec}_K(N, ad, \text{Enc}_K(N, ad, m)) = m$. Security is authenticated-encryption security (Definition 6.15): encryptions of any two equal-length messages are CPA-indistinguishable (Definition 6.4), and no adversary can produce a fresh ciphertext decrypting to anything but $\perp$ (ciphertext integrity, Definition 6.14)—confidentiality and integrity simultaneously.

Construction 7.22 (DH-based hybrid public-key encryption). Fix ECDH over $G = \langle P \rangle$ of prime order q (Construction 7.16, in the interface of Definition 7.5), a secure KDF, and a secure AEAD as above. The recipient holds a static key pair $(b, B = [b]P)$ with B published. To *encrypt* a message m to B :

1. sample an ephemeral $e \xleftarrow{\$} \mathbb{Z}/q\mathbb{Z}$ and set $E_{\text{pk}} = [e]P$;
2. compute the shared secret $S = \text{Agree}(e, B) = [eb]P$;
3. derive the symmetric key $K = \text{KDF}(S, \text{context})$, where *context* includes E_{pk} and the protocol's labels;
4. output (E_{pk}, c) with $c = \text{Enc}_K(N, ad, m)$ for a fixed or derived nonce N —fixed is safe because each K is used once.

To *decrypt* (E_{pk}, c) with b : compute $S' = [b]E_{\text{pk}}$, derive $K' = \text{KDF}(S', \text{context})$ with E_{pk} read from the ciphertext, and output $\text{Dec}_{K'}(N, ad, c)$, which is m on an honest ciphertext; on a tampered one, ciphertext integrity makes the output $\perp$ except with negligible probability. Anyone holding B can encrypt; only the holder of b can decrypt.

Theorem 7.23 (Correctness of hybrid encryption). *For every message m and every honestly generated ciphertext (E_{pk}, c) of Construction 7.22, decryption returns m .*

Proof. Correctness of ECDH (Proposition 7.17) gives $S' = [b]([e]P) = [be]P = [eb]P = [e]B = S$. The KDF is deterministic and both sides supply the same context—the recipient recovers E_{pk} from the ciphertext itself—so $K' = K$. Correctness of the AEAD (Definition 7.21) finishes: $\text{Dec}_K(N, ad, \text{Enc}_K(N, ad, m)) = m$. $\square$

Theorem 7.24 (Passive security of hybrid encryption). *Model the KDF as a random oracle (Definition 3.13) and let the AEAD be secure in the sense of Definition 6.15. Consider a passive adversary A who sees the public key B and a single challenge ciphertext: she names two equal-length messages m_0, m_1 , receives the encryption of m_β for a hidden uniform bit β , and guesses β , with CPA advantage $\text{Adv}_{\text{cpa}}(A)$ defined as in Definition 6.4. Under CDH in G no PPT adversary has non-negligible*

advantage; concretely, if A makes at most q_H random-oracle queries, then

$$\text{Adv}_{\text{cpa}}(A) \leq 2q_H \cdot \text{Adv}_{\text{cdh}}(\mathcal{B}) + \text{Adv}_{\text{ae}}(C),$$

for explicit reductions $\mathcal{B}$ against CDH and C against the AEAD's authenticated-encryption security (of which the proof consumes only the confidentiality half, so $\text{Adv}_{\text{ae}}(C)$ is the CPA advantage of Definition 6.4 read against the AEAD), each running in essentially the same time as A .

Proof. The proof is a game hop.

Game 0 is the real CPA game: the challenge ciphertext is (E_{pk}, c) with $E_{\text{pk}} = [e]P$, $S = [eb]P$, $K = \text{KDF}(S, \text{context})$, and $c = \text{Enc}_K(N, \text{ad}, m_\beta)$. Say A wins a game when her guess β' equals β .

Game 1 replaces K by an independent uniform ℓ -bit string, leaving everything else unchanged.

The hop. In the random oracle model the two games proceed identically unless the adversary queries the oracle at exactly the point $(S, \text{context}) = ([eb]P, \text{context})$: until that query, the oracle's value there is a uniform string she has never seen, which is precisely what Game 1 hands out. But producing $S = [eb]P$ from the view $(B, E_{\text{pk}}) = ([b]P, [e]P)$ is exactly solving CDH. The reduction $\mathcal{B}$ embeds its CDH challenge as (B, E_{pk}) and simulates Game 1 around it: it draws β and a uniform ℓ -bit key K itself, encrypts m_β under K to form the challenge ciphertext, and implements the oracle by lazy sampling—a fresh uniform string for each new query, the stored answer on a repeated one, so that the oracle remains a consistent function. The simulation is a perfect run of Game 1, and the critical query occurs in it with the same probability as in Game 0, the two games being identical until it happens. One obstruction remains: in a pairing-free group $\mathcal{B}$ cannot recognise which of the queries carries $[eb]P$ (Example 2.9 shows recognising it is the DDH-test a gap group would supply). So $\mathcal{B}$ picks one of the at most q_H queries uniformly at random and outputs its first component; whenever the critical query occurs, $\mathcal{B}$'s choice is correct with probability at least $1/q_H$. Hence

$$|\Pr[A \text{ wins Game 0}] - \Pr[A \text{ wins Game 1}]| \leq q_H \cdot \text{Adv}_{\text{cdh}}(\mathcal{B}).$$

The factor q_H is a genuine guessing-type security loss, in the tightness taxonomy of Definition 1.22. Assuming *gap*-CDH—CDH hardness even given a DDH-decision oracle, which would let $\mathcal{B}$ test each query—restores tightness; we state the loose bound because it is what plain CDH buys, and honesty about the loss is part of the bound.

Game 1 analysed. The AEAD key is now uniform and independent of everything else in the adversary's view, which is exactly the hypothesis of the AEAD's confidentiality game. The reduction C forwards (m_0, m_1) to its AEAD challenger, embeds the returned ciphertext as c alongside a self-generated E_{pk} , and relays A 's guess; the AEAD challenger's hidden bit plays β , so $|\Pr[A \text{ wins Game 1}] - \frac{1}{2}| = \frac{1}{2} \text{Adv}_{\text{ae}}(C)$.

Accounting. The difference-form advantage of Definition 6.4 is twice the bit-guessing advantage $|\Pr[A \text{ wins}] - \frac{1}{2}|$ of Definition 1.15, so the triangle inequality across the hop gives

$$\text{Adv}_{\text{cpa}}(A) \leq 2(q_H \cdot \text{Adv}_{\text{cdh}}(\mathcal{B}) + \frac{1}{2} \text{Adv}_{\text{ae}}(C)) = 2q_H \cdot \text{Adv}_{\text{cdh}}(\mathcal{B}) + \text{Adv}_{\text{ae}}(C),$$

the stated bound.

Alternative hypothesis. The random oracle can be traded away, but not for Definition 6.24 alone: that guarantee is stated for a uniform independent salt, which the specialised KDF of Definition 7.20 omits, and the definition denies any generic unsalted guarantee. The trade needs an explicit hypothesis on the deployed function—the narrower premise Definition 6.24 allows for unsalted use: for a *uniform* group element S , the output $\text{KDF}(S, \text{context})$ is computationally indistinguishable from a uniform ℓ -bit string given the context. Under that hypothesis and DDH in place of CDH, Theorem 7.11 first replaces the shared secret by a uniform group element at cost Adv_{ddh} , and the stated indistinguishability then replaces K by a uniform string—no oracle and no q_H factor, at the price of the stronger decisional assumption on the group and an explicit pseudorandomness assumption on the KDF. $\square$

Remark 7.25 (Every ingredient is necessary). Dropping any piece of Construction 7.22 breaks the composition.

- *Without the KDF*, one would key the cipher with the raw group element S . Its bit-representation is non-uniform—curve points do not fill the encoding space—and in non-DDH groups it is partially predictable, the Legendre-symbol leakage of Remark 7.9 being the classical instance. The KDF standardises the format, extracts uniform bits, and, by folding E_{pk} into its context, ties the derived key to this particular ephemeral share.
- *Without AEAD integrity*, a passively confidential but malleable cipher would let an active adversary tamper with c undetectably—the XOR-malleability of Section 6’s stream ciphers is total. Ciphertext integrity (Definition 6.14) makes any modification decrypt to $\perp$ except with negligible probability, giving the symmetric payload its chosen-ciphertext robustness.
- *Without the ephemeral key*—a static-static variant—every message to B would begin from the same shared secret. The fresh per-message ephemeral, with E_{pk} bound into the KDF context, gives each ciphertext separate key material, so nothing relies on nonce uniqueness under a reused key. What it does *not* supply is forward secrecy against later compromise of the recipient’s static key b : from a recorded E_{pk} , an attacker who learns b recomputes $[b]E_{pk}$ and with it the ciphertext’s key. Erasing e buys only the sender-side guarantee that the key derives from no stored secret of the sender.

7.8 Key privacy

Theorem 7.24 answers the eavesdropper and leaves the scanner untouched. Her posture is different: she holds a published ciphertext against every public key she can enumerate and asks not what it says but *to whom*. Confidentiality is silent on the question—a scheme may hide the message perfectly and still stamp the recipient’s key on every ciphertext. The property that denies her is *key privacy*, due to Bellare, Boldyreva, Desai, and Pointcheval. We state it for a generic public-key scheme, then prove it for the bare static-ephemeral flow with the message folded in—ElGamal encryption, the shape whose key privacy the upper volumes consume.

Definition 7.26 (Key privacy under chosen-plaintext attack, IK-CPA). A *public-key encryption scheme* Π over pp consists of a randomised $\text{KeyGen}(pp) \rightarrow (sk, pk)$, a randomised $\text{Enc}_{pk}(m) \rightarrow c$, and a deterministic $\text{Dec}_{sk}(c)$ with $\text{Dec}_{sk}(\text{Enc}_{pk}(m)) = m$ for every key pair in the support of KeyGen and every message m . For an adversary $\mathcal{A}$ the game $\text{IK-CPA}_{\Pi}^{\mathcal{A}}(\lambda)$ runs as follows.

1. The challenger samples $(sk_0, pk_0), (sk_1, pk_1) \leftarrow \text{KeyGen}(pp)$ independently and sends (pk_0, pk_1) .
2. The adversary outputs a message m .
3. The challenger draws a hidden bit $\beta \xleftarrow{\$} \{0, 1\}$ and sends $c \leftarrow \text{Enc}_{pk_{\beta}}(m)$.
4. The adversary outputs a bit β' .

The advantage is $\text{Adv}_{\Pi}^{\text{ik-cpa}}(\mathcal{A}) = |\Pr[\beta' = 1 \mid \beta = 0] - \Pr[\beta' = 1 \mid \beta = 1]|$; the scheme is *key-private* (IK-CPA) if every PPT adversary’s advantage is negligible in λ . The message is the adversary’s own, so nothing about it is hidden; what the game hides is the key.

Construction 7.27 (ElGamal encryption). Fix $pp = (G, P, q)$ as in Construction 7.16, written

additively, and take group elements as messages; write EG for the scheme. A key pair carries its own base H : in the *fixed-base* variant $H = P$; in the *per-key-base* variant KeyGen samples $H \xleftarrow{\$} G \setminus \{O\}$, the generalised interface of Remark 7.7.

- KeyGen: choose H as above and $x \xleftarrow{\$} \mathbb{Z}/q\mathbb{Z}$; output $\text{sk} = x$ and $\text{pk} = (H, Y)$ with $Y = [x]H$.
- $\text{Enc}_{(H,Y)}(M)$: sample $r \xleftarrow{\$} \mathbb{Z}/q\mathbb{Z}$; output $(R, C) = ([r]H, [r]Y + M)$.
- $\text{Dec}_x(R, C)$: output $C - [x]R$.

Correctness is the scalar identity in the proof of Proposition 7.17, valid for any base in G : $[x]R = [x]([r]H) = [r]([x]H) = [r]Y$, so $C - [x]R = M$. The scheme is the static-ephemeral flow of §7.2 with the shared secret $[rx]H$ spent as a one-time pad on a group element; the fixed-base variant is ElGamal's original scheme transposed from $\mathbb{F}_p^\times$ to a prime-order group.

Theorem 7.28 (ElGamal is key-private under DDH). *For ElGamal over G (Construction 7.27, either base variant), every adversary A yields two distinguishers $\mathcal{B}_0, \mathcal{B}_1$ against DDH in G (Definition 2.5, read with $[x]P$ in place of g^x as in Construction 7.16), each running in the time of A plus one key generation and four scalar multiplications, with*

$$\text{Adv}_{\text{EG}}^{\text{ik-cpa}}(A) \leq \text{Adv}_{\text{ddh}}(\mathcal{B}_0) + \text{Adv}_{\text{ddh}}(\mathcal{B}_1).$$

Under the DDH assumption ElGamal is therefore IK-CPA: a ciphertext is computationally independent of the public key it was made for.

Proof. The proof is a hybrid through a game that uses no key at all. Write H_0 and H_1 for the IK-CPA game with β fixed to 0 and to 1, and p_0, p_1 for the probability that A outputs 1 in each, so that $\text{Adv}_{\text{EG}}^{\text{ik-cpa}}(A) = |p_0 - p_1|$. Let H_* be the game whose challenger, after the same key generation and the same message M from A , answers with $(R, C) = (U_1, U_2 + M)$ for $U_1, U_2 \xleftarrow{\$} G$ independent and uniform, and let p_* be the corresponding probability. The challenge of H_* is computed without reference to either key, and the triangle inequality $|p_0 - p_1| \leq |p_0 - p_*| + |p_* - p_1|$ reduces the claim to bounding each hop by a DDH advantage.

The hop $H_0 \rightarrow H_*$. The distinguisher $\mathcal{B}_0$ receives a triple $(X, Y, Z) = ([a]P, [b]P, [c]P)$ with $c = ab$ or c uniform. It samples $s \xleftarrow{\$} (\mathbb{Z}/q\mathbb{Z})^\times$ in the per-key-base variant and sets $s = 1$ in the fixed-base variant, plants the challenge in key 0 as

$$H_0 = [s]P, \quad Y_0 = [s]X = [a]H_0,$$

generates $(x_1, (H_1, Y_1))$ honestly, and hands A the pair $((H_0, Y_0), (H_1, Y_1))$. For invertible s the base $[s]P$ is uniform on $G \setminus \{O\}$, and $Y_0 = [a]H_0$ with a uniform is an honest public key whose private key a the distinguisher never learns—nor needs, the game offering no decryption. On receiving M it replies with

$$(R, C) = ([s]Y, [s]Z + M)$$

and outputs A 's bit. If $c = ab$, then $R = [sb]P = [b]H_0$ and $[s]Z = [sab]P = [b]([a]H_0) = [b]Y_0$, so (R, C) is an honest encryption under key 0 with ephemeral $r = b$ uniform: the view of A is exactly H_0 . If c is uniform, then $[s]Z = [sc]P$ is uniform in G and independent of everything else because s is invertible, and $R = [b]H_0$ is uniform in G and independent of everything else because H_0 generates G and b is independent of a, c, s , of key 1, and of M , which A chose before seeing R : the view is exactly H_* . Hence $|p_0 - p_*| = \text{Adv}_{\text{ddh}}(\mathcal{B}_0)$.

The hop $H_* \rightarrow H_1$ is the same with the challenge planted in key 1 and key 0 generated honestly, giving $\mathcal{B}_1$ with $|p_* - p_1| = \text{Adv}_{\text{ddh}}(\mathcal{B}_1)$. Each distinguisher performs one honest key generation and the four scalar multiplications $[s]P, [s]X, [s]Y, [s]Z$ beyond running A . Summing the two hops gives the bound. $\square$

Remark 7.29 (What key privacy buys upstairs). The hybrid transfers verbatim to Construction 7.22: planting the challenge in the recipient’s key and the ephemeral share exactly as above makes the shared secret S uniform, and once it is, the pair (E_{pk}, c) with $c = \text{Enc}_{\text{KDF}(S, \text{context})}(N, ad, m)$ is computed without reference to the recipient’s key—*provided* the KDF context names no recipient key, which it must not. The deployed KDF takes $(\text{sharedSecret}, \text{epk})$ (§6.6): the shared secret is the keying material, the context is epk under a personalisation label, and no key of the recipient enters. The protocol specification requires the composed note-encryption scheme to be key-private (§“Key Derivation”): a chain scanner holding a note ciphertext against every address she can enumerate learns nothing about which one it was sent to. The second use is the group-element share of diversified-address unlinkability. A diversified address $(G_d, \text{pk}_d) = (G_d, [\text{ivk}] G_d)$ (Remark 7.30) is a per-key-base ElGamal public key, and a fresh address of the same ivk over a new base $G_{d'}$ has, when that base is modelled as a uniform group element, the distribution of an encryption of O under that key; telling which of two authorities a third address belongs to is then the IK-CPA game, and Theorem 7.28 answers it. The protocol specification sets up that correspondence, modelling the group hash behind the base as a random oracle (§“DiversifyHash^{Sapling} and DiversifyHash^{Orchard} Hash Functions”); the *Wallet Guide* states the resulting unlinkability requirement (§“Diversified addresses”).

7.9 Shielded-note delivery as deployed hybrid encryption

The chain assembled in this section is not an analogy for what the deployed protocol does; it is the deployed protocol, run in band on the public ledger.

Remark 7.30 (Note delivery is HPKE on the ledger). A shielded address embeds a static public key $\text{pk}_d = [\text{ivk}] G_d$ on the protocol’s curve (Pallas, in Orchard), where the private scalar ivk is the recipient’s incoming viewing key. The base G_d is not a protocol-wide generator but a per-address *diversified base*, the hash-to-curve image (§3.7) of the address’s diversifier—the generalised interface of Remark 7.7 exists precisely to accommodate it ($\text{DiversifiedTransmissionKey}$ is derived as $\text{pk}_d = [\text{ivk}] g_d$); the *Ironwood Guide* develops the key hierarchy (§“Viewing keys”) and address encoding (§“Diversified addresses”) behind these bases. The sender runs Construction 7.22 over that base, with one deployed refinement: the ephemeral scalar esk is not sampled directly but derived by $\text{PRF}^{\text{expand}}$ (Remark 5.14) from the note’s random seed ($\text{ZIP 212; RandomSeed}::\text{esk_inner}$ applies $\text{PrfExpand}::\text{ORCHARD_ESK}$), so that the uniform-ephemeral hypothesis of Theorem 7.24 is met through the PRF’s pseudorandomness and the recipient can later re-derive esk from the delivered seed. The sender publishes $\text{epk} = [\text{esk}] G_d$ in the transaction, derives

$$K = \text{KDF}([\text{esk} \cdot \text{ivk}] G_d, \text{epk})$$

by the personalised BLAKE2b call of §6.6, and AEAD-encrypts the note plaintext—a version byte, the recipient’s diversifier, the value, the note’s random seed, and the memo—under K with ChaCha20-Poly1305 (Construction 6.18), attaching the ciphertext to the transaction. The nonce is fixed to zero and the associated data is empty, safe because each K keys exactly one encryption (protocol specification §“Encryption (Sapling and Orchard)”); the `zcash_note_encryption` crate’s `encrypt_note_plaintext`, which chains `ka_agree_enc`, `kdf`, and `ChaCha20Poly1305` with the zero nonce; the Orchard instantiation of the `Domain` trait is in the `orchard` crate).

Chain scanners see (epk, c) in every shielded output, but lacking ivk they learn nothing (Theorem 7.24); the recipient *trial-decrypts*, deriving $K' = \text{KDF}([\text{ivk}] \text{epk}, \text{epk})$ for each incoming transaction and keeping those outputs whose AEAD tag verifies *and* whose plaintext passes two consistency checks: the note commitment recomputed from it must match the commitment on chain,

and the esk re-derived from its random seed must reproduce the published epk —the notes addressed to them (protocol specification §“Decryption using an Incoming Viewing Key (Sapling and Orchard)”). Non-interactivity is essential, and it is exactly the static–ephemeral flow of §7.2: the recipient need never have been online when the note was sent. Authenticity of the static key—the concern of §7.5—is supplied by mechanism (2) of Remark 7.15: the address is part of the recipient’s verified payment credential, so the man-in-the-middle substitution has nowhere to stand; and, as noted there, this authenticates the recipient rather than the sender. The complementary guarantees on the note’s *contents*—that the value is well-formed and conserved, that the commitment on chain matches the plaintext delivered—come not from the encryption but from the accompanying zero-knowledge proof.

The primitive toolbox now covers the delivery of a secret to a party who was never online to negotiate it. What travels alongside that ciphertext—the authorising signatures over the transaction that carries it—is the business of the next section.

8 Digital signatures

The adversary of this section is the *forger*. It watches the network accept spend after spend, each authorised by a signature; it may coax the legitimate signer into signing any messages of its choosing; and it wins if it can then exhibit one signature the signer never made—a spend of coins it does not control, an authorisation conjured from public data alone. Every verifier in a consensus network checks every signature, so the forger’s target is public and its victory is total: a single forged spend authorisation steals, and a single forged balance certificate mints value from nothing. The primitive that defeats it, the digital signature, is the last cryptographic object of this volume built directly on the discrete logarithm; it will also turn out, in its Schnorr form, to be the first *proof of knowledge* we construct—the bridge to the proof systems of Section 9.

8.1 Signatures versus MACs

The message authentication codes of §6.3 already defeat a forger, but only between two parties who share a secret key: verification runs $Vrfy_k$, so whoever can *check* a tag can also *mint* one. Authentication by MAC is therefore private and repudiable—a verifier cannot exhibit the tag to a third party as evidence, since the verifier could have produced it. A blockchain inverts every one of these constraints. A transaction’s authorisation must be checked by every full node, none of whom the signer has met and none of whom may be capable of forging it; and the check must be transferable, so that a block, once validated, convinces everyone else.

A *digital signature* meets these demands by splitting the key. Signing uses a secret key sk ; verification uses a matching *public* key pk that may be published to the world. Anybody can verify; only the holder of sk can sign; and because verification needs no secret, a valid signature is *non-repudiable* evidence that the holder of sk signed. The price is computational: where the Carter–Wegman MAC of §6.3 was information-theoretically secure, every signature scheme in this section rests on the hardness of the discrete logarithm.

Zcash’s Orchard protocol uses two closely related signature schemes, both Schnorr-type schemes over an elliptic curve, and both derived from the one identification protocol this section develops (protocol specification § 5.4.7):

- the *spend authorisation signature*, whose key pair can be *re-randomised* so that successive uses are unlinkable (protocol specification § 4.15), and
- the *binding signature*, which simultaneously certifies the consistency of a transaction’s value commitments and proves knowledge of a discrete logarithm (protocol specification § 4.14).

Conventions for this section. Throughout, $\mathbb{G}$ is a cyclic group of prime order q , written additively, with a fixed generator G ; scalar multiplication is $[x]P$ for $x \in \mathbb{F}_q$ and $P \in \mathbb{G}$. Concretely $\mathbb{G}$ is the Pallas group of §2.6, whose order is the 255-bit prime q , so scalars live in the field $\mathbb{F}_q$. Because $\mathbb{G}$ has prime order and G generates it, the map $x \mapsto [x]G$ is a bijection $\mathbb{F}_q \rightarrow \mathbb{G}$: the discrete logarithm of any $P \in \mathbb{G}$ to base G always *exists* and is *unique*. Hardness of computing it is the DLog assumption of §2.1, transcribed additively.

8.2 Syntax and security goal

Definition 8.1 (Digital signature scheme). A *digital signature scheme* is a triple of PPT algorithms $\Pi = (\text{KeyGen}, \text{Sign}, \text{Verify})$ with, for each security parameter λ , a message space $\mathcal{M}_\lambda$:

- $\text{KeyGen}(1^\lambda)$ outputs a key pair (pk, sk) ;
- $\text{Sign}(sk, m)$, for $m \in \mathcal{M}_\lambda$, outputs a signature σ ;
- $\text{Verify}(pk, m, \sigma)$ is deterministic and outputs a bit.

Correctness requires that for every λ , every (pk, sk) in the support of $\text{KeyGen}(1^\lambda)$, and every $m \in \mathcal{M}_\lambda$,

$$\Pr[\text{Verify}(pk, m, \text{Sign}(sk, m)) = 1] = 1,$$

the probability over the coins of Sign . Probability-1 correctness is called *perfect*; a scheme achieving only overwhelming-probability correctness tolerates a $\text{negl}(\lambda)$ failure rate. Every scheme in this section is perfectly correct.

The security goal transcribes the MAC forgery game of Definition 6.8 into the public-key setting: the adversary now *receives* the verification key, and its oracle access models a signer who can be induced to sign arbitrary messages.

Definition 8.2 (Existential unforgeability under chosen-message attack). For a signature scheme Π and adversary $\mathcal{A}$, the game $\text{Forge}_\Pi^{\mathcal{A}}(\lambda)$ runs as follows.

1. The challenger runs $(pk, sk) \leftarrow \text{KeyGen}(1^\lambda)$.
2. The adversary $\mathcal{A}$ runs on input pk with oracle access to $\text{Sign}(sk, \cdot)$; let Q be the set of messages it submits to the oracle, and let (m^*, σ^*) be its output.
3. The game outputs 1 iff $\text{Verify}(pk, m^*, \sigma^*) = 1$ and $m^* \notin Q$.

Scheme Π is *existentially unforgeable under chosen-message attack* (EUF-CMA) if for every PPT $\mathcal{A}$,

$$\text{Adv}_{\Pi}^{\text{forge}}(\mathcal{A}, \lambda) := \Pr[\text{Forge}_\Pi^{\mathcal{A}}(\lambda) = 1] \in \text{negl}(\lambda).$$

Two features of the definition deserve emphasis. The oracle access is *adaptive* and bounded only by the adversary's polynomial running time: it may choose each query after seeing all previous signatures. And the forgery is *existential*: the adversary wins with a valid signature on *any* message never submitted to the oracle, however meaningless—the definition does not ask the forged message to be useful, so security under it protects even messages the signer would consider absurd.

Definition 8.3 (Strong unforgeability). The *strong* variant, sUF-CMA, replaces the winning condition $m^* \notin Q$ by: the pair (m^*, σ^*) was not among the message–signature pairs returned

by the oracle.

Strong unforgeability forbids producing even a *new* signature on an *already-signed* message. Plain EUF-CMA permits a scheme in which, given one valid signature, anyone can cheaply compute a second, different valid signature on the same message; such *malleability* is harmless when a signature only certifies origin, but fatal whenever a signature doubles as a unique token—certain anti-replay mechanisms, for instance, identify a message by its signature and break if signatures are malleable.

8.3 The hash-and-sign paradigm

The textbook mechanisms—RSA, Schnorr, ECDSA—sign only fixed-size inputs: an element of $\mathbb{Z}/N\mathbb{Z}$ for RSA, a scalar in $\mathbb{F}_q$ for Schnorr and ECDSA. Arbitrary messages are accommodated by hashing first, and the composition is provably safe.

Theorem 8.4 (Hash-and-sign). *Let Π' be a signature scheme with message space $\mathcal{Y}_\lambda$, and let $H: \{0, 1\}^* \rightarrow \mathcal{Y}_\lambda$ be drawn from a collision-resistant family (Definition 3.7). Define Π with message space $\{0, 1\}^*$ by*

$$\text{Sign}(sk, m) := \text{Sign}'(sk, H(m)), \quad \text{Verify}(pk, m, \sigma) := \text{Verify}'(pk, H(m), \sigma).$$

If Π' is EUF-CMA and H is collision resistant, then Π is EUF-CMA; concretely, for every adversary A against Π there are comparably efficient $\mathcal{B}_1$ against H and $\mathcal{B}_2$ against Π' with

$$\Pr[A \text{ wins}] \leq \text{Adv}_H^{\text{col}}(\mathcal{B}_1) + \text{Adv}_{\Pi'}^{\text{forge}}(\mathcal{B}_2).$$

Proof. Let (m^*, σ^*) be a valid forgery of A on a fresh message $m^* \notin Q$, and split on the digest. *Case 1: the digest is reused*, i.e. $H(m^*) = H(m)$ for some queried $m \in Q$. Then $m^* \neq m$ but their digests agree, so the pair (m^*, m) is a collision in H ; the reduction $\mathcal{B}_1$, which generates a key pair with KeyGen' itself and answers A 's signing queries by signing digests under that secret key—the collision game grants no signing oracle—outputs it. *Case 2: the digest is fresh*, i.e. $H(m^*)$ differs from the digest of every queried message. Then $(H(m^*), \sigma^*)$ is a valid forgery against Π' on a message never submitted to its signing oracle, and $\mathcal{B}_2$ outputs it. The two cases are exhaustive, so the union bound on the winning event gives the stated inequality. $\square$

Remark 8.5 (What “hash-and-sign” hides). In all the Schnorr-type schemes below, the hash absorbs more than the message: the public key and the per-signature commitment value enter the hash alongside m . Hashing the public key prevents key-substitution (duplicate-signature) attacks, in which a signature valid under one key is exhibited as valid under another; hashing the commitment is exactly what the Fiat–Shamir transform of §8.5 requires. “Hash-and-sign” in practice therefore means hashing more than the message; Theorem 8.4 captures only the message-compression part of the discipline.

8.4 The Schnorr identification protocol

Every scheme in the rest of this section is a costume worn by a single interactive protocol, in which a prover convinces a verifier that it *knows* the discrete logarithm of a public point—without revealing anything about it. We develop the protocol self-contained, with concrete statements and proofs; the general vocabulary it exemplifies (*Sigma-protocols*, of which it is the canonical instance: three-move, public-coin, honest-verifier zero-knowledge (HVZK) proofs of knowledge) is deliberately deferred to Section 9, where it is developed for arbitrary relations.

The object of the protocol is the discrete-log relation

$$R_{\text{dl}} = \{ (X, x) \in \mathbb{G} \times \mathbb{F}_q : X = [x]G \} :$$

the *statement* is a public point X , the *witness* its discrete logarithm x .

Construction 8.6 (Schnorr identification). Fix $\mathbb{G} = \langle G \rangle$ of prime order q . The prover holds a secret key $x \in \mathbb{F}_q$; both parties hold the public key $X = [x]G$. The protocol has three moves.

1. *Commitment*. The prover samples $r \xleftarrow{\$} \mathbb{F}_q$ uniformly and sends $R = [r]G$. We call R the *commitment* (or *nonce point*) and r the *nonce*.
2. *Challenge*. The verifier samples $c \xleftarrow{\$} \mathbb{F}_q$ uniformly and sends c .
3. *Response*. The prover sends $s = r + cx$ in $\mathbb{F}_q$.

The verifier accepts iff

$$[s]G = R + [c]X. \quad (2)$$

A triple (R, c, s) satisfying (2) is an *accepting transcript* for X . The verifier's only message is a uniformly random value chosen independently of R , so the protocol is *public-coin*: the verifier keeps no secrets and exercises no discretion.

Three properties make this protocol what it is: honest runs always accept; answering two distinct challenges on one commitment is possible only for a party who knows x ; and a transcript reveals nothing about x . We prove each in turn.

Proposition 8.7 (Perfect completeness). *If prover and verifier are honest, the verifier accepts with probability 1.*

Proof. With $R = [r]G$, $X = [x]G$ and $s = r + cx$, linearity of scalar multiplication gives

$$[s]G = [r + cx]G = [r]G + [cx]G = R + [c]([x]G) = R + [c]X,$$

which is exactly (2). The computation holds identically for every choice of r and c . $\square$

Theorem 8.8 (2-special soundness). *There is an efficient extractor that, given two accepting transcripts (R, c, s) and (R, c', s') for the same statement X with the same commitment R but distinct challenges $c \neq c'$, outputs the discrete logarithm x of X .*

Proof. Both transcripts satisfy (2); subtracting the two equations in $\mathbb{G}$ cancels R :

$$[s - s']G = [c - c']X = [(c - c')x]G.$$

Since G generates a group of prime order q , equality of the two multiples forces the equality of scalars $s - s' = (c - c')x$ in $\mathbb{F}_q$. The hypothesis $c \neq c'$ makes $c - c'$ a nonzero element of the field $\mathbb{F}_q$, hence invertible, so the extractor outputs

$$x = (s - s') (c - c')^{-1} \in \mathbb{F}_q.$$

Verification that $X = [x]G$ is direct, and the whole computation costs a single field inversion. $\square$

Theorem 8.8 is the central mechanism of this section. It makes the protocol a proof of *knowledge*: below, it converts a convincing prover into an extractable witness; after the Fiat–Shamir transform it is the source of unforgeability; and in §8.6 it reappears with its sign reversed, as the reason a signer must never reuse a nonce.

Remark 8.9 (From special soundness to knowledge, by rewinding). Suppose a (possibly dishonest) prover makes the verifier accept with probability ε noticeably larger than $1/q$ —the probability of simply guessing the single challenge for which a prepared response works. Run the prover once: record its commitment R , feed it a uniform challenge c , and note whether its response accepts. Now *rewind* it to the moment just after it sent R —restore its internal state, which is possible because we run the prover as a subroutine—and feed a fresh independent challenge c' . A standard averaging argument shows that with probability roughly ε^2 , both runs accept and $c \neq c'$; the argument is the *heavy-row lemma*, an instance of Markov’s inequality applied to the matrix whose rows are the prover’s random coins and whose columns are challenges (*Math Guide*, §“Random variables and expectation”). The two accepting transcripts share R , so the extractor of Theorem 8.8 recovers x . Making the verifier accept with non-trivial probability is therefore, up to rewinding, the same as *knowing* x . The same rewinding argument, sharpened into the forking lemma, carries the signature analysis of §8.6.

Theorem 8.10 (Special honest-verifier zero knowledge). *There is an efficient simulator Sim that, on input a statement $X \in \mathbb{G}$ and a challenge $c \in \mathbb{F}_q$, outputs an accepting transcript (R, c, s) distributed identically to an honest execution conditioned on the challenge being c . Consequently, for a uniform challenge, the distribution of simulated transcripts coincides exactly with the distribution of real ones.*

Proof. The simulator works *backwards*: it samples $s \xleftarrow{\$} \mathbb{F}_q$ uniformly, sets

$$R := [s]G - [c]X,$$

and outputs (R, c, s) —accepting by construction, since (2) rearranges to exactly this definition of R .

Compare distributions with the challenge fixed to c . In a real honest transcript, r is uniform on $\mathbb{F}_q$, $R = [r]G$, and $s = r + cx$; the map $r \mapsto r + cx$ is a translation of $\mathbb{F}_q$, hence a bijection, so s is uniform on $\mathbb{F}_q$, and $R = [s - cx]G = [s]G - [c]X$ is the same deterministic function of s that the simulator uses. The real transcript *is*: sample s uniform, set $R = [s]G - [c]X$ —precisely the simulated distribution. The two distributions are identical, not merely statistically close, and averaging over a uniform c preserves the identity. $\square$

Remark 8.11 (Why honest-verifier zero knowledge suffices). The simulator of Theorem 8.10 matches the real distribution only when the challenge is sampled independently of R . A cheating verifier could instead choose c as a function of R , and against such a verifier the simulator—which commits to R only *after* seeing c —no longer obviously works. For our purposes honest-verifier zero knowledge is exactly enough: after the Fiat–Shamir transform the challenge is a *hash* of R , and in the random oracle model that hash behaves like an honest, R -independent coin. The deeper point of the theorem is what the simulator’s existence *proves*: since anyone, without x , can manufacture transcripts with the true distribution, a transcript carries no information about x beyond what the statement X already reveals.

8.5 From identification to signature via the Fiat–Shamir transform

An identification protocol convinces one verifier, once, interactively. A signature must convince everyone, forever, on its own. The *Fiat–Shamir transform* removes the interaction by making the prover compute the challenge itself—as a hash of everything the verifier would have seen before issuing it, with the message folded in so that the resulting object certifies m . The transform is generic: it converts *any* Sigma-protocol into a non-interactive proof, and, with a message in the hash, into a signature. This subsection owns only the Schnorr instance; the general transform, its subtleties and its failure modes are the business of Section 9 (§“The Fiat–Shamir transform: from interactive to non-interactive”).

Construction 8.12 (Schnorr signature). Fix $\mathbb{G} = \langle G \rangle$ of prime order q and a hash function $H: \{0,1\}^* \rightarrow \mathbb{F}_q$. Messages are arbitrary bit strings.

- $\text{KeyGen}(1^\lambda)$: sample $x \xleftarrow{\$} \mathbb{F}_q$, set $X = [x]G$, output $(pk, sk) = (X, x)$.
- $\text{Sign}(x, m)$: sample $r \xleftarrow{\$} \mathbb{F}_q$, set $R = [r]G$, compute the challenge

$$c = H(R \parallel X \parallel m),$$

set $s = r + cx$ in $\mathbb{F}_q$, and output $\sigma = (R, s)$.

- $\text{Verify}(X, m, (R, s))$: recompute $c = H(R \parallel X \parallel m)$ and accept iff $[s]G = R + [c]X$.

Proposition 8.13 (Perfect correctness). *The Schnorr signature scheme is perfectly correct.*

Proof. For an honest signature (R, s) with $R = [r]G$, $c = H(R \parallel X \parallel m)$ and $s = r + cx$, the verifier recomputes the same c —the challenge is a deterministic function of (R, X, m) , all of which the verifier holds—and the check $[s]G = R + [c]X$ holds by the completeness computation of Proposition 8.7. $\square$

Remark 8.14 (Two conventions). Two presentational choices in Construction 8.12 deserve note.

1. The signature transmits (R, s) and omits c , which the verifier recomputes. The space-saving variant transmits (c, s) instead: the verifier recovers $R = [s]G - [c]X$ (the simulator’s equation, read as reconstruction) and checks $H(R \parallel X \parallel m) = c$. The two are interchangeable; the deployed schemes below use the (R, s) form.
2. The hash takes the public key X alongside R and m —*key-prefixing*. The single-key unforgeability proof of §8.6 does not require it, but it forecloses attacks that relate signatures under different keys (Remark 8.5), it is standard in modern instantiations such as EdDSA, and it earns its keep in the re-randomised setting of §8.8.

8.6 Security in the random oracle model and the forking lemma

What must a forger of Schnorr signatures accomplish? A valid pair (R^*, s^*) on a fresh message is an accepting transcript of the identification protocol whose challenge is $H(R^* \parallel X \parallel m^*)$ —a value the forger cannot choose freely if the hash is honest. The security argument makes “honest” precise by idealising H , then turns the rewinding of Remark 8.9 into a quantitative tool.

Definition 8.15 (Random oracle, general range). The random oracle of Definition 3.13 generalises verbatim to an arbitrary finite range: a random oracle with range Y is a function $O: \{0,1\}^* \rightarrow Y$ drawn uniformly from the set of all such functions, realised lazily by sampling each new answer $O(z) \xleftarrow{\$} Y$ on first query. For Schnorr signatures the range is the challenge space $Y = \mathbb{F}_q$, and the ROM for signatures models $H(\cdot) = O(\cdot)$, with every party—signer, verifier, forger, and reduction—sharing oracle access (Definition 1.26).

The model remains a heuristic, with the uninstantiability caveat and the working stance of Remark 1.29: a ROM proof rules out all attacks that treat the hash as a black box—the overwhelming majority—and it is the standard yardstick for Fiat–Shamir signatures. The reductions below exploit both powers of Proposition 3.14: *programmability* (the reduction chooses oracle answers

on the fly, provided they appear uniform) and *observability* (the reduction sees every query the adversary makes).

The first tool disposes of the signing oracle: in the ROM, the zero-knowledge simulator of Theorem 8.10 answers signing queries without the secret key.

Lemma 8.16 (Signature simulation). *In the ROM there is an efficient simulator that, given only the public key X and the ability to program H , answers signing queries with signatures distributed identically to genuine ones, except that it aborts with probability at most $q_s(q_s + q_h)/q$ over an attack making q_s signing and q_h hash queries.*

Proof. To sign m without x : sample $c \xleftarrow{\$} \mathbb{F}_q$ and $s \xleftarrow{\$} \mathbb{F}_q$ uniformly, set $R := [s]G - [c]X$ —the HVZK simulator of Theorem 8.10—then program

$$H(R \| X \| m) := c,$$

and output (R, s) . The signature verifies by construction, and by Theorem 8.10 its distribution matches a real signature's exactly: in both cases s is uniform, and c is uniform because a fresh oracle answer is.

Programming fails only if the entry $H(R \| X \| m)$ is already defined—by an earlier adversary hash query or an earlier simulated signature. The point $R = [s]G - [c]X$ is a fixed translate of $[s]G$ for uniform s , hence uniform over the q elements of $\mathbb{G}$; at most $q_h + q_s$ oracle entries exist at any point of the attack, so a given signing query collides with probability at most $(q_h + q_s)/q$, and the union bound over q_s signing queries gives the stated abort probability. Conditioned on no abort, the adversary's view is identical to the real game. $\square$

Lemma 8.16 reduces forgery to a pure discrete-log problem: the reduction can play the whole EUF-CMA game knowing only X . What remains is to convert one forgery into *two* accepting transcripts on the same commitment—the input that Theorem 8.8 demands. The quantitative engine is the forking lemma, in the game-based form due to Bellare and Neven.

Theorem 8.17 (General forking lemma). *Fix an integer $Q \geq 1$, a finite set C with $|C| = h \geq 2$, and a randomised input generator Gen . Let B be a randomised algorithm that on input y and challenges $c_1, \dots, c_Q \in C$ returns a pair (I, out) with $I \in \{0, 1, \dots, Q\}$, where $I = 0$ means failure, and let*

$$\text{acc} := \Pr[I \geq 1 : y \leftarrow \text{Gen}; c_1, \dots, c_Q \xleftarrow{\$} C; (I, \text{out}) \leftarrow B(y, c_1, \dots, c_Q)].$$

Define the forking algorithm F_B on input y : pick coins ρ for B and $c_1, \dots, c_Q \xleftarrow{\$} C$; run $(I, \text{out}) \leftarrow B(y, c_1, \dots, c_Q; \rho)$; if $I = 0$, return $\perp$; otherwise resample $c'_1, \dots, c'_Q \xleftarrow{\$} C$ afresh and run $(I', \text{out}') \leftarrow B(y, c_1, \dots, c_{I-1}, c'_1, \dots, c'_Q; \rho)$ with the same coins ρ ; if $I' = I$ and $c_I \neq c'_I$, return $(I, \text{out}, \text{out}')$, else $\perp$. Then

$$\text{frk} := \Pr[F_B(y) \neq \perp : y \leftarrow \text{Gen}] \geq \text{acc} \left(\frac{\text{acc}}{Q} - \frac{1}{h} \right),$$

the probability also over the coins of F_B .

Proof idea. Condition on the coins ρ and the prefix $c_1, \dots, c_{I-1}$ up to the forking point. For each fixed prefix there is (averaging over the suffix) a set S of good values of c_I on which B succeeds with index I ; write $p = |S|/h$. The fork succeeds at this prefix iff the first draw c_I and the independent re-draw c'_I both land in S and differ: two independent uniform draws land in S together with probability p^2 , and subtracting the diagonal of equal draws costs at most $p \cdot 1/h$, so the per-prefix fork probability is at least $p(p - 1/h)$. Taking expectations over prefixes, $\mathbb{E}[p^2] \geq (\mathbb{E}[p])^2$, since

$\mathbb{E}[p^2] - (\mathbb{E}[p])^2 = \mathbb{E}[(p - \mathbb{E}[p])^2] \geq 0$ by linearity of expectation (*Math Guide*, §“Random variables and expectation”); this squares the average success probability. The index I ranging over Q possible positions introduces the factor $1/Q$, and collecting terms yields $\text{frk} \geq \text{acc}^2/Q - \text{acc}/h$. The squaring step is where the *squared* probability—and hence the tightness loss examined below—enters. $\square$

Theorem 8.18 (EUF-CMA security of Schnorr signatures in the ROM). *Model H as a random oracle with range $\mathbb{F}_q$. If an adversary A runs in time t , makes at most q_h hash queries and q_s signing queries, and forges with probability ε , then there is an algorithm $\mathcal{D}$ running in time approximately $2t$ that computes discrete logarithms in $\mathbb{G}$ with probability*

$$\varepsilon' \geq \frac{\varepsilon^2}{q_h + q_s + 1} - \frac{q_h + q_s + 1}{q} - \frac{q_s(q_h + q_s)}{q}.$$

In particular, if DLog is hard in $\mathbb{G}$, the Schnorr signature scheme is EUF-CMA in the ROM.

Proof structure. Wrap A as the algorithm B of Theorem 8.17, with input $y = X$ —the DLog challenge, so Gen outputs $X = [x]G$ for uniform x —challenge set $C = \mathbb{F}_q$ (so $h = q$), and $Q = q_h + q_s + 1$ prepared challenges. Algorithm B runs A on $pk = X$, answering the j -th *distinct* hash query with c_j from its challenge list and answering signing queries by the simulator of Lemma 8.16, whose programming aborts cost the $q_s(q_h + q_s)/q$ term.

Suppose A outputs a valid fresh forgery $(m^*, (R^*, s^*))$, and let $c^* = H(R^* \| X \| m^*)$. The critical hash input $R^* \| X \| m^*$ must actually have been queried during the run: otherwise c^* is a uniform value A has never seen, and the verification equation $[s^*]G = R^* + [c^*]X$ then holds only with probability $1/q$ over the oracle’s choice—the $(q_h + q_s + 1)/q$ term collects these guessing events (the $+1$ covering the query B itself makes to verify the forgery). So B outputs the index I of the critical query together with $\text{out} = (R^*, c^*, s^*, m^*)$, and $\text{acc} \geq \varepsilon$ minus the loss terms above.

Now fork. Both runs of F_B share the coins ρ and the challenge prefix $c_1, \dots, c_{I-1}$, which together determine A ’s behaviour up to the moment it produces the critical query; both runs therefore query the *same* critical input $R^* \| X \| m^*$ at index I , but receive different answers $c^* \neq c'^*$. Success of the fork yields two accepting transcripts (R^*, c^*, s^*) and (R^*, c'^*, s'^*) with the same commitment and distinct challenges, whereupon the extractor of Theorem 8.8 computes

$$x = (s^* - s'^*)(c^* - c'^*)^{-1},$$

the discrete logarithm of X . Substituting acc , $h = q$ and $Q = q_h + q_s + 1$ into $\text{frk} \geq \text{acc}^2/Q - \text{acc}/h$ gives the stated bound; the running time is two runs of A plus bookkeeping. $\square$

Example 8.19 (The tightness loss, in numbers). The dominant term of Theorem 8.18 is $\varepsilon' \gtrsim \varepsilon^2/q_h$: the reduction runs the forger twice and *squares* its success probability, divided by the number of hash queries. The reduction is therefore far from tight (Definition 1.22). Concretely, a forger succeeding with probability $\varepsilon = 2^{-40}$ after $q_h = 2^{60}$ hash queries yields a guaranteed DLog-solver advantage of only about

$$\varepsilon^2/q_h = 2^{-80}/2^{60} = 2^{-140},$$

a factor 2^{100} weaker than ε itself. The guarantee runs in the useful direction—any forger yields *some* DLog solver, so DLog hardness still implies security—but the quantitative content is feeble. Two lessons follow, and they must be kept apart. The conservative lesson (Remark 1.23): a designer who wants *this theorem* to underwrite a target signature-security level must provision $\mathbb{G}$ with roughly twice as many bits of DLog security as the target, since the reduction hands back only the square of the forger’s advantage; and the squaring is intrinsic to the rewinding route—it enters at the squaring step of Theorem 8.17, so any analysis that runs the forger twice

and extracts by special soundness pays it. What the theorem does *not* establish is a matching attack: the loss belongs to the reduction, not provably to the scheme, so doubled parameters are the price of this proof, not a demonstrated necessity, and parameter selection in practice weighs concrete attack models and query bounds as well. Tighter reductions exist once the rewinding route is abandoned—under the *one-more* discrete logarithm assumption (that no efficient adversary, given $\ell + 1$ random elements of $\mathbb{G}$ and ℓ queries to a discrete-logarithm oracle, outputs all $\ell + 1$ logarithms), under DDH (Definition 2.5) together with key-prefixing, or in the idealised algebraic models of Remark 1.30.

Proposition 8.20 (Nonce reuse is fatal). *Suppose a signer produces two Schnorr signatures (R, s) and (R, s') on distinct messages $m \neq m'$ with the same commitment R —a nonce r reused outright, or drawn twice from a faulty source. Then, except with probability $1/q$ over the challenge hash, anyone can compute the secret key from the two signatures.*

Proof. The challenges $c = H(R \parallel X \parallel m)$ and $c' = H(R \parallel X \parallel m')$ differ except with probability $1/q$, the chance that the random oracle answers the two distinct inputs identically. Given $c \neq c'$, the pairs (R, c, s) and (R, c', s') are two accepting transcripts sharing a commitment with distinct challenges, and the extractor of Theorem 8.8—run now by the *adversary*—outputs $x = (s - s')(c - c')^{-1}$. $\square$

Special soundness, the source of unforgeability, is thus also its dark mirror: the security of the scheme rests entirely on the nonce being fresh and unpredictable for every signature. This is not a theoretical worry—it is the classic operational failure of deployed Schnorr-family schemes—and it motivates deriving nonces *deterministically* from the secret key and message, the defining idea of EdDSA, to which we now turn.

8.7 RedDSA: re-randomisable Schnorr for Orchard

Construction 8.21 (EdDSA, schematically). EdDSA retains the Schnorr signature algebra of Construction 8.12 unchanged and hardens two implementation surfaces.

1. *Deterministic nonces.* The nonce is derived as a hash of a secret prefix (part of the expanded secret key) and the message, rather than sampled fresh: no faulty random-number generator can repeat a nonce across distinct messages and trigger Proposition 8.20.
2. *Key-prefixing.* The challenge hash absorbs the public key alongside R and m , as in Construction 8.12 and for the reasons of Remark 8.14.

In the ROM the deterministic nonce is, to anyone ignorant of the secret prefix, a fresh uniform value per message, so the unforgeability proof of Theorem 8.18 carries over essentially unchanged.

Zcash’s signature scheme generalises EdDSA in two directions, and the generalisations are the whole point: the base point becomes a parameter, and the key becomes shiftable.

Construction 8.22 (RedDSA). RedDSA is the Schnorr/EdDSA scheme with the base point $P \in \mathbb{G}$ a *parameter* of the scheme: different uses sign with respect to different generators.

- **KeyGen:** sample $x \xleftarrow{\$} \mathbb{F}_q$, output secret key x and public key $X = [x]P$.

- $\text{Sign}(x, m)$: sample a fresh random byte string T and derive the nonce $r = H(T \parallel X \parallel m)$. This is *randomised*, hash-based nonce generation: unlike EdDSA's derivation, it is not deterministic from the signing key and message, so its safety rests on T being fresh and unpredictable, the hashing ensuring only that a repeated T cannot repeat a nonce across distinct messages. Set $R = [r]P$, $c = H(R \parallel X \parallel m) \in \mathbb{F}_q$, $s = r + cx$, and output $\sigma = (R, s)$.
- $\text{Verify}(X, m, (R, s))$: accept iff $[s]P = R + [c]X$ with $c = H(R \parallel X \parallel m)$.

For a fixed base point, RedDSA is Schnorr/EdDSA, and its EUF-CMA security in the ROM is Theorem 8.18 with G replaced by P .

Remark 8.23 (RedDSA as deployed). The deployed scheme follows Construction 8.22 line for line (protocol specification § 5.4.7). The randomiser T is 80 uniform bytes, and both hashes are the function H° : BLAKE2b-512 personalised with `Zcash_RedPallasH`, its 64-byte output reduced to $\mathbb{F}_q$ by wide reduction—the expand-then-reduce discipline of Construction 3.18, with the 512-bit width making the modular bias negligible. The nonce hash absorbs $T \parallel X \parallel m$ and the challenge hash $R \parallel X \parallel m$, key-prefixed in both cases.

Orchard instantiates RedDSA *twice*, both times as *RedPallas*—RedDSA over the Pallas curve—and the two instantiations differ *only* in the base point:

- the *spend authorisation* base point, derived by hashing to the curve with domain `z.cash:Orchard` and input G , for the re-randomisable signature of §8.8; and
- the *value-commitment randomness* base point, derived with domain `z.cash:Orchard-cv` and input r , for the binding signature of §8.9. This is byte-for-byte the randomness base of Orchard's value commitment—an identity that §8.9 shows is the entire design.

The curve varies only across protocol generations: Sapling's *RedJubjub* is the same scheme over the Jubjub curve; the curve, its two base points and the H° personalisation (`Zcash_RedJubjubH` against `Zcash_RedPallasH`) are all that distinguish the two.

8.8 Key re-randomisation and unlinkability

A shielded protocol faces a linkage problem that no ordinary signature scheme addresses. Spending a note requires proving spend authority, and the obvious design—publish the spender's public key and a signature under it—would stamp every spend by the same wallet with the same key, linking them all. Orchard's answer exploits the algebraic structure of the RedDSA public key: because the key is a homomorphic image of the secret, it can be *shifted* into a fresh disguise for every spend.

Definition 8.24 (Key re-randomisation). Fix a base point P . Given a secret key $x \in \mathbb{F}_q$ with public key $X = [x]P$, and a *randomiser* $\alpha \in \mathbb{F}_q$, the re-randomised signing and verification keys are

$$rsk := x + \alpha \in \mathbb{F}_q, \quad rk := X + [\alpha]P \in \mathbb{G}.$$

Orchard publishes a fresh rk per spend, signs with the shifted secret key rsk , and proves in zero knowledge that rk is a valid re-randomisation of an authorised public key.

Proposition 8.25 (Re-randomisation consistency). *The pair (rsk, rk) is a valid RedDSA key pair: $rk = [rsk]P$, so a RedDSA signature under rsk verifies under rk .*

Proof. Linearity of scalar multiplication gives

$$[rsk]P = [x + \alpha]P = [x]P + [\alpha]P = X + [\alpha]P = rk.$$

The pair is thus exactly KeyGen-shaped, and the correctness of signing and verification (Proposition 8.13) applies verbatim. $\square$

The enabling structure deserves a sentence of its own. The public key is a homomorphic image $[\cdot]P$ of the scalar, so shifting the secret by α shifts the public key by $[\alpha]P$ —a quantity computable from α and P alone, *without* the secret key. A verifier, or a circuit, can therefore check the relationship between X and rk given α , while only the spender who knows x can sign under the shifted key.

Theorem 8.26 (Perfect unlinkability of re-randomised keys). *Fix a base point P . For any public key $X \in \mathbb{G}$, if $\alpha \xleftarrow{\$} \mathbb{F}_q$ is uniform then $rk = X + [\alpha]P$ is uniformly distributed over $\mathbb{G}$. Consequently, in the game where an adversary—of unbounded computational power, and permitted to choose two public keys X_0, X_1 itself—receives $rk = X_b + [\alpha]P$ for a challenger’s uniform bit b and fresh uniform α , and must guess b : the distributions of rk for $b = 0$ and $b = 1$ are identical, and no adversary guesses b with probability better than $1/2$.*

Proof. The point P generates $\mathbb{G}$, which has prime order q , so $\alpha \mapsto [\alpha]P$ is a bijection $\mathbb{F}_q \rightarrow \mathbb{G}$; for uniform α , the point $[\alpha]P$ is uniform over $\mathbb{G}$. Translation by the fixed element X is a bijection of $\mathbb{G}$ and carries the uniform distribution to itself, so $rk = X + [\alpha]P$ is uniform over $\mathbb{G}$ —for every X , hence independently of X . The two conditional distributions of rk coincide exactly, so the adversary’s view is statistically independent of b ; no test distinguishes identical distributions, and the best any distinguisher can do is guess, succeeding with probability exactly $1/2$. $\square$

The guarantee is information-theoretic—perfect, resting on no computational assumption—but its *precondition* matters: α must be sampled uniformly and freshly per use, and must be kept secret. In Orchard the transaction builder samples $\alpha \xleftarrow{\$} \mathbb{F}_q$ afresh for each spend and reveals it only to the proving circuit and the signer.

Remark 8.27 (Scope of unlinkability). Theorem 8.26 covers only the re-randomised *key*. Full unlinkability of a spend further requires that the accompanying signature and zero-knowledge proof leak nothing beyond their statements, that the randomiser α stays secret, and that no other transaction field correlates two spends. In Orchard the protocol publishes rk with each Action (the unit that spends one note and creates one) while proving, inside the Halo 2 circuit, knowledge of a witness for $rk = X + [\alpha]P$ with X a public key whose holder is authorised to spend the note (Orchard’s *spend validating key*)—the circuit constrains rk as a public input—and the network verifies the spend authorisation signature against the public rk . Signing under rk requires $rsk = x + \alpha$, that is, knowledge of *both* x and α ; only the legitimate key-holder in possession of this spend’s α can sign.

Remark 8.28 (Unforgeability under re-randomised keys, and the key-prefixing subtlety). Does unforgeability survive re-randomisation? A forgery (R, s) under rk satisfies $[s]P = R + [c]rk$ with $c = H(R \parallel rk \parallel m)$. One is tempted to subtract $[\alpha]P$ from the verification equation: $[s - c\alpha]P = R + [c]X$, apparently a forgery under the original key X . But the identity holds with $c = H(R \parallel rk \parallel m)$, while the base verifier recomputes $H(R \parallel X \parallel m)$ —the plain subtraction converts forgeries only for the *non-key-prefixed* variant of Schnorr. For key-prefixed RedDSA one argues in the ROM by either of two routes. (i) The pair $(rsk, rk) = (x + \alpha, X + [\alpha]P)$ is itself a base-scheme key pair (Proposition 8.25) whose public key is uniform (Theorem 8.26), so a forger against rk is *literally* a forger against the base scheme at the key rk ; the reduction, which knows α , recovers $x = rsk - \alpha$ from the extracted rsk . (ii) When the adversary sees signatures under several randomisers α_i , one re-runs the forking argument of Theorem 8.18 on the critical query $H(R^* \parallel rk \parallel m^*)$, simulating signing under the remaining keys by oracle programming (Lemma 8.16). A pitfall marks the boundary: a naive translation that programs $H(R \parallel rk \parallel m) := H_{\text{base}}(R \parallel X \parallel m)$ must be a carefully injective do-

main swap, and it fails outright with multiple randomisers—distinct rk_i -prefixed inputs would collide on a single X -prefixed point, a detectable deviation from a random oracle.

8.9 Binding signatures as a proof of knowledge of a discrete logarithm

The second RedPallas instantiation guards a different treasure: not who may spend, but whether value is conserved. A shielded transaction hides its amounts inside commitments; the forger of this subsection is a counterfeiter who would arrange commitments whose hidden values do not balance—minting money—while still producing whatever certificate the protocol demands. The defence consumes the homomorphism established in §4.6 and turns this section’s central lesson—a Schnorr signature is a proof of knowledge of a discrete logarithm—from analysis into design.

The value-commitment setup. Shielded protocols commit to value with the homomorphic Pedersen commitment

$$\text{cm}(v, \rho) = [v]V + [\rho]P,$$

where V and P are *independent* generators of $\mathbb{G}$ —independent meaning that no party knows the discrete logarithm of one to the base of the other, arranged by deriving both by hashing to the curve (Construction 3.24)—and ρ is a random blinding scalar. (In the notation of the orchard crate and of Remark 4.21, the randomness base here called P is the generator R ; this section reserves R for the signature nonce point. The base P is exactly the binding base point of Remark 8.23.) By Theorem 4.20, the group sum of commitments commits to the sum of values with the sum of blindings.

Sapling commits *per note*: a transaction carries input commitments $\text{cm}(v_i^{\text{in}}, \rho_i^{\text{in}})$ and output commitments $\text{cm}(v_j^{\text{out}}, \rho_j^{\text{out}})$, and the net commitment is

$$B = \sum_i \text{cm}(v_i^{\text{in}}, \rho_i^{\text{in}}) - \sum_j \text{cm}(v_j^{\text{out}}, \rho_j^{\text{out}}).$$

Orchard has *no* per-note value commitments: each Action—the unit that simultaneously spends a note of value v^{old} and creates a note of value v^{new} —commits once to its *net* value change, publishing

$$\text{cv}^{\text{net}} = \text{cm}(v^{\text{old}} - v^{\text{new}}, \rho) = [v^{\text{old}} - v^{\text{new}}]V + [\rho]P,$$

and $B = \sum_i \text{cv}^{\text{net}, i}$ over the transaction’s Actions (protocol specification § 5.4.8.3). The Action structure itself—how spends and outputs are fused into these units and assembled into a transaction—is the business of the *Ironwood Guide*. Either way, the homomorphism collapses the pile:

$$B = [\Delta_v]V + [\bar{\rho}]P,$$

where Δ_v is the net value— $\sum_i v_i^{\text{in}} - \sum_j v_j^{\text{out}}$ in Sapling, $\sum_i (v_i^{\text{old}} - v_i^{\text{new}})$ in Orchard—and $\bar{\rho}$ is the net blinding. The transaction balances precisely when Δ_v equals the public balancing value v^{bal} ; for a transaction with internal balance $\Delta_v = 0$, the V -component vanishes and $B = [\bar{\rho}]P$. Three observations set up the construction. Every element of $\mathbb{G}$ is some multiple of P , since P generates the prime-order group, so what distinguishes a balanced transaction is not the *existence* of a representation $B = [\beta]P$ but the ability to compute one. When the values balance, the signer knows the multiplier: it is the net blinding $\bar{\rho}$, assembled from the individual blindings. When they do not, computing any P -multiple representation of B from the published commitments would require knowing $\log_P V$, assumed hard.

Definition 8.29 (Binding signature). With B and $\bar{\rho}$ as above and the transaction balanced, so that $B = [\bar{\rho}]P$, set the *binding verification key* $\text{bvk} := B$ and the *binding signing key* $\text{bsk} := \bar{\rho}$.

The binding signature is a RedDSA signature over the base point P , with signing key $\bar{\rho}$ and verification key B , on a message that is the hash of the transaction (the *sighash*). The verifier recomputes B from the publicly listed value commitments and the public balancing value, then checks $[s]P = R + [c]B$ with $c = H(R \parallel B \parallel \text{sighash})$.

Theorem 8.30 (The binding signature is a proof of knowledge of $\log_p B$). *In the ROM, a valid binding signature on a transaction is a proof of knowledge of the discrete logarithm of the net commitment B to the base P ; combined with the soundness of the accompanying zero-knowledge proofs (per Action in Orchard, per note in Sapling), which fix and range-check the committed values, it forces the transaction to balance.*

Proof. Part 1: proof of knowledge. The binding signature is exactly a Fiat–Shamir Schnorr proof for the relation R_{dl} with base P and statement B . By special soundness (Theorem 8.8) together with the forking argument of Theorem 8.18—specialised to a single statement and *no* signing oracle—any algorithm producing a valid signature under B with non-negligible probability can be rewound to two accepting transcripts (R, c, s) , (R, c', s') with $c \neq c'$, from which the extractor recovers $\bar{\rho} = (s - s')(c - c')^{-1}$ with $B = [\bar{\rho}]P$. Producing a valid binding signature is therefore equivalent, up to the standard rewinding loss, to *knowing* $\log_p B$.

Part 2: knowledge forces balance. Write $B = [\Delta_v]V + [\bar{\rho}']P$ for the *true* net value and net blinding that the commitments imply; these are well defined once the accompanying proofs fix the committed openings—in Orchard the per-Action circuit fixes v^{old} , v^{new} and the opening of cv^{net} , in Sapling the per-note proofs fix each v_i, ρ_i . If the signer knows $\bar{\rho}$ with $B = [\bar{\rho}]P$, subtracting the two expressions for B gives

$$[\Delta_v]V = [\bar{\rho} - \bar{\rho}']P,$$

where Δ_v enters as a scalar, that is, as its residue in $\mathbb{F}_q$. Whenever $\Delta_v \not\equiv 0 \pmod{q}$, that residue is a nonzero element of $\mathbb{F}_q$, hence invertible, so

$$V = [(\bar{\rho} - \bar{\rho}') \Delta_v^{-1}]P$$

—exhibiting $\log_p V$ and contradicting the independence of V and P . Therefore $\Delta_v \equiv 0 \pmod{q}$.

Part 3: from congruence to equality. Balance in $\mathbb{F}_q$ alone is not yet economic balance: an integer net value equal to a nonzero multiple of q would pass the test above while minting q units at a stroke. The gap is closed by the *range checks* in the same accompanying proofs: in Orchard the per-Action circuit constrains v^{old} and v^{new} to $\{0, \dots, 2^{64} - 1\}$ (the Action statement, protocol specification § 4.18.4), and in Sapling the per-note proofs constrain each v_i likewise. A transaction with n Actions (count notes instead for Sapling) therefore has $|\Delta_v| \leq n(2^{64} - 1)$ as an integer, vastly below the 255-bit q for any transaction that can be serialised, and the only multiple of q in that range is 0. Hence $\Delta_v = 0$ over $\mathbb{Z}$, not merely in $\mathbb{F}_q$. $\square$

In deployment the transaction is not internally balanced: it declares a public balancing value v^{bal} , and the verifier signs off against $B - [v^{\text{bal}}]V$, folding the public value into the verification key—the orchard crate’s `Bundle::binding_validating_key` subtracts a commitment to v^{bal} with zero blinding from the sum of the Actions’ cv^{net} (protocol specification § 4.14). The argument of Theorem 8.30 applies verbatim to the folded key and forces $\Delta_v \equiv v^{\text{bal}} \pmod{q}$; the balancing value is encoded as a signed 64-bit integer, so the range checks promote the congruence to the integer equality $\Delta_v = v^{\text{bal}}$ exactly as in Part 3.

Remark 8.31 (Why a signature, and not just a proof). The single pair (R, s) accomplishes three things at once. First, it proves knowledge of $\bar{\rho}$, certifying balance, by Theorem 8.30. Second, because the Fiat–Shamir challenge hashes the transaction *sighash*, the same signature *authenticates* the transaction: altering any signed field changes c and invalidates (R, s) , so the binding signature doubles as an integrity check binding the shielded components together—the origin of the

name. Third, the construction needs no trusted setup and no separate circuit for the balance check: it is pure elliptic-curve arithmetic that anyone can verify. The homomorphism of the commitment turns the arithmetic statement “the values balance” into the algebraic statement “ B is a known multiple of P ”, and the Schnorr signature is the off-the-shelf proof of knowledge for precisely that statement—the purest illustration in the protocol that a digital signature *is* a proof of knowledge of a discrete logarithm.

Remark 8.32 (One scheme, two meanings). Set the two RedPallas uses side by side. The spend authorisation signature verifies under a re-randomised key $rk = X + [\alpha]P$, chosen to *hide* which long-term key authorised the spend: the signing key $rsk = x + \alpha$ encodes authority, and α buys unlinkability (Theorem 8.26). The binding signature verifies under a key $B = [\bar{\rho}]P$ that the spender does *not* choose: the transaction’s own commitments force it, the “secret key” $\bar{\rho}$ is the net blinding factor, and signing proves balance (Theorem 8.30). Both are the same three lines of Schnorr arithmetic; what differs is the *meaning* assigned to the discrete logarithm—authority in one case, value-conservation in the other. The unifying lesson of this section: once a public key is the image of a secret under $x \mapsto [x]P$, a Schnorr signature is a compact, non-interactive proof of knowing that secret, and the designer chooses what knowing it *means*.

8.10 The transparent layer’s schemes: ECDSA and BIP-340 over secp256k1

Two signature schemes in the series live outside the Pallas group: the transparent layer authorises spends with ECDSA, and the issuance bundle of Zcash Shielded Assets is authorised with a BIP-340 Schnorr signature. Both run over secp256k1, and both are registered here at the level a consumer needs—syntax, deployed conventions, and where their security stands—rather than constructed from scratch.

The curve secp256k1 is the short Weierstrass curve $y^2 = x^3 + 7$ over the prime field $\mathbb{F}_p$ with $p = 2^{256} - 2^{32} - 977$, with a fixed base point G of prime order n (a 256-bit prime) and cofactor 1; every constant is a named parameter of SEC 2. In SEC 2’s terminology it is a *Koblitz curve*, one with $a = 0$, a choice that buys implementation speed and plays no role in what follows. The group law, point encodings, and the discrete-logarithm problem on such a curve are the general machinery of the *Math Guide*, §“Elliptic curves” and §“The elliptic-curve discrete logarithm problem”; the hardness assumption is Definition 2.1 in the group $\langle G \rangle$.

Construction 8.33 (ECDSA over secp256k1, schematically). Let $e \in \mathbb{Z}/n\mathbb{Z}$ be the integer read from the 32-byte digest of the message, and write $x(R)$ for the x -coordinate of a point, read as an integer; the secret scalar is written d , since x is taken.

- KeyGen: sample $d \xleftarrow{\$} [1, n-1]$, set $X = [d]G$, output (X, d) .
- Sign(d, e): sample $k \xleftarrow{\$} [1, n-1]$, set $R = [k]G$ and $r = x(R) \bmod n$, then $s = k^{-1}(e + rd) \bmod n$; restart if $r = 0$ or $s = 0$; output (r, s) .
- Verify($X, e, (r, s)$): require $r, s \in [1, n-1]$, compute $R' = [es^{-1}]G + [rs^{-1}]X$, and accept iff R' is not the identity and $x(R') \equiv r \pmod{n}$.

Correctness: $R' = [s^{-1}(e + rd)]G = [k]G = R$. The nonce k must be fresh and secret, for the reason of Proposition 8.20 in different algebra: two signatures (r, s) and (r, s') under one k on distinct digests $e \neq e'$ give $s - s' = k^{-1}(e - e')$, hence $k = (e - e')(s - s')^{-1}$ and then $d = (sk - e)r^{-1}$.

Where Schnorr’s challenge is a hash of R , ECDSA’s is the coordinate r itself, so the scheme is not a Fiat–Shamir transform and the forking-lemma route to Theorem 8.18 does not apply. Unforgeability analyses exist—in the generic group model, and in the ROM under additional as-

sumptions on the map $R \mapsto x(R) \bmod n$ —but they lie outside this volume’s path: the transparent layer inherits ECDSA’s security from Bitcoin’s track record rather than from a reduction.

Remark 8.34 (Deployment conventions: the sighash, DER, and low s). Three conventions fix how the transparent layer uses Construction 8.33, which protocol specification § 4.1.7 names as the script signature scheme.

1. *The message is the sighash.* The digest e is the 32-byte SIGHASH transaction hash of protocol specification § 4.10 (ZIP 244 for version-5 transactions), so the sighash algorithm plays the role of the hash in Theorem 8.4. Signer and verifier both feed it in as a digest.
2. *DER encoding and the sighash-type byte.* On the wire a signature is the DER encoding of (r, s) —an ASN.1 SEQUENCE of two minimally encoded INTEGERS—followed by one trailing byte, the *sighash type*, selecting which parts of the transaction the sighash commits to (SIGHASH_ALL for every signature the transaction builder emits; a signer working from a partially created transaction honours whichever valid type the input records). The signer appends the byte; the verifier splits it off and enforces strict DER unconditionally, the BIP-66 rule, so a malformed encoding fails script evaluation. The public key travels as its 33-byte compressed encoding, the bytes that hash160 commits to (§3.9).
3. *Low- s normalisation.* If (r, s) verifies then so does $(r, n - s)$, so a third party can re-encode a valid signature and change a pre-ZIP-244 transaction id; the normalisation takes the representative $s \leq (n - 1)/2$ (BIP-62), which the deployed signer always emits (secp256k1 crate, `sign_ecdsa`, whose libsecp256k1 backend documents the created signature as always in lower- s form) and which script evaluation checks only under the LowS flag, one the consensus check does not set (zcash_script crate, `src/external/pubkey.rs`, `check_low_s`; zebra-script crate, `src/lib.rs`).

Construction 8.35 (BIP-340 Schnorr over secp256k1, schematically). Write $H_\tau(m) = \text{SHA256}(\text{SHA256}(\tau) \parallel \text{SHA256}(\tau) \parallel m)$ for the *tagged hash* with ASCII tag τ —domain separation in the sense of Definition 3.16, the doubled tag digest making the prefix a whole 64-byte SHA-256 block—and read digests as integers modulo n .

- *Keys.* Sample $d \xleftarrow{\$} [1, n - 1]$ and set $X = [d]G$. The public key is the 32-byte coordinate $x(X)$ alone (*x-only*): the verifier reconstructs X as the point with that abscissa and *even* y , and the signer replaces d by $n - d$ when $y(X)$ is odd, keeping the pair consistent.
- *Sign(d, m):* derive $k = H_{\text{BIP0340/nonce}}(t \parallel x(X) \parallel m) \bmod n$ with $t = d \oplus H_{\text{BIP0340/aux}}(a)$ for auxiliary randomness a ; set $R = [k]G$, negating k when $y(R)$ is odd; compute $e = H_{\text{BIP0340/challenge}}(x(R) \parallel x(X) \parallel m) \bmod n$ and $s = k + ed \bmod n$; output the 64 bytes $(x(R), s)$.
- *Verify($x(X), m, (r, s)$):* lift $x(X)$ to the even- y point X , recompute e , set $R' = [s]G - [e]X$, and accept iff R' is not the identity, $y(R')$ is even, and $x(R') = r$.

Remark 8.36 (BIP-340 against Construction 8.12; the ZSA consumer). The algebra is the Schnorr signature of Construction 8.12 in its (R, s) form with key-prefixing (Remark 8.14), under $\mathbb{G} = \langle G \rangle$ of order n and $H = H_{\text{BIP0340/challenge}}$; the unforgeability proof of §8.6 carries over with that tagged hash modelled as the random oracle. What differs is convention: points travel by abscissa alone with parity fixed by the even- y rule, saving a byte per key and the parity of R per signature and letting the same 32 bytes serve as key encoding and as hash input; the challenge hash is domain-separated by tag; and the nonce is deterministic with optional auxiliary randomness, as in Construction 8.21. The in-series consumer is the issuance authorisation signature of

Zcash Shielded Assets: ZIP 227 (*specified*, in a draft ZIP) instantiates IssueAuthSig as BIP-340 over secp256k1, prefixes both the validating-key encoding and the signature with a scheme byte 0x00, and signs the transaction's sighash. The reference implementation, not yet deployed, is the QED-it fork of the orchard crate (commit cf801a5), `src/issuance/auth.rs`, whose ZSASchnorr signs through the secp256k1 crate's BIP-340 routine with zeroed auxiliary randomness. The issuance protocol itself is the business of the *ZSA Guide*.

9 Interactive proofs, zero knowledge, and SNARKs

The adversary of this section wears two faces, one on each side of a conversation. Facing the verifier stands the *cheating prover*: a party with no witness—no spendable note, no authorisation, no satisfying assignment—who would nevertheless talk the verifier into accepting, and thereby mint value, spend another's coins, or certify a false computation to a network in which every validator takes the proof's word for the hidden data. Facing the prover stands the *curious verifier*: a party who follows the protocol, or deviates from it arbitrarily, in order to squeeze from the interaction more than the single bit it is owed—to distinguish which witness the prover holds, and so unmask the spender, link the transactions, or read the amounts a shielded protocol exists to hide. The two threats pull in opposite directions: defeating the first means the transcript must pin the prover down; defeating the second means the transcript must reveal nothing. That both can be defeated at once—and, more remarkably still, by a proof *shorter than the statement it certifies*—is the subject of this section, and the reason a private transaction can be checked by millions of verifiers who learn nothing from it but its validity.

9.1 Three questions

Three questions organise everything that follows.

Can a prover convince a sceptical verifier that a statement is true? Not by assertion—the verifier trusts nothing—but by a protocol at whose end the verifier is justified in accepting, because no cheating strategy could have survived it. The answer is the theory of *interactive proofs* and *arguments*, refined into *proofs of knowledge*: protocols after which the verifier is entitled to conclude not merely that a witness exists but that the prover *possesses* one.

Can the prover convince without revealing why the statement is true? The witness—the factorisation, the colouring, the discrete logarithm, the transcript of the computation—must stay hidden; the verifier must finish the protocol knowing the statement holds and nothing else. The answer is *zero knowledge*, made precise by the simulation paradigm.

Can the proof be small—dwarfed by the statement it certifies—and cheap to check? A blockchain verifier cannot afford to re-execute every computation it accepts, nor even to read a proof as long as the computation's description. The answer is the *SNARK*: the succinct non-interactive argument of knowledge.

The development answers the questions in order and culminates in the architecture underlying the Halo 2 proving system deployed in Orchard: a *polynomial interactive oracle proof*, compiled into an argument by a *polynomial commitment scheme* (§4.9) and made non-interactive by the *Fiat-Shamir transform*. Every earlier section of this volume feeds in: the games and reductions of §1, the discrete logarithm of §2, the random oracle of §3, the commitments of §4, and the Schnorr protocol of §8.4, which will turn out to have been this section's canonical example all along.

Conventions are those of §1 throughout: λ is the security parameter, adversaries are PPT (non-uniform where it matters), and negl denotes a negligible function. Proof systems introduce one re-indexing: the objects of study are indexed by an *instance* x rather than by 1^λ , and asymptotic statements are measured in the instance length $|x|$ —for the instances this volume cares about, $|x|$ and λ are polynomially related, so the two scales agree up to the polynomial slop the definitions already absorb.

9.2 Languages, relations, and witnesses

What is a “statement”, and what does it mean to hold the “why” of one? The vocabulary is that of languages and relations over bit strings.

Definition 9.1 (Language, relation, witness). Fix the alphabet $\{0, 1\}$. A *language* is a set $L \subseteq \{0, 1\}^*$. A *binary relation* is a set $R \subseteq \{0, 1\}^* \times \{0, 1\}^*$ of pairs (x, w) , where x is called the *instance* (or *statement*) and w a *witness* for x . The relation *induces* the language

$$L_R := \{x : \exists w \text{ with } (x, w) \in R\},$$

and for each instance x the (possibly empty) *witness set* is $R(x) := \{w : (x, w) \in R\}$. The relation R is an *NP-relation* if it is

1. *polynomially balanced*: there is a polynomial p with $|w| \leq p(|x|)$ for every $(x, w) \in R$; and
2. *polynomial-time decidable*: a deterministic algorithm decides, in time polynomial in $|x|$, whether a given pair (x, w) belongs to R .

The class NP is exactly the class of languages induced this way: a language L lies in NP if and only if $L = L_R$ for some NP-relation R . This certificate characterisation of NP we take from complexity theory without proof; it is the bridge over which every “statement with a short checkable reason” enters the theory. Membership in L_R is existentially quantified—*some* witness exists—while holding an element of $R(x)$ is possession of the reason itself. The gap between the two quantifiers is where this section lives.

Example 9.2 (Three running relations). Three relations recur through the section, in increasing order of expressive power.

1. *Discrete logarithm*. In the setting of §8.4—a cyclic group $\mathbb{G} = \langle G \rangle$ of prime order q , written additively—the relation is

$$R_{\text{dl}} = \{(X, x) \in \mathbb{G} \times \mathbb{F}_q : X = [x]G\}.$$

Because $x \mapsto [x]G$ is a bijection $\mathbb{F}_q \rightarrow \mathbb{G}$, every instance $X \in \mathbb{G}$ has exactly one witness: the induced language is *trivial*—all of $\mathbb{G}$ —and membership is vacuous. What is hard, under the DLP assumption of §2.1, is to *produce* the witness. This is the prototype of why proving membership differs from proving knowledge.

2. *Graph three-colouring*. With instances encoding finite graphs $G = (V, E)$,

$$R_{\text{3col}} = \{(G, c) : c : V \rightarrow \{1, 2, 3\} \text{ and } c(u) \neq c(v) \text{ for all } \{u, v\} \in E\},$$

the relation of *proper* three-colourings. The induced language L_{3col} is NP-complete—every NP language reduces to it in polynomial time, a notion and fact we take from complexity theory without proof—so a zero-knowledge protocol for it yields one for *every* NP language by reduction (§9.8).

3. *Circuit satisfiability*. With instances encoding circuits,

$$R_{\text{sat}} = \{(C, w) : C \text{ a Boolean (or arithmetic) circuit and } C(w) = 1\}.$$

This is the universal target of practical proof systems: a computation is first *arithmetised* into the satisfiability of an arithmetic circuit over a finite field $\mathbb{F}$, and the prover proves knowledge of a satisfying assignment (§9.8).

The discrete-logarithm example already shows that “ $x \in L$ ” is too weak a target for cryptography: every group element is some multiple of G , so a protocol establishing mere membership establishes nothing. The content of the Schnorr protocol is that the prover *knows* the scalar—and formalising “knows” for an algorithm, an object with no mental states, is the central conceptual move of this section (§9.4).

9.3 Interactive protocols, proofs, and arguments

Definition 9.3 (Interactive protocol). An *interactive protocol* is a pair (P, V) of algorithms, the *prover* and the *verifier*. Both receive a common input x ; the prover additionally receives a private input w (a witness, or an arbitrary auxiliary string); each party holds its own private random tape. The parties alternate sending messages $a_1, a_2, \dots$ over a number of rounds polynomial in $|x|$; after the last message the verifier outputs a single bit, written

$$\langle P(w), V \rangle(x) \in \{0, 1\},$$

with 1 for *accept*—a random variable over both parties’ coins. The *transcript* of an execution is the full sequence of exchanged messages (together with x , and with the verifier’s coins when relevant). The protocol is *public-coin* if every message the verifier sends is a fresh uniformly random string that it also reveals—the verifier’s messages *are* its coins—and its final decision is a deterministic function of the transcript. The verifier is required to run in probabilistic polynomial time, always.

The last clause is the asymmetry on which everything turns. The verifier is the party being *protected*, so its power is fixed at PPT once and for all: a guarantee that only holds against verifiers with super-polynomial resources would protect nobody. The prover is the party whose power we *vary*, and that variation is exactly the distinction between proofs and arguments.

Definition 9.4 (Interactive proof and interactive argument). Let $L \subseteq \{0, 1\}^*$ be a language, and let $c, s: \mathbb{N} \rightarrow [0, 1]$. The protocol (P, V) is an *interactive proof system* for L with *completeness error* c and *soundness error* s if:

1. *Completeness*. For every $x \in L$ there is a prover input w with

$$\Pr[\langle P(w), V \rangle(x) = 1] \geq 1 - c(|x|).$$

2. *Soundness*. For every $x \notin L$ and *every* prover strategy P^* —of *unbounded* computational power, with arbitrary auxiliary input z —

$$\Pr[\langle P^*(z), V \rangle(x) = 1] \leq s(|x|).$$

The soundness clause is a game in the sense of §1.4: the honest verifier is the challenger, the prover strategy is the adversary, and the win condition is an accept on a false instance. The two thresholds must be bounded apart,

$$c(|x|) + s(|x|) \leq 1 - \frac{1}{\text{poly}(|x|)},$$

and in the protocols of this volume both errors are negligible (often $c = 0$). The class of languages admitting such proof systems—with the honest prover unbounded—is IP. Every protocol this volume constructs is for a language $L = L_R$ induced by an NP-relation R , with the

honest prover required to run in polynomial time given a witness $w \in R(x)$ as its private input; the unbounded honest prover matters only for the complexity-theoretic statements below.

An *interactive argument system* (or *computationally sound proof*) relaxes only the soundness quantifier: the guarantee holds against every PPT cheating prover P^* , that is, for $x \notin L$ and PPT P^* with arbitrary auxiliary input z , $\Pr[\langle P^*(z), V \rangle(x) = 1] \leq s(|x|)$, where s is required to be negligible. Soundness of an argument typically rests on a computational hardness assumption.

Remark 9.5 (Proofs versus arguments: neither dominates). A proof resists an unbounded cheating prover: its soundness is information-theoretic and unconditional. An argument resists only efficient adversaries, and an unbounded prover can typically break one outright—by breaking the binding of a commitment the protocol relies on (Theorem 4.8 showed perfectly hiding commitments have this exposure), or simply by brute-forcing a discrete logarithm. In exchange, arguments achieve properties provably impossible for proofs, chief among them *succinctness*: a sound interactive proof for an NP-complete language whose communication is much smaller than the witness would let an unbounded prover compress NP certificates below what complexity theory is believed to allow, collapsing classes believed distinct, whereas succinct *arguments* exist under standard assumptions (§9.8). For cryptographic applications—all of the SNARK literature included—the prover is a real machine with real resource bounds, and the argument notion is exactly right: bounded-prover soundness is the price of succinctness, and its gateway.

Why interaction, and why randomness? Strip both away and see what is left. With a deterministic verifier and a single prover message, the system proves exactly the class NP: the one message is a certificate, the verifier’s check is the relation’s decision procedure—and there is no hope of zero knowledge, since the certificate is, in spirit, the witness itself, and the verifier could replay the prover’s analysis at leisure. Verifier randomness changes the game: a cheating prover must commit to its messages *before* knowing the challenge, so a random spot-check catches what a predictable one never would. Interaction lets the verifier probe adaptively, round after round. Together they buy real power: the equality $IP = PSPACE$ —interactive proofs capture exactly the languages decidable with polynomial memory and unrestricted running time, a class believed far larger than NP—is a landmark of complexity theory (Lund, Fortnow, Karloff, and Nisan; Shamir) that we cite without proof. For this section the sharper point is qualitative: interaction and randomness are what make zero knowledge and knowledge extraction possible at all. A single static certificate can be neither hidden nor rewound.

Proposition 9.6 (Soundness amplification by repetition). *Let (P, V) be an interactive proof (or argument) for L with completeness error c and soundness error s , and let $k = k(|x|)$ be polynomially bounded. The protocol that runs k independent instances and accepts only if all k accept has completeness error at most $k \cdot c$ and soundness error at most s^k . Sequential repetition achieves this always; parallel repetition achieves it for proofs and, with care, for many arguments. In particular, if $s \leq 1 - \delta$ for a noticeable function δ , then choosing k with $\delta(|x|) \cdot k(|x|) = \omega(\log|x|)$ —always possible with k polynomial—drives the soundness error below every inverse polynomial, so a base protocol needs only a noticeable soundness gap.*

Proof. For completeness, the honest prover fails at least one of the k runs with probability at most $k \cdot c$ by the union bound (*Math Guide*, §“The union bound and a birthday calculation”). For soundness under sequential repetition, fix $x \notin L$ and any cheating strategy; conditioned on the transcripts of the first $i - 1$ runs, the i -th run is an execution against a fresh independently-random verifier, so it accepts with probability at most s whatever state the prover carries forward; multiplying the conditional bounds gives s^k . (For parallel repetition of proofs the same bound holds by a more careful argument, which we omit; for arguments parallel repetition can fail to amplify in general, and we use only the sequential form.) Finally $s^k \leq (1 - \delta)^k \leq e^{-\delta k}$, which is negligible once $\delta k = \omega(\log|x|)$. $\square$

9.4 Knowledge soundness and extractors

Soundness, as defined, is silent exactly where cryptography speaks. For the relation R_{dl} every instance $X \in \mathbb{G}$ lies in the induced language, so the soundness clause of Definition 9.4—quantified over $x \notin L$ —holds *vacuously*: a verifier that always accepts is “sound” for the trivial language. The assurance actually wanted is that a convincing prover *possesses* the discrete logarithm. But possession is not a property an algorithm wears on its sleeve; the definition reads it operationally. A prover knows a witness if an efficient procedure, granted full control over the prover—the ability to run it, feed it messages, and *rewind* it to an earlier state to try different continuations—can compute the witness from it. The procedure is called the *extractor*, and what it extracts is the meaning of “knows”.

Definition 9.7 (Knowledge soundness; proof of knowledge). Let (P, V) be an interactive protocol and R an NP-relation. The protocol is a *proof of knowledge* for R with *knowledge error* $\kappa: \mathbb{N} \rightarrow [0, 1]$ if there exist a probabilistic oracle algorithm $\mathcal{E}$ (the *extractor*) and a polynomial q such that for every prover strategy P^* —possibly cheating, possibly computationally unbounded—and every instance x , writing

$$\varepsilon(x) := \Pr[\langle P^*, V \rangle(x) = 1]$$

for the probability that P^* convinces the honest verifier, the following holds whenever $\varepsilon(x) > \kappa(x)$: the extractor $\mathcal{E}$, given x and oracle access to P^* *with the ability to rewind*—to run P^* from any point with the same coins, feeding verifier messages of $\mathcal{E}$ ’s choosing—outputs a witness $w \in R(x)$ within expected running time

$$\frac{q(|x|)}{\varepsilon(x) - \kappa(x)}.$$

If the guarantee is required only for *PPT* prover strategies P^* , the protocol is an *argument of knowledge*.

Three consequences can be read straight off the definition. The extraction cost may blow up as ε approaches κ : the bound $q(|x|)/(\varepsilon - \kappa)$ is useful precisely when the prover’s success is noticeably above the knowledge error. A prover succeeding with probability at most κ need yield nothing at all— κ is the “free” success rate, typically the probability of blind guessing, below which conviction carries no evidence of knowledge. And any prover noticeably above κ surrenders the witness: whatever code and coins produce conviction can be mined, efficiently, for the very object the conviction asserts. Knowledge is thereby defined as *efficient computability relative to the prover’s own code and coins*—an algorithm knows what can be efficiently computed from it. Note the expected-time clause: extractors are the standing reason this volume’s adversary class admits expected polynomial time (Remark 1.4).

Remark 9.8 (The rewinding justification). No live counterparty could rewind the prover, and none ever does: the extractor is a *thought experiment*, never executed during an honest run of the protocol. It exists to give the security claim its meaning—and, in security proofs, its teeth. In a reduction the extractor *is* the reduction: the cheating prover is a subroutine the reduction runs through its input/output interface alone (Definition 1.20), and black-box access already includes choosing the subroutine’s random tape and restarting it from the top with the same coins—so replaying it under different challenges is not science fiction but the ordinary right of a machine over its own subroutines, no reading of its code required. The same device runs in both directions across this section: the *simulator* of §9.5 rewinds a verifier to fake transcripts without a witness, and the extractor rewinds a prover to mine real witnesses from conviction. Rewinding is the single most important technical device in the theory.

Proposition 9.9 (Knowledge soundness implies soundness). *A proof of knowledge for R with knowledge error κ is an interactive proof for L_R with soundness error $s(x) \leq \kappa(x)$. The converse fails: a protocol can be sound without being a proof of knowledge.*

Proof. Fix $x \notin L_R$, so that $R(x) = \emptyset$. If some prover strategy P^* achieved $\varepsilon(x) > \kappa(x)$, the extractor run on P^* would halt in finite expected time and output some $w \in R(x)$ —impossible, as the witness set is empty. Hence $\varepsilon(x) \leq \kappa(x)$ for every P^* , which is the soundness bound. For the failure of the converse: soundness constrains only false instances, so a protocol for a trivial language—such as R_{dl} ’s, where no false instances exist—is automatically sound however it behaves, while nothing forces a witness to be extractable from it. One may be convinced of a true statement by a prover who does not know why it is true; knowledge soundness is strictly stronger, and it—not mere soundness—is the notion cryptographic applications actually rely on (§9.8). $\square$

9.5 The simulation paradigm and zero knowledge

What should “the verifier learned nothing” mean? The definitional move—often called the most influential idea in modern cryptography—is the *simulation paradigm*: the verifier learned nothing beyond the truth of the statement if it could have produced, entirely on its own and without ever talking to the prover, a transcript indistinguishable from the real one. Whatever a party can manufacture for itself it did not learn from the conversation; if the whole conversation is reproducible without the witness, the conversation taught nothing about the witness. The three grades of indistinguishability—perfect, statistical, computational—carry over verbatim from §1.2 (Definitions 1.6 and 1.9, with the unbounded-distinguisher bound of Proposition 1.7 and the hierarchy of Proposition 1.10) under one re-indexing: ensembles are indexed by instances x (and auxiliary inputs) rather than by the security parameter alone, and negligibility is measured in $|x|$.

Definition 9.10 (Zero knowledge). Let (P, V) be an interactive proof or argument for $L = L_R$. For a verifier strategy V^* , an instance x , a witness $w \in R(x)$, and an auxiliary input z , the *view*

$$\text{View}_{V^*}(P(w), V^*(z))(x)$$

is the random variable consisting of x , the random tape of V^* , and every message V^* receives—everything the verifier sees. The protocol is *zero knowledge* if there exists a PPT simulator Sim such that for every $x \in L$, every witness $w \in R(x)$, and every auxiliary input z , the ensembles

$$\{\text{Sim}(x, z)\} \quad \text{and} \quad \{\text{View}_{V^*}(P(w), V^*(z))(x)\}$$

are indistinguishable; the grade of indistinguishability—perfect, statistical, computational—gives *perfect* (PZK), *statistical* (SZK), or *computational* (CZK) zero knowledge. Two regimes for the quantifier over V^* :

1. *Honest-verifier zero knowledge* (HVZK): the guarantee is required only for $V^* = V$, the verifier that follows the protocol, with z empty; the simulator must reproduce the distribution of honest-execution transcripts.
2. *Malicious-verifier (auxiliary-input) zero knowledge*: the guarantee holds for *every* PPT verifier strategy V^* , deviating arbitrarily, with arbitrary auxiliary input z . Here Sim may use V^* as a subroutine *and rewind it*, and is permitted expected polynomial running time (Remark 1.4).

Why does the simulator’s existence suffice? The simulator receives x —and z —but *not* the

witness w . If its output is nevertheless indistinguishable from the real view, then the real view carries no efficiently extractable information about w beyond the fact $x \in L$: any property of w an efficient verifier could compute from the interaction, it could compute equally well from the simulator's output, which was manufactured without w —otherwise the computation would itself be a distinguisher. The simulator wields a power the prover lacks: it may rewind V^* , retrying until the verifier's challenges happen to align with a transcript preparable without the witness. That power is legitimate for exactly the reason rewinding always is (Remark 9.8): the simulator is the analyst's tool, holding the verifier as a subroutine, not a participant in any real execution.

The auxiliary input z models the prior knowledge a malicious verifier may bring to the conversation—the history of earlier protocol runs, partial information about the witness, anything. Quantifying over all z is what makes zero knowledge *compose*: a protocol that is zero knowledge for every auxiliary input remains zero knowledge when run as a subroutine of a larger system, whose surrounding state simply becomes the z of the analysis. Practice therefore adopts the auxiliary-input formulation, and so does this volume.

9.6 Sigma-protocols

The general definitions are now in place, and one protocol shape realises them with remarkable economy: three moves, public coins, and algebra doing all the work. The shape is named for the capital Σ its message flow traces on the page.

Definition 9.11 (Sigma-protocol). Let R be an NP-relation. A *Sigma-protocol* for R with *challenge space* C is a three-move public-coin protocol on common input x , the prover holding w with $(x, w) \in R$:

1. *Commitment*. The prover sends a first message a (the *commitment*, or announcement), computed from (x, w) and fresh randomness.
2. *Challenge*. The verifier sends $e \xleftarrow{\$} C$, uniformly random.
3. *Response*. The prover sends z , computed from (x, w, a, e) and its randomness.

The verifier decides by a deterministic predicate $\text{Verify}(x, a, e, z) \in \{0, 1\}$. Three properties are required.

- *Completeness*. An honest prover holding $(x, w) \in R$ convinces the honest verifier with probability 1.
- *Special soundness*. There is a PPT algorithm that, given x and any two accepting transcripts (a, e, z) and (a, e', z') with the *same* commitment a but *distinct* challenges $e \neq e'$, outputs a witness $w \in R(x)$.
- *Special honest-verifier zero knowledge* (sHVZK). There is a PPT simulator that, on input x and *any* challenge $e \in C$, outputs a pair (a, z) such that $\text{Verify}(x, a, e, z) = 1$ and the triple (a, e, z) is distributed exactly—or statistically, or computationally, close to—a real honest transcript conditioned on the challenge being e . The simulator chooses a *after* fixing e ; reversing the honest order is what makes simulation possible without w .

Two accepting transcripts sharing a commitment but differing in challenge are called a *collision*. Special soundness says a collision *is* a witness: it is the local, two-transcript reformulation of knowledge extraction, and the engine of the theorem that follows.

Theorem 9.12 (Special soundness yields knowledge soundness). *Let (P, V) be a Sigma-protocol for R with challenge space C of size $N := |C|$. Then (P, V) is a proof of knowledge for R with knowledge error $\kappa = 1/N$.*

Proof. We construct an extractor establishing the slightly weaker knowledge error $2/N$, with expected running time $O(1/(\varepsilon - 2/N))$ prover invocations whenever the prover's success probability satisfies $\varepsilon > 2/N$; the closing paragraph addresses the gap to the optimal $1/N$.

The prover as a matrix. Fix x and a prover strategy P^* ; fold any auxiliary input into its code. The behaviour of P^* is a deterministic function of its random tape ρ : fixing ρ fixes the commitment $a(\rho)$, and then each challenge e determines the response $z(\rho, e)$. Form the Boolean matrix

$$H[\rho, e] := \begin{cases} 1 & \text{if } \text{Verify}(x, a(\rho), e, z(\rho, e)) = 1, \\ 0 & \text{otherwise,} \end{cases}$$

with one row per random tape and one column per challenge; then $\varepsilon = \Pr_{\rho, e}[H[\rho, e] = 1]$, and a collision is two 1-entries in a single row.

The extractor. *Step 1.* Run P^* with fresh ρ and a uniform challenge e , repeatedly, until an accepting transcript (a, e, z) appears. The number of trials is geometric with success probability ε , so this costs $1/\varepsilon$ invocations in expectation—the geometric waiting time (*Math Guide*, §“Beyond finite spaces: countable additivity, limits, and densities”). *Step 2.* Alternate two kinds of trial: (i) rewind P^* to the moment just after it emitted a —the tape ρ stays fixed—and feed one fresh uniform challenge $e' \neq e$; (ii) make one entirely fresh probe, as in Step 1. If a kind-(i) trial accepts, a collision $(a, e, z), (a, e', z')$ is in hand: go to Step 3. If a kind-(ii) trial accepts, *abandon* the current row and continue Step 2 from the new accepting transcript (a fresh *phase*). *Step 3.* Apply the special-soundness algorithm of Definition 9.11 to the collision and output the witness.

Analysis. Call a row ρ *heavy* if its acceptance density $\delta_\rho := \Pr_e[H[\rho, e] = 1]$ is at least $\varepsilon/2$. The heavy-row lemma—a Markov-style averaging—states that an accepting entry lies in a heavy row with probability at least $1/2$: the total accepting mass contributed by light rows is less than $\varepsilon/2$ (each light row contributes less than $\varepsilon/2$ of its own row measure), so conditioned on acceptance the light rows carry less than half the mass. Every phase opens with a fresh accepting transcript distributed exactly as Step 1's (the opening transcript is either Step 1's or a kind-(ii) success, both conditioned draws from the accepting entries), so the lemma applies at every phase, not only the first.

Each alternation of Step 2 contains a kind-(ii) trial, which accepts with probability ε ; the number of alternations in a phase is therefore dominated by a geometric variable with mean $1/\varepsilon$, and a phase costs $O(1/\varepsilon)$ invocations in expectation. This interleaving is not an optimisation but a necessity: a row whose only accepting challenge is the already-used e —any row with $\delta_\rho \leq 1/N$ —contains no collision at all, and an extractor that only rewound would sit in such a row for unbounded expected time. The fresh probes cap every phase.

Now condition on the phase's row being heavy. A kind-(i) trial draws e' uniformly from $C \setminus \{e\}$ and accepts with probability at least

$$\frac{\delta_\rho N - 1}{N - 1} \geq \delta_\rho - \frac{1}{N} \geq \frac{\varepsilon}{2} - \frac{1}{N},$$

which is positive precisely when $\varepsilon > 2/N$. The phase ends at its first success of either kind, and the chance that this first success is a collision rather than a row-switch is at least

$$\frac{\varepsilon/2 - 1/N}{(\varepsilon/2 - 1/N) + \varepsilon} = \Omega\left(\frac{\varepsilon - 2/N}{\varepsilon}\right).$$

Un-conditioning costs the heavy-row factor $1/2$, so each phase yields a collision with probability $\Omega((\varepsilon - 2/N)/\varepsilon)$, and the number of phases is dominated by a geometric variable with mean

$O(\varepsilon/(\varepsilon - 2/N))$. By Wald’s identity—the expected total is the expected number of phases times the expected cost per phase (*Math Guide*, same section)—the extractor’s expected number of prover invocations is

$$O\left(\frac{\varepsilon}{\varepsilon - 2/N}\right) \cdot O\left(\frac{1}{\varepsilon}\right) = O\left(\frac{1}{\varepsilon - 2/N}\right),$$

of the form $q(|x|)/(\varepsilon - 2/N)$ demanded by Definition 9.7 with $\kappa = 2/N$.

Closing the factor of two. The threshold $\delta_\rho \geq \varepsilon/2$ defining heaviness is an artefact of this exposition, and it is exactly what costs the extra $1/N$ in the knowledge error. Sharper accounting removes it: Damgård’s refined heavy-row analysis in his Sigma-protocol lecture notes, and, in full generality, the k -special-soundness extractor of Attema, Cramer, and Kohl, which achieves knowledge error exactly $(k - 1)/N$ —here $k = 2$, giving the optimal $\kappa = 1/N$ that matches the soundness error. We cite these rather than reproduce them; every use in this volume tolerates the factor of two. $\square$

Remark 9.13 (Knowledge error equals soundness error, and how to size the challenge space). The knowledge error $1/N$ of Theorem 9.12 is no accident of the analysis: an explicit attack meets it. A prover with no witness guesses the challenge in advance: it picks $e^* \in C$, runs the sHVZK simulator on (x, e^*) to obtain an accepting pair (a, z) —for a protocol such as Schnorr’s the simulator’s algebra works for *every* instance—and sends a . If the verifier’s challenge happens to equal e^* , which occurs with probability exactly $1/N$, the prepared z is accepted. One challenge is thus always answerable without a witness, so $1/N$ is also the protocol’s soundness error, and Theorem 9.12 says the two coincide. One caveat keeps the lower bound honest: under Definition 9.7 an extractor facing this guessing prover ($\varepsilon = 1/N$) is granted expected time $q(|x|)/(1/N - \kappa)$, so for exponential N a claimed κ below $1/N$ is not formally refuted—the definition would hand the extractor a super-polynomial budget, enough to brute-force a witness unaided. The attack does rule out any noticeably smaller κ once the extractor is required to run in strict polynomial time and witnesses are genuinely hard to compute: such an extractor, run on this witness-free prover, would otherwise mine a witness from nothing—for R_{dl} , solving the discrete-logarithm problem itself. To make the error negligible there are two routes: take C exponentially large—for instance $C = \mathbb{F}_q$ for a λ -bit prime q , so $N = q$ —or repeat a small-challenge protocol $\omega(\log \lambda)$ times (Proposition 9.6). Large challenge spaces are why a *single* round of Schnorr already has negligible knowledge error, while a protocol with a polynomially small challenge space needs many repetitions.

Multi-round protocols generalise the collision to a tree. The succinct arguments of §9.8 run many challenge rounds, and their extractors need more than two transcripts.

Definition 9.14 (Tree special soundness). Let (P, V) be a $(2\mu + 1)$ -move public-coin protocol for R in which the prover speaks first and the verifier issues μ uniform challenges from a space C , and let $k_1, \dots, k_\mu$ be positive integers. A $(k_1, \dots, k_\mu)$ -*tree of accepting transcripts* for x is a tree of depth μ in which every node at level i has k_i children labelled by *pairwise distinct* challenges for round i , every root-to-leaf path carries the prover messages of one accepting transcript, and transcripts passing through a common node share their prefix up to that node. The protocol is $(k_1, \dots, k_\mu)$ -*special-sound* if a PPT algorithm computes a witness $w \in R(x)$ from any such tree. A Sigma-protocol is the case $\mu = 1, k_1 = 2$.

Extraction generalises phase by phase: under the hypotheses of a tree-extraction theorem—the Attema–Cramer–Kohl analysis cited in the proof of Theorem 9.12 is the general form—rewinding the prover round by round assembles the required tree of $\prod_i k_i$ accepting transcripts: the protocol is a proof of knowledge with knowledge error growing from $1/|C|$ to the order of $(\sum_i (k_i - 1))/|C|$, and with extraction cost polynomial provided the tree size $\prod_i k_i$ is. Both degradations are benign in the regime the compiled protocols of this section occupy: C exponentially large, μ logarithmically bounded, and the k_i constant, so the error stays negligible

and $\prod_i k_i$ polynomial. This is the notion under which the inner-product argument’s extractor operates (Proposition 4.39): each of the $\log_2 d$ folding rounds contributes one level of branching to the tree, three accepting children at every node. Each round’s accepting relation is linear in three unknown parent terms with coefficients that are powers of the challenge, so three suitably distinct challenges make that system invertible. The *Halo 2 Guide* carries out that extraction for the deployed protocol, whose IPA normalisation involves each round’s challenge through the coefficients u^{-1} , 1 , u : the extractor accordingly needs pairwise distinct *nonzero* challenges at every node—a negligible-fraction exclusion absorbed into the knowledge error—and not the pairwise distinct squares ($u \neq \pm u'$) that the alternative u^{-2} , 1 , u^2 normalisation would demand.

Proposition 9.15 (sHVZK implies HVZK). *A Sigma-protocol satisfying special honest-verifier zero knowledge is honest-verifier zero knowledge in the sense of Definition 9.10, with the same grade (perfect, statistical, or computational).*

Proof. The honest view is a triple (a, e, z) in which e is uniform on C . The HVZK simulator draws $e \xleftarrow{\$} C$ itself, invokes the sHVZK simulator on (x, e) to obtain (a, z) , and outputs (a, e, z) . For each fixed e , the simulated pair (a, z) is distributed as a real run conditioned on that challenge—exactly, or up to the stated grade of closeness—and the simulator’s e has the same uniform marginal as the honest verifier’s; averaging the conditional distributions over uniform e therefore reproduces the honest view, with any statistical or computational slack preserved under the averaging. $\square$

Example 9.16 (Schnorr: the canonical Sigma-protocol). The Schnorr identification protocol of §8.4 is a Sigma-protocol for the discrete-logarithm relation R_{dl} of Example 9.2, and the properties proved there, self-contained, are exactly the clauses of Definition 9.11. The commitment is $R = [r]G$; the challenge is a uniform $c \in \mathbb{F}_q$; the response is $s = r + cx$; the predicate is $[s]G = R + [c]X$. Perfect completeness is Proposition 8.7; 2-special soundness is Theorem 8.8, whose collision algebra yields the witness $x = (s - s')(c - c')^{-1}$; special HVZK is Theorem 8.10, whose simulator samples s and solves $R = [s]G - [c]X$ —choosing the commitment after the challenge, as the definition prescribes. The challenge space is $\mathbb{F}_q$, so $N = q$ and the knowledge error $1/q$ is negligible in a single round (Remark 9.13). By Theorem 9.12 and Proposition 9.15, Schnorr’s protocol is a perfect-HVZK proof of knowledge of a discrete logarithm.

Remark 9.17 (From honest to malicious verifiers: three remedies). Sigma-protocols are honest-verifier zero knowledge only. A malicious V^* may choose e as a function of the commitment a , and the sHVZK simulator—which fixes e before producing a —does not obviously cope: nothing lets it predict which challenge the verifier’s function will return for a commitment it has yet to manufacture. Three standard remedies close the gap. (1) *Guess and rewind*, for small challenge spaces: the simulator guesses $e^* \xleftarrow{\$} C$, runs the sHVZK simulator on (x, e^*) , and feeds the resulting a to V^* ; if the verifier’s challenge equals e^* , the round is simulated, and otherwise the simulator rewinds V^* and guesses afresh. The honest prover fixes a before any challenge exists, so the simulated a carries (almost) no information about the guess and each attempt succeeds with probability close to $1/N$: for polynomially large N the expected $O(N)$ attempts stay polynomial, and sequential repetition of the small-challenge protocol—simulated round by round, the auxiliary input carrying the history—is full malicious-verifier zero knowledge, at the price of many rounds. For exponentially large N the guess never lands, and repetition does not generically upgrade honest-verifier to malicious-verifier zero knowledge. (2) *Challenge commitment*: the verifier commits to e before seeing a (a coin-tossing step using the commitments of §4), so e cannot depend on a ; under the additional conditions such a compiler imposes for simulation, the simulator rewinds past the opening. (3) *The Fiat–Shamir transform*: remove the verifier’s choice altogether by deriving e from a hash of the transcript—under the hypotheses of Theorem 9.20, in the programmable random oracle model, the resulting *non-interactive* protocol is zero knowledge against *all* verifiers, since there is no live challenge left to skew. For the SNARK pipeline route (3)

is the one that matters, and honest-verifier zero knowledge is exactly the property the transform requires. It is next.

9.7 The Fiat–Shamir transform: from interactive to non-interactive

In a public-coin protocol the verifier contributes nothing but fresh randomness: its messages are coin tosses, its decision a deterministic function of the transcript (Definition 9.3). Every such message can therefore be replaced by a value that *anyone* can compute—a hash of everything sent so far—and the interaction collapses into a single proof string, verifiable by recomputing the hashes. The prover no longer talks to a live, possibly malicious verifier but to a fixed public function, which is why honest-verifier zero knowledge upgrades to full non-interactive zero knowledge. The cost is that the hash must be idealised: the analysis lives in the random oracle model, with $H: \{0,1\}^* \rightarrow C$ a random oracle whose range is the challenge space (Definition 8.15), and it exploits both powers established in Proposition 3.14: *observability*—the reduction sees every query the adversary makes—and *programmability*—the reduction may fix the oracle’s output at a freshly queried point to any value, provided answers stay consistent and uniformly distributed.

Definition 9.18 (The Fiat–Shamir transform). Let (P, V) be a public-coin protocol for R with challenge space C , and let H be a random oracle with range C . The *Fiat–Shamir transform* $\text{FS}[(P, V), H]$ is the non-interactive system defined, in the Sigma case, by:

- *Prover*. Compute the commitment a as in (P, V) ; set $e := H(x, a)$; compute the response z ; output the proof $\pi := (a, z)$ (equivalently (a, e, z)).
- *Verifier*. Given x and $\pi = (a, z)$, recompute $e := H(x, a)$ and accept iff $\text{Verify}(x, a, e, z) = 1$.

A multi-round public-coin protocol applies the rule per round: the i -th challenge is

$$e_i := H(x, a_1, e_1, \dots, a_i),$$

each challenge hashing the *entire* transcript prefix. To bind a message m to the proof—turning it into a signature—one includes m in the hash: $e := H(x, a, m)$.

Remark 9.19 (The pitfall: weak Fiat–Shamir). The definition’s insistence that each challenge hash *all* prior messages, the instance x included, is load-bearing, and omitting any part of it has broken deployed systems. Hashing $e := H(a)$ without x —“weak Fiat–Shamir”—lets an attacker fix (a, e, z) first and choose the instance *afterwards*, searching for an x that the frozen challenge happens to validate: the interactive protocol’s soundness quantified over provers who commit before the challenge, and an unhashed x re-opens exactly that order of quantifiers, yielding proofs of false statements. In multi-round protocols, hashing only the latest message rather than the running prefix breaks soundness the same way, one round at a time. The maxim: *the challenge must be a hash of everything the verifier would have seen up to the moment it speaks*.

Theorem 9.20 (Fiat–Shamir security in the ROM). Let (P, V) be a Sigma-protocol for R with challenge space C of size N , special soundness, and special HVZK; assume further that, conditioned on (x, e) , the first message output by the sHVZK simulator has min-entropy $\omega(\log \lambda)$ (min-entropy H_∞ : Definition 6.23). In the random oracle model, $\Pi = \text{FS}[(P, V), H]$ is a non-interactive argument for R that is:

1. perfectly complete, inheriting the protocol’s completeness;
2. knowledge sound: from any PPT prover P^* that makes at most Q random-oracle queries and

outputs an accepting proof for x with probability ε , an extractor obtains a witness $w \in R(x)$ with probability at least $\varepsilon(\varepsilon/Q - 1/N)$ up to constant factors—non-negligible whenever ε is non-negligible and N is super-polynomial; and

3. non-interactive zero knowledge: a PPT simulator that controls the random oracle produces, without the witness, proofs indistinguishable from real ones—for all verifiers, including malicious ones.

Proof. Completeness. The honest prover computes $e = H(x, a)$ and an honest response; the verifier recomputes the very same e from the same inputs and runs the same predicate, so the protocol’s probability-1 completeness transfers verbatim.

Zero knowledge, by programming. The simulator, on input x , draws $e \xleftarrow{\$} C$, runs the sHVZK simulator on (x, e) to obtain an accepting pair (a, z) , and then *programs* the oracle:

$$H(x, a) := e.$$

Programming fails only if (x, a) was queried before this moment; by hypothesis the simulated first message has, conditioned on (x, e) , min-entropy $\omega(\log \lambda)$, so a given prior query collides with a with probability $2^{-H_\infty(a)}$, and a union bound over the adversary’s prior queries makes the failure probability negligible. Conditioned on no failure, the programmed answer is a fresh uniform value—exactly what an unprogrammed oracle would have returned—so the oracle’s joint distribution is untouched, and by sHVZK the output (a, z) is distributed as a real proof. The programming is invisible to every verifier, and there is no live challenge for a malicious verifier to skew: its malice is moot, and honest-verifier simulation delivers full non-interactive zero knowledge.

Knowledge soundness, by forking. Let P^* output an accepting proof (a, z) for x with probability ε . With overwhelming probability a successful P^* actually queried H at (x, a) : otherwise the challenge $e = H(x, a)$ used by the verifier is a uniform value the prover never saw, and the proof verifies with probability at most $1/N$. Say the critical query is the j -th of the at most Q queries. The extractor runs P^* once, recording the transcript of oracle answers and the index j of the critical query; it then *rewinds* P^* to just before the j -th query and re-runs it with the same coins, answering the first $j - 1$ queries identically—so the commitment a is unchanged—and the j -th and subsequent queries with fresh uniform values, the j -th being $e' \xleftarrow{\$} C$. This is precisely the experiment of the general forking lemma (Theorem 8.17, with $h = N$): if P^* succeeds with probability ε over Q query slots, then with probability at least $\varepsilon(\varepsilon/Q - 1/N)$ both runs succeed at the same index and $e \neq e'$. Conditioned on that event, the two runs yield accepting transcripts (a, e, z) and (a, e', z') with a common commitment and distinct challenges—a collision—and the special-soundness algorithm of Definition 9.11 outputs $w \in R(x)$. $\square$

Remark 9.21 (Caveats, and the refinements SNARKs need). The random oracle is an idealisation: a real hash function is fixed, public, and not random, and there exist contrived protocols secure in the ROM yet insecure under *every* concrete instantiation. The transform is nonetheless trusted in practice, with the working stance of Remark 1.29, and it underlies essentially every deployed non-interactive proof and signature—Fiat–Shamir-compiled SNARKs included. Two refinements matter for the compiled arguments of §9.8. First, Theorem 9.20 covers the one-challenge Sigma case only, and does not by itself prove security for a multi-round protocol such as Halo 2’s: there the extractor must fork the transcript at *each* round, its success probability degrading with the round count and with the protocol’s soundness structure, and special soundness must generalise to tree special soundness (Definition 9.14)—a separate multi-round theorem, whose hypotheses must match the protocol at hand. Second, when the underlying interactive protocol is itself only computationally sound—an argument—the ROM extraction *layers on top of* the computational soundness reduction rather than replacing it. There is also a gap between proofs that merely observe the oracle and proofs that program it (the non-programmable ROM

is strictly weaker as a proof technique); deployed systems adopt the programmable model—the zero-knowledge clause above, which programs the oracle outright, already shows why—and an analysis must state which programmable-oracle (and, where used, algebraic) model it works in.

Example 9.22 (Schnorr signatures, recovered). Applying Definition 9.18 to Schnorr’s Sigma-protocol (Example 9.16) with a message m folded into the challenge hash yields, symbol for symbol, the Schnorr signature scheme of Construction 8.12; and its EUF-CMA security in the ROM (Theorem 8.18) is established by exactly the forking argument in the proof of Theorem 9.20, augmented with a signing-oracle simulation. The signatures section proved the instance; this section supplies the general statement it instantiates. The slogan: *proof of knowledge + Fiat–Shamir = signature*.

9.8 SNARKs: succinct non-interactive arguments of knowledge

The Sigma-protocol paradigm reaches all of NP: a commit–challenge–reveal proof of knowledge for the NP-complete relation $R_{3\text{col}}$ of Example 9.2—zero knowledge under one-way functions alone, a result of Goldreich, Micali, and Wigderson taken without proof—extends to every NP language by reduction. The generic reduction is hopelessly inefficient, however, its proofs polynomially *longer* than the computation they certify, so practical systems abandon it and arithmetise the computation directly.

A blockchain verifier operates under constraints the classical theory never contemplated. It cannot interact—a transaction is broadcast once and verified by parties who will never speak to its author, years later, from a proof string sitting in a block. It cannot afford to re-execute—every full node checks every proof, so verification must cost less than the computation being certified, by orders of magnitude. And it must resist a prover with every incentive to cheat: an accepting proof *is* money. The object meeting all three constraints at once is the SNARK.

Definition 9.23 (Non-interactive argument). A *non-interactive argument* for an NP-relation R is a triple of PPT algorithms:

- $\text{Setup}(1^\lambda, R) \rightarrow \text{srs}$, producing a *structured reference string*—the public parameters. In the random oracle model all three algorithms additionally access one globally sampled oracle; the oracle is not part of the *srs*—a PPT setup cannot output an infinite object—though public parameters may be derived through it. A setup holding no secret trapdoor is *transparent*: no secret is ever held, so none must be trusted to have been discarded (Remark 4.3).
- $\text{Prove}(\text{srs}, x, w) \rightarrow \pi$, for $(x, w) \in R$.
- $\text{Verify}(\text{srs}, x, \pi) \rightarrow \{0, 1\}$.

The system is *complete* if honestly generated proofs always verify. A *preprocessing* argument lets Setup depend on the circuit or relation and output, alongside the proving parameters, a short *verification key*, amortising the verifier’s per-statement work across all statements of that shape.

Definition 9.24 (SNARK). A *succinct non-interactive argument of knowledge* (SNARK) for an NP-relation R is a non-interactive argument that is additionally:

1. *Knowledge sound*. For every PPT prover P^* there is a PPT extractor $\mathcal{E}$ —in the ROM, ob-

serving P^* 's oracle queries and rewinding or forking it—such that, on the same setup,

$$\Pr[\text{Verify}(srs, x, \pi) = 1 \wedge (x, \mathcal{E}^{P^*}) \notin R] \leq \text{negl}(\lambda),$$

the probability over the setup and both algorithms' coins: an accepting proof implies an extractable witness, except with negligible probability.

2. *Succinct*. The proof size $|\pi|$ and the verifier's running time are *sublinear* in the size of the computation proved—typically $|\pi| = O(\text{poly}(\lambda))$ or $O(\text{poly}(\lambda) \log|C|)$, and verification $O(\text{poly}(\lambda) \log|C|)$ for a circuit C , independent of, or polylogarithmic in, the witness and circuit sizes. (Some of the literature reserves “succinct” for polylogarithmic proofs; the constant-size proofs of pairing-based schemes are the strongest form.)

With zero knowledge—the non-interactive, ROM form of Definition 9.10, as delivered in the Sigma case by Theorem 9.20, clause 3—it is a *zk-SNARK*.

Remark 9.25 (Why these systems are arguments). The “AR” in SNARK is earned by necessity, not by preference. Remark 9.5 showed that a succinct *proof* for an NP-complete language is essentially ruled out: a transcript shorter than the witness cannot information-theoretically pin the witness down, and an unbounded prover could otherwise compress certificates below what complexity theory is believed to allow. Succinct *arguments* escape because a computationally bounded prover cannot *find* a convincing-but-false short proof even though one may exist—succinctness and computational soundness arise together, by necessity. The constructions below realise the necessity at a specific spot: their polynomial commitments are binding only against *efficient* adversaries, under cryptographic assumptions, and an unbounded prover could equivocate on a commitment and defeat extraction. It is at that commitment layer that the information-theoretic core turns into an argument.

What does *knowledge* soundness buy an application? Consider a private transaction whose statement reads: *there exists a note I am authorised to spend, and input and output values balance*. Plain soundness guarantees only that *some* satisfying witness exists—and for such existential statements about a large anonymity set, witnesses exist in abundance: other people's notes satisfy the existential quantifier perfectly well. Soundness alone would not prevent a prover holding no spendable note from proving the existential statement, should she somehow come by a proof. Knowledge soundness closes the gap: whoever produced the accepting proof *actually holds* a concrete witness—a real spendable note, a real authorisation—that the extractor could recover from her. The application may then act—release funds, grant access—exactly as though the witness had been presented in the clear, while it stays hidden. Soundness protects against false statements; knowledge soundness protects against provers who do not know what they claim. Almost every cryptographic use, and certainly every monetary one, requires the latter (Proposition 9.9).

The pairing-based route

The first SNARKs to reach deployment, and the ones the deployed Sapling protocol still uses, verify with a bilinear pairing.

Definition 9.26 (Bilinear pairing). Let $\mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T$ be cyclic groups of prime order r , the first two written additively with generators G and $\hat{G}$. A *bilinear pairing* is a map $e: \mathbb{G}_1 \times \mathbb{G}_2 \rightarrow \mathbb{G}_T$ that is

1. *bilinear*: $e([a]P, [b]Q) = e(P, Q)^{ab}$ for all $P \in \mathbb{G}_1, Q \in \mathbb{G}_2$, and $a, b \in \mathbb{F}_r$;

2. *non-degenerate*: $e(G, \hat{G}) \neq 1$, so that $e(G, \hat{G})$ generates $\mathbb{G}_T$; and
3. *efficiently computable*.

Bilinearity lets a verifier test one multiplication of hidden scalars—whether $c = ab$, given $[a]G$, $[b]\hat{G}$, and $[c]G$ —without learning any of them (Example 2.9); the KZG check of Construction 4.35 is such a test. The groups come from *pairing-friendly* elliptic curves, chosen with a small embedding degree (the quantity of Definition 2.24, there required large) so that $\mathbb{G}_T$ lives in a manageable extension field; the curve of the deployed Sapling protocol is BLS12-381, a Barreto–Lynn–Scott curve of embedding degree twelve (protocol specification § 5.4.9.2). The deployed system treats the pairing as a black box, and so does this volume.

The lineage is short. The BCTV14 family (Ben-Sasson, Chiesa, Tromer, and Virza) is a preprocessing argument in the sense of Definition 9.23: the setup encodes one fixed arithmetic circuit into a structured reference string of group elements, the prover’s work is a handful of multiscalar multiplications over that string, and the proof is a constant number of group elements—eight, in the deployed encoding (protocol specification § 5.4.10.1)—checked by a constant number of pairing equations, whatever the circuit’s size. Groth’s refinement (Groth16) cuts the proof to *three* group elements, $\pi_A, \pi_C \in \mathbb{G}_1$ and $\pi_B \in \mathbb{G}_2$, and verification to a single pairing-product equation: three pairings— π_A against π_B , π_C against a fixed verification-key element, and a public-input-weighted combination of verification-key elements against another—compared with one pairing precomputed once per circuit. Deployed Zcash uses Groth16 over BLS12-381 for every Sapling spend and output proof (protocol specification § 5.4.10.2; proofs produced and verified by the bellman crate’s `groth16` module, consumed through the `zcash_proofs` crate). PLONK, the third generation, is the recipe of Theorem 9.28 below with the KZG scheme of Construction 4.34 as its commitment: the gate-and-permutation arithmetisation Halo 2 also uses, each polynomial opening checked by two pairings, so proof size and verification cost stay constant. Constant-size proofs and constant-time verification, independent of the circuit, are the family’s signature—the strongest form of the succinctness clause of Definition 9.24.

The price is the setup. The reference string is produced from secret scalars—the evaluation point τ of Construction 4.34 and, in Groth16, further secrets that tie the proof elements to one circuit—which must be *destroyed* once the parameters are published: they are the *toxic waste*. Whoever retains them forges accepting proofs of false statements at will, exactly as the trapdoor τ forges KZG openings (Remark 4.37), and no verifier can tell: knowledge soundness collapses silently. For Groth16 the waste is *per circuit*—every new statement shape needs a fresh setup—whereas PLONK’s KZG string is universal, one powers-of- τ string serving every circuit up to a size bound, but a trusted one still. The mitigation is a *multiparty ceremony*: participants contribute secret randomness in turn, each re-randomising the running parameters and destroying its own share, so that the waste stays unknown as long as *one* participant was honest. Zcash ran such ceremonies for the parameters of its earlier proof systems—the BCTV14 Sprout circuit and the Groth16 Sapling circuits (protocol specification §§ 5.7–5.8)—and each remains sound only on that one-honest-participant assumption.

The other route through the design space keeps the discrete logarithm and nothing else. The inner-product commitment of Construction 4.38 needs no pairing and no trusted setup—its generators are hashed into existence, with no secret behind them—at the price of logarithmic proofs and a linear-time opening check (Proposition 4.39); it is the road the *Halo 2 Guide* takes, and the one the rest of this subsection paves.

Polynomial interactive oracle proofs

Modern SNARKs are built in two layers—one information-theoretic, one cryptographic—and the information-theoretic layer speaks the language of polynomials.

Definition 9.27 (Polynomial interactive oracle proof). Fix a finite field $\mathbb{F}$. A *polynomial interactive oracle proof* (Poly-IOP) for a relation R over $\mathbb{F}$ is a public-coin interactive protocol in which, in place of ordinary messages, the prover in each round sends an *oracle* for a polynomial $p_i \in \mathbb{F}[X]$ of some bounded degree; the verifier sends uniformly random field-element challenges, and may *query* each polynomial oracle at points of its choice, receiving the evaluation $p_i(\zeta)$. After the interaction the verifier accepts or rejects as a function of its challenges and the few evaluations it queried. Completeness and (knowledge) soundness are as for interactive proofs, with the soundness analysis purely algebraic—resting, invariably, on the Schwartz–Zippel lemma: a nonzero polynomial of degree at most d over $\mathbb{F}$ vanishes at a uniformly random point with probability at most $d/|\mathbb{F}|$ (*Math Guide*, §“The Schwartz–Zippel lemma”). A good Poly-IOP verifier makes only $O(1)$ or $O(\log)$ queries, each a single evaluation.

The oracle is an idealisation—no real prover can hand over an opaque box answering evaluation queries honestly—but a productive one: it isolates the algebra from the cryptography. Soundness of the Poly-IOP layer is *unconditional*, resting on no assumption beyond the rigidity of low-degree polynomials; the cryptography enters only when the oracles are instantiated.

Arithmetisation is how a computation becomes a Poly-IOP statement. The claim “I know w with $C(w) = 1$ ” (the relation R_{sat} of Example 9.2) is encoded as a small set of polynomial identities over $\mathbb{F}$ that hold if and only if the computation is correct. The computation is first expressed as a *constraint system*—the customisable *PLONKish* gate-and-permutation form used by Halo 2—whose satisfaction is equivalent to a handful of polynomials satisfying identities on a chosen evaluation domain, a fixed finite set of points with one point per constraint row. In the PLONKish form, polynomials encoding the wire values a, b, c and fixed selector polynomials q_L, q_R, q_M, q_O, q_C chosen at circuit-definition time must satisfy the gate identity

$$q_L \cdot a + q_R \cdot b + q_M \cdot ab + q_O \cdot c + q_C = 0$$

at every point of the domain—selectors switching each row among addition, multiplication, and custom behaviours—while a *permutation argument* enforces the copy constraints wiring one gate’s output to another’s input. One more step turns the on-domain requirement into a spot-checkable one: the identity is demanded only on the domain H , and a polynomial vanishes on all of H exactly when it is divisible by the vanishing polynomial $Z_H := \prod_{h \in H} (X - h)$ (*Math Guide*, §“Roots and the factor theorem”), so the prover also sends an oracle for the quotient t with

$$q_L \cdot a + q_R \cdot b + q_M \cdot ab + q_O \cdot c + q_C = t \cdot Z_H$$

as an identity of polynomials—one that must now hold at *every* point of $\mathbb{F}$. Checking it at a single random point is the Poly-IOP: the verifier queries the oracles at a random ζ , checks the displayed equation there, and by Schwartz–Zippel a false statement survives with probability at most $d/|\mathbb{F}|$, with d bounding the degree of either side.

The *polynomial commitment scheme is the cryptographic half*. The PCS of Definition 4.32 supplies exactly what the idealised oracle promised: a short, binding commitment to a polynomial, with short proofs—verifiable against the commitment alone—that $p(\zeta) = v$; evaluation binding forbids two accepted values at one point, and the *extractability* the definition names is what SNARK compilation consumes: a prover whose evaluation proofs verify must “know” an underlying polynomial. Three instantiations mark out the design space: KZG (Construction 4.34)—pairing-based, constant-size, trusted setup; the inner-product-argument commitment over a single elliptic curve (Construction 4.38)—transparent, no trusted setup, logarithmic-size openings, the choice made by Halo 2; and the hash-based transparent schemes, outside this volume’s deployed path and mentioned only to bound the territory.

Theorem 9.28 (The compilation recipe, informal). *Let PIOP be a public-coin polynomial IOP for R , complete and knowledge-sound, and let PC be a polynomial commitment scheme that is correct, evaluation-binding, and extractable. Compile as follows: wherever the prover would send a polynomial oracle p_i , it sends $\text{Commit}(p_i)$ instead; wherever the verifier would query $p_i(\zeta)$, the prover sends the claimed value together with a PC evaluation proof, which the verifier checks against the commitment. Under compatible composition hypotheses the result is a public-coin interactive argument of knowledge for R , and a suitable multi-round Fiat–Shamir theorem in the ROM (Definition 9.18 gives the transform: every challenge a hash of the entire transcript so far) makes it non-interactive. Its proof is succinct; under the strict Definition 9.24 it is a SNARK only if verification is also sublinear in the proved computation. If, additionally, the Poly-IOP is properly blinded—random masking terms on the prover’s polynomials, hiding commitments (Remark 4.41)—and the corresponding simulation theorem applies, the result is zero knowledge.*

Proof sketch: layered extraction and the succinctness account. Completeness transfers verbatim: honest commitments open to honest evaluations, and the honest Poly-IOP verifier accepts.

Knowledge soundness is layered, one extractor feeding another. Given a successful prover against the compiled argument, the SNARK extractor first invokes PC *extractability* on each commitment, recovering actual polynomials $\hat{p}_i$ of the right degrees consistent with every evaluation the prover got accepted; this step is where the cryptographic assumption lives, and why the compiled object is an *argument*—a prover that could equivocate on its commitments would defeat it, and only computational binding rules that out. The extractor then hands the $\hat{p}_i$ to the Poly-IOP’s *unconditional* knowledge extractor, which is sound against *any* polynomials—in particular the extracted ones—and outputs $w \in R(x)$; evaluation binding guarantees the prover could not have answered queries inconsistently with the committed $\hat{p}_i$, so the algebraic soundness analysis applies to the extracted polynomials as run. Fiat–Shamir then converts the public-coin interactive argument to a non-interactive one in the ROM by the multi-round generalisation of Theorem 9.20: challenges become transcript hashes (Remark 9.19 governing what they must absorb), observability and forking supply the rewinding that the layered extractor needs—tree-style, per Definition 9.14—and the honest-verifier zero knowledge of the blinded protocol becomes full non-interactive zero knowledge by oracle programming.

Succinctness is an accounting exercise, and the account is paid entirely by PC. The proof consists of a constant or logarithmic number of commitments and evaluation proofs—short by construction, sublinear in the circuit, independent of the witness—so the compiled *proof* is succinct whenever the commitment scheme’s objects are. The *verifier* is sublinear only when the evaluation checks are themselves succinct, as for KZG’s two pairings; with the inner-product commitment each evaluation check costs $O(d)$ group operations (Proposition 4.39), so the compiled verifier runs in time linear in the circuit while the proof stays succinct. The compiled object therefore satisfies every clause of Definition 9.24 when the PC openings verify succinctly, and otherwise yields succinct proofs with linear-time verification—the trade Halo 2 accepts and then recovers through batching. $\square$

Remark 9.29 (Halo 2 is the recipe, instantiated). The proving system deployed in Orchard (protocol specification § 5.4.10.3) instantiates the recipe of Theorem 9.28, and each ingredient can be pointed to in the deployed source. The Poly-IOP is a PLONKish arithmetisation—custom gates declared as polynomial expressions, lookup arguments asserting that witness values lie in a declared table, and a permutation argument enforcing copy constraints. The polynomial commitment is the inner-product-argument scheme over a single elliptic curve, *transparent*—the generators are derived by hashing to the curve from the fixed domain string `Halo2-Parameters`, so the setup holds no trapdoor—with logarithmic-size openings. Fiat–Shamir over a concrete hash makes the argument non-interactive: the transcript is a running BLAKE2b state into which every prover message is absorbed, domain-separated by type, and out of which every challenge is squeezed—the entire-prefix discipline of Definition 9.18 enforced by construction.

The linear cost of each IPA opening check (Proposition 4.39) makes the deployed verifier’s work linear in the commitment dimension: Halo 2 has succinct *proofs* but does not meet the strict sublinear-verifier clause of Definition 9.24, and it is called a zk-SNARK under the broader convention that asks succinctness of the proof alone. The system’s signature innovation—the technique that named *Halo*—is *accumulation*: the one expensive step, the $O(d)$ final check of each IPA opening, is deferred and passed along rather than paid inside a circuit, one proof attesting to the verification of another—which is what makes *recursive* composition possible without a trusted setup. Deployed Zcash does not use recursion: each Orchard proof is verified directly, and what the deployed verifier takes from the Halo idea is the deferral alone, exploited for *batching*—many proofs’ deferred final checks collapsing into one multiscalar multiplication (consumed for Orchard bundles as described in Remark 4.40). These claims rest on model and composition hypotheses beyond the generic recipe (Remark 9.21).

Three load-bearing ideas from this section carry forward into everything the upper volumes build. *Rewinding* (Remark 9.8) is the universal device: the same cloning of an algorithm’s state powers the extractor that defines knowledge and the simulator that defines zero knowledge. *The layered split* of Theorem 9.28—an unconditionally sound information-theoretic core, wrapped in a cryptographically enforced commitment layer—is the architecture of every modern proof system, and the right frame for auditing one: algebraic bugs and cryptographic bugs live in different layers. And *knowledge soundness*, not mere soundness, is what an application stakes its security on: a shielded protocol does not care that a spendable note exists; it cares that the prover has one.

10 Merkle trees and commitments to sets

The adversary of this final section tells lies about a set. Her first lie is *inclusion*: she claims that a value sits in a committed collection it never entered—a payment out of thin air, spent against a ledger that never recorded it. Her second lie is *equivocation*: she publishes one short commitment and later produces two different collections that both explain it, so that the commitment pinned down nothing. Her third role is not a lie but a gaze: as an *observer* she watches an honest member prove that it belongs, and learns *which* member spoke—enough, in a payment system, to trace every coin. This section defeats all three with a single construction, built from nothing beyond the collision-resistant hash of §3, and closes the volume by spending everything the earlier sections banked: the commitments of §4 become the leaves, the algebraic hashes of §3 and §4 become the internal nodes, and the zero-knowledge arguments of §9 hide the path from leaf to root.

The plan runs from public to private. We first fix the two statements a verifier might want proved about a large collection, and recall the compression function and its collision game. The binary Merkle tree follows, with its root as a binding commitment to the whole list; a worked depth-two tree keeps every index honest. Authentication paths then give logarithmic-size membership proofs, complete and sound; the soundness proof introduces the one technique this section reuses until the end—extracting a hash collision from the first level at which two disagreeing computations agree. Layer and domain separation keep that extraction honest against structural confusion between leaves and nodes, with a deployed failure as the cautionary tale. Append-only trees and the frontier make the structure grow in logarithmic time; the vector-commitment abstraction names exactly what the root achieves; and zero-knowledge paths convert the public proof into a private one. The capstone assembles an Orchard shielded spend end to end.

10.1 Two statements about a set

The setting is asymmetric. A prover holds a large collection of values—transactions in a block, public keys in a directory, records in a log; for this volume, the ever-growing set of note commitments in a shielded payment system. A verifier wants conviction about that collection without storing it, indeed without ever seeing most of it. Two statements cover the uses that matter here.

- *Membership*. “Value v occurs at position i of the list”—or, forgetting the position, “ v belongs to the set.”
- *Integrity*. “The collection is exactly the one committed to earlier, unaltered.”

What is wanted is a single short *commitment* that determines the whole collection, together with short *membership proofs*: “short” meaning of size independent of the number of elements, or at worst logarithmic in it. A commitment of that shape makes integrity a one-comparison check, and membership a small certificate that travels with the value it certifies. The construction achieving this—the *Merkle tree*, due to Ralph Merkle—needs nothing more than a collision-resistant hash function, and it is the membership backbone of Orchard: every shielded spend proves, against a published root, that some note commitment lies in the tree.

Notation is as fixed in §3.1 and carried throughout the volume: $\{0, 1\}^*$ is the set of finite binary strings, $\{0, 1\}^n$ the strings of length exactly n , $x \parallel y$ concatenation, and λ the security parameter. All algorithms are PPT in λ , and $\text{negl}(\lambda)$ denotes a negligible function (*Math Guide*, §“Polynomial, exponential, and negligible functions”).

10.2 The compression function and collision resistance

The tree consumes one ingredient, which we recall in the keyed form that makes its security a meaningful asymptotic statement.

Definition 10.1 (Compression function). Let $H: \mathcal{K} \times \{0, 1\}^* \rightarrow \{0, 1\}^\ell$ be a keyed hash family exactly as in Definition 3.2: a key $s \in \mathcal{K}$ is sampled publicly at random and fixed, the digest length is $\ell = \ell(\lambda)$, and $H_s := H(s, \cdot)$ is the family member in force. When the input length is restricted to a fixed multiple of the output length, the family is a *compression function*; a *two-to-one* compression function has signature

$$h_s: \{0, 1\}^{2\ell} \rightarrow \{0, 1\}^\ell,$$

taking two digests to one.

The key s models the choice of a concrete function from a family, which is what rescues collision resistance from the hardcoded-collision triviality discussed in §3.1: for a single fixed function a colliding pair exists and some small program outputs it, so only the keyed formulation supports a clean quantifier. In practice one deploys a fixed standardised function—or, inside arithmetic circuits, an algebraic hash in the sense of §3.8 and §4.8—and we suppress s from the notation, writing h and H plain.

The security property the tree needs is collision resistance, exactly as in Definition 3.7 (§3.2): the game COL_H^λ hands the adversary the key s , the adversary outputs a pair (x, x') , and it wins if $x \neq x'$ while $H_s(x) = H_s(x')$; the family is collision resistant if every PPT adversary wins with probability $\text{negl}(\lambda)$. This is the strongest of the three classical hash notions (Proposition 3.8), and the generic birthday attack caps an ℓ -bit digest at $\ell/2$ bits of collision security (Lemma 3.11)—whence the 256-bit outputs used at the 128-bit level here. Merkle trees require *full* collision resistance, not merely second-preimage resistance, for a reason the soundness theorem (Theorem 10.12) makes exact: the party who assembles the tree may itself be the adversary, choosing every leaf, which is precisely the both-inputs-adversarial regime that Remark 3.10 assigns to collision resistance.

10.3 Binary Merkle trees

The construction hashes the collection pairwise, level by level, until one digest remains.

Definition 10.2 (Binary Merkle tree). Fix a two-to-one compression function $h: \{0, 1\}^{2\ell} \rightarrow \{0, 1\}^\ell$, a depth $d \in \mathbb{N}$, and an ordered list $L = (L_0, \dots, L_{2^d-1})$ of 2^d leaf values in $\{0, 1\}^\ell$. The *Merkle tree of depth d over L* is the complete binary tree with 2^d leaves whose nodes carry labels in $\{0, 1\}^\ell$, defined level by level from the leaves (level 0) to the root (level d) by

$$N_{0,j} = L_j \quad (0 \leq j < 2^d), \quad N_{t+1,j} = h(N_{t,2j} \parallel N_{t,2j+1}) \quad (0 \leq t < d, 0 \leq j < 2^{d-t-1}). \quad (3)$$

Level t has 2^{d-t} nodes, and $N_{t,j}$ is the j -th node from the left at level t . The unique level- d node is the *root*: $\text{root}(L) := N_{d,0}$.

Example 10.3 (A depth-two tree). With $d = 2$ and leaves L_0, L_1, L_2, L_3 , the tree has seven nodes: the four leaves, the two level-one nodes

$$N_{1,0} = h(L_0 \parallel L_1), \quad N_{1,1} = h(L_2 \parallel L_3),$$

and the root

$$\text{root}(L) = N_{2,0} = h(h(L_0 \parallel L_1) \parallel h(L_2 \parallel L_3)).$$

The single ℓ -bit root summarises the whole list of four ℓ -bit strings; the example carries through the next two subsections.

A list whose length m is not a power of two is handled by fixing the depth d with $2^d \geq m$ and padding the remaining $2^d - m$ leaf slots with a canonical value—the all-zero string, say, or the hash of the empty string. Padding must not create ambiguity: two different lists must never yield the same padded leaf sequence, else the root fails to determine the list before any cryptography is consulted. Binding the true length m into the root’s computation—one further compression of the top node with an encoding of m —removes any such ambiguity. An alternative is the *unbalanced* tree, in which a node with a single child passes its child’s label upward unchanged; it is convenient but subtle, and we return to it alongside second-preimage attacks in Remark 10.16.

The root is short; the claim that it *commits* to the list is a security property, stated as a game in the tradition of Definition 1.15.

Definition 10.4 (Merkle commitment; binding). The *Merkle commitment* to a list L of 2^d leaves is $\text{root}(L) \in \{0, 1\}^\ell$. The game $\text{BIND}_{H,d}^A(\lambda)$ runs as follows.

1. The challenger samples $s \xleftarrow{\$} \mathcal{K}$, fixing the compression function $h := H_s$, and sends s to the adversary.
2. The adversary outputs two lists L, L' , each of 2^d leaves in $\{0, 1\}^\ell$.
3. The game outputs 1 iff $L \neq L'$ and $\text{root}(L) = \text{root}(L')$.

The commitment is *binding* if every PPT adversary wins with probability $\text{negl}(\lambda)$.

Proposition 10.5 (Binding of the root). *If h is collision resistant, the Merkle commitment is binding. Concretely, there is a deterministic procedure that, from any two distinct equal-length lists $L \neq L'$ with $\text{root}(L) = \text{root}(L')$, extracts a collision for h in time linear in the tree size—one root-to-leaf walk of $O(d)$ hash comparisons. The reduction is tight: an adversary winning BIND with probability ϵ yields a collision finder succeeding with the same probability ϵ .*

Proof. Build both trees. The lists differ at some leaf index j_0 , so $N_{0,j_0} \neq N'_{0,j_0}$, writing N and N' for the labels of the two trees. Follow the leaf-to-root path

$$N_{0,j_0}, N_{1,\lfloor j_0/2 \rfloor}, N_{2,\lfloor j_0/4 \rfloor}, \dots, N_{d,0}$$

in both trees. The labels differ at level 0 and agree at level d —the common root—so there is a *smallest* level t at which they first agree: with $j = \lfloor j_0/2^t \rfloor$,

$$N_{t,j} = N'_{t,j}, \quad \text{yet} \quad N_{t-1,2j} \neq N'_{t-1,2j} \quad \text{or} \quad N_{t-1,2j+1} \neq N'_{t-1,2j+1},$$

since one of the two children at level $t-1$ is the path node, at which the labels still disagree by minimality of t . By the recurrence (3),

$$h(N_{t-1,2j} \parallel N_{t-1,2j+1}) = N_{t,j} = N'_{t,j} = h(N'_{t-1,2j} \parallel N'_{t-1,2j+1}),$$

while the two 2ℓ -bit inputs differ in at least one half: a collision for h , located after at most d comparisons along one path. The reduction wrapping this extractor runs the BIND adversary once and extracts whenever it wins, so collision resistance forces the winning probability to be negligible. $\square$

The proof's engine deserves its name now, because the section reuses it almost verbatim twice more: *a discrepancy at the leaves and agreement at the root must, by this pigeonhole-like descent, manifest a collision somewhere on the path between them*. There is a smallest level at which two disagreeing computations agree; the compression call producing that first agreement has two differing preimages. All three security theorems of this section are this one sentence wearing different inputs.

10.4 Authentication paths and membership proofs

The root alone certifies integrity; membership needs a certificate relating one leaf to the root. The certificate is the list of siblings along the leaf's path upward—everything the verifier needs to recompute the root from the leaf, and nothing more.

Definition 10.6 (Authentication path). Let a depth- d tree over L be given, and let i be a leaf index, $0 \leq i < 2^d$, with binary expansion $i = b_{d-1}b_{d-2} \cdots b_0$, so that bit b_t records whether the level- t node on the leaf-to-root path is a left child ($b_t = 0$) or a right child ($b_t = 1$). The *authentication path* (also *Merkle proof* or *membership witness*) for index i is

$$\pi_i = ((s_0, b_0), (s_1, b_1), \dots, (s_{d-1}, b_{d-1})),$$

where s_t is the label of the *sibling* of the level- t node on the path. Explicitly, with $j_t = \lfloor i/2^t \rfloor$ the path node's index at level t ,

$$s_t = \begin{cases} N_{t,j_t+1} & \text{if } b_t = 0 \quad (\text{sibling on the right}), \\ N_{t,j_t-1} & \text{if } b_t = 1 \quad (\text{sibling on the left}). \end{cases}$$

The path has exactly d entries.

Definition 10.7 (Path verification). The verifier $\text{Verify}(rt, i, v, \pi_i)$, on a root rt , an index i , a claimed leaf value $v \in \{0,1\}^\ell$, and a path $\pi_i = ((s_0, b_0), \dots, (s_{d-1}, b_{d-1}))$, proceeds in two stages. First the *index-consistency check*: reject unless, for every t , the bit b_t carried in π_i equals bit t of the binary expansion of i . Then set $a_0 := v$ and fold the path upward: for $t = 0, \dots, d-1$,

$$a_{t+1} := \begin{cases} h(a_t \parallel s_t) & \text{if } b_t = 0, \\ h(s_t \parallel a_t) & \text{if } b_t = 1. \end{cases} \quad (4)$$

Accept iff $a_d = rt$. We call a_d the *root recomputed from* (i, v, π_i) . The bit b_t places the running

value on the side matching the argument order of the tree-building recurrence: a left child is hashed on the left.

Proposition 10.8 (Completeness). *For every list L , every index i , and π_i the genuine authentication path for i in the tree over L ,*

$$\text{Verify}(\text{root}(L), i, L_i, \pi_i) = \text{accept}.$$

Proof. The index-consistency check passes by construction, the genuine path carrying exactly the bits of i . We show by induction on t that $a_t = N_{t, \lfloor i/2^t \rfloor}$, the label of the level- t node on the leaf-to-root path. The base is $a_0 = L_i = N_{0,i}$. For the step, the level- t path node and its sibling s_t are precisely the two children of the level- $(t+1)$ path node, and the fold (4) places a_t on the side dictated by b_t with s_t on the other side—reproducing exactly the two arguments of h in the recurrence (3), so $a_{t+1} = N_{t+1, \lfloor i/2^{t+1} \rfloor}$. At $t = d$ this reads $a_d = N_{d,0} = \text{root}(L)$, and the verifier accepts. $\square$

Proposition 10.9 (Logarithmic proof size). *An authentication path in a depth- d tree consists of d sibling labels and d direction bits, $d\ell + d$ bits in all, and verification performs exactly d evaluations of h . Choosing the minimal sufficient depth $d = \lceil \log_2 m \rceil$ for a list of $m \geq 2$ leaves (any $2^d \geq m$ suffices) yields proofs of $\lceil \log_2 m \rceil (\ell + 1)$ bits—logarithmic in the length of the list.*

Proof. One sibling label, one direction bit, and one compression per level, over the d levels above the leaves. $\square$

The concrete scale is worth pausing on. For a set of a billion elements— $m = 2^{30} = 1\,073\,741\,824$ —a membership proof consists of 30 hash values and 30 direction bits. With a 256-bit hash the sibling labels occupy $30 \cdot 256 = 7680$ bits, which is 960 bytes; direction bits included, 964 bytes—under a kilobyte, however the billion elements are arranged. (The byte counts are the output of the volume’s companion script.) This logarithmic succinctness is the central objective: the verifier’s storage is one digest, the certificate fits in a packet, and neither grows meaningfully as the set does.

Example 10.10 (A depth-two membership proof). Continue Example 10.3, and prove membership of L_2 at index $i = 2 = (10)_2$, so $b_0 = 0$ and $b_1 = 1$. At level 0 the path node $N_{0,2} = L_2$ is a left child, with sibling $s_0 = N_{0,3} = L_3$. At level 1 the path node $N_{1,1} = h(L_2 \parallel L_3)$ is a right child, with sibling $s_1 = N_{1,0} = h(L_0 \parallel L_1)$. The path is

$$\pi_2 = ((L_3, 0), (h(L_0 \parallel L_1), 1)).$$

Verification folds upward: $a_1 = h(L_2 \parallel L_3)$, then $a_2 = h(h(L_0 \parallel L_1) \parallel a_1) = \text{root}(L)$, and the verifier accepts. Note what the verifier never saw: neither L_0 nor L_1 themselves, only their digest $h(L_0 \parallel L_1)$ —yet it is convinced that L_2 sits at index 2 of the committed list.

10.5 Soundness: forging a path implies a collision

Completeness says honest paths verify; the security question is whether dishonest ones can. The threat is the inclusion lie of the section’s opening: an adversary who convinces the verifier, against the genuine root, that position i holds a value it does not.

Definition 10.11 (Path forgery). Fix a list L of 2^d leaves with root $rt = \text{root}(L)$. The game $\text{PATHFORGE}_{H,L}^A(\lambda)$ runs as follows.

1. The challenger samples $s \xleftarrow{\$} \mathcal{K}$ to fix $h := H_s$, builds the tree over L , and sends s together with L —equivalently, the whole tree—to the adversary.
2. The adversary outputs an index i and a pair (v^*, π^*) .
3. The game outputs 1 iff $\text{Verify}(rt, i, v^*, \pi^*) = \text{accept}$ and $v^* \neq L_i$.

A winning output is a *path forgery at index i* : the verifier accepts, against the honestly computed root, a leaf value different from the genuine one at position i .

Theorem 10.12 (Soundness of Merkle membership proofs). *Let h be collision resistant. Then no PPT adversary, given the list L and the whole tree, produces a path forgery except with probability $\text{negl}(\lambda)$. In constructive form: there is a deterministic extractor that from any list L , index i , and accepting forgery (v^*, π^*) with $v^* \neq L_i$ outputs a collision for h , in time linear in the tree size—recomputing the genuine leaf-to-root labels from L dominates; given the built tree, one verifier run plus a level-by-level comparison, $O(d)$ work in all, completes the extraction.*

Proof. Run the verifier's fold (4) on (i, v^*, π^*) , obtaining the chain $a_0^* = v^*, a_1^*, \dots, a_d^* = rt$ (the final equality because the forgery is accepting). Alongside it place the genuine chain $a_t = N_{t, \lfloor i/2^t \rfloor}$, the labels of the leaf-to-root path in the honest tree; by the induction of Proposition 10.8 this is the chain the fold produces on the genuine inputs, and $a_d = rt$.

The two chains disagree at the bottom, $a_0^* = v^* \neq L_i = a_0$, and agree at the top, $a_d^* = a_d = rt$. Hence

$$t^* := \min \{ t : a_t^* = a_t \} \in \{1, \dots, d\}$$

exists, and $a_{t^*-1}^* \neq a_{t^*-1}$ by minimality. Because the forgery is accepting, the index-consistency check of Definition 10.7 forces the direction bits carried in π^* to equal the bits of i —the same bits the genuine path carries—so at level $t^* - 1$ both chains place their running value on the same side of the compression call: both compute $h(\cdot \parallel \text{sibling})$ if $b_{t^*-1} = 0$, both $h(\text{sibling} \parallel \cdot)$ if $b_{t^*-1} = 1$. Form the two 2ℓ -bit preimages accordingly,

$$X^* = \begin{cases} a_{t^*-1}^* \parallel s_{t^*-1}^* & \text{if } b_{t^*-1} = 0, \\ s_{t^*-1}^* \parallel a_{t^*-1}^* & \text{if } b_{t^*-1} = 1, \end{cases} \quad X = \begin{cases} a_{t^*-1} \parallel s_{t^*-1} & \text{if } b_{t^*-1} = 0, \\ s_{t^*-1} \parallel a_{t^*-1} & \text{if } b_{t^*-1} = 1, \end{cases}$$

where $s_{t^*-1}^*$ is the sibling claimed in π^* and s_{t^*-1} the genuine sibling. Then

$$h(X^*) = a_{t^*}^* = a_{t^*} = h(X), \quad X^* \neq X,$$

the inequality because the two preimages differ in the half occupied by the running value. The pair (X^*, X) is a collision for h , found deterministically in $O(d)$ work once the genuine chain is at hand—a read-off from the built tree, or $2^d - 1$ hashes from the bare list L . $\square$

The reduction wrapping the extractor is immediate. If a PPT adversary $\mathcal{A}$ produces path forgeries with probability $\varepsilon(\lambda)$, the collision finder $\mathcal{B}$ that receives the key, samples or receives L , runs $\mathcal{A}$, and applies the extractor outputs a collision whenever $\mathcal{A}$ succeeds—hence with probability $\varepsilon(\lambda)$, at an overhead of building the tree over L , which $\mathcal{B}$ does in any case to play the challenger, plus one verifier run and $O(d)$ comparisons. Collision resistance of h therefore forces $\varepsilon(\lambda) \in \text{negl}(\lambda)$, and the reduction is tight: no advantage is lost in the translation.

Remark 10.13 (Scope: a fixed honest root). Theorem 10.12 is a statement about a *fixed honest root*: relative to a root that genuinely is $\text{root}(L)$, the only leaf value admitting an accepting path at position i is L_i , up to a collision in h . It does not by itself prevent an adversary who is free to *choose* the root from committing to a maliciously crafted tree; that threat is what binding (Proposition 10.5)

rules out. With an untrusted party supplying the root, both properties are necessary and they divide the labour cleanly: binding ensures the root determines at most one list, and path soundness ensures openings are faithful to it. Together they make the root a sound, binding commitment with succinct openings—the abstraction §10.8 names, where a single argument will deliver both faces at once.

10.6 Layer and domain separation

The soundness theorem has a structural blind spot, inherited from an innocent-looking convenience. Suppose the leaves are arbitrary-length data, hashed into $\{0,1\}^\ell$ by the *same* function used internally. Then an internal label $h(A \parallel B)$ is itself a well-formed leaf value—it has the right length—and one string can play both roles. An adversary exploits the ambiguity directly: she presents the internal node $N_{1,0} = h(L_0 \parallel L_1)$ as if it were a *leaf* near the top of a shallower tree, equipped with the correspondingly shorter authentication path, and a verifier who does not know the depth accepts. The root now has a second list explaining it—a genuine second-preimage attack, with no collision of h anywhere in sight. Absent a way for a label to assert “I am a leaf” or “I am an internal node”, the leaf-to-root structure that Theorem 10.12 relies on collapses.

The repair is the domain-separation discipline of §3.5, applied *within* a single structure: distinct structural positions get distinct tags.

Definition 10.14 (Layer and domain separation). A Merkle construction has *domain separation* if the function applied at distinct structural positions is forced to differ, by prepending a position-dependent tag to every hash input. Concretely, from a variable-input-length hash H define

$$\text{LeafHash}(v) := H(0x00 \parallel v), \quad \text{NodeHash}_t(A, B) := H(0x01 \parallel \langle t \rangle \parallel A \parallel B),$$

where the byte tags $0x00$ and $0x01$ distinguish the *layer kind*—leaf versus internal node—and $\langle t \rangle$ is a fixed-length encoding of the level index, distinguishing internal nodes by depth. The tree recurrence becomes

$$N_{0,j} = \text{LeafHash}(L_j), \quad N_{t+1,j} = \text{NodeHash}_{t+1}(N_{t,2j}, N_{t,2j+1}),$$

and the verifier’s fold applies the same tagged functions at the same positions.

Proposition 10.15 (Separation forces within-domain collisions). *With the tagging of Definition 10.14, any collision extracted by the binding proof (Proposition 10.5) or the soundness proof (Theorem 10.12) is a collision for H between two inputs sharing the same leading tag: a leaf preimage against a leaf preimage, or a level- t node preimage against a level- t node preimage. In particular, no accepting forgery can substitute an internal node for a leaf, or a node at one level for a node at another, without exhibiting a within-domain collision of H —which collision resistance forbids.*

Proof sketch. The extractors of both proofs locate the smallest level at which two disagreeing computations agree, and collide the two preimages of the compression call at that position. Under the tagged recurrence, the verifier’s recomputation applies the position’s prescribed function exactly as the honest tree does: at the leaves both preimages pass through LeafHash , at level $t \geq 1$ both pass through NodeHash_t . Both colliding preimages therefore begin with the same tag— $0x00$, or $0x01 \parallel \langle t \rangle$ —and the collision lies within a single domain. A cross-domain confusion, by contrast, would require two inputs with *different* leading tags and equal H -images, which is again an H -collision; either way the adversary has broken collision resistance, and the shallower-tree

attack above is foreclosed because a leaf preimage and a node preimage can no longer be the same string. $\square$

Remark 10.16 (Unbalanced trees, padding, and a deployed failure). The subtleties are not hypothetical. Bitcoin’s transaction Merkle tree handles an odd level by *duplicating* its last node before hashing pairwise, and the duplication rule is not injective: a list of transactions and the same list with its tail duplicated can yield the same root. This is CVE-2012-2459, a consensus-splitting vulnerability—an attacker could present two *different* transaction lists sharing a root, one valid and one invalid, and nodes could be made to cache the invalid one as permanently rejected (documented at length in Zcash’s inherited validator: `zebra-chain crate, src/block/merkle.rs`, which reproduces the upstream warning that the flawed algorithm must not be reused; the deployed defence—rejecting any block whose transaction hashes are not unique—lives in `zebra-consensus, src/block/check.rs`). The lesson generalises to three rules. First, layer-separate leaves from internal nodes, as Definition 10.14 prescribes. Second, make every padding or duplication rule *injective* on admissible lists—for instance by binding the true leaf count into the root’s computation. Third, feed `NodeHash` only fixed-length inputs, so that no length-extension structure of the underlying hash (§3.5) can intervene between layers.

Remark 10.17 (Algebraic hashes in the tree; Orchard’s MerkleCRH). When the Merkle path is verified inside a zero-knowledge circuit (§10.9), arithmetic gates over $\mathbb{F}_p$ evaluate the compression function, and the choice of h dominates the cost. A bit-oriented hash such as SHA-256 costs tens of thousands of constraints per call; one therefore uses an *algebraic* hash with compression $h: \mathbb{F}_p^2 \rightarrow \mathbb{F}_p$. Poseidon (§3.8) is a low-degree permutation-based map computable in few constraints; Sinsemilla (§4.8) is cheap in constraints through lookups and incomplete curve additions, *not* through low degree. For Poseidon, layer and domain separation come not from byte tags but from distinct field constants—initial capacity values or round constants—fed into the permutation; the tree’s security argument is identical, with “collision” read as a collision of the field-valued compression function.

Orchard’s Merkle hash MerkleCRH retains the tag mechanism of Definition 10.14 in encoded form. Each internal call prepends the 10-bit encoded layer index to the two 255-bit child encodings—a single fixed input length of 520 bits (a total the volume’s companion script checks), exactly the fixed-length-per-personalisation regime in which the deployed Sinsemilla hash is collision resistant (§4.8)—with *all* layers sharing one Sinsemilla instance (`MerkleHashOrchard::combine`, prepending `i2lebsp_k(level)` to two `L_ORCHARD_MERKLE = 255`-bit child encodings; protocol specification § 5.4.1.3). Separation between *distinct* Sinsemilla instances comes from the personalisation string—here `z.cash:Orchard-MerkleCRH`—fed to hash-to-curve to derive the instance-specific base point Q , while the generator table S is shared across all instances (`HashDomain::new` derives Q from the personalisation via the `z.cash:SinsemillaQ` hash-to-curve domain; the table derives from the fixed `z.cash:SinsemillaS`). Sinsemilla is collision resistant under a discrete-logarithm assumption (Proposition 4.29), so the tree’s hashing and its commitment role fuse into one Pedersen-like map: the same structure that compresses the children also binds them.

10.7 Append-only and incrementally updatable trees

A shielded pool’s set of note commitments only ever grows, and it grows one leaf at a time. Rebuilding a depth-32 tree from scratch on every insertion would cost 2^{32} hashes; the structure of the tree permits the update in d hashes and $O(d)$ state, and the state has a name.

Definition 10.18 (Append-only Merkle tree). Fix a maximum depth d , giving capacity 2^d , and the level-indexed `NodeHash` functions of Definition 10.14. Leaf values here enter as level-0 labels directly, with no `LeafHash`: the leaves are already digests, and the fixed depth together

with the level tags (Remark 10.17) carries the layer separation—the convention of deployed Orchard, whose leaf is the extracted coordinate itself (Example 10.31). An *append-only* (or *incremental*) Merkle tree maintains a position counter m —the number of leaves inserted so far, $0 \leq m \leq 2^d$ —and supports one mutating operation $\text{Append}(v)$, which places v at leaf index m and increments m . Empty leaf slots $m, \dots, 2^d - 1$ carry a fixed *empty-leaf* value $\perp_0$, and the labels of entirely empty subtrees follow by precomputation:

$$\perp_{t+1} := \text{NodeHash}_{t+1}(\perp_t, \perp_t), \quad 0 \leq t < d.$$

The *root after m appends*, written root_m , is the root of the depth- d tree, under this recurrence, whose first m level-0 labels are the inserted values and whose remaining ones are $\perp_0$.

Definition 10.19 (Frontier). The *frontier* of an append-only tree holding m leaves is the minimal state needed to (i) compute the current root and (ii) update incrementally on the next Append . For each level $t \in \{0, \dots, d-1\}$ it holds at most one *carry*: the label of the left child of the next node to be completed at level $t+1$, retained precisely when leaf index m has a 1 bit at position t —a completed left sibling awaiting its right neighbour. Equivalently, writing m in binary as $b_{d-1} \dots b_0$, the frontier stores, for each t with $b_t = 1$, the already-finalised left-sibling label f_t at level t along the rightmost filled path. Its size is at most d labels—one per set bit of m , hence $O(\log m)$ state.

The right picture is arithmetic. The binary representation of m decomposes the inserted leaves into completed perfect subtrees of distinct heights—one of height t for each set bit b_t —and the frontier holds exactly the roots of those subtrees: the pieces that are “complete and waiting for a right neighbour”. An append is then the Merkle analogue of incrementing a binary counter: a carry at level t corresponds to combining two height- t subtrees into one of height $t+1$, and the chain of carries is as long as the run of trailing 1-bits of m .

Proposition 10.20 (Incremental append in logarithmic time). *From the frontier of an append-only tree holding $m < 2^d$ leaves, $\text{Append}(v)$ computes the new frontier and the new root root_{m+1} using at most d evaluations of NodeHash and $O(d)$ additional work, while storing only $O(d)$ labels throughout.*

Proof. The algorithm is carry propagation followed by an upward fold. Set the running label $c \leftarrow v$ at level $t \leftarrow 0$. While bit t of m is 1, the new path node at level t is a right child whose left sibling is the stored carry f_t : set $c \leftarrow \text{NodeHash}_{t+1}(f_t, c)$, clear the carry f_t , and increment t . When bit t of m is 0, store c as the new carry f_t and stop. Writing z for the number of trailing 1-bits of m , the carry phase costs exactly z hashes and deposits f_z at level z —the root of the just-completed height- z subtree containing the new leaf, matching the binary increment $m \mapsto m+1$, which clears z trailing ones and sets bit z .

To obtain root_{m+1} , fold upward from level z with the just-deposited f_z as the running value—it is the path label at level z , not a sibling. For $t = z, \dots, d-1$, combine the running value with the stored carry f_t as left sibling when bit t of m is 1—those carries lie above the cleared run and survive the append untouched—and with the precomputed empty-subtree label $\perp_t$ as right sibling otherwise; at level z itself bit z of m is 0, so the first combination is with $\perp_z$, never with the deposited carry. One hash per level makes $d - z$ hashes in all (zero if $z = d$, when the tree has just filled). The total is exactly $z + (d - z) = d$ evaluations of NodeHash , and the retained state is the at-most- d carries: precisely the frontier for $m+1$ leaves. $\square$

Example 10.21 (The frontier as a binary counter). Take depth $d = 3$ (capacity 8) and insert v_0, v_1, v_2 in turn; the frontier evolves exactly as the binary counter of the leaf count m .

- After v_0 , $m = 1 = (001)_2$: one completed height-0 subtree; the frontier is $\{f_0 = v_0\}$.

- After v_1 , $m = 2 = (010)_2$: leaf v_1 at index 1 is a right child, triggering one carry $c = \text{NodeHash}_1(v_0, v_1)$ —a completed height-1 subtree; the frontier is $\{f_1 = c\}$, with f_0 cleared.
- After v_2 , $m = 3 = (011)_2$: leaf v_2 at index 2 is a left child and becomes a new height-0 carry; the frontier is $\{f_1 = \text{NodeHash}_1(v_0, v_1), f_0 = v_2\}$ —exactly the two set bits of 3.

Each append touched only the carry chain to the new leaf, never the whole tree. (The trace involves no numeric hash values and follows mechanically from the algorithm of Proposition 10.20; the volume’s companion script replays it symbolically and checks the occupied frontier levels against the set bits of m at every step.)

Appending changes the root, which raises a question the definition of soundness so far does not answer: which root is a proof valid against?

Definition 10.22 (History binding). An append-only tree is *history-binding* when the surrounding protocol treats every historical root as an acceptable commitment in its own right: each membership proof verifies against the root it was formed for, and appending further leaves never invalidates a previously valid $(\text{root}, \text{leaf}, \text{path})$ triple for that root. The soundness of each individual proof is exactly Theorem 10.12 applied to the corresponding fixed root.

Deployed Orchard is history-binding by construction. The chain records each block’s note-commitment-tree root—its *anchor*—and a spend may prove membership against *any* recorded historical anchor: consensus requires the anchor named by a transaction to be the root of the Orchard note-commitment tree as it stood at the end of some earlier block, checked by membership in the set of recorded anchors (protocol specification § 3.7, ‘Action Transfers and their Descriptions’). Per-proof soundness is then the fixed-root soundness theorem, once per anchor. The frontier is likewise the deployed representation of the growing tree (incrementalmerkletree crate, types `Frontier` and `NonEmptyFrontier`; the empty-subtree labels are the `EMPTY_ROOTS` table of `orchard`).

Remark 10.23 (Why append-only, and what it costs). Deletion or in-place update could be supported at $O(\log m)$ per operation, given the relevant siblings; the restriction to appends is a design choice, made for a monotonicity it buys. Once a leaf is inserted it is never removed, so a membership proof valid against the root at insertion time remains valid against *that historical root* forever. This is exactly what a shielded pool requires—a note, once added, can always be spent later by proving against some past anchor—and it spares the spender from tracking a moving target as the tree grows under everyone else’s insertions. The cost falls on the verifier, who must accept a window of historical roots rather than a single current one; the previous paragraph is that window, deployed.

10.8 Vector commitments

The section has been proving properties of one construction; it pays to name the abstraction the construction inhabits, both to state precisely what the root achieves and to mark the boundary with its relatives. The abstraction is a commitment to an *ordered vector* that opens one position at a time. (Its unordered sibling—the *cryptographic accumulator*, which certifies set membership with no notion of position—exists in the literature, but Orchard never needs it: notes occupy positions, and the position is part of what the spend circuit witnesses. We do not develop it.)

Definition 10.24 (Vector commitment). A *vector commitment* is a tuple of PPT algorithms $(\text{Setup}, \text{Commit}, \text{Open}, \text{Vf})$:

- $\text{Setup}(1^\lambda, n) \rightarrow pp$ produces public parameters for vectors of length n ;

- $\text{Commit}(pp, (v_0, \dots, v_{n-1})) \rightarrow (C, aux)$ outputs a short commitment and auxiliary state;
- $\text{Open}(pp, i, aux) \rightarrow \pi_i$ produces a proof that position i holds v_i ;
- $\text{Vf}(pp, C, i, v, \pi) \rightarrow \{\text{accept}, \text{reject}\}$ verifies an opening.

The scheme is *correct* if honest openings always verify, and *succinct* if $|C|$ and $|\pi_i|$ are bounded by a fixed polynomial in λ and $\log n$. It is *position-binding* if every PPT adversary wins the following game with probability $\text{negl}(\lambda)$: given pp , the adversary outputs (C, i, v, v', π, π') and wins iff $v \neq v'$ while both $\text{Vf}(pp, C, i, v, \pi)$ and $\text{Vf}(pp, C, i, v', \pi')$ accept. Note that the adversary chooses C itself: binding must hold even for maliciously formed commitments, with no honest committer anywhere in the game.

Theorem 10.25 (Merkle trees are succinct position-binding vector commitments). *Instantiate the syntax by: Setup samples h (and fixes d with $n = 2^d$); Commit computes the tree and outputs $C = \text{root}$ with aux the full tree; Open returns the authentication path; and $\text{Vf} = \text{Verify}$. If h is collision resistant, this is a correct, succinct, position-binding vector commitment, with commitment size ℓ and opening size $O(\ell \log n)$.*

Proof. Correctness is completeness (Proposition 10.8), and the size bounds are Proposition 10.9. For position-binding, suppose an adversary outputs two accepting openings (v, π) and (v', π') at the same index i under the same root C , with $v \neq v'$. Run the verifier's fold (4) on each, obtaining two recomputed chains that disagree at the bottom ($v \neq v'$) and agree at the top (both reach C , since both openings accept). The index-consistency check forces both paths to carry the direction bits of i , so at every level both chains place their running value on the same side of the compression call. The first-level-of-agreement extraction of Theorem 10.12 now applies verbatim to this pair of chains and yields two distinct 2ℓ -bit inputs with the same h -image: a collision. The argument needed *no honest reference tree*—it pits the two forged openings against each other—so it binds even a maliciously chosen root, which is the strong form of position-binding the definition demands. $\square$

Remark 10.26 (Position-binding versus path soundness). The two theorems answer different verifiers. Path soundness (Theorem 10.12) fixes an *honest* root and forbids opening a position to anything but its true value; it speaks to a verifier who trusts where the root came from. The vector-commitment theorem (Theorem 10.25) forbids opening a single position two *different* ways even under an *adversarial* root; it speaks to a verifier who received the root from an untrusted source, and it is what “commitment” should mean— C determines at most one value per position, whoever made C . Both reduce to the same first-level-of-agreement collision extraction; the only difference is which pair of chains is collided, forged-against-genuine in the one case and forged-against-forged in the other.

10.9 Privacy-preserving membership via zero-knowledge paths

The Merkle membership proof is succinct but not private—and for a shielded payment system, not private is fatal. Revealing the authentication path of §10.4 discloses the leaf value itself (here, the note commitment being spent) and the index i at which it sits, and even the sibling labels can leak structural information about the neighbourhood of the leaf. An observer correlating spends with the appends that created their leaves links payments end to end: spending a note must not reveal *which* note is spent. The remedy is to prove the *existence* of a valid path in zero knowledge, treating the entire path as a private witness and publishing only the root.

Definition 10.27 (The membership relation). For the domain-separated Merkle construction of depth d , with leaves entering as level-0 labels under the convention of Definition 10.18, define

$$R_{\text{mem}} := \{ (rt; (v, i, \pi)) : \text{Verify}(rt, i, v, \pi) = \text{accept} \},$$

with the *statement* the public root rt and the *witness* comprising the leaf v , its index i , and the authentication path π . A zero-knowledge argument for R_{mem} lets a prover convince a verifier that *some* leaf lies in the tree with root rt while revealing none of v , i , or π . We assume a zk-SNARK as §9 develops it (Definition 9.24): *completeness* (honest provers convince), *knowledge soundness* (an accepting prover “knows” a witness, formalised by an extractor), and *zero knowledge* (the proof reveals nothing beyond the truth of the statement, formalised by a simulator).

Construction 10.28 (In-circuit path verification). The Merkle-path circuit C_{mem} of depth d takes as public input the root rt , and as private (witness) inputs the leaf v , the direction bits $b_0, \dots, b_{d-1}$, and the siblings $s_0, \dots, s_{d-1}$. It implements the d compression calls of the fold (4)—the level- t call being the position’s prescribed function, NodeHash_{t+1} in the domain-separated construction, with no LeafHash call since the leaf enters as the level-0 label (Definition 10.18)—each preceded by the conditional swap that selects the argument order (a_t, s_t) or (s_t, a_t) according to b_t , and enforces the single output constraint $a_d = rt$. Its size is d times the cost of one compression call plus $O(d)$ constraints for the swaps. The position enters *only* through the bits $b_0, \dots, b_{d-1}$ —the binary decomposition of i —so the index-consistency check of Definition 10.7 is enforced structurally: there is no separate index to disagree with its bits. Deployed Orchard matches this shape exactly: the circuit’s Merkle gadget decomposes the witnessed position into little-endian bits and applies a conditional swap per layer (protocol specification § 4.9, ‘Merkle Path Validity’).

Theorem 10.29 (Privacy-preserving membership). Let h be collision resistant and let (P, V) be a zk-SNARK for R_{mem} —equivalently, for satisfiability of C_{mem} . Then the protocol “the prover sends a SNARK proof for the statement rt ; the verifier checks it” is a membership argument that is:

1. *complete*: a prover holding a genuine leaf and its path produces an accepting proof;
2. *sound*: any prover producing an accepting proof for an honest root $rt = \text{root}(L)$ knows—via the SNARK extractor—a witness (v, i, π) with $\text{Verify}(rt, i, v, \pi) = \text{accept}$, and v equals the genuine leaf L_i unless a collision in h is thereby found: the proven leaf is a real member of the committed list;
3. *private*: the proof reveals nothing about v , i , or π beyond the fact that some valid leaf exists under rt .

Proof idea. Completeness composes SNARK completeness with tree completeness (Proposition 10.8): the genuine (L_i, i, π_i) satisfies C_{mem} , and the honest prover proves it. Soundness composes the SNARK knowledge extractor with the path-forgery-to-collision extractor of Theorem 10.12: from an accepting prover, the SNARK extractor produces (v, i, π) accepted by Verify ; if $v \neq L_i$, that output is precisely a path forgery, hence yields a collision for h , hence occurs with probability $\text{negl}(\lambda)$. Zero knowledge carries over verbatim from the SNARK simulator, which produces from rt alone a proof indistinguishable from the real one—so the real proof cannot be conveying v , i , or π . $\square$

Remark 10.30 (Public root, private path; deployment trade-offs). The asymmetry in R_{mem} is the essential design point. The root must be *public* so that the verifier checks the proof against a state

everyone agrees on—a previously published note-commitment-tree anchor—and soundness is relative to that public root, exactly as Remark 10.13 delimits. The path must be *private* so that observers cannot link the spend to a particular leaf. SNARK succinctness adds the final ingredient: the verifier checks a proof whose size is essentially independent of d —polylogarithmic—even though the witness contains d siblings, so the construction *compresses* the membership proof while simultaneously *hiding* it. This is the precise sense in which a Merkle root over note commitments, opened in zero knowledge, is a privacy-preserving membership primitive.

Deployment chooses where to pay for that verification. Pairing-based schemes such as KZG (§4.9) achieve constant-size proofs with fast verification, at the price of a trusted setup. Halo 2’s IPA-based proofs grow logarithmically in size, but their verification is *linear* in the circuit size; the deployment amortises it by batch-verifying many proofs in a single multiexponentiation. The accumulation technique of the Halo lineage—which would make per-proof verifier work sublinear by deferring the linear check across recursively composed proofs—is not deployed: each Orchard proof is verified, in batch, directly.

10.10 A shielded spend, end to end

The volume closes by assembling the pieces. Every primitive below has been constructed in a previous section or in this one; the shielded spend is their composition, and the description follows the deployed Orchard sources.

Example 10.31 (An Orchard shielded spend). Each shielded output of a transaction appends a *note commitment* cm —itself a hiding, binding Sinsemilla commitment to the note’s value, recipient, and randomness (§4.8)—as a leaf of an append-only Merkle tree built with the Sinsemilla compression function of Remark 10.17. Strictly, the leaf is the extracted x -coordinate

$$cm_x = \text{ExtractP}(cm),$$

a 255-bit Pallas base-field element (*Math Guide*, §“Pallas and Vesta assembled”) matching the child encodings of Remark 10.17. The protocol periodically publishes the current root—the *anchor*—on chain, one per block.

To spend a note, its owner:

1. locates the leaf cm_x in the public tree and reads off its authentication path π and position i (`MerklePath::root` recomputes the anchor from exactly these data);
2. chooses a recent anchor $rt = \text{root}_m$ whose tree already contains the leaf—any recorded historical anchor serves, per §10.7;
3. proves in zero knowledge, per Theorem 10.29, the statement rt with witness (cm_x, i, π) : membership of the note commitment under the anchor, well-formedness of the note opening, and correct derivation of a *nullifier*—a deterministic tag whose on-chain uniqueness prevents double-spending, computed from the note and the spender’s nullifier key on $\text{PRF}^{\text{nfOrchard}}$ of §3.8 by a construction of the *Ironwood Guide*, §“The nullifier, and the loop back to minting” (the recomputed root is constrained to equal the public anchor, and the instance carries the anchor and the nullifier; protocol specification § 4.9, ‘Merkle Path Validity’, and § 4.18.4, ‘Action Statement’);
4. publishes the proof and the nullifier—but neither cm , nor i , nor π .

By Theorem 10.29, the spend convinces the network that the spender owns *some* note genuinely added to the tree—soundness resting on the collision resistance of Sinsemilla (Proposition 4.29) and the knowledge soundness of the proof system—while the network learns nothing about *which* note. (One refinement of the deployed circuit is worth recording: the root-equals-anchor constraint is enforced for notes of nonzero value, while zero-valued dummy spends may witness

an arbitrary path—by design, so that every action carries a spend whether or not a real note backs it.) The Merkle root is thus a succinct, binding, privacy-preserving commitment to the set of all notes ever created.

The spend consumes the whole volume. The note commitment is the hiding and binding envelope of §4; the tree hash and its personalised domains are the discipline of §3 instantiated by Sinsemilla; the nullifier is derived under the spender’s nullifier key from a PRF of §5, instantiated by Poseidon (§3.8), by the construction of the *Ironwood Guide*, §“The nullifier, and the loop back to minting”; the proof is a zk-SNARK of §9 for the relation R_{mem} extended with the note’s opening; and the authorising signatures over the spend—re-randomised RedDSA for spend authority, the binding signature for value balance—are those of §8. How the pieces sit inside a transaction, per Action, with the key hierarchy and note-delivery encryption around them, is the business of the *Ironwood Guide*.

Part III

Halo 2 Guide: The Halo 2 proof system, from arithmetisation to recursion

Abstract

Assuming the mathematics of the *Math Guide* and the cryptographic primitives of the *Crypto Guide*, this volume of *The Zcash Arboretum* develops the Halo 2 proof system: the polynomial-IOP design pattern, a rigorous account of knowledge soundness and the algebraic group model, recursive proof composition and accumulation schemes, and finally Halo 2 itself — PLONKish arithmetisation, the permutation and lookup arguments, the inner-product polynomial commitment, the Halo accumulation trick, and the complete protocol. A closing worked example carries one concrete statement from computation to verifier acceptance, with every number computed in $\mathbb{F}_{97}$.

1 Introduction

This is the *Halo 2 Guide*, the third volume of *The Zcash Arboretum*, developing the cryptography behind Zcash’s Orchard protocol. The first two volumes — the *Math Guide* and the *Crypto Guide* — constructed, from the ground up, the mathematics (finite fields, elliptic curves, polynomials and evaluation domains, probability) and the cryptographic primitives (hash functions, commitments, Σ -protocols and the Fiat-Shamir transform, zero knowledge, polynomial commitment schemes) that this volume assumes. The present volume assembles those ingredients into a *proof system*.

The formal development proceeds in four steps. First, we make the polynomial-IOP design pattern precise: the recipe “encode a computation as polynomial identities over an evaluation domain, then compile to a succinct argument with a polynomial commitment.” Next, we treat knowledge soundness rigorously, including the algebraic group model and the tree-of-transcripts extractor that the inner-product argument requires. We then motivate recursive proof composition and develop the accumulation schemes that make it practical. Finally, we assemble Halo 2 itself: PLONKish arithmetisation, the permutation and lookup arguments, the inner-product polynomial commitment with a worked example and its accumulation (the “Halo trick”), and the complete seven-phase protocol. Orchard uses this proof system to construct its shielded payment relation.

A reader who wants the concrete picture before the formal machinery may begin with Section 6. It carries the claim “I know x with $x^3 + x + 5 = 35$ ” through a grid of constraints, polynomials, a random-point check, commitments, wiring, lookups, Fiat-Shamir, and accumulation. Every number is small enough to verify by hand; Sections 2–5 then supply the definitions and security arguments that the example deliberately sketches.

2 Polynomial IOPs and the SNARK design pattern

The modern SNARK design pattern factors the protocol into two clean layers: a *polynomial interactive oracle proof* (PIOP) for the relation, and a polynomial commitment scheme (the *Crypto Guide*) that realises the PIOP’s oracles. This factoring, due in its current form to Bünz, Fisch, Szepieniec, and others, has substantially clarified the nature of SNARKs.

2.1 The polynomial IOP, formally

Definition 2.1 (Polynomial IOP). A *polynomial interactive oracle proof* for a relation $\mathcal{R}$ is a public-coin interactive protocol where in each round:

- The prover sends one or more *polynomial oracles* $p_i \in \mathbb{F}[X]$ (each of bounded degree).
- The verifier sends a random challenge $c \in \mathbb{F}$.

At the end of the protocol, the verifier may query each oracle at a small number of points, learning the polynomial’s evaluations at those points. The verifier’s final accept/reject is a deterministic function of the challenges, evaluations, and the public input x .

We define completeness, soundness, and knowledge soundness as usual.

A PIOP differs from a vanilla interactive argument in one essential respect: the verifier does not read the prover’s messages in full — it queries only a few evaluations of each polynomial oracle. This is what renders the verifier “succinct” even when the polynomials themselves have large degree.

2.2 Compiling a PIOP into a SNARK

Construction 2.2 (PIOP-to-argument compiler). Given a PIOP Π for $\mathcal{R}$, a PCS scheme (Setup, Commit, Open, Verify) for $\mathbb{F}[X]$, and a random oracle H , the compiler produces a non-interactive argument:

1. Setup: run the PCS setup to obtain pp .
2. Prover: simulate Π , replacing each polynomial-oracle message p_i by its PCS commitment C_i . Use Fiat–Shamir (the *Crypto Guide*) to derive verifier challenges from the transcript, and supply PCS openings for the PIOP’s oracle queries.
3. Verifier: parse the transcript, recompute Fiat–Shamir challenges, verify each PCS opening, and run the PIOP’s final accept/reject check.

A PIOP is *round-by-round* knowledge sound, informally, when partial transcripts admit a “doomed” labelling that a prover message cannot escape except with bounded probability over the next verifier challenge. A related *state-restoration* definition lets a prover restore an earlier verifier state and redraw its challenge. Precise equivalence or implication theorems between these notions require additional protocol and extractor conditions; we do not identify them definitionally here.

Theorem 2.3 (Compilation principle, informal). Suppose Π has a round-by-round or state-restoration knowledge-soundness theorem, the PCS supplies compatible online extraction and binding properties, and the transcript satisfies the hypotheses of an appropriate multi-round Fiat–Shamir theorem. Then the compiled protocol is a non-interactive argument of knowledge in the random-oracle model. The exact error and query loss depend on those definitions and on the extraction model; they are not determined solely by two scalar errors $\kappa_{\Pi}, \kappa_{\text{PCS}}$.

Fiat–Shamir exposes a query-budget loss because changing a commitment redraws the later challenges. Plain nested forking can lose $Q^{O(r)}$ for r rounds; stronger round-by-round or state-restoration notions are designed to obtain tighter bounds. Claiming a single factor Q nevertheless requires the precise theorem’s state-restoration, transcript, challenge, and extraction hypotheses.

This factoring is of considerable utility: research on PIOPs (PLONK, Marlin, Spartan, HyperPlonk, ...) proceeds independently of research on polynomial commitment schemes, and any

compatible pair combines into a new SNARK that inherits the PIOP’s relation and the commitment scheme’s cryptographic profile. This document develops the one PCS that Halo 2 uses — the inner-product argument (§5.5) — and treats the PIOP/PCS split only insofar as it clarifies the Halo 2 construction.

2.3 A worked example: a PIOP for univariate evaluation

To render the abstraction concrete, consider a minimal PIOP for the simplest non-trivial relation: “the prover knows a polynomial p of degree $\leq d$ such that $p(z) = y$ for a public (z, y) .”

Construction 2.4 (PIOP for univariate evaluation). On input (z, y) and prover witness p :

1. Prover sends the degree- $\leq d$ polynomial oracle p and the degree- $\leq d - 1$ quotient oracle $q := (p - y)/(X - z)$.
2. Verifier samples a random challenge $r \in \mathbb{F}$.
3. Verifier queries p and q at r .
4. Verifier accepts iff $p(r) - y = q(r) \cdot (r - z)$.

Completeness is immediate (q is a polynomial since $p(z) - y = 0$). Soundness follows from Schwartz–Zippel: if $(X - z) \nmid (p(X) - y)$, then $p(X) - y - q(X)(X - z)$ is a non-zero polynomial of degree $\leq d$, and its random evaluation is zero with probability $\leq d/|\mathbb{F}|$.

Compiling this PIOP with the IPA-PCS and applying Fiat–Shamir yields a non-interactive argument for the same relation. Its opening proof is logarithmic in the degree bound, but the final IPA MSM makes verification linear unless it is deferred by an accumulation scheme. The PCS handles polynomial hiding and binding; the PIOP handles the relation logic. This separation is precisely the modular proof-system design pattern.

2.4 PLONK as a PIOP

PLONK (§5.1) is a more elaborate PIOP. Its oracles are:

- One advice polynomial per advice column.
- A permutation running-product polynomial Z_{perm} .
- Two permuted lookup columns A', S' per lookup.
- One lookup running-product polynomial per lookup.
- A quotient polynomial h (split into chunks).

Its challenges are, in order: θ (lookup column compression), β, γ (the permutation and lookup products), y (the gate-equation combination), and x (the final evaluation point). Its final check is the PLONK gate equation $C(x) = h(x) \cdot Z_H(x)$, which the verifier checks from the evaluations of the oracles at x (and possibly ωx).

This is precisely the PIOP that phases 1–5 of the Halo 2 protocol of §5.7 compile, phase 5’s claimed evaluations answering its oracle queries; phases 6–7 bind those answers to the commitments — the multipoint reduction (the challenges $x_1, \dots, x_4$ and the combined polynomial f^*) followed by a single IPA-PCS opening of f^* .

3 A rigorous treatment of knowledge soundness

We now develop knowledge soundness with the care needed for arguments about modern SNARKs. The treatment combines the rewinding-extractor framework (the *Crypto Guide*) with the algebraic group model.

3.1 Definitional refinements

The bare definition of the *Crypto Guide* suffices for Σ -protocols but is awkward for compound systems. We record the following standard refinement.

Remark 3.1 (Witness-extended emulation). An interactive argument $(\mathcal{P}, \mathcal{V})$ has *witness-extended emulation* if there is a PPT emulator $\mathcal{E}$ such that for every PPT prover $\mathcal{D}^*$ and every input x :

- Run $\mathcal{D}^*$ honestly with $\mathcal{V}$; let τ be the transcript and b the verifier’s accept/reject bit.
- Run $\mathcal{E}^{\mathcal{D}^*}(x)$; it outputs (τ', b', w) .

The distributions of (τ, b) and (τ', b') are computationally indistinguishable, and moreover whenever $b' = 1$, $(x, w) \in \mathcal{R}$ except with negligible probability.

WEE captures simultaneously two properties: “the extractor produces a witness whenever the protocol accepts” and “the extractor’s external behaviour resembles a normal protocol execution.” It is the appropriate notion for arguments embedded in larger protocols, such as the batched verification of many proofs at once (§5.6); plain knowledge soundness, however, suffices for this volume’s statements.

3.2 The algebraic group model

The IPA’s knowledge-soundness proof proceeds through the tree extractor of §5.5, with knowledge error $O(d/|\mathbb{F}|)$. The interactive tree extractor costs $3^{\log_2 d} = d^{\log_2 3}$ transcripts and runs in expected polynomial time (§3.3); the looseness arises for the non-interactive argument, where the naive Fiat–Shamir analysis of the $\log d$ -round protocol loses a factor $q^{O(\log d)}$ in the adversary’s random-oracle query count q — quasi-polynomial for polynomial q . The *algebraic group model* (AGM; Fuchsbauer, Kiltz, Loss; CRYPTO 2018) enables tighter analyses for appropriate protocols by restricting the adversary’s group operations; it is not by itself a theorem eliminating every Fiat–Shamir loss.

Definition 3.2 (Algebraic adversary). An *algebraic adversary* against a group $\mathbb{G}$ is a PPT algorithm that, whenever it outputs a group element $P \in \mathbb{G}$, additionally outputs an explicit linear combination $P = \sum_i [a_i]Q_i$ expressing P as a combination of previously-seen group elements Q_i .

The AGM sits between the standard model (no restriction on the adversary) and the generic group model (group operations available only through an oracle). Most natural adversaries are algebraic — they compute group elements by known scalar multiplications and additions — so the heuristic is regarded as conservative. In an AGM analysis, an algebraic prover outputs every group element together with an explicit linear combination of the elements it has seen, so the extractor reads candidate witnesses directly off these representations instead of reassembling them from forked transcripts. Ghoshal and Tessaro (CRYPTO 2021) carry out this analysis for Bulletproofs-style inner-product arguments, proving tight state-restoration soundness and deriving Fiat–Shamir security bounds under their stated hypotheses. Applying those bounds to the deployed Halo 2 transcript requires matching the precise folding, masking, challenge distribution, and composition assumptions; we do not prove that instantiation theorem here. The

accumulation analysis of BCMS20 (Bünz, Chiesa, Mishra, Spooner; TCC 2020), by contrast, does not invoke the AGM: it proves security in the random oracle model under the DL assumption (Theorem 5.8).

3.3 Tree-extractor analysis

To render the tree extractor concrete, consider the key step at each round of the IPA halving (see §5.5):

Lemma 3.3 (Single-round algebraic recovery). *Consider one node of a recursive IPA tree extractor. Suppose extraction below each child has already exposed the child’s folded scalar data, and the group messages have the algebraic representations required by the extraction model. In the symmetric normalisation of Construction 5.3, three branches whose nonzero challenges have pairwise distinct squares give an invertible system with rows $(u_j^{-2}, 1, u_j^2)$. In the deployed normalisation of Remark 5.5, the corresponding rows are $(u_j^{-1}, 1, u_j)$ and merely require pairwise distinct nonzero challenges. Solving the system recovers the parent representation and its cross-term consistency.*

Sketch. After the child data have been extracted, the three accepting relations are linear equations in the three parent coefficient terms. Multiplying a deployed row by u_j gives $(1, u_j, u_j^2)$, an ordinary Vandermonde row; its determinant is nonzero for distinct u_j . Multiplying a symmetric-normalisation row by u_j^2 gives $(1, u_j^2, u_j^4)$, so that version requires distinct squares. The raw group equations alone do not reveal discrete logarithms or witness halves; the conclusion uses the recursively extracted child data and the stated algebraic-representation or special-soundness hypotheses. $\square$

The full extraction requires three accepting transcripts at each of the $k = \log_2 d$ rounds, giving a tree of $3^k = d^{\log_2 3} \approx d^{1.585}$ leaves—polynomial in d , though more than the $O(d)$ a binary tree would cost. In the AGM the extractor can read algebraic representations directly. A standard-model or Fiat–Shamir extractor additionally needs the appropriate rewinding or state-restoration theorem; its cost cannot be obtained by simply multiplying every tree node by an independent $1/\varepsilon$ factor.

3.4 Concrete versus asymptotic security

The literature on knowledge soundness contains numerous tightness gaps. The more important ones include:

- Bulletproofs’ naive Fiat–Shamir knowledge-soundness analysis loses a factor $q^{O(\log d)}$, exponential in the round count $\log d$ with base the query count q ; the interactive standard-model extraction itself is polynomial ($d^{\log_2 3}$ transcripts, expected polynomial time).
- Security analyses for Halo 2’s deployed parameters rely on the AGM and/or the random-oracle model.

This reflects the practical reality of deployed SNARKs: tight reductions in idealised models, loose reductions in standard models, and deployment selecting the former.

4 Recursive proof composition and accumulation schemes

We now turn to recursive proof composition: proving, inside one proof, that another proof verifies. This is what makes transparent Halo-style recursion efficient. Orchard’s mainnet deployment does not recurse. Its verifier can batch independent proof equations, but that randomized MSM batching is distinct from the recursive accumulation scheme developed here.

4.1 Recursion and its two classical obstacles

Consider a computation iterated many times, $z_n = f^{(n)}(z_0) := f(f(\dots f(z_0) \dots))$ — Valiant’s *incrementally-verifiable computation* (TCC 2008) — or, more generally, computations that branch and merge, each consuming the proofs of its predecessors — *proof-carrying data* (Chiesa, Tromer; ICS 2010). The natural construction is *recursive SNARK composition*: at step $i + 1$ the prover proves “ $z_{i+1} = f(z_i)$, and there exists a proof π_i that the verifier circuit for step i accepts.” One proof then attests to the entire history, and the per-step proving cost is independent of the history’s length.

For this to work, the SNARK’s verifier must itself be efficiently expressible as a circuit, and here the classical instantiations meet two obstacles:

1. Classical pairing-based systems such as Groth16 and KZG-backed PLONK have constant-size proofs and succinct verifiers, but their verifiers evaluate pairings—costly to express inside a circuit because they require extension-field arithmetic—and these particular systems use trusted setup.
2. Transparent SNARKs built on the IPA need no pairing and no trusted setup, but their verifiers run in linear time — the $O(d)$ MSM of §5.5.2. A linear-time check dominates the verifier circuit and negates the per-step efficiency.

Halo (Bowe–Grigg–Hopwood 2019) defeats obstacle 2 directly — deferring the linear-time part of verification instead of recomputing it at every step — and sidesteps obstacle 1 by using the IPA, which involves no pairing. BCMS20 formalised the deferral as an *accumulation scheme*.

4.2 Accumulation schemes

Definition 4.1 (Accumulation scheme, informal). For a predicate Φ (for example, “this IPA opening is correct”), an *accumulation scheme* consists of:

- An *accumulator*, a compact data structure representing a batch of claims about Φ .
- A *folding* algorithm: given an old accumulator and a new claim about Φ , it produces a new accumulator that represents both — the new accumulator is valid only if the old one was valid and the new claim holds. Folding is fast: $O(\log d)$ in the IPA instantiation below.
- A *decider*, which is expensive (linear-time) but which the scheme invokes only once at the very end to validate the accumulator. Under the scheme’s computational assumptions, an accepting decider implies that every folded claim is true, except with the stated soundness error.

For the IPA-PCS such a scheme exists, and it resolves the recursion problem in the IPA setting: the in-circuit verifier checks only the $O(\log d)$ “succinct” part of an IPA proof and folds the $O(d)$ “deferred” MSM check into a running accumulator. The decider pays the deferred work once, at the end of the entire chain. We state the scheme precisely as Theorem 5.8, once the IPA construction of §5.5 is in hand.

4.3 A cycle of curves for efficient native recursion

The accumulation scheme’s in-circuit folding involves arithmetic on group elements of the underlying curve. For this to be efficient inside the circuit, the group’s base field must equal the circuit’s native field for native coordinate arithmetic. Since the circuit’s native field is the scalar field of the curve on which the prover commits the SNARK — which in turn is generally different from the curve’s own base field — a difficulty arises: an E_p -IPA proof’s group elements have coordinates in $\mathbb{F}_p$, so verifying it natively would require a circuit whose native field is $\mathbb{F}_p$. But a SNARK’s verifier circuit naturally operates over the curve’s *scalar* field $\mathbb{F}_q$ (where $q = |E_p|$): the

Fiat–Shamir challenges and folding scalars are $\mathbb{F}_q$ elements. Curves with $p = q$ do exist — the *anomalous* curves, with trace of Frobenius 1 — but on them the discrete logarithm is computable in polynomial time (Smart; Semaev; Satoh–Araki, 1998–99), so no *secure* curve has $\mathbb{F}_p = \mathbb{F}_q$, and no one curve E_p can host both the proof’s coordinates and its own verifier circuit natively.

An efficient native resolution is a 2-cycle of elliptic curves, where $|E_p| = q$ and $|E_q| = p$. Then:

- A circuit committed on E_q verifies a proof generated over E_p ; that circuit’s native field is the scalar field of E_q , namely $\mathbb{F}_p$ (since $|E_q| = p$) — which coincides with the coordinate field of E_p , enabling native manipulation of E_p ’s group elements.
- A circuit over $\mathbb{F}_q$ verifies the next proof, generated over E_q .
- Proofs alternate between the two curves.

The cycle makes the preceding curve’s *coordinate* arithmetic native; it does not identify both scalar fields. Challenge and scalar computations still need compatible encodings, range checks, and, where required, non-native scalar arithmetic. Nor is the cycle a logical necessity: non-native coordinate arithmetic is possible, at substantial cost. It is the efficient native design used by the Pasta cycle of the *Math Guide*, where the two curves are Pallas (over $\mathbb{F}_{p_{\text{Pallas}}}$, order p_{Vesta}) and Vesta (over $\mathbb{F}_{p_{\text{Vesta}}}$, order p_{Pallas}).

5 The Halo 2 proof system

We now assemble the abstractions of the *Crypto Guide*–§4 into the concrete proof system Halo 2: a transparent PLONKish argument whose polynomial commitment uses one curve at a time. Orchard proves over the Pallas base field and commits on Vesta; the full Pasta cycle is needed only by the optional native-recursion construction above. This section develops Halo 2 to the level of detail required to state *what an Orchard Action proof actually is*.

5.1 PLONKish arithmetisation

Definition 5.1 (PLONKish constraint system). Fix a prime field $\mathbb{F}$ and an integer k defining a multiplicative subgroup $H := \{\omega^0, \dots, \omega^{n-1}\} \subset \mathbb{F}^*$ of size $n := 2^k$, where ω is a primitive n -th root of unity. A *PLONKish circuit* consists of:

- Columns partitioned into *fixed* (preprocessed), *advice* (prover-witness), and *instance* (public-input).
- A maximum constraint degree d .
- A finite sequence of *custom gates* $g_t(\vec{x}_0, \vec{x}_1, \dots) = 0$ — multivariate polynomials of degree $\leq d$ in column values at the current row and at offsets $\omega^r X$ for fixed small r .
- A subset $S \subseteq H \times [\text{\#cols}]$ of *equality-constraint cells* controlled by a permutation σ .
- A set of *lookup arguments*: input tuples $(A_0, \dots, A_{m-1})$ and table tuples $(S_0, \dots, S_{m-1})$, with the assertion that for every row $i \in [0, n)$, $(A_0(\omega^i), \dots, A_{m-1}(\omega^i))$ matches some row of the table.

Vanilla PLONK (Gabizon–Williamson–Ciobotaru, 2019) employs a fixed five-selector universal gate

$$q_M(X)a(X)b(X) + q_L(X)a(X) + q_R(X)b(X) + q_O(X)c(X) + q_C(X) \equiv 0 \pmod{Z_H(X)},$$

where $Z_H(X) := X^n - 1$ is the *vanishing polynomial* of the domain H (it is zero exactly on H , so “ $\equiv 0 \bmod Z_H$ ” means “zero at every row”). “PLONKish” generalises this by permitting designer-chosen gates of higher arity and degree, each gated by a selector $q_i(X)$ vanishing on rows where the gate is inactive. The Orchard Action circuit uses ten advice columns plus a small fixed-column lookup table for Sinsemilla; its custom gates encode the algebraic constraints of the Action statement together with the complete- and incomplete-addition arithmetic for in-circuit elliptic-curve operations (the *Math Guide* names the distinction).

Polynomial representation. The Lagrange basis $\ell_j(X)$ for H , where $\ell_j(\omega^j) = 1$ and $\ell_j(\omega^{j'}) = 0$ for $j' \neq j$, interpolates each column into a polynomial of degree $< n$ over H . The prover commits to each polynomial via the IPA-PCS (§5.5). All subsequent arguments are *polynomial identities* on these committed polynomials, verifiable by querying them at random points.

5.2 From statement to circuit: arithmetisation in practice

Definition 5.1 states what a PLONKish circuit *is*; it does not say how a statement *becomes* one. The deployed code is organised into *chips*, reusable circuit components each packaging a set of columns, the gates and lookups declared on them, and the synthesis code that fills their cells. This subsection traces that passage: a real custom gate (the ECC chip’s incomplete point addition), two real lookup declarations (the ten-bit range check and the Sinsemilla generator table), and the synthesis machinery that turns witness data into the column vectors on which every argument of this section operates. The source versions audited here are `halo2_proofs` 0.3.2, `halo2_gadgets` 0.5.0, and `orchard` 0.15.3, with file paths relative to each crate’s `src/`. This audited Orchard release includes both the post-NU6.2 circuit correction and the post-NU6.3 circuit version.

Shape first, data second. A circuit, to `halo2_proofs`, is a type implementing the `Circuit` trait, whose two principal methods split the construction into phases. The static method `configure` receives a mutable `ConstraintSystem` and declares the *shape*: it mints fixed, advice, and instance columns, allocates selectors, registers gates and lookups, and marks columns as equality-enabled. No witness exists at this point; `configure` is not even handed an instance of the circuit type. The method `synthesize` runs later, with the data, and fills cells. The split is the proof system’s division of labour: everything `configure` declares is public and determines the verifying key — the verifier’s entire view of the circuit — while everything `synthesize` writes into advice cells is the prover’s witness. (The verifying-key and proving-key generation routines also run `synthesize`, against a witness-free backend that records fixed-cell values, selector activations, and copy constraints while ignoring advice; the prover runs the same code against a backend that records only advice. One synthesis procedure, two observers: `Assembly` and `WitnessCollection`.)

Gates are queried identities. A call `meta.create_gate(name, closure)` defines one custom gate. Inside the closure, each `query_advice(column, Rotation(r))` — and its fixed and instance analogues — yields a symbolic handle to “the value of this column r rows below the gate’s own row”: the offsets $\omega^r X$ of Definition 5.1, realised at evaluation time by querying the column polynomial at $\omega^r x$. The closure combines the handles by field arithmetic into an expression tree (`Expression`: constants, selector nodes, queried cells, negations, sums, products, scalings), one tree per constraint, and the helper `Constraints::with_selector` multiplies every constraint c by the gate’s selector expression, storing $q \cdot c$. This helper is the mechanical origin of the “selector times polynomial” shape of every deployed gate but one — the generalisation, gate by gate, of the fixed vanilla-PLONK equation of §5.1. (The exception, the ECC chip’s witness-point gate, multiplies by hand: its $(q \cdot x) \cdot (y^2 - x^3 - 5)$ and the helper’s $q \cdot (x \cdot (y^2 - x^3 - 5))$ are equal polynomials but different expression trees, and the pinned verifying key freezes the

tree; halo2_gadgets, ecc/chip/witness_point.rs.) A selector is virtual at this stage: a named boolean column, off everywhere by default, enabled row by row during synthesis. At key generation, `compress_selectors` packs the selectors into genuine fixed columns — several to a column wherever the circuit’s degree bound leaves headroom — and rewrites every selector node into a fixed-column expression. The deployed Action circuit declares 56 selectors, whose compressed columns are among the 29 fixed columns pinned in its verifying key; the pinned count is the post-compression one.

Regions, the layout, and wiring. Data enters at synthesis through a Layouter. A chip requests a *region* — a small rectangular window of cells addressed by relative offsets — fills it cell by cell, and leaves the choice of the region’s absolute starting row to the *floor planner*. Values flow between regions not by position but by *copy constraints*: the call `enable_equality(column)` enrolls a column in the single global permutation of §5.3, and `constrain_equal` — or the assign-and-constrain combination `copy_advice` — merges the two cells’ cycles in that permutation (the halo2 book’s *Permutation argument* chapter documents the cycle-merging). Public inputs bind the same way, an advice cell copy-constrained to an instance cell (§5.8). The deployed circuit uses the measuring V1 floor planner, which records every region’s shape in a witness-free pass, sorts the regions by advice area, and first-fits the largest ones first; the orchard crate additionally pins the planner’s sort to a vendored Rust 1.56.1 `pdqsort` (feature `floor-planner-v1-legacy-pdqsort`) so that the layout — and with it the consensus verifying key — is identical across platforms and toolchains.

A deployed gate: incomplete point addition. The ECC chip’s incomplete-addition gate (halo2_gadgets, ecc/chip/add_incomplete.rs) computes $P + Q = R$ on Pallas under the promises $P, Q \neq O$ and $x_p \neq x_q$. Its configuration is one selector $q_{\text{add-incomplete}}$ and four equality-enabled advice columns x_p, y_p, x_q, y_q ; the gate queries x_p, y_p, x_q, y_q at the current row and x_r, y_r at the next row of the *same* two columns x_{qr}, y_{qr} (hence their names: Q on the selector row, R below it). The two constraints, named x_r and y_r in the source, are

$$q_{\text{add-incomplete}} \cdot ((x_r + x_q + x_p)(x_p - x_q)^2 - (y_p - y_q)^2) = 0, \quad (1)$$

$$q_{\text{add-incomplete}} \cdot ((y_r + y_q)(x_p - x_q) - (y_p - y_q)(x_q - x_r)) = 0. \quad (2)$$

These are the affine chord formulas (the *Math Guide*) with the slope eliminated: for $x_p \neq x_q$ they are equivalent to $x_r + x_q + x_p = \lambda^2$ and $y_r + y_q = \lambda(x_q - x_r)$ with $\lambda := (y_p - y_q)/(x_p - x_q)$, multiplied through by $(x_p - x_q)^2$ and $(x_p - x_q)$ respectively to clear the denominators (the halo2 book’s *Addition* chapter records the same clearing). The raw degrees are three and two; with the selector factor, four and three — well inside the deployed circuit’s constraint-degree bound of nine (below). The *complete*-addition companion gate, which branches over the identity, doubling, and inverse cases that this gate excludes, needs twelve constraint polynomials of degree up to six and five auxiliary witness cells per invocation; incompleteness is a bargain wherever its promises can be kept.

Synthesis for one invocation (`assign_region`, same file) makes the witness path concrete. The chip enables the selector at the region’s row i ; copies the four input coordinates into that row with `copy_advice`, which both assigns each cell and copy-constrains it back to its source; computes, *out of circuit*,

$$\lambda = \frac{y_q - y_p}{x_q - x_p}, \quad x_r = \lambda^2 - x_p - x_q, \quad y_r = \lambda(x_p - x_r) - y_p;$$

and assigns x_r, y_r into row $i + 1$ (Table 1). The slope λ is never written to any cell: the prover uses it to compute the answer, and the committed identities (1)–(2) verify the answer without it. The division is not even performed here — coordinates are carried as deferred fractions, and every

denominator falls to a single batch inversion when synthesis ends. The output cells are returned as a new non-identity point with no further on-curve check — sound because, once the call site guarantees distinct x -coordinates on non-identity inputs, the two identities force $R = P + Q$, a non-identity curve point.

row	x_p	y_p	x_qr	y_qr	$q_{\text{add-incomplete}}$
i	x_p	y_p	x_q	y_q	1
$i + 1$	—	—	x_r	y_r	0

Table 1: The incomplete-addition region: two rows by four advice columns. The four input cells x_p, y_p, x_q, y_q on the selector row i are copy-constrained to the cells they came from; the result cells x_r, y_r occupy the x_{qr}, y_{qr} columns of row $i + 1$, where the gate’s `Rotation::next()` queries find them. Dashes mark cells the gate does not use.

Incompleteness, and where it is discharged. Reading the two polynomials at the excluded inputs shows exactly what “incomplete” costs. If $Q = -P \neq O$ (so $x_p = x_q$ and $y_p = -y_q \neq 0$), constraint (1) collapses to $-(2y_p)^2 \neq 0$: no assignment satisfies the gate — a completeness failure, never a soundness one. If $Q = P$, both polynomials vanish identically and (x_r, y_r) is *unconstrained*: a reachable doubling would let the prover claim any result. If an input were the identity in the gadget’s $(0, 0)$ affine encoding, the constraints would accept a garbage output. The deployed circuit discharges the three exclusions in three different places. Non-identity travels with the type: the gate’s inputs are `NonIdentityEccPoint` values. A fresh point entering the chip is witnessed under the gate

$$q_{\text{point-non-id}} \cdot (y^2 - x^3 - 5) = 0,$$

which $(0, 0)$ cannot satisfy because 5 is not a square and -5 is not a cube in $\mathbb{F}_{p_{\text{Pallas}}}$ (the halo2 book’s *Addition* chapter) — but no fresh witness ever reaches incomplete addition. In the deployed circuit the incomplete-addition gate fires only inside fixed-base scalar multiplication (`ecc/chip/mul_fixed.rs`), which sums precomputed windowed multiples $[(k + 2) \cdot 8^w]B$ of a fixed base B . There the addends are assembled by the type’s unchecked constructor (`from_coordinates_unchecked`) and pinned instead to the precomputed window table by `mul_fixed`’s own coordinate gates (`coords_check`, same file): the x -coordinate must equal a degree-seven polynomial in the window value whose fixed coefficients interpolate the window’s eight table x -coordinates, and the y -coordinate must satisfy $y + z_w = u^2$ against a per-window fixed z_w and a witnessed u , alongside an on-curve check — together pinning the point to the table entry selected by the window digit, and no entry of the table is the identity. The running accumulator is a sum of such entries under this very gate; the type carries the non-identity guarantee from either construction site to the gate, which re-checks nothing. The distinctness $x_p \neq x_q$ is enforced by no constraint at all; it too is structural: the window tables store $[(k + 2) \cdot 8^w]B$ rather than the naive $[(k + 1) \cdot 8^w]B$ precisely because the naive offset admits a window collision that produces the fatal doubling (the halo2 book’s *Fixed-base scalar multiplication* chapter). Prover-side, `assign_region` rejects a *known* bad witness with an error — a developer diagnostic, not a constraint. (The circuit’s other incomplete additions — the variable-base ladder’s fused double-and-add rounds and Sinsemilla’s two per chunk — are separate gates built on the same discipline.)

Non-algebraic relations become lookups. A gate can assert only polynomial identities. “This value has ten bits” and “this point is the j -th Sinsemilla generator” are not identities but set memberships, and the circuit declares them as lookups: a call `meta.lookup(closure)` returns pairs of an input *expression* and a table column, and pushes one lookup argument onto the constraint

system. Each declaration is one instance of the argument of §5.4, the input expressions playing $A_0, \dots, A_{m-1}$ and the table columns $S_0, \dots, S_{m-1}$. The deployed circuit carries exactly three: one shared range-check declaration, and the Sinsemilla generator declaration twice — once per Sinsemilla chip instance, each on its own half of the ten advice columns.

All three consume the same table. Loaded once at synthesis, it holds, for $0 \leq j < 2^{10}$,

$$\text{row } j: \quad (j, X(S[j]), Y(S[j])), \quad S[j] := \text{GroupHash}(\text{"z.cash:SinsemillaS"}, \text{LE}_{32}(j)),$$

in three table columns $(t_{\text{id}_x}, t_x, t_y)$; the generators ship as precomputed constants that the sinsemilla crate's own tests re-derive from the hash. After the assignment, the layouter fills every remaining usable row of each table column with that column's row-0 value; a defaulted input row therefore still matches a genuine table row, and the type `TableColumn` hides its underlying fixed column exactly so that no chip can bypass this default-filling.

The range check is one declaration serving two modes (`halo2_gadgets`, `utilities/lookup_range_check.rs`):

$$q_{\text{lookup}} \cdot (q_{\text{running}} \cdot (z_i - 2^{10} z_{i+1}) + (1 - q_{\text{running}}) \cdot z_i) \mapsto t_{\text{id}_x}, \quad (3)$$

with z_i and z_{i+1} the running-sum advice column at the current and next rotations. With $q_{\text{running}} = 1$ the input is a running-sum step: the prover witnesses $z_{i+1} = (z_i - a_i) \cdot 2^{-10}$ on successive rows, the expression recovers the limb $a_i = z_i - 2^{10} z_{i+1}$, and membership in $t_{\text{id}_x} = \{0, \dots, 2^{10} - 1\}$ asserts that each limb of the decomposition has ten bits; a *strict* decomposition additionally copy-constrains the final z to the constant 0, range-constraining the decomposed element to $10w$ bits for w limbs. With $q_{\text{running}} = 0$ the cell itself is looked up (checks of fewer than ten bits add a small bit-shift gate in the same file). Every range check in the Action circuit funnels through this one declaration, on a single advice column.

The Sinsemilla declaration is the three-column instance:

$$(q_{S1} \cdot m, q_{S1} \cdot x_p + (1 - q_{S1}) \cdot X(S[0]), q_{S1} \cdot y_p + (1 - q_{S1}) \cdot Y(S[0])) \mapsto (t_{\text{id}_x}, t_x, t_y), \quad (4)$$

where $m = z_i - 2^{10} \cdot q_{\text{run}} \cdot z_{i+1}$ recovers the ten-bit message word from the hash's own running-sum decomposition (with $q_{\text{run}} := q_{S2} - q_{S2}(q_{S2} - 1)$, an expression in a fixed column $q_{S2} \in \{0, 1, 2\}$ that evaluates to 1 on the interior rows of a message piece and to 0 on final rows, where the trailing running sum is implicitly zero), and where y_p is not a witnessed cell at all but the derived expression $Y_A \cdot 2^{-1} - \lambda_1(x_A - x_p)$ over the chip's double-and-add columns (`ecc/chip/mul/incomplete.rs` supplies $Y_A := (\lambda_1 + \lambda_2)(x_A - x_R)$ and $x_R := \lambda_1^2 - x_A - x_p$). Lookup inputs, in other words, may be arbitrary expressions over queried cells, not merely cells; the chip proves that the generator it is adding is $S[m]$ without ever spending an advice cell on $y_{S[m]}$. The $(1 - q_{S1})$ branches default every row on which the chip is inactive to table row 0, so the membership assertion of §5.4 holds on all rows. One API restriction is visible here: the selectors gating lookup inputs ($q_{\text{lookup}}, q_{\text{running}}, q_{S1}$) must be declared as `complex_selectors` — simple selectors are reserved for gates, where the key-generation compression is licensed to rewrite them.

The handoff. When `synthesize` returns, the construction has produced nothing exotic: every column — advice, fixed, instance — is a vector of $n = 2^k$ field elements; in the advice columns the deferred fractions are resolved by one batch inversion and the reserved trailing rows filled with blinding randomness (§5.9, Remark 5.2). The prover commits to each advice column directly in the Lagrange basis (`commit_lagrange` — the same group element as committing the interpolated coefficients, so the commitment even precedes interpolation) and then interpolates each column over H by an inverse FFT (`lagrange_to_coeff`): these are precisely the column polynomials of §5.1. Each gate's stored expression tree is then evaluated with the committed column polynomials substituted for the queried cells — a query at rotation r reading the polynomial at $\omega^r X$ — and the results are folded with powers of the challenge y and divided by $Z_H(X) = X^n - 1$. The

gate identities declared in `configure` have become the quotient $h(X)$ of Phase 4 (§5.7). This is the point where the worked example of §6.3 resumes: it interpolates a four-row trace and assembles exactly this $G(X)$ whose divisibility by Z_H the rest of the protocol certifies — and where Orchard’s deployed gadget stack bottoms out: each gadget is a bundle of `configure`-time gates and lookups plus `synthesize`-time regions, and after synthesis nothing remains of the stack but rows in these same columns. The scale-up from the toy is numerical, not conceptual. The deployed Action circuit has $k = 11$ ($n = 2048$ rows), the ten advice columns of §5.1, one instance column, and twenty-nine fixed columns, with a constraint-degree bound of nine across its gates and permutation and lookup expressions — so the quotient splits into $9 - 1 = 8$ chunks, and the prover’s FFTs run over an extended domain of size $2^{14} = 8 \cdot 2^{11}$. The next two subsections supply the arguments this construction leans on: the permutation argument behind every copy constraint, and the lookup argument behind declarations (3) and (4).

5.3 The permutation (equality) argument

A PLONK-style permutation argument, which Halo 2 generalises to multiple columns, enforces equality constraints between distinct cells. Each cell (i, j) (column i , row j) receives a unique label $\delta_i \cdot \omega^j$ with δ_i chosen so all products are distinct as (i, j) ranges over the trace. Preprocessing fixes σ -permuted polynomials $s_i(X)$ with $s_i(\omega^j) = \delta_{i'} \cdot \omega^{j'}$ when $\sigma(i, j) = (i', j')$; the verifying key contains commitments to them.

Given Fiat–Shamir challenges $\beta, \gamma \in \mathbb{F}$, the prover commits to a running-product polynomial $Z_{\text{perm}}(X)$ asserting, where $v_i(X)$ denotes the Lagrange interpolation (§5.1) of the i -th column participating in the permutation,

$$\prod_{i,j} (v_i(\omega^j) + \beta \delta_i \omega^j + \gamma) = \prod_{i,j} (v_i(\omega^j) + \beta s_i(\omega^j) + \gamma).$$

The assignment satisfies every equality constraint (cells related by σ carry equal values) precisely when the denominator factors are a permutation of the numerator factors, in which case the two products agree for every β, γ (and their ratio is 1 whenever its denominator is nonzero); conversely, if some constraint is violated, the two products at the sampled β, γ are equal with probability at most $O(N/|\mathbb{F}|)$ by Schwartz–Zippel, where N is the number of cells participating in the permutation. In the one-product presentation the verifier checks this by opening Z_{perm} at x and ωx , verifying the boundary $Z_{\text{perm}}(\omega^0) = 1$ and a cross-multiplied row-by-row update, which remains defined when a factor is zero. Excluding the final IPA MSM, this argument contributes $O(m + \log n)$ verifier work, where m is the number of participating columns; the complete deployed verifier still has the $O(n)$ commitment check of §5.5.2.

In the deployed degree-bounded construction, participating columns are split into chunks of at most $d - 2$ columns, each with its own running product; boundary constraints link successive products. Orchard’s fifteen equality columns (ten advice, one instance, and four fixed) at $d = 9$ therefore use three products, not one. This is implemented in `halo2_proofs 0.3.2` (`chunk_len = cs_degree - 2`); the pinned Orchard circuit description lists the participating columns.

5.4 The lookup argument

Halo 2’s lookup argument is structurally simpler than `plookup` (Gabizon–Williamson 2020) and supports arbitrary (possibly non-fixed) tables.

Objective. Establish that every entry of column $A(X)$ on the active domain appears as some entry of column $S(X)$.

Construction. The prover supplies polynomials $A'(X), S'(X)$ where (i) A' is a permutation of A with like values grouped into contiguous runs, and (ii) S' is a permutation of S in which the first row of each run of like values in A' has the matching value in S' . The prover commits to A' and S' ; after receiving challenges β, γ , it commits to a running-product column $Z_{\text{lookup}}(X)$ and proves:

$$\begin{aligned} \ell_0(X)(1 - Z_{\text{lookup}}(X)) &= 0, \\ Z_{\text{lookup}}(\omega X)(A'(X) + \beta)(S'(X) + \gamma) - Z_{\text{lookup}}(X)(A(X) + \beta)(S(X) + \gamma) &= 0, \\ \ell_0(X)(A'(X) - S'(X)) &= 0, \\ (A'(X) - S'(X))(A'(X) - A'(\omega^{-1}X)) &= 0. \end{aligned}$$

The first two constraints, with the boundary conditions of Remark 5.2, certify with the usual Schwartz–Zippel error that A' and S' are permutations of A and S respectively; the third aligns row 0 ($A'_0 = S'_0$), where the wrap-around reference $A'(\omega^{-1}X)$ deprives the fourth constraint of force. The fourth constrains every subsequent row either to start a new run aligned with S' ($A'_i = S'_i$) or to continue the previous one ($A'_i = A'_{i-1}$). Inductively, every value of A equals some value of S' , hence of S .

A random-linear combination with a challenge θ reduces multi-column lookups to single-column: $A_c(X) := \sum_j \theta^{m-1-j} A_j(X)$. Variable tables (where S is itself an advice column) and range checks (where S enumerates the allowed range) are special cases.

Remark 5.2 (Active rows and blinding). The lookup constraints above, like the permutation constraints of §5.3, hold on the *active* rows only: the deployed system reserves the trailing rows of every column for random blinding values (§5.9). Each product-update constraint is therefore multiplied by the factor $(1 - (\ell_{\text{last}}(X) + \ell_{\text{blind}}(X)))$, where ℓ_{blind} and ℓ_{last} are Lagrange-basis selectors of the blinding rows and of the boundary row immediately before them — computed directly from the domain by both parties, not committed circuit columns — and boundary constraints such as $\ell_{\text{last}}(X)(Z(X)^2 - Z(X)) = 0$ pin the running product to $\{0, 1\}$ at the boundary row. The random rows consequently do not violate the active-row constraints. Soundness still excludes the bad challenges for which product factors vanish: the boundary value zero is not an unconditional certificate of multiset equality. Their probability belongs in the full algebraic error bound.

5.5 The inner-product polynomial commitment

The Halo 2 polynomial commitment, instantiating the abstract PCS of the *Crypto Guide*, is the inner-product argument of Bootle–Cerulli–Chaidos–Groth–Petit (EUROCRYPT 2016) as refined in Bulletproofs (Bünz et al., 2017). We give the construction in detail. Write $d := 2^k$ for the exclusive degree bound (the number of generators, supporting degree at most $d - 1$); the round count $k = \log_2 d$ reuses the letter k of Definition 5.1, harmlessly, because in the assembled protocol of §5.7 the committed polynomials have degree $< n$, so $d = n = 2^k$ there. The letter H likewise does double duty, inherited from two literatures: $H \subset \mathbb{F}$ is the PLONK evaluation domain (as in Z_H), while $H \in \mathbb{G}$ is the Pedersen blinding generator of the *Crypto Guide*; the ambient set ($\mathbb{F}$ versus $\mathbb{G}$) disambiguates every occurrence. Two further letters carry double duty: p denotes the group order (as in $\mathbb{F}_p$) and, with an argument or in $\deg p$, the committed polynomial $p(X)$; and the permutation polynomials $s_i(X) \in \mathbb{F}[X]$ of §5.3 are distinct from the IPA structured scalars $s_i \in \mathbb{F}_p$ of §5.5.2. Type and argument disambiguate every occurrence.

Construction 5.3 (IPA-based polynomial commitment). Let $\mathbb{G}$ be a cyclic group of prime order p in which DL is hard. Choose $d = 2^k$ and sample, via hash-to-curve, $\vec{G} = (G_1, \dots, G_d)$ and $H \in \mathbb{G}$ with no known non-trivial linear relation among them.

For $p(X) = \sum_{i=0}^{d-1} a_i X^i \in \mathbb{F}_p[X]$ and blinder $r \in \mathbb{F}_p$,

$$\text{Commit}(p; r) := \langle \vec{a}, \vec{G} \rangle + [r]H \in \mathbb{G}.$$

To open p at $x \in \mathbb{F}_p$ with claimed value v , the relation is

$$\mathcal{R}_{\text{IPA}} := \left\{ ((P, x, v); (\vec{a}, r)) : P = \langle \vec{a}, \vec{G} \rangle + [r]H, v = \langle \vec{a}, \vec{b}(x) \rangle, \deg p \leq d-1 \right\},$$

with $\vec{b}(x) := (1, x, x^2, \dots, x^{d-1})$.

5.5.1 The recursive halving protocol

The argument runs in $k = \log_2 d$ rounds. Initialise $\vec{a}^{(k)} := \vec{a}$, $\vec{G}^{(k)} := \vec{G}$, $\vec{b}^{(k)} := \vec{b}(x)$, and let $U \in \mathbb{G}$ be a challenge group element independent of the commitment generators, with unknown discrete-log relations (in Fiat–Shamir, derived from the transcript, excluding the identity). The prover absorbs $[v]U$ so that the relation becomes

$$P + [v]U = \langle \vec{a}, \vec{G} \rangle + [r]H + [\langle \vec{a}, \vec{b} \rangle]U.$$

In round j (descending from k to 1), sample independent uniform $\ell_j, r_j \in \mathbb{F}_p$, split each vector into halves, and let

$$\begin{aligned} L_j &:= \langle \vec{a}_{\text{lo}}^{(j)}, \vec{G}_{\text{hi}}^{(j)} \rangle + [\ell_j]H + [\langle \vec{a}_{\text{lo}}^{(j)}, \vec{b}_{\text{hi}}^{(j)} \rangle]U, \\ R_j &:= \langle \vec{a}_{\text{hi}}^{(j)}, \vec{G}_{\text{lo}}^{(j)} \rangle + [r_j]H + [\langle \vec{a}_{\text{hi}}^{(j)}, \vec{b}_{\text{lo}}^{(j)} \rangle]U. \end{aligned}$$

The verifier returns a uniform $u_j \in \mathbb{F}_p^*$. Both parties fold:

$$\begin{aligned} \vec{a}^{(j-1)} &:= u_j \vec{a}_{\text{lo}}^{(j)} + u_j^{-1} \vec{a}_{\text{hi}}^{(j)}, \\ \vec{G}^{(j-1)} &:= u_j^{-1} \vec{G}_{\text{lo}}^{(j)} + u_j \vec{G}_{\text{hi}}^{(j)}, \\ \vec{b}^{(j-1)} &:= u_j^{-1} \vec{b}_{\text{lo}}^{(j)} + u_j \vec{b}_{\text{hi}}^{(j)}. \end{aligned}$$

After k rounds each vector has length 1. The prover sends the final scalar $a := \vec{a}^{(0)}$ and the aggregated blinder

$$r^* := r + \sum_{j=1}^k (u_j^2 \ell_j + u_j^{-2} r_j).$$

The verifier accepts iff

$$P + [v]U + \sum_{j=1}^k ([u_j^2]L_j + [u_j^{-2}]R_j) \stackrel{?}{=} [a]G^{(0)} + [r^*]H + [a \cdot b^{(0)}]U, \quad (5)$$

where $G^{(0)} = \langle \vec{s}, \vec{G} \rangle$ and the challenges determine $\vec{s}$ as below.

Remark 5.4 (Folding alone is not zero knowledge). The symmetric presentation above proves the inner-product relation, but does not by itself hide the witness: the final scalar a reveals a challenge-dependent linear form in the original coefficients. Blinding the commitment and the round points hides those points, not this scalar. The deployed protocol first masks the polynomial by an independently random polynomial vanishing at the opening point; its commitment S and subsequent challenge are essential to the full opening's zero-knowledge argument (Remark 5.5).

5.5.2 The structured scalars and the polynomial g

Write the binary expansion $i = \sum_{\ell=0}^{k-1} b_\ell 2^\ell$ for $i \in [0, d)$. By induction on the folding,

$$G^{(0)} = \langle \vec{s}, \vec{G} \rangle, \quad b^{(0)} = \langle \vec{s}, \vec{b}(x) \rangle = g(x; u_1, \dots, u_k),$$

where

$$s_i := \prod_{\ell=0}^{k-1} u_{\ell+1}^{(-1)^{1-b_\ell}}, \quad g(X; u_1, \dots, u_k) := \prod_{\ell=1}^k (u_\ell^{-1} + u_\ell X^{2^{\ell-1}}).$$

The essential asymmetry is the following: $g(X)$ has degree $d-1$ and the verifier can evaluate it at x in $O(\log d)$ field operations; but computing $G^{(0)} = \text{Commit}(g; 0)$ requires an $O(d)$ -size multi-scalar multiplication (MSM). This asymmetry is the seed of the accumulation scheme (§5.6).

Remark 5.5 (The deployed IPA normalisation). The deployed opening (halo2_proofs 0.3.2, poly/commitment/prover.rs and verifier.rs) uses the rescaled Halo convention, not the Bulletproofs one above: the folding weights are $(1, u_j^{-1})$ on the coefficient vector and $(1, u_j)$ on $\vec{b}$ and $\vec{G}$, the verifier accumulates $[u_j^{-1}]L_j + [u_j]R_j$, and the deferred polynomial is $g(X) = \prod_{i=0}^{k-1} (1 + u_{k-1-i} X^{2^i})$. The opening also begins with an extra blinding commitment S and two challenges ξ, z , and ends with two scalars: the collapsed coefficient c and a synthetic blinder f . The two conventions are algebraically related by per-round rescaling and a formal relabelling of the challenge. This does *not* identify their challenge distributions: a uniform field element in the code is not the square of a uniform field element over all of $\mathbb{F}_p$. Consequently extraction and soundness conditions must be stated for the deployed coefficients $u_j^{-1}, 1, u_j$, as in Lemma 3.3, rather than transferred automatically from the square-weighted presentation. The code squeezes full-field challenges without rejection; a zero u_j makes the required inversion fail, and analogous denominator collisions occur elsewhere with negligible probability. Thus its implementation-level completeness is overwhelming rather than literally perfect.

5.5.3 Worked example: an IPA opening at $d = 4$

Take $d = 4$, so $k = 2$ rounds. Let $p(X) = a_0 + a_1X + a_2X^2 + a_3X^3$ with $\vec{a} = (a_0, a_1, a_2, a_3)$. The verifier holds $P = \langle \vec{a}, \vec{G} \rangle + [r]H$ (we suppress r to declutter; the H -part propagates identically) and requires $v = p(x)$ at some $x \in \mathbb{F}_p$.

Set $\vec{b}^{(2)} := (1, x, x^2, x^3)$ and $\vec{G}^{(2)} := \vec{G}$. After the prover absorbs $[v]U$, the relation is $P + [v]U = \langle \vec{a}, \vec{G} \rangle + \langle \vec{a}, \vec{b}^{(2)} \rangle U$.

Round 2. Split into halves of size 2. The prover sends

$$\begin{aligned} L_2 &= a_0G_3 + a_1G_4 + (a_0x^2 + a_1x^3)U, \\ R_2 &= a_2G_1 + a_3G_2 + (a_2 + a_3x)U. \end{aligned}$$

The verifier picks $u_2 \in \mathbb{F}_p^*$ and both parties fold to length-2 vectors as above.

Round 1. Split the length-2 vectors. The prover sends L_1, R_1 analogously; the verifier picks u_1 . After folding, all three vectors have length 1.

Final check. With $i = b_0 + 2b_1$ ranging over $\{0, 1, 2, 3\}$:

$$s_0 = u_1^{-1}u_2^{-1}, \quad s_1 = u_1u_2^{-1}, \quad s_2 = u_1^{-1}u_2, \quad s_3 = u_1u_2.$$

The verifier evaluates $g(x; u_1, u_2) = (u_1^{-1} + u_1x)(u_2^{-1} + u_2x^2)$ in four field multiplications (u_1x, x^2, u_2x^2 , and the product of the two factors), and computes $G^{(0)} = \sum_{i=0}^3 [s_i]G_{i+1}$ as a length-4 MSM. The check (5) becomes

$$P + [v]U + [u_1^2]L_1 + [u_1^{-2}]R_1 + [u_2^2]L_2 + [u_2^{-2}]R_2 \stackrel{?}{=} [a]G^{(0)} + [a \cdot g(x; u_1, u_2)]U.$$

The folding rule for $\vec{G}$ ensures that $G^{(0)}$ is a specific linear combination of the original $\vec{G}$ with coefficients $\vec{s}$; that the s_i match $\prod_{\ell}(u_{\ell}^{-1} \text{ or } u_{\ell})$ according to the binary expansion of i is exactly the closed-form identity. The verifier can evaluate g in $O(\log d)$ field operations (about $3k - 2$: k coefficient products, $k - 1$ squarings of x , $k - 1$ factor multiplications) but recomputing $G^{(0)}$ takes $O(d)$ MSM — the asymmetry the accumulator defers.

5.5.4 Knowledge soundness via rewinding

Theorem 5.6 (Knowledge soundness of IPA). *Under the DL assumption on $\mathbb{G}$, the interactive opening protocol of Construction 5.3 is knowledge-sound for $\mathcal{R}_{\text{IPA}}$ with knowledge error $O(d/|\mathbb{F}_p|)$, via an expected-polynomial-time tree extractor. The Fiat–Shamir-compiled argument inherits the bound only up to the naive $q^{O(\log d)}$ random-oracle query loss; the tight non-interactive bound needs the AGM (§3).*

Sketch of the tree extractor. The extractor recursively obtains the folded scalar data below each child, then rewinds the prover $\mathcal{P}^*$ to obtain three accepting branches that agree on (L_j, R_j) . Lemma 3.3 then inverts the resulting linear system to recover the parent representation. For Construction 5.3, the three nonzero challenges need pairwise distinct squares because the coefficients are $u_j^{-2}, 1, u_j^2$; for the deployed normalisation, pairwise distinct nonzero values suffice because the coefficients are $u_j^{-1}, 1, u_j$. Three raw group equalities alone would not reveal the scalar halves: recursive child extraction and the protocol’s special-soundness or algebraic-representation hypotheses are essential.

Extracting the full witness $\vec{a} \in \mathbb{F}_p^d$ requires three sibling transcripts at the deepest level, which yield one round-1 witness; three such triples, hung off pairwise distinct challenges at the next-shallower round, yield one round-2 witness; and so on, each shallower round tripling the requirement. The tree thus has $3^k = d^{\log_2 3}$ leaves — polynomial in d (§3) — yielding polynomial expected extraction time when $d = \text{poly}(\lambda)$. The theorem’s knowledge error of $O(d/|\mathbb{F}_p|)$ does not come from a tree-wide union bound. Each of the $k = \log_2 d$ rounds excludes $O(1)$ bad challenges (equal squares in the symmetric presentation, or equal values in the deployed one), contributing $O(\log d/|\mathbb{F}_p|) \leq O(d/|\mathbb{F}_p|)$; Schwartz–Zippel terms of degree d account for the rest. Collisions among the resampled challenges inside the extractor’s 3^k -leaf tree affect only its expected running time — the extractor rewinds and resamples — not the knowledge error; a naive union bound over the roughly 3^k sibling challenge draws would bound the tree-wide collision probability by $O(d^{\log_2 3}/|\mathbb{F}_p|)$, still negligible, but that is not the source of the theorem’s figure.

This is the recursive generalisation of special soundness for Schnorr (the *Crypto Guide*): a single transcript reveals nothing, but a few transcripts on the same first message permit inversion of a small linear system; in the IPA the system is 3×3 per round and the recursion stacks k rounds. $\square$

5.6 Accumulation and the Halo trick

The verifier’s check (5) runs in $O(\log d)$ scalar operations and $\log d$ point additions, *except* for the single evaluation $G^{(0)} = \langle \vec{s}, \vec{G} \rangle$ — an MSM of length d . For optional recursive composition, the accumulation idea (Bowe–Grigg–Hopwood 2019; formalised by BCMS20) is to *defer* this MSM and amortise it across many recursively composed proofs.

Definition 5.7 (Atomic accumulator for IPA). An IPA *accumulator instance* is a tuple $\text{Acc} := (G^{(0)}, u_1, \dots, u_k) \in \mathbb{G} \times \mathbb{F}_p^k$. It is *valid* iff $G^{(0)} = \langle \vec{s}(\vec{u}), \vec{G} \rangle = \text{Commit}(g(X; u_1, \dots, u_k); 0)$ — the deferred claim concerns the unblinded commitment, the blinder having been discharged by the $[r^*]H$ term of (5). Subsequent unary $\text{Commit}(\cdot)$ abbreviates $\text{Commit}(\cdot; 0)$.

Theorem 5.8 (IPA accumulation, informal BCMS20 interface). *BCMS20 constructs an accumulation scheme $(\text{AccVerifier}, \text{AccDecider})$ for IPA with:*

- *AccVerifier, on input a new IPA instance and an old accumulator, runs in $O(\log d)$ and outputs a new accumulator Acc' .*
- *AccDecider, run once at the end of a chain, performs an MSM of length d to check $G^{(0)} = \text{Commit}(g)$.*
- *Soundness reduces to DL on $\mathbb{G}$ in the random-oracle model.*

Operational interpretation. A Halo 2 verifier circuit, expressed over the scalar field of one Pasta curve and verifying a proof produced over the *other* curve, can:

1. Run the $O(\log d)$ “succinct” checks of the IPA verifier natively.
2. Witness the next-step claimed $G^{(0)'}$ and the next challenges $u'_1, \dots, u'_k$ as public inputs to the next proof, deferring the MSM check.
3. Fold the deferred MSM checks: with a random challenge ρ , the two claims $G^{(0)} = \text{Commit}(g)$ and $G^{(0)' } = \text{Commit}(g')$ reduce to the single check $G^{(0)} + [\rho]G^{(0)' } = \text{Commit}(g + \rho g')$, with both challenge tuples retained so that the combined length- d coefficient vector $\vec{s}(\vec{u}) + \rho \vec{s}(\vec{u}')$ is recomputable; a false claim survives the combination with probability at most $1/|\mathbb{F}_p|$. In the recursive instantiation the combined commitment is opened at a fresh random point inside the next proof’s multipoint argument — each g_i there is evaluable in $O(\log d)$ — so the deferred claim emerging from the next IPA opening is again a single tuple of the form of Definition 5.7; the tuple type is closed under accumulation only through this prover-assisted step, not under verifier-side linear combination.

A single final MSM at the end of the chain validates the entire history. The Pasta cycle (§4.3) renders each step native: a Vesta-IPA proof’s circuit, whose native field is the Vesta scalar field $\mathbb{F}_{p_{\text{Pallas}}}$, verifies the previous Pallas-IPA proof, whose group elements (the column and quotient commitments, the (L_j, R_j) pairs, the accumulator’s $G^{(0)}$) have coordinates in exactly that field; the next Pallas proof verifies the Vesta proof by the symmetric argument. This cross-curve alternation avoids non-native emulation of the preceding curve’s coordinate field; it does not eliminate all scalar-field conversion work. The same coincidence is what lets Orchard’s $\mathbb{F}_{p_{\text{Pallas}}}$ -native Action circuit manipulate the Pallas points cm , rk , and cv^{net} directly (§5.8).

Orchard’s deployment does not recurse: the verifier of an Action proof pays the deferred MSM itself. The library’s batch verifier separately random-scales and adds the complete MSM equation produced for each proof, then evaluates one combined MSM. This is randomized batch verification; it neither emits an accumulator for a later proof nor invokes the BCMS recursive accumulation scheme. Both techniques use linear combinations, but their interfaces and soundness statements are different.

The deferred equation $G^{(0)} = \text{Commit}(g(X; u_1, \dots, u_k); 0)$ is a commitment-equality (equivalently, an MSM) claim for a publicly structured coefficient vector. It is not itself an evaluation opening. The accumulation protocol reduces and carries this claim using additional prover-assisted steps.

5.7 The complete Halo 2 protocol

We now assemble the pieces into the seven-phase protocol implemented in the halo2 codebase. Throughout, $\mathbb{F}$ denotes the constraint field (Orchard: $\mathbb{F}_{p_{\text{Pallas}}}$), $\mathbb{G}$ the IPA group (Orchard: $\text{Vesta}(\mathbb{F}_{p_{\text{Vesta}}})$), and H the multiplicative subgroup of $\mathbb{F}$ of order n .

Setup (preprocessed, one-time). The circuit designer specifies columns, gates, lookup tables, and the equality permutation σ . Selector compression materialises selectors in fixed columns, and assigned lookup tables are fixed-column data; they are not two additional disjoint classes of committed columns. Key generation commits to the resulting fixed-column polynomials and to the permutation polynomials. The transparent IPA parameters separately contain the domain data and generator bases. At the generic `halo2_proofs` API boundary, verification receives those parameters, a PLONK verifying key, the instance values, and the proof. Orchard’s higher-level `VerifyingKey` wrapper contains both the IPA parameters and the PLONK key, so its public interface can present those as one argument. All are reproducible from public algorithms and the circuit; no secret setup trapdoor exists.

Phase 1 — advice commitments. Both parties first absorb a digest of vk and the (unblinded) instance commitments into the transcript (§5.8), so every subsequent challenge depends on the circuit and the public input. The prover then constructs the advice column values $\vec{w}_i \in \mathbb{F}^n$ satisfying all gates and equality constraints; fills the reserved trailing rows with uniformly random field elements (the blinding rows of §5.9); interpolates each $\vec{w}_i$ to a polynomial $w_i(X)$ of degree $< n$ on H ; commits via IPA-PCS with a fresh blinder; sends the commitments.

Phase 2 — lookup compression: challenge θ . The verifier (via Fiat–Shamir, here and throughout) returns $\theta \in \mathbb{F}$. For each lookup the prover compresses multi-column inputs and tables with powers of θ , computes the permuted columns A', S' (§5.4), and commits to them.

Phase 3 — running products: challenges β, γ . The verifier returns $\beta, \gamma \in \mathbb{F}$. The prover commits to the permutation running products (one per column chunk; §5.3) and to one lookup running product Z_{lookup} per lookup (§5.4). The order matters for soundness: the permuted columns A', S' must be fixed *before* the challenges β, γ that randomise the products are drawn.

Phase 4 — quotient challenge y . Before y is drawn, the prover commits to the random polynomial used to blind the quotient opening. The verifier then returns a fresh $y \in \mathbb{F}$. The prover assembles the *gate equation*: a single polynomial $C(X)$ that is a y -weighted linear combination of (i) each custom gate, (ii) permutation constraints, (iii) lookup constraints. By construction $C(X)$ vanishes on H when all gate, permutation, lookup, and boundary constraints hold, so the quotient $h(X) := C(X)/Z_H(X) = C(X)/(X^n - 1)$ is a polynomial. Its degree, however, is bounded by $d(n-1) - n = (d-1)(n-1) - 1$ for maximum gate degree d — a gate of degree d multiplies up to d column polynomials of degree $< n$, and division by Z_H removes n — which, for the deployed $d = 9$, exceeds the IPA’s degree bound n . The prover therefore uses a fixed proof shape of $d-1$ chunks $h_0, \dots, h_{d-2}$ of degree $< n$, retaining leading zero chunks if the actual degree is smaller, with $h(X) = \sum_i X^{ni} h_i(X)$, and commits to each chunk separately; the verifier recombinesthe chunk commitments, weighted by powers of x^n once the evaluation point x is drawn. The prover’s FFTs correspondingly run over an *extended* domain whose size accommodates the *quotient degree*, not necessarily $\deg C$. It evaluates the constraint expression pointwise on a coset disjoint from H , divides there by Z_H , then interpolates the quotient. For $d = 9$, $8n$ points suffice for h even though C can have degree near $9n$.

Phase 5 — evaluation challenge x . The verifier returns $x \in \mathbb{F}$. The prover evaluates every committed polynomial required by the verifier’s queries, except the quotient chunks, at x , and at the rotated points the constraints reference — ωx for the running products (and for any gate reading an adjacent row), $\omega^{-1}x$ for the permuted lookup input A' — and sends these claimed evaluations as field elements. No evaluation of h is sent: the verifier computes the expected quotient evaluation $C(x)/(x^n - 1)$ from the sent evaluations and enforces it against the collapsed chunk commitment inside the multipoint opening of phases 6–7, where the claimed evaluations themselves also bind to their commitments.

Phase 6 — multipoint opening. The claimed evaluations must now be bound to the commitments, and Halo 2 reduces all the required openings to a single IPA opening. The committed polynomials are grouped by their set of query points: most are queried at $\{x\}$ alone, while the running products are also queried at ωx (and A' at $\omega^{-1}x$), so only a handful of distinct point sets arise. Within each group, a challenge x_1 folds the polynomials into a single combination $q_i(X)$ by powers of x_1 . (The letters q and f here follow the halo2 book’s multipoint-opening notation and are unrelated to the selector and fixed-column polynomials of the Setup paragraph, which are themselves among the polynomials being folded; the index i now ranges over query-point sets, not columns.) For each group, both parties interpolate the low-degree polynomial $r_i(X)$ that agrees with the claimed evaluations on the group’s point set; the quotient $f_i(X) := (q_i(X) - r_i(X)) / \prod_z (X - z)$, with z ranging over the group’s points, is a polynomial precisely when the claimed evaluations are correct. The prover combines the f_i by powers of a second challenge x_2 into $f(X)$ and commits to it. The verifier returns a fresh evaluation point x_3 , and the prover sends the evaluation of each q_i at x_3 ; the verifier reconstructs the expected value of $f(x_3)$ from the $q_i(x_3)$, the $r_i(x_3)$, and the vanishing factors — if each f_i is a genuine polynomial this value matches the committed f , and by Schwartz–Zippel a match at random x_3 convinces the verifier that every claimed evaluation was correct. A final challenge x_4 folds f and the q_i into one polynomial $f^*(X)$, whose expected evaluation at x_3 the verifier assembles from the same data; a single opening of f^* at x_3 then discharges every original opening at once.

Phase 7 — IPA opening. Prover and verifier execute the deployed recursive halving IPA of Remark 5.5 on f^* . The prover sends the blinding commitment S , then $\log_2 n$ pairs (L_j, R_j) , and finally the scalars c, f . The verifier accepts if the resulting deployed-normalisation MSM equation holds; Equation (5) is the symmetric presentation, not the code’s literal equation.

Optional accumulation (recursion; not deployed by Orchard). The next proof’s verifier circuit *defers* the MSM check $G^{(0)} = \text{Commit}(g(X; u_1, \dots, u_k))$ at the end of phase 7 into an accumulator and folds it with prior accumulators (§5.6). A single decider at the very end validates the entire chain.

The concrete transcript. The deployed implementation instantiates Fiat–Shamir with a transcript built on the BLAKE2b hash function. Every prover message — commitments as curve points, claimed evaluations as field elements — is absorbed into the hash state in the order the protocol fixes, and each verifier challenge is squeezed from the state at its prescribed point, after everything it must depend on has been absorbed. Both parties also absorb the data they share in advance — the vk digest and the unblinded instance commitments — which therefore never appear in the proof. The proof *is* the ordered sequence of the prover’s messages; the verifier replays the absorptions and recomputes every challenge.

Costs. The prover’s work is dominated by FFTs over the extended domain and by one $O(n)$ -size MSM *per committed polynomial* — several tens in all: each advice column, the permuted lookup columns, the running products, the quotient chunks, and the multipoint polynomial f (typical

Orchard Action circuit: $n = 2^{11}$, a few seconds on commodity hardware). The verifier performs $O(\log n)$ field and group operations per proof—recomputing the challenges, the gate-equation field evaluation, and the $\log n$ rounds of phase 7—plus the single deferred $O(n)$ MSM. A batch verifier may combine complete MSM equations using fresh random scalars; that batching is distinct from the optional accumulation scheme of §5.6.

Proof size. Two kinds of data make up a proof. The IPA of phase 7 contributes $2 \log_2 n + 1$ group elements (a blinding commitment and the pairs (L_j, R_j) ; Remark 5.5) plus two final scalars — for Orchard’s $n = 2^{11}$, twenty-two points in the pairs; this is the only part that scales with n , and only logarithmically. Everything else scales with the *circuit structure*, independently of n : one commitment per advice column, two permuted-column commitments and one running product per lookup, the permutation-product and quotient-chunk commitments, the multipoint commitment, and one field element per claimed evaluation. Each Pasta point or field element serialises to 32 bytes. For the Orchard Action circuit the total is fixed by consensus (ZIP 225): a bundle’s proof occupies $2720 + 2272 \cdot a$ bytes for a Actions — just under 5 kB for a single-Action bundle — the per-Action term covering the per-circuit commitments and evaluations, the constant term the evaluations of the verifying key’s fixed-column and permutation polynomials together with the shared quotient, multipoint, and IPA data.

5.8 Instance columns and binding the public statement

Definition 5.1 lists *instance* columns alongside fixed and advice columns, but the arguments of §5.3–§5.7 never single them out. We single them out now, because what an Orchard Action proof *asserts* is precisely a statement about its instance values — the anchor, nullifier nf , net value commitment cv^{net} , randomised key rk , extracted new-note commitment cmx , the `enableSpends/enableOutputs` flags, and—for the post-NU6.3 circuit—the `disableCrossAddress` flag.

The instance polynomial. An instance column is a public-input column: its n entries $\vec{\pi} = (\pi_0, \dots, \pi_{n-1}) \in \mathbb{F}^n$ are not part of the witness but both parties agree on them before verification. Like every other column it is interpolated over H in the Lagrange basis,

$$\pi(X) := \sum_{j=0}^{n-1} \pi_j \ell_j(X) \in \mathbb{F}[X], \quad \deg \pi < n,$$

where $\ell_j(\omega^j) = 1$ and $\ell_j(\omega^{j'}) = 0$ otherwise (§5.1). The prover writes the public values into a handful of designated rows; all remaining π_j are 0.

Binding by copy constraints. Orchard binds the public values through the permutation argument of §5.3: the instance column participates in the permutation σ , and the circuit equality-constrains each instance cell to the advice cell carrying the corresponding in-circuit value (`halo2’s constrain_instance`). A satisfying assignment therefore exists *only* when the advice cells holding the anchor, nf , cv^{net} , rk , cmx , and the applicable enable/disable flags equal the public values the verifier supplied. (An alternative design binds public inputs with a dedicated gate $q(X)(a(X) - \pi(X)) = 0$, where the selector q is active exactly on the designated rows; Orchard does not use it — the copy constraint reuses the existing permutation machinery at no extra gate degree.)

Where it enters the protocol. The verifier never takes $\pi(X)$ on trust from the prover. In the deployed IPA backend, *both* parties compute the commitment to each instance polynomial — unblinded, since the values are public, so the two computations agree — and absorb it into the transcript before the first challenge (Phase 1, §5.7); every Fiat–Shamir challenge thereby depends

on the public input. Thereafter the instance column is treated like any other: its claimed evaluation $\pi(x)$ appears among the Phase 5 evaluations, enters the gate equation and the permutation expressions, and the multipoint opening of Phase 6 binds it to the instance commitment. Because the verifier computed that commitment itself, from the instance values it intends to verify against, the prover has no freedom over π : an evaluation inconsistent with the verifier’s own values fails the opening argument.

Consequence. A prover who targets different public values $\vec{\pi}'$ must open against the instance commitment the *verifier* computes from the genuine $\vec{\pi}$, and must satisfy the copy constraints against those values. Knowledge soundness (§3) then extracts a witness for the relation instantiated at exactly those public values; a prover that does not know such a witness — whatever witnesses it holds for other instances — convinces the verifier with at most negligible probability. This binds the proof to its public statement: an Action proof that a node verifies against an anchor it accepts, a nullifier it will mark spent, and a value commitment it will sum into the net balance check can only verify if the prover knew a witness consistent with *those* values.

5.9 Zero knowledge: hiding the witness in Halo 2

Knowledge soundness (§3) establishes that a convincing prover *knows* a witness; it remains to explain why the proof *reveals* no more than that. Halo 2 is zero-knowledge through two devices — hiding commitments and blinded polynomials — assembled into a simulator.

Hiding commitments. The polynomial commitment of §5.5 is *perfectly hiding*. The prover commits a polynomial p as

$$\text{Commit}(p; r) = \langle \vec{p}, \vec{G} \rangle + [r]H, \quad r \xleftarrow{\$} \mathbb{F}_p,$$

with $\vec{p}$ the coefficient vector of p and H an independent generator of no known discrete-log relation to $\vec{G}$. The $[r]H$ term makes the commitment a *uniform* group element for every p , so a commitment by itself leaks nothing about what the prover committed (the perfect hiding of a Pedersen commitment, *Crypto Guide*). Every commitment the prover sends in phases 1–6 — the advice columns, the permuted lookup columns A', S' , the permutation and lookup running products, the quotient chunks, and the multipoint polynomial f — has this form. The verifying-key and instance commitments are deliberately unblinded: they commit public data both parties recompute (§5.7, §5.8), so there is nothing to hide.

Blinding the witness polynomials. The prover eventually *opens* a commitment, and an opening reveals evaluations. The prover interpolates the advice columns over the domain H to polynomials of degree $< n$; to prevent those evaluations from leaking the witness, the prover does not use all n rows for circuit data. It reserves a small fixed number of rows at the bottom of the table — the *blinding rows*, never covered by an active gate — and fills them with uniform random field elements, at least one more than the number of times the verifier will open any single column. Its n evaluations pin down a degree- $(< n)$ polynomial, so with enough independent random values planted on the blinding rows, evaluations of a single such polynomial at sufficiently few distinct points outside H are jointly uniform and independent of its fixed active-row data. This marginal linear-algebra fact is not yet a proof about the joint transcript of related polynomials. The prover blinds the permutation and lookup running products the same way; Remark 5.2 describes how the constraint system exempts the blinding rows, so the random values never violate an argument. The quotient pieces admit no such treatment — C fully determines them — so their commitment blinders, together with the blinding-row randomness they inherit through C , protect them instead; their single collapsed opening at x is the publicly computable value $C(x)/(x^n - 1)$. (The

deployed implementation also commits a uniformly random polynomial alongside the quotient and opens it at the same point, keeping the multipoint argument zero-knowledge.)

Why the openings leak nothing. In each phase the verifier sees only (i) hiding commitments — uniform group elements — and (ii) a bounded number of evaluations of the committed polynomials at the Fiat–Shamir challenge points. The blinding rows supply randomness that a full composition proof must show masks the joint evaluations subject to the public relations. Correlations between advice, products, quotient, and multipoint polynomials cannot be discarded. The final inner-product opening uses the deployed initial vanishing-polynomial mask as well as commitment and round blinders (Remark 5.4); a final commitment blinder alone would not hide the folded coefficient.

The simulator. Formally, zero knowledge requires a simulator for the *complete joint transcript*, given only the public input. The usual proof samples suitably blinded commitments and evaluations satisfying the verifier equations, then programs fresh Fiat–Shamir queries to the required challenges. Perfect hiding of each commitment is one ingredient, not by itself a proof that this joint distribution matches a real execution: the blinding dimension, transcript order, opening simulation, and freshness of programmed queries must all be covered by the composed protocol theorem. Under those hypotheses the interactive protocol is honest-verifier zero knowledge and its Fiat–Shamir compilation is NIZK in the programmable random-oracle model.

The three properties together. Subject to the full PIOP–PCS composition and Fiat–Shamir hypotheses just stated, Halo 2 targets three guarantees for the relation fixed by a PLONKish circuit: *overwhelming completeness* (an honest prover holding a satisfying witness convinces the verifier except on the negligible transcript degeneracies noted in Remark 5.5), *knowledge soundness* (§3: from any convincing prover an extractor pulls a satisfying witness, so an accepted proof certifies one is *known*), and *zero knowledge* (a witness-free simulator reproduces the proof, so it leaks nothing else). It is exactly this combination — “the prover knows a satisfying witness, and the proof discloses nothing more” — on which the Orchard Action proof rests: knowledge soundness yields balance and nullifier integrity, while zero knowledge yields privacy.

6 Worked example: one computation, carried from claim to conviction

6.1 What we are going to prove, and what “prove” means

Pick a claim small enough to hold in one hand:

“I know a secret x such that $x^3 + x + 5 = 35$.”

The secret is $x = 3$ (indeed $27 + 3 + 5 = 35$). This worked example follows that one sentence until a sceptic is convinced of it — without ever being told that $x = 3$.

Three demands make this nontrivial, and a Halo 2 proof meets all three. The proof has *overwhelming completeness*: an honest prover who knows x convinces the verifier except on negligible transcript degeneracies. It is *sound* (more precisely, *knowledge sound*): a prover who does *not* know a valid x is rejected, except with negligible probability — and “knowledge” is meant literally, in the sense that a satisfying x could in principle be extracted from any prover who succeeds. And it is *zero-knowledge*: the verifier finishes convinced yet learns nothing about x beyond the bare truth of the claim. Completeness is easy; the content of the subject is *soundness with zero-knowledge*, and that is what we will watch being built.

Why this toy? Because it is the Orchard spend statement in miniature. When Alice spends a note she proves “I know a note, a key, and a Merkle path (*Crypto Guide*, §“Merkle trees and commitments to sets”) such that the nullifier — the unique tag whose publication marks the

note spent — is computed correctly, the net value commitment (a Pedersen commitment to the old-minus-new value; *Crypto Guide*, §“The Pedersen commitment”) opens, and the spent note sits in the tree” — a conjunction of a few thousand small arithmetic facts over $\mathbb{F}_{p_{\text{Pallas}}}$ in the deployed Orchard Action relation. Our claim is a conjunction of *four* small arithmetic facts. The proof system does not care which; it is the same machine at two scales. Master $x^3 + x + 5 = 35$ and the Action circuit is only larger, not different.

The journey. The route has five legs, and each is a translation that loses nothing:

1. **Computation → grid** (§6.2). Rewrite the claim as a table of rows, each row one elementary gate ($\times$ or $+$), with wires between rows forcing shared values.
2. **Grid → polynomials** (§6.3). Interpolate each column of the table into a polynomial. “Every row obeys its gate” becomes “one polynomial $G(X)$ is zero at every row.”
3. **Vanishing → identity** (§6.4). “ G is zero on all rows” becomes the clean algebraic identity $G(X) = t(X) Z_H(X)$ for some quotient t . A correct computation *is* a polynomial identity; an incorrect one cannot be.
4. **Identity → one random check** (§6.5). The verifier tests the identity at a single random point z . Here lives the magic: a false identity survives a random point with vanishing probability (Schwartz–Zippel).
5. **Trust the openings** (§6.6–§6.9). A binding *commitment* stops the prover from changing her polynomials after seeing z ; the permutation argument enforces the wiring; an instance column pins the public “35”; and blinding makes the whole transcript reveal nothing.

Section 6.12 threads the legs back together into one sentence: why the verifier, having checked a few numbers, is right to be convinced.

We work throughout in the prime field $\mathbb{F}_{97}$. This is a toy size chosen so the reader can check every product by hand. Over Orchard’s roughly 255-bit field, a *fixed* false identity of degree around 2^{14} survives one fresh uniform evaluation challenge with probability around 2^{-240} instead of our visible $3/97$. This is one Schwartz–Zippel term, not the security level of the complete proof system.

6.2 From computation to a grid of constraints

A verifier cannot re-run Alice’s computation: that would need her secret. So we first dismantle the computation into *local* facts, each so simple it involves only a handful of values, and arrange them so that “all the local facts hold, and the values are consistently shared” is equivalent to “the whole computation ran correctly.” This rearrangement is *arithmetisation*; the Halo 2 dialect of it is *PLONKish* (§5.1).

Break the computation into gates. Evaluate $x^3 + x + 5$ the way a machine would, naming each intermediate:

$$x^2 = x \cdot x, \quad x^3 = x^2 \cdot x, \quad s = x^3 + x, \quad s + 5 = 35.$$

Four elementary operations, two multiplications and two additions. Each becomes one row of a table. A row has three *wires* — call them a, b, c , the two inputs and the output — and a bank of *selectors* q_M, q_L, q_R, q_O, q_C that switch on whichever gate this row performs. The universal gate equation, evaluated at every row, is

$$q_M a b + q_L a + q_R b + q_O c + q_C + \text{PI} = 0, \tag{6}$$

where PI carries any public input on that row (here, the target 35). A multiplication gate $a \cdot b = c$ is the choice $q_M = 1$, $q_O = -1$, the rest zero; an addition gate $a + b = c$ is $q_L = q_R = 1$, $q_O = -1$.

Filling in our four operations gives the *execution trace*, Table 2. (The deployed system generalises this fixed selector bank to designer-chosen gates and lookups, declared once and filled row by row at proving time; Section 5.2 shows that machinery on the real Orchard gates.)

row	a	b	c	q_M	q_L	q_R	q_O	q_C	PI	gate performed
0	3	3	9	1	0	0	-1	0	0	$x \cdot x = x^2$
1	9	3	27	1	0	0	-1	0	0	$x^2 \cdot x = x^3$
2	27	3	30	0	1	1	-1	0	0	$x^3 + x = s$
3	30	0	0	0	1	0	0	5	-35	$s + 5 = 35$

Table 2: The execution trace for $x = 3$. Each row satisfies the gate equation (6): e.g. row 0 gives $1 \cdot 3 \cdot 3 + (-1) \cdot 9 = 0$, and row 3 gives $1 \cdot 30 + 5 + (-35) = 0$. The selector columns $q_\bullet$ are fixed circuit data and PI is the public input (both known to the verifier); the wires a, b, c are the prover’s *advice* (her secret trace).

Wire the rows together: copy constraints. The table as printed has a fatal loophole. Nothing yet forces the x in row 0 to be the same x in rows 1 and 2, nor the output $x^2 = 9$ of row 0 to be the input of row 1. A cheating prover could put $x = 3$ in row 0 but $x = 8$ in row 2 and satisfy each gate *in isolation* while computing nonsense overall. We must assert that certain cells are *equal*. These are the *copy constraints* (or *equality constraints*):

$$\underbrace{a_0 = b_0 = b_1 = b_2}_{\text{all are } x=3}, \quad \underbrace{c_0 = a_1}_{x^2=9}, \quad \underbrace{c_1 = a_2}_{x^3=27}, \quad \underbrace{c_2 = a_3}_{s=30}.$$

They are what turn a list of unrelated gates into a single dataflow. Enforcing them is a small argument in its own right (§6.7); for now simply note what they say.

What the grid achieves. The claim is now equivalent to a purely combinatorial assertion: *there exist values for the advice cells a, b, c such that (i) every row satisfies the gate equation (6) with the fixed selectors, and (ii) every copy constraint holds*. Knowing x is exactly knowing how to fill the advice columns. The verifier knows the fixed columns and the public input; the prover must convince him that satisfying advice exists, while revealing none of it. Everything from here is machinery for doing precisely that.

6.3 From the grid to polynomials

The grid has a finite number of rows, and we want to make statements about “all rows at once” that a verifier can check by looking at *one* place. Polynomials are the tool: a low-degree polynomial is rigid — pinned down everywhere by its behaviour at a few points — and that rigidity is what we will weaponise.

An index set for the rows. Our table has $n = 4$ rows. Choose four distinct field elements to name them. The inspired choice is the n -th roots of unity: a value ω with $\omega^4 = 1$ and no smaller power equal to 1. In $\mathbb{F}_{97}$ we may take $\omega = 22$, for which $\omega^2 = 96 = -1$ and $\omega^4 = 1$, so the row labels are

$$H = \{\omega^0, \omega^1, \omega^2, \omega^3\} = \{1, 22, 96, 75\} = \{1, 22, -1, -22\}.$$

Row i now lives at the point ω^i . (The roots of unity are chosen not for elegance but for speed: they make the polynomial arithmetic below an FFT (*Math Guide*, §“The fast Fourier transform”). Any 4 distinct points would do for correctness.)

Columns become polynomials. Each column of the table is four numbers, one per row. *Interpolate*: for the advice column a there is a unique polynomial $a(X)$ of degree < 4 with

$$a(\omega^0) = 3, \quad a(\omega^1) = 9, \quad a(\omega^2) = 27, \quad a(\omega^3) = 30,$$

and likewise $b(X), c(X)$, and the (publicly known) selector polynomials q_M, q_L, q_R, q_O, q_C and the public-input polynomial PI , each a polynomial in X of degree < 4 . The column i is the polynomial, sampled at the rows. A polynomial of degree < 4 is exactly four coefficients, so no information is gained or lost — only re-encoded into a form that bends to algebra.

All gates at once, as one polynomial. Now assemble the left-hand side of the gate equation (6), but with the *polynomials* in place of the per-row numbers:

$$G(X) := q_M(X) a(X) b(X) + q_L(X) a(X) + q_R(X) b(X) + q_O(X) c(X) + q_C(X) + PI(X). \quad (7)$$

This G has degree at most 9 (the term $q_M ab$ multiplies three degree-3 polynomials). Here is the point of the whole construction. Evaluate G at the label of row i : because each polynomial returns that row's value at ω^i , $G(\omega^i)$ is *exactly* the gate equation (6) for row i . Therefore

$$G(\omega^i) = 0 \text{ for every row } i \iff \text{every gate is satisfied.}$$

For our honest trace this is true by construction: G vanishes at all four points of H . “Every gate is satisfied” has become “ G vanishes on H ”; copy constraints remain separate. We have replaced four separate checks with a single statement about one polynomial — and it is a statement we can now make checkable in one shot.

6.4 From vanishing to a single identity

“ G vanishes on the set H ” is still a statement about four points. One more step compresses it into a statement with *no* reference to the individual rows at all — a bare algebraic identity — and this is the technical heart of arithmetic proving.

The vanishing polynomial. The polynomial that is zero exactly on H , and nowhere else, and as simply as possible, is

$$Z_H(X) := \prod_{i=0}^3 (X - \omega^i) = X^4 - 1.$$

(The product of $(X - \zeta)$ over all n -th roots of unity ζ telescopes to $X^n - 1$; that is the only reason roots of unity are convenient here.) The elementary fact we lean on is the factor theorem, applied n times:

$$G \text{ vanishes on all of } H \iff (X - \omega^i) \mid G \text{ for each } i \iff Z_H(X) \mid G(X). \quad (8)$$

Divisibility is the same as the existence of a quotient. So:

$$\text{every gate holds} \iff G(X) = t(X) Z_H(X) \text{ for some polynomial } t(X).$$

Satisfaction of the *gate* constraints is precisely the existence of this quotient t ; correct wiring still requires the copy constraints. For our honest trace the division comes out even: G has degree 9, Z_H has degree 4, and

$$t(X) = \frac{G(X)}{X^4 - 1}$$

is an honest polynomial of degree 5 — the remainder is zero, as it must be.

Why a cheat cannot produce t . Suppose the prover's advice is *wrong* — some gate fails. Then G does *not* vanish on all of H , so by (8) Z_H does *not* divide G , so *no* polynomial t satisfies $G = t Z_H$. The dividing line is exact: the quotient exists if and only if every gate is satisfied. This is the lever the verifier will pull. Concretely, take a prover who claims $x^2 = 10$ and writes $c_0 = 10$ in row 0 (Table 2) instead of 9. Now row 0's gate reads $3 \cdot 3 - 10 = -1 \neq 0$, so the tampered G^* has $G^*(\omega^0) = -1 = 96$ while still vanishing at the other three rows. Dividing G^* by Z_H leaves a nonzero remainder. For this example, choose the Euclidean-division quotient t^* ; this is an illustrative choice, not a claim that it maximises the chance of cheating. Then

$$D(X) := G^*(X) - t^*(X) Z_H(X) = 24(1 + X + X^2 + X^3)$$

is a *nonzero* polynomial of degree 3. The identity she needs is false, and D measures exactly by how much. Keep D in view: the next section is about catching it cheaply.

6.5 Why checking one random point is a proof

The prover wishes to convince the verifier that $G(X) = t(X) Z_H(X)$ as polynomials, with G built from her secret advice. The verifier will not read the polynomials — they encode the witness, and there are too many coefficients to be cheap anyway. Instead:

1. The prover *commits* to her polynomials a, b, c and the quotient t (Section 6.6 says what a commitment is and why it binds her); the selector and public-input polynomials the verifier already knows.
2. The verifier picks a point $z \in \mathbb{F}$ *uniformly at random* and sends it.
3. The prover *opens* her commitments at z : she returns the field elements $a(z), b(z), c(z), t(z)$ together with short proofs that these are the true evaluations of the committed polynomials.
4. The verifier forms $G(z)$ from the openings and the (self-evaluated) public polynomials via (7), computes $Z_H(z) = z^4 - 1$, and checks the *single field equation*

$$G(z) \stackrel{?}{=} t(z) Z_H(z). \quad (9)$$

He accepts if and only if (9) holds. That is the entire test: one equation between numbers.

The honest case (completeness). If the prover is honest then $G = t Z_H$ holds as polynomials, hence at every point, hence at z . With our trace and $z = 20$:

$$Z_H(20) = 20^4 - 1 = 46, \quad t(20) = 67, \quad G(20) = 75 = 67 \cdot 46 = t(20) Z_H(20). \quad \checkmark$$

The verifier accepts, every time, for every z .

The dishonest case (soundness), and the one idea that makes it work. Suppose the trace is invalid — some gate fails, say at row i , so $G^*(\omega^i) \neq 0$ and G^* does *not* vanish on all of H . Take *any* polynomial t^* the cheating prover might commit to (the honest quotient, or anything else) and set

$$D(X) := G^*(X) - t^*(X) Z_H(X).$$

Were D the zero polynomial, then $G^* = t^* Z_H$ would be divisible by Z_H and so vanish on all of H — which it does not. Hence D is a *nonzero* polynomial, however cleverly t^* was chosen. And D is fixed *before* z is drawn: committing to a, b, c pins down the polynomial G^* (the verifier never receives G^* itself — from the openings at z he reconstructs only the single *number* $G^*(z)$), and committing to t^* pins down the rest, so the entire polynomial D is frozen by the commitments before the challenge is revealed. The check (9) passes at z exactly when $D(z) = 0$ — that is,

exactly when the random z happens to be a root of D . And here is the lever: *a nonzero polynomial of degree d has at most d roots* (Schwartz–Zippel, in one variable just the factor theorem). A degree- d polynomial is pinned down by any $d+1$ of its values, so it has no freedom to vanish at more than d points without collapsing to zero everywhere; a nonzero low-degree polynomial is therefore *mostly* nonzero, its roots a sparse set that a random probe almost surely misses. So

$$\Pr_z[\text{cheat accepted}] = \Pr_z[D(z) = 0] \leq \frac{\deg D}{|\mathbb{F}|}.$$

A liar is caught with overwhelming probability over the verifier’s single coin toss. *This* is what “proving” means here: not that the verifier re-checks the work, but that a false statement is an algebraic object so rigid it cannot hide — it disagrees with the truth almost everywhere, and one random probe almost surely lands where it disagrees.

Watch it happen. For our $x^2 = 10$ cheat, $D(X) = 24(1 + X + X^2 + X^3)$ has degree 3, so at most 3 roots — and in $\mathbb{F}_{97}$ its roots are exactly $\{22, 96, 75\}$, the three rows where the tampered gate still held. This particular quotient choice slips past only if the verifier’s random z lands in that three-element set:

$$\Pr[\text{cheat accepted}] = \frac{3}{97}.$$

At the verifier’s actual draw $z = 20 \notin \{22, 96, 75\}$, we get $G^*(20) = 20$ but $t^*(20) Z_H(20) = 64$, and $20 \neq 64$: caught. The fraction $3/97$ is large only because our field is a toy. Over Orchard’s roughly 255-bit field, a fixed nonzero gap polynomial has at most $\deg D$ roots. For $\deg D$ around 2^{14} , one independently uniform z contributes a Schwartz–Zippel term around 2^{-240} . The complete proof has other randomized identities and cryptographic assumptions, and Fiat–Shamir grinding multiplies a per-draw term by the adversary’s query budget. In particular this isolated figure is not an overall 240-bit security claim; the Pasta-group discrete-log layer has about 126 bits of classical concrete security.

Bounding the degree. The argument needs D to have *low* degree relative to $|\mathbb{F}|$ — otherwise “ $d/|\mathbb{F}|$ roots” is no constraint. Here any admitted t^* of degree at most 5 gives $\deg D \leq 9$, so the general gate-test bound is $9/97$, not the illustrative $3/97$. This is why a circuit’s gates are kept to low degree and why t , of degree ≤ 5 here (and up to a few times n in general), is split into bounded pieces in the deployed system. Low degree is not an efficiency footnote; it is the source of soundness.

6.6 Why the prover cannot wriggle: binding commitments

The random-point argument has an obvious gap: it assumed the prover’s polynomials were *fixed* before she saw z . If she could choose a, b, c, t after learning z , she would simply pick numbers making (9) hold and reveal nothing real. A *polynomial commitment* closes exactly this gap.

What a commitment must do. A commitment to a polynomial f is a short string $C = \text{Commit}(f)$ with two properties (the *Crypto Guide* develops both):

- **Binding.** Having published C , the prover cannot later open it as two different polynomials; C pins down f . So committing first, then receiving z , then opening, leaves her no freedom: she is answering for the *one* polynomial she fixed in advance — the situation Section 6.5 assumed.
- **Hiding.** C reveals nothing about f itself. (This is what will let the proof be zero-knowledge, §6.9.)

Add an *evaluation proof*: given C and a point z , the prover can prove “ $f(z) = v$ ” for the committed f without exposing f . With binding plus trustworthy evaluation proofs, the four-step protocol of Section 6.5 is exactly as sound as its idealised version: the opened numbers $a(z), \dots, t(z)$ are forced to be the true evaluations of fixed polynomials, so the check (9) tests a genuine, pre-committed identity at a genuinely random point.

How Halo 2 commits, in one breath. Halo 2 uses the *inner-product argument* (IPA). Encode f by its coefficient vector $\vec{f} = (f_0, \dots, f_m)$ and fix a public list of independent curve points $\vec{G} = (G_0, \dots, G_m)$ with no known relationship. The commitment is the single curve point

$$C = \langle \vec{f}, \vec{G} \rangle = \sum_i f_i G_i,$$

a *Pedersen* vector commitment: binding because finding two openings would solve a discrete-log relation among the G_i , and hiding once a random blinding term is added. To prove $f(z) = v$ is to prove a statement about the inner product of $\vec{f}$ with the vector $(1, z, z^2, \dots)$, and the IPA does this in a logarithmic number of rounds, each *halving* the vectors, until the claim is trivial — so the evaluation proof is $O(\log m)$ group elements and, decisively, needs *no trusted setup* (§6.11 says why, and what the fold’s one expensive step costs). The full fold and its soundness are developed in §5.5. For our purposes one property is enough: *the commitment binds*, so the prover is honestly answering for polynomials she could not change once z was on the table.

6.7 Enforcing the wiring: the permutation argument

One promise from Section 6.2 is still unpaid: the copy constraints. The gate identity of Sections 6.3–6.5 checks each row *in isolation*; it never says that a_0, b_0, b_1, b_2 are the same value x , or that row 1’s input is row 0’s output. Without that, a prover could satisfy every gate with an incoherent dataflow. The *permutation argument* adds the missing constraint, and it does so with the same philosophy: turn “these cells are equal” into “a certain product is 1,” then check the product at a random point.

Equalities as a permutation. Group the cells into classes that must share a value — here $\{a_0, b_0, b_1, b_2\}, \{c_0, a_1\}, \{c_1, a_2\}, \{c_2, a_3\}$ — and let σ be the permutation that cycles each displayed class in the order written. Give every cell a distinct *label*: cell in row j of the a -, b -, c -column gets $\omega^j, k_1\omega^j, k_2\omega^j$ respectively, where k_1, k_2 place the three columns in disjoint cosets so no two labels collide (in $\mathbb{F}_{97}$ we may take $k_1 = 2, k_2 = 3$). Asserting “cells in a class are equal” is then equivalent to asserting that the map sending each cell’s *value-with-label* to its *value-with- σ -permuted-label* is consistent — a balance between two multisets.

The grand product. With verifier challenges β, γ , the prover builds a running product whose endpoint enforces

$$\prod_{\text{cells } (i,j)} (v_i(\omega^j) + \beta(\text{label}) + \gamma) = \prod_{\text{cells } (i,j)} (v_i(\omega^j) + \beta(\sigma\text{-label}) + \gamma).$$

If the wiring is honest, numerator and denominator run over the *same* multiset of factors, so the equality holds identically. If a copy constraint is false, the resulting nonzero polynomial in β, γ can still vanish for a particular challenge pair, but only with probability $O(n/|\mathbb{F}|)$ by Schwartz–Zippel. Thus the converse is probabilistic, not an “if and only if” for fixed challenges. The actual circuit enforces cross-multiplied row updates, not literal field divisions, so a zero factor does not make the constraint undefined. The prover commits to the running-product polynomial and the verifier checks its boundary and row-to-row update at random z .

A two-cell glimpse. Strip the product down to a single wired pair, $c_0 = a_1$ (both should be 9), carrying labels L_{c_0}, L_{a_1} that σ swaps. Their factors satisfy

$$(9 + \beta L_{c_0} + \gamma)(9 + \beta L_{a_1} + \gamma) = (9 + \beta L_{a_1} + \gamma)(9 + \beta L_{c_0} + \gamma),$$

because the shared value 9 makes the two numerator factors a mere reordering of the two denominator factors — the labels may permute under σ precisely because the values agree. Now let the prover cheat the copy, $a_1 = 8 \neq 9$: the factor $8 + \beta L_{a_1} + \gamma$ on one side generally has no match on the other, and the two products differ. Equal values let labels permute; one unequal value strands a factor, and the whole product feels it.

Watch it happen. For our honest trace, with $\beta = \gamma = 2$, the grand product comes out to exactly 1. Tamper one wire — set $b_1 = 4$, breaking the copy $b_1 = x = 3$ — and the same product becomes $61 \neq 1$: the inconsistent dataflow is exposed without any value being revealed.

6.8 Proving a value sits in a table: lookups

Gates handle algebra; some constraints are not algebra. “This chunk indexes an entry of the Sinsemilla generator table” or “this cell is a ten-bit number” has no low-degree formula. Written as one polynomial gate it would need degree about the size of the table — which in our toy field would eat the soundness budget outright (§6.5), and at any scale balloons the quotient polynomial the prover must split, commit, and open (§6.4). Yet Orchard’s circuit needs such range and table facts constantly.

The statement has a familiar shape. “Every entry of my column A appears somewhere in the table S ” is, like the copy constraints, a statement about *multisets*. One wrinkle: membership is not a permutation — A may repeat one table entry and ignore the rest — so the prover first tidies both sides. She commits to A' , her column reshuffled so equal values sit in contiguous runs, and S' , a reshuffle of the table aligned so that each run in A' begins beside its matching table value. Two *row-local* checks then pin membership: each row of A' either *starts a run* ($A'_i = S'_i$: the value is vouched for by the table) or *continues one* ($A'_i = A'_{i-1}$: equal to the row above, already vouched for). What certifies the two reshuffles is the §6.7 machinery with one deliberate change: *no labels*. Labelled factors pinned one particular wiring σ ; here the reshuffles are the prover’s own choice, so a single running product with unlabelled factors — row by row, $(A' + \beta)(S' + \gamma)$ against $(A + \beta)(S + \gamma)$ — certifies exactly that each primed column is *some* reshuffle of its original, which is all membership needs. Induction up the rows does the rest.

On the toy’s own four-row grid: constrain cells to two-bit values with the table $S = (0, 1, 2, 3)$ and the column $A = (2, 3, 2, 2)$. Grouped, $A' = (2, 2, 2, 3)$; align $S' = (2, 0, 1, 3)$, a reshuffle of S placing 2 beside the first run and 3 beside the second. Row 0 starts a run ($2 = 2$), rows 1 and 2 continue it (2 above), row 3 starts a run ($3 = 3$) — every check row-local, every polynomial low-degree, however large the table. A forged $A = (2, 7, 2, 2)$ dies: its 7 lands at the start of a run in A' , and no reshuffle of a table without a 7 can put a 7 beside it.

This is how the deployed circuit affords its ten-bit range checks and its 2^{10} -entry Sinsemilla generator table: a table fact costs a few committed columns and one more running product, and the gates stay low-degree. The deployed declarations of both are exhibited in §5.2; the exact constraints, the boundary row, the padding of short columns, and multi-column tables are developed in §5.4.

6.9 Binding the public statement, and hiding the witness

Two finishing pieces complete the picture, one for soundness and one for secrecy.

Pinning the public “35”: the instance column. A proof must attest to *this* statement, not some other the prover finds convenient. The target 35 is public, so it is not advice but *instance* data: it sits in the public-input polynomial $PI(X)$ of (7), which the verifier builds himself (it is -35 on row 3 and 0 elsewhere). Because PI enters the very identity being checked, a prover who tried to prove “ $x^3 + x + 5 = 36$ ” would be checking a *different* G and her honest trace would no longer vanish on H . The public input is thus welded into the same polynomial identity that soundness already guards (§5.8). The verifier proves something about the claim he chose, not one the prover swapped in.

Revealing nothing: zero-knowledge. So far the prover has handed over commitments and a few evaluations $a(z), b(z), \dots$ at a random z . Do these leak the witness? They could: an evaluation is a linear equation in the secret coefficients, and enough of them would pin x down. Halo 2 closes this the standard way — *blinding*. The witness trace reserves a few independent random rows, and secret polynomial commitments carry independent random blinding terms. Adding arbitrary random coefficients would change the active-row values and is not equivalent. The gate and permutation arguments are arranged to ignore those random rows (Remark 5.2), so they do not disturb soundness; in the full zero-knowledge argument, the openings at z are masked by fresh randomness and their joint distribution reveals nothing about x . Formally one exhibits a *simulator* that, under the composition hypotheses of §5.9, matches the transcript distribution knowing only that the statement is true. The simulator discussion appears in §5.9; the intuition is simply: *commitments hide, and blinded openings are noise pinned to one honest constraint*.

6.10 Removing the verifier: the Fiat–Shamir transform

One awkwardness remains. Everything above needs a *live* verifier for exactly one service: once the commitments a challenge tests are pinned, somebody the prover cannot predict must choose that challenge — z here, β and γ for the permutation argument, the coins inside each opening round (§6.6). Nothing else about him matters: his messages are coin tosses, and his final act is a computation anyone could run. A payment protocol cannot work this way. Alice would have to sit in a live session with every full node that will ever validate her transaction.

The fix is almost impudent. Wherever the verifier would send a fresh challenge, the prover instead computes it herself as

$$\text{challenge} := H(\text{everything sent so far}),$$

the digest of the entire conversation to that point, public statement and every commitment included, under a fixed hash function H . The ordering that §6.6 fought for is now enforced by construction: a commitment is an *input* of the hash, so it is fixed before the challenge that tests it exists at all.

Why is a hash as good as a coin? Because the prover’s only route to a favourable challenge is to vary some input of H — and every input she controls is something the finished proof must stand behind: the statement she is claiming, or a commitment she must open. Each variation re-rolls the challenge blindly: one hash evaluation buys one ticket in a lottery she wins only with the relevant per-draw soundness error. If that error is ϵ and she makes at most Q distinct transcript queries, a union bound gives at most $Q\epsilon$ for this grinding route; for the illustrative identity check, ϵ is around 2^{-240} . Thus the query budget must be included rather than declared irrelevant. Two fine points matter. The hash must digest the *whole* transcript — leave the statement or any commitment out, and she aims precisely at what the hash cannot see. And the guarantee is proved with H modelled as a random oracle, a strong but standard heuristic; the *Crypto Guide* (§“The Fiat–Shamir transform: from interactive to non-interactive”) gives the theorem and its caveats.

What this buys is the “N” in SNARK: *non-interactive*. The proof becomes a single static object — commitments and openings, nothing else — and verification becomes recomputing the challenges by the same hash and running the same checks. No session, no coordination: a full node that has never met Alice verifies her Action years later, exactly as the deployed system does (§5.7 describes the transcript it uses).

6.11 The “Halo” in Halo 2: deferring the expensive check

One cost has stayed quietly non-trivial. The opening proofs of §6.6 shrink by halving, so the verifier’s rounds are logarithmic — except at the very end, where checking the final fold means recomputing one combination of *all* the generators $G_0, \dots, G_m$: a wall of group arithmetic proportional to the whole circuit’s size. When checking a batch, the deployed library random-combines the complete per-proof MSM equations and evaluates one combined MSM. That is a standard randomized batch check, not the recursive accumulation operation of §5.6. Where the cost would be truly ruinous is *recursion*. For a circuit to verify a *proof* — the step that would let one proof attest to a whole chain of others — everything the verifier does must be re-expressed as gates, and a circuit-sized wall of group arithmetic inside a circuit defeats the point.

For optional recursion, Halo’s namesake trick is to defer payment. The expensive claim has a special shape — “this point equals that known combination of the generators” — and two claims of this shape can be *folded* into one by the move this worked example keeps meeting: commit, draw a challenge, take a random combination, a few group operations in place of a wall. So the recursive verifier checks everything cheap, and instead of paying the expensive step it folds the claim into a running *accumulator* carried along the chain. One final check, run once at the chain’s end and outside any circuit, pays a single wall of group arithmetic for the entire history. The bookkeeping’s soundness reduces to the discrete logarithm, exactly like the commitments it manages (§5.6); the cycle of curves that keeps the recursive arithmetic in the right fields is developed in §4.3.

A last remark completes the picture begun in §6.6: none of this ever needed a ceremony. The generators $G_0, \dots, G_m$ are nothing-up-my-sleeve points, hashed into existence, with no secret behind them that anyone ever knew — nothing to leak, nothing to destroy. The pairing-based KZG family buys constant-size opening proofs and constant-time verification at the price of a *trusted setup*: secret randomness contributed through a ceremony, whose compromise permits silent forgeries (the *Crypto Guide*, §“Polynomial commitment schemes”, constructs that scheme and the forgery its trapdoor permits). Sapling and the later Groth16 version of Sprout likewise needed parameter ceremonies for their proof systems (Sprout initially used a different system), though that is a separate construction. Halo 2’s inner-product commitment trades proof size for exactly this freedom; being rid of ceremonies was a stated reason Orchard chose it.

6.12 Why the verifier is convinced

Stand back and read the chain in the direction the verifier experiences it. He accepts a proof. What has he thereby learned?

1. The check $G(z) = t(z) Z_H(z)$ passed at his random z . Because the commitment *binds* (§6.6), the polynomials behind the openings were fixed before z ; because z was random and a false identity disagrees with the true one at all but $\leq \deg D$ points (§6.5), passing supports the conclusion that $G(X) = t(X)Z_H(X)$ holds as polynomials, except with the corresponding $\deg D/|\mathbb{F}|$ Schwartz–Zippel error for this fixed check.
2. $G = t Z_H$ means $Z_H \mid G$, which means G vanishes on all of H (§6.4), which means *every gate equation holds at every row* (§6.3).

3. The permutation grand product passed (§6.7), so, except with its random-challenge soundness error, the *copy constraints* hold too: the per-row gates are wired into one coherent computation, not four disconnected coincidences.
4. The public-input polynomial fixed the target at 35 (§6.9), so the computation he has verified is the one he asked about.

Putting these together — and recalling from §6.2 that filling the advice columns *is* knowing x , since those cells are the wires carrying x and its powers — there exists an assignment to the advice columns, a value of x , making the entire arithmetisation of “ $x^3 + x + 5 = 35$ ” hold. Under the composed PIOP, PCS-extraction, and Fiat–Shamir hypotheses of §3, an extractor for the *full* protocol can obtain such an x . The IPA opening extractor alone establishes knowledge of committed polynomial data; it does not by itself prove that those polynomials encode a satisfying circuit witness. And by zero-knowledge (§6.9) the verifier learns no more than this single bit of truth. The verifier has checked a small transcript and a logarithmic-size opening proof; its confidence is governed by the sum of all algebraic error terms and the PCS, Fiat–Shamir, and discrete-log assumptions, not by the isolated 2^{-240} term alone.

The same machine, at Orchard scale. Nothing above used that our statement was small. Replace the four gates by the few thousand constraints of the Orchard Action — nullifier integrity, the value-commitment opening, the Merkle path recomputation, and key re-randomisation (*Crypto Guide*, §“Key re-randomisation and unlinkability”) — and the columns grow from four rows to 2^k , the gate polynomial gains custom high-degree terms, and a Sinsemilla lookup table (§6.8; the *Crypto Guide*, §“Sinsemilla: an algebraic hash-based commitment”) joins the permutation. But the spine is identical: arithmetise into a grid, interpolate to polynomials, compress “all rows correct” into one identity $G = t Z_H$, commit, and test at a random point. When a Zcash full node verifies a shielded spend, randomized polynomial-identity tests of the kind in Section 6.5 form its algebraic core. The full verifier also checks permutation and lookup relations, multiple openings, transcript challenges, and the final IPA equation before concluding that the asserted Action relation holds.

What the formal treatment supplies. The earlier sections explain the deferred mechanisms and state their remaining model hypotheses: the inner-product argument and why its openings are trustworthy (§5.5); the algebraic-group-model extractor behind “the prover knows x ” (§3); the soundness of the accumulation fold sketched in §6.11 (§5.6); and the cycle of curves that makes recursion efficient (§4.3). This worked example has supplied the picture those formalisms formalise: *how random evaluations of committed polynomials contribute to knowing that an unseen computation was done right.*

Part IV

Consensus Guide: Nakamoto consensus: the double-spend race, the arithmetic of confirmations, and the deployed most-work rule

Abstract

Assuming the *Math Guide*'s probability foundations and the hash and commitment foundations of the *Crypto Guide*, this volume of *The Zcash Arboretum* constructs Nakamoto consensus from first principles and quantifies the security it buys. It states the block tree and the most-work rule as the protocol specification defines them and as the deployed zebra implements them; builds, on the *Math Guide*'s foundations, the race-specific probability the analysis needs (memoryless clocks, the biased random walk, the negative binomial law); develops the exact first-seen-model double-spending analysis of Rosenfeld's *Analysis of hashrate-based double-spending* — the catch-up theorem, the confirmation formula, and the economics of the attack — with every derived number recomputed by script and the results instantiated for Zcash's proposed 25-second blocks; explains the draft NU7 transition to 25-second blocks and halving-preserving reissuance; and explains the protocol and implementation safety rails (difficulty adjustment, reorg windows, coinbase maturity, wallet confirmation policy) that frame the model's assumptions. Where the specification, the paper, and the deployed nodes disagree, the divergence is stated.

1 Scope, sources, and the two double-spends

The protocol layers above this one take a working ledger for granted. This volume documents the mechanism that selects that ledger: Nakamoto consensus — proof of work, the most-work rule, and the probabilistic finality they purchase. The treatment is quantitative throughout. Its spine is Meni Rosenfeld's *Analysis of hashrate-based double-spending* (11 December 2012; latest version 12 February 2014; arXiv:1402.2009), which gives an exact counterpart to the approximate analysis in Satoshi Nakamoto's whitepaper; we reprove its results in full, recompute its tables by script, and instantiate its conclusions for the proposed NU7 parameters.

Two distinct attacks are both called “double-spending”, and this volume treats exactly one of them. Two spends of the same shielded note reveal the same nullifier, and the protocol specification's §“Nullifier Sets” forbids a repeated nullifier within a transaction or valid block chain; such a same-chain attempt is therefore invalid under the consensus rules. What consensus must prevent is the chain-level attack: paying a merchant on the chain everyone sees, then replacing that chain wholesale with another in which the payment never happened. No zero-knowledge proof (*Crypto Guide*, §“Interactive proofs, zero knowledge, and SNARKs”) rules this out; under the model developed here, the proof-of-work race gives the attack its probabilistic bound.

The quantitative argument assumes the *Math Guide*'s probability section (§“Probability, asymptotics, and computation”), which constructs probability from its definition — finite spaces, conditioning, independence, random variables, expectation — and extends it to the three rungs used here: countable additivity, continuity along monotone events, and density-defined laws (§“Beyond finite spaces: countable additivity, limits, and densities”). The block and anchor vocabulary uses the *Crypto Guide*'s hash and commitment sections. On that base, Section 3 constructs the race-specific instruments no lower volume owns — memoryless clocks, random walks, the negative binomial law; the algebra used is elementary. Readers wanting only the results can read Sections 2, 6, and 8 and take the theorems on faith.

1.1 Sources and their standing

Rosenfeld’s analysis supplies the probability and economic model, not consensus rules; its two printed tables are reproduced by exact arithmetic before instantiating the model here. The protocol specification and the relevant ZIPs define the protocol.

This volume targets the *draft NU7 rules* as of 23 September 2026: ZIP-259 selects ZIP-218, ZIP-237, and ZIP-2003. Activation heights remain unassigned, and some proposed limits and issuance rules are incomplete. These boundaries are stated where they affect a calculation; no unassigned constant is treated as a deployed rule.

Remark 1.1 (The paper’s own condition). The arXiv v1 PDF of the paper is lightly broken: it contains exactly two citations, both rendered as “[?]” (in the abstract and in §1), and no References section at all. Both plainly intend the Nakamoto whitepaper, and we read them so. The paper numbers only two of its displayed equations — its equation (1) is our Theorem 5.2, its equation (2) our equation (3) — and sets its two model assumptions as a bare numbered list; the *Assumption* environments of Section 4 are ours.

2 The block tree and the most-work rule

Zcash transactions are grouped into blocks, and every block names its parent by including the parent header’s hash (*Crypto Guide*, §“Hash functions and the random oracle model”); the genesis block alone has no parent. Blocks therefore form a tree rooted at genesis, and each root-to-leaf path — each *branch* — is one internally consistent version of history. Branches need not be consistent with one another: one branch can contain a transaction that conflicts with a transaction in a sibling branch, and that freedom is exactly the attack surface this volume quantifies.

2.1 Work, and the best chain

A single coherent history requires a rule every node can apply locally and use to converge once one valid branch is strictly preferred. The rule is not simply “longest”: it weighs each block by the computational effort represented by its header target.

Definition 2.1 (Work of a block). Let `ToTarget` be the conversion from the header’s `nBits` field to its 256-bit target threshold (specification §7.7.4). The *work* of a block is

$$\text{work} = \left\lfloor \frac{2^{256}}{\text{ToTarget}(\text{nBits}) + 1} \right\rfloor$$

(specification §7.7.5, “Definition of Work”). A lower target admits fewer valid headers, so a block meeting it certifies proportionally more expected hashing; work is that certificate made additive.

The deployed Zebra validator computes this quantity with a 256-bit in-word trick: zebra evaluates $(\text{!expanded.0} / (\text{expanded.0} + 1)) + 1$; the retired `zcashd` computed the identical expression on `arith_uint256`, with `~` spelling the complement. The form agrees with Definition 2.1 exactly: writing t for the target and $A = 2^{256}$, the complement is $\neg t = A - 1 - t$, and

$$\left\lfloor \frac{A - 1 - t}{t + 1} \right\rfloor + 1 = \left\lfloor \frac{A - (t + 1)}{t + 1} \right\rfloor + 1 = \left\lfloor \frac{A}{t + 1} \right\rfloor,$$

so the in-word computation equals the out-of-word quotient (checked by script over random targets).

Definition 2.2 (Best valid block chain). Specification §3.3 (“The Block Chain”): a node “sums the work, as defined in §7.7.5, of all blocks in each valid block chain, and considers the valid block chain with greatest total work to be best. To break ties between leaf blocks, a node will prefer the block that it received first.”

Rosenfeld’s paper states the same rule for Bitcoin — “the branch representing the most proof of work”, with first-seen tie-breaking — so the model uses the same selection rule. The shorthand “longest chain” is exact only while difficulty is constant, which is precisely the paper’s Assumption 4.2 and not a property of the deployed chain (Section 8.2). Note also what the rule does *not* say: nothing marks a chain as permanently chosen. A node that learns of a heavier branch switches to it, however deep the fork point — subject only to the node-local rails of Section 8.4. Consensus is a standing competition, not an election with a winner.

Definition 2.3 (Confirmations). A transaction has n *confirmations* if it is included in a block of the best chain and there are n blocks on the path from that block to the chain’s leaf, inclusive. The block containing the transaction is its first confirmation.

Different nodes may briefly disagree on the best chain when two blocks of equal cumulative work arrive in different orders. Absent a concurrent extension of the other branch, a later block makes one branch strictly heavier, and nodes converge on it once that block propagates (Figure 1). Such ties are the benign case. The attack of Section 5 is the malicious one: a fork manufactured in private and released only when it wins.

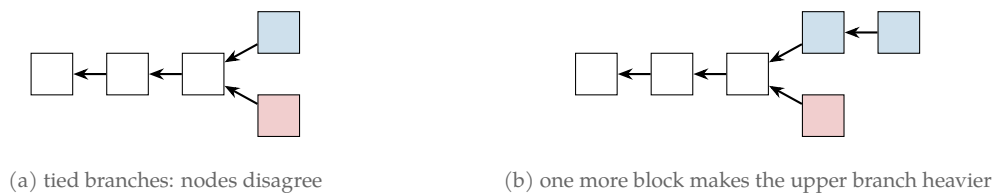


Figure 1: Branch selection. Arrows point from child to parent. In panel (a) two blocks extend the same parent and the branches carry equal work; each node keeps the one it saw first. In panel (b) a new block lands on the upper branch, which now has strictly more cumulative work; during propagation, every node that knows this state selects the upper branch, unless the lower branch has meanwhile been extended. At constant difficulty “more work” and “longer” coincide, which is how the paper, and this volume’s figures, may draw block counts.

2.2 The attack

The double-spend attack, as the paper lays it out (§3):

1. broadcast a transaction paying the merchant;
2. secretly mine a branch from the last pre-transaction block, containing instead a conflicting transaction paying the attacker himself;
3. wait until the merchant’s transaction has n confirmations and the merchant, satisfied, hands over the goods;
4. in the paper’s first-seen tie model, keep extending the secret branch until it carries more work than the public one, then broadcast it. Every node that sees it switches to it, the payment to the merchant is no longer part of history, and the attacker has both the goods and the coins.

Nothing in step 4 violates any validity rule — the released branch is a perfectly well-formed chain, merely a different one. The only question is probabilistic: with what probability does the secret branch ever get ahead? Answering it exactly in that first-seen model is the business of Sections 4 and 5; the toolkit comes first.

3 A probability toolkit

The analysis ahead needs four tools: memoryless clocks (to model block discovery), a reduction from continuous time to a coin sequence, the negative binomial distribution (to count the attacker's progress), and one normalisation lemma about races. Their foundations are the *Math Guide's*: probability spaces, conditioning, independence, random variables, and expectation are constructed there from the definition of probability onward (§“Probability, asymptotics, and computation”), and the extension beyond finite spaces — countable additivity, continuity along monotone events, density-defined laws, and infinite sequences of independent trials — is its §“Beyond finite spaces: countable additivity, limits, and densities”. What no lower volume owns are the four instruments themselves; each is constructed here on that base, only as far as the critical path requires.

3.1 Memoryless clocks and the attribution sequence

Definition 3.1 (Exponential clock). A random variable X is *exponential with rate $\lambda > 0$* if $\Pr[X > t] = e^{-\lambda t}$ for all $t \geq 0$ — a law defined by the density $\lambda e^{-\lambda t}$ on $t \geq 0$ in the sense of the *Math Guide's* density definition, with mean $1/\lambda$.

Lemma 3.2 (Memorylessness). *An exponential X satisfies $\Pr[X > s + t \mid X > s] = \Pr[X > t]$ for all $s, t \geq 0$: having waited without success teaches nothing about the remaining wait.*

Proof. $\Pr[X > s + t \mid X > s] = e^{-\lambda(s+t)} / e^{-\lambda s} = e^{-\lambda t}$. □

Lemma 3.3 (Race of two clocks). *Let X and Y be independent exponentials with rates λ and μ . Then*

$$\Pr[X < Y] = \frac{\lambda}{\lambda + \mu},$$

and the winning time $\min(X, Y)$ is exponential with rate $\lambda + \mu$.

Proof. Integrating over the time at which X fires,

$$\Pr[X < Y] = \int_0^\infty \lambda e^{-\lambda t} \Pr[Y > t] dt = \int_0^\infty \lambda e^{-(\lambda+\mu)t} dt = \frac{\lambda}{\lambda + \mu}.$$

For the minimum, $\Pr[\min(X, Y) > t] = \Pr[X > t] \Pr[Y > t] = e^{-(\lambda+\mu)t}$. Moreover, the joint density that X wins at time t factors as

$$\lambda e^{-(\lambda+\mu)t} = \frac{\lambda}{\lambda + \mu} (\lambda + \mu) e^{-(\lambda+\mu)t},$$

and similarly for Y . Thus the winner is independent of the winning time. □

Proposition 3.4 (The attribution sequence). *Model the honest network and attacker as independent exponential clocks with rates p/T_0 and q/T_0 , restarting when either fires (Section 4). Successive block attributions form an infinite sequence of independent, identically distributed Bernoulli trials in the Math Guide’s sense: attacker with probability q , honest network with probability p , independent of all earlier attributions and all elapsed times.*

Proof. By Lemma 3.3 the first block is the attacker’s with probability $(q/T_0)/(p/T_0 + q/T_0) = q$. When a block is found, the finder’s clock restarts by construction, and the loser’s remaining wait is distributed as a fresh clock by Lemma 3.2; the state after each block is therefore probabilistically identical to the start, independent of the past. The density factorisation in Lemma 3.3 also makes each winner independent of that race’s elapsed time. The claim follows by induction. $\square$

Proposition 3.4 is the paper’s bridge (its §3, with footnote 1 supplying the clock rates) from physical time to combinatorics: since the question “does the attacker ever get ahead?” concerns only the *order* of block discoveries, not their times, every result below is a statement about a p/q coin sequence, and the time constant T_0 disappears from the answers. Where the exponential model itself comes from — and how well Zcash’s Equihash fits it — is a deployment question, taken up in Section 8.3.

3.2 The negative binomial law

Proposition 3.5 (Progress of the loser). *In an attribution sequence with attacker probability $q = 1 - p$, let m be the number of attacker blocks found strictly before the honest network’s n -th block. Then*

$$P(m) = \binom{m+n-1}{m} p^n q^m, \quad m = 0, 1, 2, \dots$$

— the negative binomial distribution.

Proof. The event fixes the first $m+n$ trials exactly this far: the $(m+n)$ -th trial is honest (the n -th honest success), and among the first $m+n-1$ trials exactly m are the attacker’s, in any of $\binom{m+n-1}{m}$ orders. Each such order has probability $p^{n-1} q^m \cdot p$. $\square$

Lemma 3.6 (First-to- n race). *In the same sequence, let H_n be the event that the honest network reaches n blocks before the attacker does, and A_n the reverse. Then H_n and A_n partition the sample space, and*

$$\Pr[H_n] = \sum_{m=0}^{n-1} \binom{m+n-1}{m} p^n q^m, \quad \Pr[A_n] = \sum_{h=0}^{n-1} \binom{h+n-1}{h} q^n p^h,$$

$$\Pr[H_n] + \Pr[A_n] = 1.$$

Proof. Among the first $2n-1$ trials one side already has n successes, so the race terminates; a tie is impossible because block counts advance one at a time. The event H_n says the n -th honest block arrives while the attacker holds $m \leq n-1$; summing Proposition 3.5 over those m gives the first sum, and A_n is the same computation with the roles of p and q swapped. $\square$

Remark 3.7 (From hash trials to exponential clocks). The exponential model is itself a limit, not an axiom. A miner’s work is modelled as a stream of candidate evaluations, each an independent Bernoulli trial with a tiny success probability; the number of trials to the first success is geometric, and a geometric law with small success probability, viewed at the scale of its mean, converges to the exponential of Definition 3.1. The step that matters is *progress-freeness*: a candidate that

fails must carry no information usable by the next one. Whether Zcash’s Equihash satisfies this is examined with the deployed parameters in Section 8.3.

4 Playing catch-up

The attacker of Section 2.2 finds himself, at the moment the merchant ships, some number of blocks behind the public chain, and wins if his branch ever overtakes it. This section computes the probability of that event as a function of the deficit; the next section supplies the distribution of the deficit itself.

Both computations run under the paper’s two standing assumptions, stated here nearly verbatim (the environment form is ours, Remark 1.1):

Assumption 4.1 (Constant hashrates). The total hashrate of the honest network and the attacker is constant. Combined they have a hashrate of H , of which pH belongs to the honest network and qH to the attacker, where $0 < p, q < 1$ and $p + q = 1$.

Assumption 4.2 (Constant difficulty). The mining difficulty is constant, and such that with hashrate H the average time to find a block is T_0 .

Under Assumption 4.2 every block carries equal work, so “most work” and “longest” coincide and the race can be scored in block counts. Section 8.2 explains how the deployed chain departs from these assumptions — difficulty adjusts after every block and hashrate drifts — and why the formulas should be read as an equal-difficulty baseline rather than an exact deployment model.

Definition 4.3 (The deficit walk). Let z denote the honest chain’s lead in blocks over the attacker’s branch. By Proposition 3.4, each newly found block moves z up by 1 with probability p (honest block) or down by 1 with probability q (attacker block), independently of the past:

$$z_{i+1} = \begin{cases} z_i + 1 & \text{with probability } p, \\ z_i - 1 & \text{with probability } q. \end{cases}$$

The attack succeeds the moment z reaches -1 : the secret branch is then strictly ahead, and releasing it rewrites history.

Theorem 4.4 (Catch-up probability; Rosenfeld §3). Let a_z be the probability that the walk of Definition 4.3, started at deficit z , ever reaches -1 . Then

$$a_z = \min(q/p, 1)^{\max(z+1, 0)} = \begin{cases} 1 & \text{if } z < 0 \text{ or } q \geq p, \\ (q/p)^{z+1} & \text{if } z \geq 0 \text{ and } q < p. \end{cases} \quad (1)$$

Proof. For $z < 0$ the walk has already succeeded, so $a_z = 1$; assume $z \geq 0$. Fix an integer $N > z$ and absorb the walk at both -1 and N ; let $h_N(z)$ be the probability of reaching -1 before N . Conditioning on the first step,

$$h_N(z) = p h_N(z+1) + q h_N(z-1) \quad (0 \leq z \leq N-1), \quad h_N(-1) = 1, \quad h_N(N) = 0,$$

a second-order linear recurrence. The trial solution $h_N(z) = t^z$ satisfies it exactly when t is a root of the characteristic polynomial $pt^2 - t + q = p(t-1)(t-q/p)$.

If $q \neq p$ the roots 1 and q/p are distinct, so $h_N(z) = A + B(q/p)^z$; the boundary conditions give

$$h_N(z) = \frac{(q/p)^z - (q/p)^N}{(q/p)^{-1} - (q/p)^N}.$$

If $q = p$ the root is double and $h_N(z) = A + Bz$, giving $h_N(z) = (N - z)/(N + 1)$.

Now let $N \rightarrow \infty$. The events $E_N = \{\text{reach } -1 \text{ before } N\}$ are nondecreasing in N , and their union is the event that the walk ever reaches -1 : a walk that does so does it at a finite time, having visited finitely many states, so E_N holds for every N above the largest state visited. By continuity along monotone events (*Math Guide*, §“Beyond finite spaces: countable additivity, limits, and densities”), $a_z = \lim_{N \rightarrow \infty} h_N(z)$. For $q < p$ the term $(q/p)^N$ vanishes and the quotient tends to $(q/p)^{z+1}$; for $q = p$, $(N - z)/(N + 1) \rightarrow 1$; for $q > p$, dividing numerator and denominator by $(q/p)^N$ sends both to -1 , so the quotient tends to 1. $\square$

Corollary 4.5 (Half or more always catches up). *When $q \geq p$, the attacker overtakes the public chain with probability 1 from any finite deficit. No confirmation count defends when the attacker controls at least half of the hashrate.*

Corollary 4.6 (Geometric decay in the deficit). *When $q < p$, each additional block of deficit multiplies the catch-up probability by q/p . Concretely (script-computed): at $q = 10\%$, $a_0 = 1/9 \approx 0.1111$ and $a_5 \approx 1.88 \times 10^{-6}$; at $q = 30\%$, $a_0 = 3/7 \approx 0.4286$ and $a_5 \approx 6.20 \times 10^{-3}$.*

Corollary 4.7 (Blocks, not time). *Equation (1) contains neither T_0 nor any other time scale: the security of a deficit is a function of block counts alone. Two chains with different block spacings but the same q offer identical security per confirmation — and therefore different security per minute (Section 6.1).*

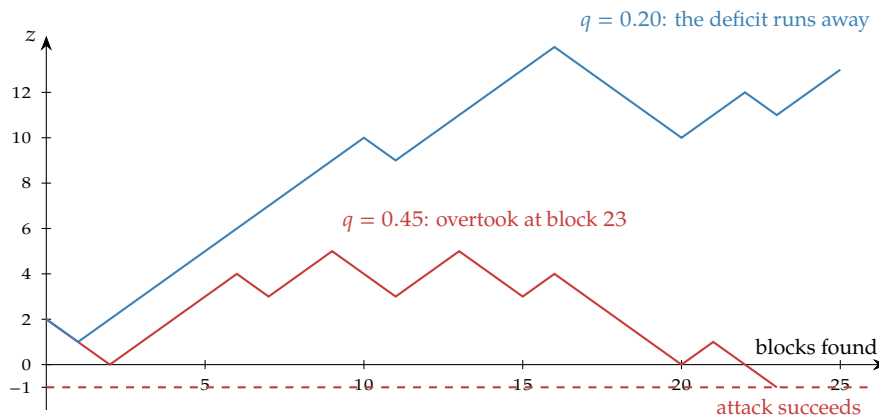


Figure 2: Two runs of the deficit walk from $z = 2$, simulated by script with fixed seeds. At $q = 0.45$ the drift is weak and this run reaches -1 after 23 blocks — the attack succeeds; at $q = 0.20$ the deficit grows roughly linearly, and by Theorem 4.4 the chance of ever returning from deficit 13 is $(1/4)^{14} \approx 3.7 \times 10^{-9}$.

5 Waiting for confirmations

The merchant's defence is patience: ship only after the paying transaction has n confirmations (Definition 2.3). This section computes exactly how much that patience buys in the paper's first-seen model. The paper's model of the attack timeline has one convention worth surfacing before the theorem.

Remark 5.1 (The pre-mined block). The paper assumes the attacker banks one block before the race is scored: “we assume one block was pre-mined by the attacker before commencing the attack” (§4). The attacker forks from the public leaf at the moment the merchant's transaction is broadcast, and a patient attacker times the broadcast to coincide with a private block he has just found but not announced. At the moment the honest chain reaches n confirmations, the attacker who has found m further blocks therefore holds $m + 1$, and the deficit walk of Definition 4.3 starts at

$$z = n - m - 1.$$

The convention is a modelling choice, not a law: an attacker with no banked block starts at $z = n - m$, and every probability below shrinks accordingly. It is also, as Theorem 5.2 shows, precisely the choice that makes the final formula collapse into a symmetric race.

By Proposition 3.5, the number m of attacker blocks found while the honest network finds its n is negative binomial. Averaging the catch-up probability of Theorem 4.4 over m gives the central result for the specification and Rosenfeld first-seen tie rule. Section 8.1 explains the additional success opportunity supplied by Zebra's tie-break. The model is not an exact deployed rollback probability.

Theorem 5.2 (Double-spend success probability; Rosenfeld eq. (1)). *Let the merchant wait for $n \geq 1$ confirmations, under Assumptions 4.1 and 4.2 and the convention of Remark 5.1, with a private branch accepted only after it becomes strictly heavier. The probability that the attacker's branch eventually overtakes the public chain is $r(q, n) = 1$ if $q \geq p$, and otherwise*

$$r(q, n) = 1 - \sum_{m=0}^{n-1} \binom{m+n-1}{m} (p^n q^m - p^m q^n) = 2 \sum_{m=0}^{n-1} \binom{m+n-1}{m} p^m q^n. \quad (2)$$

Proof. Conditioning on m and applying Theorem 4.4 at $z = n - m - 1$,

$$r = \sum_{m=0}^{\infty} P(m) a_{n-m-1} = \underbrace{\sum_{m=0}^{n-1} P(m) \left(\frac{q}{p}\right)^{n-m}}_{\text{behind: must catch up}} + \underbrace{\sum_{m=n}^{\infty} P(m)}_{\text{already ahead}},$$

valid for $q < p$; when $q \geq p$ every $a_{n-m-1} = 1$ and $r = \sum_m P(m) = 1$ immediately. For the first sum, each term transforms exactly:

$$\binom{m+n-1}{m} p^n q^m \left(\frac{q}{p}\right)^{n-m} = \binom{m+n-1}{m} p^m q^n,$$

which by Lemma 3.6 is the probability that the attacker's n -th block arrives while the honest network holds m ; summed over $m \leq n - 1$, the first sum is $\Pr[A_n]$, the probability that the attacker wins the first-to- n race. The second sum is $\Pr[m \geq n] = 1 - \Pr[H_n] = \Pr[A_n]$ by the same lemma. Hence $r = 2\Pr[A_n]$, the right-hand form of (2); the middle form follows from $\Pr[A_n] = 1 - \Pr[H_n]$ applied to one of the two copies. The printed upper limit in the paper's equation (1) is n rather than $n - 1$; the $m = n$ term is identically zero, so the forms agree. $\square$

Remark 5.3 (The race doubling). The proof exposes a structure the paper’s algebra passes through silently: under the one-pre-mined-block convention, the *catch-up half* — the honest chain reaches n first and the attacker later overtakes — has exactly the same probability as the *already-ahead half*, in which the attacker reaches n first. A first-to- n winner may have fallen behind and recovered along the way; neither half asserts otherwise. Thus the double-spend probability is twice the chance of winning a first-to- n race. The identity also absorbs the boundary case: at $q = p$ each race half is exactly $\frac{1}{2}$, giving $r = 1$ with no case split, continuously with the $q > p$ regime. The algebra here is the paper’s; reading its two sums as the two halves of one race, and the observation that the pre-mine convention is what makes them equal, are this volume’s presentational additions (verified by script over random (q, n) ; the two sums agree to floating-point precision).

Example 5.4 (A full race, exactly). Take $q = 1/5$ (a twenty-percent attacker) and $n = 6$. The deficit starts at $z = 5 - m$, the catch-up factor is $(q/p)^{6-m} = 4^{-(6-m)}$, and every quantity below is an exact rational, printed to six decimals (script-computed):

m	$P(m)$	$a_{5-m} = 4^{-(6-m)}$	product
0	0.262144	0.000244	0.000064
1	0.314573	0.000977	0.000307
2	0.220201	0.003906	0.000860
3	0.117441	0.015625	0.001835
4	0.052848	0.062500	0.003303
5	0.021139	0.250000	0.005285
catch-up half (sum of products)			0.011654
already-ahead half, $\Pr[m \geq 6]$			0.011654
$r(0.2, 6)$			0.023308

The two halves agree to every printed digit — they are equal exactly, by Remark 5.3 — and the closed form (2) evaluates to the same 0.023308. A merchant facing a possible twenty-percent adversary who ships after six confirmations is assigned a 2.33% strict-overtake probability by this model.

Remark 5.5 (Satoshi’s approximation). The whitepaper’s §11 computes the same quantity with one approximation and two offsetting conventions: the attacker’s progress is approximated as Poisson with mean $\lambda = nq/p$ (the law $\Pr[k] = e^{-\lambda} \lambda^k / k!$ on the nonnegative integers); no block is pre-mined, but success is scored at *breakeven* rather than strictly ahead, and those two conventions cancel exactly — the whitepaper’s catch-up factor for m attacker blocks is $(q/p)^{n-m}$, term for term the factor of Remark 5.1. The Poisson approximation is the entire gap. In each of the following script-computed comparisons it pushes the estimate down: at $q = 10\%$, $n = 6$, the whitepaper’s formula gives 0.0243% against the exact 0.0591% — an understatement by a factor of about 2.4; at $q = 30\%$, $n = 6$: 13.211% against 15.645%; at $q = 10\%$, $n = 10$: 0.0001% against 0.0008%. The direction is not uniform over all (q, n) ; the reason to prefer the negative binomial model is that it is exact under the stated assumptions, not that the approximation is always one-sided.

q	$n=1$	2	3	4	5	6	7	8	9	10
2%	4.000	0.237	0.016	0.0011	$8 \cdot 10^{-5}$			$< 10^{-5}$		
5%	10.000	1.450	0.232	0.039	0.0066	0.0012	$2.1 \cdot 10^{-4}$	$4 \cdot 10^{-5}$	$< 10^{-5}$	
10%	20.000	5.600	1.712	0.546	0.178	0.059	0.020	0.0067	0.0023	0.0008
20%	40.000	20.800	11.584	6.669	3.916	2.331	1.401	0.848	0.516	0.316
30%	60.000	43.200	32.616	25.207	19.762	15.645	12.475	10.003	8.055	6.511
40%	80.000	70.400	63.488	57.958	53.314	49.300	45.769	42.621	39.787	37.218
45%	90.000	85.050	81.375	78.342	75.716	73.375	71.252	69.299	67.487	65.793
48%	96.000	94.003	92.508	91.264	90.177	89.201	88.307	87.478	86.703	85.972
50%					100 for every n					

Table 1: The double-spend success probability $r(q, n)$, in percent, computed by script from Theorem 5.2. On the rows the paper’s Table 1 also prints ($q \in \{2, 10, 20, 30, 40, 48, 50\}\%$), the script reproduces its values cell-for-cell; the 5% and 45% rows are this volume’s additions.

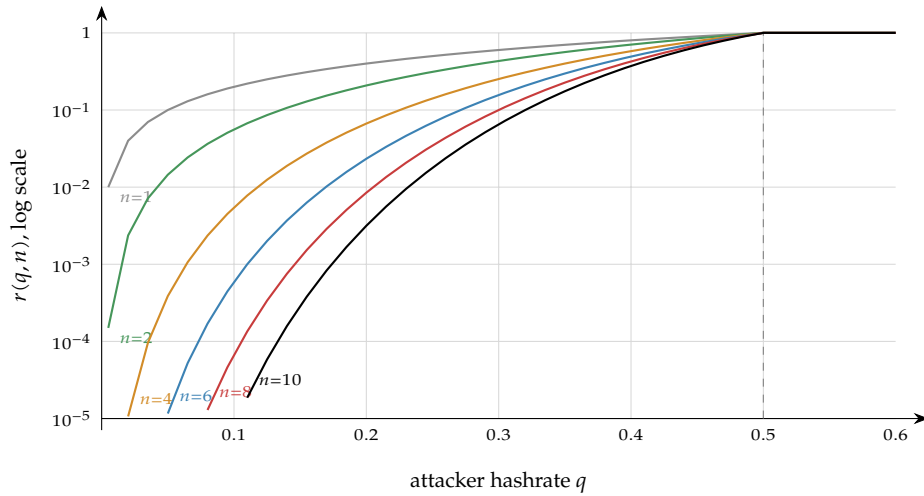


Figure 3: The success probability $r(q, n)$ against the attacker’s hashrate share q , for $n \in \{1, 2, 4, 6, 8, 10\}$ confirmations, logarithmic vertical scale clipped at 10^{-5} ; coordinates computed by script from Theorem 5.2. Every curve reaches 1 at $q = \frac{1}{2}$ (dashed) and stays there: beyond the majority line, confirmations are worthless. Below it, each added confirmation drops r by a roughly constant factor — the curves are near-equally spaced in log scale — and the factor improves as q falls.

6 What the numbers say

Theorem 5.2 compresses the security of Nakamoto consensus into one two-parameter function, and its qualitative lessons are all visible in Table 1 and Figure 3. This section states them precisely, then instantiates the one lesson that is specific to Zcash.

Proposition 6.1 (No finite count is final). *For every $0 < q < p$ and every n , $r(q, n) \geq 2q^n > 0$.*

Proof. By Theorem 5.2, $r = 2\Pr[A_n]$, and one way for the attacker to win the first-to- n race is to find the first n blocks outright, an event of probability q^n . $\square$

Proposition 6.2 (Divergence at the majority line). *Fix a target $0 < \varepsilon < 1$. For $0 < q < \frac{1}{2}$, let $n_\varepsilon(q)$ be the least n with $r(q, n) < \varepsilon$. Then $n_\varepsilon(q) \rightarrow \infty$ as $q \uparrow \frac{1}{2}$.*

q	$r < 10\%$	$r < 1\%$	$r < 0.1\%$
2%	1	2	3
5%	2	3	4
8%	2	3	5
10%	2	4	6
15%	3	6	9
20%	4	8	13
25%	5	12	20
30%	9	20	32
35%	15	36	58
40%	34	82	133
45%	135	331	539

Table 2: The least number of confirmations pushing $r(q, n)$ below three targets, computed by script. The growth is logarithmic in the inverse target risk but explosive in q : a 10% attacker is answered by 6 confirmations, a 45% attacker by 539, and at 50% by no number at all.

Proof. First, $n_\varepsilon(q)$ exists. The event A_n that the attacker reaches n blocks first is the event that at least n of the first $2n - 1$ block discoveries are the attacker's. Since $q < p$, each such length- $(2n - 1)$ sequence has probability at most $q^n p^{n-1}$, and there are fewer than 2^{2n-1} sequences. Hence

$$r(q, n) = 2\Pr[A_n] \leq \frac{(4pq)^n}{p} \rightarrow 0,$$

because $4pq < 1$.

For fixed n the function $q \mapsto r(q, n)$ is a polynomial on $[0, \frac{1}{2}]$, hence continuous, and $r(\frac{1}{2}, n) = 1$: at $q = p$ each half of the race of Remark 5.3 has probability exactly $\frac{1}{2}$ by Lemma 3.6 and symmetry. Given any N , for each $1 \leq n \leq N$ there is a $\delta_n > 0$ with $r(q, n) > \varepsilon$ whenever $q > \frac{1}{2} - \delta_n$. Taking $\delta = \min_{1 \leq n \leq N} \delta_n > 0$ excludes every $n \leq N$ throughout that neighbourhood, so $n_\varepsilon(q) > N$. Since N was arbitrary, the claimed divergence follows. $\square$

Remark 6.3 (Nothing is special about six). The paper (§5): “There is nothing special about the default, often-cited figure of 6 confirmations. It was chosen based on the assumption that an attacker is unlikely to amass more than 10% of the hashrate, and that a negligible risk of less than 0.1% is acceptable. Both these figures are arbitrary” — six confirmations are “overkill for casual attackers, and at the same time powerless against more dedicated attackers”. Table 2 is the honest replacement: pick the adversary you defend against and the risk you accept, and read off n ; the pair (q, ε) is a policy, not a law of nature.

6.1 Zcash time: 25-second blocks

Corollary 4.7 says security is purchased in blocks. The wall-clock price is set by the target spacing: 25 seconds in ZIP-218. Since public discoveries arrive at rate p/T_0 , waiting for n confirmations takes nT_0/p seconds in expectation in this model. Three confirmations cost a nominal 75 seconds and ten cost 250 seconds when all hashpower publishes on the public chain. With a 10% withholding attacker, the expected wait is instead divided by 0.9.

For example, a nominal thirty-minute wait buys 72 confirmations at 25-second spacing, compared with three at Bitcoin's 600-second spacing. The corresponding strict-overtake probabilities against a 10% attacker are approximately 9.35×10^{-34} and 1.712%, respectively; either count takes $30/0.9 \approx 33.3$ minutes in expectation under withholding. These are model calculations, not guarantees about either network.

The comparison holds q fixed. It contains no network and therefore does not price honest blocks racing one another during propagation. Quantifying that lost work requires a propagation

distribution and mining topology. Moreover, a policy waiting for a fixed amount of accumulated miner reward asks a different question: decreasing the reward per block in proportion to the target spacing requires proportionally more blocks to reach that amount. Ignoring transaction fees (value paid for transaction inclusion) and rounding, the shorter spacing then cancels out of the mean wait. The monetary and fixed-hashrate models are not interchangeable.

Remark 6.4 (Where the model bends). Three assumptions do not survive contact with the deployed chain; they determine how cautiously the calculations may be transferred to it. (i) *Difficulty is not constant*: Zcash retunes the target after every block (Section 8.2). The race is scored in work, not block counts, and q may be interpreted approximately as the attacker's share of work production while the branches' block weights remain close; once their difficulty schedules diverge, the equal-step random-walk formula is no longer exact. (ii) *Hashrate is not constant*: q in the theorems is whatever share the attacker sustains for the attack's duration; the paper's own caveat is that T_0 re-enters only if the attacker cannot sustain his hashrate long enough, which it judges unlikely for a serious attacker. (iii) *Propagation is not instantaneous*: honest self-orphaning wastes honest work and thereby raises the attacker's effective share; its size is not quantified here. A fourth gap is not the model's: the attacker of this volume follows the protocol's validity rules perfectly and attacks only the *choice* between valid histories. Attacks on the rules themselves are other volumes' subjects.

7 The economics of the attack

The probabilities of Section 5 answer the adversarial merchant, who assumes the customer is an attacker. The paper's §6 answers the realistic one: when is the attack worth mounting at all? Throughout this section $q < p$ — “[o]therwise, all bets are off” (§6) — and the model is deliberately stylised; its own caveats close the section.

The attacker targets k merchants simultaneously with payments of v each, all invalidated by the same secret branch; the goods are worth αv to him, $0 < \alpha \leq 1$; he gives up after mining o blocks in vain; and each of his blocks would earn the block income B if it ended up in the accepted chain. If the attack succeeds his blocks are all accepted (he keeps goods *and* coins *and* rewards); if it fails he has paid kv , keeps goods worth αv , and his o orphaned blocks forfeit oB . For an exact finite-stop profit calculation, r would have to be the success probability of that stopping policy. Following the paper, the tables below instead insert the unlimited-horizon $r(q, n)$ as a heuristic proxy. They are not an exact evaluation of a strategy that always stops after $o = 20$ private blocks.

Proposition 7.1 (Profitability; Rosenfeld §6). *Given success probability r and these stipulated pay-offs, the attacker's expected profit over not attacking is*

$$k\alpha v - (1 - r)(kv + oB) = kv(\alpha + r - 1) - (1 - r)oB,$$

positive if and only if

$$v > \frac{(1 - r)oB}{k(\alpha + r - 1)}$$

(when $\alpha + r > 1$; when $\alpha + r \leq 1$ the modeled attack is never strictly profitable). This modeled attack is not strictly profitable whenever

$$v \leq \frac{o(1 - r)B}{k(\alpha + r - 1)} \stackrel{\alpha \equiv 1}{=} \frac{o}{k} \cdot \frac{B(1 - r)}{r}, \quad (3)$$

which with the paper's choices $o = 20$, $k = 5$, $\alpha = 1$, $B = 25$ BTC is its equation (2), $v \leq 100(1/r - 1)$ BTC.

Proof. The gain $k\alpha v$ is certain (the goods are obtained either way); the loss $kv + oB$ is paid ex-

actly when the attack fails, with probability $1 - r$. Rearranging the positivity condition gives the threshold; substituting the paper’s constants gives $20 \cdot 25 \cdot (1 - r)/(5r) = 100(1 - r)/r$. $\square$

7.1 Reward-normalised exposure

The *block subsidy* is value issued by the protocol in a block, distinct from fees transferred by its transactions. Its *scheduled* component is base issuance determined by height, before any reserve payout. A *funding stream* is a required allocation of a share of that subsidy to a designated recipient. The *coinbase* is the block’s first transaction, which distributes the available subsidy and fees according to the applicable rules. The *reserve* is the accounting balance of funds removed from circulation and awaiting reissuance.

The income B forfeited by an orphaned block is the amount its miner would have retained: miner subsidy plus retained transaction fees, not the subsidy allocated to other recipients. Under NU7 this depends on scheduled issuance, the prior reserve balance, and the applicable fee-removal rule (Section 8.8). A universal ZEC figure would therefore hide assumptions about both chain state and unresolved draft rules.

Instead, Table 3 measures transaction value in units of the chosen block income B . With the paper’s $o = 20$, $k = 5$, and $\alpha = 1$, equation (3) becomes

$$\frac{v}{B} \leq 4 \frac{1 - r}{r}.$$

Once a particular threat model supplies B , multiplying by it gives the corresponding monetary threshold. Treating B as constant over the attack remains a modelling assumption.

q	$n=1$	2	4	6	8	10
5%	36	271	10,327	344,743	10,931,506	336,737,801
10%	16	67	729	6,759	59,475	508,917
20%	6	15	55	167	467	1,262
30%	2	5	11	21	35	57
40%	1	1	2	4	5	6

Table 3: The largest whole-number multiple of block income B with non-positive heuristic expected profit: $\lfloor 4(1 - r)/r \rfloor$, computed by script. The table is dimensionless: it neither fixes a ZEC subsidy nor predicts future fee income. Values inherit every assumption of the model; the caveats of Remark 7.3 apply in full.

Within the model, a handful of confirmations makes the attack unprofitable against small attackers at any plausible payment size; against a 30% attacker, six confirmations make modeled expected profit non-positive only up to about 21 times the assumed block income B . The required count grows only logarithmically in the value at stake, by the geometric decay of Theorem 5.2. These are neither exact finite-stop profit calculations nor bounds on deployed rollback risk; the deployed tie-break, changing difficulty, and rollback rails require separate modelling.

Remark 7.2 (Majority is a different problem). At $q = p$, eventual strict overtake has probability 1, but the walk has zero drift; that fact alone does not give the attacker a lasting monopoly. When $q > p$, the economics change character entirely: the paper notes a strict majority attacker can double-spend at no cost beyond normal mining, reject every other miner’s blocks to take the whole issuance, and exclude transactions at will, driving honest miners out and entrenching himself — so a majority attack “is best seen as an attempt to destroy” the currency, motivated from outside it (a short position, a competing system), not an attempt to profit inside it. Confirmation policy is not the defence at that point; the defence is the cost of assembling the majority, which lies outside this model.

Remark 7.3 (The model’s own caveats). The paper flags each stylisation itself. The give-up point $o = 20$ “is a significant simplification” (an optimal attacker balances completion against compounding losses); the safe values “should be taken with a grain of salt, because of the many modeling assumptions”; and a model resting on mining *costs* rather than forfeited *rewards* would make the required confirmations linear, not logarithmic, in the transaction value — “very poor security, hence in a situation where this is relevant, we have already lost anyway” (§6). The bound’s role is the shape it reveals — logarithmic patience buys exponential safety — not its third significant digit.

8 NU7: 25-second blocks and bounded shielded work

The ZIP-259 draft combines three changes: shorter blocks with limits on shielded work (ZIP-218), reissuance that preserves the halving schedule — periodic divisions of scheduled subsidy by two (ZIP-237) — and rejection of version 4 transactions (ZIP-2003). This section describes that proposal, not a deployed upgrade. Its activation height, denoted A , is unassigned on each network at the source boundary in Section 1.1.

NU7 introduces no new transaction format: versions 5 and 6 remain available. Their signatures nevertheless commit to a *consensus branch identifier*, a value identifying the network upgrade’s rules. An unchanged encoding therefore does not make a pre-NU7 signature valid under NU7. Disabling version 4 also disables spending funds in Sprout, an older shielded pool, because the newer formats cannot represent its transfers; it does not transfer their value to a reserve or remove it from the supply.

8.1 Chain selection and the tie-break

The most-work rule remains Definition 2.2: compare the sum of the blocks’ work, not their count. Changing target spacing changes neither the work definition nor this ordering.

Remark 8.1 (The implementation tie-break). The specification and Rosenfeld’s model prefer the first-seen block at equal cumulative work. The examined Zebra implementation instead uses the larger tip hash. This distinction concerns local chain choice, not the validity of a block. The divergence matters to the attack model. At equal work, an attacker can compare its private tip hash with the public tip and reveal immediately when its hash wins the deployed deterministic tie-break; the strict-overtake boundary $z = -1$ used above does not count that success opportunity. Therefore $r(q, n)$ is exact for the stated strict-overtake model and is a lower bound if only its tie-break is replaced by Zebra’s, keeping equal difficulty, unlimited rollback, and the other assumptions fixed. It is not thereby a lower bound on the full deployed process: changing difficulty and rollback limits also change the success event. Quantifying the extra probability, including any scope to choose among candidate private tips, requires a different stopping rule and is outside this volume.

8.2 Difficulty in motion

Assumption 4.2 freezes difficulty; Zcash retargets after every block. In ZIP-218, the NU7 target spacing is 25 seconds and the target-averaging window is 102 blocks. The timestamps determining the adjustment still come from the branch being evaluated, including on an attacker’s private branch.

Write $N = 102$ for the target-averaging window and $T = N \cdot 25 = 2,550$ seconds for its nominal timespan.

The specification’s §7.7.3 defines the rest of the calculation. For a block height h , let $m(h)$ be the median timestamp of the eleven preceding blocks: their sixth-smallest timestamp. Let $\bar{t}(h)$

be the mean target of the N blocks preceding h : the sum of their targets divided by N . Define the observed and damped timespans by

$$\begin{aligned} a(h) &= m(h) - m(h - N), \\ d(h) &= T + \text{trunc}\left(\frac{a(h) - T}{4}\right), \end{aligned}$$

where trunc discards the fractional part towards zero. The bounded timespan $c(h)$ is $d(h)$ clamped between $[0.84T]$ and $[1.32T]$: values below the lower endpoint are replaced by that endpoint, and similarly above the upper endpoint. The maximum target PoWLimit is $2^{243} - 1$ on Mainnet, the production network, and $2^{251} - 1$ on Testnet, the test network (§5.3). The target calculation is

$$t(h) = \min\left(\text{PoWLimit}, \left\lfloor \frac{\bar{t}(h)}{T} \right\rfloor c(h)\right),$$

followed by the compact encoding defined in the specification’s §7.7.4. The order of the floor and multiplication matters. At NU7 the clamp endpoints are 2,142 and 3,366 seconds; the eleven-timestamp median, fourfold damping, and target cap are unchanged.

Activation does not reset the accumulated work or multiply the target by three in a single operation. The first NU7 blocks still look back over pre-NU7 history and adjust through the same bounded rule. Likewise, the timestamp rules do not shrink with the target spacing: a timestamp must exceed the median of the preceding eleven and be at most 90 minutes above it (§7.6). A further non-consensus receipt check rejects blocks more than two hours ahead of the validator’s clock. Testnet’s exceptional minimum-difficulty rule continues to use six target spacings: its delay threshold becomes $6 \cdot 25 = 150$ seconds. This exception does not apply to Mainnet.

Remark 8.2 (What per-block retargeting changes). The averaging window, damping, and clamp constrain the adjustment but do not prove that two forked branches retain close targets. Work per block moves inversely with the target; a branch’s timestamps can change both its future block rate and the work carried by each block. Equal expected work production does not make the probability of an eventual overtake independent of these jump sizes. Consequently, the probability tables are an equal-work baseline, not an exact analysis of private branches whose difficulty schedules diverge.

8.3 Equihash and the memoryless model

Zcash’s proof of work is Equihash with parameters $n = 200, k = 9$ (specification §7.7, §7.7.1; the algorithm is Biryukov and Khovratovich’s, ePrint 2015/946), but the quantity the race counts is gated one step later: “[t]he difficulty filter is unchanged from Bitcoin, and is calculated using SHA-256d on the whole block header (including `solutionSize` and `solution`)”, required to be at most $\text{ToTarget}(n\text{Bits})$ (§7.7.2). Each candidate Equihash solution therefore faces a fresh SHA-256d lottery. Treating distinct headers as independent trials is the usual random-function model for SHA-256d, not a property asserted by the specification; under that model these are the Bernoulli trials of Remark 3.7.

The specification nowhere asserts progress-freeness — the argument is this volume’s, not a citation. What Equihash changes is the granularity: candidates arrive in batches, one batch per memory-hard solver run. Progress within a run is discarded when a competing block arrives. Under independent identically distributed runs, the number of runs until an accepted header is geometric; replacing those discrete completion times by an exponential clock is a further approximation, coarse within one run and good only when a run is short relative to the target spacing: 25 seconds in the NU7 draft. Run duration is implementation- and hardware-dependent and is not quantified here. A proof-of-work whose solve time were comparable to the spacing would make block discovery visibly non-exponential and pull the attribution sequence away from Proposition 3.4.

8.4 Reorg rails, maturity, and the finality floor

Consensus does not impose a maximum reorganisation depth. ZIP-218 recommends a node-local 600-block rollback window: at the target spacing this is $600 \cdot 25 = 15,000$ seconds, or 250 minutes. Such a limit bounds the history that a node automatically reverses; it is not a proof that older blocks cannot be replaced by a higher-work valid chain. Following a branch outside that retained history requires different local state or operator intervention. Actual node implementations may choose a different window.

Remark 8.3 (What the rails buy). A finite rollback policy truncates some paths to eventual overtake, so the unlimited-rollback probability $r(q, n)$ is not the exact failure probability of a node enforcing that policy. A limit is also not global agreement: different nodes may stop on different sides of a long split. The specification’s §3.3 adds a separate requirement for validators that risk Mainnet funds or display transaction information: their chain must include the activation block of the most recent settled network upgrade. That socially anchored checkpoint is distinct from probabilistic finality.

A *transparent output* records its value and destination in cleartext. The consensus maturity rule prevents spending such an output of a coinbase until the spending block is at least 100 heights later (§7.1.2). ZIP-218 keeps that count, a nominal $100 \cdot 25 = 2,500$ seconds. Coinbase outputs placed directly in a shielded pool do not have this transparent maturity restriction. A 600-block rollback window can therefore reverse an already mature coinbase: maturity permits spending, not finality.

8.5 Wallet confirmation policy

A wallet can distinguish *trusted* outputs that its own transaction handling can recreate after a rollback from an *untrusted* external payment that may require the sender to pay again. The examined wallet backend uses three confirmations for the former and ten for the latter, with zero-confirmation shielding of transparent funds allowed by its default policy. These choices are not consensus rules.

Table 1 quantifies their modelled trade-off: three confirmations assign a 10% attacker risk 1.712%, while ten assign the same attacker about 0.0008% and a 20% attacker about 0.316%. The count remains a (q, ϵ) policy, not a guarantee.

The *anchor* is the note-commitment-tree root against which a spend proves membership (*Crypto Guide*, §Privacy-preserving membership via zero-knowledge paths). ZIP-218 keeps the recommended anchor depth at three blocks. Its recommended transaction-expiry delta is 120 blocks, a nominal $120 \cdot 25 = 3,000$ seconds, without overriding an explicitly configured expiry delta. There is no instruction to divide every confirmation count by three.

8.6 A block budget for shielded work

A byte limit does not directly bound the computational work inside a block. The 2,000,000-byte limit must therefore be considered alongside the 25-second target spacing and the costs of shielded validation and wallet scanning. ZIP-218 adds limits on the shielded components that drive this work.

An Orchard *Action* pairs one spend slot with one output slot; dummy slots allow either side to carry no real transfer. Ironwood is the newer shielded pool using the same Action structure (ZIP-229). The older Sapling shielded pool counts spends and outputs separately; its cryptographic construction is not needed for this accounting.

Let O and I count Orchard and Ironwood Actions in a block, and let S count Sapling spends plus outputs. With the Ironwood coverage proposed in PR 1361, the complete shielded budget is

$$O \leq 330, \quad I \leq 330, \quad S \leq 300, \quad O + I + S \leq 330.$$

The final inequality is shared across pools: their separate maxima cannot all be filled in one block. Sprout transfers cannot occur because NU7 rejects the version 4 format that represents them.

The Ironwood term remains under review rather than already specified by the merged ZIP-218 draft. For example, $O = 100$, $I = 200$, $S = 30$ exhausts the shared budget; increasing I to 201 violates it even though each per-pool limit still holds. At two Actions per transaction, a block devoted to one Orchard-protocol pool fits at most 165 such transactions under this budget, or a nominal $165/25 = 6.6$ transactions per second. This is a capacity calculation under a specified transaction shape, not a measured throughput guarantee. Transparent components remain subject to the byte limit rather than this shielded-action budget.

8.7 Scheduled issuance with shorter blocks

A *zatoshi* is the atomic currency unit: 1 ZEC equals 10^8 zatoshi. A *halving* divides the scheduled subsidy by two. Retaining its nominal wall-clock interval while tripling block frequency requires both three times as many blocks per interval and one third of the subsidy per block. ZIP-218 therefore changes the full interval from 1,680,000 to $L = 5,040,000$ blocks. These earlier intervals are needed to compute progress at activation, not to supply an alternative rule. The change preserves expected timing, not exact calendar dates, because actual block times remain random.

The progress already made towards the next halving must survive activation. Let H_B be the Blossom activation height: 653,600 on Mainnet and 584,000 on Testnet. Before Blossom, halvings were separated by 840,000 blocks; the initial slow-start schedule, which ramps up the subsidy during the first 20,000 blocks, contributes a shift of 10,000 blocks. These constants and the maximum subsidy $M = 1,250,000,000$ zatoshi are defined in the specification's §5.3 and §7.8. For a height $h \geq A$, the draft's halving count and scheduled subsidy s_h , measured in zatoshi, are

$$k(h) = \left\lfloor \frac{H_B - 10,000}{840,000} + \frac{A - H_B}{1,680,000} + \frac{h - A}{5,040,000} \right\rfloor, \quad (4)$$

$$s_h = \left\lfloor \frac{M}{2 \cdot 3 \cdot 2^{k(h)}} \right\rfloor. \quad (5)$$

The factors 2 and 3 account separately for Blossom's and NU7's spacing changes. If activation occurs during the second-halving era, $k(h) = 2$ gives $s_h = 52,083,333$ zatoshi, or 0.52083333 ZEC, until the next halving; $k(h) = 3$ gives 26,041,666 zatoshi.

The old Mainnet third-halving height $H_3 = 4,406,400$ is therefore not the NU7 third-halving height if $A < H_3$. Converting the remaining interval gives

$$H_3^{\text{NU7}} = A + 3(H_3 - A) = 13,219,200 - 2A.$$

No numerical activation height is assigned here. Nor does retiming the halving function automatically retime a funding stream whose endpoint is specified as a literal block height. The draft does not provide such a change, so the funding-stream endpoints and the rescaled halving heights must be treated separately.

8.8 The NSM reserve and delayed reissuance

The *Network Sustainability Mechanism* (NSM) separates temporarily removed funds from the still-unissued scheduled subsidy. In the merged ZIP-237 draft, the *issued supply* V_h is the total value recorded in the chain's value pools after block h . Each pool is a separately tracked balance for one part of the protocol; tracking a shielded pool's total does not reveal the values of its individual notes. The *NSM reserve* R_h records value removed from circulation and not yet reissued. The reserve is public consensus state, not a spendable pool or an account with a spending key. In particular, it is not 21 million ZEC minus the issued supply: that larger difference also includes future scheduled issuance.

The initial reserve recovers the historical subsidy and fees that miners left unclaimed before NU6. Since ZIP-236, a coinbase must account for its full available value, so that historical shortfall is fixed. The draft defines the seed R_{A-1} as total scheduled issuance through the last pre-NU6 block minus issued supply there. The exact integer seeds for Mainnet and Testnet remain unspecified.

Let $D \geq A$ be the separately assigned height at which reissuance starts, and let d_h be the non-negative number of zatoshi removed from circulation in block h by whatever removal mechanisms have been deployed. With $L = 5,040,000$ blocks per halving interval, the draft selects

$$f = \frac{\lfloor 6,931,680,000/L \rfloor}{10^{10}} = \frac{1375}{10^{10}}.$$

Its additional subsidy u_h , total block subsidy b_h , and reserve update are

$$u_h = \begin{cases} 0, & h < D, \\ \lfloor fR_{h-1} \rfloor, & h \geq D, \end{cases} \quad (6)$$

$$b_h = s_h + u_h, \quad R_h = R_{h-1} - u_h + d_h. \quad (7)$$

All amounts are integers in zatoshi. Reissuance uses the previous block's reserve, so a removal in the current block cannot finance its own additional subsidy. Before D , removals accumulate without payout. After D , a positive reserve pays at least one zatoshi: because R is an integer and $0 < f < 1$, one has $1 \leq \lfloor fR \rfloor \leq R$ for $R > 0$. The draft rejects a negative reserve; it does not silently clamp it to zero.

For a small exact example, take $R_{h-1} = 100,000,000$ zatoshi (1 ZEC) and $h \geq D$. The additional subsidy is $\lfloor 13.75 \rfloor = 14$ zatoshi. Without a new removal the reserve becomes 99,999,986; with a same-block removal of another 1 ZEC it becomes 199,999,986, but the current payout remains 14. This example does not use the unassigned historical seed.

Why this preserves supply and halvings. Transfers between ordinary value pools cancel in their total. A block increases that total by its subsidy and reduces it by its removals:

$$V_h - V_{h-1} = s_h + u_h - d_h, \quad (V_h + R_h) - (V_{h-1} + R_{h-1}) = s_h.$$

The second identity follows by adding the reserve update; reissuance moves value between the two ledgers without creating a second copy. Scheduled issuance retains its halvings, with the reserve paying an additional amount on top. Funding streams, wherever active, take their prescribed fractions of the *total* subsidy b_h , not only s_h . The resulting subsidy depends on prior chain state as well as height.

To understand the chosen rate, temporarily ignore integer rounding and new removals. The recurrence becomes $R_{h+j} = (1 - f)^j R_h$, so keeping about half the reserve after L blocks requires $f = 1 - 2^{-1/L} \approx (\ln 2)/L$. The draft implements an integer-scaled approximation with numerator 1375. Indeed, $(1 - 1375/10^{10})^{5,040,000} \approx 0.50007$. This is a half-life of the reserve without new contributions, not a new halving rule for the scheduled subsidy; small reserves depart from the smooth curve because every positive payout rounds up.

Unresolved deployment and fee details. ZIP-259 selects a reissuance date corresponding to February 2031, but the network-specific height D remains unassigned. The drafts do not yet specify a complete, consistent NU7 removal rule: ZIP-235 requires removal of at least 60% of fees through an explicit coinbase amount, while ZIP-259 describes a block-level contribution requiring no new transaction data. The reserve equations therefore leave d_h as a defined input. Neither a particular fee-removal mechanism nor an exact NU7 miner income is inferred from these incomplete rules.

9 Layer boundary

Every theorem above rests on the *Math Guide*'s probability section: the definition of probability, its extension beyond finite spaces, and the continuity theorem used in the catch-up proof. This volume supplies the best-chain model and the quantitative bounds in Tables 1, 2, and 3. Policies that decide which chain view to trust or when to treat a transaction as final lie above that boundary.

Part V

Ironwood Guide: The Zcash Orchard protocol and its Ironwood pool

Abstract

This volume constructs the Zcash Orchard shielded-payment protocol as carried by its current value pool, Ironwood. The presentation follows value through its lifecycle: a recipient prepares keys and an address, a note is minted and delivered, it rests in the commitment tree, it is spent — retiring its nullifier, balancing its value commitment, authorised by its re-randomised signature — change returns, and value crosses the boundary between the sealed Orchard pool and Ironwood. Each piece of machinery is constructed at the lifecycle stage that first demands it and assembled into the Action statement every shielded transfer proves; the volume then descends to the circuit beneath the statement, pays the deferred security debts in end-to-end reductions, and closes with the proof-system lineage and the quantum horizon. It assumes the *Math Guide*, the *Crypto Guide*, the *Halo 2 Guide*, and the *Consensus Guide*, and is non-normative throughout: the protocol specification and the ZIPs remain authoritative.

1 Introduction: the life of a shielded zatoshi

Halo 2 is a general-purpose proving system, and it has no notion of money. It can attest that a prover knows a witness satisfying a fixed arithmetic relation, but nothing in it says what a coin is, who owns one, or when one has been spent twice. The Orchard application turns that general instrument into a shielded-payment protocol by one act of specification: it fixes a single relation — the *Action statement* — that every shielded transfer must prove, and lets the consensus rules of a public chain enforce the rest. This volume constructs that relation and everything around it, deriving each cryptographic object as the answer to one sub-problem, tracing a complete payment from Alice to Bob end to end, and only then stating the Action relation formally (§8). It assumes four earlier Arboretum volumes: the *Math Guide* for the underlying mathematics, the *Crypto Guide* for hash functions, commitments, signatures, key agreement, zero knowledge, and the rest of the primitive toolbox, the *Halo 2 Guide* for the proof system, and the *Consensus Guide* for chain selection and reorganisations. Here those tools are assembled into the Orchard protocol as carried by its current pool, Ironwood — and the protagonist of the assembly is one *zatoshi*, the atomic unit of ZEC, Zcash’s native currency, followed from the moment it turns shielded.

1.1 Two pools, one protocol

Zcash changes its consensus rules by *network upgrades*: scheduled rule changes, each activating at a block height this volume never needs numerically and carrying a name — NU5, NU6.3 — that serves only to order events. A *Zcash Improvement Proposal* (ZIP) is the document form in which protocol changes are specified; ZIP-229 fixes the vocabulary used throughout. The *Orchard protocol* is the shared cryptographic construction — the curves, Sinsemilla, the Action circuit, notes, commitments, nullifiers, keys, and note encryption, an inventory of names the roadmap (§1.3) assigns to the sections that construct them — and it survives the pool transition. The *Orchard pool* is the legacy value pool: it has run the Orchard protocol since NU5, and at NU6.3 it is sealed to spend-only. The *Ironwood pool* is the current pool, carrying the same protocol. One protocol, two carriers: when a statement holds for the shared cryptography we say “Orchard”, and when it depends on which pool a note lives in we name the pool.

The next proposed upgrade, NU7, does not introduce another shielded pool or replace this cryptography. Its deployment document, ZIP 259, remains Draft at the 2026-09-23 source check: neither network has an assigned activation height. The changes relevant here concern which

transactions a block may contain and how quickly blocks are targeted, not a new note lifecycle. This volume uses those proposed rules where they affect the construction.

Value reaches a shielded pool through three doors, each public in amount. A *shielding* transaction spends transparent value into the pool; *shielded coinbase* lets the block reward — freshly issued coins — arrive shielded directly; and *migration* moves value from another shielded pool, which for this volume’s two pools means out of sealed Orchard into Ironwood. All three doors are formalised in §9; their publicity is by design, for what a pool hides is who holds which note, never how much the pool as a whole contains.

1.2 The four requirements on a shielded payment

A transparent ledger — Bitcoin, or Zcash’s transparent pool — records, for every unspent output, its spending conditions and denomination, and a payment publicly links spent outputs to new ones. Addresses and scripts need not identify their holders in the real world. A shielded payment must achieve the same effects — transfer of value, prevention of double-spending, conservation of total supply — while concealing the private note data and spend-to-note links. Four requirements must hold simultaneously:

- (R1) represent value so that its recipient and amount stay hidden, and public commitments do not identify real versus dummy notes, yet the owner can later prove possession — solved by notes and note commitments (§4);
- (R2) prove possession of a previously committed nonzero note without identifying which note — solved by the commitment tree with a membership proof (§6);
- (R3) prevent double-spending without a public list of which notes an owner has spent — solved by nullifiers (§7);
- (R4) prove that no one created value without revealing any amount — solved by value commitments and the binding signature (§7).

Each stage of the journey ahead opens by naming the requirement it discharges, and the loop is closed at the end: §12 proves that the assembled protocol meets all four.

1.3 Method and roadmap: the journey

A protocol this dense can be presented from its symbols or from the problems it solves; we take the second way, and give it a narrative spine. The volume follows value through the system — a recipient prepares to be paid, a note is minted and delivered, it rests in the commitment tree, it is spent, change returns, and at the end of its shielded life value exits back across the pool boundary, in public amount — with each piece of protocol machinery constructed at the lifecycle stage that first demands it.

After the curve-level preliminaries and the Sinsemilla hash are packed once (§2), each object arrives as the answer to one sub-problem. The recipient’s keys and addresses come first (§3). Notes and note commitments represent value (§4); note encryption and trial decryption deliver a note privately and let its owner detect it (§5); the commitment tree and its membership path establish a nonzero note’s prior creation without revealing which (§6); nullifiers prevent double-spending, value commitments and the binding signature conserve value without disclosure, and the RedPallas signature scheme authorises the spend (§7). The unit these pieces assemble into is the *Action* — one Action spends one old note and creates one new one — and §8 walks one payment end to end, shows how a node verifies it without seeing any note, and states the relation the proof carries. The two pools and their doors follow (§9), then the seal and the turnstile that empty the legacy pool (§10), the circuit beneath the statement (§11), and the end-to-end security reductions that pay the volume’s deferred debts (§12). A closing pair of sections looks

backward and forward: the lineage Sprout → Sapling → Orchard — three generations of shielded proof, and within Orchard three circuit versions ending at Ironwood (§13) — and the quantum horizon (§14). Figure 1 draws the map.

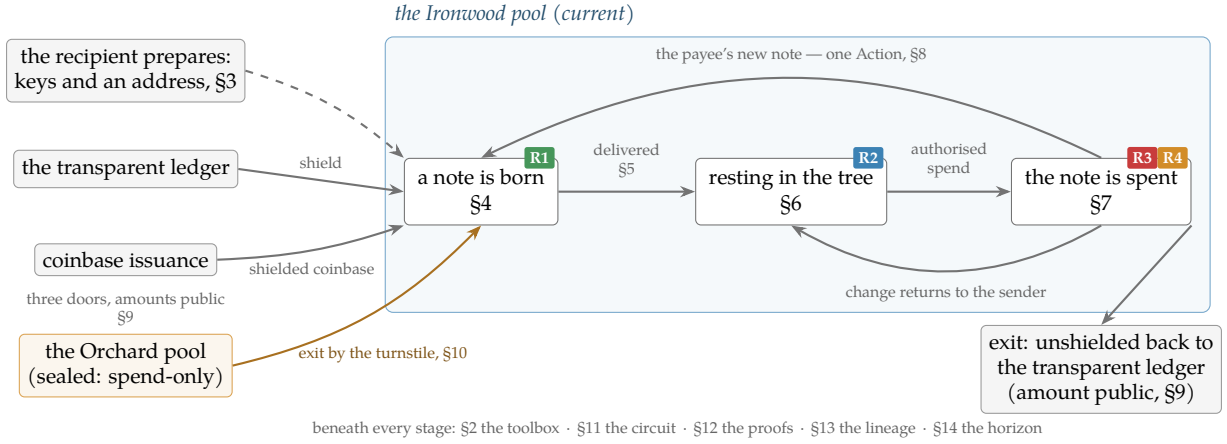


Figure 1: The volume's map: one zatoshi's lifecycle across the two pools. Value enters the Ironwood pool through three doors, each public in amount (§9); a recipient who has prepared keys and an address (§3) is paid with a freshly minted note (R1, §4), which is delivered in band (§5) and rests as a leaf of the commitment tree (R2, §6). Its spend (R3 and R4, §7) is one half of an Action (§8) whose products — the payee's new note and the sender's change — re-enter the cycle. The sealed Orchard pool moves value in one direction only, out through the turnstile (§10). At the end of its shielded life, value exits as it entered: unshielded back to the transparent ledger, in public amount (§9). Each stage is tagged with the section that constructs its machinery and the requirement of §1.2 it discharges.

2 Provisions: hashing on Pallas and Sinsemilla

Every stage of the journey ahead draws on the same two workhorses: a small kit of curve-level encodings and hash-to-curve maps, and the Sinsemilla hash built from them. Rather than interrupt the narrative to construct these where each stage first needs them, we pack them here, once — the one deliberate non-narrative cost of the volume, paid up front and kept tight. All curve-level primitives of this section operate on the Pallas group $\mathcal{E}(\mathbb{F}_{p_{\text{Pallas}}})$: the prime-order group of points of the Pallas curve, one half of the Pasta cycle of curves and fields constructed in the *Math Guide*, §“Elliptic curves”.

2.1 Points to field elements: Extract, the star encoding, and LE_n

The coordinate map Extract. The map $\text{Extract} : \mathcal{E}(\mathbb{F}_{p_{\text{Pallas}}}) \rightarrow \mathbb{F}_{p_{\text{Pallas}}}$ sends a curve point to its x -coordinate, $\text{Extract}(x, y) := x$, with $\text{Extract}(O) := 0$ by convention. It appears wherever a later object must be a *field element* rather than a curve point — the Merkle tree (§6), the nullifier set (§7), and the proof's public inputs (§8) are all field elements. The map is not injective ($\pm(x, y)$ share an x), which is harmless here: the protocol only ever requires the x -coordinate as a binding digest, never to recover the point.

The point encoding $\star$. The starred form $P^* \in \{0, 1\}^{256}$ is the canonical bit-encoding of a point $P \in \mathcal{E}(\mathbb{F}_{p_{\text{Pallas}}})$: its x -coordinate (an element of $\mathbb{F}_{p_{\text{Pallas}}}$, which fits in 255 bits since $p_{\text{Pallas}} < 2^{255}$) packed into a 256-bit string together with one sign bit, placed in the otherwise-unused top bit, recording which of the two square roots $\pm y$ is meant. Unlike Extract, this encoding is injective — it determines P completely — and it is what is fed, as a bit-string, into a hash or commitment.

Integer encodings LE_n . The string $\text{LE}_n(m) \in \{0,1\}^n$ is the n -bit little-endian encoding of an integer $m \in \{0, \dots, 2^n - 1\}$: the bit string $b_0 b_1 \dots b_{n-1}$ with $m = \sum_{i=0}^{n-1} b_i 2^i$ (a field element encodes via its integer representative). Three widths recur: LE_{32} indexes the Sinsemilla generator table (§2.3), LE_{10} tags the Merkle-tree layers (§6), and LE_{64} , LE_{255} encode note values and field elements (§4).

2.2 GroupHash, domain separation, and nothing-up-my-sleeve generators

The hash-to-curve GroupHash. The map $\text{GroupHash}^{\mathbb{G}} : \{0,1\}^* \times \{0,1\}^* \rightarrow \mathbb{G}$ takes a pair (domain separator D , message M) and returns a point of the group $\mathbb{G}$. It is *deterministic* (the same input always yields the same point), *public* (anyone can evaluate it), and *efficiently computable*. It instantiates the IETF hash-to-curve standard: `hash_to_field` expands (D, M) — via `expand_message_xmd` over BLAKE2b-512, with D embedded in the domain-separation tag — into *two* base-field elements; each passes through the simplified SWU map onto a curve isogenous to $\mathbb{G}$; and the sum of the two resulting points is carried through the isogeny to $\mathbb{G}$ (the *Crypto Guide*, §“Hashing to a curve point”, develops this construction in full). Everything that follows uses only that GroupHash is deterministic, public, and behaves like a random choice of point.

That last property is the purpose of the construction. Because every stage is public and deterministic, the public strings D, M pin down the output $\text{GroupHash}^{\mathbb{G}}(D, M)$, so no party can have chosen it to satisfy a hidden relation such as $P_2 = [s]P_1$ for a secret s ; such points are called *nothing-up-my-sleeve* generators. For the security arguments we *model* GroupHash as a random oracle into $\mathbb{G}$ (the *Crypto Guide*, §“The random oracle model”): its outputs are treated as independent uniform points, so the family it produces has no efficiently-findable non-trivial discrete-log relation $\sum_i [a_i]P_i = O$ (discrete logarithms: the *Crypto Guide*, §“The discrete logarithm problem”) — precisely the assumption the Sinsemilla binding and collision-resistance reductions invoke (§12). As with all random-oracle arguments this is a strong, standard, but unprovable modelling heuristic about the concrete hash, not a theorem.

Domain separators. A *domain separator* $D \in \{0,1\}^*$ is a fixed ASCII byte string naming the use site, fed to GroupHash so that logically distinct uses draw independent, unrelated generators (a collision found in one context cannot be transported to another). The strings are part of the protocol specification; the ones relevant to this volume are

<code>z.cash : Orchard</code>	fixed generators $G^{\text{SpendAuth}}, G^{\text{nf}}$ (§3, §7),
<code>z.cash : Orchard-gd</code>	the diversified base g_d (§3),
<code>z.cash : Orchard-NoteCommit</code>	note commitments (§4),
<code>z.cash : Orchard-CommitIvk</code>	the ivk commitment (§3),
<code>z.cash : Orchard-MerkleCRH</code>	the Merkle-tree node hash (§6),
<code>z.cash : Orchard-cv</code>	the value-commitment bases (§7),
<code>z.cash : SinsemillaQ, z.cash : SinsemillaS</code>	the Sinsemilla Q point and S table (§2.3).

2.3 Sinsemilla: hash, commitment, and short forms

Two objects the journey will build — the note commitment (§4) and the Merkle-tree node hash (§6) — require the same primitive: a commitment/hash that the spender must recompute *inside the zero-knowledge circuit*, the arithmetised form (the *Halo 2 Guide*, §“PLONKish arithmetisation”) of the spend statement (§8). In-circuit cost is therefore the binding constraint, and it is what Sinsemilla (Bowe and Hopwood) is designed to minimise. Its collision resistance (the *Crypto Guide*, §“Security notions: preimage, second-preimage, and collision resistance”) reduces to discrete-log hardness on a prime-order curve, so it introduces no assumption beyond the DL hardness that Orchard’s algebraic core already requires. And it builds directly on the provisions above:

its generators are GroupHash outputs (§2.2), and its short form returns an Extracted coordinate (§2.1).

Construction 2.1 (Sinsemilla). Fix a prime-order group $\mathbb{G}$ (Orchard: $\mathcal{E}(\mathbb{F}_{p_{\text{Pallas}}})$), a chunk width $k = 10$, and a maximum chunk count $c = 253$. From the domain separator D , derive the $2^k + 1$ Sinsemilla generators:

$$\begin{aligned} Q(D) &:= \text{GroupHash}^{\mathbb{G}}(\text{z.cash} : \text{SinsemillaQ}, D) \in \mathbb{G}, \\ P[m] &:= \text{GroupHash}^{\mathbb{G}}(\text{z.cash} : \text{SinsemillaS}, \text{LE}_{32}(m)) \in \mathbb{G}, \quad 0 \leq m < 2^k. \end{aligned}$$

The point $Q(D)$ is the domain-dependent starting accumulator; the 2^k points $P[0], \dots, P[2^k - 1]$ form a fixed table indexed by a k -bit chunk value, shared across all Sinsemilla domains. For a message M of at most $kc = 2530$ bits, split into k -bit chunks $m_1, \dots, m_\ell$ with $\ell \leq c$ (so each $m_i \in \{0, \dots, 2^k - 1\}$ indexes the table),

$$\text{Sinsemilla}_D(M) := \text{Acc}_\ell, \quad \text{Acc}_0 := Q(D), \quad \text{Acc}_{i+1} := (\text{Acc}_i \dot{+} P[m_{i+1}]) \dot{+} \text{Acc}_i,$$

with $\dot{+}$ *incomplete* short-Weierstrass addition (the *Math Guide*, §“Elliptic curves”). The output $\text{Acc}_\ell \in \mathbb{G}$ is a curve point. The deployed hash zero-pads M on the right to a multiple of k bits before chunking (protocol specification §5.4.1.9); every use in this volume fixes the input bit-length, and the padding is why Theorem 2.2 below is restricted to fixed-length inputs.

Theorem 2.2 (Sinsemilla collision resistance). *For fixed-length inputs, Sinsemilla_D is collision-resistant assuming DL on $\mathbb{G}$ and modelling the generator derivation as a random oracle.*

The proof is deferred: Theorem 2.2 is proved in §12, by reducing a collision to a non-trivial discrete-log relation among the random-oracle-derived generators — the assumption named in §2.2. Every later security property of the volume follows the same pattern: stated where its object is constructed, proved once in §12.

Definition 2.3 (Sinsemilla commitment and short forms). The *Sinsemilla commitment* adds a hiding term to the hash:

$$\text{SinsemillaCommit}_r(D, M) := \text{Sinsemilla}_{D \parallel -M}(M) + [r]H_D \in \mathbb{G},$$

where $r \in \mathbb{F}_{p_{\text{Vesta}}}$ is the commitment randomness (trapdoor; the Pasta cycle identifies $\mathbb{F}_{p_{\text{Vesta}}}$ with the scalar field of Pallas), the hash runs in the suffixed domain $D \parallel -M$, and $H_D \in \mathbb{G}$ is a fixed blinding generator for the domain D , derived as the GroupHash output

$$H_D := \text{GroupHash}^{\mathbb{G}}(D \parallel -r, \varepsilon)$$

— the suffix $-r$ appended to the domain string, with the empty message ε (a nothing-up-my-sleeve point, §2.2). The *short commitment* returns only its x -coordinate, a base-field element rather than a curve point:

$$\text{ShortCommit}_r(D, M) := \text{Extract}(\text{SinsemillaCommit}_r(D, M)) \in \mathbb{F}_{p_{\text{Pallas}}}.$$

The unkeyed *short hash* omits the blinding term altogether:

$$\text{ShortHash}_D(M) := \text{Extract}(\text{Sinsemilla}_D(M)) \in \mathbb{F}_{p_{\text{Pallas}}},$$

the unblinded analogue of ShortCommit — no randomness argument, collision-resistant but not hiding. Collision resistance of the two short forms needs one case beyond Theorem 2.2: the theorem binds curve points, while Extract identifies a point with its negation (§2.1), so an x -coordinate collision may also be a pair of opposite points. Both cases are discharged together in §12.

Why is the commitment binding and hiding? Because H_D is a GroupHash output in its own sub-domain $(D \parallel -r)$, no party knows a discrete-log relation between it and the generators $Q(D \parallel -M), P[m]$ that the hash uses; this independence is exactly what keeps the commitment binding — a party knowing such a relation could trade the blinding term against the message and open one commitment to two distinct messages. Hiding needs no assumption at all: for uniform r the term $[r]H_D$ is uniform on $\mathbb{G}$ and masks the hash completely. This is the same division of roles the independent base H supports in a Pedersen commitment (the *Crypto Guide*, §“The Pedersen commitment”).

For independent uniform r , the commitment primitive is therefore *perfectly hiding* and *computationally binding* (the *Crypto Guide*, §“Hiding and binding”) — the two properties §4 requires of NoteCommit. Indeed the note commitment $\text{NoteCommit}_{\text{rcm}}(\cdot)$ is exactly $\text{SinsemillaCommit}_{\text{rcm}}$ under the domain $\text{z.cash} : \text{Orchard-NoteCommit}$. The short forms appear wherever the consumer requires a field element instead of a point: ShortHash in the Merkle-tree node hash (§6), ShortCommit in the incoming viewing key ivk (§3). The latter use enters through a named instance that fixes the domain and the message encoding:

$$\text{Commit}_{\text{r}_{\text{ivk}}}^{\text{ivk}}(x, y) := \text{ShortCommit}_{\text{r}_{\text{ivk}}}(\text{z.cash} : \text{Orchard-CommitIvk}, \text{LE}_{255}(x) \parallel \text{LE}_{255}(y)),$$

taking two field elements and concatenating their 255-bit little-endian encodings into the 510-bit message. Its output lies in $\{1, \dots, p_{\text{Pallas}} - 1\}$: the zero output cannot occur for a valid key, so ivk is a usable scalar.

2.4 Why Sinsemilla: in-circuit cost

Each round of Construction 2.1 consumes a whole k -bit chunk via one table lookup of $P[m]$ and two incomplete additions, so a 510-bit message costs 51 rounds rather than 510 bit-operations. In a PLONKish circuit with lookups (the *Halo 2 Guide*, §“PLONKish arithmetisation” and §“The lookup argument”) this is dramatically cheaper than the bit-by-bit Pedersen hash (the *Crypto Guide*, §“Commitment schemes”) of the previous shielded generation, Sapling — which is the reason the protocol changed hash between generations. The provisions are packed; the journey can now set out.

3 The recipient: keys and addresses

The journey begins with someone able to receive. Before a single zatoshi moves, the recipient must hold the material that later stages will lean on, so we build the Orchard machinery bottom-up, beginning with the keys: everything later — addresses, note commitments, nullifiers — derives from them. A practical requirement shapes the key hierarchy: different parties should be able to perform different operations. The owner can spend. An auditor should be able to *see* incoming and outgoing payments without being able to spend. A point-of-sale terminal should be able to *detect* incoming payments without seeing outgoing ones. Each of these is a distinct capability, and the key tree makes each capability a separate derived key, so the owner can hand out a lower-level key without surrendering the powers above it.

3.1 From sk to the spend-side secrets

Definition 3.1 (Spending key). The root secret of the Orchard key hierarchy is a 32-byte *spending key* sk . Key generation repeatedly samples a candidate uniformly from $\{0, 1\}^{256}$ and rejects it if it derives $ask = 0$, or if either the external or internal $\text{Commit}^{\text{ivk}}$ output is 0 or $\perp$. In a deployed wallet the candidate typically derives from a seed phrase via ZIP-32 hierarchical-deterministic derivation. Every other key is a deterministic function of an accepted sk .

Deriving sub-secrets from one root requires a pseudorandom function. A *pseudorandom function* (PRF) is a keyed family $\{f_K\}_K$ whose outputs are, for a secret uniform key K , computationally indistinguishable from those of a truly random function (the *Crypto Guide*, §“Pseudorandom functions and permutations”).

Definition 3.2 (Expansion function and reduction maps). The *expansion function* $\text{PRFexpand} : \{0, 1\}^{256} \times \{0, 1\}^* \rightarrow \{0, 1\}^{512}$ is

$$\text{PRFexpand}(sk, t) := \text{BLAKE2b-512}(\text{``Zcash_ExpandSeed''}, sk \parallel t),$$

unkeyed BLAKE2b-512 — a conventional, non-arithmetisation-friendly hash with a 512-bit (64-byte) digest (the *Crypto Guide*, §“PRFs from hash functions: length extension and keyed BLAKE2”) — with the fixed 16-byte personalisation string ```Zcash_ExpandSeed''` and input $sk \parallel t$. The protocol obtains its PRF not from BLAKE2b’s native keyed mode but as *personalised, unkeyed* BLAKE2b over the concatenation $sk \parallel t$: it is required to behave as a PRF of t for a secret uniform sk . The secret sk acts as the PRF key, and the leading bytes of t are a domain-separation tag identifying which sub-secret the call produces; we write the key as a subscript, $\text{PRFexpand}_{sk}(t)$, when convenient.

The 512-bit output is reduced into the required field or scalar set by one of two *reduction maps*, which read it as an integer and reduce modulo the relevant prime (p_{Pallas} and p_{Vesta} the Pallas base- and scalar-field orders of §2):

$$\begin{aligned} \text{ToScalar}(x) &:= \text{LEOS2IP}_{512}(x) \bmod p_{\text{Vesta}} \in \mathbb{F}_{p_{\text{Vesta}}}, \\ \text{ToBase}(x) &:= \text{LEOS2IP}_{512}(x) \bmod p_{\text{Pallas}} \in \mathbb{F}_{p_{\text{Pallas}}}, \end{aligned}$$

where LEOS2IP_{512} is the little-endian octet-string-to-integer map sending a 64-byte string to the integer $\sum_{i=0}^{63} x_i 256^i \in \{0, \dots, 2^{512} - 1\}$.

The width is deliberate. Because the value being reduced is 512 bits wide while each modulus is just over 2^{254} (of bit length 255), the modular reduction is statistically close to uniform on the target set: the modular bias is $O(2^{254}/2^{512}) = O(2^{-258})$, negligible. Under the PRF assumption, the distinct tag bytes 0x06, 0x07, 0x08 below computationally domain-separate the three derived outputs.

Construction 3.3 (Spend-side secrets and the spend validating key). From sk derive the three spend-side secrets

$$\begin{aligned} ask &:= \text{ToScalar}(\text{PRFexpand}_{sk}(0x06)) \in \mathbb{F}_{p_{\text{Vesta}}} && \text{(the spend authorising key),} \\ nk &:= \text{ToBase}(\text{PRFexpand}_{sk}(0x07)) \in \mathbb{F}_{p_{\text{Pallas}}} && \text{(the nullifier key),} \\ r_{\text{ivk}} &:= \text{ToScalar}(\text{PRFexpand}_{sk}(0x08)) \in \mathbb{F}_{p_{\text{Vesta}}} && \text{(the CommitIvk randomness),} \end{aligned}$$

so that ask and r_{ivk} are scalars in $\mathbb{F}_{p_{\text{Vesta}}}$ (the scalar field of Pallas, i.e. the integers mod the Pallas group order p_{Vesta}), while nk is a base-field element in $\mathbb{F}_{p_{\text{Pallas}}}$. The two targets differ only

because the protocol uses nk as a field element — it keys the PRF at the heart of the derivation of the *nullifier*, the base-field element whose publication marks a note spent, constructed in §7 — whereas ask and r_{ivk} are scalars multiplying group elements.

Let $G^{\text{SpendAuth}} := \text{GroupHash}^{\text{Pallas}}(z.\text{cash} : \text{Orchard}, G) \in \mathcal{E}(\mathbb{F}_{p_{\text{Pallas}}})$, a fixed, publicly-known generator of the Pallas group (a nothing-up-my-sleeve point, §2). The *spend validating key* is

$$ak := [ask] G^{\text{SpendAuth}} \in \mathcal{E}(\mathbb{F}_{p_{\text{Pallas}}}).$$

Key generation canonicalises its sign: if $[ask] G^{\text{SpendAuth}}$ has odd y -coordinate, ask is replaced by $-ask$, so that ak always has even y and the x -coordinate $\text{Extract}(ak)$ (§2) alone determines the point.

3.2 Viewing keys

Construction 3.4 (Viewing keys). The *full viewing key* bundles the three quantities needed to recognise and decrypt one’s own notes:

$$fvk := (ak, nk, r_{ivk}) \in \mathcal{E}(\mathbb{F}_{p_{\text{Pallas}}}) \times \mathbb{F}_{p_{\text{Pallas}}} \times \mathbb{F}_{p_{\text{Vesta}}}.$$

From fvk derive

$$\begin{aligned} ivk &:= \text{Commit}_{r_{ivk}}^{ivk}(\text{Extract}(ak), nk) \in \{1, \dots, p_{\text{Pallas}} - 1\} \subset \mathbb{F}_{p_{\text{Pallas}}}, \\ K &:= \text{LE}_{256}(r_{ivk}) \in \{0, 1\}^{256}, \\ R &:= \text{PRFexpand}(K, 0x82 \parallel \text{I2LEOSP}_{256}(ak) \parallel \text{I2LEOSP}_{256}(nk)) \in \{0, 1\}^{512}, \\ (dk, ovk) &:= (R_{[0..256]}, R_{[256..512]}), \end{aligned}$$

the incoming-viewing scalar ivk , the *diversifier key* dk , and the *outgoing viewing key* ovk . In protocol terminology the full *incoming viewing key* is the pair (dk, ivk) : ivk alone performs trial decryption, while dk is additionally needed to derive diversified addresses. Here Commit^{ivk} is the named `ShortCommit` instance of §2; LE_{256} is the 256-bit little-endian bit encoding of §2, giving K the type `PRFexpand` requires of its key; $\text{Extract}(ak)$ is the x -coordinate of ak , which after the sign canonicalisation of Construction 3.3 determines the point; and I2LEOSP_{256} denotes the 32-byte little-endian encoding of a field element or point.

Precision on the dk/ovk derivation, because the formula is easy to misread: a *single* 512-bit `PRFexpand` output R , keyed by r_{ivk} over the input $0x82 \parallel ak \parallel nk$ (not over an empty message), is split into its two 256-bit halves — dk the first, ovk the second. The diversifier d of the next subsection is *not* part of this split; it derives afterwards from dk .

3.3 Diversified addresses

Construction 3.5 (Diversified addresses). For any index $d_{\text{plain}} \in \{0, 1\}^{88}$, define

$$\begin{aligned} d &:= \text{FF1-AES256}_{dk}(d_{\text{plain}}) \in \{0, 1\}^{88} && \text{(the diversifier),} \\ g_d &:= \text{GroupHash}^{\text{Pallas}}(z.\text{cash} : \text{Orchard-gd}, d) \in \mathcal{E}(\mathbb{F}_{p_{\text{Pallas}}}), \\ pk_d &:= [ivk] g_d \in \mathcal{E}(\mathbb{F}_{p_{\text{Pallas}}}), \end{aligned}$$

where FF1-AES256_{dk} is the NIST FF1 format-preserving encryption mode built on AES-256 (the *Crypto Guide*, §“Format-preserving encryption and FF1”), keyed by dk : a keyed *permuta-*

tion of the 88-bit index space, so each d_{plain} maps to a distinct 88-bit diversifier and different d_{plain} give unlinkable-looking addresses, all derived from the one key. The *diversified address* is the pair $\text{addr} := (d, \text{pk}_d) \in \{0, 1\}^{88} \times \mathcal{E}(\mathbb{F}_{p_{\text{Pallas}}})$.

Choosing the index. The receiving wallet chooses the diversifier index freely: *any* of the 2^{88} values yields a valid, spendable address for the same ivk ; no derivation fixes it and no index is forbidden — any 88-bit string is a valid diversifier index. The wallet allocates indices on demand, conventionally sequentially from $d_{\text{plain}} = 0$ (index 0 is the *default diversifier*), assigning a fresh index per payee or per invoice to keep addresses mutually unlinkable. A randomly drawn index would yield an equally valid address, but the specification states that diversifier indices SHOULD NOT be chosen at random: ZIP-32 specifies their usage in hierarchical-deterministic wallets, whose seed-only recovery of issued addresses depends on deterministic allocation.

Every index works for a reason specific to Orchard: the hash-to-curve $\text{GroupHash}^{\text{Pallas}}$ is total (§2) — it returns a point for every input — so every index yields a well-defined g_d ; in the negligible-probability event that the two summed SWU outputs (§2) cancel to the identity O , the specification substitutes the fallback $\text{GroupHash}^{\text{Pallas}}(\text{z.cash} : \text{Orchard-gd}, "")$, the same domain separator with the empty message (protocol specification §5.4.1.6). This is a deliberate improvement over Sapling, the previous shielded generation, where the diversifier hash failed on roughly half of all indices and the wallet had to increment d_{plain} until it found a valid one; in Orchard no such rejection loop exists.

Rationale for routing the index through AES. The step from d_{plain} to d is a keyed *permutation* (format-preserving encryption) for three reasons that no simpler map satisfies together. (i) *Unlinkability*: wallets allocate indices in order $0, 1, 2, \dots$, and consecutive diversifiers would reveal a shared owner and an issue count; encryption under dk scatters them pseudorandomly over $\{0, 1\}^{88}$. (ii) *Bijection*: a permutation never collides two indices onto one d and is invertible — with dk the wallet recovers the index behind any of its addresses, storing no per-address secret; a plain hash is neither collision-free nor invertible. (iii) *Format preservation*: the address format fixes the diversifier at exactly 88 bits, and FF1 maps $\{0, 1\}^{88} \rightarrow \{0, 1\}^{88}$ bijectively, which a truncated hash cannot. The derivation runs *wallet-side*, never inside the proof circuit, so a conventional fast cipher (AES-256 under NIST FF1) is appropriate; arithmetisation-friendliness, the concern of §2, is irrelevant here.

3.4 The capability ladder

Figure 2 draws the hierarchy as it deserves to be read: as a deliberate capability ladder, each rung conferring strictly less power than the one above.

- *Spend* — sk / ask . Whoever holds ask can authorise spends; it sits at the top and nothing below it can recover it.
- *Full viewing* — fvk (hence (dk, ivk) and ovk). Detects incoming notes and their spends, and recovers outgoing notes encrypted with its ovk . Custom or discarded outgoing viewing keys limit that recovery (§5). It cannot spend: in particular, recovering ask from ak requires solving DL.
- *Detect incoming* — the scalar ivk alone. Sufficient to recognise notes addressed to the holder and decrypt their values and memos, but not derive their nullifiers or authorise spends — a sensitive key that can be delegated to an always-online wallet server.
- *Receive* — a diversified address (d, pk_d) . A public handle anyone can pay; one incoming viewing key (dk, ivk) generates 2^{88} mutually unlinkable addresses — one per diversifier index (Construction 3.5), practically inexhaustible — all spendable by the same key.

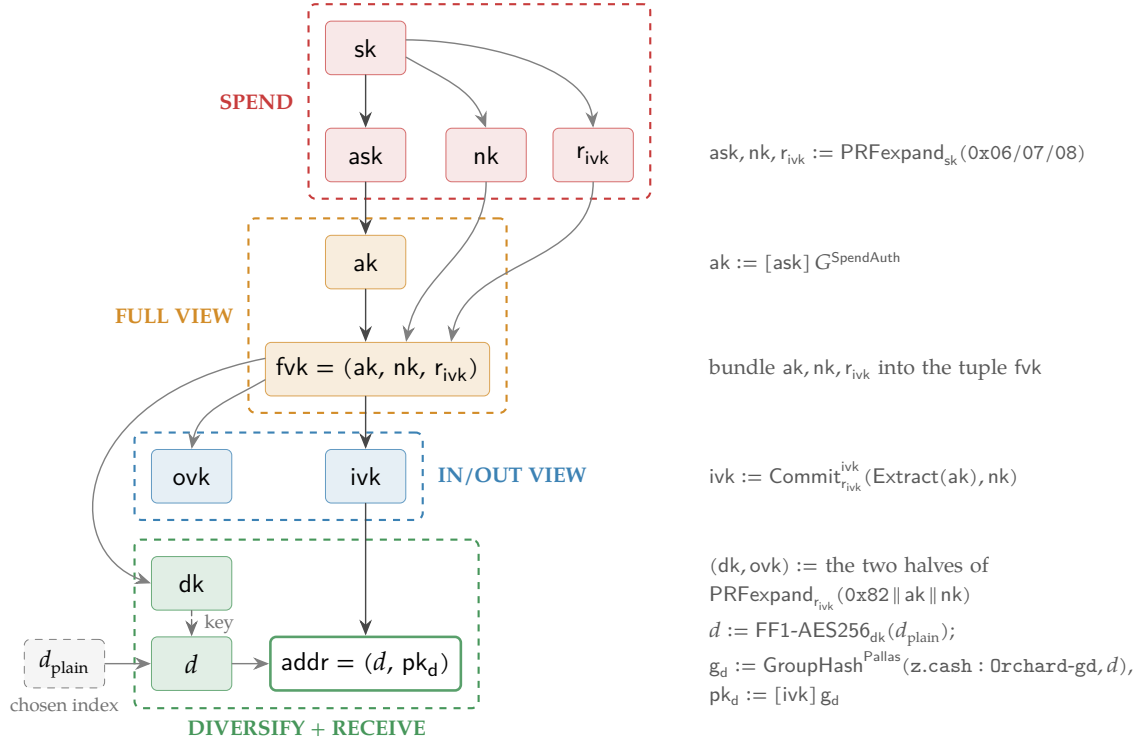


Figure 2: The Orchard key hierarchy as a capability ladder, read top to bottom. Each arrow is a derivation; the right-hand gutter shows the operation that produces the child from its parent(s). Not every edge is one-way: the bundling edges (ak, nk, r_{ivk} into the tuple fvk ; d into the address) invert by projection, and the FF1 edge is a keyed permutation the dk holder inverts — the tiers are separated by the one-way steps $sk \rightarrow (ask, nk, r_{ivk})$, $ask \rightarrow ak$, $fvk \rightarrow (ivk, dk, ovk)$, and $ivk \rightarrow pk_d$. The pair (dk, ivk) is the incoming viewing key. The dk and ovk keys are the two halves of one $\text{PRFexpand}_{r_{ivk}}$ output (tag $0x82$, input $ak \parallel nk$); they play different roles and land in different tiers — ovk is the *outgoing* viewing key, grouped with the incoming-viewing scalar ivk as the IN/OUT viewing tier, whereas dk is the *diversifier* key, which sits in the address layer because it generates diversified addresses. Concretely d is the format-preserving encryption FF1-AES256 of a chosen index d_{plain} under the key dk (the dashed edge marks dk entering as the cipher key), and $pk_d = [ivk] g_d$, so the address binds both keys. *The shaded nodes are wallet-held rather than on-chain values; the exception within them is ak , a public validation key ordinarily kept inside the full viewing key. The chosen index d_{plain} is not secret, while the diversified address (d, pk_d) is the value given to payers (shading marks tier membership, not secrecy) — in particular the viewing keys fvk , (dk, ivk) , ovk are secret: “viewing” names what they let the holder do, not that the protocol publishes them. Each dashed tier holds a strictly weaker capability than the one above, so a holder may receive a lower-tier key without gaining the powers above it. Drawn to match Constructions 3.3–3.5.*

Of these, only the diversified address is *public*. The viewing keys fvk , (dk, ivk) , ovk are *secret* key material — “viewing” names what they let the holder *do* (see transactions), not that one may publish them. Disclosing a viewing key confers the corresponding visibility on the recipient — incoming notes and their spends for fvk , incoming plaintexts for ivk , and outputs encrypted with the disclosed ovk for outgoing recovery — so the owner shares one only deliberately (with an auditor, say) and otherwise guards it like any other secret. The ladder is useful precisely because these capabilities are *separable*: one can delegate “scan incoming” (the scalar ivk) without delegating full viewing (fvk), and neither confers the ability to spend.

Two questions of motivation that the bare formulas do not answer. *Why is ivk a commitment*, $\text{Commit}_{\text{ivk}}^{\text{ivk}}(\text{Extract}(\text{ak}), \text{nk})$, *rather than a plain hash*? Because ivk must both bind *and* hide. It binds ak (spend authority) and nk (nullifier derivation) to one identity, and the Action statement proves the linkage in-circuit by recomputing the commitment from the witnessed ak , nk , r_{ivk} (the CommitIvk check, §8) — but binding alone a collision-resistant hash would supply equally, and in-circuit recomputation works for any hash: the membership check of §8 recomputes the unblinded Sinsemilla Merkle-node hash $\text{MerkleCRH}^{\text{Orchard}}$ (§6), and Sapling even proved its BLAKE2s-based CRH^{ivk} in-circuit. What the trapdoor r_{ivk} adds is *hiding*: for independent uniform r_{ivk} , the blinded ShortCommit is perfectly hiding (§2). In the deployed hierarchy, however, r_{ivk} also keys the PRFexpand call that derives dk and ovk ; when those outputs are exposed, hiding relies on a joint assumption about that derivation and Pallas scalar multiplication, not on perfect hiding alone. Under that assumption, ivk and public $\text{pk}_d = [\text{ivk}] \text{g}_d$ hide ak and nk . This keeps the IN/OUT tier below FULL VIEW: an ivk holder cannot extract nk or ak . *Why is nk separate from ask*? So that the owner can delegate the capability to compute nullifiers — which a viewing key needs to determine which of the holder’s notes are already spent (§7) — *without* conferring the ability to spend: the viewing key holds nk but never ask.

3.5 One ladder, both pools

The hierarchy just built carries one further property, worth stating at the source rather than discovering later: it is scoped to the *protocol*, not to a value pool. Orchard spending-key and viewing-key material grants authority to spend or view notes in *both* of the pools this volume follows — the legacy Orchard pool and the Ironwood pool (§1) — and a diversified address (a *receiver*, in the vocabulary of the ZIPs), with its corresponding ivk , is likewise scoped to the Orchard protocol, not to a pool (ZIP-229; ZIP-326). The entire ladder of this section therefore survives the pool split unchanged: an existing wallet’s keys and addresses work in Ironwood exactly as they stand.

4 Minting: a note is born

The recipient of §3 now holds an address; a sender wishes to pay it. This stage answers the first requirement of §1, transfer of value (R1): value must change hands on a public chain that discloses neither payer, payee, nor amount. The vehicle is the *note*, and the first design problem is representational — the note must stay invisible on-chain yet remain provable by its owner. The protocol’s answer is never to place the note on the chain at all: what the chain receives is only a *hiding commitment* to it.

4.1 The note and its commitment

Definition 4.1 (Orchard note). An *Orchard note* — the private object representing an amount of value — is a tuple

$$n := (\text{addr}, v, \rho, \psi, \text{rcm}),$$

where $\text{addr} = (d, \text{pk}_d)$ is the recipient’s diversified address (Construction 3.5); $v \in \{0, \dots, 2^{64} - 1\}$ is the value in zatoshi; $\rho \in \mathbb{F}_{p_{\text{Pallas}}}$ is a per-note uniqueness seed, fixed at note creation (§7 constructs its provenance); $\psi \in \mathbb{F}_{p_{\text{Pallas}}}$ is randomness the sender fixes through the note seed rseed , from which it derives deterministically (§4.2); and $\text{rcm} \in \mathbb{F}_{p_{\text{Vesta}}}$ is the commitment trapdoor.

Construction 4.2 (Note commitment). The protocol never publishes the note itself. What it records on-chain is the *note commitment*

$$\text{cm} := \text{NoteCommit}_{\text{rcm}}(\text{g}_d^* \parallel \text{pk}_d^* \parallel \text{LE}_{64}(v) \parallel \text{LE}_{255}(\rho) \parallel \text{LE}_{255}(\psi)) \in \mathcal{E}(\mathbb{F}_{p_{\text{Pallas}}}),$$

where $\text{NoteCommit}_{\text{rcm}}$ is the Sinsemilla commitment $\text{SinsemillaCommit}_{\text{rcm}}$ under the domain $\text{z.cash} : \text{Orchard-NoteCommit}$ (Definition 2.3). The chain stores the extracted coordinate

$$\text{cmx} := \text{Extract}(\text{cm}) \in \mathbb{F}_{p_{\text{Pallas}}},$$

the x -coordinate of the commitment point (Extract is constructed in §2.1), because the Merkle tree in which the commitment will come to rest (§6) consumes a field element, not a curve point.

The primitive is perfectly hiding for an independent uniform trapdoor. In an actual note, rcm is derived from rseed by PRFexpand , so hiding is computational (with negligible reduction bias) under the PRF assumption: the chain learns no feasible information about addr , v , ρ , or ψ from cm . The commitment is also *computationally binding* — the sender cannot later claim the commitment was to a different note. These are the two properties of a commitment scheme from the *Crypto Guide*, § “Hiding and binding”, and §2.3 established both for the Sinsemilla commitment. The scheme is Sinsemilla for the reason given there: it is inexpensive to recompute *inside the circuit*, which the spender must one day do (§2.4).

The input encoding is worth a second look, because every byte of it recurs later in the section. The three address-related quantities are exactly the key material of §3: the diversifier $d \in \{0, 1\}^{88}$, the diversified base $\text{g}_d = \text{GroupHash}^{\text{Pallas}}(\text{z.cash} : \text{Orchard-gd}, d) \in \mathcal{E}(\mathbb{F}_{p_{\text{Pallas}}})$ — the per-address generator — and the *transmission key* $\text{pk}_d = [\text{ivk}] \text{g}_d$ identifying the recipient (Construction 3.5); $\text{addr} = (d, \text{pk}_d)$. Under the star encoding of §2.1, $\text{g}_d^*, \text{pk}_d^* \in \{0, 1\}^{256}$. The NoteCommit input is the bit-string concatenation ($\parallel$) of these two point encodings with the little-endian integer encodings $\text{LE}_{64}(v) \in \{0, 1\}^{64}$ and $\text{LE}_{255}(\rho), \text{LE}_{255}(\psi) \in \{0, 1\}^{255}$ (§2.1).

Why these five fields? The address and the value are the note’s *content*: who is paid, and how much. The other three serve the machinery. The trapdoor rcm is what makes the commitment hiding (§2.3), while ρ and ψ exist solely to render the note’s eventual *nullifier* — the token whose publication will one day mark this note spent — unique and unpredictable, a role that becomes apparent only when the note is spent (§7). A note carries randomness whose purpose is downstream.

4.2 The note seed: Ironwood derivations (lead byte 0x03)

Definition 4.1 left two debts: the origins of ψ and rcm . The sender does not draw them independently; it draws a single uniform note seed $\text{rseed} \in \{0, 1\}^{256}$ (the key type PRFexpand requires, Definition 3.2) and derives all per-note randomness from it — alongside ψ and rcm , an *ephemeral secret key* esk , the sender’s key half of the note encryption constructed in §5. The derivation is versioned: a note travels to its recipient as a *note plaintext* — the byte serialisation, also constructed in §5, that the sender encrypts to the address — and the plaintext opens with a *lead byte* naming its format. Which derivation applies is the lead byte’s decision.

An Ironwood note is the same tuple $n = (\text{addr}, v, \rho, \psi, \text{rcm})$ of Definition 4.1, committed by the same NoteCommit of Construction 4.2 (§4.1); ZIP-2005 (Proposed; activation bound to the NU6.3 network upgrade) carries the pool distinction on the note plaintext, through its lead byte. What changes is that byte — and, through it, the derivation of rcm .

The format is specified and deployed. ZIP-2005 §4.7.3 gives the derivation, landed in protocol.tex at zips commit 69610984, and the released orchard 0.15.0 — the version librustzcash, the deployed Zcash wallet library, pins — implements it: `enum NoteVersion (V2/V3)` carries the lead byte, `Note::from_parts` takes it as a fifth argument, and `Note::rcm()` dispatches on it (`src/note.rs`). The bef8a27 development tree predates the release: its `Note::from_parts` (`src/note.rs:182`) still takes the four components (recipient, value, ρ , rseed) with no version tag, and its `Note::rcm()` is the legacy single-tag trapdoor of Remark 4.5.

Definition 4.3 (Ironwood note derivations). For a lead-byte-0x03 note with note seed rseed and uniqueness seed ρ (Definition 4.1), the sender derives, in this order,

$$\begin{aligned} \text{esk} &:= \text{ToScalar}(\text{PRFexpand}_{\text{rseed}}(0x04 \parallel \text{LE}_{256}(\rho))) \in \mathbb{F}_{p_{\text{Vesta}}}, \\ \psi &:= \text{ToBase}(\text{PRFexpand}_{\text{rseed}}(0x09 \parallel \text{LE}_{256}(\rho))) \in \mathbb{F}_{p_{\text{Pallas}}}, \\ \text{rcm} &:= \text{ToScalar}(\text{PRFexpand}_{\text{rseed}}(\text{pre}_{\text{rcm}})) \in \mathbb{F}_{p_{\text{Vesta}}}, \end{aligned}$$

where

$$\text{pre}_{\text{rcm}} := 0x0B \parallel g_d^* \parallel pk_d^* \parallel \text{LE}_{64}(v) \parallel \text{LE}_{256}(\rho) \parallel \text{LE}_{256}(\psi),$$

each field in its canonical byte encoding — the points star-encoded (32 bytes each), v as 8 and ρ, ψ as 32 little-endian bytes — so pre_{rcm} is a 137-byte string ($1 + 32 + 32 + 8 + 32 + 32$). In the rare event $\text{esk} = 0$ the sender redraws a fresh rseed . (ZIP-2005 §4.7.3 change; protocol.tex at zips commit 69610984, where the hash is named $H_{\text{rseed}}^{\text{rcm}, \text{Orchard}}$.)

Two observations about Definition 4.3. First, the order is forced: rcm now comes last because its preimage includes ψ (ZIP-2005). Second, esk and ψ are byte-identical to their legacy lead-byte-0x02 counterparts — only rcm changes (Remark 4.5) — and the same g_d^*, pk_d^* encodings feed both pre_{rcm} and the NoteCommit input of Construction 4.2: the trapdoor now commits, through PRFexpand, to the same material the commitment does. The purpose of this detour through BLAKE2b is *post-quantum binding*: every note field now traverses PRFexpand on its way into the trapdoor, so an adversary constrained to treat it as a random oracle cannot vary any field of a committed note without changing rcm — the property that makes every Ironwood-pool output note, and no Sapling- or Orchard-pool output note, a *recoverable* note, whose security analysis, recovery statement, and specified-but-undeployed qsk/qk key branch belong to §14.5 (ZIP-2005).

4.3 Lead bytes across pools and the legacy derivation

Which lead bytes a note plaintext may carry depends on the value pool and the block height — the lead byte is a consensus-constrained format version, ordinarily encrypted inside the note plaintext, not a free choice.

Definition 4.4 (Allowed lead bytes, pool-parameterised). Let pool be the chain value pool of a note received at block height h — SaplingPool, the value pool of the preceding Sapling shielded generation, or OrchardPool or IronwoodPool, the formal names for the Orchard and Ironwood pools of §1 — and let h_{Canopy} be the activation height of the Canopy network upgrade, which deployed ZIP-212's rseed -derived note-plaintext format and its 0x02 lead byte; the constant ZIP212GracePeriod is the fixed length, set by ZIP-212, of the window after h_{Canopy} in which

both formats remain valid. Then

$$\text{allowedLeadBytes}^{\text{pool}}(h) := \begin{cases} \{0x01\} & h < h_{\text{Canopy}}, \\ \{0x01, 0x02\} & h_{\text{Canopy}} \leq h < h_{\text{Canopy}} + \text{ZIP212GracePeriod}, \\ \{0x02\} & \text{afterwards, pool} \neq \text{IronwoodPool}, \\ \{0x03\} & \text{afterwards, pool} = \text{IronwoodPool}. \end{cases}$$

For an Ironwood-pool note plaintext the only allowed lead byte is 0x03 (ZIP-2005 §3.2.1 change; protocol.tex at zips commit 69610984).

Coinbase outputs — the outputs of the transaction that mints each block’s subsidy — obey the same discipline: an Ironwood coinbase output must carry lead byte 0x03 (ZIP-258), the Ironwood analogue of the Sapling/Orchard rule that coinbase outputs use the post-ZIP-212 0x02 format (ZIP-213).

Remark 4.5 (The legacy derivation). Lead-byte-0x02 notes — all Orchard-pool notes, before and after NU6.3 — derive

$$\text{rcm} := \text{ToScalar}(\text{PRFexpand}_{\text{rseed}}(0x05 \parallel \text{LE}_{256}(\rho))),$$

with esk and ψ exactly as in Definition 4.3. The specification packages both derivations under one dispatcher, $\text{Derive}_{\text{rcm}, \text{rseed}}^{\text{Orchard}}(\text{leadByte}, \dots)$, selecting $0x05 \parallel \text{LE}_{256}(\rho)$ for 0x02 and the hashed pre_{rcm} for 0x03 (ZIP-2005 §4.7.3 change). The dev-tree $\text{Note}::\text{rcm}()$ at bef8a27 implements only this legacy branch (src/note.rs:132–136), with $\text{PrfExpand}::\text{ORCHARD_RCM} = 0x05$ (zcash_spec 0.2.1, src/prf_expand.rs:88), no version dispatch, and no 0x03/hashed- pre_{rcm} branch; the released orchard 0.15.0 that librustzcash pins adds the NoteVersion dispatch — V2 to rcm_v2 , V3 to rcm_v3 over the 0x0B-separated pre_{rcm} (src/note.rs) — deploying the derivation of Definition 4.3.

5 Delivery: the note reaches its owner

The mint succeeded too well. The on-chain cmx (§4) conceals the note entirely, leaving the question of how the recipient learns of the note with which they were paid. The sender must deliver the note to them, privately, over the same public chain. Orchard achieves this with *in-band secret distribution*: every Action carries, beside its commitment, a ciphertext recoverable by the recipient’s viewing key; the sender also knows the plaintext, and outgoing recovery can disclose it to an auditor. Coinbase recovery is deliberately public (§4.3). This section constructs the two ciphertexts the sender attaches, the trial decryption by which the recipient finds and verifies the note, and the synchronisation procedure — a wallet restoring from a seed — that the same machinery supports.

5.1 Sender: one-shot Diffie–Hellman and the AEAD

Alice holds Bob’s address (d, pk_d) , with

$$\text{pk}_d = [\text{ivk}] g_d, \quad g_d = \text{GroupHash}^{\text{Pallas}}(\text{z.cash} : \text{Orchard-gd}, d)$$

(Construction 3.5, §3). She forms the *note plaintext*

$$\text{np} := (0x03, d, v, \text{rseed}, \text{memo}),$$

where 0x03 is the Ironwood note-plaintext lead byte, the 32-byte note seed rseed — drawn uniformly at random, fresh for each output note — is the master randomness from which the note’s rcm and ψ were derived at minting (Definition 4.3, §4) and from which the ephemeral secret

esk below derives as well, and memo is a fixed 512-byte field of sender-chosen data. Each of these derivations is one PRFexpand call keyed by rseed and distinguished by a leading domain-separator byte (the one derivation display, Definition 4.3); esk and ψ take as input $\rho := \text{nf}_{\text{old}}$, encoded as 32 little-endian bytes — which binds the note’s secret randomness to its Action, nf_{old} being the base-field nullifier of the note spent alongside (constructed in §7) — and rcm, derived last, takes every note field.

Construction 5.1 (Note encryption). Alice runs a one-shot Diffie–Hellman key agreement to the diversified base (the *Crypto Guide*, §“The Diffie–Hellman protocol” and §“Static and ephemeral keys”), yielding a *fresh* key:

$$\text{epk} := [\text{esk}] g_d, \quad s := [\text{esk}] \text{pk}_d, \quad K_{\text{enc}} := \text{KDF}(s, \text{epk}),$$

with epk the *ephemeral public key*. She then seals the plaintext under an AEAD — authenticated encryption with associated data, instantiated by ChaCha20-Poly1305 with the fixed all-zero 96-bit nonce and empty associated data (the *Crypto Guide*, §“The ChaCha20-Poly1305 AEAD”); we write $\text{Enc}_K/\text{Dec}_K$ for its encryption and decryption under the key K :

$$C_{\text{enc}} := \text{Enc}_{K_{\text{enc}}}(\text{np}).$$

The key-derivation function KDF (the *Crypto Guide*, §“Key-derivation functions”) is BLAKE2b-256 with personalisation Zcash_OrchardKDF over $s^* \parallel \text{epk}^*$, where $*$ is the point encoding of §2.

Why exactly the intended recipient? Because $\text{pk}_d = [\text{ivk}] g_d$, Bob can reach the same shared secret from his side as

$$[\text{ivk}] \text{epk} = [\text{ivk}][\text{esk}] g_d = [\text{esk}] \text{pk}_d = s,$$

so, beside the sender (who knows esk), a party that can recover the Diffie–Hellman secret — in particular the holder of ivk — can recover K_{enc} from the published pair $(\text{epk}, C_{\text{enc}})$.

5.2 The outgoing ciphertext

One ciphertext serves the recipient; a second can serve the sender’s own books. When an outgoing viewing key is supplied, the pair $(\text{pk}_d, \text{esk})$ is sealed under the *outgoing cipher key*,

$$\text{ock} := \text{BLAKE2b-256}(\text{Zcash_Orchardock}, \text{ovk} \parallel \text{cv}^{\text{net}} \parallel \text{cmx} \parallel \text{epk}), \quad C_{\text{out}} := \text{Enc}_{\text{ock}}(\text{pk}_d, \text{esk}),$$

points in $*$ -encoding, where cv^{net} is the Action’s net value commitment — a Pallas-point commitment to the Action’s value change, constructed in §7 — and ovk is the outgoing viewing key of Construction 3.4. The ciphertext C_{out} carries $(\text{pk}_d, \text{esk})$ so that the *sender* — or an auditor holding ovk — can re-derive $s = [\text{esk}] \text{pk}_d$ and read the note as well. Because ock binds the Action’s own cv^{net} , cmx, and epk, a C_{out} cannot be transplanted onto another Action without detection under the hash and AEAD assumptions. The protocol also permits $\text{ovk} = \perp$; in that case the sender chooses a random ock and random outgoing plaintext, deliberately making C_{out} unrecoverable by an outgoing viewing key.

The Action publishes

$$(\text{epk}, C_{\text{enc}}, C_{\text{out}}) \quad \text{alongside} \quad \text{cv}^{\text{net}}, \text{nf}, \text{rk}, \text{cmx},$$

with nf the Action’s base-field nullifier and rk its re-randomised validating key — both, like cv^{net} , public fields of the Action constructed in §7.

5.3 Recipient: trial decryption and re-derivation

Nothing on chain marks which Action pays Bob, so his wallet *trial-decrypts every* Orchard-protocol output: for each it recomputes

$$s := [\text{ivk}] \text{epk}, \quad K_{\text{enc}} := \text{KDF}(s, \text{epk}), \quad \text{np} := \text{Dec}_{K_{\text{enc}}}(C_{\text{enc}}).$$

The AEAD tag rejects an incorrect key except with negligible probability. A successful open is a candidate detection, not yet acceptance of a valid note: the commitment and ephemeral-key checks below must also pass.

After a successful open Bob re-derives the full note from (d, v, rseed) — taking $\rho := \text{nf}_{\text{old}}$ from the same Action — and performs the two checks that close the obvious forgeries: (i) that the sender formed the ephemeral key honestly, $[\text{esk}] g_d = \text{epk}$ for the esk re-derived from rseed (the ZIP-212 check, defeating a mauled epk); and (ii) that the note is indeed the one the Action committed on chain,

$$\text{Extract}(\text{NoteCommit}_{\text{rcm}}(\dots)) = \text{cmx}.$$

Only when both hold does he accept. Under commitment binding and AEAD integrity, a sender cannot feasibly make Bob accept a note inconsistent with the commitment.

The re-derivation fixes an order. Because the lead-byte-0x03 trapdoor rcm depends on g_d , pk_d , and ψ (Definition 4.3), the specification reorders both the ivk and the fvk decryption procedures: recover g_d and pk_d first, then ψ , then rcm (ZIP-2005, §§ 4.20.2–4.20.3 changes). The routing is deployed: the released orchard 0.15.0 routes the two lead bytes to two note-encryption domains, OrchardDomain (0x02) and IronwoodDomain (0x03) (src/note_encryption.rs), which zcash_client_backend scanning consumes (src/scanning.rs); the orchard bef8a27 development tree still carries a single 0x02-only OrchardDomain. And because keys are scoped to the protocol rather than to a pool (§3.5), the same ivk trial-decrypts both pools — one key, two domains (ZIP-326).

5.4 Synchronisation, including from cold storage

Trial decryption scales from one payment to the whole chain: it is the operation a wallet performs when it restores from a seed. Restoring, the wallet walks the chain forward from the account’s “birthday” height and trial-decrypts every Action. A *light wallet* — one that delegates block storage to a server it need not trust with keys — does so against *compact blocks*, blocks stripped to the fields trial decryption needs. Each compact output carries only the first 52 bytes of C_{enc} — those covering the *compact note plaintext*: the lead byte (1 byte; 0x03 for Ironwood outputs, the only allowed value in that pool — 0x02 appears only when scanning legacy Orchard-pool outputs), d (11 bytes), v (8 bytes), and rseed (32 bytes) — together with epk, cmx, and the Action’s nullifier nf, which the light wallet needs twice: it supplies $\rho := \text{nf}_{\text{old}}$ for the cmx recomputation that detects the note, and it is the source of the chain’s nullifier set (§7) against which spentness is checked below (ZIP-307; zcash_client_backend proto/compact_formats.proto).

The wallet records each note that opens *with its leaf position* — the index at which its cmx was appended, derived arithmetically from the tree size at the block boundary and the commitments’ order — and that position is precisely what the commitment tree consumes: the wallet marks the position in its shard tree and can then build the note’s authentication path on demand, by root-only assembly, without replaying every commitment (§6 constructs the tree, the shard store, and this assembly). To determine which recovered notes remain spendable it additionally requires the nullifier key nk (§3): it computes each note’s nullifier (§7) and discards those whose nullifier already appears in the chain’s nullifier set. The end state is a set of unspent notes, each with a value, a position, an authentication path, and a nullifier — exactly what the Action prover of §8 consumes.

The light-client surface follows the pool split. Compact blocks gain an ironwoodActions list (the same CompactOrchardAction message) and an ironwoodCommitmentTreeSize, and

the synchronisation just described runs per pool, against that pool’s own commitment tree (`zcash_client_backend/proto/compact_formats.proto`; §6). The note has reached its owner, who can detect it, verify it, and place it in the tree; there it now rests.

6 Resting: the note in the commitment tree

Minted (§4) and delivered (§5), the note now rests, and the journey reaches its second requirement (R2, §1): when the note is eventually spent, its owner must prove it *valid* — that someone actually created it — without revealing which note it is, for otherwise Alice could spend a note she invented. On a transparent chain this is trivial: the output being spent is present in the UTXO set (the set of unspent transparent outputs) that every node maintains — but pointing at the output would deanonymise the payment. The shielded solution is this section’s object. The network maintains one append-only Merkle tree — a binary hash tree, each parent a hash of its two children, so the single root digests every leaf — whose leaves are every note commitment ever published, in creation order; Alice proves *membership*, by a Merkle authentication path from her leaf `cmx` to a root the network trusts. Inside the circuit the path is a private witness (the membership check the Action statement imposes, §8), so the verifier learns that her note is one of the leaves but not which one: the proof’s candidate set is the tree at the chosen anchor. Publicly known dummy or spent leaves and other transaction metadata can reduce the effective anonymity set; tree size alone does not measure it.

6.1 The tree, its hash, and its anchors

Definition 6.1 (Orchard note commitment tree). The *note commitment tree* is an append-only Merkle tree of fixed depth 32 — room for 2^{32} notes — whose leaves are the commitments `cmx` (§4) in creation order; positions not yet used hold the *uncommitted* sentinel value 2. Number the layers from the root, layer 0, down to the leaves, layer 32. The parent at layer ℓ of a left child l and right child r at layer $\ell + 1$ is the layer-tagged Sinsemilla short hash

$$\text{MerkleCRH}_\ell^{\text{Orchard}}(l, r) := \text{ShortHash}_{\text{z.cash:Orchard-MerkleCRH}}(\text{LE}_{10}(31 - \ell) \parallel \text{LE}_{255}(l) \parallel \text{LE}_{255}(r)) \in \mathbb{F}_{p_{\text{Pallas}}},$$

the unkeyed short form of Definition 2.3 over the $10 + 255 + 255 = 520$ -bit message (52 ten-bit Sinsemilla chunks), under the same little-endian input convention as `Commitivk` and `NoteCommit`. The ten-bit tag is the fold *height* counted from the leaves, $31 - \ell$: 0 for the leaf-level combine, 31 for the combine that yields the root (protocol specification §5.4.1.3, $\text{MerkleCRH}^{\text{Orchard}}$). Leaves, internal nodes, and the root all live in $\mathbb{F}_{p_{\text{Pallas}}}$. Appending a block’s note commitments yields a new root: the root of the tree as it stands after block h is the *anchor* of block h , a single field element, so every main-chain block defines exactly one anchor. A spend proves membership against the anchor of any main-chain block, and all Actions of one bundle prove against a single shared anchor (§8).

Every node of the tree is a single base-field element by necessity, not convention. The spender recomputes the hash layer by layer *inside* the Action circuit to prove membership (the membership check, §8), and that circuit is arithmetic over $\mathbb{F}_{p_{\text{Pallas}}}$ (§2): every wire carries one such field element. A curve-point node would carry two coordinates and an on-curve constraint at each of the 32 layers; reducing each node to its x -coordinate halves the path data the circuit threads and removes that bookkeeping. This is precisely why `MerkleCRH` ends in `Extract`: Sinsemilla yields a point, and `Extract` returns it to the field, so that each layer’s output is directly the next layer’s input and the one hash composes all the way to the root. Dropping the y -coordinate at every

node is harmless for the reason §2 gave for Extract itself: the tree uses each node only as a binding digest, never to recover a child point, and a collision in the x -coordinate alone yields either a collision in the underlying Sinsemilla hash or a pair of *opposite* points (Extract identifies P with $-P$, §2.1) — both non-trivial discrete-log relations, the second by one extra case; §12 discharges the two together alongside Theorem 2.2. The layer tag, finally, is replay protection: because the tag fixes the fold height, a hash computed at one height cannot be replayed at another.

Nor is the sentinel arbitrary. At $x = 2$ the Pallas curve equation $y^2 = x^3 + 5$ (the *Math Guide*, §“Elliptic curves”) gives $y^2 = 2^3 + 5 = 13$, and 13 is a non-square modulo both Pasta primes (verified in the *Math Guide*, §“Number theory for cryptography”), so 2 is not the x -coordinate of any Pallas point: it can never equal a valid cmx, and an empty position can never be silently confused with a genuine commitment.

6.2 The authentication path from public data

Construction 6.2 (Authentication path). Alice’s commitment occupies a fixed *position* pos — its index in creation order — which her wallet records the moment trial decryption succeeds (§5). Fix an anchor, that of any block at or after the one that mined her note, and let n be the number of leaves in that block’s tree. Write $\text{pos} = b_{31} \cdots b_1 b_0$ in binary, and count levels $i = 0, \dots, 31$ from the leaf upward (the level- i fold applies the layer- $(31 - i)$ hash, whose tag is therefore i , Definition 6.1). The level- i ancestor of her leaf is a left child when $b_i = 0$ and a right child when $b_i = 1$, and the path’s level- i entry is that ancestor’s *sibling* in the tree of size n — one node per level, 32 in all. The membership check (§8) reverses the construction: starting from cmx it folds in one path entry per level — placing the running node on the side b_i dictates, left when $b_i = 0$, right when $b_i = 1$, and the sibling opposite — and asserts that the final value equals the public anchor; the bits b_i and the siblings stay inside the witness.

Each path entry is one of two kinds, and the difference organises everything a wallet does. When $b_i = 1$ the sibling subtree lies to the *left* of her leaf: it spans commitments older than hers, so every position in it is filled — a complete subtree — and, because the tree appends only on the right, its root is final, identical at every anchor. When $b_i = 0$ the sibling subtree lies to her *right* and its value depends on the anchor: its root hashes the commitments it holds at size n , unfilled positions taking the sentinel 2; a sibling subtree still entirely empty at size n contributes a per-level *constant*, the sentinel folded up through the layer-tagged hash. The path is therefore a deterministic function of pos and the first n public leaves, and of nothing else; the only secret is which leaf is hers. Figure 3 draws the computation at depth 3.

6.3 The wallet’s witness: shardtree, shards, and the cap

In practice the wallet neither rebuilds the path from the leaves at each spend nor retains the whole tree. As it scans the chain (§5) it appends every published cmx — its own and everyone else’s — but keeps only what a future path needs: the nodes from each of its own notes up to the root, the complete-subtree roots flanking those paths, and checkpoints at represented block boundaries; blocks that append no commitment need not create a distinct tree checkpoint. The ordinary recent set is bounded (currently 100), while explicitly retained durable anchors — currently selected every 288 blocks after activation — are exempt from that pruning. This is the design of the *shardtree* crate that `zcash_client_backend` drives. The wallet marks its note’s position the moment trial decryption succeeds; a *checkpoint* stores a tree state (empty or the position of the last leaf) together with the set of marks removed since the preceding checkpoint. The anchor itself is recomputed from the retained nodes on demand rather than stored in the checkpoint. To produce the path for a spend against a chosen block, the wallet descends its retained tree to the marked leaf and, on the way back to the root, reads off each sibling subtree’s root: a complete subtree contributes its stored root, a subtree lying wholly beyond the leaf count

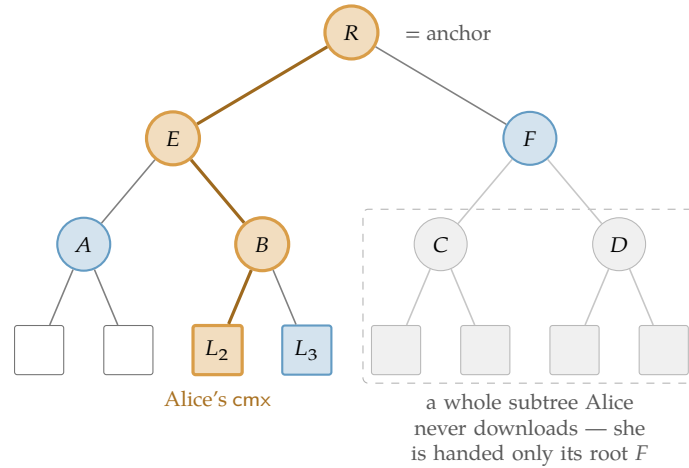


Figure 3: Computation of a Merkle authentication path (drawn at depth 3; Orchard’s tree has depth 32). The leaves are the public note commitments *cmx*, in order; Alice’s note sits at leaf L_2 (highlighted). To prove membership she computes the authentication path of L_2 — the siblings L_3 , A , F (shaded) — then recomputes the bold route from L_2 up through B and E to the root R , the anchor, folding in the sibling hash at each level. Efficiency is the only reason not to hash the whole tree: a far subtree (dashed box) enters her path solely through its single root, here F , so she never downloads its leaves (faded) — the `GetSubtreeRoots` shortcut (§6.3) that lets even a freshly restored wallet witness a years-old note.

n contributes the per-level empty root, and the single subtree straddling position n is rehashed from its retained nodes. The leaf count enters only as this cutoff — every position $\geq n$ counts as empty — so one retained tree reproduces the path against any block it has checkpointed.

Nothing in Construction 6.2 requires Alice to have been online: its inputs are public, so a freshly restored wallet — a seed phrase and nothing else — rebuilds the same state by replaying the chain’s note commitments. Nor need it touch all 2^{32} positions. Cut the tree at half its depth. Below the cut lie the *shards*, the height-16 subtrees — 2^{16} leaves apiece, up to 2^{16} of them — and above it a height-16 *cap* whose own leaves are the shard roots. (The backend pins the shard height to half the tree depth, $32/2 = 16$.) A leaf’s 32 siblings divide along the same cut: the low 16, levels 0 through 15, lie inside its own shard, and the high 16, levels 16 through 31, are cap nodes, each the root of a subtree spanning whole shards. The two halves reach Alice by entirely different routes, and that asymmetry is the economy of the scheme.

Her own shard she must hash from leaves: its 16 low siblings are the within-shard subtrees of the path descent — complete to her left, straddling or empty to her right — so she scans that shard’s commitments block by block, her own and everyone else’s, and folds them; this is the one subtree whose 2^{16} leaves she handles individually. Every other shard she takes as a single 32-byte root (the dashed far subtree of Figure 3): the cap’s leaves are exactly those roots, which a light-client server streams one per completed shard (the `GetSubtreeRoots` call of the light-client protocol, §5; `zcash_client_backend, proto/service.proto`); she folds them upward with the same layer-tagged MerkleCRH, one application per cap level — the upper half of the tree’s 32 layers — and reads her high siblings off the result. A shard that does not yet exist (the empty right of the cap) contributes the per-level empty constant rather than a download, and her own shard she holds from leaves whether or not it has completed, so she never needs its root from the server. Her level-16 sibling is thus a neighbouring shard’s root, and each sibling above it a fold of 2, 4, 8, ... roots she already holds.

Example 6.3 (Witnessing the newest shard). Let Alice’s note be the last leaf of the newest shard, shard k . Below the cut, all 16 low siblings are complete left subtrees, tiling the shard’s earlier $2^{16} - 1$ positions, so she ingests essentially the whole shard. Above the cut, the cap’s right side is

empty — no shard succeeds k , so those siblings are the empty constant — while its left side folds the roots of shards $0, \dots, k-1$, each supplied by `GetSubtreeRoots`. Stitching the halves — her leaf folds up to the shard- k root, which folds up the cap to the anchor — gives the full 32-level path.

The 2^{16} throughout bounds the cap's *capacity* — 2^{32} leaves over 2^{16} per shard — not Alice's traffic. At leaf count n , the service exposes one root for each completed shard, $\lfloor n/2^{16} \rfloor$ in all; the current partial shard is reconstructed from its leaves. The whole history above her shard thus arrives as a few kilobytes of roots in place of millions of leaves. And she performs the fold locally, never asking a server for one particular note's path — which would reveal which note is hers; the finished path is the private witness of the membership check (§8), and only the anchor is public.

6.4 Anchor choice, reorgs, and the anonymity set

By the time Alice's transaction reaches a block, the live tree has advanced past her chosen anchor, and this costs nothing: appending leaves creates new roots and alters no old one, so the anchor still names one fixed historical tree, her path still folds to exactly that root, and the membership check asserts exactly that equality — collision resistance of the layer-tagged hash (Theorem 2.2) makes a path to that root for a non-member infeasible. Consensus is equally generous: it accepts the anchor of *any* main-chain block from Orchard activation (the NU5 network upgrade) onward — this for the legacy Orchard-pool tree; the Ironwood tree's anchors run from its own activation (§6.5). In neither pool is there a sliding window of admissible anchors, so proving is fully decoupled from broadcasting: Alice need not race the chain tip.

The one recency caveat runs in the opposite direction. A reorganisation — the replacement of the chain's most recent blocks by a competing branch (the *Consensus Guide*, §"Reorg rails, maturity, and the finality floor") — can strike the anchoring block from the main chain, invalidating the anchor and forcing Alice to rebuild the transaction; deeper anchors make the event rarer, but no depth rules it out. The Draft ZIP 218 recommends a 600-block node rollback window for NU7. This is local recovery policy, not a consensus limit on reorganisations or a bound on admissible anchor age, as the *Consensus Guide* section just cited develops.

Deployed wallets nevertheless anchor close to the tip, by deliberate choice rather than necessity. The anchor's tree is the anonymity set in which the membership check conceals the spent note — the proof reveals only that the note is *some* leaf committed by that anchor — so the most recent admissible anchor, naming the largest such tree, maximises that set, while an idiosyncratically old or deep anchor would instead fingerprint the spender. These are candidate-set observations, not ZIP-315's stated rationale: its fixed depth balances input availability and hiding the input's trusted status. The wallet backend selects the newest shared checkpoint at or below $H-3$ by default, where H is the next-block target height. The trusted confirmation count fixes this bound independently of whether a selected note is trusted; untrusted notes separately require 10 confirmations. Checkpoint gaps can make the anchor older — balancing reorg robustness, set size, and uniformity. ZIP 218 recommends keeping the three-block anchor depth at NU7's 25-second target spacing. The shorter spacing changes the elapsed-time interpretation, not the membership statement or the consensus admissibility of older main-chain anchors. Consensus mandates none of this; it is wallet policy. Nor is an old anchor a soundness hole: membership in an older, smaller tree implies the note was committed even earlier, which is precisely what Alice must establish — preventing a *second* spend of the same note is not the anchor's job but the nullifier's, the base-field element published with every spend and constructed in §7.

6.5 Two pools, two trees

The volume's two pools (§1) are distinguished by *state*, not identity. Each keeps its own, fully independent note commitment tree and nullifier set (ZIP-229), hence its own anchors, and its own chain value pool balance — the consensus-tracked net value the pool holds. The Ironwood tree is built by the identical depth-32 MerkleCRH^{Orchard} construction of Definition 6.1, capacity

2^{32} (ZIP-258); it starts *empty* at the pool’s activation (the NU6.3 network upgrade), and an Ironwood Action proves membership against anchorIronwood, the anchor of the Ironwood tree, admissible from that activation onward. The Orchard tree, meanwhile, keeps growing after the split: every Orchard-pool Action still appends a cmx, because spends under the *cross-address restriction* — the post-NU6.3 consensus rule confining each Orchard-pool Action’s created note to the spent note’s own receiver — are paired with fabricated or *dummy* zero-valued output notes (rule, fabrication, and the dummy treatment are all taken up in §10), and its anchors remain live for spend proofs. The wallet machinery of §6.3 carries over unchanged — the backend pins IRONWOOD_SHARD_HEIGHT to the same 16, in the same file, as its Orchard twin. Where a note rests, and against which tree it may be spent, are henceforth facts about *which pool* — state the rest of the journey must thread through every proof.

7 Spending: the nullifier, the balance, and the signatures

The note has rested long enough; its owner now spends it, and the two remaining requirements of §1 come due at once: R3, that no note can be spent twice, and R4, that no transaction creates value — each to be achieved while disclosing neither which note is spent nor what it is worth. This section constructs the machinery the spend contributes to an Action (the unit of shielded transfer named in §1: one Action spends one old note and creates one new one; §8 assembles it in full): the nullifier that retires the old note, the value commitment and binding signature that prove conservation, the transaction digest that both signatures sign, and the re-randomised signature that proves spend authority. Each security property is stated where its object is constructed and proved once in §12, the volume’s standing division of labour.

7.1 The nullifier, and the loop back to minting

The third requirement: to prevent the owner from spending the same note twice, without a public registry of spent notes — which would leak exactly what is being hidden. Each note yields a single deterministic tag, its *nullifier*, that appears only when its owner spends the note; the network maintains a set of all revealed nullifiers and rejects any repeat. The nullifier must satisfy two properties that pull in opposite directions: it must be *deterministic* (the same note always yields the same tag, making a second spend detectable) yet *unlinkable* (publishing it reveals nothing about which note or address it came from).

Definition 7.1 (Orchard nullifier). For a note n with commitment cm (§4) and the owner’s nullifier key nk (§3),

$$nf := \text{Extract}([F_{nk}(\rho) + \psi]G^{nf} + cm) \in \mathbb{F}_{p_{\text{Pallas}}},$$

where $F_{nk} : \mathbb{F}_{p_{\text{Pallas}}} \rightarrow \mathbb{F}_{p_{\text{Pallas}}}$ is a circuit-efficient PRF — Poseidon over the Pallas base field, keyed by nk (the *Crypto Guide*, §“Arithmetisation-friendly hashing: Poseidon”); $G^{nf} := \text{GroupHash}^{\text{Pallas}}(\text{z.cash} : \text{Orchard}, K) \in \mathcal{E}(\mathbb{F}_{p_{\text{Pallas}}})$ is a fixed nothing-up-my-sleeve Pallas generator (§2.2); the scalar $F_{nk}(\rho) + \psi$ lives in $\mathbb{F}_{p_{\text{Pallas}}}$, a valid Pallas scalar because $p_{\text{Pallas}} < p_{\text{Vesta}}$; and Extract takes the x -coordinate (§2.1). We write G^{nf} for this base to distinguish it from the spend-auth generator $G^{\text{SpendAuth}}$ of §3.

We read the construction against the two requirements. *Determinism*: every ingredient (nk, ρ, ψ, cm) is fixed once the note exists, so nf is a fixed function of the note — spending it twice yields the same nf twice, and the network rejects the second. *Unlinkability*: the term $[F_{nk}(\rho) + \psi]G^{nf}$ masks the commitment with a value that appears random to anyone without nk , so no one can tie the published nf back to cm or to the address — this is where the note’s ρ and ψ , carried since §4, perform their function. Three security properties sharpen this reading:

- *No double-spend* (the *Orchard Book* calls this *balance*, since double-spending is how one would create value): each note has exactly one nullifier, and forging a colliding nullifier for a different note reduces to the discrete-logarithm problem on Pallas.
- *Spend unlinkability*: without nk , the published nf is unlinkable to the note or address, under the Decisional Diffie–Hellman (DDH) assumption on Pallas — that $([a]G, [b]G, [ab]G)$ is computationally indistinguishable from $([a]G, [b]G, [c]G)$ for independent uniform a, b, c (the *Crypto Guide*, §“Diffie–Hellman: computational and decisional”) — *or*, independently, the PRF assumption on F (§3): the design offers either assumption as a route to nullifier unlinkability, with the note-generation and commitment-hiding conditions of the corresponding argument still in force. This is not an unconditional privacy guarantee against disclosed keys or correlated public metadata. The disjunction is the design rationale for the nullifier’s group structure — defence in depth, unlinkability surviving the failure of one assumption (the *Orchard Book*, `design/nullifiers.md`).
- *Forward consistency* (*Faerie resistance*): a freshly created note takes as its uniqueness seed the nullifier of the note spent alongside it, $\rho_{\text{new}} := nf_{\text{old}}$; since the network rejects duplicate nullifiers within each pool, this forces accepted notes’ ρ values to be unique within that pool. Equality across pools is not checked: the guarantee is pool-local, not global. In particular, reusing the same fabricated zero-value dummy input in both pools can repeat its nullifier without violating either pool’s duplicate check. This pool-local uniqueness closes an attack from earlier designs in which forged notes could be made to share a nullifier.

The last item closes the loop opened at minting: the ρ that §4 could only type as a base-field element fixed at note creation now acquires its provenance. Every note is born from a spend, and inherits as ρ the nullifier that spend reveals — unique among accepted nullifiers in its pool by the double-spend rule this section establishes.

Proposition 7.2 (Nullifier uniqueness). *Each note determines exactly one nullifier, and any efficient adversary that produces two distinct notes sharing a nullifier yields a solver for the discrete-logarithm problem on Pallas (modelling the derivation of G^{nf} as a random oracle, §2.2).*

The proposition is proved in §12; here we only record what the construction is asked to deliver. The unlinkability claim above is a design-level guarantee, not a deferred theorem: it enters the volume’s privacy statement as one assumption of the informal end-to-end theorem that closes §12.

Both pools of this volume use this derivation unchanged: an Ironwood note’s nullifier is Definition 7.1 verbatim, and a note’s nullifier is checked only against its *own* pool’s nullifier set — the two pools keep separate, independent nullifier sets (ZIP-229; the split’s state is laid out in §9).

7.2 Conservation without disclosure: value commitments and the binding signature

The fourth requirement: to guarantee that a transaction creates no value, while concealing every amount. The instrument is the additive homomorphism of Pedersen commitments, constructed in the *Crypto Guide*, §“The Pedersen commitment”: commit to each amount, and verify that the commitments *sum* to the publicly declared net flow — the individual values remain hidden, while their balance is publicly verifiable.

Definition 7.3 (Net value commitment). Each Action commits to its own value change with a Pedersen commitment

$$cv^{\text{net}} := [v_{\text{old}} - v_{\text{new}}] V^{\text{Orchard}} + [rcv] R^{\text{Orchard}} \in \mathcal{E}(\mathbb{F}_{p_{\text{Pallas}}}),$$

where v_{old} is the spent value, v_{new} the created value, $\text{rcv} \in \mathbb{F}_{p_{\text{Vesta}}}$ the commitment randomness (a Pallas scalar), and

$$\begin{aligned} V^{\text{Orchard}} &:= \text{GroupHash}^{\text{Pallas}}(\text{z.cash} : \text{Orchard-cv}, "v"), \\ R^{\text{Orchard}} &:= \text{GroupHash}^{\text{Pallas}}(\text{z.cash} : \text{Orchard-cv}, "r") \in \mathcal{E}(\mathbb{F}_{p_{\text{Pallas}}}) \end{aligned}$$

are nothing-up-my-sleeve points (§2.2), hence independent Pallas generators.

The spender draws each rcv uniformly at random and afresh for every Action, at the moment of bundle assembly. In contrast to the note's rcm , ψ , and esk — all derived deterministically from the note seed rseed (§4) — the trapdoor rcv belongs to no note and is never transmitted; its uniformity and secrecy are precisely what make cv^{net} hiding, and what keep the aggregate bsk introduced next known only to the transaction author.

We now apply the homomorphism. Summing over the m Actions of a transaction,

$$\sum_{i=1}^m \text{cv}^{\text{net},i} = \left[\sum_i (v_{\text{old},i} - v_{\text{new},i}) \right] V^{\text{Orchard}} + [\text{bsk}] R^{\text{Orchard}}, \quad \text{bsk} := \sum_i \text{rcv}_i \in \mathbb{F}_{p_{\text{Vesta}}},$$

with bsk the *binding signing key*. The transaction declares its net flow in a signed cleartext field $v_{\text{balance}}^{\text{Orchard}}$, with the convention that a positive value is net value flowing *out* of the pool (paying fees or transparent outputs, say). The network subtracts the declared part:

$$\text{bvk} := \sum_i \text{cv}^{\text{net},i} - [v_{\text{balance}}^{\text{Orchard}}] V^{\text{Orchard}} \in \mathcal{E}(\mathbb{F}_{p_{\text{Pallas}}}).$$

Proposition 7.4 (Balance equals key knowledge). *If and only if the transaction balances — the summed value component equals the declared $v_{\text{balance}}^{\text{Orchard}}$ — does the V^{Orchard} -component cancel, leaving $\text{bvk} = [\text{bsk}] R^{\text{Orchard}}$: a public key whose secret bsk the transaction author knows.*

The idea is already visible in the displays: by the homomorphism the aggregate commitment splits into a V -component carrying the net value difference and an R -component carrying bsk ; subtracting the declared flow leaves a pure R -multiple exactly when the books balance, and only the author — who chose the rcv_i — knows their sum. The full argument, including why an unbalanced author cannot know a discrete logarithm of bvk to base R^{Orchard} , is deferred: Proposition 7.4 is proved in §12.

Proving knowledge of bsk — via a *binding signature* under the key bvk — is therefore precisely proving that the accounts balance, with no amount revealed (the *Crypto Guide*, §“Binding signatures as a proof of knowledge of a discrete logarithm”). The signature scheme is RedPallas , constructed in §7.4, here instantiated on the base R^{Orchard} — the value-commitment randomness base, distinct from the spend-authorisation base $G^{\text{SpendAuth}}$. A single signature over the whole transaction (concretely, over the ZIP-244 *signature digest* of §7.3) additionally binds all its Actions together, so that no one can remove or transplant any of them.

From NU6.3 — the upgrade that opens the Ironwood pool (§1) — the construction runs once *per pool*: a v6 transaction (§9) carries independent $v_{\text{balance}}^{\text{Orchard}}$ and $v_{\text{balance}}^{\text{Ironwood}}$ fields, and each pool's Actions sum to their own bvk , checked by their own binding signature over the same bases $V^{\text{Orchard}}, R^{\text{Orchard}}$ (ZIP-229). The Ironwood component's conservation is thus this subsection verbatim, run over the Ironwood Actions' cv^{net} values.

7.3 What the signatures sign: ZIP-244 digests and their v6 form

The message both Orchard signatures sign is ZIP-244's *signature digest*, not the raw transaction. ZIP-244 first defines the closely related transaction-identifier digest as a tree of personalised

BLAKE2b-256 hashes, one node per transaction component. Its root is

$$\text{txidDigest} := \text{BLAKE2b-256}(\text{ZcashTxHash_} \parallel \text{branchid}, d_{\text{hdr}} \parallel d_{\text{trans}} \parallel d_{\text{Sap}} \parallel d_{\text{Orch}}),$$

where *branchid* is the 4-byte identifier of the active network upgrade and each $d_{\bullet}$ is a personalised digest of one component's data, an absent component hashing the empty string. The signature digest has the same root and shielded children but replaces d_{trans} with a d_{transSig} that commits to the applicable transparent input and sighash flags. For a coinbase transaction or a transaction without transparent inputs, the two transparent children coincide, so the signature digest equals *txidDigest*; otherwise they need not. Orchard spend-authorisation and binding signatures sign the signature digest (ZIP-244).

The Orchard branch commits to the entire bundle,

$$d_{\text{Orch}} := \text{BLAKE2b-256}(\text{ZTxIdOrchardHash}, \\ d_{\text{C}} \parallel d_{\text{M}} \parallel d_{\text{N}} \parallel \text{flagsOrchard} \parallel \text{LE}_{64}(v_{\text{balance}}^{\text{Orchard}}) \parallel \text{anchor}),$$

through three digests that partition every Action's fields: the *compact* d_{C} over *nf*, *cmx*, *epk* and the first 52 bytes $\text{C}_{\text{enc}}[0:52]$ of the note ciphertext (§5); the *memo* d_{M} over the 512-byte $\text{C}_{\text{enc}}[52:564]$; and the *non-compact* d_{N} over cv^{net} , the re-randomised validating key *rk* (constructed in §7.4), the tail $\text{C}_{\text{enc}}[564:]$ and C_{out} . Beside them sit the bundle-shared fields: the one-byte flag field *flagsOrchard* (its bit assignments are laid out in §10; the v6 renaming in §9), the declared balance, and the anchor of §6.

Folding in every Action's *nf*, *cmx*, cv^{net} , *rk* and ciphertexts beside the shared *flagsOrchard*, balance and anchor, these child digests pin the exact bundle: no Action can be transplanted into another transaction nor any field altered without changing the digest (ZIP-244). The spend-authorisation signature of §7.4 under each *rk* and the binding signature under *bvk* both sign the signature digest.

The v6 form. From NU6.3 the tree gains an Ironwood branch (Figure 4):

$$\text{txidDigest} := \text{BLAKE2b-256}(\text{ZcashTxHash_} \parallel \text{branchid}, d_{\text{hdr}} \parallel d_{\text{trans}} \parallel d_{\text{Sap}} \parallel d_{\text{Orch}} \parallel d_{\text{Irw}}),$$

with five new Ironwood personalisations — *ZTxIdIronwd_H_v6* for the branch, *ZTxIdIrnActCH_v6*, *ZTxIdIrnActMH_v6*, *ZTxIdIrnActNH_v6* for its compact, memo, and non-compact digests, and *ZTxAuthIrnwdH_v6* on the authorizing-data side — and four revised v6 personalisations for the Sapling and Orchard branches (*ZTxIdSSpendNH_v6*, *ZTxAuthSapliH_v6*, *ZTxIdOrchardH_v6*, *ZTxAuthOrchaH_v6*); an absent component hashes the empty string under its personalisation (ZIP-229).

This digest tree is deployed. The released crate *orchard* 0.15.0 (the `librustzcash` pin) defines the seven Orchard and Ironwood_v6 personalisations and selects them per pool and transaction version (`bundle/commitments.rs`, `enum BundleCommitmentFormat`), and *librustzcash* at `e30517e4` assembles the v6 digests in production: `digest_orchard` and `digest_ironwood` in `zcash_primitives/transaction/txid.rs` call `Bundle::commitment`, the file itself defines only the two Sapling personalisations *ZTxIdSSpendNH_v6* and *ZTxAuthSapliH_v6* (lines 44 and 54), and `transaction/tests.rs` re-derives the empty-component digests as test vectors. The `bef8a27` dev tree predates this work: there `bundle/commitments.rs` (:7--11) defines only the five non-versioned personalisations (*ZTxIdOrchardHash*, *ZTxIdOrcActCHash*, *ZTxIdOrcActMHash*, *ZTxIdOrcActNHash*, *ZTxAuthOrchaHash*), with no per-pool or per-version dispatch.

One structural change rides along with v6: it moves the Sapling, Orchard, and Ironwood *anchors* from effecting data (under the *txid*) to authorizing data (under the auth digest, committed only when the component spends). A transaction can then be re-anchored — its membership proofs rebased onto a later root — without changing its *txid*, a design ZIP-229 credits to Penumbra.

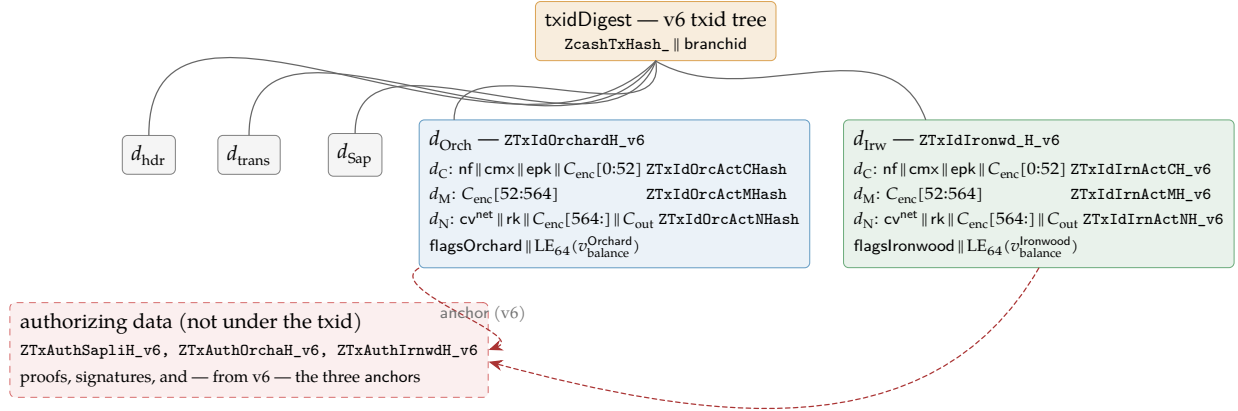


Figure 4: The v6 transaction-identifier digest as a tree of personalised BLAKE2b-256 nodes (ZIP-244; ZIP-229), with the Orchard and Ironwood branches expanded. Each node is one BLAKE2b-256 call whose personalisation (typewriter, right-aligned) separates its domain; the three Action digests d_C, d_M, d_N run over the concatenation of the named fields across all of the branch’s Actions. An absent component hashes the empty string under its personalisation. The Ironwood branch reuses the Orchard partition byte-for-byte under its own personalisations — flagsIronwood being the pool’s own flag byte, the v6 renaming of flagsOrchard — closing the format split with §9. From v6 the anchors leave the txid branches for the authorizing-data digest (dashed), so a transaction can be re-anchored without changing its txid.

7.4 Spend authorisation: RedPallas re-randomisation

One capability remains unproven: that the spender is *authorised* to spend the note — that they hold ask (§3) — established without linking this spend to any other spend by the same key. RedPallas, a re-randomisable Schnorr signature, provides exactly this. It is the Schnorr template of the *Crypto Guide*, §“RedDSA: re-randomisable Schnorr for Orchard”, instantiated on Pallas, with hash H^* — BLAKE2b-512 reduced mod p_{Vesta} .

The novel ingredient is *key re-randomisation*: for every Action, the spender generates a fresh $\alpha \in \mathbb{F}_{p_{Vesta}}$. RedPallas’s GenRandom hashes 80 uniform bytes with H^* . In the ideal-hash model its 512-bit hash output reduces to a nearly uniform scalar, with reduction bias bounded by $O(2^{254}/2^{512}) = O(2^{-258})$. The spender sets

$$\text{rsk} := \text{ask} + \alpha \in \mathbb{F}_{p_{Vesta}}, \quad \text{rk} := \text{ak} + [\alpha] G^{\text{SpendAuth}} = [\text{rsk}] G^{\text{SpendAuth}} \in \mathcal{E}(\mathbb{F}_{p_{Pallas}}).$$

The spender publishes the *randomised* validating key rk , not the real ak , and signs the signature digest with rsk .

Unlinkability is retained alongside authority. Because α is fresh per spend, the key rk is uniformly random and unlinkable across spends (the *Crypto Guide*, §“Key re-randomisation and unlinkability”), yet remains a valid key for the underlying authority. The circuit enforces that rk is a re-randomisation of the witnessed ak (the key re-randomisation check, §8) and that this ak is the key bound into the note’s address (the CommitIvk check, §8), so a valid Action proves the spender knows ask — without ever revealing ak .

Proposition 7.5 (Unforgeability under re-randomisation). *RedPallas is existentially unforgeable under chosen-message attack (EUF-CMA; the Crypto Guide, §“Syntax and security goal”) in the random oracle model under discrete-logarithm hardness on Pallas, and remains so under key*

re-randomisation: signatures under re-randomised keys $rk = ak + [\alpha] G^{\text{SpendAuth}}$, for randomisers α known to the adversary, give no advantage in forging under ak or any of its re-randomisations.

Proposition 7.5 is proved in §12.

The spend’s contributions are now complete. Alongside the delivery fields ($epk, C_{\text{enc}}, C_{\text{out}}$) of §5, an Action publishes the nullifier nf , the net value commitment cv^{net} , the randomised validating key rk with its spend-authorisation signature, and the new note’s cmx (§4); the transaction adds one binding signature per pool. What remains is to assemble these pieces into the Action and its zero-knowledge statement — the work of §8.

8 The Action assembled: walkthrough, verification, and the statement

The pieces are on the table: an address (§3), a note and its commitment (§4), the delivery ciphertexts (§5), the tree (§6), and the spend’s nullifier, value commitment, sighash, and signatures (§7). This section assembles them: one payment end to end, the network’s side of it, and the formal statement carrying both. The unit of shielded transfer is the *Action*: one Action simultaneously spends one old note and creates one new note — a deliberately fixed shape that conceals whether a given Action is “really” a spend, an output, or both.

8.1 One Action, one payment: the walkthrough

Consider the simplest possible payment: Alice pays Bob 5 ZEC ($v = 5 \cdot 10^8$ zatoshi, at 10^8 zatoshi per ZEC) from a note worth exactly 5 ZEC. This is an unpadded, zero-fee one-Action illustration; ordinary wallet construction adds fees and padding. (The names A1–A10 are the checks of Definition 8.1 below.)

1. **Bob hands Alice an address.** Bob derives a diversified address (d, pk_d) from his incoming viewing key (dk, ivk) (Construction 3.5, §3) and gives it to Alice. It reveals that receiver, but is computationally unlinkable to his other diversified addresses without further key material or side information.
2. **Alice builds the Action.** Alice owns an old note n_{old} worth 5 ZEC, whose commitment cmx_{old} is a leaf of the tree. She computes its nullifier nf_{old} (Definition 7.1), which its publication will retire; constructs Bob’s new note $n_{\text{new}} := (\text{addr}_{\text{Bob}}, 5 \cdot 10^8, \rho_{\text{new}}, \psi_{\text{new}}, rcm_{\text{new}})$ with $\rho_{\text{new}} := nf_{\text{old}}$, and its commitment cmx_{new} (Construction 4.2); forms the net value commitment cv^{net} with $v_{\text{old}} - v_{\text{new}} = 5 \cdot 10^8 - 5 \cdot 10^8 = 0$ (Definition 7.3); selects a randomiser α and forms rk (§7.4); and reads off a recent anchor and her note’s Merkle path (Construction 6.2, §6).
3. **Alice proves the Action statement.** She runs the Halo 2 prover on the Action relation, with public inputs ($\text{anchor}, cv^{\text{net}}, nf_{\text{old}}, rk, cmx_{\text{new}}$), together with the bundle flags, and her note, keys, path, and randomness as private witness. Beyond those public inputs, the proof reveals no additional witness information: the old note is in the tree (A3), its nullifier follows correctly (A5), her validating key is linked to its address (A6, A7), the notes are well-formed (A1, A2), and the value commitment is consistent (A4).
4. **Alice assembles and signs the transaction.** She encrypts n_{new} to Bob (§5), attaches a *spend-authorisation signature* (SpendAuthSig) under rk (§7.4), and signs the bundle with the binding signature under bvk (§7.2), which proves balance. Both signatures sign the ZIP-244 signature digest (§7.3), welding the Action into this transaction.
5. **The network verifies.** Consensus checks that it recognises the anchor, that nf_{old} is not already in the nullifier set (no double-spend), that the Halo 2 proof verifies, and that the SpendAuthSig under rk and the binding signature under the recomputed bvk verify (authority; balance). If all pass, it adds nf_{old} to the nullifier set and appends cmx_{new} as a new tree leaf.

6. **Bob receives.** Scanning with his ivk , Bob trial-decrypts the ciphertext (§5), recovers n_{new} , and now owns a 5-ZEC note he can later spend by repeating the entire procedure.

That sequence sketches the payment’s cryptographic lifecycle; the formal Action statement of §8.3 is precisely the list of in-circuit checks step 3 invokes. Equal values are the special case: a more valuable old note would return the difference to Alice as change in a further note, subject from NU6.3 to the routing rule constructed in §10.

After the payment, the chain has gained one nullifier and one commitment. Its cryptographic contents do not reveal sender, recipient, or hidden amount, but the transaction still exposes its action count, bundle structure, timing, fees, transparent components, and any public pool balance. The privacy claim rests on the proof’s zero knowledge and the composition of every piece of the journey; §12 states its scope.

8.2 What a node verifies, without ever observing the note

Nothing in the walkthrough handed the network a note. A validating node need not hold a viewing key or decrypt the note to validate it. For an ordinary honestly encrypted private transfer, the cryptographic checks reveal no private note contents beyond the public fields. This excludes disclosures and shielded coinbase outputs, whose plaintexts are publicly recoverable (§4.3). The proof checks the hidden relation; signatures and chain-state checks supply separate authority, balance, and spentness conditions.

What the node receives is an Orchard-protocol *bundle*, the unit into which the transaction format groups one pool’s data: the per-Action public fields; *one* anchor shared by every Action in the bundle; a *single* aggregate Halo 2 proof covering all of the bundle’s Actions; and one SpendAuthSig per Action (ZIP-225). The declared net flow $v_{\text{balance}}^{\text{Orchard}}$ and the binding signature (§7.2) appear once per bundle, and the flags enableSpends, enableOutputs, and — from NU6.3 — enableCrossAddress (bit 2 of the flags byte flagsOrchard of §7.3, reserved as 0 in v5 transactions before NU6.3) are likewise bundle-level fields (ZIP-225 reserves the bit; ZIP-229 assigns it).

Using the public Action and bundle fields, proof, signatures, and its chain state, the node checks:

- each encoded field element and curve point parses canonically, and the randomised validating key rk is not the identity;
- the bundle’s *Halo 2 proof* of the Action statement (§8.3) — this forces each spent note of nonzero witnessed value to be a genuine tree leaf (a zero-value *dummy* spend, the shape by which an Action pads with no real input, skips the membership check by A3’s gate, by design), the nullifiers and new commitments to follow correctly, and the keys to align;
- the *anchor* is a Merkle root the node has seen on its accepted main chain (§6.4);
- the nullifier nf_{old} is *not already* in the nullifier set — the one check that cannot sit inside a single proof, because it concerns the global history (this prevents a double-spend);
- the *spend-authorisation signature* verifies under rk and the *binding signature* under the recomputed bvk (§§7.2–7.4), settling authority and balance.

If every Action passes, the node performs the two state changes of step 5 — recording each nf_{old} , appending each cmx_{new} — which permit the next spender to prove membership and make a second spend of the same note fail. Thus the SNARK together with the two signatures enforces a payment’s validity, never the inspection of the note: the node confirms that the accounts balance and that the spend is authorised without requiring disclosure of private notes. Knowledge soundness supplies the hidden-relation part of that composition: an accepting proof has an extractable witness (§12), while signatures and state enforce the rest.

8.3 The Action statement, formally

The relation itself remains. The proof the node verifies is a Halo 2 proof (the *Halo 2 Guide*, §“The Halo 2 proof system”); how the relation becomes a circuit is §11’s subject, and what its soundness yields is §12’s. Every check is typed over primitives the *Crypto Guide* constructs — commitments (§“Commitment schemes”), PRFs (§“Pseudorandom functions and permutations”), Schnorr keys (§“Digital signatures”) — as instantiated on Pallas in §§2–7, and closes with the requirement it discharges.

Definition 8.1 (Orchard Action statement, Ironwood circuit (NU6.3)). The Action SNARK proves, with public inputs

$$(\text{anchor}, \text{cv}^{\text{net}}, \text{nf}_{\text{old}}, \text{rk}, \text{cmx}, \text{enableSpend}, \text{enableOutputs}, \text{disableCrossAddress})$$

— eight named inputs; the two Pallas points enter by affine coordinates, so the instance vector has ten elements — and with the two notes’ fields, the keys, the Merkle path, the randomiser α , and the value-commitment trapdoor rcv as private witness, the conjunction below. Witness types additionally constrain ak , g_{dold} , g_{dnew} , pk_{dold} , and pk_{dnew} to nonidentity Pallas points; the implementation enforces these with `NonIdentityPoint`. These typing constraints are part of the relation, not optional checks omitted by A1–A10.

[A1] **Old-note commitment integrity** — an equation between Pallas points, over the Sinsemilla commitment of Construction 4.2:

$$\text{cm}_{\text{old}} = \text{NoteCommit}_{\text{rcm}_{\text{old}}}(\text{g}_{\text{dold}}^* \parallel \text{pk}_{\text{dold}}^* \parallel \text{LE}_{64}(v_{\text{old}}) \parallel \text{LE}_{255}(\rho_{\text{old}}) \parallel \text{LE}_{255}(\psi_{\text{old}}))$$

— or `NoteCommit` returns $\perp$, the specification’s escape hatch for Sinsemilla’s incomplete-addition exceptions (Remark 8.2). The spent object really is a note (R1’s representation).

[A2] **New-note commitment integrity**, with $\rho_{\text{new}} := \text{nf}_{\text{old}}$: over the same commitment and the coordinate extraction of §2.1,

$$\text{Extract}(\text{NoteCommit}_{\text{rcm}_{\text{new}}}(\dots)) \in \{\text{cmx}, \perp\} \quad (\text{with } \text{Extract}(\perp) := \perp).$$

The created note is well-formed (R1’s representation again), and its uniqueness seed chains to the spent note’s nullifier — the Faerie resistance of §7.1.

[A3] **Membership, gated by v_{old}** — a Merkle recomputation over the layer-tagged hash of Definition 6.1: recomputing the root from $\text{Extract}(\text{cm}_{\text{old}})$ up the witness path of length 32 gives root, and $v_{\text{old}} \cdot (\text{root} - \text{anchor}) = 0$ in $\mathbb{F}_{p_{\text{Pallas}}}$. The gate permits a zero-value dummy spend to skip the membership check — how an Action pads with no real input. For nonzero v_{old} , the spent note is a real tree leaf (R2).

[A4] **Net-value commitment integrity** — the Pedersen commitment of Definition 7.3: with witnessed (magnitude, sign),

$$v_{\text{old}} - v_{\text{new}} = \text{magnitude} \cdot \text{sign}, \quad \text{cv}^{\text{net}} = [v_{\text{old}} - v_{\text{new}}] V^{\text{Orchard}} + [\text{rcv}] R^{\text{Orchard}},$$

with $v_{\text{old}}, v_{\text{new}} \in [0, 2^{64})$ and $\text{sign} \in \{-1, +1\}$. The amounts are consistent and in range; the binding signature completes value conservation (R4).

[A5] **Nullifier integrity** — the PRF-plus-group derivation of Definition 7.1:

$$\text{nf}_{\text{old}} = \text{Extract}([F_{\text{nk}}(\rho_{\text{old}}) + \psi_{\text{old}}] G^{\text{nf}} + \text{cm}_{\text{old}}).$$

The published nullifier is indeed this note's; with the chain's nullifier set, this prevents double-spending (R3).

[A6] Spend authority — a Schnorr-key re-randomisation (§7.4): $rk = ak + [\alpha] G^{\text{SpendAuth}}$. The published validating key belongs to the note owner's spend authority.

[A7] Address integrity — the short commitment $\text{Commit}^{\text{ivk}}$ and the scalar multiplication of Construction 3.4: $\text{ivk} = \perp$, or

$$\text{ivk} = \text{Commit}_{r_{\text{ivk}}}^{\text{ivk}}(\text{Extract}(ak), nk) \quad \text{and} \quad pk_{\text{dold}} = [\text{ivk}] g_{\text{dold}}.$$

The keys, the address, and the nullifier key all belong to one key hierarchy; with A6 and the external spend-authorisation signature, this establishes spending authority (R1's possession clause).

[A8] / [A9] Spend / output gating — field products against the bundle flags of §8.2: $v_{\text{old}} \cdot (1 - \text{enableSpend}) = 0$ and $v_{\text{new}} \cdot (1 - \text{enableOutput}) = 0$. The transaction builder sets these bundle-level flags and consensus rules constrain them: coinbase transactions (those minting each block's subsidy, §4.3) must set $\text{enableSpend} = 0$ (ZIP-229). Spending Ironwood notes is otherwise permitted — the $\text{enableSpend} = 0$ requirement is coinbase-scoped, not a pool-wide prohibition (protocol specification §7.1, transaction consensus rules).

[A10] Cross-address gating — when $\text{disableCrossAddress} \neq 0$, the created note pays the spent note's own receiver: $g_{\text{dnew}} = g_{\text{dold}}$ and $pk_{\text{dnew}} = pk_{\text{dold}}$ as full curve points, enforced by four product constraints of the A3 shape, one per affine coordinate. The flag is a verifier-supplied boolean public input — the tenth, at instance index 9; its semantics, its instance and wire encodings, and the consensus requirement that every post-NU6.3 Orchard-pool Action set it are all constructed in §10.

Remark 8.2 (The $\perp$ -weakening). Checks A1, A2, and A7 take the specification's $\perp$ -weakened forms because the circuit cannot exclude Sinsemilla's incomplete-addition exceptional cases: the relation proved is the disjunction, not the plain conjunction. The weakening costs only a negligible soundness term — producing a $\perp$ case is itself a non-trivial discrete-logarithm relation among the GroupHash generators (the closing step of the proof of Theorem 2.2, given in §12), so no efficient prover exhibits one.

Each of A1–A7 is one entry from the four-requirement map of §1: A1–A2 represent notes, A3 proves membership, A5 (with the chain's nullifier set) prevents double-spending, A4 conserves value, and A6–A7 authorise the spender; A8–A10 stand outside the map as consensus-level gating constraints. This is the relation $\mathcal{R}_{\text{Orchard}}$, and every Action is an instance of it. Halo 2 produces one aggregate proof *per bundle*, covering all of the bundle's Actions, while each Action carries its own SpendAuthSig. Figure 5 draws the statement in one place: every public field and every witness component, the checks each feeds, and, per check, the object it constrains and the section that constructed it. How the Action circuit arithmetises this relation is §11's subject; how its knowledge soundness yields the protocol's security properties is §12's.

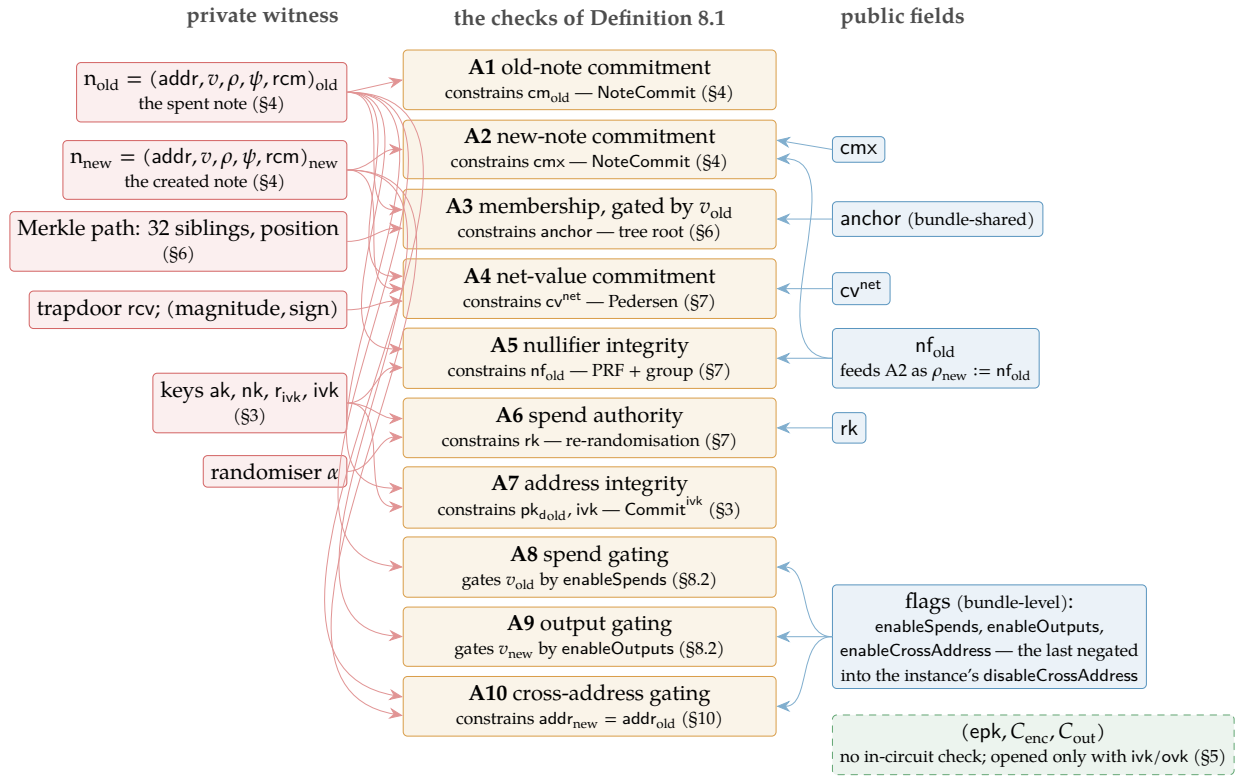


Figure 5: The Action statement as the volume’s hub. Centre: the ten checks of Definition 8.1, each annotated with the object it constrains and the section that constructed it. Left: the private witness; right: the Action’s public fields, with the bundle-level flags below. Arrows run from each input to the checks it feeds; the edge $\rho_{\text{new}} := \text{nf}_{\text{old}}$ is the Faerie chain of §7.1, tying the created note’s uniqueness seed to the spent note’s nullifier. The delivery ciphertexts (dashed) feed no check: the proof never touches them, and only the holder of ivk or ovk can open them (§5).

9 Two pools, one protocol: doors and the v6 transaction

Every stage so far has moved value *inside* a shielded pool: note to note, owner to owner, with the Action of §8 as the unit of transfer. Value also crosses the pool boundary — entering from the transparent ledger, arriving as fresh issuance, or migrating between shielded pools — and each crossing is public in amount, by design: what a pool hides is who holds which note, never how much the pool as a whole contains. This section formalises the three doors through which value enters, the ledger that meters every crossing, and the transaction format that carries a two-pool bundle; it closes with the piece of the split most easily missed — that the two pools share one circuit and one verifying key.

The split itself is already in hand. The NU6.3 network upgrade (ZIP-258, Draft) seals the value pool that has carried Orchard payments since NU5 and starts a second pool, *Ironwood*, running the same cryptography. ZIP-229 (Draft) fixed the vocabulary and §1 adopted it: one *Orchard protocol* — the shared design, its Action circuit as modified by ZIP-2006 and its note encryption as modified by ZIP-2005 — and two value pools of that one protocol, the sealed *Orchard pool* and the current *Ironwood pool*, each with its own note commitment tree, anchor, and chain value pool balance (§6.5). Nothing in the preceding sections is discarded; what changes is which pool new value may enter — this section’s subject — and how an Ironwood note derives its trapdoor, already constructed as Definition 4.3 (§4).

The volume’s deployed-behaviour citations are pinned to a fixed baseline, stated here once: the released crate `orchard 0.15.0` (the version `librustzcash` pins; the Ironwood note-format, note-encryption, and v6-digest work ships in it), the `orchard` git tree at `bef8a27` (a post-0.14.0 development tree predating the release, cited for the circuit and cross-address work it carries), `librustzcash` at commit `e30517e4`, and `protocol.tex` and the ZIPs at `zips commit 69610984`. The later deployment and migration status of ZIPs 258 and 318 is rechecked against `zips commit dd1667ff` (2026-09-01). The NU7 proposals are checked against `zips commit afa086bd` and open pull requests on 2026-09-23; this does not advance the pinned implementation baseline. Forward-looking claims carry one of three epistemic bins: *specified*, *designed but unspecified*, or *open problem*.

9.1 The doors: shielding, coinbase, migration

Value enters a shielded pool through three doors, each public in amount.

The first door is the *shielding* transaction, and its instrument is the sign of the balancing value: by the convention of §7.2, a positive $v_{\text{balance}}^{\text{Orchard}}$ is net value leaving the pool, so a shielding transaction is one whose transparent inputs fund a *negative* balancing value. Its Actions spend *dummy* notes — zero-value placeholders, $v_{\text{old}} = 0$, for which the Action statement gates off the membership check (check A3, §8) — and create real ones, and the binding signature of §7.2 proves the books balance. An observer learns the entering amount — the balancing field is a cleartext transaction field — but not the receiving note or address. The deployed construction path is function `propose_shielding`: it sweeps the given source addresses’ UTXOs (§6), at zero confirmations under the default policy, and wallets auto-shield transparent receipts to *internal* keys — address material the wallet derives for itself and never publishes. From NU6.3 this door opens into the Ironwood pool alone: a shielding transaction funds a negative $v_{\text{balance}}^{\text{Ironwood}}$, the Orchard entry being sealed (§10).

The second door is *shielded coinbase* (ZIP-213). Since the Heartwood network upgrade, new issuance may enter a shielded pool directly — “[c]oinbase transactions MAY contain Sapling outputs” — extended at NU5 to the Orchard pool and re-scoped at NU6.3: from then coinbase may carry “Sapling-pool and/or Ironwood-pool outputs”, and “it is only possible to mine to an Orchard address by sending the funds to the Ironwood pool” (ZIP-213). Two riders keep this auditable and consistent. First, coinbase *maturity* — the rule that a block’s newly minted outputs may not be spent until a fixed depth has passed — is amended to bind transparent outputs only; the amended rule is the one the *Consensus Guide* reads through its theorem in §“Reorg rails, ma-

turity, and the finality floor”. Second, every shielded coinbase output “MUST have valid note commitments” when recovered under the *all-zero* outgoing viewing key *ovk* (§3): coinbase shielding is publicly decryptable by construction, so anyone can verify what issuance entered which pool, while the notes spend exactly like any others afterwards. An Ironwood coinbase output also carries lead byte 0x03, the rule of Definition 4.4 (§4.3).

The third door is *migration*: value moving from one shielded pool into another without touching the transparent ledger. Between this volume’s two pools, migration passes through the one-way turnstile — NU6.3’s rules confining the sealed Orchard pool to outward flow — constructed in §10. Here it is enough that migration, like the other two doors, is public in amount, because every door writes to the ledger we now examine.

9.2 Pool balances: ZIP-209 and its Ironwood analogue

Every entry and exit is metered by ZIP-209’s pool balances. For each shielded pool, the *chain value pool balance* is the negation of the sum of that pool’s balancing fields over the whole chain — for the earliest shielded pool, Sprout, which declared entry and exit in a pair of cleartext fields, it is the sum of `vpub_old` – `vpub_new` — and a block is invalid if it would drive any pool balance negative — shielded, transparent, or the deferred fund (ZIP-2001’s *lockbox*, a chain value pool accruing a portion of each block subsidy for later disbursement) — or total supply past the protocol’s fixed cap `MAX_MONEY` (ZIP-209; ZIP-2001). Entry, transfer, and exit therefore reconcile publicly at pool granularity while individual notes stay hidden.

From NU6.3 the ledger gains one row. The Ironwood pool keeps its own chain value pool balance (§6.5), the negation of the summed $v_{\text{balance}}^{\text{Ironwood}}$ fields of §7.2, under the same non-negativity rule. The doors of §9.1 write to that row — a shielding transaction or a shielded coinbase output raises the Ironwood balance, value leaving through a positive balancing field lowers it — while the sealed Orchard pool’s balance can only fall: the accounting shape of the turnstile, whose rules §10 states.

9.3 The v6 transaction and the Ironwood component

The upgrade has a concrete identity. NU6.3 carries consensus branch identifier 0x37A5165B; ZIP-258 now records both networks’ activation heights and sets minimum network protocol version 170160 on both. The deployed validator fixes the same values, with `librustzcash` hardcoding the same values. ZIP-258 remains Draft, but these deployment parameters are no longer to-be-determined. From activation onward, the ZIP expects Zebra to be the network’s sole full-validator implementation.

A two-pool bundle needs a carrier, and the current carrier is the *v6 format* (ZIP-229): the v5 format’s Orchard component (ZIP-225) plus a separate Ironwood component holding that pool’s bundle (`flagsIronwood`, `valueBalanceIronwood`, `anchorIronwood`, `proofsIronwood`, `vSpendAuthSigIronwood`, `bindingSigIronwood`). Concretely, a v6 transaction (`V6_TX_VERSION` = 6, version group 0xD884B698) is the v5 format — with `enableSpendsOrchard/enableOutputsOrchard` renamed to `enableSpends/enableOutputs`, and bit 2 of `flagsOrchard` now carrying `enableCrossAddress`, the flag whose semantics §10 constructs — plus an Ironwood component that reuses the `OrchardAction` encoding byte-for-byte (820 bytes per Action; ZIP-229), laid out field by field in Table 1. NU7 retains v5 and v6, with v6 required for an Ironwood component. The Orchard-pool sealing rules bind regardless of the permitted transaction version (§10).

Two pointers close the format. At the note level, ZIP-229 states the situation exactly: every Ironwood-pool output note uses the quantum-recoverable note plaintext format — lead byte 0x03 with the hashed trapdoor of Definition 4.3 (§4) — no Orchard-pool output note does, and this is “the only note-level distinction between the two pools”. At the digest level, the v6 transaction-digest tree gains an Ironwood branch mirroring the Orchard one, already constructed alongside

Field	Size (bytes)	Content
nActionsIronwood	compactSize	Action count n
vActionsIronwood	$820 \cdot n$	OrchardAction encoding
flagsIronwood	1	same bit layout as flagsOrchard
valueBalanceIronwood	8	net flow $v_{\text{balance}}^{\text{Ironwood}}$
anchorIronwood	32	a root of the <i>Ironwood</i> tree
sizeProofsIronwood	compactSize	length of proofsIronwood
proofsIronwood	$2720 + 2272 \cdot n$	aggregate Halo 2 proof
vSpendAuthSigIronwood	$64 \cdot n$	one SpendAuthSig per Action
bindingSigIronwood	64	the pool's binding signature

Table 1: The Ironwood component of a v6 transaction (ZIP-229). Sizes marked compactSize use Bitcoin's variable-length integer encoding. The fields after the count are present iff $n > 0$. The proof size formula is the same $2720 + 2272n$ as the v5 Orchard component's proofsOrchard (ZIP-225).

the spend's signatures (Figure 4, §7.3). Everything NU6.3 changes is collected, one row per change with its owning section, in Table 2 (§10).

9.4 NU7: the same protocol under new block rules

The Draft ZIP 259 preserves both the v5 and v6 formats: it adds no Ironwood component field, note format, Action circuit, or verifying key. It proposes consensus branch identifier 0x77190AD9, but leaves the activation heights and minimum network protocol versions unassigned. Because signatures commit to the branch identifier (§7.3), an unchanged format does not make a pre-NU7 transaction valid after activation. Nor does v6's ability to replace anchors and proofs without changing its effecting data remove that branch commitment.

The proposed block target spacing is 25 seconds (ZIP 218). The same proposal limits shielded work per block; it does not shrink either commitment tree or discard its existing notes. The open pull request 1361 proposes including both pools in one shared 330-unit block budget: each Orchard or Ironwood Action consumes one unit, and work in other shielded pools consumes the remaining budget. It does not grant each pool an independent allowance of 330 Actions in the same block. This is *specified in an open proposal*, not an active limit. The complete accounting is constructed in the *Consensus Guide*, §“NU7: 25-second blocks and bounded shielded work”.

The separate ZIP 2003 removes v4 transactions at NU7. Only v4 can carry Sprout transfers, so remaining Sprout funds would become unspendable; their balance would remain issued, not be burned or transferred into Ironwood. This is different from the Orchard seal: Orchard-pool notes remain spendable through v5 or v6 under the existing one-way migration rules. The proposed issuance and reserve accounting belongs to block consensus, not the Action relation, and is developed in the *Consensus Guide*, §“NU7: 25-second blocks and bounded shielded work”.

9.5 One circuit, one verifying key

For all this new state, nothing new is proved. Both pools share the post-NU6.3 Action circuit and its proving and verifying keys; the pools are distinguished by their trees, nullifier sets, balances, and component position — not by separate circuits. The released orchard 0.15.0 encodes the pairing precisely: a BundleVersion pairs a ValuePool (Orchard or Ironwood) with a ProtocolVersion (InsecureV1, V2, or V3); circuit_version() maps V3 to OrchardCircuitVersion::PostNu6_3 for *both* pools; note_version() maps Orchard to V2 and Ironwood to V3 — the lead-byte split of §4 — and permits_cross_address_transfers() is false exactly for the pair (V3, Orchard) (src/lib.rs, src/bundle.rs). The in-circuit gate is supports_cross_address_restriction() on OrchardCircuitVersion, true only for

PostNu6_3 (src/circuit.rs); the bef8a27 development tree’s BundleFormat enum and Ironwood circuit-version name were reworked into these before release.

Remark 9.1 (The three Action-circuit versions). The deployed implementation distinguishes three Action-circuit versions: the insecure NU5 original, the NU6.2 correction (ZIP-257, Final), and the NU6.3 Ironwood circuit — and only the last constrains the cross-address flag. The legacy NU5/NU6 statement is the relation A1–A9 alone (§8), over the nine-element public-input tuple ending at enableOutputs. Consensus selects the verifying key by branch (ZIP-258), so from NU6.3 every Action — in either pool, whatever the transaction version — verifies against the Ironwood key. How the stack came to have three versions — what the original got wrong and what the correction changed — is §13’s story; this remark is the operational list that story cites.

One door remains open. The shielding and coinbase doors lead into the Ironwood pool; leaving the sealed Orchard pool — the migration the first subsection deferred — passes through a turnstile with rules of its own: the seal, the cross-address restriction the verifying-key selection just made binding, and the one-way flow the pool balances record. Constructing that turnstile is the next section’s work (§10).

10 The seal and the turnstile: leaving the Orchard pool

Every door of §9 opens inward, and from NU6.3 all of them open into the Ironwood pool. This section constructs the outward passage that remains: how value held in the sealed Orchard pool leaves it. Consensus arranges the exit as two instruments — a *seal*, three rules that stop value entering the pool or circulating inside it, and a *turnstile*, the one-way, publicly metered gate through which the sealed pool drains. We take them in order: the three rules, the in-circuit restriction that is the deepest of them, what spending under that restriction looks like, the turnstile and the migrating transaction, and the wallet posture built on top — closing with the one-place table of everything NU6.3 changes (Table 2).

10.1 The seal: three rules

From NU6.3 activation, three consensus rules seal the Orchard pool to spend-only, and all three bind every transaction *regardless of its version* (ZIP-258):

1. **No issuance.** A coinbase transaction must carry an *empty* Orchard component: the shielded-coinbase door of §9.1 no longer opens into the Orchard pool.
2. **No circulation.** Every Orchard-pool Action must have cross-address transfers disabled, enableCrossAddress = 0 (§10.2). The rule is enforced by the Action-circuit verifying key, which consensus selects by branch (Remark 9.1, §9.5), so a v5 transaction cannot bypass it — ZIP-229 states this version-independence explicitly.
3. **No entry.** Per transaction, $v_{\text{balance}}^{\text{Orchard}} \geq 0$: net value may flow out of the pool, never in (§10.4).

The deepest of the three is the second, because it lives in the circuit: from NU6.3, value inside the Orchard pool can no longer move between addresses — an Orchard-pool spend can pay only its own receiver. The motivation is the recoverability guarantee whose analysis §14.5 owns: legacy lead-byte-0x02 notes (§4.3) are unrecoverable, so consensus stops their circulation and funnels the pool’s value out through the turnstile (§10.4), while still letting owners spend what they hold.

10.2 The cross-address restriction (A10)

Definition 8.1 (§8) listed check A10 and deferred its semantics to this section. We give the constraint in full, then its two encodings — in the proof’s instance vector and on the wire — and the backward compatibility both preserve.

Definition 10.1 (The A10 constraint in full). The tenth public input of Definition 8.1, `disableCrossAddress`, sits at instance index 9 — after anchor, the two affine coordinates of cv^{net} , nf_{old} , the two coordinates of rk , cmx , `enableSpends`, and `enableOutputs`. Writing $\delta := \text{disableCrossAddress}$, and $x(\cdot), y(\cdot)$ for a Pallas point’s affine coordinates (§2.1), constraint A10 (cross-address gating) is the conjunction

$$\begin{aligned} \delta \cdot (x(g_{\text{dold}}) - x(g_{\text{dnew}})) &= 0, & \delta \cdot (y(g_{\text{dold}}) - y(g_{\text{dnew}})) &= 0, \\ \delta \cdot (x(pk_{\text{dold}}) - x(pk_{\text{dnew}})) &= 0, & \delta \cdot (y(pk_{\text{dold}}) - y(pk_{\text{dnew}})) &= 0 \end{aligned}$$

in $\mathbb{F}_{p_{\text{Pallas}}}$: four coordinate equalities which, whenever $\delta \neq 0$, force $g_{\text{dold}} = g_{\text{dnew}}$ and $pk_{\text{dold}} = pk_{\text{dnew}}$ as full curve points — equality of receivers, not merely of x -coordinates. (Crate orchard at bef8a27, `src/circuit.rs`: instance offsets ANCHOR through DISABLE_CROSS_ADDRESS; rows assigned in `synthesize_cross_address_checks`.)

The constraint shape is deliberately unoriginal: it is the product form check A3 already uses — $v_{\text{old}} \cdot (\text{root} - \text{anchor}) = 0$ — reused on four extra rows of the same custom gate (the rows just cited from Definition 10.1’s implementation). For $\delta = 0$, the added coordinate preserves the commitment to the historical nine-element instance column, which supports compatibility with existing instance data. The changed circuit nevertheless has its own verifying key; A10 does not leave pre-NU6.3 proofs or verifying keys unchanged (crate orchard at bef8a27, `src/circuit.rs`).

On the wire the flag travels in the opposite polarity: bit 2 of the component’s flags byte (§9.3) carries `enableCrossAddress` — the *enabled* sense, negated into the instance’s δ — with the bit assignments `spends 0b001`, `outputs 0b010`, `cross-address 0b100`. Deployed encoders carry both canonical flags values, `0b0000_0011` for an Orchard component and `0b0000_0111` for an Ironwood one.

The polarity is backward-compatible by design. Bit 2 = 0 — the restricted state consensus requires of post-NU6.3 Orchard-pool spends — is exactly what v5 signers treating the bit as reserved-zero already produce, so existing hardware signers (ZIP-229 names the Keystone) keep signing v5 Orchard unshielding transactions — ones whose positive balancing value pays out of the pool (§9.1) — without a firmware update. The Ironwood component carries the same flag and *sets* it: cross-address transfers stay enabled there.

10.3 Spending under the restriction

An Action spends one note and creates one (§8.1); under the restriction the created note must pay the spent note’s own receiver. An owner therefore pays anyone by *exiting the pool*: the real value leaves through a positive $v_{\text{balance}}^{\text{Orchard}}$ (§10.4), and the Action’s output slot is filled with a *fabricated* zero-valued note to the spent note’s own receiver — the fabricated outputs that keep the Orchard tree growing after the split (§6.5).

The fabrication must be thorough. The wallet fills the fabricated output’s ciphertext C_{enc} (§5) with *random bytes*: a real encryption would let any *ivk* holder — including a quantum adversary who recovered *ivk* from a published address, the very adversary §14.2 confronts — decrypt it and link the Action’s nullifier to the address (ZIP-326).

Between construction and signing, the pretence is dropped. In a PCZT — a *partially created Zcash transaction*, the format that carries an in-progress transaction between its constructor,

prover, and signers — the fabricated output carries its recipient, value, and rseed explicitly, so a signer can classify it as a tolerable dummy rather than an opaque payment (ZIP-326).

One gap remains open by construction. Paying into an Orchard-pool receiver post-NU6.3 requires a spend at that same receiver, hence the spending key, so consensus cannot fully prevent *self-sends*; ZIP-326’s wallet rules close the gap by forbidding sends to external Orchard-pool receivers (external as against the internal address material a wallet derives for itself, §9.1).

A final oddity of provenance: the cross-address restriction is deployed and enforced by consensus, but its owning ZIP does not yet exist as text — ZIP-2006 (“Restricting Transfers into the Orchard Pool”) is a nine-line Reserved stub, owner Daira-Emma Hopwood, created 2025-06-20, discussion [zcash/zips#1305](#). In the bins of §9, *specified* (ZIP-229; ZIP-258) are the flag’s encoding, its consensus rule, and the verifying-key selection by branch; *designed but unspecified* are the restriction’s normative semantics — ZIP-326 phrases it as equality of “expanded receivers” and carries a [TODO define] in its own text — together with the enableCrossAddress → disableCrossAddress instance mapping and the dummy-spend treatment, reconstructable only from ZIP-229, ZIP-258, ZIP-326, and crate orchard at bef8a27, as Definition 10.1 does.

10.4 The turnstile and migration

The exit itself is older than the pool it now drains.

Definition 10.2 (Turnstile). The *turnstile* is the mechanism by which value leaves the Orchard pool: a one-way, publicly metered gate, the same instrument Zcash used for the Sprout retirement (§13). Its consensus content consists of exactly two rules, both encountered already: the per-transaction non-negativity rule (the seal’s third, §10.1) and the Ironwood pool-balance rule (§9.2), made precise below.

The first rule, once more and exactly: from NU6.3 activation, $v_{\text{balance}}^{\text{Orchard}} \geq 0$ holds per transaction, for *any* transaction version (ZIP-258). Value may exit the Orchard pool; it may never enter.

The second rule gives the destination pool its ledger row. Writing

$$\text{IronwoodPoolBalance} := - \sum_{\text{tx}} v_{\text{balance}}^{\text{Ironwood}}(\text{tx}),$$

the chain maintains an Ironwood analogue of every pool-balance rule of §9.2 (the ZIP-209 extension): the balance is zero before NU6.3, may never go negative, and the sum of all pool balances is bounded by MAX_MONEY (ZIP-209; ZIP-258).

A migrating transaction, then, has a fixed shape: it spends Orchard-pool notes with $v_{\text{balance}}^{\text{Orchard}} > 0$ — a positive balancing field being net value flowing *out* of the pool (the sign convention of §7.2), into the transaction’s transparent funds, from which fees and transparent outputs are also paid — and absorbs that value into an Ironwood component with $v_{\text{balance}}^{\text{Ironwood}} < 0$. Both balancing fields are cleartext: the turnstile *reveals the amounts crossing between pools in every transaction*, exactly as ZIP-229’s privacy analysis states. Consensus specifies no migration-specific cap; ordinary transaction and MAX_MONEY bounds still apply, and ZIP-318’s wallet policy recommends migration outputs no larger than 10,000 ZEC.

The remaining Ironwood consensus rules are this volume’s Orchard rules re-instantiated (ZIP-258). The anchor anchorIronwood must be an earlier main-chain block’s final Ironwood treestate — the Ironwood tree as that block left it (§6.5) — and the Ironwood commitment tree has the same capacity 2^{32} as its Orchard twin (§6.5). Beyond the trees, the ZIP-221 *history tree* — the Merkle mountain range over the chain’s history whose root every block header commits to — gains three Ironwood fields, the pool’s earliest root, latest root, and transaction count, aggregated exactly like the Orchard trio (ZIP-221; ZIP-244; ZIP-258, §“ZIP 221 changes”).

10.5 Change and wallet posture

Consensus built the gate; wallets decide the traffic. ZIP-2005’s posture is unambiguous: wallets SHOULD proactively move *all* funds — transparent, Sprout, and Sapling included, not merely Orchard — into the Ironwood pool, the only pool whose output notes are recoverable (§4; the analysis in §14.5), and SHOULD treat the sweep as an ongoing process rather than a one-off event. For Sprout, that recommendation has a proposed deadline: the v4 prohibition in ZIP 2003 would prevent further migration at NU7 activation (§9.4); it does not impose the same prohibition on Sapling or Orchard-pool spends.

Deployed change-routing logic implements the turnstile’s grain, and is the routing rule the walkthrough of §8.1 deferred: change from Ironwood value stays in Ironwood; after activation, change may return to the Orchard pool only when the transaction spends Orchard notes *and* strictly less value returns than the spends remove; otherwise change is diverted to Ironwood. The strict inequality keeps every such transaction on the right side of the seal’s third rule: change never asks new value to enter the pool.

Scanning closes down symmetrically (ZIP-326). External Orchard receivers scan from the wallet birthday (§5.4) through the block immediately before NU6.3 activation — from the activation block onward, compliant wallets no longer pay external Orchard receivers. Internal Orchard receivers scan until the wallet observes a *zero* Orchard balance after activation — change may still trickle back under the routing rule above. Ironwood receivers scan from the later of birthday and activation to the chain tip. The zero-balance watermark is sound because two seal rules act together, not because of non-negativity alone. The non-negativity rule closes the entry doors — no transaction moves new value into the pool — but an ordinary intra-pool payment satisfies it while raising its recipient’s balance; what blocks that channel is the cross-address restriction: paying the wallet’s Orchard receiver post-NU6.3 requires a spend at that same receiver, hence the wallet’s own notes — no positive-valued notes remain at balance zero — or its spending key (§10.3). With entry and circulation both shut, a wallet whose Orchard balance is known to be zero at a post-activation block cannot receive new Orchard value in that block’s descendants under these assumptions (ZIP-326; ZIP-258). This means the total observed balance, not merely currently spendable funds. A reorg removing the zero-balance block invalidates that watermark.

10.6 Scheduled Orchard-to-Ironwood migration (ZIP-318)

ZIP-318 now supplies the wallet playbook that the consensus rules leave open. Its Draft recommends decomposing value into canonical denominations of the form $m 10^k$ ZEC, where $m \in \{1, 2, 5\}$, from 0.01 through 10,000 ZEC, leaving any sub-0.01 ZEC residual unmigrated. For mobile wallets, preparation transactions SHOULD contain exactly 16 Orchard Actions. Each scheduled migration transaction then has one Orchard spend padded to two Actions, one unpadded Ironwood output, no transparent inputs or outputs, and the canonical fee (ZIP-318, “Creating Migration Transactions”).

The Draft also coordinates timing so that individual wallets do not choose distinctive anchors and expiry heights. It provisionally draws anchors from a shared 144-block boundary grid, with a recency-weighted truncated distribution, and places expiry heights on a 34,560-block grid displaced by 69,120 blocks (ZIP-318, “Anchor Heights” and “Expiry Heights”). Wallets schedule and randomise broadcasts, offer network-layer privacy options, and obtain user consent before beginning. These wallet rules reduce the information disclosed by the public crossing amounts of §10.4; ZIP-229 and ZIP-258 remain the consensus-level authority for the crossing itself.

The upgrade is now fully on the table, and Table 2 collects it: one row per change, each with the place in the volume that constructs it. With the turnstile in place, the thesis journey of §1 closes — a zatoshi minted into the Ironwood pool lives the lifecycle of §§3–8, and one still resting in the sealed pool leaves through the gate just built. What the whole journey stands on — the circuit beneath the Action statement and the proofs behind each deferred proposition — is the

work of §11 and §12.

NU6.3 change	Constructed in
Ironwood note format: lead byte 0x03, hashed trapdoor rcm	Defs. 4.3, 4.4
Ironwood pool state: own tree (capacity 2^{32}), nullifier set, anchors	§6.5
Per-pool balancing field and binding signature	§7.2
Sighash: Ironwood branch, $_v6$ personalisations, anchors to authorizing data	§7.3
Action statement: tenth public input and check A10	Defs. 8.1, 10.1
One circuit for both pools; verifying key selected by branch	§9.5
Shielding and coinbase doors re-scoped to Ironwood; maturity binds transparent outputs only	§9.1
The $v6$ format: Ironwood component; flags bit 2	§9.3
Ironwood chain value pool balance row	§§9.2, 10.4
Orchard pool sealed: three version-independent rules	§10.1
Fabricated outputs with random C_{enc} for restricted spends	§10.3
Ironwood anchor rule; three new history-tree fields	§10.4
Wallet: sweep posture, change routing, scan ranges	§10.5

Table 2: Everything NU6.3 changes, one row per change, with the part of the volume that constructs it (ZIP-229; ZIP-258). Consensus branch identity and activation constants: §9.3.

11 The circuit beneath the journey

Every stage of the journey deposited checks into the Action statement; this section descends one level, from statement to prover. It is deliberately a map, not a territory.

11.1 From relation to circuit

The Action statement of Definition 8.1 (§8.3) is a *relation*: it declares which (public input, witness) pairs are acceptable and prescribes no computation. What the prover executes is the *Action circuit* — a single fixed PLONKish circuit (the *Halo 2 Guide*, §“PLONKish arithmetisation”), identical for every Action, whose satisfiability Halo 2 proves in zero knowledge. The circuit encodes the conjunction A1–A10 and is supplied the statement’s private witness: the note, the keys, the Merkle path, the randomiser α , and the value-commitment trapdoor rcv . What follows is a sketch of this encoding; gate-level detail is delegated to the *halo2 Book* and the *Orchard Book*.

The circuit is assembled from a small set of reusable *gadgets* (in the halo2 API, *chips*), each a pre-wired block of custom gates and lookups realising one operation. The dependency runs backwards through the volume: the Orchard designers chose the journey’s primitives so that these gadgets would be inexpensive under PLONKish arithmetisation — why the journey met Sinsemilla and Poseidon rather than a bit-oriented hash (§2.4).

How any gadget is *built* — how its gates, lookups, and cell assignments are declared, and how synthesis turns the filled columns into the polynomials the proof commits to — is the subject of the *Halo 2 Guide*, §“From statement to circuit: arithmetisation in practice”. This volume records only what each gadget *enforces*.

11.2 The gadget set

Five gadget families carry the ten checks.

ECC. The ECC gadget implements the Pallas group law: point addition by the complete addition formulas, with no exceptional cases (the *Math Guide*, §“Elliptic curves”); variable-base scalar

multiplication as a double-and-add ladder of cheaper incomplete-addition rounds finished by a complete tail; fixed-base scalar multiplication from precomputed tables; and witnessing that a point lies on the curve. (The j -invariant-0 endomorphism — the *Math Guide*, §“The j -invariant and the GLV endomorphism” — serves only the recursion layer (the *Halo 2 Guide*, §“Accumulation and the Halo trick”), cheapening multiplication by verifier challenges; the Action circuit does not use it.) The gadget realises every group operation in the statement: $rk = ak + [\alpha] G^{\text{SpendAuth}}$ (A6); $pk_d = [ivk] g_d$ (A7); $cv^{\text{net}} = [v_{\text{old}} - v_{\text{new}}] V^{\text{Orchard}} + [rcv] R^{\text{Orchard}}$ (A4); and the point $[F_{nk}(\rho) + \psi] G^{\text{nf}} + cm$ inside the nullifier (A5). The derivation $ak = [ask] G^{\text{SpendAuth}}$ occurs outside the circuit: ask is not an Action witness; the signature supplies that authority.

Sinsemilla. The Sinsemilla gadget implements Construction 2.1 by realising the generator table $P[m]$ as a Halo 2 lookup (the *Halo 2 Guide*, §“The lookup argument”): each 10-bit chunk costs one table lookup and two incomplete additions (§2.4). On it rest the two composed commitment gadgets the *Orchard Book* names: NoteCommit, proving the note commitments of A1 and A2, and Commit^{ivk}, the in-circuit short commitment proving $ivk = \text{Commit}_{r_{ivk}}^{\text{ivk}}(\text{Extract}(ak), nk)$ in A7 — each bundling the Sinsemilla rounds with the bit-decomposition and canonicity checks of its field inputs.

Merkle path. The Merkle-path gadget stacks 32 layer-tagged MerkleCRH^{Orchard} hashes — themselves Sinsemilla short hashes (Definition 6.1). At each layer a conditional swap orders the running node and its witnessed sibling by one bit of the leaf position; the gadget asserts the recomputed root equal to the public anchor — check A3, gated by v_{old} so that a dummy spend may skip it.

Poseidon. The Poseidon gadget implements the Poseidon permutation (the *Crypto Guide*, §“Arithmetisation-friendly hashing: Poseidon”), whose only non-linearity is $x \mapsto x^5$, to evaluate the nullifier PRF $F_{nk}(\rho) = \text{PoseidonHash}(nk, \rho)$ of Definition 7.1 cheaply in-circuit; its output feeds the ECC gadget for A5.

Decomposition. Lookup-based decomposition, range, and boolean gadgets split field elements into the bit-chunks the other gadgets consume and range-check witnessed values: $v_{\text{old}}, v_{\text{new}} \in [0, 2^{64})$ and $\text{sign} \in \{-1, +1\}$ (A4), and canonical point and field-element encodings.

11.3 Flags, public inputs, and circuit parameters

The flags `enableSpend`, `enableOutputs`, and `enableCrossAddress` (bit 2) need no gadget at all. They are verifier-supplied public inputs: ZIP-229’s flags bytes, `flagsOrchard` and `flagsIronwood`, fix them to bits outside the circuit, and they enter it only through the plain multiplication gates of A8–A10 — the third negated, wired as the Ironwood circuit’s tenth public input `disableCrossAddress` (Definition 8.1; semantics in §10.2).

The gadgets compose into one fixed circuit whose native field is the Pallas base field $\mathbb{F}_{p_{\text{Pallas}}}$ — the field of Pallas coordinates, so the ECC gadget runs natively — proved with the Halo 2 inner-product argument on Vesta, whose scalar field the Pasta cycle makes exactly $\mathbb{F}_{p_{\text{Pallas}}}$ (the *Halo 2 Guide*, §“The inner-product polynomial commitment”). The circuit occupies $2^{11} = 2048$ rows — the *Halo 2 Guide*’s circuit-size exponent $k = 11$, not Sinsemilla’s chunk width $k = 10$ (Construction 2.1) — with ten advice columns beside the fixed columns and the Sinsemilla lookup table (the *Halo 2 Guide*, §“From statement to circuit: arithmetisation in practice”). Every Action, whatever it spends or creates, is one instance of this same circuit; a bundle’s Actions are proved together in the one aggregate proof a node checks from the public inputs alone (§8.2); and batch verification — the deferred linear multi-scalar multiplications merged across proofs (the *Halo 2 Guide*, §“Accumulation and the Halo trick”) — amortises the shared multi-scalar-multiplication

work across many proofs, while retaining proof-specific logarithmic work. The machinery the journey met one stage at a time is, beneath it all, a single two-thousand-row table whose satisfiability is the payment’s validity.

12 End-to-end security

The journey’s debts now come due. Each property was stated where its object was constructed and deferred here: Sinsemilla collision resistance (Theorem 2.2, §2.3), nullifier uniqueness (Proposition 7.2, §7.1), RedPallas unforgeability under re-randomisation (Proposition 7.5, §7.4), and the balance equivalence (Proposition 7.4, §7.2). This section pays them in order, adds the two properties with no single construction site — zero knowledge and the soundness of the composed whole — and closes with the volume’s final theorem.

12.1 Guarantees provided by a valid Action proof

A SNARK π accepted on the public inputs

(anchor, cv^{net} , nf_{old} , rk, cmx, enableSpend, enableOutputs, disableCrossAddress)

testifies, with overwhelming probability, that the prover *knows* a witness satisfying the checks A1–A10 of Definition 8.1. This is the knowledge soundness of Halo 2 (the *Halo 2 Guide*, §“A rigorous treatment of knowledge soundness”): from any prover that convinces the verifier, an extractor recovers such a witness. Everything on the soundness side — nullifier uniqueness, spend authority, balance — then follows from one or more conjuncts of A1–A10 together with algebraic properties of the primitives those conjuncts invoke, and §12.2 supplies the primitive reduction they all share. Privacy rests instead on the zero knowledge of the SNARK (§12.6) together with the DDH, Poseidon-PRF, and KDF/AEAD assumptions that Theorem 12.3(iv) adds.

12.2 Sinsemilla collision resistance

Proposition 12.1 (Sinsemilla collisions are discrete-log relations). *For distinct equal-length input bitstrings $M \neq M'$, a collision $\text{Sinsemilla}_D(M) = \text{Sinsemilla}_D(M')$ yields an explicit non-trivial discrete-log relation among the random-oracle-derived generators $Q(D), P[0], \dots, P[2^k - 1]$ of Construction 2.1; so does an x -coordinate collision $\text{ShortHash}_D(M) = \text{ShortHash}_D(M')$, and, with the blinding base H_D joining the generator set, two distinct fixed-length openings of ShortCommit or of the extracted commitment cmx. Consequently Theorem 2.2 holds — for fixed-length inputs, Sinsemilla_D is collision-resistant assuming DL on $\mathbb{G}$ — and the fixed-length short forms of Definition 2.3 inherit it.*

Sketch. Equal bit length ensures that padding produces distinct chunk vectors of the same chunk count $\ell \leq 253$. Suppose first that no incomplete-addition exception occurs. Unfolding the update rule of Construction 2.1,

$$\text{Sinsemilla}_D(M) = [2^\ell] Q(D) + \sum_{i=1}^{\ell} [2^{\ell-i}] P[m_i]. \quad (1)$$

A repeated chunk value uses the same table generator, so collect its coefficient. For $0 \leq j < 2^k$, let

$$\chi_j(m) := \sum_{i=1}^{\ell} 2^{\ell-i} [m_i = j] \pmod{p_{\text{Vesta}}}.$$

The vector $\chi(m)$ is injective: before reduction, the binary digits of its j th coordinate mark exactly the positions at which $m_i = j$, and the bound $\ell \leq 253$ prevents wraparound modulo p_{Vesta} . A collision therefore gives

$$\sum_{j=0}^{2^k-1} [\chi_j(m) - \chi_j(m')] P[j] = O.$$

Because $m \neq m'$ implies $\chi(m) \neq \chi(m')$, this is a non-trivial linear relation over $\mathbb{F}_{p_{\text{Vesta}}}$ on the table generators. Since hash-to-curve modelled as a random oracle samples the generators independently (§2.2), finding such a relation is the multi-generator discrete-log-relation problem, and no efficient adversary produces one without solving DL — the reduction the *Crypto Guide* gives in its Pedersen-hash collision analysis, §“Sinsemilla: an algebraic hash-based commitment”. (The fixed-length hypothesis is load-bearing, not a convenience: the deployed hash zero-pads its input to a chunk multiple, Construction 2.1, so two messages of different bit-lengths agreeing after padding collide with no discrete-log relation at all — variable-length collision resistance is simply false.) An x -coordinate collision adds one further case: Extract identifies P with $-P$ (§2.1), so $\text{Extract}(\text{Sinsemilla}_D(M)) = \text{Extract}(\text{Sinsemilla}_D(M'))$ forces $\text{Sinsemilla}_D(M) = \pm \text{Sinsemilla}_D(M')$. The $+$ case is the point collision just treated; in the $-$ case, moving everything to one side of equation (1) gives

$$[2^{\ell+1}] Q(D) + \sum_{j=0}^{2^k-1} [\chi_j(m) + \chi_j(m')] P[j] = O,$$

whose $Q(D)$ -coefficient $2^{\ell+1}$ is non-zero modulo p_{Vesta} : a non-trivial relation whether or not any chunk differs. For the blinded forms, subtracting two openings adds $[r - r']H_D$ to the $+$ relation (and $[r + r']H_D$ to the $-$ relation); the nonzero message coefficient vector or the nonzero $Q(D)$ coefficient keeps the relation non-trivial. Finally, the incomplete-addition exceptions — inputs on which some $\dot{+}$ is undefined — are equalities between accumulator points, themselves non-trivial discrete-log relations among the same generators, so exhibiting one is negligible too — the step that priced the $\perp$ -weakening of Remark 8.2. $\square$

Collision resistance is the primitive ingredient of note-commitment binding: since the blinding base H_D of NoteCommit is independent of the hash generators (§2.3), opening one commitment to two distinct notes — even at the level of the extracted cm_x , by the $\pm$ case above — would again exhibit a relation Proposition 12.1 excludes: two distinct notes cannot share a commitment, nor a cm_x . Checks A1 and A2 are therefore genuine integrity statements about cm_{old} and the new note’s cm_x , and with knowledge soundness they entail that the prover knows the *specific* opening of each commitment — the fact every argument below consumes.

12.3 Nullifier uniqueness

The extraction just established discharges the double-spend property: two distinct notes share a nullifier only with probability negligible in the DL hardness of Pallas, with the derivation of G^{nf} modelled as a random oracle (§2.2) — exactly the hypotheses Proposition 7.2 names. No pseudo-randomness of F is invoked, nor could it be: the adversary who forges the colliding notes holds the PRF key nk itself, and every coefficient below, $F_{\text{nk}}(\rho)$ included, is known to the reduction.

Sketch of Proposition 7.2. A collision $\text{nf}(n) = \text{nf}(n')$ for notes $n \neq n'$ is an equality of x -coordinates (Definition 7.1), hence

$$[F_{\text{nk}}(\rho) + \psi] G^{\text{nf}} + \text{cm} = \pm ([F_{\text{nk}'}(\rho') + \psi'] G^{\text{nf}} + \text{cm}'),$$

a non-trivial linear relation among G^{nf} , cm , and cm' . Expanding cm and cm' through their known openings — extracted by knowledge soundness, or supplied by the adversary who forged the

notes — over the Sinsemilla generators and the blinding base H_D turns this into a *known* non-trivial discrete-log relation among the independent GroupHash outputs $\{G^{\text{nf}}, Q, P[0], \dots, P[2^k - 1], H_D\}$: distinctness of the notes forces a coefficient mismatch somewhere. Producing such a relation reduces to DL on Pallas exactly as in Proposition 12.1. $\square$

Check A5 (nullifier integrity) plus knowledge soundness then ensures that the nf_{old} among the public inputs is genuinely *the* nullifier of a specific committed note, not an arbitrary value; the chain’s duplicate-nullifier rejection (§8.2) does the rest.

12.4 RedPallas unforgeability

The signature scheme’s debt, Proposition 7.5, carries a sharpened reading: a forger that, given ak and signatures under re-randomised keys, produces a valid signature under $\text{rk} = \text{ak} + [\alpha] G^{\text{SpendAuth}}$ for an α of its own — possibly ak -dependent — choice yields a DL solver for ak .

Sketch of Proposition 7.5. The standard forking argument (the *Crypto Guide*, §“Security in the random oracle model and the forking lemma”, via the Bellare–Neven general forking lemma) reduces a Schnorr forger to a DL adversary, and it covers re-randomisation because RedPallas is *key-prefixed*: the challenge $H^*(R \parallel \text{rk} \parallel m)$ hashes the verification key, so shifting responses cannot transport a signature from one key to another. The reduction embeds the DL challenge as $\text{ak} := X$. It answers signing queries under any $\text{rk}_i = \text{ak} + [\alpha_i] G^{\text{SpendAuth}}$ without knowing any secret, via the honest-verifier zero-knowledge simulator with random-oracle programming: sample c, s uniformly, set $R := [s] G^{\text{SpendAuth}} - [c] \text{rk}_i$, and program $H^*(R \parallel \text{rk}_i \parallel m) := c$, aborting on programming collisions exactly as in the Schnorr EUF-CMA theorem of the same section. It then forks the adversary at the forgery’s critical query $H^*(R^* \parallel \text{rk}^* \parallel m^*)$, obtaining two accepting transcripts that share R^* — and, since both forks share the critical query’s input, the same rk^* , hence the same α^* — with distinct challenges. Two-special soundness (the *Crypto Guide*, §“The Schnorr identification protocol”) extracts rsk^* , the discrete logarithm of rk^* , and the output $\text{ask} = \text{rsk}^* - \alpha^*$, with α^* the adversary-supplied randomiser, solves DL for ak . $\square$

A separate, weaker statement covers honest signers only: for uniform signer-chosen α , sampling α at random and back-deriving $\text{ak} := \text{rk} - [\alpha] G^{\text{SpendAuth}}$ reproduces the joint view of $(\text{ak}, \alpha, \text{rk})$ exactly, so honest re-randomisations are distributed as plain Schnorr keys — the unlinkability statement of the *Crypto Guide*, §“Key re-randomisation and unlinkability”, which does not by itself cover adversary-chosen α ; the forking reduction above is what does.

Check A6 (spend authority) plus knowledge soundness ensures the prover knows the α relating rk to ak ; combined with the SpendAuthSig under rk at the transaction layer (§7.4), this enforces that the spender knows the ask behind ak — and hence, through check A7’s binding of ak into the address, the authority over pk_{old} .

12.5 Balance preservation

Proposition 12.2 (Balance). *Any accepted Orchard bundle satisfies*

$$\sum_i (v_{\text{old},i} - v_{\text{new},i}) = v_{\text{balance}}^{\text{Orchard}},$$

except with probability negligible in the DL hardness of the Pasta curves in the random-oracle model — Vesta for the knowledge soundness of the Action SNARK supplying the A4 openings, Pallas for the binding-signature extraction and the independence of the value-commitment bases.

Sketch. By A4 and knowledge soundness, each $cv^{\text{net},i}$ commits to the true per-Action difference $v_{\text{old},i} - v_{\text{new},i}$, and by the Pedersen homomorphism the bundle sum $\sum_i cv^{\text{net},i}$ splits into a V^{Orchard} -component carrying $\sum_i (v_{\text{old},i} - v_{\text{new},i})$ and an R^{Orchard} -component carrying $\sum_i rcv_i$ (§7.2). Consensus computes bvk and checks the binding signature; a valid binding signature is a Fiat–Shamir proof of knowledge of $\log_{R^{\text{Orchard}}} bv_k$ (the *Crypto Guide*, §“Binding signatures as a proof of knowledge of a discrete logarithm”), so the forking extraction from the proof of Proposition 7.5 recovers bsk with $bvk = [bsk] R^{\text{Orchard}}$. Combined with the extracted A4 openings, a non-zero V^{Orchard} -component of bvk would yield a discrete-log relation between the independent bases V^{Orchard} and R^{Orchard} , contradicting DL. Hence the value component equals $[v_{\text{balance}}^{\text{Orchard}}] V^{\text{Orchard}}$ and the total value flow matches the declared transparent balance modulo the group order. This is also integer equality: each value is below 2^{64} , each bundle has fewer than 2^{16} Actions, and the declared balance is bounded by $\text{MAX_MONEY} = 21,000,000 \cdot 10^8$ zatoshi, so their difference has magnitude far below the approximately 2^{254} group order (protocol specification, §7.1.2; ZIP-229, transaction consensus rules). This also completes Proposition 7.4: the “if” direction was algebraic at the construction site, and the extraction just given shows that an unbalanced author cannot know a discrete logarithm of bvk to base R^{Orchard} . $\square$

Balance preservation is the fundamental conservation property: shielded transactions cannot mint zatoshi out of nothing.

12.6 Zero knowledge

The statistical zero-knowledge claim is conditional on the full composition and random-oracle simulation conditions developed in the *Halo 2 Guide*, §“Zero knowledge: hiding the witness in Halo 2”. Its simulator receives the instance $(\text{anchor}, cv^{\text{net}}, nf_{\text{old}}, rk, cmx, \dots)$, not the witness, and must reproduce the *joint* distribution of commitments, evaluations, challenges, and the opening proof. Polynomial blinding and the deployed IPA’s initial masking commitment are ingredients, not a complete simulator on their own. Programming fresh random-oracle queries, preserving evaluation correlations, and accounting for exceptional challenges and insufficient blinding entropy are necessary parts of that argument; independently sampling every challenge and final scalar does not establish it.

Zero knowledge supplies the proof’s share of Orchard’s privacy: the SNARK transcript reveals nothing beyond the public inputs of each Action. That the published data leak nothing *further* is a separate, computational matter: the nullifier — itself a public input (§7.1) — hides its note under the PRF security of Poseidon or, independently, DDH on Pallas (the disjunction stated there), and the epk and ciphertexts published alongside the public inputs (§5) hide recipient and value under the DDH and KDF/AEAD assumptions that Theorem 12.3(iv) adds.

12.7 Composition: the end-to-end theorem

The full soundness of the Action SNARK composes three ingredients.

- *PIOP soundness.* Each polynomial-identity check of the underlying polynomial interactive oracle proof (PIOP; the *Halo 2 Guide*, §“Polynomial IOPs and the SNARK design pattern”) carries a Schwartz–Zippel error (the *Math Guide*, §“Polynomials over a field”), summed across constraints. For $\mathbb{F} = \mathbb{F}_{p_{\text{Pallas}}}$ with $|\mathbb{F}| \approx 2^{254}$ and constraint degree at most d_{max} , the error per check is at most $d_{\text{max}} n / |\mathbb{F}| \approx d_{\text{max}} \cdot 2^{-243}$: the gate polynomial composes degree- d_{max} gates with column polynomials of degree $n - 1$ for $n = 2^{11}$ rows, so a cheating prover’s recombined quotient $h \cdot Z_H$ has degree below $d_{\text{max}} n$. The bound is overwhelmingly small.
- *PCS extractability.* Extraction for the inner-product commitment reduces to DL on the IPA curve, Vesta (the *Halo 2 Guide*, §“The inner-product polynomial commitment”), with the accumulation analysis (the *Halo 2 Guide*, §“Accumulation and the Halo trick”) covering the

deferred-MSM batching by which a node amortises verification across proofs (§11); the same analysis would carry soundness across recursion, which Orchard’s deployment does not use.

- *Fiat–Shamir soundness in the random-oracle model (ROM)*. The Fiat–Shamir theorem of the *Crypto Guide*, §“The Fiat–Shamir transform: from interactive to non-interactive”, covers Σ -protocols, losing roughly a factor Q in the prover’s random-oracle query budget; for the $\log d$ -round Halo 2 transcript the naive multi-round analysis loses $Q^{O(\log d)}$, a bound vacuous already at $Q = 2^{80}$. The negligible-error conclusion rests on the tighter treatments the *Halo 2 Guide* presents in §“The algebraic group model”: the direct ROM-plus-DL analysis of BCMS20, or Ghoshal–Tessaro’s tight Fiat–Shamir bound in the AGM.

Each step’s error is negligible at Orchard’s parameters: $|\mathbb{F}| \approx 2^{254}$, security parameter $\lambda = 127$ under the convention that the group order is about $2^{2\lambda}$ — effectively about 126 bits after the automorphism speedups (the *Crypto Guide*, §“Instantiation on elliptic curves; the Pasta curves”) — and the union bound across the steps is itself negligible. Combined with the four application-level reductions — Propositions 12.1, 7.2, 7.5, and 12.2 — the full chain yields the volume’s final theorem.

Theorem 12.3 (Orchard end-to-end security, informal). *Under DL hardness on both Pasta curves (Pallas for the application-level primitives, Vesta for the IPA commitment) in the random-oracle model — and, for property (iv), additionally DDH on Pallas, the PRF security of Poseidon (§7.1), and the KDF/AEAD securing the note ciphertext (§5) — no efficient adversary can achieve the following, except with negligible probability. Property (iv) concerns honestly generated keys and notes, proper encryption, no disclosed viewing keys or plaintexts, and comparisons with the same public leakage; shielded coinbase outputs are excluded. The adversary cannot:*

- (i) *produce an accepted Action without knowing a witness satisfying A1–A10; in particular, any Action whose witnessed v_{old} is non-zero must spend a genuine leaf of the commitment tree (zero-value dummy spends, admitted by A3’s gate, are accepted by design);*
- (ii) *spend the same nonzero committed note twice in accepted Actions in one pool and chain history: its fixed nullifier would repeat, and consensus rejects repetitions;*
- (iii) *cause a bundle’s declared transparent balance $v_{\text{balance}}^{\text{Orchard}}$ to differ from its true net value flow $\sum_i (v_{\text{old},i} - v_{\text{new},i})$ — in particular, to exceed it, which would create value (Proposition 12.2);*
- (iv) *use the Action’s cryptographic fields to link it to any specific recipient or hidden value beyond what is explicit in the public data.*

The four properties are, respectively, the soundness of the Action SNARK, nullifier uniqueness, balance preservation, and zero knowledge — and they discharge the four requirements of §1: property (i) covers R1 and R2 (value moves only through well-formed notes with proven membership), (ii) is R3, (iii) is R4, and (iv) is the disclosure clause over them all. The first three reduce, informally, to DL on the Pasta curves in the random-oracle model — Pallas for the application-level reductions, Vesta for the IPA extraction. For the fourth, the zero knowledge of the SNARK transcript is *statistical* and does not depend on application-commitment hiding or on DL. Full cryptographic unlinkability of an Action, however, additionally rests on the DDH assumption on Pallas, the PRF security of Poseidon (§7.1), and the KDF/AEAD securing the note ciphertext (§5), and is therefore computational — the typical asymmetry of the discrete-log SNARK family: soundness is computational (a cheating prover is bounded, never impossible), while the zero knowledge of the transcript is information-theoretic. This is not a claim that the whole transaction hides its public balance, component and Action counts, transparent data, timing, address reuse, voluntary disclosures, or wallet/network metadata. The journey’s debts are paid; §13 sets these proofs in their history.

13 Lineage: three generations of shielded proof

The machinery whose security §12 has just established did not spring up whole: Zcash has carried shielded payments through three generations of proof system, and concrete trade-offs drove each transition — trusted-setup risk, recursion compatibility, post-quantum posture, and on-chain efficiency. This closing survey sets the volume’s stack in that history.

13.1 Sprout (2016–present, deprecated)

The original shielded protocol proved its statements with a *BCTV14* SNARK (Ben-Sasson, Chiesa, Tromer, Virza) over the pairing-friendly curve BN254 — a scheme of the pairing-based family the *Crypto Guide* surveys in §“SNARKs: succinct non-interactive arguments of knowledge” — with constant-size proofs (296 bytes) and constant verification time. Its parameters came from a per-circuit trusted setup: the 2016 ceremony, a six-party multi-party computation generating the BCTV14 proving and verification keys for the fixed Sprout circuit, sound provided at least one participant destroyed their secret share. Sprout demonstrated that zk-SNARK shielded payments were feasible at scale, but it remained a transitional design: the prover was slow, and the circuit hashed with SHA-256 — a bit-oriented hash whose boolean operations are non-native to arithmetic constraints, hence costly in-circuit. Decisively, a flaw in BCTV14 discovered in 2018 would have allowed undetected counterfeiting; the pool was deprecated, and users were enabled and encouraged to migrate shielded Sprout value to later pools. Residual Sprout value may remain. NU7’s proposed v4 prohibition would disable spending that value, without treating it as unissued (ZIP 2003; §9.4).

13.2 Sapling (2018–present)

The second network upgrade, Sapling, replaced this stack with Groth16 — 192-byte proofs, three-pairing verification — over BLS12-381, with the embedded curve Jubjub serving the in-circuit group operations and a Pedersen hash (the *Crypto Guide*, §“Commitment schemes”) serving the Merkle tree. The result was a far faster prover, a smaller proof, and cheaper in-circuit hashing. Two structural limitations remained. Groth16 still requires a per-circuit trusted setup — the 2018 Sapling MPC, whose universal first phase was the Powers of Tau ceremony with many participants — and no known elliptic-curve cycle extends BLS12-381 (the *Halo 2 Guide*, §“A cycle of curves for efficient native recursion”), so the proving system does not offer the fully native curve-cycle recursion used by Halo. Recursion using non-native arithmetic is not ruled out by that fact.

13.3 Orchard and the Pasta cycle (2022–present)

Orchard, activated by NU5 in 2022 and carried into the Ironwood pool from NU6.3 with a new note format, Action circuit, and verifying key, replaced the Sapling stack with the Halo 2 / Pasta stack this volume has used throughout. Four choices are principal.

- *Halo 2*: a PLONKish PIOP compiled with the IPA polynomial commitment, accumulation-friendly, and with no trusted setup (the *Halo 2 Guide*, §“The Halo 2 proof system”).
- *The Pasta cycle*: Pallas as the application curve, Vesta as the proving curve in the IPA layer, designed for recursion (the *Halo 2 Guide*, §“Recursive proof composition and accumulation schemes”).
- *Sinsemilla*: an algebraic hash whose collision resistance reduces to DL on Pallas — an assumption the protocol already requires elsewhere — and whose chunk-table structure maps directly onto Halo 2’s lookup argument (§2.3).

- *Poseidon*: serves the nullifier PRF (§7.1), where a keyed in-circuit PRF rather than a collision-resistant hash is required; unlike Sinsemilla, its security rests on cryptanalytic confidence in algebraic-attack resistance rather than on a hardness assumption such as DL.

The shift followed one consistent design principle: “transparent SNARK, recursion-ready” was the target, and the designers engineered the Pasta cycle to fit it.

The stack has since carried one correction and one split. Its primitive suite and Halo 2/Pasta proving stack remain, while the Action circuit exists in three versions — NU5’s original, the NU6.2 emergency correction (ZIP-257, Final), and the NU6.3 Ironwood circuit with the cross-address input (§10.2) — named `InsecurePreNu6_2`, `FixedPostNu6_2`, and `PostNu6_3` in the released crate orchard 0.15.0 (`src/circuit.rs`; the `bef8a27` development tree names the third variant Ironwood); the operational list is Remark 9.1 (§9). And from NU6.3 the one protocol carries two value pools, the sealed Orchard pool and the current Ironwood pool (ZIP-229, Terminology; §9). The Halo 2 / Pasta stack is unchanged throughout.

13.4 Comparative summary

Table 3 gathers the three generations.

	Sprout	Sapling	Orchard
SNARK	BCTV14 (original)	Groth16	Halo 2
Curve	BN254	BLS12-381 + Jubjub	Pallas + Vesta
Trusted setup	Per-circuit (six-party MPC)	Per-circuit (large MPC)	None
Recursion-ready	No	No	Yes (via Pasta cycle)
Proof size	296 B	192 B	~5 kB (one Action)
Verification time	Constant (pairing)	Constant (3 pairings)	$O(\log n + n/b)$ amortised

Table 3: Three generations of Zcash shielded-payment proof systems. The Sprout column describes its original proof system; Sprout switched to Groth16 at Sapling activation. For fixed Action count, Orchard verification costs $O(\log n + n/b)$ per proof in a batch of b proofs, where $n = 2^{11}$ is the row count, not the number of Actions. Batching shares the linear multi-scalar multiplication; the often-quoted $O(\log n)$ amortised bound requires $b = \Omega(n/\log n)$. Public-input and per-Action processing still grows with Action count (the *Halo 2 Guide*, §“Accumulation and the Halo trick”). The Orchard column covers three circuit versions — the NU5 original; the NU6.2 correction (ZIP-257, Final); the NU6.3 Ironwood circuit with the cross-address input — and, from NU6.3, two value pools (§9).

The proof size grew between Sapling and Orchard, and the growth is the cost of transparency: a one-Action bundle’s proof occupies $2720 + 2272 = 4992$ bytes in the v5/v6 encodings (Table 1, §9), including the $2 \log_2 2^{11} = 22$ group elements of the IPA rounds. The benefit, beyond the removal of trusted setup, is the foundation for recursion: future generations — the Tachyon scalability programme (§14.6), and eventual designs that aggregate many transactions under a single succinct proof — can build on Orchard’s foundation without revisiting the cryptographic stack. Of the opening list’s four trade-offs, the post-quantum posture alone remains unpriced; it is the horizon of §14.

14 The horizon: quantum adversaries and recoverability

The lineage of §13 looked backward; this closing section looks forward, and asks the one question the thesis journey of §1 leaves open: whether the value survives its owner’s adversaries. A *cryptographically relevant quantum computer* (CRQC) — one large and reliable enough to run Shor’s algo-

rithm at cryptographic parameters (the *Crypto Guide*, §“The quantum caveat: Shor’s algorithm”) — computes discrete logarithms in polynomial time, and with them falls every group-structured assumption of the end-to-end theorem: DL on both Pasta curves and, for property (iv), DDH on Pallas (Theorem 12.3). The theorem’s remaining assumptions — the PRF security of Poseidon and the security of the KDF/AEAD — are symmetric, and face only the generic quantum speedups of §14.4.

14.1 The reversibility taxonomy and the epistemic bins

Since the assumptions all fail together, asking *whether* they fail is unproductive. The productive question is when the failure matters, and the axis that answers it is reversibility.

Definition 14.1 (Reversibility of a quantum break). A quantum break of a deployed protocol is *retroactive* when data already recorded on chain becomes exploitable on the day the adversary exists: no later consensus change can help, because the data is public and permanent. A break is *prospective* when its exploitation requires a validator to *accept* a forged object after the adversary exists: disabling the affected protocol before that day removes acceptance of new transactions through that mechanism. Whether forged historical objects can still matter depends on chain-work and retained-history assumptions (§14.3).

Within the generic discrete-log channels catalogued by ZIP-2005, Orchard — in both pools — has one identified retroactive break: harvest-now-decrypt-later against the note encryption (§14.2). The other cryptographic breaks considered here are prospective — the discrete-log integrity stack of §14.3 — or face the known generic speedups — the symmetric primitives of §14.4. The section is organised accordingly, and its scope is the Orchard core seen from inside.

Remark 14.2 (Method). Every forward-looking claim below sits in one of the three epistemic bins of §9: *specified* — a published artefact pins the construction down, and is cited; *designed but unspecified* — an informal source describes the design, and the text says what a specification would still need to fix; *open problem* — no artefact exists. Deployed-behaviour claims instead cite implementation and specification, as throughout the volume. The ground truth is pinned: `protocol.tex` and the ZIPs at `zips commit 69610984 (2026-07-09)`; `crate orchard` at `git commit bef8a27e (0.14.0-18-gbef8a27, 2026-06-16)` for circuit claims; the released orchard 0.15.0 — the `librustzcash` pin — for the lead-byte-0x03 note path. Each artefact cited below was verified at these pins.

14.2 The retroactive break: harvest now, decrypt later

Each Action publishes the pair $(\text{epk}, C_{\text{enc}})$ (§5). The confidentiality of C_{enc} rests on exactly one discrete-log-dependent link: the one-shot Diffie–Hellman on Pallas between esk and pk_d (Construction 5.1; protocol §5.4.5.5). The KDF and the AEAD above that link are symmetric, and are not the weak layer.

The attack this admits is *harvest now, decrypt later* (HNDL): record the chain today, break the link when the machine arrives. A CRQC that knows a recipient’s address (d, pk_d) computes $\text{ivk} = \log_{g_d} \text{pk}_d$ by Shor — one discrete logarithm for the lifetime of the incoming-viewing-key scope — and then runs the recipient’s own trial-decryption loop of §5.3 over the recorded chain: for every Action, $[\text{ivk}] \text{epk}$ recovers K_{enc} , and the AEAD tag identifies which ciphertexts belong to that key. The adversary obtains, for every note ever sent to any diversified address sharing that ivk , the full plaintext $(d, v, \text{rseed}, \text{memo})$: every amount, every memo, the complete receiving history — and equally every *future* ciphertext to the same key scope. Learning the external ivk alone does not derive the separate internal-scope ivk .

Two limits bound the damage. First, ivk is a viewing capability, not a spending one (the ladder of §3.4): no spend authority is conferred, and ask is not exposed — its exposure is a prospective

matter, taken up in §14.3. Second, the attack is keyed by the address: without (d, pk_d) the adversary has neither the base g_d nor the target pk_d , and no known efficient generic discrete-log attack recovers this channel from the ciphertext alone. This statement excludes public metadata and auxiliary information about the recipient. ZIP-2005 records the same boundary: the exposure requires “both the ciphertext ... and the receiver address”.

The break is retroactive by construction. The pair (epk, C_{enc}) is public consensus data that archival observers can retain indefinitely; its secrecy was fixed at recording time by one specific Diffie–Hellman instance; and any protocol upgrade alters only ciphertexts not yet created (ZIP-2005, Non-requirements). The adversary needs no quantum computer today — only storage — and every Orchard output recorded since NU5 activation is already in that corpus, which accrues Ironwood outputs from NU6.3 activation onward.

Bin: *open problem*. Nothing is deployed, and nothing is specified, that protects even future ciphertexts. ZIP-2005 (Proposed) excludes privacy by an explicit Non-requirement, states that the exposure persists for all pools — Ironwood included — and says only that mitigations for future transfers are “under consideration”. The tracking issues `zcash/zips#1133` (open since 2022) and `zcash/zips#1134` (open since 2016) carry discussion, not drafts; and the zips repository at commit 69610984 contains no occurrence of ML-KEM or its pre-standardisation name Kyber (the NIST post-quantum key encapsulation), of ML-DSA (the corresponding signature standard), or of any lattice scheme. A future hybrid key encapsulation would in any case protect future outputs only; the recorded ciphertexts are unrepairable.

14.3 Prospective breaks: the discrete-log integrity stack

Shor’s algorithm invalidates the group assumptions used by the security reductions of §12; it does not turn those reductions into attacks automatically. The prospective-break inventory below therefore gives a separate concrete attack or consequence for each affected construction.

Sinsemilla binding. Collision resistance reduces to the discrete-log relation problem among the GroupHash-derived generators (Proposition 12.1). A CRQC computes every pairwise logarithm, after which each Sinsemilla commitment opens to arbitrary valid messages by adjusting the commitment blinder: a second note under an on-chain `cmx` — breaking balance through NoteCommit non-binding — and alternative pairs (ak', nk') opening the same Commit^{ivk} — the “Spendability” attacks ZIP-2005 catalogues, forgeries of the kind the Faerie resistance of §7.1 excludes. A balance violation committed this way is bounded only by the ZIP-209 turnstile (§10.4) and need not be detected at all. The unblinded Merkle hash has no freely adjustable commitment blinder; knowing generator logarithms alone does not supply an efficient generic method to fabricate a membership path under an arbitrary anchor.

Value commitments and the binding signature. The binding of cv^{net} is the unknown discrete-log relation between the bases $V^{Orchard}$ and $R^{Orchard}$ (Definition 7.3, Proposition 7.4), and the binding signature is Schnorr under DL in the random-oracle model. Knowledge of $\log_{V^{Orchard}} R^{Orchard}$ reopens any recorded or fresh `cv` to any value. A CRQC can separately solve the discrete logarithm of a target `bvk` to obtain its `bsk` and forge the binding signature: silent inflation, again turnstile-bounded.

Spend authorisation. RedPallas unforgeability is DL in the random-oracle model (Proposition 7.5). One logarithm — of `ak`, or of any published `rk` — forges spend authorisation at will. Theft additionally requires a satisfying witness or a successful proof forgery; a broken soundness assumption alone is not such a forgery.

Proof soundness. The Action SNARK is knowledge-sound under DL on Vesta with Fiat–Shamir in the random-oracle model (§12.7). A CRQC breaks the discrete-log binding/extractability assumption of the IPA polynomial commitment, invalidating the stated knowledge-soundness reduction. That permits commitment equivocation; a universal method for forging every false Action statement does not follow from the reduction alone and requires an attack on the full proof protocol.

None of the four is retroactive, and the reason deserves spelling out. Each attack must place a forged object — a proof, a signature, a commitment opening — before a validator and have it *accepted*, and acceptance happens at current height under current consensus rules. Disabling the DL-dependent pools before a CRQC exists therefore removes acceptance of new transactions under those rules. Historical blocks can still be revalidated under their original rules, so this argument also presumes that accepted pre-transition history is anchored by the chain’s work and settled-upgrade policy, not freely rewritten (*Consensus Guide*, §“Reorg rails, maturity, and the finality floor”). A timely transition addresses prospective integrity failures on that retained history — with two residues. First, switch-off must precede the first forgery: an attack landed inside the exposure window is not repaired afterwards. Second, honest funds inside a disabled pool need some other way out — which is precisely the problem ZIP-2005 exists to solve (§14.5).

Recorded cv values are perfectly hiding for uniform independent value-commitment randomness; the randomiser α nearly uniformly masks the validating key in the ideal-hash model (§7.4); and proof transcripts are statistically simulatable without the witness (§12.6). Nullifiers additionally rely on the secret nk and Poseidon PRF security for unlinkability. Thus no known efficient generic discrete-log attack extracts these secrets merely from the recorded cryptographic fields without the receiver address. This deliberately narrower claim excludes public transaction meta-data, side information, and failures of the symmetric assumptions.

14.4 Primitives facing only generic speedups

Two primitive families in Orchard carry no discrete-log structure. BLAKE2b serves PRFexpand and with it every key and note-randomness derivation (Definition 3.2, §3; Definition 4.3), the KDF and the outgoing cipher key (§5), the RedPallas challenge hash, and the ZIP-244 transaction/signature digests (§7.3); the ZIP-32 (Final) Orchard key tree is hardened-only, built entirely from these derivations. Deriving child secret material uses no group operation, although validity checking the resulting keys does. Poseidon serves the nullifier PRF, keyed by nk (§7.1).

For these primitives the assessment assumes no additional structural attack beyond the known generic quantum speedups. Grover’s search algorithm halves generic preimage security: 2^{256} classical evaluations become about 2^{128} quantum ones, a margin the parameters absorb (the *Crypto Guide*, §“The quantum caveat: Shor’s algorithm”). For collisions the picture is better still: the Brassard–Høyer–Tapp (BHT) algorithm finds collisions in $2^{n/3}$ oracle queries but requires random access to a quantum memory of about 2^{92} bits at these output sizes, and the best *known* implementable generic quantum collision attack remains the classical parallel one — the assessment ZIP-2005 adopts, flagged there as best-known rather than provably best.

One caveat attaches to Poseidon: its PRF security is a cryptanalytic heuristic with no standard-problem reduction, classically and quantumly alike (the *Crypto Guide*, §“Arithmetisation-friendly hashing: Poseidon”). A quantum adversary gains no identified structural advantage. Recorded-nullifier unlinkability nevertheless depends on Poseidon behaving as the required PRF; keeping nk off chain does not preserve that claim if the PRF assumption itself fails.

The survey’s load-bearing consequence: the symmetric backbone survives a discrete-log break intact. A seed holder can still prove, by re-executing the hardened derivations, that their keys descend from their seed — ownership remains *provable* after the assumption that made it *enforceable* has fallen. This is the fact ZIP-2005’s quantum recoverability builds on.

14.5 ZIP-2005: quantum recoverability

ZIP-2005 (“Ironwood Quantum Recoverability”; status Proposed, created 2025-03-31) is the deployed-track response to the prospective breaks. It does not make Orchard quantum-secure, and says so; what it arranges is that funds survive the eventual switch-off. Its activation is bound to the NU6.3 network upgrade (ZIP-258, Draft; consensus branch 0x37A5165B, activation constants by pointer in §9.3), a scheduling that follows the NU6.2 emergency circuit correction

(ZIP-257, Final) — which is why the originally planned earlier deployment slipped. The consensus half of the response — the seal and the turnstile — was constructed in §10; this subsection owns the note format and its security analysis.

The format half is the Ironwood note: note plaintexts carry lead byte 0x03 (§5.1), and with it the hashed trapdoor of Definition 4.3 — rcm derived from the 0x0B-domain-separated pre_{rcm} concatenating every note field, in place of the legacy $0x05 \parallel \rho$ input. Every note field thereby traverses BLAKE2b on its way into the trapdoor. The purpose was promised in one sentence at the construction site (§4.2); here is the argument in full.

Under a CRQC, NoteCommit alone is not binding: its binding is Sinsemilla collision resistance, first on the integrity stack to fall (§14.3). The Ironwood derivation restores binding *conditionally*. Model PRFexpand as a random oracle. An adversary who would open one commitment to two distinct notes, both openings respecting the derivation, cannot vary any note field freely: every field enters pre_{rcm} , so any change re-randomises rcm , and landing two honestly-derived note-trapdoor pairs on the same commitment point is a collision-type event in the oracle — generically hard, with no group structure for Shor to exploit, in the sense Theorem 14.3 makes precise. The *composite* of the derivation and NoteCommit is therefore post-quantum binding — *provided a verifier checks the derivation*. No deployed Action constraint checks that derivation: the circuit checks the commitment, not the derivation that produced its trapdoor. The binding claim is conditional on that future check, which is exactly the Recovery Statement’s job.

The key half is a rederivation path for $\text{Commit}^{\text{ivk}}$. The randomness r_{ivk} descends from sk through PRFexpand (Construction 3.4) — or, where no single sk serves — for FROST-generated keys, threshold signing in which ask arises from distributed shares rather than from sk , and for hardware-held keys, where proof authority must remain separate from the on-device spend key — ZIP-2005 derives r_{ivk} from a *quantum key* qk , produced by one BLAKE3 `derive_key` compression from a secret qsk , with signatures of knowledge — proofs of knowledge of a secret, bound to a message the way a signature is — over the transaction sighash tying the claimed keys to the spend. A caveat: the ZIP types that sighash only as a message hash, and pins no sighash algorithm.

The *Recovery Statement* (non-normative) is what a holder would one day prove, in a post-quantum knowledge-sound proof system, over a re-hashed commitment tree: the key derivations — BindKeys, ZIP-2005’s name for the derivations of ask , ak , and nk from sk , plus the r_{ivk} branch above — then $\text{ivk} = \text{Commit}^{\text{ivk}}$, $\text{pk}_d = [\text{ivk}] g_d$, the ψ and rcm derivations, the note commitment, a membership path, the nullifier, and the α -re-randomisation tying the public rk to ak . The cost: 7 BLAKE2b-512 compressions, 1 BLAKE3 compression, 2 Sinsemilla commitments, 2 Pallas scalar multiplications, and a Merkle path, with the BLAKE2b compressions dominating.

The binding arguments carry published bounds, stated in the *quantum random-oracle model* (QROM): the random-oracle model of the classical proofs (the *Crypto Guide*, §“Security in the random oracle model and the forking lemma”), with the adversary permitted superposition queries to the oracle.

Theorem 14.3 (Ironwood binding bounds, as published). *In the QROM, with q oracle queries and $N \approx 2^{254}$ the oracle output-space size (the Pallas scalar order, p_{Vesta} in this volume’s notation),*

$$\epsilon_{\text{kb}}^{\text{QROM}} \leq O(q^3/N), \quad \epsilon_{\text{Spend}}^{\text{QROM}} \leq O(q^3/N) + \epsilon_{\text{kb}}^{\text{QROM}},$$

for the key-binding game — that the derivations bind the recovered keys — and the Spendability game of the attack catalogue of §14.3, respectively (ZIP-2005).

Idea. The quantum analogue of collision resistance is Unruh’s *collapsing* property: measuring the adversary’s superposition of hash inputs is undetectable to it once the outputs are measured. Fehr’s theorem for HAIFA-mode hashes — the counter-carrying iteration mode BLAKE2b instantiates — reduces collapsing of the rcm derivation to modelling BLAKE2b’s compression function

as a random oracle, the same modelling the classical proofs already use. Both classical reductions — key binding and Spendability — are straight-line, running the adversary once without rewinding, and never reprogram the oracle, so they lift to the QROM unchanged; by Zhandry’s compressed-oracle technique the classical $O(q^2/N)$ collision bound becomes $O(q^3/N)$. $\square$

Evaluated at $q = 2^{60}$ queries, the cubic term is about 2^{-74} and the quadratic comparison is about 2^{-134} ; for $q \ll 2^{60}$ both bounds are correspondingly smaller. Saturating the cubic bound needs the memory-infeasible BHT-style algorithm of §14.4.

The scope is deliberately narrow: binding only, Orchard only. ZIP-2005 repairs binding, not privacy — the HNDL exposure of §14.2 persists unchanged for every pool, Ironwood included. Sapling is excluded, with the rationale quoted: “to reduce overall protocol complexity and analysis effort” — the ZIP-32 Sapling derivations differ significantly from Orchard’s hardened-only tree and would require their own analysis. The consequence is forced migration: Sapling, Sprout, and pre-Ironwood Orchard (lead byte 0x02) notes are unrecoverable, hence permanently unspendable once their protocols switch off — whence the wallet posture of §10.5: move all funds into Ironwood notes, and treat the sweep as ongoing, since non-recoverable notes may arrive at any time.

The bins of Remark 14.2, applied:

- *Specified and deployed.* The lead-byte-0x03 derivation and the consensus rules sealing the Orchard pool (§10.1) are enforced from NU6.3 activation, so every Ironwood output uses the recoverable note format (ZIP-229; ZIP-258).
- *Specified but undeployed.* ZIP-2005 (Proposed) specifies the qsk/qk key path, but it is not a consensus rule or an enforced part of NU6.3.
- *Designed but unspecified.* The Recovery Protocol itself. ZIP-2005 presents the Recovery Statement in outline and states that its details are “subject to change”; a specification would still need to pin down the post-quantum proof system (a stated requirement is that none be assumed now), the collapsing hash for the re-hashed commitment tree, the linking commitments between circuits, and the concrete signature-of-knowledge instantiations.
- *Open problem.* Two items. A post-quantum proof system and signature scheme for *ongoing* spends — as opposed to one-shot recovery — exist nowhere as drafts (zcash/zips#1134 carries discussion only). And the switch-off schedule: every recoverability guarantee is conditional on legacy switch-off preceding the first quantum forgery, and attacks landed in the exposure window — Spendability roadblocks that permanently block recovery, spend-authority theft, turnstile-bounded inflation (§14.3) — are not repaired retroactively.

14.6 Strategy: recoverability, not resistance

One more artefact completes the picture. Ragu, the recursive proof system under development for project Tachyon — a scalability programme beyond this volume’s scope — is deliberately based on the elliptic-curve discrete-log problem over the same Pasta cycle, its authors citing “reduced concern (in the medium term)” for post-quantum soundness risks (ragu-tachyon book, src/protocol/index.md, commit 830bbcd, 2026-07-05). Bin: designed but unspecified.

Read together, Ragu’s choice and ZIP-2005 fix Zcash’s quantum strategy precisely: *recoverability, not resistance*. Keep discrete-log cryptography while it stands, because its integrity failures are prospective and fork-mitigable (§14.3); pre-position notes so that funds provably survive the eventual transition (§14.5). The one cost this strategy cannot answer is the retroactive privacy channel: the HNDL corpus of §14.2 accrues damage now.

The strategy should also be read against the publicity. Public claims that Zcash will be “fully post-quantum” on a twelve-to-eighteen-month horizon, with ML-KEM and ML-DSA under test (CoinDesk, 2026-05-08), correspond to no ZIP, no draft, and no repository artefact at the pins of

Remark 14.2; ZIP-2005 itself disclaims making the protocol secure against quantum adversaries. The record supports a narrower claim. Without a receiver address there is no known efficient generic discrete-log route to recorded note plaintexts, and closing the affected pools before the first forgery would prevent later acceptance of those forgeries. Ironwood’s note format is designed to support a future proof of ownership, but the recovery protocol and switch-off schedule remain unspecified. That is where the journey of this volume ends.

Part VI

Wallet Guide: The Zcash wallet layer: keys, transactions, and light-client synchronisation

Abstract

Assuming the cryptographic primitives of the *Crypto Guide*, the best-chain model of the *Consensus Guide*, and the Orchard protocol of the *Ironwood Guide*, this volume of *The Zcash Arboretum* documents the deployed wallet layer above them: hierarchical key derivation (ZIP-32), diversified and unified addresses (ZIP-316), the conventional fee (ZIP-317), the transaction-construction pipeline of *librustzcash*, partially created transactions (PCZT), the transaction lifecycle from broadcast through expiry (ZIP-203), payment requests (ZIP-321), and light-client synchronisation. The synchronisation chapters cover the compact-block format (ZIP-307) and its deployed divergences, the light-client service surface as *lightwalletd*, *Zaino*, and *Zebra* implement it, the scanning pipeline of *librustzcash* with its verify-first reorg discipline, commitment-tree synchronisation, and what the server learns. Every derived quantity carries its exact derivation.

The compact-block wire format is deployed but underspecified; where the draft ZIP and the deployed implementations disagree, this volume documents the deployment and flags the divergence. The protocol target is draft NU7, whose ZIP 259 leaves both activation heights unassigned as of 23 September 2026. Proposed rules and implementation-specific behaviour are distinguished where they affect the construction.

1 Hierarchical key derivation (ZIP-32)

The *Ironwood Guide* constructs every Orchard key — ask , nk , r_{ivk} , the full viewing key, ivk , dk , ovk , and the diversified addresses — from a single 32-byte spending key sk , which it takes from a uniformly sampled 256-bit candidate after validity checks, deferring hierarchical derivation to ZIP-32 (§“The recipient: keys and addresses” there). This section supplies that origin. A deployed wallet does not draw an independent sk per account: it draws one *seed* and derives every account’s sk from it deterministically, so that a single backup recovers every key the wallet will ever hold. We develop ZIP-32’s Orchard instantiation: the hardened-only derivation tree, the standard account path $m_{\text{Orchard}}/32'/\text{coin_type}'/\text{account}'$, the *internal* keys that stand in for a change path level, the fingerprints that name keys and seeds, and the serialised encodings — together with where the deployed stack (*orchard*, *zip32*, *zcash_keys*, *zcash_client_backend*, *zcash_client_sqlite*) enforces each rule, and where its documentation and its code disagree.

1.1 Wallet seeds

The root secret of the whole wallet is a byte string S , the *seed*. ZIP-32 imposes two requirements (ZIP-32, “Wallet seeds”): the seed **MUST** be 32 to 252 bytes long, and it **MUST** be generated with at least 256 bits of entropy — a requirement that extends to any mnemonic phrase from which the seed is obtained (ZIP-315 gives further guidance). The rationale is the seed’s position in the hierarchy: every key of every account derives deterministically from S , so a lower-entropy seed is a weak link for all of them at once, and — unlike an attack on one key — a brute-force search over low-entropy seeds runs against many wallets simultaneously.

Remark 1.1 (Where the deployed stack enforces the seed bounds). Enforcement is patchier than the **MUST** suggests, and one doc comment is wrong. The derivation function itself does not check: neither `ExtendedSpendingKey::master (orchard, src/zip32.rs)` nor `HardenedOnlyKey::master (zip32 0.2.0, src/hardened_only.rs)` inspects the seed length — BLAKE2b hashes whatever it is given — although the former’s doc comment claims “Panics if the seed is shorter than 32 bytes or longer than 252 bytes”; the claimed panic does not exist. One layer up, `UnifiedSpendingKey::from_seed (zcash_keys, src/keys.rs)`

panics on seeds shorter than 32 bytes and checks no upper bound. The trait documentation of `WalletWrite::create_account` and `import_account_hd` (`zcash_client_backend`, `src/data_api.rs`) declares a panic for seeds outside 32..252 bytes, and `zcash_client_sqlite` enforces both bounds — indirectly, because it computes a seed fingerprint (Definition 1.13) via `SeedFingerprint::from_seed`, which returns `None` outside 32..252 bytes (zip32 0.2.0, `src/fingerprint.rs`; `zcash_client_sqlite`, `src/lib.rs`). The ZIP requirement therefore holds at the top of the deployed stack, not in the derivation function beneath it.

1.2 Hardened-only derivation

ZIP-32 obtains Orchard keys by instantiating its generic *hardened-only key derivation* process with two domain parameters: the master-key personalisation `Orchard.MKGDDomain = ``ZcashIP32Orchard``` (16 bytes) and the child-derivation lead byte `Orchard.CKDDomain = 0x81` (ZIP-32, “Orchard key derivation”). Non-hardened derivation — the BIP-32 mode in which a public key derives child public keys — is not defined for Orchard at all. That choice fixes the shape of the extended keys: an *Orchard extended spending key* is the pair (sk, c) of a normal 32-byte spending key and a 32-byte *chain code* c , and ZIP-32 defines *no* Orchard extended full viewing key, because with only hardened derivation a full viewing key confers no child-derivation capability and so has no use for a chain code (ZIP-32, “Orchard extended keys”; contrast the Sapling sections of the same ZIP, which do define extended FVKs). Deployed, `ExtendedSpendingKey` carries the pair inside a `HardenedOnlyKey` together with bookkeeping fields `depth`, `parent_fvk_tag`, and `child_index`.

Definition 1.2 (Orchard master key generation). Given a seed S of 32 to 252 bytes, compute

$$I := \text{BLAKE2b-512}(\text{``ZcashIP32Orchard``, } S), \quad I = I_L \parallel I_R$$

with I_L, I_R the first and last 32 bytes — unkeyed BLAKE2b-512 with the 16-byte personalisation above — and set $sk_m := I_L$, $c_m := I_R$. The *master node* of the Orchard tree is $m_{\text{Orchard}} := (sk_m, c_m)$ (ZIP-32, “Hardened-only master key generation” instantiated per “Orchard master key generation”). Deployed master generation additionally requires sk_m to be valid in the sense of Definition 1.9, else it fails with `Error::InvalidSpendingKey`.

Definition 1.3 (Orchard child key derivation). Write $i' := i + 2^{31}$ for the *hardened* index derived from $0 \leq i < 2^{31}$ (ZIP-32). Child derivation is defined only for hardened indices: for $i \geq 2^{31}$,

$$\begin{aligned} \text{CKDsk}((sk_{\text{par}}, c_{\text{par}}), i) : \quad & I := \text{PRFexpand}_{c_{\text{par}}} (0x81 \parallel sk_{\text{par}} \parallel \text{I2LEOSP}_{32}(i)), \\ & sk_i := I_L, \quad c_i := I_R, \end{aligned}$$

where $\text{I2LEOSP}_{32}(i)$ is the 4-byte little-endian encoding of i and `PRFexpand` is the BLAKE2b-512 expansion function of the *Ironwood Guide* (personalisation ```Zcash_ExpandSeed```); derivation for $i < 2^{31}$ fails (ZIP-32, “Hardened-only child key derivation” and “Orchard child key derivation”).

Note the division of labour in the PRF call: the parent *chain code* c_{par} is the PRF key, and the parent spending key is message data. Continuing the tree therefore requires the pair — the 32-byte sk alone is a leaf capability, which is exactly the property the account path below exploits when it discards the chain code. Deployed derivation also caps the depth: the `depth` field is a byte incremented with `checked_add`, so a child of a depth-255 key cannot exist and the attempt fails with `Error::MaxDerivationDepth`.

Remark 1.4 (The general form: triples, and sibling instantiations). ZIP-32’s hardened-only child derivation in full generality takes a triple $(i, \text{lead}, \text{tag})$ and appends $\|[\text{lead}]\| \text{tag}$ to the PRF message; each distinct triple yields an independent output. For $\text{lead} = 0$ and $\text{tag} = []$ the encoding omits both fields, making it byte-compatible with the earlier definition of CKDh that lacked them; Orchard child derivation always uses $\text{lead} = 0$, $\text{tag} = []$, which is why Definition 1.3 shows no such suffix (ZIP-32, “Hardened-only child key derivation” and “Orchard child key derivation”). ZIP-32 defines no path notation for non-zero lead or non-empty tag. The deployed encoder implements the omission literally: `PrfExpand::with` skips $[\text{lead}]\| \text{tag}$ when the lead is zero and the tag empty. The same hardened-only process has two sibling instantiations outside the Orchard tree: *registered* key derivation (personalisation `ZIPRegistered_KD'`, lead byte `0xAC`, subtree rooted at `ZipNumber'` from an IKM of $[\text{len}]\| \text{ContextString}\| [\text{len}]\| S$) and the deprecated *ad-hoc* derivation (personalisation `ZcashArbitraryKD'`, lead byte `0xAB`) (ZIP-32, “Specification: Registered key derivation” and “Specification: Ad-hoc key derivation (deprecated)”). They matter here only as neighbours in the domain-separation registry of Table 1.

1.3 The standard account path

Definition 1.5 (Orchard account path). The account-level key for account number $\text{account} \in \{0, \dots, 2^{31} - 1\}$ is derived along the path $m_{\text{Orchard}}/32'/\text{coin_type}'/\text{account}'$:

$$(\text{sk}_{\text{acct}}, c_{\text{acct}}) := \text{CKDsk}(\text{CKDsk}(\text{CKDsk}(\text{MKGh}^{\text{Orchard}}(S), 32'), \text{coin_type}'), \text{account}'),$$

after which the wallet keeps sk_{acct} and discards c_{acct} . The *purpose* level is $32' = 0x80000020$, registered per BIP 43; the *coin type* follows the SLIP 44 registry — 133 on Mainnet, 1 on Testnet and Regtest (all testnets share coin type 1) — and accounts are numbered sequentially from 0. Wallets MUST support this path for any account below 2^{31} (ZIP-32, “Key path levels” and “Orchard key path”).

The path terminates at the account. Addresses are not further path children: each account exposes 2^{88} diversified addresses per scope, images of the diversifier indices under FF1 (ZIP-32, “Orchard key path”; the construction is in the *Ironwood Guide*). Nor is there a change level between account and address, where BIP-44-style transparent paths put one: shielded change returning to the originating account is indistinguishable on-chain from any other output, so a change subtree would separate nothing an observer can see, and the separation the wallet itself needs — telling its own change apart from external receipts — is achieved one level lower by the *internal key derivation* of §1.5 (ZIP-32, “Key path levels”).

Deployed path derivation. Function `SpendingKey::from_zip32_seed(seed, coin_type, account)` rejects $\text{coin_type} \geq 2^{31}$ with `Error::InvalidChildIndex`, runs `ExtendedSpendingKey::from_path` over the three hardened indices of Definition 1.5, and returns only the 32-byte account spending key — the chain code is discarded, so the wallet layer retains no Orchard child-derivation capability below the account level. Above it, `UnifiedSpendingKey::from_seed` stores that plain `SpendingKey`, and the unified full viewing key’s Orchard component is the `FullViewingKey` computed from it. The zip32 crate types the path levels: `AccountId` is a 31-bit integer whose `TryFrom<u32>` rejects values at or above 2^{31} and which always enters paths hardened, and `ChildIndex` is hardened-only — `from_index` returns `None` unless the hardened bit is set, and `hardened(v)` asserts $v < 2^{31}$ and stores $v + 2^{31}$. A distinguished index `ChildIndex::PRIVATE_USE = 0x7fffffff'` is reserved for private-use subtrees; `zcashd`’s legacy Sapling account uses it, and we find no Orchard use (ZIP-32, “Sapling key path”; zip32 0.2.0, `src/lib.rs`).

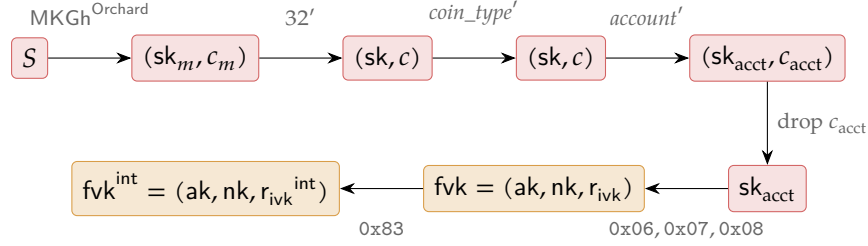


Figure 1: The deployed Orchard key path. The first horizontal arrow is master-key generation; each subsequent arrow in the top row is one CKDsk call (Definition 1.3): a PRFexpand invocation with lead byte 0x81, keyed by the parent chain code. The chain code travels only as far as the account node (sk_{acct}, c_{acct}) ; the wallet layer keeps the bare sk_{acct} , from which the external full viewing key fvk (*Ironwood Guide*, lead bytes 0x06/0x07/0x08) and the internal fvk^{int} (Definition 1.6, lead byte 0x83) both derive. Red marks spend capability, orange full-viewing capability, per the series palette.

1.4 From the account key to addresses: the layer already built

Everything below sk_{acct} is the hierarchy the *Ironwood Guide* constructs in “The recipient: keys and addresses”, and we do not re-derive it: ask, nk, r_{ivk} from PRFexpand lead bytes 0x06, 0x07, 0x08 with the ToScalar/ToBase reductions and their bias analysis; the sign-canonicalised $ak = [ask]G^{SpendAuth}$; $fvk = (ak, nk, r_{ivk})$; $ivk = \text{Commit}_{r_{ivk}}^{ivk}(\text{Extract}(ak), nk)$; the (dk, ovk) split of a single PRFexpand output with lead byte 0x82; and the FF1 diversified addresses with default index 0 (protocol specification §4.2.3, “Orchard Key Components”).

What ZIP-32 adds at the wallet level is the capability boundary of the full viewing key. The holder of fvk obtains, for *both* the external and the internal scope of §1.5: the incoming viewing keys, the diversifier keys and outgoing viewing keys, and hence every diversified address of the account; the *index* behind any of its addresses, since the diversifier map is a keyed permutation and $\text{FF1-AES256}_{dk}^{-1}(d)$ — FF1 decryption under the empty tweak — recovers the index; and the scope of any of its addresses, by trying external then internal. Two contrasts with Sapling sharpen the picture (ZIP-32, “Orchard diversifier and OVK derivation”): the diversifier key derives from the *full viewing key* rather than the extended spending key, so viewing capability suffices to locate an address’s position in the sequence; and all 2^{88} diversifiers are valid, so no rejection loop exists and the default diversifier is d_0 . What the full viewing key can *not* do: derive ask , or derive any child account — the former by DL hardness for $ak = [ask]G^{SpendAuth}$, the latter because only hardened derivation exists and no extended FVK is defined (§1.2). Finally, no mechanism derives an outgoing viewing key per *diversified address*: payments are sent from accounts, with external or internal scope, not from individual addresses (ZIP-32, “Orchard diversifier and OVK derivation”, citing the ZIP-316 rationale “Usage of Outgoing Viewing Keys”).

1.5 Internal keys: change and auto-shielding

The path of Definition 1.5 reserved no change level, so the separation between externally visible receiving and wallet-internal operations — change outputs, and the auto-shielding of transparent funds — happens below the account instead: every account key yields a second, *internal* full viewing key. The mechanism was implemented in zcashd as part of ZIP-316 support and applies to keys that are children of the account level, i.e. account keys (ZIP-32, “Orchard internal key derivation”).

Definition 1.6 (Orchard internal key derivation). Given an external full viewing key $fvk =$

(ak, nk, r_{ivk}) , let $K := \text{LE}_{256}(r_{ivk})$ and

$$r_{ivk}^{\text{int}} := \text{ToScalar}(\text{PRFexpand}_K(0x83 \parallel \text{I2LEOSP}_{256}(ak) \parallel \text{I2LEOSP}_{256}(nk))) \in \mathbb{F}_{p_{\text{Vesta}}},$$

$$\text{fvk}^{\text{int}} := (ak, nk, r_{ivk}^{\text{int}}),$$

i.e. $\text{DeriveInternalFVK}^{\text{Orchard}}$ modifies *only* r_{ivk} (ZIP-32, “Orchard internal key derivation”; protocol specification §4.2.3). The downstream keys then follow the standard FVK-level rules of the *Ironwood Guide*, applied to fvk^{int} : $\text{ivk}^{\text{int}} := \text{Commit}_{r_{ivk}^{\text{int}}}^{\text{ivk}}(\text{Extract}(ak), nk)$ and $(dk^{\text{int}}, ovk^{\text{int}})$ from the $0x82$ split keyed by $\text{LE}_{256}(r_{ivk}^{\text{int}})$. The *expanded internal spending key* is $(ask, nk, r_{ivk}^{\text{int}})$ — the same ask — and no 32-byte internal spending key exists (ZIP-32, “Orchard internal key derivation”).

The shape is deliberately that of the (dk, ovk) derivation: the same key K , the same message body $\text{I2LEOSP}_{256}(ak) \parallel \text{I2LEOSP}_{256}(nk)$, a different lead byte. Its consequences: ak and nk are unchanged (the protocol specification notes $ak^{\text{int}} = ak$ and $nk^{\text{int}} = nk$, §4.2.3), so spend authorisation and nullifier derivation are common to both scopes, while ivk , dk , and ovk are re-derived from r_{ivk}^{int} . The construction does not explicitly enforce that the two scopes’ derived keys or addresses differ; equality would be a negligible cryptographic collision, not an impossible case. Except with that negligible probability, a note addressed under the internal scope does not trial-decrypt under the external ivk (*Ironwood Guide*, “Delivery: the note reaches its owner”), which is precisely the separation a wallet wants when it hands its external incoming viewing key to a payment processor. Note that the internal FVK is *computable from* the external FVK — the derivation consumes only (ak, nk, r_{ivk}) — so disclosing the full viewing key disclosed both scopes all along; the boundary runs at the incoming-viewing tier, not the full-viewing tier. Unlike its Sapling counterpart, the construction does not rest on non-hardened derivation, which is undefined for Orchard (ZIP-32, “Orchard internal key derivation”).

Deployed, the `zip32` crate’s `Scope` enum $\{\text{External}, \text{Internal}\}$ names the two scopes and `orchard` re-exports it; `FullViewingKey::rivk(Scope)`, `to_ivk`, `to_ovk`, and `derive_internal` implement Definition 1.6. The change path is concrete: `zcash_client_backend` sends Orchard change to `ufvk.orchard().address_at(0u32, Scope::Internal)` — the internal-scope address at diversifier index 0.

The transparent pool needs the same external/internal ovk split — its auto-shielding transactions carry shielded outputs whose outgoing ciphertexts want a viewing key — but P2PKH keys (pay-to-public-key-hash: the output script pays to the hash of a single public key) have no protocol-level ovk to scope, so ZIP-316 synthesises the pair.

Definition 1.7 (Transparent internal and external OVK). For a transparent P2PKH account with BIP-32 account-level extended public key (c, pk) — chain code c , keying PRFexpand as a 256-bit key, and $\text{ser}(pk)$ the 33-byte compressed public key —

$$I_{ovk} := \text{PRFexpand}_c(0xd0 \parallel \text{ser}(pk)), \quad ovk_{\text{external}} := I_{ovk}[0..32), \quad ovk_{\text{internal}} := I_{ovk}[32..64)$$

(ZIP-316, *Deriving Internal Keys*; lead byte in Table 1).

Remark 1.8 (Byte-level normalisation between ZIP and code). ZIP-32 writes the PRF key as the bit sequence $\text{I2LEBSP}_{256}(r_{ivk})$ where PRFexpand consumes bytes; the code passes `rivk.to_repr()`, the 32-byte little-endian encoding. The two are identical under the LE_{256} convention this series uses, and we write LE_{256} throughout, as the *Ironwood Guide* already does. Likewise the ak bytes fed to the $0x82$ and $0x83$ messages are `SpendValidatingKey::to_bytes()`, the 32-byte RedPallas point encoding — which, because key generation forces the sign bit to zero (the canonicalisation of the *Ironwood Guide*), coincides with $\text{I2LEOSP}_{256}(\text{Extract}(ak))$, matching the specification’s

$I2LEOSP_{256}(ak)$ for $ak \in \{0, \dots, p_{\text{Pallas}} - 1\}$ (protocol specification §4.2.3 and the `AuthSignPublic` type). The nk bytes are the base-field representation.

1.6 Key validity and derivation failure

Definition 1.9 (Valid Orchard spending key). A 32-byte string sk is a *valid* spending key iff all three hold: (i) $ask \neq 0$; (ii) the external $ivk = \text{Commit}_{r_{ivk}}^{ivk}(\text{Extract}(ak), nk) \notin \{0, \perp\}$; (iii) the internal $ivk^{\text{int}} = \text{Commit}_{r_{ivk}^{\text{int}}}^{ivk}(\text{Extract}(ak), nk) \notin \{0, \perp\}$. Function `SpendingKey::from_bytes` returns `None` exactly when one of the three fails, and the specification’s key-generation algorithm mandates exactly these three discard conditions — “discard this key and repeat with a new sk ” (protocol specification §4.2.3, “Orchard Key Components”). `FullViewingKey::from_bytes` re-checks both scopes when parsing an FVK from bytes.

Remark 1.10 (Derivation failure has no specified recovery). ZIP-32 notes that a child spending key may yield an invalid external or internal full viewing key “with small probability” (ZIP-32, “Orchard child key derivation”) but prescribes no wallet behaviour when it does; the specification’s “repeat with a new sk ” addresses a freshly random key, not a fixed HD path, where there is nothing to redraw. The deployed stack hard-errors: `master` and `derive_child` return `Err(Error::InvalidSpendingKey)`, `from_path` propagates it, and `zcash_keys` surfaces it as `DerivationError::Orchard` (`orchard`, `src/zip32.rs`; `zcash_keys`, `src/keys.rs`); no layer implements a skip-to-next-index convention. This behaviour is normative in practice: an account index at which derivation fails simply has no key, and no deployed convention reassigns it. The event is negligibly rare — neither ZIP-32 nor the specification quantifies “small probability”, so Proposition 1.11 derives a bound — and the gap is theoretical.

Proposition 1.11 (Probability of an invalid key). *Model the PRFexpand outputs and the GroupHash-derived Sinsemilla generators as uniform and independent (the random-oracle model in which the Ironwood Guide analyses both primitives). A uniformly random 32-byte sk then fails the conditions of Definition 1.9 with probabilities*

$$\Pr[ask = 0] < 2^{-254}, \quad \Pr[ivk \in \{0, \perp\}] < 307 \cdot 2^{-254} < 2^{-245} \quad (\text{per scope}),$$

and is invalid with probability below 2^{-244} .

Proof. The scalar ask is the `ToScalar` image of the `0x06` expansion (the conditional negation that canonicalises ak preserves zeroness), so $ask = 0$ iff a uniform 512-bit integer reduces to 0 modulo p_{Vesta} . Exactly $\lceil 2^{512}/p_{\text{Vesta}} \rceil$ of the 2^{512} inputs do, and $p_{\text{Vesta}} > 2^{254}$ gives $\lceil 2^{512}/p_{\text{Vesta}} \rceil < 2^{258}$, hence $\Pr[ask = 0] < 2^{258} \cdot 2^{-512} = 2^{-254}$.

For ivk , the two memberships separately. *Zero*: conditioned on the Sinsemilla hash not returning $\perp$, the commitment point is the hash point plus $[r_{ivk}]H_D$ — the blinding term of the *Ironwood Guide*’s “Sinsemilla: hash, commitment, and short forms” — uniform on the p_{Vesta} -element Pallas group when the blinding scalar is uniform and $H_D \neq O$. Here r_{ivk} is the `ToScalar` image of a uniform 512-bit string, so the probability of any particular scalar is at most $\lceil 2^{512}/p_{\text{Vesta}} \rceil / 2^{512} < 2^{-254}$. The extractor sends only the identity to 0, because $y^2 = 0^3 + 5$ has no solution in $\mathbb{F}_{p_{\text{Pallas}}}$ — 5 is a non-square there (protocol specification §5.4.9.7, note) — so $\Pr[ivk = 0] < 2^{-254}$. *Bottom*: the $\perp$ output arises only inside the Sinsemilla hash, since the blinding addition is complete (§5.4.8.4). On the 510-bit $\text{Commit}_{r_{ivk}}^{ivk}$ message — two 255-bit field encodings, $\lceil 510/10 \rceil = 51$ ten-bit chunks — the hash performs $2 \cdot 51 = 102$ incomplete additions (§5.4.1.9). Each fails only when one operand is the identity or shares the other’s x -coordinate. To avoid assuming uniformity after conditioning on prior success, consider each round’s unconditioned complete-law expression

$(A + P) + A$. Conditional on the message and table points, its accumulator is $A = [2^i]Q + C_i$, uniform as the independent generator Q varies. For $P \neq O$, exceptional incomplete additions require $A \in \{O, P, -P, -P/2\}$; the separate event $P = O$ costs at most $1/p_{\text{Vesta}}$. Thus the 51 rounds cost at most $255/p_{\text{Vesta}}$ by a union bound, without independence between rounds. Including the negligible event that the blinding base is the identity still fits the conservative bound $\Pr[\text{ivk} \in \{0, \perp\}] < 307 \cdot 2^{-254} < 2^{-245}$ per scope. The internal scope repeats the argument with the independent $r_{\text{ivk}}^{\text{int}}$ (the 0x83 expansion). In total, $\Pr[\text{sk invalid}] < (1 + 2 \cdot 307) \cdot 2^{-254} < 2^{10} \cdot 2^{-254} = 2^{-244}$. $\square$

The bound makes ZIP-32’s “small probability” concrete: below 2^{-244} per derived key, and below 2^{-242} for the union over the four nodes of the standard account path (master included, Definition 1.2). This is a random-oracle-model bound, including randomness over the generators, not a measured frequency for the one deployed set of fixed generators. Moreover, a concrete input exhibiting $\perp$ would constitute a nontrivial discrete-log relation among the Sinsemilla generators (*Ironwood Guide*, the Sinsemilla collision-resistance proof), so none is expected ever to be observed.

1.7 Fingerprints and tags

Two identifiers name objects of this section without revealing them; both are BLAKE2b-256 digests with dedicated personalisations.

Definition 1.12 (Orchard FVK fingerprint and tag). For a full viewing key $\text{fvk} = (\text{ak}, \text{nk}, r_{\text{ivk}})$, the *fingerprint* is

$$\begin{aligned} \text{FVFP}(\text{fvk}) &:= \text{BLAKE2b-256}(\text{``Zcash0rcharDFVFP''}, \text{fvk}_{\text{raw}}), \\ \text{fvk}_{\text{raw}} &:= \text{I2LEOSP}_{256}(\text{ak}) \parallel \text{I2LEOSP}_{256}(\text{nk}) \parallel \text{I2LEOSP}_{256}(r_{\text{ivk}}), \end{aligned}$$

the digest of the 96-byte raw FVK encoding fvk_{raw} (protocol specification §5.6.4.4). The *tag* is its first 4 bytes and MUST NOT be assumed unique; the master key’s parent tag is the 4 zero bytes (ZIP-32, “Orchard Full Viewing Key Fingerprints and Tags”). Deployed as `FvkFingerprint` and `FvkTag`; child derivation stamps each child with `parent_fvk_tag` = the tag of the parent’s FVK.

Definition 1.13 (Seed fingerprint). For a seed S of 32 to 252 bytes, with $[\text{length}(S)]$ its single length byte,

$$\text{SeedFP}(S) := \text{BLAKE2b-256}(\text{``Zcash_HD_Seed_FP''}, [\text{length}(S)] \parallel S);$$

the string form is Bech32m (the checksummed base-32 string encoding of BIP-350: a human-readable part, a separator 1, and base-32 data characters ending in a six-character checksum) with human-readable part `zip32seedfp`, and no short tag is defined; the length prefix was chosen to match `zcashd`’s implementation (ZIP-32, “Seed Fingerprints”). Deployed, `SeedFingerprint::from_seed` returns `None` outside the 32..252-byte bounds, and `zcash_client_sqlite` uses the fingerprint to bind each stored account to the seed that produced it.

1.8 Extended-key encoding and test vectors

Definition 1.14 (Orchard extended spending key encoding). An extended spending key at depth depth , child index i (hardened offset included), with parent tag tag_{par} , serialises as the 73-byte string

$$\text{xsk} := \text{I2LEOSP}_8(\text{depth}) \parallel \text{tag}_{\text{par}} \parallel \text{I2LEOSP}_{32}(i) \parallel c \parallel \text{sk} \quad (1 + 4 + 4 + 32 + 32 \text{ bytes}),$$

with the master key at depth 0, tag $0x00000000$, index 0. The Bech32 human-readable parts are `secret-orchard-extsk-main` (Mainnet) and `secret-orchard-extsk-test` (Testnet) (ZIP-32, “Orchard extended spending keys”).

ZIP-32 defines this encoding “for completeness” and expects most wallets to expose only the plain `sk` encoding of the protocol specification (§5.6.4.5) — consistent with the deployed stack, which discards the chain code at the account level (§1.3). Indeed we find *no* deployed `librustzcash` code path that emits or parses the Bech32 form: the 73-byte layout is exercised only by the orchard crate’s embedded test-vector checks, and `zcash_keys` stores the raw spending key inside the unified spending key encoding (`orchard`, `src/zip32.rs`, `src/test_vectors/zip32.rs`; `zcash_keys`, `src/keys.rs`). We classify the encoding as specified but unused in the scanned stack.

Remark 1.15 (Test vectors). The ZIP-32 Orchard vectors live in `zcash-test-vectors` under `orchard_zip32`; the orchard crate embeds four of them — the nodes $m, m/1', m/1'/2', m/1'/2'/3'$ from the 32-byte seed $0x00, \dots, 0x1f$ — each giving `sk`, `c`, the 73-byte `xsk`, and the FVK fingerprint (ZIP-32, “Test Vectors”; `orchard`, `src/test_vectors/zip32.rs`, `src/zip32.rs`). They do *not* cover the standard path of Definition 1.5. Separate key-component vectors ($\text{sk} \rightarrow \text{ask}, \text{ak}, \text{nk}, \text{r}_{\text{ivk}}, \text{ivk}$ and the internal-scope components) sit in `src/test_vectors/keys.rs`, exercised by the `keys.rs` tests.

Example 1.16 (The standard path, worked). No published vector covers the path of Definition 1.5, so we compute one, from the embedded vectors’ seed $S = 0x00, \dots, 0x1f$, for Mainnet (`coin_type = 133`) and account 0:

m	<code>sk</code>	<code>7eee3c1017870990a3dd6891b82f80be8976c1e7dc20d60817a5e88e8b2cd4b8</code>
	<code>c</code>	<code>ab8b7a00509ef20e469b5292b61d474b7cfffcb1657924cda720250ae40526677</code>
$m/32'$	<code>sk</code>	<code>93af0a87c2e4704d8825e145557c6a5d4dcbc9016170e276a2080a2dc9eb0f87</code>
	<code>c</code>	<code>6e1e808d410d6d1f84e62e67c8b60e80e5ee4de5021d575c0433a625d609c45c</code>
$m/32'/133'$	<code>sk</code>	<code>149fcee594a2a03de277ab919f22611d0ef6aa8e8e9a0bef32f01a0bcb0e8c4c</code>
	<code>c</code>	<code>07d13fdeccd21da16b467f8e785edc32f74ceeb541bede0394647f10fdb7114f</code>
$m/32'/133'/0'$	<code>sk</code>	<code>b67d8d87cab9189500afb45dbca9f92c924c1de9d9ee1451be4a78313cb223b4</code>
	<code>c</code>	<code>7ff0a6392e144d4c8f45bca21fb7827b508446ac069890aefbebbe388b7a84c0</code>

The bottom `sk` is exactly what `SpendingKey::from_zip32_seed` returns for $(S, 133, 0)$; the bottom `c` is what it discards. The account key’s FVK fingerprint (Definition 1.12) is

`da8dab741dc4f6ff8964fbaa8840270caf47d286e9a0722e6c89e7f2c38189d5`

(tag `da8dab74`), its parent’s fingerprint begins `7010c19b`, and the account’s 73-byte encoding of Definition 1.14 is therefore

$$\text{xsk} = 03 \parallel 7010c19b \parallel 00000080 \parallel c \parallel \text{sk}, \quad 00000080 = \text{I2LEOSP}_{32}(0'),$$

with `c` and `sk` the bottom table row. The numbers are script-computed per the series conventions: a standalone reimplementation of $\text{MKGh}^{\text{Orchard}}$ and `CKDsk` first reproduces all four embedded vectors byte for byte (`sk`, `c`, `xsk`); the orchard crate, driven as a library, reproduces their FVK fingerprints, returns the same account `sk` from `from_zip32_seed`, and validates every node above per Definition 1.9.

1.9 The PRFexpand lead-byte registry

One PRF underlies every derivation above and in the *Ironwood Guide*, but domain separation is per PRF *key*, not protocol-wide: under a fixed key, distinct lead bytes provide domain-separated pseudorandom outputs, while separate consumers may reuse a byte under different keys. Sapling derives rcm and esk from lead bytes 0x04 and 0x05 with no ρ suffix — the assignment swapped relative to Orchard’s, which the specification attributes to an oversight (protocol specification §4.7.3, note) — and the Sapling instantiation of ZIP-32 intentionally reuses 0x00–0x02 from the Sapling key components under a different key. Table 1 collects the accounting for the Orchard, Orchard-ZIP-32, and ZIP-316 bytes this series constructs; the Sapling-only bytes (0x00–0x03, 0x10–0x18, of which §2.2 cites 0x10 and 0x16) are accounted in the specification’s list (protocol specification, the PRF^{expand} accounting under “Pseudo Random Functions” and §4.2.3; ZIP-32). The Sprout instantiation of ZIP-32 reserves the lead byte 0x80, the personalisation “ZcashIP32_Sprout”, and the HRPs zxsprout/zxtestsprout — values the ZIP says SHOULD NOT be used in future Zcash-related specifications (ZIP-32, “Values reserved due to previous specification for Sprout”). All Orchard and ZIP-32 constants of those derivations match between the specification text and zcash_spec 0.2.1 — the version locked by both the orchard and librustzcash workspaces (Cargo.lock in each; likewise zip32 0.2.0). The additional Ironwood 0x0B derivation is implemented directly in orchard 0.15.0, src/note.rs, not in that older zcash_spec registry.

Lead byte	PRF key	Message after the lead byte	Output
0x04	rseed	ρ	esk
0x05	rseed	ρ	legacy rcm
0x06	sk	—	ask
0x07	sk	—	nk
0x08	sk	—	r_{ivk}
0x09	rseed	ρ	ψ
0x0B	rseed	pre_rcm[1..137)	Ironwood rcm
0x80	Sprout ZIP-32 child (reserved; SHOULD NOT reuse)		
0x81	c_{par}	$sk_{par} \parallel I2LEOSP_{32}(i)$	$sk_i \parallel c_i$
0x82	$LE_{256}(r_{ivk})$ (per scope)	$ak \parallel nk$	$dk \parallel ovk$
0x83	$LE_{256}(r_{ivk})$	$ak \parallel nk$	r_{ivk}^{int}
0xAB	ad-hoc ZIP-32 child (deprecated)		
0xAC	registered ZIP-32 child		
0xD0	c (chain code)	$ser(pk)$	$ovk_{ext} \parallel ovk_{int}$

Table 1: Lead-byte registry of PRFexpand messages. Field arguments use $I2LEOSP_{256}$; points use their compressed encodings (the canonical ak encoding coincides with its x -coordinate encoding). Here ρ is the 32-byte nullifier of the Action’s spent note (*Ironwood Guide*, “Delivery: the note reaches its owner”). Rows 0x04–0x09 are constructed in the *Ironwood Guide* and carry the Orchard meanings of 0x04/0x05. The 0x0B row uses the 137-byte pre_rcm defined in that volume’s “The note seed: Ironwood derivations (lead byte 0x03)”, with its first byte removed because the column already lists it (ZIP-2005). Sapling reuses both bytes, without the ρ suffix and with the outputs swapped (0x04 $\rightarrow$ rcm, 0x05 $\rightarrow$ esk), under its own per-note rseed keys (protocol specification §4.7.3, note). Rows 0x81 and 0x83 are constructed in Definitions 1.3 and 1.6; 0xD0 in Definition 1.7; 0x80, 0xAB, 0xAC belong to other subsystems (ZIP-32) and appear for the completeness of the accounting.

2 Diversified and unified addresses (ZIP-316)

An address layer must reconcile two demands that pull apart. Each payee should receive their own address, mutually unlinkable with every other — yet all of them spendable by one key and recoverable from one seed; each shielded protocol solves this with *diversified addresses*. And Zcash is several protocols at once: transparent, Sapling, and Orchard receivers coexist, each with its own format, so a recipient wants to hand a sender *one* string from which the sender’s wallet picks the best receiver both sides support; ZIP-316 solves this above the protocols with the *unified* container encodings. The *Ironwood Guide* (“The recipient: keys and addresses”) already constructs the Orchard-side machinery — the key tiers, the FF1-AES256 diversifier map $d := \text{FF1-AES256}_{\text{dk}}(d_{\text{plain}})$, the diversified base $g_d = \text{GroupHash}^{\text{Pallas}}(\text{z.cash} : \text{Orchard-gd}, d)$, and the rationale for a keyed permutation over the index space — and we cite it throughout. This section adds what the wallet layer builds on top: the index discipline shared across Sapling, Orchard, and transparent derivation, the raw and unified encodings, viewing-key containers, and the deployed address-generation code.

At zips ad30e59, ZIP-316 is organised into revisions: Revision 0 is Active and is what the deployed stack implements; Revision 1 was withdrawn; Revision 2 is a Draft (ZIP-316, *Revisions*). This volume documents Revision 0 and states Revision 2 in §2.8, classified *specified, not deployed*; its reference implementation is librustzcash PR #2199, absent from both the baseline and the 2026-08-30 91f448b5 snapshot. Deployed-behaviour claims below are verified against librustzcash commit e30517e4 — the volume’s baseline (§7.7) — and the crates-io releases zcash_spec 0.2.1, zip32 0.2.0, sapling-crypto 0.7.0, and orchard 0.15.0 (exact versions pinned by the librustzcash workspace Cargo.lock); ZIP text is quoted from the zips repository at commit c6b7035.

2.1 Accounts: ZIP-32 key paths and wallet seeds

Every address below hangs off an *account*, a node of the ZIP-32 hierarchical-deterministic key tree. Sapling and Orchard share the path shape

$$m / \text{purpose}' / \text{coin_type}' / \text{account}',$$

with $\text{purpose} = 32$ (hardened index 0x80000020), the SLIP-44 coin type (133 on mainnet, 1 on all testnets), the account index counting from 0, and every level hardened (ZIP-32, *Key path levels*; *Sapling key path*; *Orchard key path*; deployed constants in zcash_protocol, constants/). Two absences are deliberate. There is no change level: shielded change pays the wallet’s own address and is indistinguishable from any other output, so the transparent external/internal split buys nothing (ZIP-32, *Key path levels*). And there is no non-hardened tail: Orchard ZIP-32 defines hardened-only derivation, instantiated with $\text{MKGDomain} = \text{"ZcashIP32Orchard"}$ and $\text{CKDDomain} = 0x81$ over the extended spending key (sk, c) ; a child may, with small probability, yield a spending key whose external or internal full viewing key is invalid, in which case derivation at that index simply fails — ZIP-32 prescribes no recovery, and the deployed stack surfaces an error (§1.6; ZIP-32, *Specification: Orchard key derivation*). Wallets MUST support this path for accounts 0 to $2^{31} - 1$ and MUST support generating each account’s default payment address; a stream of diversified addresses MAY be supported (ZIP-32, *Key path levels*; *Sapling key path*; *Orchard key path*). The seed feeding the tree MUST be 32–252 bytes with at least 256 bits of entropy: an attack on a lower-entropy mnemonic can be run against many wallets at once, by matching derived addresses and viewing keys against observed ones (ZIP-32, *Specification: Wallet seeds*).

2.2 Diversified addresses

A diversified address multiplies addresses without multiplying keys: one incoming viewing key ivk , one diversifier key dk , and one 88-bit *diversifier* d per address. The *Ironwood Guide* (“The recip-

ient: keys and addresses”) constructs the map and its design rationale — a keyed permutation of the index space, rather than a hash or a random draw; the wallet layer fixes the index discipline that ZIP-32 shares between Sapling and Orchard.

Definition 2.1 (Diversifier derivation, Sapling and Orchard). For an account’s diversifier key dk , the diversifier at *diversifier index* j is

$$d_j := \text{FF1-AES256}.\text{Encrypt}(dk, "", \text{LE}_{88}(j)), \quad j \in \{0, \dots, 2^{88} - 1\},$$

where FF1-AES256 is NIST SP 800-38G FF1 with a 256-bit AES key, radix 2, $\text{minlen} = \text{maxlen} = 88$, the tweak always the empty string, and $\text{LE}_{88}(j)$ (the ZIP’s I2LEBSP_{88}) the index’s 88-bit little-endian bit encoding; input and output are 88-bit sequences. The protocol specification abbreviates the keyed permutation as $\text{PRP}_K^d(d) := \text{FF1-AES256}_K("", d)$ (ZIP-32, *Sapling diversifier derivation; Orchard diversifier and OVK derivation; protocol specification §5.4.4, Pseudo Random Permutations*).

The Orchard diversified payment address at index j is then

$$d := \text{PRP}_{dk}^d(\text{LE}_{88}(j)), \quad g_d := \text{DiversifyHash}^{\text{Orchard}}(d), \quad \text{pk}_d := [\text{ivk}] g_d,$$

the pair (d, pk_d) , with the address at index 0 the *default* diversified payment address; the last step is the key-agreement $\text{KA}^{\text{Orchard}}.\text{DerivePublic}$ of the protocol specification, and the *Ironwood Guide* constructs all three maps. Diversifier indices SHOULD NOT be chosen at random, and an address generator MAY encode recoverable information in the index — the dk holder recovers it by FF1 decryption (protocol specification §4.2.3, *Orchard Key Components*).

The deployed cipher is the fpe crate’s $\text{FF1}\langle\text{Aes256}\rangle$ at radix 2; the orchard crate pins fpe 0.6 in its `Cargo.toml`. Both protocols encrypt the index bytes $j.\text{as_bytes}()$ wrapped as a `BinaryNumeralString` via `from_bytes_le` — the 11-byte little-endian index with bits least-significant-first within each byte, which is exactly $\text{I2LEBSP}_{88}(j)$ — under the empty tweak, and convert the 88-bit output back with `to_bytes_le` into the 11-byte diversifier, exactly LEBS2OSP_{88} .

The permutation earns its keep quantitatively. An 88-bit diversifier is too short to be cryptographically unguessable at the 128-bit security level, and diversifiers chosen at random begin to collide by the birthday bound as the number of draws approaches 2^{44} ; a keyed permutation reaches the full 2^{88} range with no repetition, provided the input index never repeats for a given node (protocol specification §5.4.1.6, notes). The remaining rationale — per-account sequences pseudorandom and mutually uncorrelated, and the issue-count leak a bare counter would cause — is the *Ironwood Guide*’s diversifier construction, cited above.

Remark 2.2 (FF1 parameter security). The property the design requires is that FF1-AES256, with the tweak fixed to the empty string, is a secure pseudo-random permutation. As parameterized here — radix 2, an 88-bit domain, 10 rounds per NIST SP 800-38G — it is unaffected by the DKLS2020 attacks on small-domain FF1: at their parameters $r = 5$ (half the number of rounds) and $n = 44$ (half the domain size in bits), the distinguishing attack has data complexity $2^{2n((r-1)-1/2)-n} = 2^{264}$ and time complexity $2^{2n((r-1)-1/2)} = 2^{308}$, no better than brute-forcing the 256-bit key (protocol specification §5.4.4).

Definition 2.3 (Diversifier validity and default diversifiers). For Sapling,

$$\text{DiversifyHash}^{\text{Sapling}}(d) := \text{GroupJHash}^{\text{URS}}(\text{"Zcash_gd"}, \text{LEBS2OSP}_{88}(d)),$$

valued in the prime-order subgroup of Jubjub — Sapling’s elliptic curve, carrying its keys and

commitments as Pallas carries Orchard’s — excluding zero, or $\perp$; the diversifier d_j is *valid* iff the result is not $\perp$, which for a given dk holds for approximately half of all indices j . The Sapling *default diversifier* is d_j for the least nonnegative j yielding a valid diversifier, and a Sapling account has approximately 2^{87} payment addresses (ZIP-32, *Sapling diversifier derivation*; *Sapling key path*). For Orchard, with $P := \text{GroupHash}^{\text{Pallas}}(\text{z.cash} : \text{Orchard-gd}, \text{LEBS2OSP}_{88}(d))$,

$$\text{DiversifyHash}^{\text{Orchard}}(d) := \begin{cases} \text{GroupHash}^{\text{Pallas}}(\text{z.cash} : \text{Orchard-gd}, "") & \text{if } P = O, \\ P & \text{otherwise,} \end{cases}$$

which never returns $\perp$: every one of the 2^{88} Orchard diversifiers is valid, the default diversifier is d_0 , and an account has exactly 2^{88} payment addresses (protocol specification §5.4.1.6; ZIP-32, *Orchard diversifier and OVK derivation*; *Orchard key path*).

The totality of the Orchard hash — and why it removes Sapling’s try-the-next-index rejection loop — is discussed in the *Ironwood Guide* (“The recipient: keys and addresses”). The residual Sapling failure rate of $\frac{1}{2}$ per index is not a curiosity: it resurfaces when a single index must serve several protocols at once (§2.7).

The two protocols also place dk at different depths of the key tree. In Sapling the diversifier key is a component of the *extended spending key*, derived down the ZIP-32 tree beside the spending authority:

$$dk_m := \text{truncate}_{32}(\text{PRFexpand}_{sk_m}([0x10])), \quad dk_i := \text{truncate}_{32}(\text{PRFexpand}_{I_L}([0x16] \parallel dk_{par})),$$

where I_L is the ZIP-32 child-derivation entropy (ZIP-32, *Sapling master key generation*; *Sapling child key derivation*). Orchard instead derives dk directly from the full viewing key (ak, nk, r_{ivk}) , as one half of a single PRFexpand output whose other half is ovk :

$$R := \text{PRFexpand}_{\text{LE}_{256}(r_{ivk})}(0x82 \parallel \text{I2LEOSP}_{256}(ak) \parallel \text{I2LEOSP}_{256}(nk)), \\ dk := R[0..32), \quad ovk := R[32..64),$$

the derivation the *Ironwood Guide* types in full ($\text{PRFexpand}_K(t)$ is BLAKE2b-512 with personalisation Zcash_ExpandSeed over $K \parallel t$; ZIP-32, *Conventions*; protocol specification §4.2.3). The consequence is capability parity with a simpler key structure: any holder of the Orchard full viewing key can determine the position of any diversifier in the account’s sequence, as in Sapling only a holder of the *extended* full viewing key can (ZIP-32, *Orchard diversifier and OVK derivation*).

Remark 2.4 (Reuse of r_{ivk}). The randomizer r_{ivk} serves twice: as the Commit^{ivk} trapdoor and as the PRFexpand key deriving dk and ovk . If dk or ovk is known to an adversary, this reuse prevents proving *perfect* hiding for Commit^{ivk} in this context, and modelling PRFexpand merely as a PRF is insufficient; the specification instead makes a joint assumption on Pallas scalar multiplication and PRFexpand (protocol specification §4.2.3, security note).

Inverting the permutation recovers the index:

$$j = \text{FF1-AES256.Decrypt}(dk, "", d).$$

Decryption alone proves nothing about ownership: every valid diversifier decrypts to *some* index under *every* diversifier key (sapling-crypto 0.7.0, `src/zip32.rs`, doc comment: “all valid diversifiers can be produced by all diversifier keys”), so the ownership test re-derives the address at the decrypted index under the candidate ivk and compares it with the given address. Deployed: the method `DiversifierKey::diversifier_index` performs the raw FF1 decryption in both protocols, and the checked variants are the Orchard `IncomingViewingKey::diversifier_index` and the Sapling `IncomingViewingKey::decrypt_diversifier`.

Remark 2.5 (Unlinkability, and an oracle to avoid). The security requirement: given two randomly selected shielded payment addresses of *different* spend authorities and a third address derivable from either, all three using different diversifiers, it is not possible to tell from which of the two authorities the third address derives. For Sapling this holds under DDH on the Jubjub prime-order subgroup when GroupJHash is modelled as a random oracle, via IK-CPA key privacy of ElGamal; the specification applies the same property and notes to Orchard (protocol specification §5.4.1.6). One pitfall is normative: an implementation **SHOULD** avoid providing a *chosen-diversifier* oracle — deriving a new address for a given ivk at an adversary-supplied diversifier — since such an oracle trivially breaks unlinkability for the other diversified addresses of that ivk (same section, final note).

2.3 Raw encodings

An Orchard shielded payment address encodes raw in 43 bytes,

$$\text{LEBS2OSP}_{88}(d) \parallel \text{pk}_d^*,$$

the 11-byte diversifier followed by the 32-byte star-encoding of pk_d ; a decoder rejects the encoding if the point fails to parse or is the zero point (protocol specification §5.6.4.2, *Orchard Raw Payment Addresses*). A Sapling address is likewise $11 + 32 = 43$ bytes under Jubjub ctEdwards compression, rejected if parsing fails, the encoding is non-canonical, or pk_d lies outside the prime-order subgroup minus zero — the exclusion of zero dates from specification version 2025.6.0 (protocol specification §5.6.3.1, *Sapling Payment Addresses*).

The Orchard raw *full viewing key* is 96 bytes,

$$\text{I2LEOSP}_{256}(\text{ak}) \parallel \text{I2LEOSP}_{256}(\text{nk}) \parallel \text{I2LEOSP}_{256}(\text{r}_{\text{ivk}}),$$

invalid if any component is a non-canonical field element, if ak is not a valid affine x -coordinate of a Pallas point, or if either the external or the internal ivk derived from it is 0 or $\perp$; the deployed `FullViewingKey::from_bytes` enforces exactly these checks, including both scopes' ivk validity (protocol specification §5.6.4.4, *Orchard Raw Full Viewing Keys*). The Orchard raw *incoming viewing key* is 64 bytes, $\text{dk} \parallel \text{I2LEOSP}_{256}(\text{ivk})$, where ivk **MUST** be a nonzero canonical Pallas base field element; deployed as `IncomingViewingKey::to_bytes, from_bytes` over a `NonZeroPallasBase` (protocol specification §5.6.4.3, *Orchard Raw Incoming Viewing Keys*).

None of these raw Orchard encodings has a string form of its own: no Bech32 or Bech32m encoding is defined for an individual Orchard address, incoming viewing key, or full viewing key — the unified encodings below are their only string carrier (protocol specification §5.6.4.2). Sapling, by contrast, retains standalone address strings (mainnet HRP `zs`, testnet `ztestsapling`; protocol specification §5.6.3.1), and the deployed renderer honours the asymmetry: the method `Receiver::to_zcash_address` emits a Sapling receiver as a bare `zs` address and a transparent receiver as a `t`-address, but an Orchard receiver only as a single-receiver unified address.

2.4 Unified addresses

A unified address (UA) bundles at most one *receiver* per type into one string, so the recipient distributes a single address and the sender's wallet chooses the pool. The choice is normative, by the *priority list* of Table 2: the Sender of a payment to a UA **MUST** use the receiver of the most preferred type it supports — Orchard over Sapling over P2SH (pay-to-script-hash: the output pays the hash of a script, revealed and satisfied when spending) over P2PKH. A wallet **MAY** let its user configure which receiver types it can send to, but **MUST NOT** allow the priority order itself to be changed, except via experiments (ZIP-316, *Encoding of Unified Addresses*; mirrored in protocol specification §5.6.4.1, *Unified Payment Addresses and Viewing Keys*).

Typecode	Item type	Receiver	UFVK item	UIVK item
0x00	transparent P2PKH	20	65	65
0x01	transparent P2SH	20	—	—
0x02	Sapling	43	128	64
0x03	Orchard	43	96	64

Table 2: Revision 0 item typecodes with byte lengths of the receiver, UFKV-item, and UIVK-item encodings. *Priority* is descending typecode among these four — 0x03 is most preferred — while encoding order is *ascending* typecode, deliberately not priority order. P2SH viewing-key items do not exist in Revision 0 (ZIP-316, *Encoding of Unified Addresses; Encoding of Unified Full/Incoming Viewing Keys*).

Definition 2.6 (Unified container, raw encoding). The raw encoding of a UA, UFKV, or UIVK is the concatenation, over its items in *ascending* typecode order, of

$$\text{compactSize}(\text{typecode}) \parallel \text{compactSize}(\ell_{\text{item}}) \parallel \text{item},$$

where `compactSize` is Bitcoin’s variable-length integer encoding — one byte for values below 253, otherwise a marker byte followed by the value in 2, 4, or 8 little-endian bytes. The deployed decoder additionally rejects typecode or length values above 0x02000000 (`MAX_COMPACT_SIZE` in `zcash_encoding, src/lib.rs`); ZIP-316 does not state that implementation limit as a normative container rule. A transparent receiver is the bare 20-byte hash — the P2PKH validating-key hash or the P2SH script hash — excluding the two version bytes of the base58check raw encoding (ZIP-316, *Encoding of Unified Addresses*).

Encoding in a fixed, non-priority order makes the byte encoding canonical; the deployed container preserves the parse order in `items_as_parsed` for round-trip serialization and separately offers `items()` sorted by preference (ZIP-316, *Rationale for Item ordering*).

Revision 0 imposes containment rules. A UA or UVK MUST contain at least one non-metadata item; a Revision 0 UA MUST contain at least one *shielded* item (typecode 0x02 or 0x03), and a Revision 0 UVK likewise (Revision 2 drops the requirement for UVKs). A consumer MUST reject: duplicate typecodes; both P2SH and P2PKH present; only metadata items; items out of ascending typecode order; trailing bytes; or any constituent item failing its own validation. A consumer MUST ignore items with unrecognized typecodes (except the MUST-understand metadata of §2.8), and every wallet must still parse a container holding unrecognized items (ZIP-316, *Requirements for both Unified Addresses and Unified Viewing Keys; Receivers*). There is intentionally no typecode for Sprout addresses or viewing keys: ZIP-211 disabled sending funds into the Sprout pool at Canopy (ZIP-316; protocol specification §5.6.4.1).

The deployed parser implements these rules once, identically for all three container types (`Address`, `Ufvk`, `Uivk`): the function `try_from_items_internal` rejects out-of-order typecodes (`InvalidTypecodeOrder`), duplicates (`DuplicateTypecode`), the P2PKH/P2SH combination (`BothP2phkAndP2sh`), and only-transparent containers (`OnlyTransparent`); typecodes 0x04 through 0x02000000 parse as `Typecode::Unknown`, and larger values fail as `InvalidTypecodeValue`.

Remark 2.7 (Unknown items versus the shielded requirement). The deployed code renders “at least one shielded item” as “not only transparent items”, and deliberately counts unknown typecodes as *not* transparent (comment in `zcash_address, kind/unified.rs`): a UA containing only a P2PKH receiver plus one unknown-typecode item parses successfully, although it contains no item of typecode 0x02 or 0x03. The ZIP’s parenthetical “currently Typecodes 0x02 and 0x03” suggests this forward-compatible reading is intended, but the ZIP nowhere states that an unknown item may satisfy the shielded requirement. We flag the divergence and do not resolve it.

Within a single container, all items obtained by hierarchical deterministic derivation SHOULD represent the same account — the ZIP-32 and BIP-44 account level (ZIP-316, *Requirements for both Unified Addresses and Unified Viewing Keys*). Above the parser, the shielded requirement is enforced at construction as well: `UnifiedAddress::from_receivers` returns `None` without a Sapling or Orchard receiver, and address generation fails with `ShieldedReceiverRequired`.

The deployed sendable-address surface is the enum `zcash_keys::address::Address`: a bare Sapling `PaymentAddress`, a `TransparentAddress` (P2PKH or P2SH), a Unified address, or a `Tex` address — the ZIP-320 transparent-source-only form wrapping a 20-byte P2PKH hash. A unified address can receive a memo iff it has a Sapling or Orchard receiver.

2.5 String encoding: F4Jumble and Bech32m

The raw container still needs a string form, and the obvious choice — Bech32m directly over the raw bytes — has a human-factors flaw at UA lengths. The Bech32m checksum does not *cascade*: a user who visually verifies only the beginning and end of a long string can miss an adversarial substitution of an internal receiver. ZIP-316 therefore passes the whole payload through *F4Jumble*, an unkeyed four-round Feistel construction approximating a random permutation of the entire byte string, before the checksummed encoding: any local change diffuses over the whole string, giving near-second-preimage resistance and nonmalleability — an adversary cannot cheaply alter an internal receiver while preserving a sufficiently long independently checked prefix. This is not authentication: anyone can jumble and checksum a wholly substituted address; a user must still obtain or verify it through a trusted channel. Three rounds would not suffice: a controlled xor-difference attack fixes the input to H_0 . The HRP, zero-padded to 16 bytes, is appended *inside* the jumbled payload, extending the same anti-malleation protection to the HRP itself and defending against cross-network confusion and address-versus-viewing-key confusion (ZIP-316, *Jumbling*).

Definition 2.8 (Unified string encoding). Let *raw* be the encoding of Definition 2.6 and *padding* the container’s HRP in US-ASCII, padded to exactly 16 bytes with zero bytes; the total length must satisfy $\ell(\text{raw}) \leq \ell_M^{\text{MAX}} - 16$. The string encoding is

$$\text{Bech32m}(\text{HRP}, \text{F4Jumble}(\text{raw} \parallel \text{padding})),$$

ignoring the BIP-350 length restrictions (Bech32m was chosen over Bech32 for better checksum behaviour at variable lengths). Decoding MUST: Bech32m-decode, rejecting bad checksums; reject out-of-range lengths; apply F4Jumble^{-1} ; verify and strip the trailing 16-byte padding; then parse the items, rejecting the whole encoding on any failure (ZIP-316, *Encoding of Unified Addresses*).

Container	Mainnet	Testnet	Regtest
Unified address	u	utest	uregtest
Unified FVK	uview	uviewtest	uviewregtest
Unified IVK	uivk	uivktest	uivkregtest

Table 3: Revision 0 human-readable parts, as deployed. ZIP-316 defines the mainnet and test-net columns (ZIP-316, *Revisions*); the regtest column is a `librustzcash` extension, present only in `zcash_protocol`, `constants/regtest.rs`.

Definition 2.9 (F4Jumble). Let M be a message of ℓ_M bytes with $38 \leq \ell_M \leq \ell_M^{\text{MAX}} := (2^{16} + 1) \cdot 64 = 4194368$, and let $\ell_H := 64$. Set $\ell_L := \min(\ell_H, \lfloor \ell_M/2 \rfloor)$ and $\ell_R := \ell_M - \ell_L$, and split $M = a \parallel b$ with $\ell(a) = \ell_L$. Then, with all letters local to this definition,

$$x := b \oplus G_0(a), \quad y := a \oplus H_0(x), \quad d := x \oplus G_1(y), \quad c := y \oplus H_1(d),$$

and $\text{F4Jumble}(M) := c \parallel d$. The round functions are

$$H_i(u) := \text{BLAKE2b-}(8\ell_L)(\text{"UA_F4Jumble_H"} \parallel [i, 0, 0], u),$$

$$G_i(u) := \text{first } \ell_R \text{ bytes of } \parallel_{j=0}^{\lceil \ell_R/\ell_H \rceil - 1} \text{BLAKE2b-512}(\text{"UA_F4Jumble_G"} \parallel [i] \parallel \text{I2LEOSP}_{16}(j), u),$$

each personalisation filling the 16-byte BLAKE2b personal field. The inverse applies the four XOR rounds in reverse order (ZIP-316, *Jumbling*).

The deployed implementation keeps the state as *left* $\parallel$ *right* with *left* the first $\min(64, \lfloor \ell_M/2 \rfloor)$ bytes, and runs $\text{g_round}(0)$ (*right* $\oplus= G_0(\text{left})$), emulating the XOF (extendable-output function; *Crypto Guide*, § “Hash functions and the random oracle model”) chunkwise in 64-byte blocks), $\text{h_round}(0)$, $\text{g_round}(1)$, $\text{h_round}(1)$, with the inverse running the same rounds in reverse; this matches the formulas above under the identification of (a, b, x, y, c, d) with the successive half-states. One bound differs: the crate accepts message lengths $48 \leq \ell_M \leq 4194368$ (VALID_LENGTH) — the Revision 0 minimum of 48 rather than Revision 2’s 38. ZIP-316 states the difference is unobservable for Revision 0-valid item sets, since no such set parses to 22–31 content bytes, and Revision 2 consumers MAY apply either minimum to Revision 0 containers (ZIP-316, *Rationale for length restrictions*; `f4jumble`, `src/lib.rs`). On the checksum side, the deployed encoder uses a custom checksum type `Bech32mZip316`, identical to `Bech32m` except that `CODE_LENGTH = 4194368 = \ell_M^{\text{MAX}}` lifts the BIP-350 length cap; decoding rejects any string whose HRP does not match a known network (`main`, `test`, `regtest`) for the container type.

The jumble is cheap at address lengths: 4 BLAKE2b compressions for $\ell_M \leq 128$ bytes — which covers both common UA shapes, since a Sapling-plus-Orchard UA has $\ell_M = 45 + 45 + 16 = 106$ bytes and a transparent-plus-Sapling-plus-Orchard UA $\ell_M = 22 + 45 + 45 + 16 = 128$ — rising to 6 for $\ell_M \leq 192$, 10 for $\ell_M \leq 256$, and 196608 at $\ell_M = \ell_M^{\text{MAX}}$. A consumer unable to handle maximum-length inputs MUST still support $\ell_M \geq 256$ bytes: two conformance levels (ZIP-316, *Efficiency*).

Display is normative too: if a UA or UVK is abridged, at least the first 20 characters MUST be shown. The worst case — the longest mainnet HRP, `uview`, plus the separator — leaves 14 characters, about 70 bits of jumbled data, below the 2^{125} security target, so 20 is an absolute minimum; and only *initial* characters count, since different displays may otherwise show disjoint substrings of the same string (ZIP-316, *Requirements for both Unified Addresses and Unified Viewing Keys*, and its rationale).

Example 2.10 (A unified address, end to end). We trace the pipeline on the smallest deployed shape — an Orchard-only mainnet UA — using the test vector derived from the 32-byte seed `000102...1f` at account 9, diversifier index $j = 1$ (`zcash_address`, `kind/unified/address/test_vectors.rs`). Every value below is script-computed; the key half runs the deployed `SpendingKey::from_zip32_seed` and `FullViewingKey::address_at`.

Receiver. The account key at `m/32'/133'/9'` yields the full viewing key, whose `PRFexpand` image under domain byte `0x82` gives `dk`; `FF1` encryption of the index bytes (Definition 2.1) gives `d1`, and the key agreement of §2.2 gives `pkd`:

<code>dk</code>	<code>82cc9d79742fe5ae9a142b9336a98677b154fe20401eb18998dbed915b0453ce</code>
<code>I2LEOSP₈₈(1)</code>	<code>010000000000000000000000</code>
<code>d₁</code>	<code>3fadf8edb20a3301e8260a</code>
<code>pk_d*</code>	<code>a311f4cbd54d7d6a76baac88c244b0b121c6dc22a8bcce15898e267829fc1e01</code>

Decryption inverts the permutation: `FF1-AES256.Decrypt(dk, "", d1)` returns the index bytes above, recovering $j = 1$ — subject to the ownership caveat of §2.2. The neighbouring diversifiers $d_0 = \text{e340636542e1c81285ed}$ and $d_2 = \text{987fd74a2256c596a66f83}$ share no visible structure with d_1 : the sequence is pseudorandom in the index.

Container. The UA has one item, so the raw encoding of Definition 2.6 is the two header bytes `compactSize(0x03) || compactSize(43) = 032b` followed by the 43-byte receiver $d_1 || \text{pk}_d^*$, 45 bytes in all. Appending the padded HRP, 75 ("u") then fifteen zero bytes, gives the 61-byte F4Jumble input M with $\ell_L = 30$, $\ell_R = 31$. The four rounds of Definition 2.9, letters as there:

$a = M[0..30)$	032b3fadbdf8edb20a3301e8260aa311f4cbdb54d7d6a76baac88c244b0b121
$b = M[30..61)$	c6dc22a8bccce15898e267829fc1e0175000000000000000000000000000000
$x = b \oplus G_0(a)$	e6f1529b03a0ad0a4209da8e4365ca14a5988bf1d03084a1c326f929c574ca
$y = a \oplus H_0(x)$	37b358beb784bb2b315fd8235c595628cfa9aa8045ef271728ddcfbf84dc81a
$d = x \oplus G_1(y)$	ad26d7fb043e09cbf18e8f4d80d2423ecb071719170463d644be4c76daab6b
$c = y \oplus H_1(d)$	98979f9a7d2dfce7b1616a8ad2b997f5f8787171b6fcf33b303f770b4aa64

The plaintext structure is legible in a and b — the TLV header 032b, the diversifier, the padding — and none of it in $\text{F4Jumble}(M) = c \parallel d$.

String. Regrouping the 61 jumbled bytes into 98 five-bit symbols (the last padded with two zero bits) and appending the 6-symbol Bech32m checksum yields, with the HRP and separator, $1 + 1 + 98 + 6 = 106$ characters:

u1nztelxna9h7w0vtpd2xjhxt4lpu8s9cmdl8n8vcr7actf2ny45nd07cv8cvuhuvw3axcp545v0ktg9cezuzx84jvhex8dk4tdvwhu4dl

Decoding reverses each stage per Definition 2.8, the inverse jumble running the same rounds in reverse order. The generating scripts reproduce this string through the deployed encoder (`zcash_address, unified::Address::encode`), re-encode all 60 vectors of the file above, and match all 8 vectors of `f4jumble, src/test_vectors.rs`, in both directions.

2.6 Unified viewing keys

A unified full viewing key (UFVK) and unified incoming viewing key (UIVK) reuse the container wholesale: the same item layout (Definition 2.6), the same containment rules, and the same jumbled Bech32m pipeline under their own HRPs (Table 3), with viewing-key items in place of receivers at the same typecodes (Table 2; ZIP-316, *Encoding of Unified Full/Incoming Viewing Keys*). The items:

- *Orchard* (0x03): the UFVK item is the 96-byte raw full viewing key of §2.3; the UIVK item is the 64-byte raw incoming viewing key $dk \parallel I2LEOSP_{256}(ivk)$.
- *Sapling* (0x02): the UFVK item is 128 bytes, $EncodeExtFVKParts(ak, nk, ovk, dk) = ak^* \parallel nk^* \parallel ovk \parallel dk$ under Jubjub compression, and SHOULD be derived from the account-level ZIP-32 extended full viewing key (ZIP-316; the encoding function is defined in ZIP-32). The UIVK item is 64 bytes, $dk \parallel I2LEOSP_{256}(ivk)$ with a zero *ivk* invalid (per protocol specification version 2025.6.0) — deliberately *more* than the protocol specification’s Sapling incoming viewing key, which is *ivk* alone: carrying *dk* is what makes UAs derivable from a UIVK.
- *Transparent P2PKH* (0x00): the UFVK item is 65 bytes, the chain code *c* followed by the 33-byte compressed public key — the last 65 bytes of the BIP-32 extended public key — at the account level $m/44'/coin_type'/account'$; the UIVK item has the same shape one level down, at the external change level $m/44'/coin_type'/account'/0$ (the deployed doc comment pins $m/44'/133'/account'/0$ for mainnet). Items of type P2SH (0x01) MUST NOT appear in Revision 0 UFVKs or UIVKs (ZIP-316, *Encoding of Unified Full/Incoming Viewing Keys*).

The deployed item types mirror the sizes exactly: `unified::Fvk::{Orchard([u8; 96]), Sapling([u8; 128]), P2pkh([u8; 65])}` and the `unified::Ivk` analogues (64/64/65).

Remark 2.11 (Viewing-key items are pruned). The transparent items deliberately omit the BIP-32 extended-key metadata — depth, parent fingerprint, child number — because the child number would reveal the wallet account index; deserialization reconstitutes dummy attributes (depth 3 for the account key, depth 4 for the IVK, parent fingerprint 0xffffffff). The Sapling item is pruned as well: the deployed `DiversifiableFullViewingKey` is exactly (ak, nk, ovk, dk) with no chain code, so round-tripping a UFVK loses the ability to perform non-hardened Sapling child derivation below the account — consistent with the ZIP, which only says the item SHOULD be derived from the account-level extended full viewing key.

The priority list of Table 2 does not govern viewing keys the way it governs receivers: a consumer of a UVK normally takes the *union* of the information in all items rather than selecting one, and although the current FVK and IVK types reuse the receiver typecodes, future viewing-key types need not correspond to receiver types (ZIP-316, *Encoding of Unified Full/Incoming Viewing Keys*). The deployed container nevertheless returns `items()` sorted in preference order for UVKs just as for UAs, keeping `items_as_parsed` for round-trips; the one place preference genuinely orders viewing-key material is the selection of outgoing viewing keys, which ZIP-316 specifies separately.

Namely, ZIP-316 refines which ovk a sender uses for its outgoing ciphertexts (the C_{out} construction of the *Ironwood Guide*, “Delivery: the note reaches its owner”): the Sender determines a sending account — preferably from the wallet API, else when all inputs belong to a single account — and a *preferred sending protocol*, the most preferred receiver type among the funds being sent, and SHOULD use the external or internal ovk (according to the transfer type) of that protocol from the account-level full viewing key; if no account can be determined, it SHOULD pick one source address of the preferred protocol and use its FVK’s ovk (ZIP-316, *Usage of Outgoing Viewing Keys*). Deployed: `UnifiedFullViewingKey::select_ovk(scope, input_sources)` picks the OVK of the highest-preference pool represented among the inputs.

The external/internal split just used comes with a strict capability separation: a full viewing key holder can derive both the external and the internal incoming viewing keys and ovks; the external IVK holder can derive neither the internal IVK nor any ovk; the external ovk holder can derive neither the internal ovk nor any IVK. For the shielded protocols this follows from onewayness of `PRFexpand` and of $\text{CRH}^{\text{ivk}}/\text{Commit}^{\text{ivk}}$, with the internal-key derivations specified in ZIP-32 (ZIP-316, *Deriving Internal Keys*). The Orchard internal full viewing key replaces only the commitment randomizer — the 0x83 derivation of Definition 1.6, from which the internal dk, ivk, and ovk follow by the standard component derivations. Transparent P2PKH accounts, having no protocol-level ovk, get the synthetic 0xd0 pair of Definition 1.7 (ZIP-316, *Deriving Internal Keys*).

A UIVK derives from a UFVK item by item, preserving typecodes: the Sapling FVK maps to its IVK per protocol specification §4.2.2, the Orchard FVK to its external-scope IVK per §4.2.3, and the transparent P2PKH account key to the external-change key by non-hardened child derivation at index 0; items with unrecognized typecodes MUST be dropped (as they must be again when deriving a UA from a UIVK). The deployed method `UnifiedFullViewingKey::to_unified_incoming_viewing_key` maps Sapling via `to_external_ivk`, Orchard via `to_ivk(Scope::External)`, transparent via `derive_external_ivk`, and sets the unknown-item list to empty, matching the ZIP (ZIP-316, *Deriving a UIVK from a UFVK*).

2.7 Deriving a unified address from a UIVK

A UA derives from a UIVK at a *single* diversifier index j , and j must be valid simultaneously for every item: the Sapling component requires j to yield a valid Sapling diversifier (Definition 2.3); the transparent component requires $0 \leq j \leq 2^{31} - 1$, since j becomes the BIP-44 non-hardened index-level child, giving the receiver path $m/44'/\text{coin_type}'/\text{account}'/0/j$; the Orchard component imposes no constraint (ZIP-316, *Deriving a Unified Address from a UIVK*). The deployed index map makes the transparent constraint concrete: the function `to_transparent_child_index` splits the 11 little-endian index bytes into the low 4 and the high 7, requires the high 7 to be zero,

and parses the low 4 as a `NonHardenedChildIndex`, whose maximum is $2^{31} - 1$ — exactly the ZIP range.

Which receivers a generated UA carries is decided per call, by the request type `UnifiedAddressRequest`: either `AllAvailableKeys` or `Custom over ReceiverRequirements`, one setting of `Require`, `Allow`, or `Omit` for each of `Orchard`, `Sapling`, and `P2PKH`; the constructor rejects any combination in which both `Orchard` and `Sapling` are `Omit`, since no shielded receiver could result. The preset constants are `ALLOW_ALL = (Allow, Allow, Allow)`, `SHIELDED = (Allow, Allow, Omit)`, and `ORCHARD = (Require, Omit, Omit)`. Intersecting two requirements prefers `Require` and `Omit` over `Allow` and errors on `Require` meeting `Omit`.

Generation at an index, `UnifiedIncomingViewingKey::address(j, request)`, derives each requested receiver at the same j : the `Orchard` receiver via `address_at(j)`, which is total and never fails; the `Sapling` receiver via the diversifiable IVK's `address_at(j)`, which yields no receiver at an invalid diversifier — an error only when `Sapling` is `Required`, a silent omission under `Allow`; the transparent receiver via `to_transparent_child_index` and public derivation, where an out-of-range index errors under `Require` and omits under `Allow`. The result must retain at least one shielded receiver. The search `find_address(j, request)` increments j only on `InvalidSaplingDiversifierIndex`, returns `DiversifierSpaceExhausted` on overflow past $2^{88} - 1$, and aborts on any other error; `default_address` is `find_address(0)`. The deployed *default unified address* therefore sits at the smallest index valid for all required receivers.

Remark 2.12 (Whose convention the default address is). ZIP-316 does not specify address-search semantics at all; the smallest-valid-index default above is a `zcash_keys` convention, not a ZIP mandate — only ZIP-32's per-protocol default diversifiers (Definition 2.3) are normative. The search's tolerance is also narrow: it skips an index only for an invalid `Sapling` diversifier, so with a `Required` transparent receiver a search that starts or arrives at an index of 2^{31} or beyond — or a failed BIP-32 child derivation, probability about 2^{-127} — aborts generation rather than advancing (`zcash_keys, keys.rs`).

Remark 2.13 (Index 0 and cross-wallet recovery). Diversifier index 0 fails to yield a valid `Sapling` diversifier with probability approximately $\frac{1}{2}$, so a wallet requiring a `Sapling` receiver often issues its first UA at a higher index — yet some wallets always generate a transparent `P2PKH` address at index 0. All Zcash wallets MUST therefore assume there may be transparent funds at diversifier index 0 of each ZIP-32 account, even if the wallet itself would never generate a UA at that index; reliable cross-wallet fund recovery depends on this (ZIP-316, note in *Deriving a Unified Address from a UIVK*).

2.8 Revision 2 (specified, not deployed)

Revision 2 of ZIP-316 is a Draft as of the zips snapshot c6b7035; everything in this subsection is classified *specified*, and none of it exists in the deployed crates — the reference implementation is `librustzcash` PR #2199, and a search of the local `zcash_address` sources finds none of the new HRP, typecodes, or expiry logic. Revision 2 adds (ZIP-316, *Revisions*; Revision 2 changelog):

- *New HRP*s. Unified addresses split into `zu` (shielded-only; transparent items prohibited) and `tu` (transparent-enabled); UFVKs use `uvf` and UIVKs `uvi`; `testnet` forms append `test` to each HRP.
- *Metadata items*. Typecodes `0xC0–0xFC` are reserved for metadata, which MUST NOT be selected as receivers and MUST NOT confer viewing authority. The subrange `0xE0–0xFC` is *MUST-understand*: a consumer that does not recognize such an item MUST treat the whole container as unsupported. Typecodes `0xFFFA–0xFFFF` are experimental, and registry additions come via 2xxx-series ZIPs (ZIP-316, *Metadata Items*; *Typecode Registry*; *Adding new types*).
- *Address expiration*. Typecode `0xE0` carries a little-endian u32 Address Expiry Height — the address is expired once the chain height exceeds it — and `0xE1` a little-endian u64 Unix-time

Address Expiry Time. A Revision 2 Sender MUST set a nonzero `nExpiryHeight` in transactions sent to an address that defines an Address Expiry Height; MUST NOT send if its normal `nExpiryHeight` would exceed the Address Expiry Height, which would leak the address’s expiry into the transaction; and SHOULD arrange expiry within 24 hours when only an expiry time is given. Expiry metadata is retained when deriving a UIVK and must not increase when deriving a UA; because shared expiry metadata can link diversified addresses, per-address variation is RECOMMENDED (ZIP-316, *Address Expiration Metadata*).

- *Viewing keys.* The at-least-one-shielded-item requirement is dropped for UVKs, and P2SH viewing-key items (BIP-388 wallet policies) are added (ZIP-316, Revision 2 changelog).

Remark 2.14 (A retroactive rule the deployed parser does not enforce). Revision 2 also legislates for Revision 0 strings: a Revision 0 container — identified by its `u/uview/uivk` HRP — MUST NOT contain MUST-understand typecodes, and consumers MUST reject violations (ZIP-316, *Metadata Items*). The deployed parser predates this amendment and does not enforce it: every typecode from `0x04` to `0x02000000` parses as ignorable `Typecode::Unknown`, so a `u`-prefixed address carrying an `0xE0` expiry item parses successfully today (`zcash_address, kind/unified.rs`). This is correct Revision 0 behaviour under the pre-amendment text and a violation of the Revision 2-amended text; we flag the divergence rather than resolve it.

3 Fees (ZIP-317)

For an ordinary wallet-created transaction, any fee is public: it is the transparent value the transaction leaves unclaimed. Consensus also permits a zero fee, and coinbase transactions do not pay one. For a shielded protocol this publicity imposes two requirements that an ordinary fee market does not. The fee must be *computable from public transaction fields alone* — a fee that depended on private data (which notes were spent, what change was returned) would leak that data through the amount paid (ZIP-317, *Requirements*) — and it should be *uniform across wallets*, because a wallet paying an idiosyncratic fee fingerprints every transaction it builds. ZIP-317 resolves both with a single *conventional fee*: a deterministic function of the transaction’s public shape that every wallet is expected to pay.

Remark 3.1 (A convention, not a consensus rule). The conventional fee is a wallet convention, not a consensus requirement. ZIP-317 (*Fee calculation*) states: “It is not a consensus requirement that fees follow this formula; however, wallets SHOULD create transactions that pay this fee, in order to reduce information leakage, unless overridden by the user.” Enforcement is economic, not validity-level: nodes are expected to relay, and the recommended block-template algorithm to preferentially mine, transactions paying at least the conventional fee (§3.6).

ZIP-317 is organised into *revisions* (Last-Updated 2026-06-26): Revision 0 is Active and is the deployed fee mechanism; Revision 1 (a fee contribution for Ironwood-pool Actions, NU6.3) and Revision 2 (memo bundles, depending on the undeployed ZIP 248) are both Draft. A contribution defined by a revision is included in all later revisions. This volume documents Revision 0 as the base rule and states Revisions 1–2 below. Although its document status remains Draft, Revision 1 is implemented and is the fee contribution used for active NU6.3 Ironwood bundles (§3.3); Revision 2 remains *specified, not deployed*.

3.1 The conventional fee

The unit of account is the *logical action*: a pool-neutral measure of how much verification work and chain space a transaction demands. Its inputs are public transaction fields defined in the protocol specification §7.1 (*Transaction Encoding and Consensus*): the total byte sizes `tx_in_total_size` and `tx_out_total_size` of the transparent `tx_in` and `tx_out` fields, and the counts `nJoinSplit`, `nSpendsSapling`, `nOutputsSapling`, `nActionsOrchard`, and — from v6 — `nActionsIronwood`.

Definition 3.2 (Logical actions, ZIP-317 Revision 0). The transaction’s pools contribute

$$\begin{aligned} \text{contribution}_{\text{Transparent}} &= \max\left(\left\lceil \frac{\text{tx_in_total_size}}{150} \right\rceil, \left\lceil \frac{\text{tx_out_total_size}}{34} \right\rceil\right), \\ \text{contribution}_{\text{Sprout}} &= 2 \cdot n\text{JoinSplit}, \\ \text{contribution}_{\text{Sapling}} &= \max(n\text{SpendSapling}, n\text{OutputsSapling}), \\ \text{contribution}_{\text{Orchard}} &= n\text{ActionsOrchard}, \end{aligned}$$

and *logical_actions* is the sum of the contributions of the applicable revision (ZIP-317, *Fee calculation*).

Definition 3.3 (Conventional fee). The conventional fee, in zatoshis, is

$$\text{conventional_fee} = \text{marginal_fee} \cdot \max(\text{grace_actions}, \text{logical_actions}),$$

with *marginal_fee* = 5000 zatoshis per logical action and *grace_actions* = 2 (ZIP-317, *Fee calculation*).

Remark 3.4 (NU7 does not redefine the conventional fee). The Draft ZIP 259 does not replace the ZIP 317 fee formula or its 5000-zatoshi marginal fee.

Nor is a logical action the unit used for NU7’s proposed block work limit. For example, the fee counts the larger of Sapling’s spend and output counts, whereas the proposed block limit counts their sum (ZIP 218). Paying the conventional fee therefore does not guarantee that a transaction fits in a particular block. The proposed block limits and allocation of fees between miners and the sustainability mechanism are separate matters, constructed in the *Consensus Guide*, §“NU7: 25-second blocks and bounded shielded work”; they do not add a field to an ordinary v5 or v6 payment.

The two byte-size divisors are the standard P2PKH sizes, chosen so that one typical transparent input or output weighs one logical action. The standard input size $150 = 36 + 110 + 4$ decomposes as the outputpoint (36: the 32-byte transaction id and 4-byte index naming the output being spent), the script — 1 overall-length byte, 1 signature-length byte, a 72-byte signature, 1 sighash-type byte (the flag selecting which parts of the transaction the signature commits to), 1 pubkey-length byte, a 33-byte pubkey, and 1 byte of margin, totalling 110 — and the sequence field (4). The standard output size $34 = 8 + 26$ decomposes as the value (8) and the script — 1 script-length byte plus the 25-byte P2PKH script (ZIP-317, *Rationale for the chosen parameters*).

Why logical actions, and why these constants. A naïve schedule would charge per input plus per output. But an Orchard Action spends one note *and* creates one (the *Ironwood Guide*, “One Action, one payment: the walkthrough”), and Orchard bundles are padded to a minimum shape (§3.2); charging inputs and outputs separately would penalise exactly that forced padding. Logical actions count an Action once and, via the max in each contribution, do not discriminate between pools (ZIP-317, *Rationale for logical actions*). The transparent contribution is size-based rather than count-based so that large scripts cannot be undercounted: a 2-of-3 P2SH multisig input is roughly 70% larger than a P2PKH input and is charged proportionally (ZIP-317, *Rationale for the chosen parameters*). The grace window *grace_actions* = 2 exists so that padding a one-action transaction to two actions — the standard behaviour of zcashd and the mobile SDKs — costs nothing; it also makes a low-value input worth sweeping: within the window, an input worth less than the marginal fee is still economic to include when a second input covers the payment

plus fee. Two is also *min_actions*, the smallest economically relevant change-producing transaction (ZIP-317, *Rationale for the chosen parameters*). Finally, *marginal_fee* = 5000 makes the minimal two-action fee 10,000 zatoshis, deliberately restoring the conventional fee that predated ZIP-313 (which had lowered it to 1,000).

Later revisions. Revision 1 (Draft, NU6.3) adds

$$\text{contribution}_{\text{Ironwood}} = n\text{ActionsIronwood};$$

Ironwood-pool Actions use the same Action structure as Orchard, so they are counted identically (ZIP-317, Revision 1). Revision 2 (Draft; ZIP-231 memo bundles, dependent on the Draft, undeployed ZIP 248 transaction format) adds

$$\begin{aligned} \text{contribution}_{\text{Memos}} &= \max(0, n\text{MemoChunks} - \text{free_memo_chunks}), \\ \text{free_memo_chunks} &= \begin{cases} 2 & \text{if } n\text{OutputsSapling} + n\text{ActionsOrchard} + n\text{ActionsIronwood} > 0, \\ 0 & \text{otherwise,} \end{cases} \end{aligned}$$

bounding the additional fee for a ZIP-231 memo bundle by 0.0019 ZEC (ZIP-317, Revision 2). Revision 1 is implemented in the deployed crates and applies to NU6.3 Ironwood bundles (§3.3); Revision 2 is not implemented.

3.2 The fee is computed on padded counts

The counts in Definition 3.2 are those of the final, serialised transaction — *after* the builder pads each shielded bundle to its minimum shape. The deployed padding rules are those of the crates `librustzcash` builds against (commit `e30517e4`: `sapling-crypto` 0.7.0 and `orchard` 0.15.0, per its `Cargo.lock`):

- *Sapling*: a Transactional bundle with any spend or output requested (or with `bundle_required` set) uses $\text{num_outputs} = \max(\text{outputs}, 2)$, the constant 2 being `MIN_SHIELDED_OUTPUTS`. Spends are never padded under `BundleType::DEFAULT`: num_spends equals spends when $\text{spends} > 0$ and 0 otherwise, so an output-only bundle has zero spend descriptions; the floor $\max(\text{spends}, 1)$ applies only when `bundle_required` is set. An empty, non-required bundle is omitted entirely.
- *Orchard*: a Transactional bundle with $\text{requested} = \max(\text{num_spends}, \text{num_outputs}) > 0$ (or `bundle_required`) uses $\text{num_actions} = \max(\text{requested}, \text{min})$, the floor min being the bundle type’s `pad_to_minimum` count, default `DEFAULT_MIN_ACTIONS` = 2. The max form of requested applies when the bundle’s flags enable cross-address transfers, as the baseline’s pre-NU6.3 bundles do (`Flags::ENABLED`); Remark 3.5 states the sum rule that replaces it when the flags disable them.

Consequently, under the default bundle types used by the wallet, any nonempty Sapling or Orchard bundle contributes at least two logical actions, and a one-input, one-output payment within a single shielded pool pays exactly the 10,000-zatoshi minimum. The low-level builders also expose unpadded bundle types, for which that two-action conclusion need not hold.

The builder wires the padded counts into the fee call: the method `get_fee` of the transaction builder passes the raw count of Sapling spends added, the Sapling output count via `BundleType::num_outputs` (padding to 2 when the bundle is nonempty), and the Orchard action count via `BundleType::num_actions` (padding to `DEFAULT_MIN_ACTIONS` = 2 when nonempty). The proposal-stage balance computation in `zcash_client_backend` does the same — additionally counting the prospective change outputs — using `sapling::builder::BundleType::DEFAULT` and `orchard::builder::BundleType::DEFAULT`, except for the current ZIP-318 canonical-crossing Ironwood bundle’s explicit unpadded choice. Fee estimation and final construction therefore agree on the shielded counts by construction.

Remark 3.5 (The flag-dependent Orchard action count). Release orchard 0.15.0 — the version the baseline Cargo.lock pins — makes the requested-action count flag-dependent: when a bundle’s flags disable cross-address transfers, a spend and an output can never share an Action, so $requested = num_spends + num_outputs$ replaces the max. Its documentation warns explicitly that ZIP-317 fee estimation must account for the larger count. The deployed builders disable cross-address transfers only for legacy-Orchard-pool bundles under NU6.3, whose bundle version mandates the restriction; every bundle the deployed wallet builds before NU6.3 keeps the flag enabled (Flags::ENABLED) and the max, so the fee arithmetic above is unchanged.

3.3 The deployed fee rule

Module `transaction::fees::zip317` of `zcash_primitives` implements Definition 3.3 as the type `FeeRule`; the constructor `FeeRule::standard()` instantiates the four ZIP constants,

$$\begin{aligned} \text{MARGINAL_FEE} &= 5000, & \text{GRACE_ACTIONS} &= 2, \\ \text{P2PKH_STANDARD_INPUT_SIZE} &= 150, & \text{P2PKH_STANDARD_OUTPUT_SIZE} &= 34, \end{aligned}$$

and the derived $\text{MINIMUM_FEE} = 10,000$ zatoshis is $\text{MARGINAL_FEE} \cdot \text{GRACE_ACTIONS}$. Non-standard parameters are gated behind the `non-standard-fees` feature and reject zero input or output sizes. The generic trait method is

```
fee_required(params, target_height,
transparent_input_sizes, transparent_output_sizes,
sapling_input_count, sapling_output_count,
orchard_action_count, ironwood_action_count) -> Zatoshis
```

The ZIP-317 implementation ignores `params` and `target_height` and computes

$$fee = 5000 \cdot \max(2, \max(\lceil \frac{t_{in}}{150} \rceil, \lceil \frac{t_{out}}{34} \rceil)) + \max(s_{in}, s_{out}) + a + i,$$

where t_{in} is the sum of the *known* transparent input sizes, t_{out} the sum of the transparent output sizes, s_{in}, s_{out} the Sapling spend and output counts, a the Orchard action count, and i the Ironwood action count (zero in the V5 transactions the wallet builds before NU6.3); overflow yields a `BalanceError`.

One term of Definition 3.2 is absent, a scope restriction rather than a divergence. There is no Sprout term: the `FeeRule` trait has no `JoinSplit` parameter at all, because the deployed wallet stack cannot construct Sprout spends; the $2 \cdot n_{JoinSplit}$ contribution is spec-only. The Ironwood term of Revision 1 is implemented: the parameter $i = \text{ironwood_action_count}$ above enters *logical_actions*, and is zero in the V5 transactions the wallet builds before NU6.3. The deployed rule is exactly ZIP-317 Revision 1 minus Sprout.

Transparent input sizing. Transparent inputs are sized along two paths. At *proposal* stage, a UTXO (unspent transaction output, the transparent pool’s spendable coin) whose `script_pubkey` is P2PKH is counted as `InputSize::STANDARD_P2PKH = Known(150)`; any unrecognised script yields `InputSize::Unknown`. An `Unknown` input makes the fee incomputable: `fee_required` returns `FeeError::UnknownP2shInputs`. The recognised forms are P2PKH, and multisig-in-P2SH where the redeem script is known. At *build* stage, the method `TransparentInputInfo::serialized_len` computes the P2PKH size exactly: $36 + (1 + 74 + 34) + 4 = 149$ bytes — the outpoint, a `scriptSig` of one `CompactSize` byte plus a 74-byte push of a 73-byte placeholder signature ($\text{MAX_SIG_SIZE} = 73$: a 72-byte DER signature plus the sighash-type byte) plus a 34-byte push of a 33-byte pubkey, and the sequence field — one byte under the ZIP’s 150, which includes a one-byte margin; P2SH multisig inputs are computed from the redeem script.

Remark 3.6 (149 versus 150). The proposal path therefore prices a P2PKH input one byte above its serialised size. The discrepancy is deliberate: the `FeeRule` documentation states that the rule “may slightly overestimate fees in case where the user is attempting to spend a large number of transparent inputs. This is intentional and is relied on for the correctness of transaction construction algorithms in the `zcash_client_backend` crate” (`zcash_primitives`, `transaction/fees/zip317.rs`). An overestimate can only leave a gratuity, never underfund the transaction.

Transparent output sizing. The trait method `OutputView::serialized_size` returns 8 (value) plus the script bytes plus the `CompactSize` length prefix; a P2PKH output is $8 + 1 + 25 = 34$ bytes, matching the ZIP constant.

The standard rule at the wallet layer. The crate `zcash_client_backend` exposes `StandardFeeRule`, an enum whose sole variant `Zip317` delegates to `FeeRule::standard()`, and the extension trait `Zip317FeeRule`, which exposes `marginal_fee()` and `grace_actions()` to the change strategies. This is the standard, default surface. A fixed-fee rule and its `SingleOutputChangeStrategy` are also available behind the `non-standard-fees` feature.

3.4 Change and dust under the fee rule

The fee rule does not act alone; it is consulted by a *change strategy* that turns a tentative selection of inputs and payments into a balanced transaction. Two ZIP-317 strategies exist: `SingleOutputChangeStrategy`, producing one change output, and `MultiOutputChangeStrategy`, splitting change across several notes according to a `SplitPolicy` so the wallet maintains a target number of spendable notes.

Change-pool selection. Change goes to Orchard if the transaction has any Orchard input or output; otherwise to Ironwood if it has any Ironwood input or output, so change from an Ironwood spend stays in that pool; otherwise to Sapling if it has any Sapling input or output; otherwise — fully transparent flows — to a caller-specified fallback pool. Once NU6.3 is active, the turnstile overrides an Orchard-preferred result to Ironwood unless the transaction spends Orchard notes and strictly less value returns to the pool than those notes remove from it. The source marks the strategy “naive” (TODO) (`zcash_client_backend`, `fees/common.rs`, `select_change_pool`).

The balance computation. The function `single_pool_output_balance` first computes *min_fee*, the fee required with *zero* change outputs. Then:

1. if $total_in < subtotal_out + min_fee$, it fails with `ChangeError::InsufficientFunds`, reporting $available = total_in$ and $required = subtotal_out + min_fee$ — the figure the input-selection loop of §3.5 feeds back;
2. at exact equality, when all flows are transparent and no change memo is requested, it emits no change output and the fee is *min_fee*;
3. otherwise it computes *max_fee* with the target change-output count and derives $change = total_in - (subtotal_out + total_fee)$.

In the shielded case a *zero-valued* change output is deliberately retained even at exact balance: omitting it would let an adversary who knows the recipients’ outputs conclude, from the absence of change, that the inputs sum exactly to payments plus fee; the zero-valued output also carries the change memo.

Dust policy. A change value below a dust threshold is handled by a `DustOutputPolicy`. The default is `(DustAction::Reject, None)`, with the unset threshold defaulting to the marginal fee, 5000 zatoshis (`default_dust_threshold = fee_rule.marginal_fee()`); `Reject` still permits exactly-zero change per the previous paragraph. The `AllowDustChange` action emits the dust change output anyway. The `AddDustToFee` action folds the dust into the fee — with a guard: if the resulting fee exceeds the computed fee by more than $10 \cdot \text{MINIMUM_FEE} = 100,000$ zatoshis the fold is not performed and the ordinary change output is kept, protecting the user from a mis-set threshold; the documentation notes that `AddDustToFee` inherently leaks more information.

Uneconomic inputs. An input is classified as dust iff its value is *at most* the marginal fee (`zcash_client_backend`, `fees/common.rs`, `check_for_uneconomic_inputs`). Dust is admitted only where it rides along free. Per pool p ,

$$\text{allowed}_p = \min(\#dust_p, (\#outputs_p + \#change_p) - \#nondust_p) \quad (\text{saturating at } 0),$$

i.e. dust inputs that do not increase the padded action count. Then, if the baseline logical-action count is below `grace_actions`, at most *one* extra dust input may be added free — pools tried in the order transparent, Sapling, Orchard, Ironwood — admitted iff the hypothetical action count with it does not exceed the baseline, where the hypothetical count is $\max(t_{\text{in}}, t_{\text{out}}) + \max(\text{padded Sapling spends}, \text{padded Sapling outputs}) + \text{padded Orchard actions} + \text{padded Ironwood actions}$ with transparent inputs assumed P2PKH. When the presence of a change output is uncertain, the minimum allowance over both manifests is used (`possible_change` construction, same file). Dust beyond the allowance is returned as `ChangeError::DustInputs`, which the input selector converts into an exclusion list (§3.5).

Remark 3.7 (Strictness divergence from the ZIP wording). The ZIP speaks of inputs “with value *below* the marginal fee” (ZIP-317, *Requirements and Rationale for the chosen parameters*), while the deployed check classifies value $\leq \text{marginal_fee}$ — a note of exactly 5000 zatoshis included — as dust, and the SQL layer (§3.5) likewise never offers notes with value ≤ 5000 . The `ChangeError::DustInputs` documentation itself acknowledges the determination is “potentially conservative”. We flag this as conservative code behaviour against the ZIP wording, not an error in either (`zcash_client_backend`, `fees/common.rs`; `zcash_client_sqlite`, `wallet/common.rs`).

Splitting change. The strategy `MultiOutputChangeStrategy` fetches account metadata counting the wallet’s existing notes above `min_split_output_value` (default fallback `SplitPolicy::MIN_NOTE_VALUE = 500,000` zatoshis, documented as `MARGINAL_FEE · 100`); the split count shrinks until each split note is at least the minimum split value (or a quarter of the post-transaction average note value when no minimum is set); if fewer splits than targeted result, the fee is recomputed for the actual count. The division remainder is added to the first change output.

Example 3.8 (Fee of a split-change transaction). Fix a five-way split policy with minimum split value 1,000,000 zatoshis and a wallet holding no other notes. Spending a 7,500,000-zatoshi Sapling note to pay 1,000,000 incurs a fee of 30,000 zatoshis — one spend against six outputs, the payment plus five change notes, is six logical actions — and leaves five change notes of 1,294,000 zatoshis each. Spending a 6,000,000-zatoshi note under the *same* policy shrinks the split: at the five-change fee of 30,000 the change totals 4,970,000 — 994,000 per note, below the minimum — so the strategy emits four change notes; the fee, recomputed for the payment plus four change notes (five logical actions), is 25,000, and each change note carries 1,243,750. The figures are recomputed by script from the ZIP-317 formula and the split rule above, and match the strategy’s own tests. The same tests verify that the strategy refuses an exact-balance two-output transaction without change, requiring the change output’s marginal fee, so that a recipient cannot reason about the sender’s input values from the absence of change.

3.5 Input selection: the fee-escalation loop

Fee and input selection are mutually dependent: adding an input can raise the logical-action count, which raises the fee, which can require another input. The deployed selector, `GreedyInputSelector`, resolves the circularity by iteration rather than by solving for a fixed point directly. Starting from an empty selection, each round:

1. chooses source pools by the preference order of §4.2 — the first pool in order whose selected notes cover the requirement is spent alone, and pools otherwise accumulate in preference order until the total covers it;
2. calls `ChangeStrategy::compute_balance` on the current selection (the fee evaluated on padded counts, prospective change included);
3. on `Ok`, emits the `Proposal` with the balanced fee; on `Err(DustInputs)`, adds the offending notes to an exclusion list and retries; on `Err(InsufficientFunds{required})`, sets the new target to *required* — which already embeds the fee recomputed for the enlarged, padded action count — and reselects notes from scratch with `TargetValue::AtLeast(required)`.

The loop terminates when a reselection fails to *strictly* increase the total selected value, returning `InsufficientFunds`. Escalation is thus implicit: each added note can raise the logical-action count by one, raising the required total by one marginal fee, which the next round's selection must cover.

The note store enforces the dust rule below the selector: both note-selection queries in `zcash_client_sqlite` filter `rn.value > min_value` with `min_value = zip317::MARGINAL_FEE = 5000`, so notes worth at most 5000 zatoshis are never even offered to selection. Value-targeted selection scans notes oldest-first under a running-sum window, taking every note while the running sum is below the target plus the single note that crosses it. The candidate notes themselves are the product of the synchronisation described in the *Ironwood Guide* ("Delivery: the note reaches its owner"); this section takes that spendable set as given. Section 10 develops the scanning pipeline that reconstructs it.

ZIP-320 (TEX) transactions. A payment to a transparent-source (TEX) address builds two transactions: the first funds an ephemeral transparent output, which the second spends towards the TEX address. The selector prices the ephemeral input as a standard 150-byte P2PKH input and the ephemeral output as a 34-byte P2PKH output. The value the ephemeral output must carry is obtained by balancing the *second* transaction with a zero-valued ephemeral input (`EphemeralBalance::Input(0)`) and reading the `required` field of the resulting `InsufficientFunds` error. In `propose_send_max`, the second transaction's one-transparent-in, one-transparent-out fee is `fee_required([Known(150)], [34], 0, 0, 0, 0) = 10,000` zatoshis — the grace minimum.

Shielding. The function `propose_shielding` computes a trial balance over all spendable UTXOs of the source addresses; on `DustInputs` it removes exactly the offending outpoints and recomputes once; the proposal is emitted only if the total reaches the caller's shielding threshold, else `InsufficientFunds`.

3.6 Relay, mempool, and block template

The conventional fee binds economically through three node-side mechanisms, all specified in ZIP-317 and ZIP-401. None is a consensus rule. The zebra claims below are verified against its sources; the `zcashd` claims rest on the ZIP text alone.

Definition 3.9 (Unpaid actions). For a non-coinbase transaction,

$$\text{unpaid_actions}(tx) = \max(0, \max(\text{grace_actions}, tx.\text{logical_actions}) - \lfloor tx.\text{fee} / \text{marginal_fee} \rfloor),$$

and 0 for coinbase; a block’s unpaid actions are the sum over its transactions, bounded by $\text{block_unpaid_action_limit} = 50$ in template construction (ZIP-317, *Recommended algorithm for block template construction*).

Relay. Nodes are expected to relay transactions paying at least the conventional fee. A transaction with more than $\text{block_unpaid_action_limit} = 50$ unpaid actions will never be mined by the recommended template algorithm, and nodes MAY drop such transactions, or those above a configurable limit (ZIP-317, *Transaction relaying*). Concretely: zcashd v5.6.0 exposes `-txunpaidactionlimit` (default 50) and `-blockunpaidactionlimit` overriding the block limit, and its relay threshold — a separate, older mechanism — was simplified in v5.5.0 (zcash/zcash#6542) (ZIP-317, *Deployment*). The zebra node is stricter than the ZIP suggests: since v1.8.0 (ZcashFoundation/zebra#8638) it hard-codes `BLOCK_UNPAID_ACTION_LIMIT = 0`, and `mempool_checks` — run on every candidate for the mempool, the node’s pool of validated transactions awaiting mining, through `VerifiedUnminedTx::new` (zebra-consensus, `transaction.rs`) — rejects any transaction whose unpaid-action count exceeds that limit (zebra-chain, `transaction/unmined/zip317.rs`). By Definition 3.9, a transaction has zero unpaid actions iff it pays at least the conventional fee, so zebra admits to its mempool — and therefore relays — only transactions paying the full conventional fee. The same check retains a legacy fee-rate floor of 100 zatoshis per 1000 bytes, clamped to $[100, 1000]$ zatoshis; the source notes it cannot fail once the zero-unpaid-action check passes, the conventional fee being at least 10,000 zatoshis (same file).

Mempool limiting (ZIP 401, as amended). Each mempool transaction has

$$\text{cost} = \max(\text{memory size in bytes}, 10,000), \quad \text{eviction_weight} = \text{cost} + \text{low_fee_penalty},$$

with $\text{low_fee_penalty} = 40,000$ if the transaction pays less than the ZIP-317 conventional fee and 0 otherwise, so low-fee transactions are evicted preferentially under memory pressure (ZIP-401, *Specification*). The threshold was switched from the legacy fee to the ZIP-317 conventional fee in zcashd v5.5.0 and zebrad v1.0.0-rc.7 (ZIP-317, *Mempool size limiting*; ZIP-401, *Deployment*). In zebra, `method cost` returns $\max(\text{serialized size}, 10,000)$ and `eviction_weight` adds 40,000 to that eviction weight (not to the fee in zatoshi) when the fee falls below the conventional fee; eviction samples at random by that weight. Because admission already requires the full conventional fee, the penalty never fires in zebra’s own mempool.

Block template construction. The recommended algorithm assigns each candidate transaction a weight

$$tx.\text{weight_ratio} = \min\left(\frac{\max(1, tx.\text{fee})}{\text{conventional_fee}(tx)}, \text{weight_ratio_cap}\right), \quad \text{weight_ratio_cap} = 4,$$

the $\max(1, \cdot)$ avoiding an all-zero-weight candidate set. Phase 1 selects randomly, with probability proportional to weight ratio, among candidates paying at least the conventional fee, subject to the limits on block size and on the number of signature operations a block’s scripts may perform; phase 2 selects among the remaining candidates subject additionally to the block’s unpaid actions staying at most $\text{block_unpaid_action_limit}$ (Definition 3.9). Whatever an adversary submits, a block built this way carries at most 50 unpaid logical actions (ZIP-317, *Recommended algorithm for block template construction*). In zebra, the two-phase selection (`select_mempool_transactions`;

zebra-rpc, `types/get_block_template/zip317.rs`) initialises the phase-2 budget to the hard-coded limit of 0, so a zebra template carries no unpaid actions at all. Overpaying beyond 4× the conventional fee therefore buys no further priority — a deliberate flattening of the fee market that keeps the conventional fee focal. Under the proposed NU7 rules, both selection phases would also have to respect the shielded block-work budget of ZIP 218, including Ironwood if the open correction in pull request 1361 is adopted. That validity constraint is independent of the unpaid-logical-action policy just described.

Deployment. ZIP-317 fees became the wallet default in `zcashd` v5.5.0; the block-template algorithm shipped in `zcashd` v5.5.0 and `zebra` v1.0.0-rc.3 (zebra has no wallet, so it computes no fees for construction) (ZIP-317, *Deployment*).

4 Transaction construction

The synchronisation procedure of the *Ironwood Guide* (§ “Delivery: the note reaches its owner”) leaves a wallet holding a set of unspent notes, each with a value, a position, and an authentication path available on demand. This section documents how the deployed wallet layer turns that state, together with a payment request, into signed transactions. The pipeline, implemented in `zcash_client_backend`, has two phases. Function `propose_transfer` takes a ZIP-321 payment request, runs input selection and fee-and-change computation, and returns a *proposal* — a complete plan, down to the zatoshi, of every transaction to be built. Function `create_proposed_transactions` executes the proposal: one transaction per step, each proved and signed, all stored together via `store_transactions_to_be_sent`; it returns the transaction ids, and submission to the network remains the caller’s responsibility. The split places every decision that requires judgement — which notes to spend, what fee to pay, where change goes — into an inspectable value computed before any key is touched. Execution follows that plan mechanically, but it is not byte-for-byte deterministic: proofs, signatures, note randomness, and ciphertexts use fresh randomness (including `OsRng`), so repeated execution can produce different transactions and txids.

Current-head extension. Both the volume baseline `e30517e4` and the 2026-09-03 `librustzcash` snapshot `5e770a91` support Ironwood in this pipeline. A proposal carries separate Ironwood inputs, anchor, witnesses, and padding. After NU6.3, an ordinary payment to an Orchard-protocol receiver targets Ironwood; Orchard change from an Orchard spend stays in Orchard through `add_orchard_change_output`, paired with the required fabricated same-receiver spend. Conventional Ironwood bundles retain default padding, while a canonical ZIP-318 crossing uses an unpadded one-Action Ironwood bundle, the shared rolling expiry, and the separately selected anchor bucket. The builder then constructs, proves, signs, and records the Ironwood bundle beside the Orchard bundle using the same Orchard spend authority. Some backend paths still require pool-specific handling, so Ironwood support is stated explicitly below rather than inferred from Orchard support.

4.1 The proposal

The payment request. The input is a ZIP-321 `TransactionRequest`: a `BTreeMap` from payment index to payment, implemented in `librustzcash`’s `components/zip321` crate. A memo attached to a payment addressed to a transparent recipient is rejected — as `Zip321Error::TransparentMemo` at parse time and again as `Error::Payment(PaymentError::TransparentMemo)` when outputs are added to the builder — because a transparent output has no ciphertext to carry one.

Definition 4.1 (Proposal). A *proposal* carries the fee rule under which it was computed, the minimum target height (the next block height), and a non-empty sequence of *steps* (type `Proposal<FeeRuleT, NoteRef>`), each step describing one transaction: the originating transaction request; a map from payment index to the pool that will pay it; the selected transparent inputs; the selected shielded inputs together with the anchor height at which their witnesses will be computed; references to outputs of prior steps to be spent; the *balance* — the proposed change values and the fee required — and a flag marking shielding steps. Validation of a multi-step proposal rejects forward references, spends of non-ephemeral change produced by a prior step, and intra-proposal double spends.

Target and anchor heights. In `zcash_client_sqlite`, the heights are

$$\text{target} := \text{tip} + 1, \quad \text{anchor} := \min_{\text{pools}} \max\{h \leq \text{target} - c : h \text{ checkpointed}\},$$

the “mempool height” and, with c the confirmation depth and “checkpointed” meaning present in the pool’s shard tree, the highest checkpoint at least c blocks below the target, minimised across Sapling and Orchard so a single height anchors both bundles. Anchor semantics and witness derivation are those of the *Ironwood Guide* (§ “Resting: the note in the commitment tree”, in particular “Anchor choice, reorgs, and the anonymity set”, which already develops the ZIP-315 3/10 defaults); here we record only what the pipeline feeds in.

Definition 4.2 (Confirmations policy). The `ConfirmationsPolicy` of ZIP 315 carries a *trusted* depth (default 3), an *untrusted* depth (default 10), and an `allow_zero_conf_shielding` flag (default true) (ZIP 315 § “Trusted and untrusted TXOs”). The constructor enforces `trusted ≤ untrusted`. An output is spendable at the trusted depth if its transaction is user-marked trusted or it was received on an internal-scope key, and at the untrusted depth otherwise; outputs of a shielding transaction inherit the mined height and trust of their transparent source inputs; and transparent UTXOs are spendable at zero confirmations for shielding when the flag is set.

Remark 4.3 (Anchor depth versus untrusted notes). Function `propose_transfer` obtains its height pair with `minconf = trusted` — even when the notes ultimately selected are untrusted and must themselves be 10 confirmations deep. The code comment explains the choice: the shallower anchor admits the maximum number of notes, and trusted-versus-untrusted filtering is deferred to note selection, which tests each note’s own depth. The bundle-wide anchor therefore sits at $\text{depth} \geq 3$, not ≥ 10 — which is what ZIP 315 itself recommends. A wallet SHOULD choose the anchor at the trusted confirmation depth: “if the current block is at height H , the anchor SHOULD reflect the final treestate of the block at height $H - 3$ ” (ZIP 315, § “Anchor selection”). The ZIP’s stated rationale: a rollback past the anchor block invalidates the transaction while its revealed nullifiers link it to any future transaction spending the same notes; an anchor too many blocks back prevents recently received notes from being spent; and a *fixed* depth — rather than one conditioned on whether the spent notes are trusted — avoids leaking their trust status (ZIP 315, § “Rationale for anchor selection”). Neither the ZIP nor the code comment claims an anonymity-set benefit for the shallower anchor.

Remark 4.4 (The ZIP-315 floor is advisory in the API). ZIP 315 states that wallets SHOULD NOT permit spending outputs with fewer than 3 confirmations, but the deployed API exposes `ConfirmationsPolicy::MIN` (1 confirmation) and constructors accepting lower values, gated only by a rustdoc warning about transaction distinguishability (`wallet.rs`, `ConfirmationsPolicy`). We present the ZIP recommendation as the norm and the constructors as a deliberate escape hatch.

4.2 Input selection

The deployed selector is `zcash_client_backend`'s `GreedyInputSelector`; it buckets each payment by receiver type: a transparent address becomes a transparent output; a TEX address is deferred to a second ZIP-320 step (below); a Sapling address becomes a Sapling output; a unified address is paid to its Orchard receiver if present — as an Ironwood-pool output once NU6.3 is active at the target height, the turnstile forcing such payments through the Ironwood bundle, and as an Orchard-pool output before then — else its Sapling receiver, else its transparent receiver, and a unified address with no receiver the wallet supports is an error.

The greedy loop. Selection then iterates: starting with no inputs, call the change strategy's `compute_balance` (§4.3); on `InsufficientFunds{required}`, ask the backend for spendable notes covering at least `required` (`select_spendable_notes` with `TargetValue::AtLeast`, the target height, the confirmations policy, and the running exclusion list); on `DustInputs`, add the identified notes to the exclusion list and retry; on success, emit the proposal. The loop terminates because the selected value must strictly increase on every round, else the selector fails with `InsufficientFunds`. Pool preference inside the loop: the shielded pools are ordered with the family matching the payment's outputs first — for a payment to an Orchard receiver, Ironwood (once NU6.3 is active) and then Orchard, ahead of Sapling; otherwise Sapling, then Ironwood, then Orchard — and pools the caller's spend policy forbids are dropped. The first pool in order whose selected notes cover the requirement alone is spent alone; otherwise pools accumulate in preference order until the running total covers it. Spending stays in the pool family matching the outputs where possible, crossing pools only when needed, which keeps single-pool payments free of cross-pool linkage. Within a pool, the `zcash_client_sqlite` backend selects the *oldest* unspent notes first: a window-function running sum over notes ordered oldest-to-newest takes every note below the target plus the one note that crosses it.

ZIP-320 (TEX) payments. A TEX address demands transparent funds whose provenance is a single ephemeral transparent address, so the selector builds *two* steps (ZIP 320 § “Design considerations for Senders”). It sizes the second step first: probing `compute_balance` over the TEX outputs with a zero-valued ephemeral input and catching `InsufficientFunds{required}` yields

$$v_{\text{eph}} = \sum_{\text{TEX payments}} v + \text{fee}(\text{tr}_1),$$

where `fee(tr1)` is computed for exactly one standard P2PKH input (150 bytes) and the TEX payments as P2PKH outputs (34 bytes each). The change-strategy contract requires the second step to balance exactly — `total = fee_required`, no change — which the selector enforces with a runtime assertion. Step `tr0` then replaces the TEX payments with a single ephemeral transparent “change” output of exactly v_{eph} , and step `tr1` spends it and pays the TEX addresses as P2PKH. Paying a TEX address from shielded inputs in a single step is rejected.

Shielding. Function `propose_shielding` targets the `tip + 1` and collects spendable UTXOs from the given source addresses — at zero confirmations under the default policy (Definition 4.2). Funds may be drawn from at most one ephemeral address at a time and never mixed with other addresses (`ProposalError::EphemeralAddressLinkability`), since co-spending would link the ephemeral address to the wallet's others. Dust UTXOs reported by `compute_balance` are dropped and the balance recomputed once; the proposal is built only if its total reaches the caller's shielding threshold. The resulting single step has an empty transaction request, `is_shielding` set, and all value flowing to internal change.

4.3 Fees and change

Fees and change are computed *together*, by a `ChangeStrategy` whose one obligation is `compute_balance`. The deployed strategies are `SingleOutputChangeStrategy` and `MultiOutputChangeStrategy` in the `fees::zip317` module; the `fees::standard` module aliases them over `StandardFeeRule`, whose only variant is `Zip317`. Both delegate to a common routine, `single_pool_output_balance`, described below.

Definition 4.5 (ZIP-317 conventional fee, as deployed). The fee rule (ZIP 317 § “Fee calculation”) is

$$\begin{aligned} \text{fee} &= \text{marginal_fee} \cdot \max(\text{grace_actions}, \text{logical_actions}), \\ \text{logical_actions} &= \max\left(\left\lceil \frac{\text{in_size}}{150} \right\rceil, \left\lceil \frac{\text{out_size}}{34} \right\rceil\right) + \max(n_{\text{spend}}^{\text{Sap}}, n_{\text{out}}^{\text{Sap}}) + n_{\text{act}}^{\text{Orch}} + n_{\text{act}}^{\text{lrw}}, \end{aligned}$$

with `marginal_fee` = 5000 zatoshi, `grace_actions` = 2, the standard P2PKH input and output sizes 150 and 34 bytes, and hence a minimum fee of 10000 zatoshi (the ZIP 317 parameter table). The Ironwood count $n_{\text{act}}^{\text{lrw}}$ is ZIP 317 Revision 1’s contribution (§3.3), zero in any transaction without an Ironwood bundle. The shielded counts fed in are the *padded* counts of §4.5 — `BundleType::num_outputs` and `num_actions` — both here and in the final builder check.

Two deployment caveats. ZIP 317 Revision 0 additionally defines a Sprout contribution $2 \cdot n_{\text{JoinSplit}}$; the wallet never constructs `JoinSplits`, so the deployed `FeeRule` omits the term (`fees/zip317.rs`) — code and ZIP agree on every transaction the wallet can produce, and we flag the omission rather than restate the ZIP formula as implemented. Second, the rule supports P2PKH transparent inputs plus P2SH inputs of known size only; a coin with an unknown P2SH redeem script fails with `FeeError::UnknownP2shInputs`, and the rule intentionally may slightly *overestimate* the fee for many transparent inputs — an overestimate the construction algorithms in `zcash_client_backend` rely on (`fees/zip317.rs`, `FeeRule` rustdoc and `fee_required`).

Change pool. Change goes to Orchard if the transaction has any Orchard input or output flows; else to Ironwood if it has any Ironwood flows; else to Sapling if it has any Sapling flows; else — fully transparent flows, as in shielding — to the caller-supplied fallback pool, subject after NU6.3 activation to the turnstile override of §3.4. The code marks this as deliberately naive (`fees/common.rs`, `select_change_pool`).

Balance computation. Routine `single_pool_output_balance` proceeds as follows. Let `in` be the sum of transparent, Sapling, Orchard, and Ironwood inputs (including any ephemeral input) and `out` the corresponding sum of outputs; let `fee0` be the ZIP-317 fee with zero change outputs. If `in < out + fee0`, fail with `required = out + fee0` (this is the value the greedy loop feeds back into note selection). If equality holds *and* the flows are fully transparent *and* no change memo is configured, emit no change with `fee = fee0`. Otherwise recompute the fee with the target number of change outputs (re-derived if the split policy below yields fewer), set `change = in - (out + fee)`, and divide it across the split count with the remainder added to the first output, each share a shielded change value in the selected pool carrying the configured change memo.

Remark 4.6 (Zero-valued change under the default policy). Under the default change strategy, a transaction with shielded flows produces a change output even when inputs exactly balance outputs plus fee: a zero-valued shielded change output is created so that an observer cannot distinguish exact-balance transactions, and so that a shielded output always exists to carry the change memo. Zero-valued change is allowed even under the `Reject dust` action, and the code notes that zero-valued notes need no witness tracking. Only a fully-transparent-flow transaction without a change memo omits change on the exact-balance branch — the second ZIP-320 transaction is the deployed example. The optional `AddDustToFee` policy below can also omit shielded

change when no memo requires it. A configured change memo is attached to every proposed shielded change output, is deliberately discarded for the ZIP-320 step that spends the ephemeral input, and itself forces a change output to exist.

Dust. The default `DustOutputPolicy` is `DustAction::Reject` with no threshold override, delegating the threshold to the strategy, whose default is the marginal fee, 5 000 zatoshi. If the total change falls below the threshold: under `Reject`, fail with `InsufficientFunds` and the requirement raised by the shortfall (unless the change is exactly zero, which Remark 4.6 permits); under `AllowDustChange`, keep the dust output; under `AddDustToFee`, burn the change into the fee and omit the output (keeping a zero-valued one if a change memo exists) — *except* when the resulting fee would exceed the computed fee by more than $10 \cdot \text{MINIMUM_FEE} = 100\,000$ zatoshi (that is, when the folded dust exceeds 100 000), in which case the change output is kept as a defence against value loss.

Uneconomic inputs. An input is *dust* iff its value is at most the marginal fee; the check runs only when the marginal fee is positive. Some dust rides for free: per pool P , up to

$$\min(|\text{dust}_P|, (\text{outputs}_P + \text{change}_P) - \text{nondust}_P) \quad (\text{saturating})$$

dust inputs do not raise the padded action count, plus at most one extra *grace* input — tried transparent first, then Sapling, then Orchard, then Ironwood — if the baseline logical-action count is below `grace_actions` and adding the candidate does not increase the padded count (the padding rules of §4.5 are invoked directly, with and without it). The allowance is the pointwise minimum over all possible change configurations; dust beyond it is returned as `ChangeError::DustInputs`, which the greedy loop converts into exclusions.

Note management: splitting change. The `MultiOutputChangeStrategy` splits change into up to `target_output_count` equal notes (remainder to the first) whenever the account holds fewer than that many notes of at least `min_split_output_value`; the note-count metadata comes from `InputSource::get_account_metadata` filtered by that minimum, or by the fallback $\text{MIN_NOTE_VALUE} = 100 \cdot \text{marginal_fee} = 500\,000$ zatoshi. The split count is

$$\text{split} = \max(1, \text{target} - \#\{\text{qualifying notes}\}),$$

decremented (floor 1) while $\text{change}/\text{split} < \text{min_split_output_value}$; when the minimum is unset it defaults to $(\text{existing notes total} + \text{change}) / (4 \cdot \text{target})$, a quarter of the projected average note value. If fewer than the target number of outputs survive, the fee is recomputed for the actual count. The `SingleOutputChangeStrategy` is the degenerate `SplitPolicy::single_output()`.

4.4 Executing the proposal

Function `create_proposed_transactions` executes the steps in order. Only *transparent* outputs of prior steps may be spent by later steps: shielded chaining is impossible, because a note’s position in the global commitment tree — and hence its witness (the *Ironwood Guide*, § “Resting: the note in the commitment tree”) — is unknown before the note is mined. Each ephemeral output must be consumed by a later step exactly once, else `EphemeralOutputLeftUnspent`. All transactions are created first and then stored together, so a failure cannot leave a partially transmitted multi-step plan.

anchors and witnesses per step. For each pool a step involves, the anchor is the shard-tree root at the proposal’s anchor-height checkpoint (`root_at_checkpoint_id`), including for a pool with outputs but no inputs. Using a recent real anchor in that case makes an output-only bundle

indistinguishable from one containing real spends. Each selected note’s Merkle path is computed with `witness_at_checkpoint_id_caching` at that same height. The mechanics are those of the *Ironwood Guide* (§ “Resting: the note in the commitment tree”) and are not repeated here.

Builder configuration. The `zcash_primitives` builder is constructed with `BuildConfig::Standard` carrying the per-pool anchors and Orchard/Ironwood padding choices, which normally select `BundleType::DEFAULT`; the Orchard builder exists only if NU5 is active at the target height, and the Ironwood builder only for a format and branch supporting Ironwood. For NU7, ZIP 218 recommends an expiry delta of 120 blocks (§6.3). The builder must use the target height’s consensus branch; NU7 retains v5 and v6, with v6 needed for an Ironwood component. The lock time of built transactions is 0.

Spends, outputs, and keys. Sapling spends are added under the external or internal (change) full viewing key according to each note’s recorded ZIP-32 scope; Orchard spends are added under the account’s Orchard full viewing key, the scope being derived internally (`FullViewingKey::scope_for_address` inside `SpendInfo::new`). Payment outputs are added in payment-index order, then the change outputs from the proposal balance: Sapling change to `ufvk.sapling().change_address()` (internal scope), Orchard change to `ufvk.orchard().address_at(0, Scope::Internal)`, and ephemeral ZIP-320 change values to freshly reserved ephemeral transparent addresses (`reserve_next_n_ephemeral_addresses`; the reservation persists even if construction later fails, so an address is never silently reused). The signing keys marshalled for the build: transparent secret keys derived per input address; *two* Sapling expanded spending keys — `usk.sapling()` and `usk.sapling().derive_internal()` — since external and internal notes sign under different keys; but a *single* Orchard spend-authorising key (`SpendAuthorizingKey`), because Orchard internal and external addresses share `ask` (the *Ironwood Guide*, § “The recipient: keys and addresses”). Ironwood spends use that same Orchard full-viewing and spend-authorizing key, but enter the separate Ironwood builder and commitment tree; current output and change routing is summarized in the current-head extension above. All randomness for the build is drawn from `OsRng`.

Remark 4.7 (OVK policy: none for change). Under the default `OvkPolicy::Sender`, external outputs are encrypted with the `ovk` selected by `UnifiedFullViewingKey::select_ovk` over the input sources — the Orchard `ovk` if any Orchard *or* Ironwood input is spent, else the Sapling `ovk`, else the external-scope `ovk` derived from the transparent account key per ZIP-316 — while wallet-internal change outputs get *no* `ovk` at all (`internal_ovk = None`) (`wallet.rs`; `zcash_client_backend/src/wallet.rs`, `OvkPolicy`; `zcash_keys/src/keys.rs`, `select_ovk`). A change output’s C_{out} is therefore unrecoverable by *any* outgoing viewing key; the wallet finds its change through the internal-scope `ivk` during scanning (trial decryption per the *Ironwood Guide*, § “Delivery: the note reaches its owner”). This matches the `OvkPolicy::Sender` rustdoc but contradicts descriptions elsewhere of change “encrypted to the internal OVK”; the `None` is deliberate. Policy `OvkPolicy::Custom` supplies both keys explicitly; policy `OvkPolicy::Discard` uses none, so the sender cannot decrypt even its own external outputs.

4.5 Padding, dummies, and shuffling

The deployed bundle builders at the volume baseline are orchard 0.15.0 and sapling-crypto 0.7.0, per `librustzcash`’s `Cargo.lock`. Their current builder APIs give the padding rules below. In Orchard the requested-action count is $\max(s, o)$ under cross-address-enabled flags and $s + o$ when NU6.3 requires cross-address transfers to be disabled (§5.13).

Orchard. The cross-address-enabled branch uses `BundleType::DEFAULT`: flags `ENABLED` (spends and outputs both enabled) and `bundle_required = false`. With s requested spends

and o requested outputs, with cross-address transfers enabled, the action count is

$$n_{\text{act}}(s, o) = \begin{cases} 0 & \text{if } \max(s, o) = 0, \\ \max(\max(s, o), 2) & \text{otherwise,} \end{cases}$$

the floor being `MIN_ACTIONS = 2`. The spend list is extended to n_{act} with dummy spends and the output list with dummy outputs; the two indexed lists are then shuffled *independently* and zipped pairwise into Actions, so the spend–output pairing within an action is random, and `BundleMetadata` records each requested spend’s and output’s post-shuffle action index. For cross-address-disabled bundles the requested count is instead $s+o$, and pairing must preserve the address restriction; the independent-shuffle description does not apply to that branch (§5.13).

Orchard dummies. A dummy spend (protocol specification § 4.8.3, “Dummy Notes (Orchard)”) is built from a freshly random `SpendingKey`, its full viewing key’s address at index 0 (external scope), and a zero-valued note whose ρ is Extract of a random Pallas point (`Rho::from_nf_old(Nullifier::dummy)`), with a dummy Merkle path of a random u32 position and 32 random $\mathbb{F}_{p_{\text{Pallas}}}$ siblings. Anchor consistency is skipped for zero-valued notes (`has_matching_anchor` returns true), which is exactly what lets a dummy coexist with any real anchor. A dummy output is a zero-valued note to a freshly random address, with no ovk and an all-zero memo.

Per-action randomness. At `ActionInfo` construction, the value-commitment trapdoor `rcv` is sampled uniformly from the Pallas scalar field — one per action, padding actions included. When the action is built, the spend-auth randomiser α is sampled per spend, giving $\text{rk} = \text{ak} + [\alpha] G^{\text{SpendAuth}}$; the output note’s ρ is the action’s own `nf_old`, and its `rseed` is freshly sampled by `Note::new`. The cryptographic roles of `rcv`, α , and `rseed` (with its derived ψ , `rcm`, `esk`) are those of the *Ironwood Guide* (§ “Conservation without disclosure: value commitments and the binding signature” for `cv/rcv`; § “Spend authorisation: RedPallas re-randomisation” for `ak/ $\alpha$ /rk`; § “Delivery: the note reaches its owner” for the `rseed` derivations).

Sapling. Under `BundleType::DEFAULT` in `sapling-crypto 0.7.0`, a non-empty bundle pads outputs to $\max(o, 2)$ but never pads spends (`num_spends = s` when $s > 0$, else 0; the $\max(s, 1)$ floor applies only with `bundle_required` set), so an output-only bundle has zero spend descriptions; the two-output floor (`MIN_SHIELDED_OUTPUTS`) is a rule the source traces to `zcash/zcash` issue 3615. An empty, non-required bundle has zero of both. Spends and outputs are extended with dummies, independently shuffled, and the positions recorded in `SaplingMetadata`. Each spend samples its own `rcv` and α (uniform in the Jubjub scalar field) and each output its own `rcv`; the binding-signature key is $\text{bsk} = \sum \text{rcv}_{\text{spend}} - \sum \text{rcv}_{\text{output}}$.

4.6 Build order and signatures

Construction 4.8 (`Builder::build`). The build proceeds in a fixed order:

1. check version compatibility;
2. compute the fee over the actual padded counts (Definition 4.5) and require the balance invariant below;
3. build the Sapling bundle and create its zk-proofs (proofs before signatures, because V4 transactions commit to proofs in the digest);
4. build the Orchard bundle and, on current v6 paths, the Ironwood bundle, both unproven;

5. assemble the unauthorised transaction data and compute the digest parts (`TxidDigester`);
6. authorise the transparent inputs;
7. compute the shared shielded sighash;
8. apply the Sapling signatures;
9. create the Orchard proof and apply its signatures, then do the same for Ironwood when present;
10. freeze the result into the final `Transaction`.

Definition 4.9 (Balance invariant). Before building, the builder requires, exactly,

$$\text{value_balance} - \text{fee} = 0,$$

where the value balance sums the transparent, Sapling, Orchard, and — on current v6 paths — Ironwood balances; a negative difference is `Error::InsufficientFunds` and a positive one `Error::ChangeRequired`. The proposal’s change outputs are precisely what make the invariant hold.

One sighash for everything. The message for *all* shielded spend-auth signatures and every present shielded pool’s binding signature is the single 32-byte ZIP-244 sighash,

$$\text{sighash} := \text{signature_hash}(\text{tx}_{\text{unauth}}, \text{SignableInput}::\text{Shielded}, \text{txid_parts});$$

the digest tree itself is constructed in the *Ironwood Guide* (§ “Conservation without disclosure: value commitments and the binding signature”, subsection “What the signatures sign: ZIP-244 digests and their v6 form”) and not repeated. For each Orchard-protocol action (in either the Orchard or Ironwood pool), with $\text{rsk} = \text{ask} + \alpha$ and $\text{rk} = \text{ak} + [\alpha] G^{\text{SpendAuth}}$ in the notation of the *Ironwood Guide* (§ “Spend authorisation: RedPallas re-randomisation”),

$$\text{spendAuthSig} := \text{Sign}_{\text{rsk}}(\text{sighash}).$$

The binding key is $\text{bsk} = (\sum_{\text{actions}} \text{rcv})$ converted via `into_bsk`, and the builder asserts consistency with $\text{bvk} = \sum \text{cv}^{\text{net}} - \text{ValueCommit}(\text{rcv} = 0, \text{value_balance})$. In `Bundle::prepare`, the binding signature $\text{Sign}_{\text{bsk}}(\text{sighash})$ is created and every dummy spend is signed immediately with its stored random ask randomised by that action’s α ; `Bundle::sign` then signs every action whose ak matches the given authorising key, and `finalize` fails with `MissingSignatures` if any action remains unsigned.

Remark 4.10 (Proving-key cache). The 2026-08-30 `librustzcash` snapshot 91f448b5 derives the Orchard/Ironwood circuit version once and reuses a process-wide cached proving key under `std`; the no-`std` fallback builds it once for that transaction. The Sapling provers remain arguments, typically a `LocalTxProver`.

4.7 After the build

The wallet immediately decrypts its own shielded outputs to record sent-note data: Orchard outputs via `decrypt_output_with_key`, trying the internal-scope `ivk` and then the external-scope `ivk` (same-address post-NU6.3 Orchard change can be external-scoped); Ironwood outputs under the internal Orchard `ivk`; and Sapling via `try_sapling_note_decryption` under its internal `ivk`. It uses the Orchard/Ironwood `BundleMetadata` and `SaplingMetadata` to map each requested output to its post-shuffle position. Transparent outputs are *not* reordered — the source notes “there is no reason to do so because it would not usefully improve privacy” — so `vout` index n is

the n -th added transparent output. The fee recorded for each sent transaction is taken from the proposal’s balance, not recomputed from the built transaction.

The PCZT path. PCZT construction parallels the signing path: `build_for_pczt` produces the `PcztParts`, and the Creator, IO-Finalizer, and Updater roles attach ZIP-32 derivation paths (Orchard, Ironwood, and Sapling $m/32'/\text{coin}'/\text{account}'$; transparent BIP-44 $m/44'/\text{coin}'/\text{account}'/\text{scope}/\text{index}$) and the proposal metadata under the proprietary fields `zcash_client_backend:proposal_info` and `:output_info`; only single-step proposals are supported.

Orchard-to-Ironwood pool migration. The `librustzcash` snapshot 91f448b5 also contains the released `zcash_pool_migration` 0.1.0 engine and `zcash_client_sqlite` 0.22.0 store. The engine plans denominations, builds and signs the migration transactions as PCZTs, schedules them by block height, and persists its state through a wallet backend; the consuming application remains responsible for broadcast and for reporting the result. The complete privacy and scheduling rules, and the boundary between these released backend pieces and an end-user wallet flow, are in the *Ironwood Guide*, § “Scheduled Orchard-to-Ironwood migration (ZIP-318)”.

Remark 4.11 (Builder-version currentness). The volume baseline pins orchard 0.15.0. That release includes the NU6.3 bundle versions, cross-address restriction, explicit change-output handling, and version-specific proving keys described in §5.13; those features are deployed, not design-stage. As of 2026-09-05, standalone orchard head be4f6594 remains release 0.15.5, while the `librustzcash` lockfile at 5e770a91 resolves 0.15.3. Neither changes the padding construction derived above.

5 Partially created transactions (PCZT)

The *Ironwood Guide* constructs a transaction as if one machine held everything at once: the notes and their authentication paths, the full viewing key, the spend authorizing key, the proving key, and a source of randomness. Deployed wallets do not enjoy that luxury. A hardware signer holds the spending key but cannot compute a Halo 2 proof and may not fit a whole transaction in memory; a threshold of co-signers must each contribute a share; a proof may be delegated to a machine that must never see spending authority. ZIP-374 resolves this by standardising the intermediate object those parties exchange: the *Partially Created Zcash Transaction* (PCZT), a format that carries the information each participant in transaction creation needs to perform its step, and that holds the accumulated proofs and signatures while the set is still incomplete — signers can be fully offline (ZIP-374, Abstract). The ZIP is Status Draft, Revision 0, owned by Jack Grigg, created 2024-12-09 (ZIP-374, header); The ZIP fixes the v1 reference encoding to `pczt` 0.7.0. The 2026-08-30 `librustzcash` snapshot 91f448b5 ships `pczt` 0.9.3: release 0.8.0 added the v2 encoding and Ironwood support, and 0.9.0 made the convenience serializer emit the minimum encoding version that can represent the logical PCZT (`pczt`, `CHANGELOG.md`; §5.13). Thus the role semantics below describe release 0.9.3; references to 0.7.0 identify only the ZIP’s byte-exact v1 baseline. At current `librustzcash` head 5e770a91 (2026-09-03), the crate version is still 0.9.3, but unreleased changes make the default transparent Signer and Spend Finalizer accept only `SIGHASH_ALL` after checking input and signature consistency; explicit policy overrides permit other sighash types (`pczt` and `zcash_transparent` `CHANGELOG.md`, Unreleased). Those changes are not attributed to the released role descriptions below.

The Motivation section of the ZIP names four problems the format solves: (i) offline and hardware signers that cannot compute Halo 2 or Groth16 proofs and may not hold a whole transaction in memory; (ii) proof delegation, where the prover needs the note’s private data and the full viewing key but no spending key material; (iii) multiparty transactions — FROST threshold signing (RFC 9591), transparent multisig (ZIP-48), and inputs contributed by several parties,

merged deterministically; and (iv) pre-authorization and deferred proving for v6 transactions, where anchors become authorizing data (ZIP-229) (ZIP-374, Motivation).

The design deliberately reuses the role structure of Bitcoin’s PSBT (BIP 174 / BIP 370) with Zcash-specific additions. PSBT itself cannot be used: it has no representation for shielded inputs and outputs, no notion of proofs as distinct from signatures, and its key–value encoding tolerates unknown fields — undesirable for signers that must understand everything they authorize (ZIP-374, Motivation; pczt, README.md). The ZIP’s Requirements section fixes the resulting contract: the format must not require spending key material to be present (dummy-spend keys are the one exception, and must be removable before Signers see the PCZT); a signer must be able to determine from the PCZT alone exactly what it authorizes; merging must be deterministic and fail loudly on conflict; and the encoding must be compact enough for memory-constrained signing devices and strictly versioned (ZIP-374, Requirements).

Two scope restrictions follow from history. PCZTs do not support Sprout and cannot create v4-or-earlier transactions: the v4 format predates the ZIP-244 transaction digest, and the role separations — in particular proving independent of signing — do not hold for it (ZIP-374, Specification). Two PCZT versions are specified: v1 supports creation of v5 transactions (ZIP-225); v2 supports v5 and v6 (ZIP-229). A PCZT version does not pin a transaction version — a v2 PCZT may carry a v5 or a v6 transaction, a v1 PCZT only v5 (ZIP-374, Versioning). The released pczt 0.9.3 parses both: `Pczt::parse` accepts version numbers 1 and 2, rejecting any other with `ParseError::UnknownVersion`. Its `serialize` emits v1 for a representable pre-v6 PCZT with a canonical-empty Ironwood bundle and no v2-only compact field state, and v2 otherwise; callers that require a fixed wire version use `pczt::v1::Pczt` or `pczt::v2::Pczt` (pczt, src/lib.rs and CHANGELOG.md, 0.9.0).

5.1 Roles and lifecycle

Definition 5.1 (ZIP-374 roles). Transaction creation is divided among the following roles, each entitled to perform a specific step on a PCZT (ZIP-374, Roles):

Creator (a single entity). Initializes the global fields and the empty per-protocol bundles.

Constructor. Adds transparent inputs and outputs, Sapling spends and outputs, and Orchard or Ironwood actions.

IO Finalizer (anyone). Declares the input/output set complete, computes the binding-signature keys, and signs dummy spends (§5.5).

Updater (anyone). Attaches metadata later roles need: derivation paths, full viewing keys, witnesses, anchors.

Prover. Attaches zero-knowledge proofs (§5.8).

Signer. Attaches signatures (§5.7).

Combiner. Merges parallel copies of a PCZT (§5.9).

Redactor. Removes fields that later entities do not need.

Spend Finalizer. Assembles the transparent `script_sigs` (§5.10).

Transaction Extractor. Produces and verifies the final transaction (§5.11).

A *Verifier* is “not a step in the transaction lifecycle, but a way of inspecting a PCZT” and checking its internal consistency on behalf of another entity that lacks the capacity to do so itself — for example a hardware signer (ZIP-374, Roles). The checks such a verifier runs are the per-action

verification equations of Construction 5.8. Release pczt 0.9.3 exposes a public `roles::verifier` module whose `Verifier` gives access to the protocol-specific verification surfaces; the underlying equations are implemented by the protocol crates.

The lifecycle is not a fixed pipeline. Proofs and signatures slot in asynchronously: proof creation needs no spending key material, and for v5 transactions signatures commit to the anchor (through the sighash) but not to proofs, while proofs do not depend on signatures — so the Prover and Signer roles can run in either order or in parallel, their outputs merged by a Combiner (ZIP-374, Roles and Prover). From the v6 format onward the two roles become independent even of the anchor choice (§5.13).

5.2 Structure

Definition 5.2 (Field kinds). Every structure in a PCZT contains three kinds of fields (ZIP-374, Specification: Structure):

- *transaction effecting data*: required, part of the final transaction, always present;
- *transaction authorizing data*: initially empty, filled in as the PCZT is processed (proofs, signatures);
- *context data*: committed to by, or relevant to, the effecting data; present so that roles can be performed (openings, keys, paths).

The current `Pczt` struct is `{global, transparent, sapling, orchard, ironwood}` with all four bundles non-optional (pczt 0.9.3, `src/lib.rs`); the historical 0.7.0 v1 implementation carried the first three, and release 0.8.0 added the `ironwood` slot (§5.13). The bundles are not logically optional because a PCZT does not always contain a semantically valid transaction, and there may be phases where protocol-specific metadata must be stored before it is known whether that protocol has any inputs or outputs (ZIP-374, Structure).

Global fields. The `global` structure carries (ZIP-374, Global): `tx_version` (u32; MUST be 5 in PCZT v1) and `version_group_id` (u32; must be consistent with `tx_version`); `consensus_branch_id` (u32; non-optional because it commits to the consensus rules that will apply to the transaction); `fallback_lock_time` (Option<u32>; assumed 0 if omitted); `expiry_height` (u32; ZIP-203); `coin_type` (u32, SLIP 44 — not part of the transaction and without consensus effect, present so later roles can check network-specific data such as derivation paths); `tx_modifiable` (a u8 bitfield, next paragraph); and `proprietary`, a map from UTF-8 strings to byte strings.

Definition 5.3 (The `tx_modifiable` bitfield). Field `tx_modifiable` tracks which parts of the transaction are still open to change (ZIP-374, Global):

Bit	Mask	Meaning	Merge
0	0b0000_0001	Transparent Inputs Modifiable	AND
1	0b0000_0010	Transparent Outputs Modifiable	AND
2	0b0000_0100	Has SIGHASH_SINGLE	OR
3–6	—	reserved; must be zero (merge fails otherwise)	—
7	0b1000_0000	Shielded Modifiable	AND

The Creator sets bits 0, 1, and 7 and clears bit 2; a Constructor checks the relevant bit before adding an input or output and may clear it; the IO Finalizer clears bits 0, 1, and 7 if there are shielded spends or outputs, and otherwise leaves the bitfield unchanged (§5.5). A Signer

clears bit 0 when adding any signature that does not use `SIGHASH_ANYONECANPAY` — which includes *all* shielded signatures — and clears bit 1 for any signature not using `SIGHASH_NONE`; a `SIGHASH_SINGLE` signature also clears bit 1, because removing the paired output would not remove the signature. Bit 2 is set by any Signer using `SIGHASH_SINGLE` and tells Constructors to preserve the input/output pairing. Every Signer clears bit 7 (ZIP-374, Global).

The split between transparent and shielded modifiable flags is deliberate: Zcash’s sighash pins *all* non-transparent effecting data as soon as any input is signed, so separate flags keep purely-transparent PCZTs semantically close to PSBTs, where Constructor and Signer roles can interleave when inputs are not `SIGHASH_ALL` (ZIP-374, Rationale). The deep construction of the sighash itself is the ZIP-244 digest covered by the *Ironwood Guide* (the transaction digest); this section only ever cites it.

Transparent inputs and outputs. A `TransparentInput` carries the outpoint (`prevout_txid` [`u8;32`], `prevout_index` `u32`), `sequence` (`Option<u32>`, default `0xFFFFFFFF`), the lock-time constraints `required_time_lock_time` (`>= 500000000`) and `required_height_lock_time` (`> 0, < 500000000`), the `script_sig` (set by the Spend Finalizer), `value` (`u64`) and `script_pubkey` — both required, because the Transaction Extractor needs every input’s UTXO to derive the shielded sighash for the binding signatures — `redeem_script` (P2SH only), `partial_signatures` (a map from 33-byte public keys to signatures), `sighash_type` (Signers MUST use it; Spend Finalizers MUST fail on signatures that mismatch it), `bip32_derivation`, four preimage maps (RIPEMD160/SHA-256/HASH160/HASH256), and `proprietary` (ZIP-374, `TransparentInput`). A `TransparentOutput` carries `value`, `script_pubkey`, `redeem_script`, `bip32_derivation`, `user_address` (`Option<UTF8String>`, set by an Updater; Signers MUST parse it and confirm that it contains the recipient, directly or as a receiver within a Unified Address), and `proprietary` (ZIP-374, `TransparentOutput`). Transparent partial signatures are stored as DER-encoded ECDSA — the elliptic-curve digital signature algorithm over Bitcoin’s curve `secp256k1`, the transparent pool’s signature scheme — with the sighash-type byte appended, `sig_bytes = DER(sig) || sighash_type`, matching the Bitcoin/PSBT convention.

The Sapling bundle. A `SaplingBundle` carries lists of spends and outputs, a `value_sum` (`i128`, net spends minus outputs — wide enough that per-spend and per-output values can be redacted while the net remains), the `anchor` (`[u8;32]`; in PCZT v1 set by the Creator and immutable, because the v5 sighash commits to it), and `bsk` (`Option`; `None` until the IO Finalizer runs, used by the Transaction Extractor for the binding signature). Each `SaplingSpend` requires `cv`, `nullifier`, and `rk` (effecting), and optionally carries a per-spend 192-byte Groth16 zkproof (Prover), a 64-byte `spend_auth_sig` (Signer), `recipient` (43 bytes), `value`, `rcm` or `rseed` (`rcm` only for pre-ZIP-212 notes; the Prover needs one of them), `rcv` (the IO Finalizer compresses these into `bsk`; the Prover requires it; it opens `cv`, hence is redactable), `proof_generation_key` (`ak,nsk`) (Updater sets, Prover requires), `witness` (a `u32` position and a 32-level path), `alpha`, `zip32_derivation`, `dummy_ask` (a Constructor chooses it; the IO Finalizer uses and clears it; Signers MUST reject PCZTs containing it), and `proprietary`. Each `SaplingOutput` requires `cv`, `cmu`, `ephemeral_key`, `enc_ciphertext`, and `out_ciphertext` (whose length depends on the transaction version), and optionally carries `zkproof`, `recipient`, `value`, `rseed` (the Prover requires it, in preference to disclosing the shared secret), `rcv`, `ock` (lets a Signer verify `out_ciphertext`; `None` under the OVK policy `None`), `zip32_derivation`, `user_address`, and `proprietary` (ZIP-374, `SaplingBundle/SaplingSpend/SaplingOutput`).

The Orchard bundle. An `OrchardBundle` carries (`pczt`, `src/orchard.rs`; ZIP-374, `OrchardBundle/OrchardAction`):

actions. A list of `OrchardAction` structures.

flags (`u8`). Bit 0 `enableSpends`, bit 1 `enableOutputs`; the remaining bits are reserved zeros in PCZT v1.

value_sum (`(u64, bool)`). The net value of spends minus outputs. The ZIP never states the boolean’s meaning; only the reference implementation defines it (`true` = negative, via the orchard crate’s sign mapping) — a spec gap we flag in §5.14.

anchor (`[u8; 32]`). Set by the Creator; fixed for the life of a v1 PCZT.

zkproof (`Option<Vec<u8>>`). A *single* Halo 2 proof for the whole bundle, set by the Prover — unlike Sapling’s per-spend and per-output proofs.

bsk (`Option<[u8; 32]>`). The binding signing key, set by the IO Finalizer.

Each `OrchardAction` is `{cvnet, spend, output, rcv}`: the net value commitment, a spend half, an output half, and *one* optional `rcv` per action, since an action carries a single net value commitment (the *Ironwood Guide*, conservation of value).

An `OrchardSpend` carries: `nullifier` and `rk` (32 bytes each; effecting, required); `spend_auth_sig` (64 bytes; `Signer`); `recipient` (43 bytes, the raw Orchard payment address encoding; a `Constructor` sets it, the `Prover` requires it); `value` (the `Prover` requires it; a `Signer` may verify it against `cv_net` and confirm change amounts; redactable); `rho` and `rseed` (32 bytes each); `fvk` (96 bytes, `ak || nk || rivk` — the full viewing key of the note’s recipient; an `Updater` sets it, the `Prover` requires it); `witness` (a `u32` position plus a 32-level, 1024-byte authentication path; `Updater` sets, `Prover` requires); `alpha` (32 bytes; a `Constructor` chooses it, the `Signer` requires it; redactable once `zkproof` and `spend_auth_sig` are set); `zip32_derivation`; `dummy_sk` (a `Constructor` chooses it for dummy notes; the `IO Finalizer` requires and clears it; `Signers` MUST reject PCZTs containing it); and `proprietary`. An `OrchardOutput` carries the effecting fields `cmx`, `ephemeral_key`, `enc_ciphertext`, and `out_ciphertext` (all required), plus `recipient`, `value`, `rseed` (the `Prover` requires it instead of disclosing the shared secret), `ock`, `zip32_derivation`, `user_address` (`Signers` MUST parse it and confirm that it contains the recipient), and `proprietary` (ZIP-374, `OrchardSpend/OrchardOutput`). The `witness`, `anchor`, and the note-commitment machinery these fields feed are exactly those of the *Ironwood Guide* (the commitment tree and authentication paths); we do not re-derive them here.

ZIP-32 derivations. A `Zip32Derivation` pairs a *seed_fingerprint* (`[u8; 32]`, the ZIP-32 seed fingerprint) with a *derivation_path* (`List<u32>`), where each index may carry the hardened flag, bit 31 (`index | 0x80000000`). Shielded spend paths are generally fully hardened; transparent paths generally include a non-hardened section matching BIP 44 or the ZIP-320 ephemeral-address path. The orchard crate rejects any non-hardened component at parse time — Orchard does not support non-hardened derivation — and its method `extract_account_index` recognizes exactly the pattern `[32', coin_type', account']` under a matching seed fingerprint.

5.3 Encoding

Construction 5.4 (PCZT encoding). A PCZT is encoded as

$$\underbrace{[0x50, 0x43, 0x5A, 0x54]}_{\text{ASCII PCZT}} \parallel \text{I2LEOSP}_{32}(\text{version}) \parallel \text{version-specific body},$$

an 8-byte header followed by the body. A parser rejects inputs shorter than 8 bytes (`TooShort`), a wrong magic (`NotPczt`), and — in the workspace crate — any version other than 1 or 2 (`UnknownVersion`); a parser encountering an unknown version MUST NOT parse the remainder. Files use the `.pczt` extension (ZIP-374, Encoding).

The body is a profile of the Postcard wire format — byte-for-byte the serde serialization of the pczt Rust crate, with the ZIP text canonical (ZIP-374, Encoding). Its primitives:

- `varint(n)`: emit *n* seven bits at a time, least-significant group first, one byte per group; every byte except the last has bit 7 set; encoders **MUST** emit the shortest encoding. A *k*-bit integer occupies at most $\lceil k/7 \rceil$ bytes, and in a maximum-length encoding the final byte contributes only the high ($k \bmod 7$) bits — the fifth byte of a u32 is at most 0x0F; decoders **MUST** reject encodings exceeding either limit.
- Signed i128 (the Sapling `value_sum`): `encode(n) = varint(zigzag(n))` with `zigzag(n) = (n << 1) ⊕ (n >> 127)`, the right shift arithmetic and the result reinterpreted as unsigned; a non-negative *n* encodes as $2n$, a negative *n* as $2|n| - 1$; at most 19 bytes.
- Option: a tag byte 0x00 (None) or 0x01 followed by the value.
- List: `varint(count)` then the elements.
- Map: `varint(count)` then the entries in strictly ascending lexicographic key order, duplicates forbidden.
- Structs: field concatenation in the ZIP’s field order, with no framing or tags.

The v1 body is the concatenation

Global || TransparentBundle || SaplingBundle || OrchardBundle,

with the transparent bundle a pair of input and output lists, the Sapling bundle {spends, outputs, `value_sum`, `anchor`, `bsk`}, and the Orchard bundle {actions, flags, `value_sum`, `anchor`, `zkproof`, `bsk`} (ZIP-374, Encoding and Data type encodings).

A minimal PCZT, byte by byte. A v6 PCZT with no spends or outputs makes the positional encoding concrete. Use NU7’s proposed branch identifier 0x77190AD9, illustrative expiry height 10,000,000, and Mainnet coin type 133. Serialising the fields under ZIP 374’s version-2 rules yields the following 32 bytes. This is an encoding example, not a payment or an assignment of an activation height.

offset	field	bytes
0	magic "PCZT"	50 43 5a 54
4	version 2 (raw LE u32)	02 00 00 00
8	tx_version = 6 (varint)	06
9	version_group_id (varint)	98 ed 92 c4 0d
14	consensus_branch_id (varint)	d9 95 e4 b8 07
19	fallback_lock_time = None	00
20	expiry_height (varint)	80 ad e2 04
24	coin_type = 133 (varint)	85 01
26	tx_modifiable = 0b1000_0011	83
27	proprietary: empty map	00
28–31	four absent bundles	00 00 00 00

Three features of the Postcard discipline are visible at once. Integers are LEB128 varints, so `coin_type = 133` costs two bytes where 1 would cost one — and re-encoding with coin type 1 indeed shortens the stream to 31 bytes, shifting every later field: the encoding is positional, which is precisely why fields cannot be inserted within a version (cross-checked by script: single-field variants differ exactly at the annotated offsets). Absence is one 0x00 byte: the version-2 format wraps each of the transparent, Sapling, Orchard, and Ironwood bundles in an Option

that collapses when the bundle is empty, so the entire bundle machinery of Construction 5.4 above costs four bytes until a first spend or output materialises. And the initial `tx_modifiable = 0x83` sets the two PSBT-inherited modifiable bits — transparent inputs (bit 0) and outputs (bit 1) — plus ZIP-374’s Zcash-specific shielded-modifiable bit 7, per Definition 5.3.

The convenience `Pczt::serialize` chooses the minimum encoding version that can represent the logical PCZT. A v6 transaction requires PCZT v2 even when its bundles are empty; PCZT v1 represents only v5 transactions.

Extensibility is deliberately narrow. New fields cannot be added within an existing PCZT version, because the Postcard encoding is positional; unlike PSBT there are no non-proprietary arbitrary key–value maps, so a Signer’s view of a PCZT is complete. The proprietary maps are the only within-version extension point and are reserved for private use; format evolution happens only through new versions namespaced by the header’s 4-byte version (ZIP-374, Extensibility and Proprietary Use fields). Serialization is deterministic: the crate’s round-trip test checks `serialize(parse(serialize(p))) = serialize(p)`.

5.4 Creation and construction

The deployed Creator, `Creator::new(consensus_branch_id, expiry_height, coin_type, sapling_anchor, orchard_anchor)`, selects a transaction format compatible with the target branch. NU7 permits v5 (`version_group_id = 0x26A7270A`) and v6 (`version_group_id = 0xD884B698`); an Ironwood component requires v6. A Creator and Signer must recognise the NU7 branch identifier; retaining v6 encoding is not permission to retain an old branch commitment (ZIP 259). It starts with `tx_modifiable = 0b1000_0011` (bits 0, 1, 7 set), Orchard flags = `0b0000_0011` (spends and outputs enabled), and empty transparent, Sapling, Orchard, and Ironwood bundles, with Sapling `value_sum = 0` and Orchard `value_sum = (0, false)`. The Ironwood anchor starts absent and may be supplied with `with_ironwood_anchor` for v6; its flags enable spends, outputs, and cross-address transfers. Builder options also set fallback lock time and per-pool flags. The Orchard initialization is non-negative zero, matching the v2 draft’s canonical empty `OrchardBundle` and `EMPTY_ORCHARD` (§5.14).

The crate ships no Constructor implementation (“The roles currently without an implementation are: Constructor”; `pczt, src/roles.rs`). In the deployed stack the Constructor step is performed by the `zcash_primitives` transaction builder: `Builder::build_for_pczt(rng, fee_rule)` checks that the value balance minus the fee is exactly zero (failing with `InsufficientFunds` or `ChangeRequired` otherwise), then builds the per-pool PCZT bundles *without proving or signing*, returning a `PcztResult` whose `PcztParts` carry the parameters, version, branch id, `lock_time = 0`, expiry height, and the four bundles (transparent, Sapling, Orchard, and Ironwood), together with the Sapling, Orchard, and Ironwood bundle metadata. Then `Creator::build_from_parts(parts)` assembles the `Pczt`, setting `tx_modifiable = 0b0000_0000` — nothing modifiable, plus the `Has-SIGHASH_SINGLE` bit if any input uses it — `fallback_lock_time = Some(lock_time)`, and `coin_type` from the network parameters.

The Orchard side of that construction produces, per action (`orchard, src/builder.rs`, `SpendInfo::into_pczt, OutputInfo::into_pczt, build_for_pczt`):

$$cv^{\text{net}} := \text{ValueCommit}^{\text{Orchard}}(v^{\text{net}}, rcv), \quad nf_{\text{old}} := \text{the spent note's nullifier under fvk},$$

a fresh uniformly random α , and $rk := \text{Randomize}(ak, \alpha)$ — the RedPallas re-randomization of the *Ironwood Guide* (spend authorization) — populating the spend’s recipient, value, ρ , $rseed$, fvk , witness, and α (all `Some`), plus `dummy_sk` for dummy spends. The output half builds the new note with $\rho := nf_{\text{old}}$ of the paired spend, computes cmx , and runs the note encryption of the *Ironwood Guide*, populating recipient, value, and $rseed$. It currently sets `ock = None` with an in-code TODO (“Extract `ock` from the encryptor ...”), so the deployed Constructor never fills

ock even when an outgoing viewing key is used — a divergence from the ZIP’s field description (§5.14).

Bundles are padded with dummy actions and their positions randomized, so a consumer must map logical spends and outputs to bundle positions through the builders’ metadata: `spend_action_index(n)` and `output_action_index(n)` on the Orchard `BundleMetadata`, and `SaplingMetadata` for Sapling. The wallet backend uses exactly this to attach per-output metadata to the right PCZT actions.

5.5 IO finalization: the binding-signature key

Once the input/output set is complete, the IO Finalizer freezes it. The deployed role fails with `NoSpend/NoOutputs` if the transaction would have no spends or no outputs; if there are shielded spends or outputs it clears `tx_modifiable` bits 0, 1, and 7. It computes the transaction-version-aware *shielded sighash*

$$\text{sighash} := \text{signature_hash}(\text{tx_data}, \text{SignableInput}::\text{Shielded}, \text{txid_parts}),$$

where `tx_data` is the *effects-only* transaction data reconstructed from the PCZT — proofs and signatures do not affect the sighash — and `txid_parts = tx_data.digest(TxidDigester)`. The helper supports the v5/ZIP-244 and v6 transaction digests. It then calls the Sapling, Orchard, and Ironwood `per-pool finalize_io(sighash, 0sRng)`.

Construction 5.5 (Orchard IO finalization). Every action’s `rcv` must be present. The role computes

$$\text{bsk} := \sum_{a \in \text{actions}} \text{rcv}_a \in \mathbb{F}_{p_{\text{Vesta}}},$$

converted through its 32-byte little-endian representation into a RedPallas binding signing key (`ValueCommitTrapdoor::into_bsk`), and recomputes

$$\text{bvk} := \left(\sum_a \text{cv}_a^{\text{net}} \right) - \text{ValueCommit}^{\text{Orchard}}(\text{value_sum}, 0),$$

where $\text{ValueCommit}^{\text{Orchard}}(v, r) := [v] V^{\text{Orchard}} + [r] R^{\text{Orchard}}$ with

$$\begin{aligned} V^{\text{Orchard}} &= \text{GroupHash}^{\text{Pallas}}(\text{z.cash} : \text{Orchard-cv}, v), \\ R^{\text{Orchard}} &= \text{GroupHash}^{\text{Pallas}}(\text{z.cash} : \text{Orchard-cv}, r) \end{aligned}$$

— the net value commitment of the *Ironwood Guide* (conservation of value). It fails with `ValueCommitMismatch` unless $\text{VerificationKey}(\text{bsk}) = \text{bvk}$: this is the ZIP’s requirement that `value_sum` be consistent with the value commitments and trapdoors being aggregated. Finally, for each action with `dummy_sk` present, it *takes* the key (`Option::take`, guaranteeing that no spend authorizing key material remains after this role), derives `ask = SpendAuthorizingKey(sk)`, and signs the action with the shielded sighash (ZIP-374, IO Finalizer).

The Sapling analogue is $\text{bsk} := \sum_{\text{spends}} \text{rcv} - \sum_{\text{outputs}} \text{rcv}$ as a Jubjub scalar, made into a RedJubjub binding signing key, checked against $\text{bvk} = (\sum_{\text{spends}} \text{cv} - \sum_{\text{outputs}} \text{cv})$ adjusted by `value_sum` as an `i64` (`InvalidValueSum` if it does not fit); dummy spends are then signed with their `dummy_ask`, which is taken and cleared (the workspace pins sapling-crypto version 0.7, same algorithm). Ironwood uses the Orchard analogue on its separate bundle and anchor.

Remark 5.6 (Why a dedicated role). The key `bsk` cannot be computed by the Transaction Extractor without preserving every `rcv` until extraction — revealing the opening of every value commitment to all intermediate participants — and cannot be maintained incrementally by Constructors,

because with parallel construction and a Combiner the incremental update would be wrong. A dedicated role that runs once, after IO completion, resolves both. The same role signs dummy spends so that no Signer ever parses spending key material (ZIP-374, Rationale, “Adding an IO Finalizer role”).

5.6 Updating

The deployed Updater wraps the protocol crates’ updater types. Method `update_global_with` sets proprietary fields; the wrapped Orchard `ActionUpdater` exposes, per action, the setters `set_spend_zip32_derivation` and `set_spend_proprietary` on the spend half, and on the output half the setters `set_output_zip32_derivation`, `set_output_user_address`, and `set_output_proprietary` (with Sapling, Ironwood, and transparent analogues). Release 0.9.3 also exposes setters for Sapling, Orchard, and Ironwood anchors and spend witnesses. Anchor updates are restricted to the v6 format under NU6.3, must precede the relevant proof, and may fill an absent anchor or repeat the same one but may not replace a different present anchor. Witnesses likewise must be attached before the corresponding proof.

5.7 Signing

Construction 5.7 (The Signer role). The deployed Signer (`pczt` feature `signer`), `Signer::new`, parses all bundles, extracts an effects-only `TransactionData`, and computes `txid_parts` and the shielded sighash of §5.5 *once*, cached across signatures. Its methods are `shielded_sighash()` and `transparent_sighash(index)` — the latter

$$\text{signature_hash}(tx_data, \text{SignableInput}::\text{Transparent}(\text{input } i), txid_parts)$$

under the input’s declared `sighash_type` — the signing methods `sign_transparent`, `sign_sapling`, `sign_orchard`, and `sign_ironwood`, and their `append_/apply_` variants for externally produced signatures (a hardware device), and `finish()`. To sign an Orchard spend, `alpha` must be present; with the wallet’s ask:

$$\begin{aligned} rsk &:= \text{ask} + \alpha \quad (\text{the RedPallas SpendAuth randomization}), \\ &\text{require } \text{VerificationKey}(rsk) = rk \end{aligned}$$

— failing with `WrongSpendAuthorizingKey` otherwise — then

$$\text{spend_auth_sig} := \text{RedPallas.Sign}(rsk, \text{sighash}).$$

The external path `apply_orchard_signature` accepts a signature iff it verifies under the action’s `rk` over the shielded sighash (`InvalidExternalSignature` otherwise); the Ironwood path applies the same RedPallas checks to its separate action bundle. After any shielded signature all three modifiable bits are cleared; after a transparent signature the bits are updated per its sighash type as Definition 5.3 specifies.

All shielded signatures behave as committing to the whole transaction — there is no shielded `SIGHASH_ANYONECANPAY` — which is why adding a shielded signature clears all modification permissions (ZIP-374, Global, `tx_modifiable`). Before signing an Orchard or Ironwood spend, the deployed Signer calls `verify_nullifier` (Construction 5.8) and treats `Missing{Recipient, Value, Rho, RandomSeed}` as acceptable — those fields may legitimately have been redacted for this signer — but hard-fails on any actual inconsistency; likewise for Sapling.

Construction 5.8 (Per-action verification equations). The orchard crate exposes, for Signers and Verifiers:

```

verify_cv_net :  $cv^{net} \stackrel{?}{=} \text{ValueCommit}^{\text{Orchard}}(v_{\text{spend}} - v_{\text{output}}, r_{cv});$ 
verify_nullifier : rebuild the note from (recipient,  $v, \rho, r_{\text{seed}}$ ),
                    require fvk owns it (fvk.scope_for_address is Some),
                    and  $nf \stackrel{?}{=} \text{the note's nullifier under fvk};$ 
verify_rk : ak from fvk,  $rk \stackrel{?}{=} \text{Randomize}(ak, \alpha);$ 
verify_note_commitment : rebuild the output note with  $\rho := nf_{\text{old}}$  of the paired spend,
                         $cmx \stackrel{?}{=} \text{Extract}(\text{NoteCommit}_{\text{rcm}}(\dots)),$ 

```

with the nullifier, randomization, and commitment constructions those of the *Ironwood Guide*. For dummy notes ($v = 0$) an `expected_fvk` argument is ignored in favour of the spend's own `fvk` field.

A signer implementation “should only need to implement”: Pedersen commitments using Jubjub/Pallas arithmetic (note and value commitments); BLAKE2b and BLAKE2s and the PRFs/CRHs built on them; the nullifier check; the KDF plus note decryption (AEAD_CHACHA20_POLY1305); the SignatureHash algorithm; RedJubjub/RedPallas signatures; and a source of randomness — no proving circuitry (`pczt, src/roles/signer/mod.rs`, module `doc`). For dependency-constrained environments a `low_level_signer::Signer` variant (not feature-gated) exposes `sign_ironwood_with/sign_orchard_with/sign_sapling_with/sign_transparent_with` closures over the parsed bundles plus `tx_modifiable`.

5.8 Proving

The Prover attaches proofs; per the ZIP it requires, for each Orchard action, the spend's recipient, value, rho, rseed, fvk, witness, and alpha, plus the action's `rcv` and the bundle anchor; delegating proving therefore reveals note private data, but not spending authority (ZIP-374, Prover). The deployed role (`pczt` feature prover) offers `requires_sapling_proofs` (any Sapling spend or output missing its zkproof), `requires_orchard_proof`, and `requires_ironwood_proof` (actions non-empty and the corresponding bundle zkproof `None`), letting a wallet skip downloading the Sapling parameters when they are unneeded; `create_orchard_proof(pk)` parses the bundle and calls the orchard crate's `Bundle::create_proof`; `create_ironwood_proof(pk)` does the same for Ironwood.

Construction 5.9 (Orchard proof creation from a PCZT). With no actions the call is a no-op. Otherwise, for each action i the prover rebuilds the spend information ($fvk_i, note_i, merkle\ path_i$) — failing with `WrongFvkForNote` if fvk_i does not own the note — and the output note with $\rho_i := nf_{\text{old}, i}$ (`RhoMismatch` if inconsistent), then forms

$$circuit_i := \text{Circuit}(\text{spend}_i, \text{output note}_i, \alpha_i, r_{cv_i}),$$

$$instance_i := (\text{anchor}, cv_i^{net}, nf_i, rk_i, cmx_i, \text{flags}),$$

rejecting the identity `rk` (`IdentityRk`, added in orchard 0.13.0 for the consensus rule introduced in zcashd v6.12.1 / Zebra 4.3.1), and creates a *single* Halo 2 proof over all actions, stored as the bundle's zkproof. The proof commits to the anchor and the flags as public inputs; the

statement proved is the Action relation of the *Ironwood Guide*. The Ironwood path applies the same construction with its separate anchor and V3 note-plaintext version.

Because v5 signatures commit to the anchor but not to proofs, and proofs do not depend on signatures, the Prover and Signer can run in either order or in parallel; the `zcash_client_backend` documentation says to use the Combiner “if you create proofs and apply signatures in parallel” (ZIP-374, Prover).

5.9 Combining, redaction, and privacy

Construction 5.10 (Merge algebra). The deployed Combiner — `Combiner::new(Vec<Pczt>)` followed by `combine()` — left-folds pairwise merges; per-protocol bundles are merged before Global, “because each is only interpretable in the context of its own global”. The rules:

- Global effecting data (`tx_version`, `version_group_id`, `consensus_branch_id`, `fallback_lock_time`, `expiry_height`, `coin_type`) must be equal; `tx_modifiable` merges bit-by-bit per Definition 5.3.
- Optional fields: $\text{merge}(\ell, \text{None}) = \ell$; $\text{merge}(\text{None}, r) = r$; $\text{merge}(\text{Some } \ell, \text{Some } r) = \ell$ iff $\ell = r$, else FAIL. Maps merge by key union, with values required equal on collision.
- Bundle structure: flags must be equal; if either side’s `bsk` is set (the IO Finalizer has run), action counts and `value_sum` must match; otherwise a strict prefix extension appends the extra actions *only if* the shorter side’s global has Shielded Modifiable set — else FAIL. Anchors follow the optional-field rule: an absent anchor may merge with a present one, while two present anchors must be equal. The per-action required fields (`cvnet`, `nullifier`, `rk`, `cmx`, `ephemeral_key`, `enc_ciphertext`, `out_ciphertext`) must be equal.

Combining must be functionally order-independent:

$$\text{Combine}(f_A(p), f_B(p)) = f_A(f_B(p)) = f_B(f_A(p)).$$

The equality-or-fail rule intentionally differs from BIP 174, which on conflicting values says the combiner “must choose the value it considers correct” — in practice last-wins — silently discarding data. Because every PCZT field is typed and known up front, requiring equality is always possible and turns every conflict into a loud failure. A Combiner MUST NOT combine PCZTs that do not represent the same transaction; once IO is finalized, a PCZT is uniquely identified by the txid its effecting data determines (ZIP-374, Combiner and Rationale).

Redaction. The Redactor provides per-bundle, per-field `clear` methods. The Orchard `ActionRedactor` can clear `spend_auth_sig`; the spend’s recipient, value, rho, rseed, fvk, witness, alpha, zip32_derivation, and dummy_sk; the spend’s proprietary entries (by key or all); the output’s recipient, value, rseed, ock, zip32_derivation, and user_address; the output’s proprietary entries; and rcv. The bundle-level redactor can clear zkproof and bsk. The intended use is multiple independent Signers, each given a PCZT with only the information it needs — for example, without the other signers’ α values.

Privacy. A PCZT contains strictly more information than the transaction extracted from it: potentially counterparty addresses, values, note randomness, rcv (which opens cv), the spending account’s full viewing key, ZIP-32 paths, ock (which opens out_ciphertext), and user-facing address strings. Whoever receives a PCZT at a given stage learns everything present at that stage; participants SHOULD send it only to entities required for the remaining roles and SHOULD use

the Redactor to strip fields. Delegating the Prover reveals note private data — not spending authority. The extracted transaction contains none of this extra information (ZIP-374, Privacy Implications).

5.10 Transparent finalization and lock time

Construction 5.11 (Script-sig assembly). The Spend Finalizer constructs each transparent input’s `script_sig`. Two forms are supported:

- *P2PKH*: require exactly one entry (*pubkey*, *sig*) in `partial_signatures` satisfying

$$\text{RIPEMD160}(\text{SHA256}(\text{pubkey})) = \text{the address hash};$$

then

$$\text{script_sig} := \text{PUSH}(\text{sig}) \parallel \text{PUSH}(\text{pubkey}),$$

where the `partial_signatures` value *sig* already consists of the DER signature followed by its sighash-type byte.

- *P2SH-P2MS* (*m*-of-*n* multisig):

$$\text{script_sig} := \text{OP_0} \parallel \text{PUSH}(\text{sig}_1) \parallel \dots \parallel \text{PUSH}(\text{sig}_m) \parallel \text{PUSH}(\text{redeem_script}),$$

where the leading `OP_0` is the dummy for the `OP_CHECKMULTISIG` bug and the signatures are the first *m* available ones in *redeem-script pubkey order* (surplus signatures beyond the threshold are discarded; fewer than *m* is a failure; pubkeys must be compressed 33-byte keys).

After finalization the role clears each input’s required lock times, redeem script, partial signatures, BIP-32 derivations, and all preimage maps, keeping value and `script_pubkey` — the UTXO — so the Transaction Extractor can verify the final transaction.

Construction 5.12 (Lock-time determination). The final `nLockTime` follows BIP 370’s rule (ZIP-374, Determining Lock Time; `pczt, src/common.rs, determine_lock_time`). Let *r* hold if some input has a required time or height lock; let *t_u* (“time unsupported”) hold if some input requires a height lock (so a time lock cannot serve it), and *h_u* symmetrically. Then:

$$\text{nLockTime} = \begin{cases} \text{FAIL} & r \wedge t_u \wedge h_u, \\ \text{max required height locks} & r \wedge \neg h_u, \\ \text{max required time locks} & r \wedge h_u \wedge \neg t_u, \\ \text{fallback_lock_time or 0} & \neg r, \end{cases}$$

height winning when both types remain possible; an input specifying neither accepts either type. The crate fails with `IncompatibleLockTimes` when one input requires only time and another only height — and, diverging from ZIP-374/BIP 370, also when a single input specifies both lock types: `determine_lock_time` derives each type’s “unsupported” flag from any input carrying the other type’s field, so a both-specifying input sets *t_u* $\wedge$ *h_u* and fails, though the ZIP has it take the height lock.

5.11 Extraction and verification

The Transaction Extractor (pczt feature tx-extractor) checks transaction-version support and lock-time compatibility, then extracts the per-pool bundles, requiring every script_sig, every Sapling zkproof and spend_auth_sig, each Orchard and Ironwood bundle’s zkproof of canonical length for its action count, every Orchard-protocol spend_auth_sig, and each non-empty bundle’s anchor and bsk; it rejects the identity rk and an invalid ephemeral_key.

On the Orchard side, extraction converts the PCZT bundle into a regular bundle whose authorization is Unbound = {proof,bsk}; the canonical proof size is validated already at the Unbound stage, so the invariant “an Authorized bundle always has a canonical proof” holds across the binding step (this canonical-size check — Proof::expected_proof_size and NonCanonicalProofSize — entered orchard in 0.14.0 as the GHSA-2x4w-pxqw-58v9 fix), and $value_balance = i64(value_sum)$ with ValueSumOutOfRange on overflow. The binding step is:

verify every spend_auth_sig under its rk over *sighash*, $binding_sig := \text{Sign}(bsk, sighash)$,

failing with the pczt crate’s Error::SighashMismatch if any spend authorization does not verify — the crate maps the None that orchard’s apply_binding_signature returns on failure; the bsk/bvk consistency established by the IO Finalizer (Construction 5.5) guarantees the binding signature verifies under the bvk derived from the transaction’s value commitments and value balance.

Finally the extractor *verifies* the completed transaction — proofs and signatures — “so that an invalid transaction is detected before it is broadcast” (ZIP-374, Transaction Extractor). Sapling proof verification requires caller-provided SpendVerifyingKey/OutputVerifyingKey (error SaplingRequired if absent but needed); the Orchard verifying key, shared by Orchard and Ironwood proof verification, is generated on the fly if not provided via with_orchard.

5.12 The deployed wallet pipeline

The wallet layer drives the roles end to end. Function create_pczt_from_proposal — taking the wallet database, network parameters, spending account, OVK policy, and proposal (zcash_client_backend feature pczt) — supports only single-step proposals. It runs the same internal transaction construction as its non-PCZT sibling create_proposed_transactions; then chains Builder::build_for_pczt, Creator::build_from_parts, and IoFinalizer::finalize_io; and finally runs an Updater pass that:

1. stores the proposal metadata {from_account,target_height}, postcard-encoded, under the Global proprietary key zcash_client_backend:proposal_info;
2. stores a reduced recipient enum — External, EphemeralTransparent, InternalShielded, or InternalTransparent, the latter three carrying the receiving account — under the per-output proprietary key zcash_client_backend:output_info on each shielded and transparent output;
3. sets user_address on external outputs; and
4. attaches derivation paths: to Orchard and Ironwood spends and non-dummy Sapling spends the fully hardened [32',coin_type',account'] under the account’s seed fingerprint, and to transparent inputs the BIP-44 path [44 | H, coin_type | H, account | H, scope, address_index] with $H = 0x80000000$ and the last two components non-hardened.

The PCZT is then handed to the caller for the Prover, Signer, and Combiner roles — on other devices if need be.

The return path is `extract_and_store_transaction_from_pczt(wallet_db, pczt, sapling_vk, orchard_vk)`: it runs the Spend Finalizer, reads back the proprietary proposal and output metadata — failing if `proposal_info` is missing, so the PCZT must have come from `create_pczt_from_proposal` — reconstructs Sapling and Orchard output notes from the PCZT’s explicit recipient/value/rseed fields (outputs whose recipient is absent are dummies; outputs whose note fields were redacted are treated as deliberately unrecoverable and skipped), runs the Transaction Extractor, recovers memos via `try_output_recovery_with_pkd_esk` with `esk` re-derived from the note (post-ZIP-212; the derivation is the PRFexpand construction of the *Ironwood Guide*, note encryption), computes the fee from the PCZT’s transparent input values, and stores the result as a `SentTransaction`, returning the `txid`. The `sapling_vk` argument is optional so callers can first check `Prover::requires_sapling_proofs` and avoid downloading the Sapling parameters.

The same backend return path does not yet reconstruct or store Ironwood sent-output metadata: the generic Transaction Extractor extracts and verifies the Ironwood bundle, but the subsequent output collection covers only Orchard, Sapling, and transparent outputs. This is a deployed backend gap, not a limitation of the PCZT format or extractor (same source file, `extract_and_store_transaction_from_pczt`).

Feature gating. The `pczt` crate is `no_std` (alloc-based). The Creator, Verifier, Updater, Redactor, Combiner, and low-level Signer are always available; `io-finalizer`, `prover`, `signer`, `spend-finalizer`, and `tx-extractor` are cargo features; per-pool support is gated by the `transparent`, `sapling`, and `orchard` features (Ironwood uses the Orchard feature); and `zcp-builder` enables `Creator::build_from_parts` from the `zcash_primitives` builder output. The crate’s end-to-end tests exercise the full pipeline — Creator/Builder → IO Finalizer → Updater → Prover → Signer → Combiner → Spend Finalizer → Transaction Extractor — for the `transparent_to_orchard`, `transparent_p2sh_multisig_to_orchard`, `sapling_to_orchard`, and `orchard_to_orchard` flows.

ZIP-48 multisig. ZIP-48 (transparent multisig) prescribes PCZT as its transaction-construction workflow: one participant constructs a PCZT spending multisig UTXOs, with change to a P2SH address at `m/48'/coin_type'/account'/133000'/1/*` (the `133000'` component reads “Zcash P2SH”); the PCZT is conveyed out-of-band over a private authenticated channel; each participant MUST verify that change outputs are wallet-controlled, verifies that the spend is expected, adds their signature(s), and returns the signed PCZT; one participant combines a threshold of signed PCZTs and extracts the final transaction (ZIP-48, Transaction construction). The crate’s `transparent_p2sh_multisig_to_orchard` test exercises exactly this flow.

Tooling. The `zcash-devtool` CLI exposes the full role pipeline as subcommands: `pczt create / create-max / shield / create-manual / pay-manual` (Creator through IO Finalizer via `zcash_client_backend`), `inspect`; `update-with-derivation`; `redact`; `prove`; `sign`; `update-with-signature`, which applies externally created signatures; `combine`; `extract`; `send`; `send-without-storing`; and `to-qr / from-qr` — animated QR rendering and webcam scanning of PCZTs (feature `pczt-qr`), illustrating the constrained-transport hardware-wallet flow. Its `sign` command selects which spends to sign by matching the seed fingerprint and account in the PCZT’s `zip32_derivation` and `bip32_derivation` fields against the wallet’s seed (`extract_account_index`), then calls the Signer’s per-pool methods.

5.13 PCZT v2: deferred anchors and the Ironwood bundle

The draft’s v2 (version number 2 in the header) extends the format for v6 transactions (ZIP-229): it adds an `IronwoodBundle`, adds a `note_version` field (an enum {V2, V3}) to Orchard-protocol

bundles, makes shielded bundle anchors `Option<[u8; 32]>`, and omits canonically-empty bundles from the encoding (ZIP-374, Versioning and v2 Encoding). Classification per our conventions: *implemented and released* — specified in the Draft ZIP and released in pczt 0.8.0 on 2026-07-23 (the v2 module with `Option`-wrapped bundles and empty-bundle omission, the ironwood slot, `note_version`, and optional anchors). The current 0.9.3 logical model retains these fields.

For `tx_version` 6, ZIP-374 makes anchors and witnesses deferrable: signatures commit to neither, and the ZIP permits an Updater to set or replace an anchor before proving — replacing an anchor **MUST** clear the bundle’s proofs, which commit to it as a circuit public input — while witnesses must root to the anchor whenever both are present. In the specified design this enables re-anchoring fully signed transactions with the `txid` unchanged, and spending *unmined* Orchard-protocol notes: an Orchard nullifier depends only on note data, with ρ fixed by the creating action rather than by tree position — impossible for Sapling even under v6, because Sapling nullifiers depend on the note’s position. An all-zeroes anchor sentinel was rejected in favour of `Option` because 0 is a valid Pallas base element (ZIP-374, Anchors and pre-authorization; Rationale, “Optional anchors in PCZT v2”).

Release pczt 0.9.3 implements the deferred case but not anchor replacement: `set_sapling_anchor`, `set_orchard_anchor`, and `set_ironwood_anchor` may fill an absent anchor (or repeat the same value) under v6/NU6.3 before proofs, but return `ConflictingAnchor` for a different present anchor and `ProofAlreadyPresent` instead of clearing a proof. Thus the released Updater supports anchoring after signing, but not the ZIP’s fully signed re-anchoring operation (pczt, `src/roles/updater/mod.rs`).

From NU6.3, Orchard-pool actions must be created with cross-address transfers disabled — bundle flag bit 2, `enableCrossAddress`, **MUST** be 0 in PCZT v1 and in any Orchard bundle mined under NU6.3+ — and wallets pair each real spend with a *fabricated* zero-valued output to the spent note’s own receiver, whose `enc_ciphertext` is random bytes. Signers **MUST NOT** reject a zero-valued output whose ciphertext does not decrypt on that basis alone, and **SHOULD** classify it as a tolerable dummy (optionally checking `cmx` against the explicit recipient/value/rseed). Conversely, a fabricated zero-valued spend paired with a real output is a *real* spend — no `dummy_sk`; it is signed via the normal Signer flow with the receiver’s spend authorizing key; only fully-dummy actions carry `dummy_sk` (ZIP-374, Fabricated same-address outputs). The `IronwoodBundle` carries the Ironwood-pool component of a v6 transaction, structurally identical to `OrchardBundle` — ZIP-229 defines the Ironwood pool as a second value pool of the Orchard protocol reusing the Action structure — differing in that `enableCrossAddress` is normally 1 (cross-address transfers are the usual case in the Ironwood pool), the anchor refers to the Ironwood note commitment tree, and `note_version` **MUST** be V3 (the ZIP-2005 quantum-recoverable note plaintext, lead byte 0x03); a non-empty `IronwoodBundle` in a `tx_version`-5 PCZT **MUST** be rejected (ZIP-374, `IronwoodBundle` and Rationale).

The orchard 0.15 release line implements the NU6.3-era machinery — bundle-format-aware flag parsing (`Flags::from_byte(flags, format)`, `UnexpectedFlagBitsSet`); the check `verify_cross_address_restriction`, requiring every action’s output addressed to the same (`gd`, `pkd`) as its spend when flag bit 2 is 0, enforced by `finalize_io` and `create_proof` and provable only with an Ironwood-version proving key; same-receiver comparison via `Address::same_receiver`; and change outputs paired with a fabricated same-receiver spend (orchard, `src/pczt/verify.rs`, `io_finalizer.rs`, `prover.rs`, `parse.rs`, and `src/builder.rs`; `librustzcash Cargo.toml`). The current standalone head is 0.15.5; the `librustzcash` lockfile resolves 0.15.3.

5.14 Divergences between ZIP-374 and the deployed code

Per our conventions, we flag rather than resolve the following.

- *Dummy-key rejection is not implemented in the reference Signer.* ZIP-374 states that “Signers **MUST** reject PCZTs that contain `dummy_sk` values” (likewise `dummy_ask`), but no code path

in the pczt Signer, through the current 0.9.3 source, inspects those fields — only the Redactor offers `clear_spend_dummy_sk/clear_dummy_ask`, and the IO Finalizer clears the keys after use. The normative MUST remains unmet by the crate’s own Signer; an embedding wallet must enforce it separately to conform, as must external signers such as hardware wallets (ZIP-374, OrchardSpend.dummy_sk; pczt, src/roles/signer/).

- *A v4 PCZT is constructible but dead.* The ZIP forbids creating v4-or-earlier transactions, yet `Creator::build_from_parts` maps `TxVersion::V4` to a PCZT with `tx_version = 4` (Sprout/V3 return `None`). Extraction and therefore roles that parse transaction effects reject that version as `UnsupportedTxVersion`, so the state cannot progress (pczt, src/roles/creator/mod.rs; src/lib.rs, `extract_tx_version`).
- *Negative-zero initialization (resolved upstream).* The released pczt 0.5.1 had `Creator::new` set the Orchard `value_sum = (0, true)`, differing from the v2 draft’s canonical empty `OrchardBundle (0, false)`; the byte encodings differ, which would have prevented the v2 empty-bundle-omission rule from round-tripping. The workspace head now sets `(0, false)`, matching `EMPTY_ORCHARD`, so the divergence is closed (pczt, src/roles/creator/mod.rs; ZIP-374, v2 Encoding).
- *The ock field is never populated.* The ZIP describes Sapling/Orchard `ock` as `None` only under the OVK policy `None`, but the deployed Orchard builder always sets `ock = None` (in-code TODO), so the Signer capability of verifying `out_ciphertext` is not currently exercisable for wallet-created PCZTs (orchard, src/builder.rs, `OutputInfo::into_pczt`).
- *Unspecified sign convention.* The ZIP declares the Orchard `value_sum` as `(u64, bool)` without stating the boolean’s meaning; only the reference implementation defines it (`true = negative`) — a spec gap (ZIP-374, OrchardBundle; orchard, src/pczt/parse.rs).
- *Reference-version scope.* The ZIP names pczt 0.7.0 specifically as the release that defines the v1 encoding. Current pczt 0.9.3 still exposes that fixed encoding through `pczt::v1`, but also ships the ZIP’s v2 encoding and Ironwood bundle. This is no longer an implementation-versus-release skew; 0.7.0 remains a deliberate historical anchor for v1 bytes (ZIP-374, Reference implementation; pczt, CHANGELOG.md).

6 Broadcast, expiry, and the transaction lifecycle

The *Ironwood Guide* leaves off with a transaction fully proved, signed, and bound — an artefact whose internal validity is settled. This section follows that artefact through the rest of its life: serialisation and the identifier under which the network knows it, the deployed store-then-broadcast discipline, the expiry mechanism that bounds how long it can remain pending, the bookkeeping by which the wallet tracks it to one of its terminal states, the confirmation policy that governs when its outputs become spendable, and the truncate-and-rescan procedure that repairs the wallet’s view after a chain reorganisation. Throughout, we develop the shielded path; the transparent-input monitoring machinery (`TransactionsInvolvingAddress` and its filters) is gated behind the transparent-inputs feature of `zcash_client_backend` and is mentioned only in passing, so for a shielded-only wallet the monitoring surface is exactly the two status queries of §6.4.

6.1 Serialisation and the transaction identifier

A built transaction is serialised for broadcast by `Transaction::write`, which dispatches on version. Transactions v5 (version-group ID 0x26A7270A) use `write_v5`, while v6 (version-group ID 0xD884B698) use `write_v6`, which adds the separately encoded Ironwood bundle; v1–v4 use

`write_v4` (ZIP-225; ZIP-229; protocol specification §7.1, *Transaction Encoding and Consensus*). The wallet places the resulting raw bytes in the data field of a `RawTransaction` gRPC message for submission.

The identifier under which the network, the mempool, and the wallet’s own database all track the transaction depends on the version. For v5 and v6, the *txid* is the root of the ZIP-244 digest tree,

$$\text{txid} := \text{BLAKE2b-256}(\text{ZcashTxHash_} \parallel \text{branchid}, d_{\text{hdr}} \parallel d_{\text{trans}} \parallel d_{\text{Sap}} \parallel d_{\text{Orch}}),$$

for v5; v6 appends the separate Ironwood digest d_{Iron} to the same top-level preimage. Every absent shielded pool contributes its protocol-defined empty-bundle digest. Here the personalisation is the 12-byte literal `ZcashTxHash_` (one underscore) followed by `branchid = LE32(CONSENSUS_BRANCH_ID)`, the 4-byte little-endian consensus branch id, and an absent pool hashes the empty string under its own personalisation (ZIP-244, *Txid Digest*). The *Ironwood Guide* assembles this tree in full (“What the signatures sign: ZIP-244 digests and their v6 form”); we do not re-derive it. Two consequences matter for the lifecycle. First, the tree is computed over *effecting* data only: changing the transaction’s consensus effect changes the identifier, while allowed recomputation of excluded attestations such as proofs or signatures leaves both the effect and identifier unchanged. This is ZIP-244’s non-malleability property. For v1–v4 the *txid* is instead the double-SHA-256 of the serialised transaction, which *is* malleable and therefore reliable only once the transaction is mined. Second, because the personalisation bakes in the branch id, the same transaction has a *different* *txid* on parallel consensus branches — replay protection across forks (ZIP-244, *Txid Digest*). For a purely shielded transaction the sighash coincides with the *txid*, as the *Ironwood Guide* already notes.

The header node of the tree is the one this section leans on:

$$d_{\text{hdr}} := \text{BLAKE2b-256}(\text{ZTxIdHeadersHash}, \text{LE}_{32}(\text{version}) \parallel \text{LE}_{32}(\text{versionGroupId}) \parallel \text{LE}_{32}(\text{branchid}) \parallel \text{LE}_{32}(\text{lockTime}) \parallel \text{LE}_{32}(\text{nExpiryHeight})),$$

with version including the overwinter flag (ZIP-244, node T.1). The field `nExpiryHeight` of §6.3 is therefore *effecting* data, committed by the *txid*: expiry cannot be malleated. Zebra’s mempool relies on exactly this, matching and rejecting expired transactions by their non-malleable *txid*.

One convention trips up every first-time implementer: *txid* bytes are stored and transmitted in protocol order but *displayed* byte-reversed and hex-encoded. The `Display` implementation of `Txid` reverses before encoding, and the `zcash_client_sqlite` view documentation states the same rule for its stored byte vectors.

6.2 Store, then broadcast

The deployed pipeline persists a transaction *before* the network ever sees it. The function `create_proposed_transactions` constructs, proves, and signs each step of a proposal (§6.6 fixes the target height it builds against), then persists the results via `WalletWrite::store_transactions_to_be_sent`, whose contract reads “This must be called before the transactions are sent to the network”. The ordering is deliberate: a transaction the network might mine but the wallet has forgotten is unrecoverable spend history, whereas a stored transaction the network never saw is merely stale, and stored transactions “should be retransmitted while it is still possible that they could be mined”. In multi-step proposals (e.g. the ZIP-320 pairs of §6.5), storage happens only after *all* steps have been created, so that failure partway through creation cannot leave a transmittable prefix behind.

Persisting a created transaction does two things. First, it records the transaction itself: *txid*, expiry height read from `tx.expiry_height()`, raw bytes, fee, target height, and `min_observed_height := target height` (Table 4). Second, it marks every spent input as spent — Sapling, Orchard, and Ironwood nullifiers into their respective `*_received_note_spends` tables, transparent outpoints into `transparent_received_output_spends` — which locks those inputs against reselection by later proposals until the transaction is mined, or expires (§6.5).

Submission itself is one gRPC call: the `lightwalletd CompactTxStreamer` method `Send Transaction(RawTransaction)` returns (`SendResponse`), where `errorCode = 0` means accepted and any other value carries an `errorMessage` (`zcash_client_backend, proto/service.proto`). Zaino implements the same `CompactTxStreamer` service as a drop-in replacement for `lightwalletd`, so submission is identical against either (`zaino, README.md`).

Remark 6.1 (The rebroadcast responsibility boundary). The `librustzcash` documentation records the obligation to retransmit — stored transactions “should be retransmitted while it is still possible that they could be mined” (`zcash_client_backend, src/data_api.rs`), and no-expiry transactions “should be rebroadcast by the wallet until they are either mined or conflict with another mined transaction” (`zcash_client_sqlite, src/wallet.rs`) — but implements no rebroadcast loop itself. The obligation falls on the embedding wallet application; the deployed CLI reference (`zcash-devtool`) submits exactly once.

6.3 Expiry (ZIP-203)

An unmined transaction cannot be left dangling forever: an indefinitely pending payment is indeterminism for the user and dead weight for every mempool. ZIP-203 (Status Final, Category Consensus) resolves this with a consensus-enforced expiry height.

Definition 6.2 (Transaction expiry, ZIP-203). Every transaction from Overwinter onward carries a field `nExpiryHeight` of type `uint32` — always a block height, never a UNIX timestamp. A transaction t is *expired* at block height H iff

$$(t \text{ is not coinbase}) \wedge (t.\text{nExpiryHeight} \neq 0) \wedge (H > t.\text{nExpiryHeight}),$$

and the consensus rule states: if a transaction is not a coinbase transaction and its `nExpiryHeight` is nonzero, it **MUST NOT** be mined at a block height greater than its `nExpiryHeight`; a block containing an expired transaction is invalid (ZIP-203, *Specification*; protocol specification §7.1). Expiry at height N thus makes block N the last block that may include the transaction. The sentinel `nExpiryHeight = 0` disables expiry, and for non-coinbase transactions `nExpiryHeight ≤ 499999999` **MUST** hold. From NU5 the bound is removed for coinbase only, and the `nExpiryHeight` of a coinbase transaction **MUST** instead equal its block height — this keeps v5 coinbase txids unique, since the ZIP-244 txid does not commit to the coinbase script height (ZIP-203, *Changes for NU5*; protocol specification §7.1).

The NU7 expiry policy. ZIP 218, “Default expiry delta”, recommends a default of 120 blocks. Let δ denote the wallet’s chosen expiry delta, normally 120 under that recommendation. With the light-wallet convention of applying the delta to the next-block target height, construction sets

$$\text{nExpiryHeight} := \begin{cases} \text{targetHeight} & \text{coinbase,} \\ \text{targetHeight} + \delta & \text{otherwise,} \end{cases}$$

and the default target-time duration is

$$120 \cdot 25 = 3000 \text{ seconds} = 50 \text{ minutes.}$$

This is an estimate, not a guaranteed validity duration: actual block intervals vary. The recommendation is wallet policy, not a new consensus expiry predicate. ZIP 218 preserves explicitly configured overrides. Wallets should otherwise use a common delta, because idiosyncratic expiry heights fingerprint transactions just as idiosyncratic fees do (ZIP 315). Expiry estimation, pending-input locking, rebroadcast, and the unknown-expiry fallback must agree with the delta actually used in construction. The coordinated migration schedule is separate (§4).

Remark 6.3 (The policy and its implementation). The pinned wallet implementation predates NU7 and does not implement the recommended delta. The formulas here expose δ explicitly rather than claiming that its defaults have already changed. An implementation must select the applicable policy for the transaction’s target height.

Mempool eviction is node policy, mining is consensus. The consensus rule of Definition 6.2 constrains block *contents*; what nodes do with their mempools is policy layered on top. ZIP-203 specifies that when a transaction expires it is removed from nodes’ mempools, together with *all of its dependent transactions*. Zebra’s `remove_expired_transactions` evicts every mempool transaction once

$$\text{tipHeight} \geq t.\text{nExpiryHeight},$$

i.e. as soon as the tip reaches the expiry height — at which point no future block can include t — and adds the evicted transaction’s non-malleable txid to its rejection list with `SameEffectsChainRejectionError::Expired`. The two formulations agree operationally with ZIP-203’s “block N or earlier; block $N + 1$ will be too late”.

Relay policy may reject a transaction that consensus would still permit in the next block; eligibility for inclusion is not a promise of propagation. Wallets should submit before the final eligible block rather than rely on last-block acceptance.

Expiry at a network upgrade. The expiry height does not determine the consensus branch for which a transaction is valid. This matters at NU7 even though v5 and v6 retain their layouts. The Draft ZIP 259 proposes branch identifier `0x77190AD9`: signatures for the old branch do not authorise the new one. Once the activation height is assigned, a wallet must either complete a pre-upgrade payment before it or reconstruct and authorise the payment for the new branch. An expiry height extending past activation does not extend the old signature’s validity. Under ZIP 2003, v4 transactions would be invalid from that height even if their expiry were zero. Sprout funds, spendable only through v4, would then be inaccessible; this does not disable Sapling, transparent, Orchard, or Ironwood components supported by the remaining formats.

6.4 Tracking a pending transaction

Between submission and its terminal state, a transaction exists only as a row in the wallet database and an entry in remote mempools. The wallet-side record is the `zcash_client_sqlite` `transactions` table (Table 4); note that it may hold data unrecoverable from the chain, e.g. wallet-created transactions that expired unmined.

Statuses and how they are learned. The backend distinguishes three externally observable states, the enum `TransactionStatus`: `TxidNotRecognized`, `NotInMainChain` (in the mempool, or mined only on a reorged-away fork), and `Mined(height)`. Statuses arrive by evaluating `TransactionDataRequest` values the wallet emits: `GetStatus(txid)`; `Enhancement(txid)`, which downloads the full raw transaction and subsumes `GetStatus`; and, behind the transparent-inputs feature, `TransactionsInvolvingAddress`. These are typically served by `lightwalletd`’s `GetTransaction` and `GetAddressTxids` RPCs and fed back via `WalletWrite::set_transaction_status` or `decrypt_and_store_transaction`. Active polling is not optional even for a shielded wallet’s own history: fully transparent transactions, and transactions containing neither shielded inputs nor shielded outputs belonging to the wallet, are invisible to the compact-block trial decryption of the *Ironwood Guide* (“Delivery: the note reaches its owner”), so scanning alone can never confirm their fate.

The wire encoding of a status reply hides a protobuf quirk. The `RawTransaction.height` field signals: 0 = in the mempool; `0xffffffffffffffff` = mined on a fork that is not the main chain;

Column	Semantics
txid	transaction identifier, protocol byte order (UNIQUE; reverse and hex-encode for display)
created	wallet-local creation time
block	height of the containing block, set only once that block has been <i>scanned</i> (foreign key to <code>blocks(height)</code>)
mined_height	mined height, settable from a status query without scanning the block; CHECK equal to <code>block</code> when <code>block</code> is non-null
tx_index	index of the transaction within its block
expiry_height	maximum height at which the transaction may be mined, if known
raw	serialised transaction bytes, if retrieved
fee	fee paid, if known
target_height	construction target height; set only for transactions this wallet created
min_observed_height	NOT NULL; minimum of the mempool-observation height and the mined height
confirmed_unmined_at_height	maximum height with positive recorded report of unminedness; CHECK NULL when <code>mined_height</code> is non-null
trust_status	user-assigned trust flag (§6.6)

Table 4: Columns of the transactions table.

any other value = mined height in the main chain (`zcash_client_backend, proto/service.proto`).
The deployed client maps

$$\begin{aligned}
 \text{gRPC NotFound} &\mapsto \text{TxidNotRecognized,} \\
 \text{height} \notin [1, 2^{32} - 1] &\mapsto \text{NotInMainChain,} \\
 \text{otherwise} &\mapsto \text{Mined}(\text{height})
 \end{aligned}$$

Applying a status. On `TxidNotRecognized` or `NotInMainChain` for an unmined transaction, `set_transaction_status` records a report of unminedness, setting `confirmed_unmined_at_height` to the current chain tip. On `Mined(h)` it sets `mined_height` to h , lowers `min_observed_height` to at most h , clears `confirmed_unmined_at_height`, links `block` if that block is already scanned, and advances the window of unused transparent address indices the wallet derives and watches past the highest index observed used; in every case the pending `tx_retrieval_queue` entry is deleted. Independently, when chain scanning itself encounters a wallet-relevant transaction in a block, `put_tx_meta` upserts `block` and `mined_height` to the scanned height along with `tx_index`, takes the minimum for `min_observed_height`, and clears `confirmed_unmined_at_height`.

When to stop asking. A `GetStatus` request is (re)generated for every transaction with `mined_height` NULL while any of the following holds (`zcash_client_sqlite, src/wallet.rs, transaction_data_requests`):

$$\begin{aligned}
 &\text{confirmed_unmined_at_height IS NULL} \\
 &\vee (\text{expiry_height} > 0 \wedge \text{confirmed_unmined_at_height} < \text{expiry_height}) \\
 &\vee (\text{expiry_height IS NULL} \\
 &\quad \wedge \text{confirmed_unmined_at_height} < \text{min_observed_height} + \text{certaintyDepth}),
 \end{aligned}$$

where $\text{certaintyDepth} := \text{PRUNING_DEPTH} + \delta$ blocks, using the expiry delta defined in §6.3. This parameterised form separates the policy from the pinned implementation’s old default. The result is unioned with explicit `tx_retrieval_queue` entries. A transaction with known nonzero expiry is thus polled until unminedness is confirmed at or beyond its expiry height; one with unknown expiry until `certaintyDepth` blocks past first observation; and one with expiry 0

only until unminedness is first confirmed — only the first disjunct can hold for it, so a single `TxidNotRecognized` or `NotInMainChain` reply ends the polling. The in-code comment above the query promises to “continue to query indefinitely” for such transactions and expects them “re-broadcast by the wallet until they are either mined or conflict with another mined transaction” (Remark 6.1); the SQL does not implement that promise — a documentation divergence. For display, the `v_transactions` view exposes the terminal flag

```
expired_unmined := mined_height IS NULL  $\wedge$   $1 \leq \text{expiry\_height} \leq \text{max(blocks.height)}$ ,
```

i.e. nonzero expiry at or below the maximum scanned height.

Remark 6.4 (Documentation divergence: the retrieval-queue codes). The table documentation for `tx_retrieval_queue` says `query_type` is “0 for raw transaction (enhancement) data, 1 for transaction mined-ness information” (`zcash_client_sqlite, src/wallet/db.rs`), but the authoritative encoder `TxQueryType::code()` maps `Status` $\mapsto$ 0 and `Enhancement` $\mapsto$ 1 and round-trips through `from_code` (`zcash_client_sqlite, src/wallet.rs`). The documentation is inverted relative to the code; the code governs.

6.5 Expiry and fund availability

Storing a transaction locked its inputs (§6.2); expiry is what unlocks them. Note selection excludes a note appearing in `*_received_note_spends` only when the spending transaction is still *unexpired* in the following sense, evaluated at the target height H of the new proposal, with the expiry-policy parameter δ of §6.3:

$$\begin{array}{ll} \text{mined_height} < H & \text{(mined)} \\ \vee \text{expiry_height} = 0 & \text{(never expires)} \\ \vee \text{expiry_height} \geq H & \text{(unexpired)} \\ \vee (\text{expiry_height IS NULL} \wedge \text{min_observed_height} + \delta \geq H) & \text{(default-delta guess).} \end{array} \quad (1)$$

A transaction that has expired unmined (`mined_height` `NULL`, $0 < \text{expiry_height} < H$) fails every disjunct, so its inputs re-enter the spendable set for target heights beyond the expiry height. ZIP-315’s known-spendable definition matches: a TXO must not be “committed to be spent in another unexpired transaction created by this wallet” (ZIP-315, Draft).

Remark 6.5 (The unknown-expiry guess, and an inverted comment). The final disjunct of (1) is a guess that the originating wallet used the default delta. Reading the predicate directly: if the sender in fact used a *larger* delta or disabled expiry, the transaction is treated as *expired* from height `min_observed_height` + δ + 1 onward while it is still mineable — its inputs unlock prematurely, and a respond can conflict with a late mining of the original. If the sender used a *smaller* delta, the inputs merely stay locked slightly longer than necessary. Either misclassification persists only until the wallet observes the transaction mined or enhances it to learn the true expiry height. The doc comment on `tx_unexpired_condition` describes the first case as “treated as unexpired when it shouldn’t be” (`zcash_client_sqlite, src/wallet/common.rs`), which is inverted relative to its own SQL; we state the direction the predicate implies and flag the comment rather than silently following either.

Nullifiers outlive the lock. Dropping a pending spend’s inputs from the *spendable* set is not the same as forgetting them. The query `get_nullifiers(NullifierQuery::Unspent)` deliberately over-selects: a note spent in an unexpired-but-unmined transaction still contributes its nullifier, so that scanning (the *Ironwood Guide*, “Delivery: the note reaches its owner”) detects the spend if the transaction is later mined; a note’s nullifier leaves the watch set only once the spending transaction is mined or can never expire.

Aftermath of an expiry. ZIP-203 (*Wallet behavior and UI*) draws the user-facing conclusions: an expiring zero-confirmation transaction has even less inclusion guarantee, so UIs and services must never rely on zero-conf transactions in Zcash, and wallets should notify the user of expired transactions that must be re-sent. ZIP-315 (Draft) adds that sent transactions SHOULD be reported with confirmations and, while unmined with an expiry height, an estimate of time until expiry. Re-sending carries a privacy cost the wallet should recognise: ZIP-315 identifies repeated spends of the same TXOs across transactions — the typical aftermath of re-sending an expired payment — as an information leak linking the invalidated transaction to its replacement. For ZIP-320 (TEX) payments the deployed convention is to set the *same* expiry height on the first (shielded-to-ephemeral) and second (ephemeral-to-TEX) transactions, so another wallet on the same seed can infer when the second would have expired without observing it; if the second expires unmined, the ephemeral UTXO should be re-shielded before being re-spent.

6.6 Confirmations, trust, and spendability

Mining is not the end of caution: a freshly mined output can be un-mined by a reorganisation (§6.7), so the wallet imposes a confirmation policy before treating outputs as spendable.

Remark 6.6 (ZIP-315 is a Draft). ZIP-315 (“Best Practices for Wallet Implementations”) has Status Draft. The deployed stack nonetheless implements its central recommendations for the 3/10 confirmation split, zero-conf shielding, and fixed anchor depth. ZIP-315 supplies their rationale, not a finalised standard. NU7’s expiry recommendation is described in §6.3.

Definition 6.7 (Confirmations policy). A `ConfirmationsPolicy` holds two `NonZeroU32` counts, c_{tr} (*trusted*) and c_{untr} (*untrusted*), with the constructor enforcing $c_{\text{tr}} \leq c_{\text{untr}}$, plus a flag `allow_zero_conf_shielding`. The default is $c_{\text{tr}} = 3$, $c_{\text{untr}} = 10$, with zero-conf shielding allowed, per ZIP-315; the constant `ConfirmationsPolicy::MIN` is `1/1/true`. Confirmations are counted since *and including* the block in which a note was produced.

Block counts are not elapsed time. At NU7’s proposed 25-second target spacing, a ten-block interval corresponds to $10 \cdot 25 = 250$ seconds. Counts that include the mining block must still be distinguished from intervals *after* that block. ZIP 218 explicitly recommends leaving the three-block anchor depth unchanged; it does not prescribe multiplying the wallet’s trusted or untrusted confirmation counts by three. Neither an equal block count nor an equal elapsed time alone establishes equal double-spend risk: the attack model and confirmation policy belong to the *Consensus Guide*, §“NU7: 25-second blocks and bounded shielded work”. The defaults below describe the pinned wallet, not a newly standardised NU7 policy.

The trusted/untrusted split is ZIP-315’s: a *trusted* TXO is one received from a party where the wallet trusts it will remain mined in its original transaction — e.g. created by the wallet’s own internal TXO handling — and everything else is untrusted. The recommended counts rest on two observations: reorganisations are anecdotally shorter than 3 blocks, and value from a trusted TXO is always recoverable after a rollback (the wallet can just re-create the transaction), whereas an untrusted TXO may require asking the sender to re-send (ZIP-315, Draft). Users may also explicitly mark specific transactions trusted via `WalletWrite::set_tx_trust`, persisted in the `trust_status` column — ZIP-315’s “MAY enable users to mark specific external transactions as trusted” provision.

Definition 6.8 (Confirmations until spendable). Fix a target height H and write $x \ominus y$ for saturating subtraction (`BlockHeight` subtraction in `librustzcash` yields a `u32` via `saturating_sub`). Let $h_{\text{tr}} := H \ominus c_{\text{tr}}$ and $h_{\text{untr}} := H \ominus c_{\text{untr}}$, and for an output of a transaction with mined height

m (if any) let

$$\text{confs}_{\text{tr}} := \begin{cases} c_{\text{tr}} & m \text{ unknown} \\ m \ominus h_{\text{tr}} & \text{else,} \end{cases} \quad \text{confs}_{\text{untr}} := \begin{cases} c_{\text{untr}} & m \text{ unknown} \\ m \ominus h_{\text{untr}} & \text{else.} \end{cases}$$

Then `confirmations_until_spendable` returns, and the output is spendable iff the result is 0:

- *transparent*: 0 if `allow_zero_conf_shielding`; else confs_{tr} if the transaction is trusted or the receiving key scope is internal; else $\text{confs}_{\text{untr}}$;
- *shielded*: confs_{tr} if the transaction is trusted; else, for internal scope, if the maximum mined height h_s of the transparent inputs to the shielding transaction is known, $h_s \ominus h_{\text{tr}}$ when those inputs are trusted and $h_s \ominus h_{\text{untr}}$ otherwise, and confs_{tr} when h_s is unknown; else $\text{confs}_{\text{untr}}$.

The saturation is what makes the count “decrease to a floor of zero” for long-mined outputs.

The internal-scope shielded branch encodes ZIP-315’s rule that shielded funds produced by zero-conf shielding are unavailable until the *source* UTXOs have sufficient confirmations: spendability of a shielding output is computed from $h_s = \text{max_shielding_input_height}$ and the inputs’ trust status, not from the shielding transaction’s own mined height.

Remark 6.9 (Deployed shielding rule vs. the ZIP-315 text). ZIP-315 is internally inconsistent here. Its formal “Shielded TXOs” bullet requires the source UTXOs to have at least $c_{\text{untr}} - c_{\text{tr}}$ confirmations and the shielding transaction at least c_{tr} ; its later “Transparent UTXOs” paragraph instead requires the source transaction to have the full c_{untr} confirmations, again alongside c_{tr} for the shielding transaction. Write h_s for the maximum source-input height, $h_m \geq h_s$ for the shielding transaction’s mined height, and $d := h_m - h_s$. The two ZIP passages respectively imply

$$H \geq h_s + \max(c_{\text{untr}} - c_{\text{tr}}, c_{\text{tr}} + d) \quad \text{and} \quad H \geq h_s + \max(c_{\text{untr}}, c_{\text{tr}} + d).$$

The deployed rule of Definition 6.8 instead tests only the source-input height, giving $H \geq h_s + c_{\text{untr}}$ for untrusted inputs, with no separate shielding-transaction check. At the defaults and $d = 1$, the formal bullet gives $h_s + 7$, while both the later prose and code give $h_s + 10$; for sufficiently large d , the code becomes laxer than both ZIP readings. The deployed trusted-input branch ($H \geq h_s + c_{\text{tr}}$) and unknown- h_s fallback likewise have no direct ZIP counterpart. There is therefore no single unambiguous ZIP threshold against which to label the implementation stricter or laxer.

The full spendability gate. Deployed note selection admits a shielded note as spendable only when all of the following hold: the note’s transaction is in a *scanned* block (`t.block non-null`); its nullifier and commitment-tree position are known and the containing shard is fully scanned (witness data exists — the *Ironwood Guide*, “Resting: the note in the commitment tree”); its mined height is at or below the anchor height; `confirmations_until_spendable = 0` under the policy; and it is not excluded by the spent-notes clause — spent in a mined or unexpired pending transaction, condition (1).

Target and anchor heights. The heights every predicate above is evaluated against are fixed at proposal time (`zcash_client_sqlite`, `src/wallet.rs` and `src/wallet/commitment_tree.rs`; `zcash_client_backend`, `src/data_api/wallet.rs`):

```
chainTip := max(scan_queue.block_range_end) - 1  (ranges are end-exclusive),
H := chainTip + 1  (the “mempool height”),
anchorHeightpool := max { checkpoint : checkpoint ≤ H - minConf },
anchor := minpools with checkpoints anchorHeightpool  (Sapling, Orchard),
```

and `propose_transfer` uses the trusted count, $\text{minConf} = c_{\text{tr}}$ (default 3); the target height that `create_proposed_transactions` hands to `Builder::new` is the proposal's `min_target_height()`. The anchor is therefore *at least* the trusted confirmation count deep under the wallet's target-height convention. It equals that depth only when every relevant tree has a checkpoint exactly at $H - c_{\text{tr}}$; checkpoint gaps and the cross-pool minimum can make it older. This fixed upper bound on anchor height avoids leaking whether the notes being spent were trusted, the property ZIP-315's anchor-selection recommendation secures. ZIP-315 phrases its recommendation relative to the chain head: its nominal block $\text{chainTip} - c_{\text{tr}}$ is one older than the wallet's newest permitted bound $H - c_{\text{tr}} = \text{chainTip} - (c_{\text{tr}} - 1)$, but actual checkpoint selection can make the deployed anchor older, so it is not unconditionally one block shallower. The tree and anchor mechanics themselves are the *Ironwood Guide's* (its commitment-tree section); the wallet-layer content is only this choice of depth.

6.7 Reorganisations: truncate and rescan

A reorganisation invalidates the wallet's suffix of the chain: mined transactions become unmined, anchors vanish, and witness data beyond the fork point is wrong. The deployed stack detects this during scanning and repairs it by truncation.

Detection. Scanning reports continuity errors: the `ScanError` variants `PrevHashMismatch` (the parent hash of a new block does not match the current tip), `BlockHeightDiscontinuity`, and `TreeSizeMismatch` all satisfy `is_continuity_error() = true`. To surface reorganisations *promptly* rather than on the next unlucky scan, `update_chain_tip` marks the `VERIFY_LOOKAHEAD = 10` blocks above the maximum scanned height with `ScanPriority::Verify` whenever that height is at or below the stable height $\text{tip} - \text{PRUNING_DEPTH}$; ranges marked `Verify` must always be scanned before `ChainTip` ranges. At NU7's proposed 25-second target spacing, that 10-block lookahead covers 250 target seconds. It is a scan-scheduling heuristic, not a consensus rule or a guarantee that no recent wallet transaction is missed.

Recovery loop. On a continuity error e , the documented algorithm is:

1. choose a rewind height at least one block before the error height (the module docs' heuristic: `e.at_height() - 10`);
2. call `WalletWrite::truncate_to_height`, which may truncate *lower* than requested (below);
3. delete cached `CompactBlocks` above the returned truncation height, re-download, and rescan — `Verify` ranges first.

The truncation target is constrained by checkpoint availability: `truncate_to_height` truncates to the greatest height at most the requested one that has a note-commitment-tree checkpoint in *every* tree containing data, and errors with `RequestedRewindInvalid` — carrying the safe rewind height, the maximum over trees of their minimum checkpoint heights — if none exists. The constant `PRUNING_DEPTH = 100` controls an inclusive checkpoint window: the code computes its pruning floor as $\text{tip} - (\text{PRUNING_DEPTH} - 1)$, a 99-block height delta when checkpoints are contiguous. It should therefore not be read as an exact 100-block rewind depth. The variant `truncate_to_chain_state` additionally permits precise truncation from provided `ChainState` frontiers even when the target checkpoint was pruned.

What truncation does. Truncating to height h :

- clamps the scan queue so the wallet's chain-tip view returns to h ;

- resets or nulls transparent outputs' `max_observed_unspent_height` — a spending transaction may be entirely invalidated by a reorganisation that alters the anchors its shielded spends were built against;
- *un-mines* every transaction with `mined_height > h`: `block`, `mined_height`, `tx_index`, and `confirmed_unmined_at_height` are all set `NULL`, so the transaction reverts to the pending state of §6.4 and will either be re-observed as mined or expire;
- truncates all populated shielded shard trees — Sapling, Orchard, and Ironwood — to the checkpoint at h (tree and anchor mechanics: the *Ironwood Guide*, its commitment-tree section);
- deletes blocks above h and stale `tx_locator_map` entries.

Sent and received notes are deliberately *not* deleted: they may hold data, such as memos, unrecoverable from the chain. Balance computations must instead simply not count received notes whose transactions are no longer mined — which the spendability gate of §6.6 already guarantees, since an un-mined transaction has no scanned block and no qualifying mined height.

Long-run states. A wallet-created transaction may become *mined* and old enough to fall below the retained checkpoint window; it may become *expired unmined*, with its inputs released (§6.5) and its record retained (Table 4) with the user notified per ZIP-203; or, for the expiry-0 transactions a caller must explicitly request via `with_expiry_height`, it may remain *pending* indefinitely, with polling ending at the first confirmed-unmined reply (§6.4) and any rebroadcast falling on the embedding wallet (Remark 6.1).

7 Payment requests (ZIP-321)

The *Ironwood Guide* constructs everything a wallet needs once the sender knows where to pay: note encryption and detection, authentication paths, anchor selection, and the ZIP-244 transaction digest. What none of that settles is the step before any of it — how the payee communicates to the payer an address, an amount, and a memo, in a form a wallet can consume mechanically. ZIP-321 (“Payment Request URIs”) standardises that step as a URI scheme; it is Status Active, Category Standards/Wallet, created 2020-08-28, owned by Kris Nuttycombe and Daira-Emma Hopwood (ZIP-321, header). This section develops the format, its deployed parser — the `zip321` crate in `librustzcash` — and the pipeline that turns a parsed request into a transaction proposal and finally a transaction.

7.1 Requests, payments, and the single-transaction rule

Definition 7.1 (Payment; payment request). A *payment* is a transfer of funds implemented by a single shielded or transparent output of a Zcash transaction. A *payment request* is a request for a wallet to construct a single Zcash transaction containing one or more payments (ZIP-321, Terminology).

A ZIP-321 URI therefore represents a request for the construction of *one* transaction; implementations SHOULD construct a single transaction paying all the specified addresses (ZIP-321, URI Semantics). The number of payments one transaction can carry is limited only by transaction-construction restrictions: the ZIP notes that if every payment were Sapling, the value-balance limit of the protocol specification, §4.13 (Balance and Binding Signature, Sapling), implies construction may fail above 2109 distinct payments, and that the effective limit may be lower when Orchard or future address types are involved. Recomputed from the specification, the bound is a

transaction-size cap used inside that section’s overflow argument, not a direct value-range consequence: each Sapling output contributes 948 bytes to the transaction — an `OutputDescriptionV4` comprises the 32-byte `cv`, `cmu`, and `ephemeralKey` fields, a 580-byte `encCiphertext`, an 80-byte `outCiphertext`, and a 192-byte proof; an `OutputDescriptionV5` is 756 bytes plus its 192-byte entry in `vOutputProofsSapling` — and a transaction must fit within the specification’s 2 MB maximum transaction size (numerically equal to the 2 000 000-byte block-size limit), so at most $\lfloor 2\,000\,000/948 \rfloor = 2109$ outputs fit (protocol specification, §4.13; field sizes from §7.1, “Transaction Encoding and Consensus”). The specification uses the cap to show that the balance sum cannot overflow the type of committed values; ZIP-321 reuses it as the payment-count ceiling.

7.2 URI syntax

Definition 7.2 (ZIP-321 URI grammar). A ZIP-321 URI is a string derived by `zcashurn` in the following ABNF (RFC 5234), reproduced from ZIP-321 (URI Syntax); the productions `ALPHA`, `unreserved`, and `pct-encoded` are those of RFC 3986, and `base64url` is described below.

```

zcashurn      = "zcash:" ( zcashaddress [ "?" zcashparams ] / "?" zcashparams )
zcashaddress  = 1*( ALPHA / DIGIT )
zcashparams   = zcashparam [ "&" zcashparams ]
zcashparam    = [ addrparam / amountparam / assetparam / memoparam
                  / messageparam / labelparam / otherreqparam / otherparam ]
NONZERO       = %x31-39
DIGIT         = %x30-39
paramindex    = "." NONZERO 0*3DIGIT
addrparam     = "address" [ paramindex ] "=" zcashaddress
amountparam   = "amount" [ paramindex ] "=" 1*DIGIT [ "." 1*8DIGIT ]
assetparam    = "req-asset" [ paramindex ] "=" *base64url
labelparam    = "label" [ paramindex ] "=" *qchar
memoparam     = "memo" [ paramindex ] "=" *base64url
messageparam  = "message" [ paramindex ] "=" *qchar
paramname     = ALPHA *( ALPHA / DIGIT / "+" / "-" )
otherreqparam = "req-" paramname [ paramindex ] [ "=" *qchar ]
otherparam    = paramname [ paramindex ] [ "=" *qchar ]
qchar         = unreserved / pct-encoded / allowed-delims / ";" / "@"
allowed-delims = "!" / "$" / "'" / "(" / ")" / "*" / "+" / "," / ":"

```

Three global properties of the grammar deserve emphasis before the per-parameter semantics.

No authority component. A ZIP-321 URI is not a hierarchical URI: the grammar cannot derive `//` after the scheme, so there is no authority component (ZIP-321, URI Syntax note). For a single payment the recipient address **SHOULD** be placed directly in the hier-part, and a URI `zcash:<address>?... MUST be treated as equivalent to zcash:?address=<address>&... (ZIP-321, URI Semantics); the query-parameter form with addrparam and distinct indices serves multiple recipients.`

Payment grouping by paramindex. Addresses, amounts, labels, and messages sharing the same `paramindex` — including the empty index — belong to the same payment. An index **MUST NOT** have leading zeros, parameter ordering is insignificant, and index values need not be sequential (ZIP-321, URI Semantics). The grammar bounds indices to `"." NONZERO 0*3DIGIT`, that is $1 \leq i \leq 9999$; the empty index denotes a distinguished payment which the deployed crate stores at map key 0. Rendering and parsing are exact inverses on this range: index 0 renders as the empty suffix, and index $i \geq 1$ renders as `.||dec(i)`; on parse, an absent index maps to key 0, and the leading-zero strings `.0` and `.01` match no production and invalidate the URI (ZIP-321, Invalid Examples).

Percent encoding is confined to qchar. Only the values of `label`, `message`, `otherreqparam`, and `otherparam` — the `qchar` productions — may contain pct-encoded sequences. Addresses, amounts, and parameter *names* must appear literally: the ZIP’s invalid examples include `zcash:taddr?amount=1%30` (encoded amount digit), `zcash:taddr?%61mount=1` (encoded parameter name), and `zcash:%74addr...` (encoded address) (ZIP-321, URI Syntax note and Invalid Examples).

The base64url alphabet. Productions `memoparam` and `assetparam` carry binary content encoded as base64url, RFC 4648 §5, with padding omitted: values 62 and 63 encode as `-` and `_`, and implementations MUST NOT accept the standard-base64-only characters `+`, `/`, `=` (ZIP-321, URI Syntax). The deployed crate decodes with the `BASE64_URL_SAFE_NO_PAD` engine.

7.3 Validity and address semantics

The ZIP imposes structural validity rules beyond the grammar (ZIP-321, URI Syntax and URI Semantics):

- a purported ZIP-321 URI that cannot be parsed according to the grammar MUST NOT be accepted;
- if any non-address parameter carries a given `paramindex`, the URI MUST contain an address parameter with that index; and
- there MUST NOT be more than one occurrence of a given parameter name at a given index.

Implementations SHOULD check that each `zcashaddress` is a valid encoding per the protocol specification, §5.6, other than a Sprout address: a transparent address (Base58Check, §5.6.1.1), a Sapling payment address (Bech32 per ZIP-173, §5.6.3.1), or a Unified Address (Bech32m per BIP 350, §5.6.4.1 and ZIP-316). New address formats SHOULD be supported whether or not ZIP-321 is updated for them; if network context is available, each address SHOULD be checked for the correct network; and every requirement of ZIP-316 applies to payments to Unified Addresses (ZIP-321, URI Semantics).

Sprout addresses MUST NOT be supported in payment requests. The rationale is economic rather than syntactic: since Canopy, ZIP-211 restricts adding value to the Sprout pool, so senders cannot generally satisfy a request to pay a Sprout address (ZIP-321, URI Semantics). Section 7.9 flags where the deployed stack enforces this.

Forward compatibility. A parameter name prefixed `req-` that the parser does not recognise renders the *entire* URI invalid (MUST); unrecognised parameters without the prefix SHOULD be ignored. The prefix is potentially part of a future parameter’s name, not a modifier applicable to arbitrary parameters, and the original parameters `address`, `amount`, `label`, `memo`, `message` never carry it, because every conformant parser is required to understand them (ZIP-321, Forward compatibility).

Backward compatibility. Implementations of the informal pre-ZIP-321 `zcash:` URI scheme generally lacked the `req-` rule, the `paramindex` multi-payment facility, the `address=` query form for single payments, and the base64url `memo` treatment (ZIP-321, Backward compatibility).

7.4 Amounts

The `amount` parameter is a decimal number of ZEC. If a fraction is present, the separator MUST be a period and both the whole and fractional parts MUST be nonempty; grouping separators are forbidden; leading zeros in the whole part and trailing zeros in the fraction are ignored; the

fraction carries at most 8 digits; and the amount MUST NOT exceed 21000000 ZEC (ZIP-321, ZEC Transfer Amount). Thus 50, 050, and 50.00 all denote 50 ZEC; 0.5 and 00.500 denote 0.5 ZEC; and 50,000.00, 50., .5, and 0.123456789 are invalid.

Construction 7.3 (Amount parsing and rendering). Fix $\text{COIN} := 10^8$ zatoshis per ZEC and $\text{MAX_MONEY} := 21\,000\,000 \cdot \text{COIN} = 2\,100\,000\,000\,000\,000$ zatoshis.

Parse. Given a string matching $1^*\text{DIGIT} [\text{"." } 1^*\text{DIGIT}]$, let c be the whole part read as an unsigned 64-bit integer, and let $z := \text{int}(\text{rpad}_8(f, 0))$, the fraction f right-padded with 0 to 8 digits ($z := 0$ when the fraction is absent). The value is

$$v := c \cdot \text{COIN} + z$$

with checked (non-wrapping) multiplication and addition, and the parse succeeds iff $0 \leq v \leq \text{MAX_MONEY}$ (the Zatoshis range check).

Render. Let $c := \lfloor v/\text{COIN} \rfloor$ and $z := v \bmod \text{COIN}$. Emit $\text{dec}(c)$; if $z \neq 0$, append $.$ followed by $\text{dec}(z)$ left-padded with 0 to 8 digits and with trailing 0 characters trimmed.

7.5 Memos, labels, and messages

The `memo` parameter carries the content of the shielded memo field — the fixed 512-byte field of the note plaintext constructed in the *Ironwood Guide* (note encryption) — as `base64url` without padding. The decoded content MUST NOT exceed 512 bytes and, if shorter, is filled with trailing zeros to 512 bytes. A memo whose same-index address does not permit memos — a transparent address — makes the *entire* URI invalid (MUST) (ZIP-321, Query Keys).

Construction 7.4 (Memo encoding and decoding). Write $\text{b64u}(\cdot)$ for RFC 4648 §5 encoding with padding omitted.

Encode. For a 512-byte memo m , the parameter value is $\text{b64u}(\text{strip}_{0x00}(m))$, where strip_{0x00} removes trailing `0x00` bytes (`MemoBytes::as_slice`).

Decode. Let $b := \text{b64u}^{-1}(\text{value})$, rejecting any input containing `+`, `/`, or `=`; require $|b| \leq 512$; then

$$m := b \parallel 0x00^{512-|b|}$$

(`MemoBytes::from_bytes` zero-pads).

An absent memo parameter yields no memo; at output construction the wallet then substitutes the “no memo” sentinel `MemoBytes::empty() = 0xF6 \parallel 0x00^{511}`.

The `label` parameter names an address (for example, the receiver’s name): a client SHOULD display it alongside the address, and it MUST be identifiable as distinct from the address itself. The `message` parameter is descriptive text the client may display to describe the payment (ZIP-321, Query Keys). Both are `qchar` values, so arbitrary UTF-8 reaches them only through percent encoding (§7.2).

7.6 Custom assets: req-asset (specified, not deployed)

The `req-asset` parameter extends payment requests to ZSA Custom Assets (ZIP-227), encoding an Asset Identifier and an amount in a single value; absent `req-asset`, the payment is for ZEC. A URI containing both amount and `req-asset` at the same index MUST be rejected. Only Orchard receivers are valid for Custom Asset payments, so a Custom-Asset request MUST use a Unified Address encoding at least one Orchard receiver (ZIP-321, Custom Assets).

Definition 7.5 (req-asset value encoding). The parameter value is

$$\text{b64u}(0\text{x}00 \parallel \text{I2LEOSP}_{64}(\text{value}) \parallel \text{EncodeAssetId}(\text{AssetId})) :$$

a version byte 0x00, the amount as an unsigned 64-bit little-endian integer of the asset’s minimal units, and the 66-byte encoded Asset Identifier of ZIP-227 — a 75-byte payload, hence 100 base64url characters. Future versions may define different structures; implementations **MUST** dispatch on the version byte (ZIP-321, Custom Assets).

Classification per our conventions: *specified*. The parameter appears in the Active ZIP-321 text, but it depends on ZIP-227, which is Status Draft, and the deployed parser rejects every unrecognised req- parameter — including req-asset — so a Custom-Asset request is invalid to the deployed stack today (zip321, src/lib.rs, to_indexed_param).

7.7 The deployed parser: the zip321 crate

The reference implementation is the zip321 crate in librustzcash (components/zip321, a single file src/lib.rs), published at version 0.9.0; release 0.1.0 (2024-08-20) factored it out of zcash_client_backend 0.13.0, which still re-exports it as a deprecated module zcash_client_backend::zip321 (components/zip321/Cargo.toml and CHANGELOG.md; zcash_client_backend, src/lib.rs). The Guide’s baseline is the local checkout — librustzcash main at commit e30517e4 — as for every other librustzcash citation in this volume, where the crate was 0.8.0. The optional-amount semantics shipped in release 0.7.0 (2026-04-23); current 0.9.0 changes dependencies and fixes the test-dependencies feature, not the parser semantics documented below.

Definition 7.6 (Types Payment and TransactionRequest). Type Payment holds a recipient address (zcash_address::ZcashAddress), an optional amount (Option<Zatoshis>), an optional memo (MemoBytes), an optional label and message, and a list other_params of retained unrecognised parameters. An absent amount means the *sender* chooses the amount — semantics introduced in release 0.7.0 (2026-04-23); releases up to 0.6.0 instead defaulted the field to zero, a behaviour ZIP-321 leaves unspecified (zip321, src/lib.rs; CHANGELOG.md, 0.7.0 entry).

Type TransactionRequest wraps a BTreeMap<usize, Payment>, map key 0 denoting the empty paramindex. Constructor new rejects more than 9999 payments (Zip321Error::TooManyPayments) and validates every non-empty request by round-tripping it through to_uri/from_uri; constructor from_indexed rejects any index key above 9999.

Construction 7.7 (Parsing: TransactionRequest::from_uri). Parsing proceeds in four stages.

- (1) Match the literal zcash: (lowercase; nom tag), then take the characters up to ? as the hier-part address. If nonempty, it must decode as a ZcashAddress and is recorded as an address parameter at payment index 0.
- (2) If input remains, require ? followed by a list, separated by & and consuming the rest of the input, of name[.index]=value pairs, each mapped to a typed parameter (address, amount, label, message, memo, or other); any unrecognised req- name is a hard error (“Required parameter {name} not recognized”).
- (3) Group parameters by payment index, rejecting any duplicate — the same param-

eter kind, or the same other-parameter name, twice at one index (`Zip321Error::DuplicateParameter`).

- (4) For each index, build a `Payment`. An address parameter must be present, an explicit amount of 0 must not target a transparent-only address, and a memo requires `can_receive_memo()` on its address; the corresponding errors are `RecipientMissing`, `ZeroValuedTransparentOutput`, and `TransparentMemo`. Labels, messages, and other parameters are collected.

Construction 7.8 (Rendering: `TransactionRequest::to_uri`). An empty request renders as `zcash:.` A request with exactly one payment stored at key 0 renders in hier-part form, `zcash:<address>` followed by `?` and the payment’s parameters if any. Otherwise every payment renders in query-parameter form: `zcash:?` followed by, in ascending key order, an address parameter and then that payment’s amount, memo, label, message, and other parameters, the index suffix omitted for key 0.

Definition 7.9 (Payment validity predicate). Constructor `Payment::new(a, v, m, ...)` returns an error iff

$$(\text{transparentOnly}(a) \wedge v = \text{Some}(0)) \vee (m \neq \text{None} \wedge \neg \text{canReceiveMemo}(a)),$$

with `PaymentError::ZeroValuedTransparentOutput` and `PaymentError::TransparentMemo` for the respective disjuncts, and `Ok` of the payment otherwise. The two predicates live in `zcash_address`: `canReceiveMemo` is true for Sprout, Sapling, and Unified addresses containing a Sapling or Orchard receiver, false for P2PKH, P2SH, and TEX; `transparentOnly` is true for P2PKH, P2SH, and TEX, and for Unified addresses that have a transparent receiver but neither a Sapling nor an Orchard receiver.

Character sets. On parse, a raw `qchar` value may contain alphanumerics plus `- . _ ~ ! $ ' ( ) * + , ; : @ %`, and a parameter name is ALPHA followed by ALPHA/DIGIT/+/-; the value is then percent-decoded and validated as UTF-8. On render, values are UTF-8 percent-encoded over the complement of `qchar`: the ASCII controls together with space and `" # % & / < = > ? [ \ ] ^ ` { | }` (the crate’s `QCHAR_ENCODE` set). Unknown parameters *without* the `req-` prefix are retained: their percent-decoded values land in `Payment::other_params` and are re-rendered, percent-encoded, by `to_uri`.

Totals. Method `TransactionRequest::total()` sums the payment amounts with overflow checking against `MAX_MONEY`, returning `Ok(None)` when any payment lacks an amount (the total is undefined) and `Err(BalanceError::Overflow)` on an out-of-range sum.

Test vectors. The crate’s test module encodes the ZIP’s own valid and invalid examples: `amount=1` parses to 10^8 zatoshis and `amount=123.456` to 12 345 600 000; the boundary 21000000 is valid and 21000000.00000001 invalid; i64-overflow and u64-wraparound amounts are invalid; `paramindex 10000` is invalid; `amount=123. is` invalid; and `req-unknown=x` is invalid.

7.8 From request to proposal to transaction

A parsed `TransactionRequest` enters the wallet through the function `propose_transfer` in module `zcash_client_backend::data_api::wallet`. It takes the wallet database, network parame-

ters, the account to spend from, an input selector, a change strategy, the request, and a confirmations policy, and returns a `Proposal<FeeRule, NoteRef>` by delegating to the input selector's `propose_transaction`; the proposal is later executed by `create_proposed_transactions`. The convenience wrapper `propose_standard_transfer_to_address` places a single `Payment` into a one-element request and uses `GreedyInputSelector` with `SingleOutputChangeStrategy`.

The selector requires every payment to carry an amount: a missing amount yields `ProposalError::PaymentAmountMissing(idx)`, so a wallet UI must fill in user-chosen amounts before proposing.

Construction 7.10 (Payment-to-pool assignment). For each $(idx, \text{payment})$ in the request, `GreedyInputSelector` converts the recipient address to the wallet's network and assigns an output pool:

- a transparent address maps to the transparent pool, emitting a `TxOut`;
- a TEX address maps to the transparent pool of the *second* step of a two-step proposal (below), its payment re-wrapped with no memo;
- a Sapling address maps to the Sapling pool;
- a Unified Address with a supported Orchard receiver maps to Orchard before NU6.3 and to Ironwood from NU6.3 onward (the Ironwood bundle delivers to the same Orchard-protocol receiver); otherwise it maps to Sapling if a supported Sapling receiver is present, then to its transparent receiver, or fails (`GreedyInputSelectorError::UnsupportedAddress`).

The selector then chooses spendable shielded notes to at least the required amount, preferring the pool matching the outputs, computes change under the change strategy, and returns a proposal whose steps embed the originating request and the resulting map `payment_pools : idx  $\mapsto$  PoolType`.

TEX addresses (ZIP-320). A payment to a TEX address is consumed as a two-step proposal: step tr_0 excludes the TEX outputs and instead creates one ephemeral transparent change output; step tr_1 spends that ephemeral output to the TEX recipients, with the memo forced to `None`. Paying a TEX address from shielded inputs in one step is rejected (`ProposalError::PaysTexFromShielded`).

Step validation. Constructor `Step::from_parts` enforces that the keys of `payment_pools` exactly match the request's payment indices, and that each selected pool satisfies `recipient_address().can_receive_as(pool)`; an Ironwood assignment is also accepted when the recipient can receive as Orchard, because the pools share the receiver protocol. It further checks that every payment has an amount, and the balance equation

$$v_{\text{transparent}}^{\text{in}} + v_{\text{shielded}}^{\text{in}} + v_{\text{prior steps}}^{\text{in}} = \text{total}(R) + \text{total}(\Delta), \quad (2)$$

where R is the embedded request, Δ the proposed balance — change, including ephemeral outputs, plus fee — and every sum is checked in `Zatoshis`; a violation is a `BalanceError`.

Serialization. A proposal serialises its transaction request *as its ZIP-321 URI*: the protobuf field `transactionRequest` of message `ProposalStep` (string, field 1, package `cash.z.wallet.sdk.ffi`) is produced by `to_uri` and parsed back with `from_uri`, alongside `PaymentOutputPool{paymentIndex, valuePool}` entries for the pool map (`zcash_client_backend, proto/proposal.proto`). This round-trip is why the crate accepts and renders the empty request

zcash: — a shielding step has no payments — a deliberate divergence from the ZIP flagged in §7.9.

Execution. At transaction creation, for each (idx, pool) in the step’s pool map the corresponding Payment is fetched from the embedded request. A shielded output uses the payment’s memo, or the sentinel `0xF6 || 0x00`⁵¹¹ when absent (§7.5); a memo on a transparent output raises `Error::Payment(PaymentError::TransparentMemo)`. For a Unified Address, the receiver actually paid is the one recorded in `payment_pools` — the Orchard, Sapling, or transparent receiver extracted from the UA. From here the construction is that of the *Ironwood Guide*: note encryption seals each output’s plaintext, and the spend and binding signatures sign the ZIP-244 sighash.

7.9 Divergences between ZIP-321 and the deployed stack

Our conventions require flagging every point where the deployed code and the ZIP disagree; ZIP-321 accumulates several, all in the parser.

- (D1) *Empty requests.* The crate accepts `zcash:` (and `zcash:?`) as a valid empty Transaction Request and renders empty requests as `zcash: (zip321, src/lib.rs)`, but the grammar of Definition 7.2 requires either an address or `? zcashparams` after the scheme, and the ZIP defines a request as having one or more payments. The divergence is deliberate: proposal serialisation round-trips shielding steps — which have no payments — through the URI form (§7.8).
- (D2) *Sprout recipients.* ZIP-321 says Sprout addresses MUST NOT be supported (§7.3), yet the crate parses a Sprout recipient without error — `ZcashAddress::try_from_encoded` decodes Sprout (`zcash_address, src/encoding.rs`) and `to_payment` performs no Sprout check; indeed `canReceiveMemo` is true for Sprout, so even a memo to one is accepted. Rejection occurs only downstream, at proposal time, when address conversion fails with `ConversionError::Unsupported("Sprout")` because `zcash_keys::address::Address` has no Sprout variant (`zcash_address, src/convert.rs`; `zcash_client_backend, src/data_api/wallet/input_selection.rs`). The deployed stack satisfies the payment-level requirement but not URI-level rejection.
- (D3) *Case sensitivity.* ABNF literal strings (RFC 5234) are case-insensitive, so a strict reading makes `ZCASH:` — and even `AMOUNT=` — grammar-valid; the deployed parser matches only lowercase (`zip321, src/lib.rs, tag("zcash:")` and the literal parameter names). The ZIP text never addresses case, which matters for QR alphanumeric-mode URIs, which are uppercase. We adopt the deployed, lowercase-only behaviour as normative and treat the ABNF case-insensitivity as a defect of the ZIP text: an uppercased URI is unparseable to the deployed stack, so a producer wanting QR alphanumeric mode must not obtain it by uppercasing the URI.
- (D4) *Valueless parameters.* The grammar makes `=` optional for `otherparam` and `otherreqparam` (a bare nonce is derivable), but the crate requires `name=value`, rejecting such grammar-valid URIs (`zip321, src/lib.rs, zcashparam`).
- (D5) *Bare percent signs.* The grammar admits `%` in a value only as pct-encoded (`% HEXDIG HEXDIG`), but the parser accepts `%` as a raw character and its percent decoder passes invalid sequences through unchanged, so `message=100%` is accepted — laxer than the grammar (`zip321, src/lib.rs, qchars` and `percent_decode`). A direct test against the crate confirms it: the values `100%`, `100%zz`, and `100%2` all parse, the malformed sequences surviving percent-decoding verbatim, while `100%25` decodes to `100%`.

- (D6) *Empty list elements.* The production `zcashparam` can derive the empty string, making `a=1&&b=2` arguably grammar-valid, while the crate rejects it (its list elements must be nonempty) — a divergence in the opposite direction of (D4) and (D5).
- (D7) *Zero-valued transparent outputs.* The crate rejects `amount=0` to a transparent-only address (Definition 7.9); ZIP-321 has no such rule. The crate’s changelog asserts such outputs are disallowed by consensus (zip321, `CHANGELOG.md`, 0.7.0 entry). The assertion is inaccurate: consensus admits zero-valued transparent outputs — `CheckTransaction` rejects only negative values and values above `MAX_MONEY` (zcashd, `src/main.cpp`), and the protocol specification constrains transparent outputs no further, requiring only a nonnegative remaining transparent value pool. What rejects them in practice is *standardness*: a zero-valued spendable output falls below the dust threshold (zero only for unspendable scripts, which no address payment produces), so zcashd’s `IsStandardTx` (`src/policy/policy.cpp`) and Zebra’s equivalent mempool check (zebrad, `src/components/mempool/storage.rs`) refuse to admit or relay it. The crate’s rule is sound — such an output is unrelayable — but on standardness, not consensus, grounds.

8 Compact blocks (ZIP-307)

A wallet that keeps payment detection local detects its notes by trial-decrypting every shielded output on the chain (*Ironwood Guide*, §“Delivery: the note reaches its owner”). That obligation costs twice: one key agreement per output for each candidate incoming viewing key, and the bandwidth of downloading every output. ZIP 307 (*Light Client Protocol for Payment Detection*) attacks the bandwidth half. A *compact block* is “a packaging of ONLY the data from a block needed to” (1) detect a payment to one’s shielded address, (2) detect a spend of one’s notes, and (3) update witnesses for new spend proofs (ZIP 307, “Compact Stream Format”). The deployed protocol widens this beyond ZIP 307’s Sapling-only scope to Orchard and Ironwood (not Sprout), and adds a fourth purpose: detecting spends of the coins of the wallet’s transparent addresses (`compact_formats.proto`, header comment, lines 21–26). The computational half survives compaction — every compact output must still be trial-decrypting — and that residual linear scan is one of the costs Project Tachyon’s proposed proof-carrying-wallet design aims to remove (Bowe–Miers, ePrint 2025/2031; tachyon-zcash/tachyon, `book/src/proof-tree.md` @ 26fdc165). The deployed service surface that carries compact blocks is catalogued in §9.4.

8.1 Normative status and wire-format history

ZIP 307 carries Status: Draft, covers Sapling only, and nowhere mentions Orchard, Chain Metadata, or transparent compact data (`zip-0307.rst`, header and whole file). Its message definitions are stale relative to deployment: the ZIP text defines `CompactBlock{BlockID id = 1; vtx = 3}` and `CompactTx{txIndex, txHash, spends, outputs}`, with element messages named `CompactSpend` and `CompactOutput` (ZIP 307, “Compact Stream Format”), whereas the deployed messages of §8.2 use different names, field numbers, and fields. The renames to `CompactSaplingSpend/CompactSaplingOutput` date to `lightwallet-protocol v0.3.1` (2021-12-09), which also added `CompactOrchardAction` (`lightwallet-protocol`, `CHANGELOG.md`).

The canonical proto files live in the `zcash/lightwallet-protocol` repository and are vendored — git subtree or symlink — into `lightwalletd`, `librustzcash` (the files under `zcash_client_backend/proto/` are symlinks into `lightwallet-protocol/walletrpc/`), and Zaino. The repository’s README states that it “IS not a specification” and defers to ZIP 307 (`lightwalletd`, `lightwallet-protocol/README.md`) — which is itself stale. No current document therefore fully specifies the deployed wire format. Per our conventions we classify the compact-block format *deployed but underspecified*: we cite `lightwallet-protocol` tagged releases as the de facto definition, and flag each point where ZIP text and deployment disagree (§8.9).

Tag	Date	Change
v0.3.1	2021-12-09	CompactOrchardAction added; CompactSpend/CompactOutput renamed
v0.3.2	2021-12-09	CompactSaplingSpend/CompactSaplingOutput CompactOrchardAction.encCiphertext renamed ciphertext
v0.3.5	2023-07-03	ChainMetadata and CompactBlock.chainMetadata added; GetSubtreeRoots/SubtreeRoot added; nul- lifier RPCs added; CompactSaplingOutput.epk re- named ephemeralKey
v0.4.0	2025-12-03	CompactTxIn/TxOut and vin/vout added; BlockRange.poolTypes added; LightdInfo.lightwalletProtocolVersion added; CompactTx.hash renamed txid (serialization- compatible)
v0.4.1	2026-02-20	nullifier RPCs designated deprecated in the proto
v0.5.0	2026-06-30	Ironwood fields added; protoVersion removed, field number 1 and name reserved

Table 5: Wire-format history of lightwallet-protocol (CHANGELOG.md); v0.5.0 is the current tagged release.

At the pinned revisions, the three codebases vendor three different revisions: lightwalletd v0.4.1 (protoVersion present, no Ironwood); librustzcash main v0.5.0 (protoVersion removed and reserved; Ironwood fields present); Zaino an intermediate snapshot carrying *both* the Ironwood fields and protoVersion (diff of the three compact_formats.proto copies). We flag the skew, and a changelog-date contradiction, in §8.9. Deployed-behaviour claims in this section are verified against librustzcash e30517e4, lightwalletd 61fee32e, Zaino 0e057e22, and zips 6961098 (checkouts of 2026-07-14).

Since that snapshot, lightwalletd has caught up: release 0.5.0 vendored tagged lightwallet-protocol v0.5.0, removed protoVersion, and added Ironwood parsing and service data; release 0.5.4 now reports lightwalletProtocolVersion = ``v0.5.0`` (lightwalletd 09593ed, CHANGELOG.md). Detailed behaviour comparisons below retain their explicit older pins so that their line citations and divergence findings remain reproducible.

8.2 The deployed messages

The messages live in the protobuf package cash.z.wallet.sdk.rpc and declare proto3 syntax (compact_formats.proto, lines 5–11). For ordinary proto3 scalar, string, bytes, and repeated fields, absence decodes as the type’s default and an absent scalar is indistinguishable from an explicitly encoded default. Message fields retain presence information. The scanner must defend against default-zero scalar metadata — the tree sizes of §8.6 are the load-bearing case.

```

message CompactBlock {
    uint32 protoVersion = 1;
    uint64 height = 2;
    bytes hash = 3;
    bytes prevHash = 4;
    uint32 time = 5;
    bytes header = 6;

    repeated CompactTx vtx = 7;
    ChainMetadata chainMetadata = 8;
}

message ChainMetadata {
    uint32 saplingCommitmentTreeSize = 1;
    uint32 orchardCommitmentTreeSize = 2;
    uint32 ironwoodCommitmentTreeSize = 3;
}

message CompactTx {
    uint64 index = 1;
    bytes txid = 2;
    uint32 fee = 3;
    repeated CompactSaplingSpend spends = 4;
    repeated CompactSaplingOutput outputs = 5;
    repeated CompactOrchardAction actions = 6;
    repeated CompactTxIn vin = 7;
    repeated TxOut vout = 8;
    repeated CompactOrchardAction ironwoodActions = 9;
}

```

removed in v0.5.0; number and name reserved
block height
block hash, 32 bytes
parent block hash, 32 bytes
Unix epoch time
“should always be unset (empty)”
compact transactions
end-of-block tree sizes

Sapling tree size at end of block
Orchard tree size at end of block
v0.5.0 only

position within the block
32 bytes, protocol byte order
unpopulated in practice

coinbase input omitted
v0.5.0 only

The container messages, from the deployed protos (lightwalletd vendored copy, `walletrpc/compact_formats.proto`, lines 14–80; `librustzcash` copy, lines 14–86):

- Field header “should always be unset (empty)” per the proto comment (lines 27–30). ZIP 307’s block header validation — Equihash (*Consensus Guide*, §“Equihash and the memoryless model”), bits, difficulty — is therefore unimplementable against deployed servers; the ZIP itself flags the section as “only partially implemented: currently only `prevHash` is checked” (ZIP 307, “Block header validation”). Client-side, only `prevHash` and height continuity are enforced (§8.6). ZIP 221’s chain-history commitment was designed to support succinct probabilistic header-chain verification, but the pinned sync client does not consume it (ZIP 221).
- Field `protoVersion` never varies on the wire. `lightwalletd` never sets it, the assignment commented out as a TODO (`parser/block.go`, line 112); deployed Zaino serves 0 on every path (`zaino-state`, `compact_block.rs` line 285, `legacy.rs` line 1113), which for a proto3 scalar is byte-identical to `lightwalletd`’s absent field. Its lone non-zero constructor (`zaino-fetch`, `src/chain/block.rs`, line 218) is a public helper whose only caller in the pinned repository formats a parse-error Debug string; no serving path calls it. Thus that literal does not reach the audited service’s wire output. Version v0.5.0 removed the field, reserving number 1 and the name (`librustzcash` copy, lines 33–36). The `librustzcash` scanner never reads it. Flagged in §8.9.
- Message `ChainMetadata` states each pool’s note commitment tree size *as of the end of this block*, as `uint32` (`compact_formats.proto`, lines 14–17). A full depth-32 tree has size 2^{32} , which this field cannot represent. Although `ScanError::TreeSizeOverflow` exists, the compact-scanning path at the pinned and current revisions does not emit it; its end-size additions remain `u32` arithmetic. Section 8.6 shows how these sizes turn a compact block into note positions.

- Field `txid` carries the 32-byte transaction identifier in protocol byte order; the proto comment mandates that it “MUST NOT be reversed or hex-encoded” — the reversed hex form is display-only — citing protocol spec §7.1.1, “Transaction Identifiers” (`compact_formats.proto`, lines 52–57).
- The `fee` field is documented “present if server can provide”, but neither deployed server populates it: `lightwalletd` leaves it commented out with `TODD: calculate fees` (`parser/transaction.go`, line 407) and `Zaino` hardcodes `fee: 0` (`zaino-fetch, src/chain/transaction.rs`, line 369). Clients cannot rely on it. For a transaction with no transparent inputs the proto comment gives the client-side reconstruction (`compact_formats.proto`, lines 59–64):

$$\text{fee} = \text{valueBalanceSapling} + \text{valueBalanceOrchard} + \sum v_{\text{PubNew}} - \sum v_{\text{PubOld}} - \sum t_{\text{Out}},$$
with `valueBalanceIronwood` joining the sum in v0.5.0.¹

- Field `vin` omits the coinbase transaction’s single null-outpoint input; a light client recognises a coinbase `CompactTx` by `index = 0`, coinbase being always first in a block. Server `lightwalletd` implements exactly this, skipping `vin` when `index = 0` (`compact_formats.proto`, lines 70–76; `parser/transaction.go`, lines 398–403). The transparent element messages are `CompactTxIn{prevoutTxid, prevoutIndex}` and `TxOut{value, scriptPubKey}` (`compact_formats.proto`, lines 82–104).

The shielded elements. Each shielded element message keeps only the detection-relevant bytes of its full on-chain description.

Message	Fields (bytes)	Payload	Full form	Saving
<code>CompactSaplingSpend</code>	<code>nf</code> (32)	32	384	~90%
<code>CompactSaplingOutput</code>	<code>cmu</code> (32), <code>ephemeralKey</code> (32), <code>ciphertext</code> (52)	116	580 + 32	~80%
<code>CompactOrchardAction</code>	<code>nullifier</code> (32), <code>cmx</code> (32), <code>ephemeralKey</code> (32), <code>ciphertext</code> (52)	148	—	—

Table 6: Compact shielded elements (`compact_formats.proto`, lines 106–130). The Sapling comparison baselines and saving figures are ZIP 307’s historical pre-v5 accounting (“Spend Compression”, “Output Compression”); the 148-byte Orchard total is arithmetic, not stated in the sources. Payload figures count raw bytes; on the wire, a `CompactSaplingSpend` encodes to 34 bytes (36 as a `CompactTx.spends` element), a `CompactSaplingOutput` to 122 (124), and a `CompactOrchardAction` to 156 (159 — its 156-byte payload needs a two-byte length varint). Field headers add 2 bytes per 32-byte field and 2 per 52-byte ciphertext. Computed by script, proto-encoding every compact element of mainnet block 3,412,934 (3 spends, 8 outputs, 10 actions; `GetBlock`, `lightwalletd` v0.4.19, 2026-07-15); the sizes are block-independent because all field lengths are fixed.

A `CompactSaplingSpend` retains only the 32-byte nullifier of the pre-v5, 384-byte `Spend` description counted by ZIP 307 (`cv` 32, `anchor` 32, `nullifier` 32, `rk` 32, `zkproof` 192, `spendAuthSig` 64) — the one field needed to detect a spend of one’s own notes; ZIP 307 quotes this historical comparison as a 90% bandwidth improvement. In v5/v6 the anchor and bundle structure are shared, so 384 bytes is not the marginal size of each additional spend (ZIP 307, “Spend Compression”). A `CompactSaplingOutput` retains `cmu`, the ephemeral key, and the first 52 bytes of `encCiphertext`, “Total size is 116 bytes” per the proto comment, against 580 bytes of ciphertext plus the 32-byte

¹Type `uint32` additionally caps the representable fee at $2^{32} - 1$ zatoshi (≈ 42.9 ZEC); the proto comment does not remark on this.

ephemeral key streamed in full form — ZIP 307’s 80% reduction (ZIP 307, “Output Compression”; `lightwalletd compact_formats.proto`, lines 114–122; full encodings per protocol spec §§7.3–7.4). A `CompactOrchardAction` serves both roles of an Action in one message: its `nullifier` detects spends of the input note, while `cmx`, `ephemeralKey`, and `ciphertext` detect and place the output note (`compact_formats.proto`, lines 124–130; full Action encoding per protocol spec §7.5).

8.3 NU7: block frequency and scanning work

At the 25-second target spacing of ZIP 218, a target day contains $86,400/25 = 3456$ blocks. Its shielded-work limit would bound scanning work per block. The calculation below uses the proposed shared 330-unit budget including both Orchard and Ironwood, whose inclusion still awaits the open pull request 1361.

Each Orchard or Ironwood Action contributes one 148-byte compact field payload (Table 6); each Sapling spend or output uses one unit of the proposed budget and carries a smaller payload. Consequently, filling every unit with Orchard-protocol Actions gives

$$\begin{aligned}\text{Actions per target day} &= 330 \cdot 3456 = 1,140,480, \\ \text{field bytes per target day} &= 148 \cdot 1,140,480 = 168,791,040.\end{aligned}$$

This is about 168.79 decimal MB of *shielded field payload*, not a bound on total network traffic. Serialising those Actions as repeated message elements already costs 159 bytes each, or 181,336,320 bytes over the same target day, before transaction and block metadata, transparent data, transport overhead, and full-transaction fetches. Actual daily totals also vary with the number of blocks mined: target spacing is not a deterministic rate limit.

Trial decryption remains per output and per candidate incoming viewing key. The Action-only case therefore calls for up to 1,140,480 attempts per key over that target day. It is not a universal two-key budget per account: ZIP 326 allows restoration to require more candidate keys before the account’s received funds establish which derivation is in use. Faster blocks require updated throughput and retention estimates, not a new decryption algorithm or a new compact-note format.

8.4 Omissions and fetch-on-detect

Compact blocks omit the 512-byte memo, the 16-byte AEAD tag, the 80-byte `outCiphertext`, `cv`, `rk`, anchors, zkproofs, and signatures (ZIP 307, “Transaction privacy”). Recovering a memo, or spend information via the outgoing ciphertext, thus requires downloading the full transaction — deployed as the `GetTransaction` RPC. ZIP 307 attaches three requirements to that fetch: the client SHOULD obscure its interest by also downloading numerous uninteresting transactions, SHOULD download all transactions of any block it touches, and MUST convey the privacy reduction to the user (ZIP 307, “Transaction privacy”).

The deployed pipeline expresses the fetch as data: `scanning` emits `TransactionDataRequest::Enhancement(txid)` — “download of complete raw transaction data” — which the caller satisfies via `GetTransaction` and feeds to `wallet::decrypt_and_store_transaction(zcash_client_backend)`. The crate’s own sync loop documents that it does not yet perform enhancement (`zcash_client_backend, src/sync.rs`, lines 1–10).

8.5 Compact trial decryption

The 52-byte `ciphertext` field is the prefix of `encCiphertext` that encrypts exactly the *compact note plaintext*: lead byte (1) || diversifier d (11) || value v (8) || `rseed` (32), so `COMPACT_NOTE_SIZE = 52`; the full plaintext appends the 512-byte memo, and the full ciphertext a 16-byte tag, giving 580 bytes. ZIP 307’s rationale: these bytes “contain the contents and opening of the note commitment,

which is all of the data needed to spend the note and to verify that the note is spendable” (ZIP 307, “Output Compression”). The *Ironwood Guide* constructs compact decryption for Orchard in §“Delivery: the note reaches its owner”; we do not repeat the construction, and record here only the deployed mechanics and the integrity argument.

Function `try_compact_note_decryption` (doc-comment: “Implements the procedure specified in ZIP 307”) proceeds as follows:

$$K := \text{KDF}(\text{KA.Agree}(\text{ivk}, \text{epk}), \text{ephemeralKey}),$$

$$\text{np} := \text{ciphertext}[0..52] \oplus \text{ChaCha20}_K(\text{nonce} = 0^{96})[64..116],$$

a bare ChaCha20 keystream with zero nonce, sought to byte offset 64 — skipping keystream block 0, which the full AEAD reserves for the Poly1305 key. The candidate `np` is parsed as a compact note plaintext (no memo), then `check_note_validity` accepts iff

- (i) the plaintext parses, and
- (ii) the recomputed commitment matches the on-chain one, $\text{Extract}(\text{NoteCommit}_{\text{rcm}}(\dots)) = \text{cmu}$ (respectively `cmx`), and
- (iii) for ZIP-212 Sapling and for Orchard/Ironwood, the ephemeral secret `esk` re-derived from `rseed` reproduces the published key: the byte encoding of $\text{KA.DerivePublic}(\text{esk}, g_d)$ equals `ephemeralKey`, compared in constant time.

Condition (ii) is the integrity substitute: truncation at 52 bytes discards the AEAD tag, so a compact decryption cannot be authenticated via Poly1305; ZIP 307 instead recomputes the note commitment from the decrypted plaintext and compares against the on-chain `cmu`, itself authenticated by the binding signature over the transaction (ZIP 307, “Output Compression”). Condition (iii) is the deployed strengthening beyond the tag: it defeats a mauled ephemeral key (*Ironwood Guide*, §“Delivery: the note reaches its owner”). For pre-ZIP-212 Sapling notes the domain cannot re-derive `esk`, so the implementation accepts without condition (iii).

Lead byte and the ZIP-212 grace period. The lead byte is `0x01` pre-Canopy and `0x02` under ZIP-212; deployed code switches on `zip212_enforcement`: during the grace period of 32,256 blocks after Canopy activation both lead bytes are accepted, and thereafter only `0x02` (`zcash_primitives`, `src/transaction/components/sapling.rs`, lines 27–40; `zcash_protocol`, `src/consensus.rs`, line 681). ZIP 307’s pseudocode disagrees with itself here: its two “[Canopy onward]” lines are *both* conditioned on $\text{height} < \text{CanopyActivationHeight} + \text{ZIP212GracePeriod}$, the first rejecting `leadByte` $\notin \{0x01, 0x02\}$ and the second rejecting `leadByte` $\neq 0x02$ (ZIP 307, “Scanning for relevant transactions”, `zip-0307.rst`, lines 275–276). Read literally, the pair rejects `0x01` *during* the grace period and constrains nothing after it — the inverse of the intent; the second condition should read $\geq$. Deployed code implements the intended semantics. The defect is tracked in `research/upstream-defects.md`, item 1 (ZIP 307: second Canopy-onward lead-byte condition rejects `0x01` during the grace period; verified 2026-07-15 at zips 69610984). We flag the divergence in §8.9.

8.6 Scanning: spends, positions, and tree sizes

Spend detection. The scanner parses every compact nullifier field up front: each `CompactSaplingSpend.nf` and each `CompactOrchardAction.nullifier`. A malformed length yields `ScanError::EncodingInvalid` rather than a panic. The parsed nullifiers are compared against the wallet’s known-unspent nullifiers in constant time (`find_spent`); unmatched nullifiers are retained in per-block nullifier maps.

Tree building and positions. Every `cmu` and `cmx` in the block — not only the wallet’s — is appended to the wallet’s note commitment tree with `Retention` marks, and each detected note is marked at its position. The position is computed arithmetically, never by replaying the chain:

$$\text{pos}(\text{output } i \text{ of } tx) = \text{start_tree_size} + \text{offset}(tx) + i,$$

where `offset(tx)` counts the outputs of earlier transactions in the block (`PositionTracker::sapling, orchard_note_position`). The shardtree that consumes these marks is constructed in the *Ironwood Guide*, §“Resting: the note in the commitment tree”; the observation that a light wallet derives positions from boundary metadata and output order appears in §“Delivery: the note reaches its owner”.

Positions are load-bearing, not merely convenient: the `TreeSizeUnknown` doc-comment records that without the tree size it is “impossible to construct the nullifier for a detected note” — the Sapling nullifier depends on the note’s position, via $\rho = \text{MixingPedersenHash}(\text{cm}, \text{NotePosition})$; Orchard’s does not, but both pools need positions for witnesses (protocol spec, “Computing ρ values and Nullifiers” and “Dummy Notes”).

Where the tree size comes from. The start size for a block is resolved by a ladder (`tree_sizes_around`):

$$\text{start(pool)} := \begin{cases} s_{\text{prior}} & \text{prior BlockMetadata known;} \\ s_{\text{final}} - n_{\text{block}} & \text{chainMetadata present, by checked subtraction;} \\ 0 & \text{block below the pool’s activation height;} \\ \perp & \text{otherwise (ScanError::TreeSizeUnknown),} \end{cases}$$

where s_{final} is the `chainMetadata` end-of-block size and n_{block} the block’s own output count. The subtraction is checked_sub: underflow yields `ScanError::TreeSizeInvalid`, which is precisely the guard against proto3’s default-zero metadata (§8.2). After walking all `CompactTx`s the accumulated position must equal the `chainMetadata` size, else scanning fails with `ScanError::TreeSizeMismatch{given, computed}`.

The error taxonomy routes recovery: `TreeSizeMismatch`, `PrevHashMismatch` (`prevHash` against the prior block’s hash), and `BlockHeightDiscontinuity` (`height` $\neq$ `prev` + 1) report `is_continuity_error() = true` and trigger the truncate-and-rescan reorganisation path (§6.7); `EncodingInvalid`, `TreeSizeUnknown`, `TreeSizeInvalid`, and `TreeSizeOverflow` do not. The overflow variant is emitted by the full-block scanner, not by the compact path described above. The final tree size also drives checkpointing: the helpers `compact_tx_contains_last_sapling_outputs_in_block`, `..._orchard_actions_in_block`, and `..._ironwood_actions_in_block` compare the running position plus the current transaction’s output or action count against the final size, to decide when to write the block-boundary checkpoint into the commitment tree.

The server side of ChainMetadata. Server `lightwalletd` fills the tree sizes from the node’s `getblock verbosity=1` response fields `trees.sapling.size` and `trees.orchard.size` (`lightwalletd`, `common/common.go`, lines 163–175, 399–400). The deployed backend is `zebrad` (§9.5), whose `getblock` omits each subfield while the pool’s tree is empty — in which case Go’s JSON unmarshal leaves `Size = 0`, which is correct for an empty tree. `TreeSizeInvalid` catches a reported size smaller than the block’s own output count, not zero metadata unconditionally. Historically, `zcashd` (retired; last release v6.10.0, 2025-10-03) computed the sizes by anchor lookup (`GetSaplingAnchorAt/GetOrchardAnchorAt` on `hashFinalSaplingRoot/hashFinalOrchardRoot`) and omitted the subfield when the anchor was unavailable (`zcashd`, `src/rpc/blockchain.cpp`, lines 279–297; the `trees` field dates to `zcashd` commit `cf2b6c796`, “rpc: Add trees field to getblock RPC output”).

8.7 The scan pipeline: chain state, batching, caching

Function `scan_cached_blocks` requires a `ChainState` — the final Sapling, Orchard, and Ironwood tree frontiers as of `from_height - 1` — and asserts `from_height = chain_state.height + 1`; all three trees have depth 32. The `ChainState` comes from the server’s `GetTreeState` RPC via `TreeState::to_chain_state()`, which hex-decodes and byte-reverses the block hash and parses empty tree fields as empty trees. Trial decryption of compact outputs is batched through `BatchRunners` with batch size 100.

The proto accessors split their failure modes: block-level convenience accessors panic on malformed server data (`CompactBlock::hash` and `prev_hash` if not 32 bytes, `height` if outside `u32`), while per-output accessors (`cmu`, `cmx`, `nf`, ephemeral key) return `CompactFormatError::InvalidLength`, `InvalidValue`; the `From` impls that *build* compact messages from full bundles take `encCiphertext[0..COMPACT_NOTE_SIZE]`. The scanner bounds `CompactTx.index` to `u16` (the expect reads “Cannot fit more than 2^{16} transactions in a block”), and `CompactBlock.time` — `uint32` Unix epoch, “if non-zero” — flows into `ScannedBlock` and thence wallet metadata.

Compact blocks are cached as serialized protos. Crate `zcash_client_sqlite` offers `BlockDb`, a `compactblocks` SQLite table decoded on read (`CompactBlock::decode`), and `FsBlockDb`, one file per block plus a `BlockMeta` row {`height`, `blockhash`, `time`, `sapling_outputs_count`, `orchard_actions_count`} that enables range queries without decoding.

ZIP 307’s own client-operation sketch — append every `cmu` to a depth-32 incremental Merkle tree, keep a per-note incremental witness, and cache roughly 100 recent tree states because “no full Zcash node will roll back the chain by more than 100 blocks” (ZIP 307, “Creating and updating note witnesses”) — describes the pre-shardtree design. The deployed replacement is the position-marked shardtree fed by `GetSubtreeRoots` (RPC added in `lightwallet-protocol` v0.3.5), whose mechanics the *Ironwood Guide* constructs in §“Resting: the note in the commitment tree”.

8.8 Server inclusion semantics and successor surface

Which transactions a compact block contains is itself divergent. ZIP 307 states that “By default, `CompactBlocks` only contain `CompactTxs` for transactions that contain Sapling spends or outputs” (ZIP 307, “Block header validation”). Server `lightwalletd` builds and caches compact representations of *all* transactions, transparent-only ones included, with `vin/vout` populated (`parser/block.go`, lines 120–125), and `GetBlock` serves them unfiltered per the v0.4.1 contract; `GetBlockRange` filters per request — with empty `poolTypes` it prunes each transaction to its Sapling and Orchard components and drops transactions left empty (`common/common.go`, `FilterTxPool/filterBlockPool`, lines 526–653) — so the default sync stream carries shielded-relevant transactions only, as Zaino’s does. The successor Zaino filters out `CompactTxs` whose compact fields are all empty, and supports per-request pool filtering via `PoolTypeFilter` and `BlockRange.poolTypes` (`zaino-proto`, `src/proto/utls.rs`, `compact_block_with_pool_types`, lines 328–380, applied on the serving paths in `zaino-state`, `chain_index.rs`) — a filter added in v0.4.0. At the pinned v0.4.1 contract an empty `poolTypes` means Sapling and Orchard; v0.5.0 extends that legacy default to include Ironwood (`service.proto`, lines 38–41). Both pinned servers therefore default to shielded-relevant transactions on the sync stream; we flag the deployed defaults against the ZIP in §8.9.

Two `lightwalletd` RPCs, `GetBlockNullifiers` and `GetBlockRangeNullifiers`, strip actions to nullifier-only, drop outputs, `vin`, and `vout`, and zero both `ChainMetadata` tree sizes (“these are not needed (we prefer to save bandwidth)”) (`frontend/service.go`, lines 212–302). Blocks so obtained cannot in general feed ordinary scanning: they omit the commitments needed by the position ladder of §8.6, stripped Orchard actions fail compact-output parsing, and zeroed sizes can conflict with preceding tree metadata. These RPCs serve spend-status refresh, not scanning. Both were declared deprecated in the `lightwallet-protocol` v0.4.0 changelog (2025-12-03); v0.4.1 designated them deprecated in the proto itself (`option deprecated =`

true, service.proto:247,267), in favour of GetBlockRange with poolTypes (service.proto, lines 244–266; CHANGELOG.md, v0.4.0/v0.4.1).

Ironwood. The v0.5.0 fields (ironwoodCommitmentTreeSize = 3, ironwoodActions = 9) exist only in lightwallet-protocol v0.5.0; librustzcash gates the corresponding tree accounting on NetworkUpgrade::Nu6_3, and the pinned lightwalletd revision neither parses nor serves Ironwood data (zcash_client_backend, src/scanning/compact.rs, lines 687–696; proto/compact_formats.proto, lines 17, 74). This describes the pinned lightwalletd revision: current lightwalletd 0.5.4 parses and serves Ironwood and vendors the tagged v0.5.0 protocol. The pool is *deployed*, although ZIP 2005 remains Proposed and ZIP 258 remains Draft as a document. NU6.3 is active from Testnet height 4,134,000 and Mainnet height 3,428,143. The pinned Zaino implementation wires both fields through its serving builders (zaino-state, chain_index/types/db/legacy.rs), where they remain empty until NU6.3 data exists on chain.

8.9 Divergences between ZIP-307 and the deployed protocol

Per our conventions, we flag rather than resolve the following.

- *No complete normative specification of the deployed wire format.* ZIP 307 is Draft, Sapling-only, and its message definitions are stale — old names, old field numbers, no Orchard, no ChainMetadata, no vin/vout — while the lightwallet-protocol README disclaims being a specification and defers to ZIP 307 (zip-0307.rst, “Compact Stream Format”; lightwallet-protocol/README.md). The deployed format is specified only by tagged proto releases.
- *The grace-period pseudocode bug.* Both “[Canopy onward]” conditions in ZIP 307’s compact-decryption procedure test height < CanopyActivationHeight + ZIP212GracePeriod; read literally they reject 0x01 during the grace period and constrain nothing after it. Deployed zip212_enforcement accepts both lead bytes during the grace period and requires 0x02 after (zip-0307.rst, lines 275–276; zcash_primitives, src/transaction/components/sapling.rs, lines 27–40; research/upstream-defects.md, item 1, verified 2026-07-15 at zips 69610984).
- *Inclusion semantics.* ZIP 307 says Sapling-relevant transactions only; the deployed default stream includes Orchard data, transparent data is available on request (BlockRange.poolTypes), and GetBlock returns all pools unconditionally (§8.8).
- *A protoVersion that never varies on the wire.* Server lightwalletd never sets it; deployed Zaino serves 0, byte-identical to an unset proto3 scalar; and v0.5.0 removed it with field number and name reserved (issue zcash/lightwallet-protocol#25); librustzcash never reads it (§8.2).
- *Dead header-validation surface.* The proto declares header “should always be unset”; ZIP 307’s header and finalSaplingRoot checks are unimplementable against deployed servers, and only prevHash/height continuity is enforced client-side (§8.2, §8.6).
- *An advertised but unpopulated fee field.* The proto says “present if server can provide”; lightwalletd has a TODO, Zaino hardcodes zero (§8.2).
- *Vendored-revision skew at the pinned revisions.* lightwalletd vendors v0.4.1, librustzcash main v0.5.0, and Zaino an intermediate snapshot with both the Ironwood fields and protoVersion; Zaino’s vendored changelog dates v0.5.0 as 2026-07-03 without the protoVersion removal, while the canonical changelog dates it 2026-06-30 with the removal — so Zaino’s vendored proto still *declares* the field the canonical proto now reserves,

though it serves only 0 on the wire (diff of the three `compact_formats.proto` copies; `zaino-protocol/lightwallet-protocol/CHANGELOG.md`, lines 40–51).

9 The light-client service

A light wallet holds keys and no chain. The *Ironwood Guide* (§“Delivery: the note reaches its owner”) establishes what synchronisation consumes: every compact shielded output on the chain, for trial decryption; every nullifier, for spentness; the note commitment tree’s frontier and subtree roots, for witness assembly; and, for the few transactions that prove to be the wallet’s own, the full serialised bytes. The deployed layer that delivers these is a single gRPC service, `cash.z.wallet.sdk.rpc.CompactTxStreamer`, spoken between the wallet and an untrusted proxy in front of a full node. This section documents the service as deployed: the authority trail from ZIP 307 through the canonical protobuf definitions (§9.1), the compact data format (§8), the interaction model the ZIP prescribes (§9.3), the twenty RPCs and their semantics (§9.4), the server implementations — `lightwalletd`, the deployed proxy (§9.5); Zaino, its successor (§9.6); and Zebra’s newer in-process server (§9.7) — and a register of the points where the implementations and the normative texts disagree (§9.8). The linear costs this architecture accepts — trial decryption of every output, a maintained witness per note — are precisely the costs Project Tachyon’s proposed proof-carrying-wallet design targets (Bowe–Miers, ePrint 2025/2031; `tachyon-zcash/tachyon`, `book/src/proof-tree.md` @ 26fdc165).

9.1 Architecture and authority

ZIP 307, “Light Client Protocol for Payment Detection” (Status: Draft), fixes a three-component architecture: a Zcash full node as the root of trust; a *proxy server* that converts chain data into a compact format; and the light client (ZIP 307, § “High-Level Design”). The security model is deliberately narrow: the client obtains *payment detection privacy* against an honest-but-curious proxy, and trusts that proxy to provide a correct view of the best chain; network-level privacy (Tor, Nym) is assumed to be provided separately; spending privacy is out of scope (ZIP 307, § “Security Model”). The chain has carried commitments designed to let a client check that view succinctly since Heartwood (ZIP 221, via `hashLightClientRoot`, renamed `hashBlockCommitments` at NU5, where the field also binds a commitment to the block’s transaction signatures and proofs); no deployed client in the pinned `lightwalletd` or `zcash_client_backend` sources verifies those commitments (ZIP 221; ZIP 244; negative source check at the pinned revisions).

Where the contract lives. The wire contract is *not* in the ZIP. The two protobuf files that define the service — `service.proto` and `compact_formats.proto` — have their canonical home in the `zcash/lightwallet-protocol` repository, which self-describes as the canonical Protobuf definitions but explicitly not a specification: “The Zcash Light Client Protocol is described in ZIP-307” (`lightwallet-protocol`, `README.md`). The authority trail is therefore circular — the ZIP defers detail to the deployed API, and the API repository defers normative force back to the ZIP — and the circle does not close: ZIP 307’s own protobuf sketch (`CompactSpend` and `CompactOutput` messages, a `BlockID` embedded in `CompactBlock`, a `RangeFilter`, a `FullTransaction` return type, a four-RPC `CompactTxStreamer`) differs from the deployed proto (`CompactSaplingSpend`, `CompactSaplingOutput`, `CompactOrchardAction`, flat height and hash fields, `BlockRange`, `RawTransaction`, twenty RPCs), and the ZIP warns that “the details of the API described below may differ from the implementation” (ZIP 307, § “Compact Stream Format”, § “Proxy operation”). We treat the canonical proto files, versioned and change-logged, as the deployed contract, and cite the ZIP for architecture and rationale. Both `lightwalletd` and Zaino vendor the proto files by git subtree (`lightwallet-protocol`, `README.md`).

Contract version. The canonical protocol is at v0.5.0 (2026-06-30). Version 0.4.0 (2025-12-03) added transparent data to the compact format — `PoolType`, `CompactTxIn`, `TxOut`, `CompactTx.vin/vout`, `BlockRange.poolTypes` — and the `LightdInfo` fields `upgradeName`, `upgradeHeight`, and `lightwalletProtocolVersion`, and declared `GetBlockNullifiers` and `GetBlockRangeNullifiers` deprecated; v0.4.1 designated the pair deprecated in the proto itself (option deprecated = true, service.proto:247,267) and documented `GetBlock`’s transparent-data behaviour. Version 0.5.0 added the Ironwood fields and removed and reserved `CompactBlock.protoVersion` (`lightwallet-protocol`, `CHANGELOG.md`).

The proto forks at the pinned revisions. The following is the July snapshot preserved for the behaviour audit. The canonical v0.4.1 has `PoolType = {INVALID, TRANSPARENT, SAPLING, ORCHARD}` and `ShieldedProtocol = {sapling, orchard}`. The fork in `librustzcash` (`zcash_client_backend/proto/service.proto` and `compact_formats.proto`) adds `IRONWOOD = 4`, `TreeState.ironwoodTree = 7`, `ShieldedProtocol.ironwood = 2`, `ChainMetadata.ironwoodCommitmentTreeSize = 3`, and `CompactTx.ironwoodActions = 9`. Zaino’s fork (`zaino-proto`) adds `IRONWOOD = 4`, `ironwoodTree = 7`, `ChainMetadata.ironwoodCommitmentTreeSize = 3`, and `CompactTx.ironwoodActions = 9`, but not `ShieldedProtocol.ironwood`, and omits the deprecated options on the two nullifier RPCs (diff of the three proto trees; zaino, docs/internal_spec.md, § “Zaino-Proto”). Zaino’s documentation states the local copies exist “for development purposes” with a plan to rely on `librustzcash`’s. The “as deployed” wire surface therefore differs by repository; this volume cites the canonical file and flags fork-only fields where they appear.

Remark 9.1 (Sources and pinning). Deployed-behaviour claims in this section were verified against `lightwalletd` master 61fee32e (2026-06-25), `zaino` 0e057e22 (2026-07-04), and `librustzcash` e30517e4 (2026-07-10); Zaino pins `zebra-chain` 11.0.0 from `crates.io` (`Cargo.lock`). File-and-line citations below refer to these revisions.

9.2 The compact format

The proxy’s one transformation is compression: it strips from each transaction everything a detecting wallet does not need. Proofs, signatures, and value commitments verify the chain, and the client has delegated chain verification to the proxy; what remains is exactly the trial-decryption input and the nullifier.

Definition 9.2 (Compact Sapling output, ZIP 307). A compact output retains three fields of a Sapling Output description:

```
CompactSaplingOutput = (cmu[32], ephemeralKey[32],
                        ciphertext = encCiphertext[0..52]),
```

116 bytes per output against ZIP 307’s comparison baseline of a 580-byte ciphertext plus its 32-byte ephemeral key — an ~80% reduction (ZIP 307, § “Output Compression”; `compact_formats.proto:114--122`). The 52-byte ciphertext prefix covers the compact note plaintext — the contents and opening of the note commitment. Truncation discards the AEAD tag; integrity of the decrypted prefix is instead checked by recomputing the note commitment, as the *Ironwood Guide* constructs in § “Delivery: the note reaches its owner”.

Definition 9.3 (Compact Orchard action, deployed proto). The deployed format extends the

same compression to Orchard, one message per Action:

```
CompactOrchardAction = (nullifier[32], cmx[32], ephemeralKey[32], ciphertext[52])
```

(compact_formats.proto:124--130). The Orchard message carries its nullifier inline because an Action both spends and creates; on the Sapling side, spend compression is a separate message containing the 32-byte nullifier. ZIP 307's ~90% figure compares this with the pre-v5 384-byte Spend encoding, not the marginal v5/v6 spend size (ZIP 307, § "Spend Compression").

Transparent data. Since v0.4.0 the compact format can carry transparent inputs and outputs (CompactTxIn, TxOut, CompactTx.vin/vout); v0.4.1 fixes the convention that the coinbase transaction's single null-outpoint vin is omitted, and that clients identify the coinbase by testing CompactTx.index = 0 (service.proto:225--237; compact_formats.proto:70--76).

Remark 9.4 (The fee field is dead on the wire). The proto documents CompactTx.fee as "present if server can provide", with the stateless-server formula for the case of no transparent inputs,

$$\text{fee} = \text{valueBalanceSapling} + \text{valueBalanceOrchard} + \sum v_{\text{PubNew}} - \sum v_{\text{PubOld}} - \sum t_{\text{Out}}$$

(compact_formats.proto:59--64); with transparent inputs a stateless server cannot compute the fee at all. Neither deployed server ever sets the field: lightwalletd's parser leaves it at 0 with a "TODO: calculate fees" comment (lightwalletd, parser/transaction.go:397--432), and Zaino's conversions pass fee = None, encoded 0 (zaino, zaino-state/src/chain_index/types/db/legacy.rs:1397--1470). The deployed contract is therefore that the field carries no information, and a client must treat it as absent; we flag the proto's "present if" language as unimplemented on both servers.

9.3 The interaction model of ZIP 307

The ZIP prescribes a four-phase client-server interaction (ZIP 307, § "Client-server interaction"):

- (A) *Startup.* The client fetches the latest BlockID. A wallet importing a pre-existing seed with no recorded birthday starts at Sapling activation, since no shielded address it can derive predates Sapling (ZIP 307, § "Importing a pre-existing seed").
- (B) *First sync.* The client requests GetBlockRange(X, Y) and, on any subsequent request, verifies the hash of the overlap block — the last block it already holds — against its cached copy, so that a reorganisation is detected as a hash mismatch.
- (C) *Transaction fetching.* Detected notes are completed by fetching full transactions via GetTransaction. Each such fetch names a txid to the proxy, so the ZIP mandates cover traffic: clients SHOULD download decoy transactions, or all transactions of a touched block, and MUST warn users of the privacy loss otherwise (ZIP 307, § "Transaction privacy").
- (D) *Incremental sync.* Later syncs repeat (B) from the cached tip, re-checking the overlap block.

Two hardening measures described by the ZIP are not deployed. Block-header validation by the client is described as only partially implemented — only prevHash linkage is checked (ZIP 307, § "Block header validation"). The *bucketing* privacy enhancement — for bucket size N , start each range request at

$$\text{floor}(X) = X - (X \bmod N)$$

instead of at the true resume height X , hash-checking and discarding blocks in $[\text{floor}(X), X]$, so that the request leaks only the bucket and absorbs reorganisations — is proposed and not

implemented (ZIP 307, § “Block privacy via bucketing”). Both are ZIP-versus-deployed gaps a reader should not assume closed.

The reference client narrows the surface further: the sync loop in `zcash_client_backend` exercises four shielded-sync RPCs (below) plus, when built with feature `transparent-inputs`, `GetAddressUtxosStream`, issued per account per cycle by `refresh_utxos` (cf. §12.3):

- `get_latest_block`, in `update_chain_tip`;
- `get_subtree_roots`, in `download_subtree_roots`;
- `get_block_range`, in `download_blocks`;
- `get_tree_state`, in `download_chain_state`.

Transaction enhancement via `GetTransaction` is listed as not yet implemented in that module (`src/sync.rs`). The complete reference loop is reconstructed in §10.8.

9.4 The RPC surface

The service defines twenty RPCs (`service.proto:221--326`); Table 7 groups them. Three are deprecated, one is testing-only, and six serve the transparent layer, outside this section’s shielded-sync scope. The four marked ✓ are the ones the reference sync loop consumes for shielded sync (§9.3).

Group	RPC	Sync loop	Notes
Chain	<code>GetLatestBlock</code>	✓	tip height and hash
	<code>GetBlock</code>		one block, all pools
	<code>GetBlockRange</code>	✓	streamed range, pool filter
	<code>GetBlockNullifiers</code>		deprecated (v0.4.0; proto option v0.4.1)
	<code>GetBlockRangeNullifiers</code>		deprecated (v0.4.0; proto option v0.4.1)
Trees	<code>GetTreeState</code>	✓	frontiers at height/hash
	<code>GetLatestTreeState</code>		frontiers at the tip
	<code>GetSubtreeRoots</code>	✓	depth-16 subtree roots
Transactions	<code>GetTransaction</code>		full bytes by txid
	<code>SendTransaction</code>		submit raw bytes
Mempool	<code>GetMempoolTx</code>		compact mempool contents
	<code>GetMempoolStream</code>		full txs until next block
Transparent	<code>GetAddressTxids</code>		deprecated, misnamed
	<code>GetAddressTransactions</code>		
	<code>GetAddressBalance</code>		
	<code>GetAddressBalanceStream</code>		
	<code>GetAddressUtxos</code>		
	<code>GetAddressUtxosStream</code>		
Metadata	<code>GetLightdInfo</code>		server and chain metadata
	<code>Ping</code>		testing-only

Table 7: The twenty RPCs of `CompactTxStreamer` (`service.proto:221--326`, `lightwallet-protocol v0.4.1`). The ✓ column marks the shielded-sync RPCs exercised by the reference sync loop; with feature `transparent-inputs` the loop also issues `GetAddressUtxosStream` per account per cycle.

Remark 9.5 (Hash orders). Two byte orders cross the interface. On the usual block and transaction paths, byte-typed identifiers such as `BlockID.hash` and `CompactBlock.hash` carry raw bytes in little-endian (protocol) order; `lightwalletd` reverses the big-endian hex it receives

from the node's RPCs (lightwalletd, frontend/service.go:69--90). Field `TreeState.hash`, by contrast, is big-endian display hex *text* (service.proto:176--184,307--312). Requests follow the same split: `TxFilter` takes protocol-order txid bytes that the server reverses into display hex for `getrawtransaction` (frontend/service.go:403--442), and the mem-pool exclude entries are protocol-order byte suffixes the server reverses into display-order prefixes (frontend/service.go:600--747). There are two deployed byte-typed exceptions: `GetTreeState` interprets its `BlockID.hash` bytes directly as display-order hex, and `SubtreeRoot.completingBlockHash` is likewise returned in display order (Remark 11.6). String-typed hashes are display-order text.

Chain tip. The RPC `GetLatestBlock(ChainSpec)` returns a `BlockID` — height and hash of the best chain tip; lightwalletd proxies `getblockchaininfo` and returns the hash as little-endian raw bytes (lightwalletd, frontend/service.go:69--90). This is the chain tip the wallet's confirmation accounting consumes (§6.6).

Single block. The RPC `GetBlock(BlockID)` returns one `CompactBlock`, and per the v0.4.1 contract includes transaction data for *all* value pools, transparent vin/vout included — unlike `GetBlockRange`, which defaults to shielded-only (service.proto:225--237). The proxy lightwalletd serves the request from its block cache first and falls back to the node on a miss; a height above the tip returns `gRPC OutOfRange` ("block %d is newer than the latest block"), a lookup by hash returns `InvalidArgument` ("not yet implemented", referencing PR #309), and height 0 with no hash is `InvalidArgument` (lightwalletd, common/common.go:495--624, frontend/service.go:166--189). Zaino accepts lookup by hash — hash takes precedence and is resolved to a height via its chain index — and builds the block with all pools included, matching the v0.4.1 contract; the proto makes hash support optional, so both behaviours are conformant, but they diverge (zaino, zaino-state/src/backends/state.rs:1954--2058; zaino-proto/src/proto/utils.rs:310--323).

Block ranges. The RPC `GetBlockRange(BlockRange)` streams consecutive `CompactBlocks` over the closed interval $[\min(s,e), \max(s,e)]$, in increasing height order iff $s \leq e$ and decreasing otherwise — lightwalletd computes $j = \text{high} - (i - \text{low})$, Zaino tests $s \leq e$ (service.proto:250--255). At the pinned v0.4.1 contract, an empty `poolTypes` field selects the legacy behaviour: prune each block to Sapling and Orchard data only; v0.5.0 and current lightwalletd add Ironwood to that default. Clients **MUST** verify server capability before setting `poolTypes` non-empty (service.proto:28--42). Filtering prunes transaction components and then drops empty transactions, but a block whose transactions were all filtered is still returned, with empty `vtx` (lightwalletd, common/common.go:574--624). Zaino validates the request up front — start and end must be present and fit in u32; `POOL_TYPE_INVALID` or unknown pool values are `InvalidArgument` — streams through an `mpsc` channel of the configured `channel_size`, caps an ascending range at the tip and appends a trailing `OutOfRange` error after the valid blocks, and bounds production by a hard timeout of 4× the service timeout (zaino, zaino-state/src/backends/state.rs:550--683; zaino-proto/src/proto/utils.rs:53--152).

Nullifier variants. The RPCs `GetBlockNullifiers` and `GetBlockRangeNullifiers` — both deprecated (v0.4.0 changelog; proto option v0.4.1) in favour of `GetBlockRange` with `poolTypes` — return `CompactBlocks` reduced to shielded nullifier data: lightwalletd strips Sapling outputs and transparent vin/vout, reduces each Orchard action to its nullifier, and zeroes both commitment-tree sizes; `GetBlockRangeNullifiers` **MUST** ignore any `TRANSPARENT` member of `poolTypes`, and lightwalletd filters it out (service.proto:239--268; lightwalletd, frontend/service.go:191--225,259--305). Zaino's `FetchService` backend addi-

tionally resolves `GetBlockNullifiers` by hash, as for `GetBlock` (zaino, zaino-state/src/backends/fetch.rs:929--1041); its `StateService` backend ignores a supplied hash and reads only the height, silently serving height 0 — genesis — for a hash-identified request (backends/state.rs:2033--2058; research/upstream-defects.md, item 40).

Tree state. The RPC `GetTreeState(BlockID)` returns the note commitment tree state for a block specified by height or hash: network name, height, block hash as a hex string, block time, and the `saplingTree` and `orchardTree` frontiers as hex (service.proto:176--184). In `lightwalletd` the reply derives from the node's `z_gettreestate` RPC, looping along `Sapling.SkipHash` when a reply carries no `FinalState`; `GetLatestTreeState` calls the same path at `getblockchaininfo`'s tip height (`lightwalletd`, frontend/service.go:307--401). The frontier is what seeds the wallet's shard tree below its birthday, so that witness assembly never replays the chain's prefix (*Ironwood Guide*, §“Resting: the note in the commitment tree”). Zaino resolves hash-or-height, reports the BIP70 network name, and returns a seventh field `ironwoodTree` (hex, empty when absent) that exists only in Zaino's and `librustzcash`'s proto forks (§9.1); `GetLatestTreeState` reuses `GetTreeState` at the current chain height (zaino, zaino-state/src/backends/state.rs:2516--2576; zaino-proto/proto/service.proto:183). One stale cross-reference: the proto comment for `GetTreeState` cites “section 3.7 of the Zcash protocol specification”, while the current `protocol.tex` numbers note commitment trees as § 3.8.

Subtree roots. Shielded outputs of a pool are numbered from genesis, and each consecutive group of $2^{16} = 65536$ note commitments forms one depth-16 subtree of the pool's commitment tree; the stream element is

$$\text{SubtreeRoot}_i = (\text{Merkle root of subtree } i, \\ \text{hash and height of the block containing output } (i + 1) \cdot 2^{16} - 1),$$

so that, for example, Sapling output 65535 — the last of subtree 0 — falls in block 558822 (service.proto:191--200; `lightwalletd`, common/common.go:178--219). The RPC `GetSubtreeRoots(GetSubtreeRootsArg)` streams these for the chosen shielded protocol from `startIndex`, with `maxEntries = 0` meaning all. The `lightwalletd` handler calls `z_getsubtreesbyindex` and then, per subtree, fetches the completing block via `GetBlock` at its end height to obtain the hash, reversed to big-endian display-order bytes, the opposite convention from `CompactBlock.hash` (Remark 11.6; `lightwalletd`, frontend/service.go:832--907). These roots are exactly the retained internal nodes from which the wallet's shard tree assembles authentication paths without replaying every commitment (*Ironwood Guide*, §“Resting: the note in the commitment tree”). Zaino converts `startIndex` and `maxEntries` to `u16` and returns `InvalidArgument` when either exceeds `u16` range, where `lightwalletd` forwards the full `uint32` — bounded by design, since a depth-32 tree has at most 2^{16} depth-16 subtrees, but the wire contracts differ; it fills the completing block data via `z_get_block` per subtree, under a $4\times$ service-timeout deadline. One Zebra-specific caveat: during Zebra's initial subtree-index rebuild the underlying RPC returns an empty list for not-yet-rebuilt indices, where `zcashd` rebuilds before serving (zaino, zaino-state/src/indexer.rs:429--452, 754--912).

Full transactions. The RPC `GetTransaction(TxFilter)` returns the full serialised transaction. Only lookup by 32-byte `txid` is supported; a block-plus-index lookup returns `InvalidArgument` (“specify a txid, not a blockhash+num”). The `lightwalletd` handler converts the little-endian `txid` bytes to big-endian hex, calls `getrawtransaction` with `verbose=1`, and maps RPC failure to gRPC `NotFound` (service.proto:44--51, 270--271; `lightwalletd`, frontend/service.go:403--442). The privacy cost of this call — it names a `txid` of interest to the proxy — is the reason for the ZIP's decoy mandate (§9.3).

Submission. The RPC `SendTransaction(RawTransaction)` submits via the node’s `sendrawtransaction`. Success returns `SendResponse{errorCode=0, errorMessage=txid hex}`; a node rejection is parsed from the “code: message” error string into a non-zero `errorCode` plus message — node-level rejection travels in-band, not as a gRPC error, so a transport-level success is not an acceptance (`service.proto:83--89`; `lightwalletd`, `frontend/service.go:454--509`). Wallet-side obligations around this call — store-then-broadcast ordering, rebroadcast, expiry — are covered in §6.2.

Remark 9.6 (The `RawTransaction.height` sentinels). An original protobuf-definition error typed the field `uint64` rather than `int64`, forcing sentinel semantics (`service.proto:53--81`; `lightwalletd`, `common/common.go:626--661`):

$$\text{height} = \begin{cases} 0 & \text{transaction in the mempool,} \\ 2^{64} - 1 \text{ (0xffffffffffffffff)} & \text{mined on a non-main-chain fork (zcashd’s } -1\text{),} \\ h & \text{mined on the main chain at height } h. \end{cases}$$

The proto’s own comments contradict each other: the message-level comment says the field carries “the latest block height” when returned by `GetMempoolStream`, the field-level comment specifies the sentinel mapping above, and a FIXME cites `librustzcash` issue #1484. The deployed servers realise both readings — `lightwalletd` streams 0, Zaino streams the current best-tip height (§9.8). We flag the mempool-height contract as unresolved (`research/upstream-defects.md`, which tracks this as the fourth known defect, already filed as `zcash/librustzcash#1484`).

Mempool contents. The RPC `GetMempoolTx(GetMempoolTxRequest)` streams `CompactTx` for the current mempool contents, possibly a few seconds stale. The request may carry `exclude_txid_suffixes`: txid byte-suffixes of any length up to 32 bytes, in client-side protocol order; the server reverses and hex-encodes each into a display-order prefix, and a mempool transaction is excluded iff its txid matches a prefix that matches *exactly one* mempool txid — when two or more txids match, none of the matching transactions are excluded, and unmatched entries are ignored (`service.proto:157--174, 289--301`; `lightwalletd`, `frontend/service.go:600--747`). Field 2 of the request is reserved for a future `exclude_txid_prefixes`. At the pinned `lightwalletd` revision, an empty `poolTypes` defaults to Sapling-plus-Orchard pruning; current `lightwalletd` also includes Ironwood. `POOL_TYPE_INVALID` is rejected. The `lightwalletd` implementation backs the RPC with a process-global cache — `mempoolMap` from txid to `CompactTx`, plus `mempoolList` — refreshed at most every 2 s from `getrawmempool` plus per-transaction `getrawtransaction verbose=0`, reusing already-fetched entries across refreshes; mempool `CompactTx`s are built with height 0 and no fee (`lightwalletd`, `frontend/service.go:589--708`). At the pinned Zaino revision, both backends enforce the same 32-byte suffix bound and exactly-one-match exclusion rule. `StateService` validates and applies `poolTypes`; `FetchService` ignores that field and converts with its default filter, which includes Sapling, Orchard, and Ironwood. Thus requested pruning is backend-dependent (`zaino`, `zaino-state/src/backends/state.rs:2302--2429`, `backends/fetch.rs:1630--1741`).

Mempool stream. The RPC `GetMempoolStream(Empty)` streams full `RawTransactions` of the current mempool and keeps the stream open, closing it when a new block is mined (`service.proto:303--305`). The `lightwalletd` implementation is process-global state — `g_txList` in arrival order, a `g_txidSeen` set, `g_lastBlockChainInfo` — polled at most every 2 s via `getblockchaininfo` and `getrawmempool`; the first stream to observe a tip-hash change resets the shared state and all open streams terminate; each per-client loop sleeps 200 ms between sends; transactions mined since listing (`height ≠ 0`) are skipped; streamed transactions carry `height = 0` (`lightwalletd`, `common/mempool.go:14--152`, `frontend/service.go:581--587`). Zaino streams `height =` current best-tip height, closes the stream when its mempool status

flips to Syncing on a tip change, and additionally imposes a hard 6× service-timeout deadline — 180 s at defaults — where `lightwalletd` keeps the stream open until the next block regardless of duration; a long block interval can therefore end a Zaino stream with a deadline error where the proto’s documented close-on-new-block semantics would hold (zaino, zaino-state/src/backends/state.rs:2433--2510, chain_index/mempool.rs:419--480).

Server metadata. The RPC `GetLightdInfo(Empty)` returns server metadata plus chain state assembled from `getinfo` and `getblockchaininfo`: version, vendor (“ECC LightWalletD”), `taddrSupport` true, chain name, the Sapling activation height — looked up via the Sapling consensus branch ID `0x76b809bb` in the upgrades map — the chain tip’s consensus branch ID, block height, build and git fields, `estimatedHeight` (less than the tip while the node itself syncs), donation address, and the next pending upgrade’s name and height (`service.proto:97--118`; `lightwalletd, common/common.go:261--319`). At the pinned revision the proxy does *not* populate the proto’s `lightwalletProtocolVersion` field 18; current `lightwalletd` 0.5.4 reports “v0.5.0”. Zaino reports vendor “ZingoLabs ZainoD”, hard-codes `lightwallet_protocol_version = “v0.5.0”` — a version string whose fields did not match the canonical tagged protocol exactly (`research/upstream-defects.md`, item 3; zaino-state/src/backends/fetch.rs:2097, state.rs:2776) — defaults the Sapling activation height to 1 on regtest, and carries the connected Zebra node’s build information in the `zcashdBuild/zcashdSubversion` fields (zaino, zaino-state/src/backends/state.rs:2724--2793). Both sides are flagged in §9.8.

Ping. The RPC `Ping` is testing-only; `lightwalletd` disables it unless started with the flag `--ping-very-insecure`, because the call lets a client create arbitrarily many concurrent server threads — a resource-exhaustion risk (`service.proto:324--325`; `lightwalletd, frontend/service.go:920--936`). Zaino returns `gRPC Unimplemented`.

Transparent-address RPCs. The six remaining RPCs (`GetAddressTxids` — deprecated and misnamed, it returns transactions — `GetAddressTransactions`, `GetAddressBalance`, `GetAddressBalanceStream`, `GetAddressUtxos`, `GetAddressUtxosStream`) proxy the node’s insight-explorer-derived RPCs `getaddresstxids`, `getaddressbalance`, and `getaddressutxos`; they serve the transparent layer and fall outside this section’s shielded-sync scope. For these, `lightwalletd` validates t-addresses against the regex `\At[a-zA-Z0-9]{34}\z` and caps each `GetAddressTransactions` inner fetch loop with a 30-second context timeout (`lightwalletd, frontend/service.go:55--164, 511--579, 749--830`). Zaino caps the request’s address list at `UTXO_MAX_ADDRESSES = 1000` (`InvalidArgument` beyond) — a server-side fan-out bound `lightwalletd` lacks — and honours `start_height` and `max_entries` (0 meaning unlimited) as response-size bounds (zaino, zaino-state/src/backends.rs:29--45, backends/state.rs:2591--2719).

9.5 `lightwalletd`: the deployed proxy

The deployed server is a Go proxy: each gRPC handler in `frontend/service.go` translates its request into node JSON-RPC calls through the indirect function `common.RawRequest`. It supports both `zcashd` and `zebrad` backends, detecting the backend from the node’s subversion string (`/Zebra:` versus `/MagicBean:`); a `zcashd` backend must have the `lightwalletd` or `insightexplorer` experimental feature enabled or `lightwalletd` exits; the default assumed node is `zebrad` (`lightwalletd, cmd/root.go:200--263, common/common.go:27--35, 61--64`).

Ingestion. A single `BlockIngestor` goroutine polls `getbestblockhash` every 2 s; when the tip differs it fetches the next block via *two* `getblock` calls — first `verbose=1` for the txids and hash, a workaround because the built-in parser miscomputes v5 transaction IDs (issue #392), then

the raw block by that hash, so a reorganisation between the calls cannot mix two blocks. A prevHash mismatch against the cache tip is treated as a reorganisation and exactly one block is dropped ($\text{Reorg}(\text{height}-1)$) per iteration; a getblock failure retries after 8 s (lightwalletd, common/common.go:321--493). The 120 s sleep (common/common.go:483--489) runs only when the cache is empty ($\text{height} = \text{GetFirstHeight}()$) and the node returned no block at that height; with the cache now starting at height 0 (cmd/root.go:296--299) the branch is reachable only against a node that cannot yet serve genesis, and its log line (“Waiting for ... Sapling activation height”) is stale wording, not stale behaviour.

The block cache. The only persistent chain-serving state lightwalletd maintains is the compact-block cache: two append-only files per chain, `<data-dir>/db/<chain>/lengths` and `.../blocks`. With $d = \text{protobuf}(\text{CompactBlock})$ the entry for height h is

$$\text{FNV1a}_{64}(\text{LE}_{64}(h) \parallel d) \parallel d \quad \text{in blocks}, \quad \text{LE}_{32}(|d|) \quad \text{in lengths},$$

with the offset table reconstructed on startup and valid lengths constrained to $74 \leq |d| \leq 4,000,000$ bytes; corruption — a bad checksum, an impossible length, a partial entry — triggers automatic cache clearing and redownload (lightwalletd, common/cache.go:21--230). The cache now starts at height 0, previously Sapling activation, because compact blocks carry transparent data (lightwalletd, cmd/root.go:265--300). All other responses are proxied per-request; the mempool caches of §9.4 are in-memory only; no per-client state is kept.

Operational defaults. gRPC binds 127.0.0.1:9067 and HTTP 127.0.0.1:9068 (Prometheus /metrics); TLS is required — the server exits if the certificate or key is unloadable — unless `--no-tls-very-insecure` (plaintext) or `--gen-cert-very-insecure` (self-signed); the data directory defaults to `/var/lib/lightwalletd`; `--nocache` disables the disk cache; `--sync-from-height` and `--redownload` control cache rebuild; the darkside mock mode (`--darkside-very-insecure`) auto-terminates after 30 minutes (lightwalletd, cmd/root.go:139--180, 375--424, 498--499).

9.6 Zaino: the successor indexer

Zaino is the successor indexer, in Rust: `zainod` is the daemon, `zaino-serve` implements `CompactTxStreamer` over tonic, `zaino-state` holds the chain index and backends, `zaino-fetch` is the JSON-RPC client, and `zaino-proto` vendors the protos. The stated motivations are the `zcashd` deprecation, a Rust stack alongside `librustzcash` and Zebra, and a clean indexer/validator trust boundary: indexing functionality moves out of the validator into Zaino, with Zebra exposing its `ReadStateService` for direct read access (zaino, README.md, § “Motivations”, § “Project Structure”; docs/internal_spec.md).

Backends. Configuration `backend = 'fetch' | 'state'` selects one of two chain-fetch backends. Backend `StateService` is backed by Zebra’s `ReadStateService` plus a `TrustedChainSync` (`init_read_state_with_syncer`): it reads Zebra’s state store directly, and at startup blocks until the syncer’s tip height *and* hash match the validator’s JSON-RPC tip — hash compared because height alone is ambiguous during a reorganisation. Backend `FetchService` is backed by the node’s JSON-RPC interface (`zcashd` back-compatibility, or a remote Zebra) and is marked “[deprecated]: Will be eventually replaced by BlockchainSource”. Both feed the same `NodeBackedChainIndex` (zaino, zaino-state/src/backends/state.rs:1,117--260, backends/fetch.rs:1,83--100). The shipped example configuration nonetheless selects `backend = 'fetch'` — the deprecated backend — and the repository documentation does not say which backend is the recommended production deployment (zaino, docs/example_configs/zainod.toml); we flag the question as open.

The chain index. Server-side state is the `ChainIndex`, with three components (`zaino`, `docs/internal_spec.md`, § “Zaino-State”). A `Mempool` — a dashmap broadcast map keyed by `txid`, whose sync task polls `get_best_block_hash` and on a tip change clears the map and notifies subscribers. A `NonFinalisedState` — an in-memory `ArcSwap` snapshot of the top blocks of all chains. A `FinalisedState` — all blocks below the seam — served either by a versioned LMDB-backed persistent database (v1) or, with `ephemeral_finalised_state`, by a stateless passthrough to the validator; large syncs and version migrations run in the background while serving continues. The seam is

$$\text{floor}(t) = t - D_{\text{op}} \quad (\text{saturating at genesis}),$$

where t is the tip height and $D_{\text{op}} = \text{OPERATIONAL_NFS_DEPTH} = \text{MAX_BLOCK_REORG_HEIGHT} + 1 = 1001$ — Zebra’s reorganisation limit is local node policy, not consensus; the finalised database serves heights $\leq \text{floor}(t)$, the in-memory state serves heights above it, and in-memory retention is hard-capped at $\text{MAX_NFS_DEPTH} = D_{\text{op}} + 10 = 1011$ blocks below the tip.

Serving. Each gRPC handler delegates to a `LightWalletIndexer` trait implemented by both backend subscribers; streaming responses are tokio mpsc channels wrapped in typed streams. The gRPC server may bind a public address only with TLS configured — rejected at startup otherwise — and the default listen address is `127.0.0.1:8137`; Zaino additionally serves a `zcashd`-style JSON-RPC server, unencrypted and restricted to loopback or private bind addresses by default (`zaino`, `README.md`, § “Server network exposure”). Service defaults: timeout 30 s, `channel_size` 32; stream deadlines are $4\times$ the timeout (120 s), and $6\times$ for `GetMempoolStream` (180 s). Until the non-finalised snapshot exists, `GetLatestBlock` and `GetMempoolStream` return an `UnavailableNotSyncedEnough` error where `lightwalletd` would proxy the node’s answer immediately; a code TODO says this “probably shouldn’t be an error”, so the startup behaviour may change (`zaino`, `zaino-state/src/backends/state.rs:1940--1951`).

Compatibility. Zaino’s integration test suite includes a `client_rpc` partition that compares its `LightWallet` gRPC outputs against `lightwalletd`’s, and the project has committed to serving all Zcash JSON-RPC services required by non-validator clients, superseding `zcashd` in that role (`zaino`, `docs/internal_spec.md`, § “Zaino-Testutils and Integration-Tests”; `docs/rpc_api.md`).

Current-head update. The detailed account above deliberately describes the pinned July revision. At `zaino 78eec726` (2026-09-02; after the latest tag, 0.9.0 of 2026-08-29), validator support is Zebra-only and the old `StateService/FetchService` split has been replaced by one `ZebraValidator` composite. JSON-RPC is always present; direct `ZebraReadStateAdapter` database access is an optional accelerator. Configuration now names the modes `direct` and `rpc`, while retaining `state` and `fetch` only as compatibility aliases (`zaino-source-zebra/src/lib.rs`; `zainod/src/config.rs`). Consequently, the behaviour-by-backend claims and divergence table below are a reproducible snapshot, not a claim about current Zaino internals.

9.7 Zebra’s in-process server

Zebra main `d1fd7ad` (2026-09-04, v6.3.0-188-gd1fd7ad) contains a third implementation, added at `85ce9a3` on 2026-08-18 after the `zebrad` 6.3.0 release. It runs `CompactTxStreamer` inside `zebrad`: most handlers call Zebra’s JSON-RPC methods in process, while compact blocks and mempool streams read the state and mempool services directly. The surface defines handlers for all twenty RPC names; nineteen are implemented, while the testing-only `Ping` deliberately returns `Unimplemented` (`zebra-rpc`, `src/lightwalletd/methods.rs` and `server.rs`).

This server remains unreleased at the cited head, experimental, and disabled unless `rpc.lightwalletd_listen_addr` is configured. It serves plaintext HTTP/2 without TLS or authentication; Zebra therefore directs operators to bind it to loopback or a private address and terminate

TLS in a reverse proxy (`zebra-rpc`, `src/config/rpc.rs`). Its existence invalidates the former statement that `lightwalletd` and `Zaino` were the only maintained server implementations, but it does not retroactively alter the pinned two-server comparison below.

9.8 Divergence register

Table 8 collects the contract-level points at which `lightwalletd` and `Zaino` at the pinned revisions differ from each other or from the normative texts; each row is developed, with citations, in the paragraph of §9.4–§9.6 that covers the RPC. The ZIP-versus-deployment gaps — the message sketch, partial header validation, unimplemented bucketing — are flagged in §9.1 and §9.3; the proto forks in §9.1.

Contract point	lightwalletd	Zaino
<code>GetBlock</code> / <code>GetBlockNullifiers</code> by hash (optional in proto)	hash lookup is <code>InvalidArgument</code> , “not yet implemented” (PR #309)	<code>GetBlock</code> supports it; pinned <code>StateService</code> ignores the hash in <code>GetBlockNullifiers</code>
Stream deadlines (proto: <code>GetMempoolStream</code> closes on new block)	none; open until next block	hard $4\times/6\times$ timeout (120 s/180 s at defaults)
<code>RawTransaction.height</code> in <code>GetMempoolStream</code> (proto comments contradict; issue #1484)	0	current best-tip height
<code>GetSubtreeRoots</code> argument range (proto: <code>uint32</code>)	forwards full range to node	<code>InvalidArgument</code> above <code>u16::MAX</code>
<code>CompactTx.fee</code> (proto: “present if server can provide”)	always 0 (parser TODO)	always 0 (<code>fee = None</code>)
<code>lightwalletProtocolVersion</code> (<code>LightdInfo</code> field 18, v0.4.0)	never populated at 61fee32	hard-coded “v0.5.0” despite the vendored-field mismatch
Behaviour before server is synced	proxies the node immediately	<code>UnavailableNotSyncedEnough</code> on <code>GetLatestBlock</code> , <code>GetMempoolStream</code>
Ping	disabled unless <code>--ping-very-insecure</code>	Unimplemented
Transparent-address fan-out	unbounded address lists	capped at 1000 addresses

Table 8: Pinned-revision divergences across the `CompactTxStreamer` contract. Citations for each row appear in the covering paragraphs of §9.4–§9.6.

Two rows deserve emphasis. The mempool-height row is a live specification bug: the proto’s two comments cannot both be satisfied, the deployed servers realise one reading each, and the FIXME in the canonical file (librustzcash issue #1484) records the conflict without resolving it; a client must therefore treat the height of a streamed mempool transaction as carrying no portable information beyond “not yet mined”. The fee row is a silent one: nothing on the wire distinguishes “fee is zero” from “fee not provided”, so the proto’s conditional language is, in deployed terms, a constant.

10 The scanning pipeline

A light wallet holds no chain. Everything it knows — which notes it owns, which of them are spent, and where each sits in the note commitment tree — it reconstructs by scanning a stream of *compact blocks* supplied by an indexing server. This section develops the deployed reconstruction

pipeline of librustzcash’s `zcash_client_backend` and `zcash_client_sqlite` crates: the compact input format (§8), the account birthday that bounds the scan (§10.2), the priority queue that orders it (§10.3), chain-tip tracking (§10.4), the per-range scanning machinery (§10.5), persistence (§10.6), reorganisation recovery (§10.7), and the reference loop that assembles the parts (§10.8).

The cost shape of the pipeline deserves stating first. For every block in a suggested range, every compact output and action of every transaction is trial-decrypted against every prepared incoming viewing key of every tracked account — two ZIP-32 scopes per pool per account (§10.5) — so synchronisation work scales with global chain traffic multiplied by tracked keys. This trial-decryption component is independent of the wallet’s own activity; other synchronisation work, such as note enhancement and retained-witness updates, is not. Restore-from-seed replays the trial-decryption computation from the birthday. This is precisely the deployed cost centre that Project Tachyon’s proposed proof-carrying wallet is designed to remove (Bowe–Miers, ePrint 2025/2031; tachyon-zcash/tachyon, `book/src/proof-tree.md` @ 26fdc165).

10.1 The compact stream

ZIP 307 (Status: Draft) defines the light-client protocol for payment detection. Its *compact output* retains, of a full shielded output, the 32-byte note commitment, the 32-byte ephemeral key, and the first 52 bytes of the note ciphertext:

$$\underbrace{32}_{\text{commitment}} + \underbrace{32}_{\text{epk}} + \underbrace{52}_{\text{ciphertext prefix}} = 116 \text{ bytes,}$$

an ~80% reduction against ZIP 307’s 580-byte-ciphertext plus 32-byte-ephemeral-key baseline; its *compact spend* is the 32-byte nullifier alone. The quoted ~90% spend reduction uses the pre-v5 384-byte Sapling Spend encoding; v5/v6 share anchor and bundle data, so that number is not the marginal size of every current spend (ZIP 307 §§“Output Compression”, “Spend Compression”). Truncating at 52 bytes discards the AEAD tag, so a compact decryption authenticates differently from a full one: the wallet recomputes the note commitment from the decrypted compact plaintext and compares it with the transmitted commitment. This authenticates the plaintext against that commitment, not the chain: the light client still trusts its data source to supply the correct accepted transaction (ZIP 307 §“Compact Stream Format”). The cryptography of trial decryption — key agreement, KDF, the 52-byte compact note plaintext, the commitment recomputation — is the *Ironwood Guide*’s material (§“Delivery: the note reaches its owner”); this volume treats it as a black box mapping a (compact output, incoming viewing key) pair to a note or $\perp$.

The deployed wire format. The format in production has evolved well beyond ZIP 307’s message definitions. A block is

```
CompactBlock { reserved 1; height=2; hash=3; prevHash=4; time=5;
               header=6; vtx=7; chainMetadata=8 }
```

where field 1 (`protoVersion`) has been removed and `ChainMetadata` carries the running per-pool note commitment tree sizes `saplingCommitmentTreeSize=1`, `orchardCommitmentTreeSize=2`, and `ironwoodCommitmentTreeSize=3` — the field on which position tracking (§10.5) depends. Each `CompactTx` carries its index, txid, fee, Sapling spends and outputs, Orchard actions, Ironwood actions (`ironwoodActions`, field 9, reusing the `CompactOrchardAction` message), and transparent vin/vout (`zcash_client_backend`, `lightwallet-protocol/walletrpc/compact_formats.proto`, vendored at librustzcash e30517e4).

Remark 10.1 (Normative anchoring). ZIP 307’s message layout — a `CompactBlock` with an embedded `BlockID`, messages `CompactSpend` and `CompactOutput` — does not match the deployed format, and the ZIP remains Draft and Sapling-only. No ZIP fully specifies the deployed compact format or service: `ChainMetadata` tree sizes, `CompactOrchardAction`, `ironwoodActions`,

`GetTreeState`, and `GetSubtreeRoots` exist only in the `zcash/lightwallet-protocol` proto files, which we therefore cite as the specification of record. Ironwood’s compact-format extensions are likewise absent from ZIP 2005, which covers quantum recoverability, not light-client synchronisation (`compact_formats.proto`; ZIP 307; ZIP 2005).

The fee field. The fee of a `CompactTx` is calculable by a stateless server only for transactions without transparent inputs, as

$$\text{fee} = \text{valueBalanceSapling} + \text{valueBalanceOrchard} + \text{valueBalanceIronwood} \\ + \sum v_{\text{PubNew}} - \sum v_{\text{PubOld}} - \sum t_{\text{Out}};$$

with transparent inputs the fee depends on prior-transaction outputs a stateless server cannot provide (`compact_formats.proto`:63--69).

Service surface. The reference synchroniser drives four RPCs of the `CompactTxStreamer` service: `GetLatestBlock(ChainSpec) → BlockID`; `GetBlockRange(BlockRange) → stream CompactBlock`, addressed by heights only, with an optional `poolTypes` pruning filter that clients **MUST** verify the server supports; `GetTreeState(BlockID) → TreeState`, returning network, height, hash, time, and the hex-serialised Sapling, Orchard, and Ironwood commitment trees; and `GetSubtreeRoots(GetSubtreeRootsArg) → stream SubtreeRoot`, returning complete-subtree roots with their completing block hash and height, selected by `enum ShieldedProtocol { sapling=0; orchard=1; ironwood=2 }` (`lightwallet-protocol/walletrpc/service.proto`:24--321).

Remark 10.2 (Pinned-revision divergence at NU6.3). The pinned `lightwalletd` tree (commit 61fee32e, 2026-06-25) contains no Ironwood support: its `walletrpc` protos and Go code lack `ironwoodActions`, `ironwoodCommitmentTreeSize`, `ironwoodTree`, and `ShieldedProtocol.ironwood`. The protos vendored by `librustzcash` (commit e30517e4) define all four; Zaino (commit 0e057e22) defines the first three but not `ShieldedProtocol.ironwood` (§9.1), so its `GetSubtreeRoots` rejects an Ironwood request with `InvalidArgument` (`zaino-state/src/indexer.rs`:761--768). The reference synchroniser unconditionally requests Ironwood subtree roots (`zcash_client_backend/src/sync.rs`:279--296); an NU6.3-aware wallet therefore requires an updated server — at these pins neither `lightwalletd` nor Zaino serves the full surface.

10.2 The account birthday

Definition 10.3 (Account birthday). Type `AccountBirthday` pairs a `prior_chain_state`: `ChainState` with a `recover_until`: `Option<BlockHeight>`. The *birthday height* is

$$\text{birthday} := \text{priorChainState.height} + 1,$$

the height of the first block to be scanned. A `ChainState` holds a block height, a block hash, and the final note commitment tree frontier of each pool: Sapling (depth `sapling::NOTE_COMMITMENT_TREE_DEPTH = 32`), Orchard, and Ironwood — the Ironwood tree is Orchard-shaped (`MerkleHashOrchard`, the same depth) but the pool is distinct and keeps its own tree.

Constructor `AccountBirthday::from_treestate` builds the birthday from a `lightwalletd` `TreeState` for the last block *prior* to the birthday height. The conversion (`TreeState::to_chain_state`) hex-decodes the block hash — reversing the bytes, since `zcashd` renders block hashes byte-reversed in hex — parses the height, and converts the hex-serialised Sapling, Orchard, and Ironwood trees into per-pool frontiers; an empty tree field yields an empty tree.

Choosing a birthday. For a new wallet the crate documentation directs constructing the birthday from the treestate at tip – 100. Under the library’s 100-block reorganisation assumption, placing the birthday below that depth prevents a reorganisation from mining a transaction intended for the wallet *below* its birthday and causing funds to be missed. This is not a consensus guarantee; Zebra’s local rollback policy permits a deeper window (§10.7) (`zcash_client_backend`, `src/data_api.rs:3199--3211`). For a restored wallet the birthday is historical, and the `recover_until` height marks the chain tip at restoration time: the wallet is *in recovery* until no unscanned ranges remain between the birthday height and `recover_until` (exclusive). The field exists to avoid confusing shifts of balance and spendability while restore-from-seed is in flight — spendability policy itself is the `ConfirmationsPolicy` of §4 and §6, which this section does not restate. Progress reporting follows the same split: type `Progress` carries two scanned-notes/notes-on-chain ratios, one over the recovery window (birthday to recovery height) and one over the scan window (recovery height to tip); either denominator may be zero.

The wallet birthday. The birthday of the *wallet* is the minimum birthday over its accounts — literally `SELECT MIN(birthday_height) FROM accounts` in the SQLite backend — and adding an account whose birthday lies below the current chain tip triggers a rescan of all blocks at and above that height. Function `add_account` persists the birthday height together with the Sapling and Orchard frontier tree sizes, calls `rewind_to_chain_state` targeting the birthday’s prior chain state — installing a `Historic`-priority rescan range above birthday – 1 up to the wallet’s known chain tip while preserving higher-priority queue entries — and finally inserts an `Ignored`-priority range covering `[saplingActivation, birthday)`.

10.3 The scan queue

The wallet does not scan the chain linearly; it maintains a priority queue of height ranges and always works on the most urgent.

Definition 10.4 (Scan range). A `ScanRange` is a half-open range `[start, end)` of block heights paired with a `ScanPriority`, with helpers `truncate_start`, `truncate_end`, and `split_at`. The SQLite backend stores the queue in table `scan_queue` as rows (`block_range_start`, `block_range_end`, `priority`) with both bounds `UNIQUE` and a `CHECK` constraint `start < end`.

Priority	Code	Meaning (doc comment)
<code>Ignored</code>	0	Lowest priority; never scanned.
<code>Scanned</code>	10	Already scanned; not re-scanned.
<code>Historic</code>	20	Advance the fully-scanned height.
<code>OpenAdjacent</code>	30	Ranges adjacent to heights at which the user opened the wallet (vestigial; Remark 10.5).
<code>FoundNote</code>	40	Complete tree shards adjacent to found notes.
<code>ChainTip</code>	50	Complete the latest shard.
<code>Verify</code>	60	Previously scanned range that must be re-verified as still on the main chain (highest).

Table 9: The `ScanPriority` ladder, ascending, with its SQLite codes.

Remark 10.5 (Six live priorities plus one vestigial). We present the ladder as six live priorities and one vestige. Priority `OpenAdjacent` is declared and mapped to SQL code 30, but no code path in the current `librustzcash` tree ever assigns it: a search at commit `e30517e4` finds only the enum declaration, the code mapping, and two doc-comment mentions (`zcash_client_backend`,

src/data_api/scanning.rs:20--21). It still participates in the ordering, so we retain it in Table 9.

Suggesting work. Method `WalletRead::suggest_scan_ranges` returns queue entries in descending priority order. Its documented contract has two clauses: Verify ranges must be scanned before all others, to avoid continuity errors under reorganisation; and the compact-block source must supply note commitment tree size metadata. The SQLite implementation orders by priority DESC, block_range_end DESC — nearest the tip first within a priority — and filters to priorities at or above `Historic`.

Remark 10.6 (Contract, not type). The guarantee that a Verify range arrives first is a documented contract of the trait realised by the SQLite ORDER BY; nothing in the type system enforces it, and a third-party `WalletRead` implementation could violate it. We state it as a trait obligation and cite the one deployed implementation that discharges it (`zcash_client_backend`, src/data_api.rs:1995--2010; `zcash_client_sqlite`, src/wallet/scanning.rs:61--90).

Queue maintenance. Function `replace_queue_entries` performs every insertion as a spanning-tree merge: existing entries overlapping or adjacent to the query range are deleted and re-inserted together with the new entries, and overlaps are resolved by a dominance rule —

- an inserted Verify or Scanned range always replaces what it overlaps;
- otherwise an existing Scanned range wins (unless a forced rescan is requested);
- otherwise the higher priority wins, and equal priorities coalesce into one range.

Gaps between disjoint ranges are filled at `Historic` priority, so the queue always tiles a contiguous span.

10.4 Tracking the chain tip

Function `update_chain_tip` feeds the queue whenever the wallet learns a new tip height tip. Two derived heights recur, with range upper bounds exclusive throughout:

$$\text{chainEnd} := \text{tip} + 1, \quad \text{stable} := \text{tip} - \text{PRUNING_DEPTH},$$

where `PRUNING_DEPTH` = 100 blocks and heights at or below `stable` are treated by the wallet as outside its assumed reorganisation window. The SQLite implementation proceeds as follows (`zcash_client_sqlite`, src/wallet/scanning.rs:488--660):

- (a) it is a no-op if the tip is below Sapling activation;
- (b) it is a no-op if the new tip is below the prior maximum scanned height — the caller has caught the chain mid-reorganisation, and the discontinuity error path of §10.7 will resolve it;
- (c) it computes a *tip shard entry* at `ChainTip` priority covering every pool's last incomplete shard,

$$\left[\max(\text{birthday}, \min_{\text{pools}} \max(\text{subtreeEndHeight})), \text{chainEnd} \right),$$

taking the minimum over the Sapling, Orchard, and Ironwood shards tables and emitting the entry only if that minimum lies below `chainEnd`;

- (d) it computes a *tip entry*: with no scanned blocks, a `Historic` range from the wallet birthday to `chainEnd` (or `Ignored` from Sapling activation when no accounts exist); with no shard metadata (linear scanning), `Historic` from `maxScanned + 1`; if `maxScanned > stable` (steady state), `ChainTip` from `maxScanned + 1` to `chainEnd`; otherwise a `Verify` range

$$[\text{minUnscanned}, \text{min}(\text{stable} + 1, \text{minUnscanned} + \text{VERIFY_LOOKAHEAD})),$$

where `minUnscanned := maxScanned + 1`.

The `Verify` discipline is documented at the same site: because `Verify` outranks everything, the connectivity check — and any rewind it forces — runs before any `ChainTip` range from this or an earlier tip update is scanned. Constant `VERIFY_LOOKAHEAD = 10` blocks covers 250 target seconds at NU7’s proposed spacing (§6.6). This is a pinned implementation heuristic, not a specified NU7 lookahead. In all cases, the `Verify` range is clamped to the stable region so that it cannot itself be reorganised away within the wallet’s assumed depth window and interfere with later `Verify` ranges.

Remark 10.7 (Two deliberate asymmetries). First, when `maxScanned = stable` the `Verify` range above is empty: after a long offline period no `Verify` entry need exist, and connectivity is then checked only implicitly, by the previous-hash comparison of §10.7 when the adjacent range is scanned. Second, case (b) above means `update_chain_tip` deliberately does *nothing* when the tip moves backwards; reorganisation detection lives in the scanner, not in tip updates, and recovery in that case waits for a fetch failure or continuity error.

10.5 Scanning a range

Function `scan_cached_blocks` scans one contiguous range; it takes the network parameters, the block source, the wallet database, a starting height `from_height` with its prior chain state `from_state`, and a block-count limit. It panics unless `fromHeight = fromState.height + 1` — the caller must supply the chain state immediately prior to the range.

Keys. The scanner loads all tracked unified full viewing keys and builds a `ScanningKeys` set (`ScanningKeys::from_account_ufvks`). Per account it registers: the Sapling diversifiable full viewing key’s (External, Internal) pairs of incoming viewing key and nullifier-deriving key; the Orchard full viewing key’s (External, Internal) incoming viewing keys; and the *same* Orchard keys a second time under the Ironwood note-encryption domain — Ironwood notes are Orchard-shaped and decrypt under Orchard viewing keys, but carry version-3 note plaintexts, so `IronwoodDomain` is distinct from `OrchardDomain`, which accepts only version-2 plaintexts. Each incoming viewing key is tagged (`AccountId`, `zip32::Scope`) (Ironwood’s note-encryption and recoverability changes are specified by ZIP 2005 (Proposed), while ZIP 258 activates the pool at NU6.3). A scanning key derives a note’s nullifier only when it carries nullifier-deriving capability: trait method `ScanningKeyOps::nf(note, position)` returns `None` for bare incoming viewing keys, so notes detected with an IVK alone never enter the tracked-nullifier set and their spends cannot be detected.

Batched trial decryption. The scanner makes two passes over the downloaded blocks. The first pass feeds every compact output into a set of `BatchRunners` — one per note-encryption domain: Sapling, Orchard, Ironwood — and flushes. A runner accumulates (domain, output) pairs across transactions and blocks into a `Batch` shared by all prepared incoming viewing keys of its domain; when the accumulated output count reaches the batch-size threshold (100 outputs in `scan_cached_blocks`) the batch is spawned on the global rayon thread pool (`rayon::spawn_fifo`) and a fresh batch begins. Results are delivered per (block hash, txid) over flume channels, and `collect_results` blocks until the relevant batch has run. Within a batch,

`zcash_note_encryption::batch::try_compact_note_decryption` batches the cryptography: `D::batch_epk` parses and prepares all ephemeral keys at once, shared secrets are computed for every (ivk, epk) pair — at the pinned `zcash_note_encryption` 0.4.1 revision, each scalar multiplication is still performed separately — and `D::batch_kdf` derives all symmetric keys in one pass before per-output trial decryption of the 52-byte compact ciphertext; the underlying primitives are again the *Ironwood Guide*’s (§“Delivery: the note reaches its owner”).

The second pass. The scanner then reads the prior block metadata and the wallet’s unspent-nullifier set (`Nullifiers::unspent`, per pool), and calls `scan_block_with_runners` per block, accumulating summary counts, updating the in-memory nullifier set after each block (`Nullifiers::update_with`: nullifiers spent in the block are removed, nullifiers of newly received full-viewing-key-decrypted notes are added), and chaining each block’s metadata into the next — so a spend of a note received earlier in the same batch is detected before anything is persisted.

Within a block, spend detection (`find_spent`) compares each revealed nullifier against the tracked unspent set in constant time — a subtle `CtOption` fold over the whole set; an in-code TODO notes the resulting $O(|\text{nullifiers}| \times |\text{spends}|)$ cost and questions whether constant-time operation is warranted. Nullifiers matching no tracked note are recorded as *unlinked* in a per-transaction nullifier map, in case they match a note found later when scanning an earlier range (`zcash_client_backend`, `src/scanning.rs:826--870`). Note detection (`find_received`) assigns each decrypted note its absolute tree position, $\text{position}(o) = \text{treeSize}_{\text{tx start}} + \text{idx}(o)$, and computes a shardtree retention for every commitment in the block:

$$\text{retention}(c) = \begin{cases} \text{Checkpoint}\{\text{Marked}\} & \text{last in block, wallet's} \\ \text{Checkpoint}\{\text{None}\} & \text{last in block only} \\ \text{Marked} & \text{wallet's only} \\ \text{Ephemeral} & \text{otherwise,} \end{cases}$$

and a note is flagged `is_change` when the receiving account also spent notes in the same transaction.

Position tracking. Positions require tree sizes, which the compact stream must supply. Per pool, the tracker (`PositionTracker::for_compact_block`) reconstructs the block’s starting tree size: from the prior block’s metadata when available; else from the block’s `ChainMetadata` final size minus the block’s output count (error `TreeSizeInvalid` on underflow — the proto-default zero is caught here); else 0 below the pool’s activation height (Sapling, NU5, and NU6.3 for Ironwood respectively); else `TreeSizeUnknown`. The tracker advances by each transaction’s per-pool output count, and at end of block its computed size is checked against `ChainMetadata` — a mismatch is the continuity error `TreeSizeMismatch`. Malformed input from the untrusted server is likewise caught up front: wrong-length nullifier bytes and malformed compact outputs or actions yield `ScanError::EncodingInvalid` — naming pool, txid, and index — rather than a panic.

Output. Each block yields a `ScannedBlock`: height, hash, block time, the wallet-relevant `WalletTx` list — a transaction is retained only if it has at least one wallet spend or output in any pool — and per-pool `ScannedBundles` holding the final tree size, the (commitment, retention) list, and the nullifier map. Transparent data in a `CompactTx` is not yet scanned (`librustzcash` issue #2187) (`zcash_client_backend`, `src/scanning/compact.rs:240--575`; `src/data_api.rs:2520--2692`). The per-call `ScanSummary` reports the scanned range and counts of spent and received Sapling and Orchard notes; received counts may include notes already spent in later blocks (a scanning-order effect), and spent counts cover only previously detected notes.

Remark 10.8 (A gap in ScanSummary). No Ironwood counters exist in ScanSummary, although ScannedBlock and WalletTx carry Ironwood spends and outputs; Ironwood activity is silently excluded from the summary. We read this as an oversight in the deployed code rather than a design decision (zcash_client_backend, src/data_api/chain.rs:437--502,677--685; research/upstream-defects.md, item 2: “ScanSummary lacks spent/received counters for the Ironwood pool”, observed at librustzcash e30517e433).

The Ironwood pool otherwise traverses the pipeline as a full peer of Sapling and Orchard: its actions are trial-decrypted under IronwoodDomain with the account’s Orchard viewing keys, its nullifiers form a separate set — method `get_ironwood_nullifiers` is a required trait method, deliberately without a panicking default because it sits on the scan path — and its commitment tree and scan-queue shard extensions are its own.

10.6 Persistence

The scanned range is committed by a single call, `WalletWrite::put_blocks(from_state, blocks)`, which requires sequential blocks in increasing height order with `from_state` the chain state prior to the first block. The backend implementation validates, for the first block b_0 and every pool,

$$\begin{aligned} & \text{fromState.height} + 1 = \text{height}(b_0) \\ & \text{and } \text{fromState.treeSize} + |\text{commitments}(b_0)| = \text{finalTreeSize}(b_0), \end{aligned}$$

each subsequent height being the predecessor plus one; violation is `NonSequentialBlocks`. In `zcash_client_sqlite` the whole call runs in one database transaction.

Stage 1: rows. Per block, `put_blocks` persists block metadata (heights, hash, time, per-pool final tree sizes and commitment counts); per transaction it persists transaction metadata, queues the txid for full-transaction retrieval (*enhancement*), marks spent notes, and persists received notes — checking each received note’s nullifier against the *persisted* nullifier map (`detect*_spend`) to catch notes spent in a later block range that was scanned earlier, the out-of-order counterpart of the in-memory check of §10.5. It then persists each block’s unlinked-nullifier map and prunes the map beyond `PRUNING_DEPTH`.

Stage 2: trees. Note commitments are assembled into shard subtrees in parallel — chunks of 1024 commitments, built at the pool’s shard height, which is 16 for all three pools (subtrees of 2^{16} commitments; the Ironwood tree is Orchard-shaped) — and each shard tree is updated. The same checkpoint set is enforced across all three trees: each tree gains a checkpoint at every height checkpointed in any other. Checkpoints at or above the NU6.3 activation height that fall on the `ANCHOR_RETENTION_INTERVAL = 288`-block grid are retained as durable anchors exempt from checkpoint pruning. This is the boundary of the present volume: shard-tree structure, witness derivation, and anchor selection are the *Ironwood Guide*’s material (§“Resting: the note in the commitment tree”).

Completing the found notes’ shards. After the tree update, `scan_complete` marks the scanned range Scanned in the queue and, if wallet notes were found, extends the queue with `FoundNote`-priority ranges covering the blocks needed to complete each found note’s subtree — the blocks that make the note witnessable. Per pool, the shard of a note at position p is

$$\text{subtreeIndex}(p) = \lfloor p/2^{16} \rfloor \quad (\text{Address}::\text{above_position}(\text{SHARD_HEIGHT}, p)),$$

and the extension spans from the previous shard’s end height — bounded below by the wallet birthday and the pool’s activation height — to the containing shard’s end height plus one. The

same pass marks notes `witness_stabilized` once their shard’s block extent is fully Scanned and the shard’s end height is at most the pruning floor,

$$\text{floor} := \text{maxScanned} - (\text{PRUNING_DEPTH} - 1).$$

Two notions of scanned height. Out-of-order scanning leaves gaps, so the backend distinguishes `block_max_scanned` — simply `max(blocks.height)` — from `block_fully_scanned`, the height to which this and all preceding blocks above the wallet birthday have been trial-decrypted. The SQLite implementation computes the latter from the queue: it is `end - 1` of the *first* Scanned range (by ascending start), defined only when that range starts at or below the birthday.

10.7 Reorganisation detection and recovery

ZIP 307 prescribes the primitive check: on each sync, re-fetch the block at the previous sync height and compare its hash with the cached copy; a mismatch means the block was orphaned (ZIP 307 §“Client-server interaction”, Phases B and D). The deployed scanner generalises this into a per-block continuity check. Function `scan_block_with_runners` first runs `check_hash_continuity`, returning `ScanError::BlockHeightDiscontinuity` if `height(b) ≠ height(prev) + 1` and `ScanError::PrevHashMismatch` if `prevHash(b) ≠ hash(prev)`; for the first block of a `scan_cached_blocks` call the prior metadata is read from the wallet database (`block_metadata(from_height - 1)`), and for subsequent blocks it is the just-scanned block’s own metadata (`zcash_client_backend, src/scanning/compact.rs:257--289; src/data_api/chain.rs:649--693`). Of ZIP 307’s block-header validation list only this previous-hash check is implemented, and the ZIP’s “block privacy via bucketing” enhancement — rounding the sync start down to a bucket multiple — is explicitly unimplemented (ZIP 307 §§“Block header validation”, “Block privacy via bucketing”).

Classification is explicit: method `is_continuity_error` on `ScanError` returns true for `PrevHashMismatch`, `BlockHeightDiscontinuity`, and `TreeSizeMismatch`; the encoding and tree-size-availability errors of §10.5 (`EncodingInvalid`, `TreeSizeUnknown`, `TreeSizeInvalid`, `TreeSizeOverflow`) are not continuity errors.

Recovery. On a continuity error the reference loop rewinds:

$$\text{rewind} := \text{errHeight} - 10 \quad (\text{saturating_sub}),$$

where any height at least one block before the error is valid and the 10-block offset is a heuristic. It then calls `truncate_to_height(rewind)` on the wallet database, `truncate(rewind)` on the block cache, and returns control so the caller restarts from `update_chain_tip` and a fresh `suggest_scan_ranges`. Truncation semantics — resolution to a shared shard-tree checkpoint at or below the requested height (error `RequestedRewindInvalid`, carrying a safe rewind height, otherwise), un-mining of transactions above it, and the deliberate retention of note rows — are covered in §6.7. This section adds two consequences: the scan queue is trimmed to the truncation height, rolling back the wallet’s view of the chain tip so that the next `update_chain_tip` re-installs `Verify` and `ChainTip` ranges above the new maximum scanned height; and at the pinned commit truncation covers all three shard trees and the nullifier-map locators.

The recovery window. A wallet’s chosen pruning depth bounds the recent checkpoint and nullifier history available for a direct rewind. The pinned `librustzcash` implementation uses `PRUNING_DEPTH = 100`; this is an implementation assumption, not consensus finality. ZIP 218 recommends a 600-block node rollback window at 25-second spacing, while retaining 100-block transparent coinbase maturity. This is a proposed node-policy target, not a consensus finality rule or a verified change to the pinned wallet’s 100-checkpoint cache. A wallet cannot infer

rollback safety from the coinbase maturity constant or from a server’s ability to follow a deeper reorganisation. Its checkpoints, nullifier history, scan queue, and restore path must support its chosen recovery window together. Durable anchors alone do not supply all of that state.

10.8 The reference sync loop

Module `sync` of `zcash_client_backend` assembles the pieces into the reference flow, whose contract the chain module documents (`src/sync.rs:58--226, 393--467; src/data_api/chain.rs:1--152`):

- (1–2) `update_subtree_roots`: download and store complete subtree roots for Sapling, Orchard, and Ironwood via `GetSubtreeRoots` (cf. Remark 10.2);
- (3–4) `update_chain_tip` with the height from `GetLatestBlock`;
- (5) `suggest_scan_ranges`;
- (6) scan every `Verify` range first: download it via `GetBlockRange`, fetch the prior `ChainState` via `GetTreeState` at `start - 1`, scan, delete the cached blocks;
- (7) scan the remaining ranges, split into chunks of the caller-supplied batch size, restarting from step (3) whenever scanning materially changed the suggestions — a new first range outranking the range just scanned, or a truncation after a continuity error (§10.7).

Blocks pass through a `BlockCache` (an extension of `BlockSource`): `get_tip_height`, `read` — short reads allowed but contiguous from the range start — `insert` — non-contiguity permitted — `delete` and `truncate`. The loop deletes each range from the cache after scanning it, the module observing that “keeping the entire chain in `CompactBlock` files on disk is horrendous for the filesystem”; deletions are spawned concurrently and awaited before the loop returns.

The module documents its own limitations: block batches are not downloaded in parallel with scanning; detected transactions are not enhanced (the full transaction, hence the memo, is not fetched); there are no progress notifications; and there is no interruption mechanism short of process exit (`zcash_client_backend, src/sync.rs:1--10`). The chain-module documentation additionally recommends scanning `Historic` ranges in reverse height order — or from both ends inward — so that recent unspent notes are discovered sooner.

11 Commitment-tree synchronisation

Scanning tells a wallet which notes it owns (§10); spending one additionally requires the note’s authentication path against a recent anchor. The construction of that path — the depth-32 tree cut at half depth into height-16 shards under a height-16 cap, the wallet hashing its own shard from scanned leaves and taking every other shard as a single downloaded 32-byte root — is the *Ironwood Guide*’s material (§“Resting: the note in the commitment tree”, subsection “The authentication path from public data”). This section supplies what that construction presumes: the RPCs that deliver tree states and shard roots, their derivation inside the validator, and the client discipline — frontier-seeded scanning, reference-retained roots, shard-completion scan ranges — by which `zcash_client_backend` and `zcash_client_sqlite` turn a compact-block stream plus two RPCs into spendable witnesses.

11.1 The linear protocol and its limit

ZIP 307 (Status: Draft) specifies only the original, linear protocol: as compact blocks arrive in height order, the client appends every note commitment to a local depth-32 incremental Merkle tree and updates one incremental witness per unspent note; because incremental

trees cannot be rewound, the ZIP advises caching roughly 100 recent tree and witness states, on the stated ground that no full node rolls back the chain by more than 100 blocks (zips/zip-0307.rst:290--320; the same figure resurfaces as PRUNING_DEPTH, §10.7). Under this design a wallet cannot spend a note until it has scanned every block from the note's block to the anchor block — each witness update consumes every intervening commitment — so restore-from-seed implies a complete linear scan before the first spend. Removing that spend-before-sync impossibility is the purpose of the machinery below.

Remark 11.1 (Normative status). No ZIP specifies `GetTreeState`, `GetSubtreeRoots`, or the subtree-root chain index. ZIP 307's Phase A leaves tree-state acquisition an explicit unresolved TODO — “Or, it has to set `X` to the block height at which Sapling activated, so as to be sent the entire commitment tree. [TODO: Decide which to specify.]” (zips/zip-0307.rst:433--439) — and the repository holding the canonical protobufs disclaims normativity: “This repository IS not a specification of the Zcash Light Client protocol” (lightwallet-protocol/README.md, v0.4.1). As in Remark 10.1, we therefore cite the proto files and the deployed implementations as the specification of record, and classify the subtree-root protocol *deployed but unspecified*.

11.2 The tree-state service surface

Definition 11.2 (Tree-state and subtree-root RPCs). The deployed light-wallet service defines three tree RPCs (lightwallet-protocol/walletrpc/service.proto:307--316):

- `GetTreeState(BlockID)` returns `(TreeState)` — the tree state at a block specified by height or by hash;
- `GetLatestTreeState(Empty)` returns `(TreeState)` — the tree state at the server's chain tip;
- `GetSubtreeRoots(GetSubtreeRootsArg)` returns `(stream SubtreeRoot)` — a stream of information about roots of subtrees of the note commitment tree of the requested shielded protocol — Sapling or Orchard in the pinned v0.4.1 protocol; v0.5.0 adds Ironwood.

^a

^aThe proto comment refers the reader to “section 3.7 of the Zcash protocol specification” (service.proto:308). The reference is stale: at the pinned spec revision, §3.4 is “Transactions and Treestates”, §3.8 “Note Commitment Trees”, and §3.7 “Action Transfers and their Descriptions” (protocol/protocol.tex:4048, 4306, 4248), likely a pointer into an older revision's numbering. We cite the current numbering.

Definition 11.3 (TreeState message). Message `TreeState`, documented in the proto as “derived from the Zcash `z_gettreestate` rpc”, carries

```
network : string = 1 (documented as "main" or "test");  height : uint64 = 2;
hash : string = 3 (big-endian display-order hex);  time : uint32 = 4;
saplingTree : string = 5;  orchardTree : string = 6,
```

where the two tree fields are protobuf *strings* — hex encodings of the legacy-serialised commitment trees of Definition 11.4 — not bytes (lightwallet-protocol/walletrpc/service.proto:176--184). The comment gives the two production network names, not a wire-level enum: for example, lightwalletd's test-only darkside mode returns “darkside”.

At the pinned `librustzcash` revision the vendored proto adds `ironwoodTree : string = 7`, absent from the deployed v0.4.1 surface (Remark 10.2).

Definition 11.4 (Legacy commitment-tree encoding). The tree state of one pool at one block is serialised in the legacy `CommitmentTree` wire format,

$$\text{ser}(T) = \text{Opt}(\text{left}) \parallel \text{Opt}(\text{right}) \parallel \text{Vec}(\text{Opt}(\text{parents}_1), \text{Opt}(\text{parents}_2), \dots),$$

where `Opt(·)` writes `0x00` for an absent node and `0x01` followed by the 32-byte node otherwise, `Vec(·)` is `CompactSize`-prefixed (Bitcoin’s variable-length integer: counts below 253 in one byte, larger ones as a marker byte then 2, 4, or 8 little-endian bytes), and each node is the 32-byte encoding of a base-field element — Jubjub base for Sapling, Pallas base for Orchard — with non-canonical encodings rejected at parse time. The parser `read_commitment_tree` bounds the parents vector at `DEPTH - 1` entries, and the parsed tree converts losslessly to a frontier via `to_frontier()`. The format’s originator is `zcashd` (retired): its Sapling final state streamed the legacy `SaplingMerkleTree` directly and its Orchard final state streamed through the `OrchardMerkleFrontierLegacySer` wrapper, so both pools use the same wire format, hex-encoded (`zcashd, src/rpc/blockchain.cpp:1402--1406, 1432--1436`).

Definition 11.5 (Subtree-root messages). Message `GetSubtreeRootsArg` carries `startIndex : uint32 = 1`, the index of the first subtree to return; `shieldedProtocol = 2` with enum `ShieldedProtocol { sapling = 0; orchard = 1 }` in v0.4.1; v0.5.0 adds `ironwood = 2`. Field `maxEntries : uint32 = 3`, where 0 means “return all entries”. Results stream in index — equivalently height — order. Message `SubtreeRoot` carries `rootHash : bytes = 2`, the 32-byte Merkle root of the completed subtree; `completingBlockHash : bytes = 3`; and `completingBlockHeight : uint64 = 4`. Field `number 1` is absent, and no reserved statement documents the gap (`lightwallet-protocol/walletrpc/service.proto:186--200`).

Remark 11.6 (Byte-order warts). Two byte-order pitfalls attach to these messages. First, `SubtreeRoot.completingBlockHash` is sent in big-endian display order — `lightwalletd` sets it to `hash32.ReverseSlice(block.Hash)` — which is the *opposite* convention from `CompactBlock.hash`, parsed into little-endian wire order; the proto documents neither convention (`lightwalletd, frontend/service.go:896--900; parser/block.go:59--61, 115`). The reference client sidesteps the trap by ignoring `completingBlockHash` entirely, reading only `rootHash` and `completingBlockHeight` (`zcash_client_backend, src/sync.rs:247--253`). Zaino constructs the field as the block hash’s `bytes_in_display_order()` (`zaino-state/src/indexer.rs:854--861`), agreeing with `lightwalletd`’s big-endian display order; whether any deployed client consumes `completingBlockHash` remains an open question (the reference client ignores it). Second, a `zcashd` release note records that the serialised `finalRoot` field of `z_gettreestate` was byte-flipped relative to `getrawtransaction` output in releases up to and including v5.2.0, later rectified (`zcashd, src/rpc/blockchain.cpp:1326--1330, help text`).

11.3 Server-side derivation

Tree states. The RPC originates in `zcashd`’s `z_gettreestate`: it accepts a block hash or height (negative heights allowed, `-1` denoting the last known valid block), requires the block to be on the main chain, and returns the block’s hash, height, and time together with one object per pool, `{skipHash, commitments : {finalRoot, finalState}}`. The `finalState` — the serialised tree of Definition 11.4 — is present only when the block’s anchor is retrievable from the coins view (`GetSaplingAnchorAt / GetOrchardAnchorAt`); otherwise `skipHash` names

the most recent prior block that has one, stopping at the pool’s activation height (zcashd, src/rpc/blockchain.cpp:1303--1455).

The proxy wraps this in a retry loop. Handler `GetTreeState` of `lightwalletd` rejects a request specifying neither height nor hash, marshals a height as a decimal string and a hash as big-endian hex, and calls `z_gettreestate`; if the reply’s Sapling `finalState` is empty and its Sapling `skipHash` is nonempty, it re-issues the RPC at the `skipHash`, iterating until a final state is found or no skip hash remains, and failing with “`z_gettreestate` did not return `treestate`” otherwise (lightwalletd, frontend/service.go:311--386). The walk terminates in practice because zcashd hard-stops it at the Sapling activation height (the `saplingActive` lambda; zcashd, src/rpc/blockchain.cpp:1408--1418). Handler `GetLatestTreeState` calls `getBlockchaininfo`, takes its `Blocks` value as the latest height, and delegates to `GetTreeState` at that height (lightwalletd, frontend/service.go:388--401).

Remark 11.7 (The skip walk keys on Sapling only). Only the Sapling `skipHash` is consulted: when the walk lands on an earlier block, the returned `TreeState` carries that earlier block’s height, hash, and Orchard tree — not the Orchard state of the requested block (lightwalletd, frontend/service.go:311--386).

Subtree roots. The originating RPC is `z_getsubtreesbyindex` “pool” `start_index` (`limit`), added in zcashd v5.6.0 and gated behind `-experimentalfeatures + -lightwalletd`: it returns roots of complete 2^{16} -leaf subtrees of the given pool’s note commitment tree, each with `end_height`, the height of the block containing the note commitment that completed the subtree, reading `SubtreeData` records from the coins view (`pcoinsTip->GetSubtreeData(pool, index)`) until data is absent or the limit is reached (zcashd, src/rpc/blockchain.cpp:1457--1539; doc/release-notes/release-notes-5.6.0.md:28). The records are maintained by the validator’s incremental tree: zcashd stores `SubtreeData{leadbyte,root,nHeight}` and `LatestSubtree{leadbyte,index,root,nHeight}` records under a `uint64_t` subtree index, fixes `TRACKED_SUBTREE_HEIGHT = 16`, and produces a completed subtree root exactly when the tree size crosses a 2^{16} boundary (zcashd, src/zcash/IncrementalMerkleTree.hpp:20--70, 409--411; IncrementalMerkleTree.cpp:913--925):

```
current_subtree_index() = size() >> TRACKED_SUBTREE_HEIGHT.
```

Handler `GetSubtreeRoots` of `lightwalletd` accepts only Sapling or Orchard and forwards to `z_getsubtreesbyindex(pool, startIndex[, maxEntries when > 0])`; for each returned subtree it fetches the block at `end_height`, populates the two completing-block fields from it, and streams the assembled `SubtreeRoot` (lightwalletd, frontend/service.go:832--907).

Example 11.8 (Mainnet Sapling subtrees 0 and 1). A worked datum recorded in `lightwalletd` source (a quoted `z_getsubtreesbyindex` transcript, common/common.go:178--210): Sapling subtree 0 — outputs 0 through 65535 — has root

```
754bb593ea42d231a7ddf367640f09bbf59dc00f2c1d2003cc340e0c016b5b13
```

with `end_height = 558822`, and subtree 1 has root

```
03654c3eacbb9b93e122cf6d77b606eae29610f4f38a477985368197fd68e02d
```

with `end_height = 670209`: measured from Sapling activation at height 419,200, the first 2^{17} Sapling outputs span roughly the chain’s first 250,000 post-activation blocks.

Deployed validator. Zebra implements both `z_gettreestate` and `z_getsubtreesbyindex` in its RPC surface, so a Zebra-backed indexer retains tree synchronisation after zcashd’s retirement. The successor indexer serves the same gRPC API: Zaino’s `get_tree_state` proxies

`z_get_treestate` to the backing validator and hex-encodes the final states (at HEAD it also populates an `ironwood_tree` field), its `get_latest_tree_state` reads the chain height and delegates, and its `get_subtree_roots` fetches each completing block via `z_get_block(end_height)`. One behavioural divergence: Zaino maps the proto's uint32 `startIndex/maxEntries` down to u16 and returns `InvalidArgument` when either exceeds 65535 — semantically safe, since a depth-32 tree has at most 2^{16} shards, but stricter than `lightwalletd/zcashd`, whose u64 subtree index yields an empty stream instead (`src/chain_index.rs:509--514` versus `zcashd src/zcash/IncrementalMerkleTree.hpp:20--70`). Both backends delegate to the validator: `NodeBackedChainIndex::get_subtree_roots` calls the blockchain source; `ValidatorConnector::State` issues `ReadRequest::`{`Sapling`,`Orchard`,`Ironwood`}`Subtrees` to Zebra's `ReadStateService`, and `::Fetch` proxies `z_getsubtreesbyindex`; only the mockchain test source self-computes.

11.4 Client state: shards, cap, and retention

The client-side container is `shardtree`: “a sparse binary Merkle tree of the specified depth, represented as an ordered collection of subtrees (shards) of a given maximum height”, plus a *cap* — the tree above the shard roots — and a checkpoint set enabling truncation. The root sits at address $(\text{DEPTH}, 0)$, shards at level `SHARD_HEIGHT`, and `max_subtree_index() = 2^{\text{DEPTH}-\text{SHARD\_HEIGHT}} - 1`. The wallet API instantiates one such tree per pool as `ShardTree<Store, DEPTH = 32, SHARD_HEIGHT = 16>` with block heights as checkpoint identifiers, via `with_sapling_tree_mut / with_orchard_tree_mut / with_ironwood_tree_mut`. The shard height is documented as chosen to conform “to the structure of subtree data returned by `lightwalletd` when using the `GetSubtreeRoots` GRPC call”: `SAPLING_SHARD_HEIGHT = ORCHARD_SHARD_HEIGHT = 16`, half the tree depth of each pool. The geometry — height-16 shards of 2^{16} leaves under the 16-level cap, at most 2^{16} shards — is the *Ironwood Guide*'s construction (§“Resting: the note in the commitment tree”); the deployed constants realise it exactly. The SQLite backend constructs each tree as `ShardTree::new(store, PRUNING_DEPTH)` with `PRUNING_DEPTH = 100`: at most 100 ordinary recent checkpoints are retained before pruning. Checkpoints occur at represented block boundaries; a block containing no note commitment in any pool need not create a distinct checkpoint. From NU6.3 activation onward, established checkpoints at heights divisible by 288 are additionally retained as durable anchors and are exempt from that automatic pruning. The 100-block depth is the pinned wallet's assumed maximum reorganisation depth and ZIP 307's historical cache-size guidance (§11.1), not a NU7 consensus limit. The shorter block spacing and proposed node rollback policy require those assumptions to be reviewed together (§10.7); no automatic tripling of these implementation constants is specified here.

What survives pruning is governed by per-node *retention*:

- `Ephemeral` — prunable together with its siblings;
- `Checkpoint{id, marking}` — the position is retained (its value prunable) while the checkpoint lives, decaying to `Marked`, `Reference`, or `Ephemeral` per the marking when the checkpoint is removed;
- `Marked` — retained until explicitly removed; the wallet's own note positions;
- `Reference` — retained until overwritten by a node with `Ephemeral`, `Checkpoint`, or `Marked` retention; downloaded subtree roots and seed frontiers.

11.5 Ingesting subtree roots

The reference sync driver begins *every* run by refreshing the shard roots: function `update_subtree_roots` downloads all subtree roots — the default `GetSubtreeRootsArg`, i.e.

startIndex = 0, maxEntries = 0 — for Sapling and, with the orchard feature, Orchard and Ironwood, converting each streamed message by

```
CommitmentTreeRoot::from_parts(completingBlockHeight, H::read(rootHash))
```

and passing the batches, each with start index 0, to put_sapling_subtree_roots, put_orchard_subtree_roots, and put_ironwood_subtree_roots. Type CommitmentTreeRoot pairs the subtree end height with the root hash, and the trait documentation fixes the meaning: each value “should be the Merkle root of a subtree ... containing $2^{\text{SHARD_HEIGHT}}$ note commitments”.

The SQLite backend implements all three puts via one generic procedure, put_shard_roots parameterised by the hash H , the tree depth, and the shard height:

- (1) no-op on empty input;
- (2) load the cap and treat it as a standalone tree of $\text{DEPTH} - \text{SHARD_HEIGHT}$ levels whose level-0 leaves are shard roots: the LevelShifter wrapper offsets the hash,

```
emptyLeaf = H.emptyRoot(SHARD_HEIGHT),
combine( $\ell, a, b$ ) = H.combine( $\ell + \text{SHARD\_HEIGHT}, a, b$ ),
```

and the downloaded roots are batch_inserted at Position::from(startIndex) with Retention::Reference;

- (3) write the cap back;
- (4) check contiguity over $[\text{startIndex}, \text{startIndex} + n)$: check_shard_discontinuity rejects a call whose shard-index range is neither overlapping nor immediately adjacent to the stored $[\text{min}(\text{shardIndex}), \text{max}(\text{shardIndex}) + 1)$ range, raising Error::SubtreeDiscontinuity — subtree roots must arrive without gaps;
- (5) upsert each root into {pool}_tree_shards(shard_index, subtree_end_height, root_hash, shard_data); on conflict only the end height and root hash are updated, so existing scanned shard data is never clobbered, and the fresh shard_data written for new rows is a single root leaf with RetentionFlags::EPHEMERAL.

The downloaded root is cached beside any persisted subtree rather than merged eagerly — “we want to avoid deserializing the subtree just to annotate its root node, so we simply cache the downloaded root alongside of any already-persisted subtree” — and the read path resolves the pair: get_shard deserialises shard_data and applies reannotate_root(Some(rootHash)).

Remark 11.9 (Idempotent full re-download). The driver fixes startIndex = 0 on every run although put_shard_roots supports arbitrary start indices; the upsert path makes the repetition idempotent, at the cost of re-transferring one 32-byte root (plus completing-block metadata) per completed shard per pool on every sync (zcash_client_backend, src/sync.rs:80--86, 228--299). As of 2026-07-15 (mainnet tip 3,413,101): 1,128 complete Sapling subtrees (last completing at block 3,390,009) plus 765 Orchard (3,388,160), so each sync re-downloads 1,893 roots — 60,576 bytes of root data, 136,378 bytes of SubtreeRoot messages, 145,843 bytes with the five-byte gRPC frame per stream element (computed by script; counts cross-check against ChainMetadata at block 3,413,000: $\lfloor 73,932,454/2^{16} \rfloor = 1,128$, $\lfloor 50,191,264/2^{16} \rfloor = 765$). No source documents the fixed startIndex = 0 as intentional.

11.6 Frontiers as scan seeds

A TreeState becomes client state through TreeState::to_chain_state(): it hex-decodes the block hash and reverses its bytes (“Zcashd hex strings for block hashes are byte-reversed”), and

parses each pool’s legacy tree (Definition 11.4) into a frontier; an *empty* tree-state string parses as the empty tree, the correct state before a pool’s activation. The result is

```
ChainState{ blockHeight; blockHash;
            finalSaplingTree : Frontier<32>; finalOrchardTree : Frontier<32>;
            finalIronwoodTree : Frontier<32> },
```

“the final note commitment tree state for each shielded pool, as of a particular block height” — the Orchard and Ironwood frontiers sit behind the orchard feature.

Every scanned batch is seeded with the chain state of the block immediately preceding it, and the seed is verified against the server’s own metadata before any tree is touched:

```
fromHeight = seed.blockHeight+1 (asserted),    seed.treeSizeP+|cmP(b0)| = finalTreeSizeP(b0)
```

per pool P , where b_0 is the batch’s first block and the final tree size is read from the block’s chainMetadata; violation yields error NonSequentialBlocks. The metadata sizes are what give the scanner absolute leaf positions when no prior metadata is available. Without positions it cannot derive Sapling nullifiers or build any pool’s witnesses (TreeSizeUnknown, a hard error); Orchard and Ironwood nullifier derivation itself is position-independent, and a size contradicted by block contents is TreeSizeMismatch, a continuity error triggering the rewind of §10.7 (lightwallet-protocol/walletrpc/compact_formats.proto:14--17,31--40).

The batch’s note commitments are then assembled into located subtrees in parallel — rayon, chunks of 1024 commitments, LocatedTree::from_iter at the shard level — starting at Position::from(seed.treeSize_P), and each pool’s tree is updated by update_tree:

- (1) insert the seed frontier (tree.insert_frontier) with retention Checkpoint{id = frontierHeight, marking = Reference} — the in-source comment: “we insert the frontier with Checkpoint retention because we need to be able to truncate the tree back to this point”;
- (2) insert_tree(subtree, checkpoints) for each built subtree;
- (3) add any cross-pool checkpoints missing above the minimum retained checkpoint (the cross-pool checkpoint-unification rule itself, §10.6).

Inside shardtree, insert_frontier splits a non-empty frontier at the shard boundary: the leaf and the ommers (the sibling hashes a frontier stores, one per filled subtree along the leaf’s path to the root) below the shard level enter the shard containing the frontier’s position, the ommers at levels $\geq$ SHARD_HEIGHT enter the cap, checkpoint retention adds a checkpoint at the leaf position, and checkpoints beyond max_checkpoints are pruned. An empty frontier with checkpoint retention records a checkpoint for the empty tree (Retention::Marked is invalid there).

Frontier seeding is also how a wallet is born and how it rewinds. An AccountBirthday wraps “the chain state prior to the birthday height” plus an optional recover_until; AccountBirthday::from_treestate builds it directly from a TreeState for the last block prior to the birthday, with height() = priorChainState.blockHeight + 1, and account creation stores the birthday tree sizes and rewinds the wallet to the birthday chain state, so no block before the birthday needs to be replayed: the prior frontier summarises the required prefix (§10.2). Symmetrically, truncate_to_chain_state inserts each pool’s frontier from the target chain state with Checkpoint{id = targetHeight, marking = None}, creating the checkpoint on which the subsequent tree truncation lands (recovery flow, §10.7).

11.7 Shard completion and spendability

Two of the scan-queue priorities of §10.3 exist purely for this layer: ChainTip is documented as “blocks that must be scanned to complete the latest note commitment tree shard” and FoundNote as “blocks that must be scanned to complete note commitment tree shards adjacent to found notes”.

The tip shard. On each tip update the backend computes per pool the maximum `subtree_end_height` over the pool’s shards table — the end of the last complete shard known from `GetSubtreeRoots` — takes the minimum across pools, and enqueues

$$[\max(\text{walletBirthday}, \text{minShardTip}), \text{tip} + 1)$$

at `ChainTip` priority, so the incomplete tip shard of every pool is scanned even before historic ranges (the full tip-update case analysis, §10.4).

Found notes. After a range is scanned, `scan_complete` maps each detected note position to its shard, $s = \lfloor p/2^{16} \rfloor$ (`Address::above_position(SHARD_HEIGHT, position)`), and extends the scanned range to that shard’s block extent — from the previous shard’s `subtree_end_height` (the pool activation height for shard 0, clamped to the wallet birthday) to the containing shard’s `subtree_end_height + 1` — enqueueing the extensions beyond the just-scanned range at `FoundNote` priority. Completing the shard’s block range is exactly what makes the note’s witness computable.

Witness derivation. The query `witness_at_checkpoint_id(position, id)` requires `position ≤ checkpoint.position()` (else `QueryError::NotContained`) and folds, from the note’s shard upward to the root, the root of each sibling address truncated at leaf count `positionas of + 1` — each sibling root drawn from the note’s own shard (scanned leaves), another shard’s cached root (`GetSubtreeRoots`, Definition 11.5), the cap, or a per-level empty-root constant beyond the tree size. A pruned node that cannot be replaced by an empty-root constant yields `QueryError::TreeIncomplete(addresses)` — the precise failure mode of an unready witness. The mathematics of the fold is the *Ironwood Guide*’s (§“Resting: the note in the commitment tree”); what this volume adds is the deployed readiness condition under which it cannot fail.

Spendability. That condition is enforced in SQL. Writing U_p for the rows of view `v_{pool}_shard_unscanned_ranges`, note selection excludes any note at position p for which some $(s, e, h_0, h_1) \in U_p$ satisfies

$$s \leq p < e \wedge h_0 \leq \text{anchorHeight} \wedge h_1 > \text{walletBirthday},$$

i.e. the note’s shard has an unscanned block range at or below the anchor; the same view drives `unscanned_tip_exists`, which invalidates anchors whose containing shard is not fully scanned. The views compute `startPosition = shardIndex << 16` and `endPositionExclusive = (shardIndex+1) << 16`, set the shard’s block extent to begin at the previous shard’s end height (the pool activation height for the first shard), join the scan-queue rows whose block ranges overlap that extent, and filter to priority above `Scanned`. Unwinding the predicate recovers the tree-readiness condition: at a given anchor, the note’s own shard must be scanned up to the anchor, every earlier shard must be covered by a downloaded root, and the blocks from the last complete shard boundary to the anchor must have been scanned (the `ChainTip` range above). Whether the note is otherwise eligible to spend, which anchor a transaction selects, and at what confirmation depth, is covered in §4.1 and §6.6.

At the pinned `librustzcash` commit — part of the NU6.3 work of Remark 10.2, not of any release deployed with `lightwalletd` v0.4.1 — notes additionally gain a `witness_stabilized` flag once their shard’s block extent is fully `Scanned` and the shard’s end height has at least `PRUNING_DEPTH` confirmations (`floor = maxScanned - (PRUNING_DEPTH - 1)`); stabilised notes bypass the confirmations check at spend time, and the tip shard by definition can never stabilise (`zcash_client_sqlite`, `src/wallet/scanning.rs:416--473`; `src/wallet/common.rs:700--724`).

What this buys, and what remains. The linear protocol’s impossibility is gone: a freshly re-stored wallet can seed its trees from the birthday frontiers instead of replaying the chain prefix, and can use one 32-byte root per completed shard. A note found during scanning becomes tree-ready once its own shard’s block extent and the tip range are scanned — not after the entire chain history. What remains is the trial decryption of every block from the birthday forward (§10) and the shard-completion scanning of this section; both are cost centres the proposed Tachyon proof-carrying wallet is designed to remove (Bowe–Miers, ePrint 2025/2031; tachyon-zcash/tachyon, book/src/proof-tree.md @ 26fdc165).

12 What the server learns

A light client outsources block storage and filtering to a server it does not trust with its keys, then reconstructs its own state by trial decryption on the device (*Ironwood Guide*, §“Delivery: the note reaches its owner”). The keys never travel; the *requests* do. This section inventories what the deployed server-side observer learns from those requests: the stated threat model, the baseline that genuinely holds, the leaks in the deployed request pattern, the operator’s instruments for recording them, and the mitigations that actually ship. Implementation references use the commits pinned above; where the code diverges from ZIP text, we flag the divergence in place.

12.1 The stated security model

ZIP 307 names one primary goal for the light-client protocol.

Definition 12.1 (Payment detection privacy). The serving proxy must not learn which transactions are addressed to a given light wallet. Under the additional assumption of network privacy (“via Tor or similar”), the proxy must also be unable to link different connections or queries as deriving from the same wallet (ZIP 307 §“Security Model”).

The adversary of ZIP 307 is the node/proxy combination, modelled as *honest but curious*: it is trusted for chain-state correctness — the client accepts the block data it serves — but assumed to record and analyse everything it sees. The ZIP explicitly does not address spending notes privately (ZIP 307 §“Security Model”); transaction submission nonetheless flows through the same server, and we treat it in §12.5.

Remark 12.2 (The de facto wire specification). ZIP 307 has Status Draft — it was never finalised — even though the protocol it sketches is the deployed synchronisation layer. The deployed wire format has moved past the sketch: the ZIP’s CompactBlock carries vtx at field 3 and a BlockID, while the deployed message carries vtx at field 7 plus (in the v0.4.1 proto lightwalletd vendors) protoVersion — removed in v0.5.0 (§8.2) — prevHash, time, an (always unset) header, and ChainMetadata; the deployed CompactTx adds txid, fee, Orchard actions, and transparent vin/vout entries (CompactTxIn outpoints and full TxOuts) (ZIP 307 §“Compact Stream Format” vs. lightwalletd, lightwallet-protocol/walletrpc/compact_formats.proto). The lightwallet-protocol proto repository, not the ZIP, is the de facto wire specification. The planned in-protocol successor is emptier still: ZIP 314, “Privacy upgrades to the Zcash light client protocol”, has Status Reserved — a title with no content (zips/zip-0314.rst). The server implementation delegates likewise: the lightwalletd README directs developers to the external wallet app threat model for “important information about the security and privacy limitations of light wallets that use lightwalletd”; the repository itself contains no privacy analysis (lightwalletd, README.md).

12.2 What never leaves the device

The baseline holds. No RPC in the deployed service carries key material: across the full `CompactTxStreamer` surface (twenty methods), requests carry only public selectors or data such as heights, hashes, txids, transparent addresses, raw transaction bytes, pool or subtree selectors, and — for the testing-only Ping RPC — a duration (`lightwalletd`, `lightwallet-protocol/walletrpc/service.proto`). Incoming viewing keys remain on the device: trial decryption of compact outputs is performed client-side by `scan_block` over locally cached `CompactBlocks`, by the mechanism of the *Ironwood Guide*, §“Delivery: the note reaches its owner”; witness maintenance over the resulting positions is that of §“Resting: the note in the commitment tree”. Nullifier matching against the `CompactSaplingSpend` and `CompactOrchardAction` nullifier fields is likewise local.

The compact form is the reason a server mediates at all: the per-output and per-spend reductions, and the recomputed-commitment integrity substitute for the discarded AEAD tag, are those of §8 (Table 6; §8.5).

Remark 12.3 (The transparent boundary). The baseline above is a shielded-layer property. The same service carries transparent-address RPCs whose requests contain cleartext addresses — `GetAddressTxids` and `GetAddressTransactions` (an address plus a block range), `GetAddressBalance(Stream)` (an address list), `GetAddressUtxos(Stream)` (an address list plus a start height) — so for the transparent layer the server learns the wallet’s addresses outright, by declaration rather than inference (`lightwalletd`, `walletrpc/service.proto`). We treat that request family as out of scope except where the shielded sync driver itself uses it (§12.3) and where deployed mitigations attach to it (§12.7).

12.3 The observable request schedule

What the server sees is a request schedule. ZIP 307 describes it in four phases: at first start, $A = \text{GetLatestBlock}$ plus `GetBlock` at the tip; on each subsequent synchronisation, $B = \text{GetLatestBlock}$ followed by `GetBlockRange(X,Y)` from the last-synced height X ; $C = \text{GetTransaction(txid)}$ for each transaction of interest after detection; and $D =$ the next `GetLatestBlock` plus `GetBlockRange(Y,Z)` (ZIP 307 §“Client-server interaction”).

The deployed driver refines this but does not change its shape. Per cycle, function run of `zcash_client_backend` (module `sync`, feature `sync`) issues: `GetSubtreeRoots` per shielded pool, `GetLatestBlock`, `GetAddressUtxosStream` over *all* transparent receivers of every account (cleartext, Remark 12.3), `GetTreeState` for the block before each scan range, and `GetBlockRange` for each suggested scan range in `batch_size` chunks. At the pinned commit the subtree-root fetch covers Sapling, Orchard, and Ironwood. Ironwood appears in no light-client ZIP — ZIP 2005 covers quantum recoverability and ZIP 258 only NU6.3 deployment — the request exists solely because the `librustzcash` proto fork defines `ShieldedProtocol.ironwood = 2` and the driver requests it (`src/sync.rs:289--296`; cf. Remark 10.1). The module’s own documentation notes that this simple implementation does not itself enhance transactions; enhancement is part of the documented flow (the crate-root diagram’s “Enhance transactions” step) and is driven by wallets via `transaction_data_requests` (`zcash_client_backend`, `src/lib.rs`, `src/sync.rs`).

The *order* of range requests is itself data. Scan ranges carry a priority (Table 9), and `suggest_scan_ranges` returns them in descending priority, then descending end height (§10.3), so `Verify` and `ChainTip` ranges near the tip are fetched first and `FoundNote` ranges pre-empt Historic backfill. The privacy consequence of that pre-emption is Remark 12.4.

12.4 Leaks in the deployed pattern

ZIP 307 asserts that compact-block synchronisation itself “leaks no information about which transactions the light client is interested in” because the compact stream is a broadcast medium

(ZIP 307 §“Transaction privacy”). The same document contradicts the claim in the immediately preceding section, for the request pattern actually deployed: “The above interaction reveals to the server at the start of each synchronisation phase (B and D) the block height which the light client had previously synchronised to. This is an information leak under our security model” (ZIP 307 §“Block privacy via bucketing”). Compact range contents alone do not reveal which outputs trial-decrypt successfully, but the requests’ bounds and timing — and, when an integrating wallet performs them, transaction-specific follow-up requests — provide additional observables. We take them in turn.

Historic backfill reveals an approximate birthday. On account creation, the scan queue marks the range from Sapling activation up to the account birthday as Ignored — never downloaded — and queues scanning from the birthday itself. The minimum historic start height visible across the request schedule therefore approximates the wallet’s earliest account birthday. Priority scheduling means this need not be the first range requested, and importing an older account can later lower it.

Range boundaries reveal scan state and progress. In ZIP 307’s linear schedule, a range start is the prior sync height, which is the leak the ZIP concedes above. The deployed priority queue can instead request a verification, shard-completion, or historic range, so its bounds expose scan state and progress but are not invariably the preceding sync height. ZIP 307’s proposed mitigation is bucketing: for a bucket size N , phases B and D would request

`GetBlockRange( $X - (X \bmod N)$ ,  $Y$ )` instead of `GetBlockRange( $X$ ,  $Y$ )`,

hiding the exact previously-synced height within an N -block bucket (ZIP 307 §“Block privacy via bucketing”). The ZIP marks this “a proposed enhancement that has not been implemented”, and the deployed code confirms: `download_blocks` issues `GetBlockRange` with start and end exactly equal to the suggested scan-range bounds, with no rounding (`zcash_client_backend, src/sync.rs`).

Wallet-driven enhancement can follow detection with a fetch. During compact-block scanning, function `put_blocks` records, per scanned block, exactly the subset of transactions relevant to this wallet, and queues a retrieval request for each. An integrating wallet that consumes `transaction_data_requests` later receives `Enhancement(txid)` or `GetStatus(txid)`, and may resolve it via the `GetTransaction` RPC. The simple sync module does not itself perform enhancement, and the API imposes no requirement that a fetch occur immediately after its containing range. When a wallet does issue these requests without cover, however, the selected txids reveal transactions it is enhancing or status-checking; that set is sensitive but can also include sent transactions and transparent-graph dependencies.

Here the code diverges from the ZIP, in the unusual direction: the ZIP is stricter than the deployment. ZIP 307’s full-transaction mitigation is normative — the client “SHOULD obscure the exact transactions of interest by downloading numerous uninteresting transactions as well”, “SHOULD download all transactions in any block from which a single full transaction is fetched”, and “MUST convey to the user that fetching full transactions will reduce their privacy” (ZIP 307 §“Transaction privacy”). No decoy, chaff, or cover-traffic code exists anywhere in `librustzcash` or `lightwalletd` (negative `grep` over both repositories at the pinned commits), and the deployed RPC surface offers no bulk full-transaction fetch to implement the second SHOULD with — `GetBlock` returns compact data (`lightwalletd, lightwallet-protocol/walletrpc/service.proto`). The protocol does not prevent an individual wallet from issuing extra `GetTransaction` requests as cover; the point is that neither pinned reference library supplies that policy automatically.

Enhancement fetches carry no timing decorrelation. The deployed request taxonomy is

```
TransactionDataRequest ∈ { GetStatus(txid), Enhancement(txid),
                           TransactionsInvolvingAddress{..., request_at, ...},
                           GetSpendingTx(·) }
```

(the last two variants are feature-gated). Only `TransactionsInvolvingAddress` carries a `request_at` field, documented as the instant chosen so as to “decorrelate this request to a light wallet server from other requests made by the same caller”, ignorable when the supplier of chain data is “private, trusted-for-privacy”. The shielded-note variants `Enhancement` and `GetStatus` have no analogous jitter: the SQLite backend emits them with no `request_at`. The one deployed timing defence (§12.7) protects transparent address checks, not the shielded fetch-on-detect path.

Remark 12.4 (FoundNote clustering — an inferred leak). When a note is found, `scan_complete` inserts scan ranges at priority `FoundNote` extending the scanned range so as to complete the note-commitment-tree shard(s) containing the found note, and `suggest_scan_ranges` orders by priority, so the server observes out-of-order, high-priority `GetBlockRange` fetches that can narrow the blocks around detected notes (`zcash_client_sqlite`, `src/wallet/scanning.rs`; `zcash_client_backend`, `src/data_api/scanning.rs`). No source documents this as a leak — neither ZIP 307 nor the code documentation — and the statement above is this volume’s own inference from the scan-queue code. It requires adversarial review — in particular, independent confirmation that `FoundNote` ranges surface to the server as distinct requests rather than being absorbed into contiguous suggested ranges — before it may be cited as fact.

Mempool polling can fingerprint the session. If a client calls `GetMempoolTx`, field `exclude_txid_suffixes` of the `GetMempoolTx` request carries suffixes of txids the client already holds, so a mempool-polling client additionally reveals which mempool transactions it has already seen, a session-linkable fingerprint (`lightwalletd`, `walletrpc/service.proto`). No source documents this as a leak; we record it for completeness.

12.5 The submission channel

Spending is out of ZIP 307’s stated scope (§12.1), but the deployed submission path runs through the same server when a wallet chooses that path. The `SendTransaction` handler hex-encodes the received raw transaction and forwards it to the backing node’s `sendrawtransaction` JSON-RPC (`lightwalletd`, `frontend/service.go`), so the operator holds the complete transaction *before* network broadcast and can link it to the submitting IP address, the TLS session, its timing, and — absent circuit isolation (§12.7) — to the same session’s sync and enhancement traffic. The backing full node also sees the transaction and the `GetTransaction` txid pattern, but without client IPs: `lightwalletd` terminates the client connections. The lifecycle the transaction then follows — store-then-broadcast, expiry, confirmation counting under the `ConfirmationsPolicy` — is described in §6.

12.6 The operator’s instruments

What the operator *can* record is a structural property, independent of any logging switch. The gRPC endpoint is operator-terminated TLS, so every connection exposes the peer IP and its open and close times — the peer may be a Tor exit or another proxy, not the wallet’s own IP. A Prometheus gauge, `grpc_server_connections_current`, tracks the live connection count, and `go-grpc-prometheus` interceptors on both the unary and streaming paths export per-method request counts and latency histograms over the HTTP endpoint (default `127.0.0.1:9068`) (`lightwalletd`, `cmd/connstats.go`, `cmd/root.go`).

Logging narrows the distance between can and does. Most handlers log request parameters at Debug level — including `GetBlockRange`, `GetTransaction`, `SendTransaction`, `GetAddressBalance`, and `GetAddressUtxos` — and ten unary return paths log full responses at Trace level; `GetLightdInfo` and the testing-only `Ping` have no equivalent handler log (`lightwalletd`, `frontend/service.go`). The default level is Info (`cmd/root.go`), so `--log-level 5` enables the Debug request logs and can record block ranges, txids, raw transactions, and transparent addresses for the named handlers, to the default log file `./server.log`, without client consent or notice. A separate unary interceptor, enabled by `--grpc-logging-insecure` (default false; the flag name itself marks the hazard), logs {peer address, method, duration, error} per request, and carries the source comment “TODO: anonymize the addresses. cryptopan?” (`lightwalletd`, `common/logging/logging.go`, `cmd/root.go`).² The project’s own architecture document is candid: “Logging may capture identifiable user data. It hasn’t received any privacy analysis yet and makes no attempt at sanitization” (`lightwalletd`, `docs/architecture.md`).

At the pinned Zaino revision, the successor indexer changes none of this structurally: it serves the full `CompactTxStreamer` surface; every handler logs the method name at Info level — the literal message is [TEST] received call, apparent leftover test instrumentation in production code — and, under the `prometheus` feature, emits per-method request counts, duration histograms, and error counters. The metric names are `zaino.grpc.requests_total`, `zaino.grpc.request_duration_seconds`, and `zaino.grpc.errors_total`, labelled by method and code, with no peer labels; no peer-address logging exists in the dispatch path (`zaino-serve`, `src/rpc/grpc/service.rs`, `src/lib.rs`). Its README acknowledges the setting: “Due to current potential data leaks / security weaknesses ... there is a need to use anonymous transport protocols (such as Nym or Tor) to obfuscate clients’ identities from Zcash’s indexing servers (Lightwalletd, Zcashd, Zaino)”, and the `serve` crate’s documentation states the ingestor “has been built so that other ingestors may be added that use different transport protocols (Nym, TOR)” — a capability anticipated, not implemented: no Nym, arti, or onion code exists in the workspace (`zaino`, `README.md`; `zaino-serve`, `src/lib.rs`; negative grep over packages/). Zaino is pre-1.0; all of the above is cited at commit `0e057e22` and may move.

12.7 Deployed mitigations

Two mitigations ship in the client stack. Neither touches the shielded fetch-on-detect leak of §12.4.

Transport: Tor. Crate `zcash_client_backend` ships an arti-based Tor client behind feature `tor`. Method `Client::connect_to_lightwalletd` tunnels the tonic gRPC channel through Tor (behind the additional feature `lightwalletd-tonic-tls-webpki-roots`), and `Client::isolated_client` returns a handle whose streams never share circuits with the parent, documented for “when you want separate parts of your program to each have a `tor::Client` handle, but where you don’t want their activities to be linkable to one another over the Tor network” (`zcash_client_backend`, `src/tor.rs`, `src/tor/grpc.rs`). Circuit isolation is a tool for removing shared-circuit linkage between the submission channel of §12.5 and sync traffic; it does not exclude timing, content, or application-level correlation. We classify this as *capability-in-librustzcash*, *deployment-status-unknown*: whether any deployed wallet routes `CompactTxStreamer` traffic through it by default — and whether `SendTransaction` uses a circuit isolated from sync — is not determinable from the local repositories.

²The interceptor is wired on the unary path only, so streaming RPCs — including `GetBlockRange`, the bulk of sync traffic — never pass through it; the flag logs a skewed subset of traffic. Whether this is intentional is not determinable from the source.

Timing: exponential jitter, transparent addresses only. The SQLite backend schedules ephemeral-address checks by sampling the next check time from an exponential distribution,

$$t_{\text{next}} := t_{\text{from}} + \Delta, \quad \Delta \sim \text{Exp}(\lambda), \quad \lambda = 1/T, \quad \mathbb{E}[\Delta] = T,$$

with the expected interval T set to $(24 \cdot 60 \cdot 60)/n$ seconds for n tracked ephemeral addresses — one expected check per address per day — over a shuffled address order, “so that we don’t always check them in the same order”. The intent is explicit in the API documentation: only one such query should be performed at a time, “to avoid linking multiple transparent addresses as belonging to the same wallet in the view of the light wallet server” (`TransparentAddressMetadata::next_check_time`). No analogous jitter exists for shielded-note enhancement (§12.4).

12.8 Summary, and the structural reading

Table 10 collects the observables.

Observable	Reveals	Mitigation status
Peer IP, connection timing	network endpoint; activity windows, not necessarily real-world identity	Tor capability in zcash_client_backend; deployment unknown
Lowest <code>GetBlockRange</code> start	earliest visible historic backfill; approximate account birthday	none
Range boundaries	scan state and progress; often a resume height	bucketing proposed (ZIP 307), unimplemented
Out-of-order <code>FoundNote</code> ranges	can narrow blocks around detected notes	none (inferred; Remark 12.4)
Wallet-driven <code>GetTransaction</code>	txids selected for enhancement or status checks	decoys normative (SHOULD); no reference-library policy
<code>exclude_txid_suffixes</code> , if mempool polling is used	mempool transactions already seen	none
<code>SendTransaction</code> , if this submission path is used	full transaction, pre-broadcast, linked to origin	circuit-isolation capability, deployment unknown
Transparent-address RPCs	addresses outright	exponential jitter and shuffling (ephemeral addresses only)

Table 10: What the server learns from the deployed light-client protocol, and the status of each mitigation. Sources: §12.4 through §12.7.

The structural reading is more limited: an unmitigated server-visible request schedule exposes connection metadata and range boundaries, while optional enhancement, mempool, and submission calls expose more when a wallet uses them. These observations are not properties of compact-block contents alone, nor are their present strengths inevitable: Tor, request bucketing, jitter, batching, and cover traffic can alter them, although the reference implementations do not supply all of those policies. Within the deployed protocol the specified upgrade path is only a stub (ZIP 314, Reserved, Remark 12.2); one proposed alternative is architectural. Project Tachyon proposes to remove exactly this layer: under its fold the wallet no longer scans, and its designers describe synchronization services as “fully oblivious to their clients’ transaction details” (Bowe–Miers, ePrint 2025/2031, §1.3; tachyon-zcash/tachyon, book/src/proof-tree.md @ 26fdc165). The ePrint gives no formal service threat model and calls timing-correlation linkage potentially fatal. The present section is the baseline against which that design must be measured.

Part VII

FlyClient Guide: Succinct chain verification: the sampling protocol, the deployed ZIP-221 commitment, and the unbuilt bridge

Abstract

Assuming the commitment foundations of the *Crypto Guide*, the deployed chain-selection rules of the *Consensus Guide*, the block structures of the *Ironwood Guide*, and the light-client layer of the *Wallet Guide*, this volume of *The Zcash Arboretum* documents succinct chain verification for Zcash: the FlyClient protocol as its paper constructs it, the ZIP-221 chain-history commitment that every post-activation Zcash block header has carried since Heartwood, and the distance between the two. The commitment is consensus-critical and required in valid post-activation headers; none of the light-client implementations examined here consumes it. Claims about the deployed tree cite the ZIPs and the implementations; claims about the sampling protocol carry the paper’s own model assumptions, and its printed defects are registered rather than silently repaired; the bridge between the two is classified plainly as unbuilt.

1 Scope, sources, and the deployment boundary

FlyClient is a light-client protocol that verifies a proof-of-work chain by sampling logarithmically many block headers under a difficulty-weighted distribution, checking each sample against a commitment that every header carries over the entire chain before it (Bünz–Kiffer–Luu–Zamani, “FlyClient: Super-Light Clients for Cryptocurrencies”, IACR ePrint 2019/226; published at IEEE S&P 2020). The commitment is a Merkle mountain range — an append-friendly Merkle tree — and the protocol’s claim is quantitative: a client holding only the genesis block can accept the head of an n -block chain after downloading $O(\lambda \log_{1/c}(n) \log_2(n) + L)$ data rather than n headers, where c bounds the validly mined fraction of an adversarial fork and L is the fork length below which the adversary is unconstrained — so the client checks the last L headers in full. Identifying c directly with a mining-power ratio requires a particular constant-difficulty interpretation; the paper does not derive it from Zcash’s retarget rule.

This volume documents two objects and the distance between them. The first is the protocol of the paper: model, commitment, sampling distribution, security theorem, and the proof structure behind it (§2–§4). The second is the consensus structure Zcash actually deployed for it: ZIP-221’s chain-history tree, a difficulty-MMR variant incorporated into every header after the Heartwood activation block, extended at NU5 by ZIP-244’s authorising-data commitment (§5–§7). The distance is the volume’s third subject (§8): Zcash commits to FlyClient’s tree substrate in every post-activation block and validates it in every full node, yet no deployed wallet, SDK, or light-client server exposes or consumes a proof; deployed light clients instead trust their server for chain state outright (*Wallet Guide*, §“Architecture and authority”).

1.1 Sources and pinning

Table 1 pins the artefacts this volume treats as ground truth. Two rigor conventions apply. Deployed behaviour cites crate, file, and line against the pinned commits; design-stage claims are classified as *specified*, *designed but unspecified*, or an *open problem*. The FlyClient paper itself is treated as a primary source with registered defects: the revision of record contains internal inconsistencies (a sign typo in a difficulty bound, a vestigial pseudocode return, conflicting evaluation constants) that are catalogued in Remark 3.5 and flagged inline where they matter, never silently repaired.

Artefact	Pin	Rigor
FlyClient paper	ePrint 2019/226, rev. 2020-08-20	peer-reviewed; errata registered
ZIP 221, ZIP 244	zips @ e753a6a3 (2026-09-03)	merged, deployed consensus
zcash_history	0.5.0 (crates.io)	deployed (Zebra dependency)
zebra	@ de14bef05 (2026-07-17)	deployed validator
zcashd	@ 16ac74376 (2025-10-03)	historical validator (retired)
lightwalletd, Zaino, librustzcash	current-branch greps, 2026-09-05	negative results (§8)

Table 1: Primary sources, pinned. Status rechecked 2026-09-05.

1.2 Relation to the sibling volumes

This volume assumes the *Ironwood Guide* for everything inside a block — notes, commitments, anchors, the NU5 transaction format — and cites its treatment of the note-commitment tree (§“Resting: the note in the commitment tree”) rather than re-deriving Merkle mechanics; the general theory of commitments and collision resistance is the *Crypto Guide’s* (§“Commitment schemes”). The *Wallet Guide* owns the deployed light-client layer whose central trust assumption — the node/proxy combination “is trusted to provide a correct view of the current best chain state” (ZIP 307, §“The stated security model”) — is precisely the assumption a FlyClient bridge would discharge; this volume owns that cross-reference.

1.3 The deployment boundary, stated at the outset

Every claim in this volume sits on one side of a line that the material itself draws sharply.

- *Deployed and consensus-critical.* The ZIP-221 chain-history tree: maintained by Zebra (and historically by zcashd), its root committed in every block header after Heartwood’s activation block (whose mainnet height is 903,000 and whose field is the mandated zero sentinel) and folded into `hashBlockCommitments` since NU5 (mainnet height 1,687,104), validated on every block connect by Zebra and, until its retirement, zcashd (§5–§7).
- *Specified but consumed by no checked light client.* The same tree as a light-client instrument: ZIP-221’s stated motivation is FlyClient, and the metadata its nodes carry exists for difficulty-weighted sampling; no deployed RPC serves an MMR proof and no deployed client requests one (§8).
- *Designed but not specified for Zcash.* The FlyClient protocol itself — sampling distribution, non-interactive transcript, proof format — exists in the paper with a security theorem in the paper’s own backbone model; no ZIP specifies a Zcash instantiation, and the paper’s model does not automatically cover Zcash’s difficulty-adjustment rule (§4).

2 The problem: verifying a chain without downloading it

2.1 The cost of header-only verification

A full node verifies a proof-of-work chain by downloading and checking every block — for Bitcoin or Ethereum in 2019, more than 200 GB (ePrint 2019/226, §1). Nakamoto’s simplified payment verification reduces this to headers alone: 80 bytes rather than roughly a megabyte per Bitcoin block. The reduction is linear, not asymptotic, and the paper’s motivating arithmetic is Ethereum’s: a 508-byte header every ~15 seconds had produced, by July 2019, more than 8.2 million blocks and hence more than 3.9 GB of headers that an SPV client must fetch and store before verifying its first payment (§1, p. 3). A device class that cannot bear this — phones, wearables, embedded clients — is exactly the class light clients exist for, and the practical alternative, a trusted wallet provider, “negates much of the benefits of these decentralized ledgers” (§1, p. 2).

An SPV client’s trust model is already weaker than a full node’s. It cannot check transaction validity or consensus rules; it assumes the most-work chain follows the rules — the paper’s Assumption 1: “The chain with the most PoW solutions follows the rules of the network and will eventually be accepted by the majority of miners” (§1, p. 2). The most-work rule itself, and the probabilistic guarantee it purchases, are the *Consensus Guide*’s subject (§“The block tree and the most-work rule”). FlyClient keeps exactly this assumption class (strengthened quantitatively, §4.1) and attacks only the cost: can a client accept the head of the chain after sublinear communication? The obstacle is that a client that no longer checks every proof of work “can be tricked into accepting an invalid chain by a malicious prover who can precompute a sufficiently-long chain using its limited computational power” (§1, p. 3).

2.2 Variable difficulty and the difficulty-raising attack

Any sampling client must contend with an attack that does not exist at fixed difficulty. Under variable difficulty the chain’s total *work*, not its length, decides validity, and an adversary may mine few but heavy blocks: Bahack’s difficulty-raising attack. The paper’s worked figure makes the threat concrete: an adversary holding one third of the honest mining power, permitted a single block of arbitrarily high difficulty, exceeds the expected weight of the honest chain with probability roughly 28% — nothing close to negligible (§3.1, p. 7). Full nodes are protected by the retargeting rule itself: difficulty moves only once per epoch and only within a clamp (the dampening filter τ ; Bitcoin operates with $\tau = 4$, epoch length $m = 2016$). A client that samples, however, never sees most transitions, so the protocol must force *every* transition it could ever rely on to be valid — the purpose of the difficulty MMR’s metadata (§3.5) and of Lemma 4 (§3.6).

2.3 Prior art: superblocks and their failure modes

The prior approach to sublinear chain proofs is the superblock: a block whose hash beats a harder target than required. Superblock counting was folklore by 2012 (the “high-value-hash highway”), was made rigorous as interactive PoPoW by Kiayias–Lamprou–Stouka, and non-interactive as NIPoPoW by Kiayias–Miller–Zindros, who also broke the interactive predecessor (ePrint 2019/226, §1, p. 3). The paper’s case against the superblock family is fourfold (§1, “Current Challenges”):

1. Fixed difficulty is assumed outright, and “it isn’t clear how to modify the super-block based protocols to handle the variable difficulties.”
2. Under variable difficulty the difficulty-raising attack of §2.2 applies, and adding transition checks to superblock proofs “is a non-obvious extension.”
3. Superblocks are identifiable and carry no extra reward, so bribing and selfish-mining strategies suppress them cheaply; NIPoPoW’s published defence reverts to full SPV, so its proofs “are therefore only succinct if no adversarial influence exists.” FlyClient distinguishes blocks only by position, leaving nothing cheap to suppress.
4. NIPoPoW transaction-inclusion proofs cost roughly 15 KB on Ethereum against FlyClient’s ~ 1.5 KB.

The comparison FlyClient claims for itself is summarised by the paper’s Table 1, reproduced as Table 2: three orders of magnitude against SPV at Ethereum’s 2019 length, under an adversary bounded by half the honest hash power and failure probability below 2^{-50} .

One reading habit matters for every number above and below. Formally, c bounds the valid-work fraction of a long adversarial fork. In its evaluation, the paper instantiates that parameter as a bound on adversarial mining power relative to *honest* mining power, not as a share of the total. Under that evaluation convention, $c = 0.9$ corresponds to $c/(1 + c) \approx 47.3\%$ of total power,

Block height	10 K	100 K	1,000 K	7,000 K
SPV (KB)	4,961	49,609	496,094	3,472,656
FlyClient (KB)	161	277	416	524
Improvement	31×	179×	1,275×	6,627×

Table 2: Proof sizes for Ethereum parameters (508-byte headers), assuming adversarial hash power at most $c = 1/2$ of the honest hash power and failure probability below 2^{-50} . Reproduced from ePrint 2019/226, Table 1 (p. 4). The 1,275× entry is the paper’s own slip: its SPV and FlyClient sizes give $496,094/416 \approx 1,193\times$ (Remark 3.5).

and the headline $c = 1/2$ rows of Table 2 correspond to one third of total power (§7.1, p. 20). The reduction between these readings is exactly the model gap discussed in Remark 4.1.

3 Merkle mountain ranges

3.1 The structure

The commitment at the heart of FlyClient is a Merkle tree variant optimised for one operation: appending a leaf without rebuilding the tree. The paper defines it recursively (ePrint 2019/226, Definition 11, p. 28); the volume restates the definition in its own numbering because everything later — ZIP-221 included — builds on it.

Definition 3.1 (Merkle mountain range). A *Merkle mountain range* (MMR) over n leaves is a binary hash tree M with root r and depth $\lceil \log_2 n \rceil$ such that, if $n > 1$, writing $n = 2^i + j$ with $i = \lfloor \log_2(n-1) \rfloor$ (so $1 \leq j \leq 2^i$): the left child subtree of r is an MMR with 2^i leaves, and the right child subtree of r is an MMR with j leaves. Each internal node holds the hash $H(\text{left} \parallel \text{right})$ of its children’s values, for a collision-resistant H .

The left subtree is therefore the largest *perfect* binary tree the leaf count admits, and the recursion repeats on the remainder: an MMR over $n = 6$ leaves splits $6 = 4 + 2$ into a perfect four-leaf subtree and a perfect two-leaf subtree; an MMR over $n = 7$ splits $7 = 4 + 3$ and the right side splits again as $3 = 2 + 1$. Every MMR is a balanced binary tree in the paper’s sense (depth at most $\lceil \log_2 n \rceil$), so ordinary Merkle inclusion proofs of at most $\log_2(n) + 1$ hashes exist for every leaf (Definitions 9–10, p. 27).

Remark 3.2 (Peaks and bagging, absent by construction). The folklore MMR presentation — Peter Todd’s, which the paper credits for the structure — keeps a *forest* of perfect trees (“peaks”), one per set bit of n , and combines (“bags”) them into a single root only when one is needed. Neither word appears anywhere in the paper: its Definition 11 is single-rooted from the start. The two presentations coincide exactly — the peaks of the classical forest are the left children of the nodes along the paper tree’s right spine together with the terminal right-spine node, and the paper’s root is the result of bagging the peaks right-to-left — but the difference in framing matters when reading ZIP-221, which specifies the same object in yet another way (§5), and when reading `zcash_history`, which stores the peaks explicitly (§7). Costs and proofs are identical between the folklore forest and the paper’s single-rooted form. ZIP-221 instead bags left-fold-leftmost (§5.2), the opposite of the paper’s implicit right fold, so with three or more peaks its bagged root and the hashes at the top of a root-anchored path differ from the paper’s; the stored mountain array and the asymptotic costs are the same.

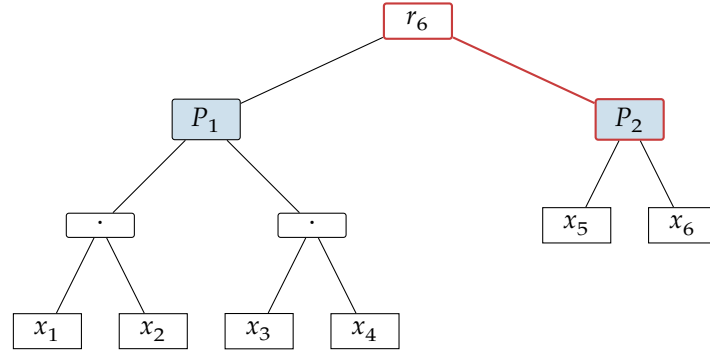


Figure 1: The MMR over six leaves $x_1, \dots, x_6$, drawn in the paper’s single-rooted form (Definition 3.1): $6 = 4 + 2$, so the root r_6 has the perfect four-leaf tree rooted at P_1 as left child and the perfect two-leaf tree rooted at P_2 as right child. The nodes P_1 and P_2 (blue fill) are exactly the *peaks* of the classical forest presentation — here the root’s left child P_1 and the terminal right-spine node P_2 (red edges) — and appending leaf x_7 rewrites only the spine: P_1 and P_2 and every node beneath them are shared unchanged with r_7 .

3.2 Appending in logarithmic time

Appending leaf x to an n -leaf MMR (the paper’s Algorithm 5, AppendLeaf, p. 29) has two cases. If the tree is perfect ($n = 2^i$), a new root is minted with the old tree as left child and x as right child, hashing $H(r \parallel x)$. Otherwise the append recurses into the right subtree and rehashes on the way out. Only the right spine is ever rewritten, so the cost is $O(\log n)$ and — decisive for a blockchain host — every perfect left subtree of the old MMR is shared, untouched, with the new one. Theorem 6 of the paper proves by induction that the result is the MMR over the old leaves with x appended rightmost.

3.3 The prefix-extension trick

An ordinary Merkle proof shows that a leaf lies under a root. FlyClient needs strictly more: when the verifier samples block k from a chain whose head commits to n blocks, it must check not only that block k is leaf k under the head’s root but that the MMR root stored *inside* block k ’s own header commits to exactly the first $k - 1$ of those same leaves — that the sampled block’s view of history is a prefix of the head’s. The MMR delivers this without any additional proof data.

Theorem 7 (ePrint 2019/226, p. 30). *For $k \leq n$, given $\Pi_{x_k \in M_n}$, the Merkle proof that leaf x_k is in the n -leaf MMR M_n , a verifier can regenerate r_k , the root of the k -leaf MMR M_k .*

The proof is an induction on the shape of Definition 3.1, and its content fits in one sentence: in this tree shape, the *left* siblings along the inclusion path of leaf k are exactly the roots of the perfect subtrees covering leaves $1, \dots, k - 1$, so folding them together bottom-up — the paper’s Get_Root routine (Algorithm 6) — reconstructs the prefix root from the very path that proved inclusion. One path, two facts: x_k is the k -th leaf under r_n , and the root over the first $k - 1$ leaves is a value the verifier can now compare against the root stored in the sampled header. The two corollaries the protocol leans on follow directly (p. 31): the MMR is position-binding for prefixes (Corollary 3: the path commits $x_1, \dots, x_k$ as the blocks of a chain that is a prefix of the head’s), and, absent a hash collision, any change to a block changes every later root (Corollary 4) — so an adversary that forks the chain can reuse no honest block from after its fork point, since the honest MMR roots inside those blocks would betray the substitution.

Remark 3.3 (Printed defects in Algorithm 6). As printed, Get_Root assigns from an undefined proof index (“ $r = \Pi[i]$ ” where the loop variable is y), ends by comparing $y = r$ and returning a bit although its specification, its call site in Algorithm 2, and Theorem 7 all require it to return

the reconstructed root, and is called with its arguments in a different order than it declares. The intended behaviour is unambiguous from Theorem 7’s proof — initialise with the first left sibling encountered, fold $r \leftarrow H(y \parallel r)$ over the remaining left siblings bottom-up, and return r . No deployed implementation examined below exposes this proof operation. Registered in Remark 3.5.

3.4 Headers as chain commitments

FlyClient’s consensus change is one field. At every height i the block producer appends the hash of B_{i-1} to the running MMR and records the new root M_i in the header of B_i (§2, pp. 5–6): the header of a block commits, through one MMR root, to exactly the sequence of blocks strictly before it. A verifier holding one trusted header therefore holds a commitment to the whole chain, and the per-sample check set is (Algorithm 2 with §2): the inclusion path hashes to the head’s root; the Get_Root fold of the same path equals the root stored in the sampled header; the sampled header’s proof of work is valid.

Remark 3.4 (Indexing drift, and the invariant that survives it). The paper is not consistent about names: §2 calls the root recorded in B_i “ M_i ”, Definition 5 and Lemma 4 call the root inside the head B_n “ M_{n-1} ”, and Algorithms 1–2 speak of a chain of $n + 1$ blocks whose head commits “the first n blocks”. Algorithm 6 says it reconstructs the root over $k - 1$ blocks, while Theorem 7 calls the same result r_k . Every variant expresses the same invariant — *a block’s header commits to all blocks strictly before it* — and this volume uses that sentence, not any one of the indexings. ZIP-221 will shift the convention once more (the root in block i covers leaves through block $i - 1$, but the tree begins at an upgrade activation rather than at genesis; §5).

3.5 The difficulty MMR

Sampling by position is the fixed-difficulty story. Under variable difficulty the verifier must sample by *weight* — the distribution of §4.4 works over relative aggregate difficulty — and must be able to reject invalid difficulty transitions it will never individually inspect (§2.2). Both needs are served by enriching every MMR node with aggregate metadata.

Definition 7 (Difficulty MMR; ePrint 2019/226, p. 18). *A difficulty MMR is a variant of the MMR with identical properties, but such that each node contains additional data. Every node can be written as $\{h, w, t, D_{\text{start}}, D_{\text{next}}, n, \text{data}\}$, where h is a λ -bit hash output, w is an integer total difficulty, t is elapsed mining time, D is the per-block difficulty level, n is the size of the subtree rooted at the node, and data is an arbitrary string. In a leaf, $n = 1$, t is the block timestamp minus its predecessor’s timestamp, and $w = D_{\text{start}}$ is the block’s difficulty level; D_{next} is the next block’s level under the difficulty-adjustment function. Each non-leaf node is $\{H(lc, rc), lc.w + rc.w, lc.t + rc.t, lc.D_{\text{start}}, rc.D_{\text{next}}, lc.n + rc.n, \perp\}$ for children lc, rc .*

Aggregation is thus fixed once per field: weights and times sum, difficulty levels propagate from the boundary children, counts add. The verifier recomputes every parent along a sampled path from its children and checks, per level: that $lc.D_{\text{next}} = rc.D_{\text{start}}$ (adjacent subtrees agree on the difficulty across their boundary); that nodes below the epoch level $\log_2 m$ carry exactly their epoch’s difficulty; that nodes spanning a transition computed D_{next} by the adjustment rule; and that a node’s aggregate (w, t) is *feasible* — some legal sequence of clamped transitions between its boundary targets could have produced them. The paper bounds feasibility explicitly: maximal weight comes from raising the difficulty by the dampening factor τ as fast as allowed and then descending, minimal weight from the reverse; a maximal difficulty decrease needs an epoch span of at least $\tau m / f$ rounds and a maximal difficulty increase at most $m / (f\tau)$. For the simplified scenario $\tau^k D_{\text{start}} = D_{\text{next}}$ over n epochs (the paper’s own worked case, pp. 18–19) the printed upper bound is

$$w \leq D_{\text{start}} \cdot \frac{(\tau + 1) \tau^{(k+n)/2} - \tau^{1+k} - 1}{\tau - 1},$$

and the printed lower bound repeats the exponent $(k + n)/2$ where evaluating its own summation gives $\tau^{-(n-k)/2}$ — a sign typo, verified against the PDF at high resolution, under which the printed bound would be negative; the summation form is authoritative (Remark 3.5).

The paper calls t a timestamp before defining it as an elapsed time, and calls D a target although its weight equations treat larger D as harder and add D into total work. The restatement above follows the equations; using the proof-of-work target T of §4.1 directly would reverse that ordering.

3.6 The paper’s transition-rewriting lemma

The transition checks would be pointless if an adversary could profit from difficulty rules the verifier fails to sample. The paper addresses this with a rewriting argument.

Lemma 4 (ePrint 2019/226, p. 19). *Let A be an adversary in the variable-difficulty backbone model that produces, with non-negligible probability p , a chain C in which k blocks are valid. Then, assuming a collision-resistant hash function, there exists an adversary A' using the same number of oracle queries that respects the retargeting rules and produces, with probability at least $p - \text{negl}(\lambda)$, a chain C' containing the same valid blocks.*

The proof’s mechanism is worth keeping, because it explains why the per-node checks of §3.5 suffice. Consider an invalid adjustment between B_i and B_{i+1} in A ’s chain. No valid inclusion proof for B_i exists, and along any claimed path there is a highest node x' at which the verifier’s recomputation fails; every proof through x' is rejected, so *every leaf under x' ’s parent* is already unsampleable-without-rejection. The rewritten adversary A' replaces the difficulty transitions inside that whole subtree with valid ones consistent with the parent’s aggregate fields — possible precisely because the parent’s own $(w, t, D_{\text{start}}, D_{\text{next}}, n)$ pass the feasibility checks — and the blocks inside need no valid proof of work at all. This is the paper’s reason for claiming that invalid transitions buy nothing: the adversary might as well have used valid transitions and mined more invalid blocks, which is exactly the adversary the sampling analysis already counts.

Remark 3.5 (Errata register for the revision of record). The ePrint revision of 2020-08-20 — the version this volume pins — contains the following printed defects, all verified against the PDF: (i) the sign typo in the difficulty-MMR lower bound (§3.5); (ii) the `Get_Root` pseudocode’s undefined index, vestigial boolean return, and call-signature mismatch (Remark 3.3); (iii) two near-duplicate evaluation paragraphs in §7 carrying conflicting MMR-node overheads (8 versus 16 bytes) — the variable-difficulty evaluation is the 16-byte one; (iv) a difficulty-bomb sentence in §7.2 asserting a proof-size increase where its own mechanism and the adjacent sentence describe a decrease; (v) `AppendLeaf`’s base case hashing $H(x \parallel r)$ in prose against $H(r \parallel x)$ in the algorithm — the algorithm is operative; (vi) Definition 7’s feasibility checks referencing $t_{\text{start}}, t_{\text{end}}$ fields the definition never declares, and drifting between D_{next} and D_{end} ; (vii) Definition 1’s target-recalculation formula uses a symbol T — in both the clamp bounds and $n(T, \Delta)$ — that its own constants list never declares; (viii) heavily overloaded symbols (δ, k, n, λ each carry two or more meanings across sections); (ix) Table 1’s 1,275× improvement entry, which its own SPV and FlyClient sizes contradict ($496,094/416 \approx 1,193\times$; the printed ratio matches a FlyClient size of ~ 389 KB) while the other three entries recompute correctly; and (x) Theorem 2 prints $c \in (0, 1]$, although its logarithms and density are undefined at $c = 1$; the operative domain is $0 < c < 1$; (xi) Definition 7 calls t a timestamp and D a target even though its leaf rule makes t an elapsed time and its weight equations use D as a difficulty level; (xii) the backbone discussion says persistence yields one chain at all honest nodes, whereas it yields a common stable prefix; and (xiii) Assumption 2’s printed clause says a c fraction of a fork’s weight “is honest”, while the surrounding analysis uses c as an upper bound on the adversary’s validly mined work (Remark 4.1); and (xiv) Lemma 1’s proof switches without definition from (μ, L) in its statement to $((1 - \delta), n)$ and never justifies replacing the stopping-time count by a Poisson variable. Under independent Poisson block clocks that count is negative binomial, and the Poisson Chernoff expression printed as the

lemma’s bound is not a valid upper bound for that stopping-time law in general; the valid clock-model bound is derived immediately after the lemma below. The acknowledgements record that CCS reviewers found “Bahack style attacks and problems with the security proof in a previous version” — the variable-difficulty machinery of §3.5 is post-review repair work, which explains some of the visible seams. These discrepancies are therefore part of the boundary of the paper’s stated result, not assumed harmless here.

4 The FlyClient protocol

4.1 The model, and what the adversary may do

The analysis lives in the variable-difficulty Bitcoin backbone model of Garay–Kiayias–Leonardos. Execution proceeds in rounds; each active player makes q sequential random-oracle queries per round; a block B with target T is valid iff $H(B) < T$; broadcast messages arrive next round, with the adversary free to reorder deliveries. Honest participation $n = \{n_r\}$ is assumed (γ, s) -respecting — in any window of fewer than s rounds, honest power varies by at most a factor γ — and targets are recalculated once per epoch of m blocks: with the last m blocks mined in Δ rounds, the raw retarget $T' = T\Delta f/m$ (aiming at m blocks per m/f rounds) is clamped so that the target moves by at most a factor τ per epoch, the *dampening filter* (ePrint 2019/226, Definition 1, §3.1, p. 7; the paper notes Bitcoin operates with $\tau = 4, m = 2016, f = 0.03$). The clamp is not a tuning detail: it is what makes the difficulty-raising attack of §2.2 survivable, and every feasibility check of §3.5 is an image of it. Backbone parameters are chosen so persistence and liveness hold. Persistence gives honest full nodes a common stable prefix; it does not require their current tips to be identical.

The adversary is *rushing* (each round it sees all honest messages before choosing its own), mines with at most a μ fraction of honest power, and wants the light client to accept a chain of at least the honest chain’s cumulative difficulty. The verifier knows only the genesis block. Two structural facts already constrain the attack. A fork can carry no honest block from after its fork point — the MMR roots inside those blocks would not match the fork (Corollary 4, §3.3) — and no adversarial work from before the fork point can be re-used after it, since the MMR forces each block to be created after everything before it. What remains is the assumption the whole analysis is priced in:

Assumption 2 ((c, L) -adversary; ePrint 2019/226, §3.2, p. 8). *There exists no adversary in the variable-difficulty model that respects the target recalculation function from Definition 1 and can, with non-negligible probability, produce a fork that contains more than L blocks such that a c fraction of the difficulty weight in these blocks is honest.*

Remark 4.1 (Reading Assumption 2, and reading c). The printed clause “a c fraction ... is honest” is loose; the meaning used throughout the paper’s own analysis is a cap on the adversary: in any fork of length at least L , at most a c fraction of the fork’s difficulty weight can be validly mined, so a fork matching the honest chain’s weight is at least a $(1 - c)$ fraction fake. In the evaluation, the paper additionally uses c as a bound on adversarial mining power relative to honest mining power; that identification is not part of Assumption 2. Under the evaluation convention, $c = 0.9$ corresponds to $c/(1 + c) \approx 47.3\%$ of total power (§7.1, p. 20). Short forks are deliberately outside the assumption: for small L the variance of mining lets any adversary win occasionally, which is why L trailing blocks are always checked in full. Connecting (c, L) formally to the backbone parameters $(\mu, \gamma, s, \dots)$ is conjectured possible and left open by the paper; the constant-difficulty case is the one bridge the paper attempts to prove:

Lemma 1 (ePrint 2019/226, p. 8). *In the constant-difficulty backbone, let X be the number of blocks mined by any adversary while the honest chain adopts L blocks, the adversary’s rate at most μ times the honest adoption rate. Then for $c > \mu$,*

$$\Pr[X \geq cL] \leq e^{L(c-\mu)} \cdot (c/\mu)^{-cL}.$$

The displayed expression is the Chernoff bound for a Poisson variable of mean μL , but the paper does not establish the domination it needs: its proof calls that Poisson variable an accurate upper-bound model and then changes (μ, L) to undefined variables $((1 - \delta), n)$. Under the usual independent Poisson block-clock model, the experiment stated in the lemma has a different law, and the printed expression is not a valid bound for that law in general. At the upper-bound rates 1 and μ for the honest and adversarial clocks, respectively, the number X of adversarial blocks found before the L th honest block is negative binomial, with moment-generating function

$$\mathbb{E}[e^{tX}] = \left(\frac{1}{1 + \mu - \mu e^t} \right)^L.$$

Markov's inequality, optimised at $e^t = c(1 + \mu)/(\mu(1 + c))$ for $c > \mu$, gives the valid bound

$$\Pr[X \geq cL] \leq \left[\left(\frac{1 + c}{1 + \mu} \right)^{1+c} \left(\frac{\mu}{c} \right)^c \right]^L.$$

Its base is below 1 when $c > \mu$, so it still gives exponential decay and recovers the qualitative constant-difficulty bridge for this explicit clock model when $L = \Theta(\lambda)$. It does not repair the paper's unproved bridge from the full variable-difficulty backbone parameters to Assumption 2.

4.2 What security means here

The object being proved about is the paper's SPV predicate: for a chain C ending in B_n , the predicate holds iff every block carries correct proof of work following the difficulty-adjustment function and each block's hash is contained in its direct successor's header (Definition 2, p. 9). Because persistence guarantees nothing about the last k blocks, proofs for chains differing only in the last k blocks count as proofs for the same chain, the highest-difficulty one representative. The verifier is a light client knowing only genesis, with oracle access to H for checking work; it receives proofs from several provers, *at least one honest* — an assumption that deliberately prices eclipse attacks out of scope — and accepts the predicate for the head of greatest cumulative difficulty. A proof protocol is *secure* if the verifier's accepted head shares a common prefix (up to the last k blocks) with the chain of every honest full node (Definition 3), and *succinct* if every proof any efficient prover can emit has size $O(\text{polylog}(N))$ in the honest chain length N (Definition 4, inherited from the NIPoPoW paper). FlyClient is thus a NIPoPoW in the primitive sense; "NIPoPoW" as a protocol name usually means the superblock construction of §2.3, which FlyClient replaces.

4.3 The interactive game

The interactive protocol (Algorithms 1–2, p. 11) is short enough to give in full. Each prover claims a chain of $n + 1$ blocks and opens with its head, whose header carries the MMR root over the first n blocks. If all provers present the same head, the client is done. Otherwise the client samples k heights per prover from the distribution of §4.4; for each sampled height the prover returns that block's header with its MMR inclusion path, and the client runs the per-sample check set of §3.4: the path folds to the head's root; the Get_Root fold of the same path equals the root stored inside the sampled header; the sampled header's proof of work is valid, at a target the difficulty-MMR metadata declares and constrains (§3.5). Any failure rejects the prover's chain; a chain surviving all samples is accepted. Provers claiming different lengths are handled by the NIPoPoW generic verifier, longest claim first.

Two consequences of the commitment structure do quiet work here. First, transaction verification decouples from synchronisation: once the head is accepted, membership of an old block is one MMR inclusion proof against the head's root, and membership of a transaction is one ordinary SPV Merkle proof inside that block — roughly 1.5 KB on Ethereum parameters against

~15 KB for superbloc NIPoPoW (§1, p. 4). Second, the protocol need not wait for a stable head. Except with its soundness error, an accepted competing head cannot omit the client’s stored stable prefix, and sampling rejects an MMR containing invalid blocks when a sample exposes one. Thus the client may run the protocol against the freshest head and still recover the stable prefix; it stores one recent *stable* block between executions (§4.2, pp. 11–12).

4.4 From uniform sampling to the optimal distribution

The sampling distribution is the information-theoretic heart of the protocol, and the paper builds it in three failed-then-fixed steps worth keeping, because each failure names a constraint the final distribution satisfies.

Uniform sampling fails on short forks. If the fork point a were known, every sample past it would be invalid with probability $1 - c$ and k samples would catch the fork with probability $1 - c^k$. But the verifier does not know a , and a fork near the tip hides from uniform samples: almost all land before the fork, where the adversary’s chain is honestly valid (§5.1, p. 12).

Binary search is interactive. With one honest prover guaranteed, the verifier can binary-search the first height at which two provers disagree — $\log n$ adaptive queries — and then sample k blocks past the fork point. This works but is multi-round and adaptive, so it does not have the one-shot public-coin form that the paper later compiles using Fiat-Shamir (§5.2, p. 13).

Dyadic suffixes get the right shape. The one-shot fix oversamples the tip: for every $j \in [0, \log n)$, sample k blocks uniformly from the last $n/2^j$ blocks, and once the interval is at most $\min(L, k)$ blocks, check all of them.

Lemma 2 (ePrint 2019/226, p. 13). *With $k \log n$ samples, the probability that the verifier fails to sample any invalid block is at most $((1 + c)/2)^k$.*

The proof needs only one of the $\log n$ intervals: whatever the fork point, some dyadic interval of size n' has it in its first half, so more than $n'/2$ of that interval is post-fork and at least $(1 - c) \cdot n'/2$ of it is fake; k uniform samples from that interval alone miss with probability at most $((1 + c)/2)^k$. The analysis discards every other interval’s samples — the paper’s own stated question, “can we achieve a better bound?”, is what the optimal distribution answers.

The optimal distribution. Reordered, the dyadic protocol is q i.i.d. draws from a staircase density; the right question is which density maximises the single-draw probability p of hitting a fake block, against an adversary who places its validly-mined blocks adversarially — then q draws catch with probability $1 - (1 - p)^q$. Model the chain as the continuum $[0, 1]$, genesis at 0, head at 1. Two structural facts pin the answer. A non-increasing density is never uniquely optimal — an adversary shifts fakes toward the lower-probability late interval, so mass should increase toward the tip (Lemma 3, pp. 14–15). Against an increasing density, the optimal adversary forking at a puts all its validly-mined weight at the end of its fork, so the segment $[a, 1 - (1 - a)c]$ is entirely fake and the single-draw catch probability at fork point a is $\int_a^{1+ac-c} f(x) dx$ (normalised). Maximising the minimum over a forces indifference — the same catch probability at every fork point — and differential analysis yields

$$f(x) = \frac{1 - c}{c(1 - x)}, \quad \int_a^{1+ac-c} f(x) dx = \frac{(c - 1) \ln c}{c} \text{ for every } a,$$

an integral independent of a , as required. The density diverges at the tip — $\int_0^1 f = \infty$ — and the repair is the protocol’s signature move: restrict sampling to $[0, 1 - \delta]$ and *always check the last δ*

fraction of the chain in full. The normalised sampling density is

$$g(x) = \frac{1}{(x-1) \ln \delta}, \quad x \in [0, 1 - \delta],$$

(both factors negative, so $g > 0$), and the single-draw catch probability becomes $p = \log_\delta(c)$: choosing $\delta = c^k$ gives $p = 1/k$ exactly, while any fork point past $(c - \delta)/c$ pushes the fake segment into the always-checked suffix and is caught with probability 1.

Theorem 2 (Optimal sampling distribution; ePrint 2019/226, p. 16). *For $0 < c < 1$ and a positive integer k , given that the last $\delta = c^k$ fraction of the chain contains only valid blocks and the adversary can create at most a c fraction of valid blocks after the fork point, the sampling distribution with PDF $g(x) = 1/((x-1) \ln \delta)$ maximises the probability of catching any adversary that optimises the placement of invalid blocks.*

Optimality is an averaging argument: any candidate density must give catch probability at least p^* to each of the k fork points $a = 1 - c^i$ for $i = 0, \dots, k-1$, and the corresponding intervals tile the support, so $1 \geq kp^*$ and no density beats $1/k$. The sample count follows from $p_m = (1 - 1/k)^m \leq 2^{-\lambda}$:

$$m \geq \frac{\lambda}{\log_{1/2}(1 - 1/\log_c(L/n))}, \quad m \approx \lambda \log_c(1/2) \ln(n) = O(\lambda \log_{1/c} n),$$

where the always-checked suffix of $L = \delta n$ blocks fixes $k = \log_c(L/n) > 1$. For $k = 1$, one sample suffices in the continuum model; the displayed logarithmic denominator is then undefined. For non-integer $k > 1$ the same density and catch-probability calculation give a valid sample bound, but the stated tiling proof establishes optimality only for integer k . The suffix length trades against the sample count and is bounded below by the (c, L) -assumption itself; the implementation optimises it numerically (about a hundred blocks at Ethereum parameters, Figure 4). Each sampled block costs a header plus a $\log_2(n)$ -hash Merkle path, giving:

Corollary 2 (ePrint 2019/226, p. 17). *Under the (c, L) -adversary assumption for any constant L , and using a collision-resistant hash function, the FlyClient proof size is $\Theta(\lambda \log(n) \log_{1/c}(n) + L)$.*

Unlike the superblock protocols, whose succinctness degrades to full SPV under adversarial influence, this bound holds against every adversary satisfying the theorem's (c, L) assumption.

4.5 Removing the verifier

The protocol is public-coin — the verifier contributes only randomness — and one-shot. The paper applies Fiat–Shamir in the random-oracle model: sampling randomness is derived by hashing the head of the chain with an ordinary secure hash (the paper suggests SHA-3), and a verifier checks the derivation along with the samples (§6.2, p. 19). Grinding is priced in proof-of-work: a cheating prover obtains fresh randomness only by re-mining the head, and the assumption bounds its total puzzle budget by $c \cdot n$, so a union bound makes the non-interactive protocol secure whenever the interactive soundness satisfies

$$p_m < \frac{2^{-\lambda}}{c \cdot n}.$$

Two deployment properties fall out (§8.1, p. 21). Proofs are *transferable*: anyone can relay a proof without recomputation. For a given head the sampled positions are deterministic; with a canonical proof encoding, one party can therefore compute a proof for the entire network.

4.6 The theorem, and staying synchronised

Theorem 1 (FlyClient; ePrint 2019/226, pp. 10, 20). *Assuming a variable-difficulty backbone protocol such that all adversaries are limited to be (c, L) -adversaries as per Assumption 2, and assuming a collision-resistant hash function H , the FlyClient protocol is a secure NIPoW protocol in the random-oracle model as per Definition 3 with all but negligible probability. The protocol is succinct with proof size $O(L + \lambda \log_{1/c}(n) \log_2(n))$.*

The paper’s proof sketch composes five steps: Assumption 2 speaks only of retargeting-respecting adversaries, and Lemma 4 (§3.6) claims to convert every other adversary into one with only negligible loss in success probability; Merkle soundness plus the prefix-extension corollaries make the MMR position- and weight-binding, so sampled answers are intended to be pinned to one chain; Theorem 2 with Corollary 2’s parameters makes evasion of sampling negligible; the paper’s random-oracle grinding argument replaces the verifier’s public coins by a hash of the head; and the transcript — L suffix blocks plus $O(\lambda \log_{1/c} n)$ sampled blocks, each with a $\log_2(n)$ -hash path — has the stated size. Remark 3.5 records why this volume treats the theorem as the paper’s conditional claim rather than as an independently completed proof.

A synchronised client is also intended to stay synchronised cheaply. The paper’s Theorem 3 (Subchain proofs, §8.2, p. 22) states that a client that accepted a chain of length n , later handed a proof for the extension from n to n' together with one MMR proof that its block n lies in block n' ’s commitment, accepts nothing it would not have accepted from a full n' -proof — a fork after n faces a fresh FlyClient instance with block n as genesis, and a fork before n fails the prefix proof outright. In practice no custom subproof is needed: the full proof is transferable, so a returning client verifies only the part past its checkpoint, and servers can precompute checkpoint proofs at chosen heights.

4.7 What the evaluation showed

The implementation created and verified every tested proof in under a second. At Ethereum’s 2019 parameters ($c = 1/2$, $\lambda = 50$, n up to seven million), the numerically-optimised trailing suffix stays near a hundred blocks, the sample count near 400–470, and the whole proof under 530 KB — against 3.4 GB of SPV headers, the 6,627× of Table 2 — with the paper’s “MMR proof optimization” (unexplained in the text, but plausibly deduplication of shared path nodes across samples) accounting for roughly a third of the size (§7, Figures 4, 6). One curiosity in the Ethereum data is instructive: proof size *falls* between three and four million blocks because the difficulty bomb concentrated so much weight near the tip that the always-checked suffix covered a larger weight fraction, cutting the sample count — difficulty growth helps the sampler. Two caveats frame all the numbers.

Remark 4.2 (Ethereum is outside the theorem). Ethereum’s moving-average difficulty rule does not fit the variable-difficulty backbone model, so Theorem 1 does not cover it; the paper is explicit that running FlyClient on Ethereum is possible “only with heuristic security guarantees” (§3.1, pp. 7–8; §7, p. 20). The point recurs for Zcash in §8: whether Zcash’s own averaging-window retarget (distinct from both Bitcoin’s and Ethereum’s) satisfies the model is exactly the kind of question a Zcash FlyClient ZIP would have to answer, and none exists.

The second caveat is historical and recorded in the paper’s own acknowledgements: CCS reviewers found “Bahack style attacks and problems with the security proof in a previous version” — so the variable-difficulty analysis is repair work, added after review. Finally, the paper closes the circle to proofs of sequential work: the chain construction is essentially Cohen–Pietrzak PoSW with block hashes as leaves, and their verifier is FlyClient with a uniform distribution — a FlyClient chain *is* a PoSW, with the stronger guarantee that no point of the chain has a constant fraction of corrupted leaves (§8.4, pp. 23–24).

5 ZIP-221: the deployed chain-history tree

Zcash deployed FlyClient’s commitment substrate as consensus. ZIP-221 — title “FlyClient — Consensus-Layer Changes”, Status Final, deployed with the Heartwood upgrade — replaces the block header’s Sapling treestate commitment with a commitment to the root of a Merkle mountain range over the chain’s history, each node of which carries the metadata a difficulty-sampling verifier needs. The header field’s lineage tells the story compactly: `hashReserved` before Sapling, `hashFinalSaplingRoot` from Sapling, renamed `hashLightClientRoot` by ZIP-221 at Heartwood, renamed again `hashBlockCommitments` by ZIP-244 at NU5 (§6). The stated motivation is the paper’s protocol verbatim: MMR proofs are to let light clients verify “the validity of a block chain received from a full node”, block inclusion, and “certain metadata of any block or range of blocks in that chain” (ZIP 221, §“Motivation”), with header downloads dropping from linear to logarithmic.

5.1 Epoch scoping: one tree per network upgrade

The single largest structural departure from the paper is what the tree covers. The paper’s MMR runs from genesis; ZIP-221’s does not:

“The protocol requires that an MMR that commits to the inclusion of all blocks since the preceding network upgrade ($B_x, \dots, B_{n-1}$) is formed for each block B_n . The root M_n of the MMR MUST be included in the header of B_n .” (ZIP 221, §“Motivation”; x is the activation height of the preceding network upgrade.)

Three consequences follow from the exact definition (§“Tree nodes and hashing”). First, the commitment is delayed one block: the root inside B_n covers leaves through B_{n-1} only, so a block never commits to itself — the ZIP’s rationale notes the corollary that a block’s final Sapling root now surfaces one block later than it did pre-Heartwood, and accepts it because anchoring on the most recent treestate “is already unsafe in reorgs”. Second, the tree *resets at every network upgrade*: the preceding-upgrade height x is the last activation height strictly below n , so each later upgrade’s activation block A carries the completed previous epoch’s root, and the very next block $A + 1$ already commits to the new epoch’s single-leaf tree over B_A . Every consensus branch thus owns a self-contained tree — which composes with the personalisation scheme below into cross-branch domain separation — and any future Zcash FlyClient must chain per-epoch proofs across upgrade boundaries, a composition the ZIP does not discuss. Third, the genesis and activation edge cases are handled by sentinel: `hashChainHistoryRoot` is undefined at genesis, and in the block that activates the ZIP the header field is all zero bytes, which “MUST NOT be interpreted as a root hash”.

5.2 Numbering, peaks, and bagging

Where the paper specifies its MMR as a single-rooted recursion (Definition 3.1) and never says “peak”, ZIP-221 specifies the same object in the classical presentation the paper omitted (Remark 3.2), adapted from Grin’s: nodes are numbered in *creation order* — each leaf followed immediately by every internal node its arrival completes — and the forest of perfect subtrees (“mountains”, leftmost peak always the highest) is *bagged* into a single root by repeatedly joining the two *leftmost* peaks: a left fold, so peaks P_1, P_2, P_3 bag as $(P_1 \parallel P_2) \parallel P_3$. Navigation is arithmetic on positions: a node at position p with altitude h has its right sibling at $p + 2^{h+1} - 1$ and its left child at $p - 2^h$, with the ZIP’s own caveat that bagging nodes fall outside these rules. The ZIP’s eleven-leaf worked example — 19 nodes at positions 0–18, peaks at positions 14, 17, 18 with altitudes 3, 1, 0 — and both navigation identities reproduce exactly under this volume’s independent implementation (script-verified 2026-07-19). Reorgs run the construction backwards: deleting the rightmost leaf pops the right spine and re-bags, $O(\log k)$ in the right subtree’s leaf count.

5.3 Node metadata: the complete field list

Every node — leaf, internal, root, treated uniformly — carries the same field vector, serialised in the exact order of Table 3. A leaf takes both members of each earliest/latest pair from its own block; an internal node inherits *earliest*-fields from its left child, *latest*-fields from its right child, and sums the additive fields. The ZIP splits the vector by purpose: seven fields — the timesamp, target, and height pairs, plus the cumulative work — are the “FlyClient Requirements”, chosen “via discussion with an author of the FlyClient paper” to let a light client reason about the difficulty-adjustment algorithm over a block range, exactly the role Definition 7 of the paper assigns its $(w, t, D_{\text{start}}, D_{\text{next}}, n)$ payload (§3.5); the remaining fields are Zcash-specific light client aids with no counterpart in the paper. Among the latter, the shielded transaction counts are the interesting design: they let a client reliably learn whether a malicious server is withholding shielded transactions, and let it skip header ranges containing none — the ZIP notes the dominant light-client cost is transmitting Equihash solutions in headers. An earlier draft counted Sprout and Sapling together; it was rejected because a Sapling-only client could not distinguish a withheld Sapling transaction from an unrequested Sprout one, and the ZIP fixes the pattern that every future pool gets its own count — which NU5’s Orchard trio and the draft Ironwood trio then follow (§6).

#	Field	Leaf value	Internal rule
1	hashSubtreeCommitment	the block’s consensus block hash, internal byte order, <i>unpersonalised</i>	personalised BLAKE2b-256 of the two children’s full serialisations
2	nEarliestTimestamp	header nTime	inherit left
3	nLatestTimestamp	header nTime	inherit right
4	nEarliestTargetBits	header nBits	inherit left
5	nLatestTargetBits	header nBits	inherit right
6	hashEarliestSaplingRoot	final Sapling treestate root	inherit left
7	hashLatestSaplingRoot	final Sapling treestate root	inherit right
8	nSubTreeTotalWork	$\lfloor 2^{256} / (\text{ToTarget}(\text{nBits}) + 1) \rfloor$	sum (modulo 2^{256})
9	nEarliestHeight	block height	inherit left
10	nLatestHeight	block height	inherit right
11	nSaplingTxCount	count of transactions with non-empty vSpendsSapling or vOutputsSapling	sum
12	hashEarliestOrchardRoot	final Orchard treestate root	inherit left
13	hashLatestOrchardRoot	final Orchard treestate root	inherit right
14	nOrchardTxCount	count of transactions with non-empty vActionsOrchard	sum
15	hashEarliestIronwoodRoot	final Ironwood treestate root	inherit left
16	hashLatestIronwoodRoot	final Ironwood treestate root	inherit right
17	nIronwoodTxCount	count of transactions with non-empty vActionsIronwood	sum

Table 3: The ZIP-221 node field vector, in serialisation order. Fields 1–11 are the V1 vector (Heartwood onward); fields 12–14 join at NU5 and are omitted before it; fields 15–17 join at NU6.3 and are omitted before it. Hash fields are char [32], timestamps and target bits uint32, work uint256, heights and counts CompactSize integers. The seven FlyClient-requirement fields are 2–5 and 8–10. Sources: ZIP 221, §“Tree Node specification”; ZIP 252; ZIP 258.

Serialised size follows from the widths: the V1 fixed portion is $32 + 4 + 4 + 4 + 4 + 32 + 32 + 32 = 144$ bytes plus three CompactSize integers of 1–9 bytes each, giving the ZIP’s stated 147–171 bytes; NU5 adds 64 fixed bytes and one more CompactSize, giving 212–244 (script-verified against both stated ranges). Two of the ZIP’s own field notes are worth carrying: nLatestTimestamp may be *smaller* than nEarliestTimestamp in subtrees narrower than PoWMedianBlockSpan, because

header timestamps are not consensus-monotonic at that scale; and within one branch, the leaf hash of B_{n-1} equals `hashPrevBlock` of the following header exactly, so the tree adds no new hashing at the leaves.

5.4 Hashing and personalisation

All internal hashing is BLAKE2b-256 with the 16-byte personalisation `ZcashHistory || CONSENSUS_BRANCH_ID` — the ASCII prefix’s twelve bytes followed by the branch ID’s four little-endian bytes, so for Heartwood’s branch `0xF5B9230B` the trailing bytes are `0B 23 B9 F5`. Baking the branch ID into the personalisation means “valid nodes from one consensus branch cannot be used to make false statements about parallel consensus branches” (ZIP 221, §“Rationale”) — composed with epoch scoping, each upgrade’s tree is domain-separated from every other’s. Two boundaries of the scheme are exact: a *leaf*’s hash field is the block’s consensus block hash itself, unpersonalised; and an *internal* node hashes the concatenation of its children’s full canonical serialisations — metadata and all, not merely their digests — so every aggregate field is hash-bound at every level. The header commitment is one more hash of the same kind: `hashChainHistoryRoot` is the personalised BLAKE2b-256 digest of the *serialised root node*, so the 32-byte header field commits to the whole epoch’s aggregates — total work, boundary timestamps and targets, boundary treestate roots, shielded transaction counts — not only to the leaf set.

Remark 5.1 (Which branch ID, at an activation block). The ZIP’s prose sets the personalisation branch ID to that of “the epoch of the block containing the commitment”, but its pseudocode hashes with the branch ID carried by the *nodes* (`make_parent` uses the left child’s, `make_root_commitment` the root’s). Within one epoch-scoped tree the two readings coincide everywhere except at an activation block, whose header carries the *previous* epoch’s finished root: there the pseudocode — and the ZIP’s own summary that “each node in the tree commits to the consensus branch that produced it” — resolve the tension in favour of the tree’s own branch ID, and the deployed implementations agree (§7).

5.5 The consensus rules at Heartwood

The consensus delta is three rules on one renamed field (ZIP 221, §“Block header semantics and consensus rules”): before activation, the Sapling rule persists under the new name (`hashLightClientRoot` is the final Sapling treestate root); in the activation block, the field **MUST** be all zero bytes, not interpreted as a root; in every subsequent block, the field **MUST** equal `hashChainHistoryRoot`. The header’s byte format and version are untouched — a deliberate boundary, argued in the rationale: ZIP-301 obliges miners to ignore jobs with unknown header versions, so a version bump risks mining-hardware breakage for a change that is semantically invisible to miners. The rationale also disposes of two alternatives: overloading is safe against a miner contriving a Sapling treestate whose root collides with an MMR root (a preimage over 32 layers of Pedersen hashing), and the seemingly cleaner home for the root — replacing `hashPrevBlock`, exactly the paper’s suggestion (§3.4) — was rejected because it would require “massive refactoring” and hurt performance, reorgs in particular being “fragile, performance-critical” and reliant on backwards iteration over the chain history.

5.6 Deltas from the paper’s construction

For a reader holding both objects, the differences reduce to six that matter and a summary judgement. The tree is epoch-scoped, not genesis-rooted, with a one-block commitment delay (§5.1). Node metadata is richer than Definition 7 of the paper: the seven `FlyClient` fields realise the paper’s $(w, t, D_{\text{start}}, D_{\text{next}}, n)$ payload — with both ends of each pair carried explicitly rather than one derived — and the pool-specific note-commitment roots and transaction counts have no paper

counterpart. Hashing is personalised, branch-separated BLAKE2b-256 over full child serialisations, where the paper hashes bare child values with an unspecified generic H . The root reaches the header through an existing repurposed field rather than a new one — and, from NU5, only as the first input of an outer commitment (§6). The bagging order is fixed left-fold-leftmost (§5.2) — the opposite of the right fold the paper’s recursion implies (Remark 3.2), so the two constructions’ bagged roots differ whenever the tree has three or more peaks. The ZIP’s own summary is accurate: “Other than the metadata commitments, the MMR tree’s construction is standard”, and its append algorithm is “adapted from” the paper’s.

Remark 5.2 (Defect ledger for a Final ZIP). The ZIP’s illustrative pseudocode — explicitly non-normative, but the only executable statement of intent the ZIP contains — has at least six defects in the checked-out text: the node class declares `nSaplingTxCount` twice where the second must read `nOrchardTxCount`; `serialize` and `get_peaks` reference instance fields without `self`./node. qualification; `make_parent` reads `sapling_root/orchard_root` attributes that exist on the block type, not the node type; `append` and `delete` read `latest_height/earliest_height` where the node fields are named `nLatestHeight/nEarliestHeight`; and a stray Rust-ism (`peaks.push`) survives in the Python. None affects the normative field table or consensus rules; all are trivially repaired by the reference implementations (§7). Separately, no ZIP-221 test vectors exist anywhere in the zips repository.

Remark 5.3 (The ZIP’s own security caveat). ZIP-221 is unusually candid about the limits of its security story, and the caveat is load-bearing for §8: the FlyClient paper “only analyses these security properties in chain models with slowly adjusting difficulty, such as Bitcoin”, leaves rapidly-adjusting chains — “such as Zcash or Ethereum” — as an open problem with “only heuristic security guarantees”, and although the difficulty-reasoning fields were chosen with a paper author, “the use of these fields has not been analysed in the academic security literature”. The ZIP concludes that “a more in-depth security analysis of FlyClient should be performed before designing a FlyClient-based light client ecosystem for Zcash”. This is the same boundary Remark 4.2 drew from the paper’s side: Zcash’s averaging-window retarget stands outside the proven model, and nobody has closed the gap in either direction since.

6 NU5: the commitment becomes one of two

6.1 Why an authorising-data root exists

NU5’s ZIP-244 made transaction identifiers non-malleable by excluding authorising data from the txid digest. The change creates a gap: `hashMerkleRoot`, now built over non-malleable txids, no longer commits to the whole serialised transaction. ZIP-244 fills it with a parallel commitment. For version 5, each transaction’s *authorising data commitment* digests the excluded material (transparent input scripts; Sapling proofs and signatures; Orchard proofs, spend-authorisation signatures, and binding signature) under fixed BLAKE2b-256 personalisations, gathered by an outer per-branch digest. ZIP-229 extends the version 6 auth digest with an Ironwood component and moves the Sapling, Orchard, and Ironwood anchors from effecting data into their corresponding authorising digests. Pre-v5 transactions are assigned a placeholder of 32 0xFF bytes. The `hashAuthDataRoot` is the root of a padded binary Merkle tree (personalisation `ZcashAuthDatHash`, null leaves `[0u8; 32]`) over those commitments, in insertion order identical to the txid tree, so one path position names the same transaction in both trees. The pair (txid tree, auth tree) restores commitment to the transaction effects and their authorising data.

6.2 hashBlockCommitments

Rather than widen the header, ZIP-244 overloads the ZIP-221 field a second time. From NU5 the field is named `hashBlockCommitments` and holds the personalised digest

$$\text{BLAKE2b-256}_{\text{zcashBlockCommit}}(\text{hashLightClientRoot} \parallel \text{hashAuthDataRoot} \parallel [\text{0u8}; 32]),$$

whose first input is the ZIP-221 chain-history root, second the authorising-data root, and third a 32-zero-byte terminator. The chain-history root therefore no longer appears directly in any NU5 header; a client checking it must carry the auth-data root (or its subtree) alongside any MMR proof. The terminator is deliberate architecture: the ZIP describes the value as “a linked list of hashes terminated by arbitrary data”, so a future upgrade can replace the zeros with the hash of a further structure ending in its own terminator; full validators must always use the entire structure of the latest *activated* protocol version they support, and servers for older light clients **SHOULD** supply the replacement value so old verification equations keep closing. Two activation details differ from Heartwood’s: there is *no* zero-byte sentinel at NU5 — the activation block already uses the new derivation, its inner chain-history input being the completed Canopy-epoch root — and the same upgrade extends the node vector with the Orchard trio (Table 3, fields 12–14), following the one-count-per-pool pattern. The draft ZIP-258 continues both patterns for NU6.3: an Ironwood trio (fields 15–17, aggregated identically to the Orchard trio, node size 277–317 bytes — the arithmetic checks against the field widths), with `hashChainHistoryRoot` changing “only through the added node fields” and `hashBlockCommitments` otherwise unaffected. That trio is the one this series’ Ironwood volume already cites at its turnstile (*Ironwood Guide*, §“The turnstile and migration”). ZIP-258 remains Draft as a document, but these rules and fields are active consensus and are implemented by Zebra; the ZIP proposes that `zcashd` will not implement NU6.3.

6.3 Activation constants

Upgrade	Branch ID	Mainnet	Testnet	Node vector
Heartwood	0xF5B9230B	903,000	903,800	V1 (fields 1–11)
NU5	0xC2D6D0B4	1,687,104	1,842,420	V2 (+ Orchard trio)
NU6.3	0x37A5165B	3,428,143	4,134,000	V3 (+ Ironwood trio)
NU7 (Draft)	0x77190AD9	TBD	TBD	V3 (unchanged)

Table 4: Chain-history activation constants. Sources: ZIP 250, ZIP 252, ZIP 258 (Draft), and ZIP 259 (Draft). Each row introduces a node format or the target NU7 epoch.

The NU7 row describes a proposal, not an activated branch. ZIP 259 assigns the new branch ID but leaves activation heights TBD. It introduces neither a transaction format nor history-node fields: the Ironwood V3 metadata remains sufficient. At activation, the existing epoch-reset rule of §5.1 still applies even though the node format is unchanged. The proposed 25-second spacing and larger difficulty-averaging window of ZIP 218 change the chain environment, not the sampling theorem’s assumptions: that theorem does not become a proof of security for Zcash’s retarget rule merely because blocks arrive faster.

7 Deployed implementations

Three codebases realise ZIP-221: the `zcash_history` crate (“to be used in zebra and via FFI bindings in `zcashd`”, its own words), Zebra’s wrapper and state service, and `zcashd`’s C++ cache with a Rust FFI bridge. Zebra pins version 0.5.0 of the crate; `zcashd` pins 0.4.0 — a skew with consequences registered in §7.4.

7.1 The `zcash_history` crate

The crate stores the MMR in its *array representation*: leaves and completed internal nodes in append order, addressed by u32 index. Two link kinds separate what persists from what does not: `Stored(u32)` indexes the persistent array; `Generated(u32)` indexes an ephemeral per-instance vector holding the *bagging* nodes that join unequal peaks into a root — and serialisation refuses entries with `Generated` children, so exactly ZIP-221’s node array persists, never the bag. An entry serialises as a one-byte tag (node or leaf), two little-endian u32 child indices for nodes, then the node data; the node data is ZIP-221’s field vector in ZIP order, `CompactSize` integers included, with maxima asserted in tests at exactly the ZIP’s figures — 171 bytes (V1), 244 (V2), 317 (V3, Ironwood), entry maximum $317 + 9 = 326$. One deliberate absence: the consensus branch ID is *context, not data* — it is never serialised, and every deserialiser must supply it.

Hashing matches §5.4 exactly: the personalisation is `ZcashHistory` $\parallel$ `LE32`(branch ID); a parent’s commitment hashes the concatenation of both children’s *full serialisations*; and the hash of the root node’s serialisation is `hashChainHistoryRoot`. A dedicated test (`v3_combine_hash_commits_to_ironwood_fields`) pins the property that parent hashes commit to child metadata, and a conformance module walks all three node versions against the `zcash-test-vectors` ZIP-221 vectors, generated by an independent Python reimplementation.

The `Tree` type is explicitly a *view*, not a container: it is instantiated per operation from a caller-supplied peak set — `Tree::new(length, peaks, extra)`, which inserts the peaks and left-folds them into generated bagging nodes — used for a few appends or truncations, and dropped. Appending pushes the leaf, collects peaks by descending from the root and stopping at complete subtrees, merges equal-sized peaks right-to-left into *stored* nodes, and re-bags the remainder left-to-right into *generated* nodes; truncation runs the inverse, consuming the caller-supplied extra nodes down the right spine. Completeness itself is derived from metadata, not shape bookkeeping: a subtree is complete iff `end_height - (start_height - 1)` is a power of two — an elegant economy with a sharp edge registered below. Notably, the crate never computes which array positions are peaks; every caller reimplements the walk — the crate’s own example 1-based, *Zebra* and *zcashd* in a matching 0-based form (highest altitude $\lfloor \log_2(n + 1) \rfloor - 1$, first candidate $2^{\text{alt}+1} - 2$, descend left while past the end, then hop right siblings) — and the same fourteen-leaf, twenty-five-node ASCII diagram annotates the walk in both *Zebra* and *zcashd*, *Zebra*’s comment attributing the code to the crate example and the explanation to *zcashd*.

7.2 Zebra

Zebra’s wrapper builds leaves from blocks: branch ID from the block’s network upgrade, subtree commitment the block hash, both ends of each pair the block’s own time, target, treestate roots and height, work from the block’s difficulty threshold, and the pool transaction counts — with a named `BlockCommitmentTreeRoots` struct whose stated purpose is that the Orchard and Ironwood roots, which share a Rust type, “cannot be silently swapped, which would corrupt the ZIP-221 chain-history commitment”. Above it, `NonEmptyHistoryTree` dispatches the node version by upgrade — Heartwood/Canopy to V1, NU5 through NU6.2 to V2, NU6.3/NU7 to V3, panic before Heartwood — and implements the epoch reset with one line: on a push whose network upgrade differs from the tree’s, “This is the activation block of a network upgrade. Create a new tree.” — the value is wholesale replaced by a fresh single-leaf tree. Every push then prunes: the peak-set walk retains only peak entries, and the in-memory crate tree is rebuilt from peaks alone, neutralising the view type’s unbounded growth. The whole object is wrapped once more as `HistoryTree(Option<NonEmptyHistoryTree>)`, empty before Heartwood — the crate itself cannot represent an empty tree (its constructor panics on an empty peak set), so emptiness lives a level up. A guard worth quoting sits on push: heights must be consecutive because “*librustzcash* assumes the heights are correct and corrupts the tree if they are wrong” — the completeness-from-heights economy, fenced.

Validation splits across two layers. Structurally, `Commitment::from_bytes` parses the header field by epoch into the five-armed enum — pre-Sapling reserved, final Sapling root, the all-zero `ChainHistoryActivationReserved` at the Heartwood activation height, `ChainHistoryRoot` through Canopy, `ChainHistoryBlockTxAuthCommitment` from NU5 — rejecting a non-zero activation sentinel outright. Contextually, the state service compares: for Heartwood/Canopy blocks, the parent chain’s tree hash against the header root; for NU5 onward, it recomputes

```
BLAKE2b-256ZcashBlockCommit(history root || auth_data_root || [0u8;32])
```

from the parent tree and the block’s own authorising-data tree and compares that. The check runs in the non-finalized state (in parallel with anchor checks and the chain push) and again for checkpointed blocks in the finalized state; zebra-consensus itself never touches the field, a division of labour the code documents — the checkpoint verifier “doesn’t bind the transaction authorizing data”, the state services do. Persistence is minimal by design: one RocksDB column family holding exactly one record — the current tip tree’s peaks — under the unit key `()`, atomically replaced per block write; a previous epoch’s tree exists nowhere in Zebra once its activation is finalized. Reorgs never truncate: the non-finalized state keeps an Arc-shared `HistoryTree` snapshot per height per chain fork, so rolling back is dropping snapshots, and the crate’s truncation machinery goes unused. The mining path closes the loop: `getblocktemplate` responses carry `chain_history_root`, “The FlyClient chain history root as of the end of the chain tip block” — producers, too, consume the tree.

7.3 zcashd

The `zcashd` node keeps the tree behind an FFI whose Rust side is thin (`src/rust/src/history.rs`): fixed-width byte arrays (244-byte nodes, 253-byte entries — version 0.4.0’s V2 maxima), three entry points (`append`, `remove`, `hash_node`), and a branch-ID dispatch that names Sprout, Overwinter, Sapling, Heartwood, and Canopy for V1 — omitting Blossom, which falls through the *everything else* arm to V2 (harmless: history trees exist only from Heartwood). The C++ side holds one `HistoryCache` *per epoch*, keyed by consensus branch ID; `PushHistoryNode` preloads the peak set with the same walk as everywhere else, appends through the FFI (at most 32 new nodes per append — the leaf plus at most 31 ancestors, the 2^{32} tree bound made concrete), and `PopHistoryNode` handles reorgs case by case, including the pleasing impossibility proof in an exception: “a history tree cannot have two nodes” — one leaf is length 1, two leaves are length 3. Validation happens in `ConnectBlock` (`src/main.cpp:3505--3614`): the chain-history root is read from the *previous block’s* epoch’s cache — which at an activation block is the previous epoch’s final root, exactly ZIP-221’s rule — and the header field is checked against the Heartwood sentinel, the bare root (pre-NU5), or `DeriveBlockCommitmentsHash`, byte-identical to Zebra’s recomputation. Persistence is the opposite pole from Zebra’s: `LevelDB` stores every epoch’s *full node array* indefinitely — per-node records keyed by (epoch, index), plus a length and root per epoch — because `zcashd`’s rollback model reopens old epochs’ trees, and its RPC exposes `chainhistoryroot` on `getblock`. A width workaround mirrors Zebra’s: V1 nodes written by pre-NU5 clients occupy 171-byte records that NU5-aware clients read and zero-pad into 244-byte buffers.

7.4 Divergence register

No ZIP-221 commitment divergence is known from the period when Zebra and `zcashd` both ran. The register below records where the two implementations differ in structure, and one place they would have differed in consensus had both crossed the NU6.3 boundary unchanged.

- *Version skew at NU6.3.* `zcashd`’s 0.4.0 dispatch sends every post-Canopy branch ID to V2 nodes, while Zebra selects V3 (Ironwood fields) for NU6.3 — as the code stands, the two would

compute divergent `hashChainHistoryRoot` values from the activation block onward. ZIP-258 resolves the tension by deployment scope rather than by changing `zcashd`: the ZIP says `zcashd` “will not implement” NU6.3, while Zebra implements V3 (§6). The skew is a documented retirement boundary, not a live consensus disagreement among the current implementations — but the `zcashd` code alone does not say so.

- *Epoch retention.* Zebra destroys a superseded epoch’s tree (tip-only, single-record persistence); `zcashd` retained every epoch’s full array, because its rollback path could reopen them. Two correct answers to the same question, shaped by each node’s reorg model.
- *Truncation asymmetry.* The crate’s `truncate_leaf` and extra-node machinery are exercised only by `zcashd`; Zebra’s per-height snapshots make deletion a non-event. A bug in truncation would be a `zcashd`-only bug.
- *A skippable check.* `zcashd` verifies the NU5 recomputation only under the `fCheckAuthDataRoot` flag; the pre-NU5 root equality has no such flag. Zebra checks unconditionally on both paths.
- *Fragile economy, fenced differently.* Subtree completeness derived from height metadata means wrong heights corrupt tree shape silently; Zebra fences with a consecutive-height assertion citing exactly this failure, `zcashd` relies on `ConnectBlock` ordering.
- *Edge-case fallback.* Zebra’s NU5 check substitutes the 32-zero-byte sentinel as the “previous root” if the NU5-style commitment appears at the Heartwood activation height itself — reachable only on `Regtest` or configured testnets; `zcashd` has no equivalent arm.

8 The deployment gap

8.1 Who computes the root, who verifies it

Every valid Zcash block after the Heartwood activation block carries a correct chain-history commitment. Consensus validators compute and check it: Zebra in its state service and, until its retirement, `zcashd` in `ConnectBlock` (§7). Template-consuming miners need not compute it: both nodes’ `getblocktemplate` hand the miner a `defaultroots` object containing `chainhistoryroot` (and, from NU5, `authdataroot` and `blockcommitmentshash`), so template consumers hash what the node computed — the state-service response type behind the template documents the field as “The FlyClient chain history root as of the end of the chain tip block”, and Zebra’s built-in miner rides the same RPC. Even this producer-facing surface has had consumer-visible bugs: a `zcashd` comment memorialises that v4.6.0’s legacy template fields “were accidentally set to always be the NU5 value” (`src/miner.cpp:462--466`).

Verification, meanwhile, is purely *reconstructive*: a validator maintains the tree itself and compares 32-byte roots. None of the checked deployed components verifies an MMR *proof* — the operation the structure exists to serve. The RPC surface tells the same story from the read side: `zcashd`’s `getblock` exposes `chainhistoryroot` per block, while Zebra’s `getblock` omits the root for queried blocks. Its code carries the literal comment “`chainhistoryroot` would be here. Undocumented. TODO: decide if we want to support it” (`zebra-rpc, src/methods.rs:3971`), and no checked RPC in either node serves MMR nodes or assembled proofs.

8.2 The light-client stack consumes none of it

The deployed light-client protocol was specified before ZIP-221 and never updated toward it: ZIP-307 does not reference ZIP-221 once, and its security model is the one the *Wallet Guide* documents at length — the node/proxy combination “is trusted to provide a correct view of the current best chain state”. The negative results are uniform across the stack (rechecked 2026-09-05): `lightwalletd` contains no FlyClient, chain-history, or MMR code and serves no related endpoint; Zaino

ships the same gRPC protocol and touches the field only to pass it through — serialising the header’s `hashBlockCommitments` bytes, and carrying a comment on its header type, “Optional fields may be added for: `hashLightClientRoot` (FlyClient proofs)...”, a capability anticipated, not implemented; Neither `zcash_client_backend` nor `zcash_client_sqlite` nor `zcash_primitives` depends on `zcash_history`; the Android SDK has no trace. What the wallet stack does instead is exactly the *Wallet Guide*’s subject: hash-chain continuity (`prevHash` and `height`) and nothing more — ZIP-307’s own text concedes its header-validation section “has been only partially implemented: currently only `prevHash` is checked”, and a grep of the wallet backend for Equihash or difficulty code returns nothing.

The consequence deserves plain statement: a Zcash FlyClient would not be “SPV, but smaller”. The checked deployed wallet stack performs no proof-of-work verification, so a FlyClient verifier would add that missing verification layer — and the seven metadata fields ZIP-221 carries for it, plus the shielded-transaction counts added specifically so light clients could detect withholding (§5.3), have waited six years without a single consumer.

8.3 What an actual Zcash FlyClient still needs

- *Prover data availability.* A proof server needs random access to the full node array. Zebra persists peaks only (§7.2); `zcashd` persists everything a prover needs (§7.3) but exposes none of it and is being retired.
- *Serving endpoints.* No RPC or gRPC in the checked node and light-client implementations serves an MMR node, an inclusion path, or an assembled proof — the entire wire surface remains to be designed and specified.
- *The sampling client.* No implementation of the verifier loop — weight-proportional sampling under g , per-sample checks, difficulty-transition feasibility over ZIP-221’s metadata — exists in any wallet or SDK.
- *Specification work.* Open in full: a proof wire format; the Fiat–Shamir transcript definition for Zcash’s header; the composition rule for chaining per-epoch trees across upgrade boundaries (§5.1 — the ZIP is silent); the security analysis Zcash’s own rapidly-adjusting difficulty rule requires (Remark 5.3); and the marriage of FlyClient header verification with ZIP-307 scanning, which replaces the header-chain trust but not the compact-block detection channel.
- *Practicality data.* The one located published end-to-end datapoint is academic and recent (web-sourced: Perazzo–Capecci, “Catching the Fly”, arXiv 2604.26736, April 2026): what its authors describe as the first practical Zcash FlyClient prover, built on a zebra fork with four added RPCs and a ~10-hour database backfill to recover the node array Zebra discards; proofs of ~1.2 MiB at three million blocks (~320 KiB in their simulated distilled variant, which requires consensus changes), with Equihash’s 1,344-byte solutions dominating per-sampled-header cost. The numbers suggest mobile plausibility and poor suitability for an on-chain verifier — and show that the retrofit cost of Zebra’s peaks-only economy is real but bounded. Their analysis substitutes an assumed adversarial work budget for the paper’s (c, L) bound; it does not derive that budget from Zcash’s retarget rule, so it does not close the original model gap.

Remark 8.1 (An inversion of readiness). The deployment gap has a neat summary: the retired node that has the prover’s data (`zcashd`, full per-epoch node arrays in LevelDB) is outside the current upgrade path, while Zebra persists only peaks. Neither has a proof interface. Meanwhile consensus keeps paying the feature’s maintenance cost — the NU6.3 upgrade extends the node vector a second time (§6), Zebra already implements it, and still no deployed consumer reads a proof from the tree.

8.4 Related work: velvet deployments

The paper proposes a *velvet fork* as an alternative deployment path (ePrint 2019/226, §8.3, p. 23). Later work shows that this path can admit *chain-sewing* attacks, which splice portions of independent forks into one proof (Nemoz–Zamyatin, IACR ePrint 2021/782). Zcash instead activated ZIP-221 by network upgrade, so that velvet-fork-specific attack does not apply; the 2026 prover paper makes the same distinction.

8.5 Closing the boundary

The volume’s three-bin summary, restated with everything above behind it. *Specified and deployed*: the chain-history tree — epoch scoped, metadata-rich, branch-separated — maintained by Zebra, formerly maintained by zcashd, and committed in each post-activation header for six years, with conformance test vectors; no divergence between the two pre-NU6.3 implementations is known. *Designed but unspecified for Zcash*: the FlyClient protocol itself, with the paper’s conditional security theorem in a model that Zcash’s difficulty rule is not known to satisfy — the ZIP says so, the paper says so, and the 2026 prototype assumes rather than derives its adversarial work budget. *Open*: the bridge — data availability, wire format, epoch composition, the sampling client, and the security analysis — of which only an academic prototype is documented, on a fork, six years after the consensus layer shipped its substrate. The chain has kept its side of an agreement whose other party has yet to arrive.

Part VIII

ZSA Guide: Zcash Shielded Assets: issuance, transfer, and the counterfeiting boundary

Abstract

Assuming the Orchard protocol of the *Ironwood Guide* and transaction construction from the *Wallet Guide*, this volume of *The Zcash Arboretum* documents Zcash Shielded Assets: per-asset value bases and asset identity, transparent issuance and finalization (ZIP-227), shielded transfer and burn (ZIP-226), and the split-note construction that guards the counterfeiting boundary. The upstream ZIPs are in flux — their transaction carrier was withdrawn and their fee terms removed upstream — so every claim is classified as specified, designed-but-unspecified, or open. Where ZIP text, fork text, and the implementation disagree, all three readings are stated and the divergence is flagged.

1 Scope, status, and sources

Zcash Shielded Assets (ZSA) is the proposed extension of Orchard that carries user-issued assets in the shielded pool. Two consensus ZIPs define it: ZIP 226, Transfer and Burn, and ZIP 227, Issuance, both drafts. The combined protocol, OrchardZSA, enables issuance (ZIP 227) and transfer and burn (ZIP 226) of Custom Assets on the Zcash chain; ZIP 227 must only be implemented in conjunction with ZIP 226, and issuance is deliberately transparent — issuance outputs are unencrypted — so that every asset’s circulating supply can be tracked on chain (zip-0226.rst lines 42–43; zip-0227.rst lines 47, 54).

Terminology is ZIP 227’s: an *Asset* is a type of note transferable on the Zcash blockchain; ZEC is the default — and currently the only defined — Asset on mainnet, TAZ on testnet; a *Custom Asset* is any Asset other than ZEC and TAZ (zip-0227.rst lines 35–39).

The design reduces to one alteration of Orchard’s value algebra. The fixed value base V^{Orchard} of the homomorphic Pedersen value commitment (*Ironwood Guide*, §“Conservation without disclosure: value commitments and the binding signature”) is replaced, per Asset, by an *Asset Base* witnessed inside the circuit; the same Asset Base must appear in the old and the new note commitments and as the base of the Action’s value commitment. Under the discrete-log binding assumption for independently derived bases, the binding signature then enforces balance separately per Asset Identifier without revealing which Assets — or how many distinct ones — a transaction transfers (zip-0226.rst lines 59–63). The transparent protocol is unchanged: unshielding is impossible for Custom Assets, burn is the only supply-reduction mechanism and sends to no address, and the Native Asset cannot be burnt (zip-0226.rst lines 206–230).

1.1 Scope and deferrals

This volume develops only the delta over Orchard: the per-asset note type and note commitment; the Asset Base and its derivation from an Asset Identifier; the per-asset value commitment, the burn mechanism, and the burn-aware balance equation; split notes, the mechanism that replaces dummy spends for Custom Assets; the changes to the Action circuit; the issuance layer of ZIP 227 — issuance keys, issue notes, reference notes, and the node-side issuance state — and the note-plaintext extension.

Everything the delta rests on is constructed in earlier volumes and is cited, never re-derived: notes and note commitments (*Ironwood Guide*, §“The note and its commitment”); value commitments, the bases V^{Orchard} and R^{Orchard} , and the binding signature (*Ironwood Guide*, §“Conservation without disclosure: value commitments and the binding signature”); the Action statement and dummy spends (*Ironwood Guide*, §“The Action statement, formally”); nullifiers (*Ironwood*

Guide, §“The nullifier, and the loop back to minting”); Sinsemilla (*Ironwood Guide*, §“Sinsemilla: hash, commitment, and short forms”); and, at the wallet layer, transaction construction, PCZT, and fees (*Wallet Guide*, §“Transaction construction”, §“Partially created transactions (PCZT)”, §“Fees (ZIP-317)”).

Deployment status is stated once, in §1.2. Subsequent sections construct their objects without requalifying each of them with it.

1.2 Status

ZSAs are not part of NU7. Its deployment draft, ZIP 259, introduces no transaction format and retains versions 5 and 6 without a ZSA carrier. ZIPs 226 and 227 remain draft proposals for a future upgrade; their mechanisms are not rules an NU7 validator enforces.

The essential distinction is between a cryptographic construction and its *carrier*: the transaction fields that encode it, together with the digests signed by its participants. ZIP 226 assumes the recoverability changes of ZIP 2005, but its carrier references the withdrawn ZIPs 230 and 246. The deployed version 6 format is ZIP 229, which does not carry ZSAs. The extensible version 7 draft, ZIP 248, lists potential ZSA bundles without specifying their encodings or digests. The experimental format below is therefore not a standardised ZSA transaction format.

Two other assignments remain incomplete: ZIP 226 leaves the note-plaintext lead byte unsigned, and the conventional fee rule in ZIP 317 defines no ZSA issuance or asset-creation terms. The fork-specific choices below are identified where used; neither is silently promoted to consensus.

1.3 Sources

ZIPs 226 and 227 define the proposed constructions. The experimental implementation is not identical to those texts; Table 1 fixes which versions supply the fork-specific formulas below. Current upgrade scope follows ZIP 259, not the experimental code’s feature names.

Artifact	Pin / date	Role
zcash/zips (upstream)	69610984, 2026-07-09	primary ZIP ground truth
QED-it/orchard, branch zsa1	cf801a5d, 2026-06-29	crate orchard 0.14.0
QED-it/librustzcash, branch zsa1	3f9b53fc, 2026-04-17	v6 carrier, fee rule
QED-it/zips, branch zsa1	fd71419a, 2026-02-11	fork baseline for diffing

Table 1: Primary ZSA sources, pinned.

Upstream versus fork. Upstream ZIP 226 diverged from the fork text at fd71419a in three substantive ways: it added the ZIP 2005 dependency paragraph; it changed the split-note ψ^{nf} from “sampled uniformly at random on $\mathbb{F}_{p_{\text{Pallas}}}$ ” to a PRFexpand derivation whose domain-separator byte 0x0A is reserved by ZIP 2005; and it replaced the lead byte 0x03 with the `{{ZSALEADBYTE}}` placeholder. ZIP 227 upstream differs from the fork only in punctuation and one reference title.

Code versus ZIP. Where the pinned code disagrees with the pinned ZIP text, this volume presents the ZIP formula and flags the divergence inline. Two content divergences head the list. First, the split-note nullifier randomizer: ZIP 226 derives ψ^{nf} with domain byte 0x0A, while orchard-zsa @ cf801a5d computes it with PRFexpand domain 0x09 over a random per-note seed (src/circuit.rs lines 316–319) — the older fork-era derivation. Second, the note-plaintext lead byte: a placeholder upstream, but 0x03 in code (src/primitives/zcash_note_encryption_domain.rs lines 46–49) — the very value ZIP 2005 now specifies differently. A third divergence is one of status rather than content: the digest

personalisations in orchard-zsa match the withdrawn ZIP 246 exactly, and librustzcash-zsa implements TxVersion::V6 with the withdrawn ZIP 230 wire format — the entire carrier layer the code implements is specified only by withdrawn ZIPs (§1.2). Further code-versus-text divergences confirmed at the same pins — the omitted explicit-fee digest field, the missing memo branch and its non-spec substitute digest, the empty-action issuance rule, and the valueBurn bound wording — are inventoried in §6.6, §3.7, and §4.4.

The two library pins are separate snapshots, not a single deployed stack. Checks of the implementation below establish library behaviour, not node-side enforcement of ZIP 227’s issuance state.

1.4 Method: the three bins for ZSA

This volume classifies every design-stage claim as *specified*, *designed but unspecified*, or an *open problem*, and keeps the bin visible in the prose. Bin (i) holds the constructions and consensus rules that the live ZIP texts state directly: the per-asset note and commitment, Asset Base derivation, value commitments and balance equation, the split-note circuit delta, issuance keys, issue and reference notes, the issuance state machine, and burn. Bin (ii) holds both the gate-level realization of that circuit delta and the carrier layer: the v6 wire format, digest tree and sighash, note-plaintext lead byte, and ZSA fee terms. Those details are fixed only by implementation, an externally hosted unversioned circuit description, withdrawn ZIPs, or the QED-it fork texts (§1.2). Bin (iii) holds assignment to a future network upgrade and activation, the eventual lead-byte assignment, and the carrier format’s future home. Formulas below identify whether their source is ZIP text, pinned code, or an explicitly labelled derivation in this volume.

2 Asset identity

An OrchardZSA note denominates value in one of many assets sharing a single shielded pool, so the protocol must attach to every note an asset identity satisfying two demands. First, uniqueness: “The Asset identification should be unique (among all shielded pools) and different issuer public keys should not be able to generate the same Asset Identifier” (zip-0227.rst:76). Second, usability inside the circuit: the identity must ultimately be a Pallas point, because it serves as the asset-specific base of the value commitment (§2.4). ZIP 227 meets both demands with one derivation chain,

$$\begin{aligned} \text{asset_desc} &\rightarrow \text{assetDescHash}; \\ (\text{issuer}, \text{assetDescHash}) &\rightarrow \text{AssetId} \rightarrow \text{AssetDigest} \rightarrow \text{AssetBase}, \end{aligned}$$

which the ZIP itself presents as a relation diagram (zip-0227.rst:210–224): the issuer-chosen description string is hashed, the hash paired with the issuer key into an identifier, the identifier hashed into a digest, and the digest mapped onto the Pallas curve. Every Asset issued under ZIP 227 is scoped to the issuer that issued it. Its Asset Identifier is intended to be globally unique and is “used across all Zcash protocols that support ZSAs” (zip-0227.rst:227–240); for two descriptions under one issuer, uniqueness is computational and rests on BLAKE2b-256 collision resistance.

This section constructs the chain end to end: the issuance key pair at its root (§2.1); the hash steps from description to identifier (§2.2); the group hash from identifier to base point (§2.3); the native asset’s special position outside the chain (§2.4); and the parts of the chain consensus sees on the wire versus recomputes (§2.5). Deployment status for the whole protocol is fixed once, in §1.2, and not restated per object.

2.1 The issuance key pair

The OrchardZSA protocol adds exactly two keys to the key material of Orchard: the *issuance authorizing key* isk , which authorizes the issuance of Asset Identifiers by a given issuer and is used only by that issuer, and the *issuance validating key* ik , which validates issuance transactions (zip-0227.rst:84–91). Neither key touches the spend path. The ZIP’s rationale: the issuance key structure is independent of the original key tree but derived analogously via ZIP 32, which keeps issuance details and Asset Identifiers consistent across multiple shielded pools, and separates issuance authority from spend authority — allowing a potential transfer of issuance authority without compromising spend authority (zip-0227.rst:524).

Definition 2.1 (Issuance authorization signature scheme). The scheme `IssueAuthSig` is instantiated as a BIP-340 Schnorr signature over the `secp256k1` curve, with constants set per the `secp256k1` standard parameters as described in BIP 340. Its associated types are: messages are arbitrary byte sequences; signatures are 64-byte sequences or $\perp$; public keys are 32-byte sequences or $\perp$; private keys are 32-byte sequences (zip-0227.rst:122–135). Signing fixes the BIP-340 auxiliary data to $a = [0x00]^{32}$: the signature is $\sigma := \text{Sign}(isk, M)$ with auxiliary data a , or $\perp$ if `Sign` fails; its scheme-level encoding is `issueAuthSig` = $[0x00] \parallel \sigma$ (zip-0227.rst:190–207).

In the implementation, type `ZSASchnorr` realizes the scheme with `ALGORITHM_BYTE` = `0x00`, secret keys as `secp256k1::SecretKey`, public keys as `XOnlyPublicKey`, and signing via `sign_schnorr_with_aux_rand` with the all-zero 32-byte auxiliary array.

Construction 2.2 (Issuance key derivation). The issuance authorizing key is generated by ZIP 32’s hardened-only key derivation — the maps `MKGh` and `CKDh`, constructed in the *Wallet Guide*, §“Hardened-only derivation”, and not re-derived here — instantiated in a dedicated Issuance context (zip-0227.rst:139–166). Let S be a seed of at least 32 and at most 252 bytes with at least 256 bits of entropy, as ZIP 32’s “Specification: Wallet seeds” requires.

- (1) *Context*. The issuance context sets `Issuance.MKGDomain` = `ZcashSA_Issue_V1` and `Issuance.CKDDomain` = `0x81` (zip-0227.rst:148–149).
- (2) *Master key*. Set $m_{\text{Issuance}} := \text{MKGh}^{\text{Issuance}}(S)$.
- (3) *Child derivation*. Set $\text{CKDsk}((sk_{\text{par}}, c_{\text{par}}), i) := \text{CKDh}^{\text{Issuance}}((sk_{\text{par}}, c_{\text{par}}), i)$. The two-argument call fixes the general form’s *lead* = 0 and *tag* = `[]`, exactly the case whose suffix the ZIP-32 encoding omits (zip-0032.rst:449–462; the *Wallet Guide*’s remark “The general form: triples, and sibling instantiations”).
- (4) *Path*. Derive along m_{Issuance} / *purpose*’ / *coin_type*’ / *account*’, with *purpose* fixed to 227 (`0xe3`; hardened `0x800000e3` per BIP 43), *coin_type* as in the ZIP 32 key path levels, and *account* fixed to index 0 (zip-0227.rst:160–164).
- (5) *Key extraction*. From the generated pair (sk, c) , set $isk := sk$ (zip-0227.rst:164–166).

The implementation realizes the context as a struct `Issuance` implementing the `zip32` crate’s hardened-only `Context` trait, with `MKG_DOMAIN` set to `ZcashSA_Issue_V1` and with `CKD_DOMAIN` reusing the Orchard byte, constant `PrfExpand::ORCHARD_ZIP32_CHILD`; the seed-level constructor `from_zip32_seed` derives along `[227', coin_type', 0']` and rejects any non-zero account with error `NonZeroAccount`. The `zip32` crate 0.2.0 implements `MKGh` and `CKDh` exactly, and its `derive_child(index)` calls `ckdh_internal(index, 0, [])` — *lead* = 0, *tag* = `[]`, omitted from the PRF message.

Remark 2.3 (Domain separation between the Orchard and Issuance trees). The byte Issuance.CKDDomain = 0x81 equals Orchard.CKDDomain (zip-0032.rst:476), and the implementation reuses `PrfExpand::ORCHARD_ZIP32_CHILD` (zcash_spec src/prf_expand.rs:119 @ d5e84264). Separation between the two ZIP-32 trees therefore rests entirely on the differing MKGDomain: distinct master secret keys and chain codes feed every subsequent PRFexpand call. Neither ZIP states whether the byte reuse is deliberate; ZIP 32 asks only that CKDDomain values be tracked as part of the global set of domains defined for PRFexpand. The derivation is *specified*; we flag the shared byte because the separation argument is easy to misattribute to it.

Definition 2.4 (Issuance validating key and issuer identifier). The issuance validating key is $ik := \text{PubKey}(isk)$, where `PubKey` is the BIP-340 algorithm and the output is in big-endian order as defined in BIP 340; the derivation returns $\perp$ if `PubKey` fails, which happens with very low probability, in which case the keys are discarded and derivation repeated with a different `isk` (zip-0227.rst:168–183). Its encoding prepends a signature-scheme byte, which **MUST** be 0x00, indicating BIP 340:

$$ik_encoding = [0x00] \parallel ik \quad (33 \text{ bytes}),$$

and the issuer identifier is defined by `issuer = ik_encoding` (zip-0227.rst:103–110, 178–179). The encoding currently appears on chain only in the `issuer` field of an issuance bundle (zip-0227.rst:178–180).

Remark 2.5 (Permanence of the issuer binding). ZIP 227 notes that “the equality of `issuer` and `ik_encoding` might not hold in future when key rotation is specified for issuance key pairs” (zip-0227.rst:103–110). No rotation mechanism exists today: an asset’s identity is permanently bound to the one `ik` under which it was issued, and the ZIP’s remedy for key compromise is to finalize the asset and reissue under a new Asset Identifier: “If an issuer identifier is compromised, the `finalize` boolean for all the Assets issued with that key should be set to 1; then the issuer should change to a new issuance authorizing key (hence a new issuer identifier), and issue new Assets, each with a new `AssetId`” (zip-0227.rst:640–645, Security and Privacy Considerations, “Issuance Key Compromise”). This is a *stated limitation* of the specified design; the rotation mechanism itself is an open design area.

2.2 From description to identifier

An issuer names each of its assets with an *asset description* `asset_desc`: a non-empty byte sequence, which **SHOULD** be a well-formed UTF-8 code unit sequence according to Unicode 15.0.0 or later (zip-0227.rst:237–241). Within the scope of an issuer, Asset Identifier uniqueness is obtained by way of the asset description (zip-0227.rst:237–240). The current ZIP imposes no maximum length — its rationale notes that a description “can be longer than 512 bytes without incurring chain costs”, the 512-byte cap being an earlier design (zip-0227.rst:528–539). The description is also global: ZIP 226 states that “Every Asset is defined by an Asset description, `asset_desc`, which is a global byte string (scoped across all future versions of Zcash)”, so that in a future upgrade the same string can carry over to a different shielded pool, with nodes transforming the identifier chain between pools without balance violations (zip-0226.rst:93–103).

Definition 2.6 (Asset Identifier). The description is hashed,

$$\text{assetDescHash} := \text{BLAKE2b-256}(\text{ZSA-AssetDescCRH}, \text{asset_desc})$$

(zip-0227.rst:243–247), and the Asset Identifier is the pair

$$\text{AssetId} := (\text{issuer}, \text{assetDescHash})$$

(zip-0227.rst:249–253), with the canonical encoding

$$\text{EncodeAssetId}(\text{AssetId}) := [0x00] \parallel \text{issuer} \parallel \text{assetDescHash} \quad (66 = 1 + 33 + 32 \text{ bytes}),$$

where the initial 0x00 is a version byte enabling future issuance protocols (zip-0227.rst:255–261). The version byte is not to be confused with the byte that follows it — the first byte of issuer, which indicates the signature scheme (Definition 2.4); every current encoding happens to begin with two 0x00 bytes of distinct meaning.

The pair structure carries the uniqueness argument. Because the issuer identifier is a component of AssetId, two distinct issuer-key encodings cannot produce the same pair; in the ZIP’s words, the Custom Asset “is described via a combination of the issuer identifier and an asset description string, to preclude the possibility of two different issuers creating colliding Custom Assets” (zip-0227.rst:525). Within one issuer, distinct descriptions separate assets via assetDescHash only subject to BLAKE2b-256 collision resistance; for every new Asset there MUST be a new and unique Asset Identifier (zip-0226.rst:93–96).

Remark 2.7 (Why a hash, not the string). ZIP 227 replaces the direct description string of an earlier design with a collision-resistant hash for five stated reasons (zip-0227.rst:528–559): a fixed 32 bytes per Issue Action costs less bandwidth than up to 512 bytes; descriptions can exceed 512 bytes without chain cost; carrying the byte string on chain would not make the user-visible description consensus-visible anyway (the string could itself be a hash of an off-chain description); absent key rotation, marking an issuer as trusted is insufficient — each Asset Identifier needs independent out-of-band verification that conveys the description; and an on-chain description is an “attractive nuisance” — hashing forces wallets to obtain the description from a trusted party, a registry, or direct user approval. Consistent with the last point, wallets MUST NOT display just the asset_desc string as the name of the Asset; the ZIP suggests a petname system mapping names to Asset Identifiers via client configuration, or the Asset Digest as a compact identifying byte sequence, e.g. in QR codes (zip-0227.rst:263–266).

Implementation. The identifier is the enum variant AssetId::V0, holding the validating key and the 32-byte description hash; encode_asset_id emits the version byte 0x00, the 33-byte key encoding, and the hash — 66 bytes, matching EncodeAssetId. The function compute_asset_desc_hash enforces non-emptiness through its argument type NonEmpty<u8> and panics if the description is not well-formed UTF-8 (orchard-zsa src/issuance.rs:135–154 @ cf801a5d). The specification says SHOULD; the implementation enforces a hard MUST at its API boundary. Since only the 32-byte hash is consensus-visible, neither variant is consensus-enforceable: we classify the UTF-8 constraint as an issuer-side convention — *specified* as a SHOULD, with the stricter behaviour an implementation choice.

2.3 From identifier to base point

Two further steps land the identifier on the Pallas curve. The full chain, collected:

$$\begin{aligned} \text{assetDescHash} &:= \text{BLAKE2b-256}(\text{ZSA-AssetDescCRH}, \text{asset_desc}), \\ \text{AssetId} &:= (\text{issuer}, \text{assetDescHash}), \\ \text{AssetDigest}_{\text{AssetId}} &:= \text{BLAKE2b-512}(\text{ZSA-Asset-Digest}, \text{EncodeAssetId}(\text{AssetId})), \\ \text{AssetBase}_{\text{AssetId}} &:= \text{ZSAValueBase}(\text{AssetDigest}_{\text{AssetId}}), \end{aligned}$$

where the digest is defined at `zip-0227.rst:268–278`, the base at `zip-0227.rst:280–290`, and the OrchardZSA instantiation of the final step is

$$\text{ZSAValueBase}(\text{AssetDigest}) := \text{GroupHash}^{\text{Pallas}}(\text{z.cash} : \text{OrchardZSA}, \text{AssetDigest})$$

(`zip-0227.rst:292–301`). The subscript `AssetId` may be omitted when clear from context (`zip-0227.rst:224`). The group hash is the protocol specification’s $\text{GroupHash}^{\mathbb{P}}$ (§5.4.9.8, “Group Hash into Pallas and Vesta”), whose construction — `expand_message_xmd` over BLAKE2b, two field elements through the simplified SWU map, summed and carried through the isogeny — the *Ironwood Guide* develops in “GroupHash, domain separation, and nothing-up-my-sleeve generators”; we reuse it as a deterministic, public, nothing-up-my-sleeve map into the Pallas group and do not re-derive it here.

ZIP 226 types the result as a *non-trivial* point: `AssetBase` is “the unique element of the Pallas group that identifies each Asset in the Orchard protocol ... a valid group element that is not the identity and is not $\perp$ ” (`zip-0226.rst:111–117`), a non-identity point of $\mathcal{E}(\mathbb{F}_{p_{\text{Pallas}}})$ in our notation. Here uniqueness can only mean that the derivation is deterministic for a given identifier, not that the map is injective: a 512-bit digest is mapped into a roughly 255-bit group, so collisions exist in principle. Asset separation at the base and global-state levels is computational and requires that collisions in the digest/group-hash chain be infeasible to find. Its byte form is

$$\text{asset_base} := \text{LEBS2OSP}_{\ell_{\mathbb{P}}}(\text{repr}_{\mathbb{P}}(\text{AssetBase})),$$

the 32-byte representation stored in note plaintexts and burn fields (`zip-0226.rst:114; zip-0230.rst:540–545`). ZIP 226 notes the encoding assumes canonicity, “true for the Pallas group, but may not hold for future shielded protocols” (`zip-0226.rst:111–117`). Table 2 collects the sizes along the chain.

Object	Bytes	Form	Source
<code>ik</code>	32	x-only BIP-340 key, big-endian	<code>zip-0227.rst:130–133</code>
<code>ik_encoding = issuer</code>	33	<code>[0x00] ik</code>	<code>zip-0227.rst:178–179</code>
<code>assetDescHash</code>	32	BLAKE2b-256 output	<code>zip-0227.rst:243–247</code>
<code>EncodeAssetId(AssetId)</code>	66	<code>[0x00] issuer assetDescHash</code>	<code>zip-0227.rst:255–259</code>
<code>AssetDigest</code>	64	BLAKE2b-512 output	<code>zip-0227.rst:268–278</code>
<code>asset_base</code>	32	$\text{LEBS2OSP}_{\ell_{\mathbb{P}}}(\text{repr}_{\mathbb{P}}(\cdot))$	<code>zip-0230.rst:540–545</code>

Table 2: Byte sizes along the asset-identity derivation chain.

Why the base must be a genuine group-hash output. The chain’s final step is not a formality. In a transfer, the output note of a split Action cannot be allowed to carry an arbitrary Pallas point as its base: “We must enforce it to be an actual output of a GroupHash computation ... Without this enforcement the prover could input a multiple (or linear combination) of an existing Asset Base, and thereby attack the network by overflowing the ZEC value balance and hence counterfeiting ZEC funds” (`zip-0226.rst:299–307`). The circuit therefore ensures the value base points of all output notes are actual GroupHash outputs “as they originate in the Issuance protocol which is publicly verified”, and ZIP 226 adds: “We prevent a potential malleability attack on the Asset Identifier by ensuring the output notes receive an Asset Base that exists on the global state” (`zip-0226.rst:104–106, 299–307`). The enforcement mechanism belongs to the transfer sections of this volume (§5).

Remark 2.8 (The identity-point corner case). The protocol specification’s $\text{GroupHash}^G(D, M)$ has the full group as codomain, so its output can in principle be the zero point (protocol.tex §5.4.9.8; the *Ironwood Guide*, §“GroupHash, domain separation, and nothing-up-my-sleeve generators”, constructs the hash-to-curve pipeline). ZIP 227’s `ZSAValueBase` never states

what happens in that event; the non-identity requirement comes from ZIP 226’s typing of the note field (`zip-0226.rst:111–117`) and from the implementation, which panics inside `AssetBase::custom` (“this will happen with negligible probability”), rejects the identity in `AssetBase::from_bytes`, and carries an `Error::AssetBaseCannotBeIdentityPoint` check in `IssueAction::verify` that the panic renders unreachable (`orchard-zsa src/note/asset_base.rs:98–138`, `src/issuance.rs:216–244 @ cf801a5d`). Whether the intended consensus behaviour is “reject the transaction” or “the issuer must pick a different description” is unspecified in the ZIP text. The event has negligible probability; the specification gap is nevertheless real, and we classify the corner case as *open*.

2.4 The native asset

ZEC is the default — and currently the only defined — Asset for mainnet, TAZ for testnet; a *Custom Asset* is any Asset other than ZEC and TAZ (`zip-0227.rst:35–39`). The native asset does not pass through the derivation chain at all. Its base is fixed by definition: “the Asset Base of the ZEC Asset will be kept as the original value base point, V^{Orchard} ” (`zip-0226.rst:98`); for ZEC,

$$\text{AssetBase}_{\text{AssetId}} := V^{\text{Orchard}},$$

“so that the value commitment for ZEC notes is computed identically to the Orchard protocol deployed in NU5. As such $\text{ValueCommit}_{\text{rcv}}^{\text{Orchard}}(v)$ as defined in ZIP 224 is used as $\text{ValueCommit}_{\text{rcv}}^{\text{OrchardZSA}}(V^{\text{Orchard}}, v)$ ” (`zip-0226.rst:182–194`). The point $V^{\text{Orchard}} := \text{GroupHash}^{\text{Pallas}}(\text{z.cash} : \text{Orchard-cv}, "v")$ is already constructed, together with its companion R^{Orchard} , in the *Ironwood Guide*, “Conservation without disclosure: value commitments and the binding signature”, Definition “Net value commitment”; we cite it and do not re-derive it.

Two consequences follow. First, indistinguishability: ZIP 230 states that “An Orchard note is essentially equivalent to an OrchardZSA note with $\text{AssetBase} = V^{\text{Orchard}}$; in particular, they have the same note commitment ... and need not be distinguished in note plaintexts” (`zip-0230.rst:531–538`), and ZIP 226’s privacy discussion states that OrchardZSA note commitments for the native asset are indistinguishable from those for non-native Assets (`zip-0226.rst:83–84`). Second, exclusion from burning: the first consensus rule for `assetBurn` requires $\text{AssetBase} \neq V^{\text{Orchard}}$ for every $(\text{AssetBase}, v) \in \text{assetBurn}$ (`zip-0226.rst:220–223`) — the native asset cannot be burnt.

Implementation. The native base is `AssetBase::zatoshi()`, computed as the Pallas hash-to-curve over personalization “z.cash:Orchard-cv” of the message “v” — exactly V^{Orchard} ; `AssetBase::is_zatoshi()` is a constant-time equality with that point; and `ValueCommitment::derive` computes $[v]V + [\text{rcv}]R$ with $V = \text{asset.cv_base}()$ and R the fixed base R^{Orchard} , so a ZEC note’s value commitment is bit-identical to NU5 Orchard’s.

2.5 Identity on the wire

Of the whole chain, consensus sees exactly two links on the issuance side: the issuer and the description hash. The v6 transaction’s issuance bundle carries `issuerLength` (`compactSize`), `issuer` (the issuer identifier per ZIP 227), `nIssueActions` (`compactSize`), `vIssueActions`, and `issueAuthSig` — the signature bytes preceded by a 0x00 byte indicating BIP 340. The first four fields are always present; `issueAuthSig` is present iff `nIssueActions > 0`, and if `nIssueActions = 0` then `issuerLength` MUST be 0 (`zip-0230.rst:166–179`, `215–217`, `430–443`). Each Issue Action carries `assetDescHash` as `byte[32]`, `nNotes` (`compactSize`, possibly zero; for an existing Asset, this permits finalization without further issuance), `vNotes` (115 bytes per note description: 43-byte recipient, 8-byte value, 32-byte ρ , 32-byte r_{seed}), and `flagsIssuance` with `finalize` in the least significant bit; the burn encoding carries `assetBase` as `byte[32]`

(zip-0230.rst:367–383, 446–494). On the transfer side, the v6 Orchard note plaintext inserts the asset between rseed and the memo key (the 32-byte key that replaces the in-band 512-byte memo under ZIP-231 and from which the recipient derives the key decrypting its memo in the transaction-level memo bundle; §4.7),

$$[\text{leadByte}] \parallel \text{LEBS2OSP}_{88}(d) \parallel \text{I2LEOSP}_{64}(v) \parallel \text{rseed} \parallel \text{asset_base} \parallel K^{\text{memo}}$$

(zip-0230.rst:523–553).

The consensus rules bind identity to validation: the `ik_encoding` used to validate `issueAuthSig` MUST be taken from the issuer field and MUST start with byte `0x00`, and for each Issue Action, the Issue Note’s `AssetBase` is derived from the `assetDescHash` field of the action and the issuer field of the bundle (zip-0227.rst:483–491). The implementation makes the recomputation structural. Function `read_bundle` decodes the issuer via `IssueValidatingKey::decode`, checking the algorithm byte; function `read_action` reads the 32-byte hash and recomputes the base as `AssetBase::custom` of `AssetId::new_v0(ik, asset_desc_hash)`. The Asset Base is never carried on the wire for issuance, always re-derived. Redundantly with construction, `IssueAction::verify` checks that every note’s base equals the one derived from the validating key and the description hash, failing with error `IssueBundleIkMismatchAssetBase` otherwise. The Asset Identifier is therefore not serialized as a standalone field: its two components are consensus-visible, and the Asset Base is a recomputable function of them, anchored to the issuer key that signs the bundle.

Classification. The entire derivation chain of this section is *specified*: every formula above appears in the ZIP texts at rigor and is realized by the pinned implementation forks at the pinned commits (orchard-zsa @ cf801a5d, librustzcash-zsa @ 3f9b53fc). Both ZIPs are Status Draft, Category Consensus, owned by Pablo Kogan, Vivek Arte, Daira-Emma Hopwood, and Jack Grigg, created 2022-05-01 and discussed in zcash/zips issue #618 and PR #680 (zip-0226.rst:1–18, zip-0227.rst:1–18); deployment status is stated in the scope section. Three reference gaps qualify the bin. First, ZIP 227’s Test Vectors and Reference Implementation sections read “LINK TBD” and its Deployment section “TBD” (zip-0227.rst:661–675); the de-facto assertion-checked vectors live in the fork: file `test_vectors/asset_base.rs`, generated from the zcash-hackworks test-vectors script `orchard_zsa_asset_base.py`, is replayed by a test that recomputes the full chain from (`ik_encoding, asset_desc`) to `asset_base` (`src/note/asset_base.rs:236–261 @ cf801a5d`), with companion vectors for the signature scheme and the ZIP-32 derivation in `issuance_auth_sig.rs` and `zip32.rs`. The vectors’ fixed `[u8; 512]` description arrays are a relic of the dropped 512-byte cap and imply no bound. Second, the corner cases flagged above remain: the identity-point behaviour is *open* (Remark 2.8), the UTF-8 constraint is a SHOULD with a stricter implementation (§2.2), and key rotation is an open design area (Remark 2.5). Third, upstream zcash/zips @ 69610984 is the text of record; the QEDit zips fork (zips-zsa @ fd71419a) is essentially identical for the asset-identity material, and the differences — upstream’s later additions of a ZIP-2005 dependency, a nullifier ψ^{nf} derivation, and a lead-byte rule — do not touch this chain.

3 Issuance and finalization (ZIP-227)

The transfer mechanism below moves Custom Assets between shielded notes; this section brings them into existence. ZIP-227 specifies issuance on the Zcash chain: a per-transaction *issuance bundle* in which a publicly named issuer creates new notes, a per-Asset *finalize* switch that ends issuance forever, and a node-maintained global supply map.

Issuance is deliberately *not* shielded. The ZIP’s Motivation states: “This ZIP only enables *transparent* issuance. As a first step, transparency will allow for proper testing of the applications that will be most used in the Zcash ecosystem, and will enable the supply of Assets to be

tracked” (ZIP-227, *Motivation*). The issuer, Asset, issued amounts, finalization status, raw recipient addresses, ρ values, and random seeds are serialized in the clear. For an ordinary recipient, the address does not reveal the nullifier key, so a later spend is designed to be unlinkable to the public issue-note description. The mandatory all-zero-key reference note is an explicit exception (§3.3).

Encoding boundary. The issuance mechanism follows ZIP 227. Its encodings and digests below follow the experimental ZIP 230/246 carrier, not ZIP 229’s deployed v6 format (§1.2). Fork-only fee rules and differences between the ZIP and the pinned libraries are marked where they affect a construction.

3.1 Issuance keys and the issuer identifier

Issuance authority is a new capability, deliberately disjoint from spend authority. The rationale in the ZIP: the issuance key tree is independent of the spending key tree but derived analogously (via ZIP 32), so an organization can delegate issuance without exposing funds, and a future scheme can replace the issuance leaf with a multisignature without touching spending keys (ZIP-227, *Rationale*, and *Issuance Key Derivation*).

The key material itself is constructed once, in §2.1: the BIP-340 signature scheme (Definition 2.1), the hardened-only ZIP-32 derivation of isk in the Issuance context (Construction 2.2), and the validating key with the issuer identifier, $ik := PubKey(isk)$ and $issuer = ik_encoding = [0x00] \parallel ik$ (Definition 2.4). We restate only the parameters the issuance layer consumes: seed S of 32 to 252 bytes, $Issuance.MKGDomain = ZcashSA_Issue_V1$, $Issuance.CKDDomain = 0x81$, hardened path $m_{Issuance} / 227' / coin_type' / 0'$, and $isk := sk$ at the leaf (ZIP-227, *Issuance Key Derivation*). No rotation exists today: the on-chain issuer name is the validating key itself, and the ZIP notes the equality might not hold in future if key rotation is ever specified (Remark 2.5; §3.8).

3.2 Asset identity: description, digest, and Asset Base

Every Custom Asset is named by its issuer plus an issuer-chosen description, and consumed by the protocol as a Pallas point. The full derivation chain — $assetDescHash$, $AssetId$, $EncodeAssetId$ with its version byte, $AssetDigest$, and $AssetBase = ZSAValueBase(AssetDigest)$ — is constructed once, in §2.2 and §2.3 (Definition 2.6); the hash-not-string rationale is Remark 2.7, and the UTF-8 SHOULD-versus-panic delta between ZIP and implementation is recorded in §2.2. The native Asset bypasses the chain entirely: its base is the Orchard value-commitment base $V^{Orchard}$ (§2.4). ZIP 226 excludes the Pallas identity from the Asset-Base type; $AssetBase::custom$ panics after hashing to it, while $AssetBase::from_bytes$ rejects it on parse (Remark 2.8). What the issuance layer adds to the chain is consensus-side recomputation: validators re-derive every Issue Note’s $AssetBase$ from the bundle’s issuer and the action’s $assetDescHash$, never reading it off the wire (§2.5; consensus rules, §3.7).

3.3 Issue notes and the derived ρ

Definition 3.1 (Issue note). An *Issue Note* is a tuple $(d, pk_d, v, AssetBase, \rho, rseed)$,

$$Note^{Issue} := \{0, 1\}^{\ell_d} \times KA^{Orchard}.Public \times \{0 \dots 2^{\ell_{value}} - 1\} \times \mathcal{E}(\mathbb{F}_{p_{Pallas}})^* \times \mathbb{F}_{p_{Pallas}} \times \{0, 1\}^{8 \cdot 32},$$

a diversifier and diversified transmission key as in Orchard, a value v with $\ell_{value} = 64$, a non-identity Asset Base, the nullifier-input field element ρ , and a 32-byte $rseed$ that MUST be sampled uniformly at random by the issuer (ZIP-227, *Issue Note*; the ZIP writes the fourth and fifth components $\mathbb{P}^*$ and $\mathbb{F}_{q_P}$, our $\mathcal{E}(\mathbb{F}_{p_{Pallas}})^*$ and $\mathbb{F}_{p_{Pallas}}$).

Issue-note commitments are not serialized: the wire format carries the note fields themselves in the clear (§3.4), and each validator computes the commitment — by the OrchardZSA NoteCommit of §4 — and appends it to the Orchard note commitment tree. For an ordinary issue note, the recipient address does not reveal the nullifier key; under the nullifier PRF assumptions, a later nullifier is computationally unlinkable to that public description. This privacy claim does not apply to the reference note, whose all-zero spending key is public (ZIP-227, *Issue Note, Reference Notes, and Issuance Protocol*). Issuance is transparent at creation; later transfers of ordinary issue notes are shielded.

Computation of ρ . An Orchard note takes $\rho := \text{nf}_{\text{old}}$ from the Action that creates it (the *Ironwood Guide*, §“Delivery: the note reaches its owner”); an issue note has no such Action, so ZIP-227 derives ρ from the transaction’s OrchardZSA transfer bundle instead:

$$\begin{aligned} \text{DeriveIssuedRho}(\text{nf}, i_A, i_N) &:= \text{ToBase}(\text{PRFexpand}_{\text{nf}}(0x84 \parallel \text{I2LEOSP}_{32}(i_A) \parallel \text{I2LEOSP}_{32}(i_N))), \\ \rho &:= \text{DeriveIssuedRho}(\text{nf}_{0,0}, \text{index}_{\text{Action}}, \text{index}_{\text{Note}}), \end{aligned}$$

with the PRF keyed by nf as 32 little-endian bytes (I2LEOSP_{256}), where $\text{nf}_{0,0}$ is the nullifier of the input note in the *first* Action of the *first* Action Group of the OrchardZSA bundle (the v6 format partitions a bundle’s Actions into one or more Action Groups, each carrying its own anchor and proof; §6.2), and the two indices are the zero-based positions of the Issue Action within the issuance bundle and of the note within its action (ZIP-227, *Computation of ρ*). The domain-separator byte 0x84 extends the PRFexpand lead-byte registry (the *Wallet Guide*, §“The PRFexpand lead-byte registry”); the patched `zcash_spec` carries it as `ORCHARD_DERIVED_ISSUE_RHO` (QED-it `zcash_spec`, `src/prf_expand.rs` line 107 @ d5e8426). The implementation computes the identical expression — PRFexpand keyed by the 32-byte little-endian nullifier over 0x84 followed by two 4-byte little-endian indices, reduced by ToBase. This derivation is why an issuance bundle cannot travel alone: requiring at least one Action Group both supplies $\text{nf}_{0,0}$ and blocks replay — a transaction containing only an issuance bundle would have a SIGHASH computed solely from that bundle, so a duplicated bundle would share the SIGHASH (ZIP-227, *Rationale*).

In the builder, notes are created with ρ unset, via `new_issue_note`; once the transfer bundle’s first nullifier is known, `update_rho_for_issuance_note` sets ρ and resamples `rseed` until both the derived ephemeral key is valid and the note commitment is not $\perp^3$ (`orchard-zsa`, `src/note.rs` lines 457–489 @ cf801a5d).

Reference notes. The first time an Asset is issued, the issuer MUST place a *reference note* as the first note of that Issue Action: a note of value $v = 0$ whose recipient (d, pk_d) is the default diversified payment address (diversifier index 0) derived from the *all-zero* Orchard spending key (ZIP-227, *Reference Notes*). The implementation hard-codes the resulting 43-byte raw address as `RAW_REFERENCE_RECIPIENT`, byte-identical to the array in the ZIP, with `ReferenceKeys::sk() = SpendingKey::from_bytes([0; 32])`; the predicate `is_reference_note` requires zero value and exactly this recipient, `get_reference_note` accepts only the *first* note of an action, and the builder inserts the reference note itself when told the issuance is a first issuance — adding a recipient for an Asset that already has notes fails with `CannotBeFirstIssuance`. The reference note is retained in global state as `note_ref` (§3.7).

The ZIP mandates reference notes and stores them in consensus state, but states no rationale for their existence. The evident function — that every issued Asset Base is guaranteed at least one note in the commitment tree, available as a ZIP-226 split-input source for a first spend — is our reading, stated nowhere in the ZIP text or its rationale sections; we classify the mechanism itself as *specified* and this purpose as *designed but unspecified*.

³Strictly, these negligible-probability rejection steps condition `rseed`, a technical deviation from the ZIP’s “sampled uniformly at random” (ZIP-227, *Issue Note*).

3.4 The issuance bundle and its encoding

An *Issuance Bundle* carries the issuer identifier, the list of Issue Actions, and one authorizing signature: issuer, which lets validators verify that each AssetId is properly associated with this issuer; vIssueActions, an array of Issue Actions; and issueAuthSig, the encoding of a signature over the transaction SIGHASH by the issuance authorizing key isk (ZIP-227, *Issuance Bundle*). An *Issuance Action* groups the notes issued for one Asset: the 32-byte assetDescHash, the note list, and a flags byte carrying finalize (ZIP-227, *Issuance Action*). Table 3 gives the v6 serialization; the encoding rigor caveat of the section introduction applies.

Field	Type	Notes
issuerLength	compactSize	MUST be 0 if nIssueActions = 0
issuer	byte[issuerLength]	= ik_encoding, 33 bytes when present
nIssueActions	compactSize	
vIssueActions	IssueAction[nIssueActions]	
issueAuthSig	IssueAuthSignature	present iff nIssueActions > 0
<i>Per IssueAction:</i>		
assetDescHash	byte[32]	
nNotes	compactSize	may be 0 (existing-Asset finalization)
vNotes	IssueNoteDescription[nNotes]	115 bytes each
flagsIssuance	byte	finalize at LSB; other bits MUST be 0
<i>Per IssueNoteDescription (43 + 8 + 32 + 32 = 115 bytes):</i>		
recipient	byte[43]	LEBS2OSP ₈₈ (d) pk _d [*]
value	uint64	little-endian
rho	byte[32]	as for Orchard notes
rseed	byte[32]	as for Orchard notes

Table 3: ZSA issuance fields of the v6 transaction format (fork ZIP-230, *Transaction Format, Issuance Action Description*, and *Issue Note Description* @ zips-zsa fd71419; upstream text identical apart from the Withdrawn status).

Three encoding facts deserve emphasis. First, the recipient field is exactly an Orchard raw payment address (the *Wallet Guide*, §“Raw encodings”), and together with the bundle’s issuer and the action’s assetDescHash, the four fields of an IssueNoteDescription determine the Issue Note completely (ZIP-230, *Issue Note Description*; ZIP-227, *Issue Note*) — the note commitment is deliberately absent, since validators recompute it (§3.3). Second, the encoding permits nNotes = 0; for an existing Asset this lets the issuer finalize without issuing more (ZIP-230, *Issuance Action Description*; the new-Asset corner case is recorded in §3.7). Third, the flags byte is strict: IssuanceFlags::from_byte returns None whenever any bit other than the LSB is set, and deserialization fails on such bytes, enforcing ZIP-230’s “remaining bits are set to 0” at parse time.

The librustzcash-zsa serializer pins down the absent-bundle corner cases: read_bundle reads issuer as a length-prefixed vector and returns None only for an empty issuer with zero actions, erroring on an empty issuer with non-empty actions and on a non-empty issuer with zero actions; write_bundle encodes an absent bundle as two zero compactSizes. The authorization serializes as a length-prefixed sighashInfo followed by the length-prefixed signature encoding, and the parser accepts exactly [0x00] as sighashInfo (transaction/components/issuance.rs lines 20–191 and sighash_versioning.rs @ 3f9b53fc; the meaning of [0x00] is §3.5).

In the orchard-zsa object model, the type IssueBundle<T: IssueAuth> holds the validating key ik (an IssueValidatingKey<ZSASchnorr>), a non-empty vector of actions, and an authorization tpestate T; an IssueAction holds the 32-byte asset_desc_hash, its Vec<Note>, and its IssuanceFlags.

3.5 Authorization: BIP-340 over the transaction sighash

The issuance authorization signature scheme `IssueAuthSig` is instantiated as a BIP-340 Schnorr signature over `secp256k1` with the standard parameters, not `RedPallas`; ZIP-227 gives no rationale for that curve choice. Batch verification MAY be used, and precomputation MAY be used if and only if it produces equivalent results (ZIP-227, *Issuance Authorization Signature Scheme*). The associated types (same section):

$$\text{Message} = \mathbb{B}^{\mathbb{Y}^{[N]}}, \quad \text{Signature} = \mathbb{B}^{\mathbb{Y}^{[64]}} \cup \{\perp\}, \quad \text{Public} = \mathbb{B}^{\mathbb{Y}^{[32]}} \cup \{\perp\}, \quad \text{Private} = \mathbb{B}^{\mathbb{Y}^{[32]}},$$

that is, arbitrary byte-sequence messages, 64-byte signatures, 32-byte x-only public keys, and 32-byte secret keys, with $\perp$ as the failure value. Signing fixes the BIP-340 auxiliary data to the all-zero block:

$$a := [0x00]^{32}, \quad \sigma := \text{Sign}(\text{isk}, M) \text{ with auxiliary data } a, \quad \text{issueAuthSig} := [0x00] \parallel \sigma,$$

returning $\perp$ on failure; validation returns 0 if $\sigma = \perp$, else 1 exactly when the BIP-340 `Verify(ik, M,  $\sigma$ )` succeeds⁴ (ZIP-227, *Issuance Authorization Signing and Validation*). With the scheme byte, the two wire encodings are 33 bytes (`ik_encoding`) and 65 bytes (`issueAuthSig`).

What is signed. The message is the transaction `SigHash` as defined by protocol specification §4.10 under the ZIP-226 modifications, with sighash type `SIGHASH_ALL`, and the consensus rule requires `IssueAuthSig.Validate(ik, SigHash, issueAuthSig) = 1` (ZIP-227, *Specification: Consensus Rule Changes*). Under ZIP-246 the signature digest includes branch S.5, the `issuance_digest`, identical to the `txid` branch — so the issuance signature commits to *all* effecting data of the transaction, including the issuance bundle’s own effecting data, though never to any signature (ZIP-246, *Signature Digest*). ZIP-246 also introduces sighash algorithm versioning: each bundle carries `sighashInfo = [sighashVersion]  $\parallel$  associatedData`, versions are numbered from 0 per transaction version, version 0 is by convention the “commit to all effecting data” algorithm, and for v6 only version 0 is defined, with empty associated data for the Transparent, Sapling, Orchard, and Issuance bundles — hence the constant `[0x00]` the parser demands (ZIP-246, *Sighash versioning*).

Implementation. The `ZSASchnorr` scheme wraps the `secp256k1` crate: signing builds a keypair from `isk` and calls `sign_schnorr_with_aux_rand(msg, keypair,  $\&[0u8; 32]$ )` — the ZIP’s $a = [0x00]^{32}$ — `ik` is the x-only public key, and `encode()` of key and signature each prepend `ALGORITHM_BYTE = 0x00`, with `decode()` rejecting any other lead byte. Bundle construction is a tpestate chain — `AwaitingNullifier` $\rightarrow$ (`update_rho`) $\rightarrow$ `AwaitingSighash` $\rightarrow$ (`prepare(sighash)`) $\rightarrow$ `Prepared` $\rightarrow$ (`sign(isk)`) $\rightarrow$ `Signed` — so a bundle cannot be signed before its notes’ ρ values and the sighash exist. On the transaction side, `v6_signature_hash` includes the `txid` `issuance_digest`, and the builder signs the issue bundle with `prepare(*shielded_sig_commitment)` where that commitment is `signature_hash(tx, SignableInput::Shielded, txid_parts)` — the same `SIGHASH_ALL` shielded digest the Sapling and Orchard signatures sign. The builder also realizes the `nf0,0` rule constructively: `first_nullifier` returns the first Action’s nullifier only for an OrchardZSA bundle, and building with an issuance builder but no such bundle fails with `BundleTypeNotSatisfiable`.

A divergence to flag. ZIP-246’s `txid` and signature digests include a sixth branch, T.6/S.6 `memo_digest` (ZIP-246, *Txid Digest* and *Signature Digest*), but `librustzcash-zsa`’s `to_hash` hashes only the header, transparent, Sapling, Orchard, and issuance branches — the fork does not implement ZIP-231 memo bundles (`transaction/txid.rs` lines 415–460 @ 3f9b53fc). The

⁴The ZIP’s `Validate` definition writes `Verify(key, M,  $\sigma$ )` for the parameter it names `ik` — an editorial slip we flag when citing (ZIP-227, *Issuance Authorization Signing and Validation*).

digest the fork code signs is therefore not byte-identical to the digest ZIP-246 specifies; whichever is corrected, the issuance signature’s coverage claim above should be read against the fork’s five-branch digest today.

3.6 Transaction digests

The v6 digest tree extends the ZIP-244 structure the *Ironwood Guide* presents in §“What the signatures sign: ZIP-244 digests and their v6 form”: the txid digest is BLAKE2b-256 with personalization `ZcashTxHash_ || CONSENSUS_BRANCH_ID` over six branches T.1–T.6 (header, transparent, Sapling, Orchard, issuance, memo), and the signature digest has the same six-branch structure with S.5 identical to T.5; a signature digest is produced for each transparent input, Sapling input, OrchardZSA Action, and additionally for each Issuance Action (ZIP-246, *Txid Digest* and *Signature Digest*). The issuance branches are three nested hashes and one authorizing hash, each over the wire encodings of Table 3 (ZIP-246, *txid_digest* T.5 and A.4: *issuance_auth_digest*):⁵

```
issuance_digest = BLAKE2b-256(ZTxIdSAIssueHash,
                               issuerLength || issuer || issue_actions_digest),
issue_actions_digest = BLAKE2b-256(ZTxIdIssuActHash,
                                   ||_actions assetDescHash || issue_notes_digest || flagsIssuance),
issue_notes_digest = BLAKE2b-256(ZTxIdIAcNoteHash,
                                   ||_notes recipient || value || rho || rseed),
issuance_auth_digest = BLAKE2b-256(ZTxAuthZSA0rHash, issueAuthSig field encoding),
```

where the A.4 preimage is the ZIP-230 `IssueAuthSignature` composite

```
sizeSighashInfo || sighashInfo || sizeSignature || signature
```

(ZIP-230, *Issuance Authorization Signature*), and a transaction without issuance components hashes the empty string under the same personalizations.

The code computes exactly these issuance-specific hashes. On the txid side, `hash_issue_bundle_txid_data` hashes

```
compactSize(|ik_encoding|) || ik_encoding || issue_actions_digest
```

under `ZTxIdSAIssueHash`; the actions digest hashes, per action, the 32-byte hash, the notes digest, and the flags byte; the notes digest hashes, per note, the 43-byte recipient, the 8-byte little-endian value, and the two 32-byte fields. On the authorizing side, `hash_issue_bundle_auth_data` hashes

```
compactSize(|sighashInfo|) || sighashInfo || compactSize(65) || [0x00] || σ,
```

asserting the 65-byte length and the 0x00 lead. The empty-bundle variants hash the empty string, and all four personalizations match ZIP-246; both digests live in `src/bundle/commitments/issuance.rs` lines 11–86 (`orchard-zsa @ cf801a5d`). For the wiring into transaction identifiers, `TxidDigester::digest_issue` maps the bundle to its `commitment()`; `to_hash` appends the digest only when the version has `_orchard_zsa()`; and `BlockTxCommitmentDigester` takes the authorizing commitment (`librustzcash-zsa, transaction/txid.rs` lines 381–383, 445–452, 595–598 @ `3f9b53fc`).

Deterministic regression digests are pinned in the tests — for a seeded two-asset bundle,

⁵Two editorial defects in ZIP-246 to note when citing: the T.5 child list labels `issuerLength/issuer/issue_actions_digest` as T.5a/T.5b/T.5c, while the following section headings call `issue_actions_digest` “T.5a” and `issue_notes_digest` “T.5a.i”; and the A.4 empty case reads “In the case that the transaction has no Orchard Actions” where “no Issuance Actions” is evidently intended.


```

issuance_digest = ee70e3b61674fd0428ac0020cc4fc5819386e39c4eb3c63357c84c998195bcd,
issuance_auth_digest = 6df77af7b5323d99376336b770e4c5b06ffc195de81ac7692d1b08b6eb19534d

```

— and cross-implementation test vectors ship in `src/test_vectors/` (`asset_base.rs`, `issuance_auth_sig.rs`), exercised by the asset-base and authorization tests (`orchard-zsa`, `src/bundle/commitments/issuance.rs` lines 156–190 and `src/test_vectors/` @ `cf801a5d`). The upstream ZIP-227 Test Vectors section, by contrast, still reads LINK TBD.

3.7 Global issuance state and the consensus rules

Supply integrity is the one property issuance cannot delegate to cryptography: no transaction field simultaneously witnesses everything issued and everything burnt for an Asset, so — along the lines of the ZIP-209 pool turnstile — every node maintains a per-Asset circulation record (ZIP-227, *Rationale for Global Issuance State*).

Definition 3.2 (Global issuance state). The map

$$\begin{aligned} \text{issued_assets} : \mathcal{E}(\mathbb{F}_{p_{\text{Pallas}}})^* &\rightarrow \{0 \dots \text{MAX_ISSUE}\} \times \{0, 1\} \times \text{Note}^{\text{Issue}}, \\ \text{AssetBase} &\mapsto (\text{balance}, \text{final}, \text{note}_{\text{ref}}), \end{aligned}$$

records, for every issued Asset, the amount in circulation (issued minus burnt), the finalization status — which can never change from 1 to 0 — and the Asset’s reference note. A missing entry ($\text{issued_assets}(\text{AssetBase}) = \perp$) is read as $\text{balance} = 0$, $\text{final} = 0$, $\text{note}_{\text{ref}} = \perp$; the constant $\text{MAX_ISSUE} = 2^{64} - 1$ caps the recorded circulating balance of any Custom Asset at any one state (ZIP-227, *Global Issuance State*).

The map threads through the chain deterministically: it MUST be updated by a node during the processing of any transaction that contains burn information or an issuance bundle; the input state for the OrchardZSA activation block is the empty map; the input state of a block’s first transaction is the final state of the preceding block; each subsequent transaction’s input state is the preceding transaction’s output state; and the block’s final state is its last transaction’s output state (ZIP-227, *Management of the Global Issuance State*).

Per-transaction rules. For a v6 transaction (ZIP-227, *Specification: Consensus Rule Changes*):

- (T1) `nActionGroupsOrchard` MUST be 0 or 1, and `nAGExpiryHeight` MUST be 0; the v6 parser of `librustzcash-zsa` enforces both at deserialization, in `transaction/components/orchard.rs` lines 107–132 @ `3f9b53fc`;
- (T2) the output state $\text{issued_assets}_{\text{OUT}}$ MUST be initialized to the input state $\text{issued_assets}_{\text{IN}}$;
- (T3) the `assetBurn` set MUST satisfy the ZIP-226 rules (§4: tuples $(\text{AssetBase}, v)$ with the Native Asset excluded, $0 < v \leq \text{MAX_BURN_VALUE} = 2^{63} - 1$, and no duplicate `AssetBase`; ZIP-226, burn rules); and
- (T4) for every $(\text{AssetBase}, v) \in \text{assetBurn}$, the node MUST check $\text{balance} \geq v$ for the entry of `AssetBase` in $\text{issued_assets}_{\text{OUT}}$ and subtract v from that balance, *prior to* processing the issuance bundle.

Issuance-bundle rules. For a transaction carrying an issuance bundle (ZIP-227, *Specification: Consensus Rule Changes*):

- (I1) the transaction MUST also contain at least one OrchardZSA Action Group (the ρ /replay rationale of §3.3);

- (I2) the encoding $ik_encoding$ MUST be taken from the `issuer` field and MUST start with the byte `0x00` (BIP-340);
- (I3) the signature MUST satisfy `IssueAuthSig.Validate(ik, SigHash, issueAuthSig) = 1`;
- (I4) for each Issue Action: every `IssueNoteDescription` MUST be a valid ZIP-230 encoding; the `AssetBase` is derived from `assetDescHash` and `issuer` (§3.2); and the Asset’s final bit in `issued_assets_OUT` MUST NOT be 1;
- (I5) if $note_{ref} = \perp$, the first Issue Note MUST be a reference note (§3.3), and the node MUST set $note_{ref}$ to it;
- (I6) for every Issue Note $note_j$: its ρ MUST equal the value computed per §3.3; the supply update $issued_assets_OUT.balance + v_j \leq MAX_ISSUE$, $issued_assets_OUT.balance += v_j$ MUST hold and be applied; and the node MUST compute the note commitment per ZIP-226;
- (I7) if `finalize = 1` in `flagsIssuance`, the node MUST set `final = 1`.

The commitments enter the tree in a fixed order: for each Action Group of the OrchardZSA bundle, the commitment of every new note per Action; then, for each Issue Action of the issuance bundle, the commitment of every Issue Note (ZIP-227, *Addition to the Note Commitment Tree*).

The rationale ties the pieces together: balances of issued Assets must never go negative, `MAX_ISSUE` is a practical limit that also lets an issuer create an Asset’s complete supply in a single transaction, and the same map records finalization so that further issuance of a finalized Asset is rejected (ZIP-227, *Rationale for Global Issuance State*). Beyond consensus, the ZIP RECOMMENDS that full validators keep a mutable `issuanceSupplyInfoMap` from `AssetId` to `(totalSupply, finalize)` for wallets and applications tracking total supply (ZIP-227, *Implementing Zcash Nodes*).

The library realization. Function `verify_issue_bundle` takes the bundle, the sighash, a caller-supplied view of global state (`get_global_records`), and the first nullifier. It rejects any sighash kind other than `AllEffecting`, verifies the BIP-340 signature over the 32-byte sighash with the bundle’s `ik`, folds the actions over a `BTreeMap<AssetBase, AssetRecord>`, and returns the updated records — `amount`, `is_finalized`, `reference_note` — for the caller to persist. The fold realizes the per-action and per-note rules as typed errors:

- `IncorrectRhoDerivation` — a note’s ρ differs from the recomputed derivation of §3.3;
- `MissingReferenceNoteOnFirstIssuance` — no record exists for the Asset, and the action’s first note is not a reference note;
- `IssueActionPreviouslyFinalizedAssetBase` — reissuance against a finalized record;
- `IssueBundleIkMismatchAssetBase` and `AssetBaseCannotBeIdentityPoint` — a note’s Asset Base differs from the one `IssueAction::verify` derives from `(ik, assetDescHash)`, or the derived point is the Pallas identity. The latter variant is intended but unreachable at this pin because `AssetBase::custom` panics first (Remark 2.8);
- `ValueOverflow` — the checked u64 value sum overflows, the code’s realization of $MAX_ISSUE = 2^{64} - 1$.

A signature-skipping variant, `check_issue_bundle_without_sighash`, exists for verifying historical blocks from a trusted checkpoint (`src/issuance.rs` lines 647–705 @ cf801a5d).

Three boundary observations, each load-bearing for anyone re-deriving consensus from these libraries:

- *Empty-action divergence.* The implementation rejects an action with no notes unless it is finalized (`IssueActionWithoutNoteNotFinalized`, `src/issuance.rs` lines 216–219), but ZIP-227’s consensus-rule list contains no such MUST, and ZIP-230 merely notes that `nNotes` may be zero “to only finalize”. For an already-existing Asset, the written consensus rules admit an empty, non-finalizing action that the reference implementation rejects; for a new Asset, the ZIP’s instruction to take the first note as the reference note is undefined when the list is empty.
- *First-issuance detection.* The ZIP keys the reference-note requirement on $\text{note}_{\text{ref}} = \perp$; the code keys it on the complete absence of an `AssetRecord` (`get_global_records` returning `None`). The two are equivalent under the invariant that note_{ref} is set exactly on first issuance — rule (I5) — and no rule ever clears it, so an entry exists if and only if its note_{ref} is set.
- *Node-side enforcement locus.* Neither `orchard-zsa` nor `librustzcash-zsa` contains the `issued_assets` chain-state maintenance or the consensus caller of `verify_issue_bundle`: the libraries return updated records for a node to persist, and the QEDit node integration is not among our local ground-truth repositories. The global-state rules are therefore *specified and library-supported*, with their node enforcement unverified locally.

3.8 Finalization

The `finalize` boolean signals that no further issuance of the specific Custom Asset is possible: every issuance action carries it in `flagsIssuance` (a requirement of the ZIP’s Requirements section), setting it sets the Asset’s final bit — rule (I7) — and transactions attempting to issue further amounts of a previously finalized Asset are rejected — rule (I4) (ZIP-227, *Requirements and Issuance Action*). The bit is monotone: consensus state never changes final from 1 to 0 (Definition 3.2).

Finalization is the ZIP’s whole lifecycle policy, and its Concrete Applications read as patterns over one bit (ZIP-227, *Concrete Applications*):

- *One-time issuance:* set `finalize = 1` on the first action; the entire supply exists from one transaction — `MAX_ISSUE` is sized to permit this (§3.7).
- *Stopping supply:* issue with `finalize = 1` in a final issuance transaction, or in one containing a single note of value 0 (the wire format admits note-less finalizing actions as well, Table 3; the library insists on the finalizing variant, §3.7).
- *NFTs:* issue multiple actions in one transaction, each with value = 1 at the Asset’s fundamental unit and `finalize = 1` — each action defines a distinct finalized Asset Identifier with supply one, subject to the collision-resistance assumptions of §2.2.

Finalization is also the entirety of the compromise story. No key rotation exists for issuance keys; on compromise of `isk`, the ZIP’s guidance is to set `finalize = 1` for all Assets issued with that key, switch to a new `isk` — hence a new issuer identifier, since `issuer = ik_encoding` — and issue new Assets with new `AssetIds` (ZIP-227, *Issuance Key Compromise*). The old Assets keep circulating and burning; they simply never grow again.

3.9 Fees

The fee treatment of issuance is the sharpest fork/upstream split in the ZSA stack, and we bin it explicitly. The QEDit fork of ZIP-317 adds two contributions to the logical-action count of the conventional fee (the *Wallet Guide*, §“Fees (ZIP-317)”, for the surrounding machinery: $\text{marginal_fee} = 5000 \text{ zatoshis}$, $\text{grace_actions} = 2$):

$$\begin{aligned} \text{contribution}_{\text{ZSAIssuance}} &= n_{\text{ZSAIssueNotes}}, \\ \text{contribution}_{\text{ZSACreation}} &= \text{creation_cost} \cdot n_{\text{AssetCreations}}, \end{aligned}$$

where $nZSAIssueNotes$ counts `IssueNote` outputs, $nAssetCreations$ counts Custom Assets newly added to the global issuance state, and $creation_cost = 100$ logical actions (fork ZIP-317, *Fee calculation @ zips-zsa fd71419*). The stated rationale: every new Asset adds a row that validators track in perpetuity (§3.7), and the two-orders-of-magnitude creation cost disincentivizes junk assets.

ZIP 317 defines neither contribution. These formulas are *fork-specified*: they describe the experimental ZSA fee policy, not the conventional fee rule for deployed transactions.

4 Transfer and burn (ZIP-226)

ZIP-226 (*Transfer and Burn of Zcash Shielded Assets*; Status: Draft; owners Pablo Kogan and Vivek Arte of QEDit, Daira-Emma Hopwood, and Jack Grigg) defines, jointly with the issuance protocol of ZIP-227 (§3), the OrchardZSA transfer and burn protocol. We develop it as four deltas against the Orchard baseline, each citing the construction it modifies in the *Ironwood Guide* rather than re-deriving it: the note gains one field, the *Asset Base* (§4.1); the note commitment becomes a case split that keeps ZEC commitments identical to NU5 (§4.2); the value commitment trades its fixed value base for the per-asset base, which turns the binding signature into a computational per-asset balance check and gives burn its consensus meaning (§4.3, §4.4); and dummy spends give way to *Split Inputs* for Custom Assets (§5). We then state the extended Action statement (§4.6) and the carrier format with its digests and fees (§4.7).

ZIP 226 assumes ZIP 2005's recoverability changes, whereas the experimental fork does not implement them. Where the difference affects a construction — the split-note ψ^{nf} (§5) and the lead byte (§4.7) — we flag it in place.

4.1 One new field: the OrchardZSA note

Definition 4.1 (OrchardZSA note). An OrchardZSA note extends the Orchard note — the tuple $(addr, v, \rho, \psi, rcm)$ of the *Ironwood Guide*, “The note and its commitment” — by exactly one field:

$$n := (addr, v, AssetBase, \rho, \psi, rcm),$$

where $AssetBase \in \mathcal{E}^* := \mathcal{E}(\mathbb{F}_{p_{Pallas}}) \setminus \{O\}$ is “a valid group element that is not the identity and is not bottom” (ZIP-226; the ZIP writes $\mathbb{P}^*$ for $\mathcal{E}^*$). All other components are as in protocol specification §3.2. Formally,

$$\begin{aligned} \text{Note}^{\text{OrchardZSA}} &:= \{0, 1\}^{88} \times \text{KA}^{\text{Orchard}}.\text{Public} \times \{0, \dots, 2^{64} - 1\} \times \mathcal{E}^* \\ &\quad \times \mathbb{F}_{p_{Pallas}} \times \mathbb{F}_{p_{Pallas}} \times \text{NoteCommit}^{\text{Orchard}}.\text{Trapdoor}, \end{aligned}$$

where the ZIP's $\mathbb{F}_{q_P}$ is this series' $\mathbb{F}_{p_{Pallas}}$ and $\ell_{\text{value}} = 64$ (ZIP-226).

The byte encoding of the new field is the star-encoding of the series' conventions: ZIP-226 sets $\text{asset_base} := \text{LEBS2OSP}_{\ell_P}(\text{repr}_{\mathbb{P}}(\text{AssetBase}))$, which is $\text{AssetBase}^* \in \{0, 1\}^{256}$ in 32 bytes.

In a transfer, an Asset Base is not an arbitrary point chosen by the transactor. Every Custom Asset's base descends from the ZIP-227 issuance chain — constructed in §3, reproduced here because the soundness argument of §5 depends on its shape:

$$\begin{aligned} \text{assetDescHash} &:= \text{BLAKE2b-256}(\text{"ZSA-AssetDescCRH"}, \text{asset_desc}), \\ \text{AssetId} &:= (\text{issuer}, \text{assetDescHash}), \\ \text{EncodeAssetId}(\text{AssetId}) &:= 0x00 \parallel \text{issuer} \parallel \text{assetDescHash}, \\ \text{AssetDigest} &:= \text{BLAKE2b-512}(\text{"ZSA-Asset-Digest"}, \text{EncodeAssetId}(\text{AssetId})), \\ \text{AssetBase} &:= \text{GroupHash}^{\text{Pallas}}(\text{z.cash : OrchardZSA}, \text{AssetDigest}) \end{aligned}$$

(ZIP-227, which names the last map ZSAValueBase). For the native Asset (ZEC/TAZ) the Asset Base is instead *defined* to be

$$V^{\text{Orchard}} := \text{GroupHash}^{\text{Pallas}}(\text{z.cash} : \text{Orchard-cv}, "v"),$$

the fixed value-commitment base of the *Ironwood Guide*, “Conservation without disclosure: value commitments and the binding signature” (ZIP-226). This single identification is what keeps every ZEC computation below identical to NU5: ZIP-226 uses $\text{ValueCommit}_{\text{rcv}}^{\text{Orchard}}(v)$ as $\text{ValueCommit}_{\text{rcv}}^{\text{OrchardZSA}}(V^{\text{Orchard}}, v)$. The implementation realizes it as $\text{AssetBase}::\text{zatoshi}() = \text{hash_to_curve}(\text{"z.cash:Orchard-cv"})(b"v")$.

4.2 The note commitment: a case split

Definition 4.2 (OrchardZSA note commitment). For an OrchardZSA note,

$$\begin{aligned} & \text{NoteCommit}_{\text{rcm}}^{\text{OrchardZSA}}(g_d^*, pk_d^*, v, \rho, \psi, \text{AssetBase}) \\ & := \begin{cases} \text{NoteCommit}_{\text{rcm}}^{\text{Orchard}}(g_d^*, pk_d^*, v, \rho, \psi) & \text{if } \text{AssetBase} = V^{\text{Orchard}}, \\ \text{cm}_{\text{ZSA}} & \text{otherwise,} \end{cases} \end{aligned}$$

where

$$\begin{aligned} \text{cm}_{\text{ZSA}} := & \text{Sinsemilla}_{\text{z.cash:ZSA-NoteCommit-M}}(g_d^* \parallel pk_d^* \parallel \text{LE}_{64}(v) \parallel \text{LE}_{255}(\rho) \parallel \text{LE}_{255}(\psi) \parallel \text{AssetBase}^*) \\ & \dot{+} [\text{rcm}] \text{GroupHash}^{\text{Pallas}}(\text{z.cash} : \text{Orchard-NoteCommit-r}, "") \end{aligned}$$

(ZIP-226; the ZIP writes I2LEBSP for the series’ LE_k). The signature is

$$\begin{aligned} \text{NoteCommit}^{\text{OrchardZSA}} : & \text{NoteCommit}^{\text{Orchard}}.\text{Trapdoor} \times \{0, 1\}^{256} \times \{0, 1\}^{256} \times \{0, \dots, 2^{64} - 1\} \\ & \times \mathbb{F}_{p_{\text{Pallas}}} \times \mathbb{F}_{p_{\text{Pallas}}} \times \mathcal{E}^* \rightarrow \text{NoteCommit}^{\text{Orchard}}.\text{Output}, \end{aligned}$$

with trapdoor and output types those of $\text{NoteCommit}^{\text{Orchard}}$, unchanged (ZIP-226).

The two branches carry the design’s two promises. In the native branch the Asset Base is not hashed at all: a ZEC note commitment under OrchardZSA *is* the NU5 commitment, bit for bit. In the ZSA branch the message appends AssetBase^* (256 bits) after ψ , against the baseline layout of the *Ironwood Guide*, “The note and its commitment”:

$$g_d^* \parallel pk_d^* \parallel \text{LE}_{64}(v) \parallel \text{LE}_{255}(\rho) \parallel \text{LE}_{255}(\psi) \mapsto \dots \parallel \text{LE}_{255}(\psi) \parallel \text{AssetBase}^*,$$

under the fresh hash domain $\text{z.cash:ZSA-NoteCommit-M}$ but the *same* randomness base $\text{GroupHash}^{\text{Pallas}}(\text{z.cash} : \text{Orchard-NoteCommit-r}, "")$. ZIP-226’s rationale says that “the nested structure of the Sinsemilla-based commitment allows us to add the Asset Base as a final recursive step”, and calls the output “still indistinguishable from the original Orchard ZEC note commitments”. More precisely, the common output type lets ZSA commitments inhabit the same note commitment tree, while their computational indistinguishability rests on the hiding and hash assumptions of the randomized Sinsemilla commitment (ZIP-226). One caveat bounds the claim: Orchard (v5) note commitments and OrchardZSA (v6) note commitments are contextually distinguishable when observed in transactions, because the two protocols use different transaction versions (ZIP-226).

Remark 4.3 (Implementation). Function $\text{NoteCommit}::\text{derive}$ computes *both* commitments. For ZEC it calls $\text{CommitDomain}::\text{new}$ on the prefix “z.cash:Orchard-NoteCommit”; for ZSA it calls $\text{new_with_separate_domains}$ on the M -prefix “z.cash:ZSA-NoteCommit” and the shared

r -prefix "z.cash:Orchard-NoteCommit". It then selects between them in constant time on `asset.is_zatoshi()`. The *sinsemilla* crate appends the `-M` and `-r` suffixes to these prefixes.

The nullifier of an ordinary (non-split) OrchardZSA note is generated exactly as in Orchard (protocol specification §4.16; *Ironwood Guide*, "The nullifier, and the loop back to minting"); only Split Inputs use the modified, randomized formula of §5 (ZIP-226).

4.3 Per-asset value: the variable-base value commitment

Definition 4.4 (OrchardZSA net value commitment). Each Action commits to its net value change in its Asset:

$$\begin{aligned} \text{cv}^{\text{net}} &:= \text{ValueCommit}_{\text{rcv}}^{\text{OrchardZSA}}(\text{AssetBase}, v^{\text{net}}) = [v^{\text{net}}] \text{AssetBase} + [\text{rcv}] R^{\text{Orchard}}, \\ v^{\text{net}} &:= v_{\text{old}} - v_{\text{new}}, \quad R^{\text{Orchard}} := \text{GroupHash}^{\text{Pallas}}(\text{z.cash: Orchard-cv}, "r"), \end{aligned}$$

where `AssetBase` is the Action's Asset Base and the randomness base R^{Orchard} is unchanged (ZIP-226).

Exactly one thing changes against the baseline $\text{cv}^{\text{net}} = [v_{\text{old}} - v_{\text{new}}] V^{\text{Orchard}} + [\text{rcv}] R^{\text{Orchard}}$ of the *Ironwood Guide*, "Conservation without disclosure: value commitments and the binding signature": the fixed-base multiplication by V^{Orchard} becomes a variable-base multiplication by the Action's `AssetBase`, while the randomness term keeps its fixed base (ZIP-226). For ZEC, $\text{AssetBase} = V^{\text{Orchard}}$ and the commitment reduces to the NU5 one (§4.1).

The prover still signs the SIGHASH with $\text{bsk} = \sum_{i=1}^n \text{rcv}_i$ over the n Actions of a transfer (ZIP-226); the verifier's key acquires burn terms — one zero-trapdoor commitment subtracted per `assetBurn` entry, each a bare multiple of its base — displayed in this volume as (2) (§5.1). The verifier checks one Pallas-group equation, not independent algebraic coordinates for the assets. Since Pallas is cyclic, every pair of non-identity bases has some discrete-log relation; the separation is computational. After the randomness terms cancel, any nonzero per-asset imbalance that still verifies yields a useful relation among the independently group-hashed Asset Bases, V^{Orchard} , and R^{Orchard} . Under the discrete-log and hash-to-curve assumptions, finding such a relation is infeasible. Subject to those assumptions, the binding signature therefore enforces that each Custom Asset's net Action value equals its `assetBurn` entry (or 0 if absent), while the ZEC component equals $v_{\text{balance}}^{\text{Orchard}}$. This is the computational content of ZIP-226's "per Asset Base" claim; its wording that Pedersen commitments add homomorphically only for the same base is not a literal group-theoretic statement. The binding-signature mechanics — `bsk/bvk` cancellation and `RedPallas` on the base R^{Orchard} — are those of the *Ironwood Guide*, "Conservation without disclosure: value commitments and the binding signature", and we do not restate them.

Remark 4.5 (Implementation: `derive_bvk`). Function `derive_bvk_raw` computes the ZIP equation verbatim: $\sum \text{cv}^{\text{net}}$ minus the zero-trapdoor commitment of `value_balance` at `AssetBase::zatoshi()` minus, over the burn set, the zero-trapdoor commitment of each value at its asset. Function `ValueCommitment::derive` computes $V \cdot \text{value} + R \cdot \text{rcv}$ with V the point `asset.cv_base()` and R the point `hash_to_curve("z.cash:Orchard-cv")` applied to `b"r"`, the value signed per the sign of the value sum.

4.4 Burn

Burning is the value-balance mechanism read as an exit door. The mechanism "does NOT send Assets to a non-spendable address, it simply reduces the total number of units of a given Custom Asset in circulation"; it is enforced at consensus level as an extension of the value-balance mechanism, and it "makes it globally provable that a given amount of a Custom Asset has been destroyed". The Native Asset (ZEC/TAZ) cannot be burnt this way (ZIP-226).

ZIP-226 attaches three consensus rules to the burn set, whose cardinality it denotes L :

1. for every $(\text{AssetBase}, v) \in \text{assetBurn}$: $\text{AssetBase} \neq V^{\text{Orchard}}$;
2. $v > 0$ and $v \leq \text{MAX_BURN_VALUE} := 2^{63} - 1$;⁶
3. every AssetBase has at most one entry in assetBurn .

Burn data lives *per Action Group*: the $\text{ActionGroupDescription}$ carries $n\text{AssetBurn}$ (compactSize) and $v\text{AssetBurn}$, an array of 40-byte entries — asset_base ($\text{byte}[32]$) followed by valueBurn (uint64 , little-endian) — so that parties assembling separate Action Groups can each burn their own Assets (withdrawn ZIP-230; §4.7).

The transaction-level field $\text{valueBalanceOrchard}$ (int64) remains the net *ZEC-only* flow, “the net value of Orchard spends minus outputs” (withdrawn ZIP-230), and the transparent protocol is unchanged, so a Custom Asset cannot be unshielded by transfer: “All Custom Assets are contained within the shielded pool” (ZIP-226). Burning is the only supply-reduction path. What a burn reveals on the wire is exactly its assetBurn entry: the 32-byte Asset Base and amount — what ZIP-226 calls “the amount and identifier of the Asset being burnt”. In a transfer, “the overall balance is checked without revealing which (or how many distinct) Assets are being transferred”; under the note-encryption and commitment-hiding assumptions, the transferred amounts and Asset Bases remain private (ZIP-226).

A burn also updates ZIP-227’s global issuance state: $\text{issued_assets}(\text{AssetBase}).\text{balance} \in \{0, \dots, \text{MAX_ISSUE}\}$ is “computed as the amount of the Asset that has been issued less the amount of the Asset that has been burnt”, and the map **MUST** be updated during processing of any transaction containing burn information (ZIP-227). Supply sufficiency — that a burn not exceed the circulating supply — is therefore a state-level rule of the issuance protocol, not one of ZIP-226’s three per-transaction rules above; we construct the state machine in §3. The implementation enforces both layers side by side: $\text{validate_bundle_burn}$ returns the per-transaction errors ZatoshiAsset , ZeroAmount , InvalidAmount , and DuplicateAsset , and the state-level errors $\text{AssetNotFoundInState}$ and $\text{InsufficientSupply}$.

4.5 Split Inputs

Padding is where the asset dimension bites. An Orchard transfer with more outputs than inputs pads with *dummy* spends: zero-value notes whose Merkle membership check is gated off by v_{old} (constraint A3 of the *Ironwood Guide*, “The Action statement, formally”). For Custom Assets that padding is unsound, because the circuit forces the output note’s Asset Base to equal the input note’s (§4.6) — and a dummy input’s Asset Base is chosen freely by the prover. ZIP-226 states the attack that must be excluded: without enforcing that every Custom Asset input exists in the note commitment tree, “the prover could input a multiple (or linear combination) of an existing Asset Base, and thereby attack the network by overflowing the ZEC value balance and hence counterfeiting ZEC funds”. The fix: dummy notes remain in use for ZEC notes only; Custom Assets always prove Merkle membership, padding instead with Split Inputs, which force the output’s Asset Base to equal a genuinely issued, GroupHash -derived base (ZIP-226).

The construction is developed once, in §5: the *Split Input* — a copy of a previously issued, tree-resident note of the target Asset, spent with $\text{split_flag} = 1$ and its value zeroed in the value-commitment computation only (Definition 5.1); its randomized nullifier with the offset point L^{Orchard} , (3); the three-way divergence over the derivation of ψ^{nf} (§5.6); and the builder mechanics of padding per asset, dummy notes for ZEC and split spends for Custom Assets (§5.8). The three wallet strategies ZIP-226 offers for sourcing the copied note are enumerated in §5.2

⁶ZIP-230’s AssetBurn table instead says valueBurn is “checked by consensus to be non-zero, and less than MAX_BURN_VALUE ”. The implementation rejects only amounts strictly greater than MAX_BURN_VALUE , i.e. permits equality, agreeing with ZIP-226 (`orchard-zsa`, `src/bundle/burn_validation.rs`, which defines $\text{MAX_BURN_VALUE} = (1 \ll 63) - 1$). A specification-text defect, not a consensus divergence.

(zip-0226.rst:287–291). Burn entry points reject duplicate Assets and amounts that do not fit in 63 bits (add_burn).

4.6 The Action statement, extended

A bin statement first. ZIP-226’s “Circuit Statement” section specifies the constraint delta in prose and formulas, while deferring “all modifications in the Circuit” to the externally hosted, unversioned document *OrchardZSA circuit*. That document describes the public and private inputs, intermediate values, and constraints, but the ZIP does not embed or pin a complete statement at the rigor of the *Ironwood Guide*’s Definition “Orchard Action statement, Ironwood circuit (NU6.3)”. The pinned implementation is therefore the reproducible source for the exact gate realization, and every gate-level claim below cites it (orchard-zsa, src/circuit/circuit_zsa.rs). Bin: *externally described and implemented, but not fixed by a versioned upstream specification*.

ZIP-226 states the deltas against A1–A9 as follows:

- (a) the Asset Base is witnessed once as an auxiliary input, and both note commitment integrity constraints (A1, A2) switch to NoteCommit^{OrchardZSA} with that same witness — one variable simultaneously enforces input/output Asset equality and commitment correctness;
- (b) an auxiliary boolean is_native_asset is constrained to equal 1 exactly when AssetBase = V^{Orchard} ;
- (c) enableZSA = 0 forces is_native_asset = 1;
- (d) the value commitment’s fixed-base multiplication (A4) becomes a variable-base multiplication by the witnessed Asset Base;
- (e) a boolean split_flag replaces the spent value inside the value commitment: the committed net value becomes $v' - v_{\text{new}}$, with $v' = 0$ if split_flag = 1 and $v' = v_{\text{old}}$ otherwise; split_flag = 1 forces is_native_asset = 0;
- (f) the Merkle path constraint (A3) becomes $(v_{\text{old}} = 0 \wedge \text{is_native_asset} = 1) \vee \text{ValidMerklePath}$ — the dummy skip survives only for the native Asset;
- (g) nullifier integrity is randomized for split notes (§5).

The circuit realizes these deltas in a single gate, q_orchard, whose twelve named constraints are stated once, in §5.5, in the code’s names (the code names the ZIP’s is_native_asset as is_zatoshi_asset); we do not restate them here.

Two gadgets carry the arithmetic weight. The in-circuit value commitment computes [magnitude] asset_base by variable-base long scalar multiplication on the witnessed non-identity point, applies mul_sign for the sign, and adds the fixed-base [rcv] R^{Orchard} (src/circuit/value_commit_orchard.rs). The in-circuit note commitment shares one Sinsemilla evaluation between the two branches of Definition 4.2: it muxes the initial point Q between Q_{ZEC} (domain z.cash:Orchard-NoteCommit-M) and Q_{ZSA} (domain z.cash:ZSA-NoteCommit-M; the hard-coded point is test-asserted in src/constants/sinsemilla.rs) on is_zatoshi_asset, hashes the common message prefix once, hashes the two suffixes (ZEC: the zero-padded piece h_2 ; ZSA: the Asset Base bits), muxes the resulting hash point, and adds the shared blinding term [rcm] R — the shared randomness base of Definition 4.2 is what makes this sharing possible (src/circuit/note_commit.rs). The Asset Base message pieces: $h_{\text{ZSA}} = (\text{bits } 249\text{--}253 \text{ of } \psi) \parallel (\text{bit } 254 \text{ of } \psi) \parallel (\text{bits } 0\text{--}3 \text{ of } x(\text{asset}))$; $i = \text{bits } 4\text{--}253 \text{ of } x(\text{asset})$; $j = (\text{bit } 254 \text{ of } x(\text{asset})) \parallel (\tilde{y}(\text{asset})) \parallel 8 \text{ zero bits (ibid.)}$.

The public instance grows from 9 to 10 field elements:

(anchor, cv^{net}.x, cv^{net}.y, nf_{old}, rk.x, rk.y, cmx,
enable_spends, enable_outputs, enable_zsa),

with the new `enable_zsa` last. In the vanilla flavor `enable_zsa` is always false; the instance-construction API pads the missing tenth element with zero, preserving the vanilla flavor’s public-input semantics. The crate carries two circuit flavors, `OrchardVanilla` and `OrchardZSA` (`orchard-zsa`, `src/circuit.rs`, `src/flavor.rs`). At the bundle level the `enableZSAs` flag is bit 2 of `flagsOrchard` (LSB order, after `enableSpendsOrchard` and `enableOutputsOrchard`; the implementation constant is `FLAG_ZSA_ENABLED = 0b0000_0100`), and coinbase transactions MUST set `enableSpendsOrchard` and `enableZSAs` to 0 (withdrawn ZIP-230).

4.7 The carrier: transaction format, digests, and fees

The following encodings belong to the experimental ZIP 230/246 carrier, not ZIP 229’s deployed v6 format (§1.2). The transfer and burn constructions are draft-specified; their transaction encoding remains *designed but unspecified* for deployment.

Note plaintext. The OrchardZSA note plaintext adds `asset_base` (32 bytes) after `rseed`. Per withdrawn ZIP-230 the v6 Orchard note plaintext is

$$\text{np} := [\text{leadByte}] \parallel \text{LEBS2OSP}_{88}(d) \parallel \text{I2LEOSP}_{64}(v) \parallel \text{rseed} \parallel \text{asset_base} \parallel K^{\text{memo}},$$

with a 32-byte memo key K^{memo} replacing the 512-byte memo field, giving `encCiphertext` of 132 bytes and an OrchardZSAAction of 372 bytes (`cv` 32 + `nullifier` 32 + `rk` 32 + `cmx` 32 + `ephemeralKey` 32 + `encCiphertext` 132 + `outCiphertext` 80). ZIP-230 notes that an Orchard note “is essentially equivalent to an OrchardZSA note with `AssetBase = V^{\text{Orchard}}`” and that the two need not be distinguished in plaintexts.

Remark 4.6 (The lead byte is unassigned upstream). Upstream ZIP-226 requires `leadByte` to be the unresolved placeholder `{{ZSALEADBYTE}}` when the protocol’s note-creation sections (§4.7.3 “Sending Notes”, §4.8.3 “Dummy Notes”) are invoked in the computation of ρ and ψ for OrchardZSA notes — a sentence with no counterpart in the fork’s ZIP-226 and no implementation counterpart. Upstream ZIP-230 replaced its original `0x03` with the placeholder `{{LEADBYTE}}` because ZIP-2005 separately claims lead byte `0x03`. The QEDit fork ZIPs still say `0x03`, and the implementation hard-codes `NOTE_VERSION_BYTE_V3 = 0x03` with `MEMO_SIZE = 512` and no K^{memo} — the pre-memo-bundle layout, giving `COMPACT_NOTE_SIZE_ZSA = 52 + 32 = 84` and a 612-byte `encCiphertext`, diverging from ZIP-230’s 132 bytes (`orchard-zsa`, `src/primitives/zcash_note_encryption_domain.rs`, `src/primitives/orchard_primitives.rs`). No lead byte is currently reserved for ZSA notes upstream. Bin: *open*.

Digests. Withdrawn ZIP-246 extends the ZIP-244 digest tree — the baseline is the *Ironwood Guide*, “What the signatures sign: ZIP-244 digests and their v6 form” — with a per-Action-Group `orchard_burn_digest` (node `T.4a.vi`) under `orchard_action_groups_digest` (personalization `ZTxId0rcActGHash`), beside the compact and non-compact action digests, `flagsOrchard`, `anchorOrchard`, and `nAGExpiryHeight`. The compact digest covers `nullifier`, `cmx`, `ephemeralKey`, and `encCiphertext` (personalization `ZTxId60ActC_Hash`); the non-compact digest covers `cv`, `rk`, and `outCiphertext` (`ZTxId60ActN_Hash`). The burn digest is BLAKE2b-256 with personalization `ZTxId0rcBurnHash` over, per `assetBurn` tuple,

$$\text{assetBase} \parallel \text{valueBurn} \quad (64\text{-bit unsigned little-endian}),$$

the empty burn set hashing to `BLAKE2b-256(ZTxId0rcBurnHash, [ ])` (ZIP-246). One disambiguation: the header-digest field `T.1g burn_amount` is ZIP-233’s network-sustainability burn, unrelated to asset burn (ZIP-246, T.1; ZIP-230, Transaction Format).

Parsing and bundle types. Function `read_v6_bundle` (gated by `zcash_unstable = "nu7"`) reads `nActionGroups`, which must be 0 or exactly 1 (“A V6 transaction must contain at most one action group”), then the actions, flags, anchor, `nAGExpiryHeight` (must be 0), the burn vector (32-byte Asset Base plus note value), the proof, per-action versioned signatures, `valueBalance`, and a versioned binding signature; the write path is symmetric via `write_burn` (`librustzcash-zsa`, `zcash_primitives/src/transaction/components/orchard.rs`). The bundle type carries the burn set as the field `burn: Vec<(AssetBase, NoteValue)>` beside `value_balance`, and `binding_validating_key()` returns `derive_bvk(actions, value_balance, burn)`; the builder collects burns in a `BTreeMap<AssetBase, NoteValue>`, canonically ordered and duplicate-free by construction (`orchard-zsa`, `src/bundle.rs`, `src/builder.rs`). One open seam: ZIP-230 permits `nActionGroupsOrchard > 1` with per-group burn fields (the multi-party rationale of §4.4), the implementation rejects more than one group, and ZIP-226’s `bvk` equation is written over a single flat action/burn set; how multi-group aggregation would interact with the binding signature is unexercised. Bin: *open*.

Tree ordering. Per Action Group, every new OrchardZSA note commitment is appended to the note commitment tree per Action; afterwards, Issue Note commitments are appended per Issue Action (ZIP-227, “Addition to the Note Commitment Tree”; §3).

Fees. The QEDit fork of ZIP-317 adds `contribution_ZSAIssuance` and `contribution_ZSACreation` terms — issue notes and asset creations respectively — to the conventional-fee formula; it adds no burn term, so a burn contributes to fees only through the Actions that fund it. Upstream ZIP-317 contains no ZSA changes. The conventional-fee baseline is the *Wallet Guide*, “Fees (ZIP-317)”.

5 Split notes and the counterfeiting boundary

An Orchard bundle pairs inputs and outputs into Actions by padding the shorter side with *dummy notes*: an Action with no real input spends a zero-valued note under a freshly sampled key, and the membership constraint waves it through because the check is gated by the spent value — constraint A3 of the Action statement, $v_{\text{old}} \cdot (\text{root} - \text{anchor}) = 0$, “how an Action pads with no real input” (the *Ironwood Guide*, “The Action statement, formally”; its node-verification section, “What a node verifies, without ever observing the note”, records that zero-value dummy spends are accepted by design). The construction is protocol §4.8.3, implemented in `orchard-zsa`, `SpendInfo::dummy`: a random spending key, a zero-valued note, and a random Merkle path that is never checked. The exemption is sound there because every Orchard value commitment rides the *fixed* base V^{Orchard} (*Ironwood Guide*, “Conservation without disclosure: value commitments and the binding signature”): a dummy spend can move no value, whatever path it fails to prove. ZIP-226 replaces the fixed base with a witnessed one, and the same exemption becomes a mint. This section develops the attack that forces the change, the *split note* that ZIP-226 substitutes for the dummy note on custom assets, the ZIP-specified circuit delta, and the implemented gate that draws the counterfeiting boundary.

Ground truth for this section: ZIP-226 and ZIP-227 at `zcash/zips` commit 69610984 (2026-07-09); the pinned implementation, crate `orchard` of the QED-it fork (`orchard-zsa`, branch `zsa1`) at commit `cf801a5d` (2026-06-29); the QED-it `zips` fork at `fd71419a` where it diverges from upstream. Split-note logic lives entirely in the `orchard` crate; the `librustzcash` fork consumes its bundle API and contains no split-note code of its own (verified by search at `3f9b53fc`).

5.1 The counterfeit construction the boundary excludes

ZIP-226 generalises the net value commitment from the fixed base V^{Orchard} to a per-asset base (ZIP-226, “Value Commitment”):

$$\begin{aligned} \text{cv}^{\text{net}} &:= \text{ValueCommit}_{\text{rcv}}^{\text{OrchardZSA}}(\text{AssetBase}_{\text{AssetId}}, v_{\text{AssetId}}^{\text{net}}) \\ &= [v_{\text{AssetId}}^{\text{net}}] \text{AssetBase}_{\text{AssetId}} + [\text{rcv}] R^{\text{Orchard}}, \end{aligned} \quad (1)$$

with $v^{\text{net}} := v_{\text{old}} - v_{\text{new}}$ and R^{Orchard} the randomness base of the *Ironwood Guide*’s net value commitment (“Conservation without disclosure: value commitments and the binding signature”), unchanged. The native asset keeps the old base: “the Asset Base of the ZEC Asset will be kept as the original value base point $\mathcal{U}^{\text{Orchard}}$ ” (ZIP-226, “Asset Identifiers”; in code `AssetBase::zatoshi()` is `GroupHashPallas(z.cash:Orchard-cv, "v")`). Verification subtracts the declared native balance and the declared burns (ZIP-226, “Value Balance Verification”; implemented in `derive_bvk_raw`):

$$\begin{aligned} \text{bvk} &= \left(\sum_{i=1}^n \text{cv}_i^{\text{net}} \right) - \text{ValueCommit}_0^{\text{OrchardZSA}}(V^{\text{Orchard}}, v^{\text{balanceOrchard}}) \\ &\quad - \sum_{(\text{AssetBase}, v) \in \text{assetBurn}} \text{ValueCommit}_0^{\text{OrchardZSA}}(\text{AssetBase}, v), \end{aligned} \quad (2)$$

and the binding signature under bvk proves knowledge of $\text{bsk} = \sum_i \text{rcv}_i$ exactly as in Orchard.

Suppose now that the dummy exemption survived unchanged: a padding Action proves no membership, and — since the base of (1) is a private witness — may witness *any* point A as its asset base. Choose $A := [-1] V^{\text{Orchard}}$, witness $v_{\text{old}} = 0$ and $v_{\text{new}} = v$:

$$\text{cv}_1^{\text{net}} = [0 - v] A + [\text{rcv}_1] R^{\text{Orchard}} = [v] V^{\text{Orchard}} + [\text{rcv}_1] R^{\text{Orchard}}.$$

Pair it with a second Action that outputs a genuine v -valued ZEC note,

$$\text{cv}_2^{\text{net}} = [0 - v] V^{\text{Orchard}} + [\text{rcv}_2] R^{\text{Orchard}},$$

and declare $v^{\text{balanceOrchard}} = 0$ with no burns. The V^{Orchard} -components cancel in (2),

$$\text{bvk} = \text{cv}_1^{\text{net}} + \text{cv}_2^{\text{net}} = [\text{rcv}_1 + \text{rcv}_2] R^{\text{Orchard}},$$

the attacker knows $\text{bsk} = \text{rcv}_1 + \text{rcv}_2$, the binding signature verifies — and a v -valued ZEC note now exists that no one paid for. Suitably chosen known relations among accepted bases can be exploited analogously. (*Classification*: the construction is assembled here from the specified formulas (1) and (2); ZIP-226 states the attack only in words.) The ZIP’s own statement, verbatim: the output note of split Actions “cannot contain just any Asset Base. We must enforce it to be an actual output of a GroupHash computation... Without this enforcement the prover could input a multiple (or linear combination) of an existing Asset Base, and thereby attack the network by overflowing the ZEC value balance and hence counterfeiting ZEC funds” (ZIP-226, “Split Notes”, rationale).

The boundary to draw is therefore precise: every asset base that enters a value commitment must either be the fixed native base or descend from an accepted GroupHash output. Security then assumes that no useful discrete-log relation among that base, V^{Orchard} , and R^{Orchard} can be found. The one place an unproven base could otherwise enter is an Action with no real input.

5.2 The split input

Definition 5.1 (Split Input, Split Action). ZIP-226 defines a *Split Input* as “an Action input used to ensure that the output note of that Action is of a validly issued AssetBase when there is no corresponding real input note, in situations where the number of outputs are larger than the number of inputs”, and a *Split Action* as an Action containing a Split Input (ZIP-226, “Terminology”).

Mechanically, a Split Input is a *copy of a previously issued note* — one whose commitment is already in the note commitment tree — with exactly two changes: a boolean witness `split_flag` is set to 1, and the note’s value is replaced by 0 *in the value-commitment computation only* (ZIP-226, “Split Notes”). The copied note’s commitment cm_{old} still opens with the true value: in the circuit the value is zeroed solely in the cv^{net} equation, while the old-note commitment check consumes v_{old} unchanged (`orchard-zsa, src/circuit/circuit_zsa.rs`: the gate zeroes v_{old} via the $(1 - \text{split_flag})$ factor in the value equation, lines 127–131, whereas the `NoteCommit` call at lines 637–655 takes v_{old} as witnessed). The Action therefore neither moves value from the copied note nor publishes that note’s ordinary nullifier; instead it publishes the randomized split nullifier of §5.6. The copy drags the original note’s asset base into the proof, where an equality constraint passes it to the output note.

The split Action participates in the same computational per-asset binding argument as §4.3: a successful imbalance would require a useful discrete-log relation among the independently derived bases. For a correct transaction the custom-asset commitments satisfy

$$\sum_{j \in S_{CA}} cv_j^{net} - \sum_{(AssetBase, v) \in \text{assetBurn}} [v] AssetBase = \sum_{j \in S_{CA}} [rcv_j] R^{\text{Orchard}},$$

so the split Action’s $cv^{net} = [0 - v_{new}] AssetBase + [rcv] R^{\text{Orchard}}$ cancels against the real spending Actions of the same asset. The builder implements exactly this: `ActionInfo::value_sum` uses a spent value of 0 when `split_flag` is set, and `derive_bvk_raw` computes (2) over the resulting commitments.

For the note to copy, ZIP-226 offers the wallet three options: (1) another note of the same Asset Base being spent in the same transaction; (2) a different unspent note of that Asset Base, which is *not* thereby spent; (3) an already spent note of that Asset Base — the zeroed value prevents double spending (ZIP-226, “Split Notes”). The ZIP adds that “we do not care about whether the previously issued note copied to create a Split Input is owned by the sender, or whether it was nullified before”; §5.7 examines how far that sentence actually holds against the constraint system.

5.3 Membership as well-formedness: the issuance anchor

The fix ZIP-226 specifies, verbatim: “for Custom Assets we enforce that every input note to an OrchardZSA Action must be proven to exist in the set of note commitments in the note commitment tree. We then enforce this real note to be unspendable in the sense that its value will be zeroed in split Actions and the nullifier will be randomized... the proof itself ensures that the output note is of the same Asset Base as the input note. In the circuit, the split note functionality will be activated by a boolean private input (aka the `split_flag` boolean)” (ZIP-226, “Split Notes”, rationale).

Why does tree membership prove that an asset base is a GroupHash output? Because for custom assets the tree is fed exclusively from two sources that both enforce it. Transfer outputs inherit their base from a proven input by the in-circuit equality constraint (§5.4). Issuance outputs are checked publicly: a consensus rule obliges validators to recompute each Issue Note’s AssetBase from the issuance bundle’s issuer key and the IssueAction’s `assetDescHash` (ZIP-227, “Specification: Consensus Rule Changes”), along the derivation chain constructed in §2.3 (ZIP-227, “Asset Bases”), and then appends the issue-note commitments to the *same* Orchard note commitment tree — transfer-Action commitments first, then issuance notes (ZIP-227, “Addition to

the Note Commitment Tree”). Hence, under note-commitment binding, a custom-asset note commitment resident in the tree commits, by induction along the two rules, to a genuine GroupHash output. ZIP-226 states the conclusion: “This ensures that the value base points of all output notes of a transfer are actual outputs of a GroupHash, as they originate in the Issuance protocol which is publicly verified” (“Split Notes”, rationale); and, on the identifier side, “We prevent a potential malleability attack on the Asset Identifier by ensuring the output notes receive an Asset Base that exists on the global state” (“Rationale for Asset Identifiers”).

One clarification the ZIP’s sentence invites: “every input note to an OrchardZSA Action” is scoped to *custom-asset* inputs. Native-asset (ZEC) Actions in a v6 bundle retain the dummy-note exemption unchanged — “The ZEC-based Actions will still include dummy input notes, whereas the OrchardZSA Actions will include split input notes and will not include dummy input notes” (ZIP-226, “Backward Compatibility”; the “Split Notes” rationale states the same). The constraint-level form of this scoping is the disjunction in §5.4.

5.4 The specified circuit delta

We state ZIP-226’s circuit changes (“Circuit Statement”) as a delta against the Action statement A1–A9 of the *Ironwood Guide* (“The Action statement, formally”). Two new boolean witnesses appear: `split_flag`, and `is_native_asset`, constrained to equal 1 exactly when the witnessed base is the native one, $\text{AssetBase} = V^{\text{Orchard}}$. One public input appears: the bundle-level `enableZSA` flag (a bit of the `v6 flagsOrchard` field), with the constraint $\text{enableZSA} = 0 \implies \text{is_native_asset} = 1$ — switching the flag off collapses the circuit to native-asset behaviour and, transitively via the split constraint below, disables split notes (ZIP-226, “Overview” and “Circuit Statement”, The `enableZSA` Flag; in code the flag is the tenth public input, taken from `Flags::zsa_enabled()`, `orchard-zsa`, `src/circuit.rs`, lines 476–494, 560).

- **A1, A2 (note commitment integrity).** Both commitment checks replace $\text{NoteCommit}^{\text{Orchard}}$ with $\text{NoteCommit}^{\text{OrchardZSA}}$, which takes the witnessed `AssetBase` as an additional final argument; the *same* once-witnessed base feeds the old-note check, the new-note check, and the value commitment (ZIP-226, “Circuit Statement”, Asset Base Equality). For the native asset $\text{NoteCommit}^{\text{OrchardZSA}}$ coincides with $\text{NoteCommit}^{\text{Orchard}}$; otherwise it is the Sinsemilla commitment over the extended message ending in the asset-base encoding, in the hash domain “z.cash:ZSA-NoteCommit-M” with the blinding base shared with Orchard (“z.cash:Orchard-NoteCommit-r”) (ZIP-226, “Note Structure and Commitment”). The full $\text{NoteCommit}^{\text{OrchardZSA}}$ construction is Definition 4.2; its in-circuit mux is described in §4.6.
- **A3 (membership).** The value gate becomes a disjunction (ZIP-226, “Circuit Statement”, Asset Identifier Consistency for Split Actions — Merkle Path Validity):

$$(v_{\text{old}} = 0 \wedge \text{is_native_asset} = 1) \vee (\text{Valid Merkle Path}),$$

so the dummy skip survives *only* for the native asset. Every custom-asset input — split or not, zero-valued or not — must prove membership.

- **A4 (value commitment).** Two changes. First, “the fixed-base multiplication constraint between the value and the value base point of the value commitment, `cv`, is replaced with a variable-base multiplication between the two” (ZIP-226, “Circuit Statement”, Value Commitment Correctness), the base being the witnessed `AssetBase`. Second, the input value is replaced by a generic v' (ZIP-226, “Circuit Statement”, Asset Identifier Consistency for Split Actions):

$$cv^{\text{net}} = \text{ValueCommit}_{\text{rcv}}^{\text{OrchardZSA}}(\text{AssetBase}, v' - v_{\text{new}}), \quad v' = \begin{cases} 0 & \text{if } \text{split_flag} = 1, \\ v_{\text{old}} & \text{otherwise,} \end{cases}$$

together with $\text{split_flag} = 1 \implies \text{is_native_asset} = 0$: split notes exist only for custom assets, and a zero-valued native dummy cannot masquerade as one.

- **A5 (nullifier).** The derivation changes for split notes “to prevent the identification of notes” (ZIP-226, “Circuit Statement”, Asset Identifier Consistency for Split Actions); §5.6 develops it.
- **A6, A7 (spend authority, address integrity).** Untouched by the ZIP’s change list — and, decisively for §5.7, they remain *unconditional*: no `split_flag` factor relaxes them.

5.5 The implemented gate

A naming note, once: the ZIP’s `is_native_asset` is `is_zatoshi_asset` throughout the implementation, assigned 1 exactly when the witnessed asset equals `AssetBase::zatoshi()` (`orchard-zsa`, `src/circuit/gadget.rs`, `circuit_zsa.rs`). We keep the ZIP name in prose and the code name inside code-derived formulas.

The implemented circuit concentrates the delta in a single gate, `q_orchard`, which carries twelve named constraints where the vanilla circuit’s corresponding gate has four (`src/circuit/circuit_zsa.rs`, lines 68–185, versus `src/circuit/circuit_vanilla.rs`, lines 59–100). In the code’s names:

$$\begin{aligned}
& \text{bool_check}(\text{split_flag}), \quad \text{bool_check}(\text{is_zatoshi_asset}), \\
& v_{\text{old}} \cdot (1 - \text{split_flag}) - v_{\text{new}} = \text{magnitude} \cdot \text{sign}, \\
& (v_{\text{old}} + 1 - \text{is_zatoshi_asset}) \cdot (\text{root} - \text{anchor}) = 0, \\
& v_{\text{old}} = 0 \vee \text{enable_spends} = 1, \quad v_{\text{new}} = 0 \vee \text{enable_outputs} = 1, \\
& \text{is_zatoshi_asset} \cdot (\text{asset}_x - \text{zatoshi}_x) = 0, \quad \text{is_zatoshi_asset} \cdot (\text{asset}_y - \text{zatoshi}_y) = 0, \\
& (1 - \text{is_zatoshi_asset}) \cdot (\Delta_x \cdot \Delta_x^{\text{inv}} - 1) \cdot (\Delta_y \cdot \Delta_y^{\text{inv}} - 1) = 0, \\
& (1 - \text{split_flag}) \cdot (\psi_{\text{old}} - \psi^{\text{nf}}) = 0, \quad \text{split_flag} \cdot \text{is_zatoshi_asset} = 0, \\
& (1 - \text{enable_zsa}) \cdot (1 - \text{is_zatoshi_asset}) = 0,
\end{aligned}$$

where Δ_x, Δ_y abbreviate the code’s `diff_asset_x, diff_asset_y` — the witnessed asset’s coordinates minus the native base’s — and ψ^{nf} is the code’s `psi_nf` (§5.6). Four points merit unpacking.

The folded value equation. The single constraint $v_{\text{old}} \cdot (1 - \text{split_flag}) - v_{\text{new}} = \text{magnitude} \cdot \text{sign}$ folds the ZIP’s case-split v' into one expression; the witness generator supplies the matching net value, $-v_{\text{new}}$ for split spends and $v_{\text{old}} - v_{\text{new}}$ otherwise (`circuit_zsa.rs`, lines 462–501).

The Merkle factor. The disjunction of A3 is encoded as the single product $(v_{\text{old}} + 1 - \text{is_zatoshi_asset}) \cdot (\text{root} - \text{anchor}) = 0$. The code comment argues the encoding safe from wrap-around: `is_zatoshi_asset` is boolean-checked and v_{old} is range-checked to 64 bits in the note-commitment evaluation, so over $\mathbb{F}_{p_{\text{Pallas}}}$ the first factor vanishes exactly when $v_{\text{old}} = 0$ and `is_zatoshi_asset` = 1 (`circuit_zsa.rs`, lines 132–139). The consequence bears repeating: for every custom-asset input the factor is nonzero, so `root = anchor` is mandatory — a strictly stronger membership requirement than vanilla Orchard’s v_{old} -gate, applying even to zero-valued non-split custom-asset inputs.

Proving inequality. The assignment `is_zatoshi_asset = 0` must itself be sound: the circuit must verify that the witnessed base *differs* from V^{Orchard} . The prover witnesses field inverses $\Delta_x^{\text{inv}}, \Delta_y^{\text{inv}}$ (assigned 0 when the corresponding difference vanishes), and the triple product above forces at least one difference to be exhibited invertible (`circuit_zsa.rs`, lines 160–170 for the gate, 792–841 for the assignment).

The variable-base value commitment. With the base witnessed, the vanilla circuit’s fixed-base short multiplication $[v^{\text{net}}]V^{\text{Orchard}}$ — whose 64-bit range check is implicit in the short-multiplication decomposition — is no longer available. The ZSA gadget makes the range check explicit: the magnitude is decomposed against the Sinsemilla lookup table as six 10-bit windows plus one 4-bit short check (64 bits), then promoted to a full-width scalar (`ScalarVar::from_base`) and fed to the variable-base long multiplication — the code notes it is “optimized for ~255-bit scalar” and leaves a TODO to specialise it for 64 bits — after which the sign is applied and the blind $[rcv]R^{\text{Orchard}}$ added (`src/circuit/value_commit_orchard.rs`, lines 38–127). Two supporting details: the ZSA circuit uses the extended lookup configuration `PallasLookupRangeCheck4_5BConfig` (an additional lookup-table column for 4- and 5-bit short checks), and it refuses to synthesise under `CircuitVersion::InsecureUnanchoredBase`, the pre-NU6.2 `halo2_gadgets` variable-base scalar-multiplication version whose base point was not anchored — directly load-bearing here, because the entire counterfeiting boundary rests on the multiplication being by the *witnessed* `AssetBase` and nothing else (`circuit_zsa.rs`, lines 22, 51, 190–197, 334–337; `circuit.rs`, lines 151–185).

The flag interactions are tested asymmetrically: the three legal combinations of

$$(\text{is_zatoshi_asset}, \text{split_flag})$$

— $(1, 0)$, $(0, 1)$, and $(0, 0)$ — are proven passing, while the $(1, 1)$ combination is excluded at witness-generation time by the test helper’s assertion (“We cannot create a split note with a zatoshi asset”), so the constraint $\text{split_flag} \cdot \text{is_zatoshi_asset} = 0$ is not itself exercised by a failing-proof test (`circuit_zsa.rs`, lines 1174–1312). Documented cost: the ZSA proof measures 5120 bytes for one Action and 7392 for two, against the vanilla 4992 and 7264, at the shared circuit size $K = 11$; the `enableZSA` bit is the tenth public input, at index 9 (the proof-size tests of `circuit_zsa.rs` and `circuit_vanilla.rs`; `circuit.rs`, lines 68–80, 536–563).

5.6 The split nullifier

A split input copies a note that may sit unspent in someone’s wallet; its Action must nevertheless publish *some* nullifier. Publishing the note’s true nullifier would be harmful in both directions: it would mark an unspent note spent (bricking it), and it would collide with the honest nullifier if the same note were spent for real in the same transaction. ZIP-226 therefore randomises the split note’s nullifier (“Split Notes”):

$$\text{nf}_{\text{old}} = \text{Extract}([F_{\text{nk}}(\rho_{\text{old}}) + \psi^{\text{nf}}]G^{\text{nf}} + \text{cm}_{\text{old}} + L^{\text{Orchard}}), \quad (3)$$

in the notation of the *Ironwood Guide*’s A5 (“The Action statement, formally”; ZIP-226 writes $\mathcal{K}^{\text{Orchard}}$ for G^{nf} and computes the bracketed sum in the Pallas base field). Two deltas against A5: the note’s ψ_{old} is replaced by a fresh ψ^{nf} , and a fixed offset point

$$L^{\text{Orchard}} := \text{GroupHash}^{\text{Pallas}}(\text{z.cash} : \text{Orchard}, "L")$$

is added (ZIP-226, “Split Notes”; the constant is pinned and test-asserted against the `GroupHash` evaluation in `orchard-zsa`, `src/constants/nullifier_1.rs`). The gate constraint $(1 - \text{split_flag}) \cdot (\psi_{\text{old}} - \psi^{\text{nf}}) = 0$ of §5.5 makes the substitution split-only: non-split OrchardZSA nullifiers coincide with the Orchard derivation, as the ZIP states (“Note Structure and Commitment”: the non-split nullifier “is generated in the same manner as in the Orchard protocol”).

In circuit, the nullifier gadget computes the standard point $\text{cm}_{\text{old}} + [F_{\text{nk}}(\rho_{\text{old}}) + \psi^{\text{nf}}]G^{\text{nf}}$ from the witnessed ψ^{nf} , forms the offset variant by adding the constant L^{Orchard} , and muxes the two on `split_flag` before extraction (`src/circuit/derive_nullifier.rs`, lines 54–111). Out of circuit, `Nullifier::derive` mirrors the computation and selects the offset variant in constant time exactly when the note carries a `rseed_split_note` (`src/note/nullifier.rs`, lines 73–90; `src/note.rs`, lines 415–426).

Deriving ψ^{nf} : a three-way divergence. The value of ψ^{nf} is always constrained through the A5 equation to the published nullifier. Only the equality $\psi^{\text{nf}} = \psi_{\text{old}}$ is gated off in the split case; no constraint derives ψ^{nf} from a seed. How a wallet samples it for a split note is therefore wallet-level, not consensus-level. The three extant texts disagree; per the volume’s policy (§1.3), the upstream ZIP formula comes first. Upstream ZIP-226 — written against the ZIP-2005 baseline, “expected to deploy before any ZSA activation” — specifies

$$\psi^{\text{nf}} := \text{ToBase}(\text{PRFexpand}_{\text{rseed_nf}}(0x0A \parallel \rho_{\text{old}}))$$

with `rseed_nf` uniform on 32 bytes and the domain byte `0x0A` “reserved by ZIP 2005 for this use” (`zip-0226.rst:297 @ 69610984`). The QED-it zips fork specifies neither derivation: there ψ^{nf} “is sampled uniformly at random” on the Pallas base field (fork `zip-0226.rst:293 @ fd71419a`). The implementation derives

$$\psi^{\text{nf}} := \text{ToBase}(\text{PRFexpand}_{\text{rseed_split_note}}(0x09 \parallel \rho_{\text{old}})),$$

the ordinary ψ domain byte keyed by a fresh random 32-byte `rseed_split_note` (`orchard-zsa, src/circuit.rs:316–319, src/note.rs:114–116 and 417–425`; domain byte `PSI = 0x09` per the QED-it `zcash_spec` fork, `src/prf_expand.rs:89 @ d5e84264`). Direct field sampling is uniform; the two PRF-and-reduction constructions are intended to be pseudorandom and have at most the negligible modular-reduction bias of `ToBase`. The implementation matches neither ZIP text exactly. (*Classification*: each derivation is specified in its own text; the divergence is consensus-invisible.)

Why the offset L^{Orchard} ? The randomised ψ^{nf} is intended to make the split nullifier unlinkable to the note’s true nullifier. ZIP-226 states no rationale for additionally adding L^{Orchard} . A plausible purpose is robustness against a prover who chooses $\psi^{\text{nf}} = \psi_{\text{old}}$ — the circuit does not forbid it in the split case — and would thereby publish the note’s true nullifier from a zero-valued split Action, bricking the original note or colliding with an honest spend of it elsewhere in the transaction. The constant offset makes the pre-extraction group point different even under that choice. It is not an absolute noncollision proof for the extracted nullifier, since elliptic-curve points with opposite signs have the same extracted x -coordinate. (*Classification*: designed-but-unspecified; the preceding sentence is our inference, stated in neither ZIP nor code.)

5.7 Ownership, in fact and in prose

Recall the ZIP’s claim that “we do not care about whether the previously issued note copied to create a Split Input is owned by the sender” (§5.2). Both the proof and the Action signature narrow that claim. Spend-authority consistency (A6) and address integrity (A7) are unchanged and unconditional in the ZSA circuit: the proof establishes $\text{rk} = [\alpha]G^{\text{SpendAuth}} + \text{ak}$ and $\text{ivk} = \text{Commit}_{\text{r}_{\text{ivk}}}^{\text{ivk}}(\text{Extract}(\text{ak}), \text{nk})$ with $\text{pk}_{\text{dold}} = [\text{ivk}]g_{\text{dold}}$ — for split inputs too (`orchard-zsa, src/circuit/circuit_zsa.rs, lines 554–623`; the old-note commitment equality at lines 625–659). Because the copied note’s `cm_old` must open to the actual tree leaf (Sinsemilla binding plus the mandatory Merkle check of §5.5), the witnessed pk_{dold} is the original recipient’s, and A7 then forces the prover to know $(\text{ak}, \text{nk}, \text{r}_{\text{ivk}})$ consistent with it — the full-viewing-key components or an equivalent witness. More decisively, every Action must carry a RedPallas spend authorization signature under `rk`. A split spend has `dummy_sk = None`; preparation auto-signs only true dummy spends, and `apply_signatures` finishes a split Action only after receiving a `SpendAuthorizingKey` whose validating key matches `ak`, otherwise returning `MissingSignatures` (`src/builder.rs, lines 308–315 and 1347–1432, SpendInfo::create_split_spend, Bundle::prepare, Bundle::apply_signatures, and Bundle::sign`). Thus constructing a valid transaction requires the copied note’s spend-authorizing key, absent a signature forgery or discrete-log break. The ZIP’s sentence holds

for *spentness* — a nullified note can still be copied because its value is zeroed — but not for ownership. The builder consequently copies the first real spend of that Asset in the same bundle, option (1) of §5.2 (`src/builder.rs`, `pad_spend`). (*Classification*: the circuit and signature behavior are implemented; the ZIP rationale overstates the available choices.)

One genuine exception exists, and neither ZIP states it. ZIP-227 requires the first note of any asset’s first issuance to be a *reference note*: value 0, recipient the default diversified address of the all-zero Orchard spending key — a key anyone can derive (ZIP-227, “Reference Notes”; `orchard-zsa`, `src/constants/reference_keys.rs` pins the all-zero spending key and the 43-byte raw address). Because that spending key is public, anyone can satisfy A6/A7 and produce the Action signature for the reference note, which makes it a universally available split input for its asset — the one case in which “ownership does not matter” holds literally. (*Classification*: inference; the connection between reference notes and split inputs is stated in neither ZIP nor the implementation.)

5.8 Builder mechanics: padding, per asset

The two padding regimes coexist per ZIP-226’s rule quoted in §5.3: dummy inputs for ZEC Actions, split inputs for custom-asset Actions. The builder partitions spends and outputs by asset, pads each side of each partition to equal length, shuffles within each asset, then shuffles the assembled Actions across assets (`orchard-zsa`, `src/builder.rs`, lines 1081–1164). The padding spend is asset-dependent (`pad_spend`, lines 928–944): for the native asset, `SpendInfo::dummy` exactly as in Orchard; for a custom asset, `create_split_spend` on the first real spend of that asset in the bundle. If the bundle spends none, `pad_spend` produces `BuildError::NoSplitNoteAvailable`; the current per-asset iterator unwraps that result and panics with the error rather than returning it from `bundle_for_version`. Unlike a dummy, a split input cannot be conjured from nothing. Output-side padding uses dummy outputs of the padding asset in both regimes.

The copy itself is minimal. Function `Note::create_split_note` duplicates the note and sets exactly one new field, `rseed_split_note` — a fresh random 32-byte seed, drawn by `RandomSeed::random` and resampled until the derived esk is valid — asserting the asset is not the native one; `SpendInfo::create_split_spend` keeps the original note’s full viewing key and Merkle path, sets no dummy signing key, and raises `split_flag` (`src/note.rs`, lines 436–442, 95–103; `src/builder.rs`, lines 302–318). Recipient, value, asset, ρ , `rseed` — hence `cmold` — are identical to the tree-resident original; the only witness-level differences downstream are `split_flag = 1` and the ψ^{nf} drawn from the new seed. Witness assembly routes exactly that: `Witnesses::from_action_context_unchecked` takes ψ^{nf} from `rseed_split_note` when present and from `rseed` otherwise — so $\psi^{\text{nf}} = \psi_{\text{old}}$ for every non-split spend — and packs $(\psi^{\text{nf}}, \text{asset}, \text{split_flag})$ into the ZSA witness extension that the vanilla flavour leaves unknown (`src/circuit.rs`, lines 199–244, 300–345).

The counterfeiting boundary, restated once in full: a custom-asset output note’s base equals its Action’s input-note base (the A1/A2 equality on the shared witness); that input note is a tree leaf (the mandatory Merkle factor); under note-commitment binding, every custom-asset tree leaf descends from a publicly recomputed `GroupHash` output (the issuance anchor of §5.3); and the copy that makes padding possible moves no value (the zeroed v'). Those links enforce the counterfeiting boundary of §5.1. Independently, the randomized nullifier (3) avoids publishing the original note’s ordinary nullifier except with negligible collision probability under honest randomisation; it is not an absolute noncollision guarantee (§5.6).

6 Transaction integration and outlook

The preceding sections construct the OrchardZSA objects — asset bases (§2), notes and Actions (§4, §5), issuance (§3), and burn (§4.4). This section places them in a transaction: the v6 wire for-

mat (ZIP-230), the digest tree that yields transaction identifiers, sighashes, and the authorizing-data commitment (ZIP-246), the transaction-level consensus rules of ZIP-226/227, the fee treatment (ZIP-317), the state of the QED-it implementation stack at the pins in Table 1, and the obligations the format imposes on wallets. The format is experimental; §1.2 states its deployment boundary.

6.1 Experimental and deployed carriers

The carrier below is the experimental OrchardZSA format of ZIP 230, with digests from ZIP 246. Both ZIPs are withdrawn. Its version number 6 must not be confused with ZIP 229’s deployed version 6 format, which carries Ironwood and no ZSA bundles. The two byte formats are incompatible despite the shared number.

We use the experimental format to explain how the note, issuance, burn, and signature constructions compose into one transaction. Its fields and digests are *designed but unspecified* for deployment; the remaining specification obligations are in §6.8.

6.2 The experimental wire format (ZIP-230)

Common and transparent fields. The common header carries header (f0verwintered and the version), nVersionGroupId, nConsensusBranchId, lock_time, nExpiryHeight, an explicit 8-byte fee (uint64, per ZIP-2002), and an 8-byte zip233Amount (uint64, per ZIP-233) (ZIP-230, Transaction Format). Making the fee an explicit field replaces its derivation as the residual difference between transaction input and output value and lets clients read it without resolving prior transparent outputs (ZIP-2002). Separately, signatures become versioned below. The transparent fields add vSighashInfo: one TransparentSighashInfo (a compactSize length followed by sighashInfo bytes) per transparent input (ZIP-230, Transaction Format).

Orchard fields: Action Groups. The Orchard component becomes a vector of *Action Groups*: nActionGroupsOrchard, vActionGroupsOrchard, then a transaction-level valueBalanceOrchard (int64) and bindingSigOrchard. The balance and binding signature are present iff nActionGroupsOrchard > 0; an absent valueBalanceOrchard means $v_{\text{balance}}^{\text{Orchard}} = 0$ (ZIP-230, Transaction Format). Each ActionGroupDescription contains, in order (ZIP-230, OrchardZSA Action Group Description):

- nActionsOrchard (compactSize, MUST be > 0) and vActionsOrchard, at 372 bytes per OrchardZSAAction;
- flagsOrchard, LSB-first: enableSpendsOrchard, enableOutputsOrchard, enableZSAs, remaining bits zero;
- anchorOrchard (32 bytes) — each group carries its own anchor;
- nAGExpiryHeight (uint32);
- nAssetBurn and vAssetBurn, at 40 bytes per AssetBurn entry;
- sizeProofsOrchard and proofsOrchard, one aggregated proof for the group;
- vSpendAuthSigsOrchard, one OrchardSignature per Action.

Field nAGExpiryHeight exists for forward compatibility with ZSA atomic swaps. ZIP-228 is now a Draft that uses this field to expire swap orders, but it still depends on the withdrawn ZIP-230/246 carrier; for the core OrchardZSA proposal the field is zero by consensus (ZIP-230, Rationale for nAGExpiryHeight; §6.4). This restriction concerns nAGExpiryHeight, not the transaction-level expiry field nExpiryHeight. Burn fields sit inside each Action Group rather than at transaction level so that, in a future multi-party Action Group world, each party can burn assets within its own group (ZIP-230, Rationale for including Burn fields inside OrchardZSA Action Groups).

Action and burn encodings. An OrchardZSAAction occupies 372 bytes: cv (32), nullifier (32), rk (32), cmx (32), ephemeralKey (32), encCiphertext (132), and outCiphertext (80) (ZIP-230, OrchardZSA Action Group Description). The 132-byte ciphertext is the memo-bundle-era layout: the note plaintext gains the 32-byte asset base and replaces the in-band 512-byte memo with a 32-byte memo key K^{memo} (ZIP-231), the memo chunks themselves moving to a transaction-level memo bundle (fields nMemoChunks, pruned, vMemoChunks, present iff fAllPruned = 0) whose pruned entries are replaced by their digests (ZIP-230, Transaction Format; §6.3). The v6 note plaintext is

$$\text{leadByte} \parallel \text{LEBS2OSP}_{88}(d) \parallel \text{I2LEOSP}_{64}(v) \parallel \text{rseed} \parallel \text{asset_base} \parallel K^{\text{memo}},$$

with $\text{asset_base} = \text{LEBS2OSP}_{\ell_p}(\text{repr}_p(\text{AssetBase}))$, and the lead byte mandatory for all v6 note plaintexts (ZIP-230, note plaintext encoding). Upstream, the lead byte is the unassigned placeholder `{{LEADBYTE}}`: the original assignment 0x03 was taken by ZIP-2005, and no replacement is assigned (§6.6). For the same K^{memo} reason the Sapling OutputDescriptionV6 shrinks encCiphertext to 100 bytes (ZIP-230, Sapling Output Description). An AssetBurn entry is 40 bytes: AssetBase (32, the encoding of $\text{AssetBase}^{\text{Orchard}}$) and valueBurn (uint64). ZIP-230 says the amount is non-zero and strictly less than MAX_BURN_VALUE, but ZIP-226’s consensus rule and the pinned code permit equality (§4.4).

Issuance bundle. The issuance component carries five fields: issuerLength (compactSize), issuer, nIssueActions, vIssueActions, and issueAuthSig. The signature is present iff nIssueActions > 0; the first four fields are always present, and if nIssueActions = 0 then issuerLength MUST be 0 — an empty issuance bundle is exactly two zero compactSize values (ZIP-230, Transaction Format). An IssueAction carries assetDescHash (32 bytes), nNotes and vNotes at 115 bytes per IssueNoteDescription, and flagsIssuance with finalize in the LSB; nNotes may be zero, so a finalize-only action for an existing Asset is expressible; the new-Asset corner case remains undefined in ZIP 227 (§3.7). An IssueNoteDescription is recipient (43 bytes, the raw Orchard payment-address encoding), value (uint64), rho (32), rseed (32). The IssueAuthSignature is sizeSighashInfo and sighashInfo, then sizeSignature and a signature “preceded by a 0x00 byte indicating a BIP 340 signature” (ZIP-230, issuance field encodings) — the same 0x00 scheme byte that prefixes the issuer key encoding (ZIP-227, Issuance Authorization Signature Scheme).

Versioned signatures. The SaplingSignature and OrchardSignature types prepend a compactSize-prefixed sighashInfo to the 64-byte RedJubjub or RedPallas signature (ZIP-230, signature encodings). ZIP-246 defines the versioning:

$$\text{sighashInfo} := [\text{sighashVersion}] \parallel \text{associatedData},$$

where version 0 is the “commit to all effecting data” algorithm; for v6, only version 0 is defined, with empty associatedData, for all four of Transparent, Sapling, Orchard, and Issuance (ZIP-246, Sighash versioning). The machinery buys the ability to introduce — and, critically, to retire — sighash algorithms per transaction version without a new format.

Coinbase. For coinbase transactions the enableSpendsOrchard and enableZSAs bits MUST be zero (ZIP-230, consensus rules).

6.3 Digests: txid, sighash, and authorizing data (ZIP-246)

ZIP-246 extends the ZIP-244 digest tree — constructed for vanilla Orchard in the *Ironwood Guide*, §“What the signatures sign: ZIP-244 digests and their v6 form” — from four top-level branches

to six:

$$\text{txid} := \text{BLAKE2b-256}(\text{ZcashTxHash_} \parallel \text{branchid}, d_{\text{hdr}} \parallel d_{\text{transp}} \parallel d_{\text{Sap}} \parallel d_{\text{Orch}} \parallel d_{\text{issue}} \parallel d_{\text{memo}}),$$

the nodes T.1–T.6 of the ZIP, each itself a personalized BLAKE2b-256 (ZIP-246, TxId Digest). New relative to ZIP-244 are the fee and burn fields in the header digest, the Sapling output digests, the entire Orchard Action-Group subtree, the issuance subtree, and the memo subtree. The header digest T.1 is

$$d_{\text{hdr}} := \text{BLAKE2b-256}(\text{ZTxIdHeadersHash}, \text{version} \parallel \text{versionGroupId} \parallel \text{branchid} \parallel \text{lockTime} \parallel \text{expiryHeight} \parallel \text{LE}_{64}(\text{fee}) \parallel \text{LE}_{64}(\text{burnAmount}))$$

(ZIP-246, T.1). The Orchard branch T.4 hashes the concatenated per-group digests and the balance, $d_{\text{Orch}} = \text{BLAKE2b-256}(\text{ZTxIdOrchardHash}, d_{\text{AGs}} \parallel \text{LE}_{64}(v_{\text{balance}}^{\text{Orchard}}))$, with a defined empty case; each group digest (T.4a) covers a compact digest (nf, cmx, ephemeralKey, encCiphertext), a non-compact digest (cv, rk, outCiphertext), flagsOrchard, anchorOrchard, nAGExpiryHeight, and a burn digest over the (AssetBase, valueBurn) tuples (ZIP-246, T.4a). The issuance branch T.5 hashes issuerLength || issuer and a per-action digest (per action: assetDescHash, a per-note digest over (recipient, value, ρ , rseed), flagsIssuance), each level with a defined empty case (ZIP-246, T.5). The memo branch T.6 hashes a nonce and the concatenation of per-chunk digests; the v6 pruned memo bundle stores exactly these memo_chunk_digest / memo_digest values in place of pruned chunks, so pruning preserves the txid (ZIP-246, T.6; ZIP-230, Transaction Format). Table 4 collects the personalization strings; the implementation defines the same constants (§6.6).

Node	Personalization	Content
T.1	ZTxIdHeadersHash	header, incl. fee and burn amount
T.3b.i	ZTxId6SOutC_Hash	Sapling v6 outputs, compact
T.3b.ii	ZTxId6SOutN_Hash	Sapling v6 outputs, non-compact
T.4	ZTxIdOrchardHash	Action-Group digests, balance
T.4a	ZTxIdOrcActGHash	one Action Group
T.4a.i	ZTxId60ActC_Hash	nf, cmx, ephemeralKey, encCiphertext
T.4a.ii	ZTxId60ActN_Hash	cv, rk, outCiphertext
T.4a.vi	ZTxIdOrcBurnHash	burn tuples
T.5	ZTxIdSAIssueHash	issuer, issue-actions digest
—	ZTxIdIssuActHash	per action: desc. hash, notes, flags
—	ZTxIdIAcNoteHash	per note: recipient, value, ρ , rseed
T.6	ZTxIdMemo___Hash	nonce, chunks digest
—	ZTxIdMemoCksHash, ZTxIdMemoCk_Hash	memo chunk digests
A.1	ZTxAuthTransHash	scripts, now incl. per-input sighashInfo
A.3	ZTxAuthOrchaHash	group auth digests, bindingSigOrchard
A.3a	ZTxAuthOrcAGHash	proofsOrchard, spend-auth digest
A.3a.ii	ZTxAuthOrSASHash	spendAuthSig encodings
A.4	ZTxAuthZSAOrHash	issueAuthSig encoding

Table 4: ZIP-246 digest personalizations (ZIP-246, TxId Digest and Authorizing Data Commitment). Unnumbered rows are children whose labels the ZIP text assigns inconsistently; see the closing remark of this subsection.

Signature digest. The sighash reuses the tree with the transparent branch replaced by an input-specific variant:

$$\text{sighash} := \text{BLAKE2b-256}(\text{ZcashTxHash_} \parallel \text{branchid}, d_{\text{hdr}} \parallel d_{\text{transp}}^S \parallel d_{\text{Sap}} \parallel d_{\text{Orch}} \parallel d_{\text{issue}} \parallel d_{\text{memo}}),$$

produced per transparent input, per Sapling input, per OrchardZSA Action, and — new in v6 — per Issue Action; the Sapling, Orchard, issuance, and memo branches are identical to their txid-side counterparts, so the issuance bundle and the memo bundle enter every sighash (ZIP-246, Signature Digest). On this digest ZIP-227 hangs its authorization rule: the SigHash is the §4.10 SIGHASH transaction hash with the ZIP-226 modifications, computed with SIGHASH_ALL, and the issuance authorization signature MUST satisfy $\text{IssueAuthSig.Validate}(\text{ik}, \text{SigHash}, \text{issueAuthSig}) = 1$, where ik is decoded from the issuer field and its encoding is required to begin with 0x00, the BIP-340 scheme byte (ZIP-227, Specification: Consensus Rule Changes).

Authorizing-data commitment. The commitment to authorizing data follows the same pattern: A.1 covers transparent scripts together with the new per-input TransparentSighashInfo; A.2 keeps the ZIP-244 structure with field encodings altered by sighashInfo; A.3 commits to the per-group proofs and spend-authorization signatures and to bindingSigOrchard; and a new A.4 commits to the issueAuthSig field encoding, with a defined empty case (ZIP-246, Authorizing Data Commitment).

Remark 6.1 (The reference text is itself unfinished). ZIP-246 is internally incomplete even as withdrawn text: the authorizing-data section carries “TODO: Update the Authorizing Data Commitment below for the sighashInfo changes to the tx format,” Rationale and Reference implementation are TBD, and the subtree labels are inconsistent — `issue_actions_digest` is listed as T.5c but headed T.5a with children T.5a.i, and the burn-digest children are labelled T.4b.i/T.4b.ii under node T.4a.vi (`zip-0246.rst`). Anyone respecifying the format inherits these defects as open editorial work; the implementation, not the ZIP, is currently the more precise artifact.

6.4 Transaction-level consensus rules

One Action Group. Although the format is a vector of Action Groups, ZIP-227 collapses it for OrchardZSA: per transaction, `nActionGroupsOrchard` MUST be 0 or 1, and `nAGExpiryHeight` MUST be 0 (ZIP-227, Specification: Consensus Rule Changes). The vector form is used by draft ZIP-228’s proposed ZSA swaps, which would relax the one-group rule for swap bundles; it is not active in the core ZIP-227 rules.

Issuance requires an Action Group. A transaction with an issuance bundle MUST also contain at least one OrchardZSA Action Group (ZIP-227, Specification: Consensus Rule Changes). The rule is load-bearing twice over. First, issue-note ρ derivation consumes the first Action’s nullifier: each issued note’s ρ is computed by the function ZIP-227 names `DerivelssuedRho` — constructed in §3.3 — keyed by $\text{nf}_{0,0}$, the nullifier of the input note of the first Action in the first Action Group, over the Issue Action and note indices (ZIP-227, Computation of ρ). The indices domain-separate notes within the transaction, while the nullifier-keyed PRF makes collisions across transactions negligible under its pseudorandomness assumption; it does not give literal global uniqueness. Second, without a spend in the transaction the issuance-only SIGHASH would be replayable; the mandatory Action Group ties the issuance signature to a nullifier that consensus consumes (ZIP-227, Rationale).

Global issuance state. Nodes maintain a map `issued_assets` from asset bases to triples (balance, final, note_ref), chained empty map $\rightarrow$ per-block $\rightarrow$ per-transaction. Burns decrement balance, requiring $\text{balance} \geq v$, before the transaction’s issuance bundle is processed; issuance requires $\text{final} \neq 1$, increments balance subject to the overflow bound $\text{MAX_ISSUE} = 2^{64} - 1$, and sets final when `finalize = 1`. The first issuance of an asset MUST include a *reference note*: value 0, paid to the default address of the all-zero Orchard spending key (ZIP-227, Global Issuance

State and Reference Notes). The 43-byte raw encoding of that address is fixed in the ZIP and reproduced as a constant in the implementation (§6.6).

Burn rules and the extended balance equation. For every $(\text{AssetBase}, v)$ in assetBurn : $\text{AssetBase} \neq V^{\text{Orchard}}$ (ZEC cannot be burned this way), $0 < v \leq \text{MAX_BURN_VALUE}$, and no two entries share an AssetBase ; the bound $\text{MAX_BURN_VALUE} := 2^{63} - 1$ is chosen for compatibility with signed 64-bit value-balance fields (ZIP-226, Burn Mechanism). Burns enter value balance through the binding key. Where vanilla Orchard computes bvk by subtracting the committed declared balance from the sum of net value commitments (the *Ironwood Guide*, §“Conservation without disclosure: value commitments and the binding signature”), ZIP-226 subtracts one zero-randomness commitment per burn entry as well:

$$\begin{aligned} \text{bvk} := & \left(\sum_{i=1}^n \text{cv}^{\text{net}, i} \right) - \text{ValueCommit}_0^{\text{OrchardZSA}}(V^{\text{Orchard}}, v_{\text{balance}}^{\text{Orchard}}) \\ & - \sum_{(\text{AssetBase}, v) \in \text{assetBurn}} \text{ValueCommit}_0^{\text{OrchardZSA}}(\text{AssetBase}, v), \end{aligned}$$

and the prover signs the SIGHASH with $\text{bsk} = \sum_{i=1}^n \text{rcv}_i$ (ZIP-226, Burn Mechanism). Under the unknown-discrete-log assumptions made explicit in §4.3, the signature computationally enforces that every non-ZEC asset nets to exactly its declared burns.

Commitment-tree order. Note commitments enter the global tree in a fixed order: first, per Action Group, the commitments of the new OrchardZSA notes of each Action; then, per Issue Action, the issue-note commitments (ZIP-227, Specification: Consensus Rule Changes). Wallets and nodes must agree on this order to agree on note positions.

6.5 The experimental fee policy

The fee treatment chains through three documents: ZIP-226 (Transaction Fees) defers to ZIP-317 “as described in ZIP 227”; ZIP-227 (Changes to ZIP 317) states that the conventional fee is altered for the presence of issuance actions and the creation of new Custom Assets, pointing at ZIP-317’s Fee calculation section. Fees remain payable in ZEC only (ZIP-227, rationale). All variants share the conventional-fee form documented in the *Wallet Guide*, §“Fees (ZIP-317)”:

$$\text{conventionalFee} := \text{marginalFee} \cdot \max(\text{graceActions}, \text{logicalActions}),$$

with $\text{marginalFee} = 5000$ zatoshis and $\text{graceActions} = 2$ (ZIP-317, Fee calculation).

ZIP 317 contains no ZSA fee contributions. The experimental fork adds a charge for creating an asset, to discourage permanent state growth from unused assets: the parameter $\text{creationCost} = 100$ logical actions, and two contributions summed into logicalActions alongside the base terms of the deployed rule:

$$\begin{aligned} \text{contribution}_{\text{ZSAIssuance}} &:= n_{\text{ZSAIssueNotes}}, \\ \text{contribution}_{\text{ZSACreation}} &:= \text{creationCost} \cdot n_{\text{AssetCreations}}, \end{aligned}$$

where $n_{\text{ZSAIssueNotes}}$ counts issue-note outputs and $n_{\text{AssetCreations}}$ counts Custom Assets newly added to the global issuance state (fork ZIP-317, Fee calculation). The rationale is a one-time charge for state growth: each new asset adds an `issued_assets` row tracked in perpetuity, while subsequent issuance, finalization, and burn only mutate the row. The implementation uses the same contributions. As with the deployed rule (the *Wallet Guide*, §“The deployed fee rule”), paying the conventional fee is not a consensus requirement in any ZIP-317 variant. Classification: the ZSA fee rule is *fork-specified*; whether it returns upstream as a ZIP-317 revision, as Ironwood’s did, is undecided.

6.6 Implementation status at the pinned commits

Serialization. The experimental reader parses the header, one `vSighashInfo` per transparent input (required to equal the version-0 encoding), the Sapling v6 bundle, the Orchard v6 bundle, and the issuance bundle (`zcash_primitives`, `transaction/mod.rs`). The Orchard reader rejects more than one Action Group (“A V6 transaction must contain at most one action group”), requires at least one Action, and rejects a nonzero `nAGExpiryHeight`; the writer emits exactly one group with `nAGExpiryHeight = 0` (`transaction/components/orchard.rs`) — the ZIP-227 consensus rules hard-coded at the serialization layer. Issuance-bundle serialization matches ZIP-230, including the two-zero-compactSize empty bundle and rejection of the inconsistent issuer/action combinations. The sighash implementation extends the v5 hash with the issuance digest for ZSA-capable versions, with `sighash-info` constants `ORCHARD_SIGHASH_INFO_V0 = ISSUE_SIGHASH_INFO_V0 = [0x00]` (`sighash_v6.rs`, `sighash_versioning.rs`). The orchard fork defines the ZIP-246 personalizations of Table 4, asserting the 0x00 BIP-340 byte when hashing the 65-byte issuance signature encoding.

Verification code. Function `derive_bvk_raw` implements the extended balance equation of §6.4 verbatim; `validate_bundle_burn` enforces the ZIP-226 burn rules — non-ZEC asset, non-zero amount, amount $\leq 2^{63} - 1$, no duplicates — plus presence in and sufficient supply under the global issuance state; and `verify_issue_bundle` checks the BIP-340 signature over the 32-byte sighash, the per-note ρ against `DerivelssuedRho`, non-finalization, u64-checked supply addition, and the reference note on first issuance, whose 43-byte `RAW_REFERENCE_RECIPIENT` constant equals the array in ZIP-227. The 0x84 domain separator for issued-note ρ lives in the QED-it `zcash_spec` fork (rev d5e8426). Proof sizes record the circuit cost: the ZSA aggregated proof has base size 2848 bytes and 2272 bytes per Action, against a vanilla base of 2720 (`primitives/orchard_primitives_zsa.rs`, `orchard_primitives_vanilla.rs`).

Divergences from the withdrawn ZIP text. Four are material. The first three block byte-compatibility with ZIP-230/246 as written; the fourth changes private witness construction but is not enforced by the circuit.

1. *No explicit fee field.* The v6 reader serializes no fee, and `zip233Amount` only behind the `zip-233` cargo feature; the header `txid` digest correspondingly omits ZIP-246’s T.1f fee field, with the comment “TODO: Factor this out ... when implementing ZIP 246 in full” (`transaction/mod.rs`, `txid.rs`).
2. *No memo bundles.* The forks implement the pre-ZIP-231 layout: OrchardZSA notes carry a full 512-byte memo, giving a 612-byte `encCiphertext` with an 84-byte compact prefix (52 vanilla bytes plus the 32-byte asset base), and Sapling v6 outputs are read in the v5 580-byte format (`primitives/zcash_note_encryption_domain.rs`, `transaction/components/sapling.rs`). No memo fields and no T.6 are implemented; instead the action-group `txid` digest inserts a non-spec memos digest (personalization `ZTxIdOrcActMHash`) between the compact and non-compact digests and hashes only the 84-byte compact slice into the compact digest, with TODOs “Remove once new Memo Bundles are implemented (ZIP-231)” (`primitives/orchard_primitives_zsa.rs`).
3. *Lead byte.* The implementation hard-codes `NOTE_VERSION_BYTE_V3 = 0x03`; upstream ZIP-230 withdrew that claim after ZIP-2005 took 0x03, leaving the ZSA lead byte an unassigned placeholder. The fork’s choice predates upstream ZIP-2005 assigning 0x03 to Ironwood notes and directing future proposals to pick a new value; the collision is fork-only context, with no format ambiguity on Zcash mainnet. Reassignment is a hard prerequisite for any deployment: the lead byte selects the note-plaintext parser.

4. *Split-note ψ^{nf}* . Upstream ZIP-226 now derives the split-note nullifier randomizer as $\psi^{\text{nf}} = \text{ToBase}(\text{PRFexpand}_{\text{rseed}_{\text{nf}}}(\text{0x0A} \parallel \rho^{\text{old}}))$ with rseed_{nf} sampled uniformly at random, the first-byte domain 0x0A “reserved by ZIP 2005 for this use” (ZIP-226, split-note nullifier); the fork at `cf801a5d` realizes the older fork text (uniformly random ψ^{nf}) via a separate `rseed_split_note` under the ordinary ψ domain `0x09` (`note.rs`, `circuit.rs`).

6.7 Wallet integration

The withdrawn ZIP-230 and ZIP-226 design imposes two obligations. First, all wallets SHOULD switch to sending only its v6 transactions once permitted: a v5 transaction reveals that all its Orchard notes are ZEC, and the design’s v6 OrchardZSA output format incorporates the quantum-recoverability changes (ZIP-230, Implications for Wallets; ZIP-226, Backwards Compatibility with ZEC Notes). This is a claim about the withdrawn carrier, not the different v6 format now assigned to ZIP-229. Second, wallets MUST support parsing and trial-decrypting v6 outputs — the new lead byte and the memo-bundle changes — by activation, or rescan once support lands: funds sent to an un-upgraded wallet are temporarily inaccessible, because the new lead byte selects different `rcm/ψ` derivations and a v5-decryption wallet rejects the note (ZIP-230, Implications for Wallets). Payment requests have a specified extension path: ZIP-321’s `req-asset` parameter carries an Asset Identifier and amount (the *Wallet Guide*, §“Custom assets: req-asset (specified, not deployed)”).

Multi-party and hardware signing is the open end. No PCZT structure exists for OrchardZSA bundles or issuance bundles: the `librustzcash` fork’s transaction extractor panics with “PCZT support for ZSA is not implemented” when handed an OrchardZSA bundle, the orchard fork’s PCZT module is written against the vanilla flavor only, and the released PCZT v2 targets ZIP-229’s Ironwood v6, not ZSA (the *Wallet Guide*, §“Partially created transactions (PCZT)” and §“PCZT v2: deferred anchors and the Ironwood bundle”). Classification: *open* — hardware-wallet and multi-party signing for ZSAs is unspecified in any draft.

6.8 Outlook

Between the pinned implementation and any deployment stand three cross-cutting respecification problems.

1. *A carrier*. Transaction version 6 now belongs to NU6.3’s Ironwood format (ZIP-229), whose non-requirements exclude the ZSA machinery (§6.1). The designated successor for what ZIP-230/246 deferred — an extensible transaction format — is ZIP-248, now a merged Draft for V7. Its registry lists OrchardZSA with a provisional variant and ZSA issuance without an allocated bundle type, under “Potential Future Bundle Types”; it does not specify either bundle’s encoding or digest. The ZSA-specific transaction format therefore remains incomplete: before a future ZSA deployment, the byte format of §6.2 must be mapped or respecified for V7, and §6.3 re-anchored to the resulting digest rules.
2. *A base protocol*. ZIP 226 assumes ZIP 2005’s recoverability changes, but the experimental implementation does not. No ZIP specifies whether OrchardZSA targets the restricted Orchard pool or is rebased onto Ironwood. Classification: *open*.
3. *Reconvergence of spec and code*. The divergence list of §6.6 — fee field, memo bundles, lead byte, ψ^{nf} — plus the orphaned fee treatment of §6.5 must land on one side or the other: either the respecified ZIPs adopt what the code does, or the code moves. Each divergence is identified, but resolving it still requires protocol choices and implementation work.

The summary judgment this volume can defend: a ZSA transaction layer is designed and implemented in the pinned forks, but the deployable protocol is *currently unspecified* in the sense this series requires — there is no complete merged carrier specification under which a mainnet

node could implement it. The byte-level facts of this section are fixed by withdrawn ZIP text and pinned fork code, and they will remain the diff base for any respecification; the remaining open items are enumerated above and in the scope section.

Part IX

FROST Guide: Threshold RedPallas: FROST and its Zcash integration surfaces

Abstract

Assuming Schnorr signatures from the *Crypto Guide*, RedPallas from the *Ironwood Guide*, PCZT from the *Wallet Guide*, and the finalizer model from the *Crosslink Guide*, this volume of *The Zcash Arboretum* documents threshold signing for Zcash. It constructs FROST as specified in RFC 9591, then its re-randomised form for Zcash spend authorisation and recoverability custody. The re-randomised Zcash stack is outside the completed cryptographic-library audits identified here; each design-stage claim is classified by the rigour of its source.

1 Scope, sources, and deployment surfaces

This volume documents threshold custody of Zcash spend authority: how t of n parties jointly produce, without reconstructing the signing key during signing, a signature that consensus accepts as an ordinary RedPallas spend-authorisation signature. Three layers carry the construction. The base scheme is FROST, the two-round threshold Schnorr protocol of Komlo and Goldberg, specified in RFC 9591 (§2). The Zcash layer is Re-Randomized FROST (Gouvêa–Komlo, IACR ePrint 2024/436), which shifts every share by a common randomizer so that the aggregate signature verifies under the per-Action randomised key; ZIP 312 specifies it for Zcash, and the `frost-rerandomized` and `reddsa crates` ship it (§3). The deployment picture — the shipped stack, the proposed uses, and the gaps — closes the volume (§4).

Three custody surfaces anchor the treatment, one per rigour bin (§1.3): spend-authorisation custody for Orchard, where the construction is specified and shipped; custody of the ZIP-2005 quantum spending key `qsk`, where a Proposed consensus ZIP constrains a FROST ceremony whose recovery protocol it defers; and Crosslink finalizer keys, for which no threshold design was identified in the cited Crosslink corpus. The surfaces and their bins are fixed in §1.4.

1.1 Scope

The object under study is a t -of- n signing protocol whose output has the same format and verification rule as single-party spend authorisation. ZIP 312 states the requirements directly: the signatures must verify as Sapling or Orchard spend-authorisation signatures, and should meet the protocol specification’s “Signature with Re-Randomizable Keys” criteria. Plain FROST fails the first clause — it signs for a fixed group key, and consensus verifies only per-Action randomised keys — and closing that gap is the volume’s central construction (§3).

The volume develops RedPallas only, per its title. ZIP 312 itself defines two ciphersuites — FROST(Jubjub, BLAKE2b-512) for Sapling RedJubjub and FROST(Pallas, BLAKE2b-512) for Orchard RedPallas — and we cite the Jubjub suite where the ZIP fixes both at once, but develop only the Pallas suite (§3.5).

ZIP 312’s stated non-requirements bound the scope from below: it does not remove the Coordinator role; it defines no key-generation procedure (out of scope there, exactly as in RFC 9591); and it provides no network privacy. Key generation nevertheless appears twice in this volume, both times forced by sources: once as published with the base scheme (§2.2), and once as the unspecified-but-shipped DKG surface of the `reddsa` crate, binned in §1.3.

One altitude note governs every deployment claim. Availability is library-level, not consensus-level: no consensus rule names FROST, and none needed to change. The verifier

accepts the aggregate because it is a valid RedPallas signature; as ZIP 312’s rationale puts it, the method by which the commitment R was generated is irrelevant to the verifier.

1.2 The source ladder

Table 1 pins the primary sources used for the full review. Repository heads were rechecked on 2026-09-05: the FROST-facing protocol text is unchanged at `zcash/zips e753a6a3`, and the algorithms and Zcash ciphersuite behaviour described below remain unchanged at `ZcashFoundation/frost 0966bd1` and `ZcashFoundation/reddsa 63c6ac0`. Local copies of the papers and RFC are archived with the volume’s drafts.

Artefact	Pin / date	Evidence status
RFC 9591 (IRTF CFRG)	June 2024	Informational; CFRG consensus
ePrint 2024/436 (Gouvêa–Komlo)	recv. 2024-03-13	preprint; no venue
ZIP 312	6961098410, 2026-07-09	Draft (Wallet category)
ZIP 2005	6961098410, 2026-07-09	Proposed (Consensus category)
<code>ZcashFoundation/frost</code>	2016e44ba4, 2026-04-23	shipped code; workspace 3.0.0
<code>ZcashFoundation/reddsa</code>	3792daa95e, 2026-05-22	shipped code; crate 0.5.2
<code>ZcashFoundation/frost-tools</code>	06c0dbdb, 2026-08-12	demos; not for production
ePrint 2020/852; 2022/833	SAC 2020; CRYPTO 2022	peer-reviewed antecedents

Table 1: Primary FROST sources used for the full review. Current heads rechecked 2026-09-05.

The RFC. RFC 9591, “The Flexible Round-Optimized Schnorr Threshold (FROST) Protocol for Two-Round Schnorr Signatures” (D. Connolly, Zcash Foundation; C. Komlo, University of Waterloo and Zcash Foundation; I. Goldberg, University of Waterloo; C. A. Wood, Cloudflare; IRTF CFRG stream, Category Informational, June 2024), records CFRG consensus on the base two-round protocol and leaves key generation out of scope. It explicitly says that it is not an Internet Standard and that IRTF results might not be suitable for deployment. Two of its four authors are Zcash Foundation, so the RFC and Zcash implementation share provenance. Its deltas against the SAC 2020 paper are catalogued in §2.8.

The paper. “Re-Randomized FROST”, by Gouvêa and Komlo (both Zcash Foundation), is IACR ePrint 2024/436 (<https://eprint.iacr.org/2024/436>): received 2024-03-13, approved 2024-03-15, publication info “Preprint”, no peer-reviewed venue as of 2026-09-05. We read it in full and work from it throughout §3. Its headline Theorem 1 claims unforgeability in the random-oracle model under AOMDL. That game restricts oracle queries to declared algebraic combinations, unlike the OMDL game used for plain FROST in BCKMTZ. The printed reduction has material gaps documented in §3.3; this volume therefore does not treat that claim as an established security theorem.

The ZIP. ZIP 312, “FROST for Spend Authorization Multisignatures” (owners Conrado Gouvêa, Chelsea Komlo, and Deirdre Connolly; Status Draft, Category Wallet; Discussions-To `zcash/zips#382`, pull request `zcash/zips#662`), is the Zcash specification of the construction.⁷ It is visibly unfinished — its Created field still reads 2022-08-dd, day unfilled — so we cite it as a moving object, by repository commit (`zcash/zips @ 6961098410, 2026-07-09`) and Draft status, never as a stable specification. Its threat model trusts the Coordinator and every key-share holder with transaction privacy and unlinkability, but its security requirement says that a rogue Coordinator must not be able to create a signed transaction without the approval

⁷The FROST ZIP is ZIP 312 alone. ZIP 313, its numeric neighbor, is “Reduce Conventional Transaction Fee to 1000 zatoshis” — Status Obsolete, obsoleted by ZIP 317 — and has nothing to do with FROST.

of MIN_PARTICIPANTS participants. What it specifies — the unlinkability requirement, the randomizer flow, and the two ciphersuites — is developed in §3.4 and §3.5.

The crates. Repository ZcashFoundation/frost (@ 2016e44ba4, 2026-04-23) is a Rust workspace at version 3.0.0 (rust-version 1.81): crate frost-core carries the ciphersuite-generic protocol; five ciphersuite crates instantiate the RFC (ed25519, ed448, P-256, ristretto255, and secp256k1), while the separate frost-secp256k1-tr crate implements a Taproot-oriented suite; frost-rerandomized carries the re-randomised layer, and gencode generates the per-suite code. The repository also vendors an audit report dated 2021-03-23 (zcash-frost-audit-report-20210323.pdf) and the source of the ZF FROST book (published at frost.zfnd.org), whose “FROST with Zcash” chapters (zcash-devtool demo, Ywallet demo, FROST server) and serialisation-format page — the page ZIP 312 cites for encode_signing_package — feed §4. Repository ZcashFoundation/reddsa (@ 3792daa95e, 2026-05-22; crate version 0.5.2) holds the Zcash ciphersuites: its frost feature enables frost-rerandomized 3.0.0, and src/frost/redpallas.rs defines FROST(Pallas, BLAKE2b-512) as struct PallasBlake2b512, ZIP 312’s reference implementation surface (§3.6). Crate reddsa also ships machinery ZIP 312 does not specify — DKG hash personalisations and even- γ normalisation hooks — binned in §1.3. Its README marks the frost feature experimental while ZIP 312 remains Draft and explicitly excludes that feature from the crate’s SemVer and MSRV guarantees.

The antecedent papers. FROST itself is Komlo–Goldberg, IACR ePrint 2020/852 (SAC 2020); the security analysis the re-randomised extraction argument extends is Bellare–Crites–Komlo–Maller–Tessaro–Zhu, “Better than Advertised Security for Non-Interactive Threshold Signatures” (CRYPTO 2022; ePrint 2022/833). They are the paper’s references [10] and [1] respectively; §2 works from both.

1.3 The three-bin rule

The series rule is binary for implementation behaviour and ternary for design. A claim about implementation behaviour cites the implementation — crate and file — and the specification section it realises; a claim about a design-stage protocol is classified *specified*, *designed but unspecified*, or an *open problem*, and the bin stays visible in the prose. This volume sits astride the boundary: one surface is shipped code, one is a consensus proposal, one is an acknowledged gap. Applied once, here, for the whole volume:

1. Re-Randomized FROST for spend-authorisation custody is *specified*: RFC 9591 published June 2024, ZIP 312 in Draft, ePrint 2024/436, and shipped code — frost-rerandomized 3.0.0 and reddsa 0.5.2. The latter’s FROST feature is explicitly experimental. Availability is library-level, not consensus-level (§1.1).
2. Custody of the ZIP-2005 quantum spending key is specified at the ZIP level (Status Proposed, Category Consensus), but the Recovery Protocol it exists to serve is “to be introduced, potentially, in a future upgrade” — *designed but unspecified*.
3. Crosslink finalizer threshold custody is an *open problem*: the Crosslink Guide’s own classification, which we adopt.

One surface refuses the clean split, and we bin it here once. No ZIP specifies FROST key generation: ZIP 312 declares it out of scope (as does RFC 9591), and ZIP 271’s May-2025 note confirms the gap — ZIP 312’s missing key-generation text is its stated reason not to couple disbursement of the *lockbox* (the in-protocol reserve that has accrued a portion of each block subsidy since NU6, per ZIP 2001) to that specification. Yet crate reddsa ships DKG support: hash personalisations FROST_RedPallasD (HDKG) and FROST_RedPallasI (HID), beyond ZIP 312’s H1–H5

and HR, plus even- Y normalisation hooks (`post_generate/post_dkg`, applying `into_even_y` to secret shares and the public key package) so that the group verifying key meets the encoding constraint of Orchard’s spend validating key. Deployed code with no specification above it: we classify the DKG surface *designed but unspecified* and cite the crate at commit 3792daa95e wherever we rely on it.

1.4 The three deployment surfaces

Spend-authorisation custody (specified). The product ZIP 312 promises is signatures “indistinguishable from RedPallas Orchard Spend Authorization Signatures”. The single-party object being imitated is settled in the *Ironwood Guide*: §“Spend authorisation: RedPallas re-randomisation” constructs $rsk := ask + \alpha$ and $rk := ak + [\alpha] G^{\text{SpendAuth}} = [rsk] G^{\text{SpendAuth}}$, with `GenRandom` hashing 80 uniform bytes through H° , and §“RedPallas unforgeability” proves EUF-CMA from discrete log in the random-oracle model. Per the layering rule we cite both and re-prove neither; this volume’s work begins where the single signer ends, at splitting `ask` into shares that a signing coalition need not reassemble (§3). The wallet-side integration path is likewise on the record: the *Wallet Guide*, §“Partially created transactions (PCZT)”, records ZIP 374’s motivation (iii), threshold signing (for example FROST, RFC 9591) — the multiparty transaction format FROST signing rides in §4.3.

Custody of the quantum spending key (designed but unspecified). ZIP 2005, “Ironwood Quantum Recoverability” (owners Daira-Emma Hopwood and Jack Grigg; Status Proposed, Category Consensus; created 2025-03-31), writes FROST into its requirements twice: the design should be fully compatible with FROST threshold multisignatures (citing ZIP 312), and should lose no pre-quantum security when FROST and/or hardware wallets are used. Its “Usage with FROST” section then constrains the ceremony: the point `ak` is derived jointly, via the FROST DKG, from shares of `ask`; the participants must privately agree on a value `sk` and use it with `use_qsk = true` to derive `qsk`, `qk`, and $\text{rivk_ext} = H_{qk}^{\text{rivk_ext}}(\text{Extract}_{\mathbb{P}}(ak), nk)$; all participants must obtain the same extracted key $\text{Extract}_{\mathbb{P}}(ak), nk$, and `rivk_ext`. Subject to the specified derivations, the ZIP says that this also implies equal `qsk` and `qk` under collision resistance of H^{qk} and $H^{\text{rivk_ext}}$. Each participant must hold `qsk` at least as securely as `ask`. The outlined Recovery Protocol would require a RedDSA signature under `ak` — preserving t -of- n for as long as discrete log on Pallas stands — together with a proof SoK^{qsk} of knowledge of a `qsk` with $H^{qk}(qsk) = qk$. The asymmetry is the point to retain: the key `qsk` is held by *every* participant, so a quantum adversary with `qsk` and the ability to solve discrete logarithms on Pallas no longer faces a threshold; the ZIP accordingly recommends moving funds to a fully post-quantum threshold protocol once one is available. The deferred Recovery Protocol is what parks this surface in the middle bin; the full treatment is §4.7.

Crosslink finalizer keys (open problem). The *Crosslink Guide*, §“Finalizer keys”, records the prototype state: one ed25519 key per finalizer (ed25519-zebra), a 76-byte vote format signed by appending a 64-byte ed25519 signature over those 76 bytes, and — by negative search over the four Crosslink repositories (tfl-book, crosslink-protocol-guide, crosslink-spec, zebra-crosslink), dated 2026-07-15 there — no FROST or threshold-signing design for finalizer keys sourced anywhere in the corpus. That guide’s classification bins threshold custody, key rotation, revocation, and initial-roster formation as open problems, and observes that FROST’s presence in the PCZT flow makes the absence “a visible gap rather than an exotic ask”. We adopt the classification and return to the surface in §4.8.

Adjacent surfaces, cited but not developed. Three further ZIPs touch FROST and mark how wide the surface is growing. ZIP 271 (“Dev Fund Extension and One-Time Disbursement”, Pro-

posed) anticipates spending lockbox one-time-disbursement chunks “to a FROST address”, and its May-2025 note that ZIP 312 does not specify key generation is its stated reason not to couple the two specifications. ZIP 326 (“NU6.3 Consequences for Wallets”, Draft, created 2026-06-30) cites ZIP 2005’s “Usage with FROST” section. ZIP 374 (“Partially Created Zcash Transaction Format”, Draft, created 2024-12-09) lists FROST per RFC 9591 under its multiparty-transactions motivation. Each appears in this volume exactly where it bears (§4); none is developed as a surface of its own.

2 FROST

The base object of this volume is the threshold Schnorr scheme FROST (*Flexible Round-Optimized Schnorr Threshold* signatures) of Komlo and Goldberg, published at SAC 2020 and maintained as IACR ePrint 2020/852; we work from the final revision of 22 December 2020. The scheme lets any t of n parties, each holding a Shamir share of a signing key s , produce a signature (R, z) that verifies under the ordinary single-party Schnorr equation for the group public key $Y = g^s$. The signature carries no threshold marker, and a verifier applies the same equation as it would to a single-signer signature. That compatibility is what lets the scheme use Zcash’s existing RedDSA validators: the consensus rules do not observe the threshold ceremony. The deployment surfaces this volume targets are spend-authorisation custody over RedPallas (the *Ironwood Guide*, §“Spend authorisation: RedPallas re-randomisation”), custody of the ZIP-2005 quantum spending key (whose “Usage with FROST” section constrains the key generation), and Crosslink finalizer keys (the *Crosslink Guide*, §“Finalizer keys”).

Throughout, $\mathbb{G}$ is a group of prime order q with generator g , and H, H_1, H_2 are hash functions with outputs in $\mathbb{Z}_q^*$, modelled as random oracles. The paper fixes the single-party scheme it must match — the Schnorr signature scheme of the *Crypto Guide*, §“From identification to signature via the Fiat–Shamir transform” — restated here in the paper’s multiplicative notation: a Schnorr signature over m under secret key $s \in \mathbb{Z}_q$ and public key $Y = g^s \in \mathbb{G}$ is produced by sampling a nonce $k \xleftarrow{\$} \mathbb{Z}_q$, setting $R = g^k$, $c = H(R, Y, m)$, $z = k + s \cdot c$, and outputting $\sigma = (R, z)$; verification recomputes c and accepts iff $R = g^z \cdot Y^{-c}$. FROST’s whole design problem is to compute the pair (R, z) jointly, without reconstructing the nonce or signing key during signing and without opening the concurrency attacks described in §2.4.

2.1 Shamir sharing and share conversion

Two secret-sharing mechanisms cooperate in FROST, and keeping them distinct is the key to reading the signing equation.

Shamir sharing (the long-lived key). A dealer — or, in §2.2, every participant acting as a dealer simultaneously — selects $t-1$ random coefficients $a_1, \dots, a_{t-1}$ and forms the polynomial of degree at most $t-1$

$$f(x) = s + \sum_{i=1}^{t-1} a_i x^i, \quad f(0) = s,$$

giving participant P_i the share $(i, f(i))$, with distinct nonzero identifiers in $\mathbb{Z}_q$ and $1 \leq t \leq n < q$. Coefficients are sampled independently and uniformly in $\mathbb{Z}_q$. Any t shares determine f by Lagrange interpolation and hence $s = f(0)$; fewer than t reveal nothing. For a signing set $S = \{p_1, \dots, p_\alpha\}$ of α participant identifiers ($t \leq \alpha \leq n$), the interpolation weight of participant i at zero is the Lagrange coefficient

$$\lambda_i = \prod_{j=1, j \neq i}^{\alpha} \frac{p_j}{p_j - p_i},$$

computable from the identifiers alone — the coalition can be chosen *after* shares are dealt, which is what makes FROST’s “any t of n , decided at signing time” flexibility possible.

Additive sharing and conversion (the per-signature nonce). A sum $s = \sum_i s_i$ with each summand chosen locally is an additive sharing: it needs no dealer and no interaction. The Cramer–Damgård–Ishai share conversion observation, used by FROST in its simplest case, is that additive shares convert locally to Shamir form: if $s = \sum_{i \in S} s_i$, then the values s_i/λ_i are Shamir shares of the same s (evaluations of a degree- $(\alpha - 1)$ polynomial interpolating them). FROST uses exactly this to make the group nonce k : each signer contributes an additively-shared summand non-interactively, and the λ_i weights in the signing equation perform the conversion so that the response aggregates by plain summation. No DKG runs at signing time.

2.2 Key generation: Pedersen’s DKG with a knowledge proof

FROST generates its long-lived shares with a variant of Pedersen’s DKG — n parallel executions of Feldman’s verifiable secret sharing, one with each participant as dealer — hardened with a proof of knowledge of each dealer’s free coefficient. The paper’s Figure 1, reproduced:

Round 1.

1. Every participant P_i samples t random values $(a_{i0}, \dots, a_{i(t-1)}) \xleftarrow{\$} \mathbb{Z}_q$ and defines the polynomial $f_i(x) = \sum_{j=0}^{t-1} a_{ij}x^j$ of degree at most $t - 1$.
2. Every P_i computes a proof of knowledge of a_{i0} : with $k \xleftarrow{\$} \mathbb{Z}_q$, $R_i = g^k$, $c_i = H(i, \Phi, g^{a_{i0}}, R_i)$, $\mu_i = k + a_{i0} \cdot c_i$, set $\sigma_i = (R_i, \mu_i)$, where Φ is a context string preventing replay across ceremonies.
3. Every P_i computes the public commitment $\vec{C}_i = \langle \phi_{i0}, \dots, \phi_{i(t-1)} \rangle$ with $\phi_{ij} = g^{a_{ij}}$.
4. Every P_i broadcasts $\vec{C}_i, \sigma_i$.
5. On receiving $\vec{C}_\ell, \sigma_\ell$, participant P_i verifies $R_\ell \stackrel{?}{=} g^{\mu_\ell} \cdot \phi_{\ell 0}^{-c_\ell}$ with $c_\ell = H(\ell, \Phi, \phi_{\ell 0}, R_\ell)$, *aborting on failure*; on success all σ_ℓ are deleted.

Round 2.

1. Each P_i securely sends each other participant P_ℓ the share $(\ell, f_i(\ell))$, then deletes f_i and every share except its own $(i, f_i(i))$.
2. Each P_i verifies each received share against the dealer’s commitment:

$$g^{f_\ell(i)} \stackrel{?}{=} \prod_{k=0}^{t-1} \phi_{\ell k}^{i^k \bmod q},$$

aborting if the check fails.

3. Each P_i computes its long-lived signing share $s_i = \sum_{\ell=1}^n f_\ell(i)$, stores it securely, and deletes the $f_\ell(i)$.
4. Each P_i computes its public verification share $Y_i = g^{s_i}$ and the group public key $Y = \prod_{j=1}^n \phi_{j0}$.

Three design decisions deserve comment. First, the *proof of knowledge of a_{i0}* is the delta over plain Pedersen DKG: it addresses rogue-key attacks in the $t \geq n/2$ regime, where a participant could otherwise choose its commitment as a function of others' to bias the aggregate key (the fix was suggested to the authors by Shlomovits and Turkel; the paper's changelog dates it to the 2020-07-08 revision). Second, the *complaint handling* is abort-only: where Gennaro et al. add a complaint/disqualification phase to make the DKG robust, FROST aborts on any failed share check and expects the cause to be investigated out of band; the paper notes that implementations wanting robustness can substitute the Gennaro et al. construction. Third, a *consistent view* of the broadcast commitments $\vec{C}_i$ across participants is assumed, not provided — the paper defers to whatever broadcast mechanism the deployment supplies. Each participant ends holding (i, s_i) , with $Y_i = g^{s_i}$ public for share verification during signing and Y the single key the outside world sees.

2.3 Preprocessing and the two-round signing protocol

Signing splits into a message-independent *preprocess* stage and a single online round (equivalently, run as one two-round protocol with the commitment exchange inlined). A semi-trusted *signature aggregator* SA coordinates. The paper trusts it for publication and correct misbehaviour reporting, not for unforgeability, and claims in its security model that a malicious SA can deny service or falsely accuse but cannot learn the private key or cause an unapproved message to be signed.

Preprocess(π) (paper Figure 2). Each participant P_i generates π single-use nonce pairs and publishes their commitments: for $1 \leq j \leq \pi$,

$$(d_{ij}, e_{ij}) \xleftarrow{\$} \mathbb{Z}_q^* \times \mathbb{Z}_q^*, \quad (D_{ij}, E_{ij}) = (g^{d_{ij}}, g^{e_{ij}}),$$

appending (D_{ij}, E_{ij}) to a list L_i published to a predetermined location, and storing the secret halves for later.

Sign(m) (paper Figure 3). The aggregator SA selects a signing set S of α participants ($t \leq \alpha \leq n$), fetches each member's next unused commitment pair, and forms the ordered list $B = \langle (i, D_i, E_i) \rangle_{i \in S}$, sending (m, B) to every $P_i, i \in S$. Each participant then:

1. validates m (a signer signs only messages it is willing to sign) and checks $D_\ell, E_\ell \in \mathbb{G}^*$ for every commitment in B , aborting on failure;
2. computes the *binding values*

$$\rho_\ell = H_1(\ell, m, B), \quad \ell \in S,$$

the group commitment and challenge

$$R = \prod_{\ell \in S} D_\ell \cdot (E_\ell)^{\rho_\ell}, \quad c = H_2(R, Y, m);$$

3. computes its response, converting its Shamir share to the additive aggregation domain via λ_i (over the set S):

$$z_i = d_i + (e_i \cdot \rho_i) + \lambda_i \cdot s_i \cdot c;$$

4. securely deletes $((d_i, D_i), (e_i, E_i))$, then returns z_i to SA.

Finally SA recomputes ρ_i , $R_i = D_i \cdot (E_i)^{\rho_i}$, $R = \prod_{i \in S} R_i$ and c , and verifies every share against the public data:

$$g^{z_i} \stackrel{?}{=} R_i \cdot Y_i^{c \cdot \lambda_i},$$

identifying and reporting the misbehaving participant and aborting if any check fails; otherwise it publishes $\sigma = (R, z)$ with $z = \sum_{i \in S} z_i$.

Correctness. The proof is two interpolations (paper §6.1). The key polynomial $F_1(x) = \sum_{j=1}^n f_j(x)$ from key generation has $s_i = F_1(i)$ and $s = F_1(0)$. The nonce values $(i, (d_i + e_i \cdot \rho_i) / \lambda_i)$ define a degree- $(\alpha - 1)$ polynomial $F_2(x)$ with $F_2(0) = \sum_{i \in S} (d_i + e_i \cdot \rho_i) = k$. Setting $F_3(x) = F_2(x) + c \cdot F_1(x)$, each response is $z_i = \lambda_i F_3(i)$, so $z = \sum_{i \in S} z_i$ interpolates $F_3(0) = k + c \cdot s$; with $R = g^k$ and $c = H_2(R, Y, m)$, the pair (R, z) is a correct Schnorr signature on m .

2.4 Why the binding factor exists: Drijvers and ROS

The one non-obvious ingredient is $\rho_i = H_1(i, m, B)$. The binding factor is FROST’s answer to concurrency attacks that affected or constrained the earlier two-round schemes discussed by the paper.

The Drijvers attack. Against two-round Schnorr multisignatures in which the adversary may open T parallel sessions, Drijvers et al. showed how to forge by finding a hash output that is a sum of others: $c^* = H(R^*, Y, m^*)$ with $c^* = \sum_{j=1}^T H(R_j, Y, m_j)$, using Wagner’s k -tree algorithm for the generalised birthday problem in time $O(\kappa \cdot b \cdot 2^{b/(1+\lg \kappa)})$ for $\kappa = T + 1$ and b the bitlength of the group order. The attack transfers from the n -of- n multisignature setting to a t -of- n threshold scheme with an adversary controlling up to $t - 1$ signers, and it is why the Gennaro et al. scheme must cap its signing parallelism. Benhamouda et al.’s subsequent polynomial-time ROS solver (for $\ell > \lg q$ parallel sessions) turned the same structural weakness from subexponential to practical against schemes including two-round MuSig and GJKR-DKG-based threshold Schnorr; their paper concludes that “the current version of” FROST is *not* affected.

How binding addresses it. Because each signer derives $\rho_i = H_1(i, m, B)$ before responding, its share z_i is bound to this message, this signing set, and this exact commitment list. Except with random-oracle collisions, changing those inputs alters some ρ_ℓ , hence R and c , so directly transplanting a response fails verification. The paper’s own generalisation (its Equation 1, with linear combinations γ_j) states the residual attack surface precisely: the forger must find

$$H_2(R^*, Y, m^*) = \sum_j \gamma_j \cdot H_2(R_j, Y, m_j),$$

where $R^* = g^{r^*} \cdot \prod_j \widehat{R}_j^{\gamma_j}$ is itself a function of every input on the right-hand side. Wagner’s algorithm needs the two sides of the collision to vary independently; here every candidate combination on the right moves the left-hand target, degrading the generalised birthday collision into multiple preimage attacks. The paper’s footnote 7 concedes this step is heuristic: sufficiency of the condition is immediate, necessity is not proved.

2.5 Nonce hygiene

The preprocessed pairs (d_{ij}, e_{ij}) are one-time keys in the strictest sense. The paper’s stipulations:

- each pair is used at most once, and the signer *securely deletes* it in the same signing step that returns z_i (Figure 3, step 6);
- an accidentally reused (d_{ij}, e_{ij}) can expose the long-term share s_i ;
- a SA that maliciously replays a commitment pair cannot elicit another response from an honest participant; the replay causes an abort and can therefore deny service;
- the commitment store (if a server) is trusted only for availability and freshness; possession of the public (D_{ij}, E_{ij}) lists grants no signing power.

See §2.8 for RFC 9591’s sharpening of nonce generation (hedged sampling, and an explicit warning that deterministic nonces enable complete key recovery in the multi-party setting).

2.6 The security theorem as stated

The paper gives an EUF-CMA reduction, in the random-oracle model, for an intermediate scheme under the *discrete logarithm problem* in $\mathbb{G}$ — deliberately not to one-more discrete log (the adversary must return the discrete logarithms of $\ell + 1$ challenges after at most ℓ queries to a discrete-log oracle), so that the negative result of Drijvers et al., which rules out security reductions from that assumption for a family of two-round Schnorr multisignatures, does not apply. The proof is for an intermediate scheme, *FROST-Interactive*, in which the binding value is produced by a one-time verifiable random function $\rho_i = a_{ij} + b_{ij} \cdot H_\rho(m, B)$ with committed keys.

Theorem 6.1 (ePrint 2020/852). *If the discrete logarithm problem in $\mathbb{G}$ is (τ', ϵ') -hard, then the FROST-Interactive signature scheme over $\mathbb{G}$ with n signing participants, a threshold of t , and a preprocess batch size of π is $(\tau, n_h, n_p, n_s, \epsilon)$ -secure whenever*

$$\epsilon' \leq \frac{\epsilon^2}{2n_h + (\pi + 1)n_p + 1} \quad \text{and}$$

$$\tau' = 4\tau + (30\pi n_p + (4t - 2)n_s + (n + t - 1)t + 6) \cdot t_{\text{exp}} + O(\pi n_p + n_s + n_h + 1),$$

such that t_{exp} is the time of an exponentiation in $\mathbb{G}$, assuming the number of participants compromised by the adversary is less than the threshold t .

Here n_h, n_p, n_s bound the forger’s random-oracle, preprocess, and signing queries. The architecture is Bellare–Neven generalised forking: the DL challenge ω is embedded as the honest participant’s free-coefficient commitment ϕ_{i0} in a simulated KeyGen, the simulator is run twice on the same tape with oracle answers diverging from index J , and two forgeries yield $\text{dlog}(Y) = (z' - z)/(h'_J - h_J)$. The VRF is what lets the simulator win the programming race: it can compute R before the forger does and program $H_2(R, Y, m)$ with success probability 1.

The step from FROST-Interactive to plain FROST (replace the VRF by $\rho_i = H_1(i, m, B)$) is argued *heuristically* in the paper’s §6.2, not proven — the Equation-1 gap of §2.4. This gap is what the later literature closed, differently and under a different assumption (§2.7).

2.7 The corrected and extended analyses

Three later results matter for how this volume cites FROST’s security.

Crites–Komlo–Maller (ePrint 2021/1375). This work, “How to Prove Schnorr Assuming Schnorr: Security of Multi- and Threshold Signatures” (IACR ePrint 2021/1375, received 2021-10-12, last revised 2022-08-03), introduced the optimisation now called *FROST2*: a single binding factor $d = h_1(vk, lr)$ shared by all signers, replacing the original per-signer $d_i = h_1(vk, lr, i)$ (that original is retroactively *FROST1*) and reducing the signers’ scalar multiplications from linear in the number of signers to constant.

Bellare–Crites–Komlo–Maller–Tessaro–Zhu (CRYPTO 2022). The paper “Better than Advertised Security for Non-interactive Threshold Signatures” — its FROST/BLS part maintained as Bellare–Tessaro–Zhu, ePrint 2022/833, June 2022 — gave the analysis now cited by RFC 9591. The authors define a hierarchy $\text{TS-UF-0} \leq \dots \leq \text{TS-UF-4}$ (and strong variants TS-SUF-2/3/4) grading what counts as a trivial forgery. Their results, proven in the ROM under the *one-more discrete logarithm* (OMDL) assumption, without the AGM, and in the trusted-dealer setting (the DKG is explicitly not modelled):

- FROST1 is TS-SUF-3-secure, and is *not* TS-UF-4-secure;
- FROST2 is TS-SUF-2-secure, and is *not* TS-UF-3-secure — the shared binding factor loses the guarantee that the signer set that committed is the signer set that signed;

- concretely (their Theorem 5.1, verbatim bound), for a TS-SUF-2 adversary A making at most q_s preprocessing and q_h random-oracle queries there is an OMDL adversary B making at most $2q_s + ns$ challenge queries (ns their number of parties) with

$$\text{Adv}_{\text{FROST2}[\mathbb{G}]}^{\text{ts-suf-2}}(A) \leq \sqrt{q \cdot (\text{Adv}_{\mathbb{G}}^{\text{omdl}}(B) + 3q^2/p)}, \quad q = q_s + q_h + 1,$$

and an analogous Theorem 5.4 for FROST1 at TS-SUF-3. In these source bounds, p denotes the group order and q the query count, replacing the earlier section’s use of q for the group order.

The BCKMTZ analysis for the RFC’s FROST1 signing layer is therefore based on OMDL in the ROM, static corruptions below t , and trusted-dealer keys. It does not establish composition with the DKG shipped by the implementation, and its basis differs from the original paper’s DL reduction for the interactive variant plus a heuristic bridge.

Gouvêa–Komlo (ePrint 2024/436). The preprint “Re-Randomized FROST” (received 2024-03-13) is the design ZIP 312 builds on: FROST unchanged, plus algorithms $\text{GenRand}(\alpha \xleftarrow{\$} \mathbb{Z}_p)$, $\text{RandShare}(\tilde{x}_i \leftarrow x_i + \hat{\alpha} \text{ with } \hat{\alpha} = H_r(\alpha, \text{aux}), \text{aux} \text{ an optional external-protocol transcript contribution})$, $\text{RandPKShare}(\overline{PK}_i = PK_i \cdot g^{\hat{\alpha}})$ and $\text{RandPK}(\overline{PK} = PK \cdot g^{\hat{\alpha}})$. Correctness rests on the $\sum_{i \in S} \lambda_i = 1$ identity, giving $z = r + c(x + \hat{\alpha})$. Its Theorem 1 *claims* unforgeability in the ROM under *algebraic* OMDL (AOMDL). Unlike BCKMTZ’s OMDL game, AOMDL restricts discrete-log-oracle queries to declared algebraic combinations, so these are not the “same security assumptions” claimed in the preprint’s abstract. The stated bound (its Equation 3) is

$$\text{Adv}_{\text{TS},A}^{\text{sec}} \leq \sqrt{q \cdot \text{Adv}_D^{\text{aomdl}}(\lambda) + 2q^2/p - \text{negl}(\lambda)}, \quad q = q_h + q_s,$$

via the local forking lemma. The intended extraction uses the public randomizer to strip $c^*H_r(\alpha^*, \text{aux}^*)$ from z^* , but the printed proof has material defects recorded in §3.3; the claimed bound is not independently established here. The coordinator chooses the randomizer and is trusted for privacy and unlinkability only, not for unforgeability. The full treatment of this paper belongs to the re-randomisation chapter (§3).

2.8 RFC 9591 against the paper: what the RFC changed

RFC 9591 (IRTF CFRG, Informational, June 2024; Connolly, Komlo, Goldberg, Wood) is the specification implemented by the `frost-core 3.0.0` workspace this volume cites. Reconciling it against ePrint 2020/852:

- *Two rounds only.* The RFC specifies the two-round protocol; the paper’s single-round-with-preprocessing variant (batch parameter π , commitment server) is declared out of scope.
- *No DKG.* The paper’s Figure-1 KeyGen is dropped; key generation is out of scope, with *trusted-dealer* generation given in Appendix C only. The BCKMTZ analysis the RFC cites matches this framing (it, too, models a trusted dealer).
- *Binding factor inputs widened.* The paper binds $\rho_i = H_1(i, m, B)$. The RFC computes, per participant,

$$\text{binding_factor}_i = H_1(\text{PK_enc} \parallel H_4(\text{msg}) \parallel H_5(\text{encoded commitment list}) \parallel \text{enc}(i)),$$

adding the *group public key* to the hash and pre-hashing the message and commitment list. The per-participant structure means RFC 9591 specifies FROST1: the RFC’s §7.2 rules the single-binding-factor (FROST2) optimisation NOT RECOMMENDED, citing the loss of the started-set-equals-signed-set guarantee (the TS-SUF-3 vs TS-SUF-2 separation of §2.7).

- *Hedged nonces.* The paper samples (d, e) uniformly; the RFC hedges: each nonce is $H_3(r \parallel \text{enc}(sk_i))$ with r a fresh 32-byte random string, so a weak RNG is backstopped by the key share. It warns that fully deterministic derivation enables complete key recovery in the multi-party setting.
- *Five hash roles.* H_1 (binding factor), H_2 (challenge), H_3 (nonce), H_4 (message pre-hash), and H_5 (commitment-list pre-hash) are domain-separated with a per-ciphersuite contextString, except that FROST(Ed25519, SHA-512) deliberately leaves H_2 unprefixed for RFC 8032 compatibility. The challenge input order `group_comm_enc || PK_enc || msg` equals the paper's $H_2(R, Y, m)$.
- *Roles renamed and channels stated.* The signature aggregator becomes the *Coordinator*; the channel is assumed authenticated. Reliable delivery is also assumed for protocol completion; confidentiality is not required. The Coordinator SHOULD verify the aggregate signature and MAY verify individual shares — the paper's per-share check $g^{z_i} = R_i \cdot Y_i^{c\lambda_i}$ survives as `verify_signature_share` — with identifiable abort, as in the paper, and no robustness.
- *Ciphersuites.* The RFC ships FROST(Ed25519, SHA-512), FROST(ristretto255, SHA-512) (RECOMMENDED), FROST(Ed448, SHAKE256), FROST(P-256, SHA-256), and FROST(secp256k1, SHA-256); no Zcash suite appears in the RFC itself. The two Zcash suites, FROST(Jubjub, BLAKE2b-512) and FROST(Pallas, BLAKE2b-512), are instead defined by ZIP 312, whose H_2 personalisation `Zcash_RedPallasH` makes the FROST challenge equal RedDSA's H° — the precise sense in which the RFC protocol plugs into the spend-authorisation verifier of the *Ironwood Guide*, §“Spend authorisation: RedPallas re-randomisation”.
- *Security claim.* The RFC claims EUF-CMA per BCKMTZ, citing both the original paper and BCKMTZ — i.e., the specification's claim rests on the OMDL/ROM analysis, not the paper's DL-based theorem for the interactive variant.

Classification: FROST as in RFC 9591 and its Zcash ciphersuites as in ZIP 312 are *specified*; ZIP 312 itself remains Draft status (zips commit 6961098410, 2026-07-09), with its key-generation procedure explicitly out of scope and PedPoP DKG for Zcash deployment *designed-but-unspecified* (implemented in `frost-core`'s `dkg` module but specified in no ZIP or RFC).

3 Re-Randomized FROST for RedPallas

The FROST of §2, specified in RFC 9591 (§2.8), constructs its aggregate relative to a *fixed* group public key PK . Zcash instead supplies a freshly randomised validating key for every Sapling Spend and Orchard Action, and consensus checks the spend-authorisation signature against that per-Action key — a key to which, as §3.1 makes precise, the unchanged FROST shares do not interpolate. This chapter presents the design that closes the gap: the Re-Randomized FROST of Gouvêa and Komlo (IACR ePrint 2024/436), its Zcash specification in ZIP 312, the ciphersuite FROST(Pallas, BLAKE2b-512), the published implementation in the `frost-rerandomized` and `reddsa` crates, and the custody surfaces the construction serves.

A rigour note before the material. RFC 9591 is an Informational RFC recording CFRG consensus, not an Internet Standard. The Gouvêa–Komlo paper is a preprint — received 2024-03-13, no peer-reviewed venue — and ZIP 312 carries Draft status (zips commit 6961098410, 2026-07-09). Following the series rule, every claim beyond the RFC is binned as *specified*, *designed but unspecified*, or an *open problem*; the closing classification is in §3.7.

3.1 The mismatch: consensus verifies only randomised keys

The protocol specification (§4.15, spend authorization signatures) prescribes, for every Sapling Spend and every Orchard Action, the following signer-side flow. The signer chooses a fresh spend-auth randomizer α and:

1. samples $\alpha \leftarrow \text{SpendAuthSig.GenRandom}()$;
2. derives $\text{rsk} = \text{SpendAuthSig.RandomizePrivate}(\alpha, \text{ask})$;
3. derives $\text{rk} = \text{SpendAuthSig.DerivePublic}(\text{rsk})$;
4. proves the Spend or Action statement with α in the auxiliary (secret) input and rk in the primary input;
5. signs: $\text{spendAuthSig} = \text{SpendAuthSig.Sign}_{\text{rsk}}(\text{SigHash})$.

Consensus therefore verifies the signature under $\text{rk} = \text{ak} + [\alpha] G^{\text{SpendAuth}}$. This differs from ak except in the possible identity-randomizer case $\alpha = 0$.

The randomisation algorithms are RedDSA's, from the protocol specification (§5.4.7):

$$\begin{aligned} \text{RedDSA.RandomizePrivate}(\alpha, \text{sk}) &:= \text{sk} + \alpha \pmod{r_{\mathbb{G}}}, \\ \text{RedDSA.RandomizePublic}(\alpha, \text{vk}) &:= \text{vk} + [\alpha] G, \end{aligned}$$

with identity randomizer 0; RedDSA.GenRandom chooses T uniformly among byte sequences of length $(\ell_H + 128)/8 = 80$ bytes at $\ell_H = 512$ — and returns $H^\circ(T)$, where $H^\circ(B) := \text{LEOS2IP}_{\ell_H}(H(B)) \pmod{r_{\mathbb{G}}}$. Signing draws its nonce as $r = H^\circ(T \parallel \text{vk}^* \parallel M)$ for fresh uniform T , sets $R = [r]G$ and $S = (r + H^\circ(R^* \parallel \text{vk}^* \parallel M) \cdot \text{sk}) \pmod{r_{\mathbb{G}}}$, and outputs $\sigma = R^* \parallel S^*$. Validation recomputes $c = H^\circ(R^* \parallel \text{vk}^* \parallel M)$ and accepts iff $R \neq \perp$, $S < r_{\mathbb{G}}$, R^* is canonical (post-ZIP-216), and $[h_{\mathbb{G}}](-[S]G + R + [c]\text{vk}) = 0_{\mathbb{G}}$. RedPallas specialises the template with $\mathbb{G}$ the Pallas group, $\ell_H = 512$, and $H(x) = \text{BLAKE2b-512}(\text{Zcash_RedPallasH}, x)$; $\text{SpendAuthSig}^{\text{Orchard}}$ uses the generator $G^{\text{SpendAuth}} = \text{GroupHash}^{\text{Pallas}}(\text{z.cash} : \text{Orchard}, "G")$ and “is instantiated as RedPallas with key re-randomization”. The security notion the protocol requires of the scheme is SURK-CMA (Strong Unforgeability with Re-randomized Keys), adapted from Fleischhacker et al. (PKC 2016, §3), whose oracle answers queries of the form (m, α) — message and adversary-chosen randomizer together.

The construction and the cited single-party EUF-CMA result are settled in the *Ironwood Guide*: §“Spend authorisation: RedPallas re-randomisation” builds $\text{rsk} := \text{ask} + \alpha$ and $\text{rk} := \text{ak} + [\alpha] G^{\text{SpendAuth}}$, and §“RedPallas unforgeability” proves EUF-CMA under DL in the ROM *including* adversary-chosen α , by key-prefixed forking with $\text{ask} = \text{rsk}^* - \alpha^*$. Per the layering rule we cite those results and do not re-prove them. That proposition is an EUF-CMA statement, not a separate proof of the protocol specification’s strong SURK-CMA notion.

The threshold problem is now visible. A FROST sharing of ask gives shares sk_i with $\sum_{i \in S} \lambda_i \text{sk}_i = \text{ask}$ for every signing set S ; an honest plain-FROST execution constructs a signature under ak . For a nonzero Action randomizer, however, the verifier’s key is rk . The noninteractive construction below shifts every participant by the same effective randomizer. A separately generated Shamir sharing of a randomizer could also shift the group secret, but would be a different, interactive construction.

3.2 Re-Randomized FROST: the construction

The design is due to Gouvêa and Komlo (both Zcash Foundation), “Re-Randomized FROST”, IACR ePrint 2024/436, received 2024-03-13; we work from the full text and page renders of its construction and equations. The transcription below normalises the source-level slips identified after the construction. As in §2 we quote the paper in its own multiplicative notation (g^x for $[x]G$, $PK \cdot g^{\hat{a}}$ for $PK + [\hat{a}]G$); the additive RedPallas form returns with ZIP 312 in §3.4.

Definition 3.1 (Perfectly rerandomizable threshold scheme; ePrint 2024/436, Definition 5). A threshold signature scheme

$$TS = (\text{Setup}, \text{KeyGen}, (\text{Sign}, \text{Sign}'), \text{Combine}, \text{Verify})$$

is *perfectly re-randomizable* if there exist additional PPT algorithms GenRand , RandShare , RandPKShare , and RandPK , and a randomness distribution X , such that the Rand^* algorithms take a randomizer $\alpha \in X$ and an auxiliary string $\text{aux} \in \{0, 1\}^*$.

Remark 3.2 (What “perfectly” specifies). The printed Definition 5 requires only the existence, inputs, and outputs of four algorithms. It states neither a correctness relation nor equality of any distributions. The adjective “perfectly” therefore adds no formal property in that definition. Functional correctness of the concrete scheme comes from Equation (1); no distributional guarantee follows from Definition 5 alone.

The paper’s Remark 1 explains aux : it models an optional contribution from an external protocol transcript, so that a new session changes the re-randomised key even under a repeated α .

Construction 3.3 (Re-Randomized FROST; ePrint 2024/436, Figure 5). The additions to plain FROST — the paper’s grey-boxed lines — are exactly four algorithms:

$$\begin{aligned} \text{GenRand}() : \alpha &\xleftarrow{\$} \mathbb{Z}_p; \\ \text{RandShare}(x_i, \alpha, \text{aux}) : \hat{\alpha} &\leftarrow H_r(\alpha, \text{aux}); \quad \bar{x}_i \leftarrow x_i + \hat{\alpha}; \\ \text{RandPKShare}(PK_i, \alpha, \text{aux}) : \hat{\alpha} &\leftarrow H_r(\alpha, \text{aux}); \quad \overline{PK}_i \leftarrow PK_i \cdot g^{\hat{\alpha}}; \\ \text{RandPK}(PK, \alpha, \text{aux}) : \hat{\alpha} &\leftarrow H_r(\alpha, \text{aux}); \quad \overline{PK} \leftarrow PK \cdot g^{\hat{\alpha}}. \end{aligned}$$

The FROST core runs unchanged on the shifted inputs. Algorithm $\text{Sign}()$ samples $(r_k, s_k) \xleftarrow{\$} \mathbb{Z}_p^2$ and commits $(R_k, S_k) = (g^{r_k}, g^{s_k})$. Algorithm Sign' , given the randomised share $\bar{x}_k$ and randomised key $\overline{PK}$ in place of x_k and PK , derives each binding factor a_i by H_{non} from the participant identifier, the message, $\overline{PK}$, and the full commitment list $\{(i, R_i, S_i)\}_{i \in S}$, then computes

$$R \leftarrow \prod_{i \in S} R_i \cdot S_i^{a_i}, \quad c \leftarrow H_{\text{sig}}(\overline{PK}, R, m), \quad z_k \leftarrow r_k + s_k a_k + c \lambda_k \bar{x}_k.$$

Algorithm Combine outputs $z \leftarrow \sum_{i \in S} z_i$, $\sigma \leftarrow (R, z)$; $\text{Verify}(\overline{PK}, m, \sigma)$ recomputes the challenge and accepts iff $R \cdot \overline{PK}^c = g^z$.^a

^aThe paper’s own argument order for H_{sig} is inconsistent: Sign' and Combine write $H_{\text{sig}}(PK, R, m)$, while Verify and the reduction’s table write $H_{\text{sig}}(PK, m, R)$ — and both differ from the order RFC 9591 (§4.6) and RedDSA fix, $c = H_2(R^* \parallel PK^* \parallel m)$, which ZIP 312 and the released reddsa crate use. We read the paper’s orderings as a presentational slip with no normative weight; the implemented order is RedDSA’s. The paper also twice refers to “Figure 4” where Figure 5 is meant.

The displayed construction also normalises several evident errors in Figure 5. Its Setup lists only H_{non} and H_{sig} even though every randomisation algorithm calls H_r . In Sign' , its formal argument names the local randomised share $\bar{x}_i$ although the algorithm and its body use signer index k ; we consistently write $\bar{x}_k$. Its loop also prints k as the first input to H_{non} for every a_i ; the iterated index must be i , as above, for the commitment $R = \prod_i R_i S_i^{a_i}$ and response term $s_k a_k$ to use the same per-participant binding factors. The paper’s preliminary Shamir definition has two further typographical slips: its polynomial omits the factor Z from the $a_1 Z$ term and its prose says that t coefficients are sampled while listing $a_1, \dots, a_{t-1}$. Its subsequent evaluation formula is the standard degree- $(t - 1)$ sharing used here.

The paper is explicit that this is the whole design: “We do not make any changes to the plain FROST protocol.” Its signing equations are unchanged; the added algorithms supply the shifted share and public-key inputs.

Correctness is one identity. The paper’s Equation (2):

$$z = \sum_{i \in S} r_i + \sum_{i \in S} s_i a_i + c \sum_{i \in S} \lambda_i \bar{x}_i = \sum_{i \in S} r_i + \sum_{i \in S} s_i a_i + c \left(\sum_{i \in S} \lambda_i x_i + \hat{a} \sum_{i \in S} \lambda_i \right) = r + c(x + \hat{a}) = r + c\hat{x}, \quad (1)$$

“because x is Shamir secret shared” and “ $\sum_{i \in S} \lambda_i = 1$ since it is the interpolation of the constant polynomial $f(X) = 1$ ”. The output is a regular re-randomised signature for $\overline{PK} = PK \cdot g^{\hat{a}}$ with $\hat{a} = H_r(\alpha, \text{aux})$.

The identity deserves emphasis, because it is the entire reason the construction is this simple. Every participant adds the *same* $\hat{a}$ to its share — no dealing, no per-participant shares of the randomizer — and the Lagrange weights, summing to one over any interpolating set, carry the common shift through interpolation so that the group secret moves by $\hat{a}$ rather than by $|S| \cdot \hat{a}$. Had the weights summed to anything else, or varied with S in a way the signers could not predict, the shifted shares would interpolate to a key nobody computed.

3.3 Unforgeability: TRUF, AOMDL, and Theorem 1

The game. The paper’s notion (Definition 6, game TRUF in Figure 4) allows adversarial randomizers and static corruption of fewer than t parties. Its signing oracle records only the queried messages Q , and a forgery wins only when $m^* \notin Q$. It is therefore an *existential* notion: a new signature for a previously signed message is not a winning forgery. The protocol specification instead requires SURK-CMA, whose oracle records message–signature pairs and accepts any fresh pair, including a different signature on a previously queried message. The paper therefore uses a weaker freshness condition, and Theorem 1 does not, as printed, establish the protocol specification’s strong notion. The game pseudocode also assigns $\text{RandShare}(x_k, \alpha, \text{aux})$ to $\bar{x}_i$ but later passes $\bar{x}_k$, and its call to Sign' omits the message argument that the construction requires. Those are interface errors in the printed game; the discussion here uses the evident k index and includes m .

There is a second mismatch in the unrandomised branch. The winning predicate also accepts a forgery under PK , but every signing-oracle answer first applies $H_r(\alpha, \text{aux})$; the adversary has no ordinary signing oracle under PK . Contrary to the paper’s prose, setting its outer randomizer α to zero does not produce the identity shift. The construction’s randomizer domain is $\mathbb{Z}_p$ and $H_r(0, \text{aux})$ need not be zero. Thus the separate verification branch does not embed the ordinary EUF-CMA game either.

The assumption. Intended AOMDL is the *algebraic one-more discrete logarithm* game: the adversary receives $\ell + 1$ challenges, may query a discrete-log oracle only on declared linear combinations of environment-generated elements, and should make at most ℓ queries. The algebraic restriction is absent from §2.7’s OMDL. Since it reduces the adversary’s allowed oracle queries, AOMDL hardness is formally a weaker hardness assumption than OMDL hardness, obtained at the cost of an algebraic-adversary restriction; it is neither literally the same assumption nor a stronger hardness assumption. Figure 1 is not a well-defined rendition of that game: its oracle accepts $\{\beta_i\}_{i=1}^{\ell}$ but its return expression uses β_0 as well, and it never checks that the queried element X equals the declared combination $g^{\alpha} \prod_i X_i^{\beta_i}$. After the natural repair of supplying coefficients for every challenge, a second defect remains: the game initialises its query counter $q \leftarrow 0$ but never increments it. Its final test $q < \ell + 1$ then imposes no query limit, so an adversary can request every challenge logarithm. We interpret AOMDL by its intended one-more definition below, not by that malformed pseudocode.

The theorem claim. The preprint states the following (Theorem 1 with Equation (3); the radical was checked against the rendered page):

Theorem 1 (ePrint 2024/436). *Rerandomized-FROST is unforgeable in the random oracle model, assuming the AOMDL assumption holds in $\mathbb{G}$, concretely*

$$\text{Adv}_{TS,A}^{\text{sec}} \leq \sqrt{q \cdot \text{Adv}_D^{\text{aomdl}}(\lambda) + 2q^2/p - \text{negl}(\lambda)}, \quad q = q_h + q_s, \quad (2)$$

with q_h random-oracle and q_s signing queries.

Remark 3.4 (Placement of the negligible term). The printed bound places $-\text{negl}(\lambda)$ *inside* the square root; we transcribe it as printed. The customary form of such forking-lemma bounds — including the BCKMTZ bound quoted in §2.7 — carries additive slack outside the radical, and a negative term inside can make the radicand negative. Neither the theorem statement nor its proof explains how that expression is to be interpreted in that case.

Intended proof shape, and why it is not verified. The reduction $\mathcal{B}$ receives $2q_s + 1$ AOMDL challenges, sets $PK = X_0$, and simulates verification shares in the exponent. Even the displayed public-share simulation is not well formed: its product shadows the output index i and leaves both j and k unbound. If C is the set of corrupted identifiers and $L_j(i)$ is the Lagrange basis coefficient for node $j \in \{0\} \cup C$ evaluated at the honest identifier i , the required exponent relation is

$$PK_i = PK^{L_0(i)} \prod_{j \in C} (g^{x_j})^{L_j(i)}.$$

The printed expression does not establish that simulated public shares interpolate to PK .

Round-one answers consume challenge pairs. In round two, the response equation

$$z_k = r_k + s_k a_k + c \lambda_k \bar{x}_k$$

requires the reduction to query the logarithm of

$$Z_k = R_k S_k^{a_k} \overline{PK}_k^{c \lambda_k}. \quad (3)$$

The preprint instead prints $R_i S_i^{a_k} \overline{PK}_k^c$: it mixes i and k and omits the Lagrange exponent λ_k . Its claimed “perfect” signing simulation therefore does not follow from the displayed query.

The random-oracle simulation also is not a consistent lazy sampler as printed. Its H_r table stores only the queried pair (α, aux) , not the sampled output, so a repeat query has no associated value to return. Its H_{non} query signature omits m although its lookup and table include m ; when it samples a_j for each other participant, it stores a_k rather than a_j ; and its new-query branch returns the sampled H_{sig} value c rather than the requested binding value a_k . The proof later invokes an unrandomised case $\alpha = \perp$, although neither the construction’s domain $\mathbb{Z}_p$ nor the TRUF oracle defines that sentinel.

The forking step is underspecified too. It says to reprogram $H_{\text{sig}}(PK', R^*, m^*)$ without defining PK' , then asserts a second forgery with the same R^* but potentially different (α, aux) . It does not show that the reprogrammed query is the challenge query for both accepted forgeries when their re-randomised public keys differ. Nor does the proof derive the concrete probability bound in Equation (2); it concludes only with a qualitative “non-negligible probability” invocation of the local forking lemma.

The first fork-extraction identity is algebraically sound if both forgeries share the same commitment R , verify under re-randomised keys, and have $c_0 \neq c_1$. Writing $h_b = H_r(\alpha_b, \text{aux}_b)$, it removes the public shifts and obtains

$$x = \frac{(z_0^* - c_0 h_0) - (z_1^* - c_1 h_1)}{c_0 - c_1}.$$

The subsequent nonce extraction is not sound as printed. If two response shares have the form

$$u_b = r_i + s_i a_b + c_b \lambda_i (x_i + h_b),$$

then for $a_0 \neq a_1$ the corresponding identity is

$$s_i = \frac{u_0 - u_1 - c_0 \lambda_i x_i - c_0 \lambda_i h_0 + c_1 \lambda_i x_i + c_1 \lambda_i h_1}{a_0 - a_1}.$$

The printed numerator gives the two c_1 terms the wrong signs and uses the outer randomizers in place of their H_r images. More seriously, the proof claims that when $c_0 = c_1$, $a_0 = a_1$, and $h_0 \neq h_1$, the reduction can still recover both (r_i, s_i) . After subtracting the known randomizer difference, however, the transcripts give only one linear equation in those two nonce scalars. They are underdetermined.

Together with Figure 1’s missing query-counter increment, these are material gaps, not merely notation. A repair may be possible, but this volume does not supply a complete replacement reduction. It therefore records Theorem 1 as a preprint claim and does not use Equation (2) as a verified assurance statement.

Scope even under a repaired proof. The claimed result is only existential unforgeability in the paper’s weaker game; no cited theorem establishes threshold SURK-CMA. The preprint contains no formal unlinkability definition or theorem. Corruptions are static, and the threshold-scheme game uses centralised key generation; the paper says it “can [be] adapted” to a DKG, but gives no composition proof. Finally, the paper’s §5.1 sketches how a DKG-style contribution round could remove the trusted-for-privacy coordinator (all parties contribute to the randomizer, none learns it) at the cost of extra communication rounds; ZIP 312 does not adopt this. The cost of re-randomisation itself is one fixed-base scalar multiplication per participant and per coordinator; the paper’s Table 1 benchmarks (Pallas suite, Ryzen 9 5900X) put the signing overhead at 83% with 2 signers, 45% at 7, and 8% at 67.

3.4 ZIP 312: the specification

ZIP 312, “FROST for Spend Authorization Multisignatures” (Status Draft, Category Wallet, created 2022-08; owners Gouvêa, Komlo, Connolly; zcash/zips PR #662), specifies Re-Randomized FROST as a delta against RFC 9591. Its target property beyond unforgeability is *unlinkability*, which the ZIP defines as statistical independence of signatures from the signers’ long-term keys: for uniform randomizers and absent metadata leakage, it is “impossible to determine whether two transactions were generated by the same party”.

The signing-flow changes to RFC 9591.

- *Randomizer generation.* A new helper `randomizer_generate(msg, commitment_list)`:
draw `rng_randomizer` $\leftarrow$ `random_bytes(Ns)`, form

```
randomizer_input = rng_randomizer||
    encode_signing_package(commitment_list, msg),
```

and return $H_R(\text{randomizer_input})$. Implementations MAY use another encoding “as long as all values (the message, and the identifier, binding nonce and hiding nonce for each participant) are unambiguously encoded”.

- *Round one* is unchanged.

- *Round two.* The Coordinator generates the randomizer and sends it to each signer “over a confidential and authenticated channel” — plain FROST requires reliability and authentication but not confidentiality — together with the message and the commitment list. Each participant then runs `sign` with $sk_i + \text{randomizer}$ in place of its share and $PK + [\text{randomizer}] G$ in place of the group public key.
- *Aggregation* shifts the group key the same way: $PK += [\text{randomizer}] G$.
- *Share verification* shifts both keys: $PK_i += [\text{randomizer}] G$ and $PK += [\text{randomizer}] G$ in `verify_signature_share`.

The ZIP’s “randomizer” is the paper’s *effective* randomizer $\hat{\alpha}$: per the paper’s §7 implementation notes, `GenRand` and H_r are folded into the single hash call above, and the value distributed to signers is the already-hashed scalar.

The algebra, in RedDSA notation. The ZIP’s Rationale restates the correctness identity of (1) additively, and it is worth displaying because it is the entire consensus-facing content of the design. A valid RedDSA response must satisfy $z = r + c \cdot sk$. Each FROST share contributes

$$z_i = (\text{hiding_nonce}_i + \text{binding_nonce}_i \cdot \rho_i) + \lambda_i \cdot c \cdot sk_i,$$

and under re-randomisation the key terms become $c \cdot (sk + \text{randomizer})$ on the single-signer side and, summed over the coalition,

$$\sum_{i \in S} \lambda_i c (sk_i + \text{randomizer}) = c \left(\sum_{i \in S} \lambda_i sk_i + \text{randomizer} \cdot \sum_{i \in S} \lambda_i \right) = c (sk + \text{randomizer}),$$

since $\sum_i \lambda_i sk_i = sk$ by Shamir and $\sum_i \lambda_i = 1$ (the ZIP proves the latter in its `sum-lambda-proof` footnote, citing `math.SE 1325342`). Every participant adds the same randomizer to its share; the weights-sum-to-one identity is exactly why the group secret shifts by α and not by $|S| \cdot \alpha$.

Who learns the randomizer. The threat model is the ZIP’s sharpest departure from RFC 9591. The Coordinator generates the randomizer and is “trusted with the privacy of the transaction (which includes the unlinkability property)” but *not* with unforgeability: “a rogue Coordinator ... should not be able to create signed transactions without the approval of `MIN_PARTICIPANTS` participants”. All key-share holders receive the randomizer and are likewise trusted with privacy only. The role assignment is natural: the Coordinator fits the transaction builder, who already creates the zero-knowledge proof — and the proof needs α as auxiliary input (step 4 of the flow in §3.1). The confidentiality requirement is not optional hygiene: anyone learning $\hat{\alpha}$ can un-randomise the on-chain key, $ak = rk - [\hat{\alpha}] G$, so the paper advises encrypting the randomizer “so that eavesdroppers can’t break unlinkability”. Non-requirements are stated with equal care: the coordinator-less variant of RFC 9591 §7.5 is unsupported;⁸ share holders can always link the signing operation to the eventual on-chain transaction; key generation is out of scope; network privacy is out of scope.

The message is a hash. The signed message is the `SigHash`, which by itself does not expose enough transaction semantics for a signer to know what is being authorised. Signers **MUST** therefore check the `SigHash` against transaction data sent over the same channel, or recompute it themselves; the mechanism is out of the ZIP’s scope. The same authorisation principle applies to any external signer: a signer who cannot see what it signs authorises whatever the builder chose.

⁸ZIP 312’s reference for the variant mislabels it as RFC 9591 “Section 7.3: Removing the Coordinator Role”; the section is 7.5 (§7.3 is “Nonce Reuse Attacks”), and the reference’s URL anchor targets the correct section.

Unlinkability, and where it is proven. Neither the paper nor ZIP 312 supplies a formal threshold-unlinkability proof: the paper attempts only an existential-unforgeability proof (§3.3), and the ZIP argues unlinkability in its Rationale by matching distributions — “randomizer_generate generates randomizer uniformly at random as required by RedDSA.GenRandom”, and the signing flow is compatible with RedDSA.RandomizePrivate, RedDSA.RandomizePublic, RedDSA.Sign, and RedDSA.Validate — while its Requirements defer the formal notion to the protocol specification’s SURK-CMA definition. More precisely, reducing a 512-bit random-oracle output modulo the group order is statistically close to uniform, not exactly uniform; the same qualification applies to RedDSA.GenRandom. The distribution-matching argument does connect to a proved statement: the *Ironwood Guide*’s §“RedPallas unforgeability” shows honest uniform re-randomisations are distributed as fresh Schnorr keys. What no source supplies is a threshold-specific unlinkability theorem. Nor does the preprint’s weaker existential game establish the threshold SURK-CMA requirement. Bin: the mechanism is *specified* (ZIP 312, Draft); its unlinkability guarantee rests on a distribution argument, and its requested strong-unforgeability guarantee has no matching threshold theorem among the cited sources.

3.5 The ciphersuite FROST(Pallas, BLAKE2b-512)

ZIP 312 defines the ciphersuite in RFC 9591’s template. The group is Pallas with base point $G^{\text{SpendAuth}}$ as in protocol §5.4.7.1, of order p_{Vesta} ; RandomScalar is uniform in $[0, \text{Order} - 1]$; SerializeElement is the point compression $\text{repr}_{\text{Pallas}}$ (P^* in series notation) and DeserializeElement is $\text{abst}_{\text{Pallas}}$, erroring on $\perp$ and on the identity element; SerializeScalar is little-endian over 32 bytes and DeserializeScalar rejects values $\geq$ the order. The hash is BLAKE2b-512 with 16-byte personalisation strings and $N_h = 64$; scalar-valued hashes interpret the 64 output bytes as a little-endian integer reduced mod the order. Table 2 lists the oracles.

Oracle	Role	Personalisation
H_1	binding factor	FROST_RedPallasR
H_2	challenge	Zcash_RedPallasH
H_3	nonce	FROST_RedPallasN
H_4	message pre-hash	FROST_RedPallasM
H_5	commitment-list pre-hash	FROST_RedPallasC
H_R	randomizer	FROST_RedPallasA

Table 2: FROST(Pallas, BLAKE2b-512) hash personalisations (ZIP 312). The challenge hash reuses RedPallas’s own personalisation; the FROST additions use the FROST_ prefix.

The single load-bearing row is H_2 : ZIP 312 states it is “equivalent to $H^\circ(m)$ ” of RedPallas, so the FROST challenge is the on-chain challenge, and signature verification for the suite is RedPallas validation itself. The ZIP’s compatibility rationale completes the argument that a FROST aggregate is a plain RedPallas signature: the (R, S) output split matches RedDSA.Validate’s parsing; FROST and Zcash use the same challenge input order $(R, \text{public key}, \text{message})$; and the fact that r and R are FROST-generated rather than produced by RedDSA.Sign’s nonce derivation is irrelevant to a verifier, which only parses R and checks the validation equation. A parallel suite FROST(Jubjub, BLAKE2b-512) — personalisations FROST_RedJubjub $\{R, N, M, C, A\}$ and Zcash_RedJubjubH — serves Sapling spend authorisation. ZIP 312 defines that suite separately; this volume develops only the Pallas suite.

3.6 Published library implementation: frost-rerandomized and reddsa

Claims in this subsection cite ZcashFoundation/frost at commit 2016e44 (2026-04-23, workspace version 3.0.0) and ZcashFoundation/reddsa at commit 3792daa (2026-05-22, crate version 0.5.2), rechecked against current heads on 2026-09-05. The pins are published releases —

tag `frost-core/v3.0.0` (crates.io 2026-04-23) and `reddsa 0.5.2` — so main-at-commit and the released crates coincide. ZIP 312 names `reddsa` as the reference implementation of both ciphersuites. A current-head check on 2026-09-05 found `frost 0966bd1` and `reddsa 63c6ac0`; their post-release changes update the Rust edition, minimum compiler, dependencies, benchmarks, and CI, not the re-randomized protocol derived below.

The generic layer. Crate `frost-rerandomized` implements Construction 3.3 over any `frost-core` ciphersuite. Trait `impl Randomize<KeyPackage>` computes the randomised signing share as `signing_share + randomizer` and the randomised verifying share by adding the `randomizer_element`; struct `RandomizedParams` carries the randomizer (“also called alpha” in its documentation), `randomizer_element = [randomizer] G`, and

$$\text{randomized_verifying_key} = PK + \text{randomizer_element}.$$

The current flow differs instructively from a literal reading of the ZIP: the Coordinator sends a randomizer *seed*, not the randomizer, and each participant recomputes the randomizer from that seed and the full round-one commitment map in its signing package. The derivation, its divergence from the draft text — the message is no longer a hash input — and the reconciliation in flight in zips PR #895 are treated once, in §4.2.

The RedPallas suite. Module `src/frost/redpallas.rs` in `reddsa` implements the `frost-core` `Ciphersuite` trait for `PallasBlake2b512` (contextual ID “FROST(Pallas, BLAKE2b-512)”). Its `generator()` is `orchard::SpendAuth::basepoint()`, whose constant bytes are reproducible as `pallas::Point::hash_to_curve("z.cash:Orchard")(b"G")` — the protocol’s $G^{\text{SpendAuth}}$. Oracles $H_1/H_3/H_4/H_5$ carry the FROST_RedPallas{R,N,M,C} personalisations of Table 2; H_2 goes through the crate’s HStar default, `Zcash_RedPallasH`; the `RandomizedCiphersuite::hash_randomizer` impl uses `FROST_RedPallasA` (the ZIP’s H_R); scalar reduction is the 64-byte-wide `from_bytes_wide`, whose Pallas implementation delegates to `pasta`’s `from_uniform_bytes`. Beyond ZIP 312, the crate defines two further personalisations — `HDKG = FROST_RedPallasD` and `HID = FROST_RedPallasI`, the DKG and identifier hashes — for key generation, which the ZIP does not specify at all.

Even- γ normalisation. One deployment detail appears in neither the paper nor the ZIP 312 body: the `reddsa` hooks `post_generate/post_dkg` normalise fresh key material to the even- γ representative the protocol specification requires of `ak`; the full mechanism is §4.1. The *randomised* key `rk` need not satisfy the even- γ condition — only `ak` does.

The RedPallas re-randomised API. Module `src/frost/redpallas/rerandomized.rs` re-exports the re-randomised flow for the suite — `sign_with_randomizer_seed()`, `aggregate()`, and `aggregate_custom()` — together with the type aliases `Randomizer` and `RandomizedParams` over `PallasBlake2b512`. The module README’s worked example runs the full trusted-dealer plus re-randomised signing flow and ends by verifying the aggregate under the `randomized_verifying_key()` of its `RandomizedParams` — the executable form of this chapter’s claim that a FROST aggregate is a plain RedPallas signature under `rk`.

3.7 Custody surfaces and classification

(a) Spend-authorisation custody. The primary surface: a t -of- n sharing of `ask` signing Orchard Actions (and, via the Jubjub twin suite, Sapling Spends). FROST replaces step 5 of the signer flow in §3.1; steps 1–4 move to the Coordinator-as-builder, who draws the randomizer, proves the Action statement with α as auxiliary input, and publishes `rk` — the circuit and consensus rules are untouched, per the *Ironwood Guide*, §“Spend authorisation: RedPallas re-randomisation”. Bin:

specified (ZIP 312, Draft), with key generation explicitly out of the ZIP’s scope; DKG-based key generation for Zcash remains *designed but unspecified* (implemented in `frost-core`, specified nowhere), as §2.8 already concluded for the base protocol. Related anticipations elsewhere in the ZIP corpus: ZIP 316 requires FROST unified viewing key material to follow ZIPs 312 and 2005, and ZIP 271 anticipates lockbox disbursement to “a FROST address”, noting (as of May 2025) that ZIP 312 does not specify key generation.

(b) ZIP 2005 quantum spending key custody. ZIP 2005’s “Usage with FROST” section constrains threshold deployments of its quantum-recovery hierarchy: the key ak is derived jointly, by FROST DKG, from participants’ shares of ask , and the participants privately agree on a value sk from which nk , qsk , qk , and $rivk_{\text{ext}}$ derive. The full ceremony, its derivations, and its asymmetric threat model — every participant holds qsk , so the threshold property does not survive an adversary that can also solve Pallas discrete logarithms — are developed once, in §4.7. Bin: the FROST-facing requirements are *specified* (ZIP 2005’s requirements list calls for full compatibility with ZIP 312 FROST, hardware wallets, and their combination); the combined DKG-plus-shared- sk key generation is *designed but unspecified* — no source specifies the protocol by which participants “privately agree” on sk .

(c) Crosslink finalizer keys. The *Crosslink Guide*, §“Finalizer keys”, bins threshold custody of finalizer keys as an *open problem*, and we adopt that bin. One precision for this volume: the applicable protocol there is *plain* FROST — the finalizer design has no key re-randomisation, so the natural suite is RFC 9591’s FROST(Ed25519, SHA-512), not either ZIP 312 suite. The prototype state and this observation are developed once, in §4.8.

(d) PCZT signing. The multiparty transaction flow that would host the Coordinator role is the PCZT of the *Wallet Guide*, §“Partially created transactions (PCZT)”: ZIP 374 (Partially Created Zcash Transaction Format, Draft, owner Jack Grigg) names threshold signing (“for example FROST”, per RFC 9591) in its Motivation among the multiparty use cases its Signer role must accommodate.

Classification. In the series’ three bins: base FROST and its five ciphersuites are *specified* in RFC 9591, an Informational RFC recording CFRG consensus (§2.8). Re-Randomized FROST, its threat model, and both Zcash ciphersuites are *specified* by ZIP 312 at Draft status. The preprint’s claimed existential-unforgeability proof has the gaps in §3.3, no cited theorem establishes threshold SURK-CMA, and unlinkability rests on a distribution argument rather than a theorem. Key generation — FROST DKG for Zcash, and ZIP 2005’s shared- sk agreement — is *designed but unspecified*; threshold custody of Crosslink finalizer keys is an *open problem*. Implementation-behaviour claims (seed-based randomizer derivation, even- Y normalisation, the extra DKG personalisations) are implementation facts of the pinned commits, cited as such.

4 Deployment and open questions

The preceding sections dealt in designs and proofs; this one deals in what ships. We pin the exact code paths available to a prospective Zcash threshold-custody deployment, reconcile them against the draft ZIP text where the two have diverged, then walk the three custody surfaces this volume targets — spend authorisation through PCZT, the ZIP-2005 quantum spending key, and Crosslink finalizer keys — and close by sorting every artefact into the series’ three bins (*specified*, *designed-but-unspecified*, *open problem*), each with the concrete trigger on which its classification must be revisited.

4.1 The shipped library stack

Two Zcash Foundation repositories carry the implementation. The frost workspace (commit 2016e44ba4, 2026-04-23, tagged frost-core/v3.0.0; workspace version 3.0.0, MSRV 1.81) provides the generic frost-core, the five RFC 9591 ciphersuite crates frost-ristretto255, -ed25519, -ed448, -p256, and -secp256k1, the separate frost-secp256k1-tr Taproot suite, and the generic re-randomisation layer frost-rerandomized. Beyond RFC 9591’s two-round protocol the workspace implements trusted-dealer key generation (the RFC’s appendix), a variant of the original paper’s Pedersen-with-knowledge-proof DKG that omits its context string (ePrint 2020/852, §2.2 — not in the RFC), Repairable-Threshold-Scheme share recovery (ePrint 2017/1155), share refresh, and Re-Randomized FROST (ePrint 2024/436, §3). The README declares the crates “considered stable and feature complete”, with SemVer guarantees.

The frost-rerandomized crate is no_std; its Randomize implementations add the randomizer group element to every verifying share of a PublicKeyPackage and the randomizer scalar to the signing share of a KeyPackage, whose randomize documentation warns: “You MUST NOT reuse the randomized key package for more than one signing.” Its README notes that “the main ciphersuite crates do not re-expose the rerandomization functions... The exceptions are the Zcash ciphersuites in reddsa which do expose the randomized functionality” — the Zcash ciphersuite modules are the shipped top-level suites that re-export those generic functions.

The reddsa crate (commit 3792daa95e, 2026-05-22, version 0.5.2) supplies those ciphersuites: its feature frost = ["frost-rerandomized", "alloc"] pulls frost-rerandomized 3.0.0, and its modules reddsa::frost::redpallas and reddsa::frost::redjubjub implement the two ZIP-312 ciphersuites (PallasBlake2b512, JubjubBlake2b512), each with a keys module (trusted-dealer, DKG, repairable) and a rerandomized submodule re-exporting sign_with_randomizer_seed, aggregate, aggregate_custom, Randomizer, and RandomizedParams. Function hash_randomizer instantiates ZIP 312’s H_R as H° (the code’s HStar) under the personalisation FROST_RedPallasA (respectively FROST_RedJubjubA), matching §3.5. The crate’s README calls this feature experimental because ZIP 312 is still Draft and states that neither its SemVer nor its MSRV guarantee applies to the feature.

The RedPallas module additionally enforces the even- y requirement that ZIP 2005’s amended protocol-spec §4.2.3 places on ak (§4.7): trait keys::EvenY negates the group verifying key, every verifying share, the signing shares, and the VSS commitment coefficients whenever the serialised key has $\tilde{y} = 1$ (the code’s evenness test is serialized[31] & 0x80 == 0), and the post_generate and post_dkg ciphersuite hooks call into_even_y on fresh key material, so trusted-dealer and DKG outputs alike land on the even- y representative. Negation commutes with Shamir sharing because it is linear. The RedJubjub module carries no such machinery; the constraint is Pallas/Orchard-specific.

4.2 The randomizer flow: draft ZIP against shipped crate

ZIP 312 (“FROST for Spend Authorization Multisignatures”; owners Gouvea, Komlo, Connolly; Status Draft; created 2022-08) specifies, at zips commit 6961098410, the randomizer generation

$$\begin{aligned} \text{signing_package_enc} &= \text{encode_signing_package}(\text{commitment_list}, \text{msg}), \\ \text{randomizer} &= H_R(\text{random_bytes}(N_s) \parallel \text{signing_package_enc}), \end{aligned}$$

with H_R for FROST(Pallas, BLAKE2b-512) being BLAKE2b-512 under the personalisation FROST_RedPallasA, output reduced to a Pallas scalar (an element of $\mathbb{F}_{p_{\text{Vesta}}}$); H_2 is Zcash_RedPallasH, i.e. exactly RedPallas’s H° (§3.5). For Jubjub the personalisations are FROST_RedJubjub{R,N,M,C,A} and Zcash_RedJubjubH, with base point G^{Sapling} . Round 2 then proceeds as plain FROST with three substitutions,

$$\begin{aligned} sk_i &\leftarrow sk_i + \text{randomizer}, & PK_i &\leftarrow PK_i + \text{ScalarBaseMult}(\text{randomizer}), \\ \text{group_public_key} &\leftarrow \text{group_public_key} + \text{ScalarBaseMult}(\text{randomizer}), \end{aligned}$$

the second applying in share verification. The Coordinator sends the randomizer over a “confidential and authenticated channel” — adding confidentiality to plain FROST’s reliable, authenticated-channel assumption, because the randomizer links the ceremony to the on-chain rk .

The released crate differs from this draft text. Crate `frost-rerandomized 3.0.0` deprecates `Randomizer::new(rng, signing_package)` — the earlier, message-inclusive API. Even that API is not byte-for-byte the draft flow: it samples and serialises a field scalar before hashing, whereas the ZIP starts with N_s unconstrained random bytes. The recommended API is instead `Randomizer::new_from_commitments`: the Coordinator draws an N_s -byte `randomizer_seed`, sends the *seed*, and each participant regenerates

```
randomizer = hash_randomizer(seed||encode_group_commitments(commitments))
```

locally via `regenerate_from_seed_and_commitments`. The message is no longer an input. The crate documentation says that hashing the complete commitment map hedges the Coordinator’s RNG because the map includes signer commitments. This is its implementation rationale, not a property proved by the cited RFC, ZIP, or preprint; in particular, the Coordinator receives the commitments before generating the seed and remains trusted for unlinkability. In the paper’s terms (ePrint 2024/436, Definition 5 and Remark 1), the commitment encoding plays the role of the optional contributory string *aux*, which models “the transcript of some external protocol”; the paper’s own §7 implementation notes describe the older design, with the message still included, in the same terms.

The gap between draft and crate is being closed in zips PR #895 (“[ZIP 312] Specify key generation, change randomizer handling”; opened August 2024, 21 commits, still open and mergeable as of 2026-09-05, with its latest review comments dated 2026-08-28). It adds a key-generation specification, revises the randomizer handling (the randomizer “couldn’t possibly be generated using the message as an input”), integrates the ZIP-2005 qsk material, renames the scheme consistently to “Re-Randomized FROST”, and adds Daira-Emma Hopwood as a ZIP owner. Its commit history also retires the draft’s serialisation reference — the ZF FROST book’s moving-target serialisation-format page — in favour of RFC 9591’s serialisation function. Until it merges, this volume presents the crate flow — message omission included — as the observed fact of the pinned commit and the ZIP text as a divergent Draft specification. No conformance conclusion follows while that text remains in flight.

The trust geometry is worth stating exactly. The paper’s §5.1 holds that the central randomizer-choosing party is “trusted with preserving privacy of the scheme” but “not trusted with security”, and that removing it would require a DKG-style contributory sub-protocol at extra rounds and complexity. ZIP 312’s threat model matches: the Coordinator is trusted with transaction privacy and unlinkability, while the requirements say it must not be able to forge without `MIN_PARTICIPANTS` approvals; every key-share holder is trusted with transaction privacy. Non-requirements are equally explicit: ZIP 312 does not remove the Coordinator role, does not (at this commit) provide key generation, and places network privacy out of scope.

4.3 PCZT as the integration surface

ZIP 374’s Motivation names “multiparty transactions — threshold signing (for example FROST [RFC 9591])” among the problems the PCZT format solves; the *Wallet Guide*, §“Partially created transactions (PCZT)”, documents the format and its role graph. The insertion point for FROST is the Signer role (*Wallet Guide*, §“Signing”): to sign an Orchard spend the action’s re-randomisation scalar α must be present, the local-key path computes $rsk := ask + \alpha$ — the re-randomisation of the *Ironwood Guide*, §“Spend authorisation: RedPallas re-randomisation” — and checks `VerificationKey(rsk) = rk`, and the *external* path, `apply_orchard_signature`, accepts a signature if and only if it verifies under the action’s rk over the shielded sighash (`InvalidExternalSignature` otherwise). A FROST-aggregated RedPallas signature is exactly

such an external signature; for it to verify, the effective FROST randomizer must equal the PCZT action's α , so that the re-randomised group key RandPK produces equals that action's rk.

PCZT v2 has since shipped in pczt 0.8.0. Current pczt 0.9.3 carries separate Orchard and Ironwood bundles and tags an externally produced SpendAuthSignature with its value pool and action index; `apply_orchard_spend_auth_signature` dispatches to the corresponding bundle and verifies under that action's rk. The invariant in the preceding paragraph — one shared shielded sighash and the action's own α — is unchanged for v6 Ironwood spends (librustzcash 91f448b5, pczt/src/roles/signer/mod.rs).

The end-to-end flow has been demonstrated (ZF FROST book, `zcash/devtool-demo.md`):

1. `zcash-devtool pczt create` produces the PCZT;
2. the `zcash-sign` tool (from ZcashFoundation/frost-zcash-demo, now `frost-tools`) prints a SIGHASH and a Randomizer from the transaction plan;
3. `frost-client` runs the FROST protocol over `frostd` with that SIGHASH as the message and that randomizer;
4. `zcash-sign sign` writes the aggregated signature into `frost_pczt.signed`;
5. `zcash-devtool pczt combine` merges the signed and proven PCZTs;
6. `zcash-devtool pczt send` broadcasts.

The current `frost-tools` head (06c0dbdb, 2026-08-12) does not exercise the version-3 seed flow described in §4.2. Its lockfile selects `frost-core` 2.2.0, `frost-rerandomized` 2.1.0, and an older `reddsa` 0.5.1 revision. For PCZT, the Coordinator accepts each Action's already-generated scalar randomizer and sends that value to participants through the legacy `raw-randomizer` API. This is a dated implementation boundary: the demo validates the PCZT external-signature path, but it is not evidence for integration of the pinned version-3 library stack or its seed flow.

That distinction creates an ordering requirement for any version-3 integration. The recommended seed API derives the effective randomizer from round-one commitments, so it cannot in general be made equal to an α already fixed inside a PCZT Action. The transaction builder must instead use the derived randomizer as that Action's α before constructing its proof and rk, or use a separately justified compatibility path. Neither the legacy demo nor a specification cited here demonstrates that reordered end-to-end flow.

Here `zcash-devtool` builds on `zcash_keys` with the features `"orchard"`, `"unstable"`, and `"unstable-frost"`. The `unstable-frost` feature (`= ["orchard"]`) exposes the constructor `UnifiedFullViewingKey::from_orchard_fvk` — a UFVK from an Orchard FVK alone, since a FROST wallet has no unified spending key — and the `zcash_keys` changelog (0.3.0, 2024-08-19) frames the flag as existing “to temporarily expose API features that are needed specifically when creating FROST threshold signatures”, to be removed “once key derivation for FROST has been fully specified and implemented”. Repository `librustzcash` itself, that is, classifies FROST key derivation as not fully specified.

One check remains outside every specification: ZIP 312 requires that signers “MUST check that the given SIGHASH matches the data sent from the Coordinator, or compute the SIGHASH themselves”, and explicitly leaves the mechanism out of scope. In the demonstrated flow the SIGHASH travels from `zcash-sign` to the participants through `frost-client`; by what mechanism a share-holding participant gains independent confidence in it is unspecified.

4.4 Wallet-integration constraints

The ZF FROST book's technical-details chapter records the constraints a FROST-capable Zcash wallet inherits. The documented Zcash integration cannot authorise transparent inputs, which require ECDSA; the book accordingly suggests supporting Orchard only (“Orchard is simpler to

handle”). That is historical integration advice rather than a current-pool limit: the page predates the Ironwood pool. With a DKG-generated `ak`, the remaining Orchard key components derive from `ak` plus either an `sk` used only for `nk` and `rivk`, or independently generated `nk` and `rivk`. Under this key-generation flow, a seed phrase alone does not restore the wallet: recovery also needs the key share, all verifying shares, the participant identifiers, and the FVK components. The book stores these in participant configuration files; the library also provides the Repairable Threshold Scheme (§4.1).

The Foundation’s status report (“The State of FROST for Zcash”, 2025-05-20) named wallet integration the missing piece (“The missing piece is for wallets to integrate FROST”): at publication time it reported no mainstream wallet integration, while “technical-inclined users can use FROST with Zcash today using our `frost-client` command line tool”. The same post lists the “lack of a standardized key generation specification for FROST” as the interoperability limiter, flags metadata protection (e.g. Tor) as needing work, and offers ZF as a candidate operator of a production `frostd`.

4.5 Transport: `frostd` and `frost-client`

The reference transport is `frostd`, a JSON-over-HTTPS session server: clients authenticate with key pairs, the Coordinator creates sessions naming the participants’ public keys, and all FROST messages are end-to-end encrypted. Thus the server need not be trusted with message contents or signing authority, although it remains an availability dependency. The CLI `frost-client` drives it; both live in ZcashFoundation/`frost-tools` (renamed from `frost-zcash-demo`). Least Authority, as Zcash Ecosystem Security Lead, audited `frost-client` and `frostd` (announcement 2025-05-01) with no high-severity findings, all findings addressed on main. No ZIP specifies this transport; ZIP 312 places network privacy out of scope (§4.2). The tools repository labels the command-line programs as demos and says they should not be used in production. The `frost-client` README also warns that it stores secrets unencrypted in its configuration file and recommends protecting that file or building a separate tool from the client as a base.

4.6 Audit coverage, precisely

Two cryptographic-library audits are relevant, and neither covers the code path Zcash custody would run. NCC Group audited the v0.6.0 release (commit 5fa17ed) of `frost-core`, `frost-ed25519`, `frost-ed448`, `frost-p256`, `frost-secp256k1`, and `frost-ristretto255` — including trusted-dealer and DKG key generation and FROST signing — with all findings addressed and reviewed by NCC (public report 2023-10-23). The repository README is explicit: “This does not include `frost-secp256k1-tr` and rerandomized FROST.” Earlier, the 2021 “Audit of FROST implementation” (JP Aumasson and Adrien Hamelink, Taurus; Omer Shlomovits, ZenGo; dated 2021-03-23) covered the pre-rewrite implementation in the RedJubjub repository’s `src/frost.rs` (404 source lines by the report’s count, at commit 2ebc08f, 2021-02-25) — two-round FROST with a trusted dealer only, RedJubjub only — concluding “We did not find any major security issue.” The report also records that the implementation omitted the paper’s secure deletion of nonces and commitments after signing.

The composition, therefore: `frost-rerandomized` plus the `reddsa` RedPallas/RedJubjub ciphersuites — the exact stack of §4.1 — is not covered by any completed third-party audit identified in the repository and audit announcements rechecked 2026-09-05. Of those audits, only the 2021 review touched a Zcash FROST ciphersuite, and it predates the `frost-core` rewrite. The separate Least Authority audit covers the transport tools, not this cryptographic composition.

4.7 Custody of the ZIP-2005 quantum spending key

ZIP 2005 (“Ironwood Quantum Recoverability”; owners Daira-Emma Hopwood and Jack Grigg; Status Proposed, Category Consensus; activation at the NU6.3 activation height) is the second

custody surface. Its “Usage with FROST” section derives ak jointly, from the participants’ shares of ask , via FROST DKG. It then says that participants “MUST privately agree on a value sk ” and use it with $use_qsk = \text{true}$ to derive the remaining material. The protocol must give every participant the same extracted validating-key value ak , nk , and $rivk_ext$; the specified derivations then imply equal qsk and qk under the ZIP’s stated collision-resistance assumptions. In this series, $ak = [ask]G^{\text{SpendAuth}}$ denotes the Pallas point. ZIP 2005 writes $ak^{\mathbb{P}}$ for that point and $ak := \text{Extract}_{\mathbb{P}}(ak^{\mathbb{P}})$ for its extracted field element. With that point/field-element distinction explicit, its derivations are

$$\begin{aligned} H^{qsk}(sk) &= \text{truncate}_{32}(\text{PRFexpand}_{sk}([0x0C])), \\ H^{qk}(qsk) &= \text{BLAKE3.derive_key}(\text{"Zcash ZIP 2005 qk-derivation v1"}, qsk, 32), \\ H_{qk}^{rivk_ext}(ak, nk) &= \text{ToScalar}^{\text{Orchard}}(\text{PRFexpand}_{qk}([0x0D] \parallel \text{I2LEOSP}_{256}(ak) \parallel \text{I2LEOSP}_{256}(nk))). \end{aligned}$$

The displayed key derivations use the PRFexpand domain-separator bytes $0x0C$ and $0x0D$; ZIP 2005 also reserves $0x0A$ and $0x0B$ for note-related uses. In the $use_qsk = \text{true}$ branch of the amended protocol-spec §4.2.3, the wallet obtains the point $ak^{\mathbb{P}}$ “by any suitably secure method that ensures the last bit of $\text{repr}_{\mathbb{P}}(ak^{\mathbb{P}})$ is 0” — the requirement the reddsa EvenY machinery of §4.1 enforces — and sets the ZIP’s field element $ak = \text{Extract}_{\mathbb{P}}(ak^{\mathbb{P}})$, $qsk = H^{qsk}(sk)$, $qk = H^{qk}(qsk)$, and $rivk = H_{qk}^{rivk_ext}(ak, nk)$. The ZIP’s list of admissible generation methods for $ak^{\mathbb{P}}$ explicitly includes “FROST distributed key generation [zip-0312-key-generation] in which the corresponding spend-authorization material is t -of- n shared among the participants”.

The threat model shifts accordingly. The host wallet in a multi-signer FROST setup is the ZIP-312 Coordinator; the ZIP gives it nk , the extracted validating-key value ak , and qk . With FROST and hardware wallets, spend authorisation breaks only by (1) possession of qsk *plus* finding discrete logarithms on Pallas — and “the adversary is an authorized FROST signer — they hold a copy of qsk by construction” is the ZIP’s variant 1.a — or (2) breaking RedDSA, FROST, the DKG, or the proving system. Each participant MUST treat qsk as a secret held at least as securely and reliably as ask : it is not needed until the Recovery Protocol, but losing it can lose funds once the current Orchard protocol is disabled. A quantum adversary holding qsk — which every participant holds — and able to solve discrete logarithms on Pallas may steal funds, so the ZIP RECOMMENDS that once a fully post-quantum threshold multisignature protocol exists, all FROST-held funds migrate to its pool. FROST’s key advantage — signing with shares without reconstructing ask during signing — would be retained in the outlined Recovery Protocol, which would still require a RedDSA signature.

One tension is unresolved. FROST DKG exists precisely to avoid a commonly-known secret, yet the ceremony above requires the participants to “privately agree on a value sk ” from which nk and qsk derive — and neither ZIP 2005 nor the crates give a protocol for that agreement. ZIP 2005’s Deployment section points at PR #895 (“There is no significant existing deployment of FROST, so we can write this into ZIP 312 from the start”), so the resolution is expected there: as of 2026-09-05 the PR’s commit history drafts a contributory sk -generation section (2026-03-05), still unmerged, so the shared- sk agreement stays *designed but unspecified* with #895’s merge as its trigger.

4.8 Crosslink finalizer keys

The third surface is prospective only. The *Crosslink Guide*, §“Finalizer keys”, records that the prototype uses single ed25519 keys per finalizer (ed25519-zebra), with 76-byte votes signed by appending a 64-byte ed25519 signature over those 76 bytes, and that no FROST or threshold-signing design for finalizer keys is sourced anywhere in the Crosslink corpus (negative search over *tfl-book*, *crosslink-protocol-guide*, *crosslink-spec*, and *zebra-crosslink*, dated 2026-07-15 in that guide); key rotation, revocation, threshold custody, and initial-roster formation are classified open problems in every design source.

We record one observation of our own, clearly marked as such: finalizer votes are plain ed25519, with no key re-randomisation, so the fitting ciphersuite would be RFC 9591's FROST(Ed25519, SHA-512) via the `frost-ed25519` crate, whose v0.6.0 predecessor was covered by the NCC audit, with no re-randomisation layer — finalizer keys are deliberately long-lived roster identities, not unlinkable one-time keys, so the entire apparatus of §3 is unnecessary there. No published design says this; it is this volume's inference from the vote format and the audit table.

4.9 Classification and revisit triggers

Per the series' three-way scheme:

- *Specified.* FROST as in RFC 9591 (§2.8); Re-Randomized FROST for spend-authorisation custody, per ZIP 312 at Draft status — PR #895's in-flight randomizer and key-generation changes are its named revisit trigger and do not move the bin (§4.2); the `frost-core/frost-rerandomized/reddsa` API surface at the pinned versions (§4.1); the PCZT Signer external-signature path (§4.3).
- *Designed-but-unspecified.* FROST key generation for Zcash (out of scope at the pinned ZIP-312 commit; the crates implement the PedPoP DKG of ePrint 2020/852, but no ZIP specifies it); the ZIP-2005 `use_qsk` FROST key-generation constraints (ZIP Proposed, referencing the unmerged #895; the `sk-agreement` step has no protocol, §4.7); the transport (`frostd` exists and is audited, but no ZIP covers it, §4.5).
- *Open problem.* Crosslink finalizer threshold custody (§4.8); post-quantum threshold signing for the `qsk`-era migration (§4.7); hardware-wallet FROST share custody — ZIP 2005's threat model assumes hardware-wallet FROST signers, but none was identified in the device integrations checked for this audit.

Two former revisit triggers have fired without changing these bins. NU6.3 is active, but `zcash_keys` still gates `from_orchard_fvk` behind `unstable-frost` pending a specified FROST derivation. PCZT v2 is released, and the current signer preserves the α and external-signature surface for Ironwood as described in §4.3. The remaining classifications expire on the following identifiable events, in rough order of expected impact:

- *PR #895 merges.* The randomizer flow, the key-generation specification, the `sk-agreement` protocol, and the ZIP-312/ZIP-2005 integration all re-derive from the merged text; the serialisation reference it pins replaces the ZF-FROST-book moving target (§4.2).
- *An audit of the re-randomised stack is announced.* The “no completed third-party audit” claim of §4.6 must be rechecked immediately before any printing in any case.
- *A finalizer key-management design is published.* The Crosslink bin moves, and the `frost-ed25519` observation is either confirmed or discarded (§4.8).
- *A post-quantum threshold multisignature protocol exists.* ZIP 2005's RECOMMENDED migration of FROST-held funds becomes actionable (§4.7).